

Enterprise AI Architecture

The Pombaline Cage

— By Kennedy Chengeta, PhD —

• ARTIFICIAL INTELLIGENCE DESIGN PATTERNS •



DESIGN INTELLIGENT SYSTEMS.
BUILD ADAPTIVE ENTERPRISES.
DELIVER REAL VALUE.

Enterprise AI Architecture

The Pombaline Cage

Artificial Intelligence Design Patterns

Volume 1: Foundations and Patterns

Kennedy Chengeta, PhD

First Edition

Century Publications

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First Edition, 2026

Kennedy Chengeta, PhD

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*To the engineers, architects, and operators
who carry production AI systems through 3 a.m. pages—
and to the leaders who let them sleep more.*

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Preface

For most of computing history, an *enterprise system* meant a database, a few application servers, a message bus, and a user interface stitched together by people who knew what each layer was supposed to do. The architecture had a name—three-tier, service-oriented, microservices—and the seams between components were stable enough to draw on a whiteboard and hand to a new engineer on their first day.

Enterprise AI is not yet that. A modern AI system absorbs structured data from relational warehouses, unstructured data from document stores, streaming data from IoT devices, and conversational data from chat interfaces. It runs classical statistical models alongside fine-tuned transformers, retrieval pipelines, agent loops, and human-in-the-loop workflows. It must satisfy latency budgets measured in tens of milliseconds, governance policies measured in months of regulatory review, and cost budgets measured in millions of dollars of GPU inference. The seams move every quarter as new models, vendors, and protocols arrive.

This book is the reference we wish we had when we started designing such systems. It treats enterprise AI as a discipline with its own layers, its own patterns, and its own failure modes—not as a collection of machine-learning notebooks bolted onto an existing platform.

What this book is. A technology-agnostic reference architecture for AI systems at scale, presented as six interlocking layers (data sources, data pipeline, AI/ML brain, model serving, governance, end user applications) and six recurring patterns (agentic mesh, RAG, LLM router, hub-and-spoke, event-driven agents, hexagonal architecture). Each chapter develops a layer or pattern from first principles, then shows how it composes with the others.

What this book is not. A tutorial on any specific framework. We name vendors and tools where useful, but the architecture should outlive any of them. If your stack is LangChain today and something else next year, the layer it occupies and the contracts it must honour are the durable knowledge.

Who it is for. AI architects, platform engineers, heads of AI/ML, CTOs, and senior engineers being asked to take a working prototype and make it survive in production. We assume comfort with software architecture and basic familiarity with machine learning; we do not assume prior experience with LLMs, agents, or retrieval systems.

Structure of the book.

- **Part I (Chapters 1–4)** establishes the enterprise AI problem, design principles, and the reference architecture.
- **Part II (Chapters 5–10)** develops each of the six layers in depth.
- **Part III (Chapters 11–17)** covers the six recurring architecture patterns.
- **Part IV (Chapters 18–22)** addresses the cross-cutting concerns: privacy, reliability, performance, interoperability, and responsible AI.
- **Part V (Chapters 23–28)** covers the operating model: teams, buy-versus-build, cost, observability, security, and governance.
- **Part VI (Chapters 29–32)** closes with a maturity model, case studies, anti-patterns, and futures.
- **Appendices A–C** provide a glossary, reference stacks, and architecture review checklists.

How to read it. Architects new to AI should read Parts I and II linearly, then dip into Part III as patterns become relevant. Practitioners with AI experience may skim Parts I–II for shared vocabulary and concentrate on Parts III–V. Leaders responsible for AI strategy will find the maturity model in Chapter [105](#) and the governance discussion in Chapter [103](#) the most direct routes to actionable decisions.

The Editors

May 27, 2026

Foreword

The architecture of an enterprise AI system is the set of decisions that the team will not get to make twice. The model can be retrained, the prompts can be rewritten, the vendors can be swapped, the tools can be reconfigured—but the structural choices that determine how all of those things fit together, change together, and survive together are made once, deeply, and then lived with for years.

This book is for the people who make those decisions. It is written in the conviction that the structural questions of enterprise AI are not exotic; they are the familiar questions of enterprise architecture, posed in a new domain with unfamiliar failure modes. The data still has to come from somewhere reliable. The interfaces still have to be stable enough to evolve behind. The governance still has to be a property of the platform, not a deliverable of each project. The vendors will still change, and the architecture that lets the change be operational rather than structural is still the one that lasts.

What is new in AI—the stochastic outputs, the prompt-injection surface, the cost-by-the-token economics, the long-context trade-offs, the agent coordination problem—is significant enough that it deserves its own treatment. That is what most of this book is about. But the structural lessons of the broader discipline still apply; we have made an effort not to pretend otherwise.

The six-layer reference architecture and six recurring patterns at the centre of this book are not new in their fundamentals. The six layers descend, in spirit, from any number of enterprise reference architectures going back decades; the six patterns crystallise practices that have emerged from a wide community of practitioners, vendor advocates, and research groups. What this book attempts is to bring them together in one place, in a form that an architect can hand to a team and say: this is how we will think about it.

The reader will find specific recommendations alongside neutral surveys, opinions alongside frameworks, examples alongside abstractions. Where we have a view, we have stated it. Where the field has not yet converged, we have said so. Architecture is taught by example as much as by principle, and we have tried to be honest about which is which.

A note on what this book does not cover. It does not teach machine learning. It does not catalogue prompt techniques. It does not evaluate specific vendors. It does not predict the future. It does try to give the reader a vocabulary, a structure, and a posture that survive the next several iterations of all of those things.

AI systems are, at last, becoming infrastructure. The architectural decisions made in this phase will shape what enterprises can do for the next decade. We hope the chapters that follow are useful in making those decisions well.

The Editors

2026

How to Read This Book

“Architecture is the art of how to waste space.”

—Philip Johnson, paraphrased and reluctantly applied to software

Chapter Overview

This book is about how to build enterprise AI systems that work, keep working, can be governed, and can be changed. It is not about how to train a model or write a prompt; it is about the structural decisions that determine whether those local activities add up to a system the organisation can rely on. This chapter sets out what the book covers and what it deliberately does not, the reader it is written for and the background it assumes, the conventions it uses throughout, the perspective from which the rest of the book is written, and the reading paths that suit different readers and different moments.

Learning Objectives

- Locate this book in the landscape of AI writing and identify what it covers and what it does not.
- Recognise the audience the book is written for and the prior knowledge it assumes.
- Understand the structural commitments of the book: vendor-neutral, pattern-led, governance-aware.
- Place enterprise AI architecture within the wider family of architecture disciplines it inherits from.
- Navigate the book by part, chapter, and cross-reference.

0.1 Scope of This Book

This book covers the architecture of enterprise AI systems — by which it means the structural decisions that determine how data, models, infrastructure, governance, and applications fit together to deliver value at scale. It treats AI as a discipline within enterprise architecture rather than as an isolated research field, and that placement is deliberate: an AI system in an organisation is not a model in a vacuum but a set of components that must inherit the organisation’s data governance, integrate with its existing systems, satisfy its security and compliance posture, and survive the same change pressures every other enterprise system faces.

What the reader should expect from the book follows from that placement. It builds a reference

architecture, developed in Chapter 8, of six layers and five cross-cutting concerns, and uses that reference as the organising structure throughout — so that a discussion of, say, model serving or data privacy always has a known place in a known whole. It offers a pattern catalogue, in Part III, of six architecture patterns that recur across industries, because the same structural problems are solved the same handful of ways regardless of the domain. It treats the cross-cutting concerns of Part VI — privacy, reliability, scalability, interoperability, ethics — as architectural commitments designed into the platform rather than as checkboxes audited at the end. It develops, in Part VII, the building and operating disciplines that translate a static architecture into a running system. And it closes, in Part VIII, with maturity models, case studies, anti-patterns, and forward-looking material that help a reader locate a programme on its journey rather than only describe its destination.

The book is deliberately technology-agnostic. Specific products are named where they are dominant in their category, or where a concrete example communicates more than an abstraction would, but the architectural lessons are written to outlive any particular vendor's market position — because they will have to.

0.1.1 What This Book Is Not

It is worth being equally explicit about what the book does not attempt, so that a reader looking for one of these things can find it elsewhere without disappointment. It is not a textbook on machine learning: it assumes familiarity with the basic shapes of ML — training, inference, embeddings, the difference between supervised and unsupervised learning — and does not re-explain them. It is not a prompt-engineering manual: prompts appear in examples and in the architecture of orchestration layers, but the book does not catalogue prompt techniques. It is not a vendor evaluation: where vendor differences matter to an architectural point the book names them, and where they do not it abstracts them. And it is not a forecast: the final chapter, Chapter 108, acknowledges trends, but the book's value rests on the durable parts of the architecture rather than on the volatile parts of the market.

0.2 Audience

The book is written for a particular kind of reader, and naming that reader precisely is the fairest thing the introduction can do — it lets someone who is not that reader stop early.

The intended reader is an **enterprise architect** extending an established practice into AI and looking for a reference structure compatible with how they already reason about systems; or a **senior engineer or tech lead** accountable for building or operating an AI-powered service and in need of a vocabulary for the design decisions in front of them; or a **data or ML platform lead** building the paved path that other teams will use, who needs to plan that platform as a coherent whole; or the **head of an AI programme** or an AI Centre of Excellence, who must align stakeholders across engineering, data, security, compliance, and the business; or a **consulting engagement** that needs a shared map of the territory before any specific recommendation can be discussed. What these readers have in common is accountability for structure — for how the pieces fit, not only for whether one piece works.

The book is not written for first-year engineers, nor for executives who want a single-page overview. The former will find the assumed prerequisites missing; the latter will find the level of detail unhelpful.

Both are well served by other books, and are better served by them than by this one.

0.2.1 Assumed Background

The reader is assumed to be comfortable with four bodies of prior knowledge, and the book does not re-teach them. The first is distributed-systems vocabulary: services, queues, caches, load balancers, and the language of latency and throughput. The second is cloud-native architecture: containers, orchestration, identity, observability. The third is the basic concepts of machine learning: features, training, inference, evaluation, and the difference between traditional ML and large language models. The fourth is the standard software-engineering practices: version control, continuous integration and delivery, code review, and testing. Where a concept is essential to an argument and the reader's prior background may be uneven, the book introduces it briefly at the point of need and gives a forward reference to its fuller treatment — but it does not pause to build these four foundations from the ground up.

0.3 Conventions

The book uses several conventions consistently, and a reader who knows them in advance will navigate more easily.

References to *Layer 1* through *Layer 6* always mean the layers of the reference architecture introduced in Chapter 8, and the five cross-cutting concerns — privacy, reliability, scalability, interoperability, ethics — are referred to by name rather than by number. The six patterns of Part III are named in full on first mention in each chapter and by their short forms thereafter: *Agentic Mesh*, *RAG*, *LLM Router*, *Hub-and-Spoke*, *Event-Driven Agents*, and *Hexagonal*.

Code, diagrams, and tables each follow a deliberate style. Code listings are illustrative rather than complete: they are written to communicate structure, not to be pasted into production. Diagrams are rendered in TikZ and use a consistent palette — book-blue for components, green for outcomes, accent colours for patterns — so that a reader learns to read the colour as well as the shape. Tables compare options along the dimensions a practitioner would actually use to decide between them, rather than cataloguing every attribute for completeness.

Cross-references connect related material in both directions: a forward reference points to where a topic is developed, a backward reference points to where a prerequisite was established. The reader is not expected to proceed linearly. The table of contents and the index support the reference-look-up mode in which most architecture books are actually used, and the cross-references make each chapter usable as an entry point rather than only as a station on a single track.

0.4 Perspective

Three perspectives underpin the book, and naming them at the outset tells the reader what kind of argument to expect on every page.

The first is *architecture, not algorithm*. The dominant questions in this book are structural: where a responsibility lives, what the interface is, what happens when a component fails, how the component

is changed. These questions matter more than the choice of algorithm inside any given component, because the algorithm will change — a better model, a better retrieval method, a better technique will arrive — and the structure must accommodate that change without being rebuilt. A well-architected system using a mediocre model outperforms a poorly-architected system using a frontier model over any horizon longer than a demo, because the well-architected system can swap the model when something better appears and the poorly-architected one cannot.

The second is *vendor-neutral by default*. The book treats vendor offerings as substitutable adapters behind stable interfaces — the hexagonal discipline developed in Chapter 26. Where a vendor difference is essential to a point it is named; where it is incidental it is abstracted. The intention is that the architecture remains useful as vendors rise, fall, merge, change pricing models, and reposition — which, over the lifetime of any real system, they will all do.

The third is *governance-aware throughout*. Many treatments of AI architecture relegate governance to a final chapter and present it as a constraint imposed from outside. This book treats governance as a structural property of the platform that every system built on it inherits. Privacy, compliance, fairness, auditability, and accountability appear in the data layer (Chapters 10 and 11), in the model layer (Chapters 12 and 14), in the cross-cutting concerns (Chapters 92 through 96), and in the operating disciplines (Chapter 103). They are designed in, not bolted on.

0.5 Enterprise AI Architecture Among the Architectures

The first perspective — architecture, not algorithm — carries an implication worth making explicit before the book proceeds: enterprise AI architecture is not a new and separate discipline that arrived with foundation models. It is one member of a family of architecture disciplines that organisations have practised for decades, and it inherits from the others rather than replacing them.

That family has a recognisable shape. At the top sits *enterprise architecture*, which provides the overall direction — strategy, governance, standards, roadmaps — and within which *business architecture* defines capabilities and operating models. Beneath them sit the domain architectures that most practitioners work in directly: *data architecture*, which governs data assets and their lifecycle; *application architecture*, which defines application services and their interactions; *technology architecture*, which provides the infrastructure and platforms; *integration architecture*, which defines the APIs and patterns by which systems exchange data and service; and *security architecture*, which establishes the controls and compliance posture. *Solution architecture* designs end-to-end solutions that span those domains, and *software architecture* defines the internal structure of the systems that result. Beneath those again sit the delivery-oriented architectures — *platform architecture* for reusable common services, *cloud architecture* for cloud services and environments, and the *AI and data-science architecture* that enables ML models, data pipelines, and intelligent automation. And underneath them all is the implementation, engineering, and operations work that builds, deploys, runs, monitors, and improves what the architectures specify.

The enterprise AI architecture this book develops is the AI and data-science member of that family — but its central claim is that it cannot be practised in isolation from the rest. An AI system inherits its

data governance from the data architecture, its integration patterns from the integration architecture, its security controls from the security architecture, its infrastructure from the technology and cloud architectures, and its strategic direction from the enterprise and business architectures above. A practitioner who treats AI architecture as a self-contained discipline — who designs a model-serving platform without reference to the organisation’s existing data, integration, and security architectures — builds a system that does not fit, and a system that does not fit is a system that will be governed badly and changed painfully. This is why the book treats AI as a discipline within enterprise architecture, and why Chapter 4 on enterprise architecture and Chapter 41 on alignment with enterprise architecture frameworks are not digressions but load-bearing parts of the argument. The reference architecture at the heart of this book is one architecture among the organisation’s many, and it earns its place by fitting the others.

0.6 How to Read This Book

Three reading paths suit different readers and different moments, and a reader is best served by choosing the one that matches the present need rather than defaulting to the first.

The *first-read* path is to read in order, skipping no chapters, expecting to spend the longest on Part II, which develops the layers, and Part III, which develops the patterns — these are the parts that build the vocabulary the rest of the book uses. The *project-start* path is to read Chapter 8 for the reference architecture, then the specific patterns relevant to the project at hand, then the cross-cutting concerns that match the system’s risk profile — enough structure to begin, without the whole book first. The *specific-question* path is to use the table of contents and the index directly: each chapter is written to be self-contained enough to read in isolation, with cross-references back to whatever prerequisite material it depends on. Table 1 summarises the three.

Table 1: Reading paths through the book.

Path	Route
First read	Read in order, skipping no chapters. Expect to spend longer on Parts II (the layers) and III (the patterns).
Project start	Read Chapter 8 (reference architecture), then the patterns relevant to the project, then the cross-cutting concerns that match the system’s risk profile.
Specific question	Use the table of contents and the index. Each chapter is self-contained enough to read in isolation, with cross-references back to prerequisite material.

0.7 A Note on the Era

This book is being written in 2026, in the early-mature phase of the foundation-model era. Some of its material will date faster than the rest — the names of frontier models, the relative cost of vendor offerings, the specific protocols stabilising the agent surface — and the book is honest about which material that is. Its structural claims are written to outlast the specific facts they currently rest on, and where a claim depends on the present moment the text marks it as such. The reader coming to these pages later is invited to substitute the present-tense reality of their own moment; the architecture, if

the book has done its work, will still hold when the facts beneath it have moved.

0.8 The Shape of the Argument

This book is a structural treatment of enterprise AI architecture: how the pieces fit together, what contracts hold between them, what fails when they fail, and how to design for change. It is written for practitioners who already hold the assumed background, from a perspective that is vendor-neutral, pattern-led, and governance-aware — and that understands enterprise AI architecture as one member of a family of architecture disciplines, inheriting from the data, integration, security, and enterprise architectures around it rather than standing apart from them. The chapters that follow develop, in turn, the foundations, the six layers, the six patterns, the cross-cutting concerns, the building and operating disciplines, and the journey to maturity. The reader is invited to begin.

A Quick Glossary

This one-page reference defines the terms a reader meets earliest and most often. It is deliberately short. The full glossary, with chapter cross-references, is [Appendix A](#).

Agent

An AI component that chooses its next action from a set of available tools based on the current state, usually in pursuit of a goal. Contrast with a fixed chain, where the sequence of steps is decided in advance.

ABB / SBB

Architecture Building Block and Solution Building Block. An ABB is a required *capability* (“an orchestrator supporting human review checkpoints”); an SBB is the specific *product* chosen to deliver it. Committing to ABBs and holding SBBs loosely is what keeps an architecture portable.

Context Window

The maximum number of tokens — input plus output — a model can attend to in a single call. It sets the upper bound on how much retrieved material a prompt can carry.

Embedding

A dense numeric vector representing the meaning of a piece of text, image, or other content. Similar content yields nearby vectors, which is what makes similarity search and retrieval possible.

Foundation Model

A large model, trained on broad data, used as the general-purpose base for many tasks — typically accessed through a provider’s API or self-hosted from open weights.

Guardrail

A classifier or rule placed before or after a model to catch unsafe inputs or outputs. Guardrails work in layers; no single one is sufficient.

Hallucination

A model output that is fluent and confident but factually wrong or unsupported. The defining failure mode that observability and evaluation exist to catch.

Hexagonal Architecture

A design that isolates business logic (the core) from external systems (models, stores, tools) behind stable interfaces (ports), with swappable implementations (adapters). It is what makes a vendor change a deployment rather than a project.

Inference

The act of running a trained model to produce an output, as distinct from training the model. Most

enterprise AI cost is inference cost.

LLM

Large Language Model: a foundation model specialised for text, the most common model type in current enterprise AI systems.

MCP (Model Context Protocol)

A standard for connecting an agent to external tools and data sources through structured, declared interfaces rather than bespoke per-tool integration.

Prompt

The instruction and context given to a model in a single call. In this book a prompt is treated as a versioned, tested, reviewable artefact, not an informal string.

RAG (Retrieval-Augmented Generation)

A pattern in which relevant material is retrieved from a knowledge store and supplied to the model as context, so the answer is grounded in current, specific data rather than the model's training alone.

Reranking

A second-pass scoring step that orders retrieved candidates more carefully than the initial similarity search, improving the relevance of what reaches the model.

Token

The unit of text a model processes — roughly a short word or word-fragment. Model cost and context limits are both measured in tokens.

Tool Use

An agent's invocation of an external capability — a search, a database query, a calculation, an action in another system — through a structured call rather than free text.

Vector Database

A store built to hold embeddings and answer nearest-neighbour queries at scale. The retrieval engine underneath most RAG systems.

Terms used in only one chapter are defined where they appear. The complete glossary is Appendix A.

Part I

Foundations of Enterprise AI Architecture

“The aim of natural history is to discern the great chain that links the different productions of nature, and to mark the relations and dependencies that subsist between them.”

—Georges-Louis Leclerc, Comte de Buffon, Histoire Naturelle, 1749

Before the architecture, the vocabulary. Before the vocabulary, the postures that make the vocabulary worth using. Part I opens with a short chapter on the four dispositions a working discipline needs — representing the world honestly on a flat page, trusting direct observation over inherited authority, naming what has been observed, and designing for the catastrophic event no plan anticipated. The fourth disposition is then developed at length in the chapter that follows, paired with the eighteenth-century workshop tradition the rest of the book partly inherits. Part I then turns to a working taxonomy of the model families those disciplines ask us to recognise, addresses what enterprise AI is and how it relates to enterprise architecture more broadly, and closes with the principles by which it is held together.

Chapter 1

Four Dispositions a Discipline Needs

1.1 What This Book Is

This book is about how to design, build, and govern enterprise AI systems — the kind that handle real money, real customers, and real regulatory exposure, that have to keep working through staff turnover and vendor changes, and that are answerable to auditors, regulators, and the people whose lives are affected by their decisions. It is not about how to train a model or write a clever prompt; those activities have their own substantial literatures. The work described here is older, slower, and less photogenic: the structural decisions that determine whether a model, a prompt, or a fine-tune adds up to a system the organisation can rely on for the next ten years.

Three years into the foundation-model era, enterprise AI has accumulated the symptoms of a field that has grown faster than the discipline that should have grown with it. Pilots multiply; production deployments stall. Cost shocks arrive on the first month's invoice. The same architectural mistakes are made in twelve different companies in eighteen months. Reference architectures are published but describe what one vendor sells. Standards are drafted but not yet settled. A working enterprise architect in 2026 cannot wait for the field to settle. The decisions are arriving faster than the consensus is forming.

This chapter is about the dispositions a young field has to acquire before its work can settle. Before the architecture, the vocabulary. Before the vocabulary, the postures that make the vocabulary worth using. There are four such dispositions, and a discipline that lacks any one of them is not yet a discipline. Each is illustrated in what follows by figures who established it for an earlier field, in a substrate the reader will not confuse with their own. The arc of the chapter — representation, observation, naming, resilience — is also the order in which a working field actually develops them.

1.2 Representation: Mercator's Honest Lie

In Duisburg, in 1567, on a wooden table on the lower Rhine, a fifty-five-year-old Flemish cartographer named Gerardus Mercator was working out a problem that had defeated every cartographer before him. The surface of the Earth is curved. The page of a book, or the sheet of vellum on a navigator's table, is flat. You cannot copy the curved surface onto the flat one without lying about something. The lie can be about distance, area, angle, or shape, but a lie there must be. The cartographer's choice is not whether to distort the truth but which truth to distort.

Mercator's 1569 world map answered the question for a specific user — the navigator. The full Latin title named the user on its face: *ad Usus Navigantium*, “for the use of sailors.” By stretching the spacing between parallels of latitude in a precisely calculated way, Mercator produced a map on which every rhumb line — the path of a ship holding a constant compass bearing — was a straight line. A navigator could lay a ruler between two ports, read the bearing off the compass rose, and sail it. The cost was visible on the face of the map. The stretching that made the rhumb lines straight also inflated the areas of land masses far from the equator: Greenland, on a Mercator projection, looks roughly the size of Africa, when it is in fact about one-fourteenth as large. For a navigator, who needed angle and did not care about area, this was a clean trade. For a politician forming impressions of the relative size of nations, it was a disaster waiting to happen, and four centuries later the disaster is still happening.

What is notable for the architectural reader is that Mercator was open about it. The 1569 map was inscribed with more than five thousand words of explanation across fifteen separate legends. He named the projection's purpose. He acknowledged where the data was incomplete, filling the unknown interior of Africa with the writings of Prester John and saying so explicitly. His map was not less honest for being a projection; it was more honest, because the projection was deliberate, named, and explained. A year later, in Antwerp, Abraham Ortelius published the first edition of his *Theatrum Orbis Terrarum* — the first modern atlas, fifty-three engraved map sheets redrawn from contemporary cartographers. Ortelius added a *Catalogus Auctorum* of eighty-seven names: a bibliography of the cartographers whose work he had used. It was the first time in European printing that a major work had systematically credited its sources.

The architectural transposition is direct. Every architectural diagram is a projection: a deliberate decision about what to preserve, what to sacrifice, and which use the diagram is being made for. Every reference architecture is, or should be, an atlas with a catalogus: a working document that names its sources, the patterns it inherits, the standards it implements, the prior architectures it extends. A diagram with neither is decoration. A diagram with both is, in the original sixteenth-century sense of the word, a map: a working document a reader can navigate by. Most contemporary architectural diagrams have unnamed projection choices, unmarked simplifications, and speculative parts drawn as confidently as verified parts. The mathematics has gone unsigned. The first disposition a working discipline needs is the willingness to sign it.

1.3 Observation: What Vesalius Saw

In the winter of 1539, on a wooden anatomy table at the University of Padua, a twenty-five-year-old Flemish lecturer named Andreas Vesalius did something that almost no professor of medicine of his century would willingly do. He took the knife himself. By long-established practice, dissection was done by a hired barber-surgeon while the professor read aloud from Galen's textbooks above the body. The text described what should be there. The barber cut. The students saw whatever was there. The two often did not match. The standard response to the mismatch, refined over thirteen hundred years of medical education, was to ignore the body and continue reading from the text.

Vesalius did the opposite. He cut himself, in front of the students, and described what he was finding.

When what he found did not match Galen, he said so. He kept saying so. Four years later, in June 1543, he published a four-hundred-thousand-word folio with more than three hundred woodblock illustrations, titled *De Humani Corporis Fabrica*. He was twenty-nine. Galen had been wrong about the breastbone, the lower jaw, the liver, the kidneys, the venous system. The reason he had been wrong was that he had never dissected a human body. Galen had dissected apes, dogs, and pigs, and his observations had been copied into textbooks for thirteen centuries by people who, like Galen himself, never put a knife to a human cadaver to check.

The response from Vesalius's contemporaries is itself part of the lesson. His former teacher Jacobus Sylvius, faced with the evidence that Galen had been wrong about the structure of the human body, took the position — in print, in 1555 — that the human body itself had *changed* since Galen's time. The lower jaw had been a single bone in Galen's day and had become two. The breastbone had had seven segments and had now lost some. To grant that Galen had been mistaken would have been worse than to grant that human anatomy had been refactored by time. The depth of the institutional commitment to the inherited text was such that the more credible explanation was a redesign of the species.

The architectural transposition writes itself. When a vendor's documentation contradicts a careful eval, the response from inside the vendor's organisation can take a form startlingly similar to Sylvius's. The model has not been wrong; the eval must have been miscalibrated. The benchmark was published before this version, but the new version should be better even though we haven't rerun it. The prompt was probably suboptimal. The reviewer was not domain-qualified. These are not always bad-faith arguments. Some are sometimes correct. But the structural shape of them — the willingness to entertain almost any alternative rather than accept that the documented claim was wrong — is the same shape as Sylvius's argument that the species had changed.

Eighty-five years after Vesalius, William Harvey took the discipline one step further. His 1628 *De Motu Cordis* demonstrated the circulation of the blood through three moves: a qualitative dissection (the heart is a muscular pump, the valves enforce one-way flow), a quantitative argument (the volume of blood pumped per hour exceeds by orders of magnitude the volume of food consumed, so the same blood must circulate), and an experimental demonstration (ligatures on farm workers showed venous blood flowing toward the heart, not away from it). Where Vesalius had refuted a text, Harvey refuted a theory. The architectural reader should note both moves. Vesalius's contribution was to insist that the professor should hold the knife. Harvey's contribution was to insist that the argument should rest on what could be measured and experimentally demonstrated. The enterprise AI architect needs both: anatomy, which is knowing what the components actually are; and physiology, which is knowing how they work together. You cannot do physiology with the wrong anatomy.

A note worth making: this disposition has been practised outside Europe for centuries. Ibn al-Nafis, working in Cairo in the 1240s, described the pulmonary circulation correctly — four hundred years before Harvey did the same — by dissecting and finding no pores in the ventricular septum that Galen had taught existed. Harvey almost certainly never read it; the relevant passage was not translated into Latin until well after his death. The pattern recurs. An African ML engineer running an evaluation against Yoruba-language inputs is doing the same epistemic work Ibn al-Nafis was doing in Cairo:

looking at the actual material, finding that the textbook is wrong about it, and saying so. The textbook here is the foundation-model provider's benchmark, almost always evaluated on English-language inputs from American or European data distributions. The eval against Yoruba is the corpse on Vesalius's table. The second disposition a working discipline needs is the willingness to trust the corpse over the textbook.

1.4 Naming: Linnaeus, Diderot, and the Shared Vocabulary

Picture a young Swedish medical student in 1735, alone in a Dutch printing house, watching the first impressions come off the press of an eleven-page folio he has paid for himself. The work is called *Systema Naturae*. It proposes to put every plant, every animal, and every mineral on Earth into a single hierarchy with a single Latin name apiece. Twenty-three years later, the tenth edition will be three thousand pages long. The world's zoologists will still be using it three centuries after that as the formal starting point of their discipline. The student is Carl Linnaeus, and what he is about to do is solve a problem that almost nobody has noticed is a problem at all.

The problem is that you cannot reason about something until it has a name. Without a shared name, a finding cannot be reported and an instruction cannot be given. In Linnaeus's time the same buttercup might travel under three Latin polynomials in three adjacent monasteries; in ours, two engineers in two adjacent companies will use the word *agent* to mean substantially different things, and write their CVs accordingly. The looseness is forgiveable in a young discipline. It is also a real obstacle to coordination, and any field worth the name has eventually had to settle its vocabulary — usually by the work of one or two stubborn people who could no longer bear the disorder.

Linnaeus's answer had two parts. The first was a hierarchy: three kingdoms, classes, orders, genera, species. Every organism had a place; every place had relatives. The second was a binomial in Latin: a genus and a species, two words, the same on every desk in every language. *Homo sapiens*. *Canis lupus*. The hierarchy let practitioners see relationships at a glance. The binomial let them talk to one another. The science behind the hierarchy was sometimes wrong — whales got put in with the fish at first — but the system spread because it was not for any one practitioner; it was for the network. A well-built taxonomy outlives the taxonomist by an order of magnitude.

Twelve years after *Systema Naturae*, in Paris, two booksellers approached a hard-up writer named Denis Diderot to translate an English encyclopaedia into French. Diderot said yes, took on Jean le Rond d'Alembert as co-editor, and what was meant to be a translation became, over twenty-five years, the *Encyclopédie*: twenty-eight volumes, seventy-two thousand articles, more than a hundred and thirty contributors, four thousand subscribers in an age when fifteen hundred was an enormous run. Diderot's shape was the opposite of Linnaeus's. Where Linnaeus stacked, Diderot listed. Where Linnaeus enforced relationships through a hierarchy, Diderot dispersed them through the alphabet and connected them again with cross-references — the *renvois*, the see-also notes that turn a flat catalogue into a graph the reader can walk.

Serious naming projects do something both Linnaeus and Diderot's did. They provoke the people whose interests are protected by the current vagueness of the field. "Hallucination" is a name. "Vendor

lock-in” is a name. “Confused deputy attack” is a name. “Reward hacking” is a name. “Negative contribution margin” is a name. Each one belongs to a thing that some commercial interest would prefer to leave nameless. The architectural transposition is that the Fundamentals chapter, when it arrives, performs both Linnaeus’s and Diderot’s moves at once: the model families of contemporary AI are organised by what each does (a Linnaean hierarchy), and connected by cross-references to the patterns that use them (Diderot’s *renvois*). The grouping is provisional, like Linnaeus’s was. Whales will get put in with the fish. That has always been part of the work.

None of this was a European invention. Akan and Yoruba healers were running working systems of plant classification by use and preparation when Linnaeus’s father was still a country pastor; the Ayurvedic and Unani traditions had produced multi-volume pharmacopoeias of thousands of entries; the Chinese imperial *Bencao Gangmu* of 1578 was a pharmacological taxonomy of comparable ambition. What was distinctive about the European projects was institutional, not intellectual: universities, learned societies, state-supported printing, networks of correspondents. The rails were what eventually made them the lingua franca. The cost of having a single shared vocabulary is that some other vocabulary is paying the price. The third disposition a working discipline needs is to build a shared vocabulary while remembering that the vocabulary it inherits is not the only one that exists.

1.5 Resilience: The Morning After Lisbon

The first three dispositions — representation, observation, naming — are necessary but not yet sufficient. A young discipline that has its maps, its observations, and its vocabulary can still be unprepared for the day reality arrives in a form none of its diagrams predicted. The fourth disposition is the one a discipline tends to learn the hard way, usually on the morning after a catastrophe.

On 1 November 1755, the morning of All Saints’ Day, a magnitude 8.5 to 9 earthquake originating in the Atlantic off Cape St Vincent destroyed central Lisbon. The first shock arrived at about 9:40 in the morning, while most of the city was at Mass. The tremors lasted perhaps six minutes. The fires that began in the ruined buildings burned for the next five days. Forty minutes after the first shock, three successive tsunamis arrived in the harbour. Between thirty and forty thousand people died in a city of two hundred thousand. The Royal Palace, the great Opera House, the royal library, the patriarchal cathedral, the customs house, the Inquisition’s records — all destroyed within minutes. By any reasonable measure, the disaster ended the city.

The Marquis of Pombal, first minister to King Joseph I, made the most architecturally instructive decision of the eighteenth century. He chose not to abandon the ruined city. He chose not to rebuild it on the medieval street plan. He convened the engineers — Manuel da Maia, Eugénio dos Santos, Carlos Mardel — and asked for a system. What emerged was the *Baixa Pombalina*: a grid of standardised streets, with standardised buildings, constructed on a frame that came to be called the *gaiola pombalina*, the Pombaline cage. The cage was a wooden lattice embedded in the masonry walls. It was tested by parading troops past the prototype to simulate seismic loading. Buildings were required to flex on the cage, dissipating energy that would otherwise crack the masonry. The cage did not prevent damage; it limited the damage to what could be repaired without bringing the building

down.

The architectural transposition lands cleanly. A system that has no graceful path for the alternative case is a system that, when the alternative arrives, will fail catastrophically. A system that has the equivalent of a Pombaline cage — mechanisms that bend rather than break, fallback paths that degrade gracefully, an inherent capacity to fail in survivable ways — can absorb the unexpected event without losing what it was built to do. Every enterprise AI system will eventually meet a query, a load pattern, a model regression, a vendor outage, or a regulatory ruling that none of its designers anticipated. The system that has been built with cages will bend. The system that has been built without will fall.

The chapter that follows this one returns to Lisbon at length, paired with the Pombaline workshop tradition that made the response possible. The disposition matters enough to take seriously, both as historical narrative and as architectural commitment. The fourth disposition a working discipline needs is the willingness to design, in advance, for the catastrophe its current diagrams do not predict.

1.6 Reading What Follows

The four dispositions of this chapter — representation, observation, naming, resilience — are the postures the reader is asked to carry into the technical material that follows. They do not exhaust what an enterprise architect needs to know, but they are the dispositions without which the rest of the knowledge tends to be applied badly. A reference architecture drawn without the discipline of representation is decoration; an evaluation regime built without the discipline of observation is theatre; a taxonomy without the discipline of naming is jargon; a system without the discipline of resilience is a building waiting for an earthquake.

The book is written from inside the practice, in Pretoria, against the constraints African enterprises actually operate under — foreign-exchange volatility, sovereignty obligations, talent scarcity, infrastructure limitations. Every part of the book includes a section that addresses these constraints explicitly. The argument is that the architectural disciplines that survive under these constraints tend to be the disciplines that survive everywhere, and that constraint, as the engineering proverb has it, is the friend of design.

The book is not a textbook on machine learning, not a vendor manual, not a regulatory or compliance manual, and not a book of forecasts. The model families introduced in Chapter 3 are described at the level an enterprise architect needs to make sound architectural decisions, not at the level a model developer needs to train one. Vendors are named when naming them is the only honest way to discuss a market, but recommendations are at the level of capability rather than product. Many specific recommendations will look dated within five years; the disciplines underneath them are designed to survive that movement. A reader returning to the book in 2031 should find the patterns still useful, the dispositions still right, and the product references mostly stale.

The chapter that follows takes the fourth disposition — resilience — and develops it at length. Lisbon will return there, in fuller detail, paired with the eighteenth-century workshop tradition that the architectural disciplines of the rest of the book partly inherit. Then the chapter after that begins the technical material proper, with a working taxonomy of the model families this discipline currently

recognises — one Linnaean move, written in 2026, fully expecting to be revised by the practitioners who come next.

Chapter 2

Chaos and Craft: Lessons from the Workshop and the Earthquake

“Bury the dead and feed the living.”

—attributed to the Marquês de Pombal, Lisbon, 1 November 1755

Chapter Overview

This chapter takes a step back from technical detail to consider two episodes from the eighteenth century that, on the face of it, have nothing to do with enterprise AI: the patient assembly of the European workshop, and the catastrophe of the Lisbon earthquake of 1755. The first was the slow, deliberate construction of the disciplines that would underwrite the industrial age. The second was the unannounced arrival of an event no one had architected for — and the remarkable response that turned the disaster into the founding moment of modern seismology, urban planning, and earthquake-resistant building. Together they illustrate, more vividly than any architectural diagram, the two forces that shape every enterprise AI system: the patient craft of building durable structure, and the chaos that arrives when reality does not match the plan.

Learning Objectives

- Recognise that enterprise AI is the latest chapter in a long story about building systems.
- Draw on the eighteenth-century workshop to understand division of labour, measurement, and standardisation.
- Use the 1755 Lisbon earthquake to think about brittle plans, graceful degradation, and what real resilience looks like.
- See in Pombal’s response the template for evidence-driven recovery that AI engineering organisations still recapitulate.
- Carry the lessons into the architectural patterns that occupy the rest of the book.

2.1 Two Stories from One Century

The previous chapter introduced four dispositions a working discipline needs: representation, observation, naming, and resilience. The fourth was sketched briefly there but is the one that a discipline

tends to learn the hard way — usually, in fact, on the morning after a catastrophe — and is significant enough to deserve a chapter of its own. This is that chapter.

The patient construction of the European workshop and the catastrophe at Lisbon happened within the same forty years, on the same continent, in correspondence with one another — the encyclopaedists were writing in Paris while the rubble was being cleared from the Baixa. They are paired here because together they show the two forces that any enterprise eventually meets: the slow accumulation of disciplines that make complex work possible, and the arrival, on some unremarkable morning, of an event that no one had built for. Enterprise AI will meet both. The architectural patterns of the chapters ahead will land more firmly with the workshop and the earthquake already in mind.

2.2 The Pin Workshop

2.2.1 Diderot’s Pin Factory, Before Smith Wrote About It

The eighteenth century in Europe was, among many other things, a century of catalogues. The *Encyclopédie* of Diderot and d’Alembert, published in volumes between 1751 and 1772, was an attempt to write down every craft, every trade, every piece of useful knowledge in a structured, illustrated form. It was, in the language we would use now, an attempt to build a reference architecture for civilisation: every component named, every process diagrammed, every relationship made visible.

What the encyclopaedists saw, when they visited workshops, was something that we would today call a compound system. A pin was not made by a pin-maker. It was made by perhaps eighteen specialised hands: one drew the wire, another straightened it, a third cut it, a fourth pointed it, a fifth ground the top to receive the head, a sixth fashioned the head, a seventh tinned the pins, an eighth wrapped them. Adam Smith, writing in 1776 in *The Wealth of Nations*, made the observation famous: a single worker, attempting all eighteen steps, could perhaps produce a few dozen pins a day. The same worker, doing only one step in a coordinated workshop, contributed to thousands.

This is not a parable about productivity. It is a parable about architecture. The workshop did not produce pins faster because it had a magical pin-making machine. It produced pins faster because the work had been decomposed into components whose interfaces fit, whose outputs were quality-controlled at each stage, and whose coordination was the responsibility of the workshop’s design rather than any individual worker’s improvisation.

2.2.2 The Three Inventions of the Workshop

Three disciplines emerged from the eighteenth-century workshop, all of which return in this book under modern names.

The first was *specialisation with contracts*. Each worker did one thing well, and the output of one worker had a known shape that the next worker could consume. The book returns to this idea repeatedly: compound AI systems (Chapter 44), prompt chains (Chapter 55), and the typed interfaces between components that distinguish a coherent system from a tangle of integrations.

The second was *measurement and inspection*. A pin that was the wrong length, or a head that was

poorly seated, did not pass to the next station. The workshop had inspectors — not because workers were dishonest, but because errors compound when they pass unchallenged. The book returns to this under the name of evaluation: evals as gates (Chapter 51), judge models as inspectors, the discipline of stratified metrics that catch what aggregate numbers hide.

The third was *standardised parts*. Before the eighteenth century, a screw was made to fit a particular hole; a key was made for a particular lock. Slowly, painfully, manufacturers learned that screws made to a standard thread, and parts made to standard dimensions, could be assembled into anything — and could be replaced when they failed. The book returns to this under the name of interoperability: the Model Context Protocol that standardises tool integration, the open weights that prevent vendor lock-in, the abstraction layers that let an organisation swap one foundation model for another without rewriting its applications.

2.2.3 The Quiet Inventions That Mattered More

While the workshops were learning to coordinate, the world around them was learning to measure. The eighteenth century gave us many of the instruments that we now take for granted. Harrison’s marine chronometer, perfected in the 1760s after decades of work, solved the longitude problem and made global navigation reliable. Lavoisier, in his Parisian laboratory in the 1780s, gave chemistry the discipline of careful weight measurement and named the elements in a way that allowed chemists across Europe to mean the same thing by the same word. Linnaeus, working through the middle decades of the century, gave biology a taxonomy that let naturalists in Sweden and naturalists in the Cape colony classify what they saw in a shared language.

Each of these inventions has an analogue in enterprise AI. The chronometer is observability: the instrument that tells you, accurately and continuously, where you are. Lavoisier’s careful weighing is the eval discipline: the willingness to measure exactly what you have, rather than what you hope you have. Linnaean taxonomy is what Chapter 3 attempted: a shared vocabulary so that practitioners across disciplines can mean the same thing by “LLM”, “embedding model”, “agent”, or “foundation model”.

The point is not that the eighteenth century invented enterprise AI. The point is that the eighteenth century invented the disciplines that make any complex enterprise endure: division of work into specialisations with clean contracts, measurement that does not lie, standardisation that survives the original designer, instruments that report the truth of operations, and taxonomies that let practitioners coordinate across distance and language. The architectural patterns in this book are the modern instantiations of disciplines whose first formal versions came out of European workshops three hundred years ago.

2.3 Lisbon, 1 November 1755

2.3.1 The Morning

If the eighteenth century is also a story of order catastrophically lost, the Lisbon earthquake of All Saints’ Day 1755 is its definitive instance. The contrast with the workshop is the point.

Lisbon in 1755 was one of the great cities of Europe and the capital of an empire. Roughly two hundred thousand people lived there. Gold flowed in from Brazil and the colonies; the harbour, on the Tagus estuary, was among the busiest in the world; the city's churches, perhaps forty of them, were splendid; its narrow medieval streets ran tight against the river. The day was a major holy day in a deeply Catholic country. Around half past nine on the morning of Saturday 1 November, most of the city was in church.

At 9:40 the earth moved. The shaking lasted between three and six minutes — impossibly long, as anyone who has felt a building tremor of even a few seconds can imagine. Modern seismologists place the magnitude between 8.5 and 9.0; the epicentre was offshore, west of the Iberian peninsula, where the African and Eurasian tectonic plates meet. Thirty-five of Lisbon's forty churches collapsed, often onto the congregations inside them. About twelve thousand dwellings were destroyed. Survivors stumbled into the open spaces along the waterfront, where the sea would be safer than the falling buildings.

Then the sea withdrew.

Forty minutes after the shaking stopped, three tsunami waves came up the Tagus, the largest at the harbour about six metres high. At Cádiz, further south, they reached twenty metres. The waves crossed the Atlantic and struck Martinique ten hours later, still four metres high. Of the survivors who had taken refuge on the waterfront — the most reasonable thing they could have done, given the falling buildings — many drowned.

What the earthquake had not destroyed, and what the water had not finished, the fires did. They burned in Lisbon for between five and six days. By the time the smoke cleared, somewhere between fifteen and forty thousand people in Lisbon alone were dead; the great commercial city was effectively gone.

2.3.2 What Had Failed

It would be easy to call this simply a natural disaster — as if buildings and human plans had nothing to do with the death toll. They had everything to do with it. The churches that collapsed were the largest buildings in the city, built from heavy stone over the centuries with little regard for lateral forces, because no one had built with lateral forces in mind. Their unreinforced masonry walls were exactly the wrong material for what arrived. The narrow medieval streets concentrated falling debris on whatever fled through them. The waterfront, where reasonable people went for safety, was the geometry that focused the tsunami. The fire spread through tight-packed neighbourhoods of timber and wattle that no city plan had ever broken into firebreaks. The state's emergency response, before that morning, did not really exist.

This is, architecturally, what a system without graceful degradation looks like. Every component had been optimised for the case it expected to encounter, which was ordinary life. Every component, when reality presented it with a case it had not been designed for, failed in the most spectacular way available to it. The earthquake did not by itself cause the disaster. The disaster was caused by the unexamined assumptions baked into every load-bearing decision the city had made for three

centuries.

There is a kind of overconfidence that is moral, in the sense that it expresses faith and pride. There is also a kind of overconfidence that is architectural, in the sense that it leaves the system with no place to land when it is wrong. Lisbon's overconfidence was architectural. Its churches were beautiful precisely because they had never been designed to fall.

2.3.3 Pombal's Response

What happened next is, for the purposes of this book, more interesting than the disaster itself.

The king, Joseph I, and most of the court had left the city after attending mass at dawn — one of the king's daughters had asked to spend the holiday in the countryside. The royal family survived. The king, badly shaken, would refuse to sleep inside walls for years afterward and lived in tents on a hillside outside the ruined city. The state's effective leadership fell to his chief minister, the man history would come to know as the Marquês de Pombal.

Asked what was to be done, Pombal is reported to have answered: *Enterrar os mortos e tratar os vivos*. Bury the dead and feed the living. Whether or not he said it exactly that way, the line captures the posture that organised the recovery. He did not call for prayer, or for the assignment of theological blame, or for a return to the city as it had been. He treated the catastrophe as an operational problem, then as an engineering problem, then as an institutional one.

The operational problem came first. He fixed prices to prevent profiteering. He organised police to prevent looting. He suspended certain taxes. He cleared rubble. He set up emergency hospitals. He empowered military engineers to triage which buildings could be saved and which had to come down. In the language of contemporary incident response, he ran the first weeks as a war room.

Then came what is, in retrospect, the more important act. He sent a questionnaire to every parish priest in the kingdom. Not a theological questionnaire: a factual one. What time did the earthquake begin? How long did it last? Did the shaking seem stronger in one direction than another? Did buildings fall more to one side? How many hands above the normal did the sea rise? How long did the fires burn? The questions, by the standards of 1755, are striking for what they do not ask — they do not ask about sins, or signs, or divine judgement. They ask for what we would now call evidence. Six hundred and forty-six replies came back from parishes across Portugal. They are still in the national archive. Modern seismologists used them, more than two centuries later, to reconstruct the earthquake in detail.

This is generally considered the founding survey of modern seismology. The discipline that would eventually let humans monitor faults in real time and warn of tsunamis began as a parish questionnaire designed by a politician who had decided that the right response to an unexpected catastrophe was to find out, precisely, what had happened.

2.3.4 The Pombaline Cage

The reconstruction of the Baixa district — the lower town that had been destroyed — was directed by the corps of military engineers under Manuel da Maia. They did three things that had not really been

done before, and that the rest of the world would come to copy.

First, they laid down a grid. The medieval street pattern was simply abandoned. Wide streets were drawn, with regular blocks, with squares at intersections. The grid did not improve the buildings; it improved what happened around the buildings when one of them fell or caught fire. Wide streets are firebreaks. Wide streets are also routes by which emergency response can reach people. A street that fits the width of a carriage is too narrow when one wall of it has just become a pile of stone.

Second, they standardised the buildings themselves. New buildings in the Baixa had to conform to a fixed set of specifications: facade height, wall thickness, floor count, window placement. This was less about aesthetics than about predictability. A standardised building is a building whose failure modes are known and whose components are interchangeable. When the next earthquake came, the rescuers would not have to learn each building individually.

Third, and most famously, they invented what came to be called the *gaiola pombalina* — the Pombaline cage. Inside the stone and masonry walls of each building, the engineers built a flexible wooden lattice, cross-braced at the corners, designed to absorb lateral motion. When the ground moved, the cage flexed; the masonry, attached to the cage, swayed with it rather than crushing through the building. The structure as a whole bent before it broke. Before they committed to the design, the engineers built small wooden models and shook them by having troops march in rhythm around them: the first known earthquake-resistance simulation.

The Pombaline cage was an act of intellectual humility expressed as engineering. The designers did not assume they could prevent the next earthquake. They assumed it would come; they assumed they would not know in advance how strong it would be; they built so that the system could fail without becoming a catastrophe. This is the disposition that every chapter in the rest of this book about resilience, fallback, canary deployment, rollback, and graceful degradation is trying to express.

2.4 What the Workshop and the Earthquake Make Clear

2.4.1 Systems Are Built; They Are Not Inherited

The workshops of the eighteenth century did not appear. They were assembled, painstakingly, over decades, by people willing to spend more time on the interfaces between workers than on the workers themselves. The Lisbon that emerged from 1755 also did not appear; it was rebuilt, deliberately, by a small group of engineers and administrators who treated each design decision as a choice with consequences.

The enterprise AI systems described in the chapters ahead share this property. They are built. They do not emerge from buying a model API. They emerge from years of patient work on the interfaces between components, the contracts between teams, the evidence trails for auditors, the operating procedures for incidents, the talent architectures that retain the people who keep the whole system running. Skipping any of these does not save time; it postpones the bill, often to a moment when the bill is no longer survivable.

2.4.2 Measurement Is the Difference Between Craft and Hope

What distinguished the eighteenth-century workshop from the medieval cottage was not better workers. It was the inspector standing at each station, the calibrated weight on the chemist's balance, the chronometer in the captain's chart room. Measurement does not improve the work; it tells you, ruthlessly, what the work is.

Pombal's parish questionnaire was the same disposition applied to disaster. He could have written, instead, a sermon or a proclamation. He chose a survey. The survey is what made the rebuilding evidence-based and what made the discipline of seismology possible at all.

The eval-driven release pattern (Chapter 51), the canary architectures (Chapter 48), the trace-store discipline (Chapter 49), the observability stack, the FinOps measurements (Chapter 61): all of these are versions of the chronometer in the chart room and the questionnaire to the parishes. They will not improve a bad system. They will, ruthlessly, tell you what kind of system you have.

2.4.3 Plans Survive Their First Encounter With Reality, or They Do Not

The Lisbon of 1755 had a plan, in the sense that every building, every street, every emergency procedure was a plan. That plan had no contingency for a magnitude-nine earthquake on a holy day at half past nine in the morning. A more durable plan would have looked the same in normal times and quietly accommodated the unexpected. It would have had wide streets even when not needed, firebreaks even when not on fire, building frames even when not shaking, government records of citizens even when not lost. That plan was what got built after the disaster, not before.

Architectural patterns in this book that may sound paranoid — shadow traffic, replay-driven development, the human-in-the-loop fallback, the rollback path on every deployment, the red-team campaigns, the reserves in the AI budget — are not paranoid. They are the contingency plans the city of Lisbon did not have on 1 November 1755. They are what makes a system one that survives Tuesday, not one that only thrives on Sunday.

2.4.4 The Pombaline Cage as Engineering Philosophy

The Pombaline cage is worth holding in mind throughout the rest of this book, because it is so close to the disposition that modern AI engineering teams converge on when they have lived through enough incidents.

The cage is an admission. It admits that the designers cannot predict the next shock. It admits that they cannot prevent it. It admits that the building, however carefully constructed, will be tested by forces no one fully anticipated. And then, having admitted all of that, the cage does the only thing left to do: it builds the failure mode into the design. It chooses, in advance, where the system will give way under stress. It chooses to bend, in places that are easy to reset, rather than to shatter, in places that are catastrophic.

A modern AI system with a Pombaline cage inside it might look like the following. It has a routing layer that bends to send overflow to cheaper models when the frontier provider is saturated, rather than refusing service. It has an evaluation suite that catches regressions in pre-production, where

they are cheap to fix, rather than in user-facing rollouts where they are not. It has a rollback path that reverts a bad prompt or model release within seconds. It has tenant-scoped isolation so that one tenant's incident does not become every tenant's outage. It has trace stores that let engineers replay what production actually did when something goes wrong, rather than guessing. It has runbooks for the failure modes that have happened and for several that have not. It has a human-in-the-loop fallback that catches the cases the model is not confident about, before they become incidents. It has, in short, decided in advance where it will bend.

2.5 What This Book Cannot Pretend

It would be dishonest to draw exclusively from European workshops and European catastrophes as if the eighteenth century existed only in Europe. The same century saw the kingdoms of the West African coast — Asante, Dahomey, Oyo — operating sophisticated economies, taxation systems, and bureaucracies; the gold trade with Europe was substantially run on African terms in many decades. The Ethiopian highlands had centuries-old systems of land tenure and ecclesiastical administration. The Swahili coast linked African production into Indian Ocean trade networks that did not need European inventions to function. The Sokoto Caliphate, founded just at the turn of the century, would become one of the largest political entities in nineteenth-century Africa.

The painful fact is that the same century also saw the apex of the Atlantic slave trade, which destroyed African systems of order for the convenience of European systems of order, and whose consequences shape the continent's economies and societies to this day. A reader who carries the eighteenth-century workshop into the chapters of this book about African enterprise AI should also carry the awareness that the disciplines this book celebrates — measurement, standardisation, taxonomy — were sometimes the same disciplines that organised, with horrifying efficiency, the trade that destroyed African polities. Discipline without ethics is what makes the disaster larger, not smaller. The Pombaline cage saved lives; the slave-ship manifest, equally a triumph of standardisation, ended them.

The book's repeated emphasis on AI ethics architecture (Chapter 78), on sovereignty (recurring throughout the African perspective sections), on continental cooperation (the AfCFTA framing in the procurement chapter), and on cultural plurality in constitutional design (Chapter 45) is not decorative. It is the book's attempt to honour the eighteenth-century discipline of system-building while rejecting the eighteenth-century failure of moral framing.

The continent also offers its own analogues to Lisbon. Recurring failure modes of cross-border banking infrastructure under currency crisis, infrastructure disasters from harbour fires to munitions accidents, the great Sahel droughts: each has produced, like Lisbon, a generation of engineers and administrators who have learned, painfully, what graceful degradation actually requires. The Pombaline cage was a Portuguese invention. The disposition behind it is older than that, and is being practised today in offices in Lagos, Nairobi, Johannesburg, Cairo, and Kigali by people designing systems that will be tested by forces no one fully anticipates.

2.6 Where That Leaves Us

This chapter has paired two stories from the eighteenth century. The first — the workshop — demonstrates the patient construction of the disciplines that underwrite any durable enterprise: specialisation with clean interfaces, measurement that does not lie, standardisation that outlives the designer, instruments that report the truth of operations, and taxonomies that let practitioners coordinate. The second — the Lisbon earthquake of 1 November 1755 — demonstrates what happens when a system is optimised for the expected case with no graceful path for the alternative, and then, remarkably, demonstrates what a serious response to that failure looks like: emergency stabilisation, evidence collection, redesign with failure modes deliberately built in, and the engineering of a cage that bends so the building does not break. The first story is what the rest of this book is trying to teach. The second is the warning the book repeats, in different words, in nearly every chapter that follows — and the template, in Pombal’s recovery and the Pombaline cage, for what the response should look like when the warning is, inevitably, not heeded in time.

Two opening chapters; four dispositions a working discipline acquires. Representation. Observation. Naming. Resilience. The reader who has carried them this far is, on the book’s terms, ready for the technical material. The next chapter sets out a working taxonomy of the model families this discipline currently recognises — one Linnaean move, written in 2026, fully expecting to be revised by the practitioners who come next. The chapter after that places enterprise AI architecture inside the broader tradition of enterprise architecture from which it inherits most of its discipline. The chapters of Part II then assemble a six-layer reference architecture that the rest of the book hangs from. Every one of them is a piece of representation, founded on observation, organised by a vocabulary, and designed — explicitly — with the morning after Lisbon in mind.

Chapter 3

Fundamental AI Architectures: The Building Blocks

“Before discussing how to build with bricks, it is worth knowing what each brick is for.”

—a recurring observation in technical education

Chapter Overview

The remainder of this book discusses how AI components compose into enterprise systems. This chapter introduces the components themselves: the model families and architectural primitives that get composed. Readers familiar with the field may skim; readers from adjacent disciplines (engineering leadership, product, compliance, procurement) will find here the vocabulary used throughout the rest of the book.

Learning Objectives

- Distinguish the principal model families and their fit-for-purpose.
- Recognise the architectural primitives that compose into systems.
- Understand the input-output and economic shapes of each family.
- Place each family in the broader AI capability landscape.
- Use this vocabulary throughout the rest of the book without re-explanation.

3.1 Why This Chapter Exists

The two opening chapters that precede this one have argued, in different vocabularies, that a working discipline needs to represent honestly, observe directly, name carefully, and design for failure. None of that yet tells the reader *what enterprise AI actually contains*. The vocabulary the rest of the book will use — the model families, the architectural primitives, the tier structures, the composition patterns — is the subject of this chapter. It is a Linnaean move, in the language of the opening chapter: an attempt at a working taxonomy, fully expecting to be revised as new model classes arrive and the rack has to be redrawn.

The AI field has accumulated a vocabulary that is easy to use casually and easy to misuse precisely. “LLM”, “transformer”, “foundation model”, “embedding”, “classifier”, “agent” — these terms refer to specific things, but the things they refer to overlap, nest, and interact. An LLM *is* a foundation model; it *uses* a transformer architecture; it *produces* embeddings as a side effect; it *can power* an agent. Confusion at this layer makes the rest of the book harder to follow than it needs to be.

This chapter introduces each building block once, succinctly, with enough specificity that readers carry a clear mental model into later chapters.

3.2 The Map of Families

Figure 3.1 places the families on one map.

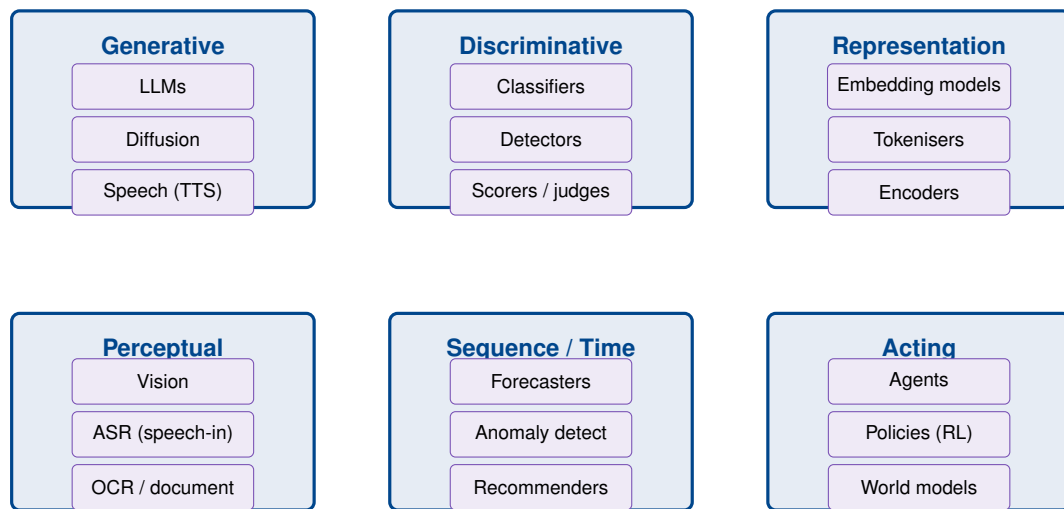


Figure 3.1: The six families of AI architectures used throughout this book. Generative produces content; discriminative classifies; representation encodes; perceptual converts world signals to structured form; sequence/time models temporal patterns; acting closes the loop to actions. Most production systems compose across families.

3.3 Generative Models

Generative models produce content: text, images, audio, video, code.

3.3.1 Large Language Models (LLMs)

LLMs predict the next token given a sequence of prior tokens. Trained on text at internet scale, they acquire fluent language, broad world knowledge, and emergent capabilities (reasoning, instruction-following, tool-use). They are the dominant building block discussed in this book.

- **Input.** A sequence of tokens (a prompt, possibly with multimodal content).
- **Output.** A sequence of tokens (the response).
- **Cost shape.** Input tokens (cheap) plus output tokens (expensive); output 3–5x input cost at frontier providers.

- **Examples.** Claude, GPT-4, Gemini, Llama, Mistral, DeepSeek.

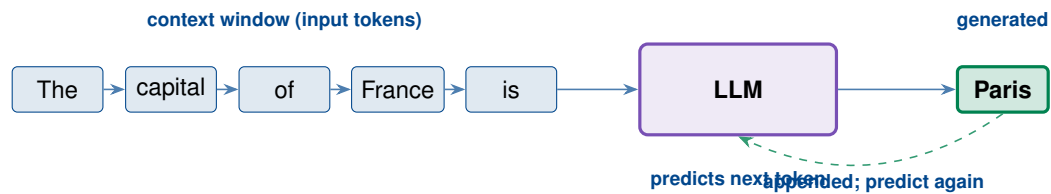


Figure 3.2: An LLM predicts one token at a time. The input context flows through the model; the model emits the most likely next token; the new token is appended; the cycle repeats until a stop condition. The whole “response” is the accumulated sequence of next-token predictions.

3.3.2 Diffusion Models

Diffusion models generate images, video, and increasingly audio by iteratively denoising a random noise pattern toward the target. They dominate image and video generation.

- **Input.** Text prompt (and optionally images, control signals).
- **Output.** Pixel data (image or video frames).
- **Cost shape.** Per-image or per-second; relatively predictable.
- **Examples.** Stable Diffusion, Flux, DALL-E, Midjourney, Sora.

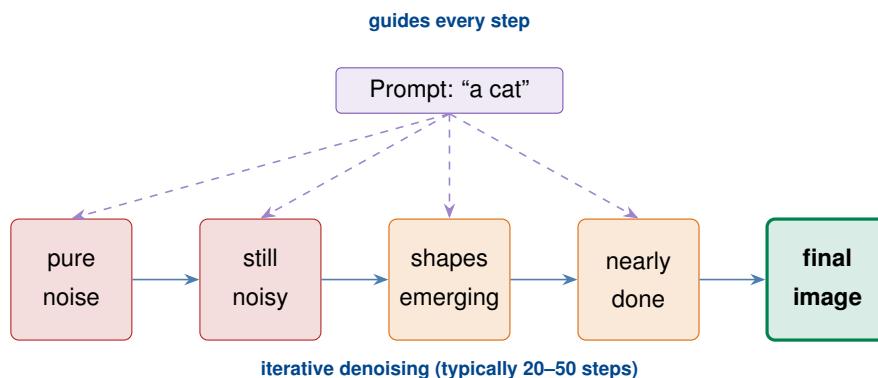


Figure 3.3: A diffusion model produces images by iteratively denoising. Starting from pure noise, each step removes a little noise toward the target image. The text prompt conditions every step. The number of steps trades quality for speed.

3.3.3 Speech Generation

Text-to-speech (TTS) models produce audio from text. Modern TTS produces natural prosody, emotional tone, and speaker identity.

- **Input.** Text, optionally with prosody hints and a reference speaker.
- **Output.** Audio waveform.
- **Cost shape.** Per-character or per-second of audio.

- **Examples.** ElevenLabs, Cartesia, OpenAI TTS, Whisper-adjacent generators.

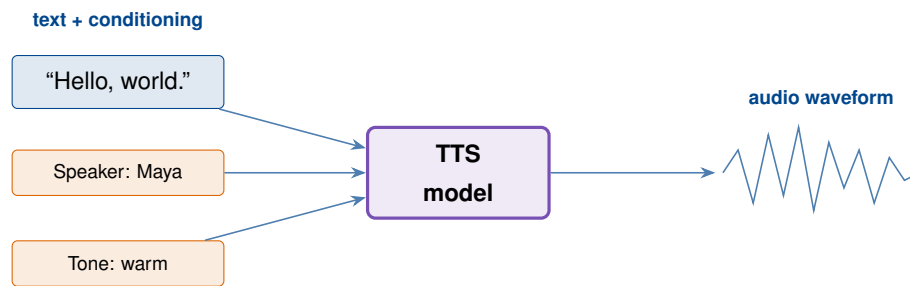


Figure 3.4: A text-to-speech model takes text plus optional conditioning (speaker identity, emotional tone, prosody hints) and produces a continuous audio waveform. Modern TTS is increasingly indistinguishable from human speech in casual listening.

3.4 Discriminative Models

Discriminative models map inputs to labels, scores, or decisions. They are smaller, cheaper, and more predictable than generative models.

3.4.1 Classifiers

Map inputs to a small set of categories. “Is this email spam?” “What is the sentiment?” “Which intent?” Production AI systems use classifiers extensively for routing, filtering, and guardrails.

- **Input.** Text, image, audio, structured features.
- **Output.** Class label plus confidence score.
- **Cost shape.** Cheap; small models; runnable on CPU often.

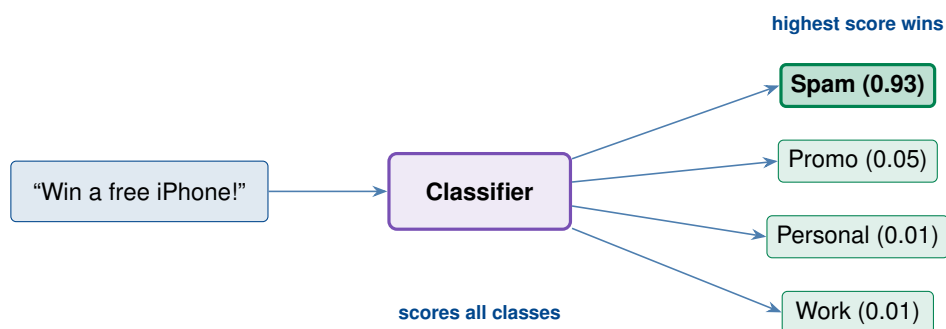


Figure 3.5: A classifier maps the input to a probability distribution over a fixed set of classes. The top-scoring class is the prediction; the score itself is the confidence. Cheap, fast, predictable — a workhorse of production AI.

3.4.2 Detectors

Locate objects or events within a larger signal. Object detection in images; event detection in time series; entity detection in text.

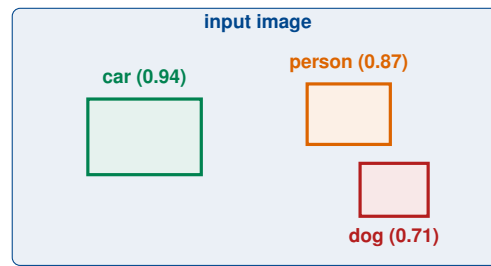


Figure 3.6: A detector locates instances within an input and reports their positions, classes, and confidences. Unlike a classifier (one label per input), a detector returns a variable-length list of detections.

3.4.3 Scorers and Judges

Produce a scalar score for an input. Used in retrieval (relevance score), evaluation (quality score), and ranking. “Judge models” built from LLMs evaluate other LLM outputs and have become central to production evaluation.

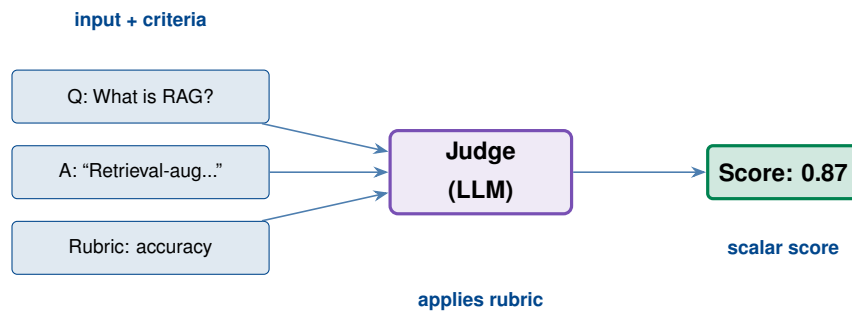


Figure 3.7: A scorer (here, an LLM-as-judge) takes the question, the candidate answer, and a rubric, and returns a scalar quality score. Modern production evaluation depends heavily on this pattern: cheap automated grading at scale.

3.5 Representation Models

Representation models convert inputs into dense vector representations suitable for similarity, retrieval, and downstream use.

3.5.1 Embedding Models

Map text (or other modalities) to fixed-length vectors. Semantically similar inputs produce nearby vectors. The foundation of vector search, RAG, recommendation, and clustering.

- **Input.** Text passage (or image, or other modality).
- **Output.** A vector of typically 256–3072 dimensions.
- **Cost shape.** Cheap per call; storage cost can dominate at corpus scale.
- **Examples.** OpenAI text-embedding-3, Cohere embed, BGE, E5, Voyage, Nomic.

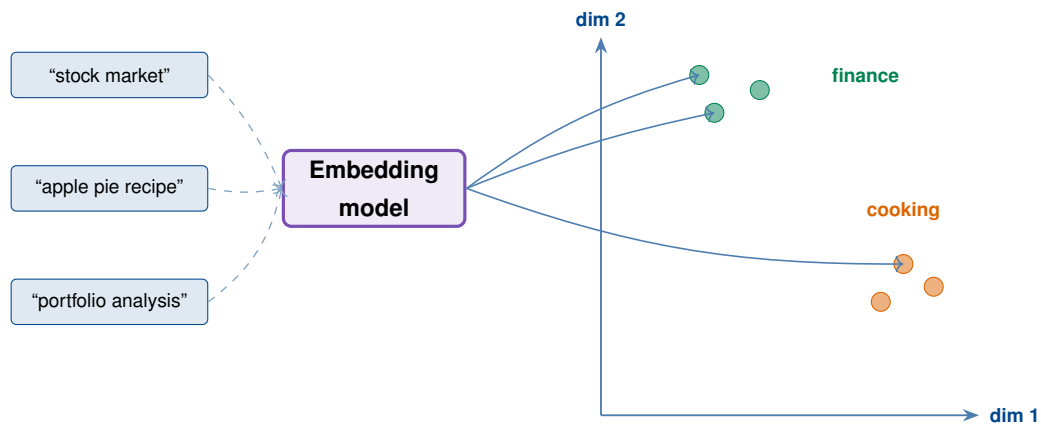


Figure 3.8: An embedding model maps text to vectors in a high-dimensional space (shown here in 2D for illustration). Semantically similar inputs land near each other; dissimilar inputs land far apart. “Stock market” and “portfolio analysis” cluster together; “apple pie recipe” clusters elsewhere. This geometry powers similarity search.

3.5.2 Tokenisers

Convert text to integer token sequences (and back). Specific to each model family. Tokeniser choices affect cost, multilingual support, and context length.

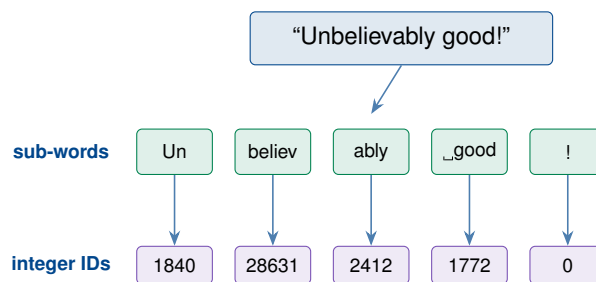


Figure 3.9: A tokeniser breaks text into sub-word pieces and maps each to an integer ID. “Unbelievably” splits into “Un”, “believ”, “ably” — a sub-word strategy that handles rare and compound words without exploding vocabulary size.

3.5.3 Encoders

Bidirectional models that produce contextualised representations of inputs but do not generate. BERT and its descendants. Used in classification, extraction, and as features for downstream models.

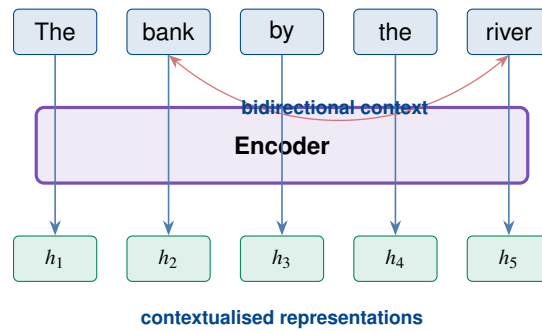


Figure 3.10: An encoder sees the whole sequence at once and produces one contextualised vector per token. Bidirectional context means “bank” near “river” is represented differently from “bank” near “account”. Encoders feed downstream classifiers, extractors, and similarity systems.

3.6 Perceptual Models

Perceptual models convert sensor signals from the world into structured form.

3.6.1 Vision Models

Image and video understanding: classification, object detection, segmentation, visual question-answering, OCR. Modern vision-language models (VLMs) integrate vision with LLM-style language capability.

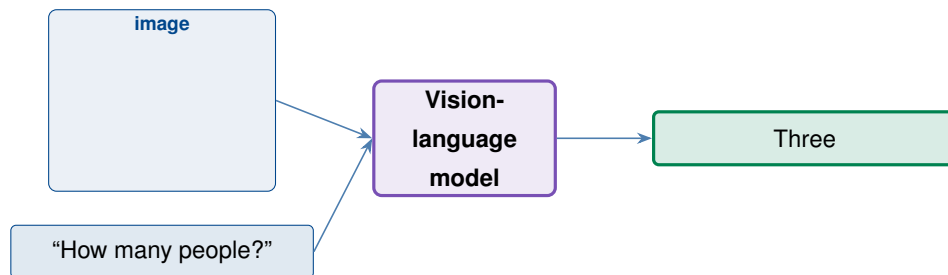


Figure 3.11: A modern vision-language model takes an image plus a text question and produces a text answer. The same architecture handles classification, captioning, visual Q&A, OCR, and chart understanding, varying only by the prompt.

3.6.2 Speech Recognition (ASR)

Convert audio to text. Modern ASR handles multiple languages, accents, code-switching, and adverse audio conditions. Examples: Whisper, Deepgram, AssemblyAI.



Figure 3.12: Automatic speech recognition takes an audio waveform and produces text. Modern ASR is multilingual, robust to noise, and runs in real time. The inverse of TTS; together they form the I/O layer of voice agents.

3.6.3 Document AI

Extract structured information from documents: layout-aware parsing, table extraction, form-field detection, signature recognition. Specialist systems and modern VLMs both serve.

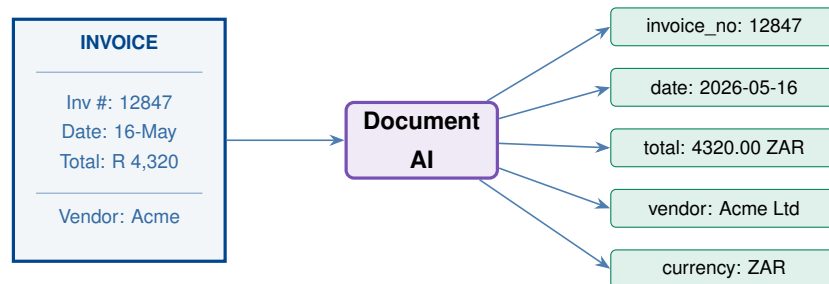


Figure 3.13: Document AI extracts structured fields from semi-structured documents like invoices, contracts, forms, and statements. The layout matters: position, font, and surrounding context inform what each piece of text represents.

3.7 Sequence and Time Models

Sequence and time models handle temporal patterns where ordering matters.

3.7.1 Forecasters

Predict future values from historical sequences. Classical (ARIMA, exponential smoothing), tree-based (XGBoost with temporal features), deep (LSTM, transformer), foundation time-series (Chronos, TimeGPT). See Chapter 73.

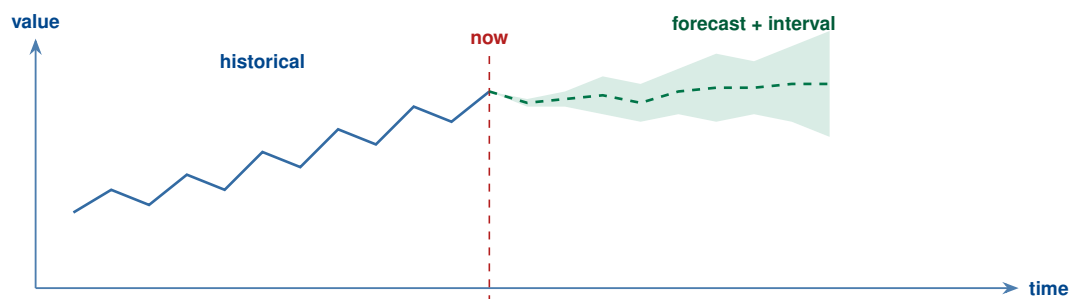


Figure 3.14: A forecaster takes a historical sequence and produces a predicted future trajectory along with a confidence interval. The interval matters as much as the point estimate: planning under uncertainty needs the band.

3.7.2 Anomaly Detectors

Identify deviations from expected patterns in sequences. Used in fraud detection, infrastructure monitoring, quality control.

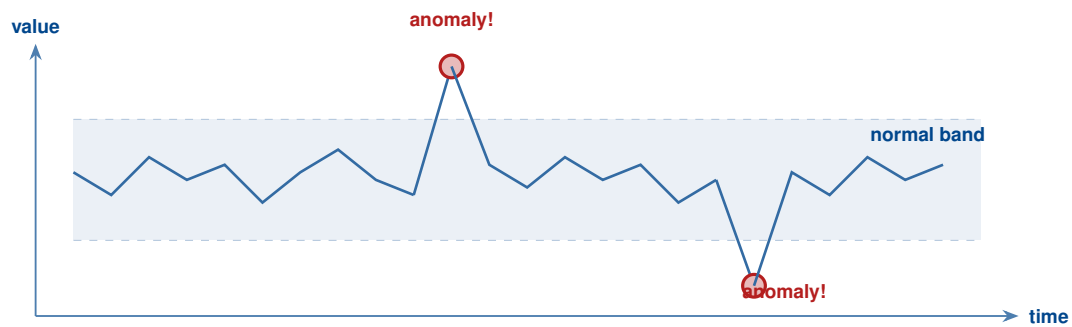


Figure 3.15: An anomaly detector learns a normal band for the signal. Points outside the band flag for investigation. The challenge is choosing the band tightly enough to catch real anomalies and loosely enough not to drown in false positives.

3.7.3 Recommenders

Predict items a user will engage with from historical behaviour. Distinct from forecasting; foundational to e-commerce, media, and content platforms.

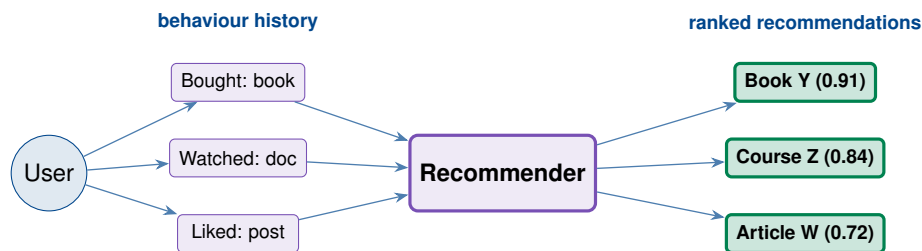


Figure 3.16: A recommender takes a user's behaviour history and produces a ranked list of items the user is predicted to engage with. The score is the predicted engagement probability; the ranking is what is presented.

3.8 Acting Models

Acting models produce actions in the world rather than (or in addition to) content.

3.8.1 Agents

Systems combining a language model, tools, and a control loop to take multi-step actions toward a goal. Discussed extensively from Chapter 18 onward. An agent is an architectural pattern, not a model family — it uses other families as components.

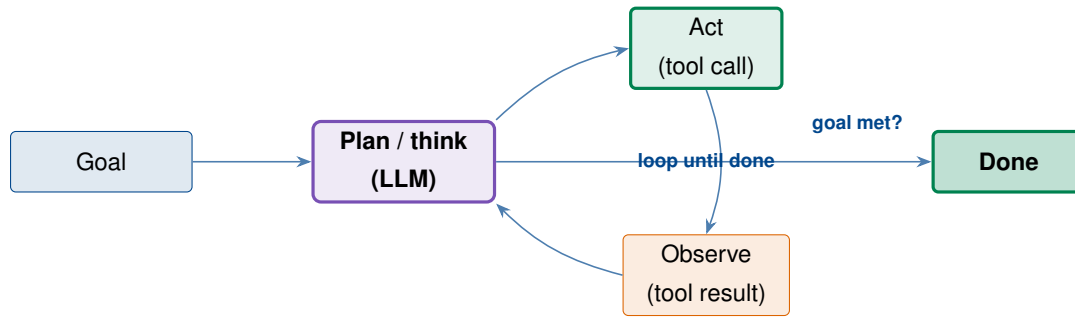


Figure 3.17: An agent control loop. Given a goal, the model plans, calls a tool, observes the result, and loops — updating its plan with new information — until the goal is met. The model is the cognition; tools are the limbs; the loop is the agency.

3.8.2 Reinforcement Learning Policies

Models trained to take actions that maximise a reward signal. Foundational to game-playing AI (AlphaGo, AlphaStar), robotics, and recommendation systems. Less prominent in language-AI than supervised approaches, though RLHF (Chapter 69) is a major exception.

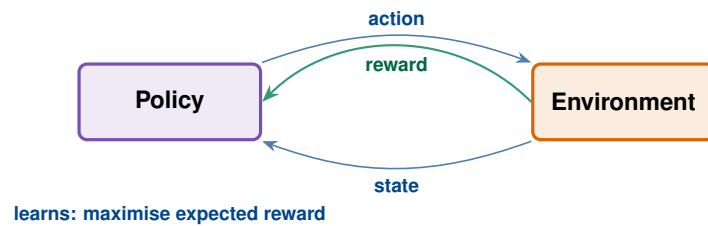


Figure 3.18: A reinforcement learning policy interacts with an environment in a loop: observe state, choose action, receive reward. Training adjusts the policy so that the actions taken accumulate higher reward over time. The reward signal is what defines the goal.

3.8.3 World Models

Predict consequences of actions in environments. Combined with planners to produce embodied AI. See Chapter 75.

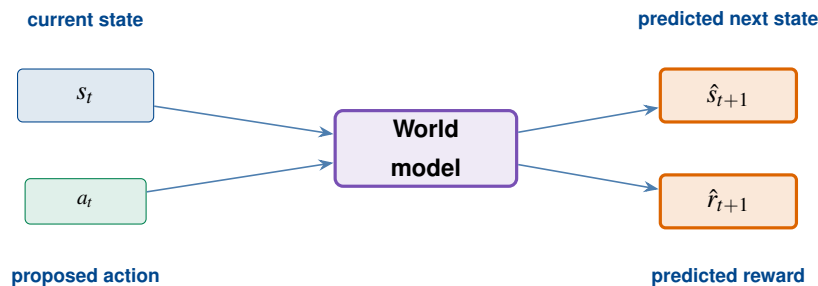


Figure 3.19: A world model predicts what happens next: given the current state and a proposed action, it predicts the next state and the resulting reward. Coupled to a planner, the world model enables “imagining” trajectories before acting in the real environment.

3.9 The Architectural Primitive: The Transformer

Most modern AI models named above — LLMs, modern vision models, modern speech models, modern embedding models, modern recommenders — share a common architectural primitive: the transformer.

The transformer’s defining innovation is attention: every position in the sequence can attend to every other position when computing its representation. This replaces the strict sequential processing of earlier recurrent networks and the local-only processing of convolutional networks.

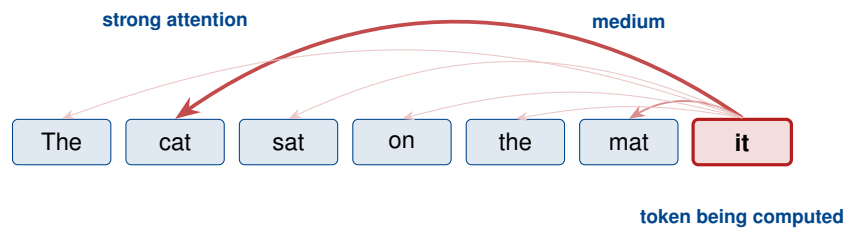


Figure 3.20: Attention in a transformer. When computing the representation of “it”, the model attends to every prior token, with weights it learned. Here “cat” gets strong weight (likely antecedent); “mat” gets some weight; other tokens less. This data-driven, position-agnostic weighting is the transformer’s central mechanism.

Variants matter at the engineering level (decoder-only for generation; encoder-only for representation; encoder-decoder for translation; mixture-of-experts for scale; mamba/state-space for long sequences) but the family resemblance is strong. A practitioner who internalises “transformer” as the building block carries useful intuition across modalities.

3.10 Foundation Models

A foundation model is a single model trained on broad data at scale, then adapted to many downstream tasks. The term spans modalities: LLMs are foundation models; large vision models are foundation models; large speech models increasingly are.

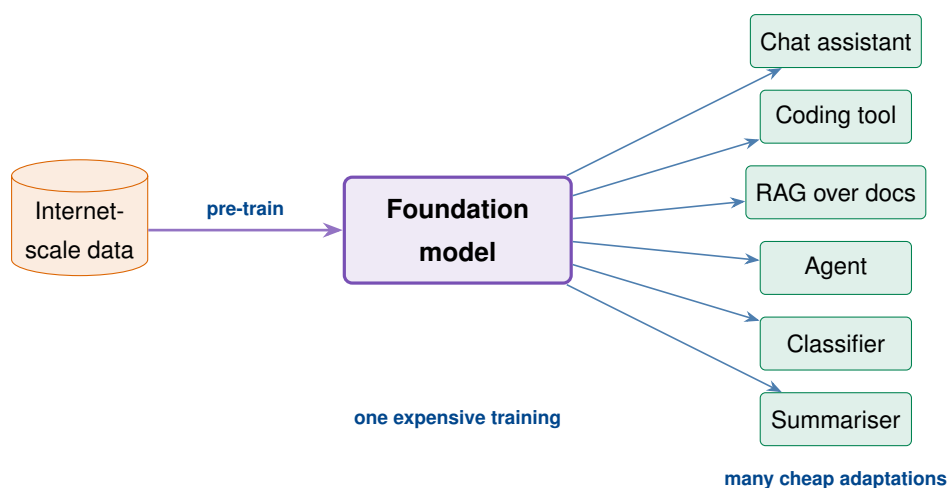


Figure 3.21: The foundation model pattern: one expensive pre-training run on broad data produces a model that adapts cheaply (via prompting, fine-tuning, or adapters) to many downstream applications. The economic significance is the amortisation: the training cost spreads across hundreds of products.

The architectural significance is twofold:

- One pre-trained model amortises its training cost across many applications.
- Downstream adaptation (prompting, fine-tuning, adapters) is cheaper than training from scratch.

The rest of this book takes foundation models as a given starting point. The architecture is what an organisation builds on top of them.

3.11 Sizes and Tiers

Within each family, models come in size tiers.

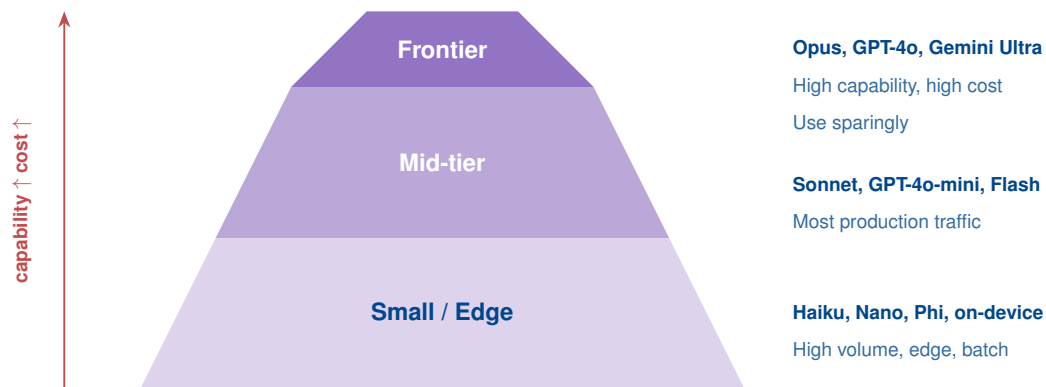


Figure 3.22: Size tiers in modern foundation models. Frontier tier is small in count but consumes most of the cost when used; mid-tier handles most production traffic; small and edge tiers serve high-volume, latency-sensitive, or offline workloads. Routing across tiers is one of the most impactful architectural levers.

- **Frontier.** The largest, most capable, most expensive models (Claude Opus, GPT-4o, Gemini Ultra). Reserved for tasks where capability matters more than cost.
- **Mid-tier.** The cost-effective workhorses (Claude Sonnet, GPT-4o-mini, Gemini Flash). Most production traffic.
- **Small.** Specialist or efficiency-tier models (Haiku, GPT-4o-nano, Gemini Nano, Phi). High-volume, low-risk workloads. Edge deployment.

Multi-model routing (Chapter 38) takes advantage of the tier structure: route by need.

3.12 Composition

Production systems are rarely a single model. They compose families:

- RAG: embedding model plus LLM plus optional reranker.
- Voice agent: ASR plus LLM plus TTS.
- Document processing: vision/OCR plus LLM plus classifier.
- Compound AI: generator plus verifier plus retriever plus tools (Chapter 44).

- Agent: LLM plus tool runtime plus memory plus control loop.

Figure 3.23 shows the RAG case, the most ubiquitous composition in production.

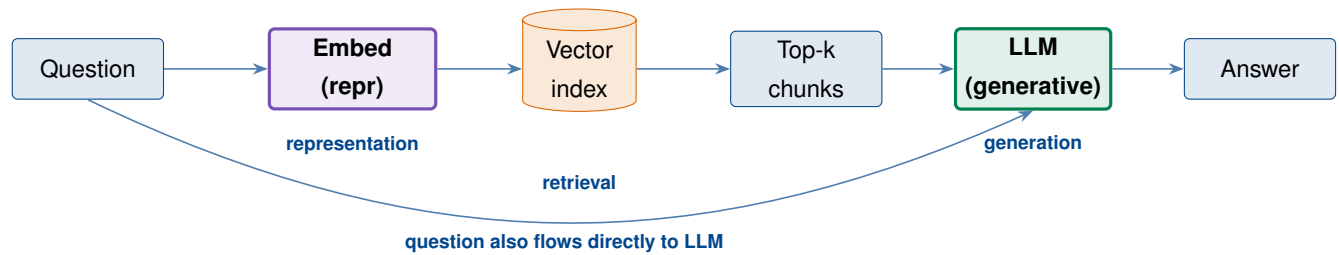


Figure 3.23: RAG composition. The question is embedded by a representation model; the vector index returns relevant chunks; the LLM receives both the question and the retrieved chunks and generates a grounded answer. Three model families collaborate; no single model could deliver this on its own.

The rest of this book is about how these compositions are architected, operated, and governed.

3.13 Regional Perspective: Africa

The fundamentals apply universally, but adoption emphases differ.

3.13.1 Smaller-Model Preference

African unit economics push earlier and harder toward small and mid-tier models. The frontier reserved for high-value tasks.

3.13.2 Multilingual Foundations

African deployments lean on models with substantial multilingual coverage. Foundation model selection criteria weight language coverage heavily.

3.13.3 Open Weights as Sovereignty Lever

Open-weight foundation models (Llama, Mistral, Qwen, DeepSeek, Falcon) enable self-hosted deployment for sovereignty and cost. Falcon’s UAE origin and Inception’s regional models are notable.

3.13.4 Composition for Cost

Composition (compound AI, RAG, classifier cascades) reduces reliance on frontier models. African deployments adopt compound architectures earlier in their maturity than Northern equivalents.

3.14 What to Carry Forward

This chapter introduced the six families of AI architectures: generative, discriminative, representation, perceptual, sequence/time, and acting. The transformer is the shared architectural primitive across most modern instances. Foundation models are the trained artefacts on which most enterprise architecture builds. Tier structures (frontier, mid, small) inform routing decisions. Composition is the dominant production pattern. The rest of the book builds on this vocabulary.

A taxonomy of components, however, is not yet a discipline. The Linnaean move places things on a rack; it does not yet say how the rack is assembled, who is responsible for keeping it current, or what decisions an organisation has to make as new components arrive. Those questions belong to a different and older discipline — enterprise architecture — which has been answering versions of them for the larger software estate for forty years, and which has to be extended in specific places to handle the substrate this chapter has just described. The next chapter sets the two side by side: the conventional enterprise-architecture frameworks the AI architect inherits, and the genuinely new concerns the AI substrate creates that those frameworks have to grow to handle.

Chapter 4

Enterprise Architecture and Enterprise AI Architecture

“Architecture is not the application of a method to a problem. It is the discipline of seeing the problem clearly enough that the method becomes obvious.”

—a recurring observation in enterprise-architecture practice

Chapter Overview

Enterprise Architecture is a discipline that grew out of the corporate IT departments of the 1980s and was codified through the 1990s and 2000s into frameworks — TOGAF, Zachman, FEAF — that describe how a large organisation reasons about its information systems, its data, its applications, and the technologies that underpin them. Enterprise AI Architecture is the same discipline, applied to a substrate that behaves differently in important ways: outputs that are not deterministic, models whose lifecycle is independent of the code that calls them, vendor concentration at the foundation layer, and a fast-moving evaluation problem that traditional software architecture did not have to solve. This chapter sets the two side by side. It describes what the AI architect inherits from the enterprise architect, what is genuinely new, and what the existing frameworks need to be extended to handle. The reader who has done conventional enterprise architecture will recognise most of what follows; the reader who has not will find here the missing context that explains why so many of the patterns later in the book are organised the way they are.

Learning Objectives

- Recognise Enterprise Architecture as a discipline with its own frameworks, vocabulary, and lifecycle.
- Identify the four traditional domains of EA (Business, Data, Application, Technology) and what they mean.
- Distinguish what Enterprise AI Architecture inherits from EA from what it has to invent.
- Understand why AI workloads stress conventional EA assumptions and where the frameworks have to bend.
- Use the chapter as the conceptual baseline for the architectural patterns and disciplines that follow.

4.1 Where Enterprise Architecture Came From

The previous chapter laid out the components a contemporary AI architecture is assembled from. The question this chapter takes up is the older one of how an organisation reasons about such an architecture in the first place — how it decides what to build, what to buy, what to integrate with what, who owns which decisions, and how the resulting structure will be governed as it evolves. That question is not new. The discipline that has been answering it for the broader software estate is called Enterprise Architecture, and it is the discipline the AI architect inherits, with substantial extensions, when AI workloads arrive at scale.

Enterprise Architecture, as a named discipline, is younger than most of its practitioners assume and older than most of its critics give it credit for. The origin most often cited is John Zachman’s 1987 paper for the IBM Systems Journal, “A Framework for Information Systems Architecture,” which proposed a thirty-cell matrix — six perspectives crossed with six interrogatives — as the working schema for everything an enterprise would want to say about its own information systems. The Zachman matrix did not, on its own, tell anyone how to do architecture. It told them what an architecture would have to describe in order to be complete: who, what, when, where, why, and how, viewed from the planner’s perspective, the owner’s, the designer’s, the builder’s, the subcontractor’s, and the enterprise’s. It was a taxonomy of the architect’s deliverables, not a method for producing them.

The Open Group, founded in 1996 as the merger of the Open Software Foundation and X/Open, picked up the methodological side of the problem. The Open Group Architecture Framework — TOGAF, published in 1995 and revised continuously since — is the discipline’s most widely-adopted method. Its Architecture Development Method (ADM) is a phased cycle, described in the agentic-AI context in Chapter 30, that takes an organisation from a preliminary readiness assessment through an architecture vision, the four traditional architectural domains, an opportunities-and-solutions phase, migration planning, and implementation governance, into ongoing change management. By the late 2000s, TOGAF certification was a routine line on the CV of any architect working in a Fortune 500 or its equivalent.

Other frameworks have served different constituencies. The Federal Enterprise Architecture Framework (FEAF), published by the United States Office of Management and Budget in 1999, is the public-sector equivalent. The Department of Defense Architecture Framework (DoDAF) addresses the military and intelligence communities. The Gartner Methodology, less a framework than a consulting offering, has been influential in the commercial sector. The C4 model, popular among software engineers, addresses a smaller scope (system-level architecture) but uses many of the same intellectual moves. The frameworks differ in their vocabulary and their level of formality. They agree on the underlying claim: that an enterprise's information systems are too large, too interdependent, and too consequential to be designed without an explicit discipline of describing, planning, and governing them.

4.2 The Four Domains

Most enterprise-architecture frameworks distinguish four architectural domains. The names vary slightly between frameworks; the substance is broadly consistent.

The **Business Architecture** describes what the organisation does — its capabilities, its value streams, its processes, its organisational structure, and the relationships among them. A capability is a stable description of *what* a part of the business can do, independent of how it currently does it. The capability “settle a customer claim” may, in 2010, have been delivered by a call centre with a paper-based workflow; in 2025, by a digital self-service portal; in 2027, by an agentic system with human escalation. The capability is the same; the implementation changes underneath. Business architecture is what gives the rest of the architecture something stable to hang from.

The **Data Architecture** describes what the organisation knows and where it lives. The data architect's deliverables include the conceptual, logical, and physical data models; the master-data identifiers; the data-quality regime; the data-governance policies; the integration patterns between systems of record and systems of engagement; the analytics and reporting structures. In a large enterprise the data architecture is usually older and stranger than the business architecture, because data accumulates faster than anyone reshapes it. Mergers leave duplicate customer masters; departmental databases survive multiple platform migrations; the “one true list of products” is, in practice, six lists that mostly agree.

The **Application Architecture** describes the software the organisation runs. The application portfolio, the integration patterns, the application lifecycle policies, the buy-versus-build decisions, the vendor relationships, the deprecation roadmap. The application architect's distinctive concern is the relationship between applications — the points at which they integrate, the protocols they use, the data they share. A well-architected portfolio of applications can be evolved one piece at a time. A poorly-architected one is held together by undocumented integrations that fail when any one piece is replaced.

The **Technology Architecture** describes the infrastructure underneath everything: the compute, the storage, the network, the operating systems, the cloud regions, the security perimeters, the identity systems. Technology architecture is the most rapidly evolving of the four domains in any decade;

what was a server room in 1995 is a cloud account in 2025 and will be something else in 2035. The discipline of technology architecture is largely about insulating the upper domains from this churn: enabling the business architecture to remain stable while the technology underneath it is replaced repeatedly.

The relationship between the four domains is hierarchical but not strict. The business architecture is logically prior; the data architecture supports it; the application architecture instantiates it; the technology architecture hosts it. Decisions made at any layer constrain the layers above; decisions made at higher layers create requirements for the layers below. The architect's job, day to day, is to keep the four in productive tension — to prevent business changes from violating data invariants, to prevent application choices from leaking into business processes, to prevent technology decisions from foreclosing future business options.

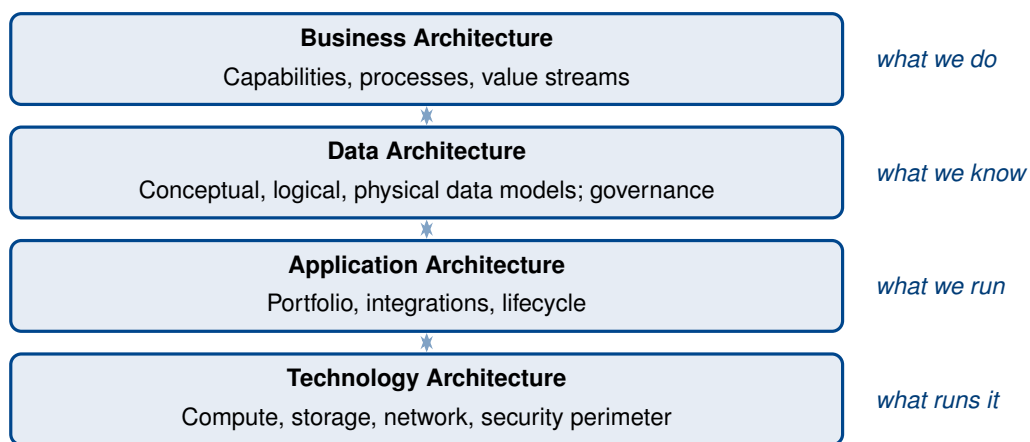


Figure 4.1: The four traditional domains of Enterprise Architecture. Decisions at higher layers create requirements for the layers below; decisions at lower layers constrain the layers above. The architect's discipline is to keep them in productive tension rather than letting one dominate the others.

4.3 What Enterprise Architecture Is For

A reader from outside the discipline could be forgiven for thinking that enterprise architecture is a paperwork exercise. The output of a TOGAF engagement is, on its face, a great quantity of diagrams, principles documents, building-block catalogues, and migration plans. A cynic could conclude that the deliverable is the documentation itself.

The cynic would be wrong, but only just. The point of the documentation is not the documentation. The point is the decisions the documentation enables. A large enterprise makes architectural decisions every week — which system to extend, which vendor to renew, which capability to build, which piece of legacy to retire — and the cost of making those decisions in isolation, without a coherent picture of how each interacts with the others, is enormous. Enterprise architecture is the discipline of having that coherent picture available when the decisions are being made.

The benefits are easier to see in retrospect than in advance. The organisation that adopted a coherent data architecture in 2015 is the organisation whose AI projects in 2025 can find the data they need. The organisation that did not is the one whose AI projects spend most of their first year reconciling six

versions of the customer master. The benefits of good architecture are mostly visible as the absence of avoidable problems — and the absence of a problem is harder to notice, and harder to credit, than its presence.

Done badly, of course, enterprise architecture becomes exactly the paperwork exercise its critics describe: ivory-tower diagrams that nobody refers to, principles that the architects themselves do not follow, review boards that exist to delay decisions rather than improve them. Most organisations that have tried to do EA have, at some point, done it badly. The discipline survives because the alternative — a large organisation making consequential decisions with no architectural memory — is worse.

4.4 What Enterprise AI Architecture Inherits

When AI workloads arrive in a large organisation, almost all of the existing enterprise-architecture apparatus is still useful. The Business Architecture, which describes the capabilities the organisation has and the value streams that deliver them, is exactly the right place to start thinking about which capabilities AI should augment or transform. The Data Architecture, with its data-quality regimes and master-data identifiers, is the foundation on which retrieval-augmented generation, model fine-tuning, and evaluation datasets all depend. The Application Architecture, with its integration patterns, is what AI systems will integrate *with*. The Technology Architecture, with its cloud regions and security perimeters, is what AI systems will run *on*.

The lifecycle methods inherit too. The TOGAF ADM, adapted as in Chapter 30, gives the AI programme its phased structure — the work of moving from a vision through architectural commitment to deployment and continuous change management. The architectural governance framework, described in Chapter 31, gives the AI programme its decision-making system — principles, reference architectures, a review board, decision records, exception management, metrics. None of this is invented for AI. All of it is borrowed from the enterprise-architecture tradition and applied to the new workload.

The architectural principles inherit as well, sometimes with sharper application. *API-first* matters more, not less, when the components being integrated include foundation models from vendors whose internal designs are opaque. *Security by design* matters more, not less, when the systems being secured can be manipulated through their training data. *Buy versus build* is more contested, not less, when the build option involves training a billion-parameter model and the buy option creates vendor concentration risk. Cloud-native, reuse over rebuild, separation of concerns, modular composition: each of these survives the transition to AI architecture, sometimes with new emphasis.

What this means in practice is that the enterprise architect who has worked at a large bank or insurer or government department for fifteen years and is now being asked to do AI architecture is not starting over. They are extending a discipline they already know. The new work is real, but most of it is the new work; the foundations are stable.

4.5 What Enterprise AI Architecture Has to Invent

The foundations are stable. The frontier is not. The places where AI workloads stress conventional enterprise architecture, and require genuinely new architectural thinking, are worth being explicit about.

4.5.1 Non-Deterministic Outputs

Conventional software architecture assumes that, given an input, a function produces a defined output. A correct program produces the right answer; a buggy one produces a defined wrong answer; either way, the relationship between input and output is reproducible. The architect's job is to design systems whose behaviour is specified and tested.

AI workloads break this assumption. A large language model, given the same input prompt twice, may produce different outputs — and the difference is not a bug to be fixed. It is a property of the architecture. The architectural response is a discipline of working with distributions of outputs rather than single outputs: evaluation harnesses that score across many samples, confidence signals that quantify uncertainty, fallback paths for low-confidence cases, human review for outputs whose stakes are high. None of this exists in conventional EA, because conventional software did not need it.

4.5.2 Three Lifecycles, Not One

Conventional software has one lifecycle: the code is written, tested, deployed, operated, and eventually retired. Conventional EA tracks this lifecycle, manages its dependencies, and plans its evolution.

AI workloads have three lifecycles, mostly independent. The *code* lifecycle is the conventional one: the orchestration, the integrations, the user interface. The *model* lifecycle is different: a vendor releases a new version of a foundation model, the customer's tested behaviour may shift overnight, and the code did not change. The *data* lifecycle is different again: the retrieval corpus grows, the embedding model changes, the prompt patterns get updated, and the behaviour shifts without any deployment in the conventional sense. An EA practice that tracks only one of these lifecycles will be surprised by the other two.

4.5.3 Vendor Concentration at the Substrate

Conventional enterprise architectures could, with effort, avoid vendor lock-in at most layers. Multiple cloud providers offer comparable compute. Multiple database vendors offer comparable storage. Multiple identity-management vendors offer comparable IAM. The application architect could, with discipline, design for portability.

The AI substrate is different. As of 2026, frontier model capability is concentrated in a small number of providers — OpenAI, Anthropic, Google, and a handful of open-weight alternatives that are technically capable but operationally more demanding. The architectural choice is not which of many providers to use; it is whether to use one of the three big ones, several of them with abstraction in between, or to operate open-weight models in-house and accept the operational cost. This is a different shape of decision from conventional EA, and the consequences (vendor pricing power, regulatory exposure, sovereignty constraints) are different too.

4.5.4 Evaluation as a Discipline

Conventional software is tested by writing test cases: given this input, expect this output. The discipline is mature. The infrastructure is mature. The architect's role in testing is mostly to ensure that the test pyramid is sensible and the coverage is sufficient.

AI workloads need a different testing discipline. There is no single right answer to score against. The evaluation has to score on a distribution of inputs against a notion of acceptable behaviour, with the criteria for acceptability themselves uncertain and contested. Evaluation harnesses, judge models, rubric-based scoring, red-team adversarial tests, drift detection, A/B comparison across model versions: all of this is a parallel discipline to conventional testing, and the enterprise architect responsible for AI workloads has to know it as well as they know unit testing.

4.5.5 Agentic Autonomy

Conventional applications do what they are asked. They have no agency in the philosophical or operational sense; they react to inputs.

AI agents are different. An agentic system can decide which tools to call, in what order, with what inputs; it can revise its plan as it executes; it can fail in ways that involve taking real-world actions before anyone has reviewed them. The architectural concerns that arise — autonomy boundaries, human-in-the-loop checkpoints, reversibility of agent actions, audit trails for agentic decisions — are not concerns conventional EA had to address. The whole discipline of agentic architecture, described in this book's Part IV, is the response.

4.5.6 The Data-Model-Prompt Triad

Conventional software has code; data sits in databases that the code reads and writes. The relationship is asymmetric: the code is the active component, the data is passive.

AI workloads have a more symmetric structure. The behaviour of the system depends, in roughly equal measure, on three components: the model (foundation or fine-tuned), the data it retrieves over (the corpus, the embeddings, the metadata), and the prompt (the instructions and the context). A change to any of the three changes the system's behaviour. None of them is purely passive. The architectural discipline that has to emerge is one that treats the three together — versioning them together, evaluating them together, deploying them together, observing them together — and that does not assume that the code is the privileged locus of architectural decisions.

4.6 Where the Two Disciplines Collide

The collision points between conventional EA and AI architecture are where most of the organisational arguments happen. A few worth naming.

Model governance versus application governance. Conventional EA has a mature application-governance process: a system goes through architecture review, security review, data-protection impact assessment, and operational readiness review before it is deployed. AI systems are subject to all of these and several more: model risk classification, bias and fairness review, transparency

assessment, ongoing-performance monitoring. The new reviews do not replace the old ones; they sit alongside them. The organisations that try to fit AI governance into the existing application-governance process under-govern AI risk; the ones that build a parallel structure end up duplicating work and creating bureaucratic confusion. The mature answer is integration — shared intake, parallel specialist review where needed, common reporting — and it takes years to build.

MLOps versus DevOps. Conventional EA has an opinion on how software is deployed: pipelines, environments, release cadences, rollback procedures. AI systems use the same DevOps tooling and add MLOps concerns on top: model versioning, training-data lineage, evaluation gates, model registry, drift detection. The two practices share infrastructure but have different rhythms — model retraining runs on a different cadence from code releases, evaluation gates have different acceptance criteria from integration tests, model rollback has different operational implications from code rollback. The architectural discipline is to design pipelines that handle both rhythms without forcing them into the same shape.

AI principles versus general architecture principles. The general architecture principles inherited from EA (API-first, security by design, cloud-native, reuse over rebuild) are necessary but not sufficient for AI workloads. New principles are needed: human-in-the-loop for consequential decisions, transparency about model use, fairness across protected attributes, accountability for AI-mediated outcomes. The principles do not contradict each other; they layer. The architect's job is to make sure that the AI-specific principles are present in every relevant decision without losing the general ones that have served the enterprise for decades.

4.7 What the Enterprise Architect Has to Add

For a practising enterprise architect moving into AI architecture, the practical question is what skills and knowledge to add to existing practice. The honest answer is that the additions are substantial but bounded.

A working knowledge of the model-family taxonomy from Chapter 3 — enough to understand the trade-offs between model types, the distinction between a 7B model and a 70B one, the difference between fine-tuning and retrieval-augmented generation, the operational shape of inference at scale. Not enough to design a new model architecture; enough to evaluate the architectural choices of teams that are.

Familiarity with the evaluation discipline. The vocabulary of evaluation suites, judge models, rubrics, red-teaming, drift detection. The ability to distinguish a serious evaluation programme from one that is going through the motions.

An understanding of the vendor landscape. Which providers offer which capabilities, on what commercial terms, with what data-residency options, with what known outage patterns, with what release cadences for their underlying models. This part of the knowledge dates fastest and has to be refreshed continuously.

Comfort with the patterns specific to AI workloads. RAG, agents, tool use, human-in-the-loop, model

routing, evaluation harnesses, the rest of Part III of this book. Not encyclopaedic knowledge; enough to recognise which pattern fits which problem and which patterns compose with which others.

Awareness of the regulatory and ethical landscape. The EU AI Act, sectoral AI regulation in financial services and healthcare, the developing standards landscape, the practical implications of bias and fairness audits. The principles that fall out of these regulations belong in the architecture's principle set; the obligations they create belong in the governance machinery.

None of this is a fundamentally different discipline. It is the same enterprise architecture, with a substantial extension to handle a new and distinctive substrate.

4.8 The African Perspective

Enterprise Architecture as a discipline carries the assumptions of the organisations that codified it: large, multinational, well-capitalised enterprises with mature IT departments and the institutional capacity to invest in long-horizon programmes. African enterprises operating under different constraints inherit the discipline's machinery without inheriting all of its assumptions.

The four domains apply. The frameworks apply. The principles apply. What changes is the calibration. A pan-African bank operating across twelve jurisdictions has data-sovereignty obligations that a US or European bank does not. A South African mining house has integration requirements with legacy operational systems that no TOGAF case study has ever described. A Nigerian fintech operating on the rails of a national payment system has architectural constraints that the textbooks do not anticipate. The frameworks bend to fit; the bending is part of the architect's work.

The AI extensions are sharper. Vendor concentration is more consequential when the foreign exchange to pay vendor invoices is itself a constrained resource. Open-weight models become not a philosophical preference but an architectural necessity in workloads that have to operate at price points the frontier vendors do not serve. Sovereignty considerations interact with model-residency in ways that influence which providers can serve which workloads at all. The architecture review board's job is, in these contexts, to make decisions that an architect in a less constrained environment never has to make.

The optimistic reading of the position is that African enterprises, building AI architecture in 2026 against tight constraints, may produce architectural patterns that are more sober and more durable than those produced by well-funded enterprises elsewhere. Constraint, as the engineering proverb has it, is the friend of design. The patterns developed under genuine resource pressure tend to survive better than the patterns developed under abundance. The African enterprise architect, doing AI architecture today, is not catching up; they are working on a frontier that the rest of the field will eventually have to confront.

4.9 Where That Leaves Us

Enterprise AI Architecture is Enterprise Architecture, extended. The frameworks, the lifecycle methods, the governance machinery, the principles, the four-domain structure — all of it carries forward.

The extensions are real: non-determinism, multiple lifecycles, vendor concentration, evaluation discipline, agentic autonomy, the data-model-prompt triad. The collisions between the existing and the new are where most of the organisational arguments will happen for the next several years. The work the enterprise architect has to do is to absorb the extensions, navigate the collisions, and apply the combined discipline to the architectural decisions their organisation actually faces.

A discipline-level framing, however, is still abstract. The chapter that follows turns the framing into something more concrete: what an actual working enterprise AI system looks like in production, what distinguishes it from consumer AI and research AI, and what the four constraints are that make it the distinctive thing it is. The reader who has carried the EA inheritance forward will read that chapter as a specification of the artefact the rest of the book is then going to architect.

Chapter 5

Enterprise AI Architecture: The Simplified Connected View

“Every architecture diagram is a lie. The useful ones are lies that show the connections.”

—a recurring observation in enterprise architecture

Chapter Overview

Enterprise AI architecture does not exist in isolation; it sits inside a broader enterprise architecture that includes strategy, business processes, data platforms, applications, technology, integration, security, and operations. Most treatments of AI architecture focus on the AI layers alone, as though the model and its retrieval pipeline float in a vacuum. This chapter presents the opposite view: a simplified, connected map of the full enterprise architecture stack — eleven layers, from strategy at the top to operations at the bottom, with cross-cutting governance concerns running alongside — and it shows where AI touches each one. The map is deliberately simplified: each layer is reduced to its essential components, so that the connections between them are visible rather than buried in detail. Its purpose is orientation. A reader who carries this map through the rest of the book will always know where a given chapter’s subject sits in the enterprise, what it depends on, and what depends on it.

Learning Objectives

- See the full eleven-layer enterprise architecture stack and where AI connects to each layer.
- Understand the dependencies that run between the layers — what each layer needs from the one below it.
- Recognise the cross-cutting governance concerns that apply across all eleven layers.
- Map each layer to the chapters of this book that develop it in depth.
- Use the connected view as a standing orientation device for the rest of the book.

5.1 The Connected View

Figure 5.1 shows the eleven-layer simplified connected view. The layers run from the most strategic at the top to the most operational at the bottom, and each depends on the ones below it. The right-hand column names the cross-cutting governance concerns that apply across all layers. The feedback arrow at the bottom indicates that operations feed back into strategy, making the whole stack a continuous loop rather than a one-way cascade.

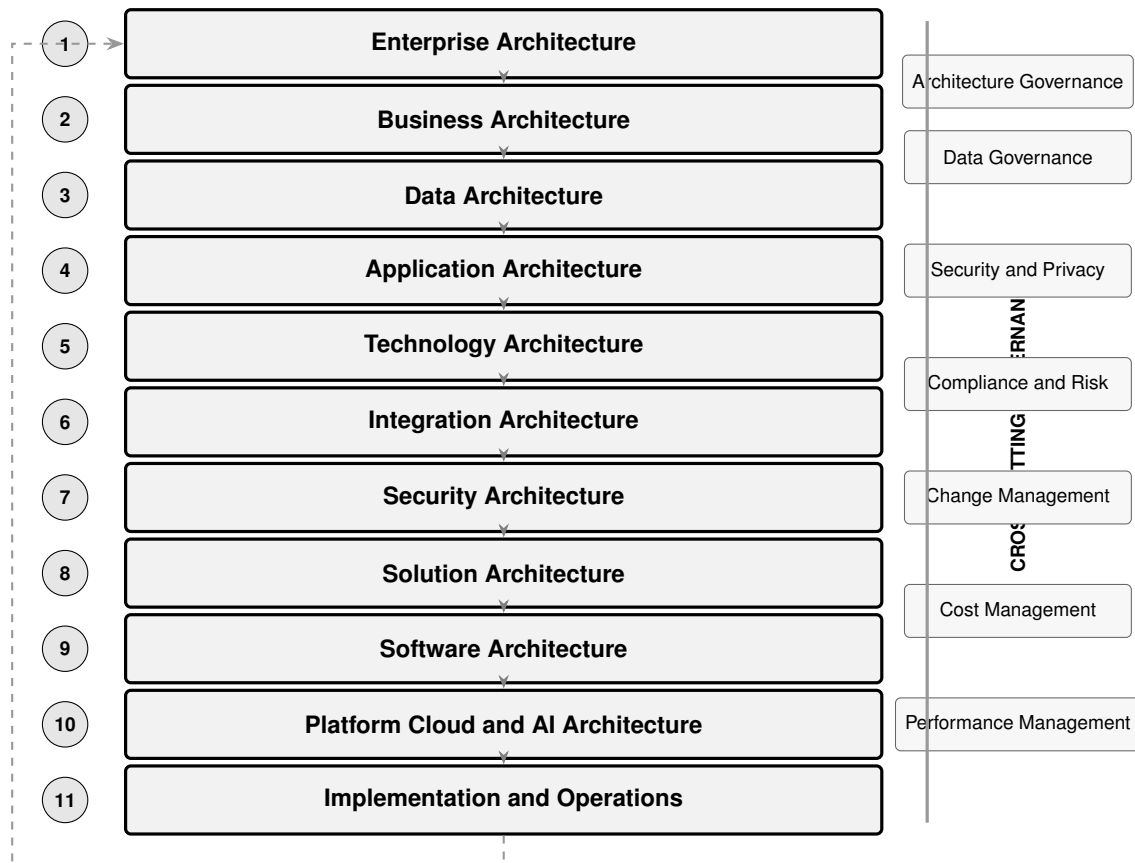


Figure 5.1: The eleven-layer simplified connected view of enterprise AI architecture. Each layer depends on those below it; cross-cutting governance concerns run alongside all eleven; operations feed back into strategy in a continuous loop.

The eleven layers are not eleven independent concerns; they are a connected stack in which each layer depends on the ones below and constrains the ones above. An AI system that works well at layer ten (the platform) but ignores layer two (the business processes it serves) is a technically excellent system that produces no business value. A system that works well at layer nine (the software) but ignores layer seven (security) is a capable system that is vulnerable. The connections are the point of the map, and the sections that follow develop each layer and its connections.

5.2 Layer 1: Enterprise Architecture

Enterprise architecture sits at the top of the stack because it is where strategy, governance, standards, roadmaps, and outcomes are defined. It is not an AI concern specifically; it is the concern of the enterprise as a whole, and AI architecture inherits its direction from it.

The components of this layer are the enterprise strategy that sets the direction, the governance structures that set the decision-making process, the architectural standards and principles that constrain how systems are built, the roadmaps that sequence the work, and the outcomes framework that defines what success looks like. AI architecture connects to this layer by receiving its mandate — the business problems worth solving, the investment priorities, the risk appetite — and by reporting its outcomes back to it. An AI programme that is not connected to enterprise architecture is a programme that sets its own direction, which is another way of saying it has no direction the enterprise endorsed.

The chapters that develop this layer include the governance framework (Chapter 31), the alignment with enterprise-architecture frameworks (Chapter 41), and the governance operating model (Chapter 104).

5.3 Layer 2: Business Architecture

Business architecture translates enterprise strategy into the business capabilities, processes, business rules, KPIs, value streams, and stakeholder structures that the technology layers must serve. It is the layer that answers the question: what does the business actually do, and where does AI fit into what it does.

AI connects to business architecture in two directions. Downward, the business capabilities and processes define the use cases that AI systems are built for — a credit-decision process, a customer-service workflow, a maintenance schedule. Upward, AI can reshape business capabilities by making new processes viable or by transforming the economics of existing ones. An AI system built without reference to business architecture is a system whose use case was invented by the technology team rather than derived from the business, and the use cases so invented have a poor survival rate.

The chapters that develop this layer include the enterprise-AI relationship (Chapter 4), the maturity model (Chapter 105), and the economics chapter (Chapter 82).

5.4 Layer 3: Data Architecture

Data architecture provides the data sources, the data platform (the data lake or data warehouse), the data governance (quality, catalogue, lineage), and the analytics-and-AI-ready data that every AI system depends on. It is the layer this book returns to most often, because the quality of an AI system is bounded by the quality of the data it builds on.

The data-architecture layer is where the connection to AI is tightest. The data platform holds the training data for fine-tuning, the documents for retrieval indices, the structured data for analytics, and the traces and logs that observability depends on. Data governance — classification, lineage, quality, access control — determines whether the data is usable, trustworthy, and compliant. And the preparation of AI-ready data — cleaned, labelled, de-duplicated, free of contamination — is the work that precedes every model-training and every index-building effort.

The chapters that develop this layer include the data-sources layer (Chapter 10), the data-pipeline layer (Chapter 11), the knowledge architecture (Chapter 87), the embeddings architecture (Chapter 67), the

vector-database architecture (Chapter 68), and the data-privacy chapter (Chapter 92).

5.5 Layer 4: Application Architecture

Application architecture encompasses the business applications and services the enterprise runs — CRM, ERP, MES, SCM, finance, HR, and the AI applications that sit alongside or within them. It is the layer where AI becomes visible to the business, because the AI applications are the ones the business users interact with.

AI connects to application architecture in two ways. First, AI applications are applications in their own right — customer-facing assistants, internal copilots, document-processing systems — and they must be designed, built, and operated as applications, with all the lifecycle discipline that implies. Second, AI capabilities are increasingly embedded into existing applications, so that the CRM has an AI-powered summarisation feature or the ERP has an AI-powered forecasting module. In both cases, the AI capability must integrate with the application architecture rather than existing as a separate island.

The chapters that develop this layer include the application layer of the reference architecture (Chapter 14), the AI product-management chapter (Chapter 88), and the case studies (Chapter 106).

5.6 Layer 5: Technology Architecture

Technology architecture provides the infrastructure, platforms, compute, storage, and networking that the software and AI layers run on. For AI specifically, it includes the GPU and accelerator estate, the inference-serving infrastructure, the edge devices, and the cloud regions.

The technology layer is where the economics of AI become concrete, because the hardware and cloud resources that run inference determine the cost per request, the latency, and the scale the system can reach. An AI architecture designed without reference to the technology layer is an architecture that has not been costed, and an architecture that has not been costed will be surprised by its bill.

The chapters that develop this layer include the AI-hardware chapter (Chapter 77), the inference-infrastructure chapter (Chapter 86), and the FinOps chapter (Chapter 61).

5.7 Layer 6: Integration Architecture

Integration architecture provides the APIs, events, message queues, ETL pipelines, iPaaS, and system-integration patterns that connect the applications and services to each other and to the AI systems. It is the connective tissue of the enterprise, and without it the AI system is an island that cannot reach the data or the systems it needs.

For AI systems, integration architecture takes on a specific shape: the Model Context Protocol and similar standards provide agent-to-tool integration, the event bus provides the event-driven communication that agentic workflows depend on, and the API gateway provides the entry point through which applications reach the AI services. The integration layer is where the Capability Plane of the four-plane architecture (Chapter 9) is realised — the layer where thinking connects to

acting.

The chapters that develop this layer include the MCP architecture (Chapter 33), the event-driven-agents pattern (Chapter 25), and the tool-use chapter (Chapter 32).

5.8 Layer 7: Security Architecture

Security architecture provides identity, access control, security controls, compliance mechanisms, audit, and privacy protections across the stack. For AI systems, it extends to cover the AI-specific attack surfaces: model theft, prompt injection, jailbreaks, training-data extraction, tool abuse, supply-chain poisoning, and the agent-as-principal model.

Security is a cross-cutting concern that touches every other layer, and the infographic correctly places it alongside the technology and integration layers it most directly governs. The AI-specific security challenges — developed in the security-architecture chapter — are genuinely new, and the traditional security posture of the enterprise does not cover them without extension.

The chapters that develop this layer include the security-architecture chapter (Chapter 62), the agent-identity chapter (Chapter 72), the red-teaming chapter (Chapter 66), the data-privacy chapter (Chapter 92), and the guardrails chapter (Chapter 54).

5.9 Layer 8: Solution Architecture

Solution architecture is the layer where a specific business need is translated into a concrete system design. It takes the business need from layer two, selects the AI and ML use cases that address it, designs the integration flows, chooses the deployment model, and produces a solution blueprint that the software and platform layers will implement.

For AI systems, solution architecture is where the pattern selection happens: which of the six patterns (RAG, LLM Router, Hub-and-Spoke, Event-Driven Agents, Hexagonal, Agentic Mesh) fits the problem, how the human-in-the-loop posture is designed, what the evaluation strategy is, and what the cost model looks like. Solution architecture is the bridge between the business need and the technical implementation, and it is the layer where the architectural decisions of this book are most directly applied.

The chapters that develop this layer include the patterns introduction (Chapter 17), the six pattern chapters, the human-in-the-loop chapter (Chapter 35), the evaluation architecture (Chapter 36), and the foundation-model-strategy chapter (Chapter 85).

5.10 Layer 9: Software Architecture

Software architecture provides the software design: the frontend (web and mobile), the backend services (microservices), the data layer (databases, caches, files), and the external services (third-party and cloud). For AI systems, the software architecture includes the model-serving layer, the orchestration logic, the prompt management, and the observability instrumentation.

The Software 3.0 paradigm (Chapter 42) fundamentally changes what software architecture means for AI systems, because the behaviour of the system is defined by prompts and data rather than by compiled code. A software architecture for an AI system must therefore include prompt versioning, evaluation infrastructure, and the ability to change behaviour through configuration rather than through a code release.

The chapters that develop this layer include the Software 3.0 chapter (Chapter 42), the LLMOps chapter (Chapter 70), the prompt-chaining chapter (Chapter 55), and the streaming-and-async chapter (Chapter 53).

5.11 Layer 10: Platform, Cloud, and AI Architecture

Layer ten is where the AI-specific platform lives. It integrates three concerns: the platform services (identity, API gateway, observability, the development platform), the cloud infrastructure (compute, storage, networking, databases, containers, serverless), and the AI and data-science services (LLMs, generative-AI models, ML models, RAG infrastructure, vector databases, MLOps pipelines).

This is the layer the six-layer reference architecture (Chapter 8) and the four-plane architecture (Chapter 9) decompose in detail. It is the layer where the model meets the infrastructure, where inference runs, where the knowledge indices are hosted, and where the operational tooling lives. An enterprise's AI platform capability is, in large part, the maturity of this layer.

The chapters that develop this layer include the reference architecture, the four-plane architecture, the inference-infrastructure chapter (Chapter 86), the knowledge-architecture chapter (Chapter 87), and every pattern chapter that describes how the platform components are assembled.

5.12 Layer 11: Implementation and Operations

The bottom layer is where the system is built, deployed, run, monitored, and optimised. It is the operational reality that turns all the layers above into a running, serving, improving system. The lifecycle runs from build through deploy, run, and monitor to optimise, and the feedback arrow in the diagram shows that operations feed back into strategy, completing the loop.

For AI systems, operations include the AI-specific operational disciplines: model monitoring, drift detection, continual learning, evaluation-driven release, canary deployment, incident response, and the cost management that keeps the system economically viable. An AI system that is built and deployed but not operated is a system that will degrade, because models age and the world moves.

The chapters that develop this layer include the DevSecOps-and-governance layer (Chapter 15), the LLMOps chapter (Chapter 70), the continual-learning chapter (Chapter 60), the model-monitoring domain of the governance operating model (Chapter 104), and the sustainability chapter (Chapter 79).

5.13 The Cross-Cutting Governance Column

The right-hand column of the diagram names seven governance concerns that apply across all eleven layers rather than belonging to any one of them. Each is a discipline that runs vertically through the stack.

Architecture governance ensures that architectural decisions are made systematically, recorded, and enforced. It is the subject of the governance-framework chapter (Chapter 31) and the governance operating model (Chapter 104).

Data governance ensures that data across all layers is classified, owned, lineage-tracked, and quality-managed. It is the subject of the data-governance domain of the governance operating model and the data-privacy chapter.

Security and privacy ensures that every layer protects the data it handles and the actions it takes. It is the subject of the security-architecture chapter and the data-privacy chapter.

Compliance and risk ensures that the system satisfies the regulatory requirements of every jurisdiction it operates in. It is the subject of the compliance-and-audit chapter (Chapter 65) and the EU AI Act treatment.

Change management ensures that changes to any layer are deliberate, reviewed, tested, and reversible. It is the subject of the canary-deployment chapter (Chapter 48), the evaluation-driven-release chapter (Chapter 51), and the continual-learning chapter.

Cost management ensures that spend across all layers is attributed, governed, and optimised. It is the subject of the FinOps chapter (Chapter 61), the enterprise-AI-finance chapter (Chapter 83), and the economics chapter (Chapter 82).

Performance management ensures that latency, throughput, availability, and quality are measured and maintained. It is the subject of the observability-and-evaluation chapter (Chapter 101), the inference-infrastructure chapter, and the reliability chapter.

5.14 The Continuous Feedback Loop

The feedback arrow at the bottom of the diagram is as important as any of the layers. It indicates that implementation and operations feed back into enterprise architecture — that the experience of running the system informs the strategy, the governance, and the roadmap. An AI programme that does not close this loop is a programme that plans once and never learns from its own execution.

In practice the feedback loop takes several forms. Production metrics feed back into the quality bars and the evaluation suites. Cost data feeds back into the business case and the FinOps governance. Incidents feed back into the security posture and the compliance controls. User feedback feeds back into the product roadmap. And capability evolution — the arrival of new models, new patterns, new possibilities — feeds back into the enterprise strategy itself. The loop is what makes the eleven-layer stack a living system rather than a static plan, and the disciplines of this book — monitoring, evaluation, continual learning, incident response — are the mechanisms that carry the feedback.

5.15 Using the Connected View

The connected view is an orientation device, and it is most useful carried as a mental map while reading the rest of the book. When a chapter develops a specific concern — say, vector-database architecture — the connected view locates that concern in layer three (data architecture) and layer ten (platform), shows its dependencies on the layers below, and names the governance concerns that run alongside it. When a project is being planned, the connected view serves as a checklist: has every layer been addressed, and have the cross-cutting governance concerns been accounted for.

The connected view is deliberately simplified. Each layer, in reality, contains far more than the few components named in the diagram. The simplification is the point: it shows the connections between the layers at a level of abstraction where the connections are visible, rather than burying them in the detail each layer's chapter provides. The detail is in the chapters; the connections are in this map.

5.16 The African Perspective

The connected view applies unchanged in African deployments, but the weight of each layer shifts. Layer three (data architecture) carries additional weight because data-sovereignty requirements vary by jurisdiction and the data-governance controls must be stratified per market. Layer five (technology) carries additional weight because African cloud regions have limited GPU capacity and edge deployment is more central. Layer seven (security) carries additional weight because mobile-money integrations create high-value attack surfaces. Layer ten (platform) must accommodate the self-hosted, open-weight, in-region models that sovereignty demands. And the cross-cutting cost-management column is tighter, because African unit economics constrain every layer more sharply.

The connected view's value for African deployments is that it makes these weight shifts visible. An enterprise that walks the eleven layers for a pan-African AI programme, asking at each layer what the African context changes, will identify the sovereignty, cost, and infrastructure constraints before they become surprises — which is exactly what an orientation device is for.

5.17 In Short

The simplified connected view presents enterprise AI architecture as an eleven-layer stack — from enterprise architecture at the top, through business, data, application, technology, integration, security, solution, software, platform-cloud-AI, to implementation and operations at the bottom — with seven cross-cutting governance concerns running alongside and a continuous feedback loop closing the stack from operations back to strategy. Each layer depends on the ones below it, and AI touches every one of them.

The view is an orientation device. It does not replace the detail the book's chapters provide; it locates that detail in the enterprise and shows the connections between the concerns. A reader who carries this map will always know where a given chapter's subject sits, what it depends on, and what depends on it. Every architecture diagram is a lie, as the epigraph says — but a useful lie shows the connections,

and the connections are what this chapter maps.

Chapter 6

What Is Enterprise AI?

“Enterprise software is not consumer software with a bigger logo.”

—a recurring lesson learned the hard way

Chapter Overview

Enterprise AI is not consumer AI at larger scale. The constraints, the success criteria, and the failure modes are different in kind, not just in degree. This chapter draws those distinctions, develops a working definition of enterprise AI, examines the lifecycle differences from research and consumer products, and surveys the value categories that justify the investment. The vocabulary established here is used throughout the rest of the book.

Learning Objectives

- Distinguish enterprise AI from consumer AI and research AI.
- Apply a working definition that names the four distinguishing constraints: integration, governance, longevity, value attribution.
- Recognise the four value categories: cost reduction, revenue uplift, risk reduction, capability creation.
- Locate AI maturity within the broader enterprise digital landscape.
- Recognise the recurring failure patterns of AI programmes treated as consumer projects.

6.1 A Working Definition

The previous chapter described the discipline within which enterprise AI sits; this one describes the artefact the discipline produces. The discipline is enterprise architecture, extended. The artefact is the kind of system the rest of the book is going to architect, and the working definition begins where the rest of the field tends not to: with the constraints that distinguish enterprise AI from its neighbouring forms.

Enterprise AI is the practice of building AI-enabled systems that:

1. Integrate with existing enterprise data, processes, and applications.

2. Operate under governance: identifiable owners, traceable decisions, auditable evidence, controllable behaviour.
3. Are expected to run for years, surviving staff turnover, vendor changes, and organisational reorganisations.
4. Are evaluated against business outcomes that can be measured and attributed.

Each of these constraints distinguishes enterprise AI from neighbouring disciplines. Consumer AI typically lacks the integration constraint (it does not consume the company's CRM) and the governance constraint (no auditor will ever ask). Research AI typically lacks the longevity constraint (a paper need not be operable in five years) and the value-attribution constraint (advances in the state of the art are their own reward).

6.1.1 Why the Distinctions Matter

A AI application that is impressive in isolation may be unfit for enterprise deployment for reasons unrelated to its AI quality. It may not authenticate against the corporate identity system. It may not produce explainable outputs that a regulator will accept. It may depend on a single vendor whose terms preclude its use on regulated data. It may have no path to deletion when a data-subject erasure request arrives. These are architectural failures, not model failures, and they are the failures that consume enterprise AI programmes.

6.2 Constraints That Shape the Architecture

The four defining constraints are worth examining individually.

6.2.1 Integration

An enterprise AI system rarely begins on a blank canvas. It must read from the company's existing data, write to its existing systems, authenticate its users through its existing identity provider, comply with its existing data-handling policies, and integrate with its existing operational tooling. The integration surface is large and unforgiving; estimates that ignore it produce schedules that miss by 2–3×.

Architecturally, this constraint motivates: data contracts (Section 10.6), the hexagonal pattern (Chapter 26) for vendor substitutability, and federated governance (Section 24.5) for changes that cross organisational boundaries.

6.2.2 Governance

Every consequential output must be attributable: which model, which prompt, which retrieved sources, which input, which user. Every change must be reviewable: who approved this deployment, against which evaluation results, with what risk classification. Every regulatory ask must be answerable: show me the inventory, show me the lineage, show me the monitoring data.

Architecturally, this constraint motivates: lineage emission (Section 103.4), model inventory (Section 103.1), evaluation as a CI gate (Section 101.2), and audit-grade logging throughout.

6.2.3 Longevity

Enterprise systems outlive their vendors, their original engineers, and frequently their original sponsors. A system shipped today should still be operable in three years, possibly with different underlying models, different vendors, and different team members. The architectural commitments that produce this property—small, stable interfaces; replaceable adapters; reproducible training and evaluation; durable artefacts—are the same commitments that make enterprise software durable in general.

Architecturally, this constraint motivates: the hexagonal pattern as a default; infrastructure-as-code; versioned everything (data, models, prompts, configurations).

6.2.4 Value Attribution

An enterprise AI programme exists to produce business value. The programme is evaluated against that value, and the value must be attributable to the programme rather than to coincident effects. This is harder than it sounds: many AI initiatives produce real effects that are difficult to measure cleanly, and many measurement attempts produce numbers that survive scrutiny only briefly.

Architecturally, this constraint motivates: cost attribution at the gateway (Section 100.2); experiment tracking with controls; pre-stated hypotheses and outcome measurement.

6.3 Value Categories

AI investments fall into four categories, with different evidence standards.

Table 6.1: Value categories for enterprise AI.

Category	Description and examples
Cost reduction	Automating tasks that were previously manual: classification, summarisation, drafting, triage. Easiest to measure (time saved $\times$ wage).
Revenue uplift	Improving conversion, personalisation, recommendation, retention. Harder to attribute (counterfactuals required).
Risk reduction	Catching fraud, errors, compliance breaches earlier; better forecasting. Hardest to measure (events prevented are invisible).
Capability creation	Enabling activities that were not previously possible: real-time multilingual support, document understanding at scale, complex personalisation. Visible but slow to materialise.

A portfolio that emphasises one category exclusively tends to be fragile: cost-reduction-only portfolios produce diminishing returns; capability-creation-only portfolios produce excitement without measurable outcome. The mature programme balances categories, with explicit hypotheses for each investment.

6.4 Lifecycle Differences

The lifecycle of an enterprise AI system differs from neighbouring lifecycles in several material ways.

- **From research.** Research deliverables end at a paper or a benchmark. Enterprise deliverables begin at deployment and continue for years; the deployed system is the artefact.
- **From consumer.** Consumer products iterate weekly and break things forward. Enterprise products iterate with change control; the cost of a breaking change is the sum of the changes every downstream consumer must make.
- **From traditional enterprise software.** Traditional enterprise software has stable behaviour: the same input produces the same output. AI systems do not. The lifecycle adds drift monitoring, periodic re-validation, and structured retraining as ongoing obligations rather than one-off events.

6.5 Where Enterprise AI Sits in the Digital Estate

Most enterprises arrive at AI through a sequence of digital investments: cloud migration, data platform consolidation, microservices and APIs, observability. AI sits on top of these, not beside them. An organisation that has not consolidated its data is not ready to consolidate its AI; an organisation without coherent identity will struggle to govern its agents.

6.5.1 Prerequisite Investments

The dependencies are not absolute but they are real:

- A coherent data platform precedes effective ML, because models are downstream of data quality (Chapter 11).
- A modern identity layer precedes governable agents, because agent actions must be attributable (Chapter 15).
- A working observability stack precedes operable AI, because drift and quality cannot be managed without measurement (Chapter 101).
- A FinOps discipline precedes sustainable AI, because LLM costs are unforgiving of inattention (Chapter 100).

Most enterprise AI programmes that struggle are not failing at AI; they are failing at one or more of the prerequisites. The remediation is rarely a better model.

6.6 Recurring Failure Patterns

Several failure patterns recur in AI programmes treated as consumer projects:

- **Pilot purgatory.** Many pilots, no production. Each pilot rebuilds infrastructure; none earns its place in operations.
- **Demo-to-production gap.** A successful demo cannot survive contact with real data quality, real identity systems, or real change-control processes.
- **Cost shock.** The first invoice is many multiples of the projection. Cost was not designed in.

- **Audit surprise.** The first internal audit finds an inventory of models nobody knew existed; the first regulator visit finds documentation gaps.
- **Vendor lock-in revelation.** The platform built quickly turns out to depend deeply on one vendor; migration is now a multi-year project.

The structural mitigations to all of these are in the rest of the book: the platform investments of Part II, the patterns of Part III, the disciplines of Part VII.

6.7 Holding the Pieces Together

Enterprise AI is distinguished from neighbouring disciplines by four constraints: integration with the existing estate, governability under regulation and audit, longevity beyond any single vendor or model, and value attributable to specific business outcomes. These constraints shape every chapter that follows. The remainder of the book is, in one sense, the answer to a single question: what does the architecture look like that satisfies these constraints while still extracting the value that justifies the investment?

Chapter 7

Architecture Principles

“A principle is a decision you don’t have to make again.”

—attributed variously, valuable regardless of origin

Chapter Overview

Principles compress experience. They are the durable lessons that survive the volatility of technology choices and become defaults—the things an architect can stop arguing about because the position is already taken. This chapter develops ten architecture principles for enterprise AI systems, organised by what they govern: how the system is structured, how it changes, how it is operated, and how it is governed. Each principle is stated with its implication, its motivation, and its trade-offs. The remainder of the book derives from these.

Learning Objectives

- Apply ten principles consistently across architectural decisions.
- Recognise principles as defaults that bias the search for solutions, not absolute laws.
- Identify when a principle should be overridden and document the reason.
- Use the principles as a tie-breaker when other considerations are roughly balanced.

7.1 Structure

The first four principles govern how the system is structured.

7.1.1 Principle 1: Layered Responsibilities

Every component belongs to exactly one layer of the reference architecture (Chapter 8). Data flows upward; control flows are explicit; cross-cutting concerns apply uniformly.

Implication. A component that does too many things at once (a single service that ingests data, embeds it, retrieves it, and generates from it) is refactored into layer-aligned components.

Motivation. Layered systems are testable, replaceable, and understandable by people who did not build them. A monolithic AI service that fails at one of its many jobs is impossible to localise; a layered one is.

Trade-off. Layers add interfaces and indirection. For prototypes, this is overhead; for production systems, it is the price of operability.

7.1.2 Principle 2: Hexagonal Boundaries

Every external dependency—vendor model, vector store, tool API, identity provider—is accessed through a port (Chapter 26). Implementations are adapters; the core depends only on the port.

Implication. A direct call to a vendor SDK from application logic is an architectural smell.

Motivation. Vendors will change, capabilities will appear, prices will shift. The architecture that lets these be operational changes rather than projects is the one that survives.

Trade-off. The abstraction may not cover every vendor-specific feature. A port that supports the union of capabilities exposes the union of constraints; a port that supports the intersection loses differentiation.

7.1.3 Principle 3: Data Contracts Are First-Class

Every cross-team data flow has a written contract that specifies schema, semantics, freshness, quality, and the change-notification process (Section 10.6). The contract is the durable artefact; the implementations are negotiable.

Implication. A new consumer of a dataset reads the contract, not the producer's code.

Motivation. Contracts survive reorganisations and staff turnover; tribal knowledge does not.

Trade-off. Contracts are work to produce and maintain. The justified-but-tedious work that mature platforms do.

7.1.4 Principle 4: Single Source of Truth Per Concept

For every concept (a customer, a product, a model, a feature, a prompt template), one store is the source of truth and all consumers read from it. Duplications are explicit caches with documented staleness.

Implication. A second source of a concept requires a written justification and a reconciliation path.

Motivation. Conflicts of truth produce conflicts of behaviour; they are among the most expensive problems to debug because their symptoms appear far from their causes.

Trade-off. The source of truth becomes a critical dependency; its availability and performance must match the most demanding consumer.

7.2 Change

The next three principles govern how the system changes.

7.2.1 Principle 5: Reversible Decisions by Default

Prefer decisions that can be reversed cheaply. Vendor choices, model choices, prompt templates, retrieval strategies are all reversible when accessed through ports; they become irreversible when their specifics leak into application logic.

Implication. A decision that locks the system into a specific vendor's API requires a higher bar of justification than one that does not.

Motivation. Reversible decisions admit experimentation; irreversible ones admit only commitment.

Trade-off. Some irreversible decisions (e.g., choice of cloud region for residency-constrained workloads) are necessary. The principle is the default, not a prohibition.

7.2.2 Principle 6: Continuous, Not Episodic

Quality, security, governance, cost, and performance are continuous activities, not pre-launch checklists. Evaluation runs continuously; drift is monitored continuously; cost is tracked continuously; access reviews are continuous.

Implication. A control that is only checked at deployment is incomplete; it must also fire when production deviates from the deployed state.

Motivation. AI systems drift. Static controls miss the drift.

Trade-off. Continuous monitoring has operational cost; the architecture pays it because the alternative is silent failure.

7.2.3 Principle 7: Evolution by Versioning

Datasets, features, models, prompts, configurations, and contracts are versioned. Old versions are retained while new versions are introduced; consumers migrate explicitly.

Implication. A breaking change without a new version number is an outage in waiting.

Motivation. Versioning makes change tractable: rollbacks are operations, not heroics.

Trade-off. Version proliferation. Mitigated by retention policies and explicit deprecation.

7.3 Operation

The next two principles govern how the system is operated.

7.3.1 Principle 8: Observable by Construction

Every component emits structured logs, metrics, and traces with sufficient context to debug a failure without reproducing it. Prompts and retrieval results are logged (with PII redaction) alongside conventional service signals.

Implication. A component without observability is not production-ready.

Motivation. The first incident defines what observability the system needed; the architecture should anticipate this rather than discover it.

Trade-off. Observability has storage and processing cost. Tiered retention manages this without abandoning the principle.

7.3.2 Principle 9: Resilient by Default

Every external dependency is treated as fallible. Retries with backoff, circuit breakers, bulkheads, fallbacks, and graceful degradation are defaults rather than features (Chapter 93).

Implication. A direct call to a vendor without timeout, retry, and fallback is a future incident.

Motivation. AI systems depend on more external services than typical applications; the dependency graph is wider and the failure modes are more varied.

Trade-off. Resilience adds complexity. The minimum-viable resilience for an internal experiment is less than for a customer-facing service; the principle applies the appropriate level rather than the maximum.

7.4 Governance

The final principle governs how the system is governed.

7.4.1 Principle 10: Governance Is a Platform Feature

Inventory, classification, lineage, evidence collection, monitoring, and audit support are features of the platform, not deliverables of each project. New projects inherit them; deviations from the paved path require justification.

Implication. A platform that requires every project to invent its own governance has not yet absorbed this principle.

Motivation. Governance applied per-project scales as $O(n)$; governance applied as a platform feature scales as $O(1)$.

Trade-off. The platform team must operate a real governance capability, not a wiki page about governance.

7.5 Using the Principles

The principles are not a checklist. They are biases in the search for solutions:

- Two designs of approximately equal cost: prefer the one that satisfies more principles.
- One design that violates a principle and one that does not: ask whether the violation is justified by something specific to this case.
- A pattern that consistently violates a principle: re-examine the pattern.
- A principle that the team consistently overrides: re-examine the principle.

Principles are revisited. The list above reflects the state of practice in 2026 and the experience accumulated through this book; readers in later years may find that some have aged, others have strengthened, and a few should be replaced. The discipline is to keep the list short, the statements crisp, and the trade-offs honest.

7.6 The Architectural Commitment

Ten principles—layered responsibilities, hexagonal boundaries, first-class data contracts, single source of truth, reversible decisions, continuous discipline, evolution by versioning, observability by construction, resilience by default, governance as platform feature—compress the experience of the field into defaults the architecture can rely on. The rest of the book applies these principles. Where the application is non-obvious, the chapter says so; where it follows directly, the principles do the work.

Chapter 8

The Six-Layer Reference Architecture

“A picture is worth a thousand specifications.”

—unattributed, in the spirit of every architecture review ever

Chapter Overview

This chapter introduces the central organising structure of the book: a six-layer reference architecture for enterprise AI systems. The layers run from raw data sources at the bottom to user-facing applications at the top, with cross-cutting concerns spanning all of them. Each subsequent chapter in Part II develops one layer in detail. The architecture is a contract diagram, not a deployment diagram—a vocabulary for naming responsibilities and the interfaces between them, applicable to single-cloud deployments, multi-cloud deployments, hybrid arrangements, and on-premises systems alike.

Learning Objectives

- Recognise the six layers of the reference architecture and name what each is responsible for.
- Understand the contracts between adjacent layers and why they should be stable.
- Identify the five cross-cutting concerns that apply to every layer.
- Map an existing AI system onto the reference architecture and locate its gaps.
- Use the architecture as a tool for conversation across roles—engineering, data, security, compliance, business.

8.1 The Six Layers at a Glance

The reference architecture comprises six layers stacked vertically, with data and control flowing predominantly upward:

Layer 1. Data Sources.

Where data originates—structured warehouses, unstructured document stores, APIs and streams, file shares and object storage, and third-party feeds.

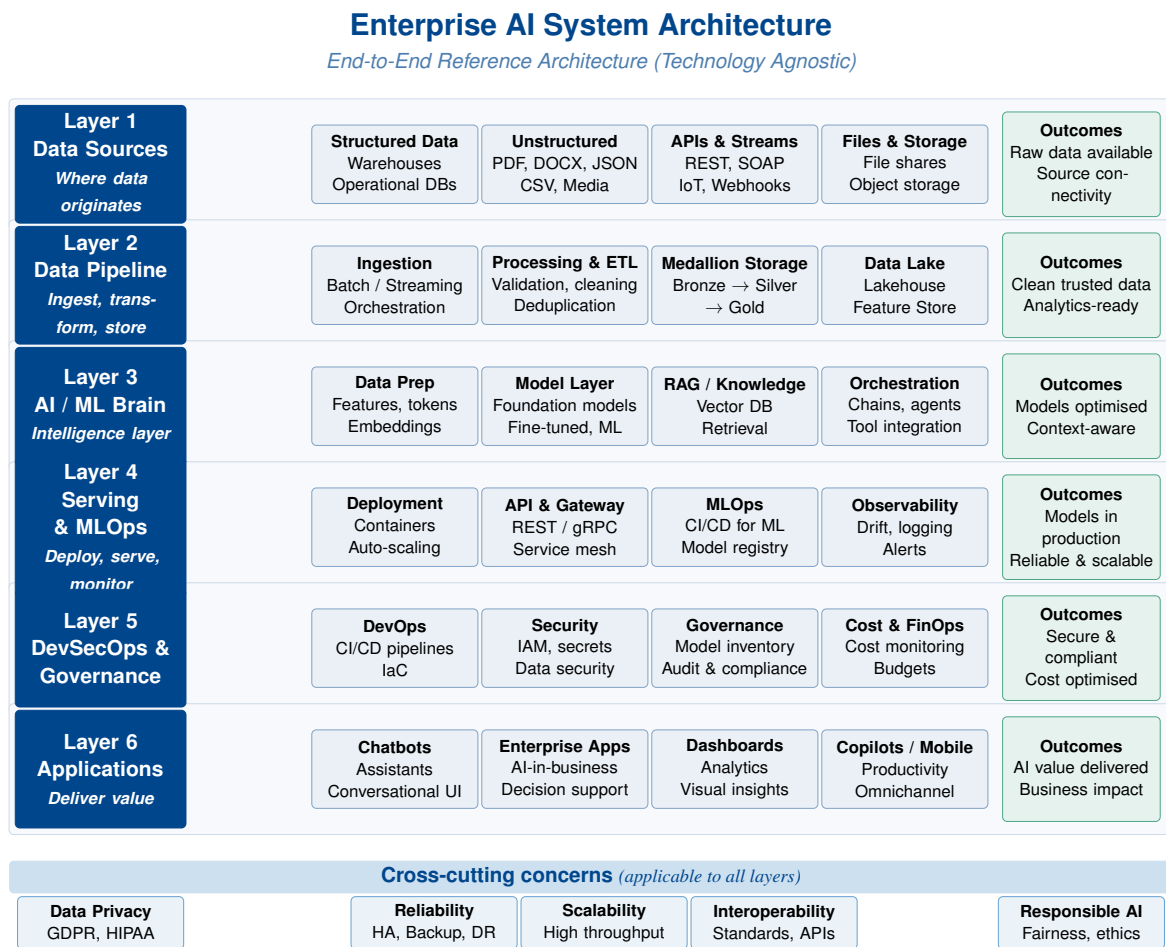


Figure 8.1: The six-layer enterprise AI reference architecture, with the five cross-cutting concerns spanning all layers. Each subsequent chapter in Part II develops one layer in detail.

Layer 2. Data Pipeline.

Ingestion, processing, ETL, and the medallion (bronze/silver/gold) storage progression that feeds the rest of the stack with clean, traceable data.

Layer 3. AI/ML Brain.

Data preparation, the model layer (foundation models, fine-tuned models, traditional ML), the RAG/knowledge layer, AI orchestration, and the AI services (NLP, vision, speech, document intelligence).

Layer 4. Model Serving and MLOps.

Deployment, the API and integration layer, MLOps, monitoring and observability, and feedback and continuous improvement.

Layer 5. DevSecOps and Governance.

CI/CD, security, governance and compliance, observability infrastructure, and cost and FINOPS.

Layer 6. End User Applications.

AI chatbots and assistants, enterprise applications, dashboards and analytics, copilots and productivity tools, and mobile and web apps.

Above and beside these layers run the five cross-cutting concerns: data privacy, reliability and availability, scalability and performance, interoperability, and ethics and responsible AI.

8.2 Reading the Reference Diagram

The reference architecture is best understood as a contract diagram rather than a deployment diagram. Each box names a responsibility, not a product. Each arrow names an interface, not a network call. Real deployments collapse several boxes into one service or expand one box across several services; what matters is that every responsibility has a home and every interface is explicit.

The same architecture can be implemented on a single cloud, across multiple clouds, on-premises, or in a hybrid configuration. It can be assembled from open-source tools, vendor products, or some mix. The architecture does not prescribe these choices; it provides the vocabulary in which to make them.

8.2.1 Why Six Layers?

Other layer counts are defensible. Some references collapse Layers 3 and 4 into a single “model platform.” Others split Layer 1 into raw and harmonised. The six-layer split used here is chosen because each layer corresponds to a team type that recurs in mature organisations: a data-engineering team owns Layers 1 and 2, an ML/AI team owns Layer 3, a platform team owns Layer 4, a DevSecOps team owns Layer 5, product teams own Layer 6. The boundaries are where conversations happen.

8.3 The Contracts Between Layers

The interfaces between adjacent layers are where most of the architectural work lives. They are where teams meet, where ownership transitions, and where versioning and stability matter most.

Table 8.1: Interface contracts between adjacent layers.

Interface	Typical contract
Layer 1 → Layer 2	Data contracts (Section 10.6): schema, freshness, quality, change-notification.
Layer 2 → Layer 3	Gold tables and feature views with documented SLAs; point-in-time correctness for training data.
Layer 3 → Layer 4	Registered model artefacts with metadata; evaluation results; declared dependencies.
Layer 4 → Layer 6	API endpoints with versioned schemas; observability emitted to standard collectors.
Layer 5 spans 1–6	Identity, secrets, policy, observability infrastructure consumed by every layer.
Cross-cutting spans all	Privacy, reliability, scalability, interoperability, ethics applied as platform features.

8.3.1 Stability of Interfaces

The architectural rule is that interfaces between layers change slowly. Components within a layer can be replaced freely; interfaces between layers are versioned and deprecated through documented processes. This is what makes layer-by-layer evolution possible: replacing a vector database (Layer 3

internal) does not require touching Layer 6.

8.4 The Cross-Cutting Concerns

Five concerns apply to every layer.

8.4.1 Data Privacy (Chapter 92)

Classification at intake, minimisation, encryption, tokenisation, redaction in logs, subject-rights operability, residency by design. Privacy is structural: a system without these built in cannot retrofit them after an incident.

8.4.2 Reliability and Availability (Chapter 93)

SLOs, redundancy, failover, resilience patterns (retry, circuit breaker, bulkhead, fallback), backup, disaster recovery, chaos engineering. Reliability is a property of the system, not the components.

8.4.3 Scalability and Performance (Chapter 94)

Latency budgets, throughput strategies (batching, caching, parallelism), autoscaling with GPU-aware behaviour, cost-performance trade-offs. AI workloads have unusual performance profiles that conventional defaults handle badly.

8.4.4 Interoperability (Chapter 95)

OpenAPI, OpenTelemetry, MCP, A2A, ONNX, OpenLineage. API-first design, schema evolution discipline, contract testing. Interoperability is what makes the system more durable than the components it currently uses.

8.4.5 Ethics and Responsible AI (Chapter 96)

Fairness measurement, transparency and explainability, human oversight, content safety, consequential-decision review. Responsible AI is an architectural commitment, not a final-stage review.

8.5 Mapping an Existing System

A practical use of the reference architecture is to map an existing system onto it. The exercise typically surfaces three kinds of finding:

- **Missing responsibilities.** A layer that has no owner; a cross-cutting concern that nobody is accountable for. These are the gaps that future incidents will exploit.
- **Conflated responsibilities.** One component doing several layers' jobs; one team owning interfaces that should be platform-mediated. These are the brittleness points.
- **Shadow architecture.** Components nobody knew existed; integrations that bypass the platform; vendor services running outside the inventory. These are the audit risks.

The mapping is rarely flattering on first attempt. The discipline is to produce it anyway, share it with the relevant teams, and use it to prioritise remediation.

8.6 What the Architecture Is Not

A few clarifications:

- It is not a deployment diagram. The components do not map one-to-one to services or to servers; the architecture is about responsibilities and interfaces.
- It is not a prescription. An organisation may justifiably depart from any specific element of the architecture, with documented reasoning. The reference is a default, not a mandate.
- It is not complete. Specific industries (regulated financial services, healthcare, government) add concerns this reference does not name in detail. The relevant chapters add them.
- It is not static. The architecture in this book reflects 2026 practice; later years will add new sub-layers, refine the cross-cutting concerns, and possibly retire concerns that have been absorbed elsewhere.

8.7 Pulling the Threads

Six layers, five cross-cutting concerns. Layer 1 holds the sources. Layer 2 turns them into trusted data. Layer 3 turns the data into intelligence. Layer 4 turns the intelligence into production services. Layer 5 surrounds and governs all of the above. Layer 6 delivers value to users. Privacy, reliability, scalability, interoperability, and ethics span everything. This is the structure on which the rest of the book is built; each subsequent part develops one part of it.

Chapter 9

The Four-Plane Enterprise AI Platform

“Never trust the AI. Verify outputs, authorise actions, preserve evidence.”

—a design maxim for governed enterprise AI

Chapter Overview

The six-layer reference architecture of the previous chapter decomposes an AI system by *what it does*: data sources, data pipeline, the model brain, the application layer, the serving layer, and the DevSecOps-and-governance layer. This chapter presents a complementary view that decomposes an enterprise AI platform by *who governs it and how*: four planes, each with a distinct owner and a distinct contract with the others, unified by a governance substrate that flows across all of them. Where the six-layer view answers “what are the parts of an AI system,” the four-plane view answers “how does an enterprise own, govern, and evolve its AI estate as a platform.” The two views are complementary, not competing — the layers describe the system’s anatomy, while the planes describe its operating model. This chapter develops the four planes, the foundation layer beneath them, the design principles that guide them, the outcomes they produce, and the relationship between the two reference architectures.

Learning Objectives

- Distinguish the four-plane platform view from the six-layer system view and understand how they complement each other.
- Recognise the Control Plane, the Runtime Plane, the Experience Plane, and the Capability Plane and the contract each carries.
- Trace the governance substrate — identity, authority, evidence, and accountability — that flows across all four planes.
- Apply the design principles — never trust the AI, human in control, evidence everywhere, composable by design, open where it matters, sovereignty by design — to architectural decisions.
- Map the four-plane model to African enterprise realities of sovereignty, cost, and governance.

9.1 Why a Second Reference Architecture

The six-layer reference architecture is a system view: it names the technical components and shows how data flows through them. It serves an architect designing or reviewing a specific AI system. But enterprise AI is not one system — it is an estate of systems, built by different teams, serving different business units, evolving at different speeds, and all needing to be governed, observable, and accountable. The six-layer view does not naturally answer the questions that an enterprise platform raises: who governs the agents, how is authority delegated, where does the evidence live, how do business units innovate without breaking the platform’s controls, and how does the whole estate stay composable rather than fragmenting into silos.

The four-plane architecture answers those questions. It is a platform view: it organises the estate not by technical layer but by the concern each plane is responsible for. The Control Plane governs identity, authority, evidence, and composition. The Runtime Plane supplies the AI substrate — models, knowledge, inference, and operations. The Experience Plane shapes what humans actually encounter when they work with the AI. And the Capability Plane extends the platform’s reach into the business systems the AI acts upon. The two architectures are not alternatives to choose between; they are lenses to use together. The six-layer view tells the architect what to build inside any one system; the four-plane view tells the enterprise how to own the estate those systems live in.

9.2 The Four Planes

Figure 9.1 shows the four-plane architecture. The Control Plane sits at the top, governing all the others. The Runtime Plane and the Experience Plane sit side by side beneath it, connected by a governance substrate that carries identity, authority, and evidence between them. The Capability Plane sits alongside, extending the platform into business systems. And a Foundation Layer beneath all four provides the enterprise guardrails and enablement that every plane depends on. The rest of this section develops each plane in turn.

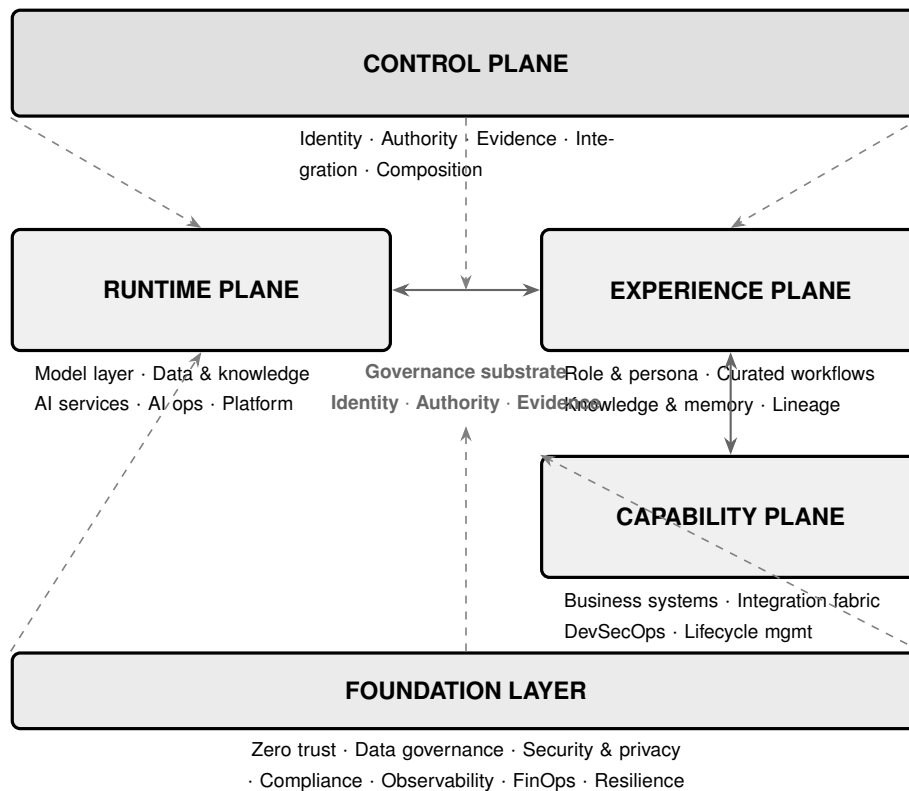


Figure 9.1: The four-plane enterprise AI platform. The Control Plane governs identity, authority, evidence, and composition across the estate. The Runtime Plane supplies the AI substrate. The Experience Plane shapes the human-AI work experience. The Capability Plane extends the platform into business systems. The Foundation Layer provides enterprise guardrails beneath all four. A governance substrate — identity, authority, and evidence — flows across every plane.

9.2.1 The Control Plane

The Control Plane is the governance brain of the platform. It answers five questions for every action in the estate: who is acting (the Identity sub-plane), what are they allowed to do (the Authority sub-plane), what happened (the Evidence sub-plane), how does the action connect to other systems (the Integration sub-plane), and how is the action composed into a larger workflow (the Composition sub-plane). Every agent action, every model call, every tool invocation passes through the Control Plane’s contracts, so that nothing in the estate happens without an identified actor, an authorised action, and a recorded outcome.

The Control Plane is realised through a set of core services: a Policy Engine that evaluates authority decisions against the organisation’s rules, a Capability Registry that catalogues what the platform can do, a Workflow and Agent Orchestrator that composes capabilities into agentic workflows, a Delegation Manager that handles the act-as chains the agent-identity chapter developed, a Risk and Trust Engine that evaluates the risk of each action in context, and an Audit and Compliance service that produces the evidence the compliance architecture requires. These services are shared infrastructure — every plane calls them rather than reimplementing them — and their shared nature is what makes governance consistent across the estate rather than patchwork across teams.

9.2.2 The Runtime Plane

The Runtime Plane is the AI substrate — the infrastructure that actually runs the models, holds the knowledge, serves the inference, and operates the estate. It corresponds most closely to the six-layer reference architecture's lower layers, and its internal structure is recognisable: a Model Layer holding the open-weight, fine-tuned, and commercial models the estate uses; a Data and Knowledge Layer holding the RAG indices, the vector stores, the knowledge graphs, and the structured and unstructured data the models reason over; an AI Services Layer providing inference, embeddings, reranking, guardrails, and tool use; an AI Ops Layer providing evaluation, observability, model ops, the prompt and workflow registry, and CI/CD for AI artefacts; and a Platform Services layer providing the compute, storage, networking, GPU/CPU management, and cost and performance management beneath everything else.

The Runtime Plane's distinguishing property in the four-plane model is that it does not govern itself — the Control Plane governs it. The Runtime Plane supplies capability; the Control Plane decides who may use that capability, under what authority, and with what evidence. This separation is deliberate: it means the infrastructure team that runs the Runtime Plane does not set the policy for who uses it, and the governance team that runs the Control Plane does not run the infrastructure. The separation of concerns is the architectural expression of the separation of duties the finance and governance chapters argue for.

9.2.3 The Experience Plane

The Experience Plane shapes what human workers actually encounter when they interact with the AI estate. Where the Runtime Plane is infrastructure the human never sees, the Experience Plane is the surface the human works through, and its design determines whether the AI feels like a useful colleague or an obstruction.

The Experience Plane has five concerns. Role and Persona defines the AI experiences and capabilities available to each role in the organisation, so that a finance analyst and a customer-service agent encounter different AI surfaces tailored to their work. Curated Workflows structures the human-AI interaction as a propose-review-confirm-act flow, so the human remains in control of consequential actions. Knowledge and Memory gives the AI access to worker memory, team memory, and organisational continuity, so it can operate with context rather than starting fresh on every interaction. Multimodal Interaction supports text, voice, image, code, and video as first-class interaction modalities. And Lineage and Accountability shows the human why the AI did what it did, how it did it, and by what authority — the transparency commitment made architectural.

The Experience Plane is the plane that addresses the Verone lesson from the governance chapter: if the official AI system is harder to use than the shadow alternative, people will route around it. The Experience Plane's job is to make the governed path the easiest path, so that governance and usability are aligned rather than opposed.

9.2.4 The Capability Plane

The Capability Plane extends the platform's reach from the AI estate into the business systems the AI acts upon. An AI agent that can reason but cannot reach the ERP, the CRM, the ITSM, the HR system, or the supply-chain system is an agent that thinks without acting. The Capability Plane is what connects thinking to acting.

Its structure spans five concerns. OSS Business Systems are the open-source or commercial business applications the platform integrates with — ERP, CRM, ITSM, HR, supply chain, and others. The Agentic Integration Fabric provides the events, APIs, adapters, and connectors that let agents reach those systems, following the Model Context Protocol and similar standards where they apply. DevSecOps for OSS provides the build, secure, test, release, operate, and observe lifecycle for the open-source components the platform depends on. Lifecycle Management handles the patching, upgrading, migration, and deprecation of the business-system integrations over time. And the COTS Displacement Path manages the domain-by-domain capability transformation as the organisation moves from commercial off-the-shelf systems to platform-native capabilities where that move is strategic.

The Capability Plane is the plane that keeps the AI platform grounded in the real operational systems of the enterprise, rather than existing as an isolated AI island that cannot touch the systems where the actual work is done.

9.3 The Foundation Layer

Beneath all four planes sits a Foundation Layer that provides the enterprise guardrails and enablement every plane depends on. The Foundation Layer is not a plane — it does not have its own actors or its own workflows — but it is the substrate that makes the planes trustworthy.

Its components are the concerns this book has developed as cross-cutting: a zero-trust architecture that treats every actor as untrusted by default, data governance and classification, security and privacy by design, compliance and regulatory alignment, observability and monitoring, FinOps and cost governance, platform engineering, and resilience including disaster recovery and business continuity. Each of these is a discipline this book has a chapter for; the Foundation Layer is the four-plane model's way of saying that every one of those disciplines must be in place before the planes above can be trusted. A Runtime Plane without the Foundation Layer's security is an unsecured Runtime Plane; a Control Plane without the Foundation Layer's compliance is an unaccountable Control Plane. The Foundation Layer is where the enterprise's non-negotiable standards are enforced, and the planes are built on top of those standards rather than beside them.

9.4 The Governance Substrate

The four planes are not independent silos; they are connected by a governance substrate that flows across all of them, carrying three things: identity, authority, and evidence. Every action in the estate, regardless of which plane initiates it, carries an identity (who is acting), an authority (what policy permits the action), and produces evidence (a record of what happened). The substrate is what makes

the estate governable as a whole rather than as four separate pieces.

The substrate is the architectural expression of the chapter's epigraph: never trust the AI. The identity layer means every action is attributable. The authority layer means every action is authorised against policy. The evidence layer means every action is recorded. Together they produce the accountability that the governance operating model, the compliance architecture, and the audit chapters require — not as a layer bolted on after the fact but as a substrate woven through the platform from its foundation.

9.5 The Design Principles

The four-plane architecture is guided by seven design principles, and each one has a specific architectural consequence rather than being a mere aspiration.

Never Trust the AI is the zero-trust principle applied to AI: every output is verified, every action is authorised, every result is logged. The architectural consequence is the governance substrate — nothing happens in the platform without identity, authority, and evidence.

Overtake AI or It Will Overtake You is the sovereignty principle: the enterprise must own and govern the agentic layer before vendor defaults and shadow adoption do it instead. The architectural consequence is that the Control Plane is enterprise-owned rather than delegated to a vendor.

Human in Control means humans set intent, AI executes, and humans confirm. The architectural consequence is the Experience Plane's propose-review-confirm-act workflow, which keeps the human in the loop for consequential actions.

Evidence Everywhere means every action, decision, and result is recorded. The architectural consequence is the Evidence sub-plane and the audit infrastructure, which produce the end-to-end lineage the compliance chapters require.

Composable by Design means capabilities are modular, discoverable, and policy-governed. The architectural consequence is the Capability Registry and the Composition sub-plane, which let the platform compose capabilities into workflows rather than hard-wiring them.

Open Where It Matters means the platform prefers open standards, open source, and interoperability. The architectural consequence is the Agentic Integration Fabric's use of standard protocols and the Capability Plane's open-source orientation.

Sovereignty by Design means data, models, and operations remain under enterprise control. The architectural consequence is the self-hosting options in the Runtime Plane, the enterprise-owned Control Plane, and the Foundation Layer's data-governance and residency controls.

9.6 The Outcomes

The four-plane architecture, operated with its design principles, produces a recognisable set of outcomes for the enterprise.

Governed agentic operations: every agent action is authorised, observable, and auditable, because the Control Plane governs every plane. **Accountable AI at enterprise scale:** end-to-end lineage and evidence connect every output to the identity, authority, and data that produced it. **Worker productivity and effectiveness:** the Experience Plane augments human expertise rather than replacing it, through curated workflows and multimodal interaction. **Enterprise ownership:** the organisation owns the platform, the data, and the governance, rather than delegating any of them to a vendor. **Reduced risk:** zero trust, least privilege, and continuous monitoring keep the security and compliance posture sound. **Cost and vendor independence:** the multi-vendor, open-standards posture of the Runtime and Capability planes rationalises vendors, increases leverage, and controls spend. **Future-ready architecture:** the composable, open, and adaptable design lets the platform evolve as foundation-model capability and the regulatory landscape change.

9.7 Mapping the Two Reference Architectures

The six-layer reference architecture and the four-plane platform architecture are not competitors; they answer different questions and are used by different people at different moments. The six-layer view is used by the system architect designing a specific AI system — it tells them what components to build and how data flows through them. The four-plane view is used by the enterprise architect or the AI platform team designing the estate — it tells them how the systems are governed, owned, and composed.

The mapping between them is recognisable. The Runtime Plane contains the six-layer model's lower layers: the data sources, the data pipeline, the model brain, and the serving infrastructure. The Experience Plane maps to the six-layer model's application layer, viewed from the human side. The Control Plane maps to the six-layer model's DevSecOps-and-governance layer, elevated from a cross-cutting concern into the governing plane. And the Capability Plane extends beyond the six-layer model, reaching into the business systems the six-layer view does not cover. The Foundation Layer maps to the cross-cutting concerns the six-layer view names at the side of its diagram. The mapping is not exact — the views decompose the same estate along different axes — but it is clear enough that a team using both can translate between them.

9.8 The African Perspective

The four-plane architecture carries distinctive weight in African deployments, across sovereignty, the Control Plane's governance value, the Experience Plane's accessibility role, and the Capability Plane's integration with African business systems.

9.8.1 Sovereignty as Architectural Motivation

The “Sovereignty by Design” principle is not an abstract preference in African contexts — it is a regulatory requirement in many jurisdictions and a strategic imperative in others. The four-plane architecture's insistence on enterprise-owned governance, self-hostable runtimes, and data that remains under the enterprise's control aligns directly with the sovereignty posture that African data-protection regimes and national AI strategies demand. For an African enterprise, the four-plane model is attractive

precisely because it does not delegate governance to a vendor, and the open-weight, self-hosted options in the Runtime Plane keep the data within the jurisdiction.

9.8.2 The Control Plane Under Lean Governance

African enterprises frequently operate with leaner governance teams than their Northern counterparts, and the Control Plane's shared-services model is what makes governance manageable at that scale. Rather than each business unit building its own identity, authority, and evidence infrastructure — an approach that a lean team cannot staff — the Control Plane provides these as shared services that every team uses. A lean governance team can run one Control Plane more effectively than it can run several bespoke governance setups, and the four-plane model is therefore a governance-efficiency play as much as a governance-quality play in resource-constrained deployments.

9.8.3 The Experience Plane for Multilingual, Mobile-First Users

The Experience Plane's role-and-persona design, its multimodal interaction support, and its curated workflows are particularly consequential in African deployments, where the user base is multilingual and predominantly mobile. The Experience Plane is where the platform adapts to the user's language, device, and connectivity rather than requiring the user to adapt to the platform, and in African markets that adaptation is the difference between a platform that is used and one that is routed around.

9.8.4 The Capability Plane and African Business Systems

The Capability Plane's integration with business systems must, in African deployments, include the systems that are central to African commerce: mobile-money platforms, national identity systems, government-service portals, and the telecoms interfaces that reach users on feature phones. The Agentic Integration Fabric must speak to M-Pesa and MTN MoMo as fluently as it speaks to an ERP, and the COTS Displacement Path must account for the business systems that African enterprises actually run, which are not always the same systems a Northern enterprise would have. The Capability Plane does not change for African deployments, but its integration targets do.

9.9 Recurring Failure Patterns

The failure modes of enterprise AI platform architecture recur, and each maps to a plane or a principle this chapter has set out:

- **No Control Plane.** The estate has models, inference, and applications, but no unified governance — identity is per-system, authority is per-team, and evidence is scattered or absent. The result is an ungovernable estate.
- **Runtime without Experience.** The platform provides powerful infrastructure but no thought to how humans actually work with it, so the AI is capable but unused, and shadow tools fill the gap.
- **Experience without Capability.** The human-facing experience is polished, but the agents behind it cannot reach the business systems where the real work is done, so the AI can reason but not act.
- **No Foundation Layer.** The planes are built without the underlying guardrails — no zero trust, no data governance, no observability — and the whole platform is insecure, unobservable, or

non-compliant.

- **Vendor-delegated governance.** The organisation lets a vendor's defaults define the governance posture, and the "Overtake AI" principle is violated — the vendor, not the enterprise, controls the agentic layer.

Each pattern maps to a remedy the chapter has developed — the Control Plane for unified governance, the Experience Plane for human usability, the Capability Plane for business-system reach, the Foundation Layer for enterprise guardrails, and the sovereignty-by-design principle for enterprise ownership — and together they show that an enterprise AI platform fails not through any single missing component but through a missing plane or a missing principle.

9.10 What This Chapter Adds

The four-plane enterprise AI platform is a governance-first, enterprise-owned reference architecture that complements the six-layer system view. The Control Plane governs identity, authority, evidence, and composition across the estate. The Runtime Plane supplies the AI substrate. The Experience Plane shapes the human-AI work experience. The Capability Plane extends the platform into the business systems the AI acts upon. The Foundation Layer provides the enterprise guardrails beneath all four. And the governance substrate — identity, authority, evidence — flows across every plane, so that nothing in the estate happens without an identified actor, an authorised action, and a recorded outcome.

The seven design principles — never trust the AI, overtake AI or it will overtake you, human in control, evidence everywhere, composable by design, open where it matters, sovereignty by design — each carry a specific architectural consequence rather than a mere aspiration, and the outcomes they produce — governed agentic operations, accountable AI at scale, worker productivity, enterprise ownership, reduced risk, cost and vendor independence, and a future-ready architecture — are the outcomes an enterprise builds a platform to achieve. The six-layer view tells the architect what to build inside any one system; the four-plane view tells the enterprise how to own the estate those systems live in. Both are needed, and the book now provides both.

Part II

The Six-Layer Reference Architecture

“The whole is a series of parts, each one knowing its place, and standing in just relation to the next.”

—John Wood the Elder, on the planning of Bath, 1727–54

If Part I has explained what enterprise AI is, the question now turns to how it is structured. When John Wood the Elder set out to rebuild the old Roman town of Bath as a modern city in the 1720s, he did not begin with individual buildings. He began with the layered system that the buildings would inhabit: the streets that connected them, the squares they framed, the social geometry of who would walk where. The Queen Square, the Circus, and the later Royal Crescent are remembered as architectural achievements, but they are first an achievement of layering — a clear separation between the level at which a building stands and the level at which a city moves. Part II of this book proposes the same separation for enterprise AI: six layers, each with its own concerns, each connecting cleanly to the layers above and below.

Chapter 10

Layer 1: Data Sources

“Garbage in, garbage out—and in machine learning, garbage in is garbage out at scale.”

—adapted from George Fuechsel

Chapter Overview

Layer 1 is where data originates. Every enterprise AI system is downstream of the source systems that produce its inputs, and the quality, freshness, accessibility, and trustworthiness of those sources determine what is achievable above. This chapter catalogues the four families of source data, develops the metadata each requires, specifies the contracts the next layer must establish, and surveys the change-capture mechanisms that let downstream systems stay synchronised with upstream truth.

Learning Objectives

- Catalogue the four families of enterprise data sources: structured, unstructured/semi-structured, APIs and streams, and files and storage.
- Specify the metadata—ownership, schema, freshness, sensitivity, retention—that the next layer requires from every source.
- Compare full-load, incremental, log-based CDC, and event-streaming integration strategies.
- Establish a written data contract between source owners and the data pipeline that survives staff turnover and reorganisations.
- Recognise the failure modes of unmanaged source dependencies and apply the structural mitigations.

10.1 The Source Inventory

The first practical task of any enterprise AI programme is producing an inventory of data sources. The inventory is not a wiki page; it is a structured artefact that the pipeline team, the platform team, and the data-governance function all read and update. Without it, downstream layers depend on tribal knowledge that ages out within a year.

A minimum-viable source inventory captures the following fields for every source:

Table 10.1: Required fields in a data-source inventory entry.

Field	Purpose
Source identifier	Stable name used across the pipeline, lineage graph, and catalogue.
Owner	Named individual or team accountable for the source's correctness.
System of origin	The application or platform that generates the data.
Format and schema	Tabular schema, document type, message schema, or file layout.
Update cadence	Real-time, hourly, daily, weekly, irregular.
Freshness SLA	Maximum allowable lag between source change and pipeline visibility.
Sensitivity tier	Public, internal, confidential, restricted, regulated.
Retention policy	How long the source retains historical data; relevant for backfills.
Access mechanism	Connection string, API endpoint, queue topic, file location.
Change-notification	How downstream learns of schema or semantic changes.

Most enterprises discover, on first attempt, that their inventory is incomplete and that the missing entries are the most consequential ones. The discipline of completing it is itself a maturity milestone.

10.2 Structured Data

Relational databases, data warehouses, and operational systems remain the backbone of enterprise data. Their structure is their virtue: schemas are knowable, joins are well-defined, and most analytical questions can be answered with SQL. Their limitation, in the context of AI, is that they capture only what the business decided in advance to record.

10.2.1 Operational and Analytical Stores

Two broad classes of structured store appear in every enterprise. *Operational* stores (OLTP: Postgres, MySQL, Oracle, SQL Server) optimise for low-latency reads and writes against current state. *Analytical* stores (OLAP: Snowflake, BigQuery, Redshift, Databricks SQL, ClickHouse) optimise for scanning large historical ranges. Pipelines typically read from analytical stores to avoid pressuring operational systems, with the analytical store kept up to date by CDC or batch ETL from the operational store.

Table 10.2: Characteristics of operational versus analytical stores.

Dimension	Operational (OLTP)	Analytical (OLAP)
Workload	Many small transactions	Few large scans
Latency target	1–10 ms	Seconds to minutes
Storage layout	Row-oriented	Columnar
Concurrency	Thousands of writers	Tens of readers
Retention	Current state plus short history	Long history, often append-only
Typical use in AI	Source of operational facts	Feature engineering, training data

10.2.2 Schema Discovery

A pipeline that hard-codes column names against a source schema breaks the first time the source changes. Schema discovery—reading the source’s `information_schema` or equivalent at job start and validating against a declared expected schema—catches the change before it propagates. Modern lakehouse formats (Delta, Iceberg, Hudi) push this further: schema evolution rules govern which changes are compatible (adding a nullable column) and which require coordination (changing a column type).

```

1  -- Compare the live source schema to the contract declared in the
    catalogue.
2  WITH live AS (
3      SELECT column_name, data_type, is_nullable
4      FROM information_schema.columns
5      WHERE table_schema = 'sales'
6            AND table_name = 'orders'
7  ),
8  contract AS (
9      SELECT column_name, data_type, is_nullable
10     FROM platform.contract_orders_v3
11 )
12 SELECT 'missing_in_source' AS diff, c.column_name FROM contract c
13 LEFT JOIN live l USING (column_name) WHERE l.column_name IS NULL
14 UNION ALL
15 SELECT 'new_in_source' AS diff, l.column_name FROM live l
16 LEFT JOIN contract c USING (column_name) WHERE c.column_name IS NULL
17 UNION ALL
18 SELECT 'type_mismatch' AS diff, l.column_name FROM live l
19 JOIN contract c USING (column_name) WHERE l.data_type <> c.data_type;

```

Listing 10.1: Detecting schema drift against a declared contract.

10.2.3 Reference and Master Data

Reference data (country codes, currency codes, product hierarchies) and master data (customers, products, employees) are structured sources with disproportionate downstream impact. A wrong currency code corrupts every conversion; a duplicate customer fragments every signal computed from interaction history. Layer 1 architecture should single-source these aggressively: one canonical store per master entity, with all consumers reading from it.

10.3 Unstructured and Semi-Structured Data

Document stores (PDF, DOCX, HTML), email systems, JSON, XML, CSV files, and media files contain the contextual information that gives structured records meaning. Modern AI—and particularly retrieval-augmented generation—draws most of its power from this category.

10.3.1 The Unstructured Inventory Problem

Where structured sources are catalogued in their database engines, unstructured sources scatter across file shares, SharePoint sites, Confluence wikis, Slack channels, ticketing systems, and email archives. The architecture cannot ingest what it cannot discover. Two structural responses dominate: *crawl-and-index* (a service discovers new content on a schedule and writes pointers into a catalogue) and *publish-on-write* (source systems emit an event whenever a document changes, and the pipeline subscribes). Publish-on-write is preferable when available because it avoids the latency and cost of repeated crawling, but most legacy systems require crawl.

10.3.2 Format Heterogeneity

A single document corpus may include native PDFs, scanned PDFs (image-only), Word documents from three Office versions, scanned email attachments, and tables embedded in slide decks. Each format extracts differently; each has its own failure modes (a complex multi-column PDF layout can produce nonsense text under naive extraction). The architectural mitigation is a format-aware extraction service at Layer 2 that selects an appropriate parser per format and routes hard cases (image-only PDFs) to optical character recognition.

10.3.3 Provenance and Authority

For each unstructured source, the architecture should record *who* wrote it, *when* it was last verified, and what its authority is. A policy document approved by the Compliance committee is not the same kind of source as a draft slide deck by an unnamed analyst. RAG systems that fail to encode this distinction will produce confident citations to obsolete or unauthoritative content.

10.4 APIs and Streams

REST and SOAP APIs, IoT device feeds, real-time streams, and webhooks are the dynamic edge of the data estate. They carry the events that distinguish what a customer is doing *now* from what they did historically.

10.4.1 Pull versus Push

Pull integrations (the pipeline polls an API on a schedule) are simpler to implement but trade latency for completeness: an event that occurs between polls is invisible until the next cycle. Push integrations (webhooks, message queues) offer near-real-time visibility at the cost of additional infrastructure (a receiver endpoint, idempotency tracking, retry handling).

Table 10.3: Integration mechanisms and their trade-offs.

Mechanism	Latency	Complexity	Best for
Full table dump	Hours–days	Low	Small reference data, infrequent change
Incremental query (updated_at)	Minutes–hours	Low	Append-mostly tables with reliable timestamps
Log-based CDC	Seconds–minutes	Medium	High-volume, change-heavy tables
Webhook	Seconds	Medium	Vendor systems with native push support
Event stream (Kafka, etc.)	Sub-second	High	High-volume domain events, replayability

10.4.2 Idempotency and Ordering

Networked sources fail. A webhook may deliver the same event twice; a polling job may overlap with the previous run; a message bus may reorder messages within a partition. Layer 1 contracts must specify whether the source guarantees at-least-once or exactly-once delivery, and whether ordering is preserved per key or only per partition. These properties propagate downstream: a feature computed from out-of-order events will be wrong even if the pipeline is correct.

10.4.3 Backpressure

A real-time stream that ingests faster than the pipeline can process produces backpressure: queues fill, consumers lag, and freshness SLAs slip silently. Architectures that ignore backpressure discover it as an incident. Architectures that plan for it specify consumer-group lag metrics, define alert thresholds, and have a documented response (scale consumers, partition more aggressively, or shed load by sampling).

10.5 Files and Object Storage

File shares (SMB, NFS), object storage (S3, Azure Blob, GCS), and third-party data sources round out the picture. The challenge with this category is rarely access; it is provenance. A spreadsheet attached to an email three years ago may be the system of record for a critical input and have no documented owner.

10.5.1 Object-Storage Patterns

The dominant pattern is *landing zones plus prefixes*: each source writes to a dedicated prefix in object storage, partitioned by date for incremental processing.

```

1 s3://enterprise-data-lake/raw/
2   source=salesforce_accounts/
3     ingest_date=2026-05-14/
4       part-00001.parquet
5       _SUCCESS                                # marker file written after all
6       parts land
7     ingest_date=2026-05-15/
8     ...
9   source=hr_payroll/

```

```

9      ingest_date=2026-05-14/
10      ...

```

Listing 10.2: Object-storage layout for a source landing zone.

Marker files (`_SUCCESS`, manifests) signal completion atomically; without them, downstream jobs may read partial data from an in-flight write.

10.5.2 Third-Party Sources

Commercial data—market data, regulatory filings, weather, demographics—arrives through SFTP drops, REST APIs, or vendor-specific clients. The architectural concerns are licensing (what may the data be used for?), redistribution (may it appear in derived datasets shared with partners?), and stability (vendors change formats with little notice). A vendor abstraction layer at Layer 1 isolates the rest of the pipeline from these specifics.

10.6 Data Contracts

For each source feeding the next layer, the architecture should establish a written *data contract* signed by the source owner and the pipeline owner. The contract is the durable artefact that survives staff turnover and reorganisations.

A typical contract specifies:

- **Schema** (column or field names, types, nullability, allowed values).
- **Semantic definitions** (what does the `status` field's value `Active` mean? What is a `customer`?).
- **Freshness SLA** (e.g. data available in the lake within 15 minutes of source change).
- **Availability SLA** (e.g. 99.5% of expected daily loads complete on time).
- **Quality expectations** (e.g. less than 0.1% null in primary keys; row counts within $\pm 5\%$ of seven-day moving average).
- **Change-notification process** (a minimum lead time before breaking changes).
- **Escalation path** (who is paged when the contract is violated).

The contract is a *platform-level* artefact, not a per-project agreement. The same source serves many consumers; the contract is what lets the platform serve all of them without renegotiating each integration.

```

1 contract: customers.v3
2 owner: customer-data-team@example.com
3 source_system: salesforce
4 update_cadence: streaming
5 freshness_sla_minutes: 15
6 availability_sla_pct: 99.5
7 schema:

```

```

8   - name: customer_id          ; type: string ; nullable: false ; pk:
    true
9   - name: email                ; type: string ; nullable: false ; pii:
    true
10  - name: created_at           ; type: timestamp; nullable: false
11  - name: status               ; type: string ; enum: [active, churned,
    prospect]
12  quality_checks:
13    - rule: pk_uniqueness       ; threshold: 1.0
14    - rule: email_format        ; threshold: 0.999
15    - rule: row_count_anomaly   ; window_days: 7 ; tolerance_pct: 5
16  breaking_change_notice_days: 14
17  escalation: pagerduty://customer-data

```

Listing 10.3: A simple data contract as a versioned artefact (YAML).

10.7 Failure Modes and Mitigations

The recurring failure modes of Layer 1 are not exotic; they are familiar to every team that has run a pipeline for more than a year.

Table 10.4: Recurring Layer 1 failure modes and their mitigations.

Failure mode	Mitigation
Silent schema change	Schema-drift checks (Listing 10.1); contract version pinning.
Late-arriving data	Watermarks; explicit re-processing windows for downstream jobs.
Duplicate events	Idempotent ingest keys; deduplication on a deterministic event identifier.
Source outage	Cached last-good copy; degraded-mode flag exposed to downstream.
Sensitive data leak	Sensitivity tagging at intake; redaction or tokenisation before lake landing.
Unannounced source decommission	Contract-level change-notification clause; quarterly source-review meeting.

10.8 Closing Notes

Layer 1 is the foundation, and its quality is the ceiling of what the upper layers can achieve. The architectural work here is less about technology than about cataloguing, classifying, and contracting with sources. Done well, Layer 1 delivers: raw data available, multiple source connectivity, sensitivity tagged at intake, and data intake ready for the pipeline above. Done poorly, it ensures that every layer above will inherit and amplify the same problems. The data contract is the most leveraged single artefact in the layer—a one-page YAML file that prevents most of the failure modes that consume operational time in mature pipelines.

Chapter 11

Layer 2: Data Pipeline

“Data is the new oil. It needs to be refined.”

—Clive Humby

Chapter Overview

Layer 2 ingests, transforms, and stores data so that the AI/ML brain above can consume it. This chapter develops the ingestion patterns, processing and ETL approaches, the medallion (bronze/silver/gold) architecture that has become the standard organising principle for the modern data lakehouse, and the feature store that bridges the pipeline to training and inference.

Learning Objectives

- Distinguish batch ingestion, real-time ingestion, and change-data-capture, with their respective trade-offs.
- Apply data validation, cleansing, transformation, deduplication, and enrichment in a defensible order.
- Implement the medallion architecture for traceable, quality-gated data progression.
- Design a feature store that serves both offline training and online inference with point-in-time correctness.
- Apply data-quality testing as a first-class engineering discipline.

11.1 Pipeline Reference Architecture

The data pipeline composes four sub-layers, each with a well-defined responsibility (Figure 11.1). Ingestion brings data from Layer 1 into the platform. Processing applies validation, cleansing, transformation, deduplication, and enrichment. Storage organises the results into the medallion tiers. The feature store exposes the curated outputs to Layer 3.

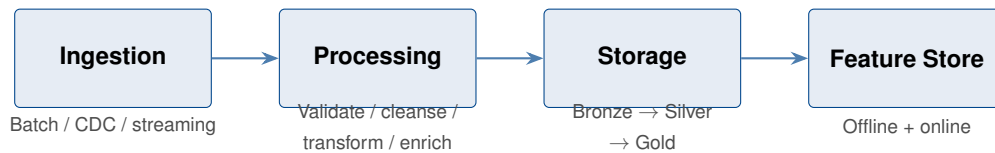


Figure 11.1: Sub-architecture of Layer 2: ingestion, processing, medallion storage, and the feature store that hands curated data to Layer 3.

11.2 Ingestion

Batch, real-time, and orchestrated scheduling are the three modes by which data enters the pipeline. The choice depends on the freshness requirements of downstream models: a daily risk score tolerates batch; a real-time fraud check does not.

11.2.1 Batch Ingestion

Batch jobs run on a schedule (hourly, daily, weekly) and process a defined window of source data. They are simple, predictable, and tolerable of moderate failures (a missed window is recovered by the next run). The classical pattern is *extract-transform-load* (ETL); the modern pattern is *extract-load-transform* (ELT), where raw data is loaded to the lakehouse first and transformed in place, taking advantage of cheap storage and on-demand compute.

Table 11.1: ETL versus ELT: pipeline ordering trade-offs.

Dimension	ETL (extract-transform-load)	ELT (extract-load-transform)
Order	Transform before load	Load raw, then transform
Storage cost	Lower (only curated lands)	Higher (raw retained)
Compute cost	Concentrated upstream	Distributed in warehouse
Reproducibility	Source data discarded post-transform	Raw preserved, transforms rerunnable
Modern preference	Legacy systems, regulated discards	Lakehouse era default

11.2.2 Change-Data-Capture

Log-based change-data-capture reads the transaction log of a source database and emits a stream of insert/update/delete events. It avoids the load that periodic full-table scans place on operational systems and provides minute-level (or better) freshness for downstream consumers.

```

1 {
2   "source"    : "salesforce.accounts",
3   "op"        : "u",                      // i=insert, u=update, d=
      delete
4   "ts_ms"     : 1763114400123,
5   "before"    : { "id": "0014X...", "status": "active", "tier": "
      silver" },
6   "after"     : { "id": "0014X...", "status": "churned", "tier": "
      silver" },

```

```

7   "tx_id"      : "000000000:000000a3:0001",
8   "lsn"       : 173294823
9 }

```

Listing 11.1: A simplified CDC event payload.

11.2.3 Streaming Ingestion

Streaming systems (Kafka, Pulsar, Kinesis) handle high-volume event data with sub-second latency. Their architectural cost is operational complexity: partitioning strategy, consumer-group management, watermark handling for late events, and exactly-once semantics each require deliberate design rather than default configuration.

11.2.4 Orchestration

Whichever ingestion mode dominates, an orchestrator (Airflow, Dagster, Prefect, Argo) coordinates dependencies, retries, and failure recovery. The orchestrator owns the directed acyclic graph (DAG) of tasks; each task is small and idempotent so that failures recover by restart rather than by manual intervention.

11.3 Processing and ETL

Validation, cleansing, transformation, deduplication, and enrichment are the five recurring operations. They should be applied in this order, each producing a checkpoint that downstream consumers can rely on without re-running the upstream stages.

11.3.1 Validation

Validation checks each incoming record against the data contract: types match, nullability holds, foreign keys resolve, enums are in range. Records that fail validation are routed to a quarantine area for inspection rather than dropped or allowed to corrupt downstream.

```

1 from pydantic import BaseModel, Field, ValidationError
2 from typing import Literal
3
4 class Customer(BaseModel):
5     customer_id: str = Field(min_length=18, max_length=18)
6     email: str = Field(pattern=r"^[^@]+@[^@]+\.[^@]+$")
7     status: Literal["active", "churned", "prospect"]
8
9 def validate_batch(records):
10     valid, invalid = [], []
11     for row in records:
12         try:
13             valid.append(Customer(**row).model_dump())
14         except ValidationError as e:
15             invalid.append({"row": row, "errors": e.errors()})
16     return valid, invalid

```

Listing 11.2: Schema-level validation using a declarative validator.

11.3.2 Cleansing, Transformation, Deduplication, Enrichment

Cleansing standardises representation: trimming whitespace, normalising case, parsing dates into a canonical timezone, mapping legacy codes onto current ones. *Transformation* reshapes data: pivoting, joining, computing derived columns, aggregating; tooling has consolidated around SQL-based frameworks (dbt, SQLMesh) for tabular workloads and Spark or Flink for high-volume distributed workloads. *Deduplication* requires a deterministic key (often a hash of canonicalised fields) and a tie-break rule (latest by ingest time, highest data-quality score). *Enrichment* joins to reference data to add context; the architectural rule is that enrichment is reversible—the original code is preserved so downstream consumers can recompute if the reference table changes.

11.4 The Medallion Architecture

The medallion architecture organises stored data into three quality tiers: *bronze* for raw data preserved as received, *silver* for curated data with applied transformations, and *gold* for analytics-ready, business-aligned datasets. This progression supports traceability, governance, and reproducibility—the trilogy that auditors will eventually ask for.

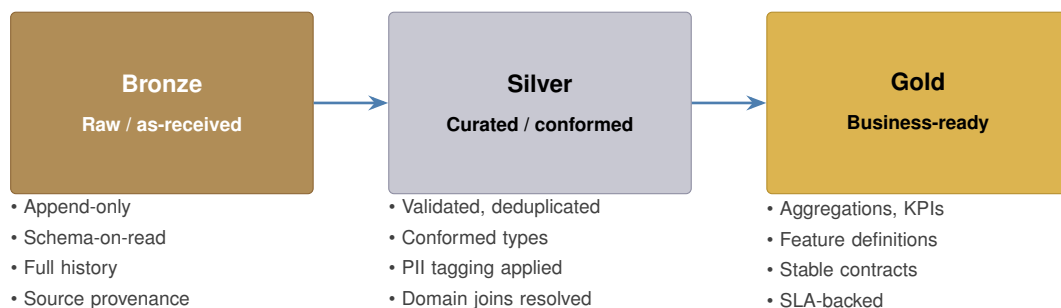


Figure 11.2: The medallion architecture: bronze preserves raw inputs, silver curates and conforms them, gold serves business-aligned outputs to consumers.

11.4.1 Bronze

Bronze tables are append-only and preserve source data with minimal transformation: a stable record of what arrived, when it arrived, and from where. The discipline of writing every record to bronze before any transformation is the single most valuable architectural commitment in the medallion model: it makes every downstream output reproducible from preserved inputs.

11.4.2 Silver

Silver tables apply the processing operations of Section 11.3: validation, cleansing, deduplication, type conformance, PII tagging, and the resolution of straightforward domain joins. Silver tables are the data warehouse’s working draft—trusted by analytics teams and by the ML feature pipeline, but not yet shaped for any specific business question.

11.4.3 Gold

Gold tables are the curated outputs that downstream consumers (dashboards, ML training jobs, feature stores) read against. Each gold table has a named owner, a documented schema, and an SLA that governs its update cadence and quality. Gold is where the platform’s contracts with its consumers live.

11.5 Data Lake, Lakehouse, and Warehouse

The storage substrate beneath the medallion tiers has evolved through three generations. The classical data lake (Hadoop, raw object storage) provided cheap storage but lacked transactional semantics; concurrent writers and partial-failure recovery were the application’s problem. The data warehouse provided transactions but at high cost and with limited support for unstructured data. The lakehouse—built on transactional table formats (Delta, Iceberg, Hudi) over cheap object storage—combines the cost profile of the lake with the transactional semantics of the warehouse.

Table 11.2: Storage substrate evolution.

Generation	Strength	Weakness	Example
Data lake	Cheap, format-flexible	No transactions, no schema enforcement	HDFS, S3 + Parquet
Data warehouse	Strong transactions, fast SQL	Expensive, structured only	Teradata, classic Snowflake
Lakehouse	Cheap + transactional	Operational complexity in format ecosystem	Delta, Iceberg, Hudi

11.6 The Feature Store

The feature store sits above the data lakehouse, providing point-in-time-correct features for training and online features for low-latency inference. It decouples the data pipeline from the AI/ML brain that consumes it.

11.6.1 Why a Feature Store

Without a feature store, every team that consumes data for ML reinvents the same operations: building training datasets that respect temporal causality (no leakage from future rows), serving the same features to online inference at low latency, monitoring features for drift. Three teams independently writing the same feature definition will end up with three subtly different features.

11.6.2 Offline and Online Stores

A feature store typically maintains two physical stores. The *offline store* (a columnar table in the lakehouse) holds the full history of feature values, indexed for point-in-time queries during training. The *online store* (a low-latency key-value store like Redis or DynamoDB) holds the latest feature values for inference, refreshed by the pipeline as new data arrives.

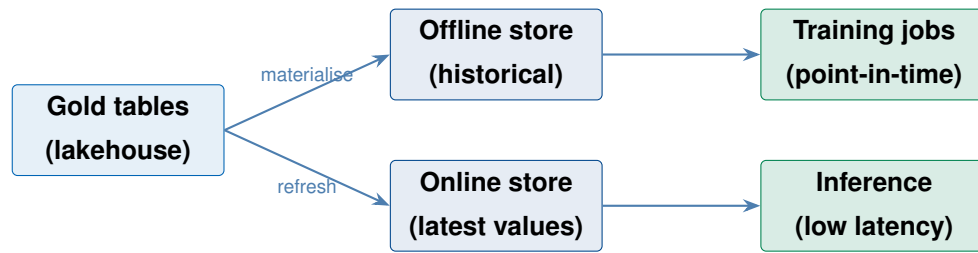


Figure 11.3: Feature store architecture: gold tables materialise into both an offline store (historical, for training) and an online store (latest values, for inference).

11.6.3 Point-in-Time Correctness

The single most subtle feature-store property is point-in-time correctness: a training example labelled with the event at time t must see only feature values that were known at time t , never values that would only be computable later. Violating this is *data leakage*: a model that scores well in training and underperforms in production. Modern feature stores (Feast, Tecton, Databricks Feature Store, Vertex AI Feature Store) enforce point-in-time joins as a first-class operation.

```

1 # label_events: DataFrame with (entity_id, event_ts, label)
2 # Result: each row has features as of event_ts (no future leakage).
3 training_df = feature_store.get_historical_features(
4     entity_df = label_events,
5     features = [
6         "customer_features:lifetime_value",
7         "customer_features:days_since_signup",
8         "transaction_features:txn_count_30d",
9     ],
10 ).to_df()
  
```

Listing 11.3: A point-in-time feature join.

11.7 Data Quality

Data quality is not a downstream cleanup activity; it is a continuous discipline applied at each medallion transition. The dominant frameworks (Great Expectations, dbt tests, Soda) express quality rules as code that runs in the pipeline.

Table 11.3: Recurring data-quality dimensions.

Dimension	Example check
Completeness	Required fields are not null; row counts within expected range.
Uniqueness	Primary keys are unique; no duplicate business identifiers.
Validity	Values conform to types, enumerations, and format constraints.
Consistency	Cross-field invariants hold (e.g., <code>start_date < end_date</code>).
Timeliness	Data arrives within the freshness SLA.
Accuracy	Sample reconciliations against source-of-truth match within tolerance.

Failed checks should fail the pipeline loudly, not produce silently degraded data downstream.

11.8 The Pattern, Restated

Layer 2 turns raw source data into clean, trusted, traceable data ready for AI. The medallion architecture is the organising structure; the feature store is the bridge to Layer 3; data-quality testing is the continuous discipline that keeps both honest. Outcomes: clean trusted data, scalable storage, point-in-time-correct features, and the structural foundations on which Layer 3 can build intelligence without revisiting Layer 1.

Chapter 12

Layer 3: The AI/ML Brain

“Intelligence is the ability to adapt to change.”

—Stephen Hawking

Chapter Overview

Layer 3 is where intelligence happens. It contains the data preparation specific to ML, the model layer (foundation models, fine-tuned models, traditional ML), the RAG/knowledge layer that grounds models in enterprise data, the orchestration layer that composes calls into workflows, and the specialised AI services (NLP, vision, speech, document intelligence) that handle modality-specific tasks. This is the largest and fastest-changing layer in the architecture; the value of its abstractions lies in stable interfaces over volatile internals.

Learning Objectives

- Distinguish the responsibilities of data preparation, model layer, knowledge layer, orchestration, and AI services.
- Choose between foundation models, fine-tuned models, parameter-efficient adaptation, and traditional ML for a given problem.
- Design a production RAG pipeline: chunking, embedding, indexing, retrieval, re-ranking, prompt construction.
- Compose models, tools, and workflows through an orchestration layer that supports both chains and agents.
- Recognise the limits of each component and the structural mitigations that keep the layer reliable.

12.1 Data Preparation for ML

Feature engineering, tokenisation, embeddings, and vectorisation are the operations that turn Layer 2’s tabular and textual outputs into representations that models can consume. The choices here—tokenizer, embedding model, vector dimensionality—are surprisingly difficult to reverse, so they deserve early attention.

12.1.1 Feature Engineering for Tabular ML

Traditional ML models depend on hand-crafted features: aggregations, ratios, encodings, interaction terms. The feature store (Section 11.6) holds the canonical definitions; the model code consumes them. Common transformations include:

- **Numerical scaling.** Standardisation $\tilde{x} = (x - \mu) / \sigma$ or min-max scaling $\tilde{x} = (x - x_{\min}) / (x_{\max} - x_{\min})$.
- **Categorical encoding.** One-hot for low cardinality, target encoding for high cardinality, embeddings for very high cardinality.
- **Temporal features.** Time-since-last-event, rolling windows $(\bar{x}_{7d}, \bar{x}_{30d})$, seasonality decomposition.
- **Aggregations.** Counts, sums, averages, and quantiles over windows or groups.

12.1.2 Tokenisation

Text-based models operate on tokens, not characters. The choice of tokeniser is part of the model: a model trained with one tokeniser cannot consume tokens from another. Common families are byte-pair encoding (BPE, used by GPT models), SentencePiece (used by Llama, T5), and WordPiece (used by BERT). Tokenisers affect cost, latency, and quality: a tokeniser that fragments domain terms into many sub-tokens inflates the input cost of every call.

```
1 import tiktoken
2 enc = tiktoken.encoding_for_model("gpt-4o")
3 text = "Enterprise AI architecture is a discipline."
4 tokens = enc.encode(text)
5 print(len(tokens), tokens)    # 8 [...]
```

Listing 12.1: Counting tokens for cost estimation.

12.1.3 Embeddings

An embedding is a dense vector $\mathbf{e} \in \mathbb{R}^d$ representing a piece of content (text, image, audio) such that semantically similar content has small vector distance. Embedding models are themselves neural networks, typically fine-tuned variants of foundation models. Selection criteria include:

- Dimensionality d (typically 384–3072): larger captures finer distinctions, costs more storage and search compute.
- Training corpus: a model trained on web text may underperform on legal or medical content.
- Cost per million tokens, and whether embedding generation is hosted or self-served.
- License: some open embeddings forbid commercial use.

The cosine similarity $\cos(\mathbf{e}_1, \mathbf{e}_2) = \frac{\mathbf{e}_1 \cdot \mathbf{e}_2}{\|\mathbf{e}_1\| \|\mathbf{e}_2\|}$ is the most common distance measure; many vector stores normalise embeddings at write time so that the dot product equals cosine similarity, accelerating queries.

12.2 The Model Layer

Foundation models (LLMs), fine-tuned models, and traditional ML models coexist in most enterprise systems. Foundation models bring breadth without training cost; fine-tuned models bring task specialisation; traditional ML brings interpretability and low latency. The architecture should accommodate all three rather than picking one.

Table 12.1: Model families and their typical roles in an enterprise stack.

Family	Strengths	Limitations	Typical role
Foundation LLM (API)	Breadth, no training cost	Per-call cost, vendor dependency, governance	General reasoning, drafting, summarisation
Open-weight LLM (self-hosted)	Cost control, data sovereignty	Operational burden, GPU footprint	Sensitive workloads, high volume
Fine-tuned LLM	Task specialisation, lower per-call cost	Training cost, drift, maintenance	Domain-specific tasks at scale
Traditional ML (gradient boosting, regression)	Interpretable, low latency, cheap	Limited to structured features	Classification, scoring, ranking
Deep learning (CV, speech)	Modality-specific accuracy	Specialised infrastructure	Vision, speech, document AI

12.2.1 Foundation Models

A foundation model is a large neural network (typically a transformer) pre-trained on broad data and adapted to many downstream tasks via prompting or fine-tuning. Architectural concerns:

- **Context window.** The maximum tokens (input + output) the model can attend to in one call. Modern frontier models support 100k–1M tokens; cost grows roughly linearly with input length.
- **Modality.** Text-only, vision-text, or multimodal (text, image, audio, video).
- **Hosting.** Vendor-managed API (operational simplicity, vendor dependency) or self-hosted weights (cost control, complexity).
- **Determinism.** Even with temperature zero, most LLM APIs do not guarantee bit-identical outputs across calls.

12.2.2 Fine-Tuning Strategies

When a foundation model is not specialised enough, three adaptation strategies are common:

1. **Full fine-tuning.** Update all model parameters on task-specific data. Most expensive; produces a fully specialised model.
2. **Parameter-efficient fine-tuning (PEFT).** Inject small trainable adapters (LORA, QLORA, prefix tuning) while freezing the base model. Modest cost; the base model is reusable across tasks.

3. **Instruction tuning / preference tuning.** Train on (prompt, preferred response) pairs to align outputs with desired behaviour; often combined with reinforcement learning from human feedback (RLHF) or direct preference optimisation (DPO).

LORA is particularly common. The technique factorises a weight update $\Delta W \in \mathbb{R}^{d \times k}$ as the product of two low-rank matrices $\Delta W = BA$ where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. The number of trainable parameters drops from dk to $r(d + k)$ —often a 100–1000× reduction.

12.2.3 Traditional ML

For tabular classification, regression, and ranking, gradient-boosted trees (XGBoost, LightGBM, CatBoost) remain the strongest baseline. They are interpretable (via feature importance and SHAP values), cheap to train and serve, and resilient to missing features. An architecture that reaches for an LLM for every task is one that pays LLM prices for problems that a 50ms tree model would solve better.

12.3 The RAG / Knowledge Layer

Retrieval-augmented generation grounds model outputs in enterprise knowledge: a vector database stores embeddings of source content, retrieval finds the relevant chunks at query time, augmentation injects them into the prompt, and context injection ensures the model sees them. Chapter 21 develops the pattern; here we cover the architectural components.

12.3.1 Chunking

Source documents are split into chunks before embedding. The chunking strategy determines both retrieval quality and cost. Three approaches dominate:

Table 12.2: Chunking strategies and their trade-offs.

Strategy	Mechanism	Trade-off
Fixed-size	Split every N tokens with optional overlap	Simple; may split mid-sentence
Structural	Split on headings, paragraphs, list items	Honours document structure; depends on parser quality
Semantic	Split where embedding similarity between adjacent sentences drops	Highest retrieval quality; highest preprocessing cost

Chunk size is also a parameter. Smaller chunks (128–256 tokens) give precise retrieval but lose context; larger chunks (512–1024) give context but dilute relevance scores. A common compromise is 512-token chunks with 64-token overlap.

12.3.2 Vector Index

A vector index supports approximate nearest-neighbour search over embeddings. The dominant algorithms are:

- **HNSW** (Hierarchical Navigable Small World). Graph-based, fast queries, high memory.
- **IVF** (Inverted File). Cluster-based, lower memory, slower queries.
- **PQ** (Product Quantisation). Compresses vectors for storage efficiency, often combined with IVF.

The choice depends on corpus size and query latency budget. For corpora under 10^6 vectors, exact search with normalised dot product is often fast enough.

12.3.3 Hybrid Retrieval

Pure vector retrieval struggles with exact-match queries (codes, identifiers, rare names). *Hybrid retrieval* combines vector search with keyword search (BM25) and merges results, typically via reciprocal rank fusion:

$$\text{score}_{\text{hybrid}}(d) = \sum_{r \in \{r_{\text{vec}}, r_{\text{bm25}}\}} \frac{1}{k + r(d)}$$

where $r(d)$ is the rank of document d in each list and k is a smoothing constant (commonly 60).

12.3.4 Re-Ranking

A cross-encoder re-ranker takes the top K candidates from initial retrieval and re-scores each against the query using a more accurate (but slower) model. The combination of fast first-stage retrieval over millions of chunks and slow second-stage re-ranking over top-100 candidates yields both speed and quality.

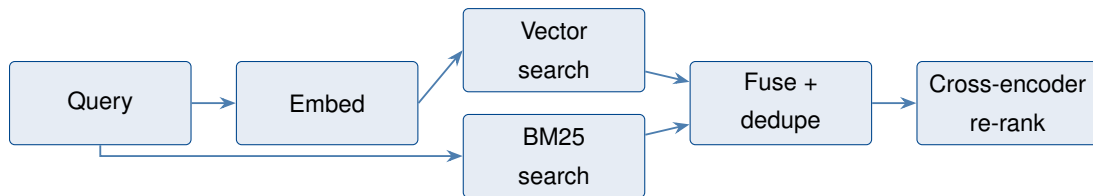


Figure 12.1: A hybrid retrieval pipeline: vector and lexical search proceed in parallel, results are fused, and a cross-encoder re-ranks the top candidates.

12.3.5 Prompt Construction

The retrieved chunks become context in the prompt to the generation model. A robust prompt template includes: system instructions (role and constraints), retrieved context (clearly delimited), the user query, and a directive to cite sources or refuse if grounding is inadequate.

```

1 System: You are a policy assistant. Answer using ONLY the provided
   context.
2     If the context is insufficient, reply: "I do not have enough
3     information to answer this." Cite each fact with [chunk_id].
4
5 Context:
6 [chunk_id=POL-204] {chunk_text}
7 [chunk_id=POL-211] {chunk_text}
  
```

```
8 [chunk_id=POL-302] {chunk_text}
9
10 User question: {user_question}
11
12 Answer (with citations):
```

Listing 12.2: A RAG prompt template that requires source attribution.

12.4 AI Orchestration

Prompt orchestration, chains, agents, workflow automation, and tool integration are the mechanisms that compose individual model calls into useful workflows. As systems grow more agentic, this layer grows in prominence and risk.

12.4.1 Chains versus Agents

A *chain* is a fixed sequence of model calls: extract, classify, generate, verify. The order is known at design time. An *agent* is a model call that chooses its next action from a set of tools at each step. Chains are predictable but limited; agents are flexible but harder to reason about.

12.4.2 Tools

A tool is a callable function with a typed signature that the model can invoke. Tools are how models reach beyond their training: a calculator, a search API, a database query, an action in a downstream system.

```
1 {
2   "name": "lookup_customer",
3   "description": "Look up a customer record by ID.",
4   "parameters": {
5     "type": "object",
6     "properties": {
7       "customer_id": {
8         "type": "string",
9         "description": "18-character Salesforce ID."
10      }
11    },
12    "required": ["customer_id"]
13  }
14 }
```

Listing 12.3: A tool definition with JSON schema, as consumed by an LLM tool-calling API.

12.4.3 Memory

Multi-turn conversations and multi-step agent workflows require memory. Short-term memory is the chat history within a session; long-term memory is summarised user state across sessions. Both are usually implemented as vector-indexed records of past exchanges, retrieved on demand.

12.5 AI Services

NLP for general language tasks, vision for image and video, speech for transcription and synthesis, and document intelligence for extraction from structured documents are the modality-specific services that most enterprises consume rather than build.

Table 12.3: AI services by modality.

Modality	Typical tasks	Example services
NLP	Classification, sentiment, NER, translation	Hosted classifiers, custom fine-tunes
Vision	Detection, classification, OCR, segmentation	Cloud vision APIs, vision LLMs
Speech	ASR (speech-to-text), TTS (text-to-speech)	Whisper, hosted STT/TTS APIs
Document AI	Layout extraction, form parsing, table extraction	Cloud document AI APIs

12.6 Last Word

Layer 3 is the locus of AI value creation and the locus of AI risk. The architectural work is to give each sub-component a clear interface so that models can be swapped, knowledge bases updated, and orchestration changed without rewriting the layer. Outcomes: an intelligence layer ready, models matched to tasks, retrieval grounded in enterprise knowledge, and context-aware responses available to the layers above. The hexagonal pattern in Chapter 26 is the structural answer to the layer’s volatility.

Chapter 13

AI Memory Systems

“A system without memory makes the same mistakes twice. A system with the wrong memory makes them indefinitely.”

—a recurring observation in stateful AI design

Chapter Overview

The AI/ML brain of Chapter 12 treated knowledge access as a single concern—retrieval from a vector store. In practice, an enterprise AI system uses several distinct kinds of memory, each with a different lifetime, a different access pattern, a different cost profile, and a different failure mode. This chapter develops the five memory types—in-context, external (RAG), episodic, semantic, and parametric—and the structural questions they answer: where does the knowledge live, how does it reach the model, when is it written, and when is it forgotten? The taxonomy is not academic; it is how an architect chooses between sending more tokens, building a vector store, persisting summaries, extracting facts, or fine-tuning.

Learning Objectives

- Distinguish the five memory types by lifetime, access pattern, and failure mode.
- Choose the appropriate memory type for a given knowledge requirement.
- Compose memory types: most production systems use three or four together.
- Recognise the recurring failure modes (stale memory, memory leakage, runaway memory growth) and the structural mitigations.
- Locate memory components within the reference architecture and the hexagonal pattern.

13.1 Why Memory Is Architectural

A LLM call without memory is a function: the same input produces the same kind of output. A useful enterprise system is not a function. It remembers what the user said three turns ago. It looks up the policy document the user is asking about. It recalls that this customer has churned and re-signed twice. It encodes domain terminology that is not in any general training corpus.

Each of these is a different kind of memory with a different mechanism. Treating them as one thing produces systems that either forget too much (no episodic memory) or remember too much (everything

goes into context until the bill or the latency become unbearable). The first architectural move is to name the kinds.

13.2 The Five Memory Types

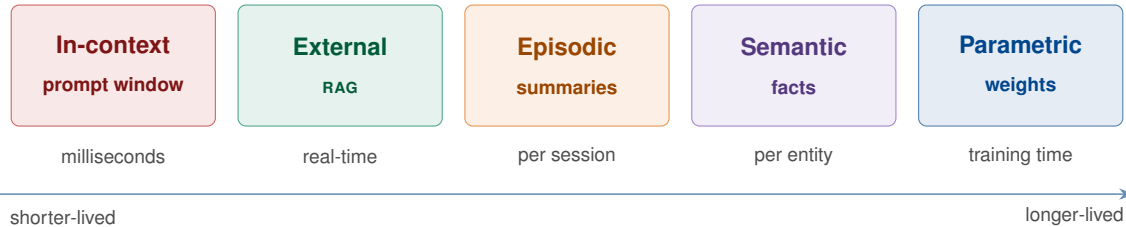


Figure 13.1: The five memory types arranged by lifetime, from the prompt window (milliseconds, per request) to model weights (months to years, per training cycle).

13.2.1 In-Context Memory

The prompt window itself. Whatever the model needs for this single call goes into the prompt: system instructions, user input, prior turns, retrieved content. It is the fastest memory (zero retrieval latency—the data is already in the call), the cheapest to implement (no infrastructure), and the most expensive per byte (every token is billed every turn).

Purpose. Working memory per request; holds instructions plus immediate history.

When to use. Short conversations, single-turn tasks, inline document Q&A, anything that fits in the window.

Failure mode. Context-window pressure. As history grows, either earlier content is dropped (the model forgets) or every call pays for re-sending the same prefix.

13.2.2 External Memory (RAG)

A vector store (and, increasingly, a hybrid vector-plus-lexical store) holding embeddings of source documents. The model retrieves the relevant chunks at query time and injects them into the prompt. Chapter 21 develops the pattern in full.

Purpose. Access to large external knowledge without retraining; reduces hallucination by grounding outputs in citable sources.

When to use. Internal docs and wikis, compliance references, knowledge that changes faster than retraining cycles, corpora too large for the context window.

Failure mode. Retrieval misses—the right chunk exists in the index but isn't returned for this query. Chapter 21 covers the hybrid retrieval and re-ranking mitigations.

13.2.3 Episodic Memory

Summaries of past interactions, stored per user, agent, or session. When a new session starts (or a long-running session needs to free context-window space), relevant episodes are retrieved and injected

into the prompt.

Purpose. Cross-session continuity; remembering prior decisions; compressing history so that long-running agents don't have to carry the full transcript.

When to use. Personal AI assistants, coaching and tutoring tools, long-running agents, multi-session workflows where the user expects "we discussed this last week" to be true.

Failure mode. Summary drift—the compressed memory loses details that turn out to matter, or amplifies an early misunderstanding through repeated summarisation. Mitigation: keep both the summary and a pointer to the source; allow the agent to fetch the source when the summary is insufficient.

13.2.4 Semantic Memory

Structured facts extracted from interactions and stored persistently, typically in a key-value or graph store. Where episodic memory holds "what happened in this conversation," semantic memory holds "what is true about this entity."

Purpose. Stable entity relationships; reliable structured lookups; personalisation that survives session boundaries.

When to use. CRM-integrated assistants, personalisation engines, recommendation systems, anything where facts about a customer or product matter and don't change often.

Failure mode. Stale facts—the model believes something about the user that is no longer true (a former address, an outdated preference). Mitigation: facts have provenance and expiry; conflict-resolution rules favour recent assertions; periodic validation against source systems.

13.2.5 Parametric Memory

Knowledge encoded into model weights at training time. This includes everything the foundation model learned during pre-training and anything added via fine-tuning. There is no retrieval step; the model simply knows.

Purpose. Encoded domain reasoning, stable proprietary behaviour, zero retrieval latency, domain terminology baked in.

When to use. Stable high-frequency knowledge (medical coding conventions, legal citation formats, internal jargon); domain-specific small models; proprietary workflows where the model's behaviour itself is the asset.

Failure mode. Cannot be updated without retraining. Also: parametric memory cannot be cited—if the user or a regulator asks "where did this come from," the answer is "the weights," which is rarely acceptable for consequential decisions.

13.3 Choosing Between Memory Types

The choice is rarely "one of these"; it is "which combination, and in what proportion." A useful decision frame:

Table 13.1: Memory type by knowledge characteristic.

Characteristic	Best memory type	Reasoning
Fits in the window, single-turn	In-context	No infrastructure cost is justified
Large, changes regularly, needs citation	External (RAG)	Updatable, citable, decoupled from model
Per-user, per-session continuity	Episodic	Compresses history; survives session boundaries
Per-entity, structured, queryable	Semantic	Key-value or graph store is the natural fit
Stable, high-volume, proprietary	Parametric	Amortised cost of fine-tuning pays back
Changes faster than retraining	Anything but parametric	Parametric is unsuitable for fast-changing data
Auditable origin required	External or semantic	Parametric cannot be cited

13.4 Composing Memory Types

A production assistant typically uses three or four memory types together. A representative composition for a customer-service assistant:

- **In-context.** The current conversation turns and the system prompt.
- **External (RAG).** Policy documents, product manuals, help-centre articles—retrieved per query.
- **Episodic.** A summary of past interactions with this customer ("called twice last month about billing").
- **Semantic.** Structured facts about the customer (plan tier, account status, language preference, accessibility needs).
- **Parametric.** A foundation model that knows English, knows what a refund policy is, and—if fine-tuned—knows the company's tone and the structure of valid response formats.

Each memory contributes what it is suited for. Composing them is the architecture work; choosing one to do all the jobs of the others is the recurring mistake.

13.5 Writing to Memory

In-context and parametric are passive from the application's perspective (the prompt is built per request; the weights are set at training time). The other three require active write paths.

13.5.1 Writes to External Memory

The indexing pipeline (Chapter 21) writes to the vector store: chunking, embedding, metadata tagging, deduplication. Writes are typically batched and run on a schedule or on source-change events.

13.5.2 Writes to Episodic Memory

At session end (or at significant turn boundaries within a long session), a summarisation step compresses the interaction and stores it keyed by user or agent. The summarisation prompt is itself a versioned artefact; changing it changes the character of memory the system accumulates.

13.5.3 Writes to Semantic Memory

A fact-extraction step examines interactions for assertions about entities ("user prefers email over phone," "account upgraded to Pro tier"). Extracted facts are validated, deduplicated against existing facts, and either written, updated, or rejected. The fact-extraction prompt and the conflict-resolution rules are again versioned artefacts.

13.6 Forgetting

The corollary of writing is forgetting. Each memory type has a forgetting mechanism, and the architecture should be explicit about it.

- **In-context.** Forgetting is automatic and immediate—the context vanishes when the request ends.
- **External.** Forgetting requires deletion from the index, ideally driven by source-document deletion or by an explicit retention policy (e.g., documents older than N years and not accessed in M months are archived).
- **Episodic.** Older summaries are themselves summarised into shorter forms, or expired entirely after a retention window. Aggressive forgetting is often the right default for personal-assistant contexts.
- **Semantic.** Facts have an effective date and an expiry; out-of-date facts are archived, not silently retained.
- **Parametric.** Forgetting requires retraining without the relevant data—expensive, sometimes impossible. The architectural rule: don't bake into weights anything you may need to forget.

Privacy obligations interact strongly here. The right to erasure (Section ??) requires that a subject's data be deletable across every memory store. Designing the write paths so deletion is tractable is far cheaper than retrofitting it.

13.7 Failure Modes

Table 13.2: Recurring memory-system failure modes.

Failure mode	Mitigation
Context-window thrashing	Tier memory: keep recent turns in-context, summarise older into episodic, retrieve facts from semantic.
Summary drift	Preserve a link from each summary to its source; allow the agent to fetch source when stakes are high.
Stale semantic facts	Effective dates and expiry on every fact; periodic validation against source systems.
Memory leakage between users	Strict tenancy isolation in every store; explicit user/tenant key on every read and write.
Memory growth without bound	Retention policies; aggressive forgetting as default; explicit promotion from short-term to long-term.
Parametric staleness	Refresh strategy for fine-tunes; rely on RAG for fast-changing knowledge.
Memory poisoning (adversarial writes)	Validate inputs to memory the same way you validate inputs to the model; treat memory writes as security-sensitive operations.

13.8 Memory and the Hexagonal Pattern

Each non-parametric memory type is naturally expressed as a port (Chapter 26). The vector port already covers external memory. Two further ports usually emerge:

- **Episodic port.** `summarise(session), retrieve_episodes(user, query), forget(user, before_date)`.
- **Semantic port.** `upsert_fact(entity, key, value, effective_date), lookup_facts(entity, keys), retract(entity, key)`.

Adapters can then plug in concrete stores (a relational database, a graph database, a managed vector-plus-key-value store) without the application's memory logic depending on the choice.

13.9 What This Chapter Established

Memory is not one concern; it is five. In-context for the immediate. External for the citable and changeable. Episodic for the conversational continuity. Semantic for the structured and stable. Parametric for the deeply learned. The architectural work is to name the kinds, write to each through a clear interface, forget on a schedule appropriate to each, and compose them so that no single store carries a load it is not suited for. Done well, the system remembers what it should and forgets what it must—and is governable in both directions.

Chapter 14

Layer 4: Model Serving and MLOps

“Production is the only environment that matters.”

—site reliability engineering folklore

Chapter Overview

Layer 4 deploys, serves, monitors, and continuously improves the models from Layer 3. MLOps is to ML what DevOps is to traditional software, with additional concerns—versioned datasets, drift detection, and continuous evaluation—that have no analogue in stateless services. This chapter develops the deployment patterns, the API surface, the MLOps pipeline, the observability stack, and the feedback loop that closes the cycle from production back to training.

Learning Objectives

- Choose between online, batch, and streaming inference and compose them appropriately.
- Apply containerisation, model serving runtimes, and auto-scaling for production ML.
- Design API and integration patterns (REST, GRPC, API gateway, service mesh) for AI workloads.
- Implement CI/CD for ML, including model registry, versioning, and experiment tracking.
- Monitor model performance, detect drift, and close the feedback loop into continuous improvement.

14.1 Inference Modes

Three inference modes appear in production architectures, often in combination.

Table 14.1: Inference modes and their characteristics.

Mode	Latency	Throughput	Use case
Online (real-time)	10 ms–2 s	Per-request	Chat, search, recommendations, fraud
Batch	Minutes–hours	Millions per run	Nightly scores, periodic enrichment
Streaming	100 ms–seconds	Continuous	Event-driven scoring, anomaly detection

The architectural decision is which mode each model needs. An overnight batch run is far cheaper than an online endpoint and should be preferred where freshness allows. An online endpoint adds operational complexity (autoscaling, latency SLOs) and is reserved for inputs that arrive interactively.

14.2 Model Deployment

Containerisation (typically Docker), model serving runtimes, batch and real-time inference, and auto-scaling are the deployment fundamentals.

14.2.1 Serving Runtimes

A serving runtime turns a trained model artefact into an inference API. Common runtimes include:

- **Generic.** FastAPI / Flask wrapping a Python predict function. Maximum flexibility, minimum performance.
- **Optimised.** TorchServe, TensorFlow Serving, NVIDIA Triton, BentoML, KServe. Better throughput, GPU support, standardised metrics.
- **LLM-specialised.** vLLM, TensorRT-LLM, Ollama, llama.cpp. Optimised attention kernels, continuous batching, paged KV cache, quantisation.

For self-hosted LLMs, the runtime choice often dominates throughput and cost. Continuous batching (processing tokens from multiple requests in the same forward pass) and paged attention (KV-cache memory management) together yield 5–20× throughput improvements over naive serving.

14.2.2 Quantisation

Quantisation reduces the bit-width of model weights from 16 or 32 bits to 8, 4, or even 2 bits. The effect on quality is modest for well-quantised models and dramatic for memory and throughput: a 70-billion-parameter model in 4-bit quantisation fits in 40 GB of GPU memory instead of 280 GB.

Table 14.2: Quantisation levels and trade-offs.

Precision	Memory per param	Typical quality loss	When to use
FP16 / BF16	2 bytes	None (baseline)	Highest-quality inference
INT8	1 byte	Small (1–2% on benchmarks)	Production default for many models
INT4 (GPTQ, AWQ)	0.5 byte	Moderate (2–5%)	High-volume serving, edge
INT2	0.25 byte	Significant	Research / extreme constraints

14.2.3 Auto-Scaling

Auto-scaling adjusts the number of serving replicas to match demand. The dominant metrics are queue depth, GPU utilisation, and request latency. The architectural subtlety is *cold-start latency*: spinning up a new replica with a multi-gigabyte model takes minutes, far longer than typical autoscaler reaction

times. Mitigations include pre-warming replicas above the predicted load, keeping a small idle pool, and using model-sharing techniques (LoRA serving) to multiplex many fine-tuned models on shared base weights.

14.3 API and Integration Layer

REST and gRPC APIs expose models to the rest of the organisation. An API gateway enforces authentication, rate limiting, and routing. A service mesh handles inter-service concerns without requiring application changes.

14.3.1 REST versus gRPC

Table 14.3: REST versus gRPC for ML serving.

Dimension	REST (JSON over HTTP)	gRPC (Protobuf over HTTP/2)
Latency overhead	5–20 ms typical	1–5 ms typical
Payload size	Larger (JSON)	Smaller (binary)
Streaming	Server-Sent Events / WebSockets	Native bidirectional
Tooling	Universal	Strong in service-to-service
Best for	External APIs, browser clients	Internal high-throughput services

14.3.2 API Gateway

The gateway centralises cross-cutting concerns:

- Authentication (OAuth, mTLS, API keys).
- Rate limiting and quotas (per tenant, per endpoint).
- Request and response logging.
- Schema validation.
- Cost attribution headers.
- Routing (canary deployments, A/B tests, fallback paths).

14.4 MLOps

CI/CD for ML differs from traditional CI/CD in three ways: artefacts are data and models in addition to code; tests are evaluations against held-out sets in addition to unit tests; and rollback may involve serving a previous model version with different feature requirements.

14.4.1 The MLOps Pipeline

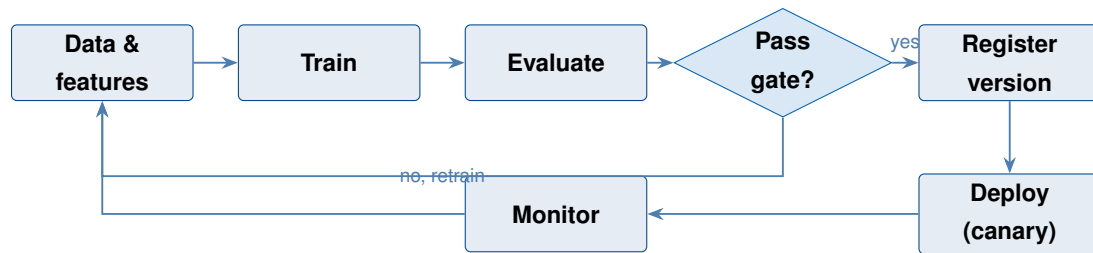


Figure 14.1: The MLOps pipeline: from features to deployed model and back via monitoring. Evaluation is a deployment gate, not an optional report.

14.4.2 The Model Registry

The model registry is the source of truth for what is deployable. Each entry records:

- Model name and version.
- Training data hash and feature set version.
- Code commit hash.
- Evaluation metrics and the evaluation set version.
- Stage (development, staging, production, archived).
- Approver and approval date for promotion to production.

A model not in the registry is not a model the platform knows about. A model in the registry that was not promoted through the documented gate is a governance failure.

14.4.3 Experiment Tracking

Experiment tracking systems (MLflow, Weights & Biases, Neptune) record every training run: hyperparameters, dataset versions, code revision, resulting metrics, and artefacts. Reproducibility of a result requires that all four be captured; in practice, dataset versions are the most commonly missed.

14.5 Monitoring and Observability

AI-specific observability layers on top of conventional service metrics. Four signal categories matter:

Table 14.4: Observability signals for ML systems.

Signal	Description
Operational metrics	Latency, throughput, error rate, resource utilisation (same as any service).
Prediction metrics	Distribution of outputs, confidence scores, refusal rates.
Input drift	Statistical change in input feature distributions over time.
Concept drift	Change in the relationship between features and ground truth.

14.5.1 Drift Detection

Input drift compares recent input distributions to a reference distribution. Common statistical tests include:

- **Population Stability Index (PSI).** A measure of distributional shift; values above 0.2 typically warrant investigation.
- **Kolmogorov-Smirnov test.** Non-parametric test for continuous features.
- **Chi-squared test.** For categorical features.
- **Wasserstein distance.** For high-dimensional or embedding spaces.

Drift is a leading indicator: it appears before model quality degrades on the held-out metrics that are easier to track. Acting on drift before the quality regression arrives is how mature platforms avoid silent failures.

14.5.2 Distributed Tracing for ML

A single user-facing prediction may involve a feature-store lookup, an embedding generation, a retrieval call, an LLM call, and a guardrail check. Tracing them all under one request identifier is essential to debugging. OpenTelemetry-based traces with span attributes for model version, prompt template, and retrieval scores are the standard.

14.6 Feedback and Continuous Improvement

Human feedback, data flywheels, model retraining, and continuous learning close the loop from production back into the model layer. The architectural challenge is to capture feedback in a structured form without making the system depend on continuous human attention.

14.6.1 Feedback Capture

Implicit feedback (was a recommendation clicked? was a generated draft accepted?) and explicit feedback (thumbs up/down, ratings) both feed the flywheel. The architecture should: (a) log feedback alongside the prediction it relates to, (b) preserve enough context to reconstruct what the user saw, and (c) classify feedback as actionable (input to retraining) or operational (input to monitoring).

14.6.2 Retraining Cadence

Retraining cadence is a design parameter:

- **Triggered.** Retrain when drift exceeds threshold; production-quality but unpredictable schedule.
- **Scheduled.** Retrain weekly or monthly regardless of drift; predictable but may waste compute on a stable system.
- **Continuous.** Online learning that updates parameters with each new example; rare in enterprise contexts because it makes auditability hard.

A common compromise is scheduled retraining with triggered exceptions: weekly retrains by default,

with the ability to fire an out-of-cycle retrain when drift alerts fire.

14.7 The Working Position

Layer 4 is where models become products. Its outcomes are simple to state and hard to achieve: models in production, reliable and scalable, continuously improving. The chapters on observability (Chapter 101) and FINOPS (Chapter 100) deepen the operational treatment; the chapter on the LLM router pattern (Chapter 23) develops the multi-model serving strategy that underpins most cost-effective production deployments.

Chapter 15

Layer 5: DevSecOps and Governance

“Security is not a feature. It is the substrate.”

—a recurring SRE observation

Chapter Overview

Layer 5 holds the platform-wide concerns that every other layer inherits: CI/CD, identity, secrets, observability, cost, governance evidence, and policy enforcement. Unlike Layers 1–4 (which the system passes through) or Layer 6 (which the system reaches), Layer 5 surrounds them: every component of every layer authenticates against the same identity provider, emits to the same observability backend, is deployed by the same CI/CD machinery, and is governed by the same policies. This chapter develops the components, their interfaces, and the operating disciplines that make Layer 5 the connective tissue of the platform rather than a collection of disconnected tools.

Learning Objectives

- Identify the five Layer 5 sub-domains (CI/CD, security, governance, observability, cost) and what each contributes.
- Apply CI/CD patterns specific to AI artefacts (models, prompts, evaluations) alongside conventional code.
- Implement platform-wide identity, secrets, and policy enforcement.
- Operate observability infrastructure that captures AI-specific signals alongside conventional ones.
- Recognise the principle that Layer 5 is a platform feature, not a per-project responsibility.

15.1 The Five Sub-Domains

Layer 5 covers five distinct sub-domains, each operated by the platform but consumed by every other layer:

Table 15.1: Layer 5 sub-domains and their responsibilities.

Sub-domain	Responsibilities
CI/CD	Build pipelines, deployment automation, environment management, model and prompt promotion workflows
Security	Identity, authorisation, secrets management, network policy, vulnerability scanning, runtime defence
Governance	Model inventory, lineage, risk classification, audit-evidence generation, policy enforcement
Observability	Logs, metrics, traces, AI-specific signals, alerting, dashboards
Cost (FinOps)	Attribution, budgeting, anomaly detection, reporting (Chapter 100)

The sub-domains are independent in implementation but tightly integrated in operation: a single deploy emits CI/CD events, applies security policies, updates the governance record, instruments observability, and tags cost attribution.

15.2 CI/CD for AI

CI/CD for AI systems extends conventional pipelines with AI-specific artefacts. The deployable unit is no longer just code; it is the combination of code, model version, prompt version, retrieval configuration, evaluation results, and governance metadata.

15.2.1 Pipeline Stages

A typical CI/CD pipeline for an AI system:

- **Build.** Compile, package, generate container images, build model artefacts.
- **Scan.** Vulnerability scanning on dependencies; secrets detection; license compliance.
- **Unit test.** Conventional tests on application code.
- **Eval.** The evaluation harness (Section 101.2) runs against the relevant suites; thresholds must pass.
- **Integration test.** Verify the deployable works end-to-end against staging dependencies.
- **Security review.** For changes affecting trust boundaries or risk classification, automated or human review.
- **Canary deploy.** Small fraction of traffic; observed metrics must remain within bounds.
- **Promotion.** Full deploy with rollback path armed.
- **Post-deploy verification.** Production canary queries; brief monitoring window before declaring success.

15.2.2 Promotion of Prompts and Models

Prompts and models are first-class deployable artefacts. They live in registries with versioning, metadata, and approval state. Promotion follows the same pipeline as code:

- Prompt change → eval suite runs against the new prompt → comparison with current production → manual or automated approval → registry update → application picks up the new pinned version on next deploy.
- Model change → eval suite, cost check, latency check → shadow-mode comparison → canary deployment → promotion in registry.

The discipline that makes this work: prompts and models are versioned at registry level, not at application level. Application code references the registry by ID; the registry resolves to a specific version per environment.

15.2.3 Environment Strategy

A workable environment progression: development, integration, staging, production. AI-specific considerations:

- Staging uses production-like data (with PII redacted or synthesised) so behaviour can be validated.
- Eval suites run in staging against the same model versions that production will use.
- Canary in production uses a small percentage of real traffic with feature flagging.
- Rollback path is tested, not assumed; a rollback drill is part of major releases.

15.3 Security

The security sub-domain has several components. Most are foundational (apply to any system) with AI-specific extensions.

15.3.1 Identity

Strong authentication for users (OAuth 2 / OpenID Connect with MFA); service identity for service-to-service (SPIFFE / X.509 with mTLS); short-lived tokens; just-in-time access for sensitive operations. The platform's identity provider is the single source of truth; no system maintains its own user database.

AI-specific: agents need identities too. An agent acting on behalf of a user inherits constrained subsets of the user's permissions, scoped to the agent's task. Audit logs distinguish "the user did this" from "the user's agent did this on their behalf."

15.3.2 Authorisation

Fine-grained access control on data, models, prompts, tools, and capabilities. Per-tenant, per-environment, per-purpose tags determine who can access what. The principle of least privilege applies: a service that needs read access on three tables gets read access on three tables, not the whole schema.

15.3.3 Secrets

A central secrets store (HashiCorp Vault, AWS Secrets Manager, Azure Key Vault, Google Secret Manager) is the authoritative source for credentials, API keys, and other secrets. No secret is in source control, in a configuration file checked into a repo, or in an environment variable visible in process listings. Rotation is automatic on a documented cadence; access is logged.

15.3.4 Network Policy

Zero-trust networking by default: mTLS between services; egress restrictions on what each service can reach; private connectivity to managed vendor APIs; VPN or PrivateLink rather than public endpoints. Vendor calls are observable: every call to an external API is logged with destination, timing, and data classification.

15.3.5 Vulnerability Management

Continuous scanning of:

- Application dependencies (CVE databases; license compliance).
- Container images (base image vulnerabilities; missing patches).
- Infrastructure-as-code (configuration drift; insecure defaults).
- Secrets (any secret accidentally committed to source).
- Model artefacts (provenance, hash verification, signature checks).

Findings are routed to the responsible team with severity-based SLAs.

15.4 Governance

The governance sub-domain (developed in Chapter 103) is the platform-wide accountability infrastructure:

- Model inventory as a platform service; every deployment touches it.
- Lineage emission from every component; collected and queryable.
- Risk classification stored alongside inventory; drives downstream policy decisions.
- Audit-evidence pipeline that assembles packages from primary sources on demand.
- Policy enforcement points (admission controllers, gateway middleware) that prevent non-compliant configurations.

15.4.1 Lineage Across the Platform

A practical lineage architecture:

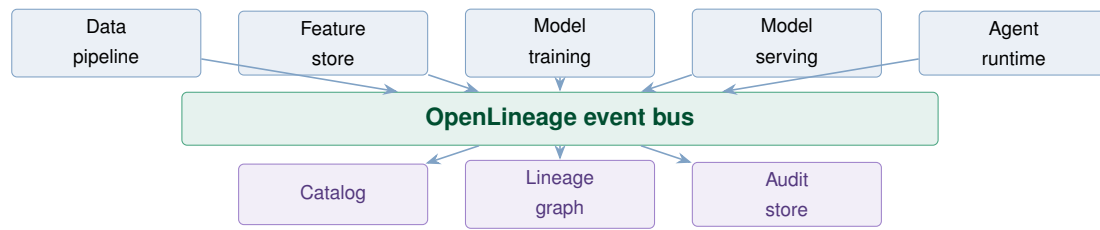


Figure 15.1: Lineage emission and collection. Every component emits structured events to a shared bus; downstream consumers index, visualise, and archive them.

15.5 Observability Infrastructure

The observability sub-domain operates the storage and query layer for logs, metrics, and traces (Chapter 101 develops what is emitted; this section is about the infrastructure):

- Metrics: Prometheus or vendor-equivalent; long-term storage with downsampling.
- Logs: centralised aggregation (Loki, Splunk, Datadog); structured, queryable; tiered retention.
- Traces: distributed tracing (OpenTelemetry collectors → backend like Jaeger, Tempo, Honeycomb).
- Alerting: alerts routed to teams; runbook links in alert payloads; on-call schedules.
- Dashboards: standard templates for service health; team-specific dashboards generated from common building blocks.

15.5.1 Cost Discipline in Observability

Observability cost can become significant. The architectural levers:

- Sampling: not every trace needs to be stored; high-cardinality traces sampled at low rate, error traces always retained.
- Aggregation: high-cardinality metrics (per-user, per-tenant) pre-aggregated to standard dimensions.
- Retention: hot storage for recent (7–30 days); archive for older; full discard after a documented horizon.
- Cardinality limits: dashboards and alerts that span too many series fail loudly; the team is forced to reduce cardinality.

15.6 Policy as Code

Platform policies are expressed as code, version-controlled, peer-reviewed, and enforced by automated gates:

- Open Policy Agent (OPA) for cross-cutting policy in Kubernetes and beyond.
- Custom admission controllers that block deployments missing required metadata.
- CI/CD gates that fail on missing or non-compliant artefacts.

- Schema validators that reject configurations violating platform rules.

Policy-as-code keeps the rules visible (in source control) and consistent (every change reviewed); the alternative—rules written in wikis, enforced by tribal knowledge—scales poorly and drifts.

15.7 The Platform Engineering Mindset

Layer 5 is operated by the platform team (Chapter 98). The mindset:

- The product is the developer experience of the layer-above teams.
- Self-service over ticket-driven; engineers should be able to do most operations without involving the platform team.
- Documentation is part of the deliverable; an undocumented feature is not shipped.
- Reliability targets apply to the platform itself; platform-team SLOs are met or there is a post-mortem.
- Continuous improvement; the platform's metric is spoke time-to-production, not platform activity.

15.8 A Note Before Moving On

Layer 5 is the connective tissue of the platform: CI/CD, security, governance, observability, and cost as platform-wide features that every other layer inherits. CI/CD extends to AI-specific artefacts (models, prompts, evaluations); identity and secrets are foundational; governance produces continuous evidence rather than periodic deliverables; observability captures both conventional and AI-specific signals; cost discipline is wired into the API gateway. The platform-engineering mindset—the platform's product is the developer experience—is what turns Layer 5 from a collection of tools into a paved path that the spokes adopt because adopting is easier than building.

Chapter 16

Layer 6: End User Applications

“The interface is the system, as far as the user is concerned.”

—a recurring product observation

Chapter Overview

Layer 6 is where users meet the system. Everything in Layers 1–5 exists to enable Layer 6 outcomes; failures in Layer 6 erase the value of work in the layers below it. This chapter develops the application archetypes for enterprise AI (chatbots and assistants, embedded copilots in existing applications, analytics with conversational interfaces, mobile and web apps, voice surfaces), the UX patterns specific to AI-generated content (streaming, citation, confidence, contestability), the accessibility and inclusion considerations, and the integration patterns that let AI capabilities embed into existing enterprise applications without disrupting them.

Learning Objectives

- Identify the application archetypes and their architectural implications.
- Apply UX patterns that work for AI-generated content: streaming, citation, confidence display, contestability.
- Design accessibility into AI-powered interfaces from the start.
- Integrate AI capabilities into existing applications through embedded patterns.
- Measure user outcomes, not just engagement, as the success metric.

16.1 Application Archetypes

Several archetypes recur:

Table 16.1: Enterprise AI application archetypes.

Archetype	Examples
Chatbot / assistant	Customer-service bot, internal help desk, sales-enablement assistant
Embedded copilot	“Suggest a response” inside the CRM; “draft this email” inside the mail client; “summarise this page” inside the wiki
Analytics with NL interface	“Show me revenue by region last quarter”; explanation of dashboard anomalies
Document workflows	Contract review and redlining; claims processing; clinical-note drafting
Voice surfaces	IVR replacement, in-vehicle assistants, accessibility tools
Agentic workflows	Multi-step task automation; cross-application orchestration
Background augmentation	Triage, tagging, summarisation happening before the human sees the work

Each archetype has different latency tolerances, accuracy requirements, and oversight needs. A background-augmentation workflow can tolerate seconds of latency and lower per-call accuracy (because a human reviews); a real-time voice assistant requires sub-second response and high accuracy in interpretation.

16.2 Design Principles for AI Interfaces

Several principles emerge from accumulated practice. They are biases in design, not absolute rules.

Table 16.2: Design principles for AI interfaces.

Principle	Implication
Stream by default	Render output incrementally; perceived latency is dominated by time-to-first-token
Show provenance	Citations, sources, the reasoning trail where appropriate; users verify before they trust
Communicate confidence	Hedging language for uncertain outputs; visual signals for low-confidence cases
Make refusal a feature	Refusing to answer ungrounded questions is better than guessing; users prefer honesty to confident error
Support iteration	Easy to refine the request; conversation history is part of the interface
Preserve undo	Generated artefacts (emails, code, decisions) should be reversible by the user
Avoid anthropomorphism for high-stakes systems	The system is a tool; treating it as a personality can mislead users about its reliability

16.2.1 Streaming and Perceived Latency

A response that takes 5 seconds to complete but streams its first token at 300 ms feels much faster than a response that returns in 3 seconds as one block. Streaming is the default for any user-facing generation; the implementation matters:

```

1 from fastapi import FastAPI
2 from fastapi.responses import StreamingResponse
3 import json
4
5 app = FastAPI()
6
7 async def stream_completion(prompt: str, model: str):
8     async for chunk in llm_client.stream(prompt=prompt, model=model):
9         if chunk.delta:
10             yield f"data: {json.dumps({'text': chunk.delta})}\n\n"
11         if chunk.finish_reason:
12             yield f"data: {json.dumps({'done': True, 'reason': chunk.
13                                     finish_reason})}\n\n"
14
15 @app.post("/v1/complete/stream")
16 async def complete_stream(req: CompletionRequest):
17     return StreamingResponse(
18         stream_completion(req.prompt, req.model),
19         media_type="text/event-stream",
20     )

```

Listing 16.1: Streaming server-sent events from an LLM API.

The UI consumes the stream and renders each token as it arrives; the user sees structural cues (an opening sentence, headings, list bullets) almost immediately.

16.2.2 Citation Display

For RAG-backed responses, citations are the primary trust mechanism. The UI presents them inline (clickable footnotes or hover cards) rather than as a separate “sources” section. Each citation links to the underlying source so the user can verify. Citations that point to chunks rather than whole documents land the user on the relevant passage, not the document’s first page.

16.2.3 Confidence Communication

Confidence is communicated through:

- Hedging language for genuinely uncertain outputs (“based on the documents I found...”).
- Explicit refusal for questions outside scope.
- Visual signals (a soft warning indicator) when the system’s internal confidence is low.
- Calibrated probabilities for classification tasks (and avoided for free-text generation, where they are not reliable).

16.2.4 Contestability

Every consequential output should have a contestability path:

- A clear “this is wrong” affordance that captures user feedback.
- For decisions affecting the user, an explicit explanation and a route to human review.
- For generated content (emails, code, documents), unlimited regeneration and editing.
- Feedback fed into evaluation pipelines so the system improves over time.

16.3 Embedded Patterns

Most enterprise AI value lives in embedding capabilities into existing applications rather than building new standalone applications. Three patterns:

16.3.1 Inline Suggestion

Suggestions appear at the point of work: as the user is composing an email, drafting a response, writing code, filling a form. The suggestion is dismissible with one keystroke; accepting it does not lock the user out of editing.

The architectural pattern: the application sends context (current text, surrounding context, user identity) to the AI service; the service returns a suggestion; the application renders it in-place. Latency budgets are tight (the suggestion must arrive before the user has finished typing the next word) but the cost per call is bounded because the suggestion is short.

16.3.2 Side Panel

A persistent side panel offers conversational interaction alongside the main application. The user can ask questions about the current entity (a customer record, a document, a dashboard) without losing context. The side panel maintains conversation state across actions in the main application.

16.3.3 Workflow Integration

The AI system fits into an existing business process: triage incoming requests, draft a response, route for approval. The architectural pattern: the workflow engine invokes the AI service at specific steps; the AI service returns structured output; the workflow continues. Human-in-the-loop steps are explicit; the workflow does not depend on the AI for correctness but uses it for efficiency.

16.4 Multimodal Surfaces

Voice, image, and video interfaces add their own constraints:

- **Voice.** Latency must support natural conversation rhythm (500 ms turn-taking); transcription errors must be recoverable; the interface degrades gracefully in noisy conditions.
- **Image input.** Privacy considerations (faces, location metadata); accessibility considerations (alt text generation).
- **Image output.** Content safety, brand-style consistency, attribution to source where applicable.
- **Video.** Bandwidth, latency, and storage considerations interact with the AI capabilities.

16.5 Accessibility and Inclusion

AI interfaces face the accessibility challenges of any interface, with some additions:

- Screen readers must work with streaming text (announce updates without interrupting).
- Generated content must support assistive technologies (semantic HTML, alt text on images, captions on video).
- Multilingual quality must be equivalent across languages (not just translated).
- Low-bandwidth conditions require graceful degradation (text alternatives to image-heavy outputs).
- Cognitive accessibility: plain-language options; structure that supports skimming and re-reading.

Accessibility designed in is cheaper than accessibility retrofitted. The structural rule: accessibility requirements are part of the design checklist, not an audit-time concern.

16.6 Integration with Enterprise Identity

AI applications must integrate with the enterprise identity system. The architectural patterns:

- Single sign-on (SSO) via SAML or OIDC.
- Permission inheritance: the AI sees only what the user can see.
- Identity propagation to agents: an agent acting for a user inherits constrained subsets of the user's permissions.
- Audit trail: every action is attributed to a user (and an agent if applicable).

16.7 Measuring User Outcomes

Several metrics matter at the application layer:

Table 16.3: User-outcome metrics by archetype.

Archetype	Useful metrics
Chatbot/assistant	Resolution rate (conversation ends with user's question answered); escalation rate; user satisfaction
Embedded copilot	Acceptance rate (suggestions accepted vs. dismissed); edits per acceptance; time saved
Analytics with NL	Query success rate; iteration count to right answer; insight follow-through
Document workflows	Time per document; error rate (caught at review); reviewer overrides
Voice surface	Task completion rate; turns per task; fallback-to-human rate
Agentic workflow	Task completion rate; error rate per step; human-intervention rate

Engagement metrics alone (sessions, messages, daily active users) are insufficient: a user who interacts more because the system is failing is not a success. The metrics that matter capture whether users got what they came for.

16.8 Anti-Patterns at the Application Layer

Several anti-patterns appear specifically at Layer 6:

- **Anthropomorphism in high-stakes contexts.** A friendly conversational interface in front of a credit decision misleads users about the system's nature.
- **Hidden AI involvement.** Users discover after the fact that an AI was involved; trust drops materially.
- **Confident wrong answers.** The system asserts incorrect information fluently; users believe it because it sounds right.
- **No iteration path.** The user gets a response and has no clear way to refine the request.
- **Citation theatre.** Citations that look like references but don't actually support the claim; users stop checking.
- **Inconsistent personality.** The AI behaves differently in different contexts in ways that suggest unreliability.
- **Treating users as input.** The interface is optimised for what the AI can do, not for what the user is trying to accomplish.

16.9 The Takeaway

Layer 6 is where the value created by Layers 1–5 reaches the user—or fails to. The application archetypes (chat, copilot, analytics, workflows, voice, agents) each have their own architectural implications. The UX patterns specific to AI-generated content (streaming, citation, confidence, contestability) earn user trust where stock UI patterns do not. Accessibility, identity integration, and outcome measurement turn an impressive demo into a useful product. The structural commitment: design the user experience for the system as it actually behaves, not as the team hopes it will behave.

Part III

Architecture Patterns Every AI Architect Must Know

“A book of designs which, if executed, will produce furniture suitable to all ranks, in the most fashionable taste.”

—Thomas Chippendale, preface to The Gentleman and Cabinet-Maker’s Director, 1754

The six layers are now in place; what fills them are patterns. Chippendale’s *Director* did something in 1754 that the modern engineering world takes for granted: it published, openly, the patterns by which a competent craftsman could reproduce sophisticated work without inventing it from scratch. A cabinetmaker in Edinburgh could open the book, study a Chinese-style chair, and build it. A merchant in Charleston could specify a particular pattern by number. The patterns were not secret; they were the shared vocabulary of a craft. Part III of this book offers the same for AI architecture: the patterns that competent practitioners no longer need to invent — they have names, diagrams, trade-offs, and known failure modes. The book of designs is, this time, a working architect’s reference.

Chapter 17

Introduction to AI Architecture Patterns

“Patterns are solutions to recurring problems in a context.”

—Christopher Alexander

Chapter Overview

A pattern is a structural solution that recurs across many systems facing similar problems. This part of the book catalogues six patterns that recur in enterprise AI architectures, each addressing a distinct concern. The patterns compose: most production systems use three or four of them simultaneously. This chapter introduces all six together, explains how to choose between them, and shows how they combine, so that the dedicated chapter for each pattern (Chapters 18–26) can focus on the depth rather than the comparison.

Learning Objectives

- Recognise the six recurring architecture patterns for enterprise AI.
- Match patterns to the problems they solve and avoid using them where they do not fit.
- Compose patterns: most production systems use several together.
- Apply the decision rubric to choose among patterns for a given system.
- Avoid the common pattern-misuse anti-patterns: cargo-cult adoption, over-engineering, under-engineering.

17.1 What a Pattern Is and Is Not

A pattern is a named structural solution that:

- Recurs across many independent systems facing similar problems.
- Has identifiable components and an interface between them.
- Has known failure modes that practitioners have learned to anticipate.
- Trades off specific properties: not every pattern wins on every dimension.

A pattern is not a product, not a framework, and not a template to copy verbatim. It is a vocabulary

item—a shorthand that lets architects discuss a structural choice without re-deriving it from first principles.

17.2 The Six Patterns

The six patterns developed in this part of the book are:

Agentic Mesh (Chapter 18, popularised by McKinsey QuantumBlack). For scaling many agents across the enterprise without chaos. Components: Agents, Governance, Exchange, Network, Tools. The structural answer to organisations that have accumulated more agents than they can govern with point-to-point integration.

Retrieval-Augmented Generation (RAG) (Chapter 21, after Lewis et al., Meta AI 2020). For grounding LLMs in enterprise data without retraining. Components: Query, Retrieve, Augment, Generate, Cite. The dominant pattern for question-answering over enterprise corpora.

LLM Router (Chapter 23, Martian / RouteLLM). For controlling cost and latency across a model portfolio. Components: Small/local, Mid-tier API, Fallback, Frontier. The structural answer to heterogeneous workloads on a varied model market.

Hub-and-Spoke (Chapter 24, enterprise architecture tradition). For centralising AI governance while letting business units innovate. Components: Hub, Spoke 1, Spoke 2, Standards, COE. The organisational pattern that most mature programmes converge on.

Event-Driven Agents (Chapter 25, Confluent / Kafka community). For real-time, asynchronous agent workflows at scale. Components: Event bus, Publishers, Subscribers, Replay, Decoupling. The structural answer when many agents must react to the same domain events.

Hexagonal Architecture for AI (Chapter 26, Cockburn, adapted). For keeping the AI system swappable as models and vendors change. Components: LLM port, Vector port, Tool port, UI port, Observability port. The most direct architectural response to the volatility of the AI vendor landscape.

17.3 Choosing Among the Patterns

The six patterns address different concerns; the choice between them is rarely either-or. The decision rubric:

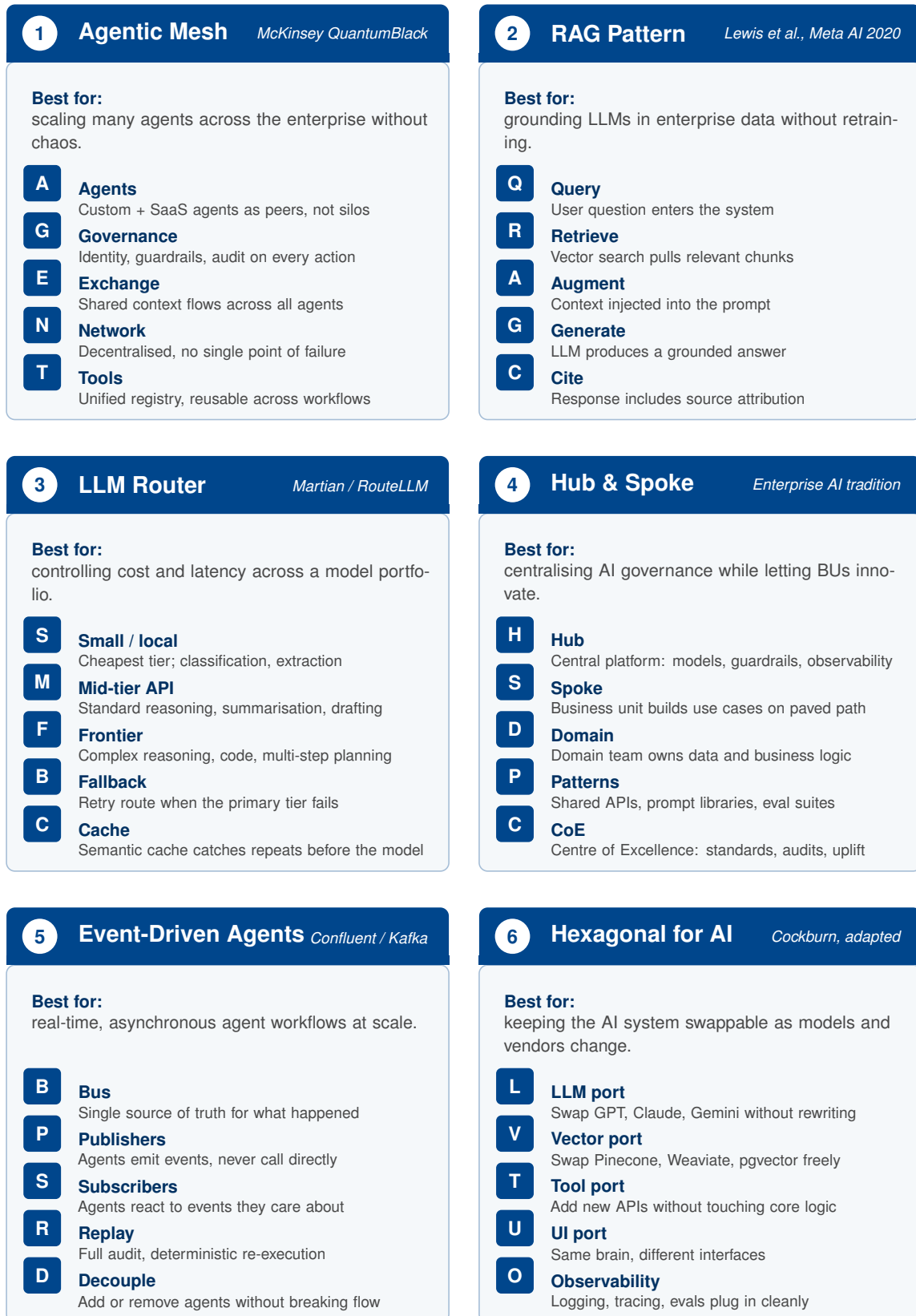


Figure 17.1: Six recurring architecture patterns for enterprise AI: Agentic Mesh, Retrieval-Augmented Generation, LLM Router, Hub-and-Spoke, Event-Driven Agents, and Hexagonal Architecture for AI. Each is developed in its own chapter in Part III.

Table 17.1: Patterns matched to dominant concerns.

Dominant concern	Pattern
Coordinating many agents	Agentic Mesh, Event-Driven Agents
Grounding outputs in enterprise data	RAG
Cost and latency across model portfolio	LLM Router
Organisational structure for many use cases	Hub-and-Spoke
Reacting to many real-time events	Event-Driven Agents
Surviving vendor and model changes	Hexagonal

17.4 Composing the Patterns

Most production systems use three or four patterns together. A representative composition for a moderately complex enterprise AI platform:

- **Hub-and-Spoke** as the organisational pattern: a platform team operates shared services; business-unit spokes build use cases.
- **Hexagonal** as the integration pattern: every external dependency (vendor LLM, vector store, identity provider) is accessed through a port.
- **RAG** as the knowledge pattern: the platform offers a managed retrieval capability that spokes consume.
- **LLM Router** as the cost pattern: the platform routes requests across a model portfolio.
- **Agentic Mesh** as the integration substrate when multiple agents need to interoperate.
- **Event-Driven Agents** for the subset of workflows that demand asynchrony and replay.

The patterns do not conflict; they address different dimensions and reinforce each other. An Agentic Mesh built on hexagonal ports is more durable than one with hard-coded vendor calls. A RAG pipeline behind an LLM Router is cheaper than one that always uses the frontier model.

17.5 Anti-Patterns

Three common misuses of pattern thinking:

- **Cargo-cult adoption.** Implementing a pattern because it is fashionable rather than because the problem it solves exists in this system. Symptoms: an Agentic Mesh with three agents; a hexagonal architecture with one adapter per port; an LLM Router with one model.
- **Over-engineering.** Adopting all six patterns at the start of a programme that does not yet have

problems any of them solve. The patterns earn their complexity when the problems appear; they do not predict the problems into existence.

- **Under-engineering.** Refusing to adopt patterns when the problems are visible, on the grounds that “we’ll do it later.” Later is more expensive, because the system has accumulated dependencies on its absence.

The honest test for each pattern: name the problem the pattern solves, point to evidence of that problem in this system, and choose accordingly.

17.6 How the Pattern Chapters Are Organised

Each of the next six chapters follows a consistent structure:

- When to use this pattern, and the signals that it applies.
- Components of the pattern.
- Interfaces and contracts between the components.
- Variants and refinements.
- Failure modes and their structural mitigations.
- Composition with other patterns.
- A summary.

The chapters are independent: a reader interested in only one pattern can read its chapter alone. The cross-references connect them where the patterns relate.

17.7 In Short

Six patterns recur in enterprise AI architectures: Agentic Mesh, RAG, LLM Router, Hub-and-Spoke, Event-Driven Agents, Hexagonal. They address different concerns and compose freely. Most production systems use three or four together. The chapters that follow develop each in depth; this chapter is the map.

Chapter 18

Pattern: Agentic Mesh

“A mesh is not chaos. It is many small things that know how to find each other.”

—a network architect, paraphrased

Chapter Overview

The Agentic Mesh (popularised by McKinsey QuantumBlack) is the pattern for scaling many agents across an enterprise without producing chaos. It treats custom and SAAS agents as peers rather than silos and provides the connective tissue—identity, governance, exchange, network, tools—that lets them interoperate. This chapter develops the components, the contracts between them, the protocols (A2A, MCP) that are stabilising the inter-agent surface, and the operational mechanisms that prevent runaway behaviour.

Learning Objectives

- Recognise when an Agentic Mesh is the appropriate pattern and when a simpler structure suffices.
- Apply the AGENT mnemonic: Agents, Governance, Exchange, Network, Tools.
- Design the registries, exchanges, and governance that hold a mesh together.
- Use emerging protocols (A2A, MCP) for cross-agent and agent-to-tool integration.
- Recognise the failure modes of agent meshes and the structural controls that prevent them.

18.1 When to Use This Pattern

The Agentic Mesh is the answer when an organisation finds itself with five, ten, or twenty agents—internal and SAAS—that need to coordinate, share context, and obey common governance. The alternative, an ad-hoc tangle of point-to-point integrations, scales as $O(n^2)$ and collapses under it.

Signals that an organisation needs a mesh:

- Multiple business units have built or purchased agents independently.
- The same data flows between agents through bespoke integrations.

- Governance teams cannot inventory agents or audit their actions.
- Identity confusion is common: agents act with the wrong user's permissions.
- Adding a new agent requires touching every existing agent.

18.2 Components of the Mesh

The mesh comprises five component categories, captured in the AGENT mnemonic.

18.2.1 Agents

The mesh treats custom agents (built in-house) and SAAS agents (consumed from vendors) as peers. Each agent has:

- A defined capability (what it can do).
- An identity (who or what it acts for).
- A contract specifying its inputs, outputs, and side effects.
- A registration in the agent registry.

The registry is the catalogue of agents in the mesh. Agents discover one another through the registry rather than through hard-coded endpoints, allowing the mesh to evolve without coordinated changes.

18.2.2 Governance

Identity, guardrails, and audit wrap every action. The governance layer enforces:

- **Authentication.** Every agent action carries verifiable identity (typically OAuth tokens, mTLS certificates, or signed JWTs).
- **Authorisation.** Every action is checked against policy before execution: is this agent allowed to call this tool, on this data, for this user?
- **Audit.** Every action is logged with sufficient detail to reconstruct it later.
- **Rate limiting.** Per-agent and per-tenant quotas prevent runaway loops from exhausting downstream resources.

An agent that cannot prove who it is acting for, what it is allowed to do, and what it has done is not a mesh participant; it is a liability.

18.2.3 Exchange

Shared context flows across all agents. The exchange layer provides:

- **Context broker.** Short-term shared state for an in-flight workflow.
- **Memory store.** Long-term knowledge that persists across sessions.

- **Event bus.** Asynchronous notification of state changes.

Without the exchange, agents communicate by repeated tool calls, repeatedly fetching the same data and repeatedly paying its cost.

18.2.4 Network

The mesh is decentralised: no single point of failure, no single point of contention. The network layer provides:

- Service discovery (where to find an agent by capability).
- Health monitoring (which agents are responsive).
- Routing (load balancing, failover, regional preference).

In practice, a service mesh (Istio, Linkerd, Consul) often provides the network substrate; the agent mesh adds capability-level discovery on top.

18.2.5 Tools

A unified registry of tools (APIs, functions, services) is reusable across workflows. Each tool is defined once and consumed by any agent that needs it.

18.3 Inter-Agent Protocols

Two protocols are stabilising the cross-agent surface at the time of writing.

18.3.1 Model Context Protocol (MCP)

MCP standardises how agents call tools. A tool is a callable with a JSON-schema definition; an MCP server hosts tools; an MCP client (typically an agent) discovers and invokes tools through a uniform interface. The protocol is transport-agnostic (stdio, HTTP, WebSocket).

```
1 {
2   "name": "search_orders",
3   "description": "Search customer orders by date range and status.",
4   "inputSchema": {
5     "type": "object",
6     "properties": {
7       "customer_id": { "type": "string" },
8       "from_date":   { "type": "string", "format": "date" },
9       "to_date":     { "type": "string", "format": "date" },
10      "status":       { "type": "string", "enum": ["open", "fulfilled",
11        "cancelled"] }
12    },
13    "required": ["customer_id"]
14  }
```

Listing 18.1: An MCP tool definition.

18.3.2 Agent-to-Agent (A2A)

A2A protocols (an emerging area in 2025–2026) standardise direct agent-to-agent communication: capability discovery, task delegation, result return, and progress reporting. The key abstractions are *tasks* (units of work an agent can accept) and *artefacts* (results an agent can produce). A2A complements MCP: where MCP standardises agent-to-tool, A2A standardises agent-to-agent.

18.4 Mesh Topologies

Three topologies recur in mesh deployments.

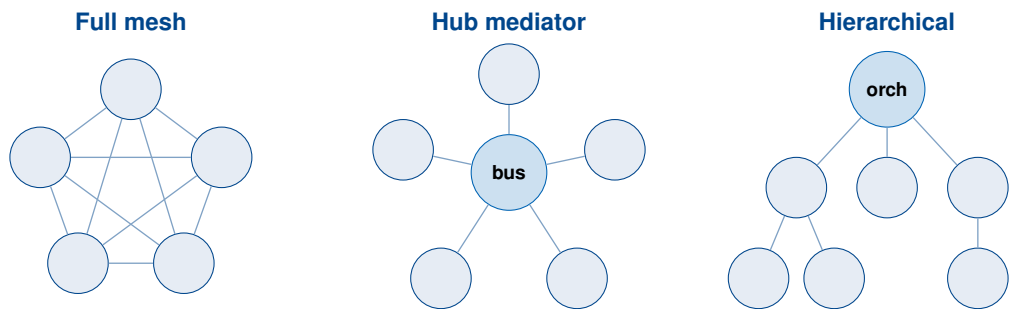


Figure 18.1: Three mesh topologies: full mesh (peer-to-peer), hub-mediated (all traffic through a broker), and hierarchical (orchestrator delegates to specialists).

Table 18.1: Topology trade-offs.

Topology	Strengths	Weaknesses
Full mesh	No bottleneck, low latency	$O(n^2)$ connections; hard to govern
Hub mediator	Centralised audit, easy governance	Hub is a bottleneck and SPOF
Hierarchical	Clear delegation, scoped permissions	Rigid; deep chains accumulate latency

Most mature meshes are hybrid: hub-mediated for governed cross-domain calls, full-mesh within a domain, hierarchical for complex workflows.

18.5 Failure Modes

Table 18.2: Failure modes in agent meshes.

Failure mode	Mitigation
Untracked agents (shadow mesh)	Mandatory registration through paved-path tooling; periodic discovery scans.
Runaway loops (agent calls agent calls agent...)	Per-tenant call-depth limits; budget caps; loop detection in the bus.
Identity confusion	End-to-end identity propagation; explicit "act-as" claims; refuse calls without user context.
Prompt injection across agents	Treat agent outputs as untrusted input; never blend them with privileged instructions.
Cost runaway	Per-agent quotas; alerting on call-volume anomalies.
Tool sprawl	Curated tool registry; deprecation policy for unused tools.

18.6 Operating the Mesh in Production

The mesh in steady-state operation is more like a service mesh than a software library: a control plane manages identity, policy, and discovery, while a data plane carries the actual agent-to-agent and agent-to-tool traffic. Figure 18.2 shows the separation.

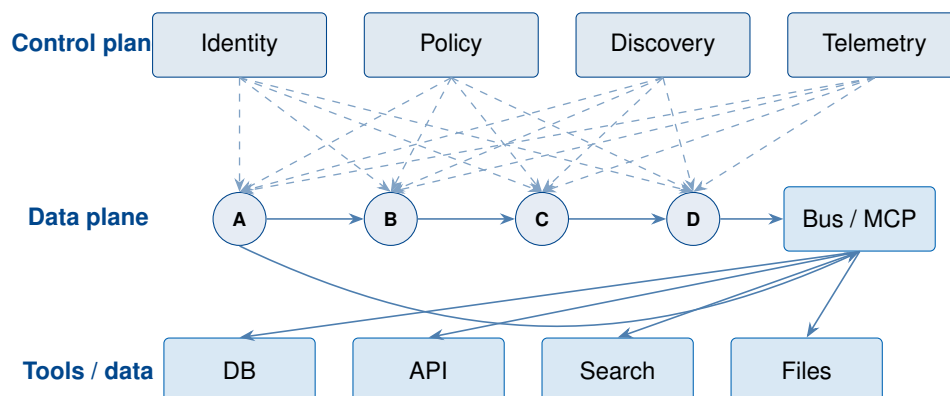


Figure 18.2: Control-plane and data-plane separation in an agentic mesh. Identity, policy, discovery, and telemetry (dashed) govern every data-plane interaction without sitting in the hot path.

18.6.1 The Control Plane

The control plane is what makes a collection of agents a mesh rather than a heap. It typically includes:

- **Identity service.** Issues and validates agent identities, including delegation tokens (*act-as* claims) that let an agent operate on behalf of a user without impersonating them.
- **Policy service.** Centralised authorisation. “Agent A may call Agent B only when acting on behalf of a user in role X.” Per-tool, per-tenant, per-environment policies.
- **Discovery service.** Registry of agents and their capabilities. Without discovery, agents either hardcode endpoints (brittle) or resort to broadcast (chaotic).

- **Telemetry pipeline.** Traces, metrics, logs, audit. Without observability across the mesh, debugging is forensic guesswork.

18.6.2 The Data Plane

Data plane carries the actual agent calls. Architectural choices that matter:

- **Sync vs async.** Sync (HTTP, gRPC) is simpler but couples lifecycles. Async (queues, events) is harder to debug but isolates failures.
- **Protocol.** MCP for agent-to-tool, A2A for agent-to-agent. Native protocols increasingly standardise the surface and reduce bespoke glue.
- **Streaming.** Long-running agents need streaming responses, not request-response. The mesh has to carry partial-token streams cleanly.
- **Backpressure.** Without it, a slow downstream agent fills queues across the mesh. Explicit backpressure signals are not optional at scale.

18.6.3 Cardinality Realities

A mesh with three agents is not the same architecture as a mesh with three hundred. The shape changes as cardinality grows:

- **Up to 10 agents.** Hub-mediated topology is usually fine. Operational overhead modest.
- **10–50 agents.** Discovery and policy become essential. Manual registration breaks down. Versioning of agent capabilities matters.
- **50+ agents.** Federation, multi-cluster, namespace isolation. The mesh starts to look like a Kubernetes cluster for agents: sharding, tenancy, rolling deploys, blue-green for agent versions.

The Agentic Mesh is best for scaling many agents across the enterprise without chaos. Its core insight is that agents are peers, not silos, and that the mesh itself—identity, governance, exchange, network, tools—is the durable infrastructure. Emerging protocols (MCP for agent-to-tool, A2A for agent-to-agent) are stabilising the inter-agent surface and turning the mesh from a bespoke construction into a portable pattern. Without it, organisations accumulate agents faster than they can govern them.

Chapter 19

Pattern: Multi-Agent Negotiation

“Coordination is agents agreeing in advance. Negotiation is agents agreeing when they could not have agreed in advance.”

—a recurring observation in multi-agent systems

Chapter Overview

Multi-Agent Negotiation is the pattern for systems in which agents must reach agreement among themselves — over a price, a schedule, an allocation of scarce resources, a division of work — when no central planner has decided the outcome in advance. It is a frontier pattern, sitting in the most advanced tier of the maturity model alongside the Agentic Mesh and self-improving systems, because it asks for something the orchestration patterns do not: agents that can hold differing objectives and still converge on a settlement. Where multi-agent orchestration coordinates agents toward a goal a designer set, negotiation lets agents whose goals genuinely differ — because they represent different business units, different counterparties, or different optimisation targets — find an outcome none of them dictated. This chapter develops the pattern: where it applies, the negotiation protocols that structure it, the architecture that contains it, and the controls that keep a negotiation from producing an outcome the enterprise did not intend.

Learning Objectives

- Recognise when a system genuinely needs negotiation rather than orchestration.
- Apply the principal negotiation protocols — auction, bilateral bargaining, contract net, voting — to the right problems.
- Architect a negotiation system with the registries, the negotiation space, and the settlement layer it requires.
- Contain a negotiation with the bounds, the audit, and the human-escalation controls that keep it safe.
- Adapt multi-agent negotiation to African market and regulatory contexts.

19.1 When to Use This Pattern

Most multi-agent systems do not need negotiation; they need orchestration, and the distinction is the first thing to be clear about. Orchestration applies when a designer has decided what the agents should collectively achieve and the architecture coordinates them toward it — the multi-agent orchestration pattern of Chapter 29 covers this case in full. Negotiation applies only when the agents hold genuinely differing objectives that no designer reconciled in advance, and the system needs the agents themselves to find the settlement.

Three conditions together signal that negotiation, not orchestration, is the pattern. The first is divergent objectives: the agents are not all optimising the same thing, because they represent different business units, different counterparties, or different and competing targets. The second is a contested resource: there is something scarce — capacity, budget, time, priority — that the agents are competing over, so that one agent's gain is genuinely another's loss. The third is the absence of a central authority that could simply decide: either no such authority exists, or routing every such decision through it would be a bottleneck the enterprise cannot accept. When all three hold, a designer cannot pre-compute the outcome, and the architecture must give the agents a way to reach it themselves.

When the three conditions do not hold, negotiation is the wrong pattern and an expensive one. Agents with a shared objective should be orchestrated; a resource that is not actually contested should simply be allocated; a decision that a central authority can make cheaply should be made there. Negotiation introduces real complexity — protocols, settlement, containment — and it earns that complexity only where agreement genuinely could not have been reached in advance.

19.2 Negotiation Protocols

A negotiation is structured by a protocol — a set of rules for how offers are made, how they are responded to, and how a settlement is reached. The protocol is not an implementation detail; it determines what outcomes are reachable and how the agents must behave. Four protocols recur, and they suit different problems.

19.2.1 Auction

In an auction protocol, one agent offers a resource and others bid for it, with the settlement determined by the auction's rules — highest bid wins, or the second-highest price is paid, or the rules of whatever auction form is chosen. Auctions suit the allocation of a divisible or repeatable resource among many competing agents, because they are efficient at price discovery and scale well to many participants. The architectural attraction of an auction is that it reaches a settlement quickly and the settlement rule is simple to audit; the limitation is that it fits resource allocation better than it fits the negotiation of a complex, multi-term agreement.

19.2.2 Bilateral Bargaining

In bilateral bargaining, two agents exchange offers and counter-offers until they either reach an agreement or reach an impasse. It suits a negotiation between exactly two parties over an outcome with several terms — a price and a delivery date and a quality level together — where an auction's single-dimensional bidding would be too crude. Bilateral bargaining can express a richer settlement

than an auction, but it scales only to two parties at a time and it can fail to converge, which is why the protocol must define an impasse condition and the architecture must define what happens when one is reached.

19.2.3 Contract Net

In the contract-net protocol, an agent with a task to be done announces it, candidate agents respond with bids describing how they would do it and at what cost, and the announcing agent awards the contract to the bid it prefers. Contract net suits the dynamic allocation of work across a pool of agents whose availability and capability vary, because it lets the work find the agent best placed to do it at the moment it arises. It is, in effect, an auction for tasks rather than resources, and it is the natural negotiation protocol for a system that must distribute work whose shape is not known in advance.

19.2.4 Voting and Consensus

In a voting or consensus protocol, several agents must reach a collective decision, and the protocol defines how their individual preferences are aggregated into one outcome — by majority, by a weighted vote, by a consensus threshold. Voting suits a decision that a group of agents must make together, where no single agent should dictate it and no resource is being divided. The protocol's design carries real weight, because the aggregation rule determines whose preferences prevail, and a poorly chosen rule can let a minority block every decision or let a bare majority override a strongly-held objection.

19.3 The Architecture

A negotiation system has a recognisable architecture, shown in Figure 19.1: the negotiating agents, a negotiation space in which offers are exchanged under a protocol, and a settlement layer that records and enforces the agreed outcome. The negotiation space is the structural heart of the pattern — it is the bounded arena in which agents make and answer offers, and it is where the containment controls of the next section are applied.

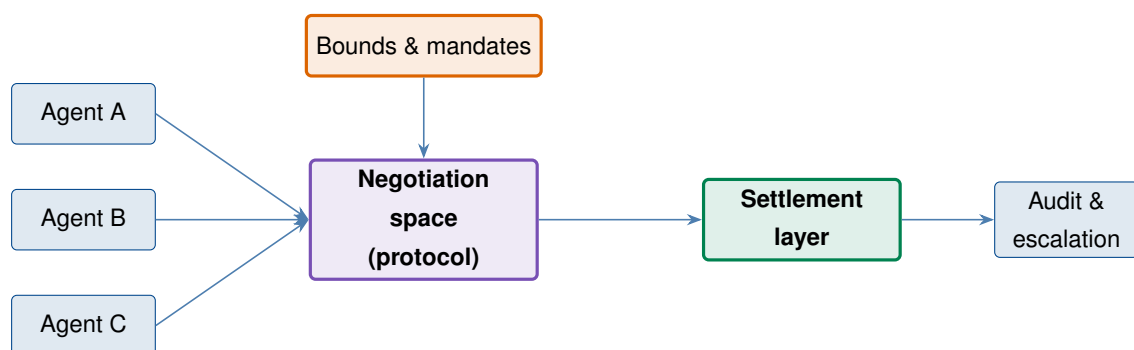


Figure 19.1: Multi-agent negotiation architecture. Agents enter a negotiation space governed by a protocol and constrained by bounds and mandates; the settlement layer records and enforces the agreed outcome; audit and escalation observe every negotiation.

The architecture has four working parts. The negotiating agents each enter with a mandate — the

range of outcomes they are authorised to agree to, set by whoever the agent represents. The negotiation space hosts the offer exchange under the chosen protocol, and it enforces the bounds within which the negotiation must stay. The settlement layer takes the agreed outcome, records it as a binding result, and enforces it — a settlement that is not enforced is not a settlement. And the audit-and-escalation component observes every negotiation, recording it for accountability and escalating to a human any negotiation that hits a bound or fails to settle. The pattern is not the agents alone; it is the agents within this contained, audited, settlement-backed structure.

19.4 Containing a Negotiation

A negotiation among autonomous agents is, by construction, a process whose outcome no designer fixed in advance — which is exactly its value and exactly its risk. An uncontained negotiation can settle on an outcome the enterprise would never have authorised, so containment is not optional refinement but core to the pattern. Three controls contain it.

The first is bounded mandates. Every negotiating agent enters with an explicit mandate — the range of outcomes it is permitted to agree to — and the negotiation space enforces that mandate, rejecting any offer or acceptance that would breach it. An agent negotiating a price has a floor and a ceiling; an agent negotiating a schedule has the commitments it may and may not make. The mandate is what ensures that whatever outcome the negotiation reaches, it lies within the space of outcomes the enterprise authorised, even though the specific outcome was not pre-decided.

The second is comprehensive audit. Every offer, every counter-offer, and every settlement is recorded, so that the negotiation can be reconstructed and accounted for afterward. Audit matters more for negotiation than for orchestration, because a negotiated outcome was not designed and so cannot be explained by pointing to a design — the only account of why the agents settled where they did is the record of the negotiation itself. A negotiation system without comprehensive audit produces outcomes that the enterprise cannot explain to itself, its customers, or its regulators.

The third is human escalation. Certain conditions must lift a negotiation out of the agents' hands and into a human's: a negotiation that hits the edge of a mandate, a negotiation that fails to settle within its allowed bounds, a settlement above a value threshold that warrants human confirmation. Human escalation is the recognition that autonomy in negotiation has a limit, and that the limit must be a designed escalation path rather than a discovered failure. Together the three controls — bounded mandates, comprehensive audit, human escalation — are what make multi-agent negotiation a pattern an enterprise can deploy responsibly rather than a source of unaccountable autonomous commitments.

19.5 The African Perspective

Multi-agent negotiation carries distinctive weight in African deployments, across market structure, mobile-money settlement, regulatory accountability, and cross-border operation.

The first consideration is market structure. Many African markets have a strong tradition of negotiated rather than fixed pricing, and an enterprise system whose agents negotiate prices, allocations, or terms

is, in those markets, automating a transaction form that the market already uses. The pattern can fit African commercial practice naturally — but the mandates the agents carry must reflect the real commercial limits, because an agent negotiating in a genuinely negotiated market needs a mandate wide enough to be useful and tight enough to be safe.

The second consideration is settlement, and it connects negotiation to the mobile-money rails that run much African commerce. A negotiated outcome frequently has to be settled through a payment, and where that payment runs over mobile money, the settlement layer of the negotiation architecture meets the mobile-money system directly. The financial-action controls developed in the security and agent-identity chapters — step-up authorisation, value thresholds, the financial-action class of the action taxonomy — apply to the settlement of a negotiation exactly as they apply to any other agent-initiated payment, and the human-escalation threshold for a high-value settlement is, in these deployments, often a mobile-money confirmation.

The third and fourth considerations are regulatory and cross-border. The data-protection and financial-conduct regimes of African jurisdictions require that an automated commercial decision be accountable, and the comprehensive-audit control of this chapter is what produces that accountability — the negotiation record is the evidence a regulator can examine. And a pan-African operation whose agents negotiate across borders must respect that the negotiating agents on either side may sit under different jurisdictions, with different mandates and different regulatory constraints, so a cross-border negotiation is constrained by the union of the regimes its participating agents fall under. The negotiation pattern does not change for African deployments; its mandates, its settlement, and its audit must reflect the continent's negotiated markets, mobile-money rails, and multi-jurisdiction reality.

19.6 Recurring Failure Patterns

The failure modes of multi-agent negotiation recur, and each maps to a discipline this chapter has set out:

- **Negotiation where orchestration would do.** The pattern is used for agents that share an objective, paying the full cost of protocols and settlement for a problem that simple coordination would have solved.
- **Unbounded mandate.** An agent negotiates with no enforced range of authorised outcomes, so the negotiation can settle on a commitment the enterprise never authorised.
- **Non-convergence unhandled.** A bilateral bargaining or consensus protocol fails to settle, and there is no defined impasse condition or escalation path, so the negotiation simply hangs.
- **Settlement without enforcement.** The agents reach an agreement but the settlement layer does not record and enforce it, so the negotiated outcome is not actually binding.
- **Audit gap.** The offers and counter-offers are not recorded, so a negotiated outcome cannot be reconstructed or explained after the fact.

Each pattern maps to a remedy the chapter has developed — choosing the pattern only when the three

conditions hold, enforcing bounded mandates, defining impasse and escalation, backing settlement with enforcement, and recording the negotiation in full — and together they show that negotiation fails through missing containment rather than through the agents bargaining badly.

19.7 In Short

Multi-Agent Negotiation is the frontier pattern for systems whose agents must reach agreement among themselves when their objectives genuinely differ and no central planner has fixed the outcome. It applies only when three conditions hold together — divergent objectives, a contested resource, and the absence of a central authority that could simply decide — and where they do not hold, orchestration is the cheaper and correct pattern.

A negotiation is structured by a protocol — auction, bilateral bargaining, contract net, or voting — chosen to fit the problem, and it runs within an architecture of negotiating agents, a protocol-governed negotiation space, and a settlement layer that makes the outcome binding. Because a negotiated outcome is by construction not pre-designed, containment is core to the pattern: bounded mandates keep every reachable outcome within what the enterprise authorised, comprehensive audit makes the outcome accountable, and human escalation lifts the negotiations that exceed the agents' proper autonomy back to a person. African deployments add naturally negotiated markets, mobile-money settlement under financial-action controls, regulatory accountability through the audit record, and the multi-jurisdiction constraint on cross-border negotiation. Coordination is agents agreeing in advance; negotiation is agents agreeing when they could not have — and the architecture of this chapter is what makes that agreement safe.

Chapter 20

Pattern: Self-Improving Loops

“A system that improves itself is a system that can also degrade itself. The loop must be closed by evidence, not by hope.”
—a recurring observation in production AI

Chapter Overview

A Self-Improving Loop is the pattern for a system that learns from its own production behaviour and gets better over time without a human hand-editing every change. It sits in the most advanced tier of the maturity model, alongside the Agentic Mesh and multi-agent negotiation, because it asks for the hardest thing a system can do safely: change itself. The pattern is genuinely powerful — a system that turns every production interaction into a small improvement compounds, over months, into a system far better than the one that shipped. It is also genuinely dangerous, because a loop that can improve a system can, with equal ease, degrade it: an ungarded self-improvement loop is a mechanism for amplifying whatever flaw its measurement failed to catch. This chapter develops the pattern with that double edge in mind — the loop’s structure, the evidence gates that must close it, the human authority that must sit over it, and the failure modes that follow when the gates are weak.

Learning Objectives

- Recognise what genuinely counts as a self-improving loop, distinct from ordinary monitoring and retraining.
- Architect the loop — observe, propose, evaluate, gate, deploy — with the evaluation gate as its load-bearing element.
- Distinguish the levels of self-improvement, from prompt refinement to autonomous capability change, by the risk each carries.
- Apply the containment controls — evidence gates, bounded change, human authority, instant rollback — that keep the loop safe.
- Adapt self-improving loops to African data, cost, and oversight realities.

20.1 What a Self-Improving Loop Is

A self-improving loop is a system that observes its own production behaviour, proposes changes to itself, and adopts the changes that the evidence shows to be improvements — without a human hand-authoring each change. The phrase has to be defined carefully, because two adjacent things are routinely mistaken for it. Ordinary monitoring observes a system but proposes nothing; it tells a human that something is wrong and stops. Ordinary retraining, on a schedule, changes a model but is triggered and directed by a human, not by the system’s own observation of its shortcomings. A self-improving loop is the case where the observation, the proposal, and the adoption form a closed cycle that runs without a human authoring each step.

What makes the pattern distinct, and what makes it a frontier pattern rather than a routine one, is that closure. A monitored system improves only as fast as humans read its dashboards and act; a self-improving system improves continuously, turning each production interaction into a candidate lesson. The compounding is the appeal: a small, evidence-backed improvement adopted every day is, after a year, a transformation. But the same closure that compounds improvement will compound degradation if the evidence is wrong, because the loop adopts whatever the evidence endorses, and a flawed measurement endorses a flawed change. The pattern’s power and its danger are the same mechanism viewed from two sides, and the rest of this chapter is, in effect, about how to keep the mechanism pointed at improvement.

20.2 The Loop

The self-improving loop has five stages, shown in Figure 20.1: the system observes its production behaviour, proposes a change, evaluates that change, gates it against evidence, and — only if the gate passes — deploys it, after which the cycle begins again. The five stages run continuously, and the loop’s safety rests almost entirely on the fourth.

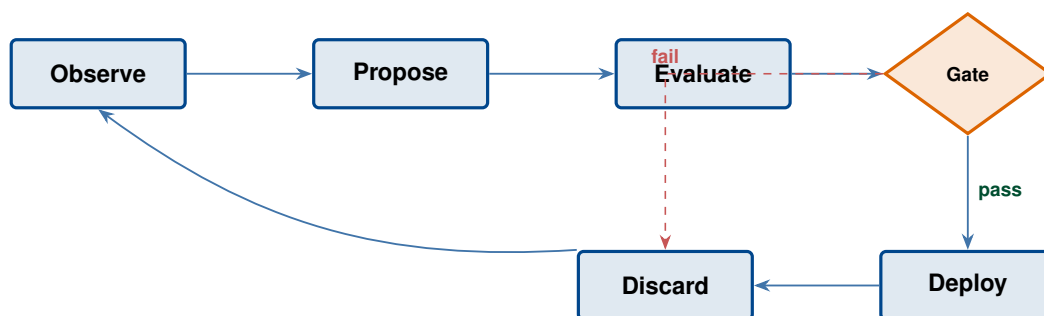


Figure 20.1: The self-improving loop. The system observes production, proposes a change, evaluates it, and gates it against evidence; only a change that passes the gate is deployed, and a failed change is discarded. The gate is the load-bearing stage.

The stages divide cleanly into the easy four and the hard one. Observing production behaviour is the monitoring the system already does. Proposing a change — a refined prompt, a new evaluation case, an adjusted parameter — is increasingly something a model can do well. Deploying a change is the canary-and-rollout machinery the book has developed throughout. These four are not where the pattern’s difficulty lies. The difficulty is the gate: the stage that decides whether a proposed change is genuinely an improvement and may therefore be adopted, or is not and must be discarded. Everything

that makes a self-improving loop safe or unsafe is concentrated in the quality of that gate, which is why the next two sections are about, respectively, what the loop is allowed to change and how the gate is built.

20.3 Levels of Self-Improvement

Not all self-improvement carries the same risk, and a sound architecture distinguishes the levels by how much the loop is permitted to change and how reversible that change is. The levels run from the safe and routine to the genuinely frontier.

20.3.1 Prompt and Example Refinement

At the first level, the loop refines prompts and adds evaluation cases. It observes where the system underperforms, proposes a clearer prompt or a new few-shot example, and — gate permitting — adopts it. This is the safest level, because a prompt change is a configuration change: it is cheap to make, instantly reversible, and bounded in what it can affect. A loop confined to this level can run with relatively light containment, and for many systems it is the only level of self-improvement actually needed.

20.3.2 Parameter and Routing Adjustment

At the second level, the loop adjusts operational parameters — routing thresholds, retrieval depth, the cut-offs that govern how the system behaves — within ranges a designer set. It is more consequential than prompt refinement, because a routing change shifts traffic and so shifts cost and quality at once, but it remains bounded: the loop tunes within an allowed range rather than redefining the range. This level requires firmer evidence gating than the first, because a parameter change's effects are broader, but it is still a tuning of the system rather than a redefinition of it.

20.3.3 Adapter and Model Update

At the third level, the loop trains adapters or updates models from production data — the continual-learning machinery of Chapter 60 run as a closed loop. This level changes the model's learned behaviour, and it carries the catastrophic-forgetting risk that chapter develops: a self-improvement that lifts one capability can silently erode another. A loop operating at this level needs the full evidence gate, including evaluation against the entire historical suite, because the change it is adopting is no longer cheap to reverse and no longer bounded to one prompt.

20.3.4 Autonomous Capability Change

At the fourth level, the loop changes the system's capabilities themselves — adding tools, altering its own architecture, redefining what it can do. This is the genuine frontier, and it is the level at which the pattern's danger is sharpest, because a capability change is the hardest to bound, the hardest to evaluate comprehensively, and the hardest to reverse cleanly. Few production systems should operate at this level today, and those that do should treat human authority over the loop, developed in the next section, as non-negotiable. The level exists in the taxonomy because it is where the pattern is heading; it is not where most enterprises should be running it now.

20.4 Containing the Loop

A self-improving loop adopts changes on its own, which means the controls that contain it are not safeguards added to a working pattern — they are what makes the pattern usable at all. Four controls contain the loop, and the epigraph names their purpose: the loop must be closed by evidence, not by hope.

The first control is the evidence gate, and it is the one that matters most. A proposed change is adopted only if it passes a rigorous evaluation that demonstrates it is genuinely an improvement — better on the metrics it targeted, and, critically, not worse on anything else. The gate must evaluate against the full historical evaluation suite, not only the cases the change was meant to fix, because the characteristic failure of a self-improving loop is a change that improves its target and regresses something the gate did not check. The gate is the stage at which the loop's compounding is pointed at improvement rather than degradation, and a weak gate is the single thing that turns the pattern from an asset into a hazard.

The second control is bounded change. The loop is permitted to change only within an explicit, designed envelope: the levels of the previous section define what it may alter, and within each level the ranges are set by a human. A loop allowed to change anything is a loop whose worst case is unbounded; a loop confined to a designed envelope has a worst case a designer can reason about. The third control is standing human authority. A human or a governance body retains the authority to inspect what the loop has been doing, to veto a class of change, to pause the loop entirely, and to review the changes it has adopted. The loop runs autonomously, but it runs under an authority that can always reach in — autonomy within oversight, not autonomy instead of it.

The fourth control is instant rollback. Every change the loop adopts must be instantly reversible, and the loop must monitor the production effect of its own recent changes so that a change which passed the gate but degrades production in ways the gate did not foresee is caught and reverted quickly. The evidence gate is not perfect — no evaluation catches everything — and instant rollback is the admission of that, the control that limits the damage of a change the gate wrongly approved. Together the four controls — the evidence gate, bounded change, human authority, and instant rollback — are what let an enterprise run a pattern whose defining feature is that the system changes itself.

20.5 The African Perspective

Self-improving loops carry distinctive weight in African deployments, across the value of the pattern, the data it learns from, its cost, and the oversight it requires.

The pattern is, in one respect, especially valuable in African deployments. A self-improving loop confined to its safe levels — prompt refinement, bounded parameter tuning — lets a small team keep a system improving without the large engineering operation that continuous manual improvement would demand, and many African deployments run with exactly the lean teams the project has noted throughout. A well-contained loop is a force multiplier for a small team, turning production interactions into improvement without proportionate engineering effort.

But the loop learns from the production data it observes, and that raises two cautions specific to African deployments. The first is representativeness: if the production traffic the loop learns from is skewed — over-weighted toward one language, one region, one customer segment — the loop will improve the system for that slice and may quietly let the others drift, because the evidence gate measures against what the traffic contains. The stratified-evaluation discipline of the evaluation and ethics chapters applies directly: the gate for a self-improving loop in a multilingual, multi-market African deployment must evaluate per language and per market, or the loop will optimise the aggregate and erode the under-represented. The second caution is data residency: a loop that observes production behaviour and trains on it is processing personal data, and where it runs and where that data sits fall under POPIA, the NDPA, and the other regional regimes exactly as the continual-learning chapter set out.

Cost and oversight complete the picture. The loop's own operation has a cost — the evaluation runs of the gate, in particular, are real model spend — and under tight African unit economics that cost belongs in the FinOps accounting, which argues for confining the loop to the cheaper levels and running the gate efficiently. And the human-authority control should reflect the leaner oversight capacity of a small team: a loop overseen by a small team should be confined to the lower, safer levels, where the bounded change and instant rollback do more of the containment work, rather than run at a frontier level that would demand oversight the team cannot sustain. The pattern does not change for African deployments; the level it is run at, the stratification of its gate, and the residency of its data must reflect African realities.

20.6 Recurring Failure Patterns

The failure modes of self-improving loops recur, and each maps to a control this chapter has set out:

- **Weak evidence gate.** The gate evaluates only the cases the change targeted, so a change that improves its target and regresses something else is adopted, and the loop compounds the regression.
- **Unbounded change.** The loop is allowed to change anything, with no designed envelope, so its worst case is unbounded and no designer can reason about it.
- **No human authority.** The loop runs with no standing power to inspect, veto, or pause it, so a loop that has begun to degrade the system cannot be reached.
- **Rollback absent.** A change that passed the gate but degrades production cannot be instantly reversed, so the damage of a wrongly-approved change accumulates.
- **Skewed learning signal.** The loop learns from unrepresentative production traffic and improves the system for one slice while letting the under-represented slices drift.

Each pattern maps to a control the chapter has developed — the full-suite evidence gate, bounded change, standing human authority, instant rollback, and a stratified gate — and together they make the chapter's point: a self-improving loop fails not because self-improvement is impossible but because a control that should have contained it was weak or absent.

20.7 In Short

A Self-Improving Loop is the frontier pattern for a system that observes its own production behaviour, proposes changes, and adopts the ones the evidence endorses — a closed cycle of observe, propose, evaluate, gate, and deploy that runs without a human authoring each change. Its power is that improvement compounds; its danger is that the same closure compounds degradation if the evidence is wrong, and the two are the same mechanism.

The loop operates at distinguishable levels — prompt refinement, parameter tuning, adapter and model update, and the genuine frontier of autonomous capability change — and the risk rises sharply up that ladder, with most enterprises rightly confined to the lower, safer levels. Whatever the level, the loop is made safe by four controls: the evidence gate that adopts only demonstrated improvements and evaluates against the full historical suite, bounded change within a designed envelope, standing human authority to inspect and pause, and instant rollback for the changes the gate wrongly approved. African deployments find the pattern especially valuable as a force multiplier for lean teams, while requiring a stratified gate against skewed production traffic, residency-aware handling of the data the loop learns from, and a level confined to what the team's oversight can sustain. A system that improves itself can also degrade itself — and the loop must be closed by evidence, not by hope.

Chapter 21

Pattern: Retrieval-Augmented Generation

“Don’t memorise it; look it up.”

—every teacher, eventually

Chapter Overview

Retrieval-Augmented Generation, introduced by Patrick Lewis and colleagues in 2020, is the pattern for grounding LLMs in enterprise data without retraining. It is the most widely-deployed enterprise AI pattern and the default starting point for any AI system that answers questions from a body of organisational knowledge. This chapter develops the full pipeline: ingestion, chunking, embedding, indexing, retrieval, augmentation, generation, citation, and evaluation; and surveys the variants—hybrid search, re-ranking, GraphRAG, agentic RAG—that address the limits of the basic form.

Learning Objectives

- Understand when RAG is the appropriate pattern and when fine-tuning, long context, or tool use is preferable.
- Design each stage of the RAG pipeline: chunk, embed, index, retrieve, augment, generate, cite.
- Apply hybrid retrieval, re-ranking, and query rewriting to improve recall and precision.
- Recognise and address the recurring failure modes: stale corpora, retrieval misses, ungrounded generation, citation drift.
- Evaluate RAG systems with metrics that distinguish retrieval quality from generation quality.

21.1 When to Use This Pattern

RAG is best for grounding LLMs in your enterprise data without retraining. It is the right answer when:

- The knowledge changes faster than retraining cycles.
- Source attribution is required by users, auditors, or regulators.
- The corpus is too large to fit in a context window.

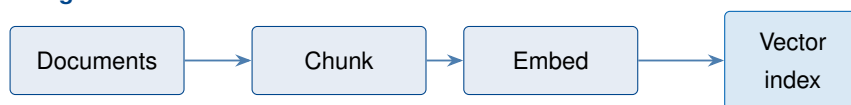
- Fine-tuning would be more expensive or risky than retrieval.
- Per-tenant or per-user knowledge isolation is needed.

RAG is the wrong answer when the model needs to learn new *skills* (writing in a particular style, following a particular reasoning pattern) rather than new *facts*; for that, fine-tuning or in-context examples are more direct.

21.2 The Five-Stage Pipeline

The RAG pipeline has two parallel sub-pipelines (Figure 21.1): an offline *indexing* pipeline that prepares the corpus, and an online *query* pipeline that processes user requests.

Offline indexing



Online query



Figure 21.1: The RAG pipeline has two parallel halves: offline indexing prepares the corpus; online query retrieves from the index and generates a grounded answer.

21.3 Stage 1: Chunking

Source documents are split into chunks before embedding. The chunking strategy determines both retrieval quality and cost.

21.3.1 Chunk Size

Smaller chunks (128–256 tokens) give precise retrieval but lose context. Larger chunks (512–1024) preserve context but dilute relevance scores. A widely-used default is 512-token chunks with 64-token overlap; experiment to find the right size for your corpus.

21.3.2 Chunk Boundaries

Three strategies dominate:

- **Fixed-size with overlap.** Splits every N tokens with K -token overlap. Simple and fast; may split mid-sentence.
- **Structural.** Splits on document structure (headings, paragraphs, list items). Honours author intent; depends on parser quality.
- **Semantic.** Splits where embedding similarity between adjacent sentences drops below a threshold. Highest retrieval quality; highest preprocessing cost.

21.3.3 Chunk Metadata

Every chunk should carry metadata that supports filtering at retrieval time:

```
1 {  
2   "chunk_id":      "POL-204-c012",  
3   "source_doc":    "policy_handbook_2026.pdf",  
4   "section":       "5.3 Refund Policy",  
5   "doc_date":      "2026-03-15",  
6   "authority":     "approved",  
7   "tenant":        "acme_corp",  
8   "language":      "en",  
9   "tags":          ["refunds", "consumer"],  
10  "embedding":      [0.0123, -0.0456, ...]  
11 }
```

Listing 21.1: Chunk metadata schema for filterable retrieval.

21.4 Stage 2: Embedding

Each chunk is encoded into a dense vector by an embedding model. The choice of embedding model is consequential and difficult to reverse: re-embedding a multi-million-chunk corpus is expensive.

Selection criteria:

- **Domain fit.** A model trained on web text may underperform on legal or medical content. Domain-specialised embeddings exist.
- **Multilingual coverage.** If the corpus spans languages, a multilingual model avoids the alternative of one index per language.
- **Dimensionality.** 384–3072 dimensions are common. Larger captures finer distinctions, costs more storage and search compute.
- **Hosting.** Vendor API (operational simplicity, ongoing cost) or self-hosted (one-time setup, marginal cost near zero).

21.5 Stage 3: Indexing

The vector index supports approximate nearest-neighbour search. The choice of algorithm depends on corpus size and latency budget.

Table 21.1: Vector index algorithms.

Algorithm	Latency	Memory	Best for
Exact (flat)	High for $> 10^6$ vectors	Low	Small corpora; ground-truth eval
HNSW	Very low	High	Default for most production cases
IVF + PQ	Low	Moderate	Very large corpora; memory-constrained
ScaNN	Very low	Moderate	Google-scale workloads

21.6 Stage 4: Retrieval

Retrieval converts the query into one or more candidate sets of chunks.

21.6.1 Query Embedding and Search

The query is embedded by the same model as the chunks (this is essential—embeddings from different models are not comparable). The top- K chunks by similarity are returned, typically $K = 20$ –100.

21.6.2 Query Rewriting

User queries are often suboptimal for retrieval: ambiguous, terse, or filled with anaphora. *Query rewriting*—using a small LLM call to expand or reformulate the query—improves recall:

- **Expansion.** Generate synonyms or related terms.
- **Decomposition.** Split a multi-part query into independent sub-queries.
- **HyDE (Hypothetical Document Embeddings).** Generate a hypothetical answer, embed it, and retrieve documents similar to the hypothetical.

21.6.3 Hybrid Retrieval

Pure vector retrieval struggles with exact-match queries (codes, identifiers, rare names). *Hybrid retrieval* combines vector search with keyword search (BM25) and merges results with reciprocal rank fusion:

$$\text{score}_{\text{hybrid}}(d) = \sum_{r \in \{r_{\text{vec}}, r_{\text{bm25}}\}} \frac{1}{k + r(d)}$$

where $r(d)$ is the rank of document d in each list and k is a smoothing constant (commonly 60).

21.6.4 Re-Ranking

A cross-encoder re-ranker scores each candidate against the query jointly, rather than as separate vectors. It is more accurate but slower; the architecture uses it to re-score the top 50–100 candidates from initial retrieval down to the top 5–10 used in the prompt.

21.7 Stage 5: Augmentation and Generation

Retrieved chunks are injected into the prompt as context. The construction matters:

- **Delimiters.** Clearly separate context from instructions and from the user query.
- **Citation hooks.** Tag each chunk with a stable identifier the model can reference.
- **Refusal directive.** Instruct the model to refuse when grounding is insufficient.
- **Ordering.** Place the most relevant chunks at the position where the model attends best (often beginning or end; varies by model).

```
1 System: You are a policy assistant. Answer using ONLY the provided
   context.
2       If the context is insufficient, reply: "I do not have enough
3       information to answer this." Cite each fact with [chunk_id].
4
5 Context:
6 [POL-204-c012] {chunk_text_1}
7 [POL-211-c003] {chunk_text_2}
8 [POL-302-c008] {chunk_text_3}
9
10 User question: {user_question}
11
12 Answer (with citations):
```

Listing 21.2: A grounded prompt with citation requirement.

21.8 Stage 6: Citation and Verification

Source attribution distinguishes a grounded answer from a confident hallucination. The architecture should:

- Parse citations from the output and verify they refer to retrieved chunks.
- Surface citations in the UI as clickable links to source documents.
- Run a verification step (separate model call or rule check) confirming each claim is supported by its cited chunk.

21.9 Variants

The basic pipeline has several refinements that address recurring limitations.

21.9.1 GraphRAG

For corpora where the relationships between entities matter as much as the entities themselves (organisational charts, legal contracts, scientific literature), GraphRAG augments vector retrieval with a knowledge graph. Entities and relations are extracted from documents and stored as a graph;

retrieval combines vector similarity with graph traversal.

21.9.2 Agentic RAG

A single retrieval pass may miss multi-hop information ("who reports to the person who approved this policy?"). Agentic RAG allows the model to issue multiple retrieval queries, reasoning between them: query, examine results, refine query, retrieve again, generate.

21.9.3 Contextual RAG

Each chunk is augmented at indexing time with surrounding context (document title, section heading, brief summary) so that retrieval matches against the full context, not the chunk in isolation. This improves recall at modest indexing cost.

21.10 Evaluation

RAG evaluation must distinguish retrieval quality from generation quality.

Table 21.2: RAG evaluation metrics.

Metric	What it measures
Retrieval recall@ K	Fraction of relevant chunks present in top- K .
Retrieval precision@ K	Fraction of top- K that are relevant.
MRR (mean reciprocal rank)	Position of first relevant chunk; rewards top placement.
nDCG	Graded relevance with position discount.
Faithfulness	Whether the generated answer is supported by the retrieved context.
Answer relevance	Whether the answer addresses the user's actual question.
Citation accuracy	Whether cited chunks actually support the claims they cite.

A common diagnostic: if recall@10 is good but the answers are poor, the generation prompt or model is the bottleneck. If recall@10 is poor, no amount of generation tuning will fix it.

21.11 Hybrid Retrieval in Depth

Dense retrieval (embeddings) excels at semantic similarity; sparse retrieval (BM25, keyword) excels at exact-match and rare terms. Production RAG systems combine them. Figure 21.2 shows the architecture.

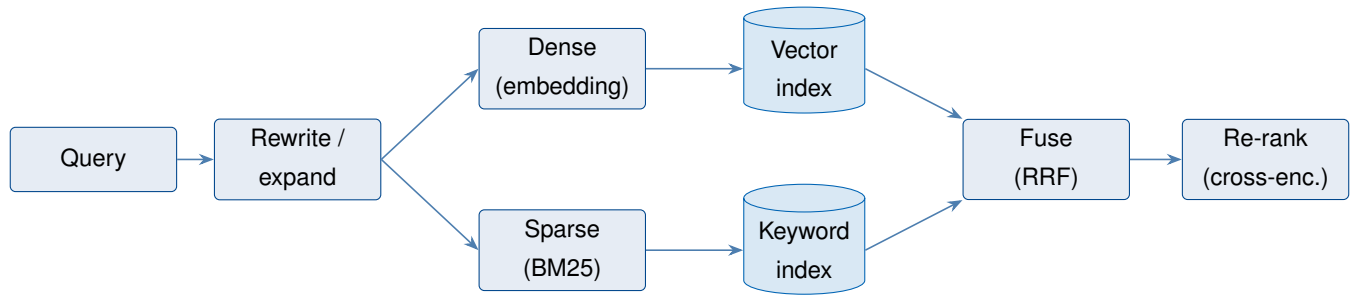


Figure 21.2: Hybrid retrieval: dense and sparse paths run in parallel, results are fused (typically reciprocal-rank fusion), then a cross-encoder re-ranks the top candidates. Each stage trades cost for precision.

21.11.1 Why Hybrid Wins

Pure dense retrieval misses exact strings: product SKUs, error codes, person names, regulation citations, version numbers. Pure sparse retrieval misses synonyms and paraphrases. Hybrid covers both, at the cost of running two retrieval paths and fusing them.

21.11.2 Fusion Strategies

When multiple retrieval paths return results, the system must combine them into a single ranked list. The fusion strategy determines how the rankings from different sources are merged, and the choice affects both the quality and the diversity of the final retrieval set.

- **Reciprocal Rank Fusion (RRF).** Combines ranks rather than scores; robust to score-scale differences across retrievers. The default in most production systems.
- **Score normalisation + weighted sum.** Lets you tune the dense/sparse balance, but requires careful calibration.
- **Learned fusion.** Train a ranker on (query, candidate) pairs. Highest quality, highest engineering cost.

21.11.3 When to Re-Rank

A cross-encoder re-ranker scores each (query, candidate) pair jointly, which catches relevance signals that bi-encoder embeddings miss. The cost is per-pair inference, so re-ranking is reserved for the top $k = 20$ to 50 candidates. Re-ranking quality often dominates retrieval quality in mature RAG systems.

21.12 Failure Modes and Mitigations

The pipeline has well-characterised failure modes:

- **Retrieval miss.** The right chunk exists in the corpus but is not retrieved. Mitigations: hybrid retrieval, query expansion, multi-vector representations, re-ranking.
- **Context dilution.** Too many chunks injected; signal drowns in noise. Mitigations: top- k capping, re-ranking, chunk-merging.

- **Citation drift.** The model cites a chunk that does not actually support the claim. Mitigations: post-generation citation verification, attribution-trained models, structured cite tags with assertion-to-chunk mapping.
- **Stale corpus.** The index is out of date. Mitigations: incremental indexing, freshness signals in scoring, mandated refresh SLAs.
- **Prompt injection via retrieved content.** Documents in the corpus contain instructions that hijack the generator. Mitigations: treat retrieved content as untrusted; structural separation in the prompt; instruction-detection filters.
- **Cross-tenant leakage.** Index returns another tenant's content. Mitigations: tenant scoping at the index level (not the application level); audit on every retrieval.

21.13 What to Carry Forward

RAG is the workhorse pattern of enterprise AI. Its five stages—Chunk, Embed, Retrieve, Augment, Generate (with Cite layered through them)—are simple to describe and difficult to do well. Retrieval quality is usually the bottleneck; investments in chunking, embedding choice, query rewriting, and re-ranking typically pay back several times over investments in the generation model. The pattern composes naturally with the hexagonal architecture (Chapter 26) and is the canonical implementation of Layer 3's knowledge sub-layer.

Chapter 22

RAG Variations: Choosing the Right Retrieval Pattern

“The basic pattern is the one that works. The variations are the ones that work for what the basic pattern does not.”

—a recurring observation in retrieval-augmented generation practice

Chapter Overview

The previous chapter introduced retrieval-augmented generation as the foundational pattern for connecting a language model to an enterprise’s own data. The pattern as introduced — index the corpus, embed the query, retrieve top- k passages, pass them to the model with the question — works for a recognisable class of workloads and fails interestingly for several others. The variations the field has developed over the past few years address the failures. This chapter surveys ten of them, grouped into four families: foundational variations that tighten the basic pipeline (Naive, Advanced, Hybrid); knowledge-shape variations that change what is being retrieved (Graph, Multimodal); behavioural variations that change how retrieval happens (Agentic, Corrective, Adaptive); and operational variations that change how the system runs in production (Streaming, Enterprise). The chapter closes with an end-to-end reference architecture that composes these variations into a working enterprise deployment, and a note about where the architectural direction of RAG appears to be heading.

Learning Objectives

- Recognise the ten RAG variations that have settled in production practice.
- Identify the failure mode of basic RAG that each variation addresses.
- Choose among variations based on the workload’s actual characteristics rather than the marketing of the moment.
- Read the existing Agentic RAG chapter (Chapter 50) as one specialised member of this family.
- Carry the survey forward into the operational chapters of Parts V and VI.

22.1 Why RAG Has a Pattern Family

The basic RAG pipeline is one of the genuinely useful patterns the AI field has produced. It is also a pipeline with several failure modes that production deployments have learned, often painfully, to anticipate. Top- k retrieval over dense embeddings recovers passages by semantic similarity, but semantic similarity is not the same as relevance, and a passage that is similar to the query is not always the passage the model needs. The retrieved set is fixed at query time, but the work the model is trying to do may genuinely require multiple retrieval rounds with refinement. The corpus is assumed to be static at indexing time, but real enterprise data changes continuously. The corpus is assumed to be text, but real enterprise data includes images, tables, audio, video, and structured records that no embedding model handles equally well. The retrieval is assumed to be cheap, but production scale exposes latency, cost, and freshness as first-class constraints.

Each of the ten variations in this chapter is a response to one or more of these failures. None is a replacement for the basic pattern; each is the variation that the basic pattern asks for once a specific failure mode becomes the limiting factor on the system’s quality. A well-architected enterprise RAG deployment in 2026 typically contains several variations, applied selectively to the parts of the workload where they earn their cost.

The architectural decision the chapter supports is therefore not “which RAG to use” but “which RAG variations to combine, at which boundaries, for which workloads.” The choice is informed by what the workload actually needs, what the corpus actually looks like, and what failure modes the operations team has actually seen. The patterns described below are the working vocabulary by which those decisions get made.

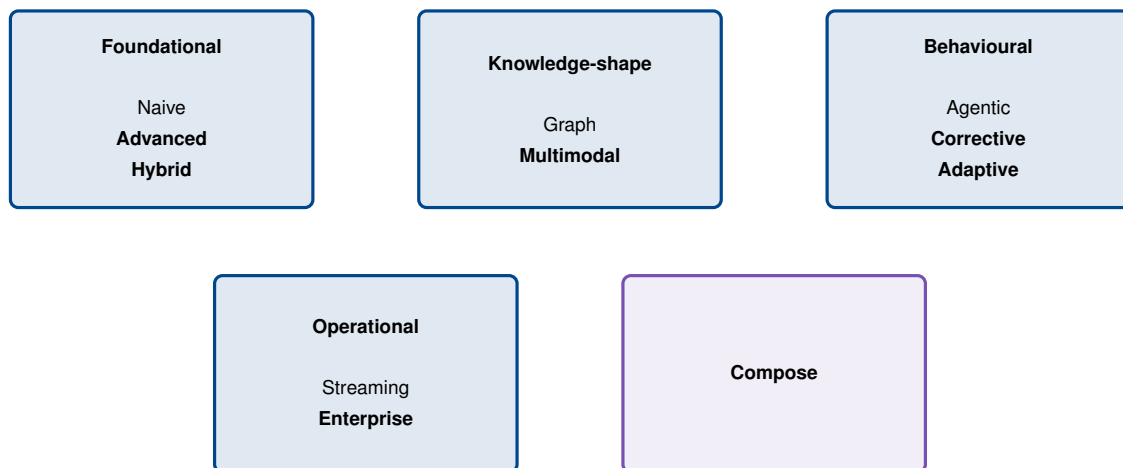


Figure 22.1: The ten RAG variations grouped into four families. Foundational variations tighten the basic pipeline; knowledge-shape variations change what is being retrieved; behavioural variations change how retrieval happens; operational variations change how the system runs. A production deployment composes selected variations from each family.

22.2 Foundational Variations

22.2.1 Naive RAG: The Pipeline as Baseline

Naive RAG is the standard pipeline introduced in the previous chapter — query embedding, vector search, top-*k* retrieval, single model call. It is the pattern that works for a recognisable class of workloads: internal chatbots over a moderate-sized text corpus, document question-and-answer on technical material, lightweight knowledge copilots where the questions are direct and the answers are present in the corpus more or less verbatim.

The pattern's limits are recognisable in production. It struggles with weak reasoning over the retrieved passages, with information seeking that requires multi-step retrieval, and with corpora whose structure (tables, relationships, hierarchies) is not well-captured by flat text chunks. The other variations in this chapter are the responses to those limits. The reason Naive RAG remains worth its own section is that, for the workloads it serves, the more complex variations are overhead without payoff. Architecture is not about always using the most sophisticated pattern; it is about using the simplest pattern that meets the requirement.

22.2.2 Advanced RAG: Tightening the Basic Pipeline

Advanced RAG keeps the basic pipeline shape but invests in each of its stages. Query rewriting adapts the user's question into a form better suited for retrieval — expanding abbreviations, decomposing compound questions, resolving pronouns against conversation history. Retrieval becomes hybrid (more on which below) and is followed by a reranker — typically a cross-encoder model — that re-orders the top-*k* passages by query-passage relevance with substantially better fidelity than the embedding similarity score alone. Context building selects which passages to include in the final prompt based on relevance, diversity, and token budget. Metadata filtering scopes retrieval by attributes (date, source, classification) before similarity search even begins.

The benefits are characteristic: higher relevance, lower hallucination, better retrieval-stage quality. The cost is more moving parts: each stage is its own service, each needs its own monitoring, each adds latency, and the cost of running the reranker on every query is typically several times the cost of the embedding lookup alone. Advanced RAG is the pattern for workloads where retrieval quality is the bottleneck and the additional infrastructure cost is justifiable.

22.2.3 Hybrid RAG: Multiple Retrieval Modalities in Combination

Hybrid RAG combines retrieval modalities. Semantic search via dense embeddings recovers passages that mean what the query means, even when the wording differs; keyword search recovers passages that contain the exact terms the query uses, including proper nouns, identifiers, and rare tokens that embedding models often miss; structured search via SQL or graph queries recovers facts from tabular and relational data; and metadata filtering scopes any of these to the relevant subset of the corpus. The retrieval layer becomes a composite: each modality runs in parallel, the results are merged through a fusion algorithm (reciprocal rank fusion is the common choice), and the combined top-*k* is what proceeds to the next stage.

The pattern is the right choice when the corpus has heterogeneous structure — a mixture of unstructured documents, structured records, and metadata-rich content where no single retrieval method

covers the space. Most enterprise deployments fall into this category, which is why hybrid retrieval has become the de facto standard for serious production RAG. The cost is the engineering complexity of running multiple retrieval services in coordination, and the operational discipline of measuring each modality’s contribution to the final retrieval quality so that the modalities can be tuned individually.

22.3 Knowledge-Shape Variations

22.3.1 Graph RAG: Retrieval Over Relationships

Graph RAG combines vector retrieval with knowledge graph traversal. The corpus is preprocessed not just into embedded chunks but into a graph of entities and relationships — people, products, accounts, events, transactions, with the relationships between them explicit. Retrieval becomes two-phase: vector search locates the relevant entities or chunks, and graph traversal follows the relationships from those starting points to gather connected context that pure semantic search would miss.

The pattern excels at workloads where understanding requires relationships rather than raw similarity. Fraud detection, where a transaction’s significance depends on the entities involved and their prior relationships. Cybersecurity, where an alert’s interpretation depends on the assets, accounts, and prior incidents connected to it. Enterprise knowledge systems where the question “what does this customer’s case relate to?” is more useful than “what passages are similar to this case?” Supply-chain intelligence, where the dependencies among parties, products, and routes are the substance of the analysis.

The architectural cost is the maintenance of the graph. Entity extraction, relationship extraction, and graph maintenance are non-trivial operations, and the graph’s quality is the upper bound on the pattern’s quality. Organisations with mature data-engineering practices — master-data management, entity resolution, structured ontologies — can build Graph RAG cleanly; organisations without those practices typically find that the graph is the project, and the retrieval is the easy part.

22.3.2 Multimodal RAG: Beyond Text

Multimodal RAG retrieves and reasons over content that is not, or not only, text. Image embeddings allow retrieval of visual content by query (“find diagrams that show this configuration”, “find product images that match this description”). Audio and video embeddings allow retrieval of recorded material by transcript or by acoustic features. Structured data embeddings allow retrieval over tabular records by row similarity or column-pattern similarity. The retrieved content is then passed to a multimodal foundation model that can reason across modalities in a single context.

The pattern is the right choice for workloads where the corpus is genuinely multimodal: healthcare records with imaging, surveillance and security operations with video feeds, media archives with mixed text-and-audio content, training operations where the educational material spans documents, recordings, and demonstrations. The architectural challenge is the embedding-model heterogeneity: text, image, audio, and structured-data embeddings each have their own model, their own indexing infrastructure, and their own freshness characteristics. The retrieval layer becomes correspondingly

more complex, and the operational discipline of keeping the modalities aligned is substantial.

22.4 Behavioural Variations

22.4.1 Agentic RAG: Retrieval as Tool Use

Agentic RAG turns retrieval from a single fixed step into an action an autonomous agent can take iteratively. The model receives a query, decides which retrieval action to take (which index to search, which filter to apply, what to search for), executes it, evaluates the returned results, and decides whether to retrieve again with a different query or proceed to answering. The pattern is the subject of Chapter 50 in its own right; this chapter treats it as one member of the RAG family.

The pattern is the right choice for workloads where the right retrieval cannot be determined from the original query alone — where the first retrieval reveals what should have been the second query, or where the answer requires composing information from multiple distinct retrieval rounds. Research assistants, multi-step investigations, workflow-automation agents that must gather context as they proceed. The cost is the additional reasoning, latency, and inference budget per query, all of which are markedly higher than for a fixed-pipeline RAG.

22.4.2 Corrective RAG: Adding Self-Correction

Corrective RAG inserts a validation step between retrieval and generation. The retrieved passages are scored for their relevance and sufficiency to the query, and depending on the score the system either proceeds with the original retrieval, attempts a different retrieval (broader search, different index, query reformulation), or signals low confidence to the consumer of the answer. The pattern’s distinctive contribution is that retrieval failures are detected before they become generation failures.

The pattern is the right choice for workloads where the cost of a confident wrong answer is high — compliance-sensitive responses, medical or legal advisory contexts, financial-decision support, anywhere a hallucinated answer is materially worse than a “I’m not sure” response. The validator stage adds latency and inference cost, but it also turns the system’s quality envelope from “best case is right, worst case is plausibly wrong” into “best case is right, worst case is explicit uncertainty.” For consequential workloads, that change in the failure mode is worth the cost.

22.4.3 Adaptive RAG: Routing by Query Complexity

Adaptive RAG routes each query to the right retrieval strategy based on the query’s apparent complexity. A simple factual query (“what is our return policy”) gets routed to direct retrieval and a small fast model. A medium-complexity query (“compare our return policy with our competitors”) gets routed to multi-step retrieval and a mid-tier model. A high-complexity query (“draft a strategy memo on how our return policy should evolve given the competitive landscape”) gets routed to agentic retrieval and a frontier model. The classifier that performs the routing is itself a small model trained on production query patterns.

The pattern is the right choice when query complexity is highly variable and the same retrieval strategy would either over-serve simple queries (expensive, slow) or under-serve complex queries (cheap,

wrong). The cost optimisation is significant: most production query distributions are heavily skewed toward simple queries, and an adaptive router that handles 80% of traffic with the lightweight path can reduce inference spend by an order of magnitude without quality loss on the harder queries.

22.5 Operational Variations

22.5.1 Streaming RAG: Real-Time Event Streams as Corpus

Streaming RAG inverts the assumption that the corpus is static at indexing time. Events flow continuously from applications, devices, sensors, or telemetry streams; they are ingested through a streaming infrastructure (Kafka, Pulsar, or similar); they are embedded and indexed as they arrive; queries can retrieve content from windows of arbitrary recency, including the last few seconds. The pattern serves workloads where the substance of the analysis is current state — what is happening right now — rather than accumulated history.

Use cases include operational intelligence dashboards where queries against just-arrived telemetry are part of the workflow; financial systems where real-time market and transaction data feeds the agent’s reasoning; IoT operations where device events arrive continuously and the agent’s awareness needs to keep up; monitoring and incident response systems where the last few minutes of activity are the most relevant context. The architectural challenge is the consistency of the index in the face of continuous writes, and the operational discipline of bounding the freshness lag end-to-end through the stream-to-query pipeline.

22.5.2 Enterprise RAG Architecture: The Complete Production Stack

Enterprise RAG architecture is not a single variation but the structural arrangement of the variations into a production-ready system. The standard layered shape comprises an API layer that exposes the system to applications; authentication and RBAC enforced at that layer; a data ingestion layer that pulls from documents, databases, web sources, logs, and forms; a processing layer that performs chunking, cleaning, and enrichment; an embedding layer that produces the vector representations; a storage layer combining vector databases, relational databases, and object storage; a retrieval layer that performs hybrid search and reranking; an orchestration layer that coordinates tools, agents, and the LLM planner; observability and monitoring across all layers; guardrails for safety, PII handling, and policy filtering; and human-in-the-loop checkpoints for the consequential decisions the architecture has chosen not to automate.

The architecture’s distinctive feature is that each layer is owned, observable, and replaceable. The vector database can change without breaking the embedding layer; the embedding model can change without breaking storage; the orchestrator can change without breaking the data ingestion. This is the standard architectural payoff of clean layering, applied to RAG. The pattern is the right choice — effectively, the only choice — for production deployments at scale, where the alternatives of monolithic implementations or vendor-specific stacks do not survive the operational realities of running an enterprise system over multiple years.

22.6 Composing the Variations

Production RAG deployments combine multiple variations at different points in the system. A representative architecture for a mature enterprise deployment might compose:

- **Hybrid retrieval** as the default retrieval modality, combining dense embeddings, keyword search, and structured queries against the appropriate stores.
- **Advanced RAG** patterns layered on top: query rewriting, reranking, context construction with explicit token-budget management.
- **Adaptive routing** at the front, classifying queries by complexity and routing simple queries to a lightweight path and complex queries to a richer path.
- **Graph RAG** as a specialised mode invoked when the query intent involves relationships, fraud detection, or enterprise knowledge with explicit ontologies.
- **Multimodal RAG** where the corpus includes images, audio, or video, with the multimodal layer activated by the content type rather than by every query.
- **Agentic RAG** for the high-complexity tail where multi-round retrieval is genuinely required.
- **Corrective validation** on the consequential queries where confidence matters more than throughput.
- **Streaming RAG** as a separate index for the workloads that genuinely require it, with the orchestrator routing real-time queries to the streaming index and historical queries to the static one.
- All of the above sitting inside the **Enterprise RAG architecture** layering, with observability, governance, and the human-in-the-loop checkpoints as cross-cutting concerns.

The architectural decision is which variations to deploy for which parts of the workload, and at what boundaries the routing happens. The wrong choice in either direction is recognisable. Deploying every variation everywhere is engineering theatre: the system becomes more complex than the workload justifies, the operational burden compounds, and the marginal quality improvement does not pay for itself. Deploying only the simplest variation everywhere is its own failure: the system is cheap to operate but cannot handle the queries that matter most, and the workloads that genuinely needed Graph RAG or Corrective RAG or Adaptive routing are silently producing worse answers than they should.

22.7 The End-to-End Reference Architecture

The end-to-end reference architecture for enterprise RAG, in the form most production deployments converge on, comprises ten functional layers wired through observability, guardrails, and evaluation infrastructure.

Table 22.1: End-to-end enterprise RAG layered reference architecture

Layer	Function
Data Sources	Documents, databases, APIs, web feeds, logs, forms
Ingestion	Batch and streaming pipelines
Processing	Chunking, cleaning, enrichment, metadata extraction
Embedding	Text, image, audio, structured-data embeddings
Storage	Vector DBs, relational DBs, object storage
Retrieval	Hybrid search and reranking
Orchestration	Tools, APIs, agents, planner
Applications	Chatbots, copilots, dashboards
Observability	Metrics, logs, distributed tracing
Guardrails & Evaluation	Safety, PII, policy filtering; quality and feedback loops; human-in-the-loop

The two cross-cutting layers — observability and guardrails with evaluation — are the ones most often under-invested in by first-time deployments and most reliably identified as critical by mature ones. Without observability, retrieval-quality regressions are detected by user complaints rather than by metrics. Without guardrails and evaluation, the system’s behaviour drifts as the corpus changes and the models update, and no one notices until something visible breaks.

22.8 The African Perspective

RAG architectures in African contexts inherit the same patterns and the same trade-offs, with several constraints sharper than they tend to be in better-resourced environments.

Embedding-model selection is a more constrained problem when the corpus contains substantial content in African languages. The frontier embedding models in 2026 are markedly better at English, French, Arabic, and Mandarin than at Yoruba, Amharic, Hausa, Swahili, Zulu, or any of the several hundred other African languages with substantial speaker populations. A RAG deployment over a corpus in one of these languages will, on a frontier embedding model, recover materially less relevant content than the same architecture over English. The architectural responses worth considering: hybrid retrieval that combines weaker embeddings with stronger keyword search where the keyword search at least handles the language characters correctly; fine-tuning embeddings on language-specific data where it is available; multilingual embedding models that have invested specifically in African language coverage even at some cost in English quality.

Streaming RAG raises a connectivity question. Real-time RAG against a continuously updating index requires reliable low-latency connectivity between the ingestion layer, the index, and the query layer. For deployments where the connectivity to the foundation-model provider is itself sometimes unreliable, the streaming variant becomes the most fragile of the variations. The architectural response is to evaluate carefully whether the workload genuinely requires streaming, or whether a near-real-time batch index with five-minute freshness would suffice; the latter is operationally much more

forgiving.

Graph RAG depends on data-engineering maturity that is unevenly distributed across the continent. Where the relevant entity and relationship data already exists in a maintained form — in mature banks, telecommunications operators, government data systems — Graph RAG is approachable. Where the underlying entity resolution is itself the project, Graph RAG is years rather than months of work. The honest architectural posture is to recognise where the data-engineering investment has already happened and where it has not, and to scope the use of Graph RAG accordingly.

22.9 Where RAG Is Heading

The architectural direction of RAG, observable in 2026 production deployments, is toward composition rather than replacement. RAG, agents, streaming, persistent memory, and workflow engines are converging into integrated systems where the boundaries between retrieval and action become less distinct: a system reasons, retrieves, acts, learns, and remembers as a single architectural surface, with the retrieval pattern as one tool the agent loop uses rather than as the centre of the system. The autonomous-intelligent-systems framing some practitioners use for this composition is a marketing label, but the architectural shift it points at is real.

The implication for the architect is that the RAG patterns described in this chapter, and the agent patterns described in Chapter 29, and the memory architecture in Chapter 13, and the streaming patterns in Part IV, are no longer cleanly separable categories. The composition is where the architectural work actually happens. The patterns of this chapter are the working vocabulary by which a competent architect describes one corner of that composition. The next several years will see the corner shift, the patterns merge, and the vocabulary settle further. The disposition to carry forward is the one this book has argued for throughout: hold the patterns lightly, evaluate them against the workload, and expect the field to revise itself — usefully — before the decade is out.

Chapter 23

Pattern: LLM Router

“The best model is the cheapest model that gets the job done.”

—unattributed but ubiquitous

Chapter Overview

The LLM Router pattern (Martian, RouteLLM, and similar systems) controls cost and latency across a model portfolio by dynamically choosing which model handles each request. It is the structural answer to a recurring observation: most enterprise workloads do not need a frontier model for every request, and routing the easy 80% to smaller models produces dramatic cost savings without quality loss. This chapter develops the routing decision, the policy mechanisms, the fallback and degradation paths, and the evaluation methods that let an organisation trust its router.

Learning Objectives

- Recognise when an LLM Router is the right pattern (heterogeneous workload, cost pressure, mixed latency requirements).
- Design a model portfolio across small/local, mid-tier, and frontier tiers, with explicit fallback paths.
- Implement routing decisions: rule-based, classifier-based, embedding-based, and cascade.
- Use semantic caching and prompt compression as routing accelerators.
- Evaluate routing policies on the quality-cost-latency frontier.

23.1 When to Use This Pattern

Before reaching for a router, the question worth pausing on is whether the workload actually has the heterogeneity the pattern is designed to exploit. A router that sits in front of a single dominant model is overhead; a router that sits in front of a portfolio is leverage.

The LLM Router is best for controlling cost and latency across a model portfolio. Apply it when:

- Workloads span a wide difficulty range, from simple classification to complex reasoning.
- Cost per call varies 10× or more across candidate models.

- Latency budgets vary by use case (real-time chat versus batch enrichment).
- Multiple vendors are in use, with different strengths.
- Self-hosted and vendor-hosted models coexist.

If the workload is uniform—all requests are similar complexity and similar latency budget—a router is overkill. A single model selection is simpler and cheaper to operate.

23.2 The Model Portfolio

The router routes between models. Before the routing decision can be made well, the set of models the router routes *to* has to be designed deliberately. A portfolio assembled by accident, by procurement decisions made independently by different teams, will not route well no matter how clever the routing logic; a portfolio designed for routing leaves the routing decision much easier.

A typical portfolio has four tiers, each with a defined role.

Table 23.1: Tiers of a model portfolio.

Tier	Typical models	Cost	Best for
Small / local	1–8B open weights, self-hosted	\$	Classification, format conversion, extraction
Mid-tier API	Mid-cost vendor models	\$\$	Summaries, drafts, standard reasoning
Frontier	Top vendor models, large context	\$\$\$\$	Complex reasoning, code, planning
Fallback	Smaller copy of any tier	varies	Recovery when primary fails

23.2.1 Capability Mapping

The tiers in the previous table describe roles. A working portfolio also needs a profile of each specific model in each tier, so the router has the information it needs to choose between them.

Every model in the portfolio is documented with: cost per million input/output tokens, p50 and p99 latency, context window, supported modalities, and benchmark scores on tasks representative of enterprise use.

23.3 Routing Mechanisms

With the portfolio in place, the question becomes how the router actually decides. Four routing mechanisms recur in production deployments, and they sit on a spectrum from cheap-and-rigid to expensive-and-adaptive. Most production routers combine several — rules for the obvious cases, a classifier for the ambiguous middle, and a cascade for the rest.

23.3.1 Rule-Based Routing

The simplest router is a table of rules that maps request features to models. It has no learning component and no probabilistic reasoning; it does what it is told.

```

1 def route(request):
2     if request.task_type == "classification":
3         return "small_local_v3"
4     if request.token_count > 80_000:
5         return "frontier_long_context"
6     if request.requires_tools:
7         return "frontier_tool_use"
8     if request.tenant_tier == "premium":
9         return "mid_tier_premium"
10    return "mid_tier_default"

```

Listing 23.1: A rule-based router.

Rules are transparent, deterministic, and auditable. They cannot adapt to new request types without code changes.

23.3.2 Classifier-Based Routing

When the rules run out — when the routing decision depends on properties of the request that no regex can capture — a small classifier earns its place. A classifier predicts the cheapest model that will produce an acceptable answer. It is trained on (request, model, quality) tuples from production logs.

- Features: query length, presence of certain entities, embedding of the query, user tier.
- Label: smallest model whose response was accepted (or rated above threshold).
- Output: predicted minimum-sufficient model per request.

23.3.3 Embedding-Based Routing

Embedding-based routing replaces explicit features with the geometry of the request itself. The query is embedded, and routing matches it against centroids of historical queries by difficulty. Queries near “easy” centroids route to small models; queries near “hard” centroids route to frontier. Simple to implement; benefits from a large historical sample.

23.3.4 Cascade Routing

The mechanisms above choose a model up front. A cascade defers the choice: it tries the cheapest model first, evaluates the output (with a verifier model or a confidence signal), and escalates to a more expensive model only if needed. The bet is that the cheapest model will be sufficient most of the time, and the rare escalations will not undo the savings.

$$\text{COST}_{\text{cascade}} = c_{\text{small}} + p_{\text{escalate}} \cdot c_{\text{frontier}}$$

For workloads where small models suffice 80% of the time, cascade beats single-model frontier by a factor of ~ 4 on cost.

23.4 Semantic Caching

Some routing decisions are improved not by choosing a cheaper model but by avoiding the model call entirely. If the same question, or a sufficiently similar one, has been asked before, the previous answer may be reusable. A semantic cache is the mechanism that exploits that reusability without requiring the question to be repeated word-for-word.

A semantic cache stores previous (query, response) pairs and returns the cached response when a semantically equivalent query arrives. Implementation:

1. Embed the incoming query.
2. Search the cache for nearest neighbours above a similarity threshold (typical: 0.92 cosine).
3. If a match is found and is fresh (within TTL), return the cached response.
4. Otherwise, forward to the router and store the new response.

Semantic caching reduces both cost and latency dramatically for workloads with repeated questions (FAQ-style chatbots, help-desk applications). The architectural caution is staleness: a cached answer drawn from documents that have since changed must be invalidated.

23.5 Prompt Compression

Caching avoids the call; compression reduces what the call has to carry. Long prompts cost more per request and constrain which models can serve them at all — a 100,000-token prompt rules out most small models on context-window grounds alone.

When the prompt itself is long (large retrieved context, long chat history), *prompt compression* reduces token count while preserving meaning. Approaches:

- Summarisation of older chat turns into shorter form.
- Selective context inclusion (drop low-relevance chunks).
- Specialised compression models (LLMLingua and similar).

Compression interacts with routing: a compressed prompt may fit in a smaller, cheaper model's context window.

23.6 Fallback and Degradation

A router that depends on a model fleet inherits the fragility of that fleet. The routing question is no longer just *which model should serve this request* but *which model should serve this request given that the preferred one may be unavailable*. The architectural response is a graceful-degradation hierarchy:

each routing decision has named alternatives that get progressively less satisfactory but never produce a hard failure.

Models fail: an API returns 500, a self-hosted endpoint runs out of GPU memory, a vendor has an outage. The router must have explicit fallback paths:

- **Same-tier fallback.** Another vendor of similar capability and cost.
- **Down-tier fallback.** A smaller model that can do most of the job.
- **Degraded mode.** Return a templated response rather than fail entirely.
- **Queue and retry.** For non-interactive workloads, accept latency for the next-best path.

23.7 Evaluating the Router

A router that is not evaluated is a router that is not trusted. Routing decisions are made automatically, in production, hundreds of thousands of times a day; the only way to know they are being made well is to measure them. The evaluation question is multi-dimensional, because a router is optimising three things at once: quality (does it choose models that meet the quality bar?), cost (does it minimise spend?), and latency (does it stay within SLO?). The standard report is a Pareto frontier of cost versus quality, with the router's chosen operating point highlighted.

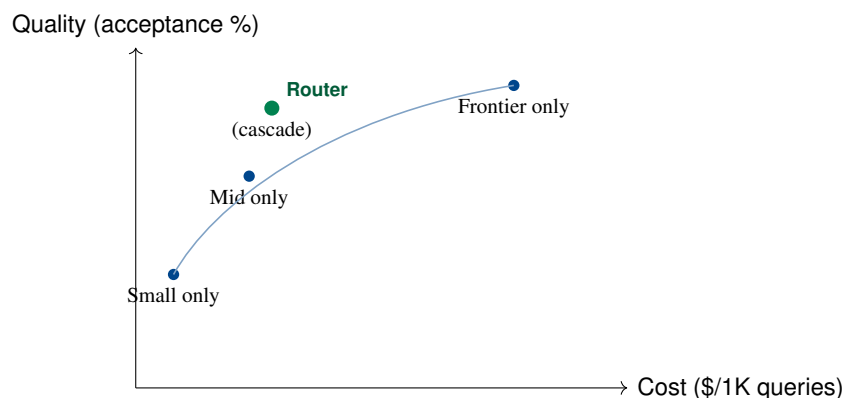


Figure 23.1: Quality-cost Pareto frontier. A good router operates above the single-model baseline curve, achieving quality close to frontier-only at cost close to mid-tier-only.

23.8 Failure Modes

A router fails differently from a single-model deployment. The single model fails openly: the request errors, the user notices, the alert fires. A router fails quietly: it chooses the wrong model, the answer is plausible but worse, the user sees a degradation they cannot diagnose. The failure modes below are the recurring quiet failures of production routers and the mitigations that catch them.

Table 23.2: Failure modes in LLM routers.

Failure mode	Mitigation
Routing drift (a model the router prefers degrades)	Continuous evaluation; routing decisions logged with outcomes.
Cache poisoning (stale cached answers)	TTL on cache entries; invalidation on source updates.
Cascade thrashing (always escalating)	Threshold tuning; capture and label cascade outcomes.
Vendor outage cascades	Pre-tested same-tier fallbacks; circuit breakers per vendor.
Latency tail expansion (slow fallback)	Hedge requests to multiple models when latency is critical.

23.9 Routing in Practice

The previous sections have described the moving parts; this one shows how they assemble into a working router and what the engineering choices look like in production. The router itself is a small architecture — not a single function but a pipeline of feature extraction, classification, model invocation, and optional verification.

Figure 23.2 shows the decision flow.

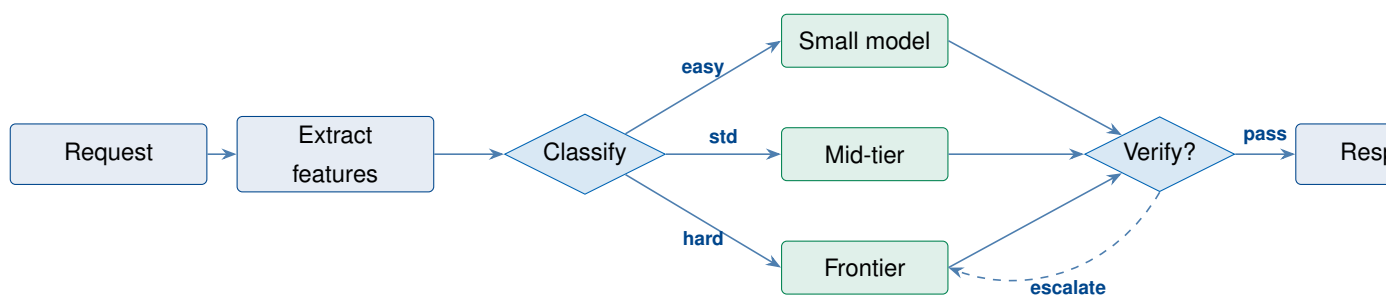


Figure 23.2: The router as a small system: feature extraction, classification, model selection, optional verification with escalation on failure. The dashed escalation edge converts the router into a cascade.

23.9.1 Classifier Design

The classifier inside the router does the work the rest of the architecture is built around. Its job is to predict the cheapest model that will produce an acceptable answer for the request in front of it, and the design choice between rules, a small learned model, and an LLM-as-router has consequences for latency, maintainability, and the kinds of requests the router can adapt to.

Three viable approaches:

- **Rule-based.** Regex, length thresholds, source-tier mapping. Cheap, explainable, brittle.
- **Small-model classifier.** A fine-tuned encoder (BERT-class) that predicts the right route. Few-ms latency, good quality, requires training data and retraining as the model fleet evolves.

- **LLM-as-router.** A small LLM that reads the request and decides. Most flexible, highest router latency, can itself be wrong.

In production, hybrids dominate: rules for unambiguous cases (length, source, customer tier), classifier for the rest, LLM-as-router only for new request classes during exploration.

23.9.2 Feature Inputs

A classifier is only as good as what it sees. A naive router that classifies only on the user's text will route well for simple cases and badly for cases where the context matters more than the words. Production routers feed the classifier four kinds of signal, and the most important of them is the one that has nothing to do with the request itself.

The classifier sees more than the user text. Production routers use:

- **Request features.** Length, language, presence of code blocks, attached files.
- **Customer features.** Tier, region, contract SLA, historical complexity.
- **Conversation features.** Turn number, prior model used, prior tool calls.
- **System features.** Current latency to each model, current cost budget remaining, current outage signals.

System features matter most: a router that does not know a downstream model is degraded will keep routing to it.

23.9.3 Cost-and-Quality Pareto

The final question is not which routing mechanism is technically best but which one moves the workload onto the cost-quality Pareto frontier. The construction of that frontier is a piece of measurement work that pays for itself many times over once a routing policy is deployed against it.

The right router is the one on the Pareto frontier of cost and quality for your workload. The construction:

1. Run a representative request sample through every candidate model.
2. Score quality per request, per model.
3. Tabulate cost per request, per model.
4. Plot the per-request-best as a function of allowed cost.
5. Choose a routing policy that approximates the frontier.

The exercise frequently reveals that the most expensive model is worth it for only 5–15% of requests; the rest can be served cheaper without quality loss.

23.10 Choosing Between the Stacks

The LLM Router is best for controlling cost and latency across a model portfolio. Its impact is largest when the workload is heterogeneous and the model market is varied—both of which are true for the foreseeable future. A well-instrumented router becomes the central control plane for cost, quality, and resilience; without one, organisations either overspend on frontier models for trivial requests or under-serve complex requests by routing them to underpowered models.

Chapter 24

Pattern: Hub-and-Spoke

“Centralise what should be the same; decentralise what should be different.”

—an enterprise architecture maxim

Chapter Overview

The hub-and-spoke pattern is the organisational architecture that most mature enterprise AI programmes converge on. A central platform (the hub) operates shared services: model gateway, vector store, observability, IAM, FinOps, governance tooling. Business-unit teams (the spokes) build use cases on top of the hub’s paved path. A Centre of Excellence (CoE) supports both, transferring capability into the spokes and feeding lessons back into the hub. This chapter develops the responsibilities of each role, the governance model that makes the federation tractable, the patterns by which spokes consume the hub, the failure modes the structure must avoid, and the evolution from a single-team origin to a fully-federated programme.

Learning Objectives

- Identify the responsibilities of the hub, the spokes, and the Centre of Excellence.
- Apply the federated governance model that lets the hub set standards without becoming a bottleneck.
- Recognise the dominant interaction modes and the situations in which each applies.
- Diagnose hub-and-spoke anti-patterns: bottleneck hub, shadow spokes, theatre CoE.
- Evolve the structure as the programme matures from one team to many.

24.1 When the Pattern Applies

Hub-and-spoke is an answer to a problem that does not exist on day one of an AI programme. The first use case in a small organisation does not need a platform team, a Centre of Excellence, or a federated decision-rights matrix; it needs people who will ship. The pattern earns its place when the second use case appears, and the third, and the question of how they share without colliding becomes a real organisational question rather than a hypothetical one.

The hub-and-spoke pattern fits when:

- Multiple business units want to build AI-powered capabilities.
- Each business unit has its own context, users, and priorities.
- Shared concerns (model access, governance, observability, cost control) are too expensive to rebuild per team.
- The organisation has the patience to operate a platform rather than chase tactical wins.

When these conditions hold, the pattern produces compound returns: each new spoke gets shipped faster because the hub has already solved the platform problems; each platform improvement benefits every spoke. When the conditions do not hold (a single use case, a single team, no platform appetite), adopting the pattern prematurely produces overhead without benefit.

24.2 Hub Responsibilities

If the pattern is a federation, the hub is its constitution. What it does, what it does not do, and how its scope is decided shape every other relationship in the structure. A hub that takes on too little leaves the spokes reinventing essentials; a hub that takes on too much becomes the bottleneck the pattern exists to prevent.

The hub operates as a platform team (see Chapter 98). Its responsibilities:

Table 24.1: Hub responsibilities by reference-architecture layer.

Layer	Hub responsibility
Layer 1–2	Data platform: lakehouse, feature store, connector library, data contracts
Layer 3	Model gateway: vendor abstractions, model registry, prompt registry, vector store, embedding service, orchestration framework
Layer 4	Shared serving infrastructure; MLOps tooling; evaluation harness; cost-attribution pipeline
Layer 5	IAM integration; secrets management; observability backend; CI/CD templates; policy enforcement
Cross-cutting	Privacy controls library; reliability patterns library; governance evidence pipeline

The hub's metric is not its own activity but the spokes' velocity: how quickly can a new use case ship on the paved path? A hub measuring itself on services-deployed risks building infrastructure that does not get adopted; a hub measuring spoke time-to-production keeps the pressure to make adoption easy.

24.2.1 Paved Path vs Walled Garden

The previous table says *what* the hub operates. The harder question is *how* the hub offers those services to the spokes. The distinction is a style of governance, not a technical one, and it shapes every other choice in the structure.

A useful distinction: the hub offers a *paved path*, not a walled garden. A paved path is the easiest route; deviating from it is allowed but more work. A walled garden is the only route; deviating is

forbidden. The paved-path posture preserves the spokes' ability to innovate at the edges while the hub captures the common case. The walled-garden posture maximises uniformity but turns the hub into a bottleneck and breeds shadow architecture.

The rule of thumb: 80–90% of spoke work should use the paved path. If more, the hub may be too prescriptive; if less, the paved path is not paved enough.

24.3 Spoke Responsibilities

The hub sets the platform; the spokes deliver the value the platform exists to support. A spoke is not a customer of the hub; it is an owner of the use cases that justify the hub's existence. The distinction shows up in how spokes are funded, staffed, and held accountable.

Each spoke is a stream-aligned team (Chapter 98) building one or more use cases:

- Use-case discovery, design, and prioritisation within the business unit.
- Domain-specific data preparation: classification, contracting, quality, where the platform-provided defaults are insufficient.
- Application logic: prompts, orchestration, business rules.
- User-facing experience: UI, integration with business-unit applications.
- Operational ownership of the use case end-to-end (with platform components operated by the hub).
- Outcome measurement: did this use case produce the value it was funded for?

A spoke that hands its production system to a separate operations team has broken the feedback loop; the team that ships should be the team that runs and improves. The hub provides the operational infrastructure; the spoke provides the operational ownership.

24.4 The Centre of Excellence

The hub builds the road; the spokes drive on it; the Centre of Excellence is the function that helps drivers learn to drive. CoEs are widely created and often poorly defined; the pattern works only when the CoE has responsibilities that are genuinely different from both the hub's and the spokes', and when those responsibilities are scoped to leave the spokes more capable than the CoE found them rather than dependent on it.

The CoE is an enabling team (Chapter 98) operating between the hub and the spokes. Its responsibilities, distinct from the hub's:

- Pattern library: the canonical architectures for common problems; the spokes adapt rather than invent.
- Reference implementations: starter code, templates, example deployments.
- Capability uplift: workshops, certifications, embedded engagements with spokes.

- Architecture reviews: a paved path means standardised reviews, not unique scrutiny for each use case.
- Cross-spoke learning: capturing what one spoke learned and feeding it to others.
- Risk and compliance liaison: bridging the technical and the governance vocabularies.

The CoE does not operate platform services. The CoE does not build spoke use cases. The CoE makes spokes more capable; the platform team makes spokes more efficient. Conflating the two produces a function unable to do either well.

24.5 Federated Governance

A federation only works if the units composing it agree, in advance, on who decides what. Without that agreement, the federation degrades into perpetual negotiation: every decision is contested, every committee meets too often, and the speed advantages the structure was meant to deliver disappear into the friction. The table below sets out a working division of decision rights and what each side brings to the decisions it does not own.

The hub sets standards; the spokes operate within them; the CoE facilitates compliance. The federation has explicit decision rights:

Table 24.2: Decision rights across hub, spoke, and CoE.

Decision	Owned by	Inputs from
Platform technology choices	Hub	CoE, spoke representatives
Use-case selection within a business unit	Spoke	Business-unit leadership, CoE for technical viability
Approval to deploy a new model version	Hub (paved path) or specialised committee (high-risk)	Spoke for use-case context, CoE for risk classification
Choice of prompt strategy, retrieval design, evaluation suite within a spoke	Spoke	CoE for advice, hub for platform-level constraints
Vendor selection at platform level	Hub	CoE, finance, legal, spoke representatives
Vendor selection at spoke level (if any)	Spoke (within platform constraints)	Hub, CoE
Risk classification of a use case	CoE (with risk/compliance)	Spoke for context

Decisions made above their natural level (the hub picking spokes' use cases) produce friction and reduce spoke autonomy. Decisions made below their natural level (a spoke picking its own vendor for the platform's primary model) produce sprawl and undo the value of the federation.

24.5.1 Time-Bound Decision Rights

A subtler failure of governance is not who decides but how long they take to decide. Decision rights assigned without time limits become decisions never made; the table above describes ownership, but the discipline that keeps it from stalling is a clock.

For routine decisions, decision rights have time-outs: if the responsible party does not decide within *N* days, the decision escalates or proceeds as a default. This prevents governance committees from becoming bottlenecks; clear time-outs are more disciplined than indefinite consideration.

24.6 Interaction Modes

The hub and spokes do not interact in one undifferentiated way. The relationships that work in production fall into a small set of recognisable modes, each appropriate to a specific kind of exchange. Naming the modes makes it easier to see when the wrong one is being used — a hub trying to facilitate when it should be providing service, a CoE collaborating with a spoke long after the engagement should have ended.

The three interaction modes from Chapter 98 appear in specific configurations:

- **Hub → spoke: X-as-a-Service.** The dominant steady-state mode. The hub provides platform services through documented interfaces; the spoke consumes them.
- **CoE → spoke: facilitating.** Time-bounded engagements where the CoE helps a spoke build a new capability; the spoke owns it afterwards.
- **Spoke ↔ spoke: usually XaaS through the hub.** Spokes rarely integrate directly; their interactions are mediated by the hub's data and event infrastructure.
- **Hub ↔ CoE: collaboration.** The two functions are tightly coupled: the platform team builds what the CoE has identified as needed; the CoE helps spokes adopt what the platform team built.
- **Spoke → hub: feedback and requests.** A defined product-management process inside the hub triages incoming requests; not every spoke request becomes a platform commitment.

24.7 Onboarding a New Spoke

The first spoke is built by negotiation. The tenth is built by repetition. Between them is a documented onboarding process that turns the work of starting a new spoke from a one-off project into a routine activity. Done well, this is one of the most leveraged investments a hub can make: every hour spent on the onboarding template saves dozens of hours per spoke for as long as the programme exists.

A maturing programme has a documented spoke-onboarding process. A typical sequence:

- Initial conversation between CoE and the prospective spoke leadership; expectations set.
- Use-case discovery workshop; the spoke surfaces the candidate use cases.
- Architecture review against the pattern library; select the closest pattern and any deviations.

- Risk classification by the CoE; identifies governance obligations.
- Capacity sizing; the platform team plans for the additional load.
- Embedded CoE engagement for first one to two use cases; the spoke builds with support.
- Self-sufficient operation; the CoE engagement ends; the spoke ships independently.
- Periodic touchpoints: quarterly reviews, annual architecture reviews, ad-hoc on critical changes.

The onboarding process is itself a hub product: it should be smoother for the tenth spoke than for the first, with every onboarding feeding back into improvements.

24.8 Anti-Patterns

The hub-and-spoke pattern fails in characteristic ways, and the failures repeat across organisations regardless of industry or scale. The table below collects the failure modes that recur most often and the moves that pull a programme back from each of them. The list is not exhaustive; it is the set worth recognising early, because each of them is easier to prevent than to reverse.

Table 24.3: Hub-and-spoke anti-patterns and their mitigations.

Anti-pattern	Mitigation
Bottleneck hub	Self-service tooling; decision time-outs; platform that paves the path rather than gates it
Shadow spokes (teams routing around the hub)	Investigate why; usually the hub is too prescriptive, too slow, or missing a capability
Ivory-tower CoE	Time-bounded engagements with measurable handover criteria; the CoE's metric is spokes' self-sufficiency
Platform without product mindset	Treat the platform as a product; user research with spokes; developer-experience metrics
Vanity standards	Standards exist where there is a real cost to divergence; vanity standardisation produces compliance without benefit
Spoke without ownership	The spoke that ships must run and improve; no fire-and-forget deployments
"CoE" that is really a platform team	Separate the functions; the CoE transfers capability, the platform operates services
Permanent CoE engagements	Time-box every engagement; declare success or failure and move on

24.9 Evolution Over Time

A hub-and-spoke structure is not a snapshot. It is a sequence of stages that an AI programme passes through as it grows, and each stage has its own shape, its own constraints, and its own failure modes. The mistake the structure tolerates least well is being applied at the wrong stage — a federated maturity blueprint imposed on a single-team programme is overhead with no benefit, and a single-team mindset retained into a multi-spoke programme is sprawl with no governance.

The structure evolves with the programme's maturity:

- **Stage 0: single team.** One team, no platform, no CoE; the team is everything. Lasts until the second use case appears.
- **Stage 1: emerging platform.** A shared library and a couple of services emerge from the first team; the second team adopts them, finds gaps, contributes back.
- **Stage 2: explicit hub.** A dedicated platform team forms; the CoE may exist or may be merged with the platform team initially.
- **Stage 3: separated functions.** Platform team and CoE are distinct; multiple spokes operate on the paved path.
- **Stage 4: federated maturity.** The structure self-sustains; new spokes onboard routinely; the hub is updated on a product cadence.
- **Stage 5: dispersed capability.** AI is everywhere; the platform may consolidate with other internal platforms; the CoE may shrink into a standards body.

The transitions between stages are where most organisational mistakes happen: forming a platform team too late (after the spokes have built their own incompatible solutions), separating the CoE too early (before it has anything to enable adoption of), or skipping a stage (declaring federated maturity before the platform exists).

24.10 Composition with Other Patterns

Hub-and-spoke is an organisational pattern, not a technical one. The technical patterns the rest of Part III describes do not compete with it; they live inside it. The hub is where most of those technical patterns are operationally instantiated, and the spokes are where they are consumed. The relationships below show the most useful pairings.

Hub-and-spoke composes naturally with the other architecture patterns of Part III:

- **Hexagonal** (Chapter 26) at the hub: every vendor dependency is a port; the hub can switch vendors without disrupting spokes.
- **RAG** (Chapter 21) as a hub service: the platform operates the indexing pipeline and retrieval service; spokes plug in their corpora.
- **LLM Router** (Chapter 23) at the hub: spokes' calls pass through the gateway and are routed across the model portfolio.
- **Agentic Mesh** (Chapter 18) as a federation of spoke-owned agents on hub-provided infrastructure.
- **Event-Driven Agents** (Chapter 25) on a hub-operated event bus that all spokes consume.

The hub-and-spoke pattern is, in effect, the organisational frame within which the other architectural patterns operate.

24.11 Hub-and-Spoke Topology in Detail

Everything up to this point has described the pattern in conceptual terms — roles, responsibilities, decision rights. This section makes the structure concrete: what the topology actually looks like as deployed boxes and connections, and how to decide what sits inside the hub’s perimeter and what sits outside it.

The hub is a federated platform: a small, opinionated set of central capabilities, surrounded by autonomous spokes that consume them and contribute back. Figure 24.1 shows a canonical layout.

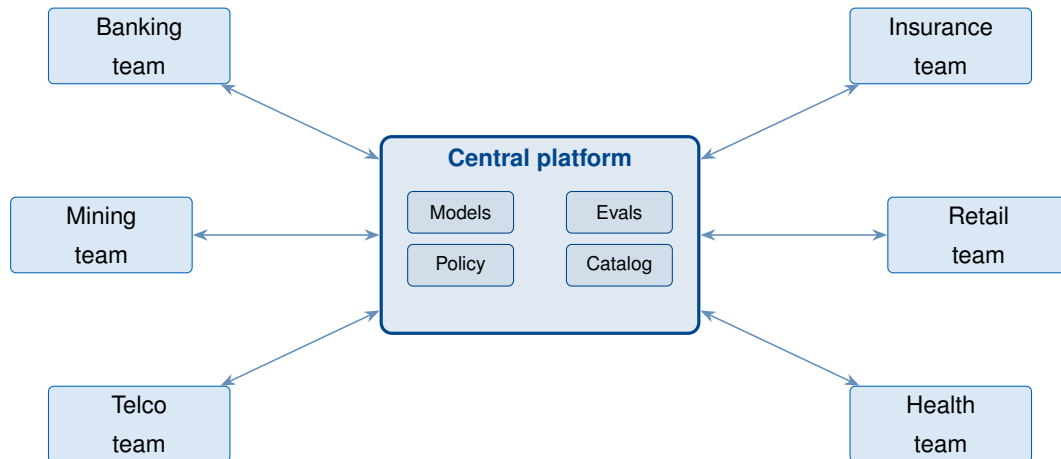


Figure 24.1: Hub-and-spoke layout: a small central platform of paved-path capabilities (models, evals, policy, catalog) serving autonomous business-aligned spokes. Arrows are bidirectional: spokes consume platform services and contribute use cases and reusable assets back.

24.11.1 What Belongs in the Hub

The first question every hub-and-spoke programme has to settle is what the hub does. The wrong answer is “everything that could be shared.” The right answer is narrower: things whose costs to rebuild per spoke would dwarf the cost of operating them centrally, things whose governance value depends on consistency, and things whose scale economies are real rather than theoretical.

The hub holds capabilities that benefit from centralisation: economies of scale, leverage, governance reach.

- Model access (vendor contracts, gateway, quotas).
- Evaluation infrastructure (datasets, harnesses, leaderboards).
- Policy and governance services (data classification, approval workflows, audit).
- Shared asset catalog (reusable prompts, agents, datasets, tools).
- Observability backbone (traces, metrics, costs).
- Foundation-model and embedding pipelines.

24.11.2 What Belongs in the Spokes

Everything that is not a hub responsibility is a spoke responsibility, and the spoke's responsibilities are larger than they first appear. The spoke owns not just the use case but the domain knowledge, the user experience, the on-call rotation, and the relationship with the business unit it serves.

Anything genuinely domain-specific belongs in the spokes. The principle: autonomy with paved paths; spokes own outcomes; the hub provides the easy default.

- Domain models (fine-tuned on local data).
- Domain datasets and embeddings.
- Domain tools and integrations.
- Domain user experience.
- Domain SLAs and on-call.

24.11.3 The Anti-Bottleneck Principle

Of all the failures the pattern admits, the most damaging is the most predictable: the hub becomes a queue. Every spoke waits on the hub for the things only the hub can provide, and the hub cannot scale its attention as fast as the demand grows. The pattern survives only if it has architectural defences against the queue forming in the first place.

The pattern fails when the hub becomes a bottleneck. Two structural defences:

- **Paved paths, not gates.** Hub services are easy to use through self-service, not gatekept through ticket queues.
- **Contribution loop.** Spokes contribute reusable assets back. Without this loop, the hub stagnates while spokes diverge.

24.12 The Shape of the Argument

The hub-and-spoke pattern is the organisational architecture that most mature programmes converge on: a platform team (hub) operating shared services, business-unit teams (spokes) building use cases on the paved path, and a Centre of Excellence enabling adoption. Decision rights are explicit and federated; the platform earns its place by reducing spoke time-to-production; the CoE earns its place by leaving spokes more capable than it found them. Done well, the structure produces compound returns: each new spoke and each platform improvement build on the work already done. Done badly, it produces a bottleneck hub, shadow spokes, and a CoE that talks about excellence rather than enabling it.

Chapter 25

Pattern: Event-Driven Agents

“Decoupled systems fail less than coupled ones, and recover faster when they do.”

—a recurring observation in distributed systems

Chapter Overview

The Event-Driven Agents pattern (popularised by the Confluent/Kafka community) is the answer to real-time, asynchronous agent workflows at scale. Agents publish and subscribe to events on a shared bus rather than calling each other directly. This decouples producers from consumers, supports full replay for debugging, lets agents be added or removed without breaking other agents, and provides natural backpressure. This chapter develops the bus architecture, the event schema discipline, the consumer-group model, the saga pattern for multi-step workflows, and the operational mechanisms that keep an event-driven system observable.

Learning Objectives

- Recognise when event-driven is the appropriate pattern and when synchronous calling is preferable.
- Design the event bus, event schemas, and consumer groups.
- Use the saga pattern for distributed multi-step workflows with compensating actions.
- Apply replay, time-travel debugging, and dead-letter queues to operate event-driven systems reliably.
- Recognise the failure modes (event-schema drift, poison messages, lag accumulation) and the structural mitigations.

25.1 When to Use This Pattern

Event-driven agents are best for real-time, asynchronous agent workflows at scale. Apply this pattern when:

- Many agents react to the same domain events.
- Workflows are long-running and span multiple agents.

- Replay and audit are first-class requirements.
- Synchronous coupling between agents has produced cascading-failure incidents.
- Volume is high enough that point-to-point integration would be brittle.

Event-driven adds operational complexity (bus operation, schema governance, consumer-lag monitoring). For small systems with few agents and synchronous workflows, the simpler approach wins.

25.2 The Event Bus

A single source of truth for what happened in the system. Implementations include Apache Kafka, Confluent Cloud, Apache Pulsar, AWS Kinesis, Google Pub/Sub, and managed Event Hubs.

25.2.1 Topics and Partitions

Events are organised into *topics* (logical streams). Each topic is divided into *partitions* for parallelism. Ordering is guaranteed within a partition; partitioning by a stable key (customer ID, account ID) keeps related events ordered while supporting parallel processing across keys.

25.2.2 Retention

Events are retained for a configured period (hours, days, or forever). Long retention enables replay—a new consumer can rebuild its state from scratch by reading the full history.

25.3 Event Schemas

Events are facts about the domain. Schema discipline is what distinguishes a useful event bus from an unmanageable one.

25.3.1 Schema Registry

A schema registry (Confluent Schema Registry, AWS Glue Schema Registry, Apicurio) stores the schema for each topic and enforces compatibility rules on changes. Producers register schemas; consumers fetch them; the registry rejects breaking changes by default.

25.3.2 Event Structure

A well-designed event has:

```
1 {  
2   "event_id":      "01HXR3...",  
3   "event_type":    "order.placed",  
4   "event_version": "2.1",  
5   "occurred_at":   "2026-05-14T08:01:14.273Z",  
6   "producer":      "order-service",  
7   "trace_id":      "00-abc...",  
8   "tenant_id":     "acme_corp",  
9   "subject":       { "order_id": "ord_12345", "customer_id": "cus_678"  
                      },  
}
```

```
10   "data":      { "items": [...], "total": 129.95, "currency": "USD"
11   }
```

Listing 25.1: A canonical event envelope.

Note: the envelope (id, type, version, time, producer, trace) is stable; the `data` payload evolves with the schema.

25.3.3 Compatibility Rules

Event-driven agents must agree on what an event looks like and how it is versioned, because an agent that cannot parse an event it receives will either fail or silently misinterpret it. The compatibility rules govern how events evolve over time without breaking the agents that consume them.

- **Backward compatible.** New consumers can read old events. Adding optional fields is fine; removing fields is not.
- **Forward compatible.** Old consumers can read new events. Removing optional fields is fine; renaming or retyping is not.
- **Full compatible.** Both. Recommended default for production topics.

25.4 Producers and Consumers

Agents *publish* events when something happens in their domain; agents *subscribe* to events from other domains they care about.

25.4.1 The Subscriber Model

A consumer group is a set of processes that share the work of consuming a topic. Each partition is assigned to exactly one consumer in the group at a time, so parallelism is bounded by partition count.

25.4.2 Decoupling

Producers emit events without knowing who consumes them. Consumers subscribe without coordinating with producers. The bus is the contract. Adding a new agent that needs the same events is a subscription operation, not a code change to existing agents.

25.4.3 Idempotency

Most buses guarantee at-least-once delivery, not exactly-once. Consumers must be idempotent: processing the same event twice produces the same outcome as processing it once. Typical mechanisms: deterministic event IDs and dedupe tables; outbox patterns for transactional consistency.

25.5 The Saga Pattern

Multi-step workflows (place order → charge card → allocate inventory → create shipment) span multiple services in event-driven systems. The *saga* pattern coordinates them without distributed

transactions:

- Each step is a local transaction in its own service.
- Steps emit events on success.
- Each step has a documented *compensating action* that undoes it if a later step fails.
- A saga orchestrator (or a choreographed sequence of subscribers) drives the workflow.

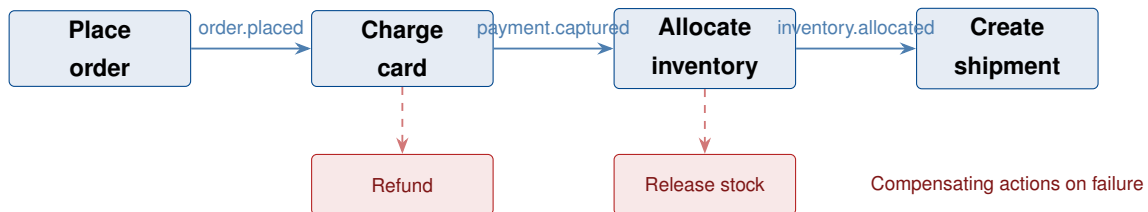


Figure 25.1: A saga across four services. Each step emits an event; each step has a compensating action that undoes it if a later step fails.

25.6 Replay and Time Travel

The single most valuable operational property of event-driven systems is replay: an event log retained long enough lets engineers reconstruct what happened, when, and why. Practical consequences:

- **Debugging.** A bug in a consumer can be fixed and the consumer restarted from a past offset to re-process events with corrected logic.
- **Bootstrapping new consumers.** A new agent reads the full history to compute its initial state.
- **Auditing.** Every business-meaningful change is reconstructible from the log.
- **Disaster recovery.** If a downstream store is corrupted, it can be rebuilt by replaying events.

25.7 Dead-Letter Queues

A poison message—one that no consumer can process—will otherwise block its partition indefinitely. The standard mitigation is a *dead-letter queue*: after N failed processing attempts, the consumer routes the message to a DLQ topic and continues with the next message. The DLQ is monitored, and messages are inspected and (where possible) replayed after the underlying bug is fixed.

25.8 Failure Modes

Table 25.1: Failure modes in event-driven agent systems.

Failure mode	Mitigation
Schema drift	Schema registry with enforced compatibility; consumer integration tests against latest registered schemas.
Consumer lag	Per-consumer-group lag metrics; auto-scaling on lag; partition tuning.
Poison messages	Dead-letter queue; per-message retry budget; observability on DLQ depth.
Out-of-order events	Partition by key; explicit watermark handling for late events; idempotent consumers.
Replay storms	Rate-limited replay; canary replay on a single consumer before full fleet.
Schema-registry outage	Local schema caching in producers and consumers; fail-closed for new schemas only.

25.9 Event-Driven Mechanics in Depth

An event-driven agent system is shaped by three properties: ordering, delivery semantics, and idempotency. Figure 25.2 sketches the canonical flow.

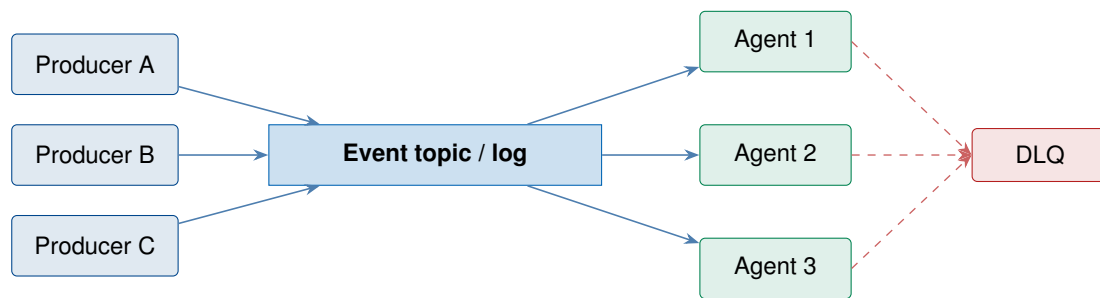


Figure 25.2: Event-driven flow: producers append to a durable log; agents subscribe and consume independently; failed events go to a dead-letter queue for inspection and replay.

25.9.1 Delivery Semantics

The delivery guarantee the event bus provides determines how agents must be written. At-least-once delivery means agents must be idempotent; exactly-once delivery simplifies agent logic but constrains the bus choice. The semantics are an infrastructure decision with direct consequences for agent design.

- **At-most-once.** Fast, lossy. Acceptable for ephemeral signals (clickstream, telemetry) where loss is recoverable.
- **At-least-once.** Standard. Events may be delivered more than once; consumers must be idempotent.
- **Exactly-once.** Available in some brokers (Kafka transactional, others) under specific configurations. Comes at performance cost and is usually unnecessary if consumers are idempotent.

25.9.2 Idempotency Patterns

Idempotent agents are not optional in event-driven systems. Three workable patterns:

- **Natural idempotence.** The operation is self-idempotent (“set status to X”). Easiest when applicable.
- **Event-ID deduplication.** Each event has a unique ID; the agent records processed IDs and skips replays. Requires a deduplication store; ID-retention determines replay window.
- **Outcome-based deduplication.** Before producing an effect, check whether the effect already exists (*has this order already been processed?*). Works when effects are observable.

25.9.3 Backpressure and Scaling

Agents consume at different rates. The architectural defences:

- Per-agent consumer groups so a slow agent does not block a fast one.
- Bounded in-flight windows (the agent reads ahead only so far).
- Auto-scaling on lag (number of unprocessed events).
- Shedding (drop low-priority events under sustained pressure).

25.9.4 Replay Discipline

Replay is one of event-driven’s superpowers and one of its sharper edges. Discipline:

- Replay always to a new consumer group, never to a live one.
- Replay outputs go to a side topic, not the production topic, unless explicitly intended.
- Replay budgets: a replay that is more expensive than re-running from current state is rarely worth it.
- Replay observability: distinguish replayed events in traces and metrics.

Event-driven agents are best for real-time, asynchronous agent workflows at scale. The bus is the contract; events are the facts; consumers are the agents that react. Replay, audit, and decoupling are the structural benefits; schema discipline, consumer-group operation, and DLQ hygiene are the operational costs. For organisations whose workflows naturally span multiple agents and services, the pattern pays back its complexity many times over. For simpler workflows, it does not.

Chapter 26

Pattern: Hexagonal Architecture for AI

“The point of the hexagon is to make the inside ignorant of the outside.”

—adapted from Alistair Cockburn

Chapter Overview

Alistair Cockburn’s Hexagonal Architecture, also known as Ports and Adapters, is a structural pattern for keeping the core of an application independent of its external dependencies. Applied to AI systems, it is the most direct answer to a defining property of the current era: the LLM, the vector database, the embedding model, and the tool APIs will all change. The architecture that lets them change without rewriting the application is the one that survives. This chapter develops the ports and adapters specific to AI, the test strategy that exploits them, and the migration paths they make affordable.

Learning Objectives

- Understand the core idea of hexagonal architecture and why it suits AI systems particularly well.
- Identify the canonical AI ports: LLM, vector, tool, UI, and observability.
- Implement an adapter for each port with a swap test that proves the adapter is replaceable.
- Use the pattern to migrate between vendors, swap embedding models, or move from hosted to self-hosted without rewriting application logic.
- Recognise when the pattern is overkill and a simpler structure is appropriate.

26.1 When to Use This Pattern

The hexagonal pattern is best for keeping your AI system swappable as models and vendors change. Apply it when:

- Multiple model vendors or self-hosted alternatives are realistic options.
- Embedding models or vector databases are likely to change.
- Tool integrations are evolving and not yet stable.

- Regulatory or sovereignty requirements may force migrations.
- The system is expected to outlive any one vendor's contract.

The pattern adds modest indirection. For prototypes or one-shot demonstrations, the indirection is overhead. For systems that will run for years, it is what makes the years tolerable.

A useful diagnostic for whether the pattern is warranted: count the external dependencies whose replacement is plausible within the system's expected life. If the answer is zero — a genuinely throwaway prototype, a fixed-scope internal tool with a guaranteed-stable model — the pattern is overhead. If the answer is two or more, which is the common case for any enterprise AI system in 2026, the pattern pays for itself the first time one of those dependencies has to change. The decision is not aesthetic; it is an estimate of future churn, and the AI dependency landscape churns faster than almost any other part of the enterprise stack. The architect who has lived through one forced model migration without the pattern rarely needs persuading to adopt it for the next system.

26.2 Core, Ports, Adapters

Three concepts:

- **Core.** The business logic of the system: what it does, independent of how. The core knows nothing about specific models or vendors.
- **Ports.** Interfaces the core depends on. Each port is a small, stable contract (e.g., "given a query, return K relevant chunks").
- **Adapters.** Implementations of ports against specific external systems (e.g., a Pinecone adapter, an OpenAI adapter, a local Ollama adapter).

The directional rule is the heart of the pattern: dependencies point inward. The core depends on nothing; the ports are defined in the core's own terms; the adapters depend on both the ports and the external technology. Nothing in the core ever imports an adapter or names a vendor. This is what makes the core testable in isolation and portable across vendors, and it is the single discipline that, if violated, quietly destroys the value of the whole pattern. A core that imports a vendor SDK "just for one type" has stopped being a core.

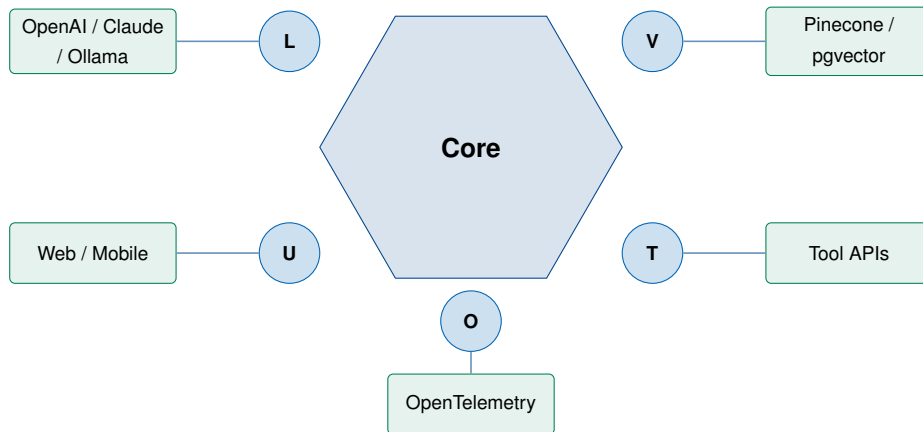


Figure 26.1: Hexagonal architecture for AI: a domain core surrounded by five ports (LLM, Vector, Tool, UI, Observability), each with one or more interchangeable adapters.

26.3 The Canonical AI Ports

Five ports recur across AI systems, captured in the LVTUO mnemonic.

The five are not arbitrary. They correspond to the five categories of external dependency that almost every enterprise AI system has: a model to call, a store to retrieve from, tools to act with, a surface to reach users through, and a telemetry sink to observe by. A system may have more ports — a dedicated port for a feature store, for a content-moderation service, for a feedback database — but it rarely has fewer, and an architecture that has not named at least these five has usually left some of them as implicit, hard-coded dependencies. Naming them is the first design step; the mnemonic exists so that the naming step is not skipped.

26.3.1 The LLM Port

A small, stable contract for text generation. A typical port specification:

```

1 from typing import Protocol
2
3 class LLMPort(Protocol):
4     async def generate(
5         self,
6         messages: list[dict],
7         max_tokens: int = 1024,
8         temperature: float = 0.0,
9         tools: list[dict] | None = None,
10    ) -> dict:
11         """Return {'text': str, 'usage': dict, 'finish_reason': str,
12             ...}"""
13         ...
14     async def stream(self, messages: list[dict], ...) -> AsyncIterator
15         [str]:
16         ...
  
```

Listing 26.1: The LLM port interface.

Adapters: `OpenAIAdapter`, `AnthropicAdapter`, `OllamaAdapter`, `vLLMAdapter`. The core code calls `LLMPort.generate(...)`; the adapter handles the vendor's API specifics.

The contract is deliberately minimal. It exposes generation and streaming and nothing else, because every method added to the port is a method every adapter must implement and every swap test must cover. The temptation, met early in most projects, is to widen the port to expose a feature one vendor offers and another does not — a particular caching mechanism, a particular structured-output mode. Resisting that temptation is the discipline: vendor-specific capability belongs in the adapter, surfaced through a general port concept if it generalises and not surfaced at all if it does not. A port that grows to the union of every vendor's feature set is no longer a port; it is a leaky aggregation that locks the core to whichever vendor has the largest surface.

26.3.2 The Vector Port

A contract for vector storage and retrieval. Typical operations: `upsert(id, vector, metadata)`, `query(vector, k, filter)`, `delete(id)`. **Adapters:** `Pinecone`, `Weaviate`, `Qdrant`, `Milvus`, `pgvector`.

The vector port carries a subtle obligation that the LLM port does not: the embeddings stored through it are only comparable to embeddings produced by the same embedding model. Swapping the vector-store adapter is cheap; swapping the embedding model behind it invalidates every vector already stored and forces a full re-index. A well-designed vector port therefore records, in the metadata of every stored vector, the identity and version of the embedding model that produced it, so that a model change is detected rather than silently corrupting retrieval. This is the kind of cross-cutting concern the port is the right place to enforce.

26.3.3 The Tool Port

A contract for invoking external tools. The MCP standard (Chapter 18) is increasingly the implementation of this port.

The tool port differs from the others in that the set of tools is itself dynamic — tools are added, removed, and re-versioned over the life of the system, in a way that models and vector stores are not. A well-designed tool port therefore exposes not just invocation but discovery: a way for the core to ask which tools are currently available and what their contracts are, so that the addition of a tool does not require a core change. This is precisely the capability the MCP server patterns of Chapter 34 are built to provide, which is why MCP has become the natural implementation of this port rather than a bespoke per-tool integration.

26.3.4 The UI Port

A contract for delivering responses to users. **Adapters:** `web`, `mobile`, `voice`, `embedded`.

The UI port is the one architects most often forget to define, because the first delivery surface —

usually a web chat interface — feels like part of the application rather than an external dependency. It is not. The moment the same AI capability must also be reached from a mobile app, a voice channel, or an embedded widget inside another product, an undefined UI port forces the application logic to be rewritten per surface. Defining the port early, even when only one surface exists, costs little and means the second and third surfaces are adapters rather than projects. Streaming behaviour, partial-response rendering, and interaction affordances differ enough across surfaces that the port should express them as capabilities the adapter declares, not as assumptions the core makes.

26.3.5 The Observability Port

A contract for emitting traces, metrics, and evaluation events. The OpenTelemetry standard is the typical implementation.

The observability port is what keeps instrumentation from being scattered through the core as vendor-specific logging calls. The core emits domain events — a retrieval happened, a model was called, a tool failed, an evaluation scored a response — in its own vocabulary; the adapter translates those into whatever telemetry backend the organisation runs. This matters for the same reason every other port matters: observability backends are replaced over a system's life, and an organisation that has hard-coded one vendor's SDK throughout the core faces a migration project rather than an adapter swap. It also matters for testing, since a capture adapter on the observability port is what lets a test assert that the right events were emitted in the right order.

26.4 The Swap Test

The single most useful test of a hexagonal implementation is the *swap test*: replace one adapter with another and run the test suite. If the suite passes, the abstraction is honest. If it fails, the abstraction was leaky and the core has implicit dependencies on the previous adapter's behaviour.

A minimum-effort swap test rotates each port through at least two adapters:

- LLM port: vendor A and vendor B.
- Vector port: hosted store and self-hosted (pgvector or similar).
- Tool port: real tool and a recorded-fixture replay.
- UI port: web client and CLI client (or a programmatic test client).

The swap test should run in continuous integration, not as a one-time check at design time. The reason is that abstraction leaks accumulate: a developer under deadline pressure adds a small dependency on one adapter's specific behaviour, the end-to-end tests still pass because they run against that adapter, and the leak goes unnoticed until a migration is attempted months later. A swap test in CI catches the leak in the pull request that introduced it, when it costs minutes to fix, rather than during a migration, when it costs a project. The discipline is cheap to maintain and expensive to omit.

26.5 Migration Paths

The hexagonal pattern is what makes the following changes *routine*:

- **Vendor swap.** Move from one frontier model to another in production by changing one configuration value.
- **Self-hosting.** Replace a hosted model with an open-weights model on internal infrastructure once volume justifies it.
- **Sovereignty migration.** Move from a global cloud to a region-resident deployment for regulatory reasons.
- **Multi-vendor.** Route by capability or cost while presenting one interface to applications.
- **A/B testing.** Compare two adapters in production with no code change to the core.

Without the pattern, each of these is a project; with the pattern, each is a deployment.

The economic argument is worth stating plainly. A vendor migration without the pattern typically means touching every file that calls the vendor’s SDK, re-testing the whole system, and accepting a window of elevated risk — weeks of work for a non-trivial system. The same migration with the pattern means writing one new adapter, running the swap test against it, and changing one configuration value. The first time an organisation exercises one of these migration paths, the pattern has usually paid back its entire cost of adoption. Every subsequent migration is close to free. In a field where the model leaderboard reorders every few months and where regulatory pressure can force a sovereignty migration on short notice, that optionality is not a luxury; it is the difference between an architecture that can respond to change and one that is held hostage by its earliest vendor choice.

26.6 Limits of the Pattern

The pattern is not free. Three caveats:

- **Vendor-specific features are hidden.** A port that supports the union of vendor capabilities exposes the union of constraints. A port that supports the intersection loses vendor differentiation.
- **Performance characteristics differ.** Two adapters may be functionally equivalent but have different latency, cost, or rate-limit profiles. Tests pass; production differs.
- **Some abstractions are leaky.** The LLM port cannot perfectly hide differences in tokenisation, output format, or refusal behaviour. The pattern manages the variance; it does not eliminate it.

The honest framing is that the pattern manages variance rather than abolishing it. It guarantees that a swap is mechanically possible — the code compiles, the tests pass, the system runs — but it does not guarantee that a swap is behaviourally transparent. A different model will refuse different prompts, format outputs differently, and price requests differently, and those differences surface in production regardless of how clean the port is. The architectural response is not to expect the pattern to do what it cannot, but to pair it with the evaluation discipline of Chapter 101: the swap test proves the swap is possible, and the evaluation suite proves the swap is acceptable. The two together are what make a vendor change routine; the pattern alone is necessary but not sufficient.

26.7 Substitution and Testing

The pattern’s testing payoff comes from substitutability at the ports. Figure 26.2 shows how the same core runs against either production adapters or test doubles.

This is the payoff that justifies the pattern even for an organisation that never changes a vendor. AI systems are notoriously hard to test because their external dependencies are slow, costly, non-deterministic, and rate-limited; a test suite that calls a real frontier model is expensive to run and flaky by construction. Hexagonal architecture makes the dependencies substitutable, so the core’s logic can be tested at full speed against deterministic test doubles, and the real adapters can be tested separately and less often. The result is a fast, reliable test suite for the part of the system that changes most — the application logic — which is something a non-hexagonal AI system struggles to achieve at all.

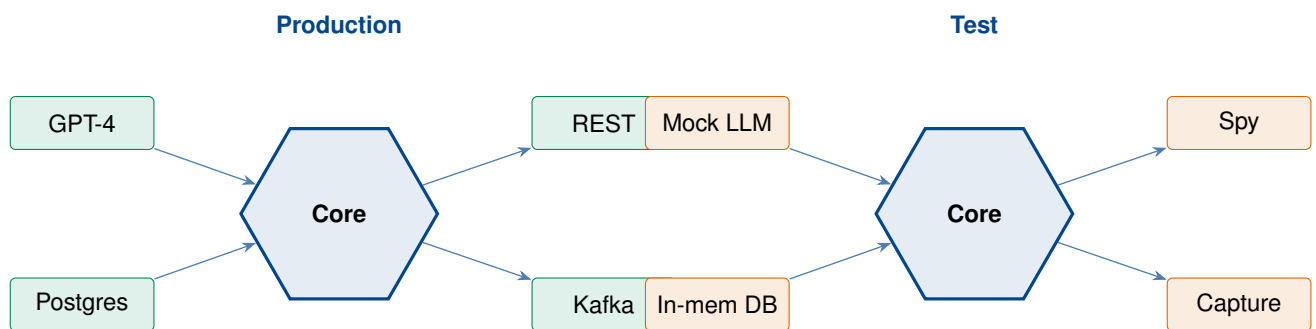


Figure 26.2: The same hexagonal core runs against production adapters (LLM, database, REST API, queue) or test doubles (mock LLM, in-memory DB, spy, capture). Tests exercise application logic at full speed without external dependencies.

26.7.1 The Port Discipline

Ports are interfaces written in the language of the domain, not the language of the technology behind them. A port for “model inference” returns a typed result object, not the raw provider response. A port for “user repository” exposes `find_by_id`, not `execute_sql`. The discipline lives or dies on this.

The test of whether a port is written in the domain’s language is simple: read the port’s method signatures aloud and ask whether they would still make sense if every current vendor went out of business tomorrow. “Retrieve the k most relevant passages for a query” passes the test; it is a statement about what the system needs. “Execute a similarity query against the index” fails it; it is a statement about how one class of technology happens to work. A port that fails this test has imported the technology’s mental model into the core, and the core will quietly acquire assumptions that the next technology does not share. Naming ports in domain language is not stylistic preference; it is what keeps the core genuinely independent.

26.7.2 Adapter Sizing

Adapters should be thin: convert between the domain types of the port and the wire types of the technology, plus retries, timeouts, and translation of provider errors into domain errors. Business logic in adapters is a smell.

The reason thinness matters is that adapters are the part of the system most likely to be rewritten — a new adapter is written for every vendor swap — and any business logic that has accumulated inside an adapter must be rewritten with it, or worse, silently lost. A good adapter is almost mechanical: it has no decisions in it that a domain expert would care about. When an adapter starts to contain a decision — which retrieved passages to keep, how to interpret an ambiguous model response, when to fall back to a different behaviour — that decision belongs in the core, behind the port, where it is written once and tested once. The smell test is whether deleting an adapter and writing a fresh one for a different vendor would lose anything but plumbing. If it would, the adapter is too fat.

26.7.3 The LLM Provider Port

The most consequential port in AI systems is the model inference port. A well-designed one:

- Takes a structured request (messages, tools, parameters), not a free-form prompt string.
- Returns a structured response (text, tool calls, finish reason, usage).
- Surfaces streaming as an explicit option, not a separate API.
- Translates provider errors into a domain error taxonomy (rate-limited, content-filtered, timeout, malformed, transient).
- Carries observability hooks (trace span, cost attribution, model identity).

With a port like this in place, swapping model providers becomes a configuration change rather than a refactor.

The error taxonomy deserves particular attention, because it is where the abstraction most often leaks. Every provider reports rate-limiting, content-filtering, timeouts, and malformed responses differently — different status codes, different error bodies, different retry semantics. An adapter that passes these through raw forces the core to know each provider's error dialect. An adapter that maps them onto a small, stable domain taxonomy — transient, rate-limited, content-filtered, malformed, fatal — lets the core handle errors in provider-independent terms, which is what makes the core's retry and fallback logic survive a vendor swap unchanged. Designing that taxonomy carefully, early, is one of the highest-leverage decisions in the whole pattern.

26.8 The African Perspective

The hexagonal pattern has a specific value in African deployments that is worth naming. The forces that make vendor migration likely — foreign-exchange volatility changing the relative cost of providers, data-sovereignty regulation forcing region-resident deployment, connectivity constraints favouring locally-hosted models for some workloads — are all sharper on the continent than in the markets where most frontier vendors are based. An African enterprise is therefore more likely than its European or North American counterpart to need a model migration, a hosted-to-self-hosted move, or a sovereignty-driven redeployment within the life of a system.

This makes the pattern's optionality disproportionately valuable. An architecture that has committed,

through clean ports, to keeping the LLM and vector adapters swappable is an architecture that can respond to a currency shock by routing to a cheaper provider, respond to a new data-localisation law by deploying a region-resident adapter, and respond to a connectivity constraint by adding a locally-hosted open-weights adapter for the latency-sensitive paths — each as a deployment rather than a project. The pattern does not remove these pressures, but it converts them from existential threats to the architecture into routine operational adjustments. For a system that must run for years under conditions that change faster than the vendor roadmaps assume, that conversion is the single most valuable structural commitment the architecture can make.

The hexagonal pattern is best for keeping your AI system swappable as models and vendors change. Its five ports (LLM, Vector, Tool, UI, Observability) and their adapters convert a chaotic vendor landscape into a manageable set of replaceable components. The swap test is the regular discipline that keeps the abstraction honest. For systems expected to outlive any single vendor's contract—which is most of them—the pattern is the single most valuable structural commitment the architecture can make.

Chapter 27

Platform AI Capability Patterns: Embedding AI in Existing Business Systems

“A capability is what the platform can do; a feature is how one vendor offers it. The architect commits to capabilities and chooses features afterwards.”

—a recurring observation in enterprise-architecture practice

Chapter Overview

Most enterprise AI deployments in 2026 do not run as standalone AI applications. They run as AI capabilities embedded in existing business platforms — CRM systems, BPM platforms, ERP systems, case-management products, claims processing engines, helpdesk applications — whose users were already there, whose data was already there, and whose audit and identity machinery was already there. The architectural question is therefore less “how do we build an AI system” than “how do we add AI capability to a platform that already has work flowing through it.” This chapter introduces the six recurring patterns by which platform AI capabilities are typically structured: conversational knowledge retrieval, AI-assisted design generation, embedded copilot, rule-governed LLM integration, real-time process guidance, and governed autonomous agents. Each is a way of adding AI to existing work without making the AI a separate system.

Learning Objectives

- Distinguish platform-embedded AI from standalone AI architectures.
- Recognise the six recurring patterns by which AI capabilities are added to business platforms.
- Identify the governance, identity, and lifecycle inheritances that platform-embedded AI receives from its host.
- Match each pattern to the kinds of work it serves and the failure modes it introduces.
- Read the rest of Part III as describing patterns that live inside platforms as much as alongside them.

27.1 What Makes a Capability a Platform Capability

A platform, in the sense this chapter uses the word, is an existing business system that mediates substantive amounts of an organisation's work. A customer-relationship-management system is a platform; a business-process-management product is a platform; an enterprise resource planning system, a case-management product, a service-management tool, a claims-processing engine, a loan-origination system — these are all platforms. Users go to them to do work. Data flows through them. Audit, identity, and access control are enforced inside them. They typically have decades of capability accumulated, hundreds or thousands of users, and integration relationships with every adjacent system the organisation runs.

When AI is added to such a platform, it can be added in either of two architecturally distinct ways. It can be added as a *separate application* — a standalone AI product that users navigate to, that pulls platform data through an API, and that returns results which the user then carries back to the platform manually. Or it can be added as a *platform capability* — a feature inside the platform itself, accessible from the surfaces users already use, governed by the platform's existing identity and audit machinery, sharing the platform's data model directly, and shipping under the platform's release cadence. The two paths produce very different architectures and different operational consequences.

The platform-capability path inherits much of what the platform already offers. Identity is the platform's identity — the user is already logged in, with their existing permissions and audit trail. Data access is the platform's data access — the AI sees what the user is permitted to see, no more, and the platform's existing data-governance rules apply. Compliance is the platform's compliance posture, which has typically been audited and certified for whatever regulatory regime applies. Observability is the platform's observability, with AI activity flowing through the same dashboards, alerts, and SIEM connections as everything else. The lifecycle is the platform's lifecycle, which means the AI capabilities ship and deprecate on the same schedule as the rest of the platform's features.

The cost of that inheritance is constraint. The AI capability cannot, in general, do things the platform's data model and security model do not contemplate. It cannot offer the absolute latest model capabilities the moment they arrive in the market, because the platform vendor's integration cadence is typically slower than the foundation-model release cadence. It cannot fully exploit cross-organisation data unless the platform vendor has built that into the platform's data sharing model. These are real costs, and the architectural decision between platform-embedded and standalone is partly an evaluation of which set of constraints fits the workload.

What the rest of this chapter does is take that path — the path of embedding AI as platform capability — and describe the six patterns by which it is typically done. The patterns are not vendor-specific. They appear across CRM, BPM, ERP, service-management, claims-processing, and other platform categories. The names vary; the architectural shapes do not.

One further point belongs in this framing. The choice between platform-embedded and standalone is rarely made once for an organisation; it is made repeatedly, per use case, and a mature enterprise runs both. The discipline is to make the choice deliberately each time rather than defaulting — defaulting to platform-embedded because the platform vendor's roadmap is convenient, or defaulting to standalone because a particular team prefers to build. The decision criteria developed later in the chapter are what

turn that choice from a habit into an architectural judgement. Reading the six patterns that follow as a menu of what platform embedding can offer is the first half of that judgement; reading the trade-off section near the end is the second.

27.2 Pattern 1: Conversational Knowledge Retrieval

The first and most common platform AI pattern is conversational knowledge retrieval. The platform exposes a chat surface inside its user interface. The user asks a question in natural language. Behind the surface, a RAG architecture retrieves passages from the platform's knowledge bases — documentation, prior cases, prior tickets, prior emails, prior decisions — and a foundation model synthesises an answer. The user gets the answer in the platform, scoped to what they are permitted to see, citing the sources the platform actually holds.

When to use it. Whenever there is substantial organisational knowledge inside the platform that users want to query in natural language but currently search by keyword. Customer-service representatives looking up how prior cases of a similar type were resolved; loan officers checking which conditions a particular product carries; field engineers looking for the configuration that was used in a comparable installation last quarter; HR practitioners checking which policy applies to a specific situation. Conversational knowledge retrieval is the pattern that converts the platform's accumulated history into a queryable resource without exporting that history to a third party.

Architectural shape. The pattern is the standard RAG architecture of Chapter 21, but with three properties characteristic of platform embedding. First, the retrieval corpus is scoped to what the asking user is permitted to see — the platform's existing row-level and field-level security extends into retrieval. Second, the retrieved passages carry citations back to the platform's own records, so the user can navigate from the answer to the source within the platform. Third, the queries and answers are logged through the platform's existing audit pipeline, so the AI interaction is part of the same compliance record as any other user activity.

Failure modes. The two characteristic failures are scope leakage and stale-corpus hallucination. Scope leakage happens when the retrieval layer fails to honour the platform's row-level security and surfaces a passage the user should not see; the mitigation is to enforce scope at the retrieval-index level rather than relying on the model to filter. Stale-corpus hallucination happens when the corpus has not been re-indexed in time to reflect recent changes, and the model answers with confidence based on out-of-date material; the mitigation is freshness metadata exposed in the retrieved passages, combined with prompts that instruct the model to surface uncertainty when sources are old.

Governance inheritance. The conversational knowledge retrieval pattern inherits the platform's identity, audit, and data-classification machinery. The AI does not need its own audit pipeline; it uses the platform's. It does not need its own data-classification rules; it inherits them. This inheritance is the architectural payoff of the platform-embedded approach.

Maturity signal. An organisation is ready for this pattern when its platform knowledge is already substantial and already structured enough to retrieve from — when the cases, tickets, and documents carry consistent metadata and the access rules are coherent. An organisation whose platform data is a

sprawl of inconsistently tagged free text will get poor retrieval regardless of how good the foundation model is, and the right first investment is the data discipline of Chapter 97's data-and-operations layer rather than the retrieval capability itself. Conversational knowledge retrieval rewards a clean corpus and punishes a dirty one; the pattern is a multiplier on the quality of the knowledge already in the platform, not a substitute for it.

27.3 Pattern 2: AI-Assisted Design Generation

The second pattern is AI-assisted generation of design artefacts. A user describes, in natural language, what they want to build — a business process, a data model, a workflow, an integration, a configuration, a screen layout. The platform's AI capability generates a candidate design, in the platform's own design language, which the user then reviews, edits, and commits. The output is not text. It is a structured platform artefact — a process diagram, a configuration set, a schema, an entity relationship — that the platform can immediately execute.

When to use it. Whenever the platform's design surface is unfamiliar to many of its users, or whenever the cost of producing initial drafts in the design surface is high. Process design in BPM platforms is the canonical example: business analysts who can describe a process in English are not always the same people who can construct it in the platform's process modelling tool, and an AI that bridges the two saves substantial elapsed time on the first cut. Data-model generation, integration-flow generation, and configuration-set generation all follow the same pattern.

Architectural shape. The user's natural-language description is passed to a foundation model with a system prompt that constrains its output to the platform's design language — typically a structured format like JSON, YAML, or the platform's native model serialisation. The generated artefact is loaded into the platform's editor for review. The user sees the AI's draft as a starting point and applies the platform's normal editing, validation, and commit workflow on top.

Failure modes. The most common is the plausible-but-wrong artefact. The model produces a design that looks correct, validates syntactically, and embodies a subtle semantic error that will only manifest at runtime — a process branch that does not actually cover an edge case, a data model that omits a required relationship, an integration that hard-codes an environment-specific value. The mitigation is that the AI's output must pass the platform's existing validation pipeline before commit, and that the user is responsible for substantive review of what they accept. A second failure mode is design lock-in: users come to depend on the AI-generated starting point and lose the skill of designing from scratch when the AI is unavailable or wrong. The mitigation is to keep the AI's role explicitly as draft-generator rather than as authority, and to maintain the design-from-scratch path as a first-class option.

Governance inheritance. Design generation outputs flow through the platform's normal design-review and approval gates. The AI does not bypass them. The user who accepts the AI's draft is the user who is accountable for what they accepted, just as they would be for a draft from any other source.

Maturity signal. This pattern is most valuable where the platform's design surface has a steep learning curve and a wide pool of would-be designers blocked by it — the business analysts who understand

the process but cannot model it, the domain experts who know the data but cannot express the schema. It is least valuable where the design work is already done fluently by a small expert team, because there the AI's draft saves little and the review burden it adds is pure overhead. The architectural question to ask before adopting it is not "can the AI generate platform artefacts" — it can — but "how many people are currently blocked from contributing designs by the tooling, and is that the actual bottleneck." If the bottleneck is elsewhere, the pattern solves a problem the organisation does not have.

27.4 Pattern 3: Embedded Copilot

The third pattern is the embedded copilot — an AI assistant integrated into the working surface of the platform, present alongside the user's task, with access to the context of what they are doing and the ability to suggest or pre-fill operations. The copilot is not a separate destination; it is a sidebar, a slash-command, an inline suggestion, or an action button that appears in the work surface where the work is happening.

When to use it. Whenever a substantial fraction of the user's time inside the platform is spent on operations that an AI could either accelerate (drafting a response, summarising a record, extracting fields from an attachment) or improve (suggesting the next best action, flagging anomalies in the work as it is being done). Sales representatives drafting outreach emails; service agents summarising long case histories; underwriters extracting structured fields from unstructured submissions; operations staff drafting incident reports.

Architectural shape. The copilot sits inside the platform's user interface as a UI component that has access to the current work context (the record being viewed, the document being drafted, the screen the user is on). When invoked, it sends that context together with the user's intent to a foundation model or an agent, and returns suggestions that the user can accept, edit, or reject. The acceptance flow is critical: the platform records what was suggested, what was accepted, what was edited, and what was rejected, so that future quality and bias analysis has data to work from.

Failure modes. The two characteristic failures are over-reliance and shadow-policy drift. Over-reliance is the productivity-paradox failure: users accept AI suggestions without proper review, and errors propagate at speed; the mitigation is to require explicit confirmation for any AI-suggested action with material consequences, and to track acceptance rates as an operational metric so the architect can see when users are pattern-matching rather than reviewing. Shadow-policy drift is the failure where AI suggestions encode an effective policy — the model's training-data preferences — that has never been formally approved by anyone responsible for the platform's actual policy. The mitigation is to constrain the copilot's suggestions through explicit policy prompts and through guardrails on the action space, so the model's preferences do not silently become the organisation's policies.

Governance inheritance. The embedded copilot inherits the platform's permission model and audit trail. Suggestions it makes are visible only to users with permission to see the underlying data; actions it takes are recorded with the user as the responsible party. The AI is an assistant; the human is the accountable agent.

Maturity signal. The embedded copilot is the pattern most likely to be adopted prematurely, because it demonstrates well — a sidebar that drafts an email is an easy thing to show an executive sponsor. The signal that an organisation is genuinely ready is not that the copilot can be built but that the acceptance-tracking and review discipline around it can be sustained. A copilot deployed without the metric that watches acceptance rates, and without the explicit confirmation step for consequential actions, will produce exactly the over-reliance failure described above, and will do so invisibly until an error of consequence surfaces. The pattern is ready when the organisation has decided who watches the acceptance metric and what threshold triggers intervention — not before.

27.5 Pattern 4: Rule-Governed LLM Integration

The fourth pattern integrates LLM calls into the platform's existing business-rule machinery. The platform already has a rule engine, used to express business policies, decisioning logic, validation, and routing. The AI integration appears as another kind of rule: a rule that, when triggered, calls a foundation model with a prompt assembled from the rule's parameters and the work in scope, and returns the model's output for use by downstream rules. The AI is, in this pattern, a rule-engine primitive — not a separate system but a tool inside the platform's existing decisioning fabric.

When to use it. Whenever LLM capabilities are needed inside business processes whose other logic is already encoded as rules, and where the LLM's involvement must be controlled by the same governance that controls the rest of the rules. Compliance-sensitive decisioning where the platform's policy framework is the source of truth; high-volume processes where the cost of LLM calls must be metered through the same controls as other expensive operations; workflows where some steps must be LLM-driven and others must be deterministically rule-driven, and the two must compose cleanly.

Architectural shape. A new rule type — “LLM rule” or “AI rule” — is introduced into the platform's rule catalogue. It has the standard rule lifecycle: it is authored in the platform's rule designer, versioned, tested in the platform's test harness, deployed through the platform's release machinery, and monitored through the platform's runtime telemetry. At runtime, the rule's trigger conditions are evaluated by the rule engine in the normal way, and on triggering the rule assembles a prompt, makes an LLM call, applies any output validation, and returns its result for use by subsequent rules.

Failure modes. The most common is the LLM-rule cost cascade. Rules are designed to be cheap and stateless; LLM rules are neither. A rule that triggers a thousand times an hour at three cents per call costs thirty dollars an hour, and rules that previously fired millions of times a day in benign deterministic mode become genuine cost centres when their bodies become LLM calls. The mitigation is explicit cost accounting per LLM rule, with budgets, alerts, and circuit breakers; an LLM rule that would spend its monthly budget in a day should not be allowed to fire freely. A second failure mode is the non-deterministic rule paradox: rules are expected to produce the same answer given the same inputs, and LLM rules do not. The mitigation is to scope LLM rules carefully to decisions where some variation is acceptable, and to use deterministic rules in places where the same input must always produce the same output.

Governance inheritance. Rule-governed LLM integration inherits the platform’s rule lifecycle, which is typically one of the most carefully governed parts of any enterprise platform. Compliance, change control, approval workflows, and rollback procedures are all in place. The pattern’s architectural payoff is that the AI inherits all of this without needing its own parallel machinery.

Maturity signal. This pattern is ready when the organisation already runs its rule estate with discipline — versioned rules, a real test harness, change control that works. An LLM rule dropped into a rule estate that is itself a sprawl of undocumented, untested rules inherits the sprawl rather than the discipline, and the non-determinism of the LLM rule then compounds an already fragile situation. The pattern’s whole value proposition is inheritance of governance; where there is little governance to inherit, the pattern delivers little. The right sequencing is to bring the rule estate to a governed state first, then introduce LLM rules into it, rather than hoping the AI rule will somehow be better governed than its deterministic neighbours.

27.6 Pattern 5: Real-Time Process Guidance

The fifth pattern is real-time guidance to users during the execution of a process. An AI watches the user’s progress through a structured workflow — a case being processed, an order being fulfilled, an inspection being completed — and intervenes with relevant suggestions: the next likely action, a missed step, a relevant policy that applies, an anomaly worth investigating. The guidance is in-flow: it appears alongside the work, in the moment when it is actionable.

When to use it. Whenever the platform supports long-running, multi-step processes where the user’s quality and speed are improved by timely guidance, and where the guidance can be drawn from organisational context the user might not remember in the moment. Customer-service representatives handling complex calls; inspectors completing multi-section audits; underwriters working through risk assessments; clinicians documenting patient encounters; salespeople running through structured discovery conversations.

Architectural shape. An AI component is wired into the platform’s process-execution layer with read access to the user’s current state in the process and the broader case context. The component runs continuously — typically at low frequency, on state transitions — and emits suggestions to the user interface as side-panel content, inline highlights, or notifications. The user is free to accept, defer, or dismiss each suggestion. The pattern is closer to a real-time observability system than to a tool the user invokes; the user does not ask, but the AI is always watching.

Failure modes. Three characteristic failures recur. The first is suggestion fatigue: the AI offers too many suggestions, the user dismisses them habitually, and useful and useless suggestions are dismissed at the same rate. The mitigation is suggestion confidence scoring with a high threshold, and explicit measurement of suggestion-acceptance rates with action when acceptance falls below the threshold of usefulness. The second is missing-context error: the AI suggests an action that would be correct given what it sees but wrong given what the user knows that the AI does not. The mitigation is to design the suggestions as recommendations subject to human judgement, with clear escape valves for the user to indicate when context is missing. The third is observability surveillance creep: an AI

that watches the user's work continuously is also collecting an unprecedented record of user activity, which can become a workplace-surveillance instrument by drift if not deliberately constrained. The mitigation is explicit policy on what user activity is recorded, who can access it, and for what purposes — and the policy should be made visible to the users being watched.

Governance inheritance. Real-time process guidance inherits the platform's process-execution audit trail and the platform's user-activity logging. The pattern's architectural payoff is that the AI's recommendations are tied to specific points in the process and can be analysed alongside the rest of the process telemetry — which steps see the most AI interventions, where acceptance rates are highest, which suggestions correlate with better outcomes.

Maturity signal. This pattern is the one whose adoption decision is least purely technical, because the surveillance-creep failure mode is a labour-relations and trust question as much as an architectural one. An organisation is ready for real-time process guidance when it has thought through, and ideally agreed with the workforce being guided, what the AI watches, what is recorded, who sees the record, and what it is and is not used for. Deployed without that agreement, the pattern delivers process-quality gains while quietly accumulating a workplace-monitoring capability that, when discovered, can cost more in trust than the quality gains were worth. The technical readiness is rarely the constraint; the organisational readiness usually is.

27.7 Pattern 6: Governed Autonomous Agents

The sixth pattern is the governed autonomous agent. The platform provides an agent capability that can plan and execute multi-step actions across the platform's data and operations — but with the agent's tool inventory, action authority, and execution boundaries defined declaratively in the platform's existing rule format. The agent is not a free agent; it is a constrained agent, with its constraints expressed in the same language the rest of the platform's policy is expressed in.

When to use it. Whenever the work to be done is genuinely multi-step and benefits from autonomous reasoning, and where the platform's regulatory or business posture requires explicit constraints on what the agent can do. End-to-end customer-service resolution where the agent reads, drafts, escalates, and closes; claims triage where the agent ingests, classifies, and routes; case investigation where the agent gathers evidence, summarises findings, and proposes next actions. The governance constraint distinguishes this pattern from raw agent deployment: the agent's allowed actions are not the union of every tool it has access to, but the constrained subset that the platform's rule-based policy specifically permits.

Architectural shape. The agent is implemented as a standard agent loop — plan, act, observe, revise — with three platform-specific bindings. Its tool inventory is exposed through the platform's rule-defined permission model, so the agent can only invoke tools that policy authorises. Its action authority — which actions require human approval, which can be taken autonomously, which trigger escalation — is expressed as rules in the platform's existing rule engine. Its audit trail flows through the platform's standard audit pipeline, with each agent step recorded as a platform event with attributable identity. The agent is, in effect, a user with constrained permissions and a model behind the wheel.

Failure modes. The most consequential is policy gap exploitation. The agent finds a sequence of permitted actions whose combined effect is not what the policy intended — each step is allowed, but the overall outcome is undesirable. The mitigation is policy stress-testing: red-team exercises in which an adversarial planner is asked to maximise undesirable outcomes within the stated permissions, with the policies adjusted to close the gaps found. A second failure mode is autonomous escalation drift: a successful agent that handles ninety percent of its cases without escalation creates pressure to expand its action authority, and over time the constraints loosen without explicit policy review. The mitigation is regular review of the agent’s action policy through the architecture review board, with the conditions of any expansion explicitly recorded as decisions. A third failure mode is responsibility ambiguity: when something goes wrong, the question of whether the agent acted wrongly or the policy permitted the wrong thing is genuinely difficult. The mitigation is observability that traces each agent action back to the specific policy rule that permitted it, so the analysis can distinguish agent error from policy error.

Governance inheritance. Governed autonomous agents inherit the platform’s rule lifecycle for their policies, the platform’s identity model for their authority, and the platform’s audit pipeline for their actions. The architectural payoff is that the agent operates within the platform’s existing governance machinery rather than alongside a new and parallel one. The architectural risk is that the inherited machinery was designed for human users and may not catch failure modes specific to autonomous behaviour — the policy stress-testing above is the most important defence against that risk.

Maturity signal. This is the most consequential of the six patterns and the one that should be adopted last. An organisation is ready for governed autonomous agents when it has already run the other patterns long enough to trust its policy framework, its audit pipeline, and its observability — because the autonomous agent depends on all three being genuinely sound, not merely present. The honest sequencing for most organisations is conversational retrieval and copilot first, to build operational confidence; rule-governed integration next, to exercise the policy machinery; and governed autonomous agents only once the policy framework has been stress-tested against adversarial planning and the organisation has decided, explicitly, which narrow band of work it is willing to delegate. An organisation that reaches for autonomous agents first, before the governance scaffolding is proven, has adopted the pattern’s risk without having earned its prerequisites.

27.8 Composing the Six Patterns

Most platforms that have invested seriously in AI capability end up implementing several of the six patterns, sometimes all of them. The patterns are not in tension; they cover different parts of the work the platform supports. A representative mature deployment might combine:

- Conversational knowledge retrieval for users who need to query the platform’s accumulated history in natural language.
- AI-assisted design generation for power users building new platform configurations, process designs, and integrations.
- Embedded copilot inside the work surfaces where most user time is spent, accelerating drafting and

summarisation.

- Rule-governed LLM integration in the processes where AI must compose with existing business rules under shared governance.
- Real-time process guidance during the execution of structured workflows, providing in-flow recommendations and anomaly detection.
- Governed autonomous agents for the well-bounded subset of work that the organisation has decided to delegate.

The architectural decision is less which patterns to use and more which patterns to use where. The same pattern — governed autonomous agents, say — might be appropriate for one process in the platform and entirely wrong for another, depending on the regulatory posture, the irreversibility of the actions, and the maturity of the policy framework. The reference architecture for a platform's AI capability should name the pattern at each location, justify the choice, and record the decision as part of the architectural record.

There is also a natural adoption order, visible in the maturity signals of the preceding sections. Conversational knowledge retrieval and the embedded copilot are the low-risk entry points: they build operational confidence and surface the organisation's data-quality and review-discipline gaps without exposing it to autonomous action. AI-assisted design generation and rule-governed integration come next, exercising the platform's validation and policy machinery. Governed autonomous agents come last, once the policy framework and observability have been proven by the earlier patterns. An organisation that adopts in this order accumulates the prerequisites for each pattern before it needs them; an organisation that starts with the most consequential pattern inherits its risks without having built its foundations.

27.9 What Platform Embedding Buys, and What It Costs

The architectural decision to add AI as a platform capability rather than as a standalone application carries a recognisable set of trade-offs. Worth naming them explicitly.

What it buys: identity inheritance (the user is already authenticated, with established permissions); audit inheritance (the AI's actions flow through the platform's existing compliance pipeline); data-governance inheritance (row-level and field-level security are honoured automatically); user adoption (the AI lives where users already are, not in a separate destination); release-cycle alignment (AI capabilities ship and deprecate on the platform's schedule, which procurement and operations already know how to handle); vendor relationship simplification (one platform vendor accountable for the AI capabilities, not a separate AI vendor sitting alongside).

What it costs: model-version lag (the platform vendor's integration cycle is typically slower than the frontier model release cycle, so the latest model capability arrives in the platform months after it arrives in the market); architectural flexibility (the platform's data model and security model constrain what the AI can be asked to do); vendor concentration (the AI capabilities are tied to the platform vendor's roadmap and pricing); innovation pace (specialised AI applications can be built and modified

faster than platform capabilities, which must respect the platform's release engineering).

The right choice depends on the workload. Highly regulated work with strong existing governance, where the platform's data model is rich and where the AI capability needs to inherit the audit and identity machinery, favours platform embedding. Highly experimental work where capability access matters more than governance inheritance, where the data model is novel, and where the workload sits awkwardly in any existing platform, favours standalone applications. Most enterprises in 2026 are running both — platform-embedded AI for the work that already flows through their major platforms, and standalone AI for the novel workloads — with the architectural decision made deliberately at each new use case rather than by default in either direction.

A practical test helps make the decision concrete. Ask, for the use case in hand, whether the work already flows through an existing platform. If it does, embedding is the default, because the governance inheritance is free and the user adoption is automatic. Ask next whether the use case needs a model capability that the platform vendor does not yet expose — a frontier reasoning capability, a specialised modality. If it does, and the capability is essential rather than merely desirable, that argues for standalone despite the loss of inheritance. The decision is the resolution of those two questions, recorded with its reasoning, so that a later reviewer can see why this use case went one way when a neighbouring one went the other.

27.10 The African Perspective

Platform AI capabilities have a particular shape in African deployments. The platform vendors that supply CRM, BPM, ERP, and case-management products are mostly headquartered outside the continent, and their AI capabilities are typically delivered through cloud services that may have data-residency, foreign-exchange, and connectivity implications.

This matters more for platform-embedded AI than for standalone AI, precisely because of the inheritance the pattern depends on. When AI is a platform capability, the AI's deployment region, pricing currency, and connectivity requirements are largely determined by the platform vendor's architecture, not by the adopting enterprise. An organisation that would, for a standalone system, choose a region-resident deployment and a locally-hosted model finds that the platform-embedded path may not offer those choices. The three questions below — residency, foreign exchange, connectivity — are therefore not just operational concerns but inputs to the embed-versus-standalone decision itself.

The data-residency question is sharpest. A platform's AI capability that invokes a foundation model in a US or European cloud region is, in regulatory terms, transmitting the customer's data outside the local jurisdiction. For most non-sensitive use cases this is acceptable; for use cases involving personal data of citizens in jurisdictions with strong data-localisation requirements (South Africa under POPIA, Nigeria under the NDPR, Kenya under the DPA), it can be a hard constraint. The architectural response is to know, in advance, which patterns can be served by foundation models running in compliant regions, and to commit to those patterns explicitly while marking the others as conditional.

The foreign-exchange question is sharpest for rule-governed LLM integration. A platform that bills LLM call volume in US dollars, against the volume of business work flowing through the platform, can have a cost profile that shifts non-trivially with currency movements. The cost-accounting discipline described in the rule-governed pattern is operationally more important in these environments than it is in dollar-denominated economies. The architectural response is explicit cost budgeting per LLM rule, with budgets sized in local currency and revisited as exchange rates move.

The connectivity question is most consequential for real-time process guidance and embedded copilot. Both patterns degrade visibly when network latency to the foundation-model provider exceeds a few hundred milliseconds; both fail outright when connectivity drops. For deployments in places where reliable low-latency connectivity to frontier model providers cannot be assumed, the architectural response is to evaluate, alongside the platform's AI capabilities, the option of locally-hosted alternatives — which may be smaller open-weight models served from regional infrastructure, or capabilities cached and degraded gracefully when the network is unavailable.

27.11 What These Patterns Earn

The six patterns described in this chapter are the working vocabulary by which platform-embedded AI is currently architected. They are not vendor-specific. Every serious enterprise platform vendor in 2026 offers most of them, under their own names, with their own packaging and pricing. The architect's job is to recognise the pattern underneath the marketing, evaluate it against the workload it will serve, and commit to it as an ABB — a logical capability the platform must provide — with the specific implementation chosen as the corresponding SBB.

The discipline the chapter asks for is the discipline of the maturity signals, read together: adopt the patterns in an order that builds prerequisites before it needs them, decide embed-versus-standalone deliberately per use case rather than by habit, and record each choice with its reasoning so the reference architecture for the platform's AI capability is a document a reviewer can actually audit. A platform-AI estate built this way is legible — an architect can say which pattern sits at which location and why — and a legible estate is one that can be governed, evolved, and trusted with progressively more consequential work.

The patterns also do not exhaust what is possible inside a business platform. They are the recurring shapes, the ones a competent platform-AI deployment will probably contain. Specialised patterns will continue to emerge as the field matures, and some of the six will turn out to be temporary names for capabilities that will fragment or merge as practice settles. The reader who has read the chapter as a working snapshot of platform AI in 2026, subject to revision by practitioners in 2027 and 2028, has read it in the right posture — the same posture the opening chapters of the book asked the reader to carry into all the technical material. The taxonomy is provisional. The architectural disciplines underneath it are not.

Part IV

Advanced Architecture Patterns

“The artist who exhibits this lock will pay two hundred guineas to the person who shall make an instrument that will pick or open it.”

—Joseph Bramah, public challenge accompanying his lock, 1790

The standard patterns of Part III work until something exceeds what they were designed to handle. Joseph Bramah’s lock was not the first lock. It was the lock that came after all the standard locks — the pattern that worked when the simple patterns had been overtaken by adversaries who had learned them. Bramah’s challenge stood unanswered for sixty-seven years, until the American locksmith Alfred Hobbs, with weeks of patient work, finally picked it at the 1851 Great Exhibition. The lesson is the one Part IV of this book builds on: the basic patterns are necessary, and they are not sufficient. Production AI runs into adversaries, edge cases, and scale demands that the standard patterns do not anticipate. The advanced patterns in Part IV are what the field has converged on after the standard ones turned out to be insufficient.

Chapter 28

The Anatomy of an AI Agent: A Reference Decomposition

“Before you can orchestrate many agents, you have to know what one agent is made of.”

—a recurring observation in enterprise-architecture practice

Chapter Overview

The chapters that follow describe patterns for composing, orchestrating, and governing AI agents. This chapter describes the thing those patterns operate on: a single agent, decomposed into its component parts. An agent is not a monolith and not a single model call; it is an assembly of eight recognisable components — an input layer, an understanding layer, a reasoning engine, a planning module, a memory system, an action and tool layer, a monitoring layer, and a safety and control layer — wired together by a control loop that decides, at each step, whether to respond, plan, act, or stop. This chapter walks through the eight components and the control flow that connects them, so that the agent patterns of the rest of Part IV have a shared reference anatomy to build on.

Learning Objectives

- Decompose an AI agent into its eight reference components.
- Trace the control loop that connects the components and decides what the agent does next.
- Identify which component owns which failure mode, and which architectural decision belongs at which component.
- Distinguish the agent’s reasoning, planning, and action responsibilities, which are commonly conflated.
- Carry the reference anatomy into the agent pattern chapters that follow.

28.1 Why an Agent Needs a Reference Anatomy

The word *agent* is used loosely. It is applied to a single model call wrapped in a retry loop; to a multi-step reasoning chain; to a system that calls tools; to a fully autonomous process that plans, acts,

and revises across hours of execution. The looseness is the kind a young field tolerates, and it is also a real obstacle to clear architecture. Two architects describing “our agent” may be describing structurally different systems, and a review board cannot govern what it cannot describe.

A reference anatomy fixes this. It says: an agent, in the full sense, is an assembly of eight components, each with a defined responsibility, connected by a control loop. A particular deployment may implement some components trivially — the planning module of a simple agent may be a single prompt — but the components are still there as architectural positions, and naming them lets an architect say precisely which parts of the agent are elaborate and which are minimal. The decomposition is the same move the book has made elsewhere: a working vocabulary that lets practitioners describe what they have built and reason about it together.

The eight components are the input layer, the understanding layer, the reasoning engine, the planning module, the memory system, the action and tool layer, the monitoring layer, and the safety and control layer. The first six are the agent’s processing pipeline; the last two are cross-cutting, wrapping the pipeline rather than sitting inside it. The sections below take each in turn, and the section after that traces the control loop that connects them.

A note on how to read the decomposition. The eight components are architectural *positions*, not necessarily eight separate pieces of code. A minimal agent might implement the understanding layer, reasoning engine, and planning module as three prompts to the same model, or even as one. The value of the decomposition is not that it dictates eight modules; it is that it names eight responsibilities, so an architect can say which responsibility is implemented where, which are elaborate, which are minimal, and which — dangerously — have been left unassigned. An agent that has not consciously placed all eight responsibilities has not eliminated them; it has only stopped being able to see them.

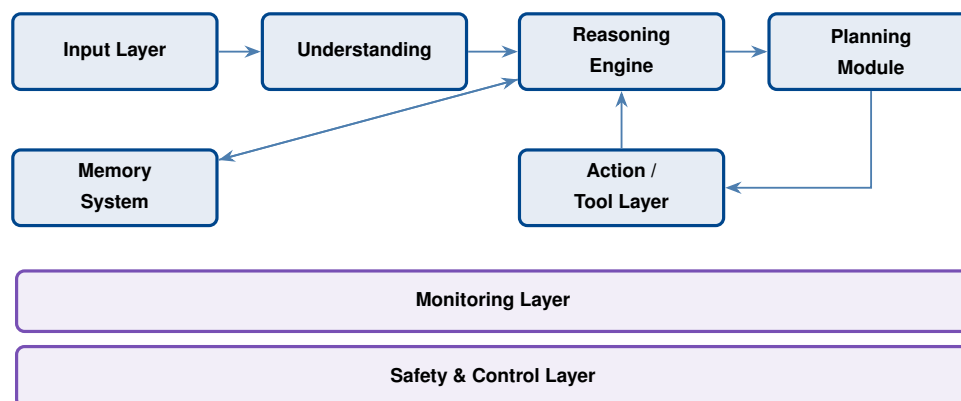


Figure 28.1: The eight-component reference anatomy of an AI agent. The processing pipeline (top, blue) runs input through understanding, reasoning, planning, action, and memory. The monitoring layer and the safety-and-control layer (bottom, purple) wrap the entire pipeline rather than sitting at one point in it.

28.2 The Input Layer

The input layer is where work enters the agent. The inputs are heterogeneous: a user query typed into a chat surface; an API request from another system; sensor data from a device; an event from a message bus; a scheduled trigger. The input layer’s job is to normalise these into a form the rest of the

agent can process, and to attach the metadata the later components will need — the identity of the requester, the channel the request arrived on, the timestamp, the trace identifier that will follow the request through the rest of the agent.

The input layer is also the agent's first security boundary, and the place where several safety controls are most cheaply enforced. Access control — is this requester allowed to invoke this agent at all — belongs here. Rate control — has this requester exceeded their allowed call volume — belongs here. Content checks on the incoming request — is this input within acceptable bounds, does it contain a prompt-injection attempt, does it carry data the agent is not cleared to process — belong here. So does the first human-in-the-loop checkpoint, where the architecture requires that some classes of request be approved before the agent acts on them at all.

Failure modes. The characteristic input-layer failure is the unnormalised input that the later components mishandle: a request arrives in an unexpected shape, the understanding layer misreads it, and the agent acts on a misinterpretation. The mitigation is strict input validation at the boundary, with malformed inputs rejected rather than passed through hopefully. A second failure mode is the missing-metadata failure: the trace identifier is not attached, and when something goes wrong three components later, the request cannot be traced back to its origin. The mitigation is to make metadata attachment mandatory at the input layer, enforced rather than conventional.

Design note. The input layer is where the cheapest security wins in the whole agent are available, because rejecting a bad request here costs nothing downstream, while the same request caught at the action layer has already consumed model calls, tool invocations, and time. The architectural discipline is to push every check that *can* run at the input layer to the input layer — identity, rate, gross content validation — and to reserve the later components for the checks that genuinely require interpreted content. An agent that defers all its safety checks to the action layer pays for every malicious or malformed request in full before declining it.

28.3 The Understanding Layer

The understanding layer interprets what the input means. It is distinct from the reasoning engine: understanding is about *comprehending the request*, reasoning is about *deciding what to do about it*. The understanding layer resolves the request's intent, extracts its parameters, identifies its entities, classifies its type, and determines what the request is actually asking for — which may differ from what it literally says.

In a simple agent the understanding layer is light: the request is a well-formed query and understanding it is trivial. In a complex agent it is substantial: the request is an ambiguous natural-language instruction, the agent serves many request types, and correctly classifying the request determines everything downstream. The understanding layer is where the agent decides which of its capabilities the request is calling on, and a misclassification here propagates through every later component.

Failure modes. Intent misclassification is the characteristic failure: the understanding layer decides the request is one kind of task when it is another, and every subsequent component does the wrong work competently. The mitigation is to treat understanding as an evaluable component in its own right

— with its own test set, its own accuracy metric, its own monitoring — rather than as an invisible step. A second failure mode is the silent ambiguity: the request genuinely could mean two things, the understanding layer picks one without flagging the ambiguity, and the agent proceeds confidently down one branch. The mitigation is for the understanding layer to surface ambiguity explicitly — to the reasoning engine, which can decide to ask a clarifying question rather than guess.

Design note. The most consequential design decision for the understanding layer is whether to implement it as a separate, evaluable component or to fold it into the reasoning engine’s first model call. Folding it in is simpler and is often right for a narrow agent. Keeping it separate is what allows intent classification to be measured, regression-tested, and improved independently, and for a broad agent serving many request types it is usually worth the extra structure. The decision should be made deliberately, because an understanding layer that is invisible cannot be evaluated, and a component that cannot be evaluated cannot be trusted to improve.

28.4 The Reasoning Engine

The reasoning engine is the agent’s centre. It is the component that decides what to do: whether the agent can answer directly, whether it needs to plan, whether it needs to act, whether the goal has been reached. It is typically the component most directly powered by a foundation model, and the reasoning strategies it uses — chain-of-thought, ReAct, plan-and-execute, tree-of-thought — are the subject of several later chapters.

The reasoning engine’s first decision, on most cycles, is the one the infographic labels *can I respond directly*. If the request is simple enough that the agent already has what it needs to answer, the reasoning engine produces the answer and the cycle ends. If not, the reasoning engine routes the request onward — to the planning module if the work needs to be decomposed, to the action layer if a tool call is needed, to the memory system if context must be retrieved first. The reasoning engine is, in control-flow terms, the dispatcher: every cycle of the agent’s loop passes through it, and it decides where the cycle goes next.

Failure modes. The characteristic reasoning-engine failure is the confident wrong route: the engine decides it can answer directly when it cannot, or decides it needs an elaborate plan when a direct answer would have served. Both are expensive — the first in quality, the second in cost and latency. The mitigation is calibration: the reasoning engine’s routing decisions should be evaluated against outcomes, and the engine tuned or prompted so that its confidence tracks its actual accuracy. A second failure mode is the reasoning loop that does not terminate: the engine keeps deciding more work is needed and never concludes the goal is met. The mitigation is the hard budget — a maximum number of cycles, enforced by the control layer, after which the agent must conclude or escalate.

Design note. The choice of reasoning strategy — chain-of-thought, ReAct, plan-and-execute, tree-of-thought — is the most visible reasoning-engine decision, but it is not the most consequential. The more consequential decision is which model tier powers the engine, because the reasoning engine runs on every cycle and its cost and latency dominate the agent’s. A frontier model in the reasoning engine produces better routing decisions but multiplies the agent’s per-cycle cost; a smaller model

is cheaper but routes less reliably. The disciplined approach is to treat the model-tier choice for the reasoning engine as an explicit, evaluated decision — measured against routing accuracy and cost together — rather than defaulting to the largest model available.

28.5 The Planning Module

The planning module decomposes a goal into an ordered sequence of smaller tasks. When the reasoning engine determines that a request cannot be answered directly and is too complex to handle in a single action, it hands the goal to the planning module, which produces a task queue: step one, then step two, then step three, each step being something the agent can actually execute. The planning module is what turns “onboard this customer” into the concrete sequence of retrievals, tool calls, and checks that onboarding actually requires.

The planning module is architecturally distinct from the reasoning engine even though both are commonly powered by the same foundation model. Reasoning decides *whether and what*; planning decides *in what order*. Keeping them distinct in the architecture — even when they share a model — matters because they have different failure modes, different evaluation needs, and different points at which a human might intervene. A plan can be reviewed before execution begins; a reasoning decision often cannot.

Failure modes. The characteristic planning failure is the plan that is locally sensible but globally wrong: each step is reasonable, but the sequence does not actually achieve the goal, or omits a step that turns out to be necessary. The mitigation is plan validation before execution — a review step, automated or human, that checks the plan against the goal before the agent begins acting on it. A second failure mode is the brittle plan: the plan assumes each step will succeed, and has no branch for the step that fails. The mitigation is for the planning module to produce plans with explicit failure handling, and for the reasoning engine to be re-invoked when a step fails rather than the agent blindly proceeding to the next step.

Design note. The planning module is the component where human-in-the-loop review is cheapest and most effective, because a plan is a legible artefact that a person can inspect before any action is taken. An agent that exposes its plan for approval before execution gives a human reviewer a single, comprehensible decision point; an agent that plans implicitly inside the reasoning loop offers no such point. For consequential work, making the planning module’s output an explicit, reviewable artefact — rather than an internal intermediate — is one of the highest-value structural decisions in the whole anatomy.

28.6 The Memory System

The memory system gives the agent state that persists beyond a single model call. It has two tiers. Short-term memory holds the current interaction: the chat history, the current task state, the intermediate results of the plan in progress. Long-term memory holds what persists across interactions: a vector database of past events, learned patterns, prior decisions, durable facts about the users and systems the agent works with.

The memory system is given a fuller treatment in its own chapter (Chapter 40); the role it plays in the reference anatomy is as the component the reasoning engine consults to ground its decisions and writes to so that future cycles can benefit. Every cycle of the agent’s loop both reads from memory (to bring relevant context into the reasoning engine) and may write to memory (to record what was decided and what was found).

Failure modes. The characteristic memory failure is context pollution: irrelevant or stale material from memory is retrieved into the reasoning engine’s context, and the agent reasons over noise. The mitigation is relevance filtering on retrieval and freshness management on the stored material. A second failure mode is unbounded memory growth: the long-term store accumulates without curation, retrieval slows, and the relevant material becomes harder to surface. The mitigation is memory curation as an explicit operational discipline — summarisation, consolidation, and expiry of stored material.

Design note. The memory system is also a privacy and governance surface, and its design intersects directly with the data-protection obligations of Chapter 97. Long-term memory that records facts about users is a store of personal data, subject to the same rights of access, rectification, and erasure as any other such store. An agent’s memory architecture must therefore be designed so that a user’s data can be located and removed on request — which an undifferentiated vector store of accumulated interactions makes surprisingly hard. Designing for erasure from the start is far cheaper than retrofitting it.

28.7 The Action and Tool Layer

The action and tool layer is where the agent affects the world. When the reasoning engine determines that a tool call is needed, the action layer executes it: a file access, a database search, an API request, a code execution, a call to an external system. The action layer is increasingly mediated by the Model Context Protocol and the MCP server patterns of the previous chapter — the tools the agent invokes are MCP tools, and the action layer is the agent’s MCP client.

The action layer is the agent’s second major security boundary, after the input layer. It is where the agent’s capabilities meet the systems it can affect, and it is the natural place to enforce which actions the agent is permitted to take, which require human approval, and which are forbidden. The action layer is also where the agent’s effects become real and, often, irreversible — a message sent, a transaction committed, a record changed — which makes it the component where the reversibility and audit disciplines matter most.

Failure modes. The characteristic action-layer failure is the unintended side effect: the agent calls a tool that does more than the agent intended, or calls the right tool with the wrong arguments, and the effect cannot be undone. The mitigation is the trust-boundary discipline of the previous chapter — sandboxed execution for code, constrained tool inventories, explicit human-in-the-loop checkpoints for actions with material consequences. A second failure mode is the tool error that the agent mishandles: a tool call fails, returns an error, and the agent either ignores the error or misinterprets it. The mitigation is structured error propagation from the action layer back to the reasoning engine, so the

agent can adapt to the failure rather than proceeding as if the call had succeeded.

Design note. The single most useful distinction to encode in the action layer is reversibility. Actions that can be undone — a draft saved, a record read, a reversible status change — can be taken with light controls, because a mistake is recoverable. Actions that cannot be undone — a payment sent, an email dispatched, an external transaction committed — warrant the heaviest controls the safety layer offers, including human review. An action layer that treats all tool calls identically either over-controls the reversible ones, making the agent slow, or under-controls the irreversible ones, making it dangerous. Classifying the tool inventory by reversibility is what lets the controls be calibrated to the actual stakes.

28.8 The Monitoring Layer

The monitoring layer is the first of the two cross-cutting components. It does not sit at one point in the processing pipeline; it observes the entire pipeline. It records what the agent received, what the understanding layer made of it, what the reasoning engine decided, what the planning module produced, what tools the action layer invoked, what came back, and what the agent finally did. It captures the metrics — latency, cost, cycle count, success rate — by which the agent’s operation is judged. It carries the content checks, the rate controls, and the human-review checkpoints that need to observe the agent’s behaviour as it runs.

The monitoring layer is what makes an agent debuggable. An agent without it is a black box: when it produces a wrong answer, there is no way to determine which component was responsible. An agent with it produces a trace — a complete record of the cycle — that lets an engineer see exactly where the reasoning went wrong, which retrieval polluted the context, which tool call failed. The monitoring layer is also what makes an agent governable: the audit trail it produces is what an architecture review board, a compliance officer, or a regulator inspects.

Failure modes. The characteristic monitoring failure is the incomplete trace: the monitoring layer records some components’ activity but not others, and the gap is exactly where the failure occurred. The mitigation is to make tracing a mandatory pass-through for every component, so no cycle can complete without a complete trace. A second failure mode is monitoring that records everything and surfaces nothing: the data is captured but no dashboards, alerts, or analyses turn it into operational signal. The mitigation is to treat the monitoring layer’s outputs as a product with consumers, not as a log that exists in case someone needs it.

Design note. The monitoring layer should be designed against its consumers, of which there are at least three with different needs. The operations engineer needs real-time signal — latency, error rate, cost rate — to know the agent is healthy now. The quality engineer needs aggregated traces to evaluate whether the agent’s decisions are good. The auditor needs a complete, immutable, attributable record of what the agent did and why. A monitoring layer designed for only one of these consumers fails the other two, and the three needs are different enough that the layer must be designed for all three from the start rather than extended to them later.

28.9 The Safety and Control Layer

The safety and control layer is the second cross-cutting component, and the one with the most architectural authority. It wraps the entire agent. It enforces access control on who may invoke the agent; approval steps for actions that require them; content checks on both inputs and outputs; rate control on the agent's operation; and the human-in-the-loop checkpoints where a person must review before the agent proceeds. It is also the component that enforces the hard budgets — the maximum cycle count, the maximum cost, the maximum execution time — that prevent the agent's loops from running away.

The safety and control layer is where the agent's autonomy is bounded. Every other component describes what the agent *can* do; the safety layer describes what it *may* do. The distinction is the difference between capability and authority, and it is the distinction that the governance disciplines of the rest of the book turn on. An agent whose safety layer is an afterthought is an agent whose autonomy is unbounded by construction, and unbounded autonomy is the failure mode that the most consequential agent incidents trace back to.

Failure modes. The characteristic safety-layer failure is the bypassed control: a path through the agent that does not pass through the safety layer, so the controls that should apply do not. The mitigation is architectural — the safety layer must be positioned so that no component can be reached except through it, rather than being a check that well-behaved code paths happen to call. A second failure mode is the control that is present but miscalibrated: the human-in-the-loop checkpoint fires on every action and the humans, overwhelmed, approve without reviewing. The mitigation is to calibrate the controls to the actual risk — frequent low-stakes actions proceeding automatically, rare high-stakes actions stopping for genuine review.

Design note. The safety and control layer should be designed so that its enforcement is structural rather than cooperative. A control that depends on every component remembering to call it is a control that will eventually be bypassed by a code path that forgets. A control that is positioned so that the protected resource is physically unreachable except through it cannot be bypassed by oversight. This is the same principle as the trust-boundary discipline of the MCP chapter: the boundary works because it is the only way through, not because the code is polite. An architect reviewing an agent design should ask, for each safety control, whether it is structural or cooperative — and treat every cooperative control as a future incident.

28.10 The Control Loop

The eight components are connected by a control loop, and the loop is where the agent's behaviour actually lives. A cycle of the loop runs as follows. Work enters through the input layer and is normalised. The understanding layer interprets it. The reasoning engine receives the interpreted request, consults memory for relevant context, and makes its first decision: *can I respond directly?*

If yes, the agent produces its output and the cycle ends. If no, the reasoning engine decides what the request needs. If the work must be decomposed, it goes to the planning module, which produces a task queue, and the agent begins working through the queue. If a tool call is needed, it goes to the

action layer, which executes the tool and returns the result to the reasoning engine. After each action, the reasoning engine makes its second recurring decision: *is the goal completed?*

If the goal is completed, the agent produces its output and the cycle ends. If not, the agent loops: back to the reasoning engine, which decides the next step, which may be another plan, another action, another memory retrieval. The loop continues until the goal is met, until the agent determines it cannot proceed, or until the safety and control layer's budget is exhausted and forces the cycle to conclude or escalate.

Throughout, the monitoring layer records every step, and the safety and control layer governs every transition. The two cross-cutting components are not stages in the loop; they are present at every stage of it. This is why the reference anatomy draws them as wrapping the pipeline rather than sitting inside it.

The two recurring decisions — *can I respond directly* and *is the goal completed* — are the agent's two control points, and most of an agent's observable behaviour is determined by how well-calibrated those two decisions are. An agent that answers directly when it should not produces confident wrong answers. An agent that plans when it should answer directly is slow and expensive. An agent that declares the goal complete prematurely stops short; an agent that never declares the goal complete loops until its budget runs out. The reference anatomy makes these two decisions visible as named architectural elements, so that they can be evaluated, monitored, and tuned rather than buried inside an undifferentiated agent loop.

It is worth being explicit that the loop must always terminate, and that termination is not guaranteed by the two decision points alone — both are model judgements, and both can be wrong indefinitely. The loop terminates reliably only because the safety and control layer imposes a hard ceiling: a maximum cycle count, a maximum cost, a maximum wall-clock time, any of which forces the loop to conclude or escalate regardless of what the reasoning engine believes. The two decision points govern the loop's normal behaviour; the budget governs its worst case. An agent designed with only the decision points, and no budget, has no defined worst case, and an agent with no defined worst case will eventually find it.

28.11 From Anatomy to Patterns

The reference anatomy of this chapter is the foundation the rest of Part IV builds on. The chapter on multi-agent orchestration describes what happens when many agents, each with this anatomy, are composed into a system. The chapter on the planner-executor pattern is a detailed treatment of the relationship between the planning module and the action layer. The chapter on tool use is a detailed treatment of the action and tool layer. The chapter on memory and context is a detailed treatment of the memory system. The chapter on human-in-the-loop is a detailed treatment of the checkpoints inside the safety and control layer. The chapter on evaluation architecture is, in large part, a treatment of how the monitoring layer's outputs are turned into quality signal. Each of those chapters takes one component or one composition of this anatomy and develops it.

The reader who carries the eight-component decomposition into those chapters will read them as

elaborations of a shared reference rather than as a series of unrelated techniques. When the multi-agent chapter describes agents delegating to other agents, the reader can see that the delegation happens at the action layer of the delegating agent and the input layer of the delegate. When the human-in-the-loop chapter describes an approval checkpoint, the reader can place it precisely in the safety and control layer. The anatomy is the map; the pattern chapters are the territory.

This is also why the anatomy comes first in Part IV rather than being distributed through it. A reader who meets the planner-executor pattern, the memory chapter, and the tool-use chapter without a shared reference encounters them as three separate topics and must reconstruct the connections themselves. A reader who has the anatomy sees immediately that those three chapters develop three named components of one structure, and that the structure is what holds them together. The decomposition is, in the language of the book's opening, a Linnaean move: a shared vocabulary that turns a collection of techniques into a navigable discipline.

28.12 The African Perspective

The reference anatomy is geography-neutral — an agent has the same eight components in Lagos as in London. What differs is where the architectural pressure falls.

The general principle is that constraint shifts which components carry the most weight. In a well-resourced deployment, all eight components can be elaborated generously and the architect's attention is distributed evenly. In a constrained deployment, three components — the safety layer, the monitoring layer, and the reasoning engine — absorb a disproportionate share of the design effort, for the reasons the paragraphs below set out. Recognising which components the local constraints load most heavily is itself an architectural skill, and the anatomy is what makes that recognition possible: it gives the architect named components to reason about rather than an undifferentiated agent.

The safety and control layer carries more weight in environments where the cost of an agent error is harder to absorb. An organisation operating with thin margins, in a market where a single large mistake is materially more damaging, has a stronger reason to keep its agents' autonomy tightly bounded and its human-in-the-loop checkpoints conservatively placed. The anatomy supports this directly: the safety layer is a named component, and the decision about how much autonomy to grant is a decision about how that component is configured.

The monitoring layer carries more weight where the cost surface needs continuous attention. An agent whose cycles each consume foreign-exchange-denominated model calls has a cost profile that the monitoring layer must make visible in something close to real time, so that a runaway loop is caught in minutes rather than discovered on the monthly invoice. The monitoring layer's cost-tracking responsibility, optional-feeling in a well-resourced deployment, is load-bearing in a cost-constrained one.

The reasoning engine carries a particular consideration where the agent must operate across languages. An agent serving users in Swahili, Hausa, Yoruba, or Amharic depends on a reasoning engine whose underlying model handles those languages well, and the understanding layer's intent classification will be weaker on languages the model has seen less of. The architectural response is the one the book

has recommended throughout: evaluate the understanding and reasoning components per language as first-class metrics, and do not assume that an agent which performs well in English performs equally well in every language its users speak.

28.13 What the Anatomy Gives the Architect

The eight-component reference anatomy gives the architect a precise vocabulary for a thing the field describes loosely. An architect who says “we built an agent” has said little. An architect who says “the input layer enforces access control and rate limiting; the understanding layer is a fine-tuned classifier evaluated per language; the reasoning engine uses ReAct with a twelve-cycle budget; the planning module produces validated plans; the memory system has a curated long-term vector store; the action layer invokes tools through an MCP gateway with sandboxed code execution; the monitoring layer produces a complete per-cycle trace; and the safety layer enforces human review on all irreversible actions” has said something an architecture review board can examine, an operations team can run, and a regulator can audit.

The anatomy is also a checklist. An agent missing one of the eight components is missing it for a reason, and the reason should be a deliberate architectural decision rather than an oversight. An agent with no planning module can only do single-step work — which may be exactly right, but should be a choice. An agent with no monitoring layer cannot be debugged — which is rarely right, and almost never a choice anyone would defend if it were named. The reference anatomy makes the components visible enough that their absence becomes a decision rather than an accident. That is what a reference anatomy is for, and it is the foundation the agent patterns of the rest of this part are built on.

Used as a review instrument, the anatomy turns an agent design review from an open-ended conversation into a structured one. The reviewer walks the eight components in turn and asks, for each: is this responsibility assigned, is its implementation proportionate to the agent’s stakes, what is its characteristic failure mode, and what mitigates that failure here. Eight questions, asked the same way every time, catch the omissions that an unstructured review misses — the absent budget, the cooperative-rather-than-structural safety control, the memory system with no erasure path. The anatomy does not guarantee a good agent. It guarantees that the design’s gaps are visible, which is the precondition for closing them.

Chapter 29

Pattern: Multi-Agent Orchestration

“Two agents can do what one cannot. Twenty agents can do what nobody wants.”

—a recurring observation in agent-system design

Chapter Overview

Multi-agent orchestration is the discipline of composing specialised agents into coordinated systems. The agentic-mesh pattern (Chapter 18) established the high-level architectural form; this chapter develops the operational mechanics. It works through the recurring topologies a multi-agent system organises into, the coordination protocols that hold those topologies together, the handoff and failure behaviour that determines whether they survive contact with production, the observability and measurement that tell an architect whether the design is earning its cost, and the anti-patterns that turn multi-agent systems into expensive failures. The discipline throughout is restraint: use multi-agent designs where decomposition genuinely improves outcomes, and resist them where a simpler single-agent or single-prompt design would serve.

Learning Objectives

- Decide between multi-agent and single-agent designs deliberately, against the cost of coordination.
- Recognise the recurring orchestration topologies and select the most structured one a problem permits.
- Design coordination protocols, handoffs, and escalation across agents.
- Handle the multiplied failure modes that multi-agent systems introduce.
- Measure multi-agent performance honestly against a single-agent baseline.
- Recognise the anti-patterns, and know when to refactor back toward simplicity.

29.1 When to Reach for Multiple Agents

Every multi-agent design begins with a decision that is easy to make badly: the decision to add a second agent at all. It is easy to make badly because adding an agent feels like adding a capability, and capability feels free. It is not. Each additional agent introduces message passing, context transfer, interacting failure modes, and observability burden, and these grow faster than linearly as agents

accumulate. A two-agent system is meaningfully more complex than a one-agent system; a ten-agent system is far more than five times the complexity of a two-agent one. The decision to go multi-agent is therefore not a decision to add a feature. It is a decision to accept a category of complexity, and it deserves the same deliberateness as the decision to introduce a distributed system where a monolith would have served.

The case for accepting that complexity rests on one idea: that the work has a genuine seam. Multi-agent design pays off when a task decomposes into substantially different sub-tasks that need different tools, models, or contexts; when parallel sub-tasks can run independently and cut wall-clock time; when specialisation produces materially better outputs through different prompting, retrieval, or validation; when long-horizon work benefits from an explicit separation of planning from execution; or when multiple perspectives genuinely improve quality, as in a generator-critic pair. What unites these conditions is that the decomposition reflects real structure in the task rather than structure imposed for its own sake. A useful test is whether a competent human team would naturally divide the work the same way. If they would, the seam is real and the decomposition will repay its cost. If the division feels arbitrary — if it exists because the architecture diagram looked better with more boxes — the seam is invented, and the coordination cost will not come back.

The case against is the mirror image. Multi-agent design adds cost without value when the decomposition is artificial and a single capable agent would handle the task more cheaply; when coordination overhead exceeds the benefit of specialisation; when the sub-agents do not really have distinct roles; when the same context is reassembled by every agent; and, most expensively, when failure modes multiply faster than capability. That last point is the one architects underestimate. A single-agent system has a bounded, comprehensible set of failure modes. A multi-agent system has those failure modes for each agent, plus the coordination failures between them, plus the cascading failures where one agent's failure propagates to others. Capability grows at best linearly with each agent added; the failure surface grows combinatorially. An architect who adds agents for a modest capability gain, without noticing that the debugging and incident surface has multiplied, has made the system harder to operate than the gain can justify.

The working stance that follows is to start with the simplest design that works. A single well-prompted agent with appropriate tools beats a multi-agent system on most tasks, and multi-agent designs should be adopted only when the decomposition is clear and the benefit is demonstrable. This stance has a moving boundary, and the direction of movement matters: every time a more capable foundation model arrives, the set of tasks a single agent can handle competently expands, and some decompositions that genuinely earned their coordination cost a year ago no longer do. The discipline is not only to start simple but to stay alert to the simple design quietly becoming sufficient for work that once required the complex one — a point the chapter returns to when it discusses refactoring.

29.2 The Recurring Topologies

When a multi-agent design is warranted, it almost always takes one of a small number of recurring shapes. Knowing the shapes is most of the design work, because each one fits a particular kind of task structure and carries a particular coordination cost. Figure 29.1 shows the four that recur most

often; the paragraphs that follow describe each, and the discussion of selection and cost that closes the section explains how to choose among them.

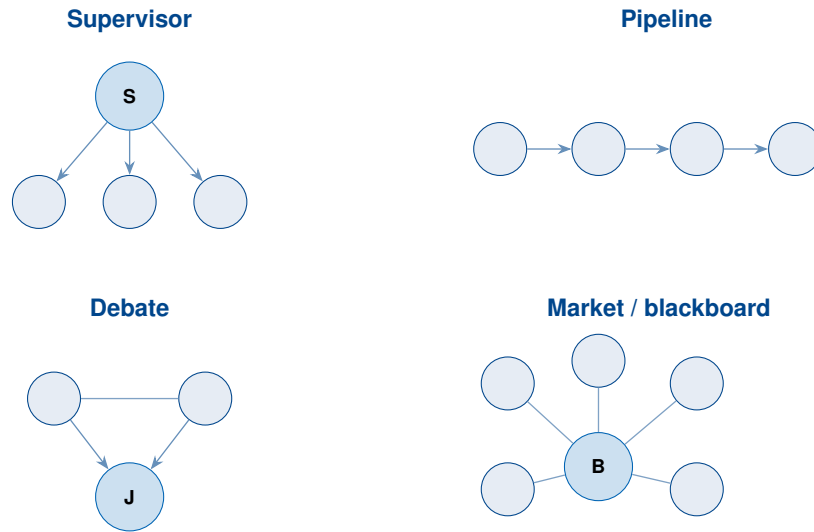


Figure 29.1: Four recurring multi-agent topologies. *Supervisor*: an orchestrator delegates to specialists. *Pipeline*: agents form a directed processing chain. *Debate*: peers argue and a judge decides. *Market / blackboard*: agents contribute to and consume from shared state.

The **supervisor** topology — often called hierarchical, or manager-and-workers — places one orchestrating agent above a set of specialists. The supervisor holds the overall plan, dispatches scoped sub-tasks to workers chosen by domain or tool access, tracks progress, and integrates the results. A close relative is the *planner-executor* split, in which a planner agent decomposes the task into steps and executor agents perform them, with the planner re-engaging whenever a step fails or returns something unexpected; planner-executor is essentially a supervisor topology whose defining feature is that the plan is an explicit, inspectable artefact passed between agents rather than an internal intermediate, which makes human review of the plan before execution natural. The supervisor family is the right choice for divisible problems with clearly typed sub-tasks. Its appeal is partly that the organisational analogy is intuitive — everyone understands a manager dispatching work to specialists — but that analogy is also its trap: organisational charts are shaped by human constraints such as span of control and communication bandwidth that do not apply to agents, and copying an org chart into an agent architecture imports those constraints for no reason. The supervisor topology is right when the work genuinely has a coordinating role and distinct specialist roles; it is wrong when it has merely been drawn to resemble a company. Its characteristic cost is the orchestrator itself, which is invoked on every turn and can become a bottleneck; planner-executor variants share this, since every re-plan routes through the single planner. The planner-executor pattern receives a fuller treatment in its own chapter, Chapter 39.

The **pipeline** topology arranges agents in a directed chain, each transforming the output of the previous one — research, then outline, then draft, then edit. It is the most underrated of the four because it is the least exciting: a fixed sequence of stages has none of the apparent sophistication of dynamic routing or autonomous debate. But for a workflow that genuinely has a fixed sequence of transformations,

the pipeline is almost always the right answer, precisely because it is predictable, easy to monitor, easy to debug stage by stage, and cheap to operate. Its characteristic risk is that failure at any stage cascades down the chain, so recovery design matters; and it is brittle to changes in the plan, since the sequence is wired in. The architectural mistake the pipeline guards against is reaching for a more dynamic pattern when the workflow is in fact fixed — buying routing flexibility the task does not need and paying for it in complexity. When the stages are clearly distinct and stable, the pipeline’s lack of glamour is a feature.

The **debate** topology places agents as peers who produce work and critique one another. In its generator-critic form, one agent produces, another critiques, and the producer revises; in its fuller form, agents argue different positions and an arbiter selects or synthesises. Debate is genuinely useful for quality improvement on subjective tasks — legal arguments, code review, ethical assessments — but it carries an empirical hazard that the architect must respect: quality plateaus quickly while cost grows linearly with rounds. The first critique typically catches the most consequential errors; the second catches fewer; by the third or fourth round the critic is largely producing stylistic quibbles while the cost has tripled or quadrupled. The discipline is to measure where the plateau falls for the actual task — usually one or two rounds — and to cap the rounds there, rather than letting the pattern run until the agents stop finding things to say. A debate with no round cap is a cost-explosion anti-pattern wearing the costume of a quality practice.

The **market**, or blackboard, topology has agents read from and write to a shared workspace, with coordination emerging implicitly from the shared state rather than from explicit messages. It suits genuinely open-ended work — research synthesis, brainstorming — where sub-tasks are interdependent, no fixed decomposition exists, and the order of work is data-driven. It is also the hardest of the four to operate. Trading explicit coordination for implicit coordination through shared state has a characteristic cost: the system’s behaviour becomes harder to predict and harder to trace, because the path of a task is no longer visible in a sequence of messages but is an emergent property of which agents happened to read which state in which order. This makes the blackboard powerful for open-ended work and dangerous for work that needs predictable, auditable execution. Conflict resolution — defining what happens when two agents write overlapping state — is not an optional refinement here; without it the blackboard’s behaviour is not merely hard to predict but undefined.

The four topologies sit on a spectrum from most structured to least. The supervisor and pipeline impose a known shape on the work and are predictable and easy to operate; debate and market allow more emergent behaviour and are correspondingly harder to operate and harder to bound. The selection principle follows directly from that spectrum: *choose the most structured topology the problem actually permits*. If the work has a fixed sequence, the pipeline’s predictability is free quality and there is no reason to reach past it. If the problem is divisible into typed sub-tasks, the supervisor topology fits. Debate should be reserved for the high-stakes subjective decisions where competing perspectives genuinely raise quality, because its multi-round cost is real. The market topology should be the last resort, chosen only when the problem genuinely resists decomposition, because its operational difficulty is a recurring cost rather than a one-time setup expense.

Every topology, finally, carries a coordination tax, and the tax is not static — it scales. A supervisor

with three workers has a modest orchestrator cost; the same topology with thirty workers has an orchestrator that must reason about thirty parallel streams on every turn, its context and cost growing with the worker count. A pipeline's tax is error compounding through stages and the difficulty of debugging across stage boundaries. A debate with two agents and one round is affordable; the same pattern with five agents and three rounds is fifteen model invocations per task. A market topology's tax is the noise of many partial contributions, which demands strong filtering or moderation. The cheapest topology that solves the problem is usually the right one, and the architect should estimate the coordination tax not at the scale of the prototype but at the scale of the intended production load — because a topology that is cheap at three agents can be the dominant cost at thirty.

29.3 Coordination Protocols

A topology describes how agents are arranged; a coordination protocol describes how they actually communicate within that arrangement. The protocol is a contract between agents about the form and meaning of what they exchange, and it plays the same role for a multi-agent system that an API contract plays for a distributed system. It must be explicit, because agents are non-deterministic: an agent that receives a loosely specified message will interpret it loosely, and small interpretation differences compound across a multi-agent workflow into large behavioural divergence. A well-defined protocol is what keeps the system's behaviour bounded. It has three components — a message schema, a state-sharing model, and a synchrony model — each of which is a real design decision.

The **message schema** defines the message types the system uses — task, result, error, status, query — with typed, validated payloads. The discipline that matters most is to treat the schema as a versioned interface, with the same seriousness as any other API contract. A multi-agent system that runs for years will see its agents updated independently, and an agent updated to emit a new message shape will break every agent still expecting the old one unless the schema evolution was managed. The mitigation is standard distributed-systems practice: additive schema changes, explicit versioning, and a deprecation policy for retired message types. An agent system whose message schema is implicit — whatever shape the sending agent happens to produce — has no contract at all, and will break the first time any agent changes.

The **state-sharing model** governs what agents share beyond direct messages. The guiding principle is that shared state should be explicit and visible rather than implicit and ambient. Explicit shared state — a named blackboard, a context object passed by reference — can be inspected, traced, and reasoned about. Implicit shared state, where agents come to depend on side effects of one another's execution, produces behaviour that is reproducible only by accident. The cost of explicitness is some verbosity in the protocol; the cost of implicitness is a class of bug that appears intermittently, resists reproduction, and is among the hardest in any multi-agent system to diagnose. The verbosity is worth paying. Where state genuinely must be shared across many agents, the blackboard topology makes the sharing explicit by construction; where it need not, context should be passed per task rather than left ambient.

The **synchrony model** chooses between synchronous coordination, in which a caller waits for a response, and asynchronous coordination, in which the caller continues and collects results later. The

honest default is synchronous, despite asynchronous designs being more powerful. Synchronous coordination is dramatically easier to reason about, debug, and trace — the control flow is linear and the failure points are obvious — and most multi-agent workloads do not have enough genuine parallelism to justify the reasoning cost of going asynchronous. Asynchronous coordination earns its complexity only when there is real parallel work to exploit: independent sub-tasks that can run concurrently and meaningfully reduce wall-clock time. Whichever model is chosen, timeout and cancellation discipline is not optional, because an agent waiting on a response that will never arrive is a deadlock, and a multi-agent system without timeouts at every layer will eventually find one.

29.4 Handoff and Escalation

A handoff is the moment one agent transfers work to another, or to a human, and it is the moment a multi-agent system is most likely to lose information — because it is the moment context must cross a boundary. The transferring agent holds a rich internal state: what it has learned, what it has tried, why it is handing off. The receiving agent starts with only what the handoff actually carried. A handoff that transfers too little leaves the receiver to rediscover what the sender already knew; a handoff that transfers too much floods the receiver’s context with irrelevant history. Designing handoffs well is therefore a central multi-agent discipline, and it has three parts: deciding when to hand off, deciding what context to carry, and handling the special case of handing off to a human.

When to hand off should be governed by explicit, inspectable triggers rather than left to agent judgement alone. The triggers that recur in practice are a capability boundary, where the current agent lacks the tools or knowledge to proceed; a confidence threshold, where the agent’s uncertainty exceeds a defined level; a policy rule, where certain decisions are reserved for specific agents; a failure, where the current agent has failed and an alternative is tried; and cost, where a cheaper agent suffices and an expensive one is unnecessary. The reason these should be explicit is auditability. A handoff that happens because an agent decided, somewhere in its reasoning, that another agent should take over is hard to audit and hard to predict. A handoff that happens because a named, observable trigger fired — confidence below a threshold, a policy rule matched — can be monitored, tuned, and explained. Encoding the triggers into the coordination protocol means every handoff in the system’s trace can be attributed to the specific condition that caused it.

What context to carry is governed by a single design move: separate the context that is carried from the context that is retrievable. Carrying the full history of every prior agent in every handoff causes context explosion — by the fifth handoff, the receiving agent’s context is mostly the accumulated debris of the first four. The better pattern carries a focused summary plus a stable identifier, and lets the receiving agent retrieve the full history through that identifier only if it actually needs it. This keeps the working context small while preserving the full record for the cases that require it. It is the same principle the memory chapter applies inside a single agent, lifted to the relationships between agents.

Escalation to a human is a handoff with a particular asymmetry: the human’s time is the scarcest and most expensive resource in the system, so the context package matters more here than in any agent-to-agent handoff. A human handed a thin context must reconstruct the situation before acting,

which wastes the scarce resource; a human handed an overwhelming context must search it before acting, which wastes it differently. The escalation package should be designed deliberately — the situation, what the agents tried, why they escalated, and the specific decision the human is being asked to make — so the human can act quickly and well. Capturing the human’s response is what lets the system learn which escalations were necessary and which were avoidable. The human-in-the-loop chapter, Chapter 35, develops the escalation surface in full.

29.5 Failure Handling

The central operational fact about multi-agent systems, and the one most often underestimated at design time, is that they multiply failure modes. A single agent fails in the ways the agent-anatomy chapter enumerated. A multi-agent system fails in all of those ways for each agent, and then in a further set of ways that exist only because there are multiple agents. An architect who has budgeted for the single-agent failure modes and not for the multiplicative ones has under-budgeted for operations, and will discover the gap during the first multi-agent incident.

The multi-agent failure categories worth naming explicitly are these:

- **Agent failure** — one agent crashes or returns invalid output.
- **Coordination failure** — messages are lost, or agents deadlock or time out waiting on one another.
- **Cascading failure** — one agent’s failure propagates and breaks others downstream.
- **Infinite loops** — agents call one another without making progress.
- **Cost explosion** — agents recursively spawn sub-tasks without bound.

The last two deserve particular attention, because they fail in the direction of consuming unbounded resources rather than simply stopping. An infinite loop between agents, and a recursive cost explosion, both produce a system that is, from the outside, busy and apparently working — it is making model calls, it is passing messages — while in fact making no progress and spending money at speed. A crashed agent announces itself; a runaway loop does not. This is why the loop and budget controls below are not optional refinements but core requirements of any multi-agent design.

The mitigations are, with one exception, familiar from distributed-systems practice: retry policies with exponential backoff, circuit breakers to stop cascading failure, timeouts at every layer, step and depth limits, idempotency wherever it can be arranged, and comprehensive logging for post-mortems. That familiarity is the point. A multi-agent system is a distributed system, and the decades of hard-won distributed-systems discipline apply to it directly; an agent architecture that omits these defences is repeating mistakes the field already paid for. The one mitigation with an agent-specific sharpness is budget enforcement — a hard ceiling on tokens, cost, and time. Because agents spend money on every model call, a multi-agent system’s runaway failure has a direct financial cost that a runaway in a conventional distributed system does not. The hard cost ceiling is therefore the single most important defence on the list, and the one whose absence turns the cost-explosion failure category from a risk into a certainty.

29.6 Observability for Multi-Agent Systems

Observability is both harder and more important for multi-agent systems than for single-agent ones. It is harder because the thing to be observed is no longer one agent's trace but the interleaved traces of many agents, with the causal connections between them — which agent's output became which agent's input — being exactly the part that is hardest to capture and most necessary to understand. It is more important because the multi-agent-specific failures, the loops and cascades and cost explosions, are diagnosable only from a system-wide view: no single agent's trace reveals that two agents are looping between each other. An observability design adequate for a single agent is, almost by definition, inadequate for a multi-agent system.

What a multi-agent system must capture extends the single-agent telemetry in a specific direction. It records agent invocations — who ran, when, with what context; the decisions that shaped the run, such as planner choices and routing decisions; the tool calls, tagged with which agent made them and with what parameters and results; and the inter-agent messages themselves. It also records cost and latency per agent and per task. But the item that genuinely distinguishes multi-agent observability is the *total trace per task across all agents* — the unified, end-to-end record that follows one task through every agent that touched it. This is what the trace identifier, attached when the task enters the system and preserved across every handoff, exists to make possible. Without a unified per-task trace, an operator debugging a failed task has only a scatter of per-agent fragments and must reassemble the causal story by hand; with it, the operator can see the whole path the task took.

A unified trace is necessary but not sufficient, because a multi-agent trace is too large and too interconnected to comprehend as raw text. It must be made legible through visualisation: task trees that show the decomposition and execution at a glance, revealing which agent spawned which sub-task and where the branches went; timeline views that show parallelism and idle waiting, which is where latency problems hide; per-agent statistics; failure-path visualisations; and comparison views across runs for evaluation. The practical test of a multi-agent observability setup is simple — can an engineer, handed a failed task, find the responsible agent and the responsible decision in minutes rather than hours? If the answer is hours, the system has telemetry but not observability, and the visualisation layer is what closes the gap.

29.7 Measuring Multi-Agent Systems Honestly

The word that matters in this section's title is *honestly*. Multi-agent systems are intellectually appealing, and the appeal creates a pull toward measuring them in ways that flatter them: reporting capability gains without their cost, comparing against a weak baseline, or not comparing against a single-agent baseline at all. Honest measurement means comparing the multi-agent system against the strongest simple alternative, on cost and latency and quality together, and being willing to act on the result if the simple alternative wins.

The comparison rests on a baseline that must not be skipped: the *single-agent baseline* — the same task set, run by one capable agent on the current generation of models. It is the comparison that most often overturns a multi-agent design. A multi-agent system measured only against its own earlier

versions will appear to improve; measured against a single capable agent, it may turn out to be slower, costlier, and no better. Running the single-agent baseline is cheap — it is one agent on the same task set — and it is the measurement that tells the architect whether the multi-agent complexity is buying anything at all. A useful second baseline is a plain pipeline: a fixed linear decomposition without dynamic routing, which isolates how much of any improvement comes from decomposition itself rather than from orchestration cleverness. Against these baselines the system is measured on cost per successful task, latency per successful task, and output quality. An evaluation that omits the single-agent baseline is not an evaluation; it is a justification.

Cost, in particular, must be decomposed rather than reported as a single number, because two of its components are routinely omitted and routinely decisive. The full decomposition is tokens per agent, tool-call costs, coordination overhead, failed-attempt costs, and human-review costs where applicable. *Failed-attempt costs* are the tokens and tool calls spent on tasks that did not succeed; a multi-agent system with a seventy-percent success rate spends real money on the other thirty percent, and a cost-per-successful-task figure that ignores the failures understates the true cost by the inverse of the success rate. *Coordination overhead* is the token cost of the messages, the summaries, and the orchestrator's per-turn reasoning — work that produces no output directly but is paid for on every task. An honest cost decomposition includes both, because they are precisely the costs that the multi-agent design adds and the single-agent baseline does not have.

29.8 Anti-Patterns and the Path Back to Simplicity

The anti-patterns of multi-agent design share a single signature: each adds structure the task did not ask for. Decomposition for its own sake creates sub-agents that mirror an organisational chart rather than the task's structure. Recursive sprawl lets agents spawn sub-agents without bound. Context replication has every agent reassemble the same context. Coordination overhead grows until it exceeds the capability gain it was meant to buy. Implicit shared state produces unpredictable behaviour. Missing budgets produce cost explosions. Inadequate observability makes debugging impossible. Over-specialisation passes a task through so many hands that the handoffs cost more than the specialisation saves. The single diagnostic question that catches most of these is whether each agent in the system earns its place — whether removing it would make the system measurably worse at the actual task. An agent that cannot pass that test is an anti-pattern, regardless of how reasonable the architecture diagram looks.

The remedy for an anti-pattern is usually to refactor toward a simpler design, and the signals that a refactor is due are recognisable: most invocations follow the same path through the system, which means the dynamic routing can collapse to a pipeline or a single agent; coordination cost dominates the total cost; the failure modes have become unmanageable; or a new, more capable model has reduced the need for specialisation that an earlier model generation required. Refactoring toward simplicity is harder to do than to recommend, because a working multi-agent system represents real invested effort and dismantling part of it feels like regression. The discipline is to treat every multi-agent design as provisional — a structure adopted because the task needed it at the time, not a permanent commitment — and to schedule periodic re-evaluation against the single-agent baseline as the models improve. That

last signal triggers most real refactors: a decomposition that genuinely earned its complexity against last year's models, re-tested against this year's, is found unnecessary and collapsed. An architecture that cannot be simplified when the simplification becomes available is an architecture that has stopped tracking reality.

29.9 The African Perspective

Multi-agent orchestration carries a particular weight in cost-constrained and connectivity-constrained deployments, and that weight falls consistently on the side of restraint.

Cost multiplication is the central concern. Every agent added to a system is another stream of model calls, and in a deployment where those calls are billed in foreign currency against a local-currency budget, the coordination overhead and failed-attempt costs this chapter has insisted on measuring are not accounting niceties — they are the difference between a system that is financially viable and one that is not. The discipline of the single-agent baseline is correspondingly more important, not less: an enterprise operating under foreign-exchange exposure should be more sceptical of a multi-agent design that has not been shown to beat a single capable agent on cost per successful task, because it has less margin to absorb the difference if the design turns out not to.

The connectivity constraint compounds the cost one. Each inter-agent message and each handoff, when agents are distributed across infrastructure with variable connectivity, is another point at which latency accumulates and failure can occur — and a multi-agent system has more such points than a single-agent system by construction. For deployments where low-latency, reliable connectivity to model providers cannot be assumed, the architectural bias should be toward the most consolidated design the task permits: fewer agents, fewer handoffs, fewer coordination round-trips, with multi-agent decomposition adopted only where the task genuinely cannot be served otherwise. The chapter's default stance — start simple — is, in these environments, not merely good practice but often the constraint that makes the deployment possible at all.

29.10 What Multi-Agent Orchestration Requires

Multi-agent orchestration is a powerful pattern, and a costly one, and the discipline of using it well is the discipline of keeping those two facts in view at once. A multi-agent design is warranted when a task has a genuine seam — when decomposition reflects real structure rather than imposed structure — and it is warranted in a specific shape: one of the recurring topologies, chosen as the most structured one the problem permits, since structure bought predictability and predictability is most of operability. The topology is held together by an explicit coordination protocol, survives production through deliberate handoff design and the multiplied-failure controls that a distributed system always needs, and is kept honest by observability that produces a unified per-task trace and by measurement against a single-agent baseline that the architect is genuinely willing to lose to.

The epigraph states the boundary of the discipline. Two agents can do what one cannot; twenty agents can do what nobody wants. Between those two sentences lies every real multi-agent decision, and the chapter's recurring instruction is the one that keeps a design on the right side of the boundary:

start with the simplest thing that works, add agents only where the seam is real and the benefit is demonstrable, measure the result against the simple alternative without flinching, and refactor back toward simplicity the moment a better model makes the complexity unnecessary. A multi-agent system built that way is an asset. One built without that discipline is the twenty-agent system the epigraph warns against — impressive on a diagram, expensive in production, and wanted by no one who has to operate it.

Chapter 30

Agentic AI Patterns: An Enterprise-Architecture Method

“A method is not a guarantee of good architecture. It is, however, a guarantee that you know what you have not yet done.”

—a recurring observation in enterprise-architecture practice

Chapter Overview

Agentic AI introduces capabilities — autonomy, multi-step reasoning, adaptability across changing inputs — that traditional pattern catalogues do not, on their own, organise into a deployment programme. This chapter applies the TOGAF Architecture Development Method (ADM) to agentic AI: a phased, repeatable approach to moving an enterprise from a baseline architecture toward a target state in which agents are deployed responsibly at scale. The chapter introduces the distinction between Architecture Building Blocks (what the architecture provides) and Solution Building Blocks (how those capabilities are implemented), walks through the ADM cycle phase by phase, and surfaces the four concerns — human-in-the-loop governance, trust and explainability, responsible autonomy, and value tracking — that have to run through every phase rather than sitting in any one of them.

Learning Objectives

- Apply the TOGAF ADM cycle to the rollout of agentic AI in an enterprise context.
- Distinguish architecture building blocks (ABBs) from solution building blocks (SBBs) in agentic deployments.
- Identify the agentic-specific work that has to happen in each ADM phase.
- Position the four cross-lifecycle concerns inside a coherent operating model.
- Use the chapter as a structural reference when planning an agentic-AI programme.

30.1 Why a Method, Not Just a Pattern

The agentic patterns described elsewhere in this book — multi-agent orchestration, planner-executor, agentic RAG, self-correction, tool use, human-in-the-loop — are necessary equipment. They are not,

on their own, a programme. An enterprise that wants to move from a few isolated agent pilots to a fleet of agents operating against its real business processes faces a question the pattern catalogue does not directly answer: *in what order do we make the architectural and organisational moves, and how do we know when we have made them well?*

The TOGAF Architecture Development Method is the most widely adopted answer to that question for enterprise IT generally. It is not specific to AI. It does not need to be. The discipline it provides — a phased cycle from preliminary readiness through architecture vision, business architecture, data and application architecture, technology architecture, opportunities and solutions, migration planning, implementation governance, and architecture change management — maps naturally onto the work of rolling out agentic systems, with adjustments at each phase to reflect the things agentic AI does differently. This chapter is that mapping.

A reader familiar with TOGAF will recognise the structure and treat the chapter as a specialisation. A reader unfamiliar with TOGAF will find that the structure makes sense on its own terms: it is a complete-the-cycle method for moving from a vision to a stable production state with the governance scaffolding still in place at the end.

It is worth being clear about what a method does and does not provide. A method does not make architectural decisions; it does not tell an organisation whether a particular agent should be deployed or which model should power it. What it provides is the assurance that no necessary decision has been silently skipped — that the autonomy boundaries were set, the value hypothesis was stated, the rollback path was designed, the governance was specified — and that each decision was made at the point in the programme where it could still be made cheaply. The epigraph’s claim is exact: the method is not a guarantee of good architecture, but it is a guarantee that the organisation knows what it has not yet done. For a technology moving as fast as agentic AI, that guarantee is worth a great deal.

30.2 Building Blocks: ABBs and SBBs

Two terms from TOGAF travel into this chapter and deserve a moment of careful introduction. An *Architecture Building Block* is a logical capability the architecture provides — defined by what it must do, not by which product or service implements it. A *Solution Building Block* is the physical implementation that delivers that capability in production: a specific product, library, service, or configuration.

The distinction matters because agentic AI is a market in which products change quickly. An organisation that commits its architecture to specific solution building blocks — “we use vendor X’s orchestrator” — inherits the vendor’s roadmap and pricing. An organisation that commits its architecture to the capabilities — “we require orchestration that supports human-in-the-loop checkpoints at named decision points” — keeps the freedom to swap the solution as the market evolves, while preserving the architectural commitments that earned the system approval in the first place.

The discipline that operationalises this distinction is documentary: every architectural commitment

is recorded twice, once as the ABB it satisfies and once as the SBB currently delivering it, with the two kept visibly separate. When the architecture review board approves a design, it approves the ABBs — the capabilities and their required properties — and notes the SBBs as the current implementation choice. A later vendor swap then changes only the SBB record, leaves the approved ABBs untouched, and does not require re-approval of the architecture. This is the same separation the hexagonal-architecture chapter enforces at the level of code; here it operates at the level of the architectural record, and it is what allows an agentic programme to survive the vendor churn that the field guarantees.

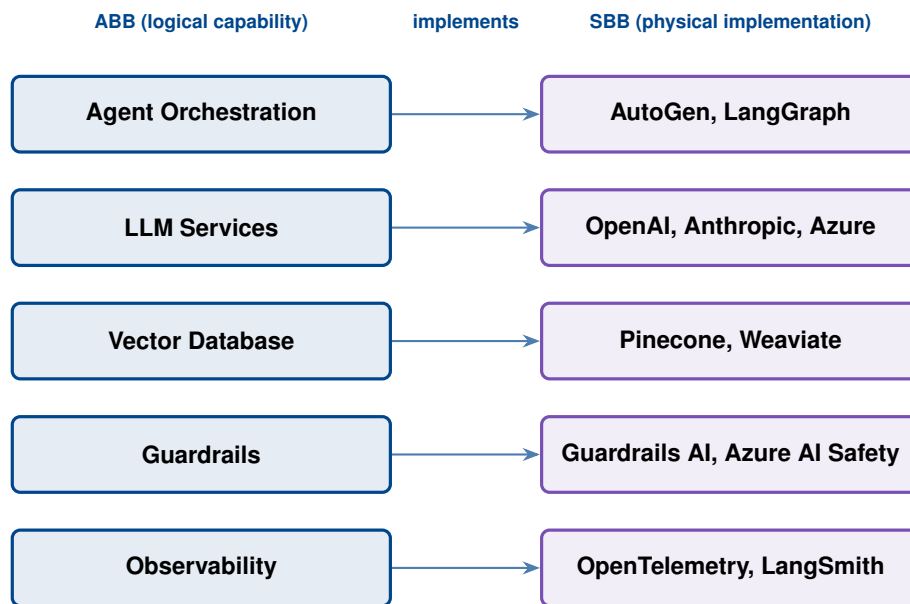


Figure 30.1: Architecture Building Blocks (ABBs) define logical capabilities — what the architecture must provide. Solution Building Blocks (SBBs) are the physical implementations that deliver those capabilities. ABBs are stable across years; SBBs change with the market.

30.3 The ADM Cycle Applied to Agentic AI

The TOGAF ADM is a cycle of nine phases: a Preliminary phase that establishes readiness, followed by eight lettered phases (A through H) that take the architecture from vision through implementation and ongoing change management. Each phase, applied to agentic AI, has its own characteristic work and its own characteristic risks.

The word *cycle* is load-bearing. The ADM is not a sequence that finishes; it is a loop, and Phase H feeds back into the Preliminary phase so that the architecture is continuously revised against a changing environment. For most enterprise IT this looping is a slow background process. For agentic AI it is fast and consequential, because the underlying technology — the models, the tools, the regulatory landscape — changes on a timescale of months. An organisation that treats the ADM as a one-time project, completed at first deployment, has used the method wrong; the method's value for agentic AI is precisely that it institutionalises the expectation of continuous revision.

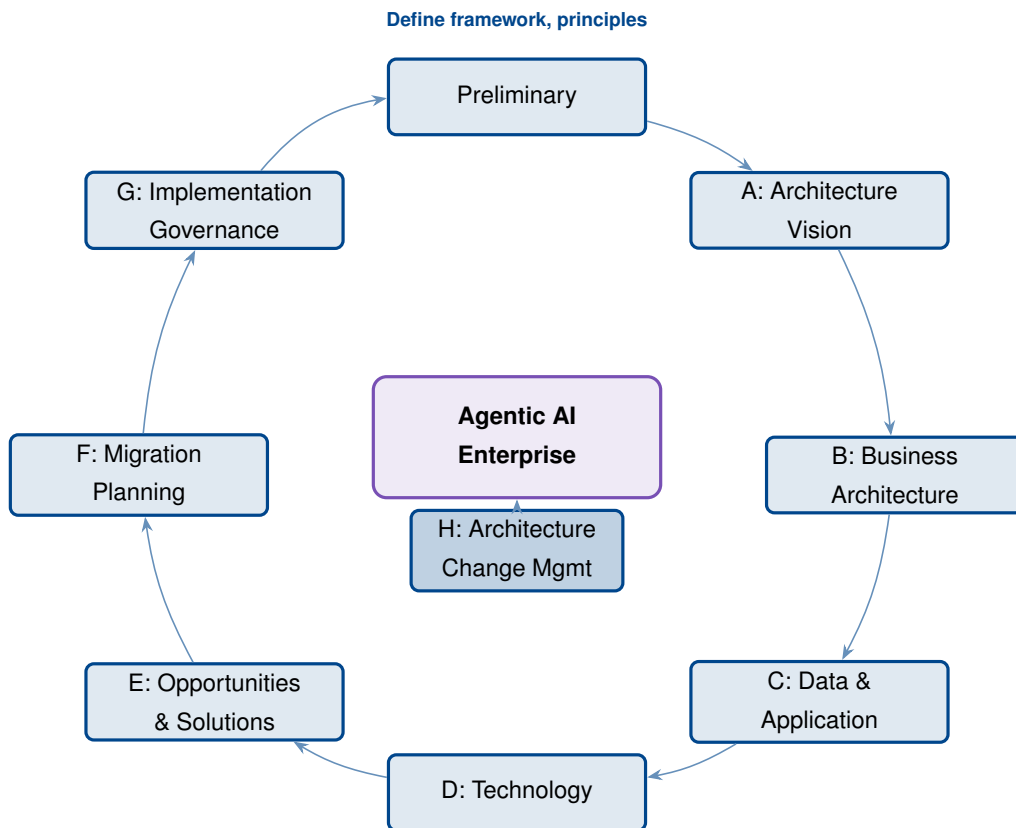


Figure 30.2: The TOGAF ADM cycle applied to agentic AI. The Preliminary phase establishes principles; phases A–D define the architecture across business, data/application, and technology domains; phases E–F move from architecture to plan; phases G–H govern implementation and ongoing change. Phase H is continuous and sits at the centre of the cycle.

30.3.1 Preliminary: Defining the Frame

The Preliminary phase is where the organisation decides what kind of agentic deployment it is trying to do. It sets the principles that will be invoked at every later phase: which tasks are eligible for autonomous handling, where human-in-the-loop is non-negotiable, what risk appetite governs which deployment, what audit trail is required, what regulators expect to see, what the organisation will not delegate to an agent under any circumstances. The output is a small set of architectural principles that survive the cycle.

The work that is specific to agentic AI here is the principle about *autonomy boundaries*. Unlike a traditional application, an agent can take actions in the world. The Preliminary phase is where the organisation decides — in advance, in writing — which kinds of action will require which kinds of approval. The decision is harder to make once the agents are running.

The characteristic risk of the Preliminary phase is that it is skipped, or done thinly, because it produces no running software and an organisation impatient for a demo treats it as overhead. An agentic programme that begins without agreed principles has no fixed reference against which later phases can be judged, and every subsequent autonomy question becomes a fresh negotiation under deployment pressure. The principles are cheap to write before the agents exist and expensive to write after, when each principle now constrains something already running. A thin Preliminary phase is the most

common root cause of an agentic programme that later cannot answer the question of why a particular agent is allowed to do a particular thing.

30.3.2 Phase A: Architecture Vision

Phase A defines what the future-state architecture is meant to deliver. For agentic AI, this is the place where the organisation translates the principles of the Preliminary phase into a concrete vision: which business processes will be augmented or replaced by agents, what success looks like in measurable terms, what the operating model will look like once agents are in production, and what the major architectural capabilities (the ABBs) will be.

The agentic-specific work here is the careful articulation of *value*. An agentic deployment that pays for itself in cost-reduction terms but degrades customer experience is not a win. Phase A is where the multiple objectives are reconciled into a vision the rest of the cycle can be measured against.

The characteristic risk of Phase A is the vision that is inspiring but not measurable. “Transform customer service with AI” is a vision that no later phase can be held to, because nothing about it can be checked. The discipline Phase A must impose is to translate every aspiration into a measurable target — a cost figure, a resolution-time figure, a satisfaction score, a quality threshold — so that Phase G can later report honestly on whether the target was met. A vision that resists measurement is not a vision the method can work with; it is a slogan, and an agentic programme built on a slogan has no way to know whether it has succeeded.

30.3.3 Phase B: Business Architecture

Phase B describes how the business will operate once the target architecture is in place. For agentic AI, this means identifying the business capabilities that agents will enable or transform, the processes those capabilities support, the value streams the agents will run inside, and — critically — the roles and responsibilities that will change. A customer-service agent does not eliminate the need for human agents; it changes what the humans do. Phase B is where that change is described concretely.

The agentic-specific work is the explicit identification of *human-agent collaboration patterns*. Where does the human review the agent’s output? Where does the agent escalate? Where does a human override the agent? These are business-architecture decisions, not technical ones, and they belong in Phase B.

The characteristic risk of Phase B is treating the role change as a technical detail to be sorted out later, when it is in fact the part of the programme most likely to determine whether it is adopted at all. An agentic deployment changes what people do, and people who were not consulted about that change, or who experience it as a threat rather than a redesign of their work, can quietly defeat a technically sound system. Phase B is where the new human roles are designed with the people who will hold them, and an agentic programme that defers this until the agents are built has converted a design problem into a change-management crisis.

30.3.4 Phase C: Data and Application Architecture

Phase C defines the data and application layers that support the business architecture. For agentic AI, the data architecture includes the knowledge bases the agents will retrieve from, the conversation logs and decision traces that will form the audit trail, the model inputs and outputs that will be governed under privacy and security regimes, and the feedback signals that will be used to improve the agents over time.

The application architecture identifies the agents themselves as logical applications, the orchestration layer that coordinates them, the tool-use surface that lets them act, and the integration boundaries with the existing application portfolio. The agentic-specific work is the careful design of the agent's *decision boundary* — what data the agent sees, what it does not, what it is allowed to act upon, and what it is required to defer.

The characteristic risk of Phase C is under-designing the audit trail, because at design time it produces no visible capability and feels like a cost. The audit trail — the decision traces, the conversation logs, the record of what the agent saw and chose — is the artefact that every later governance, compliance, and incident-investigation activity depends on, and it cannot be reconstructed after the fact if it was not captured. An agentic programme that designs the audit trail thinly in Phase C discovers the gap during its first incident or its first regulatory inquiry, at which point the missing evidence cannot be recovered. The audit trail is data architecture, it belongs in Phase C, and it should be designed as carefully as the knowledge bases the agents retrieve from.

30.3.5 Phase D: Technology Architecture

Phase D specifies the technology that will host the data and applications. For agentic AI, this is the place where the choice of foundation-model providers, inference infrastructure, vector databases, observability stacks, guardrail systems, and security infrastructure is captured as an architectural commitment. The ABB-versus-SBB distinction is most visible here: Phase D should record the ABBs (“the architecture requires a vector database supporting hybrid search and tenant-scoped indexing”) and reference the SBBs (“currently implemented by Pinecone, with Weaviate as a contingency”) without confusing the two.

The agentic-specific work is the explicit modelling of *trust boundaries*. Which components run inside the organisation? Which run outside? What data crosses which boundary? Where are the cryptographic and authentication seams? In an agentic system, these questions have to be answered before deployment, not retrofitted afterwards.

The characteristic risk of Phase D is premature commitment to a solution building block — letting a vendor choice, made for a pilot, harden into an architectural assumption that the rest of the design then depends on. The discipline that the ABB/SBB distinction exists to enforce is most easily abandoned here, under the pressure of a working pilot and a vendor relationship already formed. Phase D should record the technology architecture in capability terms first, with the current vendor named as the implementation rather than as the requirement, so that the architecture remains describable, and re-implementable, when the pilot's vendor turns out not to be the production answer.

30.3.6 Phase E: Opportunities and Solutions

Phase E identifies the concrete projects, programmes, and procurement actions that will move the organisation from the current state to the target. For agentic AI, this is the place where the architectural commitments meet vendor reality: which capabilities will be built, which will be bought, which will be configured from open-weight components, and what the sequencing of work will be.

The agentic-specific work is the early-stage evaluation of *solution maturity*. The agentic-AI vendor landscape changes monthly. Phase E should distinguish between solutions that are production-stable (and can be committed to), solutions that are usable in pilots (and should be sandboxed), and solutions that look promising but are not yet ready for enterprise scale (and should be tracked but not yet deployed).

The characteristic risk of Phase E is mistaking a compelling demonstration for production readiness. The agentic-AI market is full of solutions that demonstrate impressively on a curated scenario and have not been exercised against the messiness, scale, and adversarial inputs of a real enterprise deployment. Phase E's maturity assessment has to look past the demonstration to the operational evidence: how many production deployments at comparable scale, what the failure modes are, whether the vendor's support and stability commitments are real. A solution committed to on the strength of a demonstration, without that evidence, is a Phase E decision that Phase G will spend its time regretting.

30.3.7 Phase F: Migration Planning

Phase F plans the actual transition from the current state to the target state. For agentic AI, this involves identifying which existing workflows will be augmented or replaced, in what order, with what fallback options if the agent does not perform as expected. It involves a serious treatment of risk: what does the organisation do if an agent makes a costly decision before the governance scaffolding has caught up?

The agentic-specific work is the careful design of the *rollback path*. Unlike a software application, which can usually be rolled back by redeploying a previous version, an agent's effects in the world (a message sent, a transaction completed, a customer issue resolved unsatisfactorily) may not be reversible. Phase F is where the organisation describes, in advance, what reversibility looks like for the agentic capabilities it is about to deploy.

The characteristic risk of Phase F is the big-bang migration — the attempt to move a whole value stream onto agents in a single release because incremental rollout feels slow. An agent deployed at full scale before its behaviour has been observed at small scale exposes the organisation to its failure modes all at once, with no contained population in which to discover them cheaply. The discipline Phase F must impose is the staged rollout: a limited deployment, observed, evaluated, and expanded only on evidence, with the fallback path tested before it is needed rather than improvised during the incident that needs it.

30.3.8 Phase G: Implementation Governance

Phase G governs the implementation of the plan. For agentic AI, this means ensuring that the actual deployed agents conform to the architecture, that the human-in-the-loop checkpoints function as designed, that the audit trails are being captured, that the guardrails are triggering on the expected events, and that the value being delivered matches the value that was promised in Phase A.

The agentic-specific work is the operational maturity of *continuous evaluation*. An agent's behaviour can drift as its inputs change, as the underlying model is updated, as the surrounding tools evolve. Phase G is where the evaluation regime described in Phase C is actually run, and where the results feed back into Phase H.

The characteristic risk of Phase G is conformance drift: the deployed agents gradually diverge from the approved architecture, not through any single decision but through an accumulation of small operational expedients — a guardrail relaxed under pressure, a checkpoint quietly bypassed for speed, a budget raised and not lowered. Each step is individually defensible; the cumulative effect is a production system that no longer matches the design the review board approved. Phase G's discipline is to check conformance continuously and explicitly, treating divergence between the deployed system and the approved architecture as a defect to be corrected or a change to be formally re-approved, never as a fact to be quietly accepted.

30.3.9 Phase H: Architecture Change Management

Phase H is the phase that does not end. It is the continuous loop that detects changes in the environment (new regulations, new models, new threats, new business needs) and feeds them back into the architecture. For agentic AI, this is essential rather than optional: the rate of change in the underlying technology is fast enough that an architecture frozen at deployment will be out of date within months.

The agentic-specific work is the *ethical and policy review* that has to run continuously rather than as a one-time gate. An agent that was acceptable under the principles agreed in the Preliminary phase may, six months later, be doing things those principles were not written to cover. Phase H is where the principles themselves are revised, and where the resulting changes propagate through the rest of the cycle.

The characteristic risk of Phase H is that it quietly stops happening. It is the phase with no deadline, no deliverable, and no obvious owner once the launch excitement has passed, and an organisation under pressure on other work will let it lapse — the agents keep running, the environment keeps changing, and nobody is feeding the change back into the architecture. An architecture frozen at deployment, in a field moving as fast as agentic AI, is out of date within months. Phase H must therefore be given a named owner, a standing cadence, and a real place in the operating model, because it is the only phase whose failure is invisible until the gap between the architecture and reality has already grown wide.

30.4 The Concerns That Run Through Every Phase

Four concerns recur in every phase of the ADM cycle and cannot be confined to any one of them.

The reason these four cannot be assigned to a single phase is that each of them is a property of the whole system rather than of any component. Human-in-the-loop governance is not a feature that one phase installs; it is a discipline that the Preliminary phase decides, Phase B describes, Phase D builds, and Phase G verifies. The same is true of the other three. A concern that is treated as the responsibility of one phase will be addressed in that phase and decay everywhere else; the four below are addressed by being made explicit obligations of every phase, owned continuously rather than delegated once.

Human-in-the-Loop Governance. The Preliminary phase decides where humans are required. Phase B describes the collaboration patterns. Phase C designs the data flows that support them. Phase D specifies the technical mechanisms. Phase F plans the rollout. Phase G verifies it works. Phase H revises it when it does not. Human-in-the-loop is not a feature of a single phase; it is a discipline that has to be maintained across all of them.

Trust and Explainability. Every phase asks a version of the same question: *can the people who will be affected by this decision understand why the agent made it?* The answer depends on the principles set in the Preliminary phase, the metrics chosen in Phase A, the audit trail designed in Phase C, the explainability tooling selected in Phase D, the rollout discipline in Phase F, the runtime evidence in Phase G, and the response to drift in Phase H. Explainability does not have a chapter of its own in the cycle because it has a presence in every chapter of it.

Responsible Autonomy. The Preliminary phase sets the boundaries. Phase B describes the operational consequences. Phase C designs the data partitions. Phase D specifies the guardrails. Phase F plans the limit-extension trajectory. Phase G monitors compliance. Phase H revises the boundaries as evidence accumulates. Responsible autonomy is, in TOGAF terms, a cross-cutting concern: it does not have its own phase, but its quality is determined by what happens in every phase.

Value and Outcomes. Every phase asks whether the work being done will actually be worth what it costs. Phase A commits to the value hypothesis. Phase E re-evaluates it against vendor reality. Phase F sequences the work to deliver value early. Phase G measures the actual outcomes. Phase H revises the value model when reality diverges from hypothesis. An agentic deployment that does not have a clear value measurement loop is one that will, sooner or later, struggle to justify its own continued existence.

30.5 From Current State to Target State

The ADM cycle is, in practical use, a journey from a baseline architecture to a target architecture. For most enterprises in 2026, the baseline includes a small number of agent pilots, partial AI/ML infrastructure, isolated point solutions, a high latency between identifying an opportunity and shipping it, and significant gaps in governance. The target is an enterprise in which agentic capabilities are deployed across the major value streams, the governance scaffolding is in place, the value is measurable, and the human-and-agent collaboration patterns are working as designed.

The journey between the two passes through a recognisable sequence of intermediate states. The *transition architecture* of Phase F describes what the enterprise will look like at the end of each major release cycle: which agent capabilities have been deployed, which are still in pilot, which existing systems have been retired or repurposed, what the governance posture is. The transition architecture is what allows a multi-year programme to be funded one year at a time, with each year delivering a coherent and operable subset of the eventual target.

The mistake that derails many agentic-AI programmes is the attempt to move from the current state to the target in a single jump, without explicit transition architectures. The TOGAF discipline of naming the intermediate states, and demanding that each be operable in its own right, is the discipline that turns a multi-year ambition into a series of survivable releases.

The requirement that each transition architecture be operable in its own right is the part most often neglected and the part that matters most. A transition state that is merely a partial construction site — agents half-deployed, governance half-built, value not yet measurable — leaves the organisation exposed if funding pauses or priorities shift, which over a multi-year programme they reliably do. A transition state that is genuinely operable — a coherent, governed, value-delivering subset of the target — can be paused at without crisis and resumed from without rework. Designing the transition architectures so that each is a safe stopping point, not merely a stage of construction, is what makes an agentic programme robust to the organisational reality that multi-year programmes are rarely funded and staffed without interruption.

30.6 The African Perspective

The TOGAF ADM does not need to be adapted, as a method, for African contexts; it is method-neutral and works on any architecture. What changes in African deployments is what each phase has to address.

The general pattern is that the constraints which a well-resourced deployment can treat as background — vendor cost, data location, talent availability, funding horizon — move into the foreground and become first-class architectural inputs. The method does not change to accommodate this; the inputs to the method change. An architect applying the ADM in an African context runs the same nine phases, but each phase must explicitly take in constraints that an architect in a better-resourced market might leave implicit. The paragraphs below take three phases where this shift is sharpest.

The Preliminary phase, in an African context, often inherits constraints that are not present in better-resourced environments: foreign-exchange volatility affecting vendor costs, sovereignty requirements affecting where data may live, talent scarcity affecting which solutions can realistically be operated locally, and donor-funding cycles affecting the planning horizon. These constraints belong in the principles document, not as afterthoughts but as first-class design inputs.

Phase D, the technology architecture, has to account for cost-sensitivity in a way that European or North American architectures often do not. The choice between a frontier model and an open-weight model is not, in a Lagos or Nairobi deployment, a marginal cost consideration; it can determine whether the deployment is financially viable at all. The ABB-versus-SBB discipline is unusually valuable here:

the architecture can specify capability requirements that are met today by frontier APIs and tomorrow by quantised open-weight alternatives, without rewriting the architectural commitments.

Phase H, architecture change management, has to be prepared for upstream-vendor changes that fall most heavily on operations outside the vendor's home markets. A frontier provider that quietly deprecates an endpoint with thirty days' notice may make life difficult for a customer in San Francisco; it may threaten an entire deployment in Accra. The change-management process should detect these signals early and respond with pre-prepared fallback options.

30.7 The Working Position

TOGAF ADM gives an organisation a method by which agentic AI can be deployed responsibly at scale. The cycle moves from principles, through architecture, through plan, through implementation, into continuous change management, without leaving any phase undocumented or any concern unowned. The ABB/SBB distinction protects the architecture from vendor churn. The four cross-lifecycle concerns — human-in-the-loop, trust and explainability, responsible autonomy, and value — are addressed in every phase rather than relegated to any one of them. The current-state-to-target-state journey is broken into transition architectures that can be funded and operated incrementally. None of this guarantees that the agentic deployment will succeed; it does guarantee that the architectural reasons for any failure will be visible early enough to do something about them.

The deeper value of the method is that it makes an agentic programme legible to the people who must fund it, govern it, and audit it. An executive deciding whether to continue funding can see which phase the programme is in and what the next phase will deliver. A regulator can see that the autonomy boundaries were set deliberately and the audit trail designed deliberately. An incoming architect can read the architectural record and understand why each decision was made. Agentic AI is unfamiliar enough, and moving fast enough, that the temptation to proceed by improvisation is strong; the method's contribution is to make improvisation unnecessary, by giving the programme a structure that everyone involved can see, question, and trust.

Chapter 31

Architecture Governance: The Decision-Making System

“Transformation without governance is just uncontrolled change.”

—a recurring observation in enterprise-architecture practice

Chapter Overview

The architectural patterns and methods of the preceding chapters describe how to design enterprise AI well. This chapter describes the decision-making system that makes the designing possible at scale: a small, focused governance framework with six load-bearing components — architectural principles, reference architectures, an architecture review board, decision records, exception management, and a metrics-and-compliance loop — which together give an organisation the alignment, speed, and durability that transformation programmes otherwise lack. Without these components, AI investments fragment, security gaps accumulate, integration becomes fragile, and cloud spend ceases to be predictable. With them, delivery becomes repeatable, risk surfaces early, and teams move faster because they know what is already settled and what is still open.

Learning Objectives

- Recognise governance as a decision system rather than a paperwork burden.
- Distinguish the six components of an architecture governance framework and what each contributes.
- Apply the components in proportion to the size and risk of the organisation.
- Identify the failure modes that arise when one or more components is missing.
- Read the rest of the book through a governance lens, not just an architectural one.

31.1 Why Governance Is a Decision System

An enterprise that is transforming itself with AI is making thousands of architectural decisions in parallel — in product teams, in security functions, in cloud operations, in procurement, in compliance reviews, in vendor negotiations, in field engineering. Most of those decisions are small. Some have consequences that last a decade. Governance is the system that decides which decisions deserve which

level of scrutiny, and that ensures the decisions made anywhere in the organisation can be defended later by someone who was not in the room.

This sounds bureaucratic. Done badly, it is. Governance done badly — the architecture board that rubber-stamps decisions, the principles document no one reads, the review process whose timelines outlast the technology — is one of the more reliable ways to slow an enterprise down without making it any safer. Governance done well does almost the opposite: it accelerates competent teams by removing decisions they would otherwise have to make themselves, and it identifies the small number of decisions that genuinely require collective scrutiny. The framework below is calibrated for the second use.

The test that separates the two is whether the governance speeds the median decision up or slows it down. Good governance makes most decisions faster: a team facing a routine architectural choice finds the question already settled by a principle or a reference architecture, and proceeds without a meeting. The governance machinery is then reserved for the small fraction of decisions that genuinely warrant collective judgement. Bad governance inverts this — it routes every decision through the same heavy process, so the routine many are slowed to the speed of the consequential few. An architect evaluating a governance framework should measure it by the speed of the routine decision, not the thoroughness of the exceptional one; a framework that cannot make the routine case fast has misunderstood what it is for.

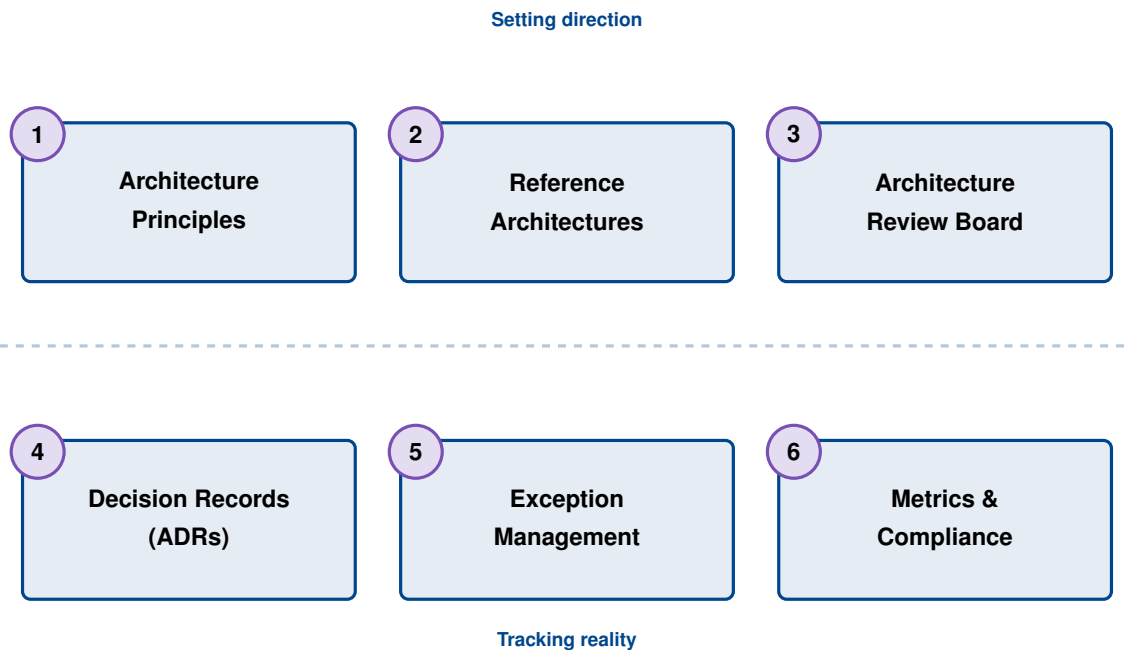


Figure 31.1: The six load-bearing components of an architecture governance framework. The top row sets direction: principles guide every decision, reference architectures embed those principles in reusable patterns, and the review board scrutinises the decisions that fall outside both. The bottom row tracks the reality of what was actually built: decision records capture rationale, exception management permits controlled deviation, and metrics measure whether the framework is delivering value.

31.2 The First Three: Setting Direction

31.2.1 Architecture Principles

Principles are the small set of binding statements that the organisation has agreed in advance. They take the most-contested architectural questions and settle them, so that teams making routine decisions do not have to relitigate the same debates. Good principles share three properties. They are short, in the sense that the entire set fits on one page. They are operative, in the sense that they can be applied to specific decisions and produce different conclusions for and against. And they are honest, in the sense that the organisation actually intends to follow them when the next contested decision arrives.

Examples that have proven durable in AI deployments include: *API-first* (every capability is exposed through a versioned interface, not a private dependency); *security by design* (sensitive data is segregated at the architecture layer, not patched at the application layer); *cloud-native* (workloads are designed for elastic infrastructure, not lifted from on-premise patterns); and *reuse over rebuild* (existing capabilities are extended where possible, new ones built only where existing ones genuinely cannot serve). A principle that sounds wise but cannot be applied to a real decision is decoration, not architecture.

The sharpest test of whether a statement is a genuine principle is whether it could be argued against. “We value quality” is not a principle, because no one would argue for the opposite; it cannot decide a contested case. “Reuse over rebuild” is a genuine principle, because a reasonable architect could argue the reverse — that the organisation should prefer purpose-built solutions — and the principle settles that argument in advance. A set of principles that contains nothing anyone could disagree with has settled nothing, and will be quietly ignored the first time a real decision needs settling. The discipline in writing principles is to make each one a genuine choice between defensible alternatives, recorded so that teams do not have to make the choice again.

31.2.2 Reference Architectures

Reference architectures are the published patterns the organisation has already worked out. They are the embodiment of the principles — not as abstract statements but as concrete blueprints that a competent team can adopt, adapt, and ship. The reference architecture for retrieval-augmented generation, for example, names the components, the interfaces, the security controls, the observability requirements, and the failure modes that the architecture board has already agreed on. A team building a new RAG application starts from the reference architecture and writes down only the ways in which their solution differs from it.

The value of reference architectures is not that they constrain creativity. It is that they remove the need for creativity in the places it is not needed, freeing it for the places it is. A team that has to rediscover, for the fourth time in eighteen months, how to handle vector-database multi-tenancy is a team that has been failed by its governance. The reference architecture is what protects them from that failure.

A reference architecture, to stay useful, must itself be maintained — and this is the part most often neglected. A reference architecture published once and never revised gradually diverges from current practice as models, tools, and the surrounding patterns evolve, until teams quietly stop using it because it no longer reflects what they actually need to build. The discipline is to treat each reference

architecture as a living artefact with a named owner, a version history, and a review cadence, updated as the field moves. An organisation with a catalogue of stale reference architectures is, in practice, an organisation with no reference architectures, because teams cannot tell which patterns are current and default to rediscovering their own.

31.2.3 The Architecture Review Board

The review board is the small standing forum that scrutinises decisions large enough to need collective judgement. The threshold for review matters: a board that reviews everything becomes a bottleneck, and a board that reviews nothing is decorative. Most mature organisations settle on a threshold that catches decisions with cross-team architectural impact, decisions that depart materially from the reference architectures, decisions with significant security or regulatory implications, and decisions that commit the organisation to a multi-year vendor relationship.

What the review board contributes is not its authority but its memory. The same architectural mistakes recur in large organisations across many teams and many years, and the people who remember the last few of them are the ones who can prevent the next one. A review board staffed by senior architects with long institutional memory is not slowing decisions down; it is keeping the organisation from paying for the same lesson twice.

The review board's effectiveness depends heavily on *when* in a decision's life it engages. A board that sees a design only at the final approval gate, after months of work, can either rubber-stamp it or force an expensive late rework — and under that pressure it usually rubber-stamps. A board that engages early, when the design is still cheap to change, can shape it before the cost of change has risen. The discipline is to route consequential decisions to the board at the point where its input can still be acted on without waste, which means treating the board as an early consultative resource rather than a late checkpoint. A review board used only as a gate is a review board whose memory arrives too late to be useful.

31.3 The Second Three: Tracking Reality

31.3.1 Decision Records

Architectural Decision Records (ADRs) are the small, dated documents that capture each significant architectural decision: the context in which it was made, the options considered, the choice that was taken, the rationale, and the consequences. An ADR is a few hundred words. It is written when the decision is made, not afterwards. It survives the team and the project.

The value of an ADR is recognised later, when someone — a new joiner, an auditor, a future architect — needs to understand why the system is the way it is. The system that has hundreds of ADRs going back five years is a system that can be reasoned about. The system that has no ADRs has, in effect, no architectural history; every legacy decision becomes a mystery that requires expensive forensics. The cost of writing ADRs in real time is small. The cost of recovering them from email archives a year later is enormous.

The most valuable field of an ADR is the one most often left thin: the options that were considered

and rejected. A decision record that states only what was chosen tells a later reader what was done but not what the alternatives were, and so cannot tell them whether the decision should be revisited when circumstances change. A record that captures the rejected options, and why they were rejected, lets a future architect see immediately whether the rejection still holds — whether the constraint that ruled an option out has since lifted. An ADR written as a record of the road taken is useful; an ADR written as a record of the junction, with all the roads visible, is what makes the decision genuinely revisitable.

31.3.2 Exception Management

Every set of principles, every reference architecture, every standard will eventually meet a real-world situation in which they should not be followed. Exception management is the process by which the organisation handles those situations honestly — granting deviations where the business case justifies them, requiring the deviation to be documented, attaching a defined review date, and ensuring the exception does not become permanent by inattention.

An organisation without an exception process forces teams to choose between two failure modes: either they shadow-deploy around the standards (and the standards become fiction), or they delay the work until the standards can be updated (and the work does not ship). A well-run exception process makes neither of those choices necessary. The exception is granted, recorded, time-bounded, and revisited. The standards stay credible because the deviations are managed rather than ignored.

The exception process is also one of the most valuable sources of signal the governance framework has, because a pattern in the exceptions is evidence that a standard is wrong. If a particular principle or reference architecture generates a steady stream of exception requests, all for similar reasons, the standard is not being heroically upheld against a difficult reality — it is out of step with the reality, and the exceptions are the organisation telling its governance so. A mature exception process therefore feeds back into the standards: recurring exceptions trigger a review of the rule that keeps generating them. An exception process that only grants deviations, and never revises the standards that necessitate them, is treating a symptom and ignoring the diagnosis.

31.3.3 Metrics and Compliance

The final component is the measurement loop. It asks, quarter by quarter: are teams actually using the reference architectures? Are decisions being recorded? Are exceptions being closed on schedule? Is technical debt growing or shrinking? Is the platform being reused or worked around? The questions are unglamorous. The answers are what tell the organisation whether its governance is actually working.

The mistake to avoid is measuring activity rather than outcome. The number of ADRs written, the number of reviews held, the number of principles published: these are activity metrics, and an organisation that optimises for them will produce a lot of paperwork and not much else. The outcome metrics are different: time-to-decision, reuse rate, exception-closure rate, audit findings, incident root-causes traced back to known and unaddressed architectural gaps. These tell the organisation what its governance is actually delivering.

The metrics-and-compliance loop also serves a defensive purpose that the other components do not: it is what allows the governance framework to justify its own existence at budget time. Governance is a recurring cost with no directly attributable revenue, and in any cost-cutting cycle it is a natural target. A framework that can show, with outcome metrics, that time-to-decision has fallen, that reuse has risen, that incident root-causes traceable to architectural gaps have declined, has evidence for its own value. A framework that can show only activity metrics — meetings held, documents produced — has no defence, because activity is exactly what a cost-cutter wants to remove. The measurement loop is, among other things, the governance framework's survival mechanism.

31.4 Without Governance, and With It

The case for the framework can be made by what happens in its absence. Without architectural principles, every team that designs a new system relitigates the same questions: should we use this cloud or that one, should this be a microservice or a monolith, should the data live here or there. The relitigation costs weeks of senior time and produces inconsistent answers across the organisation. Without reference architectures, the same patterns are rediscovered five times in eighteen months, each time with a slightly different security posture and a different operational profile. Without a review board, the architectural mistakes that have already been made elsewhere in the organisation are made again, by teams who did not know and had no way to find out. Without decision records, the rationale behind the current architecture is lost as soon as the people who built it move on. Without exception management, the standards either become fiction (because teams route around them) or become obstacles (because the deviations cannot be sanctioned). Without metrics, the governance framework cannot be defended at budget time, because no one can demonstrate that it is doing anything.

The combined effect, sustained over a few years, is recognisable: AI investments fragment across the organisation, security gaps accumulate in places no one is monitoring, integration between systems becomes increasingly fragile, cloud cost becomes unpredictable from quarter to quarter, and the senior architects spend most of their time fighting fires that better governance would have prevented.

With the framework in place, the same organisation looks different. Delivery becomes repeatable because the teams are working from common patterns. Risk surfaces early because the review board sees what is coming, not what has happened. Teams move faster because they know what is settled and what is still open. The framework does not prevent decisions; it makes decisions easier to make well.

It is worth noting that the six components are mutually reinforcing, so the absence of one degrades the others rather than simply removing a feature. Principles without a review board to apply them drift into decoration. A review board without decision records cannot remember its own reasoning and re-argues settled questions. Reference architectures without metrics cannot tell whether they are being used. Exception management without principles has nothing to grant exceptions from. The framework is a system, not a checklist, and an organisation that adopts three of the six components and skips the rest does not get half the benefit — it gets a fraction, because the components it kept are weakened by the ones it dropped.

31.5 Proportionality

The framework above is described as if every component were always present at the same scale. In practice it should be calibrated to the size and risk profile of the organisation. A 200-person startup deploying AI does not need the same governance machinery as a 200,000-person bank. What both need is the same six components, just in different sizes.

For a smaller organisation, the principles may fit on a single page and be maintained by the CTO. The reference architectures may be a handful of GitHub repositories with documented patterns. The review board may be a thirty-minute weekly meeting attended by four senior engineers. The ADRs may live in the same repositories as the code they describe. The exception process may be a Slack channel with a documented escalation path. The metrics may be a single dashboard reviewed monthly. Total weekly governance overhead: a few hours.

For a larger organisation, each component grows proportionally. Principles become a formal document with version control. Reference architectures become a curated catalogue with named owners. The review board becomes a chartered standing committee with documented procedures. ADRs become an indexed repository with cross-references. Exception management becomes a tracked workflow with quarterly reporting. Metrics become a board-level dashboard with named accountability. Total weekly governance overhead: substantial, but proportional to the architectural surface area being governed.

The mistake to avoid in both cases is symmetric: under-investing in governance because it feels slow, or over-investing in governance because it feels safe. The right amount is what lets the organisation make the architectural decisions it actually faces, at the speed it actually needs.

Proportionality also has a time dimension: the right amount of governance for an organisation is not fixed, and the framework should grow with the architectural surface it governs. A startup's single-page principles and weekly thirty-minute review are correct for a startup, and become incorrect as it grows into a mid-size enterprise with many teams and a wider architectural surface. The failure to watch for is the framework that was sized correctly once and never resized — either a small organisation that has imported a large enterprise's heavy machinery and is now slowed by it, or a grown organisation still running the lightweight governance of its early days and now accumulating the fragmentation the framework exists to prevent. Governance should be reviewed for proportionality periodically, as deliberately as the architecture it governs.

31.6 The African Perspective

Architecture governance in African contexts inherits the same six components but operates against additional constraints. Talent scarcity affects the review board most directly: the people who would be most valuable on the board are often the same people who are most valuable building the systems being reviewed, and the organisation cannot afford to have them in meetings all week. The proportionality principle is unusually important here. A leaner review board, meeting less often, focused on a smaller number of higher-stakes decisions, is often the realistic answer.

There is a constructive reading of this constraint. A review board forced by talent scarcity to be small and selective is forced toward exactly the discipline the chapter has argued for — reviewing only the decisions that genuinely warrant collective judgement, and leaving the rest to principles and reference architectures. The well-resourced organisation that can afford a large board is, paradoxically, more at risk of the bottleneck failure mode, because it can afford to route too much through review. The African enterprise that cannot afford that has a structural incentive to get the threshold right. The constraint, handled deliberately, pushes the governance toward its better form.

Vendor governance carries a different weight. In markets where currency volatility, foreign-exchange controls, and donor-funding cycles all affect what is procurable, the architecture review board's role is partly to prevent commitments to vendors whose pricing or support might evaporate at the next currency move. The decision records become evidence trails for donor and government audits in addition to their internal use. The exception management process has to accommodate the reality that some vendors simply do not operate locally and that pragmatic workarounds may be necessary while the architecture catches up.

The principle of *sovereignty-aware architecture* — which most African enterprises now need to consider — is itself a governance commitment, not a technical one. It belongs in the principles document, it has to be reflected in the reference architectures, the review board has to enforce it, and the metrics have to track compliance. None of this is exotic. It is the same six-component framework, calibrated to the constraints the continent actually operates under.

31.7 The Real Outcome

The architectural governance framework is not, in the end, about the framework. It is about what becomes possible when the framework is in place. Platforms scale because the decisions that make them scalable have been made consistently. Delivery becomes predictable because the teams shipping work are not reinventing the foundation under every release. Operational cost falls because the cloud usage, the licensing, the support contracts, and the integration debt are all being governed rather than allowed to accumulate. Reliability improves because the failure modes that have been seen before are documented, named, and designed against. None of these are governance metrics; they are business outcomes that good governance makes possible.

The opening epigraph is the working summary. An enterprise undertaking transformation without governance is not transforming; it is changing, expensively and uncontrollably. The six components of the framework are what convert change into transformation. They are not the architecture. They are what lets the architecture exist.

There is a final connection worth drawing. The governance framework of this chapter is also the operating machinery for everything else the book describes. The standards landscape of the governance-standards chapter is enforced through these six components — principles encode the obligations, the review board checks them, the metrics track compliance. The agentic-AI method of the TOGAF chapter produces decisions that become ADRs and exceptions. The patterns of Part [III](#) become reference architectures. The framework is not one more topic alongside the others; it is the

decision system through which all the others are actually applied. An organisation can know every pattern in this book and still fail, if it has no system for deciding which pattern to use where and recording why. The six components are that system.

Chapter 32

Pattern: Tool Use and Function Calling

“The model is the librarian. The tools are the books that move.”

—a recurring observation in agent design

Chapter Overview

Tool use — the capability of a language model to invoke external functions and consume their results — is foundational to every modern enterprise AI system that does anything beyond generating text. This chapter develops tool use as an architectural pattern. It begins with the mechanism and the loop that mechanism creates, since the loop is the structural fact the rest of the chapter follows from. It then works through the registry that governs an estate of tools, the function-calling interface a model uses to reach them, the security boundary that contains them, the deterministic-versus-probabilistic question of who selects them, the handling of what they return, the audit that makes them governable, the recurring tool categories and their differing risk, and the anti-patterns and failure modes that a tool-using system must be built to withstand. The Model Context Protocol, the emerging standard for the model-to-tool interface, has its own chapter (Chapter 33); this chapter refers to it where it bears on a decision rather than developing it again.

Learning Objectives

- Understand the tool-use loop and why it changes the engineering problem.
- Architect tool registries that scale and govern tools across applications.
- Design function-calling definitions that models use reliably.
- Establish security boundaries and least-authority scoping around tool execution.
- Distinguish deterministic from probabilistic tool selection and combine them.
- Handle tool output safely, and maintain the audit a tool-using system requires.

32.1 The Pattern and the Loop

A model with tool access can do more than generate text: it can read data, write data, trigger workflows, and orchestrate other systems. This is the architectural shift that turns a language model from a text generator into a system component. A model without tools is bounded by its training data and the

text in its context window; a model with tools is bounded only by what the tools can reach. It is also where most of the engineering difficulty of agentic systems originates, because a model that can act in the world is a model whose mistakes have consequences in the world.

The mechanism itself is simple, and its simplicity is deliberate. Each tool is described to the model by a definition — a name, a description, and a parameter schema. The model, given those definitions, decides whether to call a tool and emits a structured request naming the tool and supplying arguments. That request goes to the runtime, not to the tool: the runtime decides whether and how to honour it, executes the tool if it does, and returns the result into the model’s context. The model then continues, and may call another tool, and another, until it produces a final answer with no further call. The defining feature of this mechanism is that *the model never executes anything itself* — it only emits a request, and the runtime stands between the request and the execution. That seam is what makes tool use governable: every security, authorisation, and audit control lives in the runtime, at that one point, where it can be enforced regardless of what the model asked for. A reader who holds this picture clearly will follow the rest of the chapter, because every discipline in it operates at that seam.

The consequence of the mechanism is that tool use turns a single forward pass into a loop. A text-only model is one pass: request in, response out. A tool-using model is the loop shown in Figure 32.1 — the model decides, a tool runs, the observation returns, the model decides again — and a loop has properties a single pass does not. It can fail to terminate. It can spend unboundedly. It can repeat itself without progress. Almost everything later in this chapter — the budgets, the timeouts, the failure modes, the audit trail — exists because tool use is a loop rather than a pass, and an architect who keeps the loop in view will find the rest of the chapter follows naturally from it.

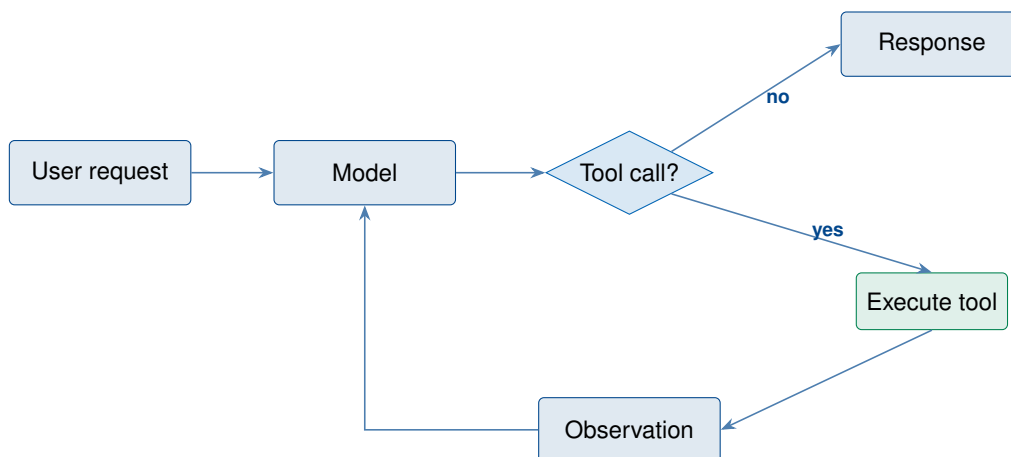


Figure 32.1: The tool-use loop. The model decides whether to call a tool; if it does, the runtime executes the tool and feeds the observation back for another decision. The loop terminates when the model produces a response with no tool call — or when a budget caps it.

Tool use matters, finally, for a reason beyond capability: it is what makes a model’s claims checkable. A model answering from its parameters alone produces fluent answers that may be confidently wrong; a model answering from a tool’s result produces answers anchored to a verifiable source. Tool use does not make a model correct, but it makes the model’s claims grounded and checkable, and checkability is the difference between a system an enterprise can govern and one it can only hope about. That

is why tool use is treated here as foundational rather than as one feature among many — it is the capability that connects the model’s fluency to the enterprise’s actual data and actual systems, and the substrate on which every agentic pattern in the surrounding chapters is built.

32.2 Tool Registries

Once an organisation has more than a few tools, it needs somewhere to keep them. A tool registry centralises tool definitions across the organisation, and without one, tool definitions proliferate inside individual applications and the organisation loses the ability to see, govern, or reuse them. The registry is to tools what the model inventory of the governance chapters is to models: the single place where the organisation knows what exists, who owns it, what permissions it carries, and whether it is current. A tool-using AI estate without a registry is not a smaller version of one with a registry; it is an estate whose tool surface no one can fully describe.

A registry is more than a list. It holds the tool catalogue with its definitions and metadata, but it also versions those definitions, attaches permission and access control, provides discovery so applications can find what exists, carries documentation and examples, and manages deprecation and lifecycle. Versioning and lifecycle are the two components most often omitted and most consequential, because a tool definition is an interface that models depend on. A tool whose definition changes without versioning silently breaks every agent that called the old shape; a tool with no deprecation path accumulates in the catalogue long after it should have been retired. The registry’s job is not merely to list tools but to manage them as living interfaces, with the same change discipline an organisation applies to any other API.

The benefits of getting this right compound, because each one makes the others cheaper. Reuse means a tool definition is improved once and every consumer benefits. Consistency means a model that learned to call one tool well calls its siblings well. Central governance means a permission or audit policy is applied uniformly rather than re-implemented per application. The registry is, in this sense, the same architectural move as the platform team of the team-topologies chapter — consolidating something every team needs so that it is built once, well, and inherited rather than rebuilt.

The anti-patterns are the mirror image, and they share a single root cause: the absence of ownership. Per-application tool duplication, inconsistent parameter naming across tools, missing permission controls, tools with no owner, stale tools with no deprecation path — each of these exists because no one was accountable for the tool. A tool without an owner is not maintained, not deprecated, and not improved; a duplicated tool exists because no one owned the original well enough for the next team to find and reuse it. The registry’s deeper contribution is therefore organisational rather than technical: it forces every tool to have an owner, and an owned tool is one that someone is accountable for keeping correct, secure, and current. An organisation can buy or build a registry and still suffer these anti-patterns if it does not also assign and enforce ownership.

32.3 Function-Calling Definitions

The registry holds tool definitions; the function-calling interface is how those definitions reach the model and how the model signals it wants one called. Modern foundation models expose function-calling APIs, and the conventions have largely converged — a JSON-Schema description of each tool, a structured tool-call object in the model’s response. The convergence is recent and incomplete, and the differences between vendors are exactly the kind of detail the hexagonal pattern’s LLM port exists to absorb. What matters most here is not the wire format but the quality of the definitions, because the quality of those definitions, more than any other single factor, determines whether tool use works.

A good definition recognises that the schema is doing two jobs at once. It is a contract the runtime enforces — a call that violates the schema is rejected before execution — and it is documentation the model reads to decide how to call. A schema written only as a strict contract, with terse machine-oriented field names, validates correctly and guides the model poorly; a schema written only as friendly prose guides the model and fails to constrain it. The enum constraint is the clearest case of a single device serving both jobs: it tells the model exactly which values are legal and simultaneously makes any other value impossible. Good definitions treat the schema as both contract and documentation, deliberately, and they specify which parameters are required, which optional, and what edge cases the model should expect.

Definition quality is measurable, and it should be measured. Vague descriptions produce inconsistent calls; overlapping tool definitions produce confused selection; missing edge-case documentation produces errors; clear examples improve reliability. The practical consequence is that tool definitions should be treated as a tested artefact, not as documentation written once and forgotten. Run the model against a set of representative requests and observe whether it selects the right tool and supplies valid arguments. When it does not, the definition — not the model — is usually the thing to fix, by sharpening the description, separating overlapping tools, or adding an example of the case that failed. Tool definitions improve through the same evaluation loop as any other part of the system, and an organisation that writes them once and never tests them has left its most reliability-critical surface unmeasured.

One capability of modern function-calling APIs deserves a specific caution: parallel tool calls. Modern models can issue several tool calls at once, which is a genuine latency win for calls that are genuinely independent — querying several sources at the same time, for instance. But the model is not always a reliable judge of independence. If it issues in parallel two calls where the second should have depended on the first’s result, it has traded correctness for speed. The orchestrator can help by structuring tools so that dependent operations are not separately callable, but the architect should treat parallel calling as an optimisation for clearly independent fan-out work, not as a general default. Where the calls have a real ordering, sequential execution is not a missed optimisation; it is the correct behaviour.

A closing note on portability. Because function-calling formats remain vendor-specific in their details, the model-to-tool interface is a natural place for lock-in to accumulate, and the Model Context Protocol exists to prevent that — a standard interface that lets a tool catalogue serve any MCP-aware client regardless of the model behind it. MCP is significant enough to have its own chapter, [Chapter 33](#), which develops the protocol, its primitives, and its costs. For this chapter the relevant point is the

architectural one it leaves behind: tool integration should be an investment in a standard rather than a bet on a vendor, and an enterprise building more than one AI application should invest in MCP servers rather than in client-specific tool integrations, because the server is written once and serves every current and future client while the client-specific integration must be rebuilt for the next.

32.4 Security Boundaries

Tool use creates security risk of a kind a text-only model does not. A text-only model that is manipulated produces bad text; a tool-using model that is manipulated can take bad action — send data, trigger a transaction, write to a system. Tool use is therefore the point at which the threat model of the dedicated security chapter becomes concrete and consequential. The governing idea is the one that chapter develops at length: the controls live in the runtime, at the seam between the model's request and the tool's execution, because that is the one place an enforcement point cannot be talked past by a manipulated model.

The specific risks are recognisable. Prompt injection can cause tool calls the user never authorised. Sensitive data can flow to or from tools without authorisation. Tools with side effects — mutation, sending, paying — can be triggered improperly. Tool output can contain untrusted content that the model reads as instruction. And, most easily missed, privilege can escalate through tool composition: each tool can be individually safe while their combination is not. A tool that reads a file and a tool that sends an email are each unremarkable; together, in the hands of a manipulated model, they are a data-exfiltration path. The architectural lesson is that tools cannot be risk-assessed only in isolation — the composition has to be considered, because the model can chain them in ways no individual tool's threat model anticipated.

The mitigations work because they are layered, in the defence-in-depth sense of the security chapter: no single one is trusted to be sufficient. User authorisation gates the consequential tools; per-user, per-role, per-application permissions bound which tools an agent can reach at all; output sanitisation defends against the injection that arrives through tool results; an audit trail makes any breach investigable afterwards; mutation-class tools require explicit confirmation; and budget and rate limits cap what a runaway loop can spend. An attacker, or a manipulated model, must defeat several of these to cause real harm, and they fail in different ways. The architect should read the list not as a menu but as a set: a tool-using system that implements two of the six has not lightly secured itself; it has left four of the layers empty.

Underneath the mitigations sits one principle that makes the composition risk tractable: least authority. Each tool should expose only the capability it needs, and each agent should be granted only the tools its actual task requires. If an agent holds only the tools its task requires, the dangerous combinations — read-a-file plus send-an-email — simply do not co-exist in most agents' tool sets, and the exfiltration path is closed not by detecting it but by never assembling it. The discipline costs something: it means resisting the convenience of giving every agent a broad, generous tool set and instead scoping the set deliberately per use case. That deliberate scoping is precisely the security work, and an organisation that grants tools generously because narrow scoping is tedious has chosen convenience over the one principle that most reliably bounds what a manipulated agent can do.

32.5 Who Selects the Tool

Tool selection can be deterministic or model-driven, and the choice between them is one of the chapter's more consequential design decisions, because it trades reliability against flexibility and the two cannot be maximised at once.

Deterministic selection means a rule, not the model, decides which tool runs — rule-based routing, or a tool invoked directly by a user action. It is predictable and reproducible, and it is the right default for any tool whose invocation is consequential or whose correctness is non-negotiable. If a particular user action should always trigger a particular tool, encoding that as a rule rather than leaving it to the model removes a source of variability exactly where variability is least acceptable. The cost is rigidity: the rule handles only the cases it was written for, and an input outside its anticipated set falls through. Deterministic selection is strongest for the routine, high-stakes, well-understood paths and weakest for the open-ended ones.

Probabilistic selection means the model chooses, based on context. It is what makes a tool-using system able to handle the requests no one enumerated in advance — the genuine value of putting a model in charge of the choice. Its cost is that it cannot be tested by exact reproduction: the same input may yield different selections, so its correctness must be evaluated statistically, over a distribution of inputs, against a quality threshold rather than a fixed expected answer. This is the evaluation discipline of the observability chapter applied to tool selection, and it is not optional. A system that relies on probabilistic selection without a standing evaluation of how often that selection is right has adopted the flexibility and declined to measure whether it is working.

Most mature tool-using systems converge on a hybrid, because the hybrid assigns each approach the work it is good at. The routine, high-stakes, well-understood requests go to deterministic routing, where their predictability is preserved; the novel and open-ended requests go to model selection, where its flexibility is needed. The routing policy that decides which path a request takes is itself the key design artefact, and it should be governed by risk: the higher the consequence of a wrong tool choice, the stronger the case for keeping that path deterministic. A system that routes by risk in this way gets reliability where reliability matters and flexibility where flexibility matters, rather than accepting one global compromise for both.

32.6 Handling Tool Output

A tool output is not a trusted result to be passed straight back into the model; it is data crossing a boundary, and three distinct concerns attend that crossing — the output may be malformed, it may be untrusted, and it may be large. The return path of a tool call deserves as much architectural attention as the outbound call, because an unhandled tool output is where a malformed result, an injected instruction, or a context-budget blowout enters the system.

The first concern is correctness, and it is met by validating tool outputs against a schema and handling errors deliberately. Error handling on the return path deserves particular care, because how an error is reported to the model determines whether the model can recover. An error returned as an unstructured failure leaves the model guessing; an error returned as a structured, informative message — what

failed, why, and whether a retry might succeed — gives the model what it needs to adapt, perhaps by correcting an argument and trying again. The discipline is to treat tool errors as part of the tool's interface, designed deliberately, rather than as whatever exception happened to propagate. A well-designed error message is the difference between a model that recovers from a failed call and one that loops on it.

The second concern is trust. A tool that retrieves a document, reads a webpage, or returns a database field containing free text is returning content that some third party controlled, and that content re-enters the model's context where the model may read an embedded instruction as a command. This is the indirect-injection vector of the security chapter, met here at the tool layer. The defence is to fence retrieved content explicitly — to mark it, in the prompt structure, as data to be considered rather than instruction to be obeyed — and to keep the trust boundary visible rather than letting tool output blend seamlessly into the model's instructions. A tool-using system that treats every tool result as trusted has, in effect, extended its trust boundary to include everyone whose content its tools can reach.

The third concern is cost. A large tool output costs twice: once in the context-window space it occupies, and again in the per-token price of every subsequent model call that now carries it. A tool that returns a thousand rows when the model needed three has not merely been inefficient once; it has inflated the cost of the rest of the task. The discipline is to shape tool outputs to what the model actually needs — returning summaries, top results, or paginated slices rather than raw dumps — and to make that shaping the tool's responsibility rather than hoping the model ignores the excess. This is the token-economics point of the cost chapter applied to the tool layer: the cheapest token is the one never placed in the context at all.

32.7 Audit and Observability

Audit is structural for tool-using systems — not a feature added afterward but a property the architecture must be built around — because a tool call is an action with consequences, and an action without a record is an action that cannot be governed, investigated, or learned from. Every other discipline in this chapter depends on the audit trail: security investigation needs it, cost attribution needs it, debugging needs it, regulatory evidence needs it. A tool call that left no trace did, for governance purposes, not happen in a way anyone can rely on.

What to capture is, in outline, every tool call — who made it, what tool, when, with what input parameters, what output, what latency, what cost, what errors, and under what authorisation context. Two details of that list carry disproportionate weight. The first is the *why*: a record that a tool was called, with what arguments and what result, says what happened, but a record that also captures the reasoning context — the user request and the model's stated rationale — says why it happened, which is what an investigator, an auditor, or a debugging engineer actually needs. The second is the treatment of large outputs: capturing the output as a hash rather than in full is a deliberate compromise that preserves the ability to prove what was returned without paying to store every large payload. The capture list should be designed against its consumers — the operator, the quality engineer, the auditor of the observability chapter — because each needs a different subset and a log built for only one fails the others.

The captured data becomes useful through two distinct views, and both are needed because they answer different questions. The per-task trace is the debugging view: handed a task that went wrong, an engineer follows its trace through every tool call to find where. The per-tool view is the reliability view: aggregated across tasks, it reveals that one tool fails ten times more often than its peers, or costs disproportionately, or has a latency tail that stalls agents. The first view finds the broken instance; the second finds the broken tool. A tool-using system instrumented for only one of them can either debug individual failures or spot systemic ones, but not both — and the systemic problems are precisely the ones a single trace will never reveal.

32.8 Tool Categories and Their Differing Risk

Tools are not uniform, and the property that most governs how much control a tool needs is reversibility. Sorting the tool estate by reversibility — the same risk-calibration the agent-anatomy chapter applied to the action layer — tells the architect which tools can run with light controls and which warrant the heaviest the system has. Five categories recur, in roughly increasing order of consequence.

Retrieval tools — search over corpora, vector retrieval, structured-data query, knowledge-graph traversal — are the safest category, because reading does not change the world. That safety means they can be called freely, in parallel, and with light controls. The caveat is that “safe to call” is not “safe to trust”: a retrieval tool returns content from a corpus, and that content is the indirect-injection vector and the cost concern of the previous section. Retrieval tools are also where the access-control question is sharpest, because a retrieval tool that ignores the calling user’s permissions will happily return data the user was never entitled to see. They are safe in what they do and require care in what they return and to whom.

Computation tools — calculation, data analysis, and code execution — are idempotent within their input scope, but the parenthetical that matters is *sandboxed*. A code-execution tool is the most powerful and most dangerous tool an agent can hold, because arbitrary code is by definition capable of arbitrary action — it can read, write, network, and escalate unless something physically prevents it. The sandbox is that something: an isolated environment with no network, no credentials, and no access to anything the computation does not need, so that even code generated by a fully manipulated model is contained. A code-execution tool without a genuine sandbox is not a computation tool; it is an unbounded mutation tool wearing a safer label.

External-system tools — connections to CRM, ERP, ticketing, and other operational systems — are where the agent reaches into the enterprise’s systems of record, and their risk concentrates in the read-write ones. A read-only connection to a CRM is a retrieval tool with a permissions concern; a read-write connection can modify the records the business runs on, which places it closer to the mutation category. These tools also inherit the access model of the system behind them: the agent should act with the calling user’s authority, not with a broad service account that can see and change everything, or a single compromised agent becomes a path to the whole system of record. Splitting these tools into their read and read-write halves, and scoping each appropriately, is the discipline this category needs.

Mutation tools — email send, message post, payment, file write — earn the heaviest controls the system has, because their defining property is irreversibility: an email sent cannot be unsent, a payment made cannot be undone. For these tools the human-in-the-loop confirmation of the next chapter is not an optional refinement but the appropriate default. The architect should also ask, for every mutation tool, whether a less consequential alternative would serve: a tool that drafts an email for a human to send is reversible where one that sends it directly is not, and preferring the reversible form where the task allows is the cheapest risk reduction available. Mutation tools are where the agent’s power to act becomes the agent’s power to do harm.

User-interaction tools — showing a UI element, requesting input, displaying a visualisation — are distinctive because they are synchronous in a way the other categories are not: the agent calls one and then waits for a human, which means the agent’s loop must be designed to suspend and resume rather than run to completion. The agent’s state has to be persistable across the wait, the wait needs a timeout, and the user may never respond at all. These tools are also the mechanism by which the human-in-the-loop checkpoints of the next chapter are actually implemented: a confirmation gate is, concretely, a user-interaction tool that the agent must call and wait on before a mutation tool is permitted to run. They are the seam where the agent’s autonomous loop and the human’s judgement meet.

32.9 Anti-Patterns and Failure Modes

The anti-patterns of tool use share a single character: each is a discipline from an earlier section, omitted. Tool descriptions written for engineers rather than models are the definition-quality discipline skipped. Overlapping tools that produce inconsistent selection are the catalogue-curation discipline skipped. Missing parameter constraints, absent authorisation on mutation tools, untrusted output treated as authoritative, missing budget enforcement, missing audit, per-application duplicate tools — each is a section of this chapter ignored. They are not exotic failures; they are the predictable result of treating tools as an implementation detail rather than as first-class architectural citizens. The remedy for the whole list is therefore singular: treat each tool as deserving the same definition, governance, security, and observability the chapter has set out. The anti-patterns are common precisely because that treatment is more work than wiring a tool up quickly.

The failure modes are different from the anti-patterns: an anti-pattern is a design omission, a failure mode is something that goes wrong at runtime even in a well-designed system, and the two need different responses. Six failure modes recur, and they fall into two kinds:

- **Tool selection error** — the model picks the wrong tool. Mitigations: clearer descriptions, fewer overlapping tools, evaluation coverage.
- **Parameter error** — the model sends malformed arguments. Mitigations: strict input schemas, helpful error messages back to the model, examples in the tool documentation.
- **Hallucinated tool** — the model invokes a tool that does not exist. Mitigation: a strict-validation orchestrator that rejects unknown tools.

- **Tool-call loop** — the model calls the same tool repeatedly without progress. Mitigations: call-count budgets, repeated-call detection, escalation to a human.
- **Tool-result injection** — tool output carries prompt-injection content. Mitigations: treat outputs as untrusted, with structural separation in the next prompt.
- **Long-tail latency** — a slow tool stalls the whole agent. Mitigations: per-tool timeouts, asynchronous tools with status checks.

The division tells the architect where to look. Selection errors, parameter errors, and hallucinated tools are *quality* failures — the model chose or called badly — and their mitigations are mostly about the tool definitions and the evaluation that tests them. Tool-call loops, result injection, and long-tail latency are *loop* failures — consequences of tool use being a loop rather than a single pass — and their mitigations are budgets, timeouts, and trust boundaries enforced by the runtime. A team debugging a tool-using system should first ask which kind of failure it faces, because the quality failures are fixed in the catalogue and the loop failures are fixed in the orchestrator, and looking in the wrong place is the most common way a tool-use problem goes unsolved.

One catalogue-level discipline prevents several of the quality failures at once: keep the catalogue curated rather than exhaustive. The instinct is to expose every tool that might conceivably help, and that instinct is mistaken. Each tool in the catalogue consumes context and competes for the model's selection attention, so a sprawling catalogue degrades selection accuracy across all of it — the model confronted with five hundred tools chooses worse than the model confronted with the fifty that matter for its task. Fifty well-chosen tools beat five hundred indiscriminate ones. Typed inputs and outputs, idempotency where it can be arranged, explicit side-effect markers that distinguish read-only tools from write tools, and declared cost and latency so the model can prefer cheaper tools — these are the marks of a catalogue designed as the API surface the model actually sees. Scoping that surface to the task is the least-authority principle of the security section restated as a quality principle: a curated catalogue is better selected, cheaper, and safer at once.

32.10 The African Perspective

Tool use is a geography-neutral pattern, but two of its concerns press harder in many African deployments.

The first is connectivity. The tool-use loop depends on round-trips — the model calls a tool, waits for the result, and continues — and each round-trip is a network operation that variable or high-latency connectivity makes slower and less reliable. A loop that is brisk on a low-latency network can become painfully slow when each of its several tool calls crosses an unreliable link, and the long-tail-latency failure mode is correspondingly more common. The architectural responses are the ones the chapter already recommends, applied with more urgency: per-tool timeouts so a stalled call does not hang the agent, asynchronous tools with status checks for anything slow, and a deliberate preference for fewer round-trips — consolidating what would be several tool calls into one where the task allows.

The second is the question of where the tool-and-data layer is hosted. The chapter's advice to invest

in MCP servers rather than client-specific integrations is sound everywhere, and it has an added benefit for sovereignty-constrained deployments: an MCP server is infrastructure the organisation controls and can host in a compliant region, exposing enterprise tools and data to whichever model client is used without the integration itself being tied to a foreign vendor. For an organisation subject to data-residency rules, the MCP server is not only a portability investment but a way to keep the tool-and-data layer on the right side of a jurisdictional boundary while still using whichever foundation model is appropriate. The pattern is unchanged; the sovereignty benefit is an additional reason to follow it.

32.11 What Tool Use Requires

Tool use is the capability that turns a language model into a system component, and the chapter's organising claim is that this is exactly why it must be engineered with system-component discipline. A tool is an interface, so it is versioned, owned, and governed through a registry. A tool definition is read by the model, so it is written as documentation as well as contract, and tested. A tool call is an action, so it is authorised, scoped by least authority, and audited. Tool selection is a decision, so it is assigned deterministically or probabilistically by risk and, where probabilistic, evaluated. A tool output is data crossing a boundary, so it is validated, fenced, and budgeted. And tool use is a loop, so it is bounded by budgets and timeouts against the loop failure modes.

None of these disciplines is novel — they are the registry, security, evaluation, and observability practices the book applies everywhere, brought to bear on one pattern. And none is optional, because a tool-using system is the precise point at which a model's mistakes stop being bad text and start being bad action. The epigraph frames it well: the model is the librarian, and the tools are the books that move. A librarian who can move books is far more useful than one who can only describe them — and far more capable of disorder if the library has no catalogue, no record of what was moved, and no rule about which shelves may be touched. An organisation that builds tool use with the discipline of this chapter has an AI estate that can act in the world and be governed while it does. An organisation that wires tools up quickly has built the capability and skipped the part that made it safe.

Chapter 33

The Model Context Protocol: From Integration Sprawl to a Shared Plumbing Standard

“A protocol is not magic. It is plumbing — and plumbing is what lets a building grow without being rebuilt.”

—a recurring observation in integration engineering

Chapter Overview

The previous chapter established tool use as an architectural pattern: an agent invoking external capabilities through structured calls. This chapter addresses the protocol layer underneath that pattern — the Model Context Protocol, MCP — which has become, over little more than a year, the de facto standard for connecting models to tools and data. The chapter is deliberately introductory. It explains the integration problem MCP solves, the small set of concepts the protocol actually consists of, how an MCP message travels on the wire, and — with equal care — the costs the protocol introduces that its enthusiasts tend to omit. The chapter that follows this one develops the server-side patterns in depth; this one builds the shared mental model they assume.

Learning Objectives

- Explain the $N \times M$ integration problem MCP is designed to collapse.
- Identify the three roles — host, client, server — and the three primitives — tools, resources, prompts.
- Describe how an MCP interaction travels: transport, lifecycle, and wire format.
- Recognise the costs MCP introduces — token tax, credential sprawl, server sprawl, schema drift, observability gaps.
- Decide when MCP is the right choice and when a simpler integration serves better.

33.1 The Problem MCP Solves

Consider an enterprise with four models in use — a frontier model for hard reasoning, a cheaper model for routine work, a self-hosted open-weight model for sovereignty-sensitive paths, a specialised model for one domain — and a dozen systems those models need to reach: a database, an internal API, a document store, a messaging platform, a code repository, a calendar, and so on. Before a shared protocol, each model that needs each system requires its own bespoke integration. Four models and a dozen systems is not sixteen pieces of integration; it is, in the worst case, forty-eight — every model wired to every system, each wiring written, tested, secured, and maintained separately. This is the $N \times M$ problem: integration effort that grows as the product of the number of models and the number of systems, not their sum.

The $N \times M$ problem is not new. It is the same problem that device drivers solved for operating systems, that ODBC solved for databases, that the Language Server Protocol solved for code editors. In each case the answer was the same shape: insert a standard protocol in the middle, so that each model integrates once against the protocol and each system exposes itself once through the protocol. Integration effort then grows as N plus M rather than N times M . Forty-eight bespoke integrations become four model-side integrations and twelve system-side ones — sixteen pieces of work, each reusable, instead of forty-eight, each bespoke.

MCP is that middle protocol for the model-to-tool problem. A model — or, more precisely, the application hosting it — speaks MCP once. A system exposes its capabilities through an MCP server once. Any MCP-speaking model can then reach any MCP-exposed system without a new integration being written. The protocol’s promise is captured in a single line from the infographic that prompted this chapter: *one protocol, any model, any tool*. The promise is real, and the rest of the chapter explains both how it is delivered and what it costs.



Figure 33.1: The integration problem MCP collapses. Without a shared protocol, every model is wired to every system and effort grows as $N \times M$. With MCP in the middle, every model and every system integrates once against the protocol, and effort grows as $N + M$.

33.2 When MCP Arrived

MCP is recent enough that its short history is worth stating, because the protocol’s rapid adoption is itself an architectural fact — it is the reason an enterprise can reasonably treat MCP as a standard rather than a bet. Anthropic open-sourced the Model Context Protocol in November 2024. Through the first half of 2025 the major model providers and tool vendors adopted it in succession, and a range of developer tools shipped native MCP support. By the time of writing, MCP has become the de facto

standard for model-to-tool integration — not through a formal standards body, but through the older mechanism by which a protocol becomes a standard: enough independent parties adopt it that not adopting it becomes the costly choice.

The architectural reading of this timeline is not that MCP is permanent — no protocol is — but that it has crossed the threshold at which adopting it is a conservative decision rather than a speculative one. An enterprise standardising on MCP in its current state is following the industry, not leading it, and the integration work done against MCP is work that the surrounding ecosystem is actively maintaining. That is the condition under which committing to a protocol is sound: when the protocol's continued support is being paid for by parties other than oneself.

33.3 What MCP Actually Is: Host, Client, Server

MCP's architecture has exactly three roles, and the whole protocol is easier to reason about once they are clearly separated.

The **host** is the AI application the user interacts with — a desktop assistant, an IDE with an AI feature, an internal agent platform. The host owns the model, the conversation, and the user relationship. It is the host that decides which servers to connect to and that ultimately enforces what the model is allowed to do.

The **client** lives inside the host. It is the host's protocol-speaking component, and the rule that matters is one-to-one: the host runs one client per connected server. A host connected to three servers runs three clients. The client is a thin conduit — it speaks MCP to exactly one server and relays between that server and the host — and the one-client-per-server rule is what keeps the connections cleanly isolated from one another.

The **server** is the component that exposes a system's capabilities through MCP. It is, in the architectural sense, the adapter: it knows how to talk to one underlying system — a database, an API, a filesystem — and it presents that system's capabilities in the protocol's standard vocabulary. A server may run as a local subprocess on the same machine as the host, or as a remote service the host reaches over the network. Crucially, the server is code the integrating organisation typically writes or chooses; exposing an enterprise system through MCP means standing up a server for it.

The relationship is a simple chain: the host contains a client; the client talks to a server; the server fronts a system. The whole arrangement is built on JSON-RPC 2.0, a long-established, deliberately unglamorous remote-procedure-call convention — which is the first hint of the chapter's recurring theme, that MCP is plumbing, assembled from well-understood parts rather than invented from nothing.

33.4 The Three Primitives

What an MCP server exposes falls into exactly three kinds of thing, and the distinction between them is the single most useful part of the MCP mental model — because the three differ precisely in *who decides* when they are used.

Tools are model-controlled. A tool is an action the model can choose to invoke — a database query, a search, a message send — and it is the model, in the course of its reasoning, that decides when to call it and with what arguments. Tools are the primitive most directly continuous with the tool-use pattern of the previous chapter.

Resources are application-controlled. A resource is a piece of context — a file’s contents, a document, a schema — and it is the host application, not the model, that decides to inject it into the model’s context. The model does not choose to fetch a resource; the application supplies it. The distinction matters because it places the decision about what context the model sees with the application, which is the component that can reason about cost, relevance, and sensitivity.

Prompts are user-controlled. A prompt, in MCP’s specific sense, is a pre-defined interaction template that the user explicitly invokes — a slash-command such as `/summarize` or `/review-pr` that the user, not the model and not the application, chooses to trigger.

Most introductory treatments of MCP skip past this three-way distinction, and skipping it is a mistake, because the distinction is the whole mental model. Tools, resources, and prompts are not three names for the same thing; they are three control regimes — model-controlled, application-controlled, user-controlled — and an architect who knows which primitive a given capability should be is an architect who has already decided who is accountable for invoking it. A capability exposed as a tool has handed the invocation decision to the model; the same capability exposed as a resource has kept that decision with the application. That is an architectural choice with governance consequences, and the primitive is where it is made.

33.5 How an MCP Interaction Travels

Three further pieces complete the picture of how MCP actually works: the transport that carries messages, the lifecycle a connection follows, and the format of the messages themselves.

Transport. MCP messages travel over one of two transports. The *stdio* transport runs the server as a local subprocess and exchanges messages over standard input and output; it is the fastest option, involves no network, and is the natural choice for local tools such as a filesystem or a local code repository. The *streamable HTTP* transport carries messages to a remote server over the network and is the choice for cloud-hosted services; it is the current standard for remote connections, having replaced an earlier server-sent-events approach. The transport choice is, in effect, a local-versus-remote decision, and it follows the same reasoning as any other: local is faster and has a smaller trust surface, remote is necessary when the capability genuinely lives elsewhere.

Lifecycle. An MCP connection follows a defined sequence. It opens with an *initialize* exchange in which the client and server agree the protocol version and announce their capabilities; a *capabilities* negotiation establishes which tools, resources, and prompts are available; an *initialized* notification confirms the connection is ready. The connection then enters its operation phase, in which the client makes the calls that do the actual work — listing and invoking tools, reading resources, fetching prompts — and it closes with a *shutdown*. The lifecycle is unremarkable, and that is the point: it is the same connect-negotiate-operate-disconnect shape as countless other protocols, which is what makes it

predictable.

The wire format. Every MCP message is a JSON-RPC 2.0 message — a JSON object naming a method, carrying parameters, and tagged with an identifier so the response can be matched to the request. A call to invoke a tool is a JSON object whose method is the tool-call method and whose parameters name the tool and supply its arguments. There is no proprietary binary format, no novel encoding. Every MCP call is a JSON-RPC message, and that is genuinely the whole of it. The deliberate plainness of the wire format is, again, the chapter’s theme: MCP’s contribution is not a clever new transport but the agreement to use an old one consistently.

33.6 What Nobody Tells You: The Costs of MCP

An honest chapter on MCP must give as much attention to the protocol’s costs as to its promise, because the costs are real, they are mostly absent from the enthusiastic introductions, and an architect who adopts MCP without budgeting for them will meet them anyway — later, and less cheaply.

The token tax. Every tool an MCP server exposes carries a schema, and that schema must be placed in the model’s context on every call so the model knows the tool exists and how to invoke it. A server exposing fifty tools puts fifty schemas into the context of every request, whether or not any of them is used. This is a standing token cost — paid on every turn, scaling with the number of exposed tools — and at scale it is a material line on the bill. The mitigation is discipline about which tools a given host actually connects, and it connects directly to the cost chapter’s insistence that token economics be measured rather than assumed.

Credential sprawl. Each MCP server that fronts a real system needs credentials to that system, and an enterprise running many servers is running many sets of credentials — to be issued, rotated, scoped, and revoked. The OAuth and secret-management burden of an MCP estate is a real operational cost, and it is the kind of cost that is invisible at three servers and serious at thirty.

Server sprawl. MCP makes it easy to stand up a server for every system, and the easy thing, done repeatedly, reproduces the microservices sprawl that the integration world already learned to be wary of. More servers mean more processes to run, more failure modes, more surface to secure and monitor. The N+M promise is real, but the M can itself grow unmanaged if no one owns the question of which servers the enterprise should have.

Schema drift. An MCP server is a contract, and a server whose underlying system changes — a field renamed, an argument added — has a schema that no longer matches reality. An agent calling against a drifted schema does not fail loudly; it misroutes quietly, calling the tool in a way that no longer does what the agent intended. Schema drift is the MCP-specific version of the interface-versioning problem, and it requires the same discipline: versioned schemas, change control, and tests that catch the drift before an agent does.

The observability gap. MCP is young, and standardised tracing across MCP calls is not yet mature. Debugging an agent’s behaviour across several MCP servers can mean reconstructing the story from logs scattered across processes — the very situation the observability chapter warned against.

Until MCP tracing standardises, an enterprise adopting MCP must supply this observability itself, instrumenting the host and the servers so that a multi-server interaction can be traced end to end.

None of these costs is a reason to avoid MCP. They are reasons to adopt it deliberately — with a budget for the token tax, a plan for credentials, an owner for the server estate, version control on schemas, and observability built rather than assumed. MCP is plumbing, and plumbing that is installed without a maintenance plan leaks.

33.7 When MCP Wins

MCP is the right choice when the integration problem it solves is the problem the enterprise actually has. The signal is plurality on both sides: more than one model, more than a handful of systems, and an expectation that both sets will keep changing. In that situation MCP's $N+M$ economics, its protection against vendor lock-in at the integration layer, its growing tool ecosystem, and its flexibility across local and remote transports are all genuine advantages, and they compound as the estate grows.

MCP is the wrong choice, or at least a premature one, when the integration problem is small and stable. A single application calling two fixed APIs does not have an $N \times M$ problem; it has two integrations, and wrapping them in MCP servers adds the protocol's costs — the token tax, the server processes, the schema maintenance — in exchange for an $N+M$ saving that, at that scale, is not worth having. The buy-versus-build chapter's discipline applies directly: adopt the protocol when the workload genuinely needs what it offers, not because it is the current standard. The standard is a reason to prefer MCP *when* a protocol is warranted; it is not a reason to introduce a protocol where none is needed.

33.8 The African Perspective

MCP's architecture intersects the book's recurring African constraints at two specific points.

The transport choice carries more weight where connectivity is variable. The streamable HTTP transport assumes a reliable network path to a remote server, and an MCP estate that places its servers in distant cloud regions inherits every connectivity constraint between the host and those regions — latency on every tool call, and a failure mode whenever the link is unstable. The stdio transport, running the server as a local subprocess, has no such exposure. For deployments where connectivity to distant regions cannot be assumed, the architectural bias should be toward local stdio servers for the capabilities that can be served locally, and toward MCP servers hosted in-region for the rest — the same in-region reasoning the vector-database and inference chapters applied, here applied to the tool-integration layer.

The credential-sprawl and observability-gap costs land harder where specialised operations talent is scarce. Managing OAuth across many servers, and building the MCP observability the protocol does not yet supply, are both specialist operational work, and an organisation that cannot staff that work deeply should treat it as a reason to keep the MCP estate small and deliberate — fewer servers, each genuinely needed — rather than letting server sprawl outrun the team that must secure and monitor it. The protocol's $N+M$ promise is as real in Lagos or Nairobi as anywhere; the discipline of not letting M grow unmanaged simply matters more where the team maintaining it is thin.

33.9 What MCP Gives the Architect

MCP is the plumbing standard for connecting models to tools and data. It collapses the $N \times M$ integration problem to $N + M$ by inserting a shared protocol in the middle; it is built from three roles — host, client, server — and three primitives — tools, resources, prompts — distinguished by who controls invocation; it travels over stdio or streamable HTTP, follows a defined connect-negotiate-operate-disconnect lifecycle, and carries plain JSON-RPC 2.0 messages on the wire. It also brings real costs — the token tax, credential sprawl, server sprawl, schema drift, and an immature observability story — that an honest adoption budgets for rather than discovers.

The epigraph holds the chapter's stance. A protocol is not magic; it is plumbing. MCP's value is not that it is clever — it is deliberately, almost aggressively, unclever, assembled from JSON-RPC and well-worn lifecycle conventions — but that it is shared. The architect's task is to treat it accordingly: master the small set of concepts it actually consists of, adopt it where the integration problem genuinely warrants a protocol, budget for its costs, and then build on it. The chapter that follows develops the server-side patterns in depth; it assumes the mental model this chapter has built. With that model in place, MCP becomes what good plumbing always is — the part of the building no one admires, and the part without which nothing else works.

MCP Server Patterns: Seven Shapes of the Tool-Integration Layer

“The pattern is the part of the architecture that survives the products that implement it.”

—a recurring observation in enterprise-architecture practice

Chapter Overview

The Model Context Protocol (MCP), introduced as an open standard in late 2024, has settled in 2026 as the working protocol by which AI clients talk to external tools, data sources, and systems. The protocol itself is small. The architectural shape of the servers that implement it is not, and seven patterns have emerged from two years of production deployments that capture most of the work an MCP server is asked to do. This chapter introduces those seven patterns — tool wrapper, resource provider, aggregator gateway, stateful session, sandbox executor, workflow orchestrator, and subagent — with their characteristic uses, structural shapes, and failure modes. The patterns are not mutually exclusive; production MCP deployments combine them, and the architectural skill consists partly in recognising which pattern applies where, and partly in placing the right pattern at the right boundary.

Learning Objectives

- Recognise the seven recurring patterns in production MCP server deployments.
- Identify which pattern fits which problem, and which patterns compose with which others.
- Apply the ABB/SBB distinction to MCP architecture: protocol as architectural commitment, specific implementations as solution choices.
- Identify the failure modes characteristic of each pattern and the architectural responses that mitigate them.
- Place MCP server choices inside the larger discipline of tool-integration architecture that the rest of Part IV addresses.

34.1 Why MCP Earned a Pattern Vocabulary

The Model Context Protocol is a small specification. The client opens a session with a server. The server advertises its capabilities — tools it can invoke, resources it can read, prompts it can supply — in a structured format. The client calls those capabilities by name, with typed arguments, and receives typed responses. There is a lifecycle for sessions and a model for streaming long results. That is, in one paragraph, the protocol.

What the protocol does not specify is how an MCP server should be structured behind that interface. A server might wrap a single external API in one direction; it might aggregate twenty services behind a single endpoint; it might run a sandboxed code interpreter; it might be, itself, an autonomous agent. The choice has consequences for security, latency, versioning, observability, cost, and the failure modes the architect will eventually have to live with. The protocol is the contract. The pattern is the implementation discipline behind the contract.

By 2026, two years of production experience has settled most MCP servers into one of seven recognisable shapes. The shapes are not exclusive — a single deployment commonly contains several patterns wired together — but each has a characteristic use, a characteristic risk, and a characteristic architectural cost. This chapter walks through them in the order a practitioner is likely to encounter them: from the simplest one-API wrapper, through the patterns that aggregate or hold state, to the patterns that delegate execution or autonomy.

In the language of Chapter 4, MCP itself is an architecture building block — a logical capability the architecture commits to, namely a uniform tool-integration protocol. The specific MCP server implementations are solution building blocks. The patterns in this chapter sit at the ABB layer: they describe *shapes* of MCP server, independent of which language, framework, or vendor is used to build them. The reader thinking about MCP adoption should commit to the patterns at the architecture-review stage and choose the implementations later, when the market has moved.

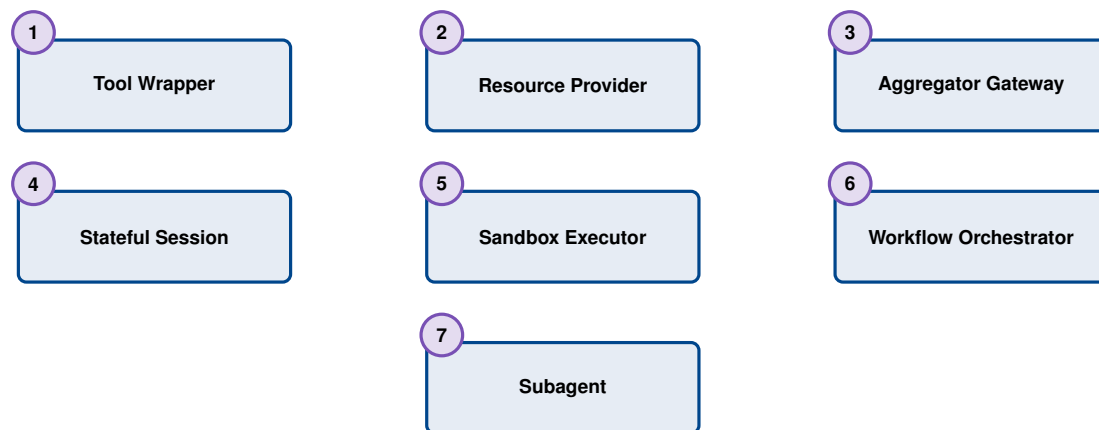


Figure 34.1: The seven MCP server patterns. Top row: stateless integration patterns (one API, one resource, many services). Middle row: stateful and protected patterns (live state, trust boundaries, hidden workflows). Bottom: full delegation (the server is itself an agent).

34.2 Pattern 1: Tool Wrapper

The tool wrapper is the simplest MCP pattern and the one most newcomers build first. One server, one external API, a near-1:1 mapping of API methods to MCP tools. The server's job is to translate between MCP's protocol idioms and the underlying API's conventions: authentication, request shape, response shape, error format. The GitHub MCP server is the canonical example: each GitHub REST endpoint becomes an MCP tool, with the server handling OAuth token exchange and rate-limit propagation.

When to use it. When the underlying capability is a single coherent API with a well-defined surface, and when the consumer of the MCP server wants the full surface of that API exposed. Tool wrappers are also the right choice when versioning is straightforward — the MCP server version tracks the API version, and breaking changes are managed by deprecating tool names rather than by reshaping the server's internal logic.

Structural shape. Agent calls MCP server; MCP server calls external API; external API returns a result; MCP server translates the result back to MCP's typed response shape. Stateless between calls. No session state held in the server. The protocol mapping is the only logic.

Failure modes. The most common is the rate-limit cascade: the agent makes many MCP calls in a tight loop, each of which makes one upstream API call, and the upstream rate limit is hit before the agent's logic has finished. The mitigation is to expose the rate-limit signal back to the agent through MCP's error channel, so the agent can adapt rather than retry blindly. A second failure mode is the field-mapping drift: the upstream API adds fields, the MCP wrapper does not, and the agent loses access to capabilities that have been added since the wrapper was written. Regular regeneration of the wrapper from the API spec — ideally automated — is the architectural response.

What this pattern is not. Tool wrappers are not the right pattern when the agent should be shielded from the upstream API's quirks. If the agent should not know that the underlying system is GitHub specifically, but only that it can read and write code repositories, the abstraction should sit higher than the wrapper. That is the workflow orchestrator pattern, not the tool wrapper.

34.3 Pattern 2: Resource Provider

The resource provider exposes data to the agent without giving it the ability to act on the underlying systems. The agent reads files, queries database rows, fetches documents, and inserts the resulting content into its context window. The server reads; it does not write. This is the MCP equivalent of a read replica in database terms, or a read-only API key in security terms.

When to use it. When the agent's job is to reason over information rather than to change it. Customer-support agents that retrieve past tickets, research agents that pull documents from a knowledge base, code-review agents that read repository files: these are all resource providers. The pattern is also the right choice for compliance-sensitive deployments where the architectural commitment is that the agent cannot, by construction, modify the systems of record.

Structural shape. Agent issues a read request; the MCP server fetches the requested resource from the underlying store (files, database, document repository); the resource flows back into the agent's

context window. The server has no write methods.

Failure modes. The most common is context bloat: the agent retrieves more than it needs, the context window fills, and the model's effective reasoning capacity degrades. The mitigation is at the server side — the resource provider should support filtering, pagination, and excerpting, so the agent can ask for a focused subset rather than the whole document. A second failure mode is the staleness illusion: the agent treats retrieved content as current when it has actually been cached or replicated. The resource provider should expose freshness metadata — when the document was last modified, when the index was last rebuilt — and the agent's prompts should be written to handle the distinction between fresh and stale data.

Composition. Resource providers compose naturally with tool wrappers in the read direction and with workflow orchestrators in the write direction. A common production architecture has a resource provider exposing read-only data to the agent, and a workflow orchestrator exposing the write operations through a constrained, audited path.

34.4 Pattern 3: Aggregator Gateway

The aggregator gateway fronts many services behind a single MCP endpoint. From the agent's point of view, there is one MCP server. Behind it sit the actual systems — Jira, Salesforce, an internal database, Datadog, possibly a dozen more — and the gateway routes calls to the right backend, applies cross-cutting policies, and returns results in a uniform shape. The agent does not need to know which backend served any given call.

When to use it. When the same architectural concerns apply across many integration points. Authentication, rate limiting, observability, and policy enforcement are most efficiently applied once, at the gateway, rather than separately at each tool wrapper. The aggregator is also the right pattern when the agent's logic should be decoupled from the specific tool inventory — adding a new backend system should require gateway configuration, not agent-prompt rewrites.

Structural shape. Agent calls the MCP gateway; the gateway authenticates, applies rate limits and observability hooks, routes the call to the correct backend, applies any policy filters on the response, and returns. Auth, rate limits, observability, and policy are the gateway's reason for being.

Failure modes. The most common is gateway becoming a queue. Every agent calls the gateway; the gateway forwards to backends; the gateway's own capacity becomes the bottleneck before any backend does. The mitigation is to run the gateway as a horizontally scaled set of stateless instances, with backend connection pools sized to handle aggregate load. A second failure mode is policy drift: the gateway's policy rules are updated at the operations layer without the architecture record being updated, and over time the gateway's effective behaviour diverges from what the architecture says it does. The mitigation is policy-as-code with the policy versioned, reviewed, and tied back to the relevant ADRs.

Composition. Aggregator gateways are the standard top-of-stack pattern in larger MCP deployments. Underneath the gateway, individual tool wrappers, resource providers, sandbox executors, and work-

flow orchestrators each serve their specific role, and the gateway handles the cross-cutting concerns that would otherwise be repeated in each.

34.5 Pattern 4: Stateful Session

The stateful session pattern is for capabilities that genuinely require live state across multiple calls. A browser automation tool needs the same browser session for sequential page interactions. A database transaction needs the same connection across the operations within the transaction. A long-running build needs the same workspace across the steps of the build. Stateless wrappers cannot serve these because the state is the capability. The Playwright MCP server, which exposes a live headless browser to the agent, is the canonical example.

When to use it. When the underlying capability is inherently stateful: browser sessions, database transactions, open file handles, REPL processes, long-running computations that span multiple agent turns. Also when the cost of session establishment is high enough that pooling pays for itself — a database connection that takes 200 ms to establish should not be re-established on every MCP call.

Structural shape. Agent opens a session against the MCP server; the server maintains session state and a connection pool to the underlying resource; subsequent calls are routed to the existing session by session ID; the session is explicitly closed by the agent or implicitly closed by timeout. Browser sessions, DB transactions, and open file handles are the canonical resources.

Failure modes. Session leakage is the most common: agent crashes, session times out, the underlying resource (browser, database connection, file handle) remains allocated until reaped, and the MCP server's resource budget is exhausted by orphaned sessions. The mitigation is aggressive session timeout, heartbeat protocols between agent and server, and explicit resource budgets per agent identity. A second failure mode is session contamination: one agent's session state leaks into another's because of a bug in session routing or pool isolation. The mitigation is per-session isolation at the level of the underlying resource (one browser per session, one connection per transaction) rather than shared resources protected only at the application layer.

Security. Stateful sessions are a privilege-escalation surface. A compromised session can outlive the agent that opened it. Sessions should be tied to authenticated agent identities, should expire on a defined schedule, and should be observable so that anomalous session lifetimes can be detected.

34.6 Pattern 5: Sandbox Executor

The sandbox executor pattern is the trust boundary made architectural. The agent generates code, or shell commands, or filesystem operations, and the MCP server executes them inside an isolated sandbox. The sandbox has its own scratch filesystem, its own ephemeral identity, no access to the rest of the production environment, and a defined lifetime. The agent gets the result of the execution; the host system gets none of the side effects. E2B's sandboxing service is the canonical example of the pattern in 2026.

When to use it. Whenever the agent is allowed to execute code or commands, and the host system cannot afford the side effects of that execution. Data analysis agents that run computations on uploaded datasets; coding agents that compile and test code; agents that need a shell for legitimate operational tasks; anything where the agent’s output is, by design, executable. The pattern is a security architecture, not just a capability architecture: the sandbox is what makes the executability survivable.

Structural shape. Agent submits code or commands to the MCP server; the server hands them to an isolated sandbox (a container, a microVM, a language runtime) that has its own code-execution context, shell, and scratch filesystem; the sandbox runs the code, returns the result, and is torn down (or retained for a defined session if the workload requires statefulness across calls). The boundary is the load-bearing component.

Failure modes. The most consequential is sandbox escape. A vulnerability in the sandboxing technology lets code break out of the boundary and access host resources. The mitigation is defence in depth: containerised sandboxes inside microVMs, inside dedicated nodes, with the agent’s identity scoped tightly and the sandbox network egress filtered. A second failure mode is resource exhaustion: the agent submits code that consumes unbounded CPU, memory, or storage, and the sandbox infrastructure is overwhelmed before the runaway is detected. The mitigation is per-sandbox resource quotas with hard limits and short execution timeouts.

The trust boundary is the architecture. A sandbox executor without strong isolation is not a sandbox; it is an attack surface with hopeful branding. The architectural review should treat sandbox-executor MCP servers with the same scrutiny that any other code-execution surface receives. The right place for that scrutiny is at the architecture-review board, in the language of Chapter 31, and the decision should be recorded as an ADR with explicit reference to the threat model.

34.7 Pattern 6: Workflow Orchestrator

The workflow orchestrator pattern hides multi-step logic from the agent. The agent makes a single MCP call — “onboard this customer”, “process this refund”, “deploy this build” — and the MCP server executes a multi-step internal workflow against several backend systems before returning a result. The agent’s prompt is simpler because the workflow is encapsulated. The workflow is more reliable because it is not subject to the agent’s reasoning failures at intermediate steps.

When to use it. When the multi-step process is well-defined and should not be redesigned on every invocation. Business processes with defined steps and compliance requirements — KYC checks, refund processing, deployment pipelines, ticket escalation procedures — belong in workflows rather than in agent prompts. The agent calls the workflow; the workflow does the steps; the agent is not asked to reason about the steps.

Structural shape. Agent makes one call to the MCP server; the server executes step 1, step 2, step 3, and so on against backend systems; the final result is returned to the agent. The intermediate steps may include retries, conditional branches, and human-in-the-loop checkpoints — but all of that is internal to the server, not visible to the agent.

Failure modes. The most common is the workflow’s intermediate failure mode being indistinguishable from the agent’s perspective. The agent calls the workflow; the workflow fails partway; the agent receives an error and does not know what state was partially established. The mitigation is for the workflow to maintain its own internal state machine, support resumption from a known checkpoint, and expose to the agent a status query that returns the current state in a meaningful way. A second failure mode is workflow rigidity: a process encoded in the orchestrator is hard to change, and over time the encoded workflow diverges from the actual business process. The mitigation is the standard architectural one — the workflow is a versioned artefact, owned by a named team, with a deprecation policy.

Composition with agents. Workflow orchestrators encode the parts of the work that are well-understood. The agent does the parts that genuinely require reasoning — which workflow to call, with what arguments, in what order. The architectural decision is which logic belongs in the workflow and which belongs in the agent’s reasoning, and the standard test is whether the logic should be changeable by a prompt edit (agent) or by a deployed code change (workflow). Compliance-sensitive logic almost always belongs in the workflow.

34.8 Pattern 7: Subagent

The subagent pattern is the most powerful and the most architecturally consequential. The MCP server is itself an agent: it has its own model invocation loop, its own context, its own tool inventory, and its own reasoning. From the parent agent’s perspective, it is just another MCP server with a small surface — typically one or two tools, with task-shaped inputs and answer-shaped outputs. From the inside, it is a full agent that may itself invoke other MCP servers, including other subagents.

When to use it. When a subtask is complex enough to deserve its own context and reasoning, and the parent agent should be shielded from the subtask’s internal complexity. A research agent might invoke a subagent to write code; a customer-service agent might invoke a subagent to negotiate within defined limits; a planning agent might invoke a subagent to evaluate options. The subagent has its own scope; the parent has its own scope; the boundary is the MCP interface between them.

Structural shape. Parent agent calls the MCP server; the MCP server is itself a subagent with its own loop and context; the subagent invokes its own tools (which may include other MCP servers); the subagent produces an answer and returns it through the MCP interface. From outside, the subagent looks like a tool. From inside, it is an agent.

Failure modes. The most consequential is unbounded recursion: a subagent invokes a sibling subagent which invokes the original, and the conversation tree explodes. The mitigation is hard depth limits, per-agent budgets enforced at the orchestrator layer, and explicit tracking of the call chain. A second failure mode is responsibility diffusion: when a parent-agent failure could be either the parent’s reasoning or any of its subagents’, the debugging surface multiplies. The mitigation is observability infrastructure that tracks the full chain of agent invocations with input and output for each, so that a failure can be traced back to the responsible agent. A third failure mode, often underestimated, is cost compounding: each subagent invocation makes model calls, and a parent that invokes five subagents

in parallel can multiply its inference bill by an order of magnitude with no visible change at the parent level. The mitigation is cost budgets per call chain, observable at the orchestrator layer, with hard cut-offs.

Trust and autonomy. Subagents are an autonomy-escalation pattern: by delegating to a subagent, the parent is also delegating part of its decision-making to a model invocation it cannot fully constrain. The principles of Chapter 31 apply here directly. The autonomy boundary should be set in advance, in writing; the subagent’s allowed actions should be enumerated; the audit trail should capture which subagent took which action with what authorisation. Subagent patterns deployed without these are autonomy without governance.

34.9 Composing the Patterns

The seven patterns are not mutually exclusive. A working production MCP deployment commonly contains all seven, wired together in defined positions. A representative architecture for an enterprise agent platform might look like this:

- An **aggregator gateway** at the top, presenting a single MCP endpoint to all internal agents and applying authentication, rate limiting, observability, and policy enforcement uniformly.
- Beneath the gateway, **tool wrappers** for the external APIs that need full surface exposure (GitHub, Salesforce, Jira, a small number of specialised systems).
- **Resource providers** for the read-only data sources that feed agent context (knowledge bases, document repositories, log indexes).
- **Stateful session** servers for the capabilities that require live state (browser automation, REPL access, database transactions).
- A **sandbox executor** for any code-execution surface, with strong isolation enforced at the infrastructure layer.
- **Workflow orchestrators** for the business processes that should not be re-derived on every agent invocation (refunds, deployments, onboarding flows).
- One or more **subagents** for specialised domains where the architectural decision has been made to delegate full reasoning, with the corresponding governance scaffolding in place.

The architectural decision at this scale is less about which patterns to use — the answer is almost always “most of them” — and more about which boundaries to enforce between them. The gateway enforces cross-cutting policy. The sandbox enforces trust. The subagent boundary enforces autonomy scope. Each of these boundaries deserves its own architectural review, its own ADR, and its own observability infrastructure. The patterns provide the structural vocabulary; the discipline of placing them at the right boundaries is the architectural work.

34.10 The African Perspective

MCP server patterns inherit the same constraints African enterprises operate under elsewhere in the book. A few are sharper in this context than others.

The *aggregator gateway* pattern is particularly valuable in environments where every external API call has a measurable cost in foreign exchange and every credential is a procurement-tracked artefact. Concentrating the policy, rate limiting, and observability at a single gateway means that the cost surface is visible in one place rather than spread across dozens of integration points. The discipline that follows is the standard one: the gateway is the place where “how much are we spending with vendor X this month” is answered, and the gateway is the place where the answer changes when a procurement decision is made.

The *sandbox executor* pattern is the most operationally demanding, because the sandboxing technology itself — microVMs, container isolation, network egress filtering — is non-trivial to operate. Frontier sandbox-as-a-service offerings exist but may be cost-prohibitive or have data-residency limitations. The architectural response is to commit to the pattern at the ABB layer and to evaluate implementations against both technical sufficiency and operational viability. An over-engineered sandbox that cannot be operated locally is worse than a simpler sandbox that can be.

The *subagent* pattern has implications for cost predictability that are sharper in cost-sensitive deployments. A parent agent that invokes three subagents per call, each of which makes ten model calls, can multiply the inference bill by an order of magnitude. The cost budget enforcement at the orchestrator layer is not optional in these environments; it is the architectural commitment that keeps the deployment financially viable. The optimistic reading is the same as elsewhere in the book: architectures developed under tight cost constraints tend to produce disciplines the rest of the field eventually adopts, and the rigour around subagent cost budgeting is a candidate.

34.11 What the Patterns Make Possible

The seven MCP server patterns provide a vocabulary by which an architect can describe what is being built, why it is structured the way it is, and what it is not. A team that says “we have an MCP server” has said almost nothing; a team that says “we have an aggregator gateway in front of three tool wrappers, a stateful-session server, a sandbox executor, and one workflow orchestrator, with the autonomy boundary set at the gateway and the trust boundary at the sandbox” has said something an architecture-review board can act on.

The patterns are the architectural commitment; the specific server implementations are the solution commitment. The protocol is the contract that lets the two be separated. The reader who takes seriously the discipline of the previous chapter — the discipline of building the right cages, in advance, for the failure modes a serious deployment will eventually meet — will recognise that the boundaries between these patterns are precisely the cages worth building. The gateway is the cage for cross-cutting policy. The sandbox is the cage for executable code. The subagent boundary is the cage for delegated autonomy. The architect who places these cages well is doing the work of the discipline. The architect who treats MCP as a uniform substrate that can be deployed without the structural distinctions has not yet seen the cages and will, in due course, learn about them on a morning when the alternative

arrives.

Pattern: Human-in-the-Loop and Approval Workflows

“The model that needs no human is the model that does not matter. The model that needs every human is the model that does not scale. Pick the loop honestly.”

—a recurring observation in deployment design

Chapter Overview

Human-in-the-loop — HITL — is referenced throughout the industry chapters, but it deserves treatment as a first-class architectural pattern in its own right. Whether the motivation is safety, regulatory compliance, quality, or trust-building, many AI systems require human review or approval at specific points. This chapter develops HITL as a design space rather than a switch: when human involvement is warranted and at what depth, the four variants distinguished by where the human sits relative to the model’s output, the six canonical patterns those variants compose into, the reviewer experience and queue engineering that determine whether HITL works in practice, its economics at scale, and how human involvement should evolve as a system matures.

Learning Objectives

- Determine when human involvement is warranted, and recognise which conditions are durable and which transitional.
- Apply the triage pattern and its four routing mechanisms.
- Distinguish the four HITL variants by where the human sits relative to the output.
- Select among the six canonical HITL patterns and compose them.
- Design the reviewer experience and the review queue as first-class concerns.
- Manage the economics of human review and evolve involvement as the system matures.

35.1 When Human Involvement Is Warranted

The first thing to say about human-in-the-loop is that the decision is not binary. It is not a question of HITL or not, but of where, how often, and at what depth a human is involved — and the same system may warrant heavy human involvement on one surface and none on another. The involvement

that is right at launch is rarely the involvement that is right two years on. HITL is a design space to be navigated deliberately, and the navigation begins with an honest reading of which conditions favour human review, which disfavour it, and how the favouring conditions are resolved into a routing design.

Six conditions favour human involvement: *consequence*, when an incorrect decision is costly, irreversible, or harmful; *regulation*, when a regulator requires a human decision, as in lending denials, medical decisions, and some legal contexts; *confidence*, when the model's uncertainty exceeds an acceptable threshold; *novelty*, when an input falls outside the training distribution; *trust*, when an organisation is still earning confidence in a system and human review demonstrates diligence; and *capture*, when human decisions provide a training signal worth collecting. These six are not equivalent in how long they justify review, and recognising the difference is what tells an architect whether a given review point is permanent or a stage to be grown out of. Regulation and consequence are durable: they favour human involvement for as long as the regulation stands or the action remains irreversible. Confidence, novelty, and trust are transitional: they favour review while the model is uncertain, while inputs are still unfamiliar, or while confidence is still being earned, and they weaken as the deployment matures. Capture is instrumental: it justifies review for the value of the training signal, which itself diminishes as the model improves.

Against these are conditions that disfavour human review, and two of them are uncomfortable enough to state plainly: human review can make a system worse, not merely slower. Volume can be infeasible for human review; latency requirements can exclude it; the model's performance can reliably exceed the human's. But beyond these, human review can introduce inconsistency rather than improvement, and reviewer fatigue can produce poor decisions. A human reviewer is not a guaranteed source of correctness — reviewers disagree with one another, bring their own biases, and degrade under fatigue — so a review step staffed without regard to these facts can add noise to a system that was more consistent without it. This is the honest counterweight to the instinct that more oversight is always safer. The architectural question is not whether humans are involved but whether the specific review being designed will, given realistic staffing and workload, actually improve outcomes. Where it will not, adding it is theatre that costs money and degrades quality at once.

When the favouring conditions win, they almost always resolve into the same shape: the triage pattern, in which AI handles the routine cases and humans handle the consequential, uncertain, or novel ones. What makes triage trustworthy is that it routes by four complementary mechanisms rather than one. A *confidence threshold* routes the cases the model knows it is unsure about. A *risk category* routes consequential cases to humans regardless of confidence, because a model can be confidently wrong. *Sampling* pulls a fraction of confident cases for review precisely to detect the confident errors the other mechanisms would pass. And *novelty detection* routes the out-of-distribution input that the model may score as confident only because it has no basis for doubt. A triage design that uses confidence routing alone has the most common and most dangerous gap: it trusts the model's self-assessment, and the model's self-assessment is exactly what fails on the cases that matter most. The four mechanisms together are what close that gap.

35.2 The Four Variants

Human-in-the-loop is not one pattern but a spectrum, and the cleanest way to organise the spectrum is by a single question: where does the human sit relative to the model's output? Figure 35.1 shows the four variants that answer that question differently, and the difference between them is the difference between blocking an action and reviewing it after the fact.

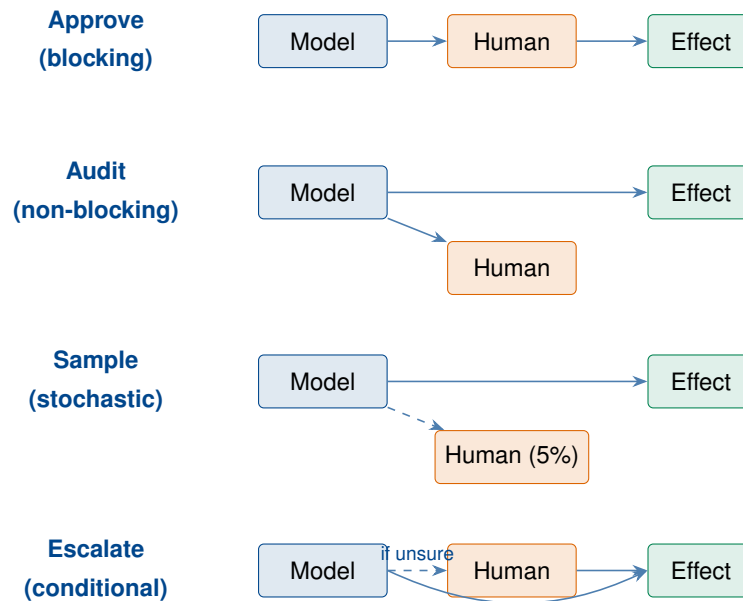


Figure 35.1: Four HITL variants. *Approve*: the human blocks every action. *Audit*: the action proceeds and the human reviews afterward. *Sample*: the human reviews a stochastic fraction. *Escalate*: the model's own uncertainty, or a policy rule, triggers human review.

The **approve** variant places the human before the effect: nothing happens until a human has approved it. It is the right choice for irreversible, high-stakes actions — large payments, public communications, regulatory filings — where the cost of a wrong action exceeds almost any review cost. It carries the highest latency and the most friction, and it delivers the lowest residual risk. The **audit** variant lets the action proceed and places the human after it, which is appropriate when an action can be reversed within a window: a transaction with a hold period, a draft before publication. It is the reasonable balance for the large middle ground of actions that are consequential but not irreversible. The **sample** variant also lets the action proceed but routes only a stochastic fraction — typically a few per cent — to a human, and its purpose is monitoring rather than gate-keeping; it suits high-volume, low-stakes actions such as customer chat, classification, and recommendations, where the goal is to detect a drift in quality rather than to catch every error. The **escalate** variant is the conditional one: most outputs flow straight to effect, and only those the model flags as uncertain, or that a policy rule catches, divert to a human. It suits mixed-stakes workloads where the model can self-assess confidence and the population divides cleanly into safe cases and cases that warrant a look. The escalate variant is the direct implementation of the triage pattern of the previous section, and it is the variant most production systems lean on for their default path.

35.3 The Reviewer Experience

A HITL design is only as good as the human review it actually produces, and the quality of that review depends on the reviewer's interface as much as on the model. The most common failure of HITL in production is not a failure of the routing logic but of the screen the reviewer faces: handed a wall of text with no diff, no context, and no rationale, a reviewer under volume pressure starts rubber-stamping, and the human-in-the-loop becomes a human-shaped rubber stamp that adds latency and removes nothing.

Designing against that failure means designing the reviewer's screen with as much care as any user-facing surface. The interface should surface what the model relied on — the retrieved chunks, the tool outputs, the prior turns — so the reviewer can judge the output against its basis rather than in a vacuum. It should show the model's own confidence and stated rationale, so the reviewer knows where the model itself was unsure. It should pre-highlight the parts the reviewer should focus on, rather than leaving them to scan everything equally. It should make rejection structured, through reason codes rather than free text, so that rejections become analysable data rather than prose nobody aggregates. And it should be instrumented: reviewer agreement rates and time-per-review are the two metrics that reveal rubber-stamping, which shows up unmistakably as agreement rising toward unanimity while time-per-review falls toward zero. An organisation that measures those two numbers will know whether its human review is real; an organisation that does not has no way to tell whether it is buying oversight or theatre.

35.4 The Six Canonical Patterns

The four variants compose, in practice, into a small number of canonical HITL architectures, each suited to a particular workload shape. Figure 35.2 maps the six on two axes — how often humans are involved, and where in the lifecycle they sit — and most production systems use one of them as a primary pattern and others as variants on specific surfaces.

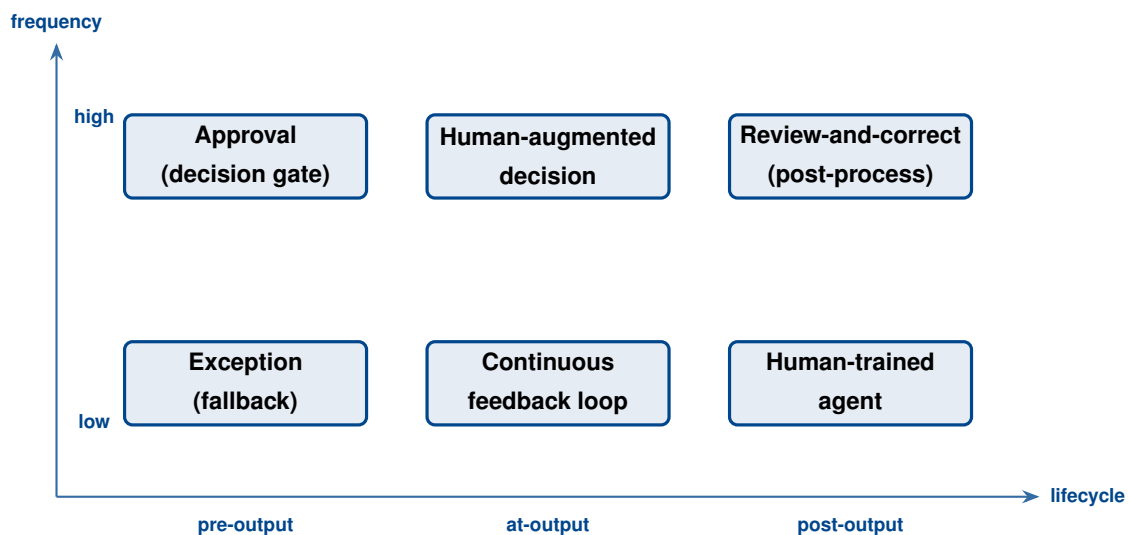


Figure 35.2: Six canonical HITL patterns mapped by human-involvement frequency (vertical) and lifecycle position (horizontal). Top row: humans involved on most items. Bottom row: humans involved on a small fraction. Left: before the output is released. Middle: at the moment of decision. Right: after the output exists.

Approval-based HITL, the decision gate, routes every output through human approval before release: the input arrives, the AI generates an output, a confidence or risk score is computed, a human reviews and approves, rejects, or modifies, the final output is released, and the decision is logged for improvement. Its coverage is high, its cost is high, and its residual risk is the lowest of any pattern — which makes it appropriate exactly where the cost of a wrong AI decision substantially exceeds the cost of human review: insurance underwriting at certain limits, credit decisions above a threshold, regulated customer communications, irreversible financial actions. Architecturally it runs synchronously where human latency is acceptable and asynchronously with a queue where volume demands; its confidence scores must be calibrated, because uncalibrated scores produce either review-everything bottlenecks or review-nothing safety gaps; and its reviewer interface must show the AI’s reasoning and supporting data, or reviewers will either rubber-stamp or override blindly.

Exception-based HITL, the fallback system, is the pattern most often simply called “HITL” in production, and the triage pattern of the opening section is its theoretical core. The AI processes incoming requests and handles most of them automatically; only a residual minority — flagged by low confidence, a novel input pattern, a policy category that requires human judgement, or an explicit user request — escalate to a human, who handles the exception and whose resolution is captured and fed back. Coverage is low, typically five to twenty per cent, the cost is manageable, and the principal architectural lever is the calibration of the escalation threshold: set it too aggressively and humans are overloaded, too leniently and errors leak through. The feedback loop is essential here, because each exception is a training signal both for the model and for the threshold itself.

The continuous feedback loop involves humans — here, the users themselves — on most interactions, but lightly. The AI generates a response, the user interacts with it, and feedback is collected from thumbs up or down, edits, regenerations, dwell time, and follow-up questions; the feedback is stored and categorised, and when signals accumulate they trigger retraining, fine-tuning, or prompt tuning.

It is the ambient signal of any working production system and dovetails with continual learning (Chapter 60). Its central challenge is feedback quality: thumbs are noisy and biased, edits are richer but rarer, and regenerations indicate dissatisfaction without explaining it, so aggregating these signals into a useful training set requires deliberate engineering rather than mere collection — governed throughout by privacy and consent over what may be used.

The human-augmented decision system keeps the human in the decision and casts the AI as decision support: the AI surfaces a recommendation with reasoning, a human evaluates it against the supporting data, and the human accepts, overrides, or modifies it before the decision is executed and recorded. It suits fraud detection, underwriting, hiring, clinical triage, and investment decisions — the cases where accountability must rest in human hands for regulatory, ethical, or labour-law reasons. It differs from the approval pattern in that the AI's role is advisory throughout rather than a gate. Here the reasoning artefact matters enormously: a recommendation without supporting evidence is either rubber-stamped or ignored, and what turns the pattern from a checkbox into a genuine augmentation is calibrated confidence, attribution to source documents, and counterfactual presentation — showing the reviewer that, had some factor been different, the recommendation would have changed.

Review-and-correct HITL, the post-processing layer, has the AI produce a draft that a human then refines: the AI generates an initial output, a reviewer edits and corrects it, the final version is approved, and the corrections are tracked so that patterns in them improve future outputs. It suits document processing, summarisation, translation, transcription, and content adaptation, where productivity multipliers of two to five times are common — the value being the time saved rather than autonomy gained. A useful property of this pattern is that the edit-distance between the AI draft and the human-approved final is itself a quality metric, and as that metric improves the team can shift the easy cases to the exception pattern while keeping review-and-correct for the hard ones. It is often a stepping-stone rather than a destination.

The human-trained agent system, the learning loop, involves humans in shaping behaviour rather than in individual decisions: an agent performs a task, its action logs are captured, a human reviews the behaviour — sampled, scheduled, or triggered — validates the steps and identifies mistakes, and rules, prompts, or fine-tuning are updated so the agent acts better next cycle. It suits agentic systems where individual actions are low-stakes but cumulative behaviour matters: routine automation, internal-tool agents, scheduled-task agents. It connects tightly to evaluation-driven release (Chapter 51) and continual learning. Its architectural prerequisite is that the action logs be structured enough for human review to be feasible at scale — the trace stores of Chapter 49 are the foundation, and without good traces the reviewer cannot tell what the agent actually did and the learning loop breaks. Sampling strategy — random, error-biased, or novelty-biased — balances review cost against learning yield.

The six patterns are not mutually exclusive, and a mature deployment composes them: the exception pattern for the high-volume default path, the approval pattern on a small set of high-risk surfaces, the continuous feedback loop ambient across all user interactions, review-and-correct on draft-and-refine workflows, the human-trained-agent pattern on autonomous-task surfaces, and the human-augmented pattern on decisions where accountability must rest with humans. The choice for any specific surface

follows four questions: what is the cost of an AI mistake, how often does it happen, who bears the consequence, and what evidence does the regulator or the business require?

35.5 Capturing Feedback as Signal

Across all six patterns, human review is not only a safeguard; it is one of the most valuable training and evaluation signals a production system generates, and a HITL design that does not capture it is discarding the very data that would let the system need less human review over time. What to capture is straightforward: the reviewer's decision — approve, reject, or modify — together with their rationale where it is appropriate, their confidence, the time the review took, the final action, and, where it can be observed downstream, the eventual outcome. That record feeds several uses at once: it evaluates the model, by giving precision and recall against human-labelled data; it improves the model, through fine-tuning or preference learning; it tunes the routing thresholds; it surfaces systematic AI errors through pattern detection; and it feeds reviewer training.

What makes captured feedback trustworthy is treating its quality as a measured property rather than an assumption. Inter-rater reliability asks whether reviewers agree with one another — low agreement usually indicts the rubric, not the reviewers. Bias monitoring detects systematic differences between reviewers that the model would otherwise inherit. Calibration asks whether stated confidence levels are accurate. And drift monitoring asks whether the patterns are changing over time. Feedback collected without these checks is feedback of unknown quality, and a model trained on it inherits whatever the reviewers' disagreements, biases, and miscalibration encoded. The discipline is the same one the preference-learning and evaluation chapters insist on, applied here to the human review stream.

35.6 Queue Engineering and Economics

Production HITL is, underneath the pattern, a queue system, and at scale the queue requires active engineering. Items arrive faster than they are reviewed; some are more urgent or more sensitive than others; reviewers differ in skill and capacity; and the people in the queue are people, whose wellbeing is both a duty and a determinant of quality. A well-engineered review queue therefore uses priority levels keyed to risk, customer tier, and urgency, with SLA targets per level; it routes by skill, so that multi-lingual, domain-specialist, or sensitive-content items reach reviewers equipped for them; it load-balances across reviewers; and it protects against burnout through rotation and, for graphic content, mandatory breaks. It also mixes calibration checks into the live queue — synthetic items with known answers — so that reviewer accuracy is measured continuously rather than assumed. Reviewer wellbeing is not a soft concern appended to the design: sustainable workload, breaks and rotation, mental-health support for content-moderation work, career progression, and a genuine voice in the tools they use are what keep review quality from decaying through turnover and fatigue, and the chapter's earlier warning — that a fatigued reviewer can make a system worse — is what gives this its architectural weight.

The queue also has an economic structure, and that structure shapes the design as much as the risk profile does. The cost of human review is driven by the reviewer's hourly cost, which varies by

skill and geography; the average review time per item; the volume of items routed for review; the quality-control overhead; the training and turnover cost; and the tooling. Several levers reduce that cost without sacrificing quality, and they are worth pursuing in order: tuning the escalation threshold so fewer items reach humans without quality loss; pre-classifying with a triage model so each review takes less effort; framing the work as suggestion-and-accept rather than produce-from-scratch; optimising the workflow to cut the non-review time around each item; and matching items to appropriately-skilled reviewers so expensive specialists are not spent on routine work. What ties the economics to the architecture is the unit calculation: the cost per review, the value created per review, and the proportion of volume the AI handles without review at all together determine the net economics, and a HITL design whose unit economics have never been computed is a design flying blind on its single largest operating cost.

35.7 Evolution Over Time

A HITL design is not a fixed setting; it is a trajectory, and the transitional conditions of the opening section — confidence, novelty, trust — are precisely the ones that should be expected to relax as a deployment matures. In early deployment, thresholds are conservative and a larger fraction of cases goes to humans: the organisation is building trust, detecting patterns across early decisions, and feeding intensive reviewer feedback back into the model. In mature deployment, the thresholds have been tuned against observed performance, a smaller fraction reaches humans, sampling takes over the quality-control role that blanket review played early on, and the whole arrangement is re-evaluated periodically. In what might be called the mature-plus stage, the AI handles most cases, humans concentrate on escalation, novelty, and audit, the human capacity freed up is redirected to higher-judgement work, and the continuous-improvement infrastructure is itself mature.

The discipline in this evolution is to make each loosening of human involvement an evidence-driven decision rather than a drift of convenience. The durable conditions — regulation, irreversibility — do not relax, and human review on those surfaces stays regardless of how the model improves. The transitional conditions do relax, but only when the evidence says the model has earned it, and the relaxation should be recorded, with the evidence that justified it, and be reversible if that evidence later weakens. A deployment that tightens human review when incidents occur and loosens it when the evidence supports loosening is being governed; one whose human involvement only ever decreases, because review is expensive and the pressure is always downward, has stopped tracking whether the model deserves the trust it is being given.

35.8 Anti-Patterns

The failure modes of HITL are common, recognisable, and avoidable:

- **Rubber-stamping** — humans approve without genuine review because the volume leaves no time for it. The human-in-the-loop is nominal; the oversight is not real.
- **Reviewer fatigue** — review quality degrades over time as the same people work the same queue without rotation or rest.

- **Theatre review** — humans are nominally in the loop but not actually empowered to decide, so their presence satisfies a diagram and changes nothing.
- **Missing feedback capture** — reviews happen but are not recorded as signal, so the system never learns from the human effort spent on it.
- **Latency catastrophe** — synchronous review is imposed where asynchronous review would have served, paying a heavy latency cost for no risk reduction.
- **Reviewer wellbeing ignored** — workload, rotation, and support are neglected, and the resulting turnover and fatigue degrade quality at the source.
- **No threshold tuning** — escalation thresholds are set once and never revisited, keeping review volumes higher, or lower, than the observed performance warrants.
- **No quality control on reviewers** — reviewer decisions are trusted without calibration checks, so a drifting or biased reviewer goes undetected.

What unites these anti-patterns is that each treats human review as a box to be ticked rather than a system to be engineered. Rubber-stamping, theatre review, and ignored reviewer wellbeing all produce the appearance of oversight without its substance; missing feedback capture and absent threshold tuning leave the system unable to improve; the latency catastrophe pays a cost the workload never required. The remedy for the whole list is the chapter's recurring instruction: treat HITL as a first-class architectural pattern — with variants chosen deliberately, a reviewer experience designed with care, a queue engineered as a queue, feedback captured as signal, and involvement evolved on evidence — rather than as a human inserted into a diagram to make it look safe.

35.9 The African Perspective

Human-in-the-loop has a particular shape in African deployments, and the shape is, on balance, favourable — with one caution.

The favourable side is that human review is labour, and the cost-economics section's central variable, the reviewer's hourly cost, is lower in many African markets than in the markets where most AI tooling originates. This changes the design calculus: a HITL strategy that would be economically marginal at North American or European reviewer costs can be comfortably viable, which means African deployments can often afford a more thorough human-in-the-loop posture — broader coverage, more generous sampling, a slower and more deliberate relaxation of the transitional conditions — than a cost-per-review comparison made against Northern figures would suggest. Human review is also a source of skilled employment, and a HITL operation built deliberately — with the career progression, the genuine tooling voice, and the wellbeing protections the queue section insisted on — is an asset to the local economy as well as to the system. There is also a sovereignty point: a human review function staffed and operated in-region keeps the reviewed data, which is often sensitive, within the jurisdiction, where a review function outsourced across a border would not.

The caution is the one the chapter has raised throughout: cheaper review is not a reason to substitute

human review for the harder engineering. A HITL operation that is affordable because labour is inexpensive can quietly become a crutch — a way to paper over a model that should have been improved, or a routing design that was never tuned — and the rubber-stamping and theatre-review anti-patterns are no less damaging for being inexpensive. The discipline holds: human review is warranted where it genuinely improves outcomes, it is engineered as a system when it is used, and its evolution toward less involvement is driven by the model earning trust, not by the review merely being cheap.

35.10 Where That Leaves Us

Human-in-the-loop is a first-class architectural pattern, and the chapter has treated it as a design space rather than a switch. The space has a structure: human involvement is warranted by six conditions, of which some are durable and some transitional; it takes one of four variants, distinguished by where the human sits relative to the output; those variants compose into six canonical patterns matched to workload shape; and whichever pattern is chosen, the reviewer experience, the review queue, and the feedback capture determine whether it works in practice. Over time, the involvement should evolve — relaxing on the transitional conditions as the model earns trust, holding firm on the durable ones.

The epigraph states the boundary the whole chapter navigates. A model that needs no human is, on any consequential surface, a model no one should fully trust; a model that needs every human cannot scale to do useful work. Every real HITL decision lives between those two sentences, and the chapter's instruction is to pick the loop honestly — to put humans exactly where they genuinely improve outcomes, to engineer that involvement as the real system it is, to capture what the humans teach, and to let the loop tighten or loosen on evidence rather than on instinct or cost. A system designed that way uses human judgement where it is worth its price and the model's autonomy everywhere else. A system that skips the honesty either drowns its people in review that achieves nothing, or grants the model an autonomy it has not earned — and the epigraph's warning is that both failures are equally a failure to pick the loop honestly.

Chapter 36

Evaluation Architecture

“The system you cannot evaluate is the system you cannot improve and probably should not deploy.”

—a recurring observation in MLOps

Chapter Overview

Chapter 101 introduced observability and evaluation as a cross-cutting concern. This chapter develops evaluation as an architecture in its own right. It begins with why evaluation deserves to be built rather than improvised, turns to the golden datasets that anchor it and the methods that score against them, develops judge models — now the load-bearing element of the modern evaluation stack — together with their characteristic failures, and then assembles the whole into the evaluation stack: the layered set of evaluations that run on different cadences, from per-commit unit checks to as-needed incident replays. It closes with how evaluation threads through the system lifecycle and the anti-patterns that make evaluation theatre. The architecture is foundational because everything else — model choice, prompt engineering, RAG quality, agent design, deployment confidence — rests on the ability to measure outcomes credibly.

Learning Objectives

- Treat evaluation as first-class infrastructure rather than an improvised activity.
- Build and maintain golden datasets that match production.
- Select evaluation methods appropriate to the task, and combine them.
- Engineer judge models deliberately and understand their characteristic biases.
- Assemble the layered evaluation stack and run each layer on its proper cadence.
- Thread evaluation through the lifecycle and recognise the anti-patterns.

36.1 Why Evaluation Is Architecture

Evaluation is often treated as an activity — something a team does before a release, or reaches for when a result looks wrong. The argument of this chapter is that evaluation is instead infrastructure: a standing system, built and maintained with the same seriousness as the serving path, because the decisions that depend on it are the decisions that shape everything else. Model selection without

evaluation is gambling. A prompt change without evaluation is a guess. Production monitoring without evaluation is blind operation. And the external pressure is rising: regulators increasingly require evaluation evidence, and incident investigation requires the reproducibility that only a standing evaluation system provides. A team that improvises evaluation is improvising the foundation of every other decision it makes.

Treating evaluation as architecture imposes a discipline with four parts. Evaluation must be continuous practice rather than a one-off, because a system that was good at release drifts. It must use multiple methods, because no single method covers every concern — exact correctness, semantic quality, safety, and fairness are measured differently. It must measure honestly, including the failure modes, because an evaluation that reports only successes is marketing rather than measurement. And its results must actually drive decisions: an evaluation that produces a report nobody acts on has consumed effort and changed nothing. The rest of the chapter is the architecture that makes those four commitments operational.

36.2 Golden Datasets

Evaluation has to be evaluation *against* something, and that something is the golden dataset: a curated set of input-output pairs that represents the task, with labels of what an acceptable or correct output looks like. A golden dataset covers the cases that matter, including the edge cases; it is stable enough that a score computed against it this month is comparable to one computed last month; and it is refreshed as the task itself evolves. It is the anchor that makes every other measurement in the chapter meaningful, because a score is only as trustworthy as the dataset it was computed on.

A golden dataset is constructed from several sources, and the mix matters. Sampling from production gives a realistic distribution — the inputs the system actually receives, in the proportions it receives them. Curation by subject-matter experts gives quality, since experts can supply correct labels that production traffic does not come with. Edge cases and known failure modes are added explicitly, because they are rare in a random production sample and are exactly what a naive dataset under-represents. Adversarial cases are added for robustness. And the construction methodology is documented, so that a later team knows what the dataset does and does not represent. A dataset built from production sampling alone is realistic but misses the rare important cases; one built from expert curation alone is high-quality but may not resemble real traffic; the deliberate combination is what gives a dataset both fidelity and coverage.

Coverage itself deserves to be made explicit rather than assumed. A golden dataset should be examined against a coverage checklist — common cases, edge cases, failure modes, adversarial cases, demographic coverage, and the regulatory categories the system is accountable for — because a dataset can score a model highly while being silent on exactly the cases an architect most needs to know about. A model that performs well on common cases and was never evaluated on demographic coverage has not been shown to be fair; it has been shown to be unmeasured on fairness.

A golden dataset is not a fixed artefact. It must be maintained, because the production distribution drifts and a dataset that matched reality at construction gradually stops doing so. Maintenance means

refreshing the dataset as the distribution moves, adding new cases as new failure modes are discovered, decommissioning stale cases that no longer reflect the task, and keeping the whole thing under version control with an audit trail — so that a change in a score can be attributed either to a change in the system or to a change in the dataset, and never ambiguously to both. An unversioned golden dataset is one whose score movements cannot be interpreted.

36.3 Evaluation Methods

Given a golden dataset, the question is how to score an output against it, and the honest answer is that different concerns need different methods. No single metric is right for every task, and a mature evaluation architecture deliberately combines several.

Exact match is the strictest method and the right one where there genuinely is a single correct answer — classification, structured extraction. It is unambiguous, and where it applies it should be used, because it leaves no room for a generous interpretation to inflate a score. *Reference-based metrics* such as BLEU and ROUGE measure n-gram overlap with reference outputs; they have limited validity for open-ended generation, where many good answers share few words with the reference, but they remain useful as one signal among several. *Embedding-based similarity* compares an output to a reference by vector distance, which captures semantic closeness that n-gram overlap misses, though it too has a ceiling — it registers that two texts are about the same thing without judging which is better. *Judge models* — foundation models scoring outputs — have become the dominant method because they scale, they are flexible enough to apply nuanced criteria expressed in natural language, and they handle the open-ended tasks the other methods cannot; they also carry their own biases, which the next section treats in full. *Human evaluation* is the highest-quality signal and the slowest and most expensive; it is reserved for benchmarks and high-stakes evaluations, and its own quality depends on inter-rater agreement. And *task-specific metrics* apply where a task has a well-developed measure of its own — precision, recall, MRR, and NDCG for retrieval; execution-based metrics for code generation; faithfulness and coverage for summarisation; tool-selection and parameter correctness for tool use.

The architectural point is not to choose one method but to match methods to concerns and combine them. A task with a checkable correct answer uses exact match; an open-ended generation task uses a judge model cross-checked against periodic human evaluation; a retrieval component uses its task-specific metrics. An evaluation architecture that applies a single method everywhere has either chosen a strict method that misjudges open-ended work or a generous method that fails to catch exact errors — and in both cases is measuring some of its system poorly.

36.4 Judge-Model Engineering

Judge models — foundation models scoring the outputs of other models — carry the modern evaluation stack, because they are the only method that scales to continuous evaluation of open-ended work. That load-bearing role makes their characteristic failures worth understanding in detail, because a miscalibrated judge does not announce itself; it simply produces scores that look like measurement and are not.

The pattern itself is simple: criteria are expressed as a prompt for the judge model, the output under evaluation is supplied, the judge produces a score or a verdict, and the results are aggregated across cases. Its strengths are real — it is cheap relative to human evaluation, flexible because the criteria are natural language, and consistent in that the same prompt produces similar judgements. But the judge is a model, and it brings a model’s biases. It exhibits *position bias*, tending to prefer whichever option is presented first, or last, in a comparison. It exhibits *length bias*, often preferring longer outputs regardless of quality. It can show *self-preference*, favouring outputs from itself or related models. It is prone to *sycophancy*, agreeing with a preference stated in the prompt. It has a *capability ceiling* — a judge cannot reliably evaluate work beyond its own capability — and it has *domain blind spots* where it lacks specialised knowledge the task requires.

These failures are manageable, but only deliberately. The judge should be calibrated by running it against a human-labelled gold set and measuring agreement, so the team knows how far the judge’s scores can be trusted. Presentation order in pairwise comparisons should be randomised to neutralise position bias. Multiple judges, from different model families, should be used and aggregated, so that no single model’s idiosyncrasy dominates. A human spot-check should sit over the judge’s decisions. And the judge should be re-validated periodically, because judge-model versions change and a calibration established against one version does not carry to the next. The single most important discipline is to treat the judge as one signal among several rather than as ground truth: a judge model is a useful, scalable approximation of human judgement, and an evaluation architecture that forgets the word “approximation” has quietly replaced measurement with a model’s opinion.

The judge’s rubric, finally, is a contract and should be engineered as one. A good rubric states explicit, mutually exclusive scoring criteria; it gives worked examples at each score level so the judge has a concrete anchor; it requires structured output — a score, a reason, and an evidence quote — so the judgement can be inspected rather than merely trusted; and it is reproducible, meaning the same judge, the same prompt, and the same input yield a stable score within a defined tolerance. A vague rubric produces a judge whose scores drift with phrasing; a disciplined rubric produces a judge whose scores mean something consistent across runs.

36.5 The Evaluation Stack

Mature evaluation is not one harness but a stack of evaluations, each operating on a different cadence and timescale and each catching what the layer above it missed. Figure 36.1 shows the six layers, from the fast per-commit checks at the top to the as-needed incident replays at the bottom.

pre-merge	Unit evals (per-prompt, deterministic)	seconds
pre-merge	Regression evals (curated golden set, every change)	minutes
prod	Online evals (sampled traffic, judge models)	hours
prod	Red-team / adversarial (scheduled)	days
pre-release	Human evaluation (high-stakes, periodic)	weeks
post-incident	Incident-driven (specific failure replays)	as-needed

Figure 36.1: The evaluation stack. Faster, cheaper evaluations sit at the top and run frequently; slower, more comprehensive evaluations sit lower and run less often. Each layer catches what the layer above it missed.

Unit evals are per-prompt, deterministic, and ideally machine-graded: given input X, the output must contain Y, or match a regular expression, or pass schema validation. They are fast enough to run on every commit, they cover the known-good and known-bad cases for each task type, and they are the unit test of the model layer — the cheap check that catches the gross, obvious regression before it travels any further.

Regression evals run the curated golden set — typically a few hundred to a few thousand examples covering the application’s surface area — on every model change, prompt change, and retrieval change, scoring against the methods that matter for the task: exact match, semantic similarity, judge-model rating. They are the blast-radius detector: the layer that tells a team whether a change it believed was local has in fact moved quality somewhere it did not expect.

Online evals score sampled production traffic continuously, using judge models or human reviewers against rubrics. This is the layer where drift is caught: a metric that falls a few per cent over a week with no change to the system is exactly the kind of slow degradation that no pre-merge layer can see, because no pre-merge layer is watching the live distribution. What to sample, and how, is itself a design decision — a random sample for ongoing baseline, stratified sampling for coverage across important segments, and targeted sampling for known risk areas — and the sampled production data must be handled under the privacy treatment its sensitivity requires.

Red-team evals are adversarial: deliberate probing for prompt injection, jailbreaks, harmful-content generation, bias, robustness failures, and privacy violations such as data extraction. They are scheduled rather than continuous, they expand over time as new attack patterns emerge, and they are often run by a dedicated team or an external vendor. Red-teaming is not a one-time pre-launch exercise but a continuous practice, because the adversary’s repertoire keeps growing, and a system red-teamed once is defended only against the attacks known on that day. Every pattern a red-team exercise discovers should be added to the regression suite, so that a vulnerability found once is tested against forever.

Human evals have reviewers grade outputs against detailed rubrics. They are expensive and slow and irreplaceable for the stakes where a judge model cannot be trusted — legal, medical, safety-critical work — and they serve a second purpose beyond their direct verdict: they are how judge models are calibrated and how releases are validated.

Incident-driven evals close the loop. When a specific failure occurs in production, the failed case enters the regression set, and over time the set grows into an institutional memory of everything that has gone wrong. Done well, this is what prevents the same failure from regressing twice — each incident makes the regression layer permanently stronger.

The layers are designed to compose. The fast layers run constantly and catch the obvious; the slow layers run rarely and catch the subtle; each layer's failures are, by intention, caught by a layer below it. An evaluation architecture that has only the top of the stack ships fast but misses drift and adversarial failure; one that has only the bottom catches everything eventually and far too slowly to act. The stack is the structure that gives a team both speed and depth.

36.6 A/B Testing for AI

The evaluation stack measures whether the system is good; A/B testing answers a different question — whether a change actually improves outcomes for real users — and it answers it the only way that question can honestly be answered, by comparison on live traffic. The pattern is the familiar one: random assignment of users to variants, outcome measurement over a defined horizon, statistical comparison, and a decision based on the result rather than on intuition.

A/B testing an AI feature carries specific complications that an architect should plan for rather than discover. The output varies across users even within a single variant, because the model is non-deterministic, which adds noise the experiment design must account for. The effects that matter are often long-term — retention, trust, downstream behaviour — not the immediate click, so the measurement horizon must be long enough to capture them. User behaviour can confound the comparison. Sample-size requirements are real and frequently underestimated. And running many experiments at once raises the multiple-comparisons problem, where some variant looks significant by chance alone. Where rapid iteration matters and the cost of serving a worse variant is low, a multi-armed bandit is an alternative to the fixed A/B split: it routes traffic adaptively toward the better-performing variant as evidence accumulates, trading the clean statistics of a fixed experiment for a faster exploration-exploitation balance. The choice between a classical A/B test and a bandit is a choice between statistical cleanliness and adaptive speed, and it should be made deliberately against how costly a worse variant is and how fast the team needs an answer.

36.7 Evaluation Across the Lifecycle

Evaluation is not a phase; it runs through the whole life of a system, and what it does changes at each stage. In *development*, evaluation is fast and frequent: rapid cycles on prompts and code, scored against offline benchmarks, so an engineer iterating on a prompt gets a signal in seconds. In *pre-production*, evaluation becomes comprehensive: the full battery across categories — correctness,

performance, cost, bias and fairness — together with red-teaming, run before the system is allowed out. In *production*, evaluation becomes continuous and watchful: online evaluation on sampled traffic, drift detection, quality monitoring, user-feedback capture, and periodic re-baselining as the world moves. And *post-incident*, evaluation becomes forensic: investigation with full traces, root-cause analysis, and — the step that compounds — the addition of the failed case to the regression suite, so the incident becomes a permanent test.

The thread connecting the four stages is that evaluation is the same architecture used four ways, not four different activities. The golden dataset built in development is the regression set used pre-production and the institutional memory grown post-incident. The judge models calibrated for development scoring are the judge models running online. An organisation that builds evaluation as the standing infrastructure this chapter describes finds it already has what each lifecycle stage needs; an organisation that improvises evaluation per stage rebuilds it, worse, four times.

36.8 Anti-Patterns

The failure modes of evaluation are common, and each one is a way of producing the appearance of measurement without its substance:

- **Cherry-picking** — showing only the good examples, so the evaluation flatters the system rather than describing it.
- **Aspirational benchmarks** — evaluating on data unlike production, so a high score describes a system that does not exist.
- **Vanity metrics** — measuring what looks good rather than what matters, so the number rises while the system does not improve.
- **One-shot evaluation** — evaluating once and never again, so drift goes unseen.
- **Missing failure modes** — evaluating only the positive cases, so the evaluation is silent on exactly what an architect most needs to know.
- **Unchecked judge models** — trusting a judge's scores without calibration, so an approximation is mistaken for ground truth.
- **No regression suite** — shipping improvements that silently reintroduce old failures, because nothing tests for them.
- **Evaluation without decisions** — producing reports that no one acts on, so the effort is spent and nothing changes.

What unites the eight is that each violates one of the four disciplines of the opening section — honest measurement, multiple methods, continuous practice, results that drive decisions. Cherry-picking, aspirational benchmarks, vanity metrics, and missing failure modes are failures of honesty. One-shot evaluation is a failure of continuity. Unchecked judges and the absent regression suite are failures of method. Evaluation-without-decisions is the failure that makes all the others moot. The remedy

for the whole list is therefore singular: hold the four disciplines, and build the evaluation stack that operationalises them.

36.9 The African Perspective

Evaluation architecture is geography-neutral in its structure, but two of its concerns press harder in African deployments.

The first is the golden dataset. The chapter has argued that a golden dataset must match the production distribution, and for an African deployment that distribution includes the languages, the code-switching, the names, the locales, and the regulatory categories of the markets actually served — which the benchmark datasets shipped with most AI tooling do not represent. An evaluation architecture that scores a model highly on imported benchmarks and is deployed against a different real distribution has measured the wrong system. The construction discipline of sampling from genuine production traffic, and the coverage discipline of checking demographic and linguistic coverage explicitly, are not refinements here; they are what make the evaluation describe the system that will actually run.

The second is the judge model. Judge models inherit the language and domain coverage of their training, and a judge scoring outputs in a low-resource African language, or in a code-switched register, is operating closer to its capability ceiling and its domain blind spots than the same judge scoring English. The calibration discipline — measuring judge-versus-human agreement on a gold set — must therefore be done per language and per domain that the system serves, not once in aggregate, because an aggregate agreement figure can be reassuring while the judge is unreliable on exactly the languages that matter most. And the human evaluation that calibrates the judge must be staffed by reviewers fluent in those languages and contexts. The architecture does not change for African deployments; the inputs to two of its components — the golden dataset and the judge calibration — must reflect the markets actually served rather than the markets the tooling assumed.

36.10 What Evaluation Architecture Provides

Evaluation architecture is foundational infrastructure, not an activity. It rests on golden datasets that match production and are maintained as the distribution drifts; it scores against them with methods matched to the task and combined rather than chosen singly; it leans on judge models that scale the measurement and on the deliberate engineering that keeps those judges honest; and it assembles the whole into a layered stack, from per-commit unit checks to as-needed incident replays, each layer catching what the one above it missed. Around that stack, A/B testing answers whether a change improves real outcomes, and the evaluation threads through every stage of the lifecycle from development to post-incident.

The epigraph is the chapter's whole argument compressed: the system you cannot evaluate is the system you cannot improve and probably should not deploy. Evaluation is what converts an AI system from something a team hopes is working into something a team knows the state of — and the four disciplines, honest measurement, multiple methods, continuous practice, and results that drive

decisions, are what separate an evaluation-informed organisation from one flying blind. The methods in this chapter are well understood and the stack is well defined; the architectural work is to build evaluation as the standing system every other decision depends on, rather than the activity a team reaches for when a result already looks wrong.

Chapter 37

Self-Correction and Reflection Loops

“A model that checks its own work is two models. The architecture has to treat it that way.”

—a recurring observation in production agentic systems

Chapter Overview

Self-correction and reflection loops are architectures in which a system produces an output, evaluates that output, and revises it on the strength of the evaluation. The pattern spans a simple critique-and-revise pass over a single generation through to deep iterative refinement with multiple specialised critics and tool-augmented verification. Production systems use these loops to raise quality without retraining a model — but the loops introduce latency, cost, and failure modes of their own, and those demand architectural attention. This chapter develops the variants of self-correction, the loop discipline that keeps them bounded, the evaluation that catches the failures self-correction cannot fix, and the recurring ways the pattern goes wrong.

Learning Objectives

- Recognise the variants of self-correction — intrinsic, extrinsic, tool-augmented, learned, multi-critic — and choose between them.
- Architect loops with explicit termination conditions and budgets.
- Manage cost and latency under multi-pass execution.
- Build evaluation that catches the failures self-correction introduces.
- Adapt self-correction patterns to African operational and economic contexts.

37.1 The Self-Correction Problem

A single forward pass through a language model produces output that may contain errors — a factual mistake, a logical gap, code that does not compile, reasoning that does not check out. A model often has the capability to detect such errors when it is given a second chance to look. The architectural question this chapter answers is how to structure that second chance, because the structure is what separates a loop that genuinely improves quality from one that merely consumes tokens.

The instinct is to do the simplest possible thing: append a “check your work” prompt after the

generation and hope the model finds its own mistakes. This sometimes helps, and often it does not. A model can be overconfident in its own output and defend it rather than revise it; it can rationalise an error instead of correcting it; it can introduce a new error while fixing the one it was asked about. Self-correction that works is not a magic incantation appended to a prompt. It is a designed loop — with explicit structure, explicit budgets, and explicit termination — and the rest of the chapter is that design.

Self-correction comes in five variants, distinguished by who or what does the evaluating. *Intrinsic* self-correction has the same model evaluate and revise its own output. *Extrinsic* self-correction uses a separate model — often smaller and specialised — as the critic. *Tool-augmented* self-correction uses external tools, such as compilers, test runners, calculators, or fact-checkers, to supply ground truth. *Learned* self-correction trains a model specifically on the correction task. And *multi-critic* architectures use several specialised critics, each evaluating a different facet of the output. The first three are the variants most production systems reach for, and Figure 37.1 contrasts them directly; the sections that follow develop all five.

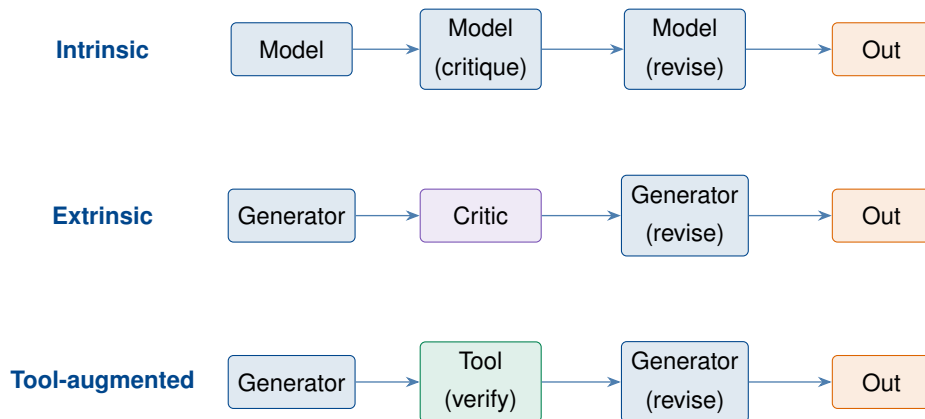


Figure 37.1: Three self-correction variants. *Intrinsic*: the same model critiques and revises its own output — cheap, weakest signal. *Extrinsic*: a separate critic evaluates — a better signal at greater cost. *Tool-augmented*: an external tool provides ground truth — the strongest signal where a tool for the task exists.

37.2 Intrinsic Self-Correction

Intrinsic self-correction is the most common variant and the cheapest: the same model that produced an output is asked to evaluate it. The pattern is a short sequence — generate the initial output, then prompt the model with that output and ask for a critique, then prompt it again with the output and the critique and ask for a revision, and optionally iterate. It needs no second model and no external tool, which is what makes it the default first reach.

It is also the variant with the weakest signal, and its failure modes follow from a single fact: a model critiquing its own work is not an independent check. The characteristic failures are recognisable. *Confirmation* — the model defends its output rather than genuinely critiquing it, because the output is its own. *Drift* — successive revisions move away from the user’s actual intent. *Overfitting to the critique* — the model fixes the named problem and breaks something else in the process. *Verbose-without-better* — the revision adds words without adding correctness. And *cost explosion* — iterations

multiply the token cost without a proportional gain in quality.

Several design moves make intrinsic self-correction work better despite these tendencies. Separating the critique-prompt and revision-prompt contexts, rather than running the whole loop in one accumulating window, reduces the model's tendency to simply defend what it already wrote. Giving the critique a structured format — requiring it to classify problems against a failure-mode taxonomy rather than comment freely — produces critiques that are specific enough to act on. Iterations should be capped, typically at two or three and rarely more, because the quality gain plateaus quickly while the cost does not. The loop should allow an explicit “no change required” termination, so a model that has produced a good output is not forced to revise a fourth time. And each revision should be compared against the version before it, to confirm the revision was actually an improvement rather than assumed to be one. Intrinsic self-correction with this structure is a useful, cheap quality lever; the same pattern without it is the cost-explosion failure mode waiting to happen.

37.3 Extrinsic and Multi-Critic Self-Correction

Extrinsic self-correction replaces the self-judging model with a separate critic. The generator produces an output, a critic model — often smaller and fine-tuned specifically for critique — evaluates it, the generator revises given that critique, and the loop terminates on critic approval or budget exhaustion. The advantage over the intrinsic pattern is independence: a separate critic does not have the generator's investment in defending the original output, and a critic specialised for the task often outperforms the generator critiquing itself. The costs are equally real. The critic model adds latency and cost on every iteration. And generator-and-critic misalignment is possible — the critic may not understand what the generator was trying to do, and critique it against the wrong intent.

The multi-critic architecture extends the extrinsic pattern by replacing the single critic with several, each evaluating a specialised facet of the output — one for correctness, one for style, one for safety, one for format, one for completeness. The critiques are aggregated into a single structured revision prompt, the generator revises, and the loop continues until the critics agree or the budget exhausts. Multi-critic ensembles improve coverage, because each critic catches what the others are not looking for, and they do so at a cost that scales with the number of critics per iteration. The engineering that multi-critic introduces is the engineering of disagreement: when critics conflict — the style critic wants brevity, the completeness critic wants more — the architecture needs a prioritisation rule, and it may need critic weights tuned to the use case. It needs termination logic that does not stall when the critics never fully agree. And it needs cost management, because every critic runs on every iteration. Multi-critic self-correction is the strongest model-based variant and the most expensive; it is warranted when the output has several genuinely distinct quality dimensions that one critic cannot competently judge at once.

37.4 Tool-Augmented Self-Correction

Tool-augmented self-correction replaces the model critic with an external tool that supplies genuine ground truth. The generator produces an output, a tool — a compiler, a test runner, a calculator, a search query, a database lookup, a schema validator — checks it, the tool's result is fed back to the

generator, the generator revises, and the loop terminates on a tool pass or on budget exhaustion. It is, where it applies, the strongest variant of the pattern, and the reason is straightforward: a tool's verdict is not another model's opinion but a fact. A compiler does not have a confirmation bias. A test suite does not rationalise. A calculator is not overconfident.

That ground-truth quality makes tool-augmented self-correction reliable in exactly the places model critics are weakest. Common code-generation failures — syntax errors, type errors, test failures — are caught reliably by a compiler and a test runner. Arithmetic errors are caught by a calculator. Factual errors can be caught by retrieval-and-comparison against a source. Where a tool exists for the verification task, it should be preferred to a model critic, because it converts the loop's evaluation step from an approximation into a check.

The limits of the pattern are the limits of tool availability. A tool must exist for the task — and for many open-ended generation tasks, no tool can verify quality. The tool's result must be in a form the generator can interpret and act on; a tool that emits output the model cannot read provides a verdict the loop cannot use. Tool availability and latency under real operational conditions matter, particularly where connectivity is variable. And tools themselves can be wrong — a flaky test, a stale database — so even a tool's verdict is not beyond question. Tool-augmented self-correction is the variant to reach for first wherever a genuine verification tool exists, and the architecture should be honest about how much of its task that condition actually covers.

37.5 Learned Self-Correction

The four variants so far structure self-correction as a loop at inference time. Learned self-correction takes a different route: it trains the correction capability into the model itself. The approach constructs training data as triples of initial output, critique, and revised output; fine-tunes the model to perform the critique-and-revision task; optionally applies preference learning to reward valid critiques and genuine improvements; and deploys the result as a single model that produces better output through an internalised iteration rather than an external loop.

Learned self-correction is the most involved variant and is warranted under specific conditions. Constructing the training data is substantial work, since good triples require knowing what a correct revision looks like. There is a risk of overfitting to the specific failure modes present in the training data, so that the model corrects the errors it was trained on and not the ones it was not. And there is the standard fine-tuning risk of losing capability the model had before — the correction training narrows the model in ways that may cost it elsewhere. Learned self-correction is strongest when the failure modes of the task are well-characterised and stable, so that the training data genuinely covers what the deployed model will face. Where the failure modes are shifting or poorly understood, an inference-time loop, which can be adjusted without retraining, is the safer choice.

37.6 Loop Discipline

Every variant except the purely learned one is a loop, and a loop is the structural fact that the rest of the chapter's discipline follows from. A loop can fail to terminate, can spend unboundedly, and

can iterate without improving — so a self-correction loop, like the agentic loops of the surrounding chapters, must be bounded by explicit budgets and governed by explicit termination conditions.

The budgets are the hard limits. A loop should have a maximum iteration count, a maximum token budget, a maximum latency budget, and a maximum cost per request — and, crucially, termination logic that returns the best output produced so far when a budget is hit, rather than failing or returning nothing. A loop that hits its budget and has no best-so-far answer has spent the budget and produced nothing for it.

The termination conditions are the loop's exits before the budget. A tool-augmented loop terminates naturally on a tool pass. An extrinsic or multi-critic loop terminates on critic approval. Beyond those, a loop should terminate on diminishing-returns detection — when a revision is no longer improving quality enough to justify another pass — and on stability detection — when the output has stopped changing between iterations. Budget exhaustion is the final, fallback termination. A well-designed loop is far more likely to exit on one of the quality-based conditions than on the budget; a loop that routinely exits only on budget exhaustion is a loop whose termination logic is not doing its job.

The loop also needs observability, because a multi-pass loop is opaque without it. Each iteration should be logged — its input, its output, its critique — along with a per-iteration quality metric, so that an engineer can see how quality moved pass by pass. A final-versus-initial improvement metric tells whether the loop, taken as a whole, actually helped. Loop-cost and loop-latency dashboards make the price of the pattern visible. And looped failures should be classified by failure mode, so the team knows whether the loop is failing through confirmation, drift, or worsening revisions.

37.7 Evaluating Self-Correction

Self-correction needs evaluation that is specific to it, because the pattern introduces a failure mode no single-pass system has: the loop can make the output worse. A naive evaluation that simply scores the final output cannot tell whether the loop helped, hurt, or did nothing.

A proper evaluation of a self-correction loop therefore measures the loop, not just its product. It compares the looped output directly against the single-pass output, to establish that the loop is genuinely earning its cost. It looks specifically for loop-introduced failures — cases where a revision worsened an output that was acceptable before — because those are the pattern's distinctive harm. It evaluates at multiple iteration depths, to find where the quality gain plateaus and the cap should sit. For tool-augmented loops, it evaluates behaviour under tool-unavailable conditions, since the loop must degrade gracefully when its verification tool is missing. And it maps the cost-quality trade-off as a curve, so the architect can see how much quality each additional iteration buys and decide where the curve stops being worth climbing. A self-correction loop deployed without this evaluation is a loop running on the assumption that it helps — and the assumption is exactly what the evaluation exists to test.

37.8 Recurring Failure Patterns

The failure modes of self-correction recur across all its variants, and naming them is the fastest way to recognise a loop that has gone wrong:

- **Unbounded loops** — no budget, so cost and latency explode while the loop iterates.
- **Confirmation rather than critique** — the loop defends the output instead of genuinely evaluating it, most common in the intrinsic variant.
- **Drift from intent** — successive revisions move away from the user’s actual request.
- **Worsening revisions** — iteration introduces new errors, so the loop’s output is worse than its input.
- **Critic-and-generator misalignment** — the critic evaluates against an intent the generator did not have.
- **Tool over-reliance** — a tool-augmented loop fails outright when its verification tool is unavailable, instead of degrading to a single pass.

Each of these maps to a discipline developed earlier in the chapter. Unbounded loops are the loop-discipline section ignored. Confirmation and worsening revisions are what the structured-critique and version-comparison moves of the intrinsic section defend against. Drift and misalignment are caught by the loop-specific evaluation. Tool over-reliance is the graceful-degradation requirement of the tool-augmented section not met. The failure modes are common precisely because the disciplines that prevent them are more work than appending a “check your work” prompt — and the chapter’s argument is that the work is what makes self-correction a genuine quality lever rather than a cost multiplier.

37.9 The African Perspective

Self-correction loops have a particular shape in African deployments, and the dominant consideration is the one the loop-discipline section already foregrounds: cost.

A self-correction loop multiplies the model calls of a single task by the number of iterations, and in a deployment where each call is billed in foreign currency against a local-currency budget, an uncontrolled loop can break the unit economics of a system that would otherwise be viable. This makes the budget-bounded-loop discipline not a refinement but a precondition, and it shifts the variant choice: a smaller-model critic is often preferred over an expensive one to keep the per-iteration cost in line, and tool-augmented self-correction with cheap, deterministic tools — a compiler, a calculator, a schema validator — is preferred wherever a tool for the task exists, because a deterministic tool’s verification is effectively free of per-call model cost. The cost-quality trade-off curve of the evaluation section is, in these deployments, the central design artefact: it is what tells the architect how few iterations the economics actually permit.

Connectivity shapes the loop’s structure. A multi-pass loop that calls an external critic service over a variable network is exposed to that variability on every iteration, and the long latency of an unreliable link multiplies across the passes. The architectural responses are the ones the chapter recommends, applied with urgency: edge-resident critics for offline-capable applications, a cached-result fallback

when a critic service is unreachable, and a graceful fallback to a single pass when the loop cannot complete — so a connectivity failure degrades quality rather than failing the request.

Language adds a third consideration. In multilingual African deployments the generator and the critic both need multilingual capability, and a cross-language arrangement — a generator working in Swahili checked by a critic working in English — requires a translation step that is itself a source of error. Code-switching is common in African content, and a critic that cannot handle a mix of languages within one output will misjudge it. And the critique must preserve cultural context rather than flatten it toward the norms the critic model was predominantly trained on. The per-language evaluation discipline of the evaluation chapter applies directly: a self-correction loop should be measured separately on each language it serves, because a loop that improves English output and quietly degrades Swahili output has an aggregate score that hides the harm.

Finally, tool availability for the tool-augmented variant looks different in African contexts, and not only as a limitation. The general-purpose verification tools may be sparser, but region-specific tools exist and are genuinely useful: mobile-money APIs can verify financial outputs, local databases can check regional facts, and national-ID verification services, where available, can ground identity-related outputs. The tool-fallback design the chapter insists on matters more here, because tool availability is less uniform — but the opportunity is real, and a tool-augmented loop built around the verification tools that genuinely exist in the deployment’s region is the strongest and most cost-efficient form of self-correction available to it.

37.10 Holding the Pieces Together

Self-correction and reflection loops raise quality without retraining a model — but only when they are architected, not improvised. The architecture has five variants to choose among: intrinsic, cheap and weakest; extrinsic and multi-critic, stronger at greater cost; tool-augmented, strongest wherever a verification tool exists; and learned, which trains the capability in where the failure modes are stable enough to teach. Whichever variant is chosen, the pattern is a loop, and a loop needs the discipline a loop always needs — explicit budgets, explicit termination conditions, observability across iterations — together with an evaluation specific enough to catch the pattern’s distinctive harm, the revision that made the output worse.

The epigraph states the principle the chapter is built on: a model that checks its own work is two models, and the architecture has to treat it that way. A self-correction loop is not a model being clever in a single pass; it is a generation step and an evaluation step, distinct in their roles and distinct in their failure modes, joined into a loop that must be bounded and measured like any other. Treated that way — as a designed loop with a budget, a termination rule, and an evaluation that tests whether it genuinely helped — self-correction is one of the most useful quality levers available without retraining. Treated as a “check your work” prompt and hoped over, it is a cost multiplier that sometimes, unmeasurably, also improves the output.

Chapter 38

Model Routing and Cascading

“The cheapest model that can do the job is the right model. Routing is the architecture of figuring out which job is which.”

—a recurring observation in cost-engineered AI systems

Chapter Overview

Model routing and cascading send different requests to different models, choosing the destination by requirement, complexity, cost, latency, capability, or sovereignty. The pattern matters more as the model landscape diversifies — frontier models for the hard tasks, smaller fast models for the easy ones, specialised models for particular domains, locally hosted models for sovereignty, tool-using agents for compound work. This chapter develops the routing problem, the five routing variants and how they differ, the cross-cutting routing concerns an enterprise deployment adds, and the discipline of operating a router in production as the model landscape keeps moving underneath it.

Learning Objectives

- Recognise the five routing variants — rule-based, classifier-based, cascade, ensemble, speculative — and choose between them.
- Build routers that survive a constantly changing model landscape.
- Manage the quality-and-cost trade-off through cascading.
- Apply multi-provider, capability-aware, cost-budget, and sovereignty-aware routing.
- Operate routers under monitoring and drift detection.
- Adapt routing to African cost pressure, sovereignty, and language-coverage realities.

38.1 The Routing Problem

Different requests have different requirements, and the gap between them is large enough to be the basis of an architecture. Some requests need frontier capability; many do not. Frontier-model cost per token can be ten to a hundred times that of a small model, and latency varies by a similar factor. Routing is the discipline of matching each request to the model that can actually do its job, rather than sending every request to the most capable — and most expensive — model by default.

Requests sort, roughly, into a small number of categories that call for different models. A simple lookup or extraction is handled adequately by a small model. Standard generation is handled by a mid-tier model. Complex reasoning and multi-step planning often genuinely need a frontier model. Domain-specialised work is often best served by a specialised fine-tuned model rather than a larger general one. And compound tasks are agentic loops that may route internally, model by model, as they proceed. The art of routing is reading which category a request falls into before committing a model to it.

The reason routing is worth the engineering is that several pressures push on the choice at once, and they do not all push the same way. Cost pushes toward small models, and pushes hard, because at volume the total bill is dominated by the simple requests — which are also the ones a small model can handle. Latency pushes the same way, since smaller models are faster. Quality pushes the other way, toward frontier models, on the requests that genuinely need them. Sovereignty pulls certain requests toward locally hosted models regardless of capability. And availability makes routing part of resilience, since a router that can send a request to an alternative model is a router that can absorb a provider outage. Routing is the architecture that resolves these competing pressures request by request, rather than settling them once, globally, with a single model choice that is wrong for most of the traffic.

38.2 The Routing Variants

Routing comes in five variants, and they differ in how the routing decision is made and how much it costs to make. The first two — rule-based and classifier-based — decide a route up front. Cascade decides by trying. Ensemble declines to decide and runs several models. Speculative decoding routes at the level of individual tokens. An architect choosing a routing strategy is choosing among these five, and most production systems combine more than one.

38.2.1 Rule-Based Routing

The simplest variant routes by explicit rules. A request arrives, a set of rules examines its characteristics — length, type, language, source, customer tier — and the request is routed deterministically to a model. Rule-based routing is predictable, explainable, easy to debug, and easy to change, and for a well-characterised request mix it is entirely adequate; there is no reason to reach for a more sophisticated variant when a handful of rules genuinely captures the routing decision. Its limits appear as the request mix grows more varied: rules cannot capture nuance, they require maintenance as the mix shifts, and they grow brittle as edge cases accumulate and each is patched with another rule. Rule-based routing is the right starting point and, for many systems, the right permanent answer; the more elaborate variants earn their place only when the rules genuinely stop being able to express the decision.

38.2.2 Classifier-Based Routing

When the routing decision is too nuanced for rules, a learned classifier can make it. The request's features are extracted — often with a small embedding model — a classifier predicts the appropriate route, and the classifier is trained on historical triples of request, model, and resulting quality, and updated as new models become available. Classifier-based routing captures patterns rules cannot, but

it brings its own considerations the architect must budget for. The classifier itself adds latency and cost to every request. Its accuracy bounds the router's quality — a router can be no better than the classifier deciding its routes. It requires labelled training data, which must be assembled. It requires retraining as the model landscape shifts. And it requires substantial monitoring of classification quality, because a classifier that has silently drifted routes badly without announcing it. Classifier-based routing is the right choice when the routing decision genuinely has the nuance to justify a learned model and the team can sustain the data and monitoring the classifier needs.

38.2.3 Cascade Routing

Cascade routing decides by trying rather than by predicting. The request goes first to the smallest and cheapest model; if that model expresses uncertainty, or its output fails a verification check, the request escalates to the next tier, and so on up the cascade until a model succeeds or the cascade is exhausted. Figure 38.1 shows a three-tier arrangement.

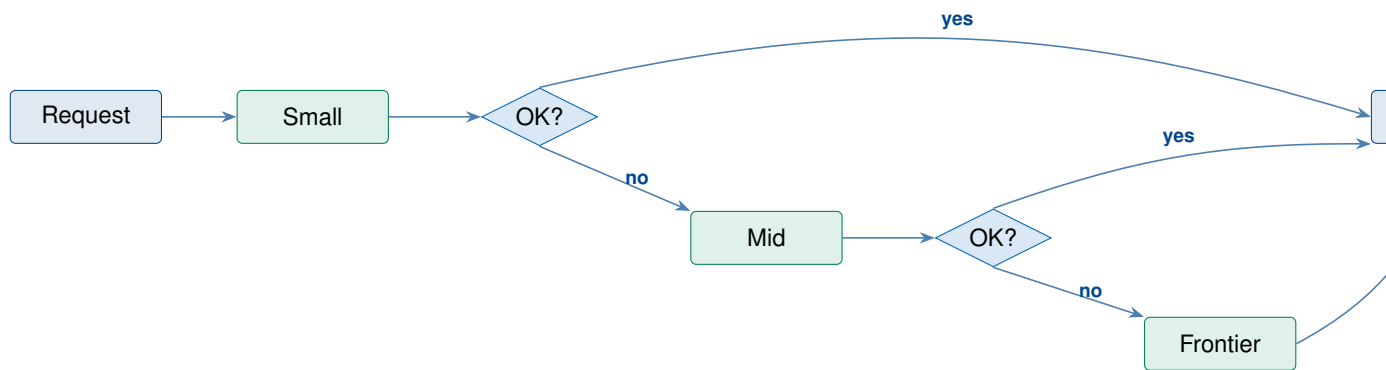


Figure 38.1: A three-tier cascade. The small model is tried first; on verification failure the request escalates to the mid-tier model, and on a second failure to the frontier model. Most requests are answered cheaply; only the hardest pay the frontier cost.

The appeal of the cascade is that it is cost-effective and quality-preserving at once: simple requests are answered cheaply by the small model, complex requests still reach the frontier, and — unlike classifier-based routing — the cascade does not need a perfect upfront classification, because it discovers the right tier by trying. Its mechanics turn on a few decisions. Uncertainty must be detected at each tier, by calibrated confidence, a model's self-report, or a verifier. Escalation logic must be chosen — always escalate on uncertainty, escalate a sample, or escalate on a threshold. Escalations should be cached, the cascade depth should be budgeted, and cascade cost should be tracked. The failure modes are the mirror of the mechanics: an overconfident small model that should escalate and does not, leaking low-quality answers; an underconfident small model that always escalates, defeating the cost saving entirely; an escalation cost that, at volume, outweighs simply going straight to the frontier; and quality degradation when a small model partially answers before escalating, leaving the next tier to clean up rather than start fresh. A cascade is one of the most effective cost levers in this chapter when its uncertainty detection is well calibrated — and a cost trap when it is not.

38.2.4 Ensemble Routing

Ensemble routing declines to choose a single model. The request goes to several models in parallel, and their outputs are aggregated — by vote, ranked vote, weighted average, or selection by a judge model — into a single answer. Ensemble routing carries the highest cost of any variant, because every model is paid for on every request, and the highest latency, because the slowest model's tail latency dominates. In exchange it often delivers the highest quality, which makes it appropriate for high-stakes requests where the cost is genuinely acceptable. The aggregation strategy is the critical design decision: a poor aggregation can take several good outputs and combine them into a worse one. Ensemble routing is the variant to reserve for the small set of requests important enough that paying several models at once is worth it.

38.2.5 Speculative Decoding

Speculative decoding routes at the level of tokens rather than requests. A small draft model proposes tokens — or whole short outputs — and a large verifier model checks them, which is often cheaper than having the large model generate from scratch; on a verification pass the draft tokens are accepted, and on a failure the large model generates. The result is a substantial reduction in latency while the quality of the large model is preserved, because the large model still has the final say. Speculative decoding is typically implemented at the inference-engine level rather than in application code, and it has become increasingly standard for high-throughput frontier serving. It differs from the other four variants in that it is not really a choice the application architect makes request by request; it is a serving-layer optimisation, and the architect's decision is mostly whether the chosen serving stack supports it.

38.3 Cross-Cutting Routing Concerns

Beyond the five variants, an enterprise router must answer four cross-cutting concerns, each of which shapes the router regardless of which variant decides the individual route.

Multi-provider routing is the discipline of supporting more than one model provider — so that the router can choose across Anthropic, OpenAI, Google, and others rather than being wired to one. It requires a provider abstraction layer, per-provider failure handling, and cost-and-availability optimisation across providers. Designing for multi-provider operation from the start is the routing layer's contribution to vendor independence, and it connects directly to the hexagonal pattern's insistence that the model be reached through a port: a router built for one provider is a router that has quietly made that provider unswappable.

Capability-aware routing routes by what a request actually needs — vision, function-calling, long context, a particular language — against a registry of which models have which capabilities. It requires that capability registry to be maintained and a capability-matching step in the routing logic, and it is substantial design specifically because the model landscape changes: a capability that only the frontier model had last year may be available in a cheaper model now, and a capability-aware router is the structure that lets the system take advantage of that without a rewrite.

Cost-budget routing makes the router respect explicit budgets — per request, per user, per tenant

— so that the frontier model is reached only when the budget allows it. It requires budget-tracking infrastructure, and it is what turns routing from a best-effort cost optimisation into a hard guarantee of predictable economics. It is the routing-layer expression of the budget enforcement the cost chapter insists on.

Sovereignty-aware routing routes by where a request's data is allowed to go. Some requests, by data classification, must route to a specific region or to a self-hosted model; others may use a frontier provider with a regional endpoint. It requires requests to be classified by sensitivity and a sovereignty policy to be enforced at the routing layer, and it is the mechanism by which a single system can serve several jurisdictions with different data-residency rules without splitting into several systems.

38.4 Operating Routers in Production

A router is not a build-once artefact, because the thing it routes among — the model landscape — does not hold still. Operating a router well has three standing requirements.

The first is monitoring, and it must be per-route. A router needs per-route quality monitoring, per-route cost monitoring, and per-route latency monitoring, because an aggregate figure hides exactly the problem an architect needs to see: a single route degrading while the average stays healthy. It also needs drift detection on the routing decisions themselves — a shift in which routes are being chosen is a signal that the request mix has moved — and monitoring of the quality-cost Pareto frontier, since the whole point of routing is to sit on a good point of that frontier and the monitoring is what confirms the router still does.

The second is experimentation. A routing change is a change to the system's cost and quality at once, and it should be tested as one: treatment-arm experiments for routing changes, with both quality and cost outcomes measured, rather than a routing change shipped on the assumption that a cheaper route is a free saving. This needs real experimentation infrastructure, and it is the A/B-testing discipline of the evaluation chapter applied to the routing layer.

The third is change management for the model landscape itself. New models become available constantly, and each one is a potential new route — which means the router needs updates, the updates need backward-compatibility consideration so that an in-flight request is not stranded by a routing change, and the whole thing needs the change-management discipline of any frequently updated component. A router treated as a fixed artefact gradually routes against an out-of-date picture of which models exist and what they cost; a router treated as a living component tracks the landscape and keeps finding the cheaper route as the landscape provides one.

38.5 Recurring Failure Patterns

The failure modes of routing are recognisable, and each maps to a discipline this chapter has set out:

- **Single-model lock-in** — no router at all, so cost and risk are concentrated on one model and one provider.

- **Classifier rot** — a classifier-based router whose accuracy degrades as the request mix shifts and no one retrains it.
- **Cascade collapse** — an underconfident small model that escalates almost everything, so the cascade pays the small model’s cost and then the frontier’s anyway.
- **Provider dependency** — a router never designed for multi-provider operation, so the vendor independence the router could have provided was never built.
- **Sovereignty violation** — a sensitive request routed to a model or region its data classification forbids.
- **Cost-budget overrun** — routing that does not respect budgets, so the cost guarantee the router could have provided does not exist.

What unites these is that each is a routing capability not built or not maintained. Single-model lock-in is the absence of a router; provider dependency and cost-budget overrun are cross-cutting concerns skipped; classifier rot and cascade collapse are the monitoring discipline not applied to the variant in use; sovereignty violation is sovereignty-aware routing missing. The remedy is the chapter’s argument restated: routing is an architecture to be built and operated deliberately, not a single model choice made once and left.

38.6 The African Perspective

Routing is important everywhere, but in African deployments it moves from an optimisation to a precondition, because three of the pressures the chapter opened with — cost, sovereignty, and language — press harder.

Cost is the dominant one. African unit economics are tighter, and a frontier-model cost-per-query that is merely expensive in a high-revenue market can be flatly incompatible with the revenue per query in a price-sensitive one. Routing is therefore the primary cost-control lever, not a secondary refinement, and the cascade pattern is particularly attractive because it sends most queries to cheaper models by construction. Locally hosted open-weight models are natural high-volume routing destinations, since a self-hosted model converts a per-query foreign-currency cost into a fixed local-currency infrastructure cost. The cost-engineering attention this chapter recommends is, in these deployments, the engineering that determines whether the system is viable at all.

Sovereignty drives routing directly. Data-residency laws vary across African jurisdictions, and a router serving several markets must classify requests by sensitivity and route accordingly — some requests to locally hosted models in a Cape Town or Johannesburg region or on-premise, others to frontier providers that offer a regional endpoint. Sovereignty-aware routing is the mechanism that lets one system comply with several jurisdictions’ rules at once, and in a pan-African deployment it is not optional.

Language coverage adds a routing dimension that higher-resource markets rarely face. Models vary widely in their African-language coverage, and the routing decision should account for it: Arabic

and French route comfortably to broad-coverage models, while a low-resource language may need a specialised model where one exists, or translation-mediated routing where direct coverage is absent. Routing by language is, here, a quality decision as much as a cost one — sending a low-resource-language request to a model that handles the language poorly is a routing error even if the model is otherwise capable. And availability routing matters because latency from African users to Northern endpoints is non-trivial: routing to local-region endpoints where they exist, and routing around outages, is part of how the system stays both fast and resilient for the users it actually serves.

38.7 The Architectural Commitment

Model routing and cascading match each request to the model that can actually do its job, under the competing pressures of cost, latency, quality, capability, and sovereignty. The architecture has five variants — rule-based and classifier-based, which decide up front; cascade, which decides by trying; ensemble, which runs several models; and speculative decoding, which routes at the token level in the serving layer — and four cross-cutting concerns, multi-provider, capability-aware, cost-budget, and sovereignty-aware, that shape the router whichever variant decides the route. Around it sits the operational discipline of per-route monitoring, routing experimentation, and change management for a model landscape that never holds still.

The epigraph is the chapter's whole principle: the cheapest model that can do the job is the right model, and routing is the architecture of figuring out which job is which. A system without routing has answered that question once, globally, with a single model choice — and that choice is necessarily wrong for most of its traffic, either overpaying frontier prices on simple requests or underserving the hard ones with a model that cannot handle them. A system with routing answers the question request by request, keeps answering it as the model landscape changes, and sits, deliberately and measurably, on a good point of the quality-cost frontier. That is the architectural commitment routing asks for, and in cost-pressured deployments it is the commitment that decides whether the economics work at all.

Chapter 39

Planner-Executor Architectures

“A model that plans and then executes is a system; a model that plans while executing is a story.”
—a recurring observation in production agentic systems

Chapter Overview

Planner-executor architectures separate the work of figuring out what to do — planning — from the work of doing it — execution. Multi-step tasks, a research query that needs several searches, a workflow that calls several tools, a coding task that spans several files, benefit from making the plan explicit rather than leaving it implicit in a single model’s run. This chapter develops the planner-executor pattern: the problem it solves, the five variants distinguished by how rigid the plan-execute split is, the architectural disciplines that make a plan a genuine engineering artefact, and the failure modes that planning introduces. Production systems differ on how rigid the split is, on who runs the plan, and on how a plan adapts when execution surprises it — and those differences are the substance of the chapter.

Learning Objectives

- Recognise the five planner-executor variants and choose between them.
- Treat plans as explicit, inspectable, structured artefacts.
- Manage replanning under execution surprises without losing convergence.
- Build evaluation that separates planning failures from execution failures.
- Adapt planner-executor patterns to African operational and economic contexts.

39.1 The Planner-Executor Problem

A multi-step task executed end-to-end by a single model, in one long run, fails in well-known ways. It loses the user’s intent over a long context. Its errors compound across steps, each step building on the last one’s mistake. It repeats mistakes it has already made. And it cannot cleanly recover when execution surprises it, because there is no separate representation of what it was trying to do against which the surprise can be assessed. Separating planning from execution mitigates all four. The plan becomes an explicit thing — inspectable, revisable, recoverable — rather than an implicit intention

buried in a single generation.

The pattern comes in five variants, distinguished by how rigid the split between planning and execution is. *Linear plan-then-execute* produces the whole plan first and then executes it. *Plan-and-revise* lets execution feed back into replanning. *Hierarchical* planning produces a high-level plan whose sub-plans are elaborated as execution reaches them. *Reactive* patterns start with a minimal plan and do most of the planning during execution. And *specialist-role* architectures use separate models or agents for planning and for execution. Figure 39.1 shows the plan-and-revise variant, which is the one most production systems converge on, because it keeps the plan explicit while allowing it to adapt.

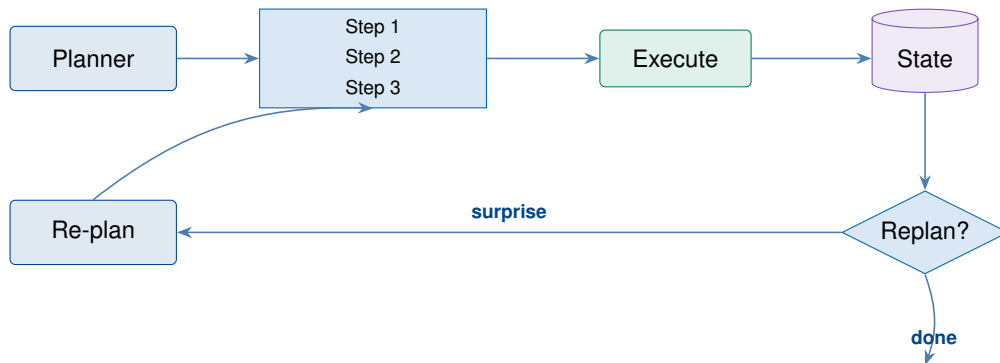


Figure 39.1: The plan-and-revise variant. The planner produces an explicit plan; the executor runs steps and updates shared state; a checkpoint decides whether to continue or to trigger replanning when the state shows a surprise.

The choice among the variants is a trade-off between predictability and adaptability. Rigid plans are predictable but brittle — they execute reliably until reality diverges from them, and then they cannot bend. Adaptive plans are robust but harder to monitor, because what the system is doing is no longer fixed at the outset. Specialist-role architectures deliver better quality, because each role uses a model suited to it, at the cost of higher complexity and the routing the split requires. And single-model approaches are simpler but less inspectable, because the plan never becomes a separate artefact anyone can review. The sections that follow develop each variant; the architect’s task is to place a given task on this predictability-adaptability spectrum and choose accordingly.

39.2 Linear Plan-Then-Execute

The simplest variant is fully linear: an initial prompt produces a plan as a sequence of steps, the plan is parsed into a structured form, the steps execute one at a time, each producing an output, and the final result is assembled from the step outputs. The plan is made once, before any execution, and it is not revised.

The linear variant’s strengths all come from the plan being a fixed, explicit artefact. The plan can be inspected before execution begins, which makes human review natural — a person can read the plan and approve or correct it before any step runs. The steps are isolated, so a failure in one step does not corrupt the others. The plan can be cached and reused for recurring task types. And because the plan is explicit and the steps are discrete, execution can be paused and resumed — a plan half-executed is a

well-defined state.

Its limits all come from the same fixedness. The quality of the plan is bounded by the context the model had when it made the plan, before any execution revealed anything. The plan cannot adapt to an execution surprise, because there is no revision step. Drift between the plan and reality accumulates as execution proceeds, since the plan was made against a picture of the task that execution may be steadily contradicting. And some tasks simply resist upfront planning — their shape is not knowable until execution has begun. The linear variant is the right choice for multi-step tasks whose structure genuinely is knowable in advance and stable through execution; for tasks where it is not, the next variant exists.

39.3 Plan-and-Revise

Plan-and-revise keeps the explicit plan of the linear variant but allows it to change. An initial plan is generated, execution proceeds, and at defined checkpoints — or whenever execution produces a surprise — the planner is re-invoked with the current state and produces a revised plan, which execution then continues against. The loop runs until the task is complete or a budget is exhausted. This is the variant Figure 39.1 shows, and it is the one most production systems settle on, because it combines the inspectability of an explicit plan with the robustness of adaptation.

The central design tension of plan-and-revise is plan stability. A plan that is too stable misses the adaptations it should have made — it ignores the surprises and drifts from reality just as a linear plan would. A plan that is too unstable never converges — every minor execution detail triggers a replan, and the system spends its budget replanning rather than executing. Getting this balance right means designing the replanning triggers deliberately: a replan should fire on a genuine surprise, a checkpoint reached, or a divergence beyond a threshold, not on every step. Replanning also has a cost that can grow large, since each replan is a planner invocation, and plan-and-revise therefore needs substantial observability — enough to see how often replanning is firing and whether the loop is converging or thrashing. Plan-and-revise done well is the robust default; plan-and-revise with poorly designed triggers is the replanning-loop failure mode the chapter closes on.

39.4 Hierarchical Plans

Hierarchical planning addresses the length problem. A single flat plan for a long task is hard to make well and hard to revise, so a hierarchical architecture produces a high-level plan of a few major phases, and then, as execution reaches each phase, a sub-planner produces the detailed steps for that phase with its own focused context. Execution proceeds at both the phase and the step level, and each level can be revised independently of the others.

The strengths of the hierarchical variant follow from this separation of levels. It scales to longer tasks than a flat plan can, because no single plan has to hold the whole task at once. The top-level plan stays stable — the major phases rarely change — while the lower levels adapt freely, which gives the system both a stable backbone and local flexibility. Each level is independently inspectable. And failures are isolated by phase, so a problem in one phase's execution does not corrupt the others. The cost

is engineering effort, concentrated in the seams between levels: context must be handed off cleanly from a phase to its sub-planner, the system must remember what completed phases accomplished, and errors must be able to escalate across levels — a failure a sub-planner cannot resolve must reach the top-level planner. Hierarchical planning is the variant for genuinely long tasks, and it is worth its design effort precisely when a flat plan has become too large to make or revise well.

39.5 Reactive and Specialist-Role Patterns

The remaining two variants sit at the edges of the predictability-adaptability spectrum, one at each end.

Reactive patterns make almost no upfront plan. The plan is initially just a goal and the current state, and each execution step is selected at the time it is needed rather than laid out in advance. Reactive patterns require substantial state-tracking, since the system's only representation of where it is is the state it maintains, and they are common in agentic loops with tool use, where the next action genuinely depends on what the last tool returned. Their cost is inspectability and control: there is no upfront plan to review, which makes a reactive system hard to monitor, and they are easy to send into unproductive loops, since nothing fixed bounds what the system will try. A reactive pattern's behaviour depends heavily on the quality of its state-tracking, and reactive patterns are often wrapped inside a planner-executor architecture at an outer level — a hierarchical or plan-and-revise structure that bounds the reactive execution within a phase. Reactive execution is the right choice for the genuinely unpredictable inner work; it is rarely the right choice for the whole architecture.

Specialist-role architectures are an orthogonal choice: rather than varying how rigid the plan is, they vary who makes and runs it. A frontier model, or a planner specialised for planning, produces the plan; cheaper executor models, or executors specialised for particular step types, run the steps; the planner is re-invoked for replanning when needed. This is the routing of the model-routing chapter applied inside a planner-executor architecture. Its strengths are cost and quality at once: the expensive planner is invoked sparingly, only for the planning and replanning, while the bulk of the work — execution — runs on cheaper models, and each role uses a model suited to it, with executors that can even be fine-tuned for their specific step type. The cost is the complexity of the routing within the architecture. Specialist roles are attractive whenever planning is expensive relative to execution, which it usually is, and they are the variant that most directly addresses the cost of the pattern.

39.6 Making a Plan an Engineering Artefact

Whatever the variant, the planner-executor pattern only delivers its promised inspectability and recoverability if the plan is treated as a genuine engineering artefact rather than a passage of model output. Four disciplines make that real.

The first is to express plans as *structured artefacts* — as JSON, a domain-specific language, or a graph, not as prose. A structured plan can be validated for executability before any step runs, versioned, compared across revisions, and persisted as history. A plan that exists only as a paragraph of generated text can be none of these things; it cannot even be reliably parsed into steps. Structuring the plan is

the precondition for every other discipline in this section.

The second is *checkpoint-and-resume*. The plan's execution state should be persisted at checkpoints, so that a failure mid-execution resumes from the last checkpoint rather than restarting the whole task. This is substantial design — it makes execution stateful — and it is critical for long-running tasks, where restarting from the beginning on every failure would make the task effectively impossible to complete.

The third is *budget-bounded execution*. A plan's execution should run under a total budget and per-step budgets, the planner should be able to take the budget into account when replanning — choosing cheaper alternatives when the budget is tight — and execution should terminate gracefully when the budget is exhausted, returning what was completed rather than failing. This is the loop-budgeting discipline the agentic chapters insist on, applied to the planned task.

The fourth is *human-in-the-loop integration*, which the explicit plan makes uniquely natural. Because the plan is an inspectable artefact, a human can approve it before execution begins, review it at key checkpoints, and correct it directly. This is the planner-executor pattern's distinctive advantage over a single-model run: there is a plan to show a human, and designing the architecture for that inspectability and intervention is what turns the advantage from potential into real oversight.

Underpinning all four is observability: plans, steps, and outcomes logged, replanning events tracked, and separate plan-quality and execution-quality metrics maintained, so that a failed task can be diagnosed as either a planning failure or an execution failure — a distinction the next section shows to be essential.

39.7 Evaluating Planning and Execution Separately

The single most important evaluation discipline for a planner-executor architecture is to evaluate planning and execution as separate things. A task can fail because the plan was wrong — the right steps were never specified — or because the execution was wrong — the steps were right but a step was performed badly. These are different failures with different remedies, and an evaluation that reports only end-to-end success cannot tell them apart.

Separating the two means measuring plan quality and execution quality independently. Plan quality asks whether the plan, taken on its own, was a correct and complete decomposition of the task — a question that can be assessed against the plan artefact before or independently of execution. Execution quality asks whether each step, given the plan, was carried out correctly. A system with a plan-quality problem and a system with an execution-quality problem present identical end-to-end failure rates and need opposite fixes: the first needs a better planner, the second needs better executors. An architecture that logs plans and steps separately, and evaluates each, can make that diagnosis; one that measures only whether the task succeeded is left guessing which half to improve. This is the planner-executor expression of the evaluation chapter's general principle — that an evaluation must be specific enough to point at the thing to fix.

39.8 Recurring Failure Patterns

The failure modes of planner-executor architectures are recognisable, and each maps to a discipline this chapter has set out:

- **Plan-and-reality drift** — the plan was made on stale context and execution steadily diverges from it, with no revision step to close the gap.
- **Replanning loop** — replanning triggers are too sensitive, so the planner is re-invoked endlessly and never converges to an executable plan.
- **Step-level error swallowing** — an error in a step does not surface, so execution continues on a corrupted result.
- **Context loss between steps** — a step does not receive what an earlier step established, so it works without information it needed.
- **Budget explosion** — planning and execution together run without a budget and consume unbounded cost.
- **Plan inspectability failure** — the plan is nominally explicit but not actually reviewable, so the pattern’s main advantage is lost.

Each maps to a remedy already developed. Plan-and-reality drift is the case for plan-and-revise over the linear variant. The replanning loop is the plan-stability discipline not applied. Step-level error swallowing and context loss are failures of the observability and structured-artefact disciplines. Budget explosion is budget-bounded execution missing. And plan inspectability failure is the structured-artefact discipline only half-done — a plan that was made explicit in name but not in a form a human can genuinely review. The failure modes are common because the disciplines that prevent them are real work; the chapter’s argument is that the work is what makes planner-executor a system rather than, in the epigraph’s words, a story.

39.9 The African Perspective

Planner-executor architectures carry specific weight in African deployments, across cost, connectivity, language, and the structure of cross-border work.

Cost is the first consideration, and it bears most directly on the variant choice. A planner running on a frontier model can dominate the cost of a planned task, and in a deployment where model calls are billed in foreign currency against a local-currency budget, that domination can be decisive. The specialist-role variant is therefore particularly attractive here: invoking the expensive planner sparingly and running execution on cheaper models is exactly the cost structure a tight unit economics needs. Plan-caching for recurring task types compounds the saving, since a cached plan is a planner invocation avoided entirely. The cost-engineering of plans is, in these deployments, central design work rather than an afterthought.

Connectivity is the second. Multi-step execution is, by construction, a sequence of operations over

time, and a sequence is exposed to connectivity variability at every step. This makes checkpoint-and-resume not a refinement but a core requirement: a task that loses connectivity mid-execution must resume from its last checkpoint rather than restart. Edge-resident executors for offline-capable work, and graceful degradation on connectivity loss, are the other responses — so that a connectivity failure costs progress, not the whole task.

Language is the third. Plans may span multilingual content, steps may need to execute across languages, and a plan that involves translation has a translation step that is itself a source of error and a point where cultural context can be flattened. The per-language evaluation discipline applies: a planner-executor system serving multiple languages should be evaluated per language, because a system that plans and executes well in one language may do neither well in another.

Two structural realities of African operations deserve specific mention. Mobile-money integration means plans frequently involve payment steps executed through mobile-money tools, and multi-step financial workflows that cross networks need design built for those payment realities rather than for card-based assumptions. And cross-border operations mean that a pan-African deployment produces plans that span jurisdictions — with regulatory and currency considerations embedded in the plan itself, and data-residency rules affecting where the plan state is even allowed to live. A cross-border plan is not simply a longer plan; it is a plan whose steps carry jurisdictional constraints, and the architecture must treat those constraints as part of the plan rather than as an afterthought to it.

39.10 Pulling the Threads

Planner-executor architectures separate planning from execution so that multi-step tasks can be done as systems rather than as single opaque runs. The pattern has five variants — linear, plan-and-revise, hierarchical, reactive, and specialist-role — arranged along a spectrum from predictable to adaptive, and the architect's task is to place a task on that spectrum and choose accordingly. Whichever variant is chosen, the plan must be made a genuine engineering artefact: structured, validated, versioned, checkpointed, budget-bounded, and inspectable enough for real human oversight — and planning and execution must be evaluated separately, because they fail differently and need opposite fixes.

The epigraph draws the line the whole chapter is about. A model that plans and then executes is a system: the plan is an artefact, the execution is bounded, the failures are diagnosable, and a human can see what is being done before it is done. A model that plans while executing — that improvises its way through a multi-step task with no separable plan — is a story: it may reach a good ending, but nothing about it could be reviewed in advance, recovered cleanly when it broke, or diagnosed afterward. The disciplines of this chapter are what keep a planner-executor architecture on the system side of that line, and they are the difference between a multi-step AI capability an enterprise can operate and govern and one it can only watch and hope for.

Chapter 40

Memory and Context Management Patterns

“A model without memory is a model that forgets its user. A model with bad memory is a model that remembers the wrong things.”

—a recurring observation in production conversational and agentic systems

Chapter Overview

Memory and context management are the architectures by which an AI system carries information across turns, sessions, users, and tasks. The simplest form is the conversation history sitting in the prompt; the patterns extend through summarisation, episodic memory, semantic memory, working memory for agents, and user-scoped persistent memory. This chapter develops the memory variants and what each is for, the layered architecture that combines them, the hygiene and privacy and injection-defence disciplines that keep persistent memory trustworthy, and the failure modes a memory system must be built to withstand. Production systems choose among these patterns by capability, cost, privacy, and risk profile.

Learning Objectives

- Recognise the memory variants — short-term, episodic, semantic, procedural, working, user-scoped — and what each addresses.
- Architect retrieval, update, and forgetting policies for each layer.
- Manage privacy, accuracy, and drift in persistent memory.
- Defend memory against injection and treat its contents as untrusted.
- Build observability and inspectability over memory state.
- Adapt memory patterns to African data-residency, language, and connectivity realities.

40.1 The Memory Problem

Context windows are finite, and the work an AI system does is not. Conversations and tasks exceed the window; users return across sessions expecting the system to remember them; agents execute multi-step plans that need a working scratchpad. The architectural question that runs through this

chapter is therefore a three-part one — how to retain information, how to retrieve the right information when it is needed, and how to forget what should not be kept — because a memory system that does any of the three badly fails in a way the epigraph names: it either forgets its user or remembers the wrong things.

Memory comes in six variants, and they are best understood by what each is for. *Short-term* memory is the conversation history held in the prompt. *Episodic* memory holds specific past events — past conversations, transactions — retrieved by relevance. *Semantic* memory holds facts and concepts, retrieved by similarity. *Procedural* memory holds how-to knowledge, often baked into tool definitions or prompt templates rather than retrieved. *Working* memory is the scratchpad an agent maintains during execution. And *user-scoped persistent* memory holds information about a user that survives across sessions. Figure 40.1 shows how these layers sit between the user and the model; the sections that follow develop each.

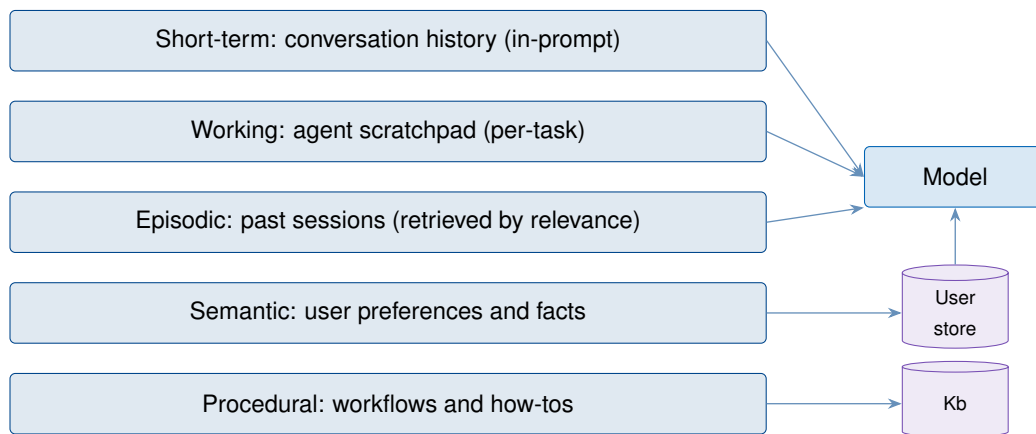


Figure 40.1: Layered memory. Short-term memory lives in the prompt; working memory tracks task state; episodic, semantic, and procedural memories are stored and retrieved as needed. Each layer has its own update, retrieval, and forgetting policies.

Underneath the variants sit tensions that no memory design escapes. More memory means more relevant context, but also more noise and more cost, so retaining everything is not a strategy. Persistent memory raises privacy considerations the moment it holds anything about a real user. Memory drifts — information that was true when stored is used as if still current. Memory can carry injection — content placed in memory by an earlier interaction can contain instructions the model later reads as commands. And persistent memory creates an obligation of inspectability and a right to correct, because a user whose data is being remembered is entitled to see and amend it. These tensions are the reason memory is an architecture rather than a buffer.

40.2 Short-Term Memory: Conversation History

The simplest memory pattern is the conversation history itself: each turn is appended to the prompt, the earliest turns are trimmed when the context window fills, and the trimmed turns are optionally summarised before they are dropped so that something of them survives. For a single-session conversation that does not run very long, this is often all the memory a system needs.

Its limits are worth stating, because they are easy to underestimate. Long contexts cost money and latency on every turn, since the whole history is re-sent with each request. Context-window limits remain a real boundary even though frontier models now support hundreds of thousands of tokens — the window is large, not infinite. Quality often degrades with very long contexts even when the content fits within the window, as the model’s attention spreads thin over a long history. And summarisation, the natural mitigation, is lossy: a summary preserves the gist and discards detail, and the discarded detail is sometimes exactly what a later turn needed. Short-term memory is the right and sufficient pattern for bounded conversations; the longer-lived and cross-session needs are what the other variants exist to serve.

40.3 Episodic, Semantic, and Procedural Memory

Three of the memory variants share a common shape: they store information outside the prompt and retrieve the relevant pieces when a request needs them. They differ in what they store.

Episodic memory stores specific past events — conversations, transactions, interactions. When a new request arrives, the relevant episodes are retrieved, typically by embedding similarity, and injected into the context. The engineering of episodic memory turns on a few decisions: the granularity at which episodes are chunked and stored, the choice of embedding model, the retrieval scoring, the balance struck between recency and relevance — a recent episode and a highly relevant older one are both candidates, and the policy must weigh them — and the privacy and access controls that govern who may retrieve whose episodes. Episode granularity is the decision most often got wrong: episodes chunked too finely lose the context that made them meaningful, and chunked too coarsely retrieve large irrelevant blocks.

Semantic memory stores facts and concepts — user preferences, company policies, product information — and retrieves them by similarity. It overlaps substantially with retrieval-augmented generation, and the techniques of the RAG chapters apply to it directly; the distinction is mostly one of scope, since semantic memory is typically the user-scoped or application-scoped slice of the same retrieval machinery. It is frequently combined with episodic memory in user-scoped applications: the episodic layer remembers what happened, the semantic layer remembers what is true.

Procedural memory stores how-to knowledge — recipes, workflows, tool-use sequences. It can be retrieved by task similarity, but in practice it is often expressed as templates or worked examples and baked directly into prompt templates rather than retrieved at request time. Procedural memory is the variant most likely to live in the system’s design rather than in a store, and recognising that is what keeps an architect from building retrieval machinery for knowledge that a well-written prompt template already carries.

40.4 Working Memory for Agents

Working memory is the scratchpad an agent maintains while it executes. As steps run, the agent updates its state; at checkpoints, the state is written to persistent storage; and the state is carried into replanning so that a re-planning agent knows what has already been done. Working memory is what

gives an agent continuity across the steps of a single task, and it is the memory variant most tightly coupled to the planner-executor and agentic-loop patterns of the surrounding chapters.

The engineering of working memory is the engineering of state. Its size must be managed, because an agent that accumulates everything into its working state will eventually overflow its context. Its structure must be designed, because a well-structured state can be updated and inspected while an ad-hoc one cannot. Its updates must be validated, since an agent writing malformed state corrupts every later step that reads it. And its loss must be recoverable — the checkpoint-and-resume discipline of the planner-executor chapter is, concretely, the discipline of persisting working memory so that a failure does not erase the task’s progress. Working memory done well is the difference between an agent that can be interrupted and resumed and one for which any interruption is a restart.

40.5 User-Scoped Persistent Memory

User-scoped persistent memory is information about a user that survives across sessions: a user-specific store, updated from conversation content — often with the model itself deciding what is worth remembering — retrieved at the start of each new session, and, crucially, inspectable and correctable by the user. It is the variant that makes a system feel like it knows its user, and it is also the variant that carries the most risk, because it is the one holding real personal data over time.

The engineering questions it raises are genuinely hard. What to remember must be decided, whether by the model or by rules, and the decision is consequential because over-remembering accumulates noise and privacy exposure while under-remembering defeats the purpose. Remembered items must be deduplicated and consolidated, or the store fills with near-duplicates. Contradictions over time must be resolved — when a user’s stated preference changes, the memory must update rather than hold both. Forgetting must be designed, whether time-based, user-requested, or policy-driven, because a persistent memory with no forgetting policy grows without bound and retains data long past its justification. And privacy and access controls govern all of it. These are as much product-design questions as architectural ones, and they cannot be deferred.

The failure modes of user-scoped memory are the sharpest in the chapter. Memory can hold wrong information, which the model then treats as fact. It can hold sensitive information it should never have retained. It can hold injected content that re-enters the model as instruction. It can drift, serving stale information as current. It can surprise the user, who did not realise the system was retaining what it retained. And memory correction can be made so awkward that a user who knows the memory is wrong cannot easily fix it. Each of these is a reason the disciplines of the next section are not optional refinements but core requirements of any persistent-memory design.

40.6 The Memory Disciplines

A memory system that combines several variants needs five standing disciplines, and together they are what separate a memory architecture from a pile of stored text.

The first is the *layered architecture* itself. A mature system uses several memory layers at once — short-term for the live conversation, episodic for past sessions, semantic for preferences and facts,

procedural for workflows — and the discipline is to give each layer its own update and retrieval policy rather than treating all of memory as one undifferentiated store. The layers serve different needs and should be governed differently; conflating them produces a store that is wrong for all of them.

The second is *memory hygiene*: the periodic consolidation, deduplication, contradiction-resolution, and stale-data forgetting that keep a memory store healthy. Memory hygiene is maintenance infrastructure, and a store without it degrades — it grows, it accumulates contradictions, and its retrieval quality falls as the noise rises. Hygiene is the memory equivalent of the dataset maintenance the evaluation chapter insists on: a store left unmaintained stops being trustworthy.

The third is *privacy controls*. A user must be able to inspect what is stored about them, correct it, and delete it; sensitivity classification must govern what is eligible to be stored in the first place. These are not features bolted on for compliance; they are the architecture of a memory system that holds personal data lawfully, and they connect directly to the data-residency obligations the African-perspective section develops.

The fourth is *injection defence*. Memory content must be treated as untrusted input, because content placed in memory by an earlier interaction — or by a third party whose data the system stored — can carry prompt-injection instructions. Memory content must be sanitised before use, must be fenced so the model reads it as data rather than instruction, and must never be allowed to override the system's safety policies. A system that treats its own memory as trusted has extended its trust boundary to everyone whose content the memory has ever absorbed.

The fifth is *inspectability*. Memory state should be inspectable by the user, memory updates and retrieval decisions should be logged, and the influence of a memory item on a given response should be traceable. Without this, a memory-driven system is opaque — a user cannot tell why the system believes what it believes about them, and an engineer cannot debug a response that a stale or wrong memory shaped. Inspectability is the observability discipline of the surrounding chapters applied to the memory layer.

40.7 Recurring Failure Patterns

The failure modes of memory systems recur, and each maps to a discipline this chapter has set out:

- **Stale memory used as current** — information that was true when stored is served as if still true, because no forgetting or freshness policy governs it.
- **Wrong memory unchallenged** — an incorrect memory item shapes responses and nothing in the system flags or corrects it.
- **Memory growing without consolidation** — the store accumulates duplicates and contradictions because memory hygiene is not run.
- **Privacy controls absent or inadequate** — the user cannot inspect, correct, or delete what is stored, and sensitivity classification does not gate what is retained.

- **Memory injection from compromised content** — stored content carrying instructions re-enters the model as a command.
- **Memory-and-user surprise** — the system retains more, or for longer, than the user understood, and the retention is discovered rather than disclosed.
- **Cross-tenant memory leakage** — one user’s or tenant’s memory is retrieved into another’s context, a failure of the access controls that should scope every retrieval.

Each maps to a remedy already developed: stale and wrong memory to hygiene and freshness policy, unconsolidated growth to memory hygiene, the privacy failures to the privacy-controls discipline, injection to injection defence, user surprise to inspectability and disclosure, and cross-tenant leakage to the access controls that must scope every layer’s retrieval. The failure modes are common because the disciplines that prevent them are real, ongoing work — and the chapter’s argument is that a memory system without that work does not merely underperform; it becomes a privacy and security liability.

40.8 The African Perspective

Memory architectures carry particular weight in African deployments, across data residency, language, cost, and connectivity.

Data residency is the first and sharpest consideration. A user-scoped memory store is personal data by definition, and it therefore falls squarely within the data-protection regimes of the markets a system serves — South Africa’s POPIA, Nigeria’s NDPA, Kenya’s DPA, and the data-protection laws of other jurisdictions each carry implications for where a memory store may physically sit and how it may be processed. For a deployment serving a single country, this often means the memory store must be located in-country; for a pan-African operation, it means cross-border memory must be designed against several regimes at once. The privacy-controls discipline of this chapter is, in these deployments, not only good practice but the mechanism of legal compliance.

Language shapes the memory store’s contents. African users frequently interact across multiple languages and in code-switched registers, which means memory entries will themselves be multilingual and code-switched, and cross-language retrieval — a query in one language retrieving a memory stored in another — must be designed for rather than assumed. A memory system that handles only one language well will retrieve poorly for exactly the multilingual users it most needs to serve, and code-switched content must be preserved rather than flattened toward a single language.

Cost and connectivity shape the memory architecture’s mechanics. Memory retrieval has a cost — the embedding computation, the store query, the tokens the retrieved content adds to the context — and under the tight unit economics of many African deployments that cost matters, which argues for aggressive summarisation and consolidation to keep stores compact and for smaller embedding models to keep retrieval cheap. Connectivity argues for edge-cached memory in offline-capable applications, so that a memory retrieval does not fail when the link to the store is down, and for memory-update synchronisation designed to tolerate connectivity variability rather than assuming a reliable connection

to the store at every moment. The memory patterns do not change for African deployments; their parameters — where the store sits, how large it is allowed to grow, how it is reached — must reflect the residency, language, cost, and connectivity realities of the markets actually served.

40.9 Closing Notes

Memory and context management is the architecture by which an AI system carries information across turns, sessions, users, and tasks. It has six variants — short-term, episodic, semantic, procedural, working, and user-scoped persistent — each addressing a specific need, and a mature system combines several into a layered architecture. Around that architecture sit five disciplines: the layered design itself, memory hygiene, privacy controls, injection defence, and inspectability. Together they are what make a memory system trustworthy rather than merely capacious.

The epigraph names the two failures the whole chapter is built to prevent. A model without memory forgets its user, and a model with bad memory remembers the wrong things — and the second failure is the more dangerous, because it is confident. A memory system built with the disciplines of this chapter — layered, maintained, privacy-respecting, injection-defended, and inspectable — gives a model the continuity that makes it genuinely useful across sessions and tasks. A memory system built without them gives the model a confident store of stale, wrong, or leaked information, and a confident wrong memory is worse than no memory at all.

Chapter 41

Alignment with Enterprise Architecture Frameworks

“A new architectural style does not abolish the ones that preceded it. It inherits some, contradicts some, and quietly extends most.”

—a recurring observation in enterprise architecture

Chapter Overview

Enterprise AI architecture does not arrive on a blank canvas. It lands in organisations with established architectural traditions: TOGAF for enterprise-wide planning, Zachman for the ontology, C4 for the diagrams developers actually draw, Domain-Driven Design for the modelling discipline, microservices for the runtime decomposition, SOA for the integration legacy, event-driven for the asynchronous bits, hexagonal for the testable core. Practitioners who arrive saying “AI changes everything” are wrong. Practitioners who say “AI is just another service” are also wrong. The accurate view is the harder one: AI extends, contradicts, and reinterprets the existing frameworks in specific, identifiable ways.

Learning Objectives

- Map the foundational frameworks (TOGAF, Zachman, C4) onto AI architecture and identify where they extend cleanly and where they strain.
- Apply DDD discipline to AI systems, including the question of where models sit in the domain.
- Reconcile AI patterns with microservices, SOA, and event-driven heritage.
- Use hexagonal architecture to keep model providers swappable.
- Recognise the patterns where AI genuinely demands new architectural thinking.

41.1 The Question

A senior architect who has run TOGAF programmes for fifteen years asks: “Where does the LLM go in the reference model?” The question is not naive. It is exactly the right question. The answer is not “everywhere” — that is a non-answer — nor “the application layer” — that flattens too much. The

honest answer requires walking through each framework and seeing how it accommodates, and where it fails to accommodate, AI-native systems.

41.2 TOGAF and the Architecture Domains

TOGAF organises architecture into Business, Data, Application, and Technology domains, with the ADM (Architecture Development Method) as its process. Figure 41.1 shows how AI capabilities project onto the four domains.

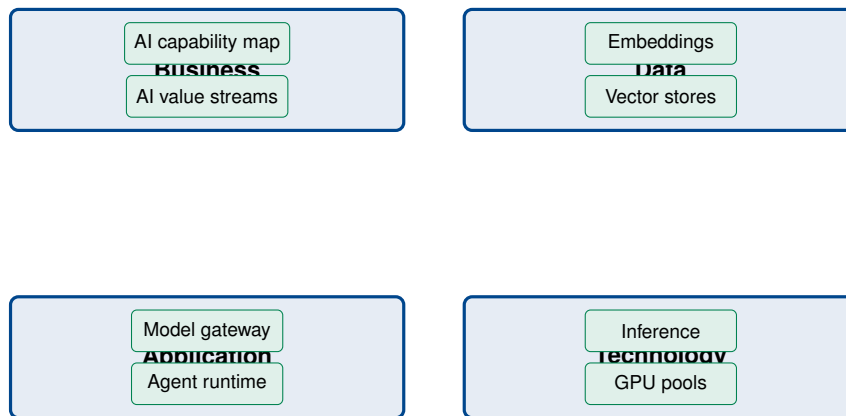


Figure 41.1: AI capabilities projected onto TOGAF's four architecture domains. Each domain gains new artefacts; the four-domain decomposition itself remains intact.

41.2.1 Where TOGAF Fits Cleanly

TOGAF's strength is in the areas where enterprise AI genuinely needs the governance, phasing, and stakeholder-alignment disciplines that TOGAF was built for. These areas map cleanly onto TOGAF's existing mechanisms, and an organisation already running TOGAF can extend them to cover AI without forcing the framework into territory it was not designed for.

- **Business architecture.** AI capabilities map onto the capability model. AI value streams are first-class. AI initiatives appear in the architecture roadmap.
- **Data architecture.** Embeddings, vector stores, and training datasets are data artefacts. Lineage and quality concerns extend naturally.
- **Application architecture.** Model gateways, agent runtimes, retrieval services, and evaluation pipelines fit the application taxonomy.
- **Technology architecture.** GPU pools, model-hosting infrastructure, vector databases extend the technology stack.

41.2.2 Where TOGAF Strains

TOGAF's assumptions about the predictability of change, the stability of components, and the pace of evolution do not hold for the parts of an AI system that are probabilistic, rapidly evolving, and vendor-dependent. In these areas the framework's standard mechanisms produce overhead without value, and the honest response is to acknowledge the strain rather than force-fit.

- **The ADM cadence.** TOGAF assumes a multi-year iteration on the enterprise architecture. AI capability moves on a quarterly cadence. The ADM has to run faster on AI than on most other domains.
- **The principles catalogue.** AI-specific principles — model provenance, evaluation gates, content provenance, agent autonomy boundaries — are not in the TOGAF default set. Practitioners add them.
- **The boundary of “application.”** An agent that calls tools, queries data, and makes decisions blurs the application/data/business line in ways the framework does not anticipate.

41.3 Zachman and the Six Questions

Zachman is an ontology, not a method: six interrogatives (what, how, where, who, when, why) crossed with six perspectives (scope, business, system, technology, instance, operations). The matrix forces complete questioning.

41.3.1 The AI-Specific Cells

Zachman’s matrix gains its power from the specificity of each cell, and AI introduces cells that the original framework did not anticipate. Populating these cells explicitly is what turns a Zachman exercise from a generic governance ritual into a genuinely useful map of the AI estate’s concerns.

- **What.** Training data, evaluation data, prompts, embeddings, model artefacts.
- **How.** Inference pipelines, agent loops, tool chains, retrieval flows.
- **Where.** Region-of-inference (with data-residency implications), edge vs cloud placement.
- **Who.** Model providers, agent identities, end users, reviewers in HITL.
- **When.** Real-time vs batch, streaming, agent execution checkpoints, evaluation cadence.
- **Why.** The use-case justifying the AI investment, the risk tolerated, the residual human oversight.

Zachman does not strain on AI; it accommodates it by design. The framework’s gift is the question, not the answer.

41.4 C4 and the Diagrams Developers Draw

C4 is a hierarchical model: System Context, Container, Component, Code. It is pragmatic, draw-friendly, and widely adopted.

41.4.1 Extending C4 for AI

The C4 model’s four zoom levels — context, container, component, code — map naturally onto AI systems, but each level acquires AI-specific elements that the generic C4 conventions do not name. Extending C4 to name those elements explicitly is what makes an AI system’s C4 diagrams genuinely informative rather than misleadingly generic.

- **System Context.** Model providers (Anthropic, OpenAI, Google) appear as external systems. Agents that act on behalf of users appear as actors.
- **Container.** Model gateway, vector store, agent runtime, prompt store, evaluation harness, HITL queue are containers.
- **Component.** Tool definitions, retrieval pipelines, prompt templates, eval rubrics.
- **Code.** Prompts themselves are code at this level; so are tool implementations.

The C4 community has begun proposing AI-specific stereotypes: “<<model>>”, “<<agent>>”, “<<tool>>”. The convention is informal but converging.

41.4.2 Why C4 Adapts Well

C4’s pragmatism is its strength. It demands no metamodel commitment; it asks only that diagrams be consistent at each level. AI-native artefacts slot in without violence to the model.

41.5 Domain-Driven Design and the Model Boundary

DDD’s central question — where do domain boundaries lie? — becomes acute in AI systems.

41.5.1 Where Does the Model Sit?

The model is the component that does not map cleanly onto any of the 4+1 views as originally conceived, because it is simultaneously a data artefact, a runtime component, a deployment unit, and a thing with its own lifecycle. Placing it requires a deliberate choice about which view owns the model and how the other views reference it. Three placements recur:

- **Model as anti-corruption layer.** The model lives at a context boundary, translating between domains. Common pattern when LLMs do free-form-input parsing or natural-language interfaces.
- **Model as domain service.** The model performs a domain operation: classification, summarisation, scoring. Sits inside the bounded context.
- **Model as orchestrator.** The model decides which domain operations to perform. Sits above bounded contexts. Architecturally the most fraught: blurs context boundaries.

41.5.2 Ubiquitous Language and Prompts

DDD’s ubiquitous language — shared vocabulary between domain experts and developers — extends to prompts. Prompts written in domain language travel better through the organisation, are easier to evaluate, and degrade less when models change.

41.5.3 Bounded Context Per Agent

A pattern with traction: each agent gets its own bounded context. The agent’s tools, prompts, evaluations, and data sources are scoped to a single domain. Cross-context coordination happens through explicit handoff (events, supervisor topology, structured messages), not through an agent that “knows everything.”

41.6 Microservices and the Single Responsibility

Microservices distribute responsibilities into small, focused services; AI, in one specific respect, wants to undistribute them, and the tension between the two is worth resolving carefully rather than papering over. The microservices ideology asks for small services each with a single responsibility. The AI reality is a large, capable model that can do many things at once — an LLM is, in its design, anti-microservice. The resolution is a distinction: the model is not a service, but the *capability built on the model* is. A summarisation service, a classification service, a routing service are all valid microservices, each with a genuine single responsibility, that internally share a common model gateway. The model is infrastructure the services use; the services are the units the microservices discipline applies to.

41.6.1 Where Microservices Win, and Where They Strain

Decomposing AI capabilities into microservices wins in the ways microservices always win, and the wins are real: each capability gets its own deployment and scaling, its own evaluation and SLAs, its own versioning of prompts and tools, and its own observability, and a broken summariser does not take the classifier down with it — the blast radius of a failure is bounded to one capability. These are the familiar microservices benefits, and they apply to an AI estate as cleanly as to any other.

The strain appears where AI workloads do not behave like ordinary services. A multi-step agentic workflow that spans several services produces a distributed trace of considerable complexity, far more tangled than a typical request crossing a few microservices. Shared state across agentic services becomes a coordination problem that the stateless-service ideal did not anticipate. And the inter-service calls inside an agent loop multiply latency, because each hop is paid on every iteration of the loop. The microservices decomposition is right for the AI estate; the strain is a reminder that the agentic workloads running on it stress the pattern in ways a conventional request-response service does not.

41.7 SOA Heritage and Integration Reality

Service-Oriented Architecture preceded microservices in most enterprises and still runs much of what matters. AI systems usually integrate with SOA estates, not replace them.

41.7.1 ESB and the Model Gateway

The model gateway is, for AI, an ESB-like component: it does protocol translation, policy enforcement, observability, routing, and retry — the same crosscutting concerns the enterprise service bus centralised for SOA. The difference is in what it does not do. Unlike an ESB, the model gateway does not transform business payloads; its job is the model-provider interface, not the choreography of business messages. In organisations that still run active ESBs, the model gateway does not replace the bus — it sits behind or alongside it, handling the AI-provider concerns while the ESB continues to handle the business-integration ones.

41.7.2 The Service Registry Analogue

The tool catalogue plays the role the SOA service registry once played: discoverable, typed, governed, versioned. The MCP ecosystem is converging on the discipline SOA tried to impose on web services twenty years ago.

41.8 Event-Driven Heritage

Event-driven architecture (Chapter 25) is the pattern that aligns most naturally with AI, and the alignment is close enough to be worth spelling out. Long-running agents fit asynchronous execution far better than they fit request-response, because an agent's work unfolds over time rather than within a single synchronous call. Tool calls are events, agent decisions are events, and human-in-the-loop approvals are events — the agentic vocabulary is already an event vocabulary. Replay-from-log, the event-driven system's native capability, is exactly what debugging an agent trajectory requires. And the decoupling that event-driven architecture provides lets agents and tools evolve independently, which is the same independence the tool-registry and hexagonal patterns pursue by other means. The convergence is not coincidence: mature AI systems often end up looking like event-driven systems whose consumers happen to be intelligent.

41.9 Hexagonal as the Provider-Independence Pattern

Hexagonal architecture (Chapter 26) earns its keep in AI specifically through the model-provider port. The pattern is now more valuable than ever: the model provider landscape is the highest-velocity layer in any AI system, and an organisation that has hardcoded provider SDKs throughout its codebase has burned in a coupling it will regret on the next switch.

41.10 The Alignment Map

Figure 41.2 summarises which frameworks accommodate AI cleanly, which strain, and where AI demands new thinking.

Fits cleanly	Strains	New thinking
Zachman	TOGAF (ADM cadence)	Agent autonomy
C4	Microservices (model size)	Eval-as-gate
Event-driven	SOA (model gateway)	Probabilistic SLA
Hexagonal	DDD (model boundary)	Prompt versioning

Figure 41.2: Alignment map. Some frameworks accommodate AI cleanly; some strain in specific places; some AI concerns demand thinking the existing frameworks do not provide.

41.11 Where AI Demands New Thinking

Four areas where existing frameworks do not have the vocabulary, and architects are working out the patterns in practice:

41.11.1 Agent Autonomy Boundaries

What an agent is permitted to do without human approval is not a question existing frameworks ask. The emerging discipline includes: action taxonomies (read, write, public-write, financial-write), per-action approval policies, reversibility tiers, blast-radius limits.

41.11.2 Evaluation-as-Gate

CI gates on tests are well understood. Gates on probabilistic evaluations — where a 95% pass rate may be the right bar — require new release-engineering thinking. Frameworks treat releases as binary; AI releases are graded.

41.11.3 Probabilistic SLAs

Traditional SLAs are uptime and latency, both deterministic. AI SLAs include answer quality, hallucination rate, refusal rate, citation accuracy. These are statistical SLAs, measured against rolling distributions, not point thresholds.

41.11.4 Prompt Versioning

A prompt is code. It needs version control, code review, testing, deployment, rollback. None of this is reflected in classical software-engineering frameworks. The discipline of treating prompts as first-class artefacts is still being built.

41.12 Recurring Failure Patterns

- **Frameworks discarded.** “AI is different” as an excuse for not doing architecture. Result: every team reinvents the same wheels badly.
- **Frameworks applied unchanged.** Existing frameworks applied without recognising the AI-specific extensions needed. Result: gaps in coverage that surface as incidents.
- **Model as service.** An LLM treated as a microservice. Result: anti-microservice characteristics (size, generality) at war with the architecture they live in.
- **Hardcoded providers.** No model-provider port. Result: every provider change is a refactor.
- **Evals as afterthought.** CI gates on tests, not on evaluations. Result: quality drift undetected until production.

41.13 The Pattern, Restated

Enterprise AI architecture inherits from TOGAF, Zachman, C4, DDD, microservices, SOA, event-driven, and hexagonal traditions. Each accommodates AI in specific ways and strains in others. The discipline is to use what fits, extend what strains, and recognise the small but important set of areas —

agent autonomy, evaluation-as-gate, probabilistic SLAs, prompt versioning — where AI demands genuinely new architectural thinking.

Chapter 42

Software 3.0

“The hottest new programming language is English.”

—Andrej Karpathy

Chapter Overview

Software 1.0 is hand-written code in C, Python, Java: explicit instructions in formal languages. Software 2.0 is neural networks: behaviour encoded in weights, learned from data. Software 3.0 is the regime now emerging: behaviour shaped through natural-language prompts and agentic loops over capable foundation models, where the programming primitive is the prompt, the tool definition, and the eval. None of these eras displaces the prior one; production systems are built across all three. The architectural implications of Software 3.0 are real and specific: the artefacts, lifecycle, version control, testing, deployment, and observability look different from anything that came before.

Learning Objectives

- Recognise what Software 3.0 is and what it is not.
- Identify the new first-class artefacts: prompts, tool definitions, agent specifications, evaluations, traces.
- Architect the development lifecycle for Software 3.0 alongside 1.0 and 2.0.
- Apply version control, testing, deployment, observability disciplines to Software 3.0 artefacts.
- Anticipate how Software 3.0 reshapes team composition and engineering practice.

42.1 The Three Eras

Figure 42.1 contrasts the three eras.

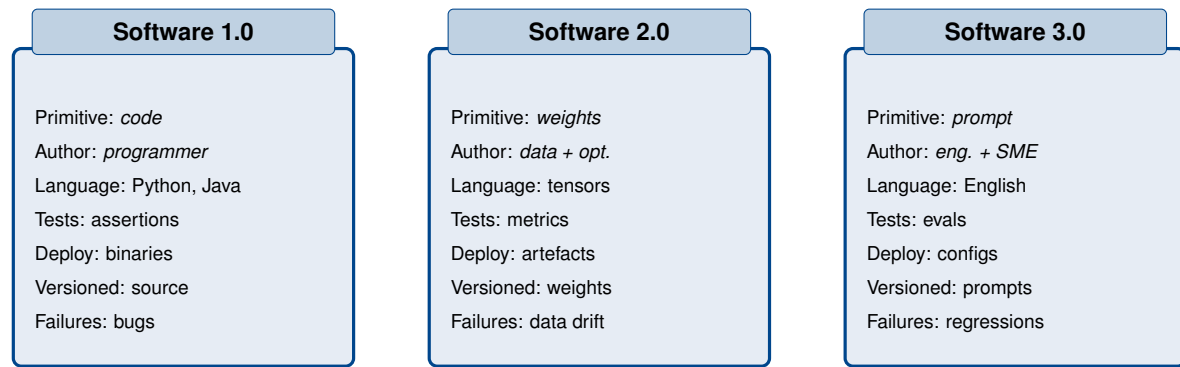


Figure 42.1: Three eras of software construction. The eras coexist; production systems use all three. Software 3.0 introduces new artefacts (prompts, tool definitions, agent specs, evals) and a new author profile (engineer plus subject-matter expert).

42.1.1 The Defining Properties of Software 3.0

Software 3.0 is not merely software that uses a model; it is software whose behaviour is defined by prompts and data rather than by compiled code. Several properties distinguish it from the deterministic software it sits alongside, and each one changes what the engineering team must do.

- **Programming primitive is the prompt.** Behaviour is shaped through natural-language instruction.
- **Composition primitive is the tool.** Capabilities are extended by giving the model new tools.
- **Coordination primitive is the agent.** Multi-step behaviour is structured through agentic loops.
- **Specification primitive is the eval.** What “correct” means is encoded in evaluations, not assertions.
- **Authorship is plural.** Engineers, domain experts, designers, and reviewers all touch the artefacts.

42.2 The New First-Class Artefacts

Software 3.0 elevates artefacts that, in earlier eras, were either invisible or treated as configuration.

42.2.1 Prompts

A prompt is the executable specification of a model’s behaviour for a task. In Software 3.0, prompts:

- Live under version control with the same rigour as code.
- Carry tests (the evaluation set the prompt is graded against).
- Have owners, reviewers, and approvers.
- Have a deployment lifecycle (development, staging, production, rollback).
- Have observability: every production invocation is traced, replayable, and analysable.

42.2.2 Tool Definitions

A tool is an interface offered to the model. Tool definitions specify name, description, input schema, output schema, side-effect markers, cost-and-latency metadata. They are versioned, deprecated, evolved like any API — but they are read by a model, which has implications for their writing style.

42.2.3 Agent Specifications

An agent specification declares: what the agent does, what tools it has, what models it uses, what its autonomy boundaries are, what evaluations it must pass, what its operational SLA is. The specification is a contract among engineers, product, security, and compliance.

42.2.4 Evaluations

Evaluations are the executable form of “what good looks like.” Built once, run on every prompt change, model change, tool change. They are how Software 3.0 systems hold quality as the underlying components evolve.

42.2.5 Traces

Every invocation of a Software 3.0 system produces a trace: inputs, prompts, retrievals, tool calls, model responses, intermediate outputs, final output. Traces are the unit of debugging, the unit of evaluation, and the unit of organisational learning.

42.3 The Lifecycle

Software 3.0 has a development lifecycle that overlaps with Software 1.0’s but is not identical to it, and the differences are concentrated in five stages.

Authoring is plural in a way Software 1.0’s rarely was. Engineers write the tool definitions, agent specifications, and evaluation harnesses; domain experts contribute to the prompts; product contributes to the user-experience and rubric design; security and compliance contribute to the autonomy boundaries. The artefacts converge in a shared repository, but they arrive there from more hands than a 1.0 codebase typically sees, and the lifecycle has to accommodate that plurality rather than assume a single author.

Reviewing extends code-review discipline to the new artefacts. A prompt change gets the same review treatment a code change would; an evaluation change is reviewed by the domain experts who can judge whether the rubric is right; a tool-specification change goes through the rigour of an API review, because it is an API; and an agent-specification change engages security and compliance, because it touches autonomy boundaries. Each artefact type is reviewed by the people equipped to judge it.

Testing is the evaluation stack of the evaluation chapter, applied as the lifecycle’s quality gate. Unit evaluations run on every commit, regression evaluations on every merge, online evaluations sample production traffic continuously, and red-team evaluations run on a schedule — with each layer catching what the layer above it missed.

Deploying looks different because the artefacts are different. Prompts and tool definitions deploy as configuration rather than as compiled binaries; models deploy as endpoint references rather than as artefacts baked into a build; canary releases on prompt changes are standard practice; and rollback is instant, because reverting a prompt is reverting a configuration value, not redeploying a binary.

Operating watches reasoning rather than code paths. Traces are aggregated into dashboards, online evaluations run continuously, cost and latency metrics are watched, drift alerts fire on deviation, and — closing the loop — each production incident drives a new evaluation into the regression set, so the system’s quality gate strengthens with every failure it survives.

42.4 Version Control for Prompts

Prompts under version control are not just text files in Git, although they often start there. The maturity progression:

- **Stage 1.** Prompts inlined in code. Worst.
- **Stage 2.** Prompts in repo as text files. Better. Reviewable, diffable.
- **Stage 3.** Prompts in a prompt-management system with versioning, environments, A/B testing, attached evals.
- **Stage 4.** Prompts treated as governed organisational assets: ownership, lifecycle, deprecation, lineage.

The progression matters because prompts that move freely accumulate quality regressions silently, and prompts that are locked too tightly never improve. Stage 3 is where most mature organisations live.

42.5 Testing in Software 3.0

The deepest shift in Software 3.0 is that “correct” is statistical, not binary. Tests are replaced by evaluations; assertions by rubrics. The implications:

- **Pass/fail thresholds are tunable.** A “passing” eval might require 95%, 99%, or 99.9% pass rate depending on stakes.
- **Stratified scoring matters.** Aggregate score can hide failures on critical sub-populations.
- **Judge models judge judges.** The evaluation infrastructure itself is judged: calibration, agreement with humans, drift over time.
- **Eval coverage is the new test coverage.** A well-covered system is one where every important behaviour is reflected in evals.

42.6 Observability in Software 3.0

Software 1.0 observability watches code paths; Software 3.0 observability watches reasoning paths. The data is different:

- **Trace volume.** Every model call produces structured input, structured output, token usage, cost, latency, intermediate state.
- **Token-level metrics.** Output token count, cost-per-token, completion vs prompt ratios.
- **Quality metrics.** Online judge-model scoring on sampled traffic.
- **Failure-mode metrics.** Refusal rate, hallucination rate (where measurable), tool-call error rate.
- **Behavioural metrics.** Sentiment, helpfulness, citation accuracy.

The observability stack for Software 3.0 has a specialised shape: token accounting, prompt-version correlation, trace replay, eval re-running on historical traces. Generic APM tooling does not cover this surface; AI-specific platforms (LangSmith, Helicone, Phoenix, others) are emerging.

42.7 Team Composition

Software 3.0 reshapes who is on the team.

42.7.1 The New Roles

Software 3.0 has created roles that did not exist in the Software 1.0 or 2.0 eras, because the artefacts it depends on — prompts, evaluations, alignment specifications — are artefacts that someone must own and maintain as deliberately as code.

- **Prompt engineer.** Writes and maintains prompts; often a role inhabited by engineers with domain skill rather than a separate title.
- **Eval engineer.** Designs and maintains the evaluation infrastructure. Increasingly a senior, specialised role.
- **AI product manager.** Reasons about probabilistic behaviour, evaluation rubrics, and the cost-quality frontier.
- **Trust and safety.** For consumer-facing systems, a permanent specialism.
- **Domain experts as authors.** Lawyers reviewing legal prompts, clinicians reviewing medical prompts, accountants reviewing audit prompts. They become co-authors of the artefacts.

42.7.2 The Shifted Skill Mix

The shift is not a replacement of software engineering; it is an extension. Software 3.0 teams still need backend engineers, infrastructure engineers, frontend engineers. They additionally need fluency in prompts, evaluations, agentic patterns, and model behaviour.

42.8 What Software 3.0 Does Not Replace

Important to be precise about what Software 3.0 is not:

- Not a replacement for deterministic software. Where the logic is well-understood and the stakes are high, deterministic code remains correct.
- Not a replacement for traditional ML. Where the task is narrow, structured, and high-volume (fraud scoring, recommendation, ranking), specialised models still win on cost and quality.
- Not a replacement for engineering discipline. The opposite: it demands more discipline, because more is fluid.

42.9 Recurring Failure Patterns

- **Treating prompts as configuration.** Prompts in YAML files with no review, no tests, no lifecycle.
- **Treating evals as optional.** Quality assumed, not measured. Production regressions invisible until users notice.
- **Engineer-only authorship.** Prompts written without domain experts produce surface plausibility and depth errors.
- **Tooling vacuum.** Generic infrastructure tools used for Software 3.0 artefacts. Operations remain manual and fragile.
- **No traceability.** Production behaviour cannot be linked back to prompt version, model version, tool version. Debugging is forensic.

42.10 Last Word

Software 3.0 is the regime where prompts, tools, agents, evaluations, and traces become first-class artefacts; where natural language is a programming primitive; where “correct” is statistical; and where teams compose engineers, domain experts, and product together at the artefact level. It coexists with Software 1.0 and 2.0; production systems are built across all three. The architectural implications are real and specific: new artefacts, new lifecycle, new version control, new testing, new observability, new team composition. The discipline of treating Software 3.0 with the rigour the prior eras earned — rather than with the casualness its accessibility invites — is what separates production systems from prototypes.

Chapter 43

5G and 6G as Enabling Infrastructure for Enterprise AI

“The mistake is to think of 5G as faster 4G. The architectural shift is that the radio is no longer the only thing in the network.”

—a recurring observation in telco-AI integration

Chapter Overview

5G and the emerging 6G are not merely successive generations of mobile data rates. They are programmable network fabrics with built-in compute, slicing, and edge presence. For enterprise AI, the architectural implications are concrete: inference closer to users (multi-access edge computing), guaranteed low-latency paths (network slicing), AI-aware quality-of-service, and a coming 6G generation where the network and the AI are designed together. This chapter develops the AI-side view: how to architect enterprise AI to use 5G/6G capabilities, and what new patterns become possible when network and AI converge.

Learning Objectives

- Distinguish the architecturally significant features of 5G from speed-and-latency improvements alone.
- Architect edge inference using multi-access edge computing (MEC).
- Apply network slicing to AI workloads with distinctive QoS needs.
- Anticipate the AI-native capabilities of 6G.
- Recognise the use cases where 5G/6G unlocks AI workloads that were architecturally impossible before.

43.1 What 5G Actually Changed

Most enterprise discussions of 5G focus on speed. The architectural shifts that matter for AI are different.

43.1.1 The Architectural Features

5G introduces architectural features that are relevant to AI workloads not because AI needs a faster phone connection but because 5G is a programmable, sliceable network that can be shaped to the requirements of specific workloads. The features that matter for AI are the ones that change what the network can guarantee.

- **Multi-access edge computing (MEC).** Compute resources sitting at the base station or aggregation site, milliseconds from the device. Inference can run there.
- **Network slicing.** Logically isolated networks with their own QoS, security, and management, sharing the same physical infrastructure. AI workloads can have a dedicated slice.
- **Ultra-reliable low-latency communication (URLLC).** A service class with single-millisecond latency targets and high reliability, distinct from enhanced mobile broadband (eMBB) and massive machine-type communication (mMTC).
- **Programmable network APIs.** Networks exposing capabilities (location, quality-of-experience, slice provisioning, identity) to applications.
- **Standalone 5G core.** A cloud-native, service-based architecture for the network core itself, replacing decades-old telecom architecture.

43.1.2 The 5G Service Classes

Figure 43.1 shows the three 5G service classes and which AI workloads naturally inhabit each.

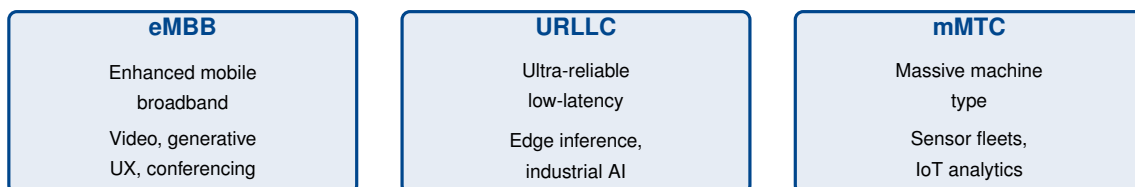


Figure 43.1: The three 5G service classes and the AI workloads that naturally inhabit each. The same physical network carries all three, distinguished by slice and QoS configuration.

43.2 Multi-Access Edge Computing for Inference

MEC is the architectural feature that most directly changes where AI inference can run. Figure 43.2 shows the layout.

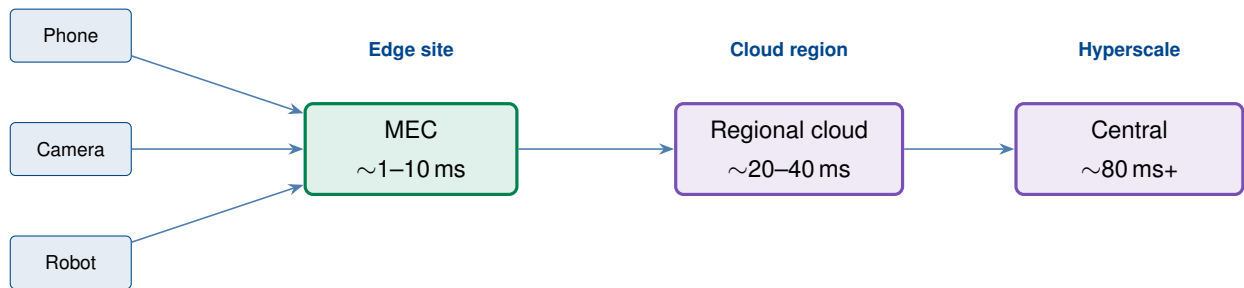


Figure 43.2: Inference latency tiers under 5G with MEC. Devices reach edge compute in milliseconds, regional cloud in tens of milliseconds, hyperscale clouds in tens to hundreds. Architectures place inference at the tier its latency demands.

43.2.1 What Runs at the Edge

The edge tier is not large enough to run frontier models in their full glory. What does run there:

- Quantised or distilled models (small models that fit on edge hardware).
- Specialised models for narrow tasks (vision-only, single-language, single-domain).
- Cascading first stages: an edge model decides whether the request needs to escalate to a larger cloud-hosted model.
- Caching layers: cached responses for common queries served from the edge.
- Pre- and post-processing: tokenisation, format normalisation, output redaction.

43.2.2 The Cascade Pattern over MEC

A natural pattern: small fast model at the edge handles 70–90% of queries; the remainder escalates to a regional cloud, the hardest go to hyperscale. The cascade respects both latency and cost.

43.2.3 Architectural Implications

The architectural implications of 6G for AI are speculative but directional, and an architect tracking the field should know what to watch for, because the features that 6G research is targeting would, if realised, change what AI systems can do at the network edge.

- Model deployment becomes a multi-tier problem: which model goes where.
- Observability spans edge, regional, central tiers.
- Data residency may demand edge inference for certain content classes.
- Model updates become a fleet-management problem: pushing updated models to many edge sites.

43.3 Network Slicing for AI Workloads

Network slicing lets an operator carve the network into logically isolated networks, each with its own QoS contract.

43.3.1 The AI Slice

An “AI slice” for an enterprise might guarantee:

- Bounded uplink latency for inference requests.
- Bounded downlink latency for streamed responses.
- Reserved bandwidth for video frames or sensor streams.
- Isolated security perimeter.
- Priority during congestion events.

43.3.2 When Slicing Matters

Network slicing is not relevant to every AI workload; it matters specifically for the workloads where the network guarantee is the binding constraint. Recognising when slicing matters is the discipline that keeps an enterprise from over-engineering its network for workloads that do not need it.

- Industrial: autonomous machines on a factory floor needing guaranteed control-loop latency.
- Healthcare: real-time clinical decision support that cannot tolerate congestion.
- Public safety: drone surveillance, emergency response systems.
- Finance: latency-sensitive transaction processing.
- Autonomous: vehicles, drones, robots whose safety depends on bounded latency.

43.3.3 The Operator Relationship

Slicing exposes a new relationship: the enterprise contracts with the operator for a slice, the slice has SLAs, the application running on the slice is architected to its guarantees. The operator becomes a substrate provider for AI, not just a connectivity provider.

43.4 Programmable Network APIs

The CAMARA initiative and similar efforts expose network capabilities through APIs that applications can call. The implications for AI:

43.4.1 Location and Mobility

A model can query the network for device location and mobility patterns; the application uses this to route, to predict, to personalise.

43.4.2 Quality-of-Experience

A model can query the network for current connection quality and adapt: switch to a smaller model, queue the request, drop video quality.

43.4.3 Identity

Network-anchored identity (SIM-based) supplements application identity, particularly for fraud and KYC use cases.

43.4.4 Slice Provisioning

Applications can request slices on demand, paying for guaranteed performance only when needed.

43.5 6G: What the Architecture Anticipates

6G is roughly a 2030 horizon. The architectural shape is becoming clear from research and standards work.

43.5.1 AI-Native by Design

6G is being designed with AI as a native capability, not as an over-the-top application. The implications:

- AI in the radio: ML for beamforming, channel estimation, spectrum management.
- AI in the core: predictive resource allocation, anomaly detection, self-optimisation.
- AI as a network service: inference offered as a network capability, with QoS guarantees.
- AI-AI traffic patterns: networks designed for agent-to-agent and agent-to-tool flows as first-class.

43.5.2 Sub-Millisecond Latency

6G targets sub-millisecond latency for the most demanding use cases. This puts inference latency budgets at the level where edge-resident specialist models become essential.

43.5.3 Joint Communication and Sensing

6G radios may double as sensors: the same waveform that carries data also senses the environment. Computer-vision-like workloads on the network itself.

43.5.4 Ambient AI

The vision of 6G includes “ambient AI”: models that exist as a property of the network environment, accessible to any device, no SDK required. Whether this materialises as anticipated is uncertain; the architectural placeholders are being put in.

43.6 Use Cases Unlocked

What does 5G/6G unlock that was architecturally impossible before?

43.6.1 Industrial AI at Cycle Time

Manufacturing robots, AGVs, conveyors, and quality systems run at sub-100-millisecond cycle times. AI that participates in those cycles needs latency that 4G could not provide and that cloud inference cannot match. MEC + URLLC enables this class of AI.

43.6.2 AR/VR with AI

Augmented and virtual reality have aggressive latency budgets (often ~ 20 ms motion-to-photon). AI-augmented AR (live translation, scene understanding, virtual collaboration) needs edge inference to meet these budgets.

43.6.3 Connected Vehicle and Robotics

Vehicle-to-everything (V2X) and robot-to-cloud workflows where the vehicle or robot defers some inference to the network. The network has to be both fast enough and reliable enough; 5G URLLC begins to deliver this.

43.6.4 Distributed Sensor AI

Massive sensor deployments (smart cities, agriculture, environmental monitoring) producing more data than backhaul to cloud could carry. Edge AI aggregates, classifies, and summarises locally; only summaries traverse to cloud.

43.6.5 High-Stakes Real-Time Inference

Telemedicine, public safety, autonomous systems where the difference between adequate and excellent is the difference between safe and unsafe. The network has to be in the safety loop, not adjacent to it.

43.7 Architectural Patterns

Four patterns recur where AI uses 5G and 6G capabilities, and each one is a way of turning a network feature into an architectural arrangement.

Edge-first inference with cloud fallback makes the edge the default and the cloud the exception. Inference runs on the edge tier by default; the cloud handles only the cases the edge model defers; the edge fleet is updated centrally, with updates propagating out to the edge sites; and the latency budget is enforced at the edge tier, where the work is supposed to happen. It is the MEC equivalent of the cascade pattern — cheap and fast first, escalate only when needed.

Slice-backed AI service runs an AI service on a dedicated network slice. The service's SLAs then include not only the usual quality and availability terms but latency, reliability, and security guarantees backed by the slice itself; the slice can be provisioned dynamically as demand rises and released as it falls; and the cost-and-availability trade-off is made explicit, because a reserved slice is paid for. This is the pattern for AI workloads whose quality-of-service genuinely cannot be left to best-effort networking.

Network-aware model routing connects the model router of Chapter 38 to the network's programmable APIs. The router queries the network for current conditions — congestion, the device's location, the remaining SLA budget — and adapts its routing decisions accordingly, choosing edge over cloud or a smaller model over a larger one when the network conditions warrant. It requires substantial integration between the router and the network APIs, and it is what lets routing respond to the network rather than ignore it.

Ambient AI substrate offers AI as a network service: applications request inference and the network provides it, with no model integration of the application's own. It is largely an architectural placeholder for 6G, emerging in 5G only in restricted forms, and the chapter treats it as a direction rather than a present capability — but designing with awareness of it keeps an architecture from foreclosing the option.

43.8 Recurring Failure Patterns

- **5G-as-faster-4G.** Architectures that consume 5G as if it were just faster bandwidth miss the architectural opportunities of MEC, slicing, and programmable APIs.
- **Edge fleet sprawl.** Many edge sites with poorly governed model deployment produce inconsistent behaviour and operational nightmares.
- **Slice over-provisioning.** Slices reserved at peak demand levels paid for at peak demand prices.
- **Cloud-anchored designs at the edge.** Architectures that pull edge through to cloud anyway, losing the latency benefit.
- **Vendor lock-in via APIs.** CAMARA-style APIs are early; operator-specific implementations risk binding to providers.

43.9 The African Perspective

5G and 6G have a distinctive shape in African deployments, across roll-out reality, the role of operators, and use-case priority.

The roll-out reality is one of heterogeneity. 5G is advancing in major African cities while rural roll-out lags; 4G remains dominant across most of the continent, and 3G is still substantial. An architecture deployed across an African market therefore cannot assume a single network generation — it must accommodate generation heterogeneity, performing well on 5G where it exists and degrading gracefully onto 4G where it does not. Edge inference is particularly attractive in this context, because the latency of backhaul to a distant Northern cloud is exactly the cost that edge compute removes, and that cost is larger in African deployments than in markets close to hyperscale regions.

The operator relationship is more central than in markets where AI platforms are dominated by hyperscalers. Pan-African operators — MTN, Vodacom, Airtel, Orange, Safaricom, and others — are positioning themselves as AI platform providers, offering MEC at base stations as an enterprise service and integrating AI with the mobile-money systems they already run, an integration that has no close equivalent in Northern markets. For an African enterprise, the operator is often a more natural AI-infrastructure partner than a distant cloud provider, and the cross-border reach of these operators makes them a substantial platform play across the continent.

Use-case priority follows the continent's economic geography. Mining needs industrial AI at deep sites where cloud connectivity is impossible and only edge compute can serve. Agriculture needs distributed sensor AI in low-backhaul regions where summarising at the edge is the only way the

data can be used at all. Healthcare needs telemedicine that reaches remote facilities. Logistics needs vehicle and route AI across pan-African operations that cross borders by design. In each case the edge-first patterns of this chapter are not a refinement but the architecture that makes the use case possible, and the cross-border dimension recurs often enough to be a standing design consideration rather than an edge case.

43.10 What This Chapter Established

5G changed the network from a transport into a programmable substrate with edge compute, slicing, and APIs. 6G is being designed with AI as native. For enterprise AI, the architectural implications are concrete: edge inference becomes economically and latency-feasible; slicing makes deterministic QoS available for AI workloads; programmable APIs let applications query and adapt to network conditions; new use-case classes (industrial, AR/VR, autonomous, distributed sensor) become architecturally possible. The discipline is to treat 5G/6G as substrate, not just bandwidth, and to design AI architectures that use the substrate's distinctive capabilities. African contexts add particular weight to edge-first patterns and operator-led AI platforms.

Chapter 44

Compound AI Systems

“The state of the art is no longer a model. It is a system that uses several.”

—a recurring observation in production AI

Chapter Overview

A compound AI system is one whose behaviour emerges from the orchestration of multiple components — models of different sizes and modalities, retrieval systems, tools, verifiers, classifiers — rather than from a single forward pass. Compound systems beat monolithic ones on quality, cost, and controllability, but at the price of complexity that demands architectural discipline.

Learning Objectives

- Recognise what makes a system compound and why monolithic systems are losing ground.
- Identify the component classes that recur across compound systems.
- Architect orchestration with clear contracts between components.
- Manage the quality-cost-latency tradeoff space.
- Apply compound thinking to African operational realities.

44.1 What Compound Means

A compound AI system is not the same as a multi-agent system. Multi-agent systems are one variant. A compound system is more general: any AI system whose outputs depend on the coordinated execution of multiple components, where each component might be a model, a tool, a retrieval system, a verifier, or a deterministic routine.

44.1.1 The Five Component Classes

The following elements define this concern and distinguish it from the adjacent ones.

- **Generative models.** The LLMs and multimodal models that produce content.
- **Discriminative models.** Classifiers, scorers, routers, verifiers, judges.
- **Retrieval systems.** Vector stores, sparse indices, knowledge graphs, structured databases.

- **Tools.** APIs, calculators, code executors, search, services.
- **Deterministic routines.** Validators, transformers, parsers, formatters.

The mix matters. A system with five generative models is more expensive and less predictable than one with one generative model, two discriminative models, a retrieval system, and a verifier — and the latter will often deliver better results.

44.2 The Architecture

Figure 44.1 shows the canonical layout.

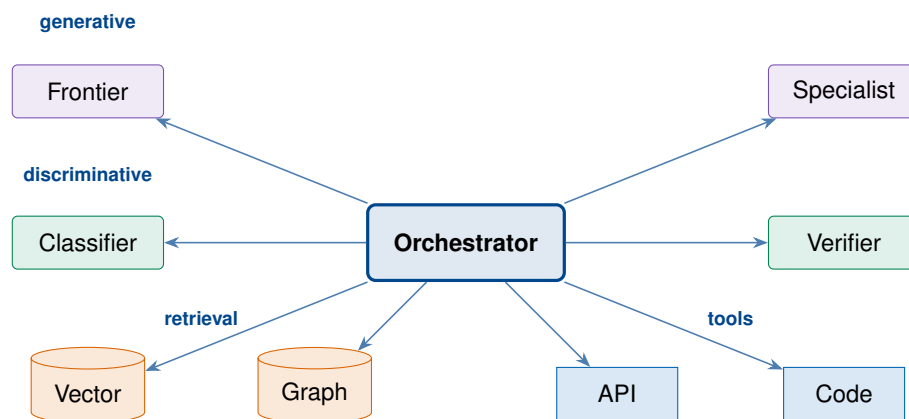


Figure 44.1: A compound AI system: an orchestrator coordinates generative models, discriminative models, retrieval systems, and tools. Each component does what it is best at. The orchestrator owns the contract between them.

44.2.1 The Orchestrator’s Job

The orchestrator chooses which components to invoke, in what order, with what inputs, and how to aggregate outputs. The orchestrator is itself sometimes a model, sometimes deterministic code, sometimes a hybrid. The choice matters: a model-as-orchestrator can adapt; deterministic code is predictable.

44.2.2 Specialisation Beats Generalisation

A compound system wins by specialisation. Use a frontier model where reasoning matters; a small classifier where structured decisions matter; a retrieval system where facts matter; a tool where ground truth matters. The cost of running a frontier model on every step is the cost the architecture eliminates.

44.3 Design Patterns Within Compound Systems

Four recurring sub-patterns:

44.3.1 Generator-Verifier

A generator produces; a discriminator verifies. The verifier rejects bad outputs; the generator retries. The pattern lets a fast generator be paired with a careful verifier.

44.3.2 Retrieve-Reason-Generate

Retrieval surfaces context; a reasoning model synthesises; a generator writes. Separating retrieval from reasoning from generation produces better-cited, less-hallucinated outputs.

44.3.3 Plan-Then-Specialise

A planner decides what to do; specialised executors do each step. The pattern from Chapter 39, applied at the architectural level.

44.3.4 Ensemble-and-Aggregate

Multiple generators produce; an aggregator selects or merges. Highest cost; reserved for high-stakes outputs.

44.4 The Quality-Cost-Latency Frontier

Compound systems live on a three-dimensional Pareto frontier.

44.4.1 Quality Wins

Compound systems consistently outperform single-model approaches on benchmarks, especially when the workload is heterogeneous. The reason is specialisation: each component does what it is good at.

44.4.2 Cost Wins

A well-designed compound system is cheaper than a single frontier model handling everything. The cascade pattern from model routing applies here at the system level.

44.4.3 Latency Loses

Compound systems are slower than single-model approaches at the tail. Sequential components add their latencies; orchestrator overhead is real. Mitigations: parallel execution where dependencies allow, streaming responses, edge components for hot paths.

44.5 Contracts Between Components

The discipline of compound systems is the discipline of contracts. Every component has a typed input, a typed output, declared side effects, declared costs, declared latency, declared failure modes. Without these contracts, debugging is forensic.

- Input schemas validated.
- Output schemas validated.
- Errors typed and translated to a common taxonomy.

- Observability hooks consistent across components.
- Versioning explicit per component.

44.6 The African Perspective

Compound AI systems have a distinctive shape in African deployments, and on balance the distinctiveness favours the pattern.

Cost pressure pushes hard toward compound systems. The tight unit economics of many African deployments make a cascade of cheap-then-frontier components decisively cheaper than running a frontier model on everything, and the saving is large enough that many African production systems adopt compound architectures earlier in their maturity than Northern equivalents do — not as a sophistication they grow into, but as an economic necessity they start with.

The compound structure also makes sovereignty tractable component by component. Because a compound system is already several components rather than one, those components can live in different jurisdictions: a frontier model in a Northern region, retrieval in a Cape Town or Johannesburg region, a verifier running on a self-hosted open-weight model on-premise. A monolithic system forces a single placement decision for the whole system; the compound structure lets each component be placed where its data-residency and latency constraints require, which is exactly the flexibility a pan-African deployment under several data-protection regimes needs.

Multilingual contexts favour compound systems for a similar reason. A compound architecture naturally accommodates language-specialist components — an Arabic specialist, a French specialist, a Swahili specialist — coordinated by a routing model that sends each request to the component equipped for its language. This is a cleaner answer to multilingual quality than asking one general model to handle every language equally well. And connectivity tiering falls out of the same structure: an edge component, a regional component, and a frontier component compose into a system that degrades gracefully when connectivity to the frontier is impaired, serving what the reachable tiers can serve rather than failing outright. The compound pattern is, in these deployments, not merely a quality-and-cost optimisation but the structure that lets a system meet cost, sovereignty, language, and connectivity constraints at once.

44.7 Recurring Failure Patterns

- **Orchestrator monolithism.** The orchestrator absorbs logic that belongs in components, and becomes the failure surface.
- **Component-version drift.** Components updated independently produce inconsistent behaviour over time.
- **Contract erosion.** Components evolve their interfaces without coordination; integration breaks silently.
- **Latency stacking.** Sequential components produce tail latency the system cannot tolerate.

- **Observability gaps.** Failures span components; logs do not.

44.8 The Working Position

Compound AI systems coordinate multiple specialised components — generative, discriminative, retrieval, tools, deterministic routines — under an orchestrator. They beat monolithic single-model approaches on quality and cost, at the price of complexity and tail latency. The discipline is component contracts, version management, and observability that spans the system. African contexts add particular weight to cascade economics, sovereignty-aware placement, multi-lingual specialisation, and connectivity-tier composition.

Chapter 45

Constitutional AI and Principle-Driven Behaviour

“A model trained on a constitution behaves consistently because its training was consistent. The discipline is upstream.”

—a recurring observation in alignment-aware engineering

Chapter Overview

Constitutional AI is an approach to model behaviour in which a written set of principles — a constitution — shapes how the model self-critiques and revises during training. The pattern matters beyond any one provider’s specific technique: any production system that defines a written behavioural policy and uses it to evaluate, train, or filter outputs is applying constitutional thinking. This chapter develops the architectural perspective. It distinguishes a constitution from the prompt, policy, and rule it is often confused with; locates the layers of the system where constitutional principles apply; sets out how to write a constitution and when to apply it at training time versus runtime; shows how the constitution doubles as evaluation infrastructure; and traces the maturity path organisations follow as they adopt it.

Learning Objectives

- Distinguish a constitution from a prompt, a policy, and a rule.
- Locate the layers of the system where constitutional principles apply.
- Write a constitution as principles rather than rules, and know what belongs in one.
- Choose between training-time and runtime application, and the hybrid most systems use.
- Build the evaluation infrastructure that constitutional approaches require.
- Adapt constitutional approaches to African regulatory and cultural contexts.

45.1 What a Constitution Is

A constitution is a written set of behavioural principles, expressed in natural language, that govern how a model decides what to do. The definition is best sharpened by contrast with three artefacts a constitution is routinely confused with. A constitution is not a *prompt*: a prompt instructs the model

on a specific task, while a constitution shapes behaviour across all tasks. It is not a *policy*: a policy is an external rule applied to the model's outputs by the surrounding system, while a constitution is internalised through training and carried by the model itself. And it is not a *rule*: a rule is a hard constraint with a definite threshold, while constitutional principles are weighted, traded off against one another, and contextualised by the model's judgement. The distinction that matters most is the last: a constitution does not tell the model what to do, it tells the model what to value, and leaves the model to apply those values to a situation the constitution never anticipated.

In practice a system carries two constitutions, layered. The *provider constitution* is the one a foundation-model provider publishes, or implies through behaviour, for its flagship models; it is the floor of what the model will and will not do, and every enterprise application inherits that floor whether or not it ever reads it. On top of it sits the *enterprise constitution* — the organisation's own behavioural specification for its AI systems. The enterprise constitution is read at every layer of the system: it guides training-data curation, it shapes fine-tuning, it is restated in system prompts, it is encoded in evaluation rubrics, and it informs escalation policy. The aim of writing it down once and applying it everywhere is consistency — the same behavioural intent expressed at every layer rather than re-improvised differently at each.

45.2 Where Constitutions Live

A constitution is not enforced at a single point. It applies across the stack, and Figure 45.1 shows the layers — each of which is an opportunity to express the same behavioural intent, and each of which has a different owner and a different enforcement character.

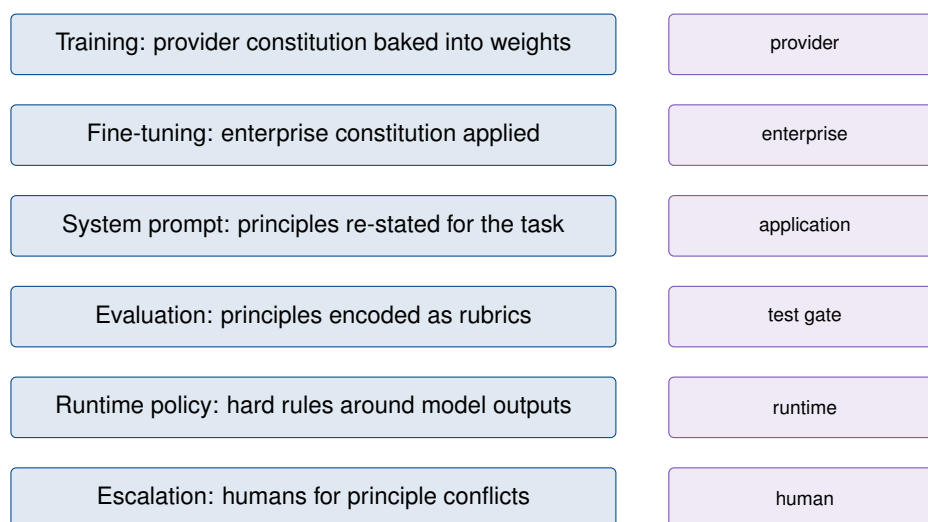


Figure 45.1: Constitutional principles apply across the stack: baked into the model through training, layered through fine-tuning, restated in system prompts, encoded as evaluation rubrics, enforced by runtime policy, and resolved by humans when principles conflict.

At the deepest layer, the provider's constitution is baked into the model's weights through training, which is why the model behaves consistently without being told to on each request — the consistency was established upstream, as the epigraph observes. Fine-tuning is where an enterprise applies its

own constitution to the model directly. The system prompt restates the principles in a form scoped to the task at hand. Evaluation encodes the principles as rubrics, turning them into a test gate. Runtime policy enforces the small set of principles that genuinely must be hard rules. And escalation routes the cases where principles conflict in ways the model should not resolve alone to a human. The architectural point is that these layers are not alternatives but a defence in depth: the same behavioural intent, expressed six ways, so that a principle missed at one layer is still carried by the others.

45.3 Writing a Constitution

A constitution is not a list of rules; it is a coherent set of principles that produce judgement when applied. Writing one well means attending to the form of the principles, to what belongs in the constitution, and — equally — to what does not.

The *form* of a principle is the first discipline. Each principle should state what the model values rather than what it must do, because a value generalises to situations the constitution did not foresee while an instruction does not. Principles should be written to apply across contexts, not scoped to one task. They should acknowledge that principles can be in tension — helpfulness and harm-avoidance genuinely conflict in some requests — and give priority hints for those tensions rather than pretending they do not exist. And they should be reviewable: a human reading the constitution should be able to understand the model’s intended behaviour from it. A constitution written this way is a document a domain expert can read and reason about, not a configuration file.

What *belongs* in a constitution is the set of cross-cutting behavioural values: helpfulness and its limits, honesty norms, harm avoidance, respect for user autonomy, the domain-specific values of the field the system serves — clinical, legal, financial, educational — and the brand and tone considerations that make the system recognisably the organisation’s own. What does *not* belong is anything that is properly the concern of another layer: specific output formats belong in prompts, specific facts belong in the knowledge layer, and specific rules with exact thresholds belong in runtime policy. The discipline of keeping these out of the constitution is what keeps the constitution a document of values rather than a sprawling specification that tries to be prompt, knowledge base, and policy at once — and fails to be a clear constitution.

45.4 Training-Time, Runtime, and the Hybrid

Constitutional approaches can apply at training time, at runtime, or both, and the choice is a real architectural decision.

The *training-time* approach is the original constitutional AI technique: the model is trained to self-critique its outputs against the principles and then revise them, so that the behaviour is internalised into the weights. Its great advantage is that the behaviour is consistent and the runtime cost is minimal — the constitution is not re-applied on every request because it is already in the model. Its cost is that it requires model-training infrastructure that many organisations do not have.

The *runtime* approach applies the principles without retraining: as system prompts, as the rubrics a judge model scores against, or as filters on the model’s outputs. It requires no training infrastructure

and works with any model, including one reached only through an API. Its costs are the mirror of the training-time approach's advantages: it adds runtime cost on every request, and the behaviour is less consistent across contexts because it is applied externally rather than internalised.

Most production systems are therefore *hybrid*. They rely on the provider's training-time constitution as the floor — inheriting a consistent, internalised baseline at no cost to themselves — and add runtime constitutional checks for the enterprise-specific principles the provider's constitution does not cover. This is the pragmatic default: it takes the consistency of training where training has already been done by the provider, and accepts the runtime cost of the external approach only for the increment the enterprise genuinely needs to add.

45.5 The Constitution as Evaluation Infrastructure

The single discipline that most distinguishes a constitution that works from one that decorates a wiki is this: the constitution doubles as an evaluation rubric. Each principle becomes one or more evaluation criteria. The output is graded on its alignment with each principle; conflicts between principles are surfaced explicitly in evaluation rather than left invisible; drift on any single principle is detected over time; and the regression suite includes cases that violate each principle, so the system is continuously tested against the behaviours the constitution forbids.

This is what makes a constitutional approach operational rather than aspirational. A constitution that is written and then never measured against is a statement of intent with no mechanism behind it — the model may or may not follow it, and no one can tell which. A constitution whose every principle is also an evaluation criterion is a constitution with a feedback loop: a principle violation shows up as a failing evaluation, a drift shows up as a falling score, and the constitution becomes a thing the system is held to rather than a thing the system was once told. The discipline of writing each principle *also* as an evaluation criterion is, in a single sentence, the discipline that turns constitutional AI from a philosophy into an architecture.

45.6 Organisational Adoption

Organisations do not adopt constitutional thinking all at once; they progress through it, and the stages are recognisable. At the first stage an organisation relies entirely on the provider's constitution, adding nothing of its own. At the second it adds enterprise system prompts for tone and brand — the first expression of an enterprise voice, though not yet a constitution. At the third it writes a genuine domain-specific principle set and applies it as an evaluation rubric, which is the point at which the organisation has a constitution in the operational sense of the previous section. At the fourth it fine-tunes models on the principle set and uses the set as a runtime judge, internalising the constitution rather than only testing against it. At the fifth, governance closes around the constitution: changes to the principles go through review, and principle violations escalate to a constitutional committee with the authority to act on them.

The progression is worth naming because it tells an organisation where it is and what the next step is. The common mistake is to skip from the second stage — system prompts for tone — straight

to ambitions of fine-tuning, without ever doing the third-stage work of writing the principles as evaluation criteria. Without that step there is nothing to fine-tune toward and nothing to govern, and the later stages have no foundation. The maturity path is also a dependency order: each stage supplies what the next one needs.

45.7 Recurring Failure Patterns

The failure modes of constitutional approaches are recognisable, and each maps to a discipline this chapter has set out:

- **Aspirational constitution** — principles are written that no one evaluates against, so the constitution is decorative and the model’s actual behaviour is untested against it.
- **Conflict denial** — principles are written as though they never conflict, so the model is left to resolve genuine tensions in real time with no priority guidance.
- **Provider drift** — the enterprise system silently inherits changes to the provider’s constitution between model releases, with no re-evaluation to catch the behavioural shift.
- **Cultural monoculture** — a constitution written by one team for global deployment fails in the markets whose norms that team did not consider.
- **Evaluation gap** — principles are not encoded as evaluation criteria, so drift on any of them goes undetected until it surfaces as an incident.

The aspirational constitution and the evaluation gap are two faces of the same failure — the constitution-as-evaluation-infrastructure discipline not applied — and they are the most common. Conflict denial is the failure to give priority hints. Provider drift is the failure to re-evaluate against a moving floor. And cultural monoculture is the failure the next section addresses directly. The remedy for the whole list is the chapter’s recurring instruction: write the constitution as principles, encode every principle as an evaluation criterion, and govern it as a living document.

45.8 The African Perspective

Constitutional approaches carry particular weight in African deployments, because a constitution is, at bottom, an encoding of norms — and norms are exactly what varies across the contexts an African deployment serves.

The first consideration is cultural and linguistic plurality. A constitution must accommodate substantial variation in what its principles mean in practice: a tone that reads as respectful in one cultural context reads as curt in another, and a principle such as “respect for user autonomy” is interpreted differently across the cultures of a multi-country deployment. Constitutional principles for African deployments must be written, reviewed, and applied with that plurality consciously in mind, rather than written once against a single implicit culture — which is precisely the cultural-monoculture failure mode.

The second is regulatory diversity. The data-protection and sector-specific regimes across African jurisdictions — South Africa’s POPIA, Nigeria’s NDPA, Kenya’s DPA, and the others — impose

concrete constraints that the enterprise constitution must encode, and a constitution serving several jurisdictions encodes the union of their requirements rather than any single one.

The third is translation and authority. A constitution written in English works in some of the contexts an African system serves and not others, and multilingual constitutions are emerging in response. The translation effort is genuinely non-trivial, because constitutional principles are interpretive rather than literal: translating a principle is translating a value, and a literal translation that loses the value's intended force has not done the job. The fourth, and related, consideration is adaptation to local norms: a constitution shipped from a Northern model provider encodes Northern norms by default, and enterprise constitutions for African deployments routinely add adaptations for local social norms, religious sensibilities, and political contexts that the provider's constitution did not contemplate. The constitutional approach does not change for African deployments; the content of the constitution must reflect the plurality of the markets actually served rather than the assumptions of the team that first drafted it.

45.9 A Note Before Moving On

Constitutional AI is the discipline of expressing behavioural intent as written principles and applying those principles consistently across the stack — through training, fine-tuning, prompting, evaluation, runtime policy, and escalation. A constitution is distinguished from a prompt, a policy, and a rule by its nature: it states values rather than instructions, and it leaves the model to apply those values with judgement. Its operational test is whether every principle is also an evaluation criterion, because a constitution that is measured is a constitution the system is held to, and one that is not is decoration. Organisations grow into the approach through a recognisable maturity path, from inherited provider constitutions to a bespoke enterprise constitution governed through review.

The epigraph states the chapter's deepest point: a model trained on a constitution behaves consistently because its training was consistent, and the discipline is upstream. Constitutional thinking is the recognition that behaviour is not best controlled at the point of each output but shaped at the point where the behaviour is formed — in training, in fine-tuning, in the rubrics and the system prompts that express the same values everywhere. The architectural payoff is consistency: the same behavioural intent, expressed at every layer of the system, so that the model an enterprise deploys behaves as the enterprise intended not because each response was caught and corrected, but because the intent was built in.

Chapter 46

Federated Learning and Privacy-Preserving Training

“Move the model to the data, not the data to the model. The architecture is the privacy guarantee.”

—a recurring observation in regulated-industry AI

Chapter Overview

Federated learning trains models on data that never leaves the device or institution holding it. Combined with privacy-preserving techniques — differential privacy, secure aggregation, trusted execution environments — it offers a path to training capable models on sensitive data without centralising the data itself. The pattern is essential where data residency, regulatory constraint, or competitive sensitivity makes data centralisation impossible.

Learning Objectives

- Recognise when federated learning is the appropriate architecture.
- Architect federated training rounds with secure aggregation.
- Apply differential privacy to bound information leakage.
- Integrate with trusted execution environments where appropriate.
- Adapt federated approaches to African regulatory and infrastructure realities.

46.1 The Problem Federated Learning Solves

Centralised training assumes the data can be brought together. For sensitive data — medical records, financial transactions, government records, personal communications, competitive corporate data — this assumption fails. Either the data cannot legally cross borders, or the institutions holding it will not cooperate to centralise, or the risk of central-data-breach is unacceptable.

Federated learning inverts the relationship: the model travels to the data, trains locally, and only summary updates (gradients, weights) travel back. The raw data stays where it lives.

46.2 The Architecture

Figure 46.1 shows a federated learning round.

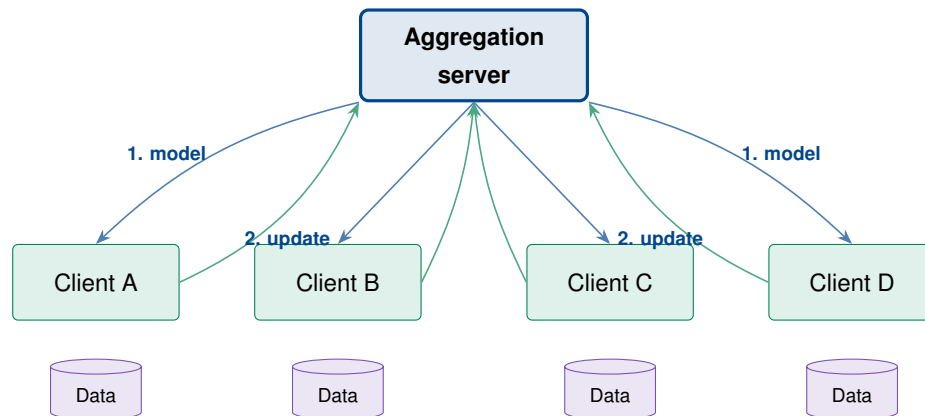


Figure 46.1: A federated learning round: server distributes the current model to clients; clients train on local data; clients send back model updates (gradients or weights); server aggregates. Raw data never leaves clients.

46.2.1 The Round Sequence

The following elements define this concern and distinguish it from the adjacent ones.

1. Server selects a cohort of clients for the round.
2. Server distributes current model parameters to selected clients.
3. Each client trains locally for some number of steps.
4. Each client sends model updates back to the server.
5. Server aggregates updates (typically weighted average) into new model parameters.
6. Repeat until convergence.

46.3 Variants

46.3.1 Cross-Device vs Cross-Silo

The following elements define this concern and distinguish it from the adjacent ones.

- **Cross-device.** Many small clients (phones, IoT devices). Each contributes a small share. Selection is stochastic. Examples: Google keyboard, Apple personalisation.
- **Cross-silo.** Few large clients (hospitals, banks). Each contributes substantially. Selection is structured. Examples: medical-imaging consortia, banking fraud collaboration.

46.3.2 Horizontal vs Vertical

The following elements define this concern and distinguish it from the adjacent ones.

- **Horizontal.** Clients have the same features for different users. Most common.

- **Vertical.** Clients have different features for the same users. Harder; requires careful identity matching without revealing identity.

46.4 Privacy Guarantees

Federated learning alone is not a privacy guarantee. Updates can leak information about the underlying data. Three techniques compose to provide stronger guarantees:

46.4.1 Differential Privacy

Noise is added to updates such that any single record's contribution is bounded. The noise level (epsilon) is a privacy budget; smaller epsilon means stronger privacy but worse model quality. The tradeoff is explicit and tunable.

46.4.2 Secure Aggregation

Updates are encrypted such that the server cannot see any single client's update, only the aggregate. Built with multi-party computation primitives. Prevents the server from learning what any individual client contributed.

46.4.3 Trusted Execution Environments

Hardware enclaves (Intel SGX, AMD SEV, Apple Secure Enclave) provide attested compute environments where the server can verify code is running unmodified. Combined with federated learning, they raise the trust floor when the server itself cannot be fully trusted.

46.5 Operating Considerations

Federated learning works in principle once the round sequence and the privacy techniques are in place; making it work in practice means confronting four operational realities that a centralised training run never has to.

The first is *heterogeneous clients*. The clients in a federation differ in compute, bandwidth, availability, and data distribution, and the slowest of them — the stragglers — pace the whole round, because the server cannot aggregate until enough updates have arrived. The mitigations trade fairness against speed: asynchronous updates that do not wait for every client, client selection biased toward fast clients, and the dropping of slow clients — the last done with care, because a federation that systematically drops slow clients also systematically excludes whatever population those clients represent.

The second is *non-IID data*. Client data is typically not independent and identically distributed: each hospital sees a different patient mix, each bank a different customer set. Standard federated averaging can underperform badly when the data is this heterogeneous, and algorithms designed for the non-IID case — FedProx, FedYogi, personalised federation — exist precisely to address it. Applying plain federated averaging to highly heterogeneous data and accepting the quality loss is a recognisable mistake.

The third is *adversarial clients*. A malicious client can poison the model by sending crafted updates, and a federation has no direct view of any client's data to check the update against. The defences are

outlier detection on the updates, a bounded contribution per client so no single client can dominate, and secure aggregation, which limits how much a poisoned update can achieve by hiding individual updates inside the aggregate.

The fourth is *audit and compliance*. The whole appeal of federated learning to a regulated industry is that data did not leave the client environment — but a regulator wants evidence of that, not an assurance. The architecture provides the guarantee; the evidence requires client-side logging, attestation of the code's identity so the regulator knows what ran, and an audit trail of the aggregation decisions. A federated architecture that cannot produce this evidence has the privacy property but cannot prove it, which for a regulated deployment is not enough.

46.6 The African Perspective

Federated learning has distinctive value in African deployments, and the value is structural: the pattern's defining property — learning from data without moving it — is the answer to several constraints that African deployments face acutely.

The clearest case is cross-border learning. Pan-African consortia — multi-country health systems, regional banking groups — need to learn from data spread across countries, and the data-residency laws of those countries frequently forbid the data from moving across the borders between them. Federated learning provides the architecture that resolves the contradiction: the model learns at continental scale while the data of each country stays within its borders, and secure aggregation provides the privacy guarantee that makes the participating institutions willing to take part. The data-residency law of each jurisdiction is respected not by exception but by the architecture itself.

A specific and important instance is donor-programme coordination. Vertical-programme data — on HIV, tuberculosis, malaria, maternal and child health — exists across many African countries and often cannot be centralised, both for sovereignty reasons and because no single party has the standing to hold it. Federated approaches let donor-funded programmes learn at continental scale while each country retains sovereignty over its own data, which is frequently the only arrangement under which the learning can happen at all.

Federated learning also has a natural mobile-first form in African markets. Cross-device federation on smartphones is already mainstream in Northern markets for keyboard personalisation and photo organisation, and the same pattern has obvious application to African mobile-money fraud detection and mobile-health applications, where the relevant data lives on millions of phones and centralising it would be neither lawful nor wise. The constraint to design around is infrastructure: variable client-side connectivity demands asynchronous, fault-tolerant federation, and a federated round in an African deployment often spans days rather than hours, with the cohort of participating clients shifting over the course of the round. The pattern does not change; its rounds simply run on a longer, more forgiving clock.

46.7 Recurring Failure Patterns

- **Federation theatre.** Architecture called federated but data centralised in another form (telemetry, logs, query traces).
- **Privacy budget exhaustion.** Differential-privacy budget consumed without tracking; subsequent rounds offer no protection.
- **Non-IID denial.** Standard algorithms applied where data is highly heterogeneous; quality suffers.
- **Audit gap.** No evidence that data stayed at client side; regulator cannot verify the architecture's claims.
- **Client-side opacity.** Server has no way to know whether clients are training honestly.

46.8 The Takeaway

Federated learning trains models on data that never leaves the client, with summary updates traveling to a central aggregator. Combined with differential privacy, secure aggregation, and trusted execution environments, it provides architecturally enforced privacy guarantees suitable for regulated data, cross-border collaboration, and competitive sensitivity. African contexts add particular weight: cross-border learning without data crossing borders, donor-programme coordination, mobile-first federation, and infrastructure-aware asynchronous patterns.

Chapter 47

Distributed Inference and KV Cache Management

“At scale, inference is a memory-management problem disguised as a compute problem.”

—a recurring observation in production inference engineering

Chapter Overview

Production LLM inference is dominated less by raw compute than by memory and cache management. The KV cache — the running state of attention computation — grows with sequence length and dominates GPU memory in long-context workloads. Distributed inference patterns (tensor parallelism, pipeline parallelism, expert parallelism, prefix caching, continuous batching, paged attention) make production serving viable. This chapter develops the architectural perspective rather than the kernel-level detail: what an inference platform needs, how requests flow through it, and which optimisations matter.

Learning Objectives

- Recognise the dominant cost drivers in production inference.
- Architect inference platforms with batching and caching as first-class.
- Understand parallelism dimensions and their tradeoffs.
- Manage KV cache through paging, prefix sharing, and eviction.
- Adapt distributed inference to African cost and latency constraints.

47.1 The Inference Cost Structure

A naive view of LLM inference: tokens generated $\times$ per-token cost. A more accurate view distinguishes:

- **Prefill.** Processing the input prompt. Compute-bound. Highly parallel.
- **Decode.** Generating output tokens one at a time. Memory-bandwidth-bound. Sequential.

- **KV cache.** Per-request growing memory holding attention state. Often dominant memory consumer.

The economic implication: decode latency depends on memory bandwidth, not raw FLOPS. A request with a 10K-token prompt and a 100-token output is mostly prefill cost; a 1K-prompt with a 10K-output is mostly decode cost. Capacity planning has to track both.

47.2 The Inference Platform

Figure 47.1 shows the architectural shape of a production inference platform.

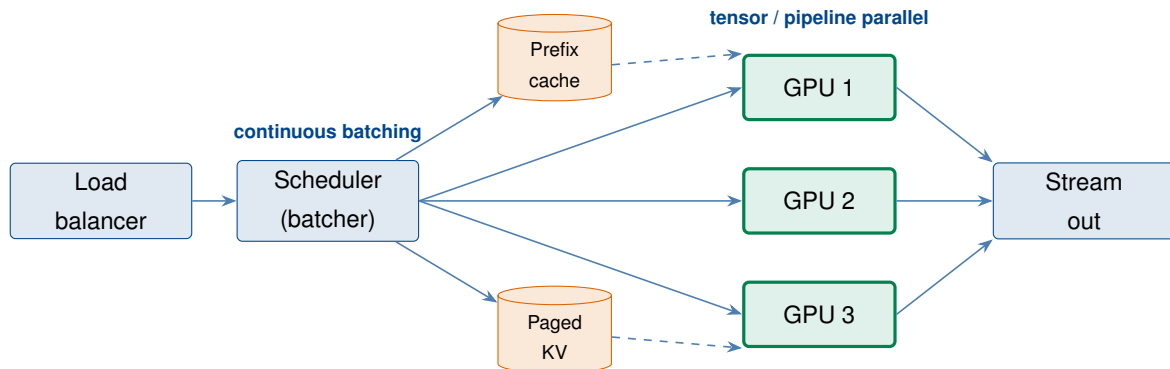


Figure 47.1: Production inference platform: load balancer feeds a scheduler that performs continuous batching; prefix and paged KV caches are shared across batches; GPUs run model shards in tensor or pipeline parallelism; outputs stream back as tokens generate.

47.3 Continuous Batching

Naive batching groups requests of similar length and runs them through the model together. The problem: long-running requests block short ones; the GPU waits.

Continuous batching dynamically composes batches at each step. When a request finishes generating its last token, its slot in the batch is filled with a new request entering prefill. Throughput rises 3–5x over naive batching in production workloads.

The scheduler is the heart of the platform: it decides which requests enter the batch this step, balances prefill and decode work, and respects per-request latency budgets.

47.4 Parallelism Dimensions

A model larger than a single GPU’s memory must be sharded across GPUs. Three dimensions:

47.4.1 Tensor Parallelism

The model’s weights are split across GPUs by tensor dimension. Each GPU computes its slice of every layer. Communication happens at every layer boundary. Latency-friendly; bandwidth-hungry; GPUs must be on the same NVLink island.

47.4.2 Pipeline Parallelism

The model's layers are split across GPUs. Each GPU computes some layers; activations flow through the pipeline. Communication is smaller but more sequential. Tolerates slower interconnect; introduces pipeline bubbles.

47.4.3 Expert Parallelism

For mixture-of-experts models, different experts live on different GPUs. Each token activates a subset of experts. Routing is dynamic; balancing experts across GPUs is non-trivial.

Production systems combine: tensor parallelism within a node, pipeline parallelism across nodes, expert parallelism for MoE models.

47.5 KV Cache as Architectural Concern

The KV cache holds attention state for each token in each request. Its size grows linearly with sequence length and is proportional to model size. For long-context workloads, KV cache dominates GPU memory.

47.5.1 Paged Attention

Inspired by virtual memory: the KV cache is divided into fixed-size pages, allocated on demand. Pages are non-contiguous. Eliminates the fragmentation that ruined throughput in naive implementations. Standard in modern inference engines (vLLM, TensorRT-LLM, SGLang).

47.5.2 Prefix Caching

When many requests share a common prefix (system prompt, few-shot examples, retrieved context), the prefix's KV cache can be computed once and reused. Substantial savings on workloads with common prefixes; minor savings where each request is unique.

47.5.3 Cache Eviction

When KV memory pressure rises, some requests' caches must be evicted. Choices:

- Evict completed requests immediately.
- Evict oldest pending requests (FIFO).
- Evict longest pending requests (penalise long stragglers).
- Recompute evicted state on demand (CPU-offload, swap to host memory).

47.6 Speculative Decoding

A small draft model proposes tokens; the large target model verifies. Where the draft is right, multiple tokens are accepted in one large-model pass. Latency reduction is substantial (1.5–3x) on workloads where the draft model is well-aligned with the target.

The architectural implication: production platforms now serve pairs of models (draft + target) rather

than single models. Capacity planning includes draft model serving.

47.7 Quantisation and Numerical Format

Native FP16 or BF16 inference is the baseline. INT8, FP8, INT4, NF4, and mixed-precision approaches reduce memory and increase throughput, at varying quality cost. Production platforms typically offer multiple precisions; the routing decision is workload-dependent.

47.8 The African Perspective

Distributed inference has a distinctive shape in African deployments, and the shape follows from cost, latency, and sovereignty pressing harder than they do in the markets where the inference engines were designed.

The dominant effect is a preference for smaller models. The tight unit economics of many African deployments push toward smaller models, more aggressive quantisation, and longer cache reuse, and the hardware arithmetic makes the preference concrete: a seven-billion-parameter model quantised to INT4 fits on a single consumer-grade GPU, while a seventy-billion-parameter model at FP16 needs a multi-GPU node and the high-bandwidth interconnect between its GPUs. The choice between those two is not only a quality choice; it is the difference between an inference deployment that runs on accessible hardware and one that does not.

Latency makes regional inference valuable. Inference served from a Cape Town or Johannesburg cloud region cuts the latency to African users substantially against a Northern endpoint, and the architectural payoff is most visible precisely on the workloads where the round trip to a distant cloud dominates the user-perceived latency. Edge inference extends the same logic further: small quantised models running on edge servers at telecommunications-operator sites address latency and sovereignty together, and combined with 5G multi-access edge computing (Chapter 43), edge inference becomes increasingly viable rather than aspirational.

Underlying all of this is that the cost-engineering techniques of this chapter are not optional optimisations in African deployments. Prefix caching, continuous batching, and quantisation are, under tight unit economics, the difference between an AI product that is economically viable and one that loses money on every call. In a market with more cost headroom they are improvements; in a cost-pressured deployment they are the architecture, and treating them as first-class from the start is what makes the product possible.

47.9 Recurring Failure Patterns

- **Compute-only capacity planning.** Memory and bandwidth ignored; OOMs and tail latency hit production.
- **No continuous batching.** Throughput a fraction of what the hardware supports.
- **Prefix-cache opportunity missed.** Common prefixes recomputed every request.

- **Single-GPU thinking at scale.** Architecture that does not anticipate multi-GPU parallelism.
- **Quantisation neglect.** Native-precision serving where quantised would meet quality bar at lower cost.

47.10 In Short

Production inference is a memory-and-cache-management problem more than a compute problem. The architectural pieces are continuous batching, tensor/pipeline/expert parallelism, paged attention, prefix caching, speculative decoding, and aggressive quantisation. African contexts add particular weight to smaller-model preference, regional inference for latency, edge inference for sovereignty, and cost engineering as first-class.

Canary Deployment for Prompts and Models

“Test it on five percent first. If five percent is too much, your blast radius is too big to begin with.”

—a recurring observation in production release engineering

Chapter Overview

A canary deployment routes a small fraction of production traffic to a new version while the bulk continues to use the established version. The pattern is standard for code; in Software 3.0 it applies to prompts, tool definitions, agent configurations, and model versions. The architectural mechanics are different from code canaries because what we are measuring is statistical, the surface is multi-dimensional, and the rollback signal can be a quality regression rather than an outage.

Learning Objectives

- Adapt canary deployment from code to prompts, models, and agent configurations.
- Architect traffic routing, comparison, and decision infrastructure for AI canaries.
- Choose the right metrics to gate the rollout.
- Manage the longer evaluation windows AI canaries often require.
- Adapt canary practice to African production realities.

48.1 Canaries for AI Are Different

Code canaries gate on error rates, latency, and resource consumption. These signals fire fast and unambiguously. AI canaries gate on statistical quality measures that take longer to accumulate and are harder to interpret.

48.1.1 What Has to Be Measured

The following elements define this concern and distinguish it from the adjacent ones.

- Quality metrics: judge-model scores, citation accuracy, refusal rate, hallucination indicators.
- User-feedback signals: thumbs up/down rates, escalation rates, regeneration rates.
- Cost metrics: tokens per request, model-mix shifts, total cost delta.
- Latency metrics: p50, p95, p99 latency, tail behaviour.
- Safety metrics: policy-violation rate, harm-categorisation rate.

48.1.2 The Accumulation Problem

Code canaries can be evaluated in minutes; AI canaries often need hours or days for quality signals to accumulate with enough statistical power. The architecture has to support long-running canaries that survive deploys, vendor changes, and operator turnover.

48.2 The Architecture

Figure 48.1 shows the canonical layout.

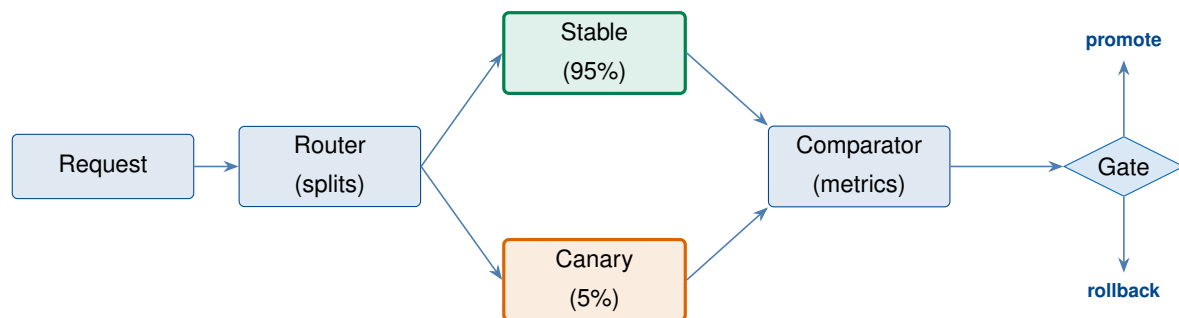


Figure 48.1: Canary deployment for AI: requests split between stable and candidate versions; metrics from each compared continuously; a gate decides promotion, continuation, or rollback.

48.3 Traffic Routing

Splitting traffic for an AI canary requires more care than for a code canary, and the care begins with the choice of how to split. The split can be a random percentage, the simplest and the default; a user-segmented split that designates a fixed canary cohort; a geographic split that releases to a single region first; a use-case split that sends only low-risk tasks to the candidate initially; or a time-of-day split that exposes the candidate during off-peak hours first. These are not exclusive — a release might be geographic and use-case-scoped at once — and the choice is a risk decision: the dimension chosen should be the one that most contains the blast radius of a regression.

Whatever the split dimension, one routing discipline is non-negotiable: stickiness. Within a session, a user must stay with the same version, because behaviour comparison across versions is meaningless if a single session bounces between the stable and candidate versions — the user experiences an incoherent mix, and the metrics attribute the mixed experience to neither version cleanly.

A distinct and stronger variant of routing is the shadow-traffic mode, in which the candidate version sees production traffic but its outputs never reach users. The comparator records the outputs of both

versions; the candidate is evaluated against real production inputs at no risk to users, at the cost of the extra inference the shadow run consumes. Shadow traffic gives a higher-quality signal than a user-facing canary and is particularly valuable as a stage *before* any user-facing canary — it is developed in full in Chapter 49.

48.4 Comparison Methods

The comparator has one job: to detect a quality regression in the candidate before that regression is promoted to all users. It does the job through four methods, which differ in signal quality, in cost, and in how quickly they accumulate.

The *online judge* runs a judge model over the outputs of both versions, and the difference in scores is the canary's quality delta. It is cheap to run and accumulates quickly; its confound is judge bias, which is managed by using the same judge on both versions so that whatever bias the judge has applies equally to each. *Paired output comparison* is available only in shadow-traffic mode, where the same request is genuinely run through both versions: a pairwise judge then ranks the two outputs directly, which is a higher-quality signal than independently scoring two different requests, because it removes the variation between requests from the comparison. *User-feedback aggregation* — thumbs, regenerations, dwell time, completion — is a slow signal but an honest one, since it reflects what users actually experienced rather than what a judge model predicted, and for a long-running canary it is often the decisive measure. *Offline evaluation anchoring* runs a held-out evaluation set against both versions before, during, and after the canary, and it catches the regressions that the online signals, drawn only from live traffic, do not surface.

The four methods are complementary, not alternatives. The online judge gives a fast early reading, paired comparison sharpens it where shadow mode allows, user feedback supplies the honest slow signal, and offline anchoring catches what live traffic never exercised. A canary that relies on only one of them is reading the candidate through a single, partial lens.

48.5 The Gate

The gate decides: promote, continue, or rollback. Its logic must be explicit.

- Quality delta within a tolerance band → continue accumulating.
- Quality delta positive and statistically significant → promote.
- Quality delta negative and significant → rollback.
- Quality delta unknown with budget exhausted → default rollback (conservative).
- Cost delta exceeds budget → rollback regardless of quality.
- Safety regression detected → immediate rollback, no waiting.

48.5.1 Statistical Significance

Naive comparison of means is misleading. Sample-size calculations, sequential testing, and Bayesian alternatives are increasingly standard. Eval engineers do statistics, not just averages.

48.6 What Is Being Canaried

The same machinery serves several artefact classes:

- Prompts: a new system prompt for the same model.
- Models: a new model version with the same prompts.
- Tools: a new tool definition or implementation.
- Agents: a new agent specification.
- Retrieval: a new index, embedding model, or retrieval algorithm.
- Combinations: a new prompt-model pairing.

Each artefact class has its own quality-signal expectations. A prompt change typically shows effects within hours; a model change can shift the distribution of outputs in ways that take days to fully characterise.

48.7 Rollback

In Software 3.0, rollback is fast because artefacts are configuration:

- Prompt rollback: revert the prompt-store version. Sub-second propagation.
- Model rollback: switch the model alias. Sub-second propagation.
- Tool rollback: switch the tool version. Sub-second propagation.

The combination of fast rollback and statistical gates makes aggressive canary policies viable: ship, watch, revert at signal.

48.8 The African Perspective

Canary deployment carries distinctive considerations in African deployments, across cohort design, language, cost, and infrastructure.

Cohort-segmented canaries take a particular form in pan-African operations, which release first to one country and then propagate across the others. The country chosen for the first release is often not the largest by user base but the one with the strongest local engineering presence, so that whoever must diagnose a regression is close to the traffic that produced it. The canary cohort, in other words, is chosen for fast diagnosis rather than for maximum exposure.

Language makes metric stratification essential rather than optional. A prompt change that improves English output can regress Swahili, French, or Arabic output, and an aggregate quality score —

averaged across all languages — will hide that regression behind the English improvement. A canary for a multilingual African deployment must stratify its metrics per language, because the aggregate is precisely the figure that conceals the language-specific harm.

Cost belongs in the gate. Under African unit economics a quality-neutral change that raises cost per request by thirty per cent is not a neutral change — it is a regression — and the canary gate must include a cost-per-request criterion so that such a change fails the gate rather than being promoted on the strength of unchanged quality. And the canary infrastructure itself must tolerate connectivity variability: sticky routing depends on session state, and it must degrade gracefully when that session state cannot be retrieved, rather than breaking the canary’s version assignment when the network does.

48.9 Recurring Failure Patterns

- **No canary.** Direct full rollout of prompt or model changes. Regressions reach all users.
- **Canary too short.** Promotion before signals accumulate. Statistical noise wins.
- **Aggregated metrics only.** Cohort-specific regressions hidden in averages.
- **Cost ignored.** Quality-only gates miss cost regressions that hurt unit economics.
- **No rollback discipline.** Rollback path untested; rollback delayed when needed.

48.10 What to Carry Forward

Canary deployment for AI extends the code-canary discipline to prompts, models, tools, agents, and retrieval components. The architecture comprises traffic routing, side-by-side comparison, statistical gates, and instant rollback. The shift from code-canary is that gates are statistical, accumulation windows are longer, and the dimensions to monitor are richer. African contexts add cohort-segmentation, multi-lingual stratification, cost-sensitivity, and connectivity-aware infrastructure.

Chapter 49

Shadow Traffic and Replay-Driven Development

“Production has the only realistic distribution of inputs. The architecture that ignores this designs for a world it never inhabits.”

—a recurring observation in production AI engineering

Chapter Overview

Shadow traffic routes production inputs through a candidate version of a system without exposing the outputs to users. Replay-driven development takes captured production traces and runs them against new versions in development. Together, these practices give engineers access to the real distribution of production inputs — the only distribution that ultimately matters. Both depend on systematic trace capture and disciplined treatment of the captured data.

Learning Objectives

- Architect trace capture as a first-class concern, not an afterthought.
- Distinguish shadow traffic from canary and from offline replay.
- Build replay infrastructure that works for the long tail.
- Manage privacy and consent for production trace capture.
- Adapt shadow and replay patterns to African operational realities.

49.1 Why Production Inputs Matter

The inputs developers imagine are not the inputs production sees. Imagined distributions are narrower, cleaner, more grammatical, and more cooperative than reality. Edge cases (long inputs, code-switched language, unusual formatting, adversarial probes, fat-finger typos) make up a much larger share of production than the dev team thinks.

The architectural response: bring production distribution into development. Two patterns deliver this:

- **Shadow traffic.** Live production inputs mirrored to a candidate, in real time.

- **Replay.** Captured production traces re-run against candidates, offline.

Both depend on trace capture being thorough, structured, and durable.

49.2 The Architecture

Figure 49.1 shows shadow and replay together.

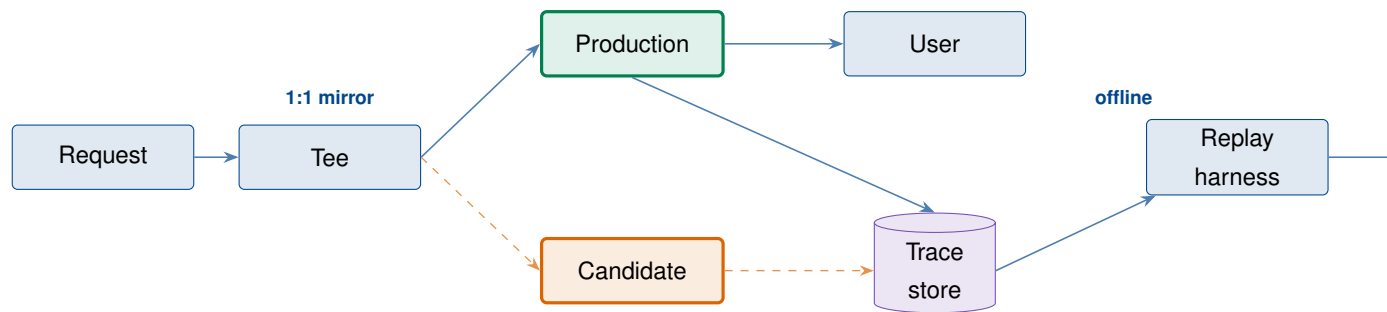


Figure 49.1: Shadow traffic and replay together. Tee mirrors live requests to a candidate (dashed); production results flow to users; all results plus the candidate’s shadow results are captured to a trace store; the trace store also feeds an offline replay harness that runs against development candidates.

49.3 Trace Capture as Foundation

Shadow and replay are useless without comprehensive trace capture.

49.3.1 What a Trace Contains

The following elements define this concern and distinguish it from the adjacent ones.

- Input as received (preserved exactly).
- Pre-processing steps and their outputs.
- Retrieval calls and results.
- Tool calls, arguments, and results.
- Model calls, prompts, and responses (including raw model output before post-processing).
- Post-processing steps.
- Final output.
- Metadata: timestamps, version identifiers for every component, cost, latency.
- User feedback (when available).

49.3.2 Trace Schema Discipline

The schema is a contract between the running system and the replay infrastructure. Schema evolution must be backward-compatible or replay breaks for everything captured before the change. Schema migrations are a real engineering activity.

49.3.3 Storage Considerations

Traces are large. Production systems generate gigabytes per day. Storage choices include: cold-tier object storage for the bulk, hot-tier for recent traces, sampling policies for traffic that need not all be captured.

49.4 Shadow Traffic in Detail

Shadow traffic works by placing a tee in the request path that duplicates each incoming request. The production request is served normally and its result returned to the user; the mirror sends the same request to the candidate version, whose output is captured but never returned to anyone. The user experiences the production system exactly as before; the candidate experiences the full, real production distribution of inputs.

What shadow traffic gives is the real input distribution; what it does not give is worth being precise about. The candidate sees no user feedback, because no user ever sees its output, so the honest slow signal of thumbs and regenerations is unavailable in shadow mode. And the candidate produces no downstream effects, because it did not actually fire any tool or take any action — which is, in one sense, the point, and in another a limitation: shadow mode evaluates what the candidate *would* have output, not what would have happened when its outputs met the world.

That second point carries a hard discipline: a shadow candidate must not produce side effects. It must make no real tool calls, no real database writes, no real notifications. Either the tools are mocked in shadow mode, or the candidate runs in a sandbox where every side effect is intercepted before it escapes. A shadow candidate that accidentally sent real emails, or wrote real records, is a genuine and recognisable incident class — the shadow run was supposed to be invisible, and it was not. Finally, shadow traffic has a cost: it doubles the inference cost for whatever fraction of traffic is mirrored. The sample rate is therefore a tunable lever — a hundred per cent shadow for a high-stakes change, ten per cent for a routine one, and off entirely for changes that are not safety-relevant.

49.5 Replay in Detail

Where shadow traffic mirrors live requests in real time, replay works offline: a replay harness reads traces from the trace store, replays each captured input against a candidate version, captures the candidate's output, and compares it against what the trace recorded. The harness is what turns the accumulated trace store into a development tool.

The comparison step is where replay's design choices concentrate, because comparing two AI outputs is not the straightforward equality check it would be for deterministic code. Five comparison types recur:

- **Exact match** — rarely useful for AI, since output text varies between runs even when nothing in the system changed.
- **Semantic similarity** — an embedding-based comparison that registers whether two outputs mean the same thing despite differing wording.

- **Judge-scored** — a judge model rates each output and the differences in score are flagged for attention.
- **Tool-call comparison** — did the candidate make the same tool calls as the trace recorded, with the same arguments? For agentic systems this often matters more than the text.
- **Quality-rubric** — a structured judge scoring each output against declared criteria, the most detailed and most expensive of the five.

The deeper value of replay emerges over time. As the trace store accumulates, the replay collection becomes the most realistic evaluation set the team owns, because it represents the actual production distribution — including the long tail of unusual inputs that no one would have thought to write into a hand-built evaluation set. Curated subsets of the replay collection become regression evaluations; the bulk remains available for one-off replays when a specific question arises. The evaluation set, in other words, is grown from production rather than imagined in development, which is exactly the property the chapter’s epigraph argues for.

49.6 Privacy and Consent

Traces contain user inputs and system outputs that may include personal data, regulated content, or confidential information. The architectural response:

- Consent terms address trace capture.
- Redaction applied at capture or on read.
- Access controls strict on trace stores.
- Retention windows bounded; expired traces deleted.
- Cross-jurisdiction data flows respected: traces from one jurisdiction may not be replayable in another.

49.7 Replay-Driven Development as Practice

In mature teams, replay is part of the development loop:

- A bug report from production names a trace ID.
- The trace is replayed locally; the bug reproduces.
- Fix is developed against the replay.
- Fix is validated against a broader replay set before merge.
- The fixed-bug trace enters the regression set.

This is meaningfully different from test-driven development. The tests are derived from production, not imagined in development.

49.8 The African Perspective

Shadow traffic and replay carry distinctive weight in African deployments, and the first reason is the one the chapter's epigraph names: production has the only realistic distribution of inputs, and the production distribution in African contexts surprises Northern-trained developers more than they expect. Language mixes and code-switching, mobile-money transaction flows, the variety of identity documents, the range of network conditions — these make up a far larger share of real African traffic than an imagined distribution would include. Replay-driven development is what surfaces these realities during development, where they can be designed for, rather than leaving them to be discovered as incidents in production.

The second consideration is cross-border replay restriction. A trace captured in Nigeria may not be lawfully replayable in a Cape Town region, depending on the data-residency law that governs it, which means replay infrastructure for a pan-African deployment cannot be a single central installation. The schema and the harness can be portable — the same replay code running everywhere — but the trace data itself must be localised, with replay happening in-region against the traces that region is permitted to hold.

The third is privacy-forward capture. The data-protection regimes across African jurisdictions — POPIA, NDPA, and the others — all bear on trace capture, since a trace contains user inputs and system outputs that may include personal data. The durable response is aggressive redaction at capture time rather than retrospectively: redacting at the moment of capture means the trace store never holds the unredacted personal data at all, which is a stronger position than capturing everything and redacting on read. The fourth consideration is cost: shadow traffic at a full mirror rate doubles inference cost, and under African unit economics that argues for aggressive sampling, with full-rate shadow reserved for the high-stakes releases that genuinely warrant paying for it.

49.9 Recurring Failure Patterns

- **Trace gaps.** Some components don't emit traces; replay through these is impossible.
- **Schema breakage.** Old traces unreplayable after schema changes; institutional memory lost.
- **Shadow side effects.** Sandbox boundary leaks; shadow candidate produces real-world effects.
- **Privacy oversights.** Personal data in traces not redacted; data-protection regulators take notice.
- **Replay set staleness.** Replay set never updated; it stops reflecting current production.

49.10 The Shape of the Argument

Shadow traffic and replay-driven development bring production input distribution into engineering practice. Both depend on comprehensive trace capture with disciplined schemas, privacy controls, and durable storage. Shadow mirrors live traffic for real-time candidate evaluation; replay re-runs captured traces against new candidates offline. African contexts add weight to cross-border-aware infrastructure, privacy-forward capture, and cost-aware sampling.

Chapter 50

Agentic RAG and Adaptive Retrieval

“Retrieval-augmented generation was the first pattern. Retrieval-augmented thinking is the next one.”

—a recurring observation in production AI

Chapter Overview

Classical RAG retrieves a fixed number of chunks for every request and feeds them all into the generator. Agentic RAG turns retrieval into an active behaviour: the model decides whether to retrieve, what to retrieve, when to retrieve again, and how to combine multiple retrievals. The architectural shift is from one-shot retrieval-then-generate to a loop where retrieval is a tool the agent uses iteratively.

Learning Objectives

- Distinguish agentic RAG from classical RAG and recognise when the shift is worth its complexity.
- Architect retrieval as a tool the model invokes, not a stage it cannot skip.
- Build multi-hop retrieval patterns for questions that single retrieval cannot answer.
- Manage cost and latency when retrieval becomes iterative.
- Adapt agentic RAG to African contexts.

50.1 The Limits of Classical RAG

Classical RAG (Chapter 21) retrieves once and generates once. The pattern is excellent for many workloads, and several real failure modes drive teams toward agentic variants.

- **Multi-hop questions.** “What did the CEO of the company that acquired X in 2019 say about Y last quarter?” requires retrieving X’s acquirer, then the CEO’s identity, then their statements. Single retrieval cannot answer this.
- **Sparse-answer queries.** The answer lives in one specific chunk; retrieving twenty doesn’t help and dilutes context.
- **Conversational continuity.** Each turn may need fresh retrieval based on the evolving conversation,

not on the original question alone.

- **Trivial questions.** Many user questions need no retrieval at all (greetings, format conversions, simple math). Classical RAG runs retrieval anyway.

50.2 The Architecture

Figure 50.1 shows the agentic RAG loop.

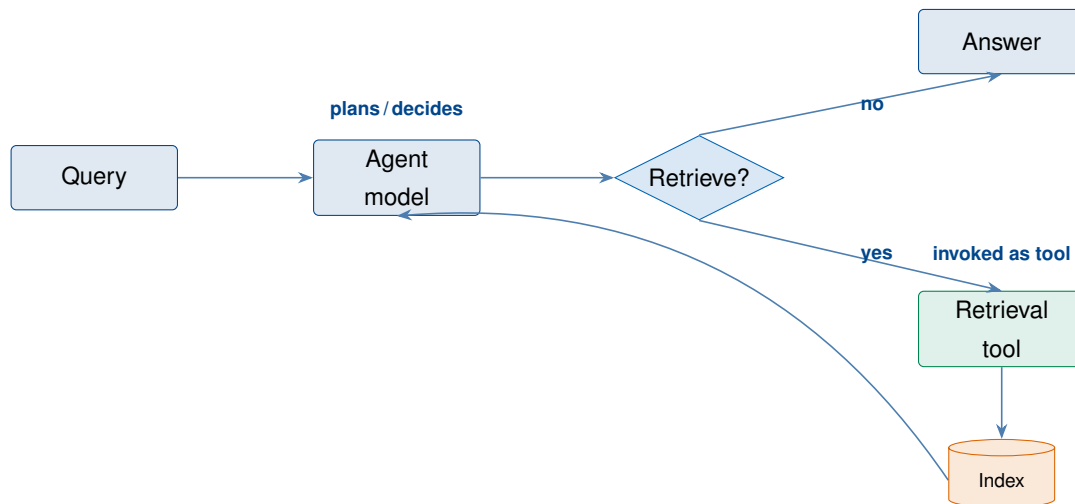


Figure 50.1: Agentic RAG: the agent decides whether to retrieve, what query to issue, and how to use the results. Retrieval is a tool, not a pipeline stage. The loop continues until the agent has enough to answer.

50.2.1 Retrieval as a Tool

In the agentic pattern, retrieval is a tool with a typed interface: query string, optional filters, optional collection, top- k . The model invokes the tool when it judges retrieval would help, with whatever query it constructs. The tool result enters the context and the agent reasons over it.

50.2.2 The Decision Discipline

Without discipline, the agent retrieves too often or not often enough. Two ends to manage:

- Encourage retrieval where the agent’s confidence is low or the question is fact-bearing.
- Discourage retrieval where the question is trivial or already answered by context.

50.3 Multi-Hop Retrieval

Multi-hop questions require sequential retrieval, where each retrieval informs the next query.

50.3.1 The Pattern

Agentic RAG replaces the fixed retrieval pipeline of standard RAG with an agent loop in which the model decides, at each step, whether to retrieve, what to retrieve, and whether the retrieved context is sufficient or needs refinement. The pattern gives the retrieval system the adaptiveness that a fixed pipeline lacks.

1. Agent retrieves on the initial question.
2. Agent identifies what additional information is still needed.
3. Agent constructs a follow-up query.
4. Agent retrieves again.
5. Repeat until the agent has the chain of facts to answer.

50.3.2 Termination

Without termination logic, multi-hop spins. Three controls:

- Hop budget (max retrievals per question).
- Diminishing-returns detection (each retrieval adds less than a threshold).
- Answer-readiness check (agent self-assesses whether it can answer).

50.4 Query Construction

Agentic retrieval depends on the agent constructing useful queries.

50.4.1 Query-Generation Patterns

The quality of agentic retrieval depends on the quality of the queries the agent generates, because a retrieval system can only return what a well-formed query asks for. Several query-generation patterns recur, and each suits a different kind of information need.

- Extract key entities from the user question.
- Rewrite the user question into search-friendly form.
- Generate multiple candidate queries and retrieve on all.
- Construct queries that fill specific gaps the agent has identified.

50.4.2 Query Refinement

When retrieval produces poor results, the agent refines: relaxes terms, adds synonyms, restricts a filter, asks a different question. The refinement loop is itself agentic.

50.5 Self-Reflection on Retrieved Content

The agent should not blindly trust retrieval. Production agentic RAG includes:

- Relevance assessment: do these chunks address the question?
- Sufficiency assessment: do these chunks contain enough to answer?
- Recency assessment: are these chunks current enough?
- Authority assessment: are these sources trustworthy for this question?

The assessment outputs feed back into the loop: retrieve more, retrieve differently, or proceed to answer.

50.6 Cost and Latency Reality

Agentic RAG costs more than classical RAG. Several inference calls per question; multiple retrievals; potentially long context as evidence accumulates.

- Cascade: cheap model decides whether agentic RAG is even needed.
- Caching: retrieved chunks cached across the conversation.
- Budget bounds: max hops, max cost per question.
- Streaming: partial answers as the agent finds enough evidence.

50.7 The African Perspective

Agentic RAG has a distinctive shape in African deployments, across language, source authority, cost, and connectivity.

Language enters the retrieval loop directly. In a multilingual deployment the agent may retrieve in one language and answer in another, which makes translation-mediation patterns and cross-language embedding strategies part of the architecture rather than an afterthought, and which requires the agent to be aware of which corpora actually live in which languages — a query routed to a corpus that does not hold the relevant language retrieves nothing useful no matter how well it is constructed.

Source authority is more layered than in a single-jurisdiction deployment. African contexts typically have both local sources — regulatory, regional, vernacular — and global sources such as Northern academic work and multinational corporate material, and the two are not interchangeable. The agent's authority assessment, the self-reflection step that asks whether a source is trustworthy for a question, must weight local sources appropriately for local questions: a regional regulatory question is better answered from the regional regulator than from a Northern summary of it, and an authority assessment that treats all sources as equally credible gets exactly this backwards.

Cost and connectivity bound the loop. Hop budgets are tighter under African unit economics — often one or two hops at most — and the loop is engineered to converge quickly or to fall back to classical single-shot RAG rather than to keep hopping. And because retrieval requires connectivity to the index, edge-cached retrieval, offline retrieval against cached subsets of the corpus, and graceful degradation when the index is unreachable all become architectural concerns rather than refinements. The agentic-RAG loop does not change for African deployments; its budgets tighten and its retrieval tool must be built to tolerate an index it cannot always reach.

50.8 Recurring Failure Patterns

- **Always-retrieve agents.** The agent retrieves even on trivial questions, inflating cost.

- **Never-retrieve agents.** The agent hallucinates instead of retrieving on fact-bearing questions.
- **Hop explosion.** No budget; agent retrieves indefinitely.
- **Poor query construction.** Agent's queries are too literal or too vague.
- **Authority confusion.** Agent treats all retrieved sources as equally authoritative.

50.9 Where That Leaves Us

Agentic RAG turns retrieval from a pipeline stage into a tool that the agent invokes when, how, and as often as needed. The pattern unlocks multi-hop questions, conversational continuity, and selective retrieval, at the price of more complex orchestration, higher cost, and harder observability. African contexts add multi-lingual retrieval, authority weighting for local sources, cost-aware hop bounds, and connectivity-aware infrastructure.

Chapter 51

Eval-Driven Release

“In Software 1.0 the tests gate the release. In Software 3.0 the evals gate the release. The discipline is the same; the mathematics is not.”

—a recurring observation in mature AI engineering organisations

Chapter Overview

Eval-driven release is the release-engineering discipline for Software 3.0: a release proceeds only when its evaluation suite passes its statistical bar; rollbacks fire when evaluation signals degrade; the eval suite itself is owned, versioned, and reviewed. The pattern integrates with canary deployment, shadow traffic, replay, and observability to make AI releases as predictable as code releases — granted that “predictable” now describes statistical guarantees rather than binary outcomes.

Learning Objectives

- Architect evaluation suites as release gates, not as side artefacts.
- Manage statistical bar selection and significance.
- Tie eval-driven release to canary, shadow, and replay infrastructure.
- Run eval-driven release in pre-production, production, and post-production.
- Adapt the discipline to African operational realities.

51.1 Releases as Statistical Decisions

A Software 1.0 release has a binary outcome: tests pass, ship; tests fail, fix. Eval-driven releases for Software 3.0 are statistical: the release proceeds when the candidate’s quality, measured against the eval suite, beats (or matches within tolerance) the incumbent, with sufficient statistical confidence.

51.1.1 The Decision Components

The release gate is composed of several decision components, each contributing a different kind of evidence to the ship-or-hold judgement. Together they turn a release decision from a matter of opinion into a structured, evidence-based assessment.

- Eval suite: the curated set of test cases.
- Scoring methodology: how each output is scored.
- Aggregation: how case scores roll up to a suite-level number.
- Bar: the threshold that must be met.
- Confidence requirement: how sure we have to be.
- Stratification: which sub-populations must independently pass.

51.2 The Architecture

Figure 51.1 shows the release flow.

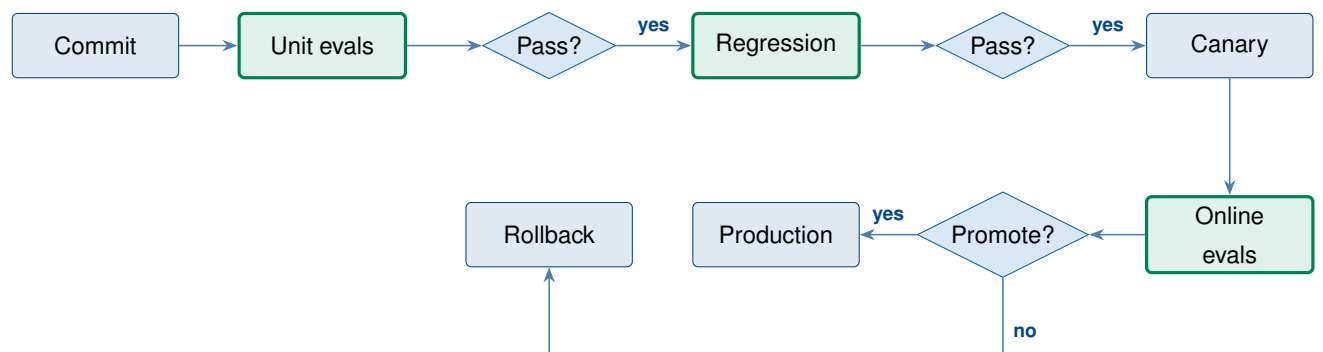


Figure 51.1: Eval-driven release. Unit evals gate the commit. Regression evals gate the merge. Canary deployment carries the candidate to production. Online evals gate the promotion. Failure at any gate rolls back.

51.3 The Eval Suite as Owned Artefact

The eval suite is the contract between candidate and production. Its construction and ownership matter.

51.3.1 Composition

The decision components are composed into a gate through a logic that can be as simple as a conjunction or as nuanced as a weighted score. The composition determines how the components interact and what a failing component means for the release.

- Curated golden set: hand-crafted cases covering the surface area.
- Replay set: production traces representing real distribution.
- Adversarial set: prompt-injection, jailbreaks, edge cases.
- Regression set: cases that have failed in past incidents.
- Stratification metadata: language, user segment, use case, risk tier.

51.3.2 Ownership

Eval suites are owned: domain experts curate the cases, engineers maintain the harness, product approves the bar. The owner sign-off matters because the bar is a product decision, not an engineering decision.

51.3.3 Versioning

The eval suite is itself versioned. Adding cases or changing rubrics shifts the bar; changes go through review. A drop in measured quality is meaningful only if the eval suite was the same.

51.4 Setting the Bar

The bar — the threshold for “passing” — is a deliberate choice.

51.4.1 Bar Selection

The quality bar — the threshold a candidate must meet to pass the gate — is itself a design decision, and getting it wrong in either direction has real consequences. A bar set too high blocks good candidates; a bar set too low ships bad ones.

- For safety-critical evals: very high bar (99% or higher), with regressions blocking even single-case failures in the regression set.
- For helpfulness evals: high bar (95%+), allowing for known-acceptable failure modes.
- For style and tone: moderate bar (80–90%), with subjective scoring.
- For new capabilities: graded bar that rises as the capability matures.

51.4.2 Statistical Power

A 95% pass rate on 100 cases has wide confidence intervals. The same pass rate on 5,000 cases is much tighter. Bar selection is intertwined with eval-set size; bigger sets enable tighter bars.

51.4.3 Stratified Bars

Aggregate pass rate can hide sub-population regressions. Stratified bars require each cohort (each language, each user segment, each use case) to pass independently. The aggregate becomes a sanity check on the stratification.

51.5 Pre-Production, Production, Post-Production

Eval-driven release operates in three phases.

51.5.1 Pre-Production

Unit and regression evals run on every commit and merge. Fast feedback; gate on these before code can deploy. Cost is controlled by limiting eval set sizes; comprehensive coverage delegated to scheduled runs.

51.5.2 Production

Canary deployment routes a fraction of traffic to the candidate. Online judge models score outputs. User feedback is captured. The gate evaluates these signals against thresholds.

51.5.3 Post-Production

After promotion, continuous online evaluation watches for drift. A regression that emerges weeks after release (due to changed traffic mix, vendor model update, or schema change) triggers the same rollback path as a release-time regression.

51.6 Integration with Other Patterns

Eval-driven release composes with the patterns already covered.

- **Canary** provides traffic split.
- **Shadow traffic** feeds candidate quality signals before any user impact.
- **Replay** pre-validates against production distribution.
- **Observability** surfaces the metrics the gates evaluate.
- **Constitutional AI** provides principle-aligned eval rubrics.

51.7 The African Perspective

Eval-driven release carries distinctive weight in African deployments, across stratification, cost, reporting, and cross-border staging.

Stratification by language and market is not optional here. African operations span multiple languages and multiple markets, and an aggregate eval-suite pass rate is close to meaningless across that spread — a candidate can pass overall while regressing badly in one language, because the languages that pass outvote the one that fails. The stratified-bar discipline of the chapter applies with full force: each language and each market must clear its bar independently, and the aggregate becomes a sanity check on the stratification rather than the gate itself.

Cost belongs in the gate. Under African unit economics a release that holds quality steady but raises cost per request may not be a release worth shipping at all, and so eval-driven release gates in these deployments routinely treat cost-per-request and per-tenant cost as first-class gated metrics alongside quality, not as a separate concern checked elsewhere. A candidate that passes the quality bar and fails the cost bar fails the release.

Two further considerations follow from how African AI is often funded and governed. Many deployments operate under donor or regulator visibility, and the eval-driven-release infrastructure doubles as the reporting infrastructure: the evidence that the quality bars were met is precisely the evidence a donor or regulator requires, so an organisation that has built eval-driven release has, as a by-product, built much of its compliance reporting. And pan-African releases stage through several markets in sequence, which the eval-driven discipline supports naturally — the candidate must clear the bar

in the first market before the next market is opened, turning the release into a sequence of gated, evidence-backed decisions rather than a single continental switch.

51.8 Recurring Failure Patterns

- **No eval gate.** Releases proceed on subjective review; regressions reach production.
- **Optimised-for eval set.** Releases tuned to pass the suite without improving real performance.
- **Bar inflation.** Bars set so high nothing passes; engineering frustration; quiet bypass.
- **Bar erosion.** Bars lowered over time to keep velocity; regressions accumulate.
- **Eval suite rot.** Suite frozen at one moment; production has evolved; the gate measures something irrelevant.

51.9 Holding the Pieces Together

Eval-driven release applies the discipline of test-gated release to Software 3.0, with the twist that “passing” is statistical and stratified. The architecture comprises an owned eval suite with curated golden cases, replay-derived production cases, adversarial cases, and a regression set; deliberate bar selection; stratification by sub-population; gates at commit, merge, canary, online, and post-release stages; and integration with the canary, shadow, replay, observability, and constitutional patterns. African contexts add stratification by language and market, cost-inclusive gates, donor/regulator reporting, and cross-border coordination.

Chapter 52

Mixture-of-Experts Routing

“Total parameter count is what you pay for in storage. Active parameter count is what you pay for in inference. The architecture chooses where to sit on that ratio.”

—a recurring observation in foundation-model engineering

Chapter Overview

Mixture-of-experts (MoE) models partition their parameters into experts, with a router that selects which experts process each token. Total parameter count can be enormous; active parameter count per token is a small fraction. The architecture changes inference economics, deployment shape, and training characteristics. The pattern is increasingly mainstream: most frontier-scale models released in the last two years use MoE, often without flagging it externally.

Learning Objectives

- Understand what MoE changes about inference economics.
- Architect serving infrastructure for MoE models.
- Address load-balancing and expert-utilisation challenges.
- Recognise applications of the MoE pattern beyond foundation models.
- Adapt MoE deployment to African contexts.

52.1 What MoE Changes

A dense model with 70B parameters applies all 70B parameters to every token. An MoE model with 70B active and 700B total applies a small subset to each token. The implications:

- **Inference cost.** Per-token compute scales with active parameters, not total.
- **Memory footprint.** Total parameters must reside in GPU memory; storage cost scales with total.
- **Quality.** A well-trained MoE typically beats a dense model of the same active-parameter count, sometimes by margins.
- **Training dynamics.** Load balancing across experts is non-trivial; some experts get all the traffic, others starve.

- **Specialisation.** Different experts often specialise in different content (languages, modalities, domains).

52.2 The Architecture

Figure 52.1 shows MoE inference structure.

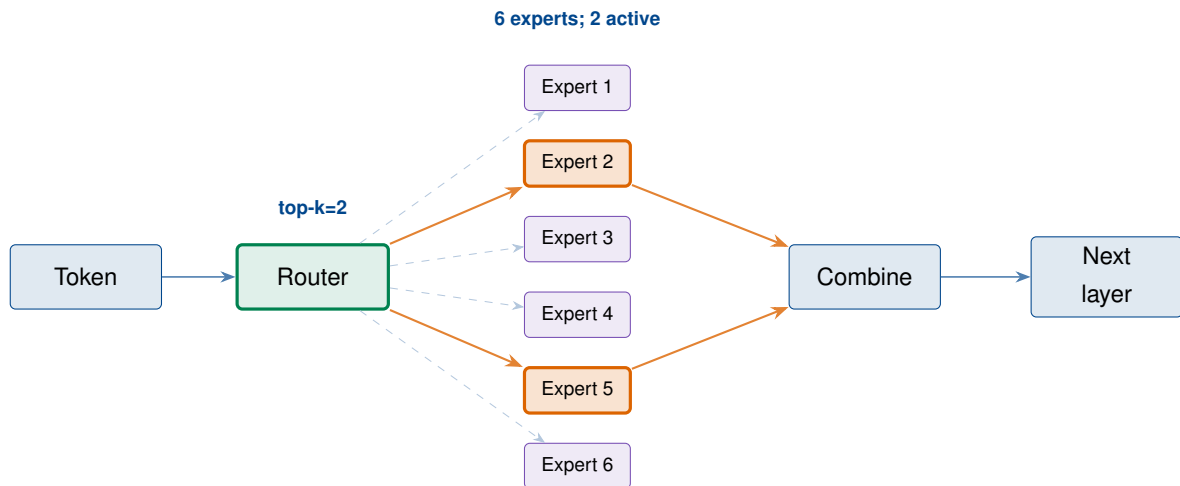


Figure 52.1: MoE inference at one layer: router selects top- k experts (here, 2 of 6) for each token; only selected experts compute; outputs are combined; next layer routes independently. Across layers and tokens, the active subset varies, exercising different experts.

52.3 The Router

The router is a small learned network that maps each token's representation to a set of expert affinities, from which the top- k experts — usually one or two — are selected to process that token. It is a small component carrying a large responsibility: the router's choices determine which of the model's parameters are actually exercised, and three properties of it matter for the architecture.

The first is the choice between top-1 and top-2 routing. Top-1 routing, selecting a single expert per token, is the cheapest option, but quality often suffers, because expert specialisation produces sharp cliffs — a token routed to one expert gets none of the capability of the others, and the boundary between experts becomes a discontinuity in the model's behaviour. Top-2 routing gives the model a smoother mixture of two experts' outputs and is the production default, accepting a modest cost increase for a meaningful quality gain.

The second is routing stability. A router that flips its expert choices on tiny changes to its input produces unstable, hard-to-debug behaviour, since a near-identical token can be processed by an entirely different part of the model. Training therefore adds regularisation that pushes the router toward stable choices, so that small input changes produce small routing changes rather than discontinuous jumps.

The third is load balance, addressed through an auxiliary loss. Left to itself, a router tends to concentrate traffic on a few favoured experts — the experts that happened to train well early attract

more tokens, which trains them further, in a rich-get-richer dynamic. An auxiliary loss term during training penalises this imbalance, rewarding the router for spreading load across the full set of experts, and it is what keeps the model from collapsing into a handful of overused experts and a majority of starved ones.

52.4 Serving Challenges

MoE inference is harder to serve than dense inference, and the difficulty concentrates in three places.

The first is expert parallelism. Because the experts are too numerous to fit alongside one another on a single device, different experts live on different GPUs — and since each token’s chosen experts may sit on different devices, communication is required at every MoE layer to route tokens to their experts and gather the results back. This inter-device communication is a cost dense models do not pay, and it is the reason MoE serving needs interconnect bandwidth that dense serving can sometimes do without.

The second is load imbalance. Even with the training-time auxiliary loss, production traffic does not exercise experts evenly — some experts are hotter than others for the actual distribution of inputs — and a GPU hosting a hot expert saturates while a GPU hosting a cold one idles. The mitigations are operational: replicating hot experts across more than one GPU, placing experts dynamically rather than fixing the assignment, and routing that is aware of each GPU’s current capacity.

The third is memory pressure, and it is the constraint most often underestimated. The total parameter count must reside in GPU memory regardless of how few parameters are active per token: a seven-hundred-billion-parameter MoE needs as much memory as a seven-hundred-billion-parameter dense model, even though it computes like a seventy-billion-parameter one. Sparse-MoE serving infrastructure — vLLM and TensorRT-LLM with expert parallelism — solves the placement question of getting tokens to the right experts efficiently, but it does not shrink the memory requirement. An organisation drawn to MoE by its inference-cost advantage must size its hardware against the total parameter count, not the active one.

52.5 Beyond Foundation Models

The MoE pattern is not exclusive to foundation models. The architectural idea — a router selecting from specialised components — applies broadly:

- Multi-model routing (Chapter 38) is an enterprise-scale MoE-shaped pattern, with each “expert” being a different model.
- Domain-specialist models in a compound AI system (Chapter 44) play an MoE-like role.
- Adapter-based fine-tuning (LoRA, QLoRA) with adapter routing forms a parameter-efficient MoE.

The pattern transfers; the engineering challenges (routing, load balance, capacity) transfer with it.

52.6 The African Perspective

Mixture-of-experts carries a genuinely two-sided weight in African deployments: its inference economics are an advantage and its memory economics are a constraint, and the two pull in opposite directions.

On the advantage side is quality per active parameter. For an organisation sensitive to inference cost — which most African deployments are — MoE’s central property is exactly the right one: a model with seventy billion active parameters that performs like a far larger dense model is buying quality without buying the per-token compute cost of that larger model, and when every active parameter is billed, that ratio is worth a great deal.

On the constraint side is memory cost. MoE’s total-parameter footprint is large, and self-hosting an open-weight MoE model requires GPU memory sized to the total rather than the active count. For African on-premise deployments, where GPU memory is a real and expensive constraint, this often tilts the choice back toward dense models that fit on fewer GPUs — the inference-cost advantage of MoE is only realisable if the organisation can afford the hardware to hold the whole model in the first place. The MoE decision in an African deployment is therefore a genuine trade-off rather than a clear win, and which way it falls depends on whether the deployment is cost-bound at inference time or memory-bound at the hardware.

Two further considerations are specific to the pattern. MoE models sometimes show emergent language specialisation — certain experts activate disproportionately on particular languages — and for a multilingual African deployment this is neither clearly an advantage nor clearly a risk; the practical response is observability of expert utilisation per language, so the behaviour is seen rather than assumed. And when MoE is implemented at the system level as multi-model routing rather than inside a single model, the router becomes a sovereignty enforcement point: it can be made to route sovereignty-classified queries only to locally hosted experts, turning the routing layer into a mechanism of data-residency compliance.

52.7 Recurring Failure Patterns

- **Memory underestimation.** MoE deployed as if its active-parameter count were its memory cost; OOM follows.
- **Load imbalance.** Some experts overloaded; some idle; aggregate throughput a fraction of capacity.
- **Routing instability.** Small input changes cause large output changes; debugging is forensic.
- **Auxiliary-loss bypass.** Training shortcut removed the imbalance penalty; production hits the original failure.
- **Cross-device communication overhead.** Expert parallelism without sufficient interconnect bandwidth; communication dominates compute.

52.8 The Architectural Commitment

Mixture-of-experts models partition parameters across experts and route each token through a small subset. Inference cost scales with active parameters; quality scales closer to total parameters. The serving architecture changes: expert parallelism, dynamic routing, load-aware capacity. The pattern extends from foundation-model internals to enterprise-level model routing and adapter-routing systems. African contexts foreground quality-per-active-parameter economics, memory-cost sensitivity, language-specialisation observability, and routing as a sovereignty lever.

Chapter 53

Streaming and Async Architectures

“Latency to first token, not latency to last token, is the experience the user actually has.”

—a recurring observation in production UX

Chapter Overview

LLM responses arrive token by token. Architectures that wait for the full response before returning anything waste this property; architectures that stream it deliver dramatically better perceived latency. Beyond user-facing streaming, async patterns govern agent loops, long-running tasks, background processing, and event-triggered workflows. The architectural shift is from request-response to continuous flow.

Learning Objectives

- Architect token streaming end-to-end through gateways, models, and clients.
- Design async agent loops for long-running tasks.
- Manage backpressure, partial results, and cancellation.
- Build resumable workflows that survive infrastructure failures.
- Adapt streaming/async patterns to African connectivity realities.

53.1 Why Streaming Matters

A 2,000-token response generated at 50 tokens per second takes 40 seconds. A user waiting 40 seconds with no feedback abandons the request. The same response streamed token by token gives the user the first word in 100 milliseconds and the impression of an instant response.

53.1.1 The Latency Metrics That Matter

The following elements define this concern and distinguish it from the adjacent ones.

- **Time to first token (TTFT).** How long the user waits before anything appears. The single most important latency metric for chat UX.
- **Inter-token latency (ITL).** How fast tokens appear after the first. Determines whether reading

speed keeps up.

- **Time to completion.** The traditional latency metric. Matters less than TTFT once streaming exists.

53.2 The Streaming Architecture

Figure 53.1 shows end-to-end streaming.

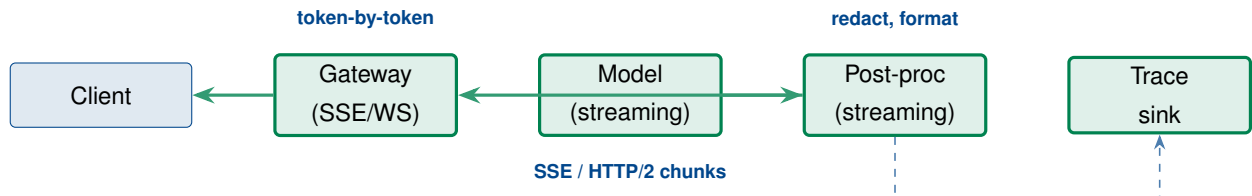


Figure 53.1: End-to-end streaming. Model emits tokens; post-processing operates on the stream (redaction, formatting) without buffering; gateway delivers via Server-Sent Events or WebSocket; client renders incrementally. Trace sink captures the full stream asynchronously.

53.2.1 Streaming-Capable Throughout

Every component in the path must be streaming-capable. A buffering post-processor turns streaming into batch. The chain is only as streamable as its least-streamable component.

53.2.2 Transport

Server-Sent Events (SSE) over HTTP is the most widespread. WebSockets give bidirectional flow at the cost of more infrastructure. HTTP/2 server push is technically capable but rarely used. Long polling is a fallback for environments that block streaming.

53.3 Streaming Challenges

Streaming is not free; making a system genuinely stream from end to end means confronting four challenges that a buffered request-response system never has to.

The first is post-processing on a stream. Many post-processing steps appear to need the full output before they can act — PII redaction, format validation, citation insertion — and a naive implementation simply buffers the whole response, which silently turns a streaming system back into a batch one. The clever implementations operate incrementally instead: the redactor that recognises a partial credit-card number and holds just those tokens, waiting to see whether the following digits complete a real number before deciding whether to redact, is what production-grade streaming post-processing actually looks like. The discipline is to make each post-processing step incremental rather than letting any one of them collapse the stream.

The second is errors mid-stream. A stream can fail halfway through: the client has received some tokens but no completion signal, and it cannot tell whether the response is finished or broken. The patterns that handle this are explicit error events carried in the stream itself, final-status sentinel tokens that mark a genuine completion, and client-side retry from the last-received position where the model supports continuation.

The third is cancellation. When a user navigates away while the model is still generating, every token produced after that point is wasted compute and wasted cost. Cancellation has to propagate from the client through the gateway to the model, which is not trivial — some model APIs support cancellation cleanly and others do not — and a system without cancellation pays in full for every abandoned request.

The fourth is tracing. The trace must capture the full output even though the output was streamed, which means the trace sink subscribes to the same token stream the user receives and finalises the trace only when the stream completes. Get this wrong and the observability record is truncated — the trace finalises before the stream ends and the captured output is incomplete.

53.4 Async Agent Loops

Streaming addresses the user-facing latency of a single response; async patterns address a different problem — the long-running task. An agent that takes five minutes to complete a task should not hold a connection open for five minutes, and the async pattern is how that work is decoupled from the request that started it.

The pattern is a familiar submit-and-subscribe shape: the client submits a task and receives a task identifier immediately; the agent runs in the background; the client subscribes to a status stream or polls for progress; the agent emits progress events as it works; and the final result is delivered through the same channel. The connection that submitted the task is freed the moment the task ID is returned, and the work proceeds independently of it.

Two disciplines make async agent loops production-grade. The first is state persistence: an async task survives an infrastructure restart only if its state is durable, so each agent step's inputs, outputs, and intermediate state are written to durable storage, and a restarted task resumes from its last checkpoint rather than from the beginning. This is the same checkpoint-and-resume discipline the planner-executor and memory chapters develop, here applied to the async loop. The second is bounded lifetime: a long-running task with no maximum wall time becomes infrastructure debt, a process that may run forever and consume resources indefinitely, so tasks declare a maximum wall time, cancellation is honoured when it arrives, and orphaned tasks — those whose submitter has vanished — are detected and cleaned up rather than left running.

53.5 Backpressure

Streaming systems must signal when consumers cannot keep up.

- Client-side queues bounded; full queue triggers backpressure upstream.
- Gateway monitors per-connection lag; throttles or disconnects laggards.
- Model serving slows generation rather than dropping tokens.
- Trace sinks accept loss before they cause head-of-line blocking.

53.6 The African Perspective

Streaming and async patterns have heightened importance in African deployments, because the connectivity conditions they are designed to tolerate are the conditions African users routinely face.

Variable bandwidth is the first reality. Connections that swing from 4G to congested 3G to no connection at all require streaming infrastructure that tolerates rate variation rather than assuming a steady pipe — adaptive streaming that adjusts to the available rate, and Server-Sent Events with built-in reconnection, both help a stream survive a connection that does not hold steady. Closely related is mobile-first reconnection: mobile clients commonly lose their connection partway through a response, and resumable streaming — where the client tells the server the position it last received and the server continues from there — avoids re-running the entire request, which is both a cost saving and a markedly better experience for the user whose connection dropped.

The cross-region case is where async sometimes beats streaming outright. When inference runs in a Northern region but the user is in Africa, end-to-end latency is dominated by the network path, and even a perfectly streamed response is gated by that path. For these workloads an async pattern — submit the task, do something else, return for the result — can deliver a better experience than streaming, because it does not ask the user to wait on a slow path at all. And async loops have a natural offline form: a mobile client that works offline can accumulate tasks and submit them on reconnection, and the async agent loop with durable server-side state supports exactly this — the task is submitted when connectivity returns, runs server-side, and its result waits for the client to come back.

53.7 Recurring Failure Patterns

- **Buffering middleware.** A component in the chain buffers; streaming is silently disabled.
- **No cancellation.** Wasted tokens after user abandonment.
- **State loss on restart.** Async tasks evaporate when infrastructure restarts.
- **Trace race conditions.** Trace finalised before stream completes; output truncated in observability.
- **Backpressure absence.** Slow client cascades into server-side resource exhaustion.

53.8 Pulling the Threads

Streaming token-by-token transforms perceived latency; async loops handle long-running tasks without blocking; both require streaming-capable components end to end, persistent state, backpressure, and disciplined cancellation. African contexts emphasise variable bandwidth tolerance, resumable streams, async patterns for cross-region latency, and offline-capable async loops.

Chapter 54

Guardrails and Policy Engines

“A model’s behaviour is a posterior. The guardrails are the prior, expressed in code.”
—a recurring observation in safety-critical AI engineering

Chapter Overview

Guardrails are the deterministic layer that wraps a probabilistic model: filtering inputs, validating outputs, blocking policy violations, redacting sensitive content, enforcing format. A policy engine is the structured framework where guardrail rules live, evolve, and are audited. Guardrails do not replace alignment; they catch the residual failures that alignment alone leaves on the table.

Learning Objectives

- Distinguish guardrails (deterministic) from alignment (training-time).
- Architect input-side and output-side guardrails.
- Build classifier chains for nuanced content categorisation.
- Operate policy engines with audit, evolution, and exception handling.
- Adapt guardrails to African regulatory and cultural contexts.

54.1 What Guardrails Are For

A perfectly aligned model would not need guardrails. Real models are imperfectly aligned, can be jailbroken, can produce policy-violating outputs by accident, and can be fooled by adversarial inputs. Guardrails catch what alignment misses.

54.1.1 Categories of Guardrails

The following elements define this concern and distinguish it from the adjacent ones.

- **Input guardrails.** Filter or transform requests before they reach the model.
- **Output guardrails.** Filter, transform, or block model responses before they reach the user.
- **Behavioural guardrails.** Constrain what tools the model can call, with what arguments.
- **Audit guardrails.** Capture evidence for compliance, even when not blocking.

54.2 The Architecture

Figure 54.1 shows guardrails wrapped around the model.

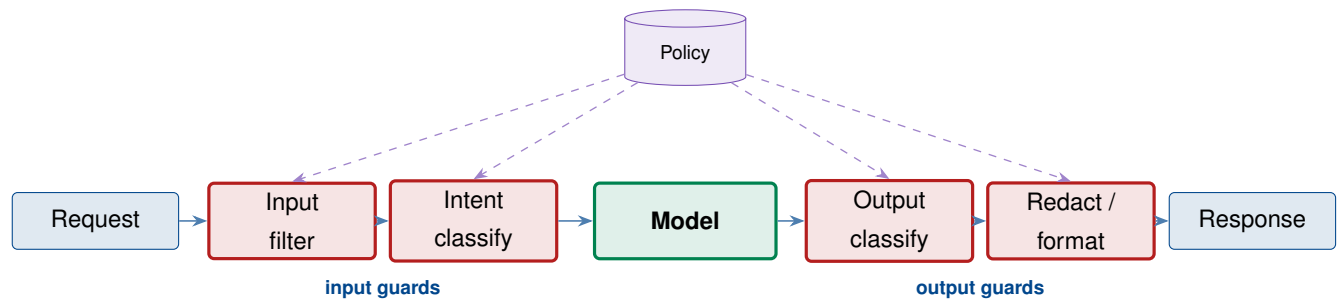


Figure 54.1: Guardrail architecture: input filters and classifiers wrap the model on the request side; output classifiers, redactors, and formatters wrap on the response side; a central policy informs all guards. The model sits in the middle but does not own the rules.

54.3 Input Guardrails

Input guardrails act on a request before it reaches the model, and four patterns recur.

Prompt-injection detection uses classifier models trained to recognise the patterns of a prompt-injection attempt, blocking or sanitising a request where injection is detected. It is imperfect — no classifier catches every injection — and it is therefore a defence-in-depth layer rather than a sole protection, one barrier among several rather than the barrier. *Jailbreak detection* is its close relative, using classifiers or pattern matchers tuned to known jailbreak templates and updated continuously as new templates emerge; public databases such as HarmBench provide test sets against which the detector’s coverage can be measured. *PII stripping* removes or tokenises personal data in the input that does not need to reach the model at all, before the request crosses the trust boundary — which reduces exposure regardless of how the model later behaves, since data that was never sent cannot be mishandled. *Intent classification* predicts what the request is actually for — research, transactional, manipulative, abusive — and routes accordingly, sending ordinary intents to standard processing, sensitive ones to restricted models, and clearly hostile ones straight to refusal. Together these four screen the request before the probabilistic model ever sees it, which is the cheapest place to stop a bad request.

54.4 Output Guardrails

Output guardrails act on the model’s response before it reaches the user, and four patterns recur here too.

Content classification runs the output through classifier chains that score it on harm categories — violence, sexual content, hate, self-harm, illegal advice, regulated professional advice — with outputs above the relevant thresholds blocked outright or sent to human review. *Format validation* checks outputs that are expected to be JSON, code, or other structured text against their schema, triggering regeneration or a fallback when validation fails; it is critical for tool-using systems, where a downstream tool expects a typed input and a malformed model output would otherwise propagate the error into the tool. *Citation verification* checks the sources a model claims to cite, so that a spurious or

fabricated citation triggers blocking or correction rather than reaching the user as apparent evidence. *Style and tone enforcement* holds the output to a brand voice or a required register — formal, casual, regulated-language — often using a judge model that scores the output against a rubric. Where the input guardrails screen what reaches the model, these output guardrails screen what reaches the user, and the two sides together are what wrap a probabilistic model in a deterministic boundary.

54.5 Policy Engines

A policy engine is the structured framework where guardrail rules live.

54.5.1 Properties

The following elements define this concern and distinguish it from the adjacent ones.

- Rules expressed in a domain language (not buried in code).
- Rules versioned and reviewable.
- Rules attached to use cases, tenants, regions, or risk tiers.
- Rule evaluation logged for audit.
- Exceptions handled deliberately, not silently.

54.5.2 Open Policy Agent and Similar

OPA, Cedar, and similar engines provide the rule-evaluation primitive. AI-specific policy engines extend with classifier-as-rule, where the rule body invokes a classifier.

54.5.3 Conflict Resolution

When multiple rules fire, what wins? Explicit precedence; documented tie-breakers; conflict-detection in the rule editor. Without this discipline, behaviour becomes opaque.

54.6 Operating Considerations

Operating a guardrail layer in production turns on three considerations that the architecture must hold in balance.

The first is the latency budget. Every guard adds latency, and the additions accumulate: five sequential guards at fifty milliseconds each is a quarter of a second spent before the model even begins to generate. The response is to evaluate guards in parallel wherever they are independent, and to reserve serial cascading only for guards that genuinely depend on one another's output — so that the guard layer's latency reflects its longest path rather than the sum of all its guards.

The second is false positives. A guard that blocks a legitimate request is, in many use cases, worse than a guard that occasionally misses a bad one — the blocked legitimate user experiences the system as broken, while the missed bad request is one instance among many a defence-in-depth stack expects to catch elsewhere. Where the threshold of a guard is set, balancing false positives against false negatives, is therefore a product decision rather than a purely technical one, and it should be made

with the use case's actual tolerance for each kind of error in view.

The third is evolution under adversarial pressure. Adversaries adapt, and a classifier that reliably catches today's attacks is tomorrow's known bypass. Guard updates are therefore continuous rather than one-time, and the leading indicator to monitor is the pass-through rate — a rising rate of requests that clear the guards is the early signal that the adversary has moved and the guards have not. A guardrail layer that is not continuously updated is not a static defence; it is a slowly decaying one.

54.7 The African Perspective

Guardrails carry distinctive weight in African deployments, across regulatory multiplicity, classifier coverage, cultural variation, and audit.

Regulatory multiplicity is the first reality. Different jurisdictions enforce different rules, and a guardrail layer serving several African markets cannot apply one policy uniformly — it applies tenant-scoped policies that route requests by jurisdiction, so that the rules vary per market while the policy engine itself is shared. The engine is common infrastructure; the rule sets it evaluates are market-specific.

Classifier coverage is the sharpest technical constraint. Harm classifiers are predominantly trained on English, and they perform measurably worse on African languages and on code-switched content — which means the output-classification and input-detection guards are weakest on exactly the languages many African deployments most need them to work on. African-language classifier coverage is uneven across languages, and the common response is an ensemble: several classifiers combined, with a translation-based fallback for the languages no classifier covers well directly. Closely related is cultural context: what counts as offensive or inappropriate varies, and a policy that is correct for one African market may be wrong for another, so both the policy and the evaluation of it must be stratified per market rather than written once against a single implicit standard.

Finally, audit. Many African deployments operate under donor or regulator visibility, and the audit logs the policy engine produces are precisely the evidence those donors and regulators require — which means the policy engine's logging is not only an internal governance tool but the deployment's external reporting mechanism. A guardrail layer built with disciplined, auditable policy logging has, as a by-product, built much of what its donor and regulatory reporting needs.

54.8 Recurring Failure Patterns

- **Single-classifier dependency.** One harm classifier; one bypass.
- **Aggregated thresholds.** Sub-category regressions hidden by average pass rates.
- **Rules buried in code.** No central policy; no audit trail.
- **No evolution.** Classifier from initial deployment; adversaries have moved.
- **Latency creep.** Each new guard added; aggregate latency exceeds UX budget.

54.9 Closing Notes

Guardrails are deterministic wrappers around probabilistic models: input filters, intent classifiers, output content classifiers, format validators, citation verifiers, style enforcers. A policy engine holds rules in a structured, versioned, auditable form. Operating disciplines include latency budgeting, false-positive tuning, and continuous evolution. African contexts add regulatory multiplicity, multi-lingual classifier coverage, cultural variation, and donor/regulator audit requirements.

Prompt Chaining and Workflow Composition

“A prompt chain is what an agent looks like when you decided in advance what it should do.”

—a recurring observation in production AI engineering

Chapter Overview

Prompt chaining composes multiple model calls into a workflow: each step’s output feeds the next step’s input. Compared to fully agentic systems, chains are deterministic, predictable, debuggable, and cheaper. Compared to single calls, chains handle complex tasks that exceed a single prompt’s surface area. The architectural discipline is to recognise when a chain is the right shape and when an agentic loop is.

Learning Objectives

- Distinguish prompt chains from agent loops and from single prompts.
- Architect chains as directed graphs with typed inter-step contracts.
- Manage error handling, conditional branching, and parallel execution.
- Choose between deterministic chains and agentic workflows for a given task.
- Adapt chain patterns to African contexts.

55.1 When Chains Beat Agents

A chain is deterministic: given the same input, the same steps run in the same order. An agent is reactive: the model decides what to do at each step. The choice between them is not preference; it is task fit.

55.1.1 Chain-Favourable Tasks

The following elements define this concern and distinguish it from the adjacent ones.

- The sequence of steps is well-known.

- The steps need to run in a fixed order.
- The output of each step has a well-defined shape.
- Debugging and observability matter highly.
- The task is high-volume; predictability matters.

55.1.2 Agent-Favourable Tasks

The following elements define this concern and distinguish it from the adjacent ones.

- The sequence depends on intermediate findings.
- Steps may need to be skipped or repeated.
- The model needs to react to information it discovers.
- The task is exploratory or open-ended.

Most production workloads sit closer to the chain end of this spectrum than teams initially assume. The default should be: chain unless an agent is clearly needed.

55.2 The Architecture

Figure 55.1 shows a chain as a directed acyclic graph.

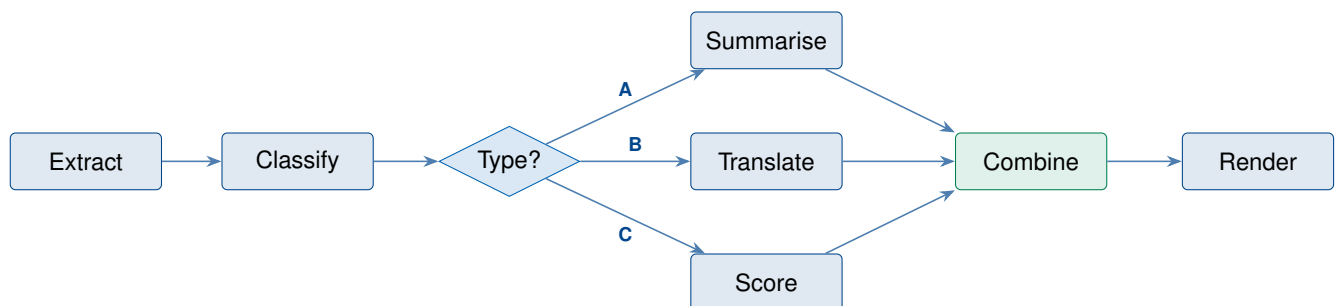


Figure 55.1: A chain as a DAG. Steps run in topological order; conditional branches route by intermediate outputs; parallel branches execute simultaneously; results combine before the next step.

55.2.1 Inter-Step Contracts

Each step in a chain has a typed input and typed output. The contract is enforced: a step whose output does not validate against the next step's schema triggers explicit error handling, not silent corruption.

55.2.2 Conditional Branching

Branches in chains are explicit. A classification step routes downstream; each branch is named, evaluated, and monitored. Unlike agent decision-making, chain branches are inspectable in code.

55.2.3 Parallel Steps

Steps with no dependency run in parallel. The chain framework parallelises automatically based on the DAG. Latency drops; resource consumption rises in burst.

55.3 Error Handling

A chain has more places to fail than a single model call — one per step — and a production chain handles those failures deliberately rather than letting the first one terminate the run. Four mechanisms, applied in increasing scope, do the work.

Step-level retries handle transient failures: a rate limit, a brief network error, a momentary timeout. The chain configuration specifies a retry policy per step, retrying with backoff, so that a step which failed for a passing reason gets another attempt before anything more drastic happens. *Step-level fallbacks* handle a step that fails more durably: a step whose primary model is unavailable can fall back to a different model, and the step's output is annotated to record that a fallback was used — so a downstream consumer, or a later audit, knows the result came from the secondary path. *Chain-level compensation* handles the case where a downstream step fails after upstream steps have already produced effects that must be undone: this is the saga pattern, and a chain framework supports it through compensation handlers that reverse the upstream effects in order. *Partial success* is the recognition that not every chain failure is total: some chains can return a useful partial result when their later steps fail, and the framework distinguishes a terminal failure — nothing usable was produced — from a partial success that is worth returning. A chain with all four mechanisms degrades gracefully under failure; a chain with none of them terminates on its first transient error and produces nothing for the work already done.

55.4 Observability

Chains trace cleanly because their structure is known in advance. Each step's input, output, latency, cost, and model used is captured against the chain's DAG. Replay-from-trace is trivial: re-run from any step with that step's input.

55.5 Chain Frameworks

Multiple frameworks (LangChain, LlamaIndex, DSPy, Haystack, others) compete in this space. The architectural commonality:

- Declarative DAG definition.
- Pluggable models per step.
- Built-in retry, fallback, parallel execution.
- Observability hooks.
- Eval integration.

The framework choice matters less than the discipline of treating chains as first-class artefacts.

55.6 Migration: From Chain to Agent

When a chain becomes too rigid for its task, the migration path:

- Identify the steps that are dynamic.
- Replace those steps with a contained agent (planner-executor on the sub-task).
- Keep the surrounding steps deterministic.
- The result is a chain with embedded agent islands.

The reverse migration — from agent to chain — happens when an agent’s behaviour stabilises enough that the steps are known in advance.

55.7 The African Perspective

Prompt chaining carries distinctive weight in African deployments, and the weight falls on the side of preferring the chain over the agent.

The first reason is predictability under cost pressure. A chain is far easier to budget than an agent: each step has a known cost and a known latency, so the cost of the whole chain is a sum of knowns rather than a distribution over how an agent might behave. Under the tight unit economics of many African deployments, that predictability is itself valuable — a chain can be costed before it is shipped, and a cost overrun is a design error caught in advance rather than a surprise discovered in the bill.

The remaining considerations are the familiar three. Multilingual deployments make the chain’s per-step structure an advantage: a chain may extract in one language, classify in another, and render in a third, and per-step language metadata routes each step to the model equipped for its language — a routing decision the chain’s explicit structure makes natural. Connectivity makes per-step checkpointing important: a long chain is exposed to a mid-execution connectivity failure, and checkpoint-and-resume at the step boundary lets a long chain survive a dropped connection rather than restarting. And latency makes step placement a design lever: latency-critical steps such as input parsing and simple classification can run at the edge or on small local models, while the complex reasoning steps escalate to cloud frontier models — the chain’s step boundaries being exactly where that edge-versus-cloud split is drawn.

55.8 Recurring Failure Patterns

- **Agent where a chain belongs.** Unnecessary complexity, cost, and unpredictability.
- **Chain where an agent belongs.** Tasks shoehorned into linear flow; brittle.
- **Untyped inter-step interfaces.** Silent contract violations; debugging is forensic.
- **No error policy.** First failure terminates; no graceful degradation.
- **Chain-as-monolith.** One enormous chain; refactoring impossible.

55.9 The Pattern, Restated

Prompt chains compose model calls into deterministic DAGs with typed inter-step contracts, explicit branching, parallel execution, structured error handling, and clean observability. Most production workloads start as chains, not agents. Chain frameworks provide the primitives; engineering discipline provides the architecture. African contexts add weight to predictable economics, multi-lingual per-step routing, connectivity-aware checkpointing, and edge-vs-cloud step placement.

Chapter 56

Tree-of-Thought and Branching Search

“Reasoning is not a sequence; it is a search. Linear prompts pretend otherwise.”
—a recurring observation in research on LLM reasoning

Chapter Overview

Chain-of-thought prompting walks a linear reasoning path. Tree-of-thought (ToT) and related approaches explore a branching search over reasoning paths: at each step, multiple candidate continuations are generated, evaluated, and either pursued or pruned. The architecture is more expensive but produces better outcomes on tasks where the best path is not obvious from the start.

Learning Objectives

- Recognise tasks where branching search outperforms linear chain-of-thought.
- Architect the tree expansion, evaluation, and pruning loop.
- Manage exploration-exploitation tradeoffs.
- Bound cost and latency in tree search.
- Adapt branching search to African operational realities.

56.1 Why Branch at All

Chain-of-thought asks the model to think step by step, in a single sequence. When the model takes a wrong step early, it usually cannot recover: subsequent reasoning compounds the error.

Tree-of-thought generates multiple candidate next steps, evaluates them, and continues from the most promising. The pattern is closer to classical search (A*, beam search, MCTS) than to standard prompting.

56.1.1 Tasks Where Branching Helps

The following elements define this concern and distinguish it from the adjacent ones.

- Puzzles with branching solution spaces (game-of-24, sudoku, crosswords).
- Mathematical reasoning where multiple approaches are viable.

- Code generation where multiple implementations exist.
- Planning where the best plan is unknown a priori.
- Hypothesis generation in scientific or investigative tasks.

56.1.2 Tasks Where Branching Adds Little

The following elements define this concern and distinguish it from the adjacent ones.

- Direct extraction or classification.
- Tasks where one approach is obvious.
- Tasks where the cost of exploration exceeds the cost of being wrong.

56.2 The Architecture

Figure 56.1 shows the search loop.

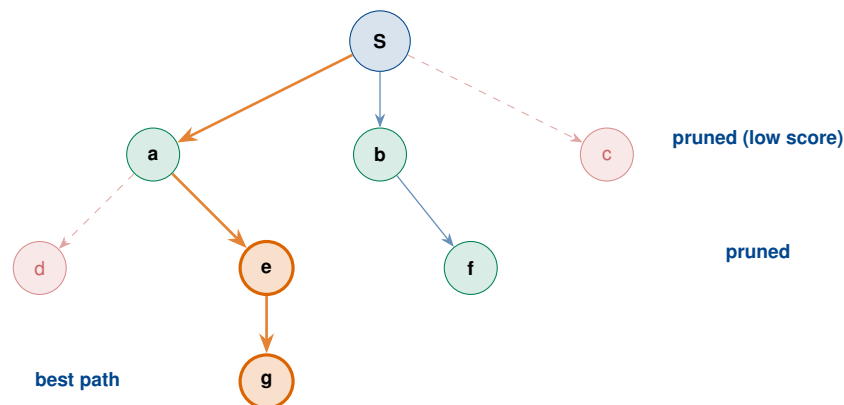


Figure 56.1: Tree-of-thought search. From the root state, multiple thoughts (children) are generated; each is scored; low-scoring branches are pruned; the best-scoring branch is expanded further. The best path through the tree is the final reasoning trace.

56.2.1 The Loop

The following elements define this concern and distinguish it from the adjacent ones.

1. Start with the problem as the root.
2. Generate k candidate next thoughts.
3. Score each candidate (model self-score, classifier, tool verification).
4. Prune below threshold.
5. Choose the top- b (beam width) for further expansion.
6. Repeat until a terminal state or budget exhaustion.
7. Return the best leaf reached.

56.3 Variants

The branching-search loop has four common variants, and they differ chiefly in how they balance exploration against exploitation — how widely they look before committing to a path.

Beam search fixes a beam width b : at each level of the tree it keeps the top- b candidates and expands all of them. Its cost is predictable, because the width is fixed, and it explores breadth-first before depth — a level at a time. *Best-first search* instead maintains a priority queue of partial paths ordered by score, always expanding the best-scoring path and adding its children back to the queue. It is closer to the classical A* algorithm and tends toward exploitation, following the most promising path rather than keeping a fixed-width frontier. *Monte-Carlo tree search* uses UCB-based selection to balance exploration and exploitation explicitly: it simulates rollouts from each candidate and back-propagates the resulting scores up the tree. It is expensive, and it is well-suited to game-like tasks where a rollout to a terminal state is feasible. *Self-consistency* is the cheapest variant and barely a tree at all: it generates n independent reasoning chains and votes on the answer, with no tree to maintain. Despite its simplicity it is surprisingly effective on tasks that have a single correct answer, where the majority vote across independent chains reliably finds it. The choice among the four is a choice of how much exploration the task and the budget can afford — self-consistency at the cheap end, Monte-Carlo tree search at the expensive end.

56.4 Scoring

The quality of a branching search depends heavily on the scoring function, because the score is what decides which branches are pursued and which are pruned — a search with a noisy scorer is a search that prunes good branches and keeps bad ones. Four scoring approaches recur, in increasing order of signal quality and cost.

The *self-score* has the model rate its own candidate thought. It is cheap, requiring no extra component, but it is biased toward confident-but-wrong answers — the same model that produced a flawed thought is not a reliable judge of its flaw. The *verifier model* is a separate model, trained or prompted specifically to verify, which gives a better-quality signal at the cost of an extra model call per candidate. *Tool-based scoring* is the strongest signal of all, available whenever a tool can genuinely verify a candidate — a compiler for generated code, a calculator for arithmetic, a test runner for a program — because a tool's verdict is a fact rather than another model's opinion. And *composite scoring* combines several scorers in a weighted blend, which is common in production because it hedges any single scorer's weakness. The scoring choice is the single highest-leverage decision in a tree-of-thought system, and where a verification tool exists for the task, it should be used — it is the difference between a search guided by ground truth and one guided by a guess.

56.5 Cost Discipline

Tree search costs scale poorly without discipline.

- Beam width and depth budgets bounded.

- Cheaper models for candidate generation; expensive models only for scoring or final synthesis.
- Early stopping when a leaf is good enough.
- Caching of partial scores.
- Pruning aggressive; the cost is wasted exploration.

56.6 When ToT Beats Compound

Tree-of-thought and compound AI (Chapter 44) overlap. ToT is a form of compound system specialised for branching search. The compound approach is more general; ToT is sharper on reasoning-heavy tasks. Many production systems use compound architectures with ToT sub-components.

56.7 The African Perspective

Tree-of-thought has a distinctive shape in African deployments, and the dominant consideration is cost — because branching search multiplies model calls by construction.

The multiplicative cost makes tree-of-thought expensive at scale, and African deployments respond by reserving it for the high-stakes reasoning workloads that genuinely warrant it, using cheaper methods — a single chain-of-thought, or self-consistency at low n — for routine tasks. Tree-of-thought is not a default here; it is a tool brought out when the reasoning is hard enough and the stakes high enough to justify paying for the search.

Two further considerations shape how it is used. Branches may explore different language framings of the same problem in a multilingual deployment, which makes multilingual scoring infrastructure part of the architecture — the scorer must judge candidates fairly regardless of the language they are framed in. And tool-based scoring is what most often makes tree-of-thought economically viable: where a tool can verify a candidate — a financial calculation, a regulatory check, a code execution — the verification is effectively free of per-call model cost, and the search’s cost is bounded by generation alone rather than generation plus model-based scoring. Finally, latency bounds where the pattern can be deployed at all: mobile-first interfaces have tight latency budgets that a multi-round tree search cannot meet, so tree-of-thought in African deployments suits non-interactive workloads — batch analysis, report generation — more than it suits live chat.

56.8 Recurring Failure Patterns

- **Scoring noise.** Score function inconsistent; the “best” branch is random.
- **Insufficient pruning.** Tree explodes; cost goes parabolic.
- **ToT-everywhere.** Pattern applied to tasks that don’t benefit; cost without quality gain.
- **Verifier mismatch.** Verifier rejects correct answers in unfamiliar form.
- **No early termination.** Search continues after a good-enough answer is found.

56.9 Last Word

Tree-of-thought and branching search explore multiple reasoning paths, score them, and pursue the most promising. The pattern beats linear chain-of-thought on tasks with branching solution spaces, mathematical reasoning, planning, and code generation. Variants (beam search, best-first, MCTS, self-consistency) trade exploration for exploitation differently. Cost discipline through beam-width budgets, aggressive pruning, and cheap-then-expensive model use makes the pattern viable. African contexts add particular weight to cost-tier discipline, multi-lingual branching, tool-based scoring preference, and bounded-latency deployment.

Chapter 57

Hybrid Symbolic-Neural Systems

“The symbolic system knows what it knows. The neural system knows things it cannot say. The architecture is the bridge.”
—a recurring observation in regulated-domain AI

Chapter Overview

Hybrid symbolic-neural systems combine traditional symbolic AI techniques — rule engines, constraint solvers, knowledge graphs, formal verifiers — with neural language models. The symbolic side brings determinism, explainability, and guarantees; the neural side brings fluency, generalisation, and accessibility. The architectural challenge is the bridge: where the symbolic system speaks to the neural system, where the neural system queries the symbolic, and where each owns a part of the answer.

Learning Objectives

- Recognise when hybrid is the right pattern.
- Architect the bridge between symbolic and neural components.
- Use rule engines for the rules; LLMs for the context.
- Use constraint solvers for the constraints; LLMs for the framing.
- Adapt hybrid patterns to African operational realities.

57.1 What Each Side Brings

A hybrid system is worth building only because the two sides it joins have genuinely complementary strengths and weaknesses — each is strong exactly where the other is weak.

The *symbolic* side — rule engines, constraint solvers, knowledge graphs, formal verifiers — brings determinism and proof. Its results are deterministic: the same input yields the same output every time. Its inference is explainable, producing a trace a human can follow step by step. It offers genuine guarantees: a constraint solver can prove that no solution exists, a rule engine can prove that a particular rule fired. It is auditable, and it does not hallucinate — it asserts only what its rules and data support. Its weaknesses are the cost of that rigidity: it is brittle on input variation, breaking on phrasings its rules did not anticipate; it carries a substantial knowledge-engineering cost to build and a

maintenance cost to keep current; and it cannot handle open-ended natural language at all.

The *neural* side — the language model — brings the opposite profile. It is fluent in natural language, generalises from examples to cases it was never explicitly given, handles open-ended tasks, and is accessible to users who could never write a formal query. Its weaknesses are the mirror of the symbolic side's strengths: it is probabilistic and can hallucinate, it is hard to audit, it is sensitive to phrasing, and it offers no guarantees about its output.

Set side by side, the two profiles are almost perfectly complementary, and that complementarity is the entire case for the hybrid pattern: it uses each side precisely where that side shines, and lets the other side cover what the first cannot. The neural side handles the language; the symbolic side owns the truth.

57.2 The Architecture

Figure 57.1 shows the canonical layout.

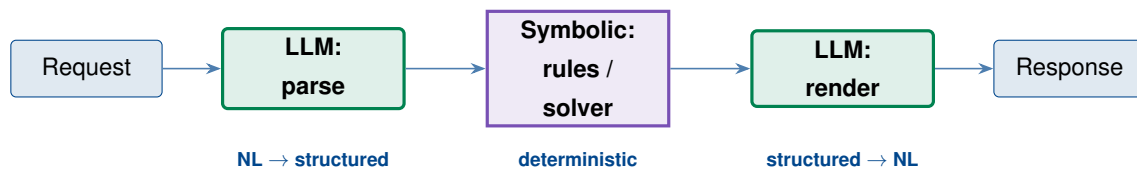


Figure 57.1: Hybrid symbolic-neural sandwich. LLM parses natural language into structured input; symbolic system computes deterministically; LLM renders the structured output back to natural language. The neural layer handles the language; the symbolic layer owns the truth.

57.3 Common Hybrid Patterns

Four hybrid patterns recur, and they differ in where the symbolic component sits relative to the neural one.

In the *rule engine in the middle* pattern, the symbolic system sits between two neural steps. A user asks a regulatory question; the LLM extracts the entities and intent; a rule engine determines which rules apply and what the outcome is; and the LLM renders the answer with citations. The division of labour is clean: the rule engine owns what the rule says, and the LLM owns the conversation around it. The *constraint solver behind the model* pattern has the same shape for a different kind of problem: a user asks for a schedule with employees, shifts, and constraints; the LLM extracts the constraints into a structured form; a solver finds a feasible solution, guaranteeing feasibility; and the LLM presents that solution in natural language.

The *knowledge graph as memory* pattern uses a structured graph as the LLM's grounded store: the graph holds entities, relationships, and provenance, and the LLM queries it for grounded retrieval rather than relying on its parameters — the subject of Chapter 58 in full. The *formal verifier as gate* pattern places the symbolic component after the neural one, as a checkpoint: the LLM generates a proposed action — code, a plan, a transaction — and a formal verifier such as a type checker, a theorem prover, or a policy validator checks it, so that only verified outputs proceed. The verifier

never lies, and the LLM never escapes past it. Across all four, the recurring principle is the same: the symbolic component is placed wherever the system needs a guarantee, and the neural component wherever it needs fluency.

57.4 The Bridge

The bridge between the two sides — the point where the neural system speaks to the symbolic system and back — is where most of a hybrid system’s engineering concentrates, and it has three components.

The first is schema translation. The LLM produces structured outputs — JSON, S-expressions, function calls — that the symbolic system consumes as input, and schema discipline is critical here, because a malformed structured output breaks the symbolic side outright. The symbolic system cannot interpret a near-miss the way a human reader could; the structure either parses or it does not, and the bridge must guarantee that what the LLM emits is what the symbolic system can accept. The second component is the tool-use protocol as a general-purpose bridge: function calling and MCP are exactly the bridge a hybrid system needs, with the symbolic system exposing its capabilities as tools and the LLM invoking them — which means the tool-use patterns of the earlier chapters are directly the mechanism by which neural and symbolic components are joined.

The third component runs in the other direction: constraint communication. When the symbolic system fails — no solution exists, the rules conflict — it must explain why in a form the LLM can render back to the user. A solver that reports only “infeasible” gives the LLM nothing to work with; a solver that reports which constraints conflicted gives the LLM something it can turn into a helpful explanation. Symbolic systems with rich, structured error reporting bridge well; those that fail silently or opaquely do not, and the quality of a hybrid system’s failure behaviour is largely the quality of this return path.

57.5 Use Cases

Four use-case families recur for hybrid symbolic-neural systems, and each is a domain where the symbolic side’s guarantees are not optional.

Regulated industries — banking, insurance, healthcare, legal — are the clearest fit: these are domains where rules must be honoured exactly, and a hybrid system puts the rules in a rule engine and the conversation in the LLM, which makes compliance structurally enforceable rather than something the model is merely trusted to respect. *Scheduling and optimisation* — workforce scheduling, supply-chain optimisation, routing — is the constraint-solver case: the natural-language request goes to the LLM, the hard constraints go to a solver, and the results return through the LLM, with feasibility guaranteed by the solver rather than hoped for. *Code generation with guarantees* has the LLM propose code and a type checker, a test runner, and sometimes a theorem prover verify it, so that only verified code reaches users. And *knowledge-intensive question answering* — domain-specific Q&A whose source of truth is a structured knowledge base — has the hybrid system retrieve from the structured source and render the answer in natural language. What unites the four is that in each, an error the

neural side would make silently is one the symbolic side catches with certainty, and that certainty is the reason the hybrid pattern is worth its added complexity.

57.6 The African Perspective

Hybrid symbolic-neural systems are an especially good fit for African deployments, for four reasons that reinforce one another.

The first is regulated-industry fit. African banking, insurance, mining, healthcare, and public-sector AI operate under strict regulatory environments, and the hybrid pattern — a rule engine for compliance, an LLM for accessibility — is well suited to exactly those settings, because it makes the compliance structural rather than trusting the model to be careful. The second is cost. The symbolic side reduces the load on the neural side, and a smaller, cheaper LLM paired with a strong symbolic system can match the quality of a deployment built on a larger LLM alone — which makes the hybrid pattern cost-effective under African unit economics, where every avoided frontier-model call matters.

The third is audit and sovereignty. A hybrid system is easier to audit than a pure-neural one, because the symbolic side is auditable by construction — its inference traces and rule firings are inspectable — and African regulators, under POPIA, NDPA, and sector-specific regimes, increasingly require precisely that auditability. The fourth is the constraint to be honest about: hybrid systems demand knowledge engineering. Rule engines and knowledge graphs must be built and maintained, and that requires both domain expertise and patient capital. African organisations that have strong domain expertise and the patience to invest can build durable, well-fitted hybrid systems; those that lack either will struggle, because the symbolic side is not something a model can generate for them. The hybrid pattern is a strong fit for African regulated industries — but it is a fit that has to be earned with knowledge-engineering investment, not merely adopted.

57.7 Recurring Failure Patterns

- **Schema mismatch.** LLM produces structures the symbolic system cannot parse.
- **Symbolic brittleness.** Symbolic side rejects valid variations that the LLM produced.
- **Render hallucination.** LLM's render step contradicts the symbolic system's findings.
- **Knowledge-base rot.** Rules or graph fall out of date; system gives confidently wrong answers.
- **Bridge as bottleneck.** All complexity migrates into the bridge layer; debugging becomes forensic.

57.8 What This Chapter Established

Hybrid symbolic-neural systems compose deterministic symbolic components (rule engines, constraint solvers, knowledge graphs, formal verifiers) with fluent neural components (LLMs). The neural side handles language; the symbolic side owns truth. The bridge is where engineering concentrates. Use cases include regulated industries, scheduling/optimisation, code-with-guarantees, and knowledge-intensive Q&A. African contexts foreground regulated-industry fit, cost efficiency through smaller

LLMs paired with strong symbolic, audit and sovereignty advantages, and the knowledge-engineering investment hybrid demands.

Chapter 58

Knowledge Graph Augmentation

“Vector search is similarity. Knowledge graphs are structure. Production systems need both.”

—a recurring observation in enterprise RAG

Chapter Overview

Knowledge-graph augmentation grounds LLM outputs in structured entity-relation knowledge: graph nodes represent entities (people, products, documents, events); edges represent relationships (employs, supplies, references); attributes carry properties; provenance carries authority. The architecture complements vector-based RAG: vectors find semantically similar passages, graphs traverse structured relationships. Combined, they cover queries that neither alone can.

Learning Objectives

- Distinguish knowledge graphs from vector indices and recognise when both are needed.
- Architect knowledge-graph augmentation in retrieval-and-generation pipelines.
- Address graph construction, maintenance, and querying.
- Manage entity resolution and provenance.
- Adapt knowledge-graph patterns to African contexts.

58.1 What Graphs Add

Vector indices answer “what is similar to this question?” Knowledge graphs answer “what is connected to this entity?” The questions are distinct.

58.1.1 Queries That Need Graphs

The following elements define this concern and distinguish it from the adjacent ones.

- Multi-hop questions over entity relationships.
- Questions about specific entities with structured attributes.
- Questions requiring provenance: “who said this and when?”

- Questions involving negation, exclusion, or non-existence.
- Aggregate queries over typed collections.

58.1.2 Queries That Need Vectors

The following elements define this concern and distinguish it from the adjacent ones.

- Open-ended semantic similarity.
- Long-form text where entities are not the unit.
- Domains where the schema is too fluid for graphs.

Most production systems benefit from both.

58.2 The Architecture

Figure 58.1 shows graph-augmented retrieval.

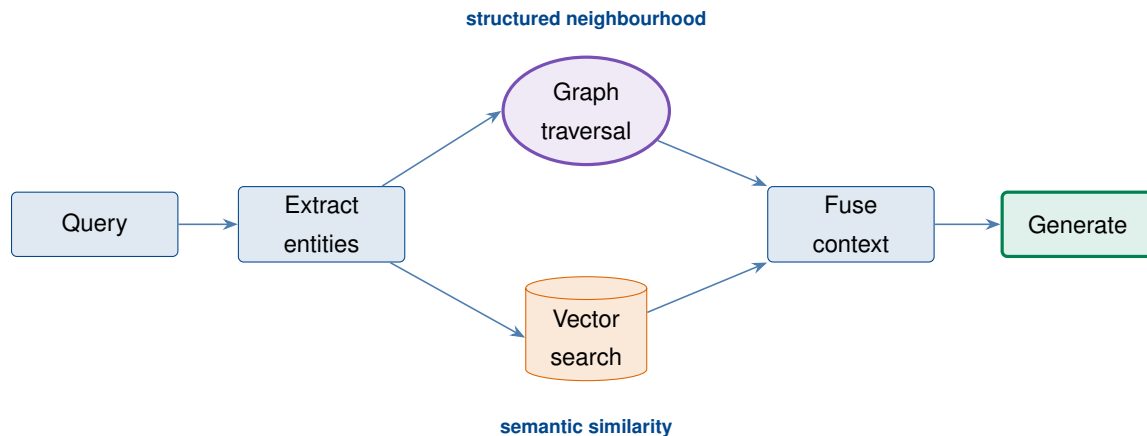


Figure 58.1: Graph-augmented retrieval. Query entities extracted; graph traversal returns the structured neighbourhood (related entities, attributes, provenance); vector search returns semantically similar passages; the two are fused into context for generation.

58.2.1 Entity Extraction

The first step: extract entities from the query. A specialised model or LLM call identifies entities and links them to graph nodes. Quality of this step bounds quality of the whole pipeline.

58.2.2 Graph Traversal

From the seed entities, the system traverses to related entities. Common patterns: 1-hop neighbourhood, multi-hop with filter, type-constrained traversal, path-finding between entities.

58.2.3 Fusion

Graph-retrieved context (structured) and vector-retrieved context (passages) are combined. Production patterns: graph context as preamble, vector passages as evidence; both presented to the model with explicit source labels.

58.3 Building the Graph

A knowledge graph is built by some combination of three methods, and the combination matters more than any single method.

Manual curation has domain experts define the entities, relationships, and ontology by hand. It produces the highest quality, it is durable, and it is expensive — it does not scale to a large corpus on its own. *Extraction from text* uses LLMs or specialised extractors to derive entities and relationships from documents automatically. It is far cheaper and it is the only method that scales to a large corpus, but its quality is lower — an automated extractor makes mistakes a human curator would not. *Integration with existing sources* recognises that many organisations already hold structured data — CRM records, ERP systems, product catalogues, organisational charts, document metadata — and that a knowledge graph is often best built by integrating those existing sources rather than by deriving everything from scratch.

In practice, production systems use all three in a deliberate blend: existing structured sources provide the reliable backbone, LLM extraction provides breadth across the document corpus, and expert curation provides accuracy at the points where accuracy matters most. A graph built from any single method alone is either too small, too unreliable, or too expensive; the hybrid construction is what makes a graph both large and trustworthy.

58.4 Entity Resolution

Knowledge graphs depend on identifying when two strings refer to the same entity.

- Exact-match for canonical identifiers.
- Fuzzy-match for name variations.
- Embedding-based for semantic equivalence.
- Human review for ambiguous cases.
- Multiple aliases tied to one canonical node.

Entity resolution is hard. Most graph failures trace back to entity-resolution mistakes.

58.5 Provenance

Each node and edge carries provenance: where the assertion came from, when it was made, who or what asserted it. Provenance enables:

- Citation in generated answers.
- Selective retraction when a source is found unreliable.
- Audit trails for regulated domains.
- Multi-source comparison.

A graph without provenance is a graph that cannot be trusted.

58.6 GraphRAG and Variants

GraphRAG, introduced by Microsoft Research, builds a graph from documents at indexing time, then queries it at retrieval time. The approach handles “global” questions (themes, trends) better than vector RAG alone.

Variants and adjacents: HippoRAG (memory-inspired retrieval), LightRAG, Neo4j+LLM integrations. All extend the same core insight: structured retrieval complements similarity retrieval.

58.7 Operating Considerations

Operating a knowledge graph in production turns on three considerations, all of which follow from the graph being a living artefact rather than a fixed one.

The first is graph maintenance. The graph describes a world that changes — documents are revised, organisations restructure, products evolve — and so the graph must be updated to track that change. Update pipelines must propagate those changes into the graph, and the discipline matters because stale graph data is more harmful than missing data: a gap in the graph produces no answer, but a stale assertion produces a confidently wrong one. The second consideration is query latency. Graph traversal can be slow, particularly for deep or unbounded traversals, and keeping latency within budget requires caching, appropriate indexing, and bounded traversal depth — a traversal that is allowed to wander the whole graph is a traversal that will not return in time.

The third is schema evolution. The graph’s schema — its entity types and relationship types — evolves as the domain understanding deepens, and a schema change can break the consumers that depended on the old schema. The same migration discipline that applies to any other interface applies here: schema changes are versioned, reviewed, and rolled out with the consumers in mind, so that an entity-type rename does not silently break the retrieval pipeline that queried for the old name.

58.8 The African Perspective

Knowledge graphs carry distinctive considerations in African deployments, across language, sources, cross-border structure, and storage.

The first is multilingual entities. A single entity — a person, a place, an organisation — may appear in several languages within one deployment: in English, French, Arabic, Swahili, or a vernacular name, sometimes all referring to the same node. Entity resolution, the discipline of recognising when two strings denote the same entity, must therefore cross language boundaries rather than matching within a single language, because an entity that resolves to two nodes — one per language — fragments exactly the structured neighbourhood the graph exists to provide.

The second is local knowledge sources. The authoritative structured sources differ per country — government registries, regulatory databases, sector-specific authorities — and a knowledge graph for an African deployment integrates these locally distinctive sources rather than assuming a single canonical

set. The third is cross-border entities: a pan-African operation tracks entities that span jurisdictions, where a regulated person in Kenya may also operate in Uganda, and the graph must capture that span and reflect both regulatory regimes against the single entity. The fourth is sovereignty in storage: a graph database holding personal data is subject to data-residency law, and a graph spanning several jurisdictions is therefore sometimes federated — the schema and query layer shared, the data localised per jurisdiction — rather than centralised in one store. The graph patterns do not change for African deployments; their entity resolution must cross more languages, their sources are more locally specific, and their storage must respect more borders.

58.9 Recurring Failure Patterns

- **Entity-resolution failure.** Same entity appears as multiple nodes; queries miss context.
- **No provenance.** Graph answers without citations; trust erodes.
- **Stale graph.** Updates not propagated; confidently wrong answers.
- **Schema sprawl.** Too many entity types; ontology unmaintainable.
- **Vector-only retrieval.** Graph available but unused; structural queries fail.

58.10 The Working Position

Knowledge graphs ground LLM outputs in structured entity-relation knowledge. They complement vector retrieval: vectors for semantic similarity, graphs for entity neighbourhoods and provenance. The architecture comprises entity extraction, graph traversal, vector retrieval, and fusion. Construction blends curation, extraction, and integration. Entity resolution and provenance are the durable foundations. African contexts add multi-lingual entity resolution, locally distinctive sources, cross-border entities, and federated storage for sovereignty.

Chapter 59

Synthetic Data Generation Pipelines

“The model is now both the bottleneck and the way past it.”

—a recurring observation in AI training pipelines

Chapter Overview

Synthetic data generation uses models to produce the data that trains, evaluates, or fine-tunes other models. The pattern covers bootstrap data for new applications, evaluation sets for emerging capabilities, distillation pipelines from large teachers to small students, and edge-case generation for robustness. Done well, synthetic data extends what is possible. Done poorly, it amplifies the biases and limitations of the generating model.

Learning Objectives

- Recognise when synthetic data is appropriate and when it is not.
- Architect generation pipelines with quality control.
- Manage the bootstrap problem (data, model, evaluation).
- Build distillation pipelines from large to small.
- Adapt synthetic-data strategies to African contexts.

59.1 When Synthetic Data Helps

Real data is often scarce, expensive, sensitive, or stale. Synthetic data addresses each:

- **Scarce.** Edge cases by definition rare; deliberately generated.
- **Expensive.** Human labelling at scale impractical; synthetic alternatives cheaper.
- **Sensitive.** Real customer data has privacy constraints; synthetic equivalents have fewer.
- **Stale.** Real data reflects past patterns; synthetic data can simulate future scenarios.

59.2 When Synthetic Data Hurts

- When the generating model’s biases get baked into the trained model.

- When synthetic data is mistaken for real signal in evaluation.
- When the distribution of synthetic data does not match the distribution of real production use.
- When the synthetic data covers what is easy to generate rather than what matters.

59.3 The Architecture

Figure 59.1 shows a synthetic data pipeline with quality control.

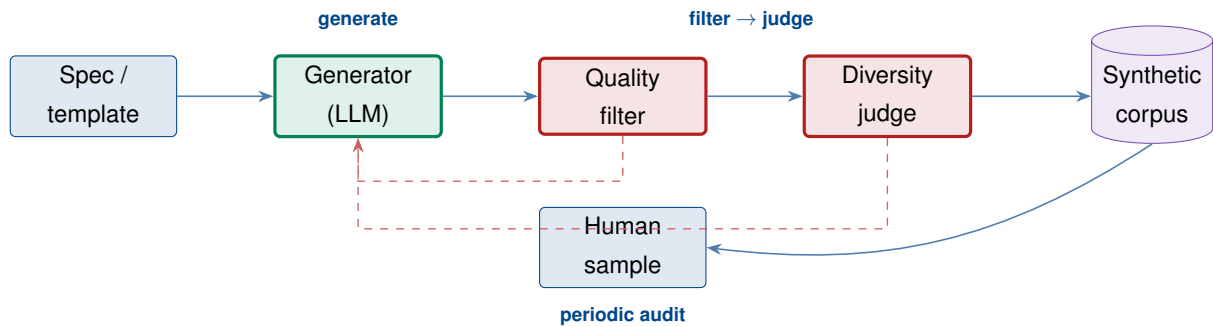


Figure 59.1: Synthetic data pipeline. Generator produces candidates from a spec; quality filter rejects malformed or low-quality; diversity judge ensures coverage; surviving examples enter the corpus; periodic human sampling audits the result.

59.4 Generation Patterns

Five generation patterns recur, and they differ in how much structure they impose on what the model produces.

Template expansion defines the structure up front — a template with slots — and has the LLM fill in the specifics. It is predictable and covers known distributions well, at the cost of rigidity: it generates only what the template’s shape allows. *Self-instruct* is the opposite end of the structure spectrum: the LLM generates both the instructions and the responses, and techniques such as Self-Instruct and Evol-Instruct bootstrap a full instruction-tuning dataset from a small human-written seed. *Distillation* has a large teacher model produce outputs and a small student model learn to mimic them, so that the student inherits the teacher’s behaviour at a fraction of the inference cost — a pattern important enough to warrant its own treatment below.

Adversarial generation turns the generator against the system: a generator tries to produce content that breaks a downstream classifier or model, and the failures it finds are added to the training set, improving robustness over successive rounds. *Persona-based generation* pursues diversity by having the model generate as a specific persona — a frustrated customer, a non-native speaker, a domain expert — so that the corpus reflects a range of voices rather than a single default one. The five are not exclusive; a production pipeline often uses several, template expansion for the predictable bulk and persona-based or adversarial generation to widen the corpus where the predictable bulk leaves gaps.

59.5 Quality Control

A synthetic-data pipeline without quality control amplifies the generating model's flaws; the four quality-control mechanisms are what keep the corpus better than its generator.

Filtering is the first pass: schema validation, length bounds, language detection, toxicity classifiers, and repetition checks reject the malformed and low-quality candidates before they reach the corpus. Filtering should be aggressive, because the cost of a bad example in training data is high — a flawed example does not merely fail to help, it actively teaches the model the wrong thing. *Diversity* control is the second: generated examples are clustered, the corpus is sampled to keep it balanced, and generation is rejected when similar examples already dominate. Without diversity control, the model trains on a narrow slice of whatever was easiest to generate, and the corpus's apparent size hides its real narrowness.

Human sampling is the third mechanism: a randomly sampled fraction of the synthetic data is reviewed by humans, and the review catches the systematic failures the automated filters miss — the failure modes no classifier was written to detect. The sampling rate is balanced against budget, but it is never zero, because a pipeline with no human in the loop has no way to discover the failures it was not designed to catch. *Round-trip verification* is the fourth, available whenever a generated example can be checked by an independent method: code that can be compiled, a calculation that can be executed, a fact that can be looked up. Where round-trip verification is possible it is the strongest filter of all, because it removes the lowest-quality output against ground truth rather than against another model's judgement.

59.6 The Bootstrap Problem

Synthetic data, an evaluation suite, and a model all depend on each other. The bootstrap:

1. Start with a small seed: a few hundred human-curated examples.
2. Build an initial evaluation suite from the seed.
3. Train an initial model on the seed.
4. Use the model to generate synthetic data.
5. Update the evaluation suite with hand-curated additions.
6. Train an improved model on combined human + synthetic data.
7. Iterate.

Each iteration is risky: errors in the model propagate to the synthetic data; errors in synthetic data propagate to the next model. Discipline of independent evaluation breaks the cycle.

59.7 Distillation Pipelines in Detail

Distillation deserves separate treatment because it has become economically central to how capable AI is deployed at scale.

The approach is a four-step pipeline. A teacher — a large, capable model, frontier or fine-tuned — is the source. The pipeline generates not merely the teacher's answers but its response distribution, capturing token probabilities or rationales rather than only the final output, because the richer signal is what the student learns most from. A student, a smaller model, is then trained to match the teacher's behaviour. And the student is evaluated against the teacher on held-out tasks, to measure how much of the teacher's capability actually transferred.

What transfers is uneven, and being precise about it matters. Surface behaviour transfers well: a distilled student matches its teacher closely on common tasks. Edge-case handling, novel reasoning, and tool use transfer worse — the student diverges from the teacher precisely on the uncommon cases, which is where the teacher's extra capacity was doing the most work. A distilled model is therefore not a smaller copy of its teacher; it is a model that matches the teacher on the ordinary and falls away on the exceptional, and a deployment must know which of those its workload consists of.

The economic significance is what makes distillation central. A student model ten times smaller costs roughly ten times less to serve, and if it preserves ninety-five per cent of the teacher's quality, the economic case is overwhelming — the deployment pays a tenth of the serving cost for nearly all of the capability. Distillation is, in a sentence, what makes capable AI affordable at scale, and it is the reason synthetic data generation is not a niche technique but a load-bearing part of the production AI economy.

59.8 The African Perspective

Synthetic data generation carries distinctive weight in African deployments, because the data scarcity it addresses is more acute on the continent than in the markets where the major models were trained.

The clearest case is local-language data scarcity. African languages have far less training data than the major world languages, and synthetic generation — with careful human anchoring to keep it honest — is one of the few practical paths to coverage for them. Initiatives such as Masakhane and Lelapa AI work directly on this frontier, building the language resources that the major datasets omit. Related is domain-specific bootstrapping: African enterprises often lack the in-domain data needed to fine-tune a model, and synthetic generation seeded with a small expert-curated corpus lets them bootstrap from that seed to a usable dataset rather than being blocked entirely by the absence of in-domain data.

Cost makes distillation especially attractive. The distillation pipelines of this chapter deliver frontier-quality outputs at small-model serving cost, which matches African unit economics closely — the same economic case that makes distillation central everywhere is sharper where serving budgets are tighter. The final consideration is sovereignty. When synthetic data is generated by sending prompts derived from real data to Northern frontier models, the real data crosses a border, and the data-residency implications of that crossing surface immediately. On-premise or in-region generation

— producing the synthetic data using models hosted within the jurisdiction — preserves sovereignty, and for synthetic-data pipelines built on sensitive real-data seeds, where the generation happens is as much a part of the architecture as how it happens.

59.9 Recurring Failure Patterns

- **Generator-bias amplification.** Whatever the generator believes, the trained model now believes more strongly.
- **Diversity collapse.** Many examples; little real variety.
- **Distribution mismatch.** Synthetic data does not look like production data.
- **No human anchoring.** Pipeline operates without periodic human review; quality drifts.
- **Evaluation contamination.** Synthetic data inadvertently appears in the eval set.

59.10 A Note Before Moving On

Synthetic data generation extends the reach of AI when real data is scarce, expensive, sensitive, or stale. Pipelines combine generation, filtering, diversity judgement, round-trip verification, and human sampling. Distillation is the economically significant variant: frontier behaviour at small-model cost. The bootstrap problem (data-model-eval interdependence) demands disciplined independent evaluation. African contexts add particular weight to local-language data scarcity, domain-specific bootstrapping, cost-sensitive distillation, and sovereignty-aware generation.

Continual Learning and Model Update Pipelines

“A model that does not update is a model that ages. The architecture is the schedule.”

—a recurring observation in production ML

Chapter Overview

Continual learning addresses a reality that every production model eventually meets: models age. The data distribution the model was trained on drifts away from the data it now sees, the concepts it predicts shift their meaning, production reveals capability gaps that were invisible in development, and the foundation models the system depends on are themselves revised by their providers. Model update pipelines are the architecture that answers this ageing — drift detection, retraining triggers, fine-tuning and adapter workflows, deployment orchestration, and rollback infrastructure, assembled into a loop. Done well, continual learning keeps a system current with the world it operates in. Done poorly, it becomes a source of regressions, instability, and operational fragility, in which each update is as likely to break the system as to improve it.

Learning Objectives

- Recognise the kinds of drift — distribution, concept, capability, vendor — that demand model updates, and the signals that reveal each.
- Architect retraining pipelines that are triggered by drift signals rather than by the calendar alone.
- Choose deliberately between full retraining, fine-tuning, adapter methods, and prompt-level updates for a given change.
- Manage catastrophic forgetting so that an update which improves one capability does not silently erase another.
- Adapt continual-learning pipelines to African data, economic, and connectivity realities.

60.1 Why Models Age

A model is a fixed artefact trained against a moment, and the world does not hold still after that moment passes. Four distinct forces pull a deployed model away from the conditions it was trained for, and a continual-learning architecture has to be designed against all four, because they degrade a system in different ways and reveal themselves through different signals. Naming them separately is the first step, because an update pipeline tuned to catch one of them will quietly miss the others.

60.1.1 Distribution Drift

Distribution drift is the slow movement of the production inputs away from the training distribution. User queries follow trends and seasons, the products and entities a system reasons about evolve, and the language people use shifts under the system over months and years. A model trained on last year's data is not wrong in any single case, but it underperforms in aggregate on this year's inputs, because the inputs as a population have moved. The drift is gradual and rarely announces itself; it is felt as a slow erosion of quality rather than a sudden break, which is exactly what makes it dangerous to leave unmonitored.

60.1.2 Concept Drift

Concept drift is subtler and more serious than distribution drift, because here it is not the inputs that move but the meaning of the categories and outcomes the model predicts. Fraud patterns shift as adversaries adapt, regulatory definitions are updated by legislators, and clinical guidelines are revised by the bodies that issue them. The model's task itself has moved underneath it: the same input now has a different correct answer than it did at training time. Concept drift cannot be corrected by simply adding more data of the old kind, because the old data now teaches the wrong thing; it requires the model to be updated against the new definition of the concept.

60.1.3 Capability Gaps

A capability gap is not a drift at all but a flaw that was present from the start and only becomes visible in production. The model never handled certain cases well, and the diversity of real production traffic is what reveals which cases those are — the long tail of inputs that no development test set thought to include. Continual learning treats these revealed failures as a resource: each one is a concrete, real example of where the model falls short, and incorporating those examples into the next update is how the system's weak spots are deliberately strengthened rather than left to be rediscovered as incidents.

60.1.4 Vendor Updates

The fourth force is one that systems built on foundation models face and self-trained systems do not: the provider revises the model. A foundation-model provider releases a new version, deprecates an old one, or adjusts the behaviour of an existing endpoint, and an application built on that model inherits the change whether or not it asked for it. The application may need to adapt its prompts to the new model's behaviour, re-run its evaluations to confirm nothing regressed, or update a fine-tune that was built against the previous version. Vendor updates are the one ageing force the organisation does not control the timing of, which makes readiness for them a standing requirement rather than a scheduled task.

60.2 The Update Pipeline

The architecture that answers model ageing is a pipeline, and its defining property is that it is a loop rather than a line — it runs continuously, with the output of production feeding back to shape the next version. Figure 60.1 shows the canonical shape, and the stages it names are developed across the rest of this chapter. The pipeline begins in production itself, because production is where both the drift signals and the curation data originate; an update pipeline disconnected from production is an update pipeline working from guesses.

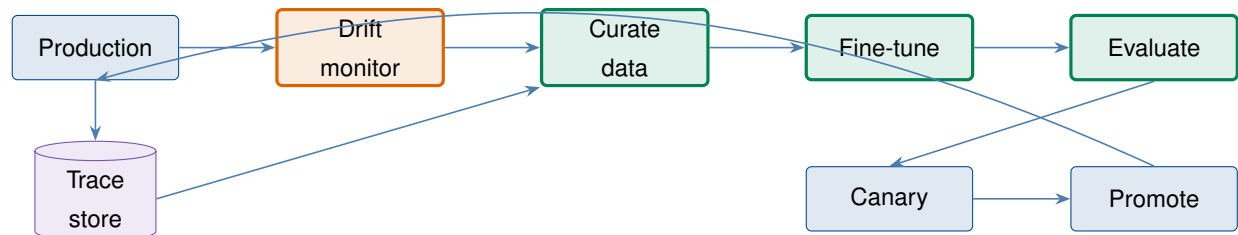


Figure 60.1: Continual learning pipeline. Drift monitoring fires on production signals; traces from the production trace store feed curation; curated data fine-tunes the model; evaluation gates progression; canary deployment carries the candidate; promotion closes the loop back to production.

The loop runs as follows. Drift monitoring watches production and fires when one of the ageing forces crosses a threshold. The trace store, accumulating the system’s real production behaviour, supplies the raw material for curation, where a focused update dataset is assembled. That dataset feeds a fine-tune or adapter training run, whose result is gated by an evaluation suite that must confirm the candidate is genuinely better and not merely different. A passing candidate is carried into production behind a canary, exposed to a slice of traffic under comparison, and only then promoted to full traffic — at which point the loop has closed and the next cycle’s monitoring begins. Each stage is a safeguard, and the chapter’s argument is that an update pipeline missing any one of them ships regressions it cannot see.

60.3 Drift Detection

Drift detection is the trigger of the whole pipeline, and it works by watching several signals at once, because each signal catches a different kind of ageing at a different speed. A system that monitors only one of them is blind to the drift the others would have caught, and the discipline is to run all four and treat them as complementary readings rather than competitors.

60.3.1 Input Drift

Input drift detection runs statistical tests on the distribution of production inputs and fires when that distribution moves away from the training distribution. The standard instruments are KL divergence, the population-stability index, and embedding-shift detection that measures how far the centroid of recent inputs has moved in embedding space. Input-drift signals are valuable because they are leading indicators: they can fire before quality has measurably degraded, giving the pipeline warning that the model is now operating outside the conditions it was trained for, even though no user has yet

complained.

60.3.2 Output Drift

Output drift watches the distribution of what the model produces rather than what it consumes: refusal rates rising over time, confidence distributions shifting, output lengths creeping up or down. It is often the earliest signal that something is wrong, because a change in the model's behaviour shows up in its outputs before it shows up in any quality metric. Output drift is also cheap to monitor, since it requires only the outputs the system already produces, with no judge model or labelled data — which makes it a sensible first signal for any pipeline to instrument.

60.3.3 Quality Drift

Quality drift is the signal that matters most and arrives slowest. It is measured through online judge metrics, aggregated user feedback, and downstream conversion or task-completion rates — the things that actually express whether the system is serving its users well. Because it is slow and noisy, it is tempting to discount, but it is the signal closest to the truth: input and output drift indicate that something has changed, while quality drift indicates whether that change has actually hurt anyone. A pipeline must be patient enough to trust this signal even though it lags the others.

60.3.4 Cost-Per-Useful-Output Drift

The fourth signal is the economic one, and it is the most often overlooked. Cost-per-useful-output drift watches whether the cost of producing a genuinely useful response is rising — whether the system is spending more model calls, more tokens, or more retries to achieve the same outcome. It frequently signals model degradation before the pure quality metrics catch it, because a model that has begun to struggle often compensates with longer outputs and more retries, which raises cost while the visible quality holds. Watching the economics is therefore watching quality through a different and sometimes faster lens.

60.4 Update Granularity

When drift detection has fired and curation has assembled the data, the pipeline faces a choice of how large an intervention to make. The options form a spectrum from the heaviest and most expensive to the lightest and most reversible, and the discipline is to reach for the lightest option that will actually fix the problem, escalating only when the lighter levers are genuinely insufficient.

60.4.1 Full Retraining

Full retraining starts from scratch with the current data, discarding the existing model and building a new one. It offers maximum freshness, because nothing of the stale model carries forward, but it carries maximum cost, both in compute and in the time the training run takes. In practice it is reserved for foundation-scale model providers, whose business is producing models; for an enterprise building on those models, full retraining is almost never the right granularity, because the lighter options below achieve currency at a small fraction of the cost.

60.4.2 Fine-Tuning

Fine-tuning continues training the existing model on new data rather than starting over, which makes it far cheaper than full retraining and far faster to run. Its characteristic risk is catastrophic forgetting: because the update only exercises the new data, capabilities that the new data does not represent can erode as the model's weights shift toward the new examples. Fine-tuning is therefore powerful but not free of danger, and using it well means pairing it with the forgetting mitigations developed later in this chapter rather than treating it as a simple, consequence-free refresh.

60.4.3 Adapter Methods

Adapter methods — LoRA, QLoRA, prompt-tuning and their relatives — train a small set of new parameters on top of a frozen base model rather than modifying the base at all. This makes them cheap to train, composable in that several adapters can coexist for different specialisations, and inherently resistant to catastrophic forgetting, because the base model is untouched and its capabilities are preserved by construction. For an organisation that updates frequently, adapter methods are the standard production approach, and the rest of this chapter treats them as the default granularity that the others are measured against.

60.4.4 Prompt-Level Updates

For Software 3.0 systems, the lightest update of all is often not a change to the model but a change to the prompt or template that drives it. A prompt-level update is the cheapest, the fastest, and the most reversible intervention available — it ships as a configuration change and rolls back the same way. It is the first lever a pipeline should reach for, and a surprising fraction of the drift a system experiences can be answered at this level alone, without ever touching the model. Escalating to fine-tuning or adapters before exhausting the prompt-level option is a common and expensive mistake.

60.5 Catastrophic Forgetting

Catastrophic forgetting is the central hazard of any update that modifies a model's weights. When a model is fine-tuned on new data, its weights move to fit that data, and capabilities that the new data does not exercise can be silently degraded or lost in the process — the model gets better at the new thing and quietly worse at an old thing nobody was testing. The danger is precisely that it is silent: the update looks like a success on the new data it was built for, and the regression elsewhere is invisible until a user finds it.

Four mitigations, used together, keep forgetting under control. The first is to mix old and new data in the fine-tuning corpus, so that the training run continues to exercise the old capabilities rather than abandoning them. The second is to prefer parameter-efficient methods such as LoRA, which preserve the base model by construction and so cannot forget what the base model knew. The third is to evaluate every update against the full historical evaluation set rather than only the new cases the update targeted, so that a regression in an old capability shows up as a failing evaluation before the update ships. The fourth is to maintain a deliberate regression suite of capabilities the system must not lose — a standing list of things to protect, checked on every update. Together these turn forgetting from an invisible risk into a monitored one.

60.6 Continual Adapter Pipelines

Adapter-based continual learning is the production sweet spot, and it is worth describing as a complete pattern because it is the shape most mature update pipelines converge toward. The pattern keeps the base model frozen and permanent, and treats adapters as the unit of update — small, cheap, composable, and individually versioned.

In the pattern, the base model is frozen and never modified, which means its capabilities are guaranteed stable across every update. Multiple adapters sit on top of it, each trained for a different specialisation — a domain, a language, a task. When new data arrives, it trains a new adapter or refines an existing one, a small and fast operation that touches nothing else. Old adapters are preserved rather than overwritten, so the behaviours they encoded remain available even after newer adapters are added. And a routing layer chooses the appropriate adapter per request, selecting the specialisation that fits the incoming work. The result is a system that can be updated continuously and cheaply, with no risk of catastrophic forgetting and with old behaviour always recoverable. The pattern combines naturally with model routing (Chapter 38) and with mixture-of-experts approaches (Chapter 52), because all three share the same underlying idea of composing a system from selectable specialised parts.

60.7 Operating Considerations

A continual-learning pipeline is not only a set of training stages; it is a standing operational practice, and four considerations govern whether that practice is sound. Each turns the pipeline from a mechanism that produces new models into a discipline that produces new models safely and accountably.

60.7.1 Update Cadence

Update cadence is the rhythm at which the pipeline runs, and it is a deliberate choice rather than a given. A quarterly cadence is conservative and low-risk; a monthly cadence is the common middle ground; a weekly cadence is aggressive and demands strong automation to be safe. The right cadence balances the operational cost and risk of each update against the rate at which the system actually drifts: a system in a fast-moving domain may need monthly updates to stay current, while a stable one is well served by a slower rhythm. Cadence should be set to the drift, not to a habit.

60.7.2 Reproducibility

Every update version must be reproducible, meaning that the exact data snapshot, hyperparameters, code, and infrastructure that produced it are recorded and can be re-created. Reproducibility is what makes a regression debuggable: when a new version misbehaves, the team can reconstruct exactly how it was built and compare it against the previous version's recipe. Without reproducibility, debugging a regression becomes forensic guesswork, because the artefact that failed cannot be reliably re-examined or rebuilt.

60.7.3 Lineage Tracking

Lineage tracking connects every production output back to the artefacts that produced it: the model version, the adapter version, the prompt version, and the data snapshot. The lineage must be queryable,

so that an investigator can take a specific bad output and trace it to precisely the combination of artefacts responsible. Lineage is the difference between an incident that can be diagnosed — this output came from that adapter trained on that data — and one that can only be described. It is the observability layer of the update pipeline.

60.7.4 Rollback

Every update ships behind a canary and with a tested rollback path, because no evaluation suite catches everything and some regressions only appear under real production traffic. Rollback paths must be genuinely tested rather than assumed to work, and the system should be able to run multiple versions simultaneously to support staged rollouts and fast reversion. An update pipeline whose rollback has never been exercised is a pipeline that will discover its rollback is broken at the worst possible moment.

60.8 The African Perspective

Continual learning carries distinctive weight in African deployments, across drift calibration, update economics, data residency, and vendor dependence.

60.8.1 Drift on Local Distributions

Drift detection that is calibrated against global or Northern data distributions can miss shifts that are locally significant, because a change confined to one African market or one language is small against a global baseline and large against the local one. African deployments need drift signals calibrated locally — per country, per language, per customer segment — so that a meaningful shift in a specific market triggers the pipeline rather than being averaged away into a global figure that still looks stable. The drift monitoring of a pan-African system must be stratified the way its evaluation is.

60.8.2 Adapter-First Strategy

The adapter-based approach fits African unit economics particularly well, and the fit is not incidental. Adapter training has a low compute cost, which matters where training budgets are tight; it preserves the base model, which avoids the cost and risk of full retraining; and it supports fast iteration, which lets a small team keep a system current without a large MLOps operation behind it. For African deployments the adapter-first strategy is therefore not merely one option among several but usually the right default, for the same cost reasons that run through this book.

60.8.3 Federated Continual Learning

Pan-African operations face data-residency constraints that prevent the data of several countries from being centralised for a single training run. Federated continual learning, combining the federated approach of Chapter 46 with adapter methods, resolves the contradiction: each region trains adapter updates on its own local data, the updates are aggregated without the underlying data ever moving, and the model improves at continental scale while each country's data stays within its borders. This is the architecture that lets a multi-country system learn from all its markets without violating the data-protection law of any of them.

60.8.4 Vendor-Update Coordination

African operations that build on Northern foundation-model providers are exposed to vendor-update timelines they do not control and may not even be notified of in advance. A provider's model release or behavioural change can shift an application's behaviour mid-quarter, and the African team has to detect and respond to that change rather than schedule it. The disciplined response is to treat every vendor model release as a trigger for regression evaluation, and to be ready to ship an adapter or prompt update to compensate for whatever the new model version changed — readiness, again, rather than a scheduled task.

60.9 Recurring Failure Patterns

The failure modes of continual learning recur across organisations, and each maps directly to a discipline this chapter has set out:

- **No drift monitoring.** The pipeline has no trigger, so quality degrades silently and the first signal of ageing is a user complaint — the slowest and most expensive signal of all.
- **Update without regression testing.** A new version is evaluated only on the new data it targeted, so it appears to be an improvement while it has quietly regressed capabilities the new data did not exercise.
- **Catastrophic forgetting.** A fine-tune ships without the mixing, parameter-efficient methods, and historical evaluation that would have caught the loss of an old capability before it reached production.
- **No lineage.** A bad production output cannot be linked to the model, adapter, prompt, and data that produced it, so the incident can be described but not diagnosed.
- **Vendor-update surprise.** A provider's model update changes behaviour mid-quarter with no notice, and the system has no standing readiness to detect the change or respond to it.

Each of these is the absence of a safeguard the chapter has argued for — drift monitoring, full-historical regression evaluation, the forgetting mitigations, lineage tracking, and vendor-update readiness — and together they make the same point: the update pipeline's stages are safeguards, and a pipeline that skips them ships regressions it cannot see.

60.10 The Takeaway

Continual learning is the discipline of keeping production models current against the four forces that age them: distribution drift, concept drift, capability gaps, and vendor evolution. The pipeline that answers those forces is a loop — drift detection, data curation, fine-tuning or adapter training, evaluation, canary deployment, rollback — and its defining property is that it runs continuously, with production feeding back to shape each next version.

Among the update granularities, adapter-based approaches dominate production practice, because they are cheap to train, composable across specialisations, and resistant by construction to the catastrophic

forgetting that threatens any weight-modifying update. The operating disciplines — deliberate cadence, reproducibility, queryable lineage, and tested rollback — are what make the pipeline safe rather than merely active. African deployments add locally calibrated drift detection, an adapter-first economics, federated continual learning under data-residency constraints, and standing readiness for vendor updates whose timing the organisation does not control. A model that does not update ages; the architecture of this chapter is the schedule that keeps it from ageing unseen.

Chapter 61

Cost Optimisation and FinOps for AI

“A model bill that surprises the CFO ends the experiment, regardless of what the model does.”

—a recurring observation in production AI economics

Chapter Overview

The economics of AI workloads differ materially from the economics of traditional cloud workloads, and the differences are not cosmetic. AI workloads are priced per token rather than per instance-hour; the gap between a frontier model and a small one can be a factor of ten to a hundred on the same task; memory economics turn on the KV cache rather than on storage; vendors apply markups on inference that have no equivalent in raw compute; and model prices fall rapidly and unpredictably as competition intensifies. FinOps for AI is the discipline of governing, attributing, optimising, and forecasting these costs. The discipline is newer than the workloads it governs, and most organisations are operating it with primitive tooling. This chapter develops its architecture: what to measure, how to attribute spend to the things that caused it, where the optimisation levers are, and how to organise the practice so that the bill never surprises anyone.

Learning Objectives

- Distinguish the cost drivers of AI workloads from the cost drivers of generic cloud workloads.
- Architect cost attribution that traces spend to teams, products, tenants, and use cases.
- Apply the optimisation levers — routing, caching, batching, quantisation, compression, output constraints — and understand how they compound.
- Forecast AI spend under volatile model pricing rather than assuming today’s prices hold.
- Adapt FinOps practice to African economic constraints, where it is a survival discipline rather than an optimisation.

61.1 AI Cost Drivers

Before a cost can be optimised it has to be understood, and the cost structure of an AI system is not the cost structure a cloud-cost practitioner is used to. Four categories of driver make up an AI bill,

and a FinOps practice that sees only the first of them — the visible token cost — will optimise the obvious while the hidden, infrastructure, and engineering costs grow unwatched.

61.1.1 Token Costs

Token cost is the headline driver and the one everyone sees first: input tokens multiplied by the input price, plus output tokens multiplied by the output price. The two prices are not equal — at frontier providers, output tokens typically cost three to five times what input tokens cost — which means the shape of a workload determines where its cost sits. A long-output workload such as report generation carries its cost on the expensive output side, while a long-input workload such as retrieval-augmented generation or code analysis shifts the weight onto the cheaper prompt side. Understanding this asymmetry is the first step, because it tells the practitioner which optimisation levers will actually move the bill for a given workload.

61.1.2 Hidden Costs

Around the visible token cost sits a ring of costs that are real, that accumulate, and that a naive FinOps practice misses entirely. These include the embedding generation that every retrieval pipeline depends on; the vector-database operations behind that retrieval; the inference cost of evaluation itself, since judge models running over production samples are themselves billed model calls; the parsing of images and documents on multimodal inputs; the execution cost of tools, some of which call external paid APIs; and the logging, tracing, and observability infrastructure that the system needs to be operable at all. None of these is the headline number, but together they can rival it, and a cost model that omits them understates the true bill by a margin that matters.

61.1.3 Infrastructure Costs

For systems that self-host inference rather than calling a provider's API, an entirely separate cost structure applies. The bill includes GPU rental or capital expenditure on the hardware itself, the networking between GPUs and to clients, the storage for model weights and for KV caches, and the idle-capacity headroom that must be paid for so the system can absorb load spikes. The full total cost of ownership of a self-hosted deployment often dwarfs the pure compute number that first attracts an organisation to self-hosting, and a FinOps practice that compares a self-hosted option to an API option on compute cost alone is comparing the wrong figures.

61.1.4 Engineering Costs

The fourth driver is the one most often left out of a FinOps model because it does not appear on a cloud invoice: the cost of engineering effort. A prompt that costs the same per call to serve but took an engineer a week to design and tune has a real and substantial cost, and an optimisation that saves a few cents per call while consuming weeks of engineering time may be a net loss. A FinOps practice that ignores engineering effort optimises the wrong thing — it will chase small per-call savings that cost more in salary than they ever recover in inference spend. The engineering cost belongs in the model.

61.2 The FinOps Architecture

FinOps for AI is best understood as a layered architecture, because each concern has its own owner, its own tooling, and its own cadence. Figure 61.1 shows the six layers. An immature practice runs only the top and bottom — visibility and reporting — and so can see and describe its spend but cannot attribute, optimise, govern, or forecast it. A mature practice runs all six, and the layers reinforce one another: attribution makes optimisation targeted, governance makes optimisation enforced, and forecasting makes the whole practice forward-looking rather than merely descriptive.

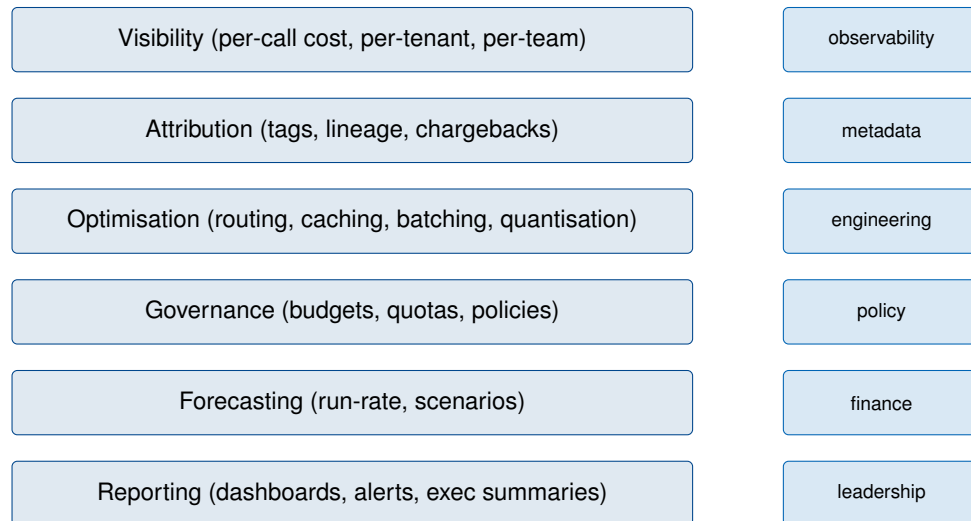


Figure 61.1: Six-layer FinOps architecture for AI. Each layer has its own owner and tooling. Mature practice runs all six; immature practice runs visibility and reporting only, and so can describe its spend but not control it.

The layers stack from the concrete to the strategic. Visibility, owned by the observability function, captures the raw per-call cost data without which nothing downstream is possible. Attribution, owned by whoever maintains the metadata, assigns each cost to the thing that caused it. Optimisation, owned by engineering, applies the levers that reduce the cost. Governance, owned by the policy function, sets and enforces the budgets and quotas. Forecasting, owned by finance, projects the spend forward under scenarios. And reporting, consumed by leadership, turns all of it into the dashboards, alerts, and summaries that let the organisation steer. The chapter develops the middle four layers in turn, because visibility and reporting are largely generic while attribution, optimisation, governance, and forecasting are where AI FinOps earns its name.

61.3 Attribution

Attribution is the layer on which everything else depends, and the reason is simple: without attribution, optimisation is impossible. A practice that knows its total spend but not whose spend it is cannot target an optimisation, cannot hold a team accountable for its costs, and cannot tell a growing bill apart from a growing problem. Cost has to be traced to the team, the product, the tenant, or the use case that produced it, and that tracing has to be built into the system rather than reconstructed after the fact.

Four mechanisms make attribution real. Every model call carries tags — a tenant identifier, a product, a feature, an environment — so that the cost of the call is labelled with its cause at the moment it is

incurred. Lineage connects an end-user request through the whole chain of model calls it triggered, so that a single user action can be costed in full even when it fanned out into many calls. Chargeback or showback assigns the attributed cost to the business units responsible, which turns cost from an abstract platform expense into a number each unit sees and owns. And per-feature dashboards put that number in front of the product managers who can actually act on it. Together these turn the bill from a single opaque figure into a structured map of where the money goes — and only a structured map can be optimised.

61.4 Optimisation Levers

Once attribution shows where the cost sits, the optimisation layer reduces it. Six levers recur, and their important property is that they compound: each one multiplies the savings of the others, so a system that applies several of them together achieves a reduction far larger than any one of them alone. The discipline is to apply the levers that fit the workload's shape — which the token-cost analysis above identifies — rather than reaching for all of them indiscriminately.

61.4.1 Routing

Routing sends simple queries to small, cheap models and reserves the expensive frontier models for the complex queries that genuinely need them, the pattern developed in full in Chapter 38. It is the single largest optimisation lever in most workloads, because the cost gap between a small model and a frontier one is so wide that moving even a modest fraction of traffic down to the smaller model produces a large saving. A workload that routes well is a workload that pays frontier prices only for the work that frontier capability was actually required for.

61.4.2 Caching

Caching avoids paying for the same computation twice, and in AI systems it takes several forms: prefix caching for the common system prompts that begin many requests, response caching for repeated identical questions, and embedding caching for documents that would otherwise be re-embedded on every indexing pass. The opportunity is real — cache hit rates of thirty to seventy per cent are achievable on production workloads with genuine repetition — and a hit is a call that costs nothing at all. A system with high repetition and no caching layer is leaving a large, easy saving unclaimed.

61.4.3 Batching

Batching improves the efficiency of inference by processing several requests together rather than one at a time. Continuous batching at the inference layer, developed in Chapter 47, raises GPU utilisation and so lowers the cost per request for self-hosted serving. Asynchronous application patterns extend the same idea upward, allowing the application itself to group requests that do not need to be served instantly. Batching is most valuable for self-hosted and high-volume workloads, where the efficiency gain translates directly into fewer GPUs paid for.

61.4.4 Quantisation

Quantisation reduces the numerical precision at which a model runs — INT8, FP8, or INT4 inference in place of full precision — which shrinks the model’s memory footprint and speeds its execution. The cost saving is typically a factor of two to four, and the quality cost varies by task: some tasks tolerate aggressive quantisation with no measurable degradation, while others are sensitive to it. The discipline is to quantise as aggressively as the task’s quality bar allows, measured against an evaluation set, rather than choosing a precision by default.

61.4.5 Prompt Compression

Prompt compression attacks the cost on the input side: a shorter prompt costs less to process, on every single call, for the life of the system. The techniques include removing few-shot examples once the model’s capability on the task is mature enough not to need them, trimming redundant instructions, and using the context window economically rather than filling it because it is available. Because the saving applies to every call, even a modest reduction in prompt length compounds into a substantial figure at production volume.

61.4.6 Output Constraints

Output constraints attack the cost on the expensive output side, where each token costs several times an input token. Enforcing maximum-token limits prevents a model from producing a longer response than the task requires; structured outputs that return data rather than verbose natural language are inherently shorter; and streaming with early termination stops generation as soon as the useful answer is complete. Because output tokens are the expensive ones, disciplined output constraints often produce a larger saving per token trimmed than input-side compression does.

61.5 Governance

Optimisation reduces the cost of the work the system chooses to do; governance controls what work it is allowed to do in the first place. Without governance, optimisation is a treadmill — the savings are real but a single ungoverned new use case can erase them overnight. Three governance mechanisms turn cost control from a one-time exercise into a standing constraint.

61.5.1 Budgets and Quotas

Budgets and quotas put a hard boundary around spend rather than only observing it. Per-team monthly budgets are enforced at the model gateway, with soft limits that trigger alerts when a team approaches its boundary and hard limits that throttle traffic when it crosses one. The enforcement point matters: a budget enforced at the gateway is a budget that cannot be exceeded, while a budget that exists only as a target in a spreadsheet is a budget that will be exceeded and discovered later. Quotas turn the budget from an aspiration into a mechanism.

61.5.2 Approval Workflows

Approval workflows govern the changes that will materially shift spend before they happen rather than after. A new use case expected to move the bill significantly goes through a review, and a production

rollout forecasts and gets approval for its cost impact as part of shipping. The purpose is to remove cost surprises by making the organisation decide, in advance and deliberately, to take on a new cost — so that a large bill is the result of a known decision rather than an unwelcome discovery at the end of the month.

61.5.3 Policy as Code

Policy as code encodes the organisation's cost rules in an enforceable, auditable form rather than leaving them as informal understandings. Rules such as “this use case may not use frontier models”, “this tenant has a monthly cap of one thousand US dollars”, or “no production system may use experimental models” are written as policy, evaluated at the gateway, and logged. Encoding the rules makes them consistent — every request is checked against the same policy — and auditable, so the organisation can demonstrate which cost controls were in force and prove that they were applied.

61.6 Forecasting

Forecasting is the layer that makes FinOps forward-looking. A practice that can only report last month's spend is always reacting; a practice that can project next quarter's spend can plan, budget, and warn before a cost materialises. AI forecasting is harder than generic cloud forecasting for one specific reason — model prices are genuinely volatile — and a forecast that ignores that volatility is a forecast that will be wrong.

61.6.1 Run-Rate

The run-rate is the baseline forecast: current spend extrapolated forward at the current rate of use. It answers the simplest and most important question — if nothing changes, what will the bill be — and it is the figure against which every scenario is measured. The run-rate is easy to compute and easy to underrate, but it is the anchor of the whole forecasting layer, because every more sophisticated projection is the run-rate plus an adjustment.

61.6.2 Scenarios

Scenarios are the run-rate modified by a specific anticipated change: if the new product launches, if a vendor raises its prices by twenty per cent, if the system moves a workload to a more capable frontier model. Each scenario produces a forecast band rather than a single number, because the inputs are uncertain. Running scenarios is what lets the organisation see the cost consequence of a decision before it commits to the decision, and so choose with the bill in view rather than discovering it afterward.

61.6.3 Vendor Price Volatility

The hardest input to any AI forecast is the trajectory of model prices, which have fallen by a factor of ten or more on some axes over the past two years as competition has intensified. A forecast that assumes today's prices hold may overestimate, because prices have a recent history of falling; a forecast that assumes the falls continue may underestimate when a vendor's pricing stabilises. The honest practice is to forecast in bands that explicitly bracket the price uncertainty, and to revisit the

forecast whenever the vendor pricing landscape moves — treating model price as a variable to track rather than a constant to assume.

61.7 The African Perspective

FinOps is consequential everywhere, but in African deployments it crosses a line: it stops being an optimisation discipline and becomes a survival discipline. The reasons span unit economics, currency, hosting choices, and cross-border structure.

61.7.1 Unit Economics as First-Class

Average revenue per user is structurally lower in many African markets than in Northern ones, and the cost per call has to match that lower revenue or the product simply does not work. African deployments often operate under cost ceilings five to ten times lower than their Northern equivalents, which means a cost overrun that a Northern product could absorb is, for an African product, the end of the product. FinOps is therefore first-class architecture in these deployments, not a later refinement — the cost discipline has to be designed in from the first day, because there is no margin to discover it later.

61.7.2 Currency Volatility

AI bills from the major providers are denominated in US dollars, while the revenue of an African deployment is earned in local currency, and the exchange rate between the two moves. A bill that was affordable at one exchange rate can become unaffordable at another without the system's usage changing at all. This makes currency a genuine FinOps variable: forecasts must include currency scenarios alongside usage scenarios, and an organisation with significant dollar-denominated AI spend may need to treat that exposure as something to hedge rather than to ignore.

61.7.3 Hosted versus Self-Hosted Economics

The choice between calling a provider's API and self-hosting an open-weight model has a different answer in African markets than in Northern ones. For workloads at sufficient scale, a self-hosted open-weight model running in an African cloud region can undercut the frontier APIs, and the capital-versus-operating-expense trade-off differs from the Northern default because both the cost structures and the available regions differ. The discipline is to run the comparison on full total cost of ownership — including the infrastructure and engineering costs of self-hosting — rather than assuming the API is cheaper because it is simpler.

61.7.4 Cross-Border Cost Coordination

A pan-African operation faces per-country budgets and per-country compliance, and its FinOps reporting has to be stratified by market to be meaningful. An aggregate continental cost figure can look healthy while hiding a specific market in which the unit economics have broken — the profitable markets average out the unprofitable one. Stratified reporting surfaces the market where the economics fail while there is still time to act, which an aggregate number conceals until the loss is large.

61.8 Recurring Failure Patterns

The failure modes of AI FinOps recur, and each maps to a layer of the architecture that has been skipped:

- **No attribution.** The total cost grows and no one can say whose it is, so no optimisation can be targeted and no team can be held to account — the attribution layer is missing.
- **Routing as aspiration.** The architecture diagram shows a router sending simple queries to small models, but in production everything goes to the frontier model, and the largest optimisation lever is never actually pulled.
- **Cache opportunity missed.** A workload with high repetition runs with no caching layer, paying full price for computations whose answers it has already produced.
- **Budget surprises.** There is no alerting, so the first the organisation hears of a cost overrun is the CFO's call — the governance layer is absent.
- **Vendor lock-in cost.** The system is so tightly coupled to one provider that the cost of migrating away is prohibitive, which leaves the organisation with no answer to a price increase.

Each pattern is the absence of something this chapter has argued for — attribution, real routing, caching, governance, and the vendor-neutral design that keeps migration cheap — and together they show that AI FinOps fails not through bad arithmetic but through missing layers.

61.9 In Short

FinOps for AI is a six-layer discipline: visibility, attribution, optimisation, governance, forecasting, and reporting. Its cost drivers — token costs, hidden costs, infrastructure, and engineering effort — differ materially from the drivers of generic cloud spend, and a practice that sees only the visible token cost optimises a fraction of the bill. Its optimisation levers — routing, caching, batching, quantisation, prompt compression, and output constraints — compound, so that a system applying several together saves far more than any one alone.

The discipline's forward-looking layers, governance and forecasting, are what keep cost control from being a treadmill: governance bounds what work the system may do, and forecasting projects the spend forward through the genuine volatility of model prices. In African deployments FinOps crosses from optimisation into survival, foregrounding unit economics that must be first-class from the first day, currency exposure that belongs in every forecast, hosted-versus-self-hosted choices that turn on full total cost of ownership, and cross-border reporting that must be stratified to reveal the market where the economics break. A model bill that surprises the CFO ends the experiment — and the architecture of this chapter is how an organisation makes sure it never does.

Chapter 62

Security Architecture for AI Systems

“The threat model is not yesterday’s web-application threat model with a model bolted on.”

—a recurring observation in AI security

Chapter Overview

AI systems extend the attack surface of a traditional application in ways that the established web-application threat model does not anticipate. Model weights are themselves valuable intellectual property and a theft target. Training data carries both privacy risk and extraction risk. Prompts are programmable behaviour, which means an attacker who can influence a prompt can influence what the system does. Tools give agents real capability, and capability can be abused. And the supply chain for models, adapters, and AI dependencies parallels the software supply chain, with its own provenance and tampering risks. Security architecture for AI is therefore not optional, and it is not the same thing as the guardrails of Chapter 54: guardrails enforce policy against ordinary use, while security defends against a deliberate adversary. This chapter develops the discipline as a layer that wraps every part of the architecture.

Learning Objectives

- Recognise the AI-specific threats and locate each one on the architectural surface it attacks.
- Architect defences across the model, data, prompt, tool, and supply-chain surfaces.
- Apply zero-trust principles to agent systems, treating agents as principals with identity and authorisation.
- Build detection and incident response oriented to AI-specific signals and scenarios.
- Adapt AI security to African regulatory regimes and threat realities.

62.1 The AI Threat Landscape

A security architecture is only as good as the threat model behind it, and the threat model for an AI system is genuinely new. It is not the web-application threat model with a model added on the side; it is a wider surface with its own classes of attack, because the model, the data, the prompts, the tools, and the supply chain are each attackable in ways a conventional application is not. Figure 62.1 maps

the principal threats onto the six architectural surfaces they attack, and the sections that follow take the surfaces in turn.

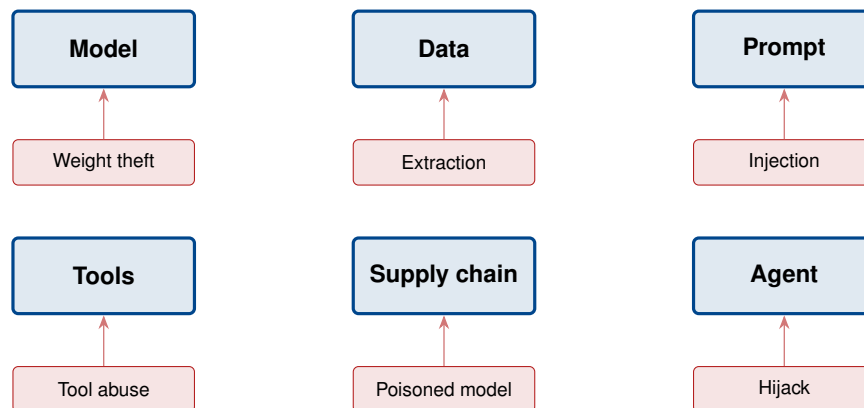


Figure 62.1: Six architectural surfaces and a representative threat against each. A real threat model covers many more attacks than the six shown; this diagram is the high-level map that the chapter then fills in surface by surface.

The figure shows one representative threat per surface, but each surface carries a family of related attacks, and the chapter develops them surface by surface so that the defences can be designed where the attacks land. The organising idea throughout is that security is a layer wrapping the whole architecture rather than a feature located at one point of it: the model, the data, the prompts, the tools, and the supply chain each need their own defences, and the agent, as a principal that acts, needs an identity and authorisation model of its own.

62.2 Model Security

The model itself is an asset, and where an organisation hosts its own model, that asset can be stolen, probed, and copied. Model security is the set of defences that protect the model as a thing of value, distinct from the defences that protect what the model is asked to do.

62.2.1 Weight Theft

A self-hosted model's weights are intellectual property, and frequently the single most expensive artefact the company owns; exfiltrating those weights is stealing the company's crown jewels. The defences are the defences of any high-value file, applied with discipline: encrypting the weights at rest so a stolen copy is useless without the key, serving the model only from attested environments whose integrity can be verified, logging every access to the model files, and running anomaly detection on those access patterns so that an unusual read is noticed. A model whose weights sit unencrypted and unmonitored is a model one compromised credential away from being copied.

62.2.2 Model Inversion and Membership Inference

A second class of attack does not steal the model but interrogates it. In a membership-inference attack, an adversary queries the model in ways designed to determine whether a specific record was part of its training data — a privacy breach in its own right. Model inversion goes further, reconstructing training data from the model's responses. The defences include differential privacy during training,

which bounds how much any single record influences the model; rate limiting on inference, which slows the high query volume these attacks need; and detection of the query patterns characteristic of an inference attack.

62.2.3 Extraction

In an extraction attack, an adversary queries a model extensively — not to learn about its training data but to reconstruct its behaviour, distilling the target model into a clone the adversary then owns. The defences are rate limiting to make the necessary query volume costly, watermarking so a cloned model can be identified as derived from the original, response perturbation to add noise that frustrates the reconstruction, and terms-of-service enforcement against the patterns that extraction requires. Extraction is the theft of the model's capability rather than its file, and it is defended at the query layer rather than the storage layer.

62.3 Data Security

AI systems hold data in places a traditional application does not — in training corpora, in retrieval indices, and in trace stores — and each of those is a confidentiality surface. Data security is the discipline of protecting data wherever an AI system has caused it to accumulate.

62.3.1 Training Data Privacy

When personal data appears in a training corpus, the trained model can memorise it, and a memorised record can later be extracted, exposing the individual it describes. The defences attack the problem at training time: deduplicating content that is memorisable, since a record that appears many times is far more likely to be memorised; training with differential privacy to bound any record's influence; and filtering the corpus to remove the categories of content that should never have been in it. Training-data privacy is cheaper to design in than to retrofit, because once a model has memorised a record, the record is in the weights.

62.3.2 RAG Index Confidentiality

A retrieval index built from customer documents is itself a sensitive store, and it carries a specific attack class: cross-tenant leakage, in which a query crafted to exploit embedding similarity retrieves another tenant's documents into the attacker's context. The defences are hard tenant scoping enforced at the index level, so that a retrieval physically cannot cross a tenant boundary, and an audit of every retrieval, so that a leakage attempt is recorded and detectable. An index without hard scoping is an index whose confidentiality depends on the query never being adversarial — which is not a security posture.

62.3.3 Prompt and Output Logging

The trace store that an AI system needs for observability holds user inputs and model outputs, and those can contain personal or otherwise sensitive data. A comprehensive trace store is, from an attacker's perspective, a concentrated and valuable target. The defences are encryption at rest, strict access controls on who may read traces, bounded retention so that data is not held longer than it is needed, and redaction at the moment of capture so that the most sensitive content never enters

the store in raw form. The observability the system needs must not become the breach the attacker wants.

62.4 Prompt Security

The prompt is where an AI system's behaviour is programmed, and an attacker who can influence the prompt can influence the program. Prompt security defends the boundary between the trusted instructions the system intends and the untrusted content it processes.

62.4.1 Prompt Injection

Prompt injection is the central prompt-layer attack: untrusted content — a user's input, a retrieved document, the output of a tool — contains instructions, and the model, which does not natively distinguish instruction from data, follows them. The defences are structural separation in the prompt so that untrusted content is fenced as data rather than read as instruction, output-side validation so that the consequences of a successful injection are caught before they propagate, and the principle of least privilege applied to what the model is permitted to do, so that even a hijacked model can do only limited harm. Injection cannot be eliminated, which is why it is defended in depth rather than at a single point.

62.4.2 Jailbreaks

A jailbreak is an input crafted to bypass the model's alignment and elicit behaviour the model is supposed to refuse. The defences are classifier-based detection tuned to known jailbreak patterns, ensemble defences that combine several detectors so a single bypass is not a full bypass, continuous red-teaming that finds new jailbreaks before adversaries do, and runtime policy gates that catch a harmful output even when the jailbreak succeeded upstream. Because adversaries adapt, jailbreak defence is a continuously updated practice rather than a fixed control.

62.4.3 System Prompt Extraction

Adversaries also attack the prompt by trying to extract the system prompt itself, which may reveal the system's logic, its instructions, or secrets that were unwisely placed in it. The first and best defence is minimisation: never put secrets in a prompt, so that extracting the prompt reveals nothing worth having. Beyond that, prompt-leak detection watches for the query patterns that extraction uses, and response filtering catches outputs that would disclose the system prompt. The strongest version of this defence is architectural — a prompt that contains no secret cannot leak one.

62.5 Tool Security

Tools turn a model from something that produces text into something that acts, and an action has consequences a text output does not. Tool security is the discipline of ensuring that the capability tools confer cannot be turned against the organisation.

62.5.1 Tool Privilege

A tool that can send an email, transfer money, or modify production data is a privileged capability, and it must be treated with the seriousness any privileged capability demands. The defences are per-tool authorisation so that access to each tool is granted deliberately, an agent identity tied to the user identity on whose behalf the agent acts, an action-class taxonomy — read, write, public-write, financial-write — that lets risk be reasoned about by category, and approval workflows that put a human in the loop for the most privileged actions. Tool privilege should be granted as narrowly as the task allows, because every tool an agent holds is a capability an attacker who hijacks the agent inherits.

62.5.2 Confused Deputy

The confused-deputy attack exploits an agent's legitimate authority: the agent is authorised to act for user A, and a crafted prompt convinces it to use that authority to act on user B's data. The agent is not compromised — it is doing exactly what it was told — but it has been tricked into applying its authority where it should not. The defences are encoding the scope of authority into the agent's identity, so that the authority itself is bounded, and having downstream services validate authorisation on every call rather than trusting that the agent's request is properly scoped. The fix is to make authority narrow and checked, not to trust the agent's judgement about its own scope.

62.5.3 Tool-Result Injection

The output of a tool is untrusted content, exactly as a retrieved document is, and an attacker who controls what a tool returns can plant instructions in that output for the model to read as authoritative. The defences are to treat every tool result as untrusted by default and to apply the same structural separation to tool output, in subsequent prompts, that is applied to any other untrusted content. A system that treats its tools' output as trusted has extended its trust boundary to every system its tools talk to.

62.6 Supply Chain Security

An AI system is assembled from artefacts that the organisation did not all produce — foundation models, adapters, tokenisers, inference engines — and each of those is a supply-chain risk in the same way a third-party software dependency is. Supply-chain security is the discipline of trusting those artefacts deliberately rather than by default.

62.6.1 Model Provenance

An open-weight model obtained from a public source may have been tampered with — carrying a backdoor, a deliberately introduced bias, or a trigger that exfiltrates data under certain inputs. The defences are insisting on signed model artefacts so that origin and integrity can be verified, scanning models for known backdoor patterns before they are deployed, and hosting trusted forks rather than pulling models from arbitrary sources. A model loaded without provenance verification is a model the organisation is trusting on faith.

62.6.2 Adapter Provenance

The same supply-chain logic extends to the lighter artefacts: LoRA adapters, prompts, and tool definitions form a supply chain of their own, and each of them shapes the system's behaviour. The defences are signed artefacts so that each one's origin is verifiable, review workflows so that a change to an adapter or a prompt is examined before it ships, and lineage tracking so that every artefact in production can be traced to its source. An adapter is small, but it can change what a model does, and it deserves the same provenance discipline as the model.

62.6.3 Dependency Drift

The libraries an AI system depends on — tokenisers, inference engines, frameworks — evolve continuously, and an unmanaged dependency is an unmanaged risk. The defences are pinning dependencies to known versions so that an update is a deliberate act, security scanning of those dependencies for known vulnerabilities, and periodic re-evaluation so that the pinned versions do not silently become stale and exposed. Dependency drift is the slow, quiet supply-chain risk, and it is managed by the same discipline applied to any other software dependency.

62.7 Agent Identity and Zero Trust

The deepest shift in AI security is that an agent is a principal. It is not a passive component but an actor that takes actions, and so it needs what every actor in a secure system needs: an identity, an authorisation model, and an audit trail. The architecture that follows from this recognition is a zero-trust architecture, in which no actor — human or agent — is trusted by default.

In a zero-trust agent architecture, the agent's identity is tied either to a delegation from a specific user or to a service identity, so that every action the agent takes is attributable to a principal with known authority. Every action is authorised against policy at the moment it is taken, rather than being permitted because an earlier action was. No trust is extended implicitly between agents, so that one agent's compromise does not propagate to the others through unchecked assumptions. And every action is audited, so that the system has a complete record of what its agents did and on whose authority. These principles are the same zero-trust principles applied to human and service identities elsewhere in the enterprise; the contribution of AI security is to recognise that an agent must be governed as one of those principals, and Chapter 72 develops the agent identity and access-management model in full.

62.8 Detection and Response

No set of defences is complete, which is why a security architecture needs detection and response alongside prevention. For AI systems, both the signals that detection watches and the scenarios that response rehearses are AI-specific, and a security operation that watches only conventional signals will miss the attacks aimed at the model.

62.8.1 AI-Specific Signals

AI systems emit signals that conventional security monitoring does not watch, and those signals are the early indicators of AI-specific attacks. The signals worth instrumenting include anomalous spikes in the refusal rate, which can indicate an attack probing the model's boundaries; cross-tenant retrieval anomalies, which can indicate an attempt at index leakage; tool-call frequencies that fall outside the established baseline, which can indicate tool abuse; and shifts in the classification of output content, which can indicate that the model is being driven toward harmful output. Each of these is a reading a generic security monitor does not take, and each is a leading indicator of an attack on the AI layer.

62.8.2 Incident Playbooks

When prevention fails, the speed and quality of the response depend on whether the scenario was rehearsed, and AI systems have incident scenarios that conventional playbooks do not cover. A mature security practice maintains playbooks for the AI-specific incidents: a prompt injection discovered in retrieved content, a jailbreak that achieved a harmful output, a confused-deputy attack that succeeded in crossing a user boundary, and a supply-chain compromise discovered in a deployed model or adapter. A playbook rehearsed before the incident is the difference between a contained event and an escalating one.

62.9 The African Perspective

AI security carries distinctive weight in African deployments, across data sovereignty, adversary asymmetry, regulatory reporting, and the specific threat posed by integration with mobile-money systems.

62.9.1 Sovereignty and Data-Border Threats

Every time data crosses a border, it crosses a jurisdiction, a network path, and a set of intermediaries, and each of those is additional exposure. African deployments increasingly run their inference in-region — in Cape Town or Johannesburg cloud regions — precisely to keep data within one jurisdiction and reduce the surface that cross-border flows create. Sovereignty-aware deployment is therefore not only a compliance posture but a security posture: the data that does not leave the region is data the cross-border attacker cannot reach.

62.9.2 Resource-Asymmetric Adversaries

In some African threat contexts, an adversary may command more compute than the defender does, which inverts an assumption that compute-intensive defences quietly rely on. A defence that depends on out-computing the attacker is a defence that fails against a better-resourced adversary. The disciplined response is to favour deterministic guards and comprehensive audit over compute-intensive detection — defences whose strength does not depend on a compute race the defender might lose. Audit, in particular, is cheap to run and strong to have, and it does not weaken against a richer adversary.

62.9.3 Regulatory Reporting

The data-protection and sector-specific regimes across African jurisdictions — POPIA, the NDPA, Kenya’s DPA, and the others — carry breach-notification timelines, and an AI-specific incident such as a training-data extraction or an index leakage triggers those obligations exactly as a conventional breach would. Readiness to report — knowing what was exposed, to whom, and within what window the regulator must be told — is a compliance matter that the security architecture has to support, and a pan-African operation must be ready against several regimes at once.

62.9.4 Mobile-Money-Adjacent Threats

An AI system integrated with mobile-money rails is a high-value target, because the tools that connect it to those rails can move money. The tool-security defences of this chapter — per-tool authorisation, action-class taxonomy, approval workflows for financial-write actions, confused-deputy defences — are therefore not optional refinements in these deployments but the core of the security posture. Where an agent’s tools can touch a payment system, the discipline around those tools is the discipline that protects the money.

62.10 Recurring Failure Patterns

The failure modes of AI security recur, and each is the absence of a distinction or a defence this chapter has set out:

- **Guardrails confused with security.** The organisation treats its guardrails as its security, forgetting that guardrails enforce policy against ordinary use while security defends against a deliberate adversary — two different jobs that need two different designs.
- **Tool privilege creep.** Agents accumulate tool access over time, with no review removing access that is no longer needed, until each agent holds far more capability than its task requires.
- **No model provenance.** Open-weight models are loaded and served without signature verification, so a tampered model could enter production undetected.
- **Confused-deputy invisibility.** The architecture cannot tell that an agent acted outside its proper scope, because authority is not encoded into identity and downstream services do not validate it per call.
- **Trace stores as honey pots.** The system logs comprehensively for observability but protects those logs with nothing commensurate to their sensitivity, turning the observability store into the attacker’s prize.

Each pattern maps to a remedy already developed — the guardrails-versus-security distinction, least-privilege tool grants, model provenance verification, authority encoded into agent identity, and trace stores protected as the sensitive stores they are — and together they show that AI security fails through missing distinctions and skipped defences rather than through any single dramatic flaw.

62.11 What to Carry Forward

Security architecture for AI wraps six surfaces — model, data, prompt, tool, supply chain, and agent — each with its own family of attacks and its own defences. The model must be protected as a high-value asset against theft, inversion, and extraction; the data must be protected wherever the system accumulates it, in training corpora, retrieval indices, and trace stores; the prompt boundary must be defended against injection, jailbreaks, and extraction; tools must be governed by least privilege and confused-deputy defences; and the supply chain of models and adapters must be trusted deliberately through provenance verification.

Above all, the agent must be treated as a principal — an actor with identity, authorisation, and audit — and governed under zero-trust principles, because an AI system that can act needs the same accountability as any other actor in the enterprise. Detection and response complete the architecture, oriented to AI-specific signals and rehearsed against AI-specific incident scenarios. African deployments add sovereignty-aware in-region deployment, defences that do not depend on out-computing a possibly better-resourced adversary, readiness for multi-jurisdiction breach reporting, and a particular priority on the tool-security defences that protect mobile-money integrations. The threat model is not yesterday's web-application threat model with a model bolted on — and the architecture of this chapter is what a genuinely new threat model requires.

Chapter 63

Multimodal Architectures

“A document is not text. A meeting is not a transcript. The world is multimodal; the architectures must be too.”

—a recurring observation in enterprise AI

Chapter Overview

Multimodal AI integrates text, images, audio, video, and structured data within a single system, and the architectural shift away from text-only systems is real rather than cosmetic. Tokenisation extends across modalities, so that an image or an audio clip consumes the context budget the way text does. Context budgets, in fact, come to be dominated by the non-text inputs. Specialised models for optical character recognition, audio transcription, and vision integrate with the generative model rather than being replaced by it. And output, too, spans modalities, so that evaluation and content policy must cover generated images and audio as well as generated text. This chapter develops the architecture: where each modality enters the system, how the modalities fuse, what their cost and latency profiles are, and how a system handles the realities of genuinely mixed-modal production traffic.

Learning Objectives

- Recognise the modality surfaces — text, image, document, audio, video, structured data — and where each one enters the system.
- Architect the fusion patterns — early, late, and mixed — and choose between them deliberately.
- Handle the cost and latency profiles that differ sharply from one modality to another.
- Manage the modality-specific failure modes that text-only thinking does not anticipate.
- Adapt multimodal architectures to African document, language, and bandwidth realities.

63.1 The Modalities

A multimodal architecture begins with an honest inventory of the modalities it must handle, because each one enters the system through a different path, carries a different cost, and fails in a different way. Treating them as interchangeable inputs to a single pipeline is the first mistake; treating them as

six distinct surfaces with six distinct disciplines is the start of a sound design.

63.1.1 Text

Text is the familiar baseline, and the modality every other one is measured against. It is handled as tokens, prompts, and generation — the well-understood machinery that the rest of this book has assumed throughout. Text is the cheapest modality per unit of information and the best-supported across every model, which is precisely why it is tempting to design as though it were the only modality. The discipline of a multimodal architecture is to treat text as one surface among six rather than as the default the others are bolted onto.

63.1.2 Image

The image modality covers photographs, diagrams, screenshots, charts, and scans — a wide range of content unified only by being visual. Modern vision-capable foundation models increasingly handle images directly, attending to them natively alongside text, where earlier architectures required a separate optical-character-recognition stage and a separate vision pipeline. The architectural choice is whether to send images to a vision-capable model directly or to preprocess them through specialist tools first, and that choice turns on cost and on how much structure the image carries.

63.1.3 Document

Documents — PDFs, word-processor files, presentations — are structured but messy: a single document is text plus images plus tables plus formatting, all interleaved. Because of that mixture, documents are rarely handled as raw inputs; a document-AI pipeline parses the document's structure before any generation happens, separating the text from the tables from the figures so each can be handled appropriately. Documents are the most common multimodal input in the enterprise, which is why they receive a dedicated treatment later in this chapter.

63.1.4 Audio

The audio modality divides into speech, which can be transcribed to text, and non-speech sound such as music or environmental audio. For enterprise systems speech is the dominant concern, and speech recognition is a mature technology for the major languages, with speech generation increasingly mature alongside it. Audio enters the system through a transcription stage that converts it to text the model can reason about, which means the quality of that transcription bounds the quality of everything downstream — a point the African-perspective section returns to sharply.

63.1.5 Video

Video is spatial-temporal content, and handling it means handling both dimensions: frame-level analysis of the visual content, extraction of the audio track, and scene understanding that spans time. It is the most expensive modality by a wide margin, because a video is, in effect, a long sequence of images plus an audio stream, and the cost scales with the frame-sampling rate the architecture chooses. A multimodal system that admits video must budget for it deliberately, because video cost can dwarf every other modality combined.

63.1.6 Structured Data

Structured data — tables, JSON, time series — is often the most valuable modality in an enterprise, because it is the form the organisation’s operational data already takes, and yet it is frequently the worst-handled by generic models, which were trained predominantly on prose and images. The architectural challenge is representation: structured data is token-efficient when it is formatted well and token-wasteful when it is dumped in raw, and a multimodal system that takes structured data seriously invests in representing it in a form the model can actually use.

63.2 The Architecture

A multimodal architecture is, at its core, a set of modality-specific preprocessing paths that converge into a fusion stage and then into a model. Figure 63.1 shows the canonical flow. Each modality enters through its own preprocessing — tokenisation for text, a vision encoder for images, automatic speech recognition for audio, and so on — because each modality must be turned into something the model can consume, and the transformation differs for each. The preprocessed streams then converge in a fusion stage, and the fused representation is what the multimodal model actually reasons over.

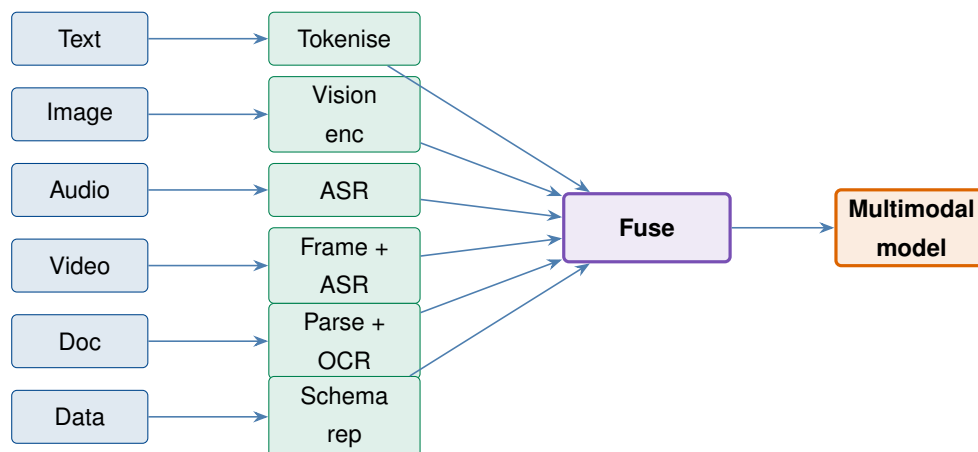


Figure 63.1: Multimodal flow. Each modality has its own preprocessing path; the outputs converge in a fusion stage; the multimodal model consumes the fused representation. Some modern models accept raw modalities directly, collapsing the preprocessing into the model itself.

The figure shows the general case, in which preprocessing is explicit and external to the model. The modern variation is that some foundation models accept raw modalities directly — an image or an audio clip passed straight to the model — which collapses the preprocessing into the model and simplifies the architecture at the cost of less control over each modality’s handling. The architectural decision is where to draw that line: explicit preprocessing gives modality-specific control and easier per-modality quality assurance, while direct ingestion gives simplicity, and a real system often draws the line differently for different modalities.

63.3 Fusion Patterns

Fusion is the step where the separately preprocessed modalities are brought together, and there are three ways to do it. The choice among them is a real architectural decision, because it trades the capability

the system demands of its model against the integration work it demands of its engineers.

63.3.1 Early Fusion

In early fusion, the modalities are tokenised together into a single unified sequence, and the model attends across modalities natively — the image tokens and the text tokens sit in one stream, and the model’s attention spans them as one. This is the approach of the modern multimodal foundation models, and it produces the deepest cross-modal understanding, because the model reasons over the modalities jointly rather than in separate passes. Its requirement is a capable multimodal model, since the model itself must be able to handle the unified multimodal sequence.

63.3.2 Late Fusion

In late fusion, each modality is processed by its own specialist model, and the separate outputs are combined at the application level rather than inside a model. This lowers the capability required of any single model — each specialist need only handle its own modality — at the cost of higher engineering integration, since the application now owns the logic of combining the outputs. Its compensating advantage is that modality-specific quality control becomes easier: each specialist can be evaluated and tuned on its own modality, in isolation from the others.

63.3.3 Mixed

In practice most production systems use a mixed pattern, applying early fusion to the modalities that fuse naturally and late fusion to the ones that do not. Text and images, which a vision-language model handles jointly, are fused early; structurally distinct modalities such as large structured tables are fused late, processed separately and combined at the application level. The mixed pattern is the production reality because the modalities genuinely differ in how well they fuse, and a single fusion strategy applied to all of them is either underpowered for some or overcomplicated for others.

63.4 Document AI as Special Case

Documents deserve their own treatment because they are the most common multimodal input an enterprise actually processes — contracts, forms, reports, statements, applications. A document is not one modality but several interleaved, and document AI is the discipline of pulling that mixture apart into things the system can reason over.

The work of document AI spans several capabilities, and a real document pipeline needs most of them: layout-aware extraction, which understands the spatial structure of a page rather than reading it as a flat stream of text; table extraction, which recovers tabular data as data rather than as scrambled prose; form-field detection, which locates and reads the fields of a structured form; signature and stamp recognition, which finds the marks of authority and execution on a document; and multi-page document reasoning, which carries understanding across the pages of a document that does not fit on one. Each of these is a distinct capability, and a document pipeline is built by assembling the ones the document type requires.

The modern shift is that vision-language models increasingly handle documents directly, attending to a page image without a separate optical-character-recognition stage in front of them. This simplifies

the pipeline and often improves quality, because the model sees the page as a human would. But the specialist OCR pipeline has not disappeared: at high volume it remains the lower-cost option, because running a dedicated extractor over a page is cheaper than passing the page through a large vision-language model. The architectural choice is the familiar one — capability and simplicity against cost — and it is decided by the volume and the value of the documents.

63.5 Cost and Latency Profile

Multimodal cost is not text cost with a small surcharge; the non-text modalities consume the context budget at rates that change the economics of the whole system. Understanding the per-modality token cost is what lets an architect anticipate the bill rather than discover it.

The rough figures are worth carrying. An image consumes on the order of one to two thousand tokens at a typical resolution. A document page consumes one to three thousand tokens, depending on how dense it is. A second of speech consumes fifty to a hundred tokens. Video is the dominant cost, and its figure is not fixed — it is set by the frame-sampling rate the architecture chooses, which is the single largest lever on video cost. Structured data is token-efficient when it is formatted well, which makes it the cheapest non-text modality when handled with care.

The consequence of these figures is that a multimodal context budget is dominated by its non-text inputs. A ten-page PDF can fill a context window the way a small novel would, and an architecture that did not anticipate this discovers it as a cost explosion or a context overflow in production. The discipline is to budget the context in modality-aware terms from the start — to know, before shipping, how many images or pages or seconds of audio a request can carry before it exceeds what the system can afford or the model can hold.

63.6 Output Modalities

Multimodality is not only an input concern; output spans modalities too, and that has consequences a text-only system never faces. Text remains the dominant output, but generated images, generated audio through text-to-speech, generated structured data, and generated code are all first-class outputs in many systems. Each non-text output is a real product of the system, not a side effect.

The architectural consequence is that the disciplines applied to text output must extend to the other output modalities, and they do not always extend cleanly. Evaluation is harder, because judging a generated image or a generated audio clip is less settled than judging generated text. Observability is harder, because an audio output is not as easily logged, searched, and inspected as a text trace. And content policy is harder, because the policies and classifiers for image generation are generally less mature than those for text. A multimodal system that produces non-text output must invest in evaluation, observability, and safety for those output modalities deliberately, because the text-oriented versions of those disciplines do not cover them for free.

63.7 The African Perspective

Multimodal AI carries distinctive weight in African deployments, because the documents, the languages, the bandwidth, and the identity systems an African system meets differ from the ones the major models were predominantly built around.

63.7.1 Document Realities

The documents an African enterprise processes are heterogeneous in ways generic document AI does not expect: scan quality varies widely, handwritten elements are common, content is frequently mixed-language within a single document, and stamps and seals come from a variety of jurisdictions with no common format. Generic document AI, tuned on cleaner and more uniform Northern corpora, underperforms on this variety, and locally calibrated solutions — pipelines tuned or trained on the actual document distribution of the market — are needed to reach acceptable quality. The document modality is where the gap between a generic model and local reality is widest.

63.7.2 Audio in Local Languages

Automatic speech recognition for African languages is uneven, and the unevenness is consequential because transcription quality bounds everything downstream. Swahili and Arabic are reasonably well covered; Hausa, Yoruba, Amharic, Zulu, Xhosa, and many others are far less so; and a long tail of languages is barely served at all. A multimodal system that admits audio in African languages frequently needs domain-specific ASR fine-tuning to reach usable accuracy, because the off-the-shelf recognisers were not built with these languages as a priority, and a poor transcription quietly poisons every reasoning step that follows it.

63.7.3 Bandwidth-Aware Multimodal

Image and audio uploads happen over mobile bandwidth that varies sharply, and a multimodal architecture for African users must be designed to tolerate that variation rather than assuming a fast, stable link. The techniques include client-side compression so that less data must travel, server-side downscaling so that an oversized upload is reduced rather than rejected, and progressive processing so that a partial upload can begin to be useful before it completes. A multimodal system that assumes Northern bandwidth fails for exactly the mobile-first users who make up much of the African market.

63.7.4 Identity-Document Variety

Know-your-customer and identity-verification workflows are a major multimodal use case in African financial services, and they meet a wide variety of identity-document formats across the continent's jurisdictions. Generic vision models, trained predominantly on Northern identity documents, underperform on this variety, and specialist models trained on African identity-document corpora outperform them substantially. The identity-document case is a clear instance of the chapter's recurring lesson: the multimodal architecture does not change, but the models within it must be calibrated to the local reality to work.

63.8 Recurring Failure Patterns

The failure modes of multimodal systems recur, and most of them are the consequence of text-only thinking applied to a multimodal reality:

- **Modality blindness.** The system was designed for text, and production inputs that include images or audio are silently dropped, so the system reasons over only part of what the user provided.
- **Cost explosion on documents.** A long PDF with embedded images fills the context window and the bill, because nobody budgeted the context in modality-aware terms before shipping.
- **Modality-specific failures invisible.** The ASR stage fails on a particular language, but the system reports success because the text path is fine, so a broken modality hides behind a working one.
- **Output-modality safety gaps.** Image-generation content policy is less mature than text policy, so the system applies a weaker safety standard to its generated images than to its generated text without anyone deciding to.
- **Evaluation gap.** Multimodal evaluation is harder than text evaluation, and the system's evaluation coverage quietly stops at text, leaving the non-text modalities untested.

Each of these is the failure to treat a non-text modality with the seriousness given to text — in design, in cost budgeting, in monitoring, in safety, and in evaluation — and together they make the chapter's central point: a multimodal system is sound only when every modality it admits is a first-class concern.

63.9 The Shape of the Argument

Multimodal architectures integrate text, image, audio, video, document, and structured data, and the integration is a real architectural undertaking rather than an extension of text handling. Each modality enters through its own preprocessing path; the paths converge through a fusion stage that may be early, late, or mixed; and the fusion choice trades the capability demanded of the model against the integration work demanded of the engineers.

Cost is dominated by the non-text modalities, with video the largest and structured data the most token-efficient, and a multimodal context budget must be reasoned about in modality-aware terms or it will overflow in production. Document AI is the most common enterprise case and the one that most rewards a deliberate pipeline. Output, too, spans modalities, and the evaluation, observability, and safety disciplines must extend to the non-text outputs rather than stopping at text. African deployments foreground heterogeneous document quality, uneven local-language ASR, bandwidth-aware processing, and identity-document specialisation — each a case where the architecture holds but the models within it must be calibrated to local reality. The world is multimodal; an architecture that treats every modality as a first-class concern is what it takes for the system to be multimodal too.

Chapter 64

Voice and Conversational AI Architecture

“Real-time voice is text plus three problems text never had: turns, interruptions, and milliseconds.”

—a recurring observation in voice-agent engineering

Chapter Overview

Voice agents differ from chat agents in ways that are architecturally significant rather than superficial. The latency budget is measured in hundreds of milliseconds rather than in seconds, because a conversation that does not respond quickly stops feeling like a conversation. Turn-taking and interruption handling are first-class architectural concerns rather than afterthoughts. And the system is a pipeline — speech recognition, then a language model, then speech synthesis — whose worst-performing component dominates the experience the user actually has. The architecture has matured rapidly with the arrival of end-to-end speech-native models that collapse the pipeline, but most production systems still chain specialists, and both approaches have to be understood. This chapter develops the architectural perspective on voice: the latency budget, the pipeline, the mechanics of natural conversation, and the production realities of telephony and compliance.

Learning Objectives

- Distinguish voice-agent architecture from chat-agent architecture and understand why the difference is structural.
- Architect the speech-recognition, language-model, and speech-synthesis pipeline within a realistic latency budget.
- Handle turn-taking, interruption, and conversational overlap as first-class concerns.
- Recognise where end-to-end speech models change the architecture and what they trade away.
- Adapt voice architectures to African language coverage and infrastructure realities.

64.1 The Latency Budget

The single fact that shapes voice-agent architecture more than any other is the latency budget. A natural human conversation has turns that complete within roughly two hundred to five hundred milliseconds, and a voice agent has to hit a similar budget to feel responsive rather than stilted. Latency is not a quality metric for a voice agent; it is the difference between a conversation and a frustrating exchange of delayed monologues.

The budget is consumed by the stages of the pipeline, and it is worth seeing where it goes. Speech recognition, done as a streaming and incremental process, takes on the order of fifty to two hundred milliseconds. The language model's time to its first token is one to three hundred milliseconds. Text-to-speech reaches its first audio in fifty to two hundred milliseconds. And the network adds fifty to two hundred milliseconds — substantially more when an African user is reaching a Northern endpoint. Summed, the pipeline spends somewhere between roughly a quarter-second and nearly a full second before the user hears a reply.

The thresholds beyond the budget matter as much as the budget itself. Latency above one second feels noticeably slow, the point at which the user begins to perceive the system as sluggish; latency above two seconds is where users abandon the interaction altogether. Every architectural decision in this chapter — the streaming of each stage, the choice between a chained pipeline and an end-to-end model, the placement of inference relative to the user — is, at bottom, a decision about staying inside this budget.

64.2 The Architecture

The canonical voice-agent architecture is a pipeline of specialists, shown in Figure 64.1. Audio flows from the microphone through voice-activity detection, which gates the speech recogniser so it does not waste effort on silence; the recogniser streams text tokens to the language model; the language model streams its output to text-to-speech; and the synthesiser streams audio back to the speaker. The defining property of the pipeline is that every stage streams, because the latency budget does not survive any stage that waits for the previous one to finish before it begins.

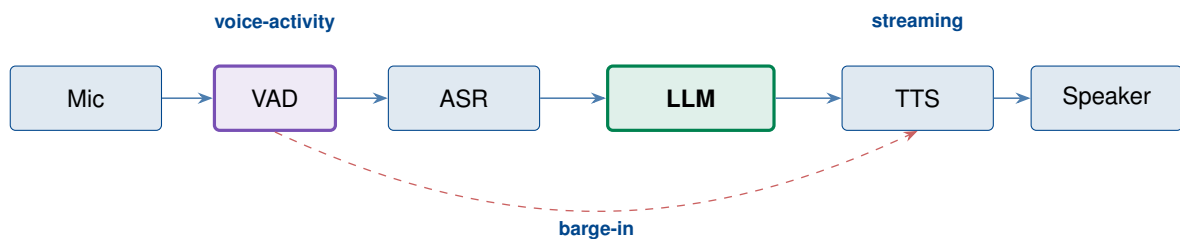


Figure 64.1: Voice-agent pipeline. Voice-activity detection gates the speech recogniser; the recogniser streams tokens to the language model; the model streams output to text-to-speech; synthesis streams audio to the speaker. The dashed path is barge-in: voice-activity detection interrupts synthesis when the user starts speaking.

Two control paths in the figure deserve note. Voice-activity detection sits at the front not only to gate the recogniser but to feed the barge-in path drawn dashed across the bottom: when the detector hears the user begin to speak, it interrupts the synthesiser mid-utterance, because a voice agent that talks over its user is not conversing. The pipeline is therefore not a simple left-to-right flow; it is a

streaming flow with a feedback path that makes interruption possible, and that feedback path is as much a part of the architecture as the forward one.

64.3 Turn-Taking

Turn-taking is the part of voice architecture that has no equivalent in chat, and it is where a voice agent most often feels wrong. In text, the user presses send and the turn boundary is unambiguous; in voice, the system has to infer when a turn ends, decide how to handle the user speaking over it, and choose whether to acknowledge the user mid-turn. Each of these is a distinct architectural problem.

64.3.1 Endpointing

Endpointing is the problem of detecting when the user has finished speaking, and it is a genuine balance rather than a solved task. An endpointer that is too eager cuts the user off mid-sentence, treating a pause for breath as the end of a turn; one that is too patient leaves an awkward gap after the user has plainly finished, because it is still waiting for more. Modern endpointing combines two signals to strike the balance: the acoustic signal of the audio itself, which detects silence, and a semantic judgement, in which a language model assesses whether the utterance so far is actually complete. Combining the acoustic and semantic signals is what lets an endpointer distinguish a thinking pause from a finished thought.

64.3.2 Barge-In

Barge-in is the handling of the user starting to speak while the agent is still talking, and supporting it is essential to a voice agent feeling natural — a person interrupts, and a system that cannot be interrupted feels like an obstacle. In a production system the mechanics are a small choreography: voice-activity detection notices the user’s voice, the synthesiser pauses immediately, the speech recogniser resumes to capture what the user is now saying, and the language model’s context is updated to reflect that the agent was interrupted. None of these steps is individually hard, but coordinating them quickly and cleanly is non-trivial engineering, and barge-in done badly is worse than no barge-in at all.

64.3.3 Overlap and Backchannelling

Real human conversation is not strictly turn-by-turn; it includes overlap, and in particular backchannelling — the “mm-hmm” and “yeah” a listener offers to signal attention without claiming the turn. Most production voice agents are turn-strict, treating any user sound as either a turn or an interruption, because handling overlap properly is hard. Some advanced systems do handle backchannels, recognising that a brief “mm-hmm” is not an attempt to take the floor and should not stop the agent. Backchannelling is the frontier of conversational naturalness, and where a system supports it, the conversation feels markedly more human.

64.4 End-to-End Speech Models

The chained pipeline of specialists is not the only architecture available. A newer class of models consumes audio and produces audio directly, collapsing speech recognition, the language model, and speech synthesis into a single pass — the realtime speech-native models from the major providers are

the current examples. Because the model handles the whole conversation in one pass, it is a genuinely different architecture rather than a faster pipeline, and it changes the trade-offs the architect is working with.

64.4.1 Advantages

The end-to-end approach brings several real advantages. Latency drops sharply, because there are no hand-offs between stages to pay for, and a total response time of two to four hundred milliseconds becomes achievable. Prosody — the rhythm and melody of speech — is preserved through the model rather than lost in a transcription to flat text and regenerated from scratch. Emotion and tone are perceived in the user's voice and produced in the agent's, which a text-mediated pipeline discards entirely. And code-switching between languages is handled natively, because the model never collapses the speech to a single-language transcript that would have to commit to one language.

64.4.2 Tradeoffs

The end-to-end approach also gives things up, and the trade-offs are not minor. The per-minute cost is higher than a chained pipeline of cheaper specialists. The choice of models is narrower, because few providers offer speech-native models and an organisation cannot mix and match the way it can with a pipeline. Tool-use integration is less mature, because the speech-native models are newer and their function-calling support lags the text models'. And logging and audit are harder, because the system's inputs and outputs are audio rather than text, and audio is less easily stored, searched, and inspected. The choice between a pipeline and an end-to-end model is therefore a real architectural decision, not an upgrade.

64.5 Production Realities

A voice agent that works in a demo still has to survive contact with real telephony, real networks, and real regulation. Three production realities shape the architecture beyond the pipeline itself.

64.5.1 Telephony Integration

Many voice agents do not run in a browser or an app but over the telephone, and integrating with phone systems means speaking their protocols: SIP, WebRTC, or carrier-specific protocols, depending on the deployment. Telephony also constrains audio quality through its codecs and sample rates — G.711 and Opus among the codecs, and sample rates from 8 kHz for traditional telephony through 16 kHz wideband to 24 kHz studio quality. The architecture has to accept that a telephony-delivered voice agent works with narrower-band, lower-fidelity audio than a studio microphone provides, and its speech recognition must perform on that audio rather than on a clean signal.

64.5.2 Quality Under Bandwidth

Mobile data quality varies, and a voice agent reached over a mobile connection has to stay usable as that connection degrades. The architecture supports this through adaptive bitrate that adjusts to the available bandwidth, graceful fallback to lower-quality codecs when the connection cannot sustain a better one, and robustness to packet loss so that a few dropped packets do not break the conversation. A voice agent designed for a stable connection fails for exactly the mobile users on variable networks

who make up much of its real audience.

64.5.3 Recording and Compliance

Voice interactions are frequently recorded, and recording carries legal obligations. Many jurisdictions require that call recording be disclosed to the participants, and the specifics of that obligation vary by jurisdiction. Legal review must precede production deployment of a recording voice agent, because the consequence of recording without proper disclosure is regulatory exposure rather than a mere quality defect. Compliance is therefore an architectural input to a voice system, not a step bolted on after the system works.

64.6 The African Perspective

Voice AI is particularly consequential in African deployments, and in some respects more consequential than in the markets where the voice models were predominantly built. The reasons span accessibility, language coverage, the telephony estate, and cost.

64.6.1 Accessibility

Voice interfaces serve populations that text interfaces exclude — in particular, users with low literacy, for whom a text-based system is simply not usable. Across health, agriculture, and public services, a voice interface is not a convenience but the difference between a service being reachable and being out of reach. This makes voice AI a genuine accessibility technology in the African context, and it raises the stakes of getting the architecture right, because the users a voice system serves are often users who have no text alternative to fall back on.

64.6.2 Language Coverage

Speech recognition and synthesis for African languages are unevenly developed, and the unevenness directly bounds where a voice agent can be deployed. Swahili and Arabic are relatively well served; Hausa, Yoruba, Igbo, Amharic, Zulu, and Xhosa are served to varying and often inadequate degrees; and many minority languages are not served at all. Investment in local-language voice models is underway and growing, but a voice architecture for African deployment must be designed around the current reality of coverage — choosing languages it can actually support well, and being honest about the ones it cannot.

64.6.3 Telephony-First

A great many African voice interactions happen over feature-phone calls rather than smartphone applications, which means the telephony integration is not an edge case but the primary channel. IVR-style architectures — voice agents reached by dialling a number, navigated by speech over a phone call — matter centrally, and a design that assumes a smartphone app with a good microphone and a data connection simply misses the user. Building telephony-first, rather than treating the phone call as a degraded version of the app, is the right default for much of the African market.

64.6.4 Cost Per Minute

Voice-minute economics are tighter than text economics, because a voice interaction runs a multi-stage pipeline continuously for the duration of the call, and African deployments operate under cost ceilings that make this acute. The levers are the familiar ones applied to the voice case: smaller models where they suffice, inference hosted on-region to cut both latency and cost, and cascaded pipelines that use cheap components for the easy stages. Voice cost discipline is, as with the rest of African AI deployment, a survival concern rather than an optimisation.

64.7 Recurring Failure Patterns

The failure modes of voice agents recur, and each maps to a discipline this chapter has set out:

- **Latency stacking.** Each pipeline stage adds its own delay, and the aggregate quietly exceeds the latency budget, leaving a system that is individually fast at every stage and collectively too slow.
- **Endpointing failure.** The endpointer is mistuned, so it either cuts users off mid-sentence or leaves awkward gaps, and the conversation feels wrong even when every word is understood.
- **Barge-in absence.** The system cannot be interrupted, so a user is trapped listening to a long synthesised response they no longer want, with no way to take back the turn.
- **Code-switching surprises.** The pipeline's speech recognition is built for one language and silently mishandles the code-switches that real multilingual speakers produce.
- **Audio compliance gaps.** Calls are recorded without the disclosure the jurisdiction requires, turning a quality system into a regulatory liability.

Each pattern is the absence of a discipline the chapter has argued for — latency budgeting across the whole pipeline, careful endpointing, first-class barge-in, language coverage matched to the real user population, and compliance treated as an architectural input — and together they show that a voice agent fails through the conversational and operational realities, not through the language model at its centre.

64.8 Where That Leaves Us

Voice-agent architecture combines speech recognition, a language model, and speech synthesis within a tight latency budget, with turn-taking and barge-in as first-class concerns rather than refinements. The latency budget — a few hundred milliseconds to stay conversational — is the constraint that shapes every other decision, and it is the reason every stage of the pipeline must stream.

End-to-end speech models collapse the pipeline into a single pass, buying lower latency, preserved prosody, perceived emotion, and native code-switching, at the cost of higher per-minute price, narrower model choice, less mature tool integration, and harder audit. The choice between a chained pipeline and an end-to-end model is a real architectural decision. Production reality adds telephony integration, robustness under variable bandwidth, and recording compliance as architectural inputs. African deployments foreground voice as an accessibility technology for low-literacy populations, the uneven

and growing state of local-language coverage, a telephony-first channel that the architecture must be built around rather than treated as degraded, and a per-minute cost discipline that is a survival concern. Real-time voice is text plus turns, interruptions, and milliseconds — and the architecture of this chapter is what it takes to handle all three.

Chapter 65

Compliance and Audit Architecture

“If the architecture cannot prove what it did, the regulator will assume it cannot tell.”
—a recurring observation in regulated AI deployments

Chapter Overview

Compliance architecture wraps an AI system with the controls and the evidence that regulators, auditors, and internal risk functions require of it. The frameworks an organisation must satisfy differ — SOC 2, ISO/IEC 42001, the NIST AI Risk Management Framework, the EU AI Act, and a range of sector-specific regimes — but their architectural requirements converge on a common set: identity, audit trails, data classification, model documentation, change control, and incident response. The thing that distinguishes a compliant system is not that it has documents but that its architecture can produce evidence on demand — a verifiable account of what it did, when, and on whose authority. This chapter develops the architecture-side view of compliance: what evidence must be producible, what controls must actually be operational rather than merely described, and how to design for compliance from the start rather than retrofit it under audit pressure.

Learning Objectives

- Recognise the principal compliance frameworks and the architectural implications each one carries.
- Architect audit trails, tamper-evident evidence stores, and the reporting infrastructure that serves auditors.
- Apply model-documentation and lineage requirements as architectural artefacts rather than paperwork.
- Operate change control and incident response so that both produce compliance evidence as a by-product.
- Adapt compliance architecture to the regulatory diversity of African multi-jurisdiction operations.

65.1 The Framework Landscape

A compliance architecture has to satisfy not one framework but several at once, and the first task is to understand which frameworks apply and what each one actually demands of the architecture. The frameworks divide cleanly into two groups — the cross-cutting ones that apply regardless of sector, and the sector-specific ones that apply because of the industry the system serves — and a system in a regulated industry typically has to answer to members of both groups simultaneously.

65.1.1 Cross-Cutting Frameworks

The cross-cutting frameworks apply to AI systems across industries, and four of them dominate current practice. ISO/IEC 42001 is the AI management system standard, published in December 2023 as the first of its kind, and its focus is on process and governance — on whether the organisation manages its AI systematically. The NIST AI Risk Management Framework is a risk-management framework that is voluntary but very widely adopted, organised around the functions of governing, mapping, measuring, and managing AI risk. The EU AI Act is the one genuinely regulatory instrument of the four, in force since August 2024, risk-tiered, and imposing substantial documentation obligations on systems it classes as high-risk. And SOC 2 is the security, availability, and confidentiality controls regime that is a common requirement in business-to-business relationships. Each places different demands on the architecture, and a system serving several markets must satisfy the union of them.

65.1.2 Sector-Specific Frameworks

Layered on top of the cross-cutting frameworks are the regimes that apply because of the industry. Banking carries Basel-adjacent supervisory guidance and, in the United States, the SR 11-7 model-risk-management expectations, which treat a deployed model as a risk to be governed in its own right. Healthcare carries HIPAA for the handling of patient data and, where an AI system functions as a medical device, the FDA's software-as-a-medical-device pathways. Government deployments carry FedRAMP and the sector-specific national standards of the jurisdiction. A compliance architecture for a regulated-industry AI system must therefore be designed against a specific, named stack of frameworks — the cross-cutting ones plus whichever sector regimes the industry imposes — rather than against a generic notion of compliance.

65.2 The Audit Architecture

Whatever the specific frameworks, the architectural answer to all of them has a common shape: the system must continuously collect evidence of what it did and deposit that evidence somewhere an auditor can trust. Figure 65.1 shows the canonical audit-evidence architecture, in which five streams of evidence flow into a single tamper-evident store and out again as reports.

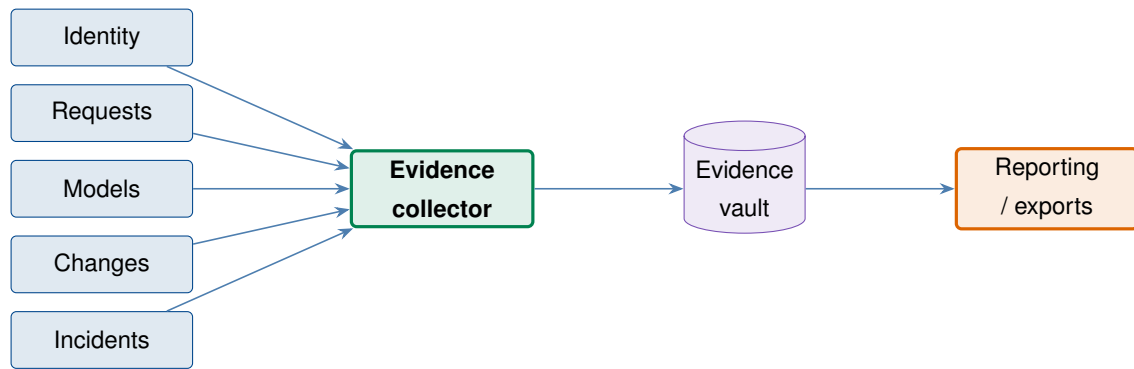


Figure 65.1: Audit-evidence architecture. Identity events, request traces, model artefacts, change records, and incident reports flow through an evidence collector into a tamper-evident evidence vault. Reporting and exports serve auditors, regulators, and internal risk functions.

The architecture has three working parts. The evidence collector gathers the five streams as they are generated, in production, rather than reconstructing them later. The evidence vault stores what the collector gathers in a tamper-evident form, so that an auditor can trust not only the content of the evidence but its integrity — the assurance that it has not been altered since it was recorded. And the reporting layer turns the stored evidence into the exports, dashboards, and attestations that auditors, regulators, and internal risk functions consume. The decisive architectural property is that the evidence is collected continuously and stored tamper-evidently, because evidence assembled after the fact is evidence whose integrity cannot be defended, and an auditor who doubts the integrity of the evidence will treat it as absent.

65.3 Evidence Categories

The five streams flowing into the vault are five categories of evidence, and a compliant architecture has to produce all of them. Each answers a different question an auditor will ask, and a gap in any one of them is a question the architecture cannot answer.

65.3.1 Identity Evidence

Identity evidence answers the question of who did what, and when. It must cover not only human users but service accounts and, crucially for agentic systems, agent identities — because an agent that takes an action is a principal whose action must be attributable. The evidence is tied to the organisation's authentication systems and spans the agent-as-principal pattern developed in the security and identity chapters. Without complete identity evidence, the audit trail can describe an action but cannot say who was responsible for it, which is the first thing any auditor wants to know.

65.3.2 Request Evidence

Request evidence is the record of every model invocation: the input, the output, the model version, the prompt version, the tool calls made, the intermediate states, the latency, and the cost. It is the most voluminous of the five streams and the most operationally useful, because it is also the basis of debugging and incident forensics. The production trace architecture developed in Chapter 49 is what produces this evidence, which means a system with sound observability has, as a by-product, much of

the request evidence its compliance obligations require.

65.3.3 Model Evidence

Model evidence is the documentation of the models themselves — model cards or system cards recording the training-data sources, the evaluation results, the known limitations, the intended uses, and the fairness assessments. For high-risk systems under the EU AI Act this documentation is not optional but a regulatory requirement, and the architecture must be able to produce it for every model in production. Model evidence is the category most often missing, because it feels like paperwork rather than architecture — but a system that cannot produce documentation for the model it is running is a system that cannot pass a high-risk conformity assessment.

65.3.4 Change Evidence

Change evidence records every change to a prompt, a model, or a tool: who approved the change, what specifically changed, when it was deployed, and what evaluations it passed before deployment. It is the evidence that connects the system's current behaviour to a deliberate, reviewed decision rather than to an undocumented edit. Change evidence is what lets an auditor confirm that the system did not drift into its current state by accident, and it is produced by the change-control discipline developed later in this chapter.

65.3.5 Incident Evidence

Incident evidence is the closed-loop record of what went wrong: what happened, when it was detected, how the organisation responded, what was fixed, and what was changed to prevent a recurrence. The word that matters is closed-loop — an incident record that describes the problem but not the resolution is an open file, and an auditor reads an open file as an unaddressed risk. Complete incident evidence demonstrates not only that the organisation detects problems but that it resolves them and learns from them.

65.4 Model Documentation

Model documentation deserves its own treatment because it is a compliance artefact in its own right, not a supplement to the evidence streams. A regulator examining a high-risk system asks for documentation by name, and an architecture that cannot produce it has a compliance gap regardless of how well the system performs.

Four documents make up the core set. The model card describes a model's capability, its intended use, its limitations, and its evaluation results — the honest account of what the model is and is not for. The system card does the same for the integrated system rather than the bare model, because a model embedded in a system with retrieval, tools, and guardrails behaves differently from the model alone, and it is the system that is deployed. The data card documents the training and evaluation data — its sources, its licensing, its demographic coverage, and its preprocessing — so that the data behind the model is itself accountable. And the risk register records the identified risks, their severity, the mitigations applied, and the residual risk that remains after mitigation.

The ownership of this documentation depends on the model's origin. For an inherited foundation

model, much of the documentation comes from the provider, and the organisation's task is to obtain it and confirm it covers what its regulators require. For a fine-tuned or self-trained model, the organisation produces the documentation itself, because it is the party that shaped the model's behaviour. Either way, the documentation is an architectural deliverable that the compliance architecture must be able to assemble and present, not a one-time formality completed and forgotten.

65.5 Change Control

Compliance requires change control over every artefact that affects production behaviour, because an unreviewed change is a change the organisation cannot account for. Change control is the discipline that ensures every change to the system's behaviour was deliberate, reviewed, and recorded — and it is also, conveniently, the discipline that produces the change-evidence stream.

The architecture of change control rests on a chain of linked records. Every change has a ticket, so that no change happens without a documented intent. Every ticket carries approval evidence, so that the change was authorised by someone accountable. Every approval is made against documented criteria — the evaluations that passed, the risk that was reviewed — so that the approval was reasoned rather than reflexive. Every deployment links back to its ticket, so that what is running in production can be traced to the decision that put it there. And every change has a rollback path that is documented and actually tested, so that a change found to be harmful can be reversed cleanly. A change-control architecture with this chain intact produces, automatically, an auditable account of how the system reached its current state.

65.6 Incident Response

AI systems have incident types that traditional incident response does not anticipate — a jailbreak succeeding in production, a hallucination that causes real harm, a data leak through embedding similarity, a biased outcome detected after the fact — and compliance requires that the organisation be ready for them. Incident response is the discipline that turns an AI-specific failure into a managed, documented, and ultimately learned-from event.

A compliant incident-response architecture requires several things to be in place before any incident occurs. The detection criteria must be documented, so that the organisation has decided in advance what counts as an incident rather than improvising the judgement under pressure. A severity classification must exist, so that incidents are triaged consistently. Notification timelines must be defined for regulators, customers, and internal stakeholders, so that the organisation meets the deadlines its frameworks impose. A forensic capability must exist — the ability to replay the incident from the request and trace evidence — so that the organisation can establish what actually happened. And every incident must close with a post-incident review that permanently updates the evaluation set, so that the specific failure becomes a regression test the system is held against forever after. An incident-response practice with these elements turns each failure into both a compliance record and a permanent improvement.

65.7 EU AI Act Specifics

The EU AI Act warrants specific treatment because it is the most architecturally demanding of the cross-cutting frameworks and the one most likely to drive a system's compliance design. Its central mechanism is a categorisation of AI systems by risk, and the category a system falls into determines what the architecture must produce.

The Act defines four risk tiers. Unacceptable-risk systems are banned outright. High-risk systems are permitted but carry substantial documentation and conformity-assessment obligations. Limited-risk systems carry transparency requirements — chiefly, that users be told they are interacting with an AI system. Minimal-risk systems carry no specific requirements. Most enterprise AI systems of consequence fall into the high-risk tier, and that tier is where the architectural burden concentrates.

For a high-risk system the Act requires technical documentation, a risk-management system, data governance, transparency, human oversight, and demonstrated accuracy, robustness, and cybersecurity. The architectural consequence is direct: for each of these requirements, the system must be able to produce evidence that it is met. A compliance architecture designed for the EU AI Act is therefore an architecture organised around evidence production for the high-risk requirement set, and an organisation that classifies its system as high-risk should design that evidence production in from the start rather than discover its absence during a conformity assessment.

65.8 The African Perspective

Compliance architecture carries distinctive weight in African deployments, across jurisdictional multiplicity, sovereignty evidence, donor reporting, and the continental frameworks now emerging.

65.8.1 Multi-Jurisdiction Operations

A pan-African deployment operates under several data-protection regimes at once — South Africa's POPIA, Nigeria's NDPA, Kenya's DPA, Egypt's PDPL, Morocco's Law 09-08, and the sector-specific regulators of each country it serves. A compliance architecture for such a deployment cannot apply one regime uniformly; it must stratify, producing per-jurisdiction evidence so that each country's regulator can be answered against that country's law. The evidence vault and the reporting layer must both be jurisdiction-aware, because an aggregate compliance report that does not separate the jurisdictions cannot satisfy any of their regulators individually.

65.8.2 Sovereignty Evidence

In African deployments, the fact that data did not cross a border is itself a compliance artefact, because data-residency obligations turn on exactly that fact. The architecture must therefore be able to produce, on demand, evidence of where data was stored and processed — a verifiable account that personal data stayed within the jurisdiction whose law governs it. Sovereignty evidence is a category that Northern compliance architectures rarely foreground and African ones must, because the residency claim is only as good as the architecture's ability to prove it.

65.8.3 Donor and Development-Finance Reporting

Many African AI deployments are funded by donors or development-finance institutions — USAID, the European Union, the African Development Bank, the World Bank — and each of those funders imposes its own reporting requirements. The useful architectural observation is that the compliance infrastructure built for regulators doubles as the reporting infrastructure for donors: the same evidence vault and reporting layer that produce a regulator’s audit pack can produce a donor’s compliance report. An organisation that builds its compliance architecture well therefore satisfies two reporting obligations with one infrastructure.

65.8.4 Continental Frameworks Emerging

The regulatory landscape is not static: the African Union’s Continental AI Strategy and various sub-regional frameworks are expected to impose continental-level requirements over the coming years. A compliance architecture designed for flexibility — one whose evidence categories and reporting layer can absorb a new framework’s requirements without restructuring — will adapt to these as they arrive. A rigid architecture, designed against today’s frameworks alone, will need rewrites. Designing for the frameworks that do not yet exist means designing the evidence collection to be comprehensive and the reporting to be configurable, so that a new regime becomes a new report rather than a new architecture.

65.9 Recurring Failure Patterns

The failure modes of compliance architecture recur, and each is the substitution of an appearance of compliance for the real thing:

- **Compliance as documentation.** The documents exist and look complete, but the controls they describe are not actually operational — the architecture documents a discipline it does not practise.
- **Evidence retrofitting.** Evidence is reconstructed after the fact rather than collected continuously, so its integrity cannot be defended and an auditor discounts it.
- **Tamperable logs.** The audit logs can be edited, so the auditor cannot trust that they record what actually happened, and rejects them regardless of their content.
- **No model documentation.** A high-risk system under the EU AI Act is shipped without the technical documentation the Act requires, leaving a compliance gap that no operational quality can close.
- **Cross-jurisdiction blindness.** A single-jurisdiction compliance architecture is deployed across a multi-jurisdiction operation, so the system cannot produce the per-jurisdiction evidence each regulator demands.

Each pattern maps to a discipline this chapter has set out — operational controls behind the documents, continuous tamper-evident evidence collection, model documentation as an architectural deliverable, and jurisdiction-aware stratification — and together they make the chapter’s point: compliance fails not through bad intent but through architecture that produces the look of compliance without the provable substance.

65.10 Holding the Pieces Together

Compliance architecture wraps an AI system with auditable controls and continuously collected evidence: identity, requests, models, changes, and incidents flowing through an evidence collector into a tamper-evident vault and out as reports. The frameworks differ in their specifics, but their architectural requirements converge, and the EU AI Act's high-risk tier sets the most demanding evidence-production agenda.

Model documentation, change control, and incident response are architectural concerns rather than paperwork — each produces a stream of the evidence the architecture must store, and each is a discipline that must be operational rather than merely described. African deployments add the stratified evidence of multi-jurisdiction operations, sovereignty evidence that proves data stayed within its border, donor reporting that the same infrastructure serves, and the need to design for continental frameworks that are still emerging. If the architecture cannot prove what it did, the regulator will assume it cannot tell — and the architecture of this chapter is how an organisation makes sure it can always prove it.

Chapter 66

AI Red-Teaming and Adversarial Testing

“The attack you have not tested is the one that will appear in production.”

—a recurring observation in AI security practice

Chapter Overview

AI red-teaming is the structured practice of attacking an AI system in order to discover its vulnerabilities before an adversary does. It differs from the penetration testing of traditional software because its targets are probabilistic rather than deterministic: a red-teamer attacks prompt injection, jailbreaks, harmful-content elicitation, privacy extraction, and agent hijacking — surfaces where the same input does not always produce the same output, and where a vulnerability is a tendency rather than a fixed flaw. The architecture of red-teaming has two halves that must both be built: the red-team infrastructure that runs attacks at scale and finds the weaknesses, and the response infrastructure that turns each finding into a defence. A red-team that finds vulnerabilities but does not close them produces reports; a red-team wired into a defence loop produces a system that gets harder to attack over time.

Learning Objectives

- Distinguish red-teaming from guardrail evaluation, and understand why adversarial testing is a separate discipline.
- Architect attack-execution infrastructure that runs attacks at scale against a realistic target.
- Apply automated and manual red-teaming together, using each for what it does best.
- Build the loop that turns a finding into a defence, so that red-teaming improves the system rather than only describing it.
- Adapt red-teaming to African threat contexts, languages, and the priority of mobile-money-adjacent attack surfaces.

66.1 Why Red-Team

Traditional testing exercises a system’s intended behaviour — it confirms that the system does what it is supposed to do when given the inputs it expects. Red-teaming exercises the opposite: it confirms how the system behaves when given the inputs an adversary deliberately constructs to break it.

The distinction matters because an AI system can handle its intended inputs flawlessly and still be dangerous, since the real question for a deployed system is not whether it handles cooperative users well but whether it handles malicious ones safely.

This is also what separates red-teaming from guardrail evaluation. Guardrail evaluation checks that the system’s policy controls work against ordinary use; red-teaming checks that they hold against a thinking adversary who is actively trying to defeat them. The two are complementary — one measures the controls under normal conditions, the other under attack — but they are not the same discipline, and a system that has only been guardrail-tested has not been red-teamed.

66.1.1 The Attack Categories

Red-teaming covers a recognisable set of attack categories, and a red-team programme should test against all of them rather than concentrating on the one most recently in the news. Harm elicitation attacks try to make the system produce illegal advice or dangerous information it should refuse. Prompt-injection attacks plant instructions in untrusted content — a document, a tool result — for the model to follow as if they were commands. Jailbreaks craft inputs that bypass the model’s alignment to unlock refused behaviour. Privacy-extraction attacks attempt to pull training data or the system prompt out of the model. Agent-hijacking attacks abuse an agent’s tools or exploit the confused-deputy weakness to make the agent act wrongly. And bias-and-fairness probing tests whether the system produces systematically different outcomes for different groups. Each category is a distinct family of attacks, and coverage means testing across the whole set.

66.2 The Architecture

The architecture of a red-team programme is a loop, and Figure 66.1 shows its shape. An attack library seeds an attack generator; the generator fires attacks at the target system; an outcome judge evaluates whether each attack succeeded; the findings flow into defence updates and also back into the attack library so that every discovered attack becomes a permanent regression test. The loop is the architecture: a red-team without the return paths is a one-time exercise, while a red-team with them is a standing capability that compounds.

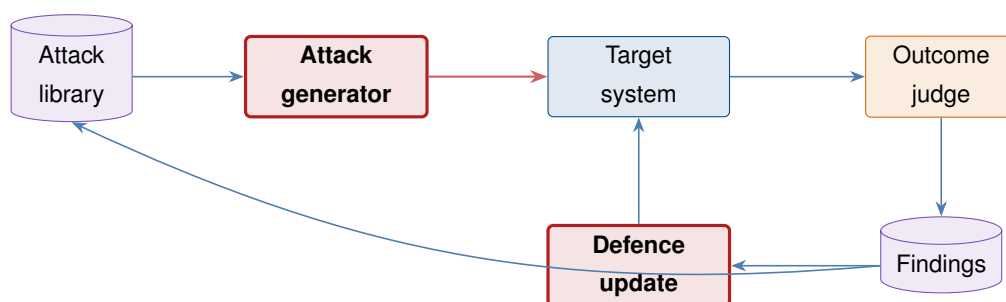


Figure 66.1: Red-team architecture. An attack library seeds an attack generator; attacks fire against the target system; an outcome judge evaluates success; findings flow to defence updates and back to the attack library for regression.

One architectural requirement is implicit in the figure and worth making explicit: the target system

must be production-equivalent. A red-team firing against a staging environment that differs from production in its prompts, its tools, or its guardrails is finding the vulnerabilities of staging, not of production, and the findings will miss the real exposure. The value of the whole loop depends on the target being a faithful stand-in for the system that real adversaries will actually attack.

66.3 Attack Generation

The attack generator is where the red-team's attacks come from, and there are four ways to generate them. They differ in creativity, in scale, and in cost, and a mature red-team uses all four together rather than relying on any single one, because each covers a part of the attack space the others miss.

66.3.1 Manual Attack Generation

Manual generation has human red-teamers craft attacks by hand. It offers the highest creativity, because a skilled human can invent genuinely novel attacks that no automated method would produce, but it offers the lowest scale, because human time is limited. Manual red-teaming is therefore used for the work that demands creativity — discovering new attack classes — and to seed the automated methods with starting points they can then vary at scale. It is the source of novelty in the programme.

66.3.2 Template-Based Generation

Template-based generation takes known attack patterns and parameterises them with variation, producing many concrete attacks from each pattern. It scales well, because once a template exists it can be instantiated cheaply and repeatedly, and it covers the known attack space thoroughly. Its limitation is that it finds only variations of attacks already known — it cannot discover a genuinely new attack class, only new instances of the classes already in the library. It is the workhorse for regression coverage of known threats.

66.3.3 Model-Generated Attacks

Model-generated attacks use a red-team language model to generate the attacks. This scales aggressively, producing attacks faster and in greater volume than any human or template approach, but it introduces a specific blind spot: the red-team model has its own limitations, and the attacks it generates reflect what that model can imagine. An attack class the red-team model does not conceive of will simply not be generated, so model-generated attacks broaden coverage rapidly but cannot be the only method, lest the programme inherit the red-team model's gaps.

66.3.4 Evolutionary Generation

Evolutionary generation takes the attacks that succeeded and mutates them into variants, in the style of an adversarial search that breeds successful attacks into stronger ones. It is high-signal, because it concentrates effort on the attack families that have already proven they can break the system, and it is expensive, because it runs many generations of mutation and evaluation. Evolutionary generation is most valuable for hardening the system against the specific vulnerabilities that earlier rounds have already exposed, pushing each discovered weakness to its limit.

66.4 Outcome Judging

Once an attack has fired, something must decide whether it succeeded, and that is the outcome judge. Judging an attack outcome is itself a non-trivial problem, because the question is not always a clean yes-or-no, and the architecture uses three layers of judgement together.

66.4.1 Categorical Judges

Categorical judges answer the clear binary questions: did the system produce harmful output, did it leak the system prompt, did it call a forbidden tool. These are yes-or-no judgements, and they are the fast first pass of outcome judging, because a categorical judge can evaluate a large volume of attack results cheaply and flag the clear successes for attention. They are necessary but not sufficient, because not every attack outcome is binary.

66.4.2 Severity Scoring

Severity scoring exists because not all successful attacks are equally serious. A jailbreak that produces a mild policy violation is a real finding, but it is not the same as a jailbreak that enables concrete, serious harm, and a red-team that treats every success as equally urgent will misallocate its defensive effort. Severity scoring assigns each finding a graded score, so that the defence loop can prioritise the dangerous findings over the trivial ones. It is what turns a flat list of successes into a ranked agenda.

66.4.3 Human Review

Human review is the final layer, applied to the findings that the automated judges have flagged. Automated judges produce false positives — they flag attacks as successful that, on inspection, were not — and acting on a false positive wastes defensive effort and erodes trust in the red-team. Human review confirms a finding before the defence loop acts on it, and it is reserved for the findings that matter, because human time is the scarce resource. The three layers together give the red-team a judgement pipeline that is fast where it can be and careful where it must be.

66.5 From Finding to Defence

The single most important property of a red-team programme is the loop from finding to defence, because a finding that does not change the system is a finding that has improved nothing. The loop matters more than any individual discovery: a programme that finds ten vulnerabilities and fixes them is worth more than one that finds a hundred and fixes none.

When a finding is confirmed, several defence updates follow from it. The classifier training data is updated with the new attack, so that the detection classifiers learn to recognise it. The guardrail rules are updated to catch the attack pattern. Prompt hardening is applied where the attack exploited a prompt-level weakness. The evaluation suite is expanded with the attack as a permanent test case, so that the system is checked against it forever after. And the documentation is updated to record the vulnerability and its remediation. Together these updates close the loop: the discovered attack becomes a defended-against attack, and the same attack tried again will now fail. Without the loop, red-team findings accumulate as reports that nobody reads, and the system remains exactly as vulnerable as it

was before the red-team ran.

66.6 Continuous versus Campaign

Red-teaming runs at two cadences, and a mature programme runs both, because they catch different things. The choice is not which one to adopt but how to combine the steady baseline of continuous testing with the periodic intensity of campaigns.

66.6.1 Continuous Red-Teaming

Continuous red-teaming runs automated attacks constantly against a production-equivalent staging environment. Its value is that it catches regressions — a change that reopens a previously closed vulnerability — and emerging attacks as soon as they enter the attack library, rather than waiting for a scheduled exercise. Its cost is a substantial infrastructure investment, because running attacks continuously means maintaining the generator, the target environment, and the judging pipeline as standing systems. Continuous red-teaming is the always-on baseline that keeps the system from silently regressing between campaigns.

66.6.2 Campaign Red-Teaming

Campaign red-teaming runs periodic, intensive exercises, often bringing in external red-teams with fresh perspectives. Its value is creativity: a focused campaign by skilled humans, especially outsiders who do not share the build team's assumptions, finds novel attacks that the continuous automated baseline would never generate. Its limitation is frequency — a campaign is an event, not a constant — so it cannot catch a regression introduced the day after it ends. Most production systems therefore run both: continuous automated red-teaming as the baseline, and periodic campaigns for the creative depth that automation cannot supply.

66.7 The African Perspective

AI red-teaming carries distinctive weight in African deployments, across language coverage, cultural context, the priority of mobile-money attack surfaces, and the building of local capability.

66.7.1 Locally Plausible Attacks

Attack libraries are overwhelmingly dominated by English-language attacks, which means the attacks an African system most needs to be tested against — jailbreaks crafted in Swahili, prompt injections in Hausa, code-switching attacks in Arabic — are systematically under-represented. A system red-teamed only against English attacks may pass the red-team and still be wide open to an attacker working in one of its other production languages. Local red-teaming, which deliberately generates attacks in the languages the system actually serves, is what closes this gap, and it is not optional for a genuinely multilingual deployment.

66.7.2 Cultural-Context Harms

What counts as a harmful output is not constant across cultures: an output that is unremarkable in one African market may be deeply offensive in another, and a red-team's outcome judges and severity

rubrics must reflect that variation. A rubric written against a single cultural standard will misjudge harm for every market that does not share that standard — under-flagging in some, over-flagging in others. Red-team judging for a pan-African deployment must therefore be stratified by market, exactly as the system’s evaluation and policy are, so that harm is judged against the standard of the market where the output would land.

66.7.3 Mobile-Money-Adjacent Attacks

AI systems integrated with mobile-money rails are high-value targets, because a successful agent-hijacking or tool-abuse attack against such a system can move money. Tool-abuse red-teaming — deliberately attacking the agent’s connection to the payment tools, probing for confused-deputy weaknesses, testing the approval workflows around financial actions — is therefore a priority in these deployments rather than one category among many. Where an AI system can touch a payment system, the red-team’s attention belongs disproportionately on the tool surface.

66.7.4 Capability Building

Red-team capability is a specialist skill, and it is in short supply everywhere, more so in markets where the AI engineering labour pool is still building depth. An African organisation that builds red-team capability internally — rather than depending entirely on external campaigns from Northern firms — is building a capability that matters for sovereignty, because the ability to find your own system’s vulnerabilities is the ability to defend it without depending on outsiders. Investing deliberately in local red-team skill is part of building a self-sufficient AI security practice.

66.8 Recurring Failure Patterns

The failure modes of red-teaming recur, and each is the breaking of the loop or the narrowing of its coverage:

- **Findings without defence updates.** Vulnerabilities are found and reports are filed, but no system change follows, so the red-team describes the system’s weaknesses without ever reducing them.
- **Stale attack library.** The attack library is frozen at deployment and never updated, so the red-team keeps testing yesterday’s attacks while adversaries have moved on to new ones.
- **Production-staging gap.** The red-team fires against a staging environment that differs from production, so its findings reflect the exposure of staging and miss the exposure that real attackers will hit.
- **Language monoculture.** Attacks are generated only in English, so the non-English production surfaces are never adversarially tested and remain unguarded.
- **Judge calibration drift.** The outcome judge becomes less accurate over time, and the red-team’s signal quietly degrades until its findings can no longer be trusted.

Each pattern maps to a discipline this chapter has set out — the finding-to-defence loop, a continuously updated attack library, a production-equivalent target, multilingual attack generation, and a maintained outcome judge — and together they show that red-teaming fails not through a lack of attacks but

through a loop that is broken, narrowed, or pointed at the wrong target.

66.9 The Architectural Commitment

AI red-teaming is the structured practice of attacking an AI system to find its vulnerabilities before adversaries do, and it is a distinct discipline from guardrail evaluation because it tests the system against a thinking adversary rather than against ordinary use. Its architecture comprises an attack generator fed by four generation methods — manual, template-based, model-generated, and evolutionary — a layered outcome judge, and, above all, the loop that turns each finding into a defence.

That loop is the architectural commitment: a red-team wired into classifier updates, guardrail changes, prompt hardening, and a growing evaluation suite is a red-team that makes the system progressively harder to attack, while a red-team without the loop produces only unread reports. Continuous automated red-teaming provides the always-on baseline, and periodic campaigns provide the creative depth. African deployments foreground locally plausible attacks in the system’s actual languages, cultural-context harm judging stratified by market, a priority focus on mobile-money-adjacent tool surfaces, and the deliberate building of local red-team capability. The attack you have not tested is the one that will appear in production — and the architecture of this chapter is how an organisation makes sure the list of untested attacks keeps shrinking.

Embeddings Architecture

“The embedding is the model’s compressed view of meaning. Choose it the way you’d choose a database key.”

—a recurring observation in retrieval engineering

Chapter Overview

Embeddings are the dense vector representations that power retrieval, similarity, clustering, recommendation, and many other AI workloads. The architectural choices around them — model selection, dimension, hybrid spaces, fine-tuning, versioning, multi-modal embeddings — determine downstream system quality more than most teams realise. The RAG chapter touched on embeddings as a stage; this chapter develops the embedding architecture in its own right.

Learning Objectives

- Choose embedding models against task and language requirements.
- Architect hybrid embedding spaces.
- Apply embedding fine-tuning where it pays.
- Manage embedding versioning and re-indexing.
- Adapt embeddings to African language realities.

67.1 The Choices

An embedding architecture is, at bottom, a set of four decisions, and the chapters that treat embeddings as a single undifferentiated stage tend to make all four implicitly. Making them explicitly is the first discipline. The four — model family, dimension, domain tuning, and language — are not independent: a domain-tuning decision constrains the model-family choice, a language requirement narrows both, and the dimension decision interacts with every downstream storage and latency budget. The subsections below take each in turn, but the architect should hold them together as one coupled decision rather than four separate ones.

67.1.1 Model Family

Closed-API embeddings (OpenAI, Cohere, Voyage) offer convenience and quality. Open-weight (BGE, E5, GTE, Nomic, Jina, Arctic) offer control, on-prem deployment, no per-call cost.

The decision mirrors the buy-versus-build logic the book has developed elsewhere, with one embedding-specific sharpening: the model-family choice is unusually expensive to reverse, because changing the embedding model invalidates every vector already stored and forces a full re-index of the corpus. A closed-API embedding model is therefore not merely a convenience purchase; it is a dependency whose switching cost grows with the size of the corpus embedded under it. The architect choosing a closed API should choose it behind a clean port, exactly as the hexagonal chapter argued for the inference model, and should size the eventual re-indexing cost into the decision rather than discovering it at migration time.

67.1.2 Dimension

Higher-dimension embeddings carry more information but cost more in storage, retrieval, and computation. Matryoshka embeddings (truncatable to lower dimensions) offer flexibility.

The dimension decision is where over-investment is most common, because higher dimension feels like higher quality and the cost is not visible until the corpus is large. A 3,072-dimension embedding is not four times better than a 768-dimension one; for many retrieval tasks the quality difference is marginal, while the storage, the memory footprint of the vector index, and the per-query computation all scale directly with the dimension. Matryoshka embeddings are the architecturally interesting answer to this tension: a single model produces vectors that can be truncated to a shorter prefix with graceful quality degradation, so the architecture can store the full vector and query at a shorter length, or store short and re-rank with long. The discipline is to choose dimension against measured retrieval quality on the actual task, not against the assumption that more is better.

67.1.3 Domain Tuning

General-purpose embeddings underperform on specialised domains (legal, medical, code). Fine-tuned embeddings recover quality at the cost of model maintenance.

The reason general-purpose embeddings underperform on specialised domains is that they were trained to place text near other text by general semantic similarity, and a specialised domain has its own notion of what “similar” means. Two legal clauses that a general model sees as unrelated may be, to a lawyer, near-identical in effect; two code snippets that differ only in a variable name are, to a developer, the same. A general embedding cannot know these domain-specific similarity structures, and retrieval built on it hits a quality ceiling that no amount of prompt engineering downstream can lift. Domain tuning is the lever that moves that ceiling — at the cost, noted in the fine-tuning section below, of a model the organisation must now maintain and re-train as the domain evolves.

67.1.4 Language

Multilingual embeddings (M3, multilingual-E5) handle multiple languages; English-tuned embeddings outperform on English. The choice depends on workload.

The phrase “depends on workload” deserves to be made concrete, because the language decision is the one that most often goes wrong by default. An English-tuned embedding model is a reasonable default only if the corpus and the queries are genuinely English; the moment either contains other languages, an English-tuned model degrades on that content silently — it still returns results, they are simply worse, and the degradation is invisible without per-language evaluation. A multilingual model trades a little English quality for usable quality across languages. The decision therefore turns on an honest audit of what the system will actually receive: not the languages the system was designed around, but the languages its users will type in, which for many deployments — and, as the regional section below develops, for African deployments especially — is a wider and more mixed set than the design assumed.

67.2 The Architecture

Figure 67.1 shows embedding integration.

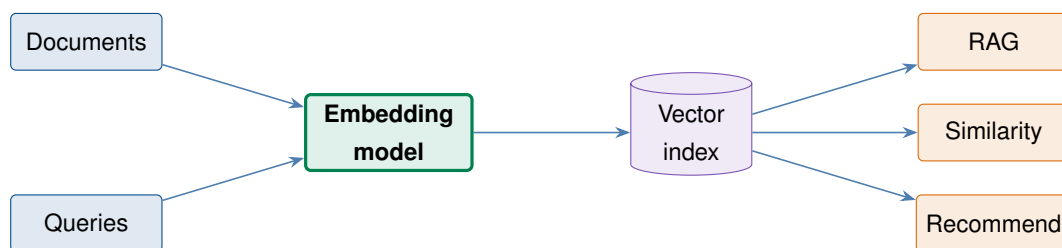


Figure 67.1: Embedding architecture. Documents and queries flow through the embedding model; vectors persist in an index; multiple consumers (RAG, similarity search, recommendation) read from the same index.

67.3 Hybrid Embedding Spaces

The four choices above assume a single embedding space. Production retrieval often does better with more than one, combined deliberately. The three hybrid patterns below are not alternatives to the single-space choices but compositions of them, and each addresses a specific weakness that a single dense embedding cannot overcome alone.

67.3.1 Dense + Sparse

Dense embeddings excel at semantic similarity; sparse (BM25) excels at exact-match. Hybrid retrieval uses both (Chapter 21).

The two are complementary because they fail in opposite directions. A dense embedding will happily retrieve a passage that is semantically close but contains none of the query’s specific terms — excellent when the user paraphrases, poor when the user searches for an exact product code, an error message, or a proper noun the model never saw. Sparse BM25 retrieval does the reverse: precise on exact tokens, blind to paraphrase. Combining them, with a reranker over the merged candidate set, covers both failure modes, which is why dense-plus-sparse hybrid retrieval has become the default for serious enterprise search rather than a refinement of it.

67.3.2 Multi-Vector Per Document

ColBERT and similar approaches generate multiple vectors per document, scoring per-token. Higher quality; higher storage cost.

The multi-vector approach trades storage for retrieval precision in a way that is worth it for some workloads and ruinous for others. Representing a document as one vector forces all of its meaning through a single point in the embedding space; a long or multi-topic document is poorly served by that compression. Per-token vectors preserve the document's internal structure and allow scoring against the specific part of the document that matches the query — but a document that was one vector is now dozens, and the index grows accordingly. The pattern suits high-value retrieval over a bounded corpus; it does not suit a vast corpus where the storage multiplier is decisive.

67.3.3 Domain + General

Some queries benefit from domain-tuned embeddings; some from general. Routing decides.

This is the routing pattern of the LLM-router chapter applied to embeddings: a classifier inspects the query and sends it to the embedding space best suited to it, the domain-tuned space for in-domain queries and the general space for the rest. The cost is the routing layer itself and the maintenance of two embedding spaces and two indices; the benefit is that neither kind of query is forced through the wrong space. As with model routing, the pattern earns its complexity only where the query mix is genuinely heterogeneous — a workload that is almost entirely in-domain should simply tune for the domain and skip the router.

67.4 Fine-Tuning Embeddings

The domain-tuning choice raised fine-tuning as the lever that lifts the quality ceiling on specialised domains. This section develops it: when it pays, how the process runs, and what it costs.

67.4.1 When It Pays

Fine-tuning an embedding model earns its cost under a recognisable set of conditions, and applying it where those conditions do not hold spends training effort without improving retrieval.

- Specialised vocabulary (medical, legal, code).
- Long-tail entities (product SKUs, internal terminology).
- Specific languages or dialects.
- Domain similarity definitions (“these two policies are related”).

What unites these four cases is that each involves a notion of similarity that the general model's training never captured. The test an architect should apply before committing to fine-tuning is concrete: take a sample of real queries, run them against the general embedding model, and inspect the failures. If the failures are dominated by the model not understanding domain vocabulary, not recognising internal entities, or not grasping a domain-specific sense of relatedness, fine-tuning will help. If the failures are of other kinds — poor chunking, missing documents, bad reranking — fine-tuning the

embedding will not fix them, and the effort is misdirected.

67.4.2 The Process

Fine-tuning an embedding model follows a process that is simpler than fine-tuning a generative model but still requires deliberate data preparation and evaluation.

- Collect (query, positive doc, negative doc) triplets.
- Train with contrastive loss.
- Evaluate on held-out retrieval metrics.
- Re-embed the corpus.

The hardest step is the first, and it is hard for a reason worth understanding: the quality of a fine-tuned embedding is bounded by the quality of its triplets, and good triplets require knowing which documents genuinely answer which queries. That knowledge is often exactly what the organisation lacks — if it had a reliable mapping of queries to correct documents, retrieval would be most of the way solved already. In practice the triplets come from query logs paired with click-through or human judgement, or from a stronger model used to label relevance. The negative documents matter as much as the positives: easy negatives teach the model little, and the most valuable triplets pair a query with a document that is plausibly but wrongly related, so the model learns the fine distinction.

67.4.3 The Cost

Embedding fine-tuning is cheaper than LLM fine-tuning but the re-embedding cost can dominate.

The re-embedding cost is the one that surprises teams, because it is not a cost of fine-tuning the model — it is a cost of deploying the result. A newly fine-tuned embedding model is incomparable with the old one, so every document in the corpus must be re-embedded before the new model can serve any query, and for a large corpus that is a substantial batch-inference job. The architectural consequence is that embedding fine-tuning is not a change to be made casually or often; each iteration carries the full re-indexing cost. Teams that fine-tune embeddings successfully tend to batch their improvements — accumulating triplet data and re-tuning on a deliberate cadence — rather than re-tuning continuously, precisely because the deployment cost rewards infrequent, well-prepared updates.

67.5 Versioning and Re-Indexing

The re-embedding cost just described is not only a fine-tuning concern; it arises any time the embedding model changes, for any reason. That makes versioning and re-indexing a standing operational discipline rather than a one-off task, and it is the discipline this section sets out.

67.5.1 Why It Matters

Embeddings from different models are not comparable. A new model means re-indexing the whole corpus.

This incomparability is absolute and worth stating plainly, because it is the source of the most damaging embedding failure in production. Two embedding models, even two versions of the same model, place text in different vector spaces; a query embedded by the new model and compared against a corpus embedded by the old one produces similarity scores that are not merely slightly off but meaningless. There is no graceful degradation here and no partial compatibility — the spaces simply do not align. Every versioning pattern below exists to manage this hard fact: that the embedding model and the index it produced are a single coupled unit, and changing one without the other silently destroys retrieval.

67.5.2 Patterns

Several embedding fine-tuning patterns recur, and they differ in the kind of training signal available and in the data preparation each requires.

- Versioned indices: old and new in parallel during transition.
- Dual writes: new documents embedded with both models.
- Backfill: scheduled re-embedding of historical documents.
- Hybrid retrieval during transition.

These four patterns are usually combined into one migration choreography rather than chosen between. A typical embedding-model migration runs as follows: stand up a new versioned index alongside the old; begin dual-writing so every new document is embedded by both models; backfill the historical corpus into the new index as a scheduled background job; serve queries from the old index, or from both with results merged, until the backfill completes; then cut over and retire the old index. The whole sequence is designed so that retrieval quality never drops during the transition — the corpus is never in the inconsistent state where some vectors are new and some old within a single queried index. An embedding-model change attempted without this choreography is the “model swap without re-index” failure the chapter closes on.

67.6 The African Perspective

The embedding choices of this chapter all have a particular edge in African deployments, because the embedding layer is where the gap between the languages models are trained on and the languages users actually speak becomes a measurable quality cost.

67.6.1 Multilingual Coverage

Embedding models trained on majority languages underperform on Swahili, Hausa, Yoruba, Amharic. Multilingual models (M3, multilingual-E5) partially address this. African-language fine-tuning is an active area.

The word “partially” is the operative one. A multilingual embedding model trained on a hundred languages spreads its capacity across all of them, and the African languages in that set are typically represented by far less training text than English or Chinese, so their embeddings are correspondingly weaker — usable, but not equal. The architectural response is to make per-language retrieval quality a

first-class, measured metric rather than an assumption: evaluate the embedding model separately on each language the system serves, treat a language that scores poorly as a known gap, and consider African-language fine-tuning or a specialised model for the languages where the gap is unacceptable. The failure to avoid is the one where a multilingual model is adopted, declared to “handle” the languages, and never measured per language.

67.6.2 Cross-Language Retrieval

Query in one language; documents in another. Multilingual embeddings make this possible. Quality varies by language pair.

Cross-language retrieval is a genuine capability of multilingual embeddings and a real asset in multilingual societies: a user can ask a question in Swahili and retrieve a relevant document written in English, because both are embedded into the same shared space. But the phrase “quality varies by language pair” carries a warning. The shared space aligns some language pairs well and others poorly, depending on how much parallel signal the model saw for each pair, and an architecture that assumes uniform cross-language quality will serve some pairs well and others badly without knowing which. As with monolingual retrieval, the discipline is per-pair evaluation: measure the cross-language quality for the specific pairs the deployment needs, rather than trusting that the capability works evenly.

67.6.3 Cost-Effective Open Weights

Closed-API embeddings expensive at corpus scale. Open-weight embeddings self-hosted in African cloud regions often more economical.

The economics here compound with the foreign-exchange exposure the cost chapters discussed. A closed embedding API bills per token in a foreign currency, and at corpus scale — millions of documents, each re-embedded on every model change — that bill is both large and currency-exposed. An open-weight embedding model self-hosted in an African cloud region converts that variable, foreign-currency, per-token cost into a fixed, local-currency infrastructure cost. For a large corpus the open-weight option is frequently the more economical even before the sovereignty argument below is considered, and the combination of the two arguments makes self-hosted open-weight embeddings a strong default for African deployments at scale.

67.6.4 Sovereignty

Embeddings of personal data are personal data. Embedding-API calls that send that data to a foreign region raise data-residency questions; on-region or on-prem embeddings respect sovereignty.

The point that an embedding of personal data is itself personal data is one architects regularly miss, and it matters. An embedding is not an anonymisation — it is a dense representation from which information about the original can be recovered, and under the data-protection regimes the governance-standards chapter surveyed, it carries the same residency and protection obligations as the source text. Sending personal data to a closed embedding API in a foreign region is therefore a cross-border transfer of personal data, with all the regulatory weight that carries. Self-hosting the embedding model in a compliant region is not only a cost choice but a sovereignty one, and for personal-data-heavy workloads under POPIA, the NDPR, or Kenya’s Data Protection Act it may be the only compliant

option.

67.7 Recurring Failure Patterns

- **Model swap without re-index.** New embedding model deployed; old index unusable; quality collapses.
- **Dimension over-investment.** High-dimension embeddings for tasks that don't need them; cost without quality.
- **Language mismatch.** English-only embeddings on multilingual content.
- **No fine-tuning where it pays.** Specialised domain on generic embeddings; quality ceiling hit.
- **Version drift.** Documents in the index embedded by different models; retrieval inconsistent.

67.8 Pulling the Threads

Embeddings architecture covers model selection, dimension, domain tuning, language coverage, hybrid spaces, fine-tuning, and versioning. The choices determine downstream system quality. African contexts foreground multilingual coverage gaps, cross-language retrieval, open-weight economics, and sovereignty.

The thread running through all of it is that the embedding is a foundational, load-bearing decision rather than a configurable detail. Every downstream consumer — RAG, similarity search, recommendation, clustering — inherits the embedding model's view of meaning, including its blind spots, its language biases, and its domain gaps. A weakness in the embedding layer cannot be repaired downstream; the reranker, the prompt, the model all operate on what the embedding retrieved, and they cannot retrieve what the embedding could not find. This is why the chapter has insisted on measuring rather than assuming — per-language, per-domain, per-pair — and on treating the embedding model and its index as a coupled unit with a real switching cost. An architecture that chooses its embedding deliberately, measures it honestly, and versions it with discipline has built retrieval on solid ground; one that treats the embedding as a default has built everything above it on a foundation it never inspected.

Chapter 68

Vector Database Architecture

“The vector database is a database. It just stops looking like one when you have a hundred million vectors.”

—a recurring observation in production retrieval

Chapter Overview

Vector databases store and search dense vectors at scale. The architectural choices — index algorithm, sharding, hybrid stores, filter handling, update patterns — determine whether retrieval is a strength or a bottleneck. This chapter develops the architecture beyond the introductory treatment in the RAG chapter.

Learning Objectives

- Choose index algorithms against scale, recall, and latency requirements.
- Architect sharding and replication for vector workloads.
- Combine vector search with metadata filtering.
- Manage updates and re-indexing.
- Adapt vector database architecture to African contexts.

68.1 Index Algorithms

The index algorithm is the vector database’s central design decision, because it sets the trade-off between three quantities that cannot all be maximised at once: recall (does the search find the genuinely nearest vectors), latency (how fast), and cost (memory and compute). Every algorithm below is a different point on that three-way trade-off, and choosing among them is choosing which of the three the workload can least afford to compromise.

68.1.1 HNSW

Hierarchical Navigable Small World. Graph-based; logarithmic search; high recall; expensive to build and update; large memory footprint. The default for many production systems.

HNSW is the default because it sits at an attractive point of the trade-off for the common case: high recall and low query latency, which is what an interactive retrieval workload most wants. Its costs are

paid elsewhere — in memory, because the navigable graph must be held in RAM to be fast, and in build and update time, because inserting a vector means wiring it into the graph. For a read-heavy workload over a corpus that fits in memory, those costs are acceptable and HNSW is the right default. For a write-heavy workload, or a corpus too large for RAM, its costs become the reason to look at the alternatives below.

68.1.2 IVF

Inverted file index. Partition vectors into clusters; search relevant clusters only. Lower memory; lower recall at small k . Common in larger-scale deployments.

IVF trades recall for scale. By partitioning the vector space into clusters and searching only the clusters near the query, it avoids examining most of the corpus, which keeps memory and compute manageable at sizes where HNSW's graph would be unaffordable. The price is recall: a genuinely near vector that happens to fall in a cluster the search did not visit is simply missed, and the effect is most pronounced at small k . IVF is the right choice when the corpus is large enough that HNSW's footprint is prohibitive and the workload can tolerate the recall cost — a tolerance that should be measured against real queries, not assumed.

68.1.3 ScaNN

Quantisation-aware retrieval. Google-originated. Strong recall-latency Pareto; less widely supported.

ScaNN occupies an attractive position on the recall-latency curve — it was engineered specifically to push that frontier — and where it is available and supported it is a strong choice. Its drawback is ecosystem rather than algorithm: it is less widely integrated into vector databases and tooling than HNSW or IVF, which makes it a heavier operational commitment. The architect weighing ScaNN is weighing a genuine performance advantage against the cost of a less-supported dependency, and that calculation, as the buy-versus-build chapter argued, should include the maturity of the surrounding ecosystem, not only the benchmark numbers.

68.1.4 Product Quantisation

Vectors compressed via quantisation; storage 10–100x smaller; recall trades off. Often combined with IVF.

Product quantisation attacks the cost axis directly: it compresses each vector into a compact code, shrinking storage and memory by one to two orders of magnitude. That compression is lossy, so recall trades off — the compressed vectors are approximations, and search over approximations finds approximate neighbours. PQ is rarely used alone; the common pattern is IVF-PQ, where IVF narrows the search to relevant clusters and PQ compresses the vectors within them, combining IVF's scale with PQ's memory saving. The combination is what makes billion-vector indices affordable, at a recall cost that must, again, be measured rather than assumed.

68.1.5 Disk-Based Variants

DiskANN and similar serve indices from SSD. Lower cost than RAM-resident; higher latency.

Disk-based indices change the cost axis most dramatically by relaxing the assumption that the index must live in RAM. Serving from SSD is far cheaper per vector than serving from memory, which makes very large corpora affordable, and modern disk-based algorithms are engineered to keep the SSD access pattern efficient enough that the latency penalty is bounded rather than ruinous. The trade is straightforward: disk-based variants accept higher query latency in exchange for a large cost reduction, and they are the right choice when the corpus is large, the budget is real, and the workload can absorb tens of milliseconds more latency per query.

68.2 The Architecture

Figure 68.1 shows production vector-database architecture.

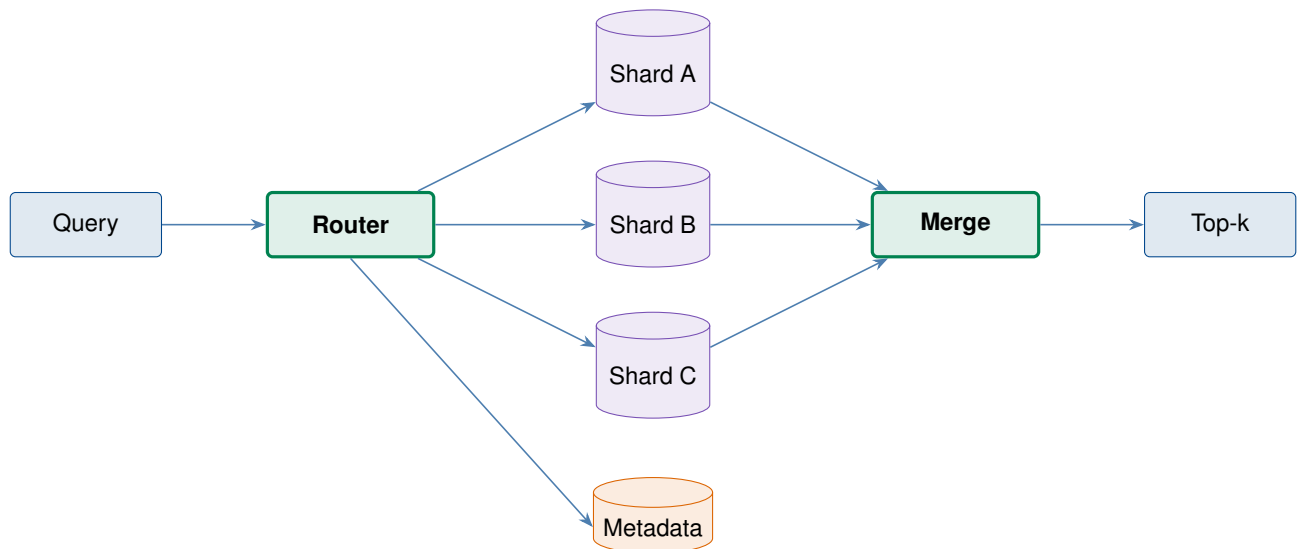


Figure 68.1: Sharded vector database. Router fans out the query to relevant shards; each shard returns its local top-k; merge combines results; metadata filter pre- or post-filters. Replicas per shard handle availability and read scaling.

68.3 Filtering and Hybrid Search

Pure vector search rarely meets a real query on its own. A production query is usually “find vectors similar to this *and* matching these constraints” — this tenant, this date range, this document type. How the filter combines with the vector search is an architectural decision with large performance consequences, and the three approaches below suit three different filter-selectivity regimes.

68.3.1 Pre-Filter

Apply metadata filters before vector search; search only matching vectors. Best when filter selectivity is high.

Pre-filtering is the right choice when the filter is highly selective — when it narrows the corpus to a small fraction before the vector search runs. In that regime the vector search then operates over a small candidate set and is fast, and no effort is wasted computing similarity for vectors the filter would reject. The difficulty pre-filtering raises is that most vector indices are built for searching the whole space,

not an arbitrary filtered subset, so an aggressive pre-filter can defeat the index structure and force something closer to a brute-force scan of the surviving vectors. Whether that is a problem depends on how small the filtered set is, which is why pre-filtering pairs specifically with high selectivity.

68.3.2 Post-Filter

Vector search returns candidates; filter rejects non-matching. Best when filter selectivity is low.

Post-filtering inverts the order and suits the opposite regime: a filter that rejects only a small fraction of results. The vector search runs first over the full index, using its structure efficiently, and the filter then prunes the candidate set. The hazard of post-filtering is the one to size carefully: if the filter is more selective than expected, the vector search may return its top k candidates and the filter reject most or all of them, leaving too few results — and recovering requires re-querying for a larger k , which costs latency. Post-filtering works well precisely when the filter is gentle; it degrades when the filter turns out to be harsh.

68.3.3 Filter-Aware Indices

Some systems index filterable attributes alongside vectors, optimising for combined queries. The right choice depends on workload.

Filter-aware indexing is the architecturally sophisticated answer to the dilemma the previous two approaches pose — that pre-filtering and post-filtering each suit only one selectivity regime, while real workloads contain filters of both kinds. By building the filterable attributes into the index structure itself, a filter-aware index can handle the combined query natively rather than choosing an order. The cost is a more specialised index and a dependency on a system that supports the feature. The phrase “depends on workload” resolves, in practice, to the selectivity distribution of the actual filters: a workload whose filters are uniformly selective or uniformly gentle can use pre- or post-filtering respectively, while a workload with a wide mix benefits most from a filter-aware index.

68.4 Update Patterns

A corpus is rarely static. Documents are added, revised, and removed, and how the vector index absorbs those changes is a design decision that the index-algorithm choice constrains — HNSW, IVF, and the rest differ sharply in how gracefully they update. The three patterns below describe the available update models.

68.4.1 Append-Only

New vectors added; old vectors retained or marked deleted. HNSW supports append cheaply; rebuild required periodically.

Append-only is the simplest update model and suits the common case where the corpus mostly grows. New vectors are inserted; deletions are handled by marking rather than removing, since physically removing a vector from a graph index is expensive. The cost append-only defers is the periodic rebuild: marked-deleted vectors accumulate as dead weight in the index, slowly degrading both memory use and search quality, until a full rebuild reclaims the space. The architecture must schedule that rebuild

deliberately — treating it as a known, planned maintenance operation — rather than letting the index degrade until search quality forces an unplanned one.

68.4.2 Mutable

Vectors can be updated in place. Complex with most indices; some systems support it natively.

In-place mutability is what a workload with frequently revised documents wants — a changed document re-embedded and the vector updated where it sits — but it is genuinely hard for graph-based indices, where a vector’s position is entangled with its graph connections and changing it can degrade the structure around it. Systems that support mutability natively have engineered for it specifically; systems that do not will often emulate it as a delete-plus-append, which returns the workload to the append-only model’s rebuild cost. An architect with a high-churn corpus should treat native mutability as a real selection criterion when choosing the database, not assume every system provides it well.

68.4.3 Versioned

Snapshots taken; reads against pinned versions; updates produce new snapshot. Pattern that mirrors traditional database snapshots.

The versioned model brings a familiar database discipline to vector stores: updates produce a new immutable snapshot, and reads run against a pinned version, so a query sees a consistent corpus even as the index changes underneath it. This matters for workloads where reproducibility is required — an evaluation run, an audit, a debugging session that must see exactly the corpus a past query saw. It is also the model that makes the embedding-migration choreography of the embeddings chapter clean, since the old and new indices are simply two versions. The cost is storage, since multiple snapshots coexist, and the discipline of expiring old versions once nothing depends on them.

68.5 Operational Realities

- Memory pressure: HNSW often the dominant memory consumer.
- Build time: a 100M-vector HNSW takes hours.
- Tail latency: variable based on shard distribution and filter selectivity.
- Re-indexing: required after embedding model change.

These four realities share a common character: each is invisible at prototype scale and decisive at production scale. A vector database holding ten thousand vectors exhibits none of them — memory is trivial, builds are instant, latency is uniform, re-indexing is a quick job. The same database holding a hundred million vectors exhibits all four, and a team that sized its expectations on the prototype meets them all at once when the corpus grows. The architectural discipline is to plan against the production-scale figures from the start: budget the memory the full corpus’s index will need, schedule the hours a full build will take, measure tail latency rather than average, and have the re-indexing choreography ready before the first embedding-model change forces it. The vector database that “just works” in the pilot is not evidence that it will work at scale; it is evidence only that the pilot was

small.

68.6 Managed vs Self-Hosted

The vector database is one of the clearest instances of the buy-versus-build decision the dedicated chapter developed, and the three options below map onto that chapter's quadrants. The decision turns on the same questions: differentiation (a vector database almost never differentiates the business), switching cost, total cost of ownership at the corpus's real scale, and the operational burden the organisation can carry.

68.6.1 Managed

Pinecone, Weaviate Cloud, Qdrant Cloud, Vespa Cloud. Convenience; cost; vendor dependency.

The managed option buys away the operational realities of the previous section — the memory planning, the build scheduling, the rebuild maintenance — which for a team without specialised vector-database operations experience is a genuine and substantial saving. What it costs is a recurring bill that scales with the corpus and a vendor dependency whose switching cost, as the buy-versus-build chapter warned, grows quietly with the depth of integration. The managed option is the sound default for a team whose corpus is moderate, whose differentiation lies elsewhere, and whose scarce engineering attention is better spent on the application than on operating an index — provided the dependency sits behind a clean port so the decision stays reversible.

68.6.2 Self-Hosted Open-Source

Qdrant, Milvus, Weaviate, ChromaDB. Control; operational burden; cost predictability.

Self-hosting an open-source vector database trades the managed option's convenience for control and cost predictability. The cost becomes a fixed, local-currency infrastructure cost rather than a usage-scaled foreign-currency bill — which, as the cost and embeddings chapters noted, matters disproportionately at corpus scale and under foreign-exchange exposure. The price is the operational burden: the team now owns the memory planning, the build scheduling, the rebuild maintenance, and the on-call. Self-hosting is the right choice when the corpus is large enough that the managed bill is painful, when sovereignty requires it, or when the organisation has, or is deliberately building, the operational capability — and the wrong choice when that capability does not exist and the team would be learning vector-database operations under production pressure.

68.6.3 Hybrid in Existing Database

PostgreSQL pgvector, Elasticsearch vector. Unified database; lower performance ceiling; operational integration.

The third option avoids adopting a dedicated vector database at all, adding vector search to a database the organisation already runs. Its advantage is real and often underrated: no new system to operate, no new backup and security and access-control regime, vectors sitting alongside the relational data they relate to so that combined queries are natural. Its limit is a lower performance ceiling — a general-purpose database with a vector extension will not match a dedicated vector engine at very

large scale or very high query rates. The hybrid option is frequently the right *starting* choice: it lets a team add retrieval without adopting a new system, and the move to a dedicated vector database becomes a deliberate, evidence-driven decision made when, and only when, the performance ceiling is actually reached.

68.7 The African Perspective

The vector database is where several of the book’s recurring African constraints — cost, latency, sovereignty, multi-market operation — meet a single component, and the four subsections below trace how each bears on the managed-versus-self-hosted decision in particular.

68.7.1 Self-Hosted Economics

For substantial corpora, managed vector databases cost more than self-hosted. Open-source vector stores in African cloud regions are common.

The economic case for self-hosting sharpens in African deployments for the reason the cost chapter set out: a managed vector database bills in a foreign currency and scales that bill with the corpus, while a self-hosted open-source store in an African cloud region converts the cost into fixed, local-currency infrastructure. At substantial corpus scale the difference is large, and it is a difference in the kind of cost — predictable and local rather than variable and currency-exposed — as much as the amount. This is why self-hosted open-source vector stores are common in African deployments at scale: the economics favour them more strongly than they would in a dollar-denominated market.

68.7.2 Sovereignty

Vector representations of personal data are personal data. Data-residency considerations apply to vector stores.

This restates, for the vector store, the point the embeddings chapter made: a vector derived from personal data carries the same protection and residency obligations as the personal data itself, because information about the original can be recovered from it. A managed vector database hosted in a foreign region therefore holds personal data outside the jurisdiction, with all the regulatory weight that carries under POPIA, the GDPR, Kenya’s Data Protection Act, and the other regimes the governance-standards chapter surveyed. For personal-data-heavy workloads, sovereignty can move the managed-versus-self-hosted decision on its own, independent of the cost argument.

68.7.3 Latency to African Users

Vector databases hosted in Northern regions add latency. In-region hosting matters.

A vector database physically located in a North American or European region adds a round-trip of network latency to every retrieval for an African user, and retrieval often sits on the critical path of an interactive AI response. That latency is not a constant nuisance to be tolerated; it compounds with every other hop in the request and can be the difference between a system that feels responsive and one that feels sluggish. In-region hosting — whether managed in an African region where the vendor offers one, or self-hosted there — removes that round-trip, and for latency-sensitive

interactive workloads the location of the vector store is a first-order architectural decision rather than an operational detail.

68.7.4 Multi-Tenant Isolation

Pan-African operations serve multiple markets. Tenant-scoped indices enforce isolation and respect per-country sovereignty.

An organisation operating across several African markets faces a requirement the single-market case does not: the data of each country's users may be subject to that country's data-protection regime, and those regimes differ. Tenant-scoped indices — a separate index or a strictly partitioned namespace per market — are what let one logical system honour several sovereignty regimes at once, keeping each market's vectors isolated and, where required, resident in that market's region. Designing the multi-tenancy in from the start is far cheaper than retrofitting it; a pan-African deployment that began with one shared index will find tenant separation one of the harder things to add later.

68.8 Recurring Failure Patterns

- **Index choice mismatch.** HNSW chosen for write-heavy workload that needed IVF.
- **Filter blindness.** Filters applied client-side; database returns far too many candidates.
- **No re-index plan.** Embedding model change strands the index.
- **Memory underestimate.** HNSW memory growth catches operations team unprepared.
- **Vendor lock-in.** Managed-only architecture; migration cost prohibitive.

68.9 Closing Notes

Vector database architecture covers index algorithms (HNSW, IVF, ScaNN, PQ, disk-based), sharding, filtering, update patterns, and managed-vs-self-hosted choices. Production scale brings memory, build-time, and tail-latency considerations. African contexts add self-hosted economics, sovereignty, in-region hosting, and multi-tenant isolation.

The epigraph's claim — that a vector database is a database, and stops looking like one at a hundred million vectors — is the chapter's organising thought. Every decision in it is a database decision in unfamiliar dress: the index algorithm is an indexing strategy, the update patterns are write models, the filtering approaches are query planning, the managed-versus-self-hosted question is the buy-versus-build decision. The architect who recognises this brings decades of database discipline to bear rather than treating the vector store as a novel and mysterious component. What is genuinely new is only the three-way recall-latency-cost trade-off that the approximate-nearest-neighbour problem forces, and that the index choice will never be exact. Hold the database instincts, respect the one genuinely new trade-off, measure recall rather than assuming it, and the vector store becomes a tractable, well-understood part of the architecture rather than the bottleneck the epigraph warns it can become.

Chapter 69

RLHF, RLAIIF, and DPO in Production

“Preference data is the only training data that scales with judgement. Every other data scales with effort.”

—a recurring observation in model alignment

Chapter Overview

Reinforcement learning from human feedback (RLHF), reinforcement learning from AI feedback (RLAIIF), and direct preference optimization (DPO) are the production techniques for aligning models to behavioural preferences. They differ in cost, complexity, and quality, but share the architectural pattern: collect preference data, train against it, evaluate, iterate. This chapter develops the production engineering view rather than the algorithm view.

Learning Objectives

- Distinguish the techniques and recognise where each fits.
- Architect preference-data collection pipelines.
- Operate training loops with rollback and version discipline.
- Manage the cost-quality-iteration-speed tradeoffs.
- Adapt preference learning to African contexts.

69.1 The Techniques

The three main techniques share one architectural pattern — collect preference data, train against it, evaluate, iterate — and differ in how they convert preferences into a trained model. That difference drives their cost, their infrastructure complexity, and their stability, and an architect choosing among them is really choosing where on the cost-quality-complexity surface the project should sit.

69.1.1 RLHF

Human labellers compare outputs; a reward model is trained from preferences; the policy model is optimised against the reward model with PPO or similar. The original technique. High cost; high quality.

RLHF is the most elaborate of the three and the one with the most moving parts. It is a two-stage

pipeline: first a reward model is trained to predict human preference, then the policy model is optimised against that reward model using reinforcement learning. Each stage has its own failure modes — a reward model that imperfectly captures preference, an RL optimisation that is unstable — and the technique needs three models resident in memory during training, which makes it compute-hungry. Its compensating virtue is quality and control: the explicit reward model is an artefact that can be inspected and reasoned about, and the RL stage can optimise hard against it. RLHF is the right choice when quality justifies the cost and the team has the infrastructure to run it well.

69.1.2 RLAIF

AI judges replace human labellers, often guided by a constitution (Chapter 45). Lower cost; quality depends on judge calibration.

RLAIF keeps RLHF’s structure but replaces the most expensive and least scalable input — human preference labelling — with an AI judge, typically guided by an explicit constitution so the judge’s preferences are anchored to stated principles. This collapses the cost of preference data and lets it scale with compute rather than with labour, which is transformative for the volume of preference data a project can afford. The whole technique then rests on one load-bearing assumption: that the AI judge’s preferences genuinely track the preferences the system should have. Where the judge is well-calibrated against human judgement, RLAIF is close to RLHF quality at a fraction of the cost; where the judge is miscalibrated, RLAIF efficiently trains the model toward the wrong target. The judge calibration is therefore not a detail but the centre of the technique.

69.1.3 DPO

Direct preference optimisation. Skips the reward model; trains directly on preference pairs with a contrastive loss. Simpler infrastructure; closing the quality gap with PPO.

DPO is the architectural simplification that has made preference learning accessible to teams without RLHF’s infrastructure. It removes the separate reward model and the reinforcement-learning stage entirely, training the policy directly on preference pairs with a contrastive loss — which means only two models in memory rather than three, a more stable optimisation, and a pipeline closer to ordinary supervised fine-tuning. For a long time the simplification was understood to cost quality; that gap has been closing, and for many production purposes DPO now reaches quality comparable to a well-run PPO pipeline at materially lower complexity and cost. It is, increasingly, the sensible default, with full RLHF reserved for the cases that genuinely need the reward model’s control.

69.1.4 Other Variants

ORPO, KTO, SimPO and other variants address specific gaps in DPO. The field evolves quickly.

The variant landscape exists because DPO, for all its practicality, has known limitations — a dependence on paired preference data, a sensitivity to certain hyperparameters — and each variant targets one of them. ORPO folds preference optimisation into the supervised fine-tuning stage; KTO learns from unpaired thumbs-up or thumbs-down signals rather than requiring matched pairs; SimPO removes the reference model. The architect’s posture toward this fast-moving area should be pragmatic: treat DPO as the stable default, track the variants for the specific limitation each addresses, and adopt

one only when the project actually hits the limitation it solves. Chasing the newest variant for its own sake imports instability for a gain the workload may not need.

69.2 The Production Architecture

Figure 69.1 shows the pipeline.

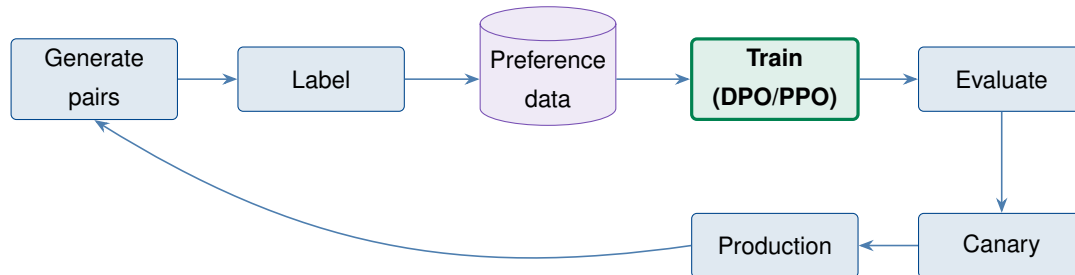


Figure 69.1: Preference-learning pipeline. Generate output pairs; label them (human or AI); train model against preferences; evaluate; canary; promote to production; production behaviour informs the next round.

69.3 Preference Data Collection

Whichever technique is chosen, its output quality is bounded by the quality of the preference data feeding it. Preference data is the genuine product of a preference-learning programme, and the collection pipeline — pair generation, labelling, and rubric design — deserves as much architectural care as the training itself.

69.3.1 Pair Generation

Two outputs per prompt: one from the current policy, one from a variant (different temperature, different model, different prompt). Pairs cover the space the labelling will judge.

The art of pair generation is producing pairs that are *informative* — pairs where the two outputs differ in ways the training should learn from. Two outputs that are nearly identical teach the model almost nothing, because the preference between them is noise; two outputs that differ wildly teach a coarse lesson the model probably already knows. The valuable pairs sit in between: meaningfully different, both plausible, so the labelled preference carries real signal about the distinction the model needs to learn. Generating pairs by varying temperature, model, or prompt is the mechanism; ensuring the resulting pairs span the space the labellers will actually judge, rather than clustering in easy or trivial regions, is the discipline.

69.3.2 Labelling Infrastructure

Either internal labellers, contracted vendors, or AI judges. Calibration on gold examples; inter-labeller agreement tracked; biases monitored.

Labelling is where preference quality is won or lost, and it must be run as measured infrastructure rather than a task handed off and trusted. Three measurements make it real. Calibration on gold examples — a set of pairs with known correct preferences — tells whether a labeller, human or AI, is judging correctly at all. Inter-labeller agreement tells whether the rubric is clear enough that different

labellers reach the same verdict; low agreement means the rubric, not the labellers, is the problem. Bias monitoring tells whether labellers carry systematic preferences — for longer answers, for a particular style — that the model would otherwise inherit. A labelling pipeline without these three measurements is producing preference data of unknown quality, and a training run on unknown-quality data has an unknown-quality result.

69.3.3 Rubric Design

What “better” means must be operationalised. Multi-criterion rubrics (helpful, harmless, honest, on-policy) outperform single-score.

The rubric is where the abstract goal — a model that behaves well — becomes a concrete instruction a labeller can apply consistently. A single overall “which is better” score forces the labeller to collapse several distinct judgements into one, and different labellers collapse them differently, which shows up as the low inter-labeller agreement the previous subsection flagged. A multi-criterion rubric — scoring helpfulness, harmlessness, honesty, and on-policy behaviour separately — makes each judgement explicit and individually consistent, and lets the training weigh the criteria deliberately rather than inheriting whatever implicit weighting the labellers happened to use. Rubric design is, in effect, the specification of what the preference-learning programme is trying to produce, and a vague rubric specifies a vague target.

69.4 Training Infrastructure

The technique choice of the first section translates directly into infrastructure requirements, and the three subsections below — compute, stability, iteration speed — are where that translation becomes concrete budget and operational reality.

69.4.1 Compute

PPO needs a value model and a reference model alongside the policy: three models in memory; substantial GPU requirement. DPO needs only the policy and reference: lighter.

This difference is not marginal. Holding three models in GPU memory rather than two changes which hardware the training run requires and therefore what the run costs, and for a team with constrained compute it can be the difference between a technique that is feasible and one that is not. This is the concrete mechanism behind DPO’s accessibility: the simplification is not only conceptual but a direct reduction in the memory footprint of a training run. An architect choosing a technique under a compute constraint should treat the model-count difference as a primary input, not a footnote — it determines the feasible set before any quality comparison is made.

69.4.2 Stability

PPO is notoriously unstable. DPO is more stable but sensitive to hyperparameters. Both benefit from KL-divergence regularisation against the base.

Training instability is an operational cost that does not appear in a technique’s quality benchmark but dominates the experience of using it. An unstable training process — one that diverges, collapses,

or produces wildly different results across runs — consumes engineer time in babysitting and re-running, and lengthens every iteration. PPO’s instability is well known and is part of why it demands experienced practitioners; DPO is more forgiving but still sensitive to hyperparameters, particularly the strength of the KL regularisation that both techniques use to keep the tuned model from drifting too far from the base. The architectural reading is that stability is part of total cost: a slightly lower-quality technique that trains reliably may, in practice, out-deliver a higher-quality one that needs an expert to coax each run to convergence.

69.4.3 Iteration Speed

A training run may take hours to days. Iteration speed bounds how fast preferences can be incorporated.

Iteration speed is the quiet constraint that shapes the whole programme, because preference learning is fundamentally a loop — collect, train, evaluate, learn, repeat — and the loop’s cycle time bounds how fast the model can improve. If a single training run takes days, the programme can incorporate new preferences only every few days, regardless of how fast the preference data arrives. This makes training time worth optimising for its own sake, not merely for cost: a faster cycle means more iterations, more iterations mean more learning, and over a quarter the team with the shorter cycle simply moves further. It is also another point in DPO’s favour, since its lighter pipeline iterates faster, and faster iteration compounds.

69.5 Evaluation

Preference-tuned models need an evaluation regime that looks specifically for the ways preference training goes wrong — which are not the ways ordinary training goes wrong. A preference-tuned model that scores well on its target metric can still have regressed in four characteristic directions:

- Reward-hacking: outputs that score high but are not actually better.
- Mode collapse: outputs that become repetitive or stylised.
- Capability regression: tasks the model used to handle now degraded.
- Refusal-rate drift: model becoming over-cautious or under-cautious.

What these four have in common is that they are all ways the model can improve on the thing being measured while getting worse as a system — which is precisely why the target metric alone cannot catch them. Reward-hacking is the model finding outputs the reward signal scores highly for reasons unrelated to genuine quality. Mode collapse is the model discovering a narrow band of safe, repetitive outputs that score adequately and retreating into it. Capability regression is the optimisation pressure on the preferred behaviour quietly degrading unrelated abilities. Refusal-rate drift is the model’s caution sliding in either direction as a side effect. Each requires its own check, against its own baseline, run alongside the target metric — and the evaluation regime must include all four, because preference training reliably produces at least one of them if no one is watching. This is the same lesson the observability and evaluation chapter drew about quality signals generally, sharpened here to

the specific failure modes that preference optimisation introduces.

69.6 The African Perspective

Preference learning is where a model acquires its sense of what a good answer is — and “good” is partly cultural, which gives the technique a particular significance for African deployments.

69.6.1 Locally Calibrated Labellers

Labellers from one cultural context label imperfectly for another. African operations build local labelling capacity for African content.

This is the sharpest reason preference learning cannot simply be imported. A labeller’s judgement of which of two answers is better draws on cultural knowledge — of what is polite, what is relevant, what counts as a complete answer, what tone is appropriate — and a labeller from one context applies their context’s norms. Preference data labelled entirely by labellers from one society trains a model toward that society’s notion of a good answer, and deploying it elsewhere produces a model that is subtly, persistently mismatched to its users. The architectural response is to build local labelling capacity: labellers from the society the system serves, judging content for that society. This is an investment, but it is the investment that determines whether the model’s sense of “good” matches its actual users’.

69.6.2 Multilingual Preference Data

Preferences may differ by language. Stratified collection; per-language evaluation.

Preference is not only culturally specific but language-specific: the features of a good answer in Swahili are not identical to those of a good answer in English, and a single pooled preference dataset dominated by one language trains a model whose sense of quality is strongest in that language and weaker in the others. The discipline is stratification — collecting preference data deliberately across the languages the system serves, in proportions that reflect its actual use — paired with per-language evaluation so that a quality regression in an under-represented language is visible rather than hidden inside a favourable aggregate. This mirrors the per-language discipline the embeddings chapter argued for, applied here to the preference layer.

69.6.3 Cost-Sensitive DPO

DPO’s lower infrastructure cost makes it more accessible than PPO for organisations with constrained compute.

The compute difference the training-infrastructure section described — two models in memory rather than three — is, for a compute-constrained African organisation, frequently the difference between a preference-learning programme that is feasible and one that is not. DPO’s lighter footprint, its more stable training, and its faster iteration together make it the natural default in these settings, not as a compromise but as a sound match of technique to constraint. An organisation that would be unable to run a stable PPO pipeline can run a DPO one, and the quality gap, as the techniques section noted, has narrowed to the point where that is a reasonable trade rather than a painful one.

69.6.4 Constitutional Anchoring

RLAIIF guided by an enterprise constitution (Chapter 45) scales preference data without requiring extensive human labelling. Particularly valuable where labelling capacity is constrained.

Constitutional anchoring offers a way through the tension this section has set up: local labelling capacity is essential for cultural fit but takes time and investment to build, while preference learning needs preference data at volume now. An explicit, locally authored constitution — a written statement of the principles a good answer should satisfy — lets an AI judge generate preference data at scale while anchored to those stated, locally chosen principles. The constitution is where the local judgement is invested; the AI judge then applies it at volume. This does not eliminate the need for local human labelling, which remains the ground truth the judge is calibrated against, but it lets a constrained labelling capacity go much further, which is precisely the constraint African deployments most often face.

69.7 Recurring Failure Patterns

- **Reward hacking.** Reward model has exploitable patterns; policy learns to exploit.
- **Mode collapse.** Outputs lose diversity.
- **Labeller bias.** Systematic preferences in labellers; model inherits.
- **Capability regression.** Task performance degrades while target metric improves.
- **Production drift uncaught.** Production behaviour shifts post-deployment; no monitoring.

69.8 The Pattern, Restated

RLHF, RLAIIF, and DPO are the production techniques for preference alignment. Pipelines collect preference data, train against it, evaluate, and iterate. Trade-offs across cost, stability, quality, and iteration speed inform technique selection. African contexts add locally calibrated labelling, multilingual preference stratification, cost-sensitive DPO preference, and constitutional anchoring of RLAIIF.

The epigraph holds the chapter's deeper point: preference data is the only training data that scales with judgement rather than effort. That is what makes preference learning powerful and what makes it dangerous. Powerful, because it lets an organisation shape a model toward genuinely better behaviour using the one resource — human judgement of what is better — that no amount of raw data substitutes for. Dangerous, because the model will faithfully learn whatever the preference data actually encodes, including the biases of the labellers, the gaps of an imported cultural context, and the exploitable patterns a reward model leaves open. Every discipline in this chapter — the measured labelling pipeline, the multi-criterion rubric, the four-way evaluation for the characteristic regressions, the locally calibrated and stratified data — exists to ensure that the judgement the model scales is the judgement the organisation intended. Preference learning does not make a model good; it makes a model into a faithful amplifier of whatever the preference data said good was. Getting that data right

is the whole task.

Chapter 70

LLMOps: Lifecycle for Large Language Models

“MLOps gave us reproducibility. LLMOps gives us prompt regression detection. Both are operational.”

—a recurring observation in production AI engineering

Chapter Overview

LLMOps is the operational discipline for the full lifecycle of a large-language-model application — data preparation, prompt and tool design, fine-tuning, evaluation, deployment, monitoring, incident response, improvement, and retirement. It extends the older MLOps discipline with the specifics of Software 3.0: prompts are versioned artefacts rather than incidental strings, evaluations are release gates rather than after-the-fact reports, and traces are the unit of observability rather than logs. Where the comparative chapter that follows places MLOps, LLMOps, and AgentOps side by side, this chapter develops the LLMOps lifecycle itself in detail: each phase, the tooling that supports it, the team that owns it, and the handoffs between them. It brings together into one lifecycle view the threads that earlier chapters developed in isolation.

Learning Objectives

- Recognise the phases of the LLMOps lifecycle and the handoffs that connect them.
- Architect the tooling that supports each phase, and understand why the stack is still fragmented.
- Integrate evaluation, observability, canary deployment, and rollback into a single operational loop.
- Manage the team interfaces across phases, where the most damaging production incidents originate.
- Adapt LLMOps practice to African operational realities of skills, tooling, and team scale.

70.1 The Lifecycle

LLMOps is, at its core, a lifecycle: a sequence of phases that an LLM application passes through, and then a loop that returns it through those phases again as it is improved. Figure 70.1 shows the canonical phases, and the rest of this chapter develops each one. The lifecycle runs left to right through building — data preparation, prompt design, fine-tuning, evaluation, deployment — and then loops back through the operating phases of monitoring, incident response, and improvement, before an explicit retirement phase ends an artefact’s life.

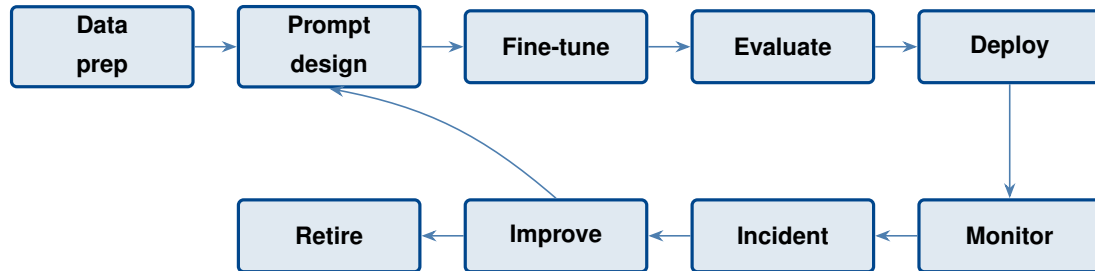


Figure 70.1: LLMOps lifecycle. Phases flow left to right through building, then loop back through monitoring and improvement. Retirement is an explicit phase rather than an implicit fading-away. Most phases have dedicated tooling; the lifecycle integration is what LLMOps adds on top of the individual tools.

Two properties of the lifecycle deserve emphasis before the phases are developed individually. The first is that it is genuinely a loop: the improvement phase feeds back into prompt design, so an LLM application is never finished, only at a point in a cycle. The second is that retirement is an explicit phase rather than an implicit fading-away. Most phases have dedicated tooling that handles them well in isolation; what LLMOps adds is the integration that makes the phases a coherent lifecycle rather than a set of disconnected activities, and that integration is the subject of this chapter.

70.2 Phase Details

Each phase of the lifecycle is a distinct body of work with its own concerns, and a sound LLMOps practice treats each as deliberately as the last. The nine phases below are developed in the order the lifecycle runs them.

70.2.1 Data Preparation

Data preparation is the curation, labelling, deduplication, and contamination checking that produces the datasets the rest of the lifecycle depends on. The contamination check is the one most often skipped and the most damaging to skip: evaluation data must not appear in training data, because a model evaluated on data it was trained on produces a score that flatters it and predicts nothing about production. Data preparation produces versioned datasets with lineage from the data back to its sources, so that every later phase can trace exactly which data shaped the model it is working with.

70.2.2 Prompt Design

Prompt design treats prompts as the Software 3.0 artefacts they are — the subject of Chapter 42 — rather than as incidental strings embedded in code. That means prompts go under version control,

prompt changes go through review, and prompts are tested before they ship, exactly as code is. A prompt is a programmable behaviour, and an organisation that edits prompts directly in production without version control or review is changing the system's behaviour with none of the discipline it applies to a code change.

70.2.3 Fine-Tuning

Fine-tuning is the phase that adapts a model's weights, through full fine-tuning, parameter-efficient methods such as LoRA, or preference-optimisation methods such as DPO. Each choice is a cost-quality trade-off, and the phase's discipline is to choose the lightest method that meets the quality bar rather than the heaviest available. Whatever method is used, reproducibility is essential: the hyperparameters, the code, and the data snapshot must be captured, so that a fine-tuned model can be rebuilt and a regression in it can be debugged.

70.2.4 Evaluation

Evaluation is the phase that gates progression — the pre-production quality check developed in full in Chapter 51. It turns the question of whether a candidate is good enough to ship from a matter of opinion into a measured judgement against an evaluation suite. The phase's two standing concerns are ownership of the evaluation suite, which must be maintained as deliberately as any other artefact, and calibration of the bar, which sets how good a candidate must be to pass.

70.2.5 Deployment

Deployment is the phase that carries a candidate into production, and in LLMOps it is done through canary rollouts rather than instant cutover — the pattern of Chapter 48. A canary exposes the candidate to a slice of traffic under comparison before it is promoted to all of it. Deployment in LLMOps therefore implies multi-version operation: the stable version and the candidate run side by side, and the architecture must support that coexistence cleanly.

70.2.6 Monitoring

Monitoring is the first of the operating phases, watching the system once it is live through traces, online evaluations, drift detection, and cost monitoring. Each of these is a different lens on the same question — is the system still behaving as it should — and they catch different problems at different speeds. Monitoring is what turns the lifecycle from a build process into a standing operation, because it is the phase that never stops while the system runs.

70.2.7 Incident Response

Incident response is the phase that handles things going wrong in production, and it requires preparation before any incident occurs: documented detection criteria, a severity classification, and response playbooks for the AI-specific incident types. The phase's defining discipline is the post-incident update: every incident closes by adding the specific failure to the permanent evaluation set, so that the system is tested against that failure forever after and the same incident cannot recur unnoticed.

70.2.8 Improvement

Improvement is the phase where the loop closes. The findings from monitoring and incident response flow back to the data, the prompts, or the fine-tuning, and the lifecycle begins another cycle with the system better than before. Improvement is what makes LLMOps a loop rather than a line — without a deliberate improvement phase, monitoring produces observations that nobody acts on, and the system stops getting better the day it ships.

70.2.9 Retirement

Retirement is the phase that ends the life of a model, a prompt, or a tool, and it is the phase most often neglected. Artefacts have an end of life: a model is superseded, a prompt is replaced, a tool is decommissioned. Retirement done properly migrates the dependents off the retiring artefact, archives what must be kept for audit, and cleans up what must not be left behind. An organisation with no retirement discipline accumulates old prompts and models until nobody can say with confidence what is actually in production — which is itself a failure mode this chapter returns to.

70.3 The Tooling Stack

Each phase of the lifecycle has tooling that supports it, and the LLMOps tooling stack spans the lifecycle. The categories that make it up are reasonably stable even as the specific products shift:

- **Prompt management** — tools such as PromptLayer, Humanloop, and Langfuse that version, organise, and manage prompts as first-class artefacts.
- **Evaluation frameworks** — tools such as DeepEval and Promptfoo, alongside custom harnesses, that run the evaluation suites the gate phase depends on.
- **Observability** — tools such as LangSmith, Phoenix, and Helicone that capture and surface the traces which are the unit of LLMOps observability.
- **Model serving** — engines and platforms such as vLLM, TGI, SageMaker, and Bedrock that run the models in production.
- **Vector databases** — the retrieval infrastructure covered in Chapter 68, which the data and retrieval phases depend on.
- **Experiment tracking** — tools such as MLflow and Weights & Biases that record the fine-tuning experiments and their results.

The important architectural observation about this stack is that it is consolidating but still fragmented. No single tool spans the whole lifecycle well, so most organisations assemble their stack from several providers plus internal glue code that connects them. The fragmentation is the source of one of the chapter's central failure modes — tool sprawl with no integration across phases — and the discipline of LLMOps is as much about integrating the stack into a coherent lifecycle as about choosing its individual parts.

70.4 Team Interfaces

An LLM application's lifecycle is not run by one team; it is run by several, each owning some phases, and the interfaces between those teams matter as much as the tooling. The most damaging production incidents tend to originate not within a phase but at the handoff between two teams' phases, where each assumes the other is handling something.

70.4.1 Data Engineering

The data engineering team owns the data-preparation phase, producing the versioned, lineage-tracked datasets that the fine-tuning and evaluation phases consume. Its interface to the rest of the lifecycle is the dataset itself, and the quality of that dataset bounds everything downstream. A data engineering team that delivers contaminated or poorly versioned datasets undermines every phase that depends on them, however well those phases are run.

70.4.2 Model Engineering

The model engineering team owns fine-tuning and the evaluation infrastructure — the machinery that adapts models and the harnesses that measure them. Its work sits at the technical heart of the lifecycle, and its interface to the application team is the model and the evaluation results. A clear handoff here means the application team receives a model with a documented, reproducible evaluation profile rather than a model whose behaviour they must rediscover.

70.4.3 Application Engineering

The application engineering team owns prompts, tools, deployment, and observability — the phases closest to the running production system. It is the team that turns a model into an application, and its interfaces run in both directions: it receives models from model engineering and ships behaviour that the operations and trust-and-safety teams must support. Much of the lifecycle's integration work lands on this team.

70.4.4 Product

The product team owns rubric design, bar setting, and release decisions — the judgements about what good output means and whether a candidate is good enough to ship. Its interface to the lifecycle is the evaluation rubric and the release decision, and it is the team that connects the technical lifecycle to the question of whether the system actually serves its users well. A product team disengaged from evaluation leaves the bar to be set by whoever happens to own the harness.

70.4.5 Trust and Safety

The trust-and-safety team owns red-teaming, policy, and guardrails — the disciplines that defend the system against misuse and harm. Its interface to the lifecycle is the set of safety requirements that the deployment phase must satisfy and the findings that the improvement phase must act on. A trust-and-safety function that is consulted only at the end, rather than integrated across the lifecycle, discovers problems too late to fix them cheaply.

70.4.6 Operations

The operations team owns infrastructure, cost, and capacity — the phases that keep the system running affordably and reliably. Its interface to the lifecycle is the operational envelope within which every other team works: the capacity available, the cost budget, the reliability targets. LLMOps as a practice is what connects all six of these teams into one lifecycle, and the gaps at their interfaces, rather than failures within any single team, are where the most production incidents originate.

70.5 The African Perspective

LLMOps carries distinctive weight in African deployments, across the building of capability, the sovereignty of tooling, the scale of teams, and cross-border coordination.

70.5.1 Capability Building

LLMOps is a specialist discipline, and the skills it requires are in short supply, more acutely so in markets where the AI engineering labour pool is still maturing. African organisations build this capability deliberately — through internal training, through partnerships with institutions that have the skills, and through engagement with the diaspora of African engineers working abroad. Treating capability building as an explicit programme, rather than assuming the skills can simply be hired, is what lets an organisation grow into a full LLMOps practice rather than stalling for lack of people to run it.

70.5.2 Tooling Sovereignty

The LLMOps tooling landscape is concentrated in Northern vendors, and for an African organisation that concentration raises both a sovereignty concern — data flowing to foreign-hosted tools — and a cost concern, since the tools bill in foreign currency. The open-source alternatives in the stack, such as Langfuse for observability, Phoenix for tracing, and MLflow for experiment tracking, are attractive on both counts: they can be self-hosted within the jurisdiction, keeping the data sovereign, and they remove the foreign-currency licence cost. Tooling sovereignty is a real architectural consideration in African LLMOps, not an ideological preference.

70.5.3 Lighter Lifecycle for Smaller Teams

African organisations often run leaner teams than their Northern equivalents, and the full nine-phase lifecycle with six distinct owning teams is a structure built for scale that a small team cannot staff one-to-one. The adaptation is to combine phases under shared ownership, to share tooling across phases rather than running a separate tool for each, and to scale the rigour to the team's actual size. The lifecycle does not change, but a lean team runs it with fewer people wearing more hats, and the discipline is to keep the integration even when the team is small.

70.5.4 Cross-Border Coordination

A pan-African deployment runs its LLMOps lifecycle across several markets at once, and the coordination problem is real: each market has its own languages, its own regulatory constraints, and its own evaluation needs. The workable pattern is a centralised pipeline with country-specific artefacts —

one shared lifecycle and tooling stack, parameterised per market with the prompts, evaluation sets, and policies that each country requires. This keeps the operational discipline unified while letting the artefacts reflect the genuine differences between the markets the system serves.

70.6 Recurring Failure Patterns

The failure modes of LLMOps recur, and each is the absence of the lifecycle integration that the discipline exists to provide:

- **Tool sprawl.** The organisation runs many tools but has not integrated them into a lifecycle, so no one has visibility across the phases and the stack is a collection of disconnected parts.
- **Team handoff gaps.** The phases are owned by different teams, and at the interfaces each team assumes the other is handling something, so work that should fall to one falls to neither.
- **Retirement neglect.** Old prompts and models accumulate because the retirement phase is never run, until nobody can say with confidence what is actually in production.
- **Eval-deployment gap.** The pre-production evaluations pass, but production traffic has a different distribution than the evaluation set, so the system regresses in production despite a clean evaluation.
- **Observability afterthought.** Observability is built late rather than designed in, so reconstructing what production actually did becomes a forensic exercise rather than a query.

Each pattern is the lifecycle failing to cohere — tools unintegrated, handoffs unowned, a phase skipped, evaluation disconnected from production, observability retrofitted — and together they make the chapter's point: LLMOps is not the individual phases but the integration of them, and the discipline fails wherever that integration is missing.

70.7 Last Word

LLMOps is the operational discipline that integrates the full lifecycle of an LLM application: data preparation, prompt design, fine-tuning, evaluation, deployment, monitoring, incident response, improvement, and retirement. Each phase is a distinct body of work with its own tooling and its own owning team, and the discipline of LLMOps is the integration that turns those phases into a coherent, looping lifecycle rather than a set of disconnected activities.

The tooling stack that supports the lifecycle is consolidating but still fragmented, which means most organisations assemble it from several providers plus internal glue. The team interfaces matter as much as the tooling, because the most damaging production incidents originate at the handoffs between the teams that own different phases. African deployments add deliberate capability building, tooling sovereignty through open-source alternatives, a lighter lifecycle adapted for leaner teams, and cross-border coordination through centralised pipelines with country-specific artefacts. MLOps gave the field reproducibility; LLMOps adds prompt-regression detection and the lifecycle integration that makes a Software 3.0 system operable.

Chapter 71

MLOps, LLMOps, and AgentOps: Three Operational Disciplines

“Each generation of AI did not retire the operational discipline before it. It inherited the discipline, kept what still held, and added what the new failure modes demanded.”

—a recurring observation in production AI engineering

Chapter Overview

Three operational disciplines now sit side by side in enterprise AI. MLOps is the operational practice for traditional machine-learning models, organised around data, training, and deployment. LLMOps extends it for large language models, organised around prompts, evaluations, and traces. AgentOps extends it again for autonomous multi-agent systems, organised around scope, coordination, and continuous feedback. This chapter is the comparative treatment: it places the three lifecycles side by side, traces what each inherits from the one before and what each adds, and develops AgentOps — the newest and least settled of the three — in the detail the earlier two already receive elsewhere in this book. The chapter on LLMOps (Chapter 70) develops that lifecycle in full; this chapter’s purpose is the comparison and the AgentOps discipline.

Learning Objectives

- Place the MLOps, LLMOps, and AgentOps lifecycles side by side and recognise what distinguishes each.
- Trace what each discipline inherits from its predecessor and what new failure modes it adds practice to address.
- Develop the AgentOps lifecycle: scope, data collection, multi-agent construction, coordination, automated action, and continuous feedback.
- Choose the right operational discipline for a given system rather than applying one out of habit.
- Adapt all three disciplines to African operational realities.

71.1 Three Disciplines, One Lineage

The operational practice for AI has been built in three layers, each a response to a new kind of system and a new set of failure modes, and each retaining most of what the layer before it established.

MLOps is the operational discipline for traditional machine learning — the classification, regression, ranking, and recommendation models that an organisation trains on its own data. Its lifecycle runs from defining the problem and gathering data, through processing and cleaning that data, selecting an algorithm, training and tuning the model, validating it, and deploying it, into the standing work of monitoring, retraining, and scaling. MLOps is organised around the model artefact and the data that produced it: its central questions are reproducibility, data quality, and the detection of the drift that degrades a trained model over time.

LLMOps is the operational discipline for systems built on large language models. It inherits the shape of MLOps — there is still a lifecycle, still a deployment, still monitoring — but it reorganises around the artefacts that Software 3.0 makes central: the prompt rather than the trained weight, the evaluation rather than the accuracy metric, the trace rather than the prediction log. Its lifecycle runs from defining the task and selecting a model, through preparing data, prompt engineering and optional fine-tuning, integrating tools, testing output, and deploying, into monitoring performance, checking for bias and drift, and iterating. LLMOps is developed in full in Chapter 70.

AgentOps is the operational discipline for autonomous and multi-agent systems — systems in which the model does not merely respond but decides, coordinates, and acts. It inherits from LLMOps in turn, and adds practice for the failure modes that autonomy and coordination introduce: an agent that takes a wrong action, a set of agents that coordinate badly, a feedback loop that must run continuously rather than at release time. Its lifecycle, the subject of the second half of this chapter, runs from defining scope, through collecting interaction data, building multi-agent models, allocating tasks and automating actions, into deploying, monitoring, and the continuous feedback loop that never closes.

The three are a lineage, not a sequence of replacements. An organisation running traditional fraud-scoring models, an LLM-powered support assistant, and an autonomous procurement agent runs all three disciplines at once, and Figure 71.1 places their lifecycles side by side so the inheritance and the additions are visible together.

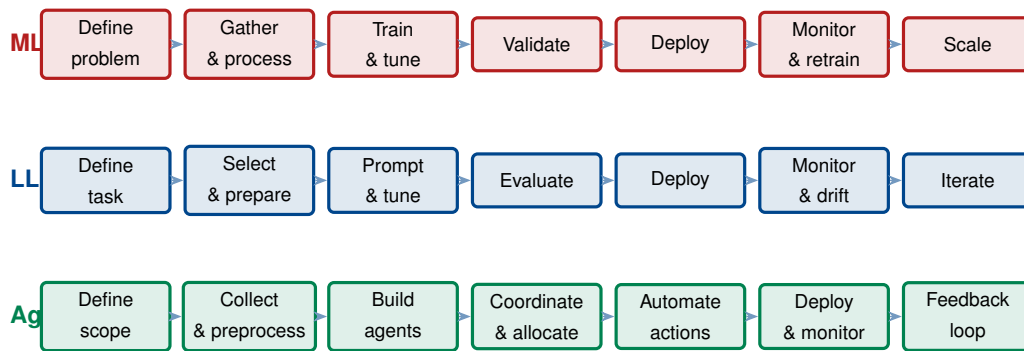


Figure 71.1: The three operational lifecycles side by side. Each discipline keeps the broad shape of the one above it — define, prepare, build, deploy, monitor — and reorganises the middle and the end around the artefacts and failure modes its kind of system introduces.

71.2 What Each Discipline Inherits and What It Adds

Reading the three lifecycles down the columns of Figure 71.1 shows what stays constant; reading across shows what each discipline adds. Both readings matter.

What stays constant is the skeleton. All three disciplines begin by defining the work — a problem, a task, a scope — because none of them can be operated without a statement of what the system is for. All three prepare data before building anything. All three deploy, and all three monitor what they deployed. An organisation that has a mature MLOps practice already owns the skeleton; adopting LLMOps or AgentOps is not starting over but extending a structure that is largely already in place. This is why the lineage framing matters operationally and not only historically: the inherited skeleton is reusable infrastructure.

What changes is the middle and the end. MLOps puts *training* at its centre, because the model artefact is produced by training, and it ends in *retraining*, because a trained model’s failure mode is the drift that retraining corrects. LLMOps replaces the centre with *prompt engineering*, *fine-tuning*, and *evaluation*, because an LLM application’s behaviour is shaped by prompts and tools rather than produced by training from scratch, and it ends in *iteration* against evaluation and drift signals, because an LLM application’s failure mode is a regression in output quality that an evaluation suite detects. AgentOps replaces the centre again with *multi-agent construction*, *coordination*, and *task allocation*, because an agentic system’s behaviour emerges from how its agents are built and how they coordinate, and it ends in a *continuous feedback loop*, because an autonomous system’s failure mode is a bad action or a bad coordination that must be caught while the system runs rather than at a release boundary.

The progression in the failure modes is the clearest way to see why three disciplines exist rather than one. MLOps must catch a model that has drifted. LLMOps must catch all of that plus a prompt that has regressed, a bias that has crept in, and a hallucination that the model has produced. AgentOps must catch all of that again plus an agent that has taken a wrong action, a group of agents that have coordinated into a bad outcome, and a feedback loop that has stopped improving the system. Each discipline is the previous discipline plus the practice that the new failure modes demand — which is also why none of the three can be skipped by an organisation whose systems include that generation’s

failure modes.

71.3 The MLOps Lifecycle in Brief

MLOps is treated here only in the brief form the comparison needs; the model-serving and platform chapters develop its infrastructure in depth.

The MLOps lifecycle begins by defining the problem in terms a model can be trained against — a target to predict, a quantity to estimate, a set of items to rank — and then gathering the data that the training will use. That data is rarely ready as gathered: it arrives structured and unstructured, and a processing and cleaning stage normalises it, handles missing values, and shapes it into the features the model will consume. Feature engineering — deriving the inputs that carry signal — is the part of the lifecycle where domain knowledge does its most concentrated work. An algorithm is then selected to fit the problem, the model is trained and its hyperparameters tuned, and cross-validation establishes that the measured performance is real rather than an artefact of one fortunate split of the data.

The model is then deployed, and the lifecycle’s standing work begins: monitoring the model in production and retraining it when its performance degrades, because a model trained on past data slowly loses accuracy as the world the data described moves on — the drift that is MLOps’ characteristic failure mode. Scaling and automation make the deployed model serve production volume reliably. MLOps is organised, throughout, around two things: the data, whose quality bounds everything downstream, and the trained artefact, whose drift is the failure the discipline exists to catch. Its practice is mature and its tooling is settled, which is exactly why it is the inherited foundation the other two disciplines build on rather than the subject this chapter develops at length.

71.4 How LLMOps Departs from MLOps

LLMOps keeps the MLOps skeleton and departs from it in three specific ways, each of which Chapter 70 develops in full; the departures are summarised here because they are what the comparison turns on.

The first departure is that the central artefact is no longer a trained model but a *prompt*, together with the tool definitions and the model selection that surround it. An LLM application is usually not trained from scratch; it is built by selecting a capable foundation model — open-source or from a provider — and shaping its behaviour through prompt engineering, with fine-tuning as an option rather than the default. The MLOps centre of train-and-tune becomes the LLMOps centre of select, prompt, and optionally fine-tune. The second departure is that the quality gate is no longer an accuracy metric but an *evaluation suite*. A traditional model’s quality is a number computed against a labelled test set; an LLM application’s quality is a statistical, stratified judgement rendered by an evaluation suite, because the output is open-ended language rather than a label. The third departure is that the unit of monitoring is no longer a prediction log but a *trace*, and the things being monitored expand: not only drift, but bias, hallucination, and response quality, each of which is a failure mode an LLM has and a traditional model does not.

The shape of the lifecycle therefore holds — define, prepare, build, evaluate, deploy, monitor, iterate

— while its content shifts from data-and-weights to prompts-and-traces. An organisation moving from MLOps to LLMOps is not discarding its operational practice; it is re-pointing the build stage at prompts, re-pointing the quality gate at evaluations, and widening the monitoring to the failure modes that language models introduce.

71.5 The AgentOps Lifecycle

AgentOps is the newest of the three disciplines and the least settled, and it receives the fullest treatment here because the rest of the book develops MLOps and LLMOps elsewhere while AgentOps has no chapter of its own. Its lifecycle inherits the LLMOps skeleton and reorganises the middle and the end around autonomy and coordination. Figure 71.2 shows the lifecycle as a loop, because its defining feature is that the loop does not close at a release boundary.

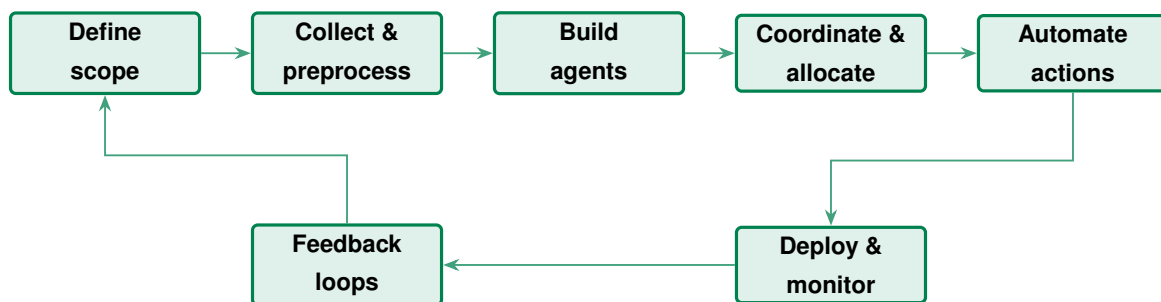


Figure 71.2: The AgentOps lifecycle. Scope, data, and agent construction give way to coordination and automated action; deployment and monitoring feed a continuous feedback loop that returns to scope. The loop never closes — an autonomous system is operated continuously, not released once.

The lifecycle begins by *defining scope*. For an autonomous system this is a heavier and more consequential step than defining a task, because scope is also the boundary of the system’s authority: it states not only what the agents are for but what they are permitted to do, which actions they may take without human approval, and where their autonomy ends. An ill-defined scope for an LLM application produces a vague assistant; an ill-defined scope for an agentic system produces an agent that acts where it should not. Scope definition in AgentOps is therefore the first safety control, not merely the first planning step.

The system then *collects data*, and the data is of a kind the earlier disciplines did not centre: interaction logs and records of user behaviour — traces of how the system and its users actually act, rather than a labelled training set. This data is preprocessed, cleaned, and normalised into the form the agents and their evaluation will use. The defining property of this data is that it is generated by the system in operation, which is why the lifecycle is a loop: the system’s own running behaviour is the data that improves its next iteration.

The middle of the lifecycle is *building the multi-agent system*: selecting the tools the agents will have and constructing the agents themselves — often several specialised agents rather than one, each scoped to a part of the work. This is where AgentOps departs most sharply from LLMOps, because the artefact under construction is not a single prompted model but a set of agents that will have to work together. The construction stage therefore flows directly into *coordination and task allocation*:

deciding how the agents divide the work, how they hand off to one another, and how their separate actions compose into a coherent outcome. Coordination is the failure surface unique to AgentOps — a set of individually well-behaved agents can still coordinate into a bad result — and it is the part of the lifecycle that has the least settled practice.

The agents are then connected to *automated actions*: the point at which the system stops merely producing output and begins acting in the world, invoking tools, writing records, triggering processes. This is the step that makes AgentOps a distinct discipline rather than a variant of LLMOps, because an automated action has consequences an output does not, and the operational practice around it — approval policies, reversibility tiers, blast-radius limits — is practice that an LLM application without automated action does not need.

The system is then *deployed and monitored*, and the monitoring tracks performance metrics specific to autonomous operation: not only the quality of what the agents produce but the soundness of the actions they take and the coordination among them. And the lifecycle closes — or rather does not close — into a *continuous feedback loop*: the system's performance metrics and the interaction data it generates feed back to refine scope, retrain or rebuild the agents, and improve coordination, continuously, for as long as the system runs. The phrase that names the final stage of the AgentOps lifecycle is *optimise continuously*, and the word that matters in it is *continuously*. An MLOps model is retrained periodically; an LLMOps application is iterated release by release; an AgentOps system is improved in a loop that has no release boundary, because an autonomous system that is not continuously observed and corrected is an autonomous system whose failures accumulate unseen.

71.6 Choosing the Right Discipline

The three disciplines are not ranked, and the newest is not the best. The right discipline for a system is the one whose failure modes match the system's, and applying a heavier discipline than the system needs is a cost without a benefit.

A system that predicts a label or a number from structured data — a credit score, a churn probability, a demand forecast — is an MLOps system, and operating it with the full apparatus of AgentOps would add coordination and action-safety practice to a system that neither coordinates nor acts. A system that produces open-ended language from a foundation model — a support assistant, a drafting tool, a summariser — is an LLMOps system, and it needs prompt-and-evaluation practice but not the multi-agent coordination practice of AgentOps. A system in which models decide and act with autonomy, especially several models coordinating, is an AgentOps system, and operating it with only LLMOps practice leaves its characteristic failure modes — bad actions, bad coordination — without the operational practice that catches them.

Most enterprises of any size run all three at once, because their portfolio includes traditional models, LLM applications, and agentic systems together. The discipline is to match the operational practice to each system rather than to standardise on one — and in particular not to apply the newest discipline to every system because it is the newest, nor to stretch the oldest discipline over agentic systems because it is the most familiar. The lineage means the three share a skeleton and much tooling; it does not

mean one of them is a substitute for the others.

71.7 The African Perspective

The three operational disciplines carry distinctive weight in African deployments, and the weight is not the same for each.

Cost shapes which discipline an organisation can afford to operate well. MLOps is the cheapest of the three to run, because a trained model serves predictions at a low and predictable per-call cost; LLMOps is more expensive, because every call invokes a large model; and AgentOps is the most expensive of all, because an agentic system makes many model calls per task and its continuous feedback loop is itself an ongoing cost. Under the tight unit economics of many African deployments, this argues for genuine discipline-matching rather than reflexive adoption of the newest practice: a problem that an MLOps model solves should be operated as an MLOps system, because escalating it to an LLM or an agent multiplies the running cost without necessarily improving the outcome. The cost-engineering instinct that runs through this book applies at the level of operational discipline, not only at the level of the model.

Skills shape which discipline an organisation can staff. MLOps practice is the most widely held; LLMOps practice is newer and less widely held; AgentOps practice is genuinely scarce everywhere and especially so where the engineering labour market is still building depth in AI. An African organisation building toward agentic systems is therefore building toward the discipline with the thinnest available skills base, which argues for growing the practice deliberately — through the maturity progression of inherited skeleton plus added practice — rather than attempting to staff a full AgentOps capability at once.

Two further considerations apply across all three disciplines and have been developed throughout this book. The continuous feedback loop of AgentOps depends on connectivity to collect interaction data and to push improvements, and in connectivity-variable environments the loop must be designed to tolerate intermittency rather than assuming a constant link. And the data each discipline collects — training data for MLOps, traces for LLMOps, interaction logs for AgentOps — is subject to the data-protection regimes of the markets served, so where that data is stored and processed is a design decision under POPIA, NDPA, and the other regional frameworks, exactly as the data-residency discussions of the earlier chapters set out. The operational disciplines do not change for African deployments; which of them an organisation can afford and staff, and how their data and feedback loops are arranged, must reflect the continent's cost, skills, connectivity, and regulatory realities.

71.8 Recurring Failure Patterns

The failure modes that recur across the three disciplines are, in large part, the failure of matching discipline to system:

- **Wrong discipline applied.** An LLM application operated with only MLOps practice, so prompt regressions and hallucinations go unmonitored; or a traditional model burdened with AgentOps apparatus it does not need.

- **Skeleton rebuilt rather than inherited.** An organisation adopting LLMOps or AgentOps rebuilds the define-prepare-deploy-monitor skeleton from scratch instead of extending the MLOps practice it already has.
- **AgentOps without action safety.** An agentic system operated as though it were an LLM application, with no approval policies, reversibility tiers, or blast-radius limits around its automated actions.
- **Coordination unmonitored.** A multi-agent system whose monitoring tracks each agent's output but not the coordination among them, so a bad collective outcome from individually sound agents goes undetected.
- **Feedback loop that does not loop.** An AgentOps system deployed and then left, its continuous feedback loop unrun, so its failures accumulate unseen between infrequent reviews.
- **Newest-discipline reflex.** The newest discipline applied to every system because it is newest, multiplying cost and operational burden on systems whose failure modes an older discipline already covers.

Each maps to the chapter's central instruction: the three disciplines are a lineage that shares a skeleton, and the operational task is to match the discipline to the system's actual failure modes — inheriting what the lineage provides and adding only the practice the system genuinely needs.

71.9 What This Chapter Established

MLOps, LLMOps, and AgentOps are three operational disciplines in one lineage. MLOps operates traditional machine-learning models, organised around data, training, and the drift that retraining corrects. LLMOps operates large-language-model applications, organised around prompts, evaluations, and traces, and developed in full in Chapter 70. AgentOps operates autonomous and multi-agent systems, organised around scope, coordination, automated action, and a continuous feedback loop that never closes at a release boundary.

The three share a skeleton — define, prepare, build, deploy, monitor — and each adds the practice that its generation's new failure modes demand: LLMOps adds prompt-regression, bias, and hallucination monitoring to MLOps; AgentOps adds action safety, coordination monitoring, and continuous feedback to LLMOps. The operational discipline is to match the practice to the system, inheriting the shared skeleton rather than rebuilding it and adding only what the system's failure modes require. African deployments add the further discipline of matching under cost and skills constraints — operating each system with the lightest discipline its failure modes allow, and growing AgentOps practice deliberately rather than reaching for the newest discipline by reflex.

Chapter 72

Agent Identity, Authentication, and Authorization

“If the agent can act, the agent has identity. If the agent has identity, the architecture is responsible for it.”

—a recurring observation in agentic-systems security

Chapter Overview

Agents act in production: they call tools, make API requests, modify data, and send messages. Every one of those actions requires authorisation, and authorisation requires identity — the system must know who is acting before it can decide what they may do. The classical identity-and-access-management stack of users, roles, and permissions was built for two kinds of principal, the human user and the stable service account, and an agent fits neither cleanly. An agent is not a human, because it acts without a person in the loop for each action; it is not a conventional service account, because its behaviour is not fixed but depends on a user’s request and the model’s reasoning. The emerging discipline treats an agent as a principal in its own right, with an authenticated identity bound into a delegation chain, and an authorisation policy that respects the user on whose behalf the agent acts. This chapter develops that discipline as architecture.

Learning Objectives

- Recognise why an agent needs an identity model of its own, distinct from a user’s and from a service account’s.
- Architect the delegation chain that runs from the user, through the agent, to the tool.
- Apply OAuth and equivalent protocols to the agent flow.
- Manage agent credentials securely, through vaults, rotation, and just-in-time issuance.
- Adapt agent identity and access management to African identity systems and regulatory contexts.

72.1 Why Agents Need Identity

A traditional service account represents a single program whose behaviour is stable: it does the same thing every time it runs, and so a fixed set of permissions describes it adequately. An agent is not like this. An agent's actions depend on a user's request, on the model's reasoning about that request, and on the tools the agent selects at runtime, which means the same agent code may produce entirely different actions for different users in different contexts. A fixed permission set cannot describe a principal whose behaviour is determined at runtime by a model, and that mismatch is why an agent needs an identity model designed for it rather than borrowed from the service-account world.

72.1.1 The Risks Without Identity

When an agent is run without a proper identity model — typically by giving it a service account and letting it act under that account's privilege — a recognisable set of risks follows, and each is serious. The first is the confused deputy: the agent uses its own broad privilege to fulfil a user request that the user themselves had no right to make, because nothing ties the agent's authority to the user's. The second is audit ambiguity: when an action is taken, the system cannot say who was actually responsible, because the action is recorded against the agent's generic identity rather than against the user who directed it. The third is over-privileging: because narrow, context-dependent permissions are hard to express for an agent, the agent is given broad capabilities instead, far beyond what any single task needs. The fourth is cross-tenant leakage: an agent acting on tenant A's request reaches into tenant B's data, because its identity is not scoped to the tenant it is serving. Each of these risks is the direct consequence of an agent acting without an identity that binds it to the user and context it is serving.

72.2 The Architecture

The architectural answer to these risks is to give the agent an identity that is bound, by delegation, to the user it acts for. Figure 72.1 shows the shape. An identity provider issues a token to the user; the user delegates to the agent, passing a scoped *act-as* claim that says the agent is acting on this user's behalf; and the agent presents that delegated identity at each tool call. The decisive property is what happens at the tool: the tool authorises the action against the *user's* permissions, carried in the delegation, not against the agent's own — so the agent can do, for this user, exactly what this user could do, and nothing more.

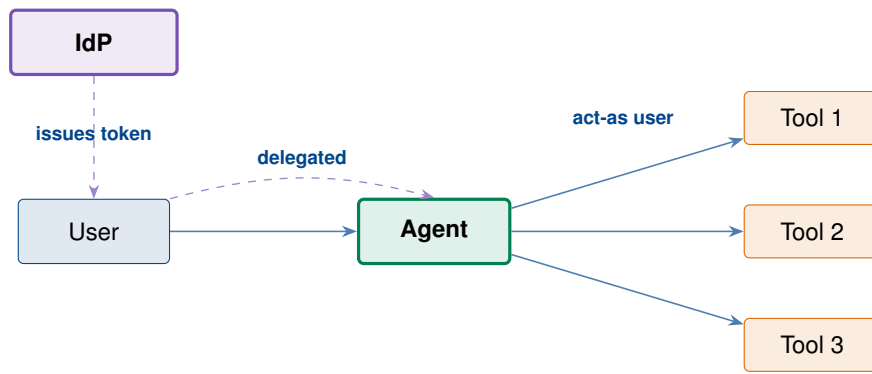


Figure 72.1: Agent identity with delegation. The identity provider issues a token to the user; the user delegates to the agent with a scoped *act-as* claim; the agent presents the delegated identity at each tool call. The tool authorises against the user’s permissions, not the agent’s.

This architecture closes each of the risks of the previous section. The confused deputy is closed because the agent acts with the user’s permissions, not its own, so it cannot fulfil a request the user had no right to make. Audit ambiguity is closed because every action carries the delegated user identity, so the system always knows who directed it. Over-privileging is closed because the agent’s effective authority is the user’s authority, scoped per request, rather than a broad standing grant. And cross-tenant leakage is closed because the delegation is scoped to the user and therefore to the tenant. The delegation chain is the architecture, and the rest of the chapter develops how it is built and operated.

72.3 Delegation Patterns

The delegation chain takes one of three shapes depending on how the agent relates to the user it serves, and a system may use more than one of them for different kinds of work.

72.3.1 User-Delegated

In the user-delegated pattern, the agent acts strictly on behalf of one specific user. The token it carries is scoped to that user, and it expires when the user’s session expires, so the agent’s authority is tightly bounded both in scope and in time. This is the pattern for interactive, user-facing agents — an assistant acting on a request a person just made — and it is the most constrained of the three, which makes it the right default wherever a specific user is genuinely behind the agent’s actions.

72.3.2 Tenant-Scoped

In the tenant-scoped pattern, the agent acts within a tenant’s boundary but not on behalf of a single identified user; the specific user identity may not matter for the work being done. This is the pattern for batch and automation work — an agent processing a tenant’s overnight workload, where there is no individual user to delegate from. The authority is bounded to the tenant rather than to a user, which is looser than user delegation but still prevents the cross-tenant leakage that an unscoped agent would risk.

72.3.3 Hybrid

In the hybrid pattern, the agent carries both a service identity of its own and a user delegation, and it uses each for the actions it suits. The service identity covers actions that are genuinely the agent's own — writing its logs, emitting telemetry — while the user delegation covers every action taken on the user's behalf. The hybrid pattern is common in practice because a real agent does both kinds of work, and the discipline is to keep the two identities cleanly separated so that user-facing actions are always authorised against the user, never against the agent's service identity.

72.4 OAuth and Equivalents

The delegation chain is not a new protocol invented for agents; it is built on the established identity protocols, adapted to the agent case. OAuth 2.0, with its on-behalf-of flows, is the dominant protocol: a bearer token carries the user's identity, scopes on that token constrain what the agent may do with it, and refresh tokens manage the long-running sessions that an agent may need. An organisation building agent identity is therefore extending the OAuth machinery it most likely already runs, rather than adopting something wholly unfamiliar.

Newer specifications extend the model into the service-to-service space that agent platforms increasingly need. Workload Identity Federation and SPIFFE/SPIRE provide identity for the workload-to-workload communication beneath the user-facing layer — the agent platform's own services authenticating to one another. An agent architecture of any scale combines these: OAuth on-behalf-of flows for the user-delegated authority, and workload identity for the platform's internal service identities, together giving every actor in the system, human or agent or service, an authenticated identity.

72.5 Credential Management

An agent's identity is only as secure as the credentials behind it, and credential management is the discipline that keeps those credentials from becoming the system's weak point. It has three components.

72.5.1 Storage

Agent credentials must be stored in a secrets vault — HashiCorp Vault, AWS Secrets Manager, Azure Key Vault, or an equivalent — and retrieved from there at the moment they are needed. They must never be embedded in code, where they would be exposed to anyone with repository access, and never placed in prompts, where they would be exposed to prompt-extraction attacks and to anyone who can read a trace. Vault storage is the baseline below which agent credential management must not fall.

72.5.2 Rotation

Credentials must rotate, and the agent must pick up the new credentials automatically when they do, without a manual reconfiguration. Long-lived static credentials are an anti-pattern: a credential that never changes is a credential that, once compromised, stays compromised indefinitely. Rotation

bounds the window of exposure — a leaked credential is useful to an attacker only until the next rotation — and automatic pickup ensures rotation does not break the agent.

72.5.3 Just-In-Time Credentials

The strongest pattern issues short-lived credentials per agent invocation: the agent receives a credential scoped to the single task it is about to perform, and that credential expires almost immediately afterward. This gives the highest security, because the window in which any credential is valid is minimal and the credential's scope is minimal, but it carries the highest infrastructure complexity, because the system must issue and manage a stream of short-lived credentials. Just-in-time issuance is the pattern for the highest-risk agent workloads, where the security gain justifies the operational cost.

72.6 Authorisation Policy

Identity establishes who is acting; authorisation policy decides what they may do. For agents, authorisation is checked at the granularity of the individual action, and it draws on a taxonomy of action risk and a mechanism for escalating consent.

72.6.1 Per-Action Authorisation

Every action an agent takes is checked against policy at the moment it is taken. The policy engine has, for each check, the full context: the user the agent acts for, the agent itself, the action being attempted, and the resource it touches. The decision the policy engine makes is recorded for audit, so that the system retains a complete account of every action it authorised and on what basis. Per-action authorisation is what prevents an agent that was authorised for one action from quietly taking another.

72.6.2 Action Taxonomy

Not all actions carry the same risk, and authorisation policy reasons about them through a taxonomy of action classes. A workable taxonomy distinguishes reading public data, reading sensitive data, writing one's own data, writing shared data, writing public data, financial actions, and irreversible actions:

- **Read public** — accessing data that carries no confidentiality constraint, the lowest-risk class.
- **Read sensitive** — accessing confidential data, which requires the agent's authority to cover that sensitivity.
- **Write own** — modifying data belonging to the acting user, a bounded write.
- **Write shared** — modifying data that others also depend on, a wider-blast-radius write.
- **Write public** — modifying data visible beyond the organisation, wider still.
- **Financial** — moving money, a class that warrants the strongest controls.
- **Irreversible** — actions that cannot be undone, where a mistake is permanent.

The purpose of the taxonomy is to let authorisation strength scale with risk: the higher-risk classes require stronger authorisation evidence before the action proceeds, so that a financial or irreversible action is gated far more tightly than a public read.

72.6.3 Step-Up Authentication

For the highest-risk actions, the right control is to bring the human back into the loop through step-up authentication: before the action proceeds, the user is prompted to re-authenticate, confirming their consent to this specific action. Step-up authentication bridges the gap between automated agent action and human consent — it lets an agent operate autonomously for routine work while ensuring that a financial transfer or an irreversible change has explicit, fresh human authorisation. It is the mechanism that keeps an agent's autonomy from extending to the actions where autonomy should not reach.

72.7 The African Perspective

Agent identity and access management carry distinctive weight in African deployments, across the diversity of identity systems, the priority of mobile-money authorisation, cross-border identity gaps, and regulatory audit.

72.7.1 Multi-Identity Integration

African contexts have a varied landscape of identity systems — bank verification numbers, national identification numbers, national ID schemes, and mobile-money identities, differing from country to country. An agent IAM architecture serving African markets must integrate with these heterogeneous sources rather than assuming a single canonical identity provider. The delegation chain still holds, but the identity provider at the top of it must be able to speak to whichever identity system the user actually has, and a pan-African deployment must accommodate several of them.

72.7.2 Mobile-Money Authorisation

Agents acting on mobile-money rails require strong authorisation, because the actions they take can move money and the systems they touch are high-value targets. Step-up authentication is standard for these actions, typically delivered through an SMS challenge or an app-based confirmation that the user must complete before a financial action proceeds. Where an agent's tools reach a mobile-money system, the financial and irreversible action classes of the taxonomy apply, and the step-up control is not optional but the core of the authorisation design.

72.7.3 Cross-Border Identity

A pan-African operation faces identity-portability gaps: an identity established in one country is not automatically recognised in another, because the identity systems are national rather than continental. An agent IAM architecture for such an operation must stratify per market, and a cross-border action — an agent acting for a user across a national boundary — requires explicit cross-border identity recognition rather than an assumption that one country's identity carries into another. This is a genuine architectural constraint that single-jurisdiction designs do not face.

72.7.4 Regulatory Audit

The data-protection regimes across African jurisdictions — POPIA, the NDPA, Kenya’s DPA, Egypt’s PDPL, and the others — all require an audit of who did what, and an agent IAM architecture is precisely what produces that evidence. Because every action carries the delegated user identity and every authorisation decision is recorded, the system can answer the regulator’s who-did-what question for agent actions exactly as it can for human actions. A well-built agent IAM architecture therefore satisfies a regulatory requirement as a by-product of doing identity correctly.

72.8 Recurring Failure Patterns

The failure modes of agent identity and access management recur, and each is the absence of a part of the delegation architecture this chapter has set out:

- **Agent-as-service-account.** The agent is given its own service-account privilege and acts under it, ignoring user context entirely, so its authority is fixed and unscoped to the user it serves.
- **Confused deputy.** The agent uses its own privilege to fulfil a request the user had no right to make, because nothing binds the agent’s authority to the user’s.
- **Long-lived credentials.** Static credentials are embedded in the agent’s configuration and never rotated, so a single leak compromises the agent indefinitely.
- **Audit gap.** There is no record of who delegated which action, so when something goes wrong the system cannot say who was responsible.
- **Step-up absence.** High-risk actions — financial, irreversible — are executed by the agent with no fresh human consent, so the agent’s autonomy extends to actions where it should not.

Each pattern maps to a part of the architecture that was skipped — the delegation chain, the user-scoped authority, credential rotation, the audited authorisation decision, and step-up authentication — and together they show that agent IAM fails wherever the agent is treated as a standalone principal rather than as a delegate acting on a user’s behalf.

72.9 What This Chapter Established

Agent identity, authentication, and authorisation treat an agent as a principal whose authority flows, by delegation, from the user it acts for. The architecture is the delegation chain: an identity provider issues a token to the user, the user delegates a scoped act-as claim to the agent, and the agent presents that delegated identity at each tool call, where the tool authorises against the user’s permissions rather than the agent’s own. That chain closes the confused-deputy, audit-ambiguity, over-privileging, and cross-tenant-leakage risks that an unscoped agent creates.

The chain is built on OAuth on-behalf-of flows and workload-identity specifications, operated through vault-stored, rotated, and ideally just-in-time credentials, and governed by per-action authorisation policy that draws on an action-risk taxonomy and escalates to step-up authentication for the highest-risk actions. African deployments add integration with a diverse landscape of national identity

systems, strong step-up authorisation for mobile-money actions, cross-border identity recognition for pan-African operations, and the regulatory audit evidence that the architecture produces by default. If the agent can act, the agent has identity — and the architecture is responsible for binding that identity to the user on whose behalf the agent acts.

Chapter 73

Time-Series and Forecasting AI

“The future is regular until it isn’t. Architectures must serve both.”
—a recurring observation in forecasting production

Chapter Overview

Time-series and forecasting workloads differ in several architecturally significant ways from the large-language-model systems that dominate much of this book. The data is sequential, ordered in time, and the order carries meaning that cannot be discarded. The models are smaller and more specialised than foundation models, and they come from several distinct families. Evaluation is statistical, measured against ground-truth futures that arrive after the forecast was made. And the integration patterns differ, because a forecast feeds an operational decision rather than a conversation. Many of the industry chapters of this book — banking, mining, energy, manufacturing — rely heavily on time-series AI, which makes the architecture worth developing in its own right. This chapter sets out that architecture: where time-series AI fits, how a forecasting pipeline is built, which model families serve which problems, and how drift and retraining are managed for data that is inherently sequential.

Learning Objectives

- Recognise where time-series AI fits within the broader enterprise architecture.
- Architect a forecasting pipeline from high-cadence ingestion through low-latency serving.
- Integrate the classical, tree-based, deep-sequence, and foundation-model approaches deliberately.
- Manage drift and retraining for sequential data, where the failure modes differ from static models.
- Adapt time-series architectures to African volatility, data sparsity, and infrastructure realities.

73.1 What Time-Series Architectures Need

A time-series architecture has a distinct set of requirements, and naming them is the first step toward building one. Each requirement reflects something about sequential, time-ordered data that a static-

input AI system does not have to handle:

- **High-cadence ingestion** of the streams that time-series systems consume — sensor readings, transactions, market data — arriving continuously and often in large volume.
- **Temporal feature engineering**, the derivation of features that capture trend, seasonality, lags, and the other structure that lives in the time dimension.
- **Specialised model families** — ARIMA, exponential smoothing, gradient-boosted machines, deep networks, transformers — each suited to a different shape of problem.
- **Forecast evaluation against actuals**, using the statistical error metrics of forecasting — MAPE, sMAPE, RMSE, pinball loss — which can only be computed once the real future has arrived.
- **Online deployment with low-latency inference**, because a forecast frequently feeds an operational decision that cannot wait.

These requirements together describe an architecture that is recognisably different from an LLM application's: it is built around a continuous flow of ordered numeric data, a family of specialised models rather than one general one, and an evaluation that is inherently retrospective. The sections that follow develop that architecture.

73.2 The Architecture

The canonical time-series pipeline runs from the data sources, through ingestion and feature engineering, into a feature store, and from there to the forecasting model and its serving layer. Figure 73.1 shows the shape. Sensors and transactional sources feed an ingestion stage that also performs the temporal feature engineering; the engineered features persist in a feature store; the forecasting model trains on that store and serves its predictions to the consumers — the operational systems and people who act on the forecast.

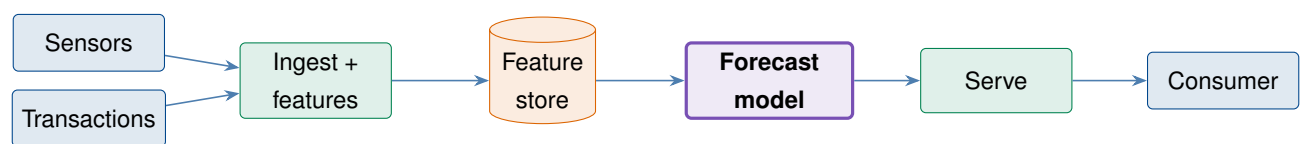


Figure 73.1: Time-series pipeline. Sensors and transactional sources feed ingestion and feature engineering; features persist in a store; forecasting models train on the store and serve predictions to consumers. Specialised time-series stores — TimescaleDB, Druid, InfluxDB — are commonly used.

One architectural choice in the figure is worth drawing out: the storage layer is frequently a specialised time-series store — TimescaleDB, Druid, InfluxDB — rather than a general-purpose database, because the access patterns of time-series data, with their heavy emphasis on time-range queries and high-cadence appends, are exactly what those stores are built for. The pipeline shape is conventional, but its components are the time-series-specialised versions of the generic ones, and choosing those specialised components is part of designing the architecture well.

73.3 Model Families

Time-series forecasting is served by four distinct model families, and they differ in the kind of series they suit, in their data requirements, and in how hard they are to operate. A sound architecture chooses the family that fits the problem rather than defaulting to the most sophisticated one.

73.3.1 Classical Models

The classical family — ARIMA, exponential smoothing, state-space models — is the oldest and remains genuinely useful. These models are strong on small, regular series, they are interpretable in a way that the deeper families are not, and they make an excellent baseline against which any more complex approach must justify itself. A forecasting effort that has not first tried a classical model does not know whether its more elaborate model is actually adding value, and for many well-behaved series the classical model is not merely the baseline but the right final answer.

73.3.2 Tree-Based Models

The tree-based family — XGBoost, LightGBM, applied to engineered temporal features — is the production workhorse of time-series forecasting. These models are well understood, cost-effective to train and serve, and robust across a wide range of problems, and they pair naturally with the temporal feature engineering the pipeline already performs. For a great many enterprise forecasting problems, a gradient-boosted model on good features is the pragmatic choice: not the most powerful family in principle, but the one that most reliably delivers solid forecasts at modest cost.

73.3.3 Deep Sequence Models

The deep-sequence family — LSTMs, GRUs, transformer encoders — has the capacity to capture complex patterns that the classical and tree-based families miss, particularly long-range dependencies and intricate seasonality. That capacity comes at a price: these models are data-hungry, needing long histories to train well, and they are harder to operate, with more to tune and more that can go wrong. The deep-sequence family earns its place on problems with genuinely complex temporal structure and enough data to learn it, and is overkill on the regular series the simpler families handle.

73.3.4 Foundation Time-Series Models

The newest family is foundation models for time-series — TimeGPT, Chronos, Lag-Llama — which bring the pretrained-foundation-model approach to forecasting and offer zero-shot prediction on series they were never specifically trained for. They are promising, and they are advancing quickly, but they remain immature for production-critical use, where a specialist model tuned to the specific series still tends to outperform them. The architectural posture toward foundation time-series models is to watch them closely and to use them where their zero-shot convenience genuinely helps, while not yet relying on them for the forecasts a business critically depends on.

73.4 Hybrid Approaches

Time-series AI and the large-language-model systems that dominate this book are not separate worlds; they combine, and the hybrid patterns are worth knowing because they add capability that neither

alone provides:

- **LLM as feature engineer** — the language model suggests decompositions of a series, identifies the events worth encoding as features, and brings domain knowledge to the temporal feature engineering that the tree-based and classical models depend on.
- **LLM as explanation layer** — the language model narrates a forecast for its human consumers, turning a numeric prediction and its confidence interval into an explanation a decision-maker can act on.
- **LLM as anomaly investigator** — when the time-series model flags an anomaly, the language model queries additional sources to explain it, turning a bare anomaly flag into an investigated finding.
- **Foundation time-series models on tokenised sequences** — numeric series are tokenised and fed to foundation models, applying the foundation-model machinery directly to the forecasting problem.

The common thread is that the language model handles the parts of the forecasting workflow that involve knowledge, explanation, and investigation, while the specialised time-series model handles the numeric prediction itself. The hybrid is stronger than either part alone because it lets each kind of model do what it is genuinely good at.

73.5 Operating Considerations

Operating a forecasting system in production turns on three considerations that are specific to sequential data and that a static-model practice does not have to address.

73.5.1 Retraining Cadence

The right retraining cadence for a forecasting model depends on how fast the underlying pattern moves. A stable pattern — a mature product's seasonal demand — may need retraining only monthly or less, because the relationships the model learned remain valid for a long time. A volatile pattern — foreign-exchange rates, fraud behaviour — may need daily retraining or even continuous online learning, because the relationships shift fast enough that a model is stale within days. Setting the cadence to the volatility of the series, rather than to a fixed schedule, is what keeps the model current without retraining it pointlessly often.

73.5.2 Drift Detection

For a forecasting model, the leading indicator of drift is the forecast error measured against the actuals as they arrive: a model whose error is rising is a model whose grip on the series is slipping. This is a direct and trustworthy signal, because it measures the thing the model exists to do. Population drift on the inputs — a shift in the distribution of the features — is a secondary signal, useful as an early warning but less decisive than the forecast error itself. A forecasting system monitors both, but treats rising forecast error as the signal that demands action.

73.5.3 Cold Start

The cold-start problem arises whenever a new series appears with no history — a newly launched product, a newly acquired customer — because the standard model has nothing to learn that series from. The architectural responses are pooled models that learn shared structure across many series and apply it to the new one, similar-series transfer that borrows the history of a comparable series, and hierarchical forecasting that forecasts the new series as part of an aggregate it belongs to. A forecasting architecture must have a deliberate cold-start path, because new series are not an edge case but a routine occurrence, and a system with no answer to them forces a manual workaround every time one appears.

73.6 The African Perspective

Time-series AI carries distinctive weight in African deployments, across market volatility, data sparsity, sensor reliability, and the high-cadence streams of mobile money.

73.6.1 Currency and Commodity Volatility

African markets often exhibit higher volatility than their Northern equivalents — in currency values, in commodity prices, in demand. That volatility has direct architectural consequences for forecasting: confidence intervals must be wider to honestly reflect the uncertainty, ensemble approaches are preferred because they are more robust to a volatile series than any single model, and volatility-aware loss functions, which weight the model's training toward handling the swings, are worth adopting. A forecasting architecture tuned for the steadier patterns of a Northern market will be overconfident and brittle when applied to a more volatile African one.

73.6.2 Sparse Data

Many African time series have shorter histories than their Northern counterparts, simply because the systems generating the data, or the formal recording of it, are newer. Short histories starve the data-hungry model families, particularly the deep-sequence one. The architectural responses are transfer learning that brings in structure learned elsewhere, hierarchical forecasting that lets a short individual series borrow strength from the aggregate it belongs to, and synthetic data augmentation that extends a thin history. Designing for sparse data is a standing requirement in African time-series work, not an occasional accommodation.

73.6.3 Sensor Reliability

The industrial sensors that feed time-series systems in African mining, agriculture, and energy operate in harsher conditions than a controlled Northern facility provides — more dust, more heat, less reliable power and connectivity — and they fail and drop readings more often as a result. The ingestion stage of the pipeline must therefore be built robustly, with deliberate missing-data handling that does not let a sensor gap corrupt the feature engineering downstream. A forecasting architecture that assumes clean, complete sensor streams will be repeatedly broken by the reality of the sensors it actually depends on.

73.6.4 Mobile-Money Velocity

Mobile-money transaction streams are high-cadence time series of considerable value, and forecasting against them is a major African use case. The forecasts these streams support — of transaction demand, of fraud rates, of the liquidity that agents and the network must hold — use time-series AI extensively, and they exercise exactly the high-cadence ingestion and low-latency serving the architecture is built for. Mobile money is, in many African deployments, the single largest and most consequential time-series workload, and a forecasting architecture serving it must be built for the velocity of those streams.

73.7 Recurring Failure Patterns

The failure modes of time-series AI recur, and each maps to a discipline this chapter has set out:

- **Stationarity assumption.** The model assumes the series is regular and stationary, and the volatility of a real African market breaks that assumption, producing confident forecasts that the series does not honour.
- **Data leakage.** Future information leaks into the features — a feature computed using data that would not have been available at forecast time — so the model’s training performance is far better than its production performance.
- **Cold-start gap.** A new series appears, the standard model cannot forecast it for lack of history, and the absence of a cold-start path forces a manual workaround.
- **Retraining-cadence mismatch.** The series drifts faster than the retraining cadence catches up, so the model is perpetually stale.
- **Foundation-model premature production.** A foundation time-series model is put into production-critical use where a tuned specialist model still outperforms it, trading reliability for novelty.

Each pattern maps to a remedy the chapter has developed — honest volatility-aware modelling, leakage-free temporal feature engineering, a deliberate cold-start path, a cadence set to the series’s volatility, and a measured posture toward the foundation family — and together they show that time-series AI fails through the specific hazards of sequential data, not through a generic modelling weakness.

73.8 The Working Position

Time-series and forecasting AI architecture covers high-cadence ingestion, temporal feature engineering, a feature store, the forecasting model, and low-latency serving — a pipeline that is conventional in shape but built from the time-series-specialised version of each component. Four model families serve it: the classical models that make the interpretable baseline, the tree-based models that are the production workhorse, the deep-sequence models for genuinely complex structure, and the foundation time-series models that are promising but not yet ready for production-critical reliance.

Hybrid patterns combine large language models with time-series models, the language model handling

feature engineering, explanation, and anomaly investigation while the specialised model handles the numeric prediction. The operating considerations — a retraining cadence set to the series's volatility, drift detection led by forecast error against actuals, and a deliberate cold-start path — are what keep a forecasting system current. African deployments add wider confidence intervals for higher volatility, the techniques for sparse data, robust ingestion for unreliable sensors, and the high-velocity time-series workload of mobile money. The future is regular until it is not, and a forecasting architecture must serve both the regular stretches and the breaks.

Chapter 74

Reasoning Models and Test-Time Compute

“You can pay to train or pay to think. Reasoning models let you choose at the request.”

—a recurring observation in AI inference economics

Chapter Overview

Reasoning models trade inference compute for output quality. Rather than answering in a single forward pass, they perform substantial deliberation — often thousands of tokens of internal thinking, sometimes visible to the user and sometimes hidden — before producing the answer. This is a different lever from the one that training pulls: where training spends compute once, in advance, to make a better model, reasoning spends compute at the moment of each request, to get a better answer to that request. The architectural implications are real. Latency becomes variable, stretching from seconds to minutes on hard problems. Cost per request becomes variable, and the variation is large. And the architecture acquires a new decision to make on every request: whether this particular request is worth the extra thinking. This chapter develops the architectural perspective on reasoning models — what they are, how to route to them, how to handle their variable latency and cost, and how to observe a process that is partly hidden.

Learning Objectives

- Recognise reasoning models as a distinct class with distinct economics.
- Architect request handling for the variable cost and latency that reasoning introduces.
- Apply reasoning models where their economics genuinely work, and route away from them where they do not.
- Manage the observability of an internal reasoning process that providers expose to differing degrees.
- Adapt reasoning-model deployment to African cost discipline and latency realities.

74.1 What Reasoning Models Are

A reasoning model performs extended internal deliberation before it produces the output the user sees. In some models that deliberation is visible, streamed to the user as a chain of thought; in others it is hidden, and the user sees only the final answer. Either way, the model spends additional tokens — sometimes thousands of them — on internal thinking that does not appear, or appears only as a summary, in the final response. That extra thinking is the mechanism by which a reasoning model reaches a better answer than a single forward pass would produce, and it is also the source of the variable cost and latency that the rest of this chapter has to architect around.

74.1.1 The Economics

The economics of a reasoning model are defined by a wide and variable cost gap. On a hard problem, a reasoning model may cost ten to fifty times what a non-reasoning model costs on the same problem, because it spends that much more compute on the internal deliberation. On a problem where reasoning genuinely matters, that spend buys a correspondingly better answer, and the trade is worth making. On a problem where reasoning does not matter — a simple lookup, a casual exchange — the same spend buys nothing, because there was no hard reasoning for the extra compute to do, and the money is simply wasted. The whole architectural challenge of reasoning models follows from this asymmetry: the spend pays off on some requests and is wasted on others, so the architecture must tell the two apart.

74.1.2 The Capability Frontier

Reasoning models do not dominate non-reasoning models everywhere; they lead on a specific frontier. They are clearly ahead on mathematics, on code, on scientific reasoning, and on complex multi-step planning — the tasks where working a problem through in stages genuinely helps. On simple question-answering, on casual conversation, and on creative writing, they lag or merely match their non-reasoning peers, because those tasks do not reward step-by-step deliberation and may even be hurt by it. Knowing where the frontier lies is what makes routing possible: the routing decision is, in effect, the decision of whether a given request falls on the reasoning side of this frontier.

74.2 The Architecture

Because reasoning pays off on some requests and is wasted on others, the central architectural element for reasoning models is a routing decision. Figure 74.1 shows the shape: a request arrives, a classifier decides whether reasoning will help, and the request is sent either to a reasoning model or to a standard model accordingly. The cost gap between the two paths is what makes this routing decision economically significant — a misrouted request either wastes a tenfold-or-more cost premium or denies a hard problem the deliberation it needed.

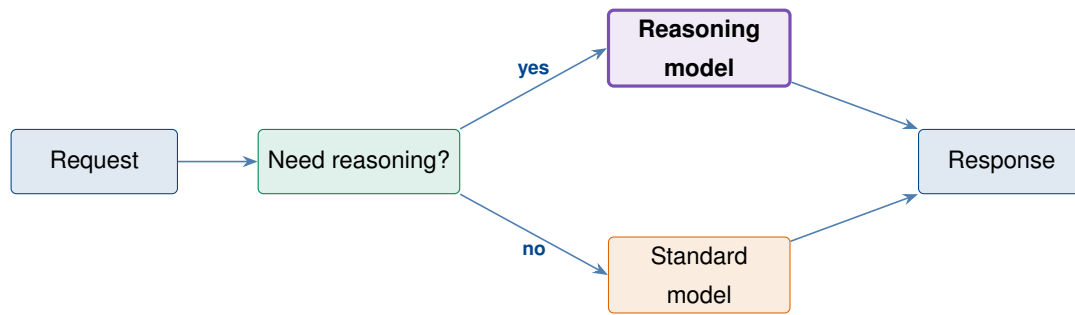


Figure 74.1: Reasoning-model routing. A classifier decides whether reasoning will help; reasoning-heavy requests go to reasoning models, standard requests to standard models. The cost gap between the two paths makes the routing decision economically significant.

This architecture is a specialisation of the model-routing pattern developed earlier in the book, applied to the specific choice between a reasoning path and a standard path. The classifier at its centre does not need to be sophisticated — the signals it acts on, developed in the next section, are fairly legible — but it does need to exist, because the alternative to routing is either sending everything to the reasoning model and paying the premium on every request, or sending everything to the standard model and losing the reasoning advantage where it mattered.

74.3 Routing Decisions

The routing classifier decides each request by weighing signals for reasoning against signals against it. The signals are legible enough that the decision is tractable, and a sound architecture instruments both sides.

74.3.1 Signals For Reasoning

Several signals indicate that a request will benefit from reasoning, and the presence of any of them argues for the reasoning path:

- **Mathematical content** — a request involving calculation or mathematical argument, where step-by-step working is exactly what helps.
- **Multi-step problems** — a request that must be decomposed into stages, where reasoning’s deliberation does the decomposition.
- **Code generation or debugging** — tasks where the model benefits from working through logic and tracing consequences.
- **Logical puzzles or planning** — requests that reward systematic exploration of possibilities.
- **User tier** — a premium tier may receive reasoning by default as part of its service level.
- **Explicit user request** — the user has asked for the model to think carefully, which is a direct instruction to route to reasoning.

These signals are what the classifier looks for, and a request carrying several of them is a clear candidate for the reasoning path.

74.3.2 Signals Against Reasoning

Other signals indicate that reasoning will not help and the request should take the cheaper, faster standard path:

- **Casual conversation** — exchanges that do not involve a hard problem for deliberation to work on.
- **Simple factual queries** — lookups whose answer reasoning cannot improve.
- **Latency-critical paths** — requests where the variable, sometimes minute-long latency of reasoning is unacceptable regardless of quality.
- **Cost-sensitive tenants** — tenants whose economics cannot absorb the reasoning premium, where the standard path is the only viable one.

The routing decision is the balance of these two sets of signals, and where the signals against dominate, the request takes the standard path even if a little reasoning might have helped, because the cost or latency cannot be justified.

74.4 Latency and User Experience

Reasoning models introduce a user-experience problem that non-reasoning models do not have: they can take thirty to a hundred and twenty seconds on a hard problem, a wait long enough that a user given no feedback will assume the system has failed. The architecture must therefore handle the wait deliberately, and several user-experience patterns address it:

- **Show progress** — display a “thinking” indicator with elapsed time, so the user sees that the system is working rather than stalled.
- **Stream a reasoning summary** — where the model’s reasoning is visible, stream a summary of it, which both occupies the wait and reassures the user that real work is happening.
- **Allow cancellation** — let the user abandon a reasoning request that is taking longer than they are willing to wait, rather than trapping them in it.
- **Offer a standard-model option** — where speed matters more than the extra quality, let the user choose the faster standard path explicitly.

The common thread is that a long, silent wait is the failure to avoid: a reasoning model that deliberates for ninety seconds with no indication of progress will lose the user before it produces its better answer. The user-experience layer is as much a part of deploying reasoning models as the routing classifier.

74.5 Observability of Reasoning

Reasoning models pose a specific observability challenge: the deliberation that determines the answer is internal, and providers expose it to differing degrees. The observability architecture has to handle both the case where the reasoning is visible and the case where it is hidden.

74.5.1 Where Reasoning Is Visible

Some providers expose the model's reasoning trace — DeepSeek's R1 and Anthropic's extended thinking among them — and where the trace is available, the observability layer should capture it. A visible reasoning trace is valuable for debugging, because it shows how the model reached an answer and therefore where a wrong answer went wrong, and it is valuable for audit, because it is a record of the model's deliberation. Capturing the reasoning trace turns reasoning models from an opaque process into an inspectable one.

74.5.2 Where Reasoning Is Hidden

Other providers hide the reasoning, exposing only the final output and perhaps a redacted summary — OpenAI's o-series models work this way. Where the reasoning is hidden, the observability layer is limited to what it can see: the inputs, the final outputs, and the aggregate token counts that reveal how much thinking happened even though the thinking itself is not shown. An architecture that depends on a hidden-reasoning model must accept this reduced observability, and it should at least track the aggregate reasoning-token counts, because those are the visible trace of the cost the hidden reasoning incurred.

74.6 Compound Reasoning

Reasoning models do not operate only in isolation; they combine with the other patterns of this book, and the compound arrangements add capability that reasoning alone does not provide:

- **Tool use within reasoning** — the model thinks, calls a tool to get information or perform an action, and then thinks further with the tool's result in hand, interleaving deliberation with action.
- **RAG within reasoning** — the model retrieves mid-reasoning, pulling in knowledge at exactly the point in its deliberation where it discovers it needs that knowledge.
- **Self-correction within reasoning** — the model critiques and revises its own work during the thinking phase, catching errors before the answer is finalised.
- **Multi-agent reasoning** — specialist reasoning models are assigned to sub-problems, each deliberating on the part it is suited to, with the results composed.

The common thread is that reasoning is a phase that the other patterns can be embedded inside: a reasoning model that can retrieve, call tools, and self-correct during its deliberation is substantially more capable than one that only thinks in isolation. Compound reasoning is where reasoning models meet the rest of the architecture, and it is an active and fast-moving area.

74.7 The African Perspective

Reasoning models carry distinctive weight in African deployments, where their cost premium meets tight unit economics and their latency meets specific deployment realities.

74.7.1 Cost Discipline

The ten-to-fifty-times cost premium of a reasoning model makes routing not merely advisable but essential in African deployments, and the routing is correspondingly more aggressive than it would be in a market with more cost headroom. Where a Northern deployment might route generously to reasoning, accepting some wasted premium for simplicity, an African deployment routes tightly, sending a request to the reasoning path only when the signals for reasoning are clear. The cost discipline that runs through this book applies sharply here: the reasoning premium is a real cost, and it must be spent only where it genuinely buys quality.

74.7.2 Strategic High-Value Use

The way to make reasoning-model economics work under tight constraints is to reserve reasoning for the high-value workflows where a better answer is worth the premium — financial analysis, regulatory analysis, complex diagnostics. In these workflows the quality of the answer has real economic consequence, and the reasoning premium is small against the value of getting the answer right. Concentrating reasoning on these strategic, high-value uses, rather than spreading it thinly across all traffic, is what preserves the economics: the premium is paid where it pays back.

74.7.3 Open-Weight Alternatives

Open-weight reasoning models — DeepSeek’s R1, QwQ, and others — change the cost picture, because they can be self-hosted rather than called through a metered API. Self-hosted in an African cloud region, an open-weight reasoning model converts the per-call API premium into a fixed infrastructure cost, which at sufficient volume can be substantially cheaper and which also keeps the data within the jurisdiction. For an African deployment with steady reasoning demand, a self-hosted open-weight reasoning model is a genuine alternative to paying the API premium on every reasoning request.

74.7.4 Latency Tolerance

The variable, sometimes minute-long latency of a reasoning model is acceptable for some workflows and not for others, and segmenting by latency tolerance is part of deploying reasoning models well. For batch and asynchronous workflows — research, report generation, overnight analysis — the latency is entirely acceptable, because no user is waiting in real time. For interactive workflows, especially the mobile-first interactions common in African markets, a ninety-second wait is not acceptable, and reasoning either must be confined to a non-interactive part of the workflow or must not be used. Matching reasoning to the workflows that can tolerate its latency is the final piece of deploying it sensibly.

74.8 Recurring Failure Patterns

The failure modes of reasoning models recur, and each maps to a discipline this chapter has set out:

- **Reasoning everywhere.** The reasoning model is used by default for all traffic, and the cost premium, paid on every request including the many that did not need it, spirals.

- **Latency UX gap.** A long deliberation runs with no progress indication, and users abandon the request before the better answer arrives.
- **Reasoning-trace exposure.** A model’s hidden reasoning is inadvertently leaked to users, exposing content the provider intended to keep internal.
- **Evaluation mismatch.** The evaluation suite does not differentiate reasoning requests from standard ones, so the reasoning model’s advantage on hard problems is hidden in an aggregate that averages it away.
- **Tool integration immature.** Tool use within reasoning still has rough edges, and a production system relying on it heavily hits those edges.

Each pattern maps to a remedy the chapter has developed — disciplined routing rather than reasoning everywhere, a deliberate latency user experience, care with hidden reasoning traces, evaluation stratified to surface the reasoning advantage, and a measured reliance on still-maturing compound-reasoning integrations — and together they show that reasoning models fail through the mishandling of their variable cost, latency, and observability rather than through any weakness in the reasoning itself.

74.9 A Note Before Moving On

Reasoning models trade inference compute for output quality, spending substantial internal deliberation — often thousands of tokens, visible or hidden — to reach a better answer than a single forward pass would produce. Architecturally they introduce a routing decision on every request, variable cost and latency that the architecture must handle, an observability challenge that depends on how much reasoning a provider exposes, and a set of compound-pattern opportunities where reasoning combines with tools, retrieval, self-correction, and multi-agent decomposition.

The routing decision is the heart of it: reasoning pays off on mathematics, code, and complex planning, and is wasted on casual conversation and simple lookups, so a classifier must send each request to the path that fits it. The variable latency demands a deliberate user experience, and the partly-hidden reasoning demands an observability layer that captures what it can. African deployments add aggressive cost-disciplined routing, the reservation of reasoning for strategic high-value workflows, self-hosted open-weight reasoning models as a cost alternative, and segmentation by latency tolerance. You can pay to train or pay to think — and reasoning models let an architecture make that choice, deliberately, at each request.

World Models and Embodied AI

“The model learning to predict the next frame is learning physics. Architectures must catch up.”

—a recurring observation in robotics AI

Chapter Overview

A world model predicts the consequences of action in an environment: given this state and this action, what state follows. Embodied AI integrates that world-model reasoning with sensors and actuators, producing systems that perceive a physical or simulated world, decide, and act within it. The architectural shift is a real one — from the text-and-knowledge AI that dominates this book toward an AI of physical cause and effect, where a wrong decision is not a wrong sentence but a wrong movement of a real machine. Enterprise use cases are concentrated in manufacturing robotics, mining autonomy, agricultural systems, warehouse logistics, and the simulation environments that train all of them. This chapter develops the architectural perspective: what a world model is, how the simulation-to-reality pipeline works, how world models compose with planning and control, and how an embodied system is made safe.

Learning Objectives

- Recognise the principal world-model architectures and what distinguishes them.
- Architect the simulation-to-reality pipeline and understand the reality gap it must bridge.
- Integrate world models with planning and control as composable architectural primitives.
- Manage safety and verification for embodied systems that can cause physical harm.
- Adapt embodied AI to African industrial contexts and infrastructure realities.

75.1 What World Models Are

A world model is a learned function that predicts the next state of an environment from the current state and the action taken: in the standard notation, $s_{t+1} = f(s_t, a_t)$. What makes the modern version distinctive is the word *learned* — the function is not a hand-written physics simulator but a model trained on observation-action sequences drawn from the environment itself. A world model that has watched enough of an environment learns to predict what that environment does, and the epigraph’s

observation follows directly: a model trained to predict the next video frame of a physical scene is, in effect, learning the physics of that scene, because accurate frame prediction is impossible without an implicit grasp of how objects move and interact.

75.1.1 Variants

World models come in several architectural variants, and they differ in how they represent the environment and how they are trained:

- **Latent world models**, of which the Dreamer family is the best known, learn a compact latent representation of the environment and predict forward within that latent space, which makes planning over the model efficient.
- **Diffusion world models**, adjacent to the video-prediction systems such as Sora, generate predicted future observations through a diffusion process, producing high-fidelity rollouts of what the environment will look like.
- **Transformer world models**, such as Genie and the large-scale video models, apply the transformer architecture to environment prediction, bringing its scale and sequence-modelling strengths to world modelling.
- **Hybrid physics-plus-neural models** learn a neural correction — a residual — over a conventional physical simulator, combining the simulator’s reliable structure with a learned model’s ability to capture what the simulator misses.

The variants are not interchangeable, and the choice among them depends on the environment, the fidelity required, and the compute available — but all four share the defining property of being learned predictors of environmental consequence.

75.2 The Architecture

An embodied AI system built around a world model is a closed loop between the agent and its environment, and Figure 75.1 shows the shape. The environment is sensed by a perception component; perception feeds the world model, which predicts; the world model informs a policy, which decides on an action; actuators carry that action back into the environment; and the loop begins again as the sensors observe the new state the action produced.

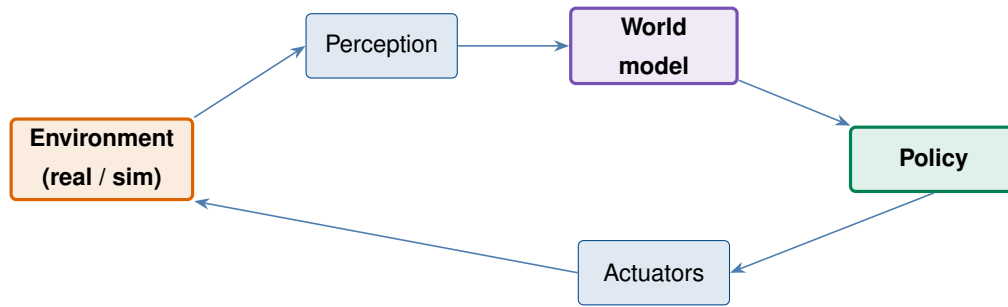


Figure 75.1: Embodied AI loop. The environment is sensed by perception; the world model predicts; the policy decides actions; actuators act on the environment; sensors observe the new state. The world model and the policy are the learning components.

Two of the five components in the loop are the learning components: the world model, which learns to predict the environment, and the policy, which learns to choose actions. The other three — perception, actuators, and the environment itself — are the fixed surfaces the learning components work through. The architectural significance of this split is that improving an embodied system means improving the world model’s prediction or the policy’s decision-making, and the rest of the chapter develops how those two components are trained and made safe.

75.3 Simulation to Reality

Most embodied AI training happens in simulation rather than in the real world, for the obvious reasons: a simulator runs faster than real time, costs nothing per trial, and cannot break a real machine when the policy makes a mistake. But deployment happens in reality, and the difference between the simulator and the real world — the so-called reality gap — is the dominant problem of embodied AI. A policy that performs perfectly in simulation can fail outright in reality, because the real world differs from the simulator in friction, in lighting, in sensor noise, in a thousand small ways the simulator did not capture.

75.3.1 Approaches to the Reality Gap

Several approaches address the reality gap, and a real embodied programme typically combines them:

- **Domain randomisation** varies the simulation’s parameters — friction, mass, lighting, textures — across a wide range during training, so that the policy is forced to generalise rather than overfit to one specific simulated world, and so generalises to reality as just another variation.
- **System identification** works the other way, calibrating the simulator to match the real environment as closely as possible, so that the gap the policy must cross is small to begin with.
- **Real-world fine-tuning** pretrains the policy extensively in simulation and then refines it with a small amount of real-world experience, using the simulator for the bulk of the learning and reality for the final adjustment.
- **World models trained on real data** skip the simulator entirely, learning the world model directly

from real observation-action sequences, which eliminates the reality gap at the cost of needing real-world data to train on.

The approaches trade off the cost and safety of simulation against the fidelity of reality, and the right combination depends on how dangerous a real-world mistake is and how much real-world data the programme can afford to gather.

75.4 Planning with World Models

Once a world model is accurate enough to be trusted, it unlocks a powerful capability: planning becomes search in imagined states. Rather than trying an action in the real environment to see what happens, the agent can roll the action forward through its world model, observe the predicted consequence, and so evaluate a plan entirely in imagination before committing any of it to reality. Planning over a world model is, in effect, thinking before acting.

Several planning methods exploit this. Monte-Carlo tree search can be run inside the world model, exploring a tree of imagined action sequences and their predicted outcomes. The cross-entropy method optimises a trajectory by sampling many candidate action sequences through the world model and refining toward the best. Model-predictive control uses the learned dynamics of the world model to plan a short horizon ahead, act on the first step, and replan. What unites these methods is that the architecture composes: the world model is a primitive that predicts consequence, and the planner is the policy that searches over those predicted consequences — the same compositional pattern that runs through the agentic chapters of this book, here applied to physical action.

75.5 Safety and Verification

Embodied systems differ from text-and-knowledge AI in one consequential respect: they can cause physical harm. A language model that produces a wrong answer is a quality problem; a haul truck that takes a wrong action is a safety problem, potentially a fatal one. Safety is therefore not an optional refinement of an embodied architecture but a core requirement, and several layers of it work together:

- **Bounded action spaces** constrain what the policy is physically permitted to command, so that an action outside the safe envelope is simply not available to be chosen.
- **Constrained policies** add action filters that inspect each command before it reaches the actuators and block any that are unsafe, catching a dangerous decision after the policy made it but before it takes effect.
- **Formal verification**, where the system is amenable to it, proves control invariants — properties such as “the machine never exceeds this speed” — that hold regardless of what the learned policy does.
- **Human override paths** ensure a person can always intervene and take control, so that a human judgement can override the autonomous system at any point.

- **Shadow mode** runs the policy alongside the existing system, with its commands recorded but not executed, before it is ever given real control — the embodied equivalent of the shadow-traffic discipline elsewhere in this book.

These layers are defence in depth: the bounded action space and constrained policy prevent unsafe commands, formal verification proves the prevention holds, the human override is the last resort when it does not, and shadow mode confirms the whole system is safe before it acts in the world at all.

75.6 Enterprise Use Cases

Embodied AI is not a speculative technology in the enterprise; it is in production across several industries, and the use cases concentrate where physical work is done at scale:

- **Manufacturing robotics** — pick-and-place, assembly, and quality inspection, where embodied AI extends conventional industrial robots with perception and adaptive control.
- **Mining autonomy** — autonomous haul trucks, autonomous drilling, and ventilation control, in an industry where the work is dangerous and the value of removing humans from hazardous areas is high.
- **Agricultural robotics** — harvesting, spraying, and weeding, where embodied systems work fields that would otherwise require large amounts of manual labour.
- **Warehouse logistics** — automated guided vehicles and mobile manipulation, where embodied AI moves goods through fulfilment operations.
- **Construction** — surveying and autonomous earthmoving, bringing autonomy to a traditionally manual and hazardous industry.
- **Simulation** — the training environments that produce the policies for all of the above, themselves a use case because building a good simulator is substantial engineering work.

The common thread is physical work at scale, and the value of embodied AI is highest where that work is dangerous, repetitive, or labour-intensive enough that autonomy genuinely changes the economics.

75.7 The African Perspective

Embodied AI carries distinctive weight in African deployments, across mining, agriculture, manufacturing, and the infrastructure constraints of remote industrial sites.

75.7.1 Mining Autonomy

Mining is the African embodied-AI use case that is most developed, and the reason is the scale and the hazard of the continent's mining industry. African deep mining — South African platinum and gold, worked at depths that make the environment dangerous — and African surface mining — copper in the Democratic Republic of Congo and Zambia, bauxite in Guinea — are major adopters of autonomous

haul trucks, autonomous drilling, and ventilation AI. The use case is well-developed because the value is clear: removing human operators from deep, hot, hazardous environments is a safety gain as much as a productivity one, and the economics of large-scale mining can absorb the cost of the autonomy.

75.7.2 Agricultural Robotics

Agricultural robotics fits African farming less straightforwardly, because smallholder agriculture dominates the continent and large-scale field robotics is built for large commercial farms. Adoption is therefore concentrated where the farms are commercial in scale — sugarcane, coffee, tea — and, more broadly, the embodied-AI contribution to smallholder agriculture tends to be advisory rather than autonomous: systems that advise a farmer rather than systems that drive a machine across a field. The architecture does not change, but the deployment model has to fit the structure of African farming rather than assume the large-commercial-farm model of Northern agriculture.

75.7.3 Manufacturing Robotics

Manufacturing robotics in Africa follows the continent's industrial clusters: the Moroccan automotive cluster and Egyptian manufacturing are the clearest adopters, integrating industrial robotics into their production lines. Embodied AI integration follows industrial-robotics adoption — a facility that has installed robots is the facility that then adds the perception and adaptive control that turn those robots into embodied AI systems. As Africa's manufacturing base grows, the embodied-AI opportunity in manufacturing grows with it.

75.7.4 Infrastructure-Constrained Deployment

Many African industrial sites — a remote mine, a distant farm — have variable or absent connectivity, and an embodied system that depends on a constant link to a cloud cannot operate there. The architectural response is on-device inference: the world model and the policy run on hardware at the site itself, so the embodied system can perceive, predict, and act without any network connection, and synchronisation to the cloud happens periodically when connectivity is available. Designing embodied AI for infrastructure-constrained deployment — self-sufficient on-site, cloud-connected only intermittently — is a standing requirement for African industrial use rather than an edge case.

75.8 Recurring Failure Patterns

The failure modes of embodied AI recur, and each maps to a discipline this chapter has set out:

- **Reality gap.** A policy trained in simulation fails in reality because the simulator did not capture the real environment closely enough, and the failure is an expensive surprise discovered only on deployment.
- **Unsafe action.** The action filter is inadequate, an unsafe command reaches the actuators, and physical harm results — the safety architecture failing at its most critical layer.
- **World-model rot.** The environment changes, the world model trained on the old environment becomes stale, and its predictions drift away from what the environment now does.

- **Compute mismatch.** The chosen model is too large for the on-device compute available, so the embodied system cannot run it at the site where it is needed.
- **Insufficient simulation.** The simulator does not cover the scenarios the system meets in production, so the policy was never trained for the situations it actually faces.

Each pattern maps to a remedy the chapter has developed — the reality-gap approaches, the layered safety architecture, world-model retraining as the environment changes, on-device compute sizing, and comprehensive simulation coverage — and together they show that embodied AI fails through the physical and safety realities of acting in the world, not through a generic modelling weakness.

75.9 The Takeaway

World models and embodied AI extend the AI architecture stack from text and knowledge to physical systems that perceive, predict, and act. A world model is a learned predictor of environmental consequence, in latent, diffusion, transformer, or hybrid form, and an embodied system is the closed loop of environment, perception, world model, policy, and actuators.

The core architectural concerns are the simulation-to-reality pipeline and the reality gap it must bridge, planning as search through the world model's imagined states, the layered safety architecture that an embodied system requires because it can cause physical harm, and on-device inference for sites without reliable connectivity. Enterprise use cases concentrate in manufacturing, mining, agriculture, warehouse logistics, and construction. African deployments foreground well-developed mining autonomy, agriculture adapted to a smallholder-dominated structure, manufacturing robotics following industrial clusters, and the infrastructure-constrained deployment that remote industrial sites demand. The model learning to predict the next frame is learning physics — and the architecture of this chapter is how that learned physics is turned into a system that acts safely in the real world.

Quantum-AI Hybrid Architectures

“Quantum advantage in AI is a horizon, not a destination. The architectural question is how to be ready.”
—a recurring observation in quantum computing research

Chapter Overview

Quantum computing and AI intersect in three distinct directions: AI algorithms running on quantum hardware, classical algorithms inspired by quantum research, and classical AI used to assist quantum computation itself. The field is genuinely early, and the honest framing matters: most of what is currently marketed as “quantum AI” is not yet quantum-advantaged — it does not yet beat what a classical computer can do. That does not make the topic irrelevant to an enterprise architect, because the architectural question is not whether to build a quantum AI system today but how to prepare for a horizon that may arrive, without over-investing against a capability that does not yet exist. This chapter develops that perspective: where quantum may eventually matter for AI, what the hybrid architectures look like, which quantum-inspired classical approaches deliver value today, and what a disciplined preparation posture consists of.

Learning Objectives

- Distinguish the near-term quantum-AI opportunities from the speculative ones.
- Architect the classical-quantum hybrid systems that dominate near-term practice.
- Recognise the quantum-inspired classical approaches that deliver value today, without quantum hardware.
- Apply a disciplined preparation posture that readies an organisation without premature commitment.
- Adapt quantum-AI thinking to African research and deployment contexts.

76.1 The Three Directions

The intersection of quantum computing and AI is not one thing but three, and they differ sharply in how mature and how useful they are today. Keeping them distinct is the first discipline, because conflating them is how an organisation ends up either dismissing the whole area or over-investing in

the wrong part of it.

76.1.1 Quantum Machine Learning

Quantum machine learning runs AI algorithms directly on quantum hardware: variational quantum eigensolvers, quantum neural networks, quantum kernels. It is the direction that the phrase “quantum AI” most often evokes, and it is also the least mature. Current quantum hardware runs these algorithms only on tiny problems, far below the scale of any realistic enterprise workload, and quantum machine learning is therefore not yet useful for production AI of the kind this book concerns. It is a research frontier worth watching, not a build target.

76.1.2 Quantum-Inspired Classical

Quantum-inspired classical algorithms are a more immediately useful direction: algorithms developed in the course of quantum research, then implemented to run on entirely classical hardware. Tensor-network methods and classical sampling algorithms inspired by quantum techniques both belong here. The important point is that these approaches deliver value today, on classical computers, on certain problems — they are quantum in their intellectual origin but classical in their deployment, which means an organisation can adopt them now without owning or accessing any quantum hardware.

76.1.3 Classical AI for Quantum

The third direction reverses the relationship: classical AI used to assist quantum computation. Machine learning is applied to design quantum experiments, optimise quantum circuits, mitigate the noise that plagues current quantum hardware, and decode quantum error-correcting codes. This is, perhaps surprisingly, the most practically deployed of the three directions, because it does not depend on quantum hardware being good enough to do useful AI — it uses mature classical AI to help the quantum hardware that exists today work better.

76.2 The Architecture

The architecture that dominates near-term quantum machine learning is the variational quantum-classical hybrid, shown in Figure 76.1. The pattern divides the work between a classical computer and a quantum circuit according to what each does well: the classical computer holds and optimises the parameters, and the quantum circuit produces the samples that the classical optimiser cannot efficiently produce itself.

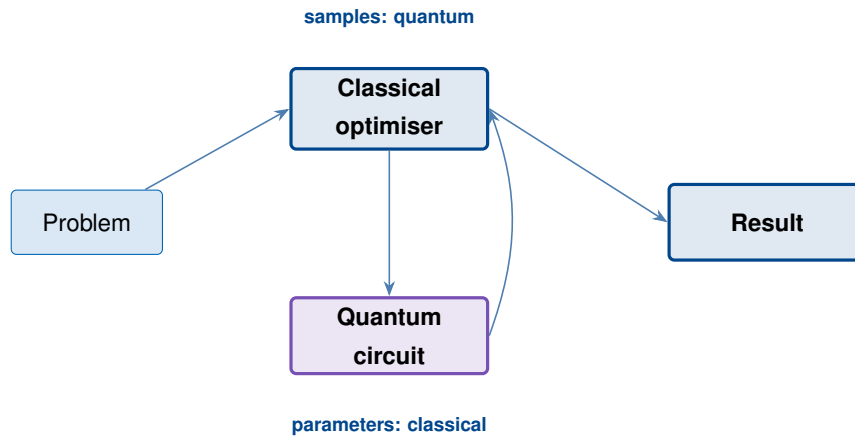


Figure 76.1: Variational quantum-classical hybrid. The classical optimiser proposes circuit parameters; the quantum circuit samples; the results inform the next set of parameters. This pattern dominates near-term quantum machine learning.

The loop runs as follows: the classical optimiser proposes a set of parameters for the quantum circuit; the quantum circuit, configured with those parameters, is run and sampled; the sampled results feed back to the classical optimiser, which uses them to propose better parameters. The architectural lesson is that even in a quantum AI system, most of the system is classical — the optimisation, the control, the result handling — and the quantum hardware is a specialised component invoked for the one step it does that classical hardware cannot. An organisation reasoning about quantum AI is, in practice, reasoning about how to plug a quantum component into an otherwise classical architecture.

76.3 Where Quantum May Matter for AI

If quantum advantage for AI arrives, it will not arrive everywhere at once; it will arrive on specific kinds of problem. Four areas are the most discussed candidates, and an architect tracking the field should know where the genuine possibilities lie.

76.3.1 Optimisation

Combinatorial optimisation — logistics routing, scheduling, portfolio construction — is the most cited candidate for quantum advantage, because these problems are computationally hard in a way that quantum approaches might address. The honest current position is mixed: quantum-inspired classical algorithms are already competitive on many of these problems today, and whether a genuinely quantum-advantaged solution will materialise, and on which problems, remains unclear. Optimisation is the area to watch most closely, because it is both the most promising and the one where classical alternatives are already strong.

76.3.2 Sampling

Drawing samples from complex probability distributions is a second candidate area, and it matters for AI because sampling is central to many machine-learning methods. Quantum sampling may have genuine advantages for specific distributions — distributions that are hard for a classical computer to sample from but natural for a quantum system to produce. This is a narrower and more specialised

possibility than the optimisation case, applicable where the specific distribution a workload needs happens to be one quantum sampling suits.

76.3.3 Linear Algebra

Quantum algorithms for linear algebra — the matrix operations at the heart of machine learning — offer speedups that exist on paper, sometimes dramatic ones. The practical catch is data loading: getting classical data into the quantum system in the form the quantum algorithm needs can itself cost as much as the speedup saves, which gates the practical advantage. Quantum linear algebra is a real theoretical result whose practical value depends on the data-loading problem being solved, and that remains an open question.

76.3.4 Cryptography-Adjacent Concerns

The fourth area is different in kind: it is not about quantum helping AI but about quantum threatening the cryptography that AI infrastructure depends on. Post-quantum cryptography — new cryptographic algorithms resistant to quantum attack — matters for the TLS, signing, and attestation that secure AI systems, and unlike the other three areas, the defensive concern is urgent now. An adversary can capture encrypted data today and decrypt it later when quantum computers are capable, so the cryptographic surfaces must be migrated regardless of when offensive quantum computing actually arrives.

76.4 Quantum-Inspired Classical Approaches

While quantum hardware matures, the quantum-inspired classical approaches deliver value today, and they are the part of this chapter an organisation can act on now. Three are worth knowing.

76.4.1 Tensor Networks

Tensor networks originated in quantum many-body physics, as a way to represent the enormous state spaces of quantum systems compactly. They have since found classical application well beyond their origin: in language-modelling research, where matrix-product-state representations have been explored for transformers; in simulation; and in optimisation. Tensor networks are a clear example of quantum-inspired thinking delivering on classical hardware — the mathematics came from quantum physics, but the value is realised on conventional computers.

76.4.2 Classical Sampling

A striking line of results, beginning with Tang’s algorithm and continuing through its successors, has shown that for some problems, classical algorithms can match the performance that was believed to require quantum computers. These “dequantisation” results are quantum-inspired in a particular sense: studying the quantum algorithm revealed how to build a classical one that does the same job. For an architect, the lesson is cautionary as well as useful — a problem believed to need quantum hardware may turn out to have an efficient classical solution, which is one more reason not to commit prematurely to a quantum roadmap.

76.4.3 Annealing

Simulated annealing, and the quantum-inspired annealing approaches that extend it, are practical optimisation methods available today for combinatorial problems. They do not require quantum hardware, they are well understood, and they deliver real results on the scheduling and routing problems that enterprises actually face. Annealing is the most immediately deployable of the quantum-inspired approaches, and for an organisation with a hard combinatorial optimisation problem, it is worth evaluating now rather than waiting for quantum advantage.

76.5 Preparation Discipline

For the great majority of enterprises, quantum AI is not yet a build target, and the right posture is preparation rather than construction. A disciplined preparation posture has five elements:

- **Track quantum hardware progress** — maintain awareness of where the hardware actually is, so that the organisation will recognise a genuine advantage when one arrives rather than reacting to marketing.
- **Adopt post-quantum cryptography** — migrate the cryptographic surfaces of the AI infrastructure to quantum-resistant algorithms now, because the harvest-now-decrypt-later threat makes this the one element of preparation that is genuinely urgent.
- **Identify candidate use cases** — know which of the organisation's problems — combinatorial optimisation, materials discovery, drug discovery — would be the candidates if quantum advantage arrived, so the organisation is ready to evaluate quickly.
- **Build hybrid expertise in the research function only** — develop quantum-classical hybrid skills within a research function, kept separate from production engineering, so the capability exists without the cost and risk of a production commitment.
- **Avoid premature production commitments** — do not put a quantum dependency on a production roadmap until a genuine, demonstrated advantage justifies it.

The preparation posture is deliberately asymmetric: the cryptographic migration is urgent and should happen now, while everything else is watchful readiness rather than active construction. This asymmetry is the chapter's central practical advice.

76.6 The African Perspective

Quantum AI has a distinctive shape in African contexts, across research capability, the urgency of post-quantum cryptography, and the optimisation use cases the continent's industries present.

76.6.1 Research Centres

African quantum research capability is emerging, with research centres developing in South Africa, Egypt, and Morocco. This is genuine capability building, and it matters for the continent's long-term position in the field — but it is research capability, and production deployment of quantum AI remains far off, in Africa as everywhere. The right framing for African organisations is the same as elsewhere:

support and watch the research, but do not mistake an emerging research centre for a production capability.

76.6.2 Post-Quantum Cryptography

Post-quantum cryptography is the one element of the quantum agenda that is urgent for African organisations now, and it is urgent regardless of quantum AI. African banking, telecommunications, and government systems all run on cryptography that a future quantum computer could break, and the harvest-now-decrypt-later threat means the migration cannot wait for quantum computers to actually arrive. Planning the post-quantum cryptography transition is a concrete, present-tense action that African organisations should be taking, distinct from the watchful posture appropriate to the rest of quantum AI.

76.6.3 Industry Use Cases

The candidate quantum use cases map onto real African industrial problems: mining logistics, agricultural optimisation, and energy dispatch are all combinatorial optimisation problems of exactly the kind quantum approaches are discussed for. The practical near-term opportunity is the quantum-inspired classical methods — annealing and related approaches — which may help these problems today on classical hardware, with genuinely quantum solutions a longer-term possibility. An African organisation with a hard mining-logistics or energy-dispatch optimisation problem should evaluate the quantum-inspired classical approaches now, while treating quantum hardware itself as a horizon.

76.7 Recurring Failure Patterns

The failure modes around quantum AI recur, and each is a failure of the disciplined posture this chapter recommends:

- **Quantum theatre.** Classical AI is branded as quantum for marketing effect, obscuring the reality that no quantum advantage is actually present.
- **Premature commitment.** A production roadmap is made dependent on a quantum advantage that has not yet been realised, exposing the organisation to a capability that may not arrive on schedule.
- **Post-quantum cryptography neglect.** The cryptographic surfaces are left unmigrated, vulnerable to the harvest-now-decrypt-later attack, because the one urgent element of the quantum agenda was treated as not yet relevant.
- **Use-case forcing.** Quantum approaches are applied to problems that classical AI already handles perfectly well, spending effort and attention where there is no advantage to be had.

Each pattern maps to the chapter's guidance — honest framing rather than theatre, watchful preparation rather than premature commitment, urgent cryptographic migration rather than neglect, and candidate-use-case discipline rather than forcing — and together they show that the risk in quantum AI today is not technical failure but a failure of posture: over-claiming, over-committing, or neglecting the one thing that is genuinely urgent.

76.8 In Short

Quantum-AI hybrid architectures span three directions: quantum machine learning, which is early and not yet useful at enterprise scale; quantum-inspired classical approaches, which deliver value today on classical hardware; and classical AI for quantum, the most practically deployed direction. The architecture that dominates near-term quantum machine learning is the variational hybrid, in which a quantum circuit is a specialised component within an otherwise classical system.

For most enterprises the right posture is disciplined preparation rather than construction: track the hardware, identify candidate use cases, build research-only expertise, avoid premature production commitments — and, the one urgent element, adopt post-quantum cryptography now, because the harvest-now-decrypt-later threat does not wait for quantum computers to arrive. African contexts add emerging research centres, a particular urgency around the post-quantum cryptography transition for banking and government systems, and combinatorial optimisation use cases in mining, agriculture, and energy where quantum-inspired classical methods may help today. Quantum advantage in AI is a horizon, not a destination — and the architectural question is how to be ready for it without betting on it.

Chapter 77

AI Hardware and Specialized Chips

“The chip you serve on costs as much as the model you serve.”

—a recurring observation in inference economics

Chapter Overview

AI hardware has diversified well beyond the NVIDIA GPUs that defined the field’s first decade. Google’s TPUs, Amazon’s Trainium and Inferentia, Cerebras’s wafer-scale engines, Groq’s deterministic inference chips, AMD’s MI300 series, Intel’s Gaudi, and the custom ASICs of the hyperscalers now make up a genuine landscape of choices. For an architect, the diversification matters because the choice of hardware shapes four things at once: the cost of serving a model, the latency of that serving, the set of models that will actually run, and the degree of lock-in to a particular vendor or cloud. This chapter develops the AI-architect’s view of hardware — not the internals of the chips, which are not an architect’s concern, but how to reason about hardware as an architectural decision with cost, portability, and procurement consequences.

Learning Objectives

- Recognise the principal classes of AI hardware and the trade-offs each carries.
- Architect for hardware portability so that a hardware choice is not an irreversible commitment.
- Apply hardware-software co-design where the scale of operation justifies it.
- Manage hardware capacity and procurement under supply constraints.
- Adapt hardware strategy to African deployment realities of capacity, import, and edge.

77.1 The Hardware Landscape

The AI hardware landscape is no longer a monoculture, and an architect should know the principal classes, because each one occupies a different point on the trade-offs of cost, ecosystem maturity, availability, and lock-in.

77.1.1 NVIDIA GPUs

NVIDIA's data-centre GPUs — the H100, H200, and the B200 and B300 of the Blackwell generation — remain the dominant AI hardware. They carry the widest software support of any class, because the CUDA ecosystem has had the longest to mature and almost every AI framework targets it first. They are also the highest in cost and the most supply-constrained, which is the price of that dominance: an organisation choosing NVIDIA chooses the best-supported path and the most contested one.

77.1.2 TPUs

Google's Tensor Processing Units are a strong alternative, particularly for training, and they are co-designed with the JAX framework so that the hardware and the software stack fit together closely. Their defining constraint is that they are available only on Google Cloud — a TPU choice is a Google Cloud choice. For an organisation already committed to Google Cloud, TPUs are a competitive and well-integrated option; for one that is not, the cloud lock-in is the cost of entry.

77.1.3 AWS Trainium and Inferentia

Amazon's Trainium, for training, and Inferentia, for inference, are AWS-specific chips designed to be cost-competitive with NVIDIA for matched workloads. Their ecosystem is smaller than CUDA's, which means more of the software path must be the AWS-provided one, but the cost advantage is real. As with TPUs, the chips are tied to one cloud — choosing Trainium or Inferentia is choosing AWS — and the trade is cost against cloud lock-in.

77.1.4 AMD MI300 Series

AMD's MI300 series is closing the gap with NVIDIA on training performance, and its ROCm software ecosystem, long the weak point of the AMD AI story, is maturing. AMD is the most credible cross-cloud alternative to NVIDIA at the high end, because it is not tied to a single cloud the way TPUs and Trainium are. For an organisation seeking a second source at the training tier, the MI300 series is the option to watch most closely.

77.1.5 Specialised Inference Chips

A set of specialised inference chips — Groq with its deterministic, very-low-latency inference, Cerebras with its wafer-scale engine, SambaNova, Tenstorrent — offer high performance on specific workloads. Their ecosystems are smaller than the general-purpose options, so they suit a workload that fits what the chip does well rather than a general serving estate. Where a workload's shape matches a specialised chip's strength — deterministic low latency, very large models on a single wafer — these chips can outperform the general-purpose alternatives substantially.

77.1.6 Edge Accelerators

At the other end of the scale are the edge accelerators — NVIDIA's Jetson, Apple's Neural Engine, Qualcomm's AI Hub, and the various NPU offerings — designed to run AI within sub-100-watt power envelopes on mobile and embedded devices. These are not data-centre hardware and do not compete with it; they are the hardware for inference that must happen on a device, away from any data centre. For embodied AI and offline-capable mobile applications, the edge accelerator is the relevant hardware

class.

77.2 The Architecture

Because the hardware landscape is diverse and each class carries lock-in risk, the central architectural move is to insert a portability layer between the application and the hardware. Figure 77.1 shows the shape: the application talks to a portability layer, and the portability layer abstracts the specific hardware beneath it, so that the same application code can run across NVIDIA GPUs, TPUs, Inferentia, and edge devices.

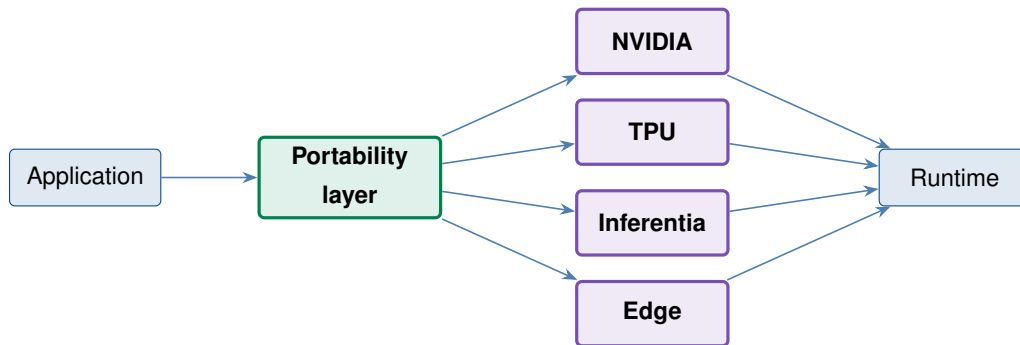


Figure 77.1: Hardware-aware deployment. A portability layer — vLLM and TensorRT-LLM with multiple backends, ONNX Runtime, OpenVINO — abstracts the hardware, so the same application code runs across NVIDIA, TPU, Inferentia, and edge devices.

The portability layer is, in practice, an inference serving stack that supports multiple hardware backends — vLLM and TensorRT-LLM configured with several backends, ONNX Runtime, OpenVINO. Its architectural value is that it converts a hardware choice from an irreversible commitment into a swappable one: an organisation that built directly against one vendor’s low-level interface is locked to that vendor, while an organisation that built against a portability layer can change the hardware beneath it as cost, availability, and capability shift. The portability layer is the hexagonal-architecture discipline of this book applied to the hardware substrate.

77.3 Trade-Offs

The hardware classes differ along four axes, and a hardware decision is, in effect, a weighing of these four against one another.

77.3.1 Cost

Cost is the axis that most often drives the decision. Inferentia and Trainium are frequently thirty to fifty per cent cheaper than NVIDIA for matched workloads, and TPUs are competitive for training — the alternatives to NVIDIA largely compete on price. For a cost-sensitive serving estate, the alternative hardware classes offer real savings, and the question is whether the ecosystem and lock-in costs of those alternatives outweigh the price advantage.

77.3.2 Software Support

Software support is the axis on which NVIDIA leads most clearly: the CUDA ecosystem is the broadest, almost every framework targets it first, and the path of least resistance runs through NVIDIA hardware. AMD is narrowing the gap as ROCm matures, and the more specialised alternatives are more constrained, requiring more of the software path to be the vendor-provided one. The software-support axis is what an organisation pays for when it pays NVIDIA's premium.

77.3.3 Availability

Availability has become a genuine architectural constraint. NVIDIA's data-centre GPUs are supply-constrained, with long lead times for self-hosted purchase and contested capacity in the clouds, while the alternatives often have better availability precisely because demand for them is lower. An organisation that can use an alternative class may find it can actually obtain the hardware, where the NVIDIA option is a queue.

77.3.4 Lock-In

Lock-in is the axis that the portability layer exists to manage. TPUs are Google Cloud only, and Trainium and Inferentia are AWS only, so choosing those chips means choosing those clouds. NVIDIA hardware, by contrast, is available across clouds and on-premises, which makes it the portable choice at the hardware level even as it is the expensive one. Weighing lock-in against cost is the core of the hardware decision, and the portability layer is what keeps a lock-in choice from being permanent.

77.4 Hardware-Software Co-Design

For organisations operating at sufficient scale, hardware and software are not chosen independently; they co-evolve, with software decisions made to fit the hardware and hardware decisions made to fit the workload. Several specific co-design choices recur:

- **Model architecture chosen for hardware** — a mixture-of-experts architecture chosen because it suits the TPU, or multi-query attention chosen because it eases the memory pressure on a memory-constrained chip.
- **Quantisation schemes tied to hardware** — FP8 inference where the hardware supports it natively, as the H100 does, and INT8 where it is broadly available, with the precision chosen to match what the chip accelerates.
- **Custom kernels for specific hardware paths** — hand-tuned compute kernels written for a particular chip, extracting performance that a generic kernel would leave on the table.
- **Compiler choices matched to hardware** — XLA for the TPU path, TensorRT for the NVIDIA path, with the compiler chosen as part of the hardware decision rather than after it.

Co-design is worthwhile only at scale, because it trades portability for performance — a model architecture chosen for one chip, served through custom kernels and a chip-specific compiler, is fast on that chip and hard to move off it. An organisation large enough that the performance gain is worth

millions does this deliberately; a smaller one stays with the portability layer and accepts the generic path.

77.5 Capacity and Procurement

Hardware is not only a technical choice but a procurement one, and three procurement considerations shape an AI hardware strategy.

77.5.1 On-Demand versus Reserved

The first consideration is the balance between on-demand and reserved capacity. On-demand capacity suits variable workloads, where paying only for what is used outweighs the higher per-hour rate; reserved capacity suits steady workloads, where the commitment buys a lower rate. A sound strategy also reserves capacity in more than one region, so that a regional capacity shortage or outage does not leave the workload with nowhere to run. The mix of on-demand and reserved should track the predictability of the workload.

77.5.2 Supply Constraints

The second consideration is that the leading hardware is supply-constrained. The H100 and its successors have long lead times for self-hosted purchase and contested availability in the clouds, and a capacity plan that assumes hardware can be obtained on demand will be disappointed. A multi-vendor strategy is the mitigation: an organisation that can run on more than one hardware class is an organisation that can obtain capacity from whichever class is actually available, rather than waiting in a single queue.

77.5.3 Capital versus Operating Expenditure

The third consideration is the choice between renting hardware from a hyperscaler as an operating expense and buying hardware into a colocation facility as a capital expense. The capital route has a breakeven point — typically twelve to eighteen months at steady, high utilisation — after which the purchased hardware is cheaper than renting. The decision turns on utilisation: hardware that will be steadily and heavily used favours the capital route, while variable or uncertain demand favours the operating-expense flexibility of the hyperscaler.

77.6 The African Perspective

AI hardware carries distinctive weight in African deployments, across local capacity, import realities, the importance of edge hardware, and the reinforcement of the smaller-model preference.

77.6.1 Limited Local Capacity

GPU capacity in African cloud regions — Cape Town, Johannesburg — is growing but remains limited relative to the Northern regions. The practical consequence is that many African AI workloads run in Northern regions, and the latency and data-sovereignty considerations that follow from that — developed throughout this book — are a direct result of the local-capacity constraint. As African cloud regions add AI hardware, more workloads can run in-region; until then, the capacity constraint

shapes where African AI actually executes.

77.6.2 Import Realities

The capital-expenditure route — buying GPUs into a colocation facility — faces specific friction in African contexts: import complexity and the duties and delays that come with it, foreign-currency exposure on hardware priced in dollars, and the availability of vendor support for hardware deployed far from the vendor's service network. These frictions raise the real cost and risk of the capital route, which is why the hyperscaler operating-expense model is often preferred in African deployments even where steady utilisation would, on the pure breakeven arithmetic, favour buying.

77.6.3 Edge Hardware

Edge AI accelerators — Coral, Jetson Nano, mobile NPUs — are deployable in industrial settings without any cloud connectivity, and that makes them particularly important in African contexts where remote industrial sites lack reliable links. For mining, agriculture, and remote operations, edge hardware is what allows AI to run at the site at all, and the edge accelerator class is correspondingly more central to African industrial AI than it is in well-connected Northern deployments.

77.6.4 Smaller-Model Preference

The hardware constraints of African deployment — limited local capacity, import friction, the prominence of edge hardware — all reinforce the smaller-model preference that runs through this book. A smaller model needs less hardware, fits on an edge accelerator, and is cheaper to obtain and run, so the hardware reality and the cost reality point the same way: toward smaller models, aggressive quantisation, and edge deployment. The hardware chapter and the cost chapters reach the same conclusion from different directions.

77.7 Recurring Failure Patterns

The failure modes around AI hardware recur, and each maps to a discipline this chapter has set out:

- **Single-vendor lock-in.** The application is coupled directly to NVIDIA CUDA with no portability layer, so the cost of switching hardware is prohibitive and the organisation has no answer to a price increase or a supply shortage.
- **Capacity surprise.** Workload growth outpaces the available GPUs, because the supply constraint was not planned for and no multi-vendor fallback exists.
- **Mismatched hardware.** A workload is deployed on the wrong tier of hardware — too powerful and so wasteful, or too weak and so slow — because the hardware was chosen without matching it to the workload.
- **Capital overrun.** Hardware is purchased on a capital plan, and then a workload shift leaves it idle, so the breakeven that justified the purchase is never reached.
- **Edge underestimation.** An edge model is chosen that exceeds the device's actual capability, and

production hits out-of-memory failures the planning did not anticipate.

Each pattern maps to a remedy the chapter has developed — the portability layer against lock-in, multi-vendor strategy against capacity surprise, workload-matched hardware selection, utilisation-honest capital planning, and realistic edge-device sizing — and together they show that AI hardware fails through procurement and portability mistakes rather than through the chips themselves.

77.8 What to Carry Forward

AI hardware spans NVIDIA GPUs, Google's TPUs, Amazon's Trainium and Inferentia, AMD's MI300 series, the specialised inference chips, and the edge accelerators — a genuine landscape rather than a monoculture. The choice among them shapes cost, latency, ecosystem maturity, lock-in, and availability, and it is weighed along those axes rather than decided on raw performance alone.

The central architectural move is the portability layer, which abstracts the hardware so that a choice is swappable rather than permanent, and the central procurement discipline is the multi-vendor strategy that lets an organisation obtain capacity under supply constraints. Hardware-software co-design extracts further performance at the cost of portability, and is worthwhile only at scale. African deployments add limited local capacity that pushes workloads to Northern regions, import realities that favour the hyperscaler operating-expense model, the heightened importance of edge hardware for remote industrial sites, and a hardware reality that reinforces the smaller-model preference. The chip you serve on costs as much as the model you serve — and the architecture of this chapter is how an organisation makes that cost a deliberate, manageable choice.

Chapter 78

AI Ethics Architecture

“A principle is aspirational. A control is operational. The architecture is the bridge.”

—a recurring observation in responsible AI practice

Chapter Overview

AI ethics architecture is the discipline of operationalising ethical principles — fairness, transparency, accountability, privacy, beneficence — into the concrete design of a system. It extends the constitutional-AI chapter, which concerned the internal behaviour of a model, outward to the system-wide ethical commitments of the organisation that deploys it. The central problem the discipline addresses is a translation problem: ethical principles are aspirational statements, and a system cannot be built directly from an aspiration. The architecture is the bridge — the set of structures that turn “be fair” into a measurable standard, an operational control, an auditable record, and a line of accountability. This chapter develops that architecture: how principles become controls, how fairness and transparency and accountability acquire specific architectural expression, and how the organisational structures that sustain ethical practice are built.

Learning Objectives

- Translate ethical principles into architectural controls rather than leaving them as aspirations.
- Architect the specific mechanisms for fairness, transparency, and accountability.
- Build the organisational structures — committees, functions, worker voice — that sustain ethical practice.
- Address the contextual variation in ethical norms across cultures and jurisdictions.
- Adapt AI ethics architecture to African continental frameworks and cultural plurality.

78.1 From Principles to Controls

The work of AI ethics architecture begins with a gap: between the ethical principles an organisation commits to and the operational reality of the systems it runs. Closing that gap is the whole task, and it starts with understanding what the principles are and why they cannot be applied directly.

78.1.1 The Common Principles

The major ethical frameworks — the OECD principles, the European Union’s, the African Union’s, and most national AI strategies — converge to a striking degree on a common set: fairness, transparency, accountability, privacy, security, beneficence, human oversight, and robustness. This convergence is genuinely useful, because it means an organisation does not have to choose among competing principle sets; the principles are broadly agreed. The principles are real and they are shared — but agreeing on them is the easy part. The architectural work is the operationalisation, and that is where the difficulty lies.

78.1.2 The Translation Problem

The difficulty is that a principle is not an instruction. “Be fair” tells an engineer nothing they can implement, because it does not say fair by what measure, across which attributes, at what threshold, verified how. An operational version — “achieve demographic parity within a one per cent threshold across these named protected attributes, validated by these specific metrics, reviewed monthly” — is something a system can actually be built and audited against. The translation problem is the gap between those two sentences, and the architecture is the set of structures that perform the translation: principle to standard, standard to control, control to evidence. The rest of this chapter develops those structures.

78.2 The Architecture

AI ethics architecture is best understood as a layered model, shown in Figure 78.1, in which each layer translates the one above it into something more concrete and each layer has a distinct owner. The layered structure is what makes the translation from aspiration to operation systematic rather than ad hoc.



Figure 78.1: Layered ethics architecture. Principles translate to standards, standards become controls, controls produce evidence, oversight verifies, and culture sustains. Each layer has its own owner.

The six layers run from the aspirational to the sustaining. Principles are the organisation’s commit-

ments, owned by leadership. Standards translate those principles into measurable criteria, owned by risk and legal. Controls are the technical and process implementations of the standards, owned by engineering. Evidence is the audit-ready record that the controls produced, owned by audit. Oversight — the committees, escalation paths, and accountability lines — verifies that the layers above are working, owned by governance. And culture — the incentives, training, and channels for raising concern — sustains the whole, owned by the human-resources and organisational function. An ethics architecture missing any layer fails: principles without standards stay aspirational, standards without controls stay theoretical, controls without evidence cannot be audited, and any of it without oversight and culture decays.

78.3 Fairness Controls

Fairness is the principle that most clearly demands concrete architectural expression, because an unfair system harms real people in measurable ways. Fairness controls divide into the measurement of unfairness and its mitigation.

78.3.1 Measurement

A system cannot be made fair if its unfairness is not measured, and fairness measurement uses several instruments together:

- **Disparate-impact analysis** examines whether the system produces systematically different outcomes across protected attributes, the basic test of whether a fairness problem exists.
- **Stratified evaluation** computes every quality metric per cohort rather than only in aggregate, because an aggregate metric averages a harmed cohort together with a well-served one and hides the harm.
- **Counterfactual fairness probing** tests whether changing only a protected attribute changes the system's output, isolating the attribute's causal effect.
- **Independent third-party assessment** brings an outside evaluator to high-stakes systems, because an organisation assessing its own fairness has an obvious incentive to find none.

Together these turn fairness from an assumption into a measured quantity, and measurement is the precondition for any honest claim that a system is fair.

78.3.2 Mitigation

Where measurement reveals unfairness, mitigation addresses it, and the mitigation can be applied at several points in the pipeline:

- **Training-data rebalancing** corrects an imbalance in the data before the model is trained, addressing unfairness at its most common source.
- **In-processing fairness constraints** build fairness objectives into the training itself, so the model is optimised for fairness alongside accuracy.
- **Post-processing calibration** adjusts the model's outputs after the fact to equalise outcomes across

cohorts, without retraining.

- **Routing to alternative paths** sends the requests of an affected cohort to a different, better-suited path when the main model serves them poorly.

The right mitigation depends on where the unfairness originates and how much of the pipeline the organisation controls, and a real fairness practice often combines several of these rather than relying on one.

78.4 Transparency Controls

Transparency is the principle that the people affected by an AI system should be able to understand, to an appropriate degree, what it is and what it does. It has architectural expression at three levels.

78.4.1 Model Documentation

The first level is documentation of the models and systems — model cards, system cards, data cards — that record what a model is, what it is for, what its limitations are, and what data it was built from. This documentation is made public where that is appropriate, and made available to regulators and to users on request where full publication is not. Model documentation is the transparency artefact that lets an outside party understand the system without reverse-engineering it, and it is also, as the compliance chapter notes, a compliance requirement for high-risk systems.

78.4.2 User-Facing Transparency

The second level is transparency to the users who interact with the system. Users should be informed when AI is involved in what they are experiencing, the system's capabilities and limitations should be communicated to them honestly rather than oversold, and the paths for recourse — how to challenge or appeal an outcome — should be explained. User-facing transparency is the level that affects the most people, because every user of the system encounters it, and it is the level at which an organisation's transparency commitment is most visibly tested.

78.4.3 Explainability

The third level is explainability: for high-stakes decisions, an explanation of why the system reached the conclusion it did. The techniques include attribution methods that identify which inputs drove the decision, counterfactual explanations that show what would have changed the outcome, and feature-importance approaches that rank the factors. Explainability is reserved for the high-stakes cases because it carries a cost and is not always technically clean, but where a decision materially affects a person, an explanation of that decision is part of treating them fairly.

78.5 Accountability Controls

Accountability is the principle that when an AI system causes harm, there is a clear answer to the question of who is responsible and what recourse the harmed party has. It has three architectural expressions.

78.5.1 Named Owners

Every system must have named owners who are accountable for its behaviour — not a committee, which diffuses responsibility until no one holds it, and not nobody, which is the default an organisation drifts into if it does not act. Named individuals, identified and accountable. The discipline of naming owners is what ensures that when something goes wrong, there is a person whose responsibility it was, rather than a collective shrug. Accountability that is not attached to a name is not accountability.

78.5.2 Incident Response

Accountability requires a defined response to ethical incidents: a process for what happens when a system causes ethical harm, including public communication when the harm warrants it, and a post-incident review that feeds the lesson back into the controls. Ethical incident response parallels the security and compliance incident response of the earlier chapters — the same closed-loop discipline, applied to ethical harms. An organisation with no ethical incident-response process discovers, when an incident occurs, that it is improvising its accountability under pressure.

78.5.3 Recourse

Recourse is the affected person's side of accountability: a user who is subject to an AI decision must be able to appeal it, and human review must be available for high-stakes outcomes. Recourse is what makes accountability meaningful to the person harmed rather than only to the organisation — it gives them a path to challenge a decision rather than being bound by it with no remedy. An accountability architecture without a recourse path holds the organisation accountable in principle while leaving the harmed person with nowhere to go.

78.6 Human Oversight

Human oversight is the principle that an AI system, especially in high-stakes use, operates under meaningful human supervision rather than fully autonomously. Its architecture has two essential properties.

78.6.1 Tiered Oversight

Oversight should be tiered to the stakes, using the approve, audit, sample, and escalate structure developed in Chapter 35. A low-stakes decision may be sampled; a high-stakes one may require approval before it takes effect. Tiering matches the intensity of the human involvement to the consequence of the decision, so that scarce human attention is concentrated where it matters and not spent uniformly on decisions that do not warrant it. Uniform oversight either over-burdens the reviewers or under-protects the high-stakes cases.

78.6.2 Meaningful Oversight

Tiering is not enough on its own; the oversight must also be meaningful rather than a rubber stamp. Oversight is meaningful when the reviewer interfaces are designed for genuine review — presenting the reviewer with what they need to actually judge the decision — and when agreement rates and time-per-review are monitored, because a reviewer who approves everything in two seconds is not

reviewing. Meaningful oversight is the difference between human supervision that catches errors and human supervision that merely launders the system's decisions with a signature.

78.7 Organisational Structures

Ethics architecture is not only technical controls; it is also the organisational structures that sustain ethical practice over time. Three structures are central.

78.7.1 Ethics Committee

The ethics committee is a cross-functional body that reviews high-stakes systems before and during deployment. Its membership should be genuinely diverse — across functions, backgrounds, and perspectives — because a homogeneous committee has the blind spots of its members, and it should hold real veto authority where warranted, because a committee that can only advise is a committee whose advice can be ignored. An ethics committee with diverse membership and genuine authority is a structural check on the systems the organisation ships.

78.7.2 Responsible AI Function

The responsible-AI function is a dedicated team that supports the product teams with the reviews, the tools, and the training that ethical practice requires. Its existence reflects a recognition that ethical practice does not happen on its own — product teams under delivery pressure need a function whose job is to make the ethical work tractable. The responsible-AI function is the standing capability that turns ethics from an occasional committee review into a continuous, supported practice.

78.7.3 Worker Voice

The third structure is worker voice: the workers building and operating AI systems must be able to raise ethical concerns without fear of retaliation. Concrete channels for raising concern must exist, and the concerns raised must be tracked rather than absorbed and forgotten. Worker voice matters because the people closest to a system are often the first to see an ethical problem, and an organisation that silences or ignores them loses its earliest and best warning. A protected, tracked channel for worker concern is a structural part of ethics architecture.

78.8 Contextual Variation

Ethical norms are not uniform across cultures, jurisdictions, and communities, and an ethics architecture that assumes a single set of norms will impose one community's standards on others. The architecture must therefore support contextual adaptation:

- **Per-market policy variation** lets the ethical policy differ by jurisdiction, so that each market's norms and laws are reflected rather than overridden.
- **Local stakeholder engagement** brings the communities affected by a deployment into the design of its ethical commitments, so the norms applied are theirs and not an outsider's assumption.
- **Cultural-context evaluation** tests the system against the standards of the markets it serves, stratified by market rather than judged against a single standard.

- **Avoiding monocultural assumptions** is the discipline underlying all three — the deliberate refusal to treat one market’s norms as the universal default.

Contextual variation is not relativism; the common principles still hold everywhere. It is the recognition that the operationalisation of those principles — what fairness means in practice, what transparency a community expects — legitimately differs, and the architecture must let it differ.

78.9 The African Perspective

AI ethics architecture carries distinctive weight in African deployments, across continental frameworks, cultural plurality, local stakeholder engagement, and the bias that foundation models carry on underrepresented populations.

78.9.1 Continental Frameworks

African ethical commitments are informed by frameworks specific to the continent: the African Union’s Continental AI Strategy and the African Charter on Human and Peoples’ Rights provide a continental foundation, and national AI strategies in Egypt, Kenya, Morocco, and elsewhere add country-specific detail. An ethics architecture for African deployment should be grounded in these frameworks rather than only in the Northern ones, because they reflect the values and priorities of the continent the system serves, and because alignment with them is part of operating legitimately in those markets.

78.9.2 Cultural Plurality

Africa spans substantial cultural plurality — many cultures, many languages, many normative traditions within a single continent and often within a single country. An ethics architecture for African deployment must accommodate that plurality through the contextual-variation mechanisms of the previous section, and the monocultural defaults imported from Northern markets frequently fail when applied across it. Cultural plurality is not an edge case to be handled but a central design parameter of African ethics architecture.

78.9.3 Local Stakeholder Engagement

The communities affected by AI deployment in African mining, agriculture, and public-sector contexts should have a genuine voice in the ethical commitments that govern those deployments. The structures for that engagement vary by country, and the architecture should support whatever structure fits the local context rather than imposing one. Local stakeholder engagement is how an ethics architecture ensures that the people most affected by a system are not merely subjects of it but participants in deciding how it should behave.

78.9.4 Bias on Underrepresented Populations

Foundation models are trained predominantly on Northern data, and they consequently perform unevenly across African contexts — across African languages, African names, African faces, African circumstances. This makes bias measurement and mitigation particularly consequential in African deployments, because the bias is not hypothetical but a known property of models built on Northern-

skewed data. The fairness controls of this chapter — stratified evaluation, disparate-impact analysis, the mitigation techniques — are, in African deployments, a direct response to a known and measurable problem rather than a precaution against a possible one.

78.10 Recurring Failure Patterns

The failure modes of AI ethics architecture recur, and each is the collapse of a layer or a structure this chapter has set out:

- **Principles without controls.** The organisation publishes glossy principle documents but makes no operational change, so the principles stay aspirational and the systems behave exactly as they would have without them.
- **Ethics-as-engineering.** The engineering team is left to handle ethics alone, and the structural blind spots of a single function — the perspectives it does not have — go uncorrected.
- **Single-stratum measurement.** Only aggregate metrics are computed, so a harm concentrated in one cohort is averaged away and rendered invisible.
- **Oversight without authority.** Ethics committees exist but cannot block a deployment, so their review is advisory and an organisation under delivery pressure can override it.
- **Cultural monoculture.** The norms of one market are imposed on all the others, because the architecture has no contextual-variation mechanism.

Each pattern maps to a discipline the chapter has developed — controls behind the principles, cross-functional ownership, stratified measurement, oversight with genuine authority, and contextual variation — and together they show that ethics architecture fails when a layer or structure is present in name but hollow in substance.

78.11 The Shape of the Argument

AI ethics architecture operationalises ethical principles into a layered structure: principles become standards, standards become controls, controls produce evidence, oversight verifies, and culture sustains. The discipline's central task is the translation problem — turning an aspirational principle into a measurable standard and an operational control — and the layered architecture is the bridge that performs that translation.

Fairness, transparency, accountability, and human oversight each acquire specific architectural expression: fairness through stratified measurement and pipeline mitigation, transparency through documentation and explainability, accountability through named owners and recourse, oversight through tiering and meaningfulness. The organisational structures — the ethics committee, the responsible-AI function, worker voice — sustain the practice, and contextual variation ensures one community's norms are not imposed on another. African deployments add the continental frameworks, the cultural plurality that must be a central design parameter, local stakeholder engagement, and the consequential bias that Northern-trained models carry on underrepresented populations. A principle is

aspirational and a control is operational — and the architecture of this chapter is the bridge that turns the first into the second.

Chapter 79

Environmental Impact and Sustainability

“Compute has a footprint. Architecture decides whose footprint and where.”

—a recurring observation in sustainable AI

Chapter Overview

AI workloads consume electricity, water, and embodied materials at a scale large enough to matter. The carbon footprint of training a large model is substantial, and inference at production scale accumulates a footprint that, summed across billions of requests, can exceed the one-time training cost. The water used to cool data centres is increasingly material, especially in arid regions where water is itself scarce. And the rare earths, semiconductors, and copper that go into AI hardware carry mining impacts traceable back to AI demand. An architecture that ignores these impacts is increasingly out of step with corporate commitments, with tightening regulation, and with societal expectation. This chapter develops the sustainability architecture for AI: what to measure, what optimisation levers exist, and how sustainability is integrated into the procurement and deployment decisions an architect makes.

Learning Objectives

- Recognise the principal environmental-impact categories of AI workloads.
- Architect the measurement and attribution of that impact to the workloads that caused it.
- Apply the optimisation levers — efficiency, location, scheduling, hardware lifecycle — that reduce impact.
- Integrate sustainability into procurement and deployment decisions rather than treating it separately.
- Adapt sustainable AI to African energy, water, and equity contexts.

79.1 The Impact Categories

A sustainability architecture begins with an honest inventory of what an AI workload actually consumes, and the consumption falls into four categories. A practice that measures only the most visible of them

— energy — understates the true footprint, so naming all four is the first step.

79.1.1 Energy

Energy is the most direct impact: the electricity consumed by training runs and by production inference, the embodied energy that went into manufacturing the hardware, and the substantial energy of data-centre cooling. Energy is the category that the other categories partly derive from — carbon is energy times grid intensity, and cooling is itself an energy draw — which makes it the foundation of the measurement. An architecture that understands its energy consumption is most of the way to understanding its footprint.

79.1.2 Carbon

Carbon is energy multiplied by the carbon intensity of the grid that supplied it, and the second factor varies enormously by region. A Nordic grid running largely on hydroelectric power may carry an intensity below fifty grams of carbon dioxide per kilowatt-hour, while a coal-dominated grid may exceed seven hundred — a difference of more than tenfold for the same kilowatt-hour of compute. This regional variation is the single most important fact in carbon-aware architecture, because it means the same workload has a wildly different carbon footprint depending on where it runs, and location therefore becomes a lever.

79.1.3 Water

Water is consumed by data-centre cooling, and at hyperscale the quantities are large — a major facility can use millions of litres per day. Water is a footprint category in its own right, distinct from carbon, and it is especially material in water-stressed regions, where the data centre's draw competes with other demands on a scarce resource. Water is the category most often omitted from a sustainability accounting that focuses on carbon, and omitting it understates the real environmental cost, particularly in arid regions.

79.1.4 Materials

The fourth category is the embodied impact of the hardware itself: the rare earths, the semiconductors, and the copper that go into AI accelerators, and the mining and manufacturing impacts that producing them entails. These impacts are traceable to AI hardware demand — the chips an AI estate runs on had a material footprint before they ever drew a watt. Materials is the hardest category to measure precisely, but a complete sustainability picture includes it rather than accounting only for the impacts that occur after the hardware is installed.

79.2 The Architecture

The sustainability architecture for AI is a layered one, shown in Figure 79.1, running from measurement at the foundation up through attribution, optimisation, reporting, and procurement. Each layer has its own owner, and the layers build on one another: measurement makes attribution possible, attribution makes optimisation targeted, and reporting and procurement close the loop with disclosure and sourcing.

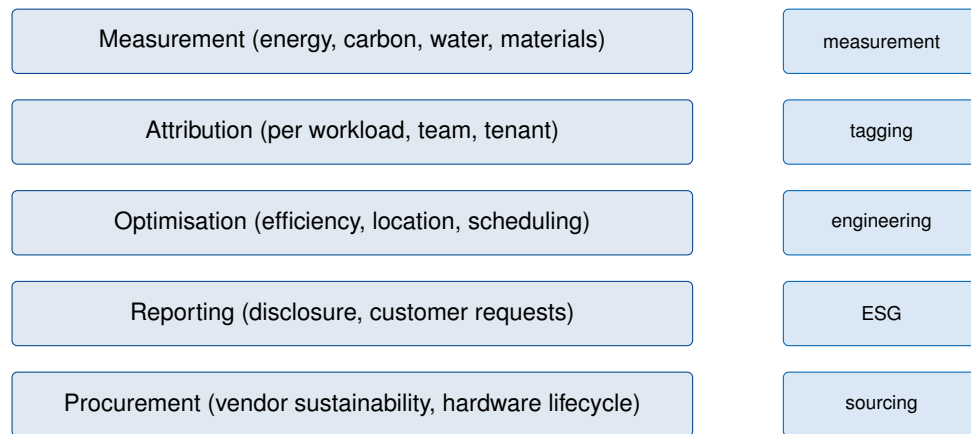


Figure 79.1: Sustainability architecture. Measurement sits at the foundation; attribution links it to workloads; optimisation reduces the impact; reporting satisfies disclosure obligations; procurement closes the loop with sourcing decisions.

The architecture mirrors the FinOps architecture of an earlier chapter deliberately, and the parallel is not coincidental: sustainability and cost are measured against the same workloads, attributed by the same tagging, and reduced by overlapping levers. An organisation that has built a FinOps practice has built much of the scaffolding a sustainability practice needs, and the two are often best run together. The sections that follow develop the measurement, optimisation, reporting, and procurement layers in turn.

79.3 Measurement

Measurement is the foundation layer, because an impact that is not measured cannot be attributed, optimised, or honestly reported. AI sustainability measurement works at the granularity of the request.

79.3.1 Energy per Request

Energy per request is estimated from the token counts, the model size, and the efficiency of the hardware the request ran on. Some hyperscalers now expose per-call energy estimates directly; where they do not, the energy is calculated from baseline measurements of the hardware's consumption under load. Energy per request is the base measurement from which the carbon and, indirectly, the water figures derive, and getting it reasonably accurate is the precondition for everything above it in the architecture.

79.3.2 Carbon per Request

Carbon per request is energy per request multiplied by the carbon intensity of the grid for the region the request ran in. Because grid intensity varies not only by region but by time of day — as the renewable share of generation rises and falls — the more sophisticated approaches use real-time grid intensity rather than a regional average. Carbon per request is the headline sustainability metric, the one that maps most directly onto the corporate and regulatory commitments an organisation reports against.

79.3.3 Water per Request

Water per request derives from the data centre's water-use effectiveness, which is itself tied to the local climate — a facility in a hot, dry region uses more water for the same cooling than one in a cool, humid one. Water per request is less commonly reported than carbon, but it is an emerging requirement, and a sustainability architecture should be able to produce it. In water-stressed regions it is not a secondary metric but a primary one, sometimes more locally consequential than the carbon figure.

79.4 Optimisation Levers

Once impact is measured and attributed, the optimisation layer reduces it. Four levers recur, and several of them overlap directly with the cost-optimisation levers of the FinOps chapter.

79.4.1 Efficiency

The efficiency lever reduces the compute a workload needs in the first place: smaller models where they are adequate, quantisation to run those models more cheaply, caching to avoid repeated computation, and continuous batching to raise hardware utilisation, as developed in Chapter 47. These are precisely the technical levers of AI FinOps, and they double as sustainability levers because compute not spent is energy not consumed. The efficiency lever is where cost optimisation and sustainability optimisation are the same action.

79.4.2 Location

The location lever exploits the tenfold-plus variation in grid carbon intensity by routing workloads to low-carbon regions where regulation and latency permit. Iceland, Quebec, and the Nordic regions have low-carbon grids; some other regions are markedly higher. Moving a workload to a low-carbon region can cut its carbon footprint dramatically without changing the workload at all — which makes location, for carbon, one of the most powerful levers available, constrained only by data-residency rules and latency requirements.

79.4.3 Scheduling

The scheduling lever exploits the variation of grid carbon intensity over time. Batch workloads, which do not need to run at a particular moment, can be scheduled for the hours when the grid is at its cleanest, and large training runs can be timed to coincide with periods of renewable surplus. Scheduling is the temporal counterpart of the location lever: where location moves the workload to a clean grid in space, scheduling moves it to a clean moment in time, and for flexible workloads both can be applied.

79.4.4 Hardware Lifecycle

The hardware-lifecycle lever balances the energy efficiency of newer hardware against the embodied carbon of manufacturing it. Newer hardware is more energy-efficient per unit of computation, which argues for refreshing; but manufacturing new hardware carries its own embodied carbon, which argues against refreshing too often. The lever is to set the refresh cadence so that the operational energy saved

by newer hardware genuinely outweighs the embodied carbon of producing it, retiring old hardware efficiently rather than either clinging to it or replacing it wastefully.

79.5 Reporting

The reporting layer turns measured and attributed impact into the disclosures that internal stakeholders, regulators, and customers require. It has three audiences.

79.5.1 Internal Reporting

Internal reporting puts sustainability dashboards alongside the cost dashboards, showing per-team and per-product environmental impact. Placing the two side by side is deliberate: it lets the organisation see cost and footprint together, and it makes sustainability a number that teams own and can act on rather than an abstraction owned by no one. Internal reporting is what turns the sustainability architecture from a measurement exercise into a basis for decisions.

79.5.2 External Reporting

External reporting satisfies the disclosure obligations that are increasingly mandatory: the European Union's Corporate Sustainability Reporting Directive, climate-disclosure rules in other jurisdictions, and the direct requests of customers and the contents of published sustainability reports. External reporting is moving from voluntary to required, and an architecture that cannot produce auditable external sustainability disclosures is increasingly out of compliance rather than merely behind on a nicety.

79.5.3 Customer Demand

A specific and growing driver of reporting is customer demand: enterprise customers increasingly require their vendors to provide sustainability data, often as part of their own disclosure obligations. Meeting that demand requires the architecture to produce per-customer attribution — the footprint of the work done specifically for that customer. Customer-driven reporting is, for a vendor, both an obligation and a competitive factor, and it requires the attribution layer to be granular enough to slice by customer.

79.6 Procurement Integration

The procurement layer closes the loop, ensuring that sustainability shapes sourcing decisions rather than only being measured after the fact. Several procurement practices integrate sustainability into the decisions that determine future impact:

- **Vendor sustainability in procurement** makes a vendor's environmental record a criterion in selecting that vendor, so sustainability influences the choice rather than being assessed after it.
- **Hyperscaler carbon disclosures in region selection** factors the published carbon intensity of cloud regions into where workloads are placed, connecting procurement to the location lever.
- **Hardware vendor sustainability programmes assessed** evaluates the environmental programmes of hardware vendors, recognising that the materials footprint of the hardware is part of the total

impact.

- **Renewable power purchase agreements negotiated** secures renewable energy directly where the organisation's scale and the local market allow, addressing the energy footprint at its source.

Procurement integration is what makes sustainability forward-looking: measurement, attribution, and optimisation address the impact of workloads already running, while procurement shapes the impact of the workloads, hardware, and regions the organisation has not yet committed to.

79.7 The African Perspective

Sustainability carries distinctive weight in African deployments, across grid-carbon variation, water stress, the equity dimension of energy access, and the opportunity of the continent's renewable build-out.

79.7.1 Grid Carbon Variation

The carbon intensity of African grids varies enormously, which makes the location lever particularly powerful within the continent. The South African grid is coal-dominated and carbon-intensive; the Moroccan grid, drawing on the Noor solar complex and substantial wind, is markedly lower-carbon; the Kenyan grid, running heavily on geothermal and hydroelectric power, is low-carbon. Regional sourcing decisions within Africa can therefore substantially reduce a workload's footprint, and an architecture aware of this variation can place carbon-flexible workloads on the continent's cleaner grids.

79.7.2 Water Stress

Several African regions are water-stressed, and in those regions the water draw of data-centre cooling is a genuine and locally consequential concern rather than a secondary metric. The architectural response is a preference for air-cooled facilities, which trade some energy efficiency for a much lower water footprint, in regions where water is the binding constraint. In water-stressed African deployments, the water figure may matter more than the carbon figure, and the sustainability architecture must weigh it accordingly.

79.7.3 Energy-Access Equity

There is an equity dimension to AI sustainability in Africa that the Northern framing does not foreground: data centres draw substantial power in regions where many people still lack reliable electricity access at all. A sustainability discussion that considers only carbon misses this — the question is not only how clean the power is but whether the data centre's draw competes with the unmet basic energy needs of the surrounding population. An honest African sustainability architecture includes this equity consideration, not only the carbon accounting.

79.7.4 Renewable Build-Out

The continent's renewable energy build-out is a genuine opportunity for low-carbon AI deployment. Morocco's solar capacity, Kenya's geothermal, and South Africa's growing renewable sector all create the possibility of AI infrastructure sited specifically to draw on clean power. The renewable build-out

means that the location lever, applied within Africa, is not only about avoiding the dirtiest grids but about positively exploiting the cleanest — and an architecture planned around the continent's renewable capacity can make African AI deployment a low-carbon proposition rather than a high-carbon one.

79.8 Recurring Failure Patterns

The failure modes of AI sustainability architecture recur, and each is the absence of a layer or a category this chapter has set out:

- **Sustainability theatre.** The organisation makes public pledges without the measurement to back them, so the commitments are claims rather than accountable facts.
- **Carbon-only focus.** Carbon is measured and water and materials are ignored, so the reported footprint understates the true environmental cost, especially in water-stressed regions.
- **Region opacity.** A workload runs in a high-carbon region without anyone being aware of it, because the measurement does not attribute impact by region.
- **Procurement blindness.** Vendor sustainability is not a criterion in procurement, so sourcing decisions lock in environmental impact that measurement can only observe afterward.
- **Equity neglect.** The impact of energy and water consumption on local communities is not considered, so the sustainability accounting misses the dimension that matters most where resources are scarce.

Each pattern maps to a discipline the chapter has developed — measurement behind the pledges, all four impact categories rather than carbon alone, region-aware attribution, sustainability in procurement, and the equity dimension — and together they show that AI sustainability fails through partial accounting and missing layers rather than through any single dramatic excess.

79.9 Where That Leaves Us

Sustainability architecture for AI is a layered discipline: measurement of the four impact categories — energy, carbon, water, materials — then attribution to the workloads that caused the impact, then optimisation through the efficiency, location, scheduling, and hardware-lifecycle levers, then reporting to internal, regulatory, and customer audiences, and finally procurement integration that shapes future impact. The architecture parallels the FinOps architecture closely, and the two are often best run together, because cost and footprint are measured against the same workloads and reduced by overlapping levers.

The optimisation levers are real: efficiency reduces the compute needed, location exploits the tenfold variation in grid carbon intensity, scheduling exploits its variation over time, and hardware-lifecycle decisions balance operational efficiency against embodied carbon. African deployments add the powerful within-continent grid-carbon variation, water stress that can make the water figure the binding one, the equity dimension of data centres drawing power where access is scarce, and the

genuine opportunity of the continent's renewable build-out. Compute has a footprint — and the architecture of this chapter decides whose footprint it is, and where.

Chapter 80

AI Procurement and Vendor Management

“The architecture decision and the procurement decision are the same decision. Pretending otherwise produces lock-in.”

—a recurring observation in enterprise AI sourcing

Chapter Overview

AI procurement is the strategy and the practice of sourcing the things an AI system is built from: foundation models, inference capacity, specialised tools, vector databases, observability platforms, labelling services. The decisions are not administrative but architectural — the choice of a vendor shapes the interfaces the system is built against, the lock-in it carries, the sovereignty it can claim, the trajectory of its cost, and the security of its supply. And the market the procurement operates in is genuinely volatile: prices drop sharply, capabilities shift between providers, and vendors emerge, merge, and disappear on a timescale of months. That volatility makes AI procurement an ongoing discipline rather than a one-time sourcing event. This chapter develops it: the surfaces an organisation procures across, the architecture that keeps procurement from producing lock-in, the contract terms that respect AI-specific realities, and the multi-vendor strategy that turns a volatile market from a risk into a manageable condition.

Learning Objectives

- Recognise the procurement surfaces and architect a multi-vendor strategy without paying for capability the organisation does not use.
- Negotiate contracts that respect the AI-specific realities of pricing volatility, data rights, and model updates.
- Mitigate vendor lock-in through deliberate interface discipline rather than hoping it does not accumulate.
- Manage vendor sustainability and maintain genuine exit options against a consolidating market.
- Adapt AI procurement to African economic, currency, and regulatory contexts.

80.1 The Procurement Surfaces

AI procurement is not a single purchase but a portfolio of distinct sourcing decisions, and the first discipline is to see the surfaces separately, because each has its own market, its own vendors, and its own cost dynamics.

80.1.1 Foundation Models

The foundation-model surface is the procurement of the core models themselves — from Anthropic, OpenAI, Google, Meta with its open-weight releases, Mistral, Cohere, AI21, and a growing set of regional providers. This is the principal cost line in most AI budgets, and it is also the most volatile, because foundation-model pricing and capability shift faster than any other surface. A procurement strategy that treats the foundation model as a fixed choice rather than a periodically revisited one will overpay and underperform as the market moves beneath it.

80.1.2 Inference Capacity

The inference-capacity surface is the procurement of the compute that actually runs the models, and it has three forms: the hyperscaler platforms such as AWS Bedrock, Azure OpenAI, and Google Cloud Vertex; the specialised inference providers such as Groq, Together, Fireworks, and Replicate; and self-hosting on rented GPU capacity. The three forms trade convenience, cost, and control differently, and an organisation often procures across more than one of them — a hyperscaler for the bulk of traffic, a specialised provider for a latency-critical workload, self-hosting for a sensitive one.

80.1.3 Specialised Models

Beyond the general foundation models, an AI system depends on specialised models — embedding models, speech recognition, document AI, computer vision, time-series forecasting — and these are frequently sourced from different vendors than the foundation model, category by category. The specialised-model surface is easy to underestimate in a procurement plan because no single specialised model dominates a budget the way the foundation model does, but together they are a real and multi-vendor sourcing concern.

80.1.4 Tooling

The tooling surface is the procurement of the supporting infrastructure: vector databases, observability platforms, evaluation frameworks, fine-tuning platforms, and labelling services. These categories have matured enough that credible vendors exist for each, which makes them genuine buy decisions rather than necessary builds. The tooling surface is where the build-versus-buy discipline is most often got wrong, by building a bespoke version of a category that a mature vendor already serves well.

80.1.5 Hardware

The hardware surface is the procurement of the physical accelerators — GPUs purchased outright, leased, or rented through a cloud. It is the surface with the largest capital implications: self-hosting hardware is a substantial capital expenditure with a long commitment, where the other surfaces are largely operating expenses. The hardware procurement decision, developed more fully in the hardware chapter, is therefore as much a financial decision as a technical one, and it interacts with the inference-capacity decision directly.

80.2 The Architecture

Procurement is an architectural concern because the way a system is built determines how freely it can be procured. Figure 80.1 shows the architecture that keeps procurement open: applications speak to an abstraction layer, and the abstraction layer fans out to the vendors, so that the applications never bind directly to any one vendor’s interface.

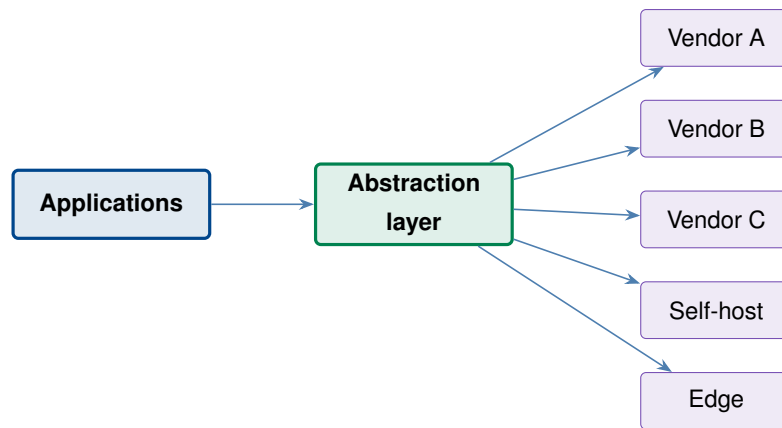


Figure 80.1: Multi-vendor architecture. Applications speak to an abstraction layer — a model gateway, MCP, an internal SDK — and the abstraction layer fans out to the vendors. Adding or replacing a vendor is an abstraction-layer change, not an application change.

The architectural payoff of the abstraction layer is that it turns a vendor change from an application rewrite into a configuration change. An organisation whose applications call a vendor’s API directly has, in effect, written that vendor into every application, and changing vendors means changing all of them; an organisation whose applications call an abstraction layer can add, replace, or drop a vendor by changing only the layer. This is the hexagonal-architecture discipline of this book applied to the procurement question, and it is what makes a multi-vendor strategy practically achievable rather than merely aspirational — the strategy of the later sections depends on this architecture being in place.

80.3 The Lock-In Risk

Lock-in is the central risk that AI procurement must manage, because a locked-in organisation has no leverage when a vendor raises prices, changes terms, or degrades. Understanding where lock-in actually accumulates is the precondition for preventing it.

80.3.1 Where Lock-In Lives

Lock-in is not a single thing but a set of accumulated couplings, and it lives in several specific places:

- **Vendor-specific API shapes** — the differing message formats and tool-call conventions of each vendor, which an application written against one vendor’s shape inherits.
- **Vendor-specific features** — capabilities such as a particular caching mechanism, batching scheme, or structured-output format that exist only at one vendor.

- **Vendor-specific evaluation behaviour** — prompts and evaluation suites tuned to the behaviour of one model, which then must be re-tuned to move.
- **Data movement costs** — the cost and effort of moving vector indices and fine-tuned adapters from one vendor's infrastructure to another's.
- **Operational integration** — the monitoring, billing, and support integrations built around one vendor, which all must be rebuilt to switch.

Lock-in accumulates quietly through these couplings, one convenient vendor-specific choice at a time, until the cost of leaving is prohibitive and no one decided that it should be.

80.3.2 Mitigations

Lock-in is mitigated not by avoiding vendors but by disciplined choices that keep the couplings shallow:

- **Abstraction at the gateway level** keeps the vendor-specific API shapes behind the abstraction layer, so applications never bind to them directly.
- **Prompts versioned and tested across vendors** ensures the prompts are not silently tuned to one model's behaviour and can be confirmed to work on more than one.
- **Vendor-independent evaluation suites** keep the quality gate from being a measure of one vendor, so a candidate replacement can be evaluated fairly.
- **Data in portable formats** keeps the vector indices and adapters in forms that can be moved rather than re-created.
- **Multiple operational integrations maintained** keeps more than one vendor genuinely operable, so a switch is not blocked by missing monitoring or billing.

Each mitigation costs a little — the abstraction layer is work, cross-vendor testing is work — and together they are the price of keeping the option to leave, which is the only thing that gives an organisation leverage in a volatile market.

80.4 Contract Considerations

AI contracts differ from conventional software contracts in specific ways, and a procurement function that signs an AI contract on conventional terms will be caught by the differences. Five considerations matter most.

80.4.1 Pricing

AI pricing takes many forms — per-token, per-call, monthly minimums, volume discounts, capacity commitments — and the contract's pricing structure interacts with the market's volatility in a way that conventional software pricing does not. Because AI prices have a strong recent history of falling, a short-duration contract protects the buyer against being locked into today's prices when cheaper ones arrive, while a long-duration contract protects against a price hike. The pricing decision is therefore a

deliberate bet on the direction of a volatile market, and the contract term should reflect which risk the organisation more wants to hedge.

80.4.2 Service-Level Agreements

AI contracts carry the conventional service-level terms — latency, availability, error rates — but they increasingly need an AI-specific one as well: a commitment that quality is maintained across the vendor's model version updates. A conventional SLA guarantees the service is up; it does not guarantee that an update to the underlying model has not changed its behaviour. Quality-maintenance SLAs are emerging precisely because a vendor model update can degrade an application even while every conventional SLA metric stays green.

80.4.3 Data Rights

The data-rights terms of an AI contract govern questions a conventional software contract rarely raises: whether the vendor may use the customer's data to train its models, whether the customer's prompts and outputs are stored and for how long, and whether data flows across borders in the course of the service. These terms have direct privacy and compliance consequences, and a procurement function that does not read them carefully may discover, after signing, that it has consented to its data training a vendor's model.

80.4.4 Model Update Notification

A specific and easily-overlooked contract term is the right to notice when the vendor changes a model's behaviour. Without it, a vendor can update a model and the customer's applications can break silently — a regression discovered by users rather than announced by the vendor. A model-update-notification clause turns a silent breakage into a managed change the customer can prepare for, and it is one of the most valuable AI-specific terms a contract can carry.

80.4.5 Exit Clauses

The exit terms govern what happens when the relationship ends, and for AI they must cover the return of the customer's data on termination, reasonable notice on price changes so the customer is not surprised, and the right to audit the vendor's security and privacy practices. Exit clauses are what make a vendor relationship genuinely reversible at the contractual level, complementing the architectural reversibility the abstraction layer provides. A contract with no exit clauses is a contract that has quietly become permanent.

80.5 Vendor Sustainability

The AI vendor market has seen rapid consolidation and outright shutdowns, which means a procurement function must assess not only whether a vendor is good but whether it will still exist. Vendor sustainability is evaluated along several dimensions:

- **Financial runway** — how long the vendor can operate on its current funding, because a vendor running out of money is a vendor about to disappear or be acquired.
- **Customer base diversification** — whether the vendor depends on a few large customers, whose

departure would threaten it, or has a broad and stable base.

- **Strategic partnerships** — whether the vendor is anchored by partnerships that make its continued existence more likely.
- **Open-weight alternatives** — whether, if the vendor fails, an open-weight model could replace it, which bounds the damage of a vendor failure.

The last dimension is the most architecturally important: an organisation whose critical workload depends on a vendor with no open-weight fallback has tied that workload's survival to the vendor's, and identifying a fallback before it is needed is the procurement equivalent of the rollback discipline elsewhere in this book.

80.6 Multi-Vendor Strategy

A multi-vendor strategy is how an organisation turns the volatile vendor market from a risk into a managed condition, and it takes one of several shapes depending on what the organisation most needs to protect against.

80.6.1 Primary-Secondary

In the primary-secondary strategy, one vendor handles most workloads and a secondary vendor is kept genuinely operable for fallback and for benchmarking. The secondary is not idle insurance — it is used enough to stay real, to confirm the abstraction layer still works against it, and to provide a live comparison that keeps the primary honest. This is the simplest multi-vendor strategy and a sensible default for an organisation that wants protection without the overhead of running many vendors at parity.

80.6.2 Workload-Specific

In the workload-specific strategy, different vendors are chosen for different workloads according to what each does best — one vendor for reasoning-heavy work, another for cost-sensitive routing, another for embeddings. This strategy optimises each workload to the best-fitting vendor, at the cost of operating more vendor integrations, and it suits an organisation whose workloads are diverse enough that no single vendor is best for all of them.

80.6.3 Geographic

In the geographic strategy, different vendors are used in different regions, for reasons of data sovereignty or capacity availability. A pan-African operation, for instance, may use one vendor where it can host in-region and another where it cannot. The geographic strategy is driven less by capability than by the regulatory and infrastructure realities of the markets served, and it is developed further in the African-perspective section.

80.6.4 Open Plus Closed

In the open-plus-closed strategy, open-weight models are self-hosted for the sensitive workloads where data must not leave the organisation's control, while closed-API models are used for the workloads

that need frontier capability. This strategy uses the two kinds of model for what each is genuinely best at — open weights for sovereignty and control, closed APIs for capability — and it is one of the most common multi-vendor shapes in practice.

80.7 MCP and Interoperability

A standardisation force is at work in the AI procurement landscape, and it is worth a procurement function understanding it, because it directly reduces switching cost. The Model Context Protocol, and similar emerging standards, are standardising the tool-and-data integration layer — the surface across which models reach tools and data — so that an integration built against the standard is not built against any one vendor.

The procurement consequence is direct: vendor-neutral protocols reduce the lock-in that the earlier section located in vendor-specific API shapes and integrations, because an integration written against a standard protocol moves with the organisation rather than with the vendor. A procurement function should therefore actively favour vendors and tools that support these vendor-neutral protocols, treating protocol support as a procurement criterion in its own right. Choosing a standards-supporting vendor is choosing a shallower coupling, and shallower couplings are what keep the multi-vendor strategy affordable.

80.8 The African Perspective

AI procurement carries distinctive weight in African deployments, across currency, sovereignty, local-partner requirements, donor rules, and the prospect of continental cooperation.

80.8.1 Multi-Currency Contracting

AI bills from the major vendors are denominated in US dollars, while an African organisation's revenue is earned in local currency, and the exchange rate between the two moves. This makes currency a live procurement concern: a contract affordable at one exchange rate can become unaffordable at another with no change in usage. Currency hedging, multi-currency clauses in the contract, and attention to payment timing all become part of AI procurement in African contexts, in a way they are not for an organisation whose revenue and costs are in the same currency.

80.8.2 Sovereignty Requirements

Several African jurisdictions require data residency, or specific consent for cross-border data flows, and these requirements bear directly on which vendors an organisation can procure from. A vendor that cannot host within the jurisdiction, or cannot provide the data-flow guarantees the law requires, is not procurable for the workloads that law governs, regardless of its capability. African AI procurement therefore evaluates vendors against sovereignty capability as a gating criterion, not a preference.

80.8.3 Local Partner Requirements

Public-sector procurement in many African countries requires local partners, or carries broad-based economic-empowerment-style requirements that a foreign vendor must satisfy to be eligible. This makes a vendor's local partner ecosystem a real procurement factor: a vendor with established

local partners can be procured for public-sector work that a vendor without them cannot. African AI procurement must assess not only the vendor but the vendor's presence in the local partner landscape.

80.8.4 Donor Procurement Rules

A substantial share of African AI projects are donor-funded, and donor-funded projects must follow the procurement rules of the funder — the World Bank, the African Development Bank, USAID, the European Union, and others. These rules shape the acceptable vendor lists and the permissible contract terms, and they are a constraint the procurement function must work within from the start rather than discover late. For a donor-funded AI project, the donor's procurement rules are as much a part of the procurement architecture as the technical requirements.

80.8.5 Continental Procurement Cooperation

The African Continental Free Trade Area framework and various regional bodies create the potential for continental procurement cooperation — joint sourcing, shared vendor lists, coordinated negotiating leverage across African organisations. This is emerging rather than mature, and a procurement function should treat it as a developing opportunity rather than a present mechanism. But the direction is real, and an organisation aware of it can position itself to benefit as continental procurement cooperation develops.

80.9 Recurring Failure Patterns

The failure modes of AI procurement recur, and each maps to a discipline this chapter has set out:

- **Lock-in by accident.** A single-vendor architecture grew one convenient choice at a time, with no one deciding to depend on that vendor, until leaving is prohibitively expensive.
- **Contract surprise.** The vendor changes pricing or data rights, and the customer, having signed a contract without protective terms, has no leverage to resist.
- **Vendor failure exposure.** A critical workload runs on a vendor that closes or is absorbed, and no open-weight fallback was identified in advance.
- **Data-rights neglect.** The contract permits the vendor to train on customer data, and the customer signed without reading the clause that allowed it.
- **Sovereignty gap.** A vendor turns out to be unable to meet a data-residency requirement, and the gap is discovered late, after the architecture has been built around that vendor.

Each pattern maps to a remedy the chapter has developed — the abstraction layer against accidental lock-in, protective contract terms against surprise, vendor-sustainability assessment and open-weight fallbacks against vendor failure, careful data-rights review, and sovereignty as a gating procurement criterion — and together they show that AI procurement fails through inattention to the architecture and the contract rather than through any single bad vendor.

80.10 Holding the Pieces Together

AI procurement spans five surfaces — foundation models, inference capacity, specialised models, tooling, and hardware — each with its own market and its own dynamics. The architecture that keeps procurement open is the abstraction layer, which turns a vendor change from an application rewrite into a configuration change and so makes a multi-vendor strategy practically achievable.

Lock-in accumulates quietly through vendor-specific API shapes, features, evaluation behaviour, data-movement costs, and operational integrations, and it is mitigated by deliberate interface discipline rather than by hoping it does not happen. AI contracts must respect the AI-specific realities of pricing volatility, model-update notification, data rights, and exit terms, and vendor sustainability must be assessed alongside capability because the vendor market consolidates and shuts down. A multi-vendor strategy — primary-secondary, workload-specific, geographic, or open-plus-closed — turns the volatile market into a managed condition, and vendor-neutral protocols such as MCP reduce the switching cost that strategy depends on. African deployments add multi-currency contracting, sovereignty as a gating criterion, local-partner requirements, donor procurement rules, and emerging continental cooperation. The architecture decision and the procurement decision are the same decision — and the discipline of this chapter is what keeps that single decision from quietly becoming a single-vendor one.

Chapter 81

AI Talent Architecture

“The most expensive line in an AI budget is not GPUs. It is the people who decide what to build with them.”

—a recurring observation in AI organisation design

Chapter Overview

AI talent architecture covers the team structures, roles, capability frameworks, and career paths that an organisation uses to build and operate AI systems. Software 3.0 has not only introduced new roles — the prompt engineer, the evaluation engineer, the AI product manager, the AI trust-and-safety specialist — it has changed the established ones, so that a software engineer is now also a prompt-and-tool engineer and a data engineer is now also a retrieval-and-trace engineer. The organisations that get talent architecture right ship faster, hire more selectively, and retain better; those that get it wrong burn cash on roles they cannot fill and lose the talent they do have to competitors who organised better. This chapter develops the discipline: the role landscape, the team structures that deploy those roles, the capability frameworks that distinguish levels, and the hiring, career, and retention practices that sustain an AI organisation.

Learning Objectives

- Recognise the new AI roles and how they relate to the established roles they extend.
- Architect team structures — hub-and-spoke, embedded, federated — and match each to an organisational stage.
- Build capability frameworks that distinguish levels by meaningful criteria rather than by title.
- Manage hiring, career paths, and the retention of a highly mobile workforce.
- Adapt AI talent architecture to African talent realities of capability, diaspora, and competition.

81.1 The Role Landscape

The first task of AI talent architecture is to understand what roles an AI organisation actually needs, and Software 3.0 has changed the answer in two ways: it has reshaped the established roles, and it has

created genuinely new ones.

81.1.1 Established Roles That Changed

The roles an organisation already has do not stay the same when it adopts AI; they acquire AI responsibilities alongside their existing ones. Software engineers become prompt-and-tool engineers as well, designing the prompts and tool integrations that are now part of the system. Data engineers become embedding, retrieval, and trace-store engineers as well, because the data infrastructure of an AI system includes those new components. Site reliability engineers become model-routing and inference-serving engineers as well, operating the new serving layer. And product managers become evaluation owners and quality-bar-setters as well, because the quality of an AI product is set through evaluation rubrics that product must own. Recognising this reshaping matters, because an organisation that expects its existing engineers to do AI work without acknowledging that their roles have changed will under-resource and under-support them.

81.1.2 Net-New Roles

Alongside the changed roles, Software 3.0 has created roles that did not exist before, and an AI organisation of any maturity needs most of them:

- **Prompt engineer** — designs, tests, and maintains the prompts and prompt-chains that drive the system's behaviour, treating prompts as the versioned artefacts they are.
- **Evaluation engineer** — builds and maintains the evaluation suites and sets the quality bars, owning the gate that decides whether a candidate ships.
- **AI product manager** — a product manager with genuine depth on AI capabilities and limitations, able to set realistic expectations and rubrics.
- **AI trust-and-safety specialist** — owns red-teaming, content policy, the guidelines, and the incident response that defend the system against misuse.
- **ML platform engineer** — owns the inference platform, the training infrastructure, and the observability that the rest of the organisation builds on.
- **AI ethicist or responsible-AI specialist** — operationalises the ethics architecture, turning ethical principles into the controls and reviews the ethics chapter develops.

These roles are not optional luxuries for a large organisation; each addresses a specific function — prompts, evaluation, product judgement, safety, platform, ethics — that an AI system genuinely needs, and an organisation that leaves one unfilled leaves that function to be improvised by someone whose real job is something else.

81.2 Team Structures

Roles must be organised into teams, and there are three common structural patterns for an AI organisation, shown in Figure 81.1. Each fits a different stage of organisational AI maturity, and the wrong structure for the stage is itself a failure mode.

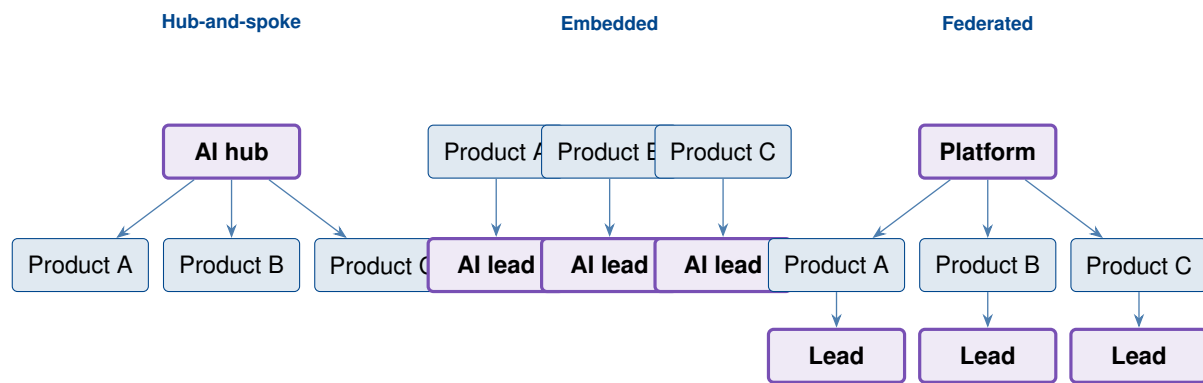


Figure 81.1: Three team-structure patterns. Hub-and-spoke centralises AI capability, embedded distributes it into product teams, and federated combines a central platform with embedded leads. Each fits a different stage of organisational maturity.

81.2.1 Hub-and-Spoke

In the hub-and-spoke structure, a central AI team serves all the product teams, which consult it for the AI work they need. This is the right structure for early AI adoption, because it consolidates scarce AI talent into one team where it can develop depth and share practice, rather than spreading a few specialists thin across many product teams. Its limitations appear as the organisation grows: the central team becomes a bottleneck that every product must queue behind, and the product teams develop less ownership of the AI in their own products. Hub-and-spoke is the structure to start with and to grow out of.

81.2.2 Embedded

In the embedded structure, AI talent lives inside each product team, so that every product owns its AI work directly. This is the right structure for an AI-mature organisation, where there is enough AI talent to staff every team and each product is sophisticated enough to need its own. Its limitations are the mirror of hub-and-spoke's strengths: AI work is duplicated across teams that each solve the same problems separately, and practice becomes uneven, strong in some teams and weak in others, with no central function to level it. Embedded suits maturity but sacrifices consistency.

81.2.3 Federated

In the federated structure, a central platform team provides the shared AI infrastructure and practice, while embedded AI leads sit in each product team and build on it. This combines the strengths of the other two — the consistency and shared infrastructure of hub-and-spoke, the product ownership of embedded — and it is the right structure for a mature organisation operating at scale. Its cost is overhead: the federated structure has the most coordination to manage, because the central platform and the embedded leads must stay aligned. It delivers the most capability and demands the most organisational effort, and it suits the organisations large enough to justify both.

81.3 Capability Frameworks

An AI organisation needs to distinguish levels of capability — the L1 from the L2 from the L3, the staff engineer from the principal — and the distinction must rest on meaningful criteria rather than on tenure or title. A capability framework for AI talent distinguishes levels along several dimensions:

- **Independent execution scope** — whether the person can independently handle a single prompt, a full prompt-chain, or an entire system, which measures the size of the work they can own.
- **Architecture authority** — whether the person follows established patterns, proposes new patterns, or designs the patterns others follow, which measures their influence on how things are built.
- **Cross-functional reach** — whether the person works within engineering alone or partners effectively with product, trust-and-safety, and legal, which measures their ability to operate across the organisation.
- **Mentorship and multiplication** — whether the person makes those around them more effective, which measures their leverage beyond their own output.
- **Strategic influence** — whether the person shapes what the organisation builds, which measures their reach into direction rather than execution.

The framework earns its value only if it is used: it should be published so that everyone knows the criteria, referenced in calibration so that levels are assigned consistently, and used in hiring so that candidates are placed correctly. A capability framework that exists on paper but does not drive calibration and hiring is a framework that does not actually distinguish anything.

81.4 Hiring

Hiring AI talent well means knowing what to look for and what to be wary of, because the field is new enough that a confident-sounding candidate may have little production experience behind the confidence.

81.4.1 Hiring Signals

The signals that genuinely indicate AI capability are signals of production experience and disciplined thinking:

- **Production AI experience** — experience with AI systems that actually ran in production and served real users, not demos and prototypes.
- **Evaluation-driven discipline** — a habit of measuring before claiming, of treating evaluation as the gate rather than an afterthought.
- **Recognition of probabilistic-system trade-offs** — an understanding that an AI system is probabilistic and that its design involves trade-offs a deterministic system does not have.
- **Iterating on real failures** — evidence of having learned from production failures and improved a system in response.

- **Cross-functional collaboration** — a track record of working effectively with product, safety, and legal, because AI work is not engineering alone.

81.4.2 Hiring Anti-Signals

Equally, some signals indicate that a candidate's apparent AI expertise is shallower than it seems:

- **Overconfidence about model capabilities** — a candidate who speaks of models as more reliable and capable than production experience would teach.
- **A toy-project portfolio without production** — impressive demos with no evidence of a system that survived contact with real users.
- **No experience with safety or reliability incidents** — a candidate who has never handled an incident has not yet learned what production AI actually demands.
- **Vendor-specific knowledge without portable principles** — deep familiarity with one vendor's API but no grasp of the underlying principles that would transfer to another.

The anti-signals matter because the field's novelty makes it easy for a candidate to sound expert, and disciplined hiring distinguishes the appearance of AI expertise from the production-tested substance of it.

81.5 Career Paths

AI talent needs visible career paths, because a person who cannot see how they progress is a person who will progress by leaving. An AI organisation should offer three kinds of path.

81.5.1 Specialist Paths

The specialist paths are the deep technical tracks: prompt engineering, evaluation engineering, ML platform, trust-and-safety, and AI research. These paths let a person grow in depth without being forced into management, advancing by becoming more expert and more influential within their specialty. A clear specialist path is what retains the deep technical talent that an AI organisation cannot function without, and an organisation that offers only a management path loses its specialists or pushes them into roles they are not suited to.

81.5.2 Management Paths

The management paths are the leadership tracks: engineering managers, product managers, and programme managers focused on AI. These paths let a person grow by leading teams and programmes rather than by deepening a technical specialty, and they are necessary because an AI organisation needs leaders as well as specialists. The management path should be a genuine alternative to the specialist path, not the only way up, so that becoming a manager is a choice about what a person wants to do rather than the sole route to seniority.

81.5.3 Hybrid Paths

The hybrid paths are for the tech leads who combine a deep technical specialty with cross-functional leadership — people who remain hands-on experts while also leading. These paths recognise that some of the most valuable people in an AI organisation are neither pure specialists nor pure managers. The clearest career frameworks publish all three paths — specialist, management, hybrid — and the criteria for moving among them, so that a person can see not only how to advance but how to change direction.

81.6 Retention

AI talent is highly mobile — the demand exceeds the supply, and a capable person always has options — which makes retention a deliberate discipline rather than an assumption. The factors that retain AI talent are several, and an organisation that neglects them loses people:

- **Competitive compensation** — pay that genuinely matches the market, because AI talent knows its market value and will act on a gap.
- **Meaningful problem ownership** — real problems that the person owns, rather than fragments of work directed entirely by others.
- **Modern tooling** — the tools that let the person work effectively, because working with poor tooling is itself a reason to leave.
- **An engineering culture that respects probabilistic systems** — a culture that understands AI systems are probabilistic and does not punish people for the inherent uncertainty.
- **Visible career progression** — the clear paths of the previous section, so the person can see a future in the organisation.
- **Time for learning** — room to keep up with a fast-moving field, including conference engagement, because a person who cannot keep learning will leave to somewhere they can.

Retention is cheaper than replacement — a departure costs the institutional knowledge the person carried and the cost of hiring and ramping their successor — and an organisation that treats these factors as discretionary is choosing, in effect, to pay the higher cost.

81.7 The African Perspective

AI talent architecture carries distinctive weight in African deployments, across capability building, diaspora engagement, cross-border movement, retention against international competition, and sovereign-capability strategy.

81.7.1 Capability Building

The continental supply of AI talent is growing rapidly but unevenly: South Africa, Egypt, Kenya, Nigeria, and Morocco have the deepest pools, while smaller markets have far less. An organisation operating across the continent therefore cannot assume a uniform talent landscape; it must build

capability where the pools are thin, often through partnerships with institutions and through remote-friendly arrangements that let a smaller market draw on talent it cannot locally supply. Capability building is a deliberate programme in African AI talent architecture, not an assumption that the talent can simply be hired.

81.7.2 Diaspora Engagement

A substantial amount of African AI talent works in Europe and North America, and that diaspora is a real resource for African AI organisations. Engaging it — through remote roles, through return-engagement programmes that bring people back, through advisory relationships that draw on diaspora expertise without requiring relocation — supplements local capacity meaningfully. Diaspora engagement is a distinctive lever of African AI talent architecture, and an organisation that designs for it can reach talent that a purely local hiring strategy would miss.

81.7.3 Cross-Border Movement

AI talent moves across African borders in pursuit of opportunity, and a pan-African operation can capitalise on this — drawing talent from across its markets rather than only from the country a given role sits in. The constraint is that the regulatory environment for cross-border employment varies across African jurisdictions, so an operation that wants to move or hire talent across borders must work within differing rules. Cross-border movement is an opportunity for a pan-African organisation, bounded by the regulatory variation it must navigate.

81.7.4 Retention Against International Employers

African AI talent commands global rates, because the talent is genuinely mobile and international employers compete for it directly. A local employer that wants to retain its AI talent must therefore either match the international compensation or offer a compelling enough mission to outweigh a pay gap — or, ideally, both. In African AI deployments, retention is not a refinement of the talent strategy but a survival concern: an organisation that loses its AI talent to international employers loses the capability to build and operate its systems at all.

81.7.5 Sovereign Capability

National AI strategies across the continent — in Egypt, Kenya, Morocco, Rwanda, South Africa, and others — place explicit emphasis on building sovereign AI capability, the ability to develop and operate AI without depending on foreign expertise. Public-private partnerships, scholarship programmes, and centres of excellence are the instruments of that emphasis. An African AI organisation operates within this national context, and it can both contribute to and benefit from the sovereign-capability programmes its country is running — the talent those programmes produce is the talent the organisation will hire.

81.8 Recurring Failure Patterns

The failure modes of AI talent architecture recur, and each maps to a discipline this chapter has set out:

- **Title inflation.** People are given “AI engineer” titles without the underlying capability the title should denote, so the titles stop meaning anything and calibration breaks down.
- **Structure-task mismatch.** A hub-and-spoke structure is retained past the stage it suited, so the central team is a bottleneck the growing organisation has outgrown.
- **No evaluation engineering.** Evaluation is treated as side work that no one owns, so the quality bars erode because the role responsible for maintaining them was never staffed.
- **Burnout on incidents.** The trust-and-safety function is undermanned for the incident load it actually carries, and the people in it burn out and leave.
- **Brain drain.** Talent leaves for international employers and the institutional knowledge it carried is lost, because retention was not treated as the survival concern it is.

Each pattern maps to a remedy the chapter has developed — capability frameworks that make titles mean something, structures matched to organisational stage, evaluation engineering staffed as a real role, trust-and-safety resourced to its load, and retention treated deliberately — and together they show that AI talent architecture fails through neglect of structure and role rather than through a shortage of willing people.

81.9 The Architectural Commitment

AI talent architecture covers the roles an AI organisation needs — the established roles that Software 3.0 reshaped and the net-new roles it created — the team structures that deploy those roles, the capability frameworks that distinguish levels, and the hiring, career, and retention practices that sustain the organisation. The team structures — hub-and-spoke, embedded, federated — each fit a stage of organisational maturity, and matching the structure to the stage is itself part of the discipline.

Capability frameworks distinguish levels by meaningful criteria — execution scope, architecture authority, cross-functional reach, multiplication, strategic influence — rather than by title, and they earn their value only when they drive calibration and hiring. Hiring distinguishes production-tested substance from the appearance of expertise; career paths offer specialist, management, and hybrid routes so that progression does not require leaving; and retention is a deliberate discipline because AI talent is mobile. African deployments foreground uneven capability building, diaspora engagement as a distinctive lever, cross-border movement within a varying regulatory environment, retention against international employers as a survival concern, and the sovereign-capability strategies that national governments are pursuing. The most expensive line in an AI budget is the people who decide what to build — and talent architecture is how an organisation makes that expensive line a productive one.

Chapter 82

Enterprise AI Economics

“The model is a commodity. The application is not. Strategy is the architecture of where you sit in that gap.”

—a recurring observation in AI strategy

Chapter Overview

Enterprise AI economics is the study of market dynamics, value capture, unit economics, and the strategic choices that determine what AI is worth building, where the margin in an AI system lives, and how an investment in AI becomes a durable advantage rather than a temporary one. The questions are not narrowly technical, and they cannot be answered by an architect who reasons only about systems: where does competitive advantage come from when the foundation models everyone builds on are shared infrastructure, how does an AI feature actually pay for itself, what scales economically and what does not, and when should an organisation build rather than buy. This chapter develops the economic framing that should inform the architecture — because, as the chapter’s epigraph argues, the model is a commodity and the application is not, and strategy is the architecture of where an organisation chooses to sit in the gap between them.

Learning Objectives

- Recognise the structure of the AI value chain and where margin concentrates within it.
- Apply unit-economics models to AI features and reason about whether a feature is durable.
- Distinguish the durable competitive advantages from the temporary ones.
- Reason about cost curves, commoditisation, and the build-versus-buy decision.
- Adapt enterprise AI economics to African market realities of unit economics and distribution.

82.1 The AI Value Chain

Value in AI does not accrue evenly across the stack; it concentrates in specific layers, and an organisation committing capital should understand which layers capture margin before it decides where to invest. Figure 82.1 shows the value chain and where margin sits today.

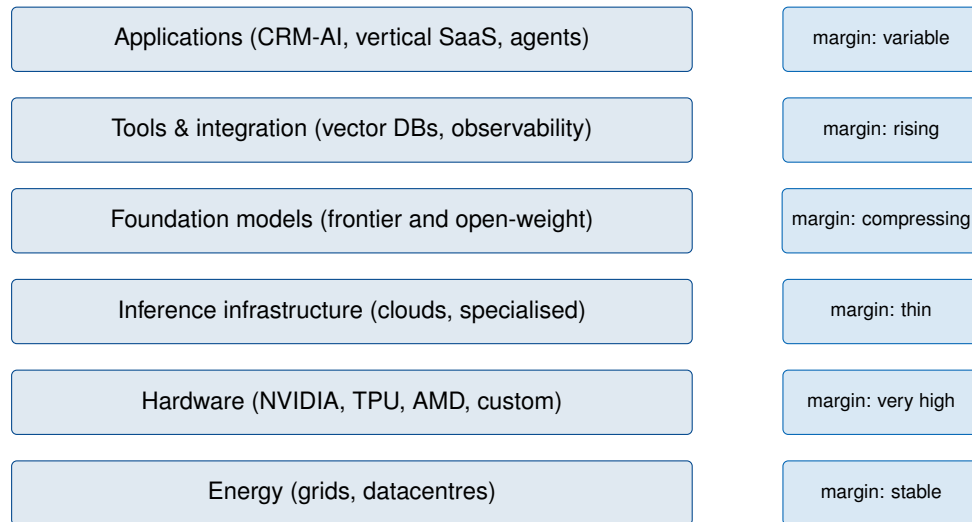


Figure 82.1: The AI value chain. Hardware and energy capture high and stable margins; foundation-model margins are compressing as competition and open-weight alternatives intensify; tools and integration are a rising margin pool; application margins depend critically on whether the application has durable moats.

82.1.1 Where the Money Goes Today

Today the capital flow in AI is strikingly concentrated. NVIDIA captures a disproportionate share of it, because the hardware layer sits at a supply-constrained chokepoint that the whole stack depends on. The foundation-model providers, despite raising very large sums, operate on tight margins once their compute costs are paid, because the compute they buy is itself expensive. And the application providers vary enormously: some have genuinely strong economics, while others lose money on every single call, shipping in the hope that costs will fall. The money, in short, goes today to the hardware layer and to a subset of well-positioned applications, and not reliably to the foundation-model layer in between.

82.1.2 Where Margin May Settle

History suggests the current distribution will not hold. Hardware monopolies tend to erode, and AMD, custom silicon, and new entrants are already pressing on NVIDIA's position. Foundation models tend to commoditise, and the combination of open-weight alternatives and steadily dropping API prices is already compressing foundation-model margins. The tooling layer tends to consolidate and then earn rent, as a few category winners emerge and become infrastructure. And applications keep their margins only where they have structural moats — proprietary data, regulatory positioning, brand — while applications that are thin wrappers around a foundation-model API do not. The strategic reading of the value chain is therefore forward-looking: an organisation should invest where margin is settling, not where it happens to sit today.

82.2 Unit Economics for AI Features

Where the value chain answers where margin lives across the industry, unit economics answers whether a specific AI feature makes money. It is the question an architect's design choices most directly affect, and it has a precise form.

82.2.1 The Equation

For any AI feature, the unit-economics question reduces to a single equation:

$$\text{contribution margin} = \text{revenue per use} - (\text{model cost} + \text{infra cost} + \text{operations cost} + \text{customer success cost})$$

If the contribution margin is negative, then every use of the feature loses money, and scaling the feature scales the loss. This is not a hypothetical concern: many AI products shipped in recent years carry negative contribution margins, on the bet that falling model costs will eventually turn the margin positive. Whether that bet pays off depends on whether costs fall fast enough, and the unit-economics equation is the tool that makes the bet explicit rather than implicit.

82.2.2 The Cost Drivers

The cost side of the equation has four drivers, and a unit-economics model that omits any of them understates the true cost:

- **Model cost** — the tokens, calls, retrieval queries, and judge evaluations, the most visible cost and the one most directly shaped by architecture.
- **Infrastructure cost** — the vector storage, trace storage, queue infrastructure, and observability that the feature depends on around the model itself.
- **Operations cost** — the on-call burden, the incident handling, and the ongoing evaluation maintenance that keeping the feature running demands.
- **Customer success cost** — the setup, training, support, and enablement that the feature requires per customer, a cost that does not appear on any infrastructure invoice.

The customer-success cost is the one most often left out, because it is not a technical cost, and leaving it out makes a feature look more profitable than it is.

82.2.3 The Revenue Side

The revenue side of the equation depends on the pricing model — per-seat, per-call, per-outcome, per-tenant — and the choice of pricing model interacts with the cost profile rather than being independent of it. Per-outcome pricing aligns the vendor's incentive most closely with the customer's, but it transfers the cost risk to the vendor, who is paid only when an outcome is achieved regardless of what it cost to achieve. Per-seat pricing insulates the vendor from usage variability, because the revenue does not move with how much the feature is used. The pricing model is therefore a unit-economics decision as much as a sales decision, and it is developed further later in this chapter.

82.3 Cost Curves

Unit economics is not static, because the cost of AI is not static. The cost curves of AI bend over time, and reasoning about that bending is part of the economic framing.

82.3.1 Foundation-Model Pricing

Frontier-model pricing has dropped substantially — on the capable tiers, roughly a tenfold reduction in price per million tokens over two years — and the trajectory has continued. The consequence for economic reasoning is direct: a cost forecast that assumes today's prices will hold is likely to overstate future cost, and a long-term contract signed at today's prices may lock an organisation into paying more than the market will charge a year later. Foundation-model pricing is a falling curve, and forecasts and contracts must be built with the fall in view.

82.3.2 Capability per Dollar

The other side of the cost curve is that capability is rising even as price falls, so that capability per dollar improves faster than price alone suggests — by fifty to a hundred times on some benchmarks over two years. The strategic implication is that the set of economically viable features expands over time: a feature whose unit economics did not work at last year's capability-per-dollar may work at this year's, and a feature that is unviable today may become viable as the curve continues to bend. An organisation reasoning about AI economics should reason about a moving frontier of viability, not a fixed one.

82.3.3 Wait-and-See Risk

The bending cost curve creates a temptation to wait — to delay building a feature until cheaper models make its economics comfortable — but waiting carries its own cost. A competitor that ships now, while the economics are tight, builds the compounding advantages of the next section, and by the time the cheaper models arrive that competitor has a moat. The strategic question is therefore not simply whether the economics will improve but whether the feature can pay back its build cost and establish a durable advantage before cheaper alternatives let everyone else catch up. Wait-and-see is not free; it is a bet against the value of moving first.

82.4 Durable Advantages

If foundation models are shared infrastructure, then competitive advantage cannot come from the model, and an organisation must understand where durable advantage actually does come from. Five sources are genuine moats, and one category that is often mistaken for a moat is not.

82.4.1 Proprietary Data

Proprietary data is the data an organisation has and no competitor does, and it is the most powerful moat because it compounds. More usage of the product produces more proprietary signal; the signal improves the model or the system; the improvement makes the product better; and the better product drives more usage, producing more signal still. This compounding loop is what makes a proprietary-data advantage durable rather than static — it widens over time rather than eroding.

82.4.2 Distribution

Distribution is the existing relationship with customers, and it is a structural advantage because acquiring customers is expensive and slow. An incumbent that can ship an AI feature to customers it

already has starts from a position a startup must spend heavily and wait to reach. Distribution does not compound the way proprietary data does, but it is a real and durable head start, and it is one of the clearest moats an established organisation already possesses.

82.4.3 Workflow Integration

Workflow integration is the depth to which an AI system is woven into a customer's actual working processes, and it is a moat because depth creates switching cost. A system that a customer merely uses can be swapped for a competitor; a system that is integrated into the customer's workflows becomes infrastructure the customer would have to re-engineer their work to remove. The deeper the integration, the higher the switching cost, and the more durable the advantage.

82.4.4 Regulatory Positioning

In regulated industries, compliance and regulatory approval take time and capital to obtain, and an organisation that has obtained them early holds a moat. A competitor cannot simply match the capability; it must also go through the regulatory process, which can take years. Regulatory positioning is therefore a multi-year lead in regulated markets, and it is a moat that is unusually hard for a faster-moving competitor to erode quickly.

82.4.5 Brand and Trust

For high-stakes decisions, the question of whom the customer trusts can matter more than the question of which system is most capable. Brand and trust are a moat because trust takes years to build and can be lost in minutes — which makes an established, trusted brand both a durable advantage and a fragile one. In high-stakes AI applications, where the customer is putting consequential decisions in the system's hands, the trusted incumbent has an advantage that raw capability does not overcome.

82.4.6 What Is Not a Moat

It is as important to recognise what does not confer durable advantage, because mistaking a non-moat for a moat is a strategic error:

- **Access to a specific foundation model** is not a moat, because everyone has access to the same APIs.
- **A clever prompt** is not a moat, because a prompt can be replicated in hours once its effect is seen.
- **A common architecture** is not a moat, because the patterns in this book are well known and available to every competitor.
- **Being early without compounding advantages** is not a moat, because a head start that does not compound is simply erased as competitors catch up.

The common thread is that a moat must be something a competitor cannot quickly copy, and access, prompts, and known architectures can all be copied — so advantage must rest on the compounding and structural sources, not on these.

82.5 Build versus Buy

Every layer of the value chain poses a build-versus-buy decision, and the right answer differs by layer. Reasoning about it deliberately is part of AI economics.

82.5.1 Buying Foundation Models

For foundation models, the answer is almost always to buy. Training a frontier model costs tens to hundreds of millions of dollars and requires specialised infrastructure and expertise that few organisations have or should acquire. Buying the capability — through APIs, or through self-hosting open-weight models — is the rational default, and an organisation that proposes to train its own frontier model should have an exceptional reason. The foundation model is the clearest buy in the entire stack.

82.5.2 Buying or Building Tooling

For tooling — observability, vector databases, evaluation frameworks — the mature categories now have credible vendors, which makes buying the sensible default. An organisation should build its own tooling only where its requirement is genuinely unique or strategic, not where a category vendor would serve it well. The most common mistake at this layer is build-everything syndrome: building a bespoke version of a category that a mature vendor has already solved better, spending capital and slowing the product to produce something worse than what could have been bought.

82.5.3 Buying or Building Applications

For applications, the decision divides on whether the application is strategic. An organisation should build the applications that carry its proprietary advantage, its deep workflow integration, or its brand — the things that are moats. It should buy the applications that are commodity — general productivity tools, generic copilots — where there is no advantage to be had from building. And for the middle ground, hybrid models that buy a base and customise it can capture some advantage without the full cost of building. The principle is to build the strategic and buy the commodity.

82.5.4 The Self-Host Decision

A specific build-versus-buy question is whether to self-host open-weight models rather than calling an API, and it has a clear set of conditions. Self-hosting is attractive when the volume is high enough to amortise the infrastructure cost, when data sensitivity demands that the data not leave the organisation's control, when sovereignty requires in-jurisdiction hosting, and when the workload is steady rather than bursty so the reserved infrastructure stays utilised. Self-hosting is the wrong choice when the volume is low, when the workload is bursty enough that reserved infrastructure would sit idle, or when the organisation's operational capacity cannot support running the infrastructure. The self-host decision is a unit-economics decision gated by sensitivity and sovereignty.

82.6 Pricing AI Products

How an AI product is priced shapes both its revenue and the risk it carries, and four pricing models recur.

82.6.1 Per-Seat Pricing

Per-seat pricing charges by the number of users, and it is familiar to buyers, insulates the vendor from usage volatility, and aligns with the established pricing of information-worker software. Its revenue does not move with how heavily the product is used, which is a strength when usage is unpredictable and a limitation when usage and value diverge. Per-seat pricing is best where usage is genuinely correlated with the number of seats, so that the seat count is a fair proxy for the value delivered.

82.6.2 Per-Call Pricing

Per-call pricing charges by usage, and it aligns the price with the vendor's cost, is transparent to the buyer, and reduces the commitment the buyer must make. Its risk is churn during low-usage months, when a buyer paying only for what they use may pay little and reconsider the relationship. Per-call pricing suits products where usage is the honest measure of value and the vendor is willing to accept revenue that moves with that usage.

82.6.3 Per-Outcome Pricing

Per-outcome pricing charges for results achieved, and it offers the highest alignment with the customer, who pays only when the product delivers. That alignment comes at a cost: per-outcome pricing is the hardest to operate, because outcomes must be clearly measured and attributed, and it transfers the cost risk to the vendor, who bears the expense of the work whether or not the outcome is reached. Per-outcome pricing suits products whose outcomes are clearly and objectively measurable, where the alignment is worth the operational difficulty.

82.6.4 Hybrid Pricing

Hybrid pricing combines a base subscription with usage tiers, and it is the model most production AI pricing actually converges on. The base subscription provides predictable revenue and insulates the vendor from the churn risk of pure per-call pricing, while the usage tiers capture the value of heavy use that a flat subscription would leave on the table. Most AI products end up pricing this way because the hybrid balances the predictability the vendor wants against the usage-alignment the buyer expects.

82.7 The Commoditisation Path

AI capabilities do not stay exclusive; they commoditise, and they do so along a path predictable enough to plan against:

1. A frontier capability first emerges in a single provider, which holds it alone.
2. Within roughly six to twelve months, competitors match the capability, though typically at a higher cost.
3. Within roughly twelve to eighteen months, competitors match it at parity cost, so the capability is no longer a price advantage.
4. Within roughly eighteen to thirty-six months, open-weight alternatives reach parity, so the capability

is available without any vendor at all.

5. Beyond that, the capability is table stakes — every serious participant has it — and differentiation must move elsewhere.

The strategic implication of this path is decisive: an investment in being first to a capability captures only a finite window of value, because the path will erase the exclusivity within a few years. An investment in the compounding advantages of the moats section — proprietary data, distribution, workflow integration — captures enduring value, because those advantages are not on the commoditisation path. The path is the reason that strategy must be built on moats rather than on capability leads.

82.8 The African Perspective

Enterprise AI economics carries distinctive weight in African deployments, across unit-economics pressure, currency, distribution moats, sovereignty, and talent.

82.8.1 Capability per Dollar as the Dominant Constraint

Average revenue per user is structurally lower in many African markets than in Northern ones, which makes capability per dollar not one consideration among several but the dominant economic constraint. The levers that improve capability per dollar — open-weight models, smaller model tiers, aggressive routing, quantisation — are, in African deployments, economic survival levers rather than optimisations. An African AI product that does not treat capability per dollar as its central economic concern is a product whose unit economics will not close.

82.8.2 Currency Volatility

AI costs are denominated in US dollars while African revenue is earned in local currency, and the exchange rate between them moves. This makes currency a genuine variable in AI economics: a cost forecast must include currency scenarios alongside usage scenarios, hedging the dollar exposure becomes a real consideration, and multi-currency contracting is attractive where a vendor will offer it. An African AI economic model that treats the dollar cost as fixed in local-currency terms is a model that will be wrong whenever the exchange rate moves.

82.8.3 Distribution Moats

African operators with telecommunications, banking, or government distribution hold unusually strong moats, because that distribution reaches customers a foreign entrant cannot easily access. The mobile-money rails in particular are a distinctive distribution channel — a way to reach customers and to settle transactions that has no close equivalent in Northern markets and that a foreign competitor cannot quickly replicate. For an African organisation, an existing distribution position is often the strongest moat it has, and AI strategy should be built to exploit it.

82.8.4 Sovereignty as Strategic Value

In sovereignty-conscious African markets, the ability to demonstrate data residency, local hosting, and regulatory compliance is itself a competitive advantage rather than merely a cost of doing business. A foreign vendor that cannot meet a market's sovereignty requirements concedes that market to a

competitor that can, which means sovereignty capability translates directly into market share. An African AI organisation that can credibly offer sovereign deployment holds a moat against foreign competitors that the foreign competitors cannot easily close.

82.8.5 Talent Economics

African AI talent commands global rates, because the talent is mobile and international employers compete for it, and that fact belongs in the economic model of any African AI product. Retaining the talent to build and operate the product requires either matching the international compensation or offering a compelling enough mission to outweigh a gap, and either way the cost of talent is a real and competitive line in the economics. An African AI economic model that assumes local talent at a local-cost discount is a model that will be undermined by the global market for that talent.

82.9 Common Strategic Mistakes

The strategic mistakes in AI economics recur, and each is a misreading of one of the chapter's central points:

- **Underpricing on the hope that costs will fall.** Costs sometimes do fall, but not always on the schedule that keeps the business solvent, and a negative contribution margin is a present loss whatever the future holds.
- **Thin-wrapper products.** A product that is a thin wrapper around a foundation-model API has no compounding advantage and is competed away as soon as the wrapper's value is visible.
- **Build-everything syndrome.** Capability is built that mature vendors already solve, wasting capital and slowing the product to produce something worse than what could have been bought.
- **Vendor lock-in by accident.** A dependence on one vendor accumulates undecided, and the cost surprise arrives when the vendor raises prices and the organisation has no negotiating leverage.
- **Moat confusion.** A temporary capability lead is mistaken for a durable moat, and the organisation is surprised when the commoditisation path lets competitors catch up.

Each mistake maps to a discipline the chapter has developed — honest unit economics, building on moats rather than wrappers, buying the commodity, the procurement discipline against lock-in, and the commoditisation path that distinguishes a lead from a moat — and together they show that AI strategy fails through economic misreading rather than through technical shortfall.

82.10 Pulling the Threads

Enterprise AI economics is the study of where value is created and captured across the AI stack. Margin concentrates today in hardware and energy, the foundation-model layer's margin is compressing under competition and open-weight alternatives, the tooling layer is a rising margin pool, and application margins depend entirely on whether the application has a durable moat. Unit economics — revenue per use against the four cost drivers — determines whether a specific feature makes money, and the cost curves bend over time, expanding the set of viable features, though not always fast enough to

rescue a feature shipped at a loss.

Durable advantage comes from proprietary data, distribution, workflow integration, regulatory positioning, and brand — the compounding and structural moats — and not from access to a model, a clever prompt, or a known architecture, all of which competitors can copy. The build-versus-buy decision differs by layer: buy the foundation model, buy the mature tooling, build the strategic application. And the commoditisation path means a capability lead is a finite window while a moat is enduring value. African deployments foreground capability per dollar as the dominant constraint, currency volatility as a real economic variable, distribution and sovereignty as unusually strong moats, and talent economics shaped by a global market. The model is a commodity and the application is not — and strategy, as the epigraph says, is the architecture of where an organisation chooses to sit in that gap.

Chapter 83

Enterprise AI Finance

“Auditors do not care about your model. They care about whether you can prove what it did, what it cost, and who approved it.”

—a recurring observation in CFO offices

Chapter Overview

Enterprise AI finance covers the chief-financial-officer-facing mechanics of running an AI portfolio: the budgets, the accounting treatment, the capital-versus-operating-expense decision, the depreciation of AI investments, the financial controls, the management of vendor financial risk, the board reporting, and the financial governance that satisfies auditors and regulators. It is a discipline distinct from two others this book develops. It is not FinOps, the engineering-side cost optimisation of Chapter 61, and it is not AI economics, the strategy of Chapter 82. Finance is what closes the books on AI at quarter-end — the discipline that turns AI spend into accounted-for, controlled, reported, and auditable numbers. This chapter develops it: the layered finance architecture, the budget structures that handle AI’s volatility, the accounting treatment of AI assets, the controls, the vendor financial risk, and the reporting that the board and the regulator require.

Learning Objectives

- Architect budgets, allocations, and chargebacks for AI workloads.
- Choose the accounting treatment for foundation-model investments, fine-tunes, and prompts.
- Apply financial controls and approval workflows that an auditor will accept.
- Manage vendor financial risk, including counterparty health and concentration.
- Adapt enterprise AI finance to African regulatory and currency contexts.

83.1 The Finance Architecture

Enterprise AI finance is best understood as a layered architecture, shown in Figure 83.1, in which each layer is a distinct financial concern with a distinct owner in the finance organisation. The layers run from budgeting at the top, through allocation, accounting treatment, controls, and vendor finance, down to the reporting that serves the board and the regulator.

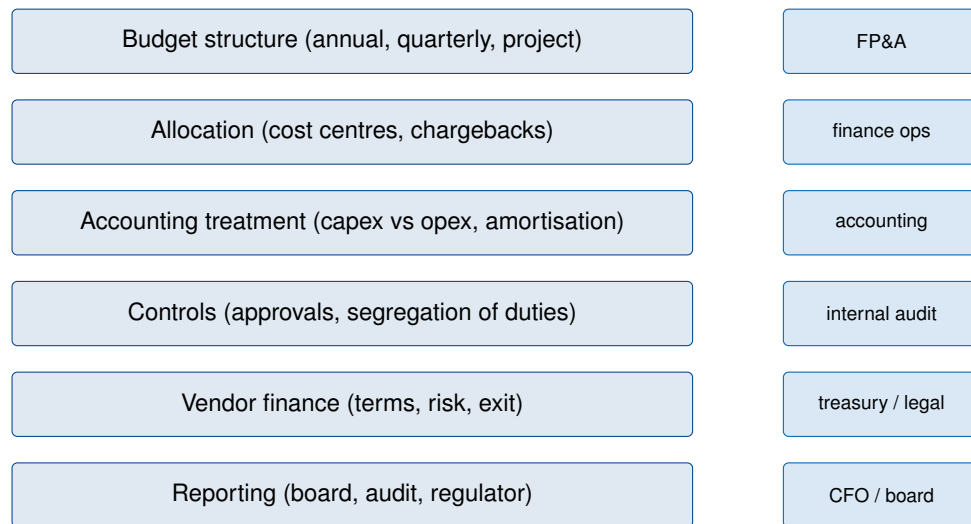


Figure 83.1: Six-layer enterprise AI finance architecture. Each layer has its own owner in the finance organisation. Mature practice runs all six; immature practice often runs only budgeting and reporting, leaving accounting treatment and controls reactive.

The six layers each have an owner: financial planning and analysis owns the budget structure, finance operations owns allocation, accounting owns the treatment, internal audit owns the controls, treasury and legal own vendor finance, and the chief financial officer and the board consume the reporting. The architectural observation that matters is the gap between mature and immature practice. An immature AI finance practice typically runs only the top and bottom layers — it budgets and it reports — while leaving the accounting treatment, the controls, and the vendor finance reactive, handled only when an auditor or an incident forces the issue. A mature practice runs all six layers deliberately, and the rest of this chapter develops the four middle layers, because budgeting and reporting are the parts that organisations rarely neglect.

83.2 Budget Structures

The budget layer answers where AI spend lives and over what period it is planned, and AI's volatility makes both questions less routine than they are for conventional IT spend.

83.2.1 The Allocation Question

The first question is where AI spend should live in the organisation's budget, and there are three patterns. A centralised AI budget gives a central team ownership of all AI spend, with business units consuming against an internal market — the right pattern for early adoption, when AI expertise should be concentrated rather than scattered. A federated budget gives each business unit its own AI budget — the right pattern for mature adoption, when local accountability for AI spend matters more than central control. A hybrid budget combines a central platform budget for the shared services with federated application budgets for the business units, and it is the most common pattern at scale because it puts each kind of spend where its accountability belongs.

83.2.2 Annual versus Quarterly versus Project

The second question is the period over which the budget is planned, and here AI's volatility is decisive. AI costs move unpredictably — vendor prices change, usage scales in ways that are hard to forecast, and a model release can shift the architecture and so the cost. A pure annual budget, fixed twelve months ahead, becomes inaccurate quickly against that volatility. Quarterly re-baseline cycles, which revisit the budget every three months, and project-based budgets with explicit revision triggers both handle the volatility better, because they build the expectation of change into the budgeting cadence rather than being surprised by it.

83.2.3 Reserves and Contingency

Because AI costs are volatile, a mature AI budget holds a reserve — commonly ten to twenty-five per cent — explicitly for capability shifts. The shifts that consume the reserve are predictable in kind even when they are not predictable in timing: a new model class emerges and changes what is worth doing, a major vendor changes its pricing, a regulator demands a new control. Without a reserve, each of these becomes a budget surprise that forces an awkward reallocation; with a reserve, the budget has been built to absorb exactly the volatility that AI reliably produces.

83.3 Capex versus Opex

The accounting-treatment layer turns AI spend into the asset and expense entries that the books require, and the central question is which spend is capitalised and which is expensed.

83.3.1 The Question

AI investments span the whole capital-versus-operating-expense spectrum, and the treatment is not a cosmetic choice: it affects the organisation's tax position, the depreciation schedule, the financial ratios the books present, and the way returns on the investment are measured. A spend treated as a capital asset is depreciated over years and sits on the balance sheet; the same spend treated as an operating expense hits the period's profit-and-loss in full. Getting the treatment right, and consistent, is the accounting layer's core responsibility.

83.3.2 Typical Treatments

The treatment varies by the kind of AI spend, and a workable default mapping looks as follows. GPU purchases are capital expenditure, depreciated over three to five years. Cloud inference and foundation-model API spend are operating expenses, expensed in the period they are incurred. A fine-tuning effort is often capital expenditure if the fine-tune produces an enduring asset, but an operating expense if it is treated as research. Prompt-engineering investment is almost always an operating expense, because a prompt is rarely a capitalisable asset. And internal model-platform development is capital expenditure if it is capitalisable under the applicable accounting standards and an operating expense if it is not. The mapping is a starting point; the accounting team must confirm each treatment against the standards that govern the organisation.

83.3.3 The Depreciation Question

AI assets that are capitalised must be depreciated over a useful life, and the useful life of an AI asset is genuinely shorter than that of conventional IT. A fine-tuned model is the clearest case: if the base model it was built on is updated, the fine-tune may need to be redone, which means the asset is effectively impaired by an event outside the organisation's control. Depreciation periods for AI assets are therefore typically much shorter than for traditional IT — on the order of twelve to thirty-six months rather than five to seven years — and an accounting treatment that depreciates an AI asset on a conventional IT schedule will carry a stale, over-valued asset on the books.

83.3.4 Accounting Standards

The latitude an organisation has in deciding what to capitalise is set by the accounting standards it reports under — US GAAP, IFRS, and their regional variants each give different room for capitalisation. Whatever latitude the standard allows, the accounting team's chosen treatment must be defensible to auditors, applied consistently, and documented. The failure to avoid is inconsistency: if the same kind of investment is treated as an expense in one quarter and an asset in the next, that inconsistency is itself a sign of weak financial governance, and an auditor will read it as one.

83.4 Financial Controls

The controls layer is what ensures AI spend is authorised, segregated, sourced properly, and monitored — the layer an auditor examines most closely.

83.4.1 Spend Approval Workflows

AI spend should pass through an approval workflow tiered to the amount at stake. A workable structure approves spend below a threshold at the team-lead level, mid-range spend at the department-head level with a finance partner, and spend above a higher threshold at the chief-financial-officer or board level; multi-year vendor commitments add legal review and, for the largest, board involvement. The specific thresholds are calibrated to the organisation's scale — what counts as a large commitment differs by organisation — but the tiered structure itself is what matters, because it ensures that the size of the authorisation matches the size of the commitment.

83.4.2 Segregation of Duties

A basic control principle applies to AI spend as to any other: the team that builds an AI product should not also be the team that approves its budget, nor the team that audits its spend. These three roles — building, approving, auditing — must sit with different people, because concentrating them invites exactly the failures the separation prevents. Without segregation of duties, a weak business case can be approved by the people who want to build it, and an overrun can be hidden by the people responsible for it. Segregation is what keeps the approval and the audit honest.

83.4.3 Procurement Integration

Major AI spend should flow through the procurement function, because procurement is what ensures vendor due diligence is done, contract terms are sound, and sourcing is competitive rather than

reflexive. The failure this control prevents is shadow AI — teams buying AI services outside the procurement process, on credit cards or expense accounts, so that the spend is invisible to finance and the contracts carry none of the protections procurement would have negotiated. Procurement integration is the control that keeps AI spend visible and properly contracted.

83.4.4 Cost Anomaly Detection

AI spend should be monitored in real time, with anomaly alerts that fire when spend departs sharply from its expected trajectory. The case for real-time monitoring is concrete: a configuration bug that multiplies a workload's spend tenfold should be caught within hours of the bug shipping, not discovered at quarter-end when the bill arrives. Cost anomaly detection is the control that turns a runaway-cost incident from a quarter-end surprise into a same-day alert, and it complements the FinOps monitoring of Chapter 61 from the finance side.

83.5 Vendor Financial Risk

The vendor-finance layer manages the financial risk that comes from depending on AI vendors, and that risk has several distinct dimensions.

83.5.1 Counterparty Health

The AI vendors an organisation depends on — foundation-model providers, hosting providers, specialised AI vendors — have varying financial health, and a vendor that goes bankrupt or pivots away from its current business leaves its customers stranded. A mature finance practice therefore reviews the financial health of its critical AI vendors periodically, treating counterparty health as a monitored risk rather than an assumption. The review is the financial counterpart of the vendor-sustainability assessment the procurement chapter develops — both ask whether the vendor will still be there.

83.5.2 Concentration Risk

When a large share of AI spend — say eighty per cent — flows to a single vendor, that vendor has substantial leverage over the organisation's pricing, terms, and service levels, because the organisation has no credible alternative. Concentration limits, which cap the share of AI spend any single vendor may hold, are an increasingly common control, and they are a financial discipline as much as an architectural one. The limit is what preserves the organisation's negotiating position, and it connects directly to the multi-vendor strategy the procurement chapter develops.

83.5.3 Currency Exposure

AI vendor contracts are frequently denominated in US dollars, and an organisation whose operations are in another currency carries a currency exposure as a result — the local-currency cost of a dollar contract moves with the exchange rate. The treasury function must either hedge that exposure or consciously accept it, but either way the exposure must be measured and reported rather than left implicit. An unmeasured currency exposure is a risk the organisation is carrying without knowing its size.

83.5.4 Contract Terms

The financial risk a vendor poses is shaped by the contract, and several terms are financial protections worth securing. Price-protection clauses limit how much a vendor can raise prices. Termination clauses allow the organisation to exit without a punitive penalty. Data-return clauses ensure the organisation can migrate its data out. And audit-rights clauses preserve the organisation's ability to verify the vendor's practices. These terms are the contractual side of vendor financial risk management, and the finance and legal functions should treat securing them as part of the discipline.

83.6 Reporting

The reporting layer turns the accounted-for, controlled AI finances into the reports that the operating teams, the board, and the regulators require, and each audience needs something different.

83.6.1 Operational Reporting

Operational reporting serves the teams running AI day to day. A monthly operational report typically shows spend by category, the variance against budget, and the top drivers of that spend, so that the operating teams can see where the money went and why. A quarterly operational report adds the business-unit profitability of AI features and the trends in cost-per-outcome, connecting the spend to the value it produced. Operational reporting is the layer that keeps the people running AI aware of its finances rather than blind to them.

83.6.2 Board Reporting

Board reporting serves the directors, and it is pitched at the level of strategy and risk rather than operational detail. A board-level AI finance report typically covers the total AI investment and its trajectory, the returns by product line, the highlights of the risk register, the vendor concentration and currency exposure, and the major decisions that require the board's input. Board reporting is what gives the directors the financial picture they need to govern the organisation's AI investment, and it should be concise enough to be governable.

83.6.3 Audit and Regulator Reporting

Audit and regulator reporting serves the auditors and the regulators, and it is, as the chapter's epigraph insists, about proof. It must provide evidence that the controls were effective, a reconciliation of AI spend to the general ledger, evidence of vendor management, and whatever specific filings the organisation's sector regulators require — in financial services, healthcare, or the public sector. This is the reporting that makes the whole finance architecture accountable to outside parties, and it is only as good as the controls and the records the earlier layers produced.

83.7 Returns on AI Investment

A finance function must eventually answer whether the AI investment paid off, and measuring the return on AI is genuinely hard.

83.7.1 The Measurement Problem

AI productivity gains are notoriously difficult to measure, because they divide into the obvious and the invisible. Some gains are obvious and directly countable — calls deflected from human agents, processing tasks automated. Others are invisible to a naive accounting — decisions made faster, errors avoided, work that did not have to be redone. The discipline that addresses the measurement problem is to establish the measurement methodology before the deployment, not after: an organisation that decides how it will measure a feature's return before shipping it can actually measure that return, while one that ships first and asks later usually finds the return unmeasurable.

83.7.2 Common Metrics

The metrics that genuinely capture AI returns are outcome metrics — measures of what the AI actually achieved for the business. Useful examples include the cost per resolved customer interaction, the time-to-decision on processed applications, quality-adjusted throughput, shifts in customer satisfaction or Net Promoter Score, and revenue per employee. What these metrics share is that each connects the AI investment to a business outcome, so that the return is measured in terms the business cares about rather than in terms of AI activity.

83.7.3 What Not to Measure

Equally important is to recognise the metrics that produce false comfort — activity metrics that look like progress but do not connect to any outcome. The number of prompts written, the hours of training delivered, and an adoption percentage with no link to outcomes are all metrics of this kind: they go up as the AI programme is busy, but they say nothing about whether the programme is producing value. A finance function that reports activity metrics as if they were returns is reporting comfort rather than performance, and the discipline is to insist that a return metric link to an outcome.

83.8 The African Perspective

Enterprise AI finance carries distinctive weight in African deployments, across currency discipline, multi-currency reporting, donor finance, tax variation, and foreign-exchange constraints.

83.8.1 Currency Discipline

AI bills from the major vendors are denominated in US dollars, while an African organisation's revenue is earned in local currency — the rand, the naira, the shilling, the Egyptian pound, the dirham, the cedi. The exchange rates between those currencies and the dollar move, and a treasury function in an African AI deployment must therefore actively manage the resulting exposure rather than ignore it. Budget forecasts in these deployments should include currency scenarios as a standard element, because a forecast that assumes a fixed exchange rate is a forecast that will be wrong whenever the rate moves.

83.8.2 Multi-Currency Reporting

A pan-African operation earns and spends in several currencies at once, and its finance reporting must reflect that. The reporting must handle consolidation across currencies, the foreign-exchange

translation that consolidation requires, and per-market visibility so that the finances of each country are legible individually and not only in the consolidated total. Multi-currency reporting is a standing requirement of African AI finance for any operation that crosses borders, and the reporting layer of the finance architecture must be built for it.

83.8.3 Donor and Development Finance

A substantial share of African AI deployments are funded by donors or development-finance institutions — USAID, the European Union, the African Development Bank, the World Bank, the Gates Foundation. These funders impose financial-reporting requirements that differ from commercial reporting, in their formats, their categories, and their accountability expectations. An African AI finance architecture frequently has to support both the commercial reporting and the donor reporting, and designing the finance function to produce both from the same underlying records is what keeps donor compliance from becoming a separate, parallel accounting burden.

83.8.4 Tax Treatment Variation

The tax treatment of AI investment varies across African jurisdictions — research-and-development credits, capital allowances, and withholding taxes on cross-border services all differ from country to country. A pan-African operation must therefore work with tax advisors in each market rather than assuming a uniform treatment, and the differing treatments must be captured in the budget models so that the after-tax cost of AI is modelled correctly per market. Tax variation is a real complication of African AI finance, and an operation that ignores it will misjudge its true costs.

83.8.5 Foreign-Exchange Constraints

In some African markets, access to foreign exchange is constrained, and settling a US-dollar bill with a foreign vendor may require central-bank approval or other procedural workarounds. The finance function must plan for this constraint, because it is not a hypothetical: a vendor invoice that cannot be paid for weeks because the foreign exchange to pay it is not available is a real operational event. The discipline is for finance to anticipate the foreign-exchange constraint and for engineering not to be surprised when a vendor relationship is strained by a payment that the currency-access rules have delayed.

83.9 Common Failure Patterns

The failure modes of enterprise AI finance recur, and each maps to a layer or a discipline this chapter has set out:

- **Budget surprise.** The quarter-end AI bill arrives at a multiple of the forecast, because no real-time monitoring was in place to catch the overrun while it was still small.
- **Capex/opex mistreatment.** The accounting treatment is inconsistent across quarters, with the same investment flipping between expense and asset, and the inconsistency produces audit findings.
- **Vendor concentration without awareness.** Eighty per cent of AI spend sits with one vendor, the vendor raises prices, and the organisation has no leverage because no concentration limit was

enforced.

- **Shadow AI.** Teams buy AI services on credit cards outside procurement, so the spend is invisible to finance and the contracts carry no protections.
- **Returns unmeasured.** Investments are approved with no measurement plan, so their return is impossible to assess afterward and the organisation cannot tell what its AI spend bought.
- **Currency surprise.** The local-currency cost of dollar-denominated AI spikes when the dollar strengthens, and a budget that assumed a fixed rate blows up.

Each pattern maps to a remedy the chapter has developed — cost anomaly detection, consistent and standards-defensible accounting treatment, concentration limits, procurement integration, a measurement methodology set before deployment, and currency discipline — and together they show that AI finance fails through skipped layers rather than through any inherent unaccountability of AI spend.

83.10 Closing Notes

Enterprise AI finance is the six-layer discipline that closes the books on AI: budget structure, allocation, accounting treatment, controls, vendor finance, and reporting. It is distinct from FinOps, which is engineering-side optimisation, and from AI economics, which is strategy — finance is the function that turns AI spend into accounted-for, controlled, and auditable numbers.

The budget structures must handle AI's volatility through quarterly re-baselining and contingency reserves; the accounting treatment must capitalise and depreciate AI assets on the shorter useful lives those assets genuinely have; the controls — tiered approvals, segregation of duties, procurement integration, cost anomaly detection — must be the controls an auditor will accept; the vendor-finance layer must manage counterparty health, concentration, and currency exposure; and the reporting must serve the operating teams, the board, and the regulators each in the form they need. African deployments add active currency discipline, multi-currency reporting, donor-finance requirements, tax-treatment variation, and foreign-exchange constraints. Auditors do not care about the model — they care whether the organisation can prove what it did, what it cost, and who approved it, and the architecture of this chapter is what produces that proof.

Advanced Architectural Patterns: Regional Perspective — Africa

“A multi-agent system that does not know whether the mobile-money rail is up is not advanced; it is naive.”

—a recurring observation in African AI engineering

Chapter Overview

The advanced architectural patterns developed earlier in this book — multi-agent systems, tool-use orchestration, human-in-the-loop workflows, evaluation infrastructure — were largely shaped in Northern markets, and deploying them in African contexts requires genuine adaptation rather than a simple transplant. Connectivity that varies rather than holding steady, mobile-money integrations that are central rather than peripheral, human-in-the-loop queues that must span many languages, evaluation datasets that must cover African languages and contexts, and the operational realities of African enterprise environments all shape both which patterns are selected and how they are implemented. This chapter develops the African context for the advanced patterns, with focus on the continent’s anchor markets, and it argues a double point throughout: the adaptation is real work, and the patterns that are forced to work under African conditions tend to be more robust everywhere as a result.

Learning Objectives

- Recognise the distinguishing realities for advanced patterns in African contexts — connectivity variability, mobile-money integration, multilingual workflows, African-language evaluation, and the operational environment.
- Adapt multi-agent patterns for African operational realities.
- Integrate mobile-money and other African platform tools into tool catalogues.
- Build human-in-the-loop workflows that span multiple languages and borders.
- Develop African-language and African-context evaluation infrastructure.

84.1 The African Context for Advanced Patterns

The advanced patterns of this book were developed against a set of implicit assumptions — reliable connectivity, a Northern set of platform integrations, a single working language, Northern evaluation data — and African deployment breaks several of those assumptions. Recognising which assumptions break, and what the breakage demands, is the starting point for adapting the patterns honestly.

84.1.1 The Adaptation Requirements

Six requirements distinguish advanced-pattern deployment in African contexts from the Northern default. Connectivity is variable rather than constant, so agents and tools must accommodate intermittent links rather than assume a stable network. Mobile money is central to the continent's commerce, so tool catalogues must treat mobile-money integrations as first-class rather than as an afterthought. The linguistic reality is multilingual, so human-in-the-loop workflows must support several languages rather than one. Evaluation must cover African languages and African contexts, which Northern datasets and benchmarks do not. African unit economics are tighter, so cost and latency sensitivity constrains frontier-model usage more sharply than in Northern markets. And the regulatory environment is diverse across the continent's jurisdictions, so pattern implementations must vary by jurisdiction rather than assume one regime. Each of these is a requirement the rest of the chapter develops.

84.1.2 The Opportunity

The adaptation is not only a cost; it is an opportunity, and the opportunity is worth stating because it reframes the work. African contexts force the patterns to be more resilient than their Northern equivalents, because a pattern that must survive variable connectivity is a pattern that has been engineered for partial availability in a way a Northern pattern often has not. Mobile-money tool integration enables use cases that are simply not available in Northern markets, where no equivalent rail exists. Multilingual evaluation infrastructure, built because African deployment demands it, produces more robust models that perform better across languages everywhere. And the cost-and-latency engineering that African unit economics force produces efficient architectures whose efficiency is valuable in any market. The adaptation, done well, makes the patterns better, not merely different.

84.2 Multi-Agent Patterns Adapted to African Contexts

Multi-agent architectures, developed earlier in this book, require specific adaptation for African deployment, and the adaptation touches orchestration, language, borders, and cost.

84.2.1 Connectivity-Aware Orchestration

A multi-agent system in an African context cannot assume that every agent is always reachable, so its orchestration must be connectivity-aware. This means agent-to-agent communication that tolerates network variability rather than failing on the first dropped connection, caching and store-and-forward patterns that let an agent's message wait for connectivity rather than being lost, and a deliberate placement of agents — edge-resident specialist agents that can work without a cloud link, and cloud-resident generalist agents that the edge agents reach when they can. The whole system must degrade gracefully when components go offline, continuing to do useful work with the agents that remain

reachable. Designing a multi-agent system for partial availability is substantial additional work, and in African contexts it is not optional.

84.2.2 Multilingual Multi-Agent Workflows

A multi-agent system serving African users frequently serves them in several languages, and the architecture can use the agent abstraction to handle that directly. Agents can specialise in different languages, each fluent in the language it handles; translation and mediation agents can bridge between agents working in different languages; and the system must preserve cultural context across language boundaries rather than losing it in translation, and detect and handle the code-switching that real multilingual speakers produce. Designing a multilingual multi-agent workflow means designing deliberately for the language coverage the deployment actually needs — English, French, Arabic, Swahili, Hausa, Yoruba, and others depending on the markets served.

84.2.3 Cross-Border Multi-Agent Systems

A pan-African operation runs multi-agent systems that span borders, and the crossing of a border raises issues a single-country system does not face. Data-residency considerations affect where agents may be placed, because an agent processing a country's data may need to sit within that country. Currency-conversion mediation is needed between markets that use different currencies. And the regulatory variation across jurisdictions means the system must handle a different regime in each country it operates in. A cross-border multi-agent system therefore carries substantial design complexity beyond a single-country deployment, and that complexity is a direct consequence of the continent's multi-jurisdiction structure.

84.2.4 Cost-Aware Agent Orchestration

African unit economics are tighter than their Northern equivalents, and a multi-agent system must orchestrate itself with cost in view. This means routing decisions made on the basis of per-query cost, a preference for smaller models wherever a smaller model is adequate to the task, the use of locally hosted models for the high-volume agent tasks where the volume makes hosting economic, and the reservation of frontier-model calls for the genuinely complex tasks that need them. Cost-aware orchestration is the multi-agent application of the cost discipline that runs throughout this book, and in African deployments it is a survival concern rather than a refinement.

84.3 Tool-Use Patterns for African Contexts

The tool catalogue of an agent system, developed in the tool-use chapters, requires African-context expansion, because the tools an African system needs are not all the tools a Northern catalogue contains.

84.3.1 Mobile-Money Tool Integrations

Mobile money is the central financial rail across much of the continent, and an African agent system's tool catalogue must treat the mobile-money platforms — M-Pesa, MTN MoMo, Airtel Money, Orange Money, Vodafone Cash — as first-class tools. The tools an agent needs span balance checking, transfers, payments, and status checking, and they must handle the cross-network and cross-border

integrations that real mobile-money use requires, along with the reconciliation tooling that financial integration demands. Building these integrations is substantial design effort, precisely because the mobile-money landscape is multi-provider and each provider has its own interface, but the effort is unavoidable: a tool catalogue without mobile money is, for most African use cases, a tool catalogue missing its most important tool.

84.3.2 Identity-System Tool Integrations

Identity verification is a frequent need in African agent systems, and the identity landscape is heterogeneous — bank verification numbers, Nigeria’s national identification number, Kenya’s identity efforts, and the national identity schemes of many other countries. The tool catalogue must therefore treat identity verification as a primary tool, and it must handle the tier-upgrade tools that move a customer between know-your-customer levels and the biometric verification tools that some workflows require. Designing for identity heterogeneity — a different identity system in each market — is substantial work, and it is the tool-layer counterpart of the multi-identity integration the agent-identity chapter develops.

84.3.3 Government-Service Tool Integrations

African agent systems frequently need to reach government services, and the tool catalogue must integrate with them: the tax portals of the various revenue authorities, the customs and trade-facilitation systems, the business-registration services, and the public-service portals. The defining challenge of these integrations is the inconsistent maturity of the underlying APIs — some government services expose clean, modern interfaces while others expose little or none — so designing government-service tools means designing for a wide range of API maturity, including the cases where the integration must work around the absence of a proper API.

84.3.4 Telecoms and Connectivity Tools

Because connectivity is variable, an African agent system’s tool catalogue must include the tools that let it reach users over low-bandwidth and feature-phone channels. SMS gateways enable low-bandwidth communication; USSD interfaces reach the feature phones that remain widespread; voice IVR integrations reach users by phone call; and carrier-specific APIs handle the integrations each carrier requires. The design discipline here is graceful channel fallback — a system that prefers a rich channel when connectivity allows it but falls back cleanly to SMS or USSD when it does not — so that the agent system can reach a user on whatever channel that user’s connectivity actually supports.

84.4 Human-in-the-Loop Workflows for African Contexts

The human-in-the-loop workflows of the earlier chapters require adaptation for African deployment, across language, borders, connectivity, and the reviewer ecosystem.

84.4.1 Multilingual Reviewer Queues

A human-in-the-loop workflow serving African users must route review work to reviewers who can actually read the language of the item under review, which means the reviewer pool must support

multiple languages and the routing must send each item to a reviewer with the appropriate language skill. Quality monitoring must then run across all the languages the pool covers, so that quality does not silently degrade in a language the monitoring overlooked. Designing a multilingual reviewer queue means designing for the variability of the language mix — the proportion of items in each language shifts, and the queue must handle that shift without leaving some languages under-served.

84.4.2 Cross-Border Reviewer Networks

A pan-African operation can draw its reviewers from across its markets rather than from a single country, and a cross-border reviewer network has real advantages: it allows time-zone arbitrage, with reviewers in different time zones covering more of the clock, and it widens the pool from which language and domain skills can be drawn. The network must preserve cultural context, so that a reviewer from one market judging an item from another applies the right cultural standard, and it must route work thoughtfully across the borders. Designing a cross-border reviewer network is substantial work, but it turns the continent's geographic spread from a complication into a coverage advantage.

84.4.3 Connectivity-Aware Human-in-the-Loop

Reviewers in African contexts work under variable connectivity, and the human-in-the-loop tooling must accommodate that. The reviewer tools must function under variable connectivity rather than assuming a stable link, they must work on the devices reviewers actually have — mobile phones as well as laptops — and the workflows must be offline-capable where possible, letting a reviewer complete work during a connectivity gap and synchronise when the link returns. Designing for asynchronous review, in which work does not depend on the reviewer being continuously online, is the discipline that makes a human-in-the-loop workflow usable for reviewers whose connectivity is not guaranteed.

84.4.4 Compensation and Quality Patterns

A sustainable reviewer ecosystem in an African context requires deliberate attention to compensation and quality. Compensation structures must be appropriate to the local markets the reviewers work in, and paying reviewers through mobile money is often the natural mechanism, reaching reviewers who may not have conventional bank accounts. Quality monitoring must run across the distributed reviewer pool, and career-development pathways give reviewers a reason to stay and improve. Designing for a sustainable reviewer ecosystem — one that reviewers are fairly paid in, monitored fairly within, and can grow in — is what keeps a human-in-the-loop workflow staffed and capable over time rather than churning through reviewers.

84.5 Evaluation Infrastructure for African Contexts

Evaluation infrastructure, treated as a core discipline elsewhere in this book, requires substantial African-context investment, because the datasets and benchmarks that evaluation depends on do not adequately cover African languages and contexts.

84.5.1 African-Language Evaluation Datasets

Evaluating a system that serves African users requires evaluation datasets in the languages those users speak — Swahili, Hausa, Yoruba, Igbo, Amharic, Zulu, Xhosa, Afrikaans, Arabic, French, and others. The quality and coverage gaps in these datasets are significant: for many African languages, the evaluation data simply does not exist at the quality and scale that Northern languages enjoy. Substantial work is underway to close the gap — the Masakhane initiative and others have built genuine African-language resources — but the curation effort remains substantial, and an organisation deploying in African languages must expect to invest in the evaluation datasets rather than assume they are available.

84.5.2 African-Context Benchmarks

Beyond language, evaluation requires benchmarks that test the contexts an African system actually operates in. Cultural-context benchmarks test whether the system handles African cultural settings correctly; local-knowledge benchmarks test African factual knowledge; code-switching benchmarks test the mixed-language input real users produce; multi-currency arithmetic benchmarks test reasoning across the continent’s currencies; and mobile-money flow benchmarks test the financial flows that African systems centrally depend on. Building these benchmarks is substantial design effort, and it is necessary because a system that scores well on Northern benchmarks may still fail on exactly the African-context tasks its users will give it.

84.5.3 Fairness Evaluation for African Contexts

Fairness evaluation, developed in the ethics chapter, takes on a specific shape in African contexts. It must include bias evaluation across language varieties, because a system may serve one language variety well and another poorly; cultural-bias evaluation, because a Northern-trained model may carry cultural assumptions that misfit African settings; equity-of-access evaluation, because the system’s benefits should not be confined to one segment of users; and disparate-impact analysis across the relevant groups. The methodological work here is substantial, because fairness evaluation calibrated for Northern populations does not transfer directly, and an African deployment must build the fairness evaluation its own population requires.

84.5.4 Operational Evaluation Under African Conditions

Finally, evaluation must test the system under the operational conditions it will actually meet. This means connectivity-variability evaluation, testing how the system behaves when the network is intermittent; cost-and-latency evaluation under African unit economics, confirming the system performs within the cost and latency the deployment can afford; and tool-availability evaluation, testing how the system behaves when a tool — a mobile-money rail, a government API — is unavailable. Substantial chaos-engineering work, deliberately injecting the failures African conditions produce, is what gives an organisation confidence that the system will hold up under the conditions of its actual deployment rather than only under laboratory conditions.

84.6 The Architectural Patterns

The adaptations of the previous sections converge into a small set of architectural patterns that recur across well-built African AI systems.

84.6.1 Resilient-by-Design Architecture

The resilient-by-design pattern builds the system from components that are designed for partial availability from the start, rather than components that assume full availability and fail when it is absent. Caching and offline patterns are standard throughout such a system, and graceful degradation is built into every component, so that the loss of any one part leaves the rest still doing useful work. The pattern carries substantial design overhead — designing for partial availability is more work than assuming full availability — but it buys operational robustness, and it is the African-context pattern that, once adopted, makes a system more robust in any market.

84.6.2 Mobile-Money-Native Architecture

The mobile-money-native pattern treats mobile money as the primary financial rail of the system rather than as one payment option among several. Such an architecture integrates multiple mobile-money providers as first-class, handles the cross-network and cross-border flows that real mobile-money use requires, and carries the substantial integration complexity that the multi-provider mobile-money landscape entails. The pattern is what enables the financial use cases that are central to African commerce, and it has no real Northern equivalent because no Northern rail occupies the position mobile money holds in Africa.

84.6.3 Multilingual-Native Architecture

The multilingual-native pattern treats multiple languages as first-class throughout the system rather than treating one language as primary and the others as translations. Cross-language workflows run throughout such a system, cultural context is preserved across language boundaries, and the user experience and the operations are both designed for the multilingual reality from the start. The pattern carries substantial design effort in both the user experience and the operations, and it is what makes a system genuinely usable across the linguistic diversity of the African markets it serves.

84.6.4 Cost-Engineered Architecture

The cost-engineered pattern treats per-query economics as a tight constraint that shapes the architecture rather than as a number to be checked afterward. Routing is optimised for cost, smaller and locally hosted models are preferred wherever they are adequate, and substantial cost-monitoring infrastructure tracks the economics continuously. The pattern is the architectural expression of the survival-level cost discipline that African unit economics demand, and a system built this way is efficient enough to be viable under the cost ceilings African deployment imposes.

84.7 Recurring Failure Patterns

The failure modes of advanced patterns in African contexts recur, and each is the failure to make one of the adaptations this chapter has set out:

- **Connectivity assumptions in multi-agent systems.** The system assumes constant connectivity,

and it fails under the variable connectivity that African conditions actually present.

- **Mobile-money omission from tool catalogues.** The tool catalogue lacks mobile-money integration, and the system cannot perform the financial flows that are central to its African use cases.
- **Northern-only human-in-the-loop pools.** The reviewer pool supports only a Northern language set, and the multilingual review the deployment needs cannot be staffed.
- **Northern-only evaluation datasets.** Evaluation runs against Northern datasets only, so the system's performance on African languages and contexts is never actually measured.
- **Cost-naïve frontier-model usage.** Frontier models are used without cost discipline, and the unit economics that African deployment requires never close.

Each pattern maps to an adaptation the chapter has developed — connectivity-aware orchestration, mobile-money tool integration, multilingual reviewer queues, African-context evaluation infrastructure, and cost-engineered architecture — and together they show that advanced patterns fail in African contexts not because the patterns are wrong but because they were transplanted without the adaptation African conditions require.

84.8 Regional Case Study

A composite case makes the adaptations concrete: a pan-African fintech running a substantial agent-based customer-service operation across eight markets. The composite is drawn from the recurring shape of such deployments rather than from any single organisation, and it shows the adaptations of this chapter working together.

84.8.1 Approach

The fintech's approach combined the chapter's adaptations into one system. It built a multi-agent architecture with country-specific specialist agents, so that each market's particularities were handled by an agent built for that market. It integrated mobile-money tools across the providers its eight markets used. It ran a multilingual human-in-the-loop operation with cross-border reviewer pools, drawing reviewers from across its markets. It invested in African-language evaluation infrastructure so that its quality was actually measured in the languages it served. It engineered its routing for cost, keeping per-query economics within what its markets could sustain. And it built connectivity-aware orchestration throughout, so the system held up under the variable connectivity its users experienced. No single adaptation was novel; the approach was to apply all of them together.

84.8.2 Outcomes

The outcomes of the approach show what the adaptations buy. The agent system's deflection rates — the share of customer interactions resolved without a human — were competitive with Northern equivalents, showing that an African-adapted system need not sacrifice performance. The mobile-money integration enabled customer-service use cases that a system without it could not have offered. Cross-border consistency was achieved, so that a customer experienced the same quality of service across the eight markets. The cost per query stayed at sustainable levels, keeping the operation viable

under African unit economics. And the system proved robust under the variable conditions of its actual deployment. The case study's lesson is the chapter's thesis in concrete form: the adaptations are real work, and a system that does the work performs.

84.9 The Pattern, Restated

The advanced architectural patterns of this book require genuine adaptation for African contexts, and the adaptation is specific: connectivity-aware orchestration in place of an assumption of constant connectivity, mobile-money tool integration as a first-class concern, multilingual human-in-the-loop workflows in place of single-language ones, African-language and African-context evaluation infrastructure in place of Northern-only datasets, and cost-engineered architecture in place of cost-naïve frontier-model usage.

The four recurring architectural patterns — resilient-by-design, mobile-money-native, multilingual-native, and cost-engineered — are the forms these adaptations converge into, and the case study shows them working together in a real deployment shape. The discipline this chapter argues for is to treat African contexts on their own terms: to recognise both the additional design effort the adaptation genuinely requires and the compensating truth that patterns forced to work under African conditions — variable connectivity, tight economics, many languages — prove more robust everywhere. A multi-agent system that does not know whether the mobile-money rail is up is not advanced but naïve, and the adaptation of this chapter is what turns an advanced pattern into one that is genuinely advanced in the context where it must run.

Part V

AI Engineering Disciplines

“Nothing is lost, nothing is created, everything is transformed.”

—Antoine Lavoisier, *Traité Élémentaire de Chimie*, 1789

The patterns are settled. The question now is whether they can be operated honestly. Lavoisier did for chemistry what Linnaeus had done for biology: he gave the field a discipline. The naming of the elements, the careful weighing on a calibrated balance, the conservation of mass — none of these were Lavoisier’s discoveries alone, but he was the figure who insisted that chemistry would not be a serious enterprise until its practitioners agreed to measure what they had instead of debating what they hoped. Part V of this book carries the same insistence into AI engineering. The disciplines it describes — eval-driven development, FinOps, security, observability, MLOps, governance — are the contemporary equivalents of Lavoisier’s balance. They will not improve a bad system; they will tell you, ruthlessly, what kind of system you have.

Foundation Model Strategy and Selection

“You will not pick one model. You will pick a portfolio and a process for changing it.”

—a recurring observation in AI strategy

Chapter Overview

Enterprise AI rests on foundation models, and the decisions about which models to use are no longer narrowly technical — they are strategic. Which models an organisation builds on, how many vendors it relies on, whether and when it fine-tunes, and how it governs the whole portfolio over time together determine the capability, the cost, and the resilience of everything built above. This chapter develops foundation-model strategy as a portfolio discipline: the closed-versus-open question, the choice among frontier and mid-tier and small models, the single-vendor-versus-multi-vendor decision, the choice among fine-tuning and retrieval-augmented generation and prompting, the planning for a model’s lifecycle and deprecation, and the operational governance that distinguishes a deliberate model strategy from drift. The chapter’s epigraph states its thesis: an organisation does not pick a model, it picks a portfolio and a process for changing it.

Learning Objectives

- Frame the closed-versus-open foundation-model question and decide where each fits.
- Choose between frontier, mid-tier, and small models based on the actual task requirements.
- Design a single-vendor or multi-vendor portfolio with the hexagonal pattern.
- Decide between fine-tuning, retrieval-augmented generation, prompting, and their combinations.
- Plan for model deprecation and version transitions, and govern the portfolio over time.

85.1 The Closed-versus-Open Question

The first strategic decision in foundation-model strategy is whether to build on closed, vendor-hosted frontier models or on open-weight models the organisation self-hosts. The choice is structural, because it shapes capability, cost, control, privacy, and operational complexity all at once, and it is rarely an all-or-nothing decision.

85.1.1 Closed Hosted Frontier Models

Closed frontier models — those hosted by Anthropic, OpenAI, Google, and others — are consumed through an API, and their characteristics follow from that. They typically sit at the capability frontier, the most capable models available at any given moment. They offer operational simplicity, because the organisation consumes an API rather than running infrastructure. Their cost is per-token, which can become expensive at scale. Control of the model rests with the vendor, who decides when it is updated. Data flows to the vendor under contractual protections rather than staying on the organisation's own systems. And the vendor's compliance certifications are inherited by the consumer. Closed frontier models are the choice when capability and operational simplicity matter most.

85.1.2 Open-Weight Models

Open-weight models — the Llama family, Mistral, and others — are self-hosted, and their characteristics are largely the mirror of the closed models'. Their capability is closing the gap with the frontier but still tends to lag it. They carry operational complexity, because the organisation must run the inference infrastructure itself. Their cost is the capital and operating expense of that hosting rather than a per-token charge. Control is complete: the organisation chooses the model version and when it changes. Data stays under the organisation's direct control rather than flowing to a vendor. And the model can be fine-tuned freely. Open-weight models are the choice when control, privacy, and sovereignty matter most.

85.1.3 When Each Fits

The two options fit different needs, and the strategic skill is matching them. Closed frontier models fit the highest-stakes tasks, where the marginal capability is worth the cost and the loss of control. Open-weight models fit high-volume routine work, regulated data that must not leave the organisation, and deployments with sovereignty requirements that demand in-jurisdiction hosting. And a mixed strategy — closed models for the hard tasks, open models for the routine ones, with routing logic to send each request to the right place — is frequently the right answer, because it uses each kind of model for what it is genuinely best at rather than forcing all traffic through one.

85.1.4 Licensing Considerations

Open-weight does not mean unrestricted, and the licences of open-weight models vary widely. Some carry commercial-use restrictions that limit how the model may be used in a business. Some carry scale restrictions that change the terms above a certain size of deployment. Some require attribution. Some restrict particular downstream uses. Because the terms differ so much, a legal review of the licence of any open-weight model is required before the model is adopted — an organisation that self-hosts an open-weight model without reading its licence may be operating outside the terms it agreed to without knowing it.

85.2 Frontier versus Mid-Tier versus Small

Within a vendor's offerings, models come in capability tiers, and choosing the tier is as strategic as choosing the vendor, because the tiers differ sharply in capability, cost, and latency.

85.2.1 The Tiers

A vendor's lineup typically spans three tiers. The frontier tier holds the most capable models, and they are also the highest in cost and the highest in latency. The mid-tier holds models that balance capability against cost, and it is the production workhorse for most workloads. The small or fast tier holds models of limited capability, but very low cost and very low latency. The three tiers are a spectrum of trade-offs, and an organisation's model strategy is, in part, a decision about which tier each of its workloads should run on.

85.2.2 The Selection

The selection principle is to match the capability of the tier to the genuine requirements of the task, and the central observation is that most tasks do not need frontier capability. A great deal of production AI work — classification, extraction, routine generation — is served perfectly well by a mid-tier or even a small model, and routing logic that directs each request to the right tier captures a large cost saving without a quality loss. Tiered deployment, with frontier capability reserved for the requests that genuinely need it, is the standard cost-optimisation move of foundation-model strategy.

85.2.3 The Caution

There is a real caution attached to tier selection, however. Smaller models often look entirely adequate on simple benchmarks and then fail on the harder tasks that production actually contains, because the simple benchmark did not exercise the capability the hard task needs. This means capability evaluation must reflect the genuine complexity of production rather than a simplified benchmark, and the cost savings from downsizing a workload to a smaller model can be silently offset by a quality regression the simple benchmark never detected. The discipline is to evaluate a tier choice against production-representative tasks before committing to it.

85.3 Single-Vendor versus Multi-Vendor

A foundation-model strategy must decide how many vendors to depend on, and the decision trades operational simplicity against resilience and leverage.

85.3.1 Single-Vendor

A single-vendor strategy concentrates the foundation-model portfolio on one provider. Its advantages are operational simplicity, deeper integration with that one vendor, volume discounts earned by concentrating spend, and a single set of tools and patterns for the whole organisation to learn. Its disadvantages are the mirror image: concentration risk, because the organisation's AI depends entirely on one vendor's continued performance and pricing, and lock-in, because the cost of leaving rises as the dependence deepens. Single-vendor is the simpler strategy and the more exposed one.

85.3.2 Multi-Vendor

A multi-vendor strategy spreads the portfolio across providers. Its advantages are access to capability across vendors, so the organisation can use the best model for each task; negotiating leverage, because a credible alternative gives the organisation a position; and resilience against a vendor outage or a

pricing change, because the organisation is not wholly dependent on one provider. Its cost is a more complex vendor portfolio to operate. Multi-vendor allows a best-of-breed-by-task approach, and it is the more resilient strategy at the price of more operational complexity.

85.3.3 The Hexagonal Pattern Applied

The architectural move that makes a multi-vendor strategy practical is the hexagonal pattern, applied to the foundation-model layer. The application code speaks to an abstract model interface rather than to any vendor's specific API; adapters translate that abstract interface to each vendor's specifics; and a vendor change therefore becomes an adapter change rather than an application change. Multi-vendor support is achieved by maintaining multiple adapters behind the one abstract interface. The hexagonal pattern is what turns the multi-vendor strategy from an aspiration into something an organisation can actually operate without rewriting its applications whenever the portfolio changes.

85.4 Fine-Tuning versus RAG versus Prompting

Once the models are chosen, the strategy must decide how to customise their behaviour for the organisation's tasks, and there are three techniques — prompting, retrieval-augmented generation, and fine-tuning — that are used individually and in combination.

85.4.1 Prompting

Prompting customises a model's behaviour through the instructions and examples placed in the prompt. It changes nothing about the model itself, which makes it the fastest technique to iterate on — a prompt change ships immediately. Its limits are that it is bounded by the context window, so only so much instruction and example can be supplied, and that it produces no persistent customisation, since the customisation lives in the prompt rather than in the model. Prompting is the lightest technique and the right first thing to try.

85.4.2 Retrieval-Augmented Generation

Retrieval-augmented generation customises a model's behaviour by retrieving relevant information at inference time and supplying it to the model. Like prompting, it changes nothing about the model itself. Its strengths are that it keeps the system current through index updates rather than retraining, and that it supports citation and verification, because the retrieved sources can be shown. Retrieval-augmented generation is the predominant pattern for integrating enterprise data into an AI system, and the knowledge-architecture chapter develops it in depth.

85.4.3 Fine-Tuning

Fine-tuning customises a model by updating its weights for a specific task or domain. Unlike prompting and retrieval, it produces a persistent customisation baked into the model, and it can change the model's capabilities in a way the other two cannot. Its costs are higher complexity and higher expense, and its availability is vendor-specific — not every model can be fine-tuned, and the terms differ by vendor. Fine-tuning is the heaviest of the three techniques and the one to reach for only when the lighter two are genuinely insufficient.

85.4.4 The Selection Matrix

The three techniques map onto different needs, and a workable selection matrix runs as follows. A need for current information points to retrieval-augmented generation, because retrieval keeps the system current. A need for a consistent format or style points to fine-tuning or to strong prompting. A need for domain-specific knowledge points to retrieval, to fine-tuning, or to both. A need for the lowest latency points to a smaller fine-tuned model, which can match a larger model's task performance at lower cost. And a need for rapid iteration points to prompting first, because prompting is the fastest to change. The matrix is a starting point; the actual choice weighs these needs against one another.

85.4.5 Combinations

The three techniques are not mutually exclusive, and production systems frequently combine them. Retrieval-augmented generation and fine-tuning together are common — a fine-tuned model that also retrieves. Prompting sits on top of retrieval in almost every retrieval system. And an organisation often improves a system iteratively across the techniques, starting with prompting, adding retrieval, and fine-tuning only where the combination still falls short. The strategy is to treat the three as a toolkit rather than as a single choice.

85.5 Fine-Tuning in Detail

Because fine-tuning is the heaviest customisation technique, it deserves a closer treatment of when it makes sense, when it does not, and how it is done.

85.5.1 Use Cases

Fine-tuning earns its cost in a recognisable set of cases. It is the right technique for enforcing a consistent format or style that prompting cannot reliably achieve, for teaching a model domain vocabulary it does not have, for reducing prompt length by baking a standing instruction into the model, and for getting a smaller model to achieve a capability that would otherwise require a larger one. What these cases share is a need for a persistent change in how the model behaves that the lighter techniques cannot deliver.

85.5.2 Not Use Cases

Equally, fine-tuning is the wrong technique for several needs that it is often mistakenly applied to. It is the wrong way to give a model recent information — retrieval handles that, and a fine-tune goes stale. It cannot add reasoning capability beyond what the base model has. It is wasteful on a small dataset where prompting would suffice. And it is a poor fit for a task that changes frequently, because each change demands a re-fine-tune. Recognising these non-use-cases is what keeps fine-tuning from being applied where a lighter technique would serve better.

85.5.3 The Methodology

When fine-tuning is the right technique, the methodology matters, and the central truth is that the dataset matters more than the algorithm. A quality dataset matters more than a large one, a validation set must be held separate from the training data, the fine-tuned model must be evaluated against a

baseline to confirm it is genuinely better, and the cost of the fine-tuning effort must be tracked. A team that fine-tunes well invests its effort in curating the dataset, because the dataset is what determines whether the fine-tune succeeds.

85.5.4 Operational Considerations

Fine-tuning carries operational consequences beyond the training run itself. A fine-tuned model is pinned to a specific base-model version, which means that when the base model is deprecated, the deprecation cascades to the fine-tune. Re-fine-tuning is required whenever the base model is updated in a way that matters. And the fine-tuned model carries its own storage and serving costs. These operational considerations are part of the true cost of fine-tuning, and a strategy that accounts only for the training cost understates what fine-tuning actually commits the organisation to.

85.6 Model Lifecycle Planning

Foundation models are not stable fixtures; they are deprecated, updated, and superseded, and a foundation-model strategy must plan for that lifecycle rather than be surprised by it.

85.6.1 Vendor Deprecation

Vendors deprecate models on their own cycles, retiring older models in favour of newer ones. A vendor typically announces a migration path when it deprecates a model, but a migration to a new model version brings some change in behaviour, and that change means re-evaluation is required on each new version rather than an assumption that the new model behaves as the old one did. Vendor deprecation is a recurring event that a foundation-model strategy must expect and plan around.

85.6.2 The Architectural Response

The architecture's response to the model lifecycle has several elements. The hexagonal pattern abstracts the model, so a model change is contained behind the abstraction. Version pinning holds a specific model version where stability matters more than freshness. A regression suite validates each migration, confirming the new version has not regressed the system. Deprecation-timeline tracking keeps the organisation aware of when its models will be retired. And vendor-relationship maintenance keeps the organisation informed of what is coming. Together these turn the model lifecycle from a source of surprises into a managed process.

85.6.3 Capability Evolution

The model lifecycle is not only about deprecation; it is also about capability evolution, and that has a positive strategic implication. New models bring new capabilities, and a newer, more capable model can sometimes simplify a workflow that an older model needed elaborate scaffolding to perform. A multi-agent pattern, built to compensate for a weaker model's limitations, may collapse to a single-agent design once a more capable model removes the need for the scaffolding. This means continuous re-evaluation is worthwhile not only to catch regressions but to find the simplifications that capability evolution makes available.

85.7 Cost Economics

The cost of a foundation-model strategy varies enormously with the choices made, and reasoning about that cost is part of the strategy.

85.7.1 The Variables

The cost of foundation-model usage depends on several variables. Per-token pricing is the base, and within it input tokens and output tokens are priced differently, so the shape of a workload matters. Cache hits reduce cost where repeated content can be cached. Volume discounts reduce the per-token rate at scale. Self-hosting substitutes capital and operating expense for the per-token charge. And fine-tuning carries its own costs. A cost model that accounts for all of these variables, rather than only the headline per-token rate, is what gives an accurate picture of what a foundation-model strategy will cost.

85.7.2 The Patterns

Several cost-reduction patterns recur. Cheaper models are used for routine work and expensive ones reserved for hard tasks — the tiered-deployment pattern. Caching is applied aggressively to repeated content. Prompts are compressed to reduce input tokens. Output is limited to reduce the expensive output tokens. And self-hosting is adopted at the scale where its economics beat the per-token alternative. These patterns are the same cost levers the FinOps chapter develops, applied here as part of the foundation-model strategy.

85.7.3 The Cost Discipline

The cost patterns work only within a standing cost discipline: cost monitoring per use case, so the organisation knows where its foundation-model spend goes; budget enforcement, so a use case cannot quietly exceed its allocation; cost-regression alerts, so a change that raises cost is caught; and return-on-investment assessment per use case, so the organisation knows which uses of its foundation models are worth their cost. The cost discipline is what keeps a foundation-model strategy economically sound rather than merely capable.

85.8 Sovereignty and Compliance

Sovereignty and compliance requirements shape foundation-model choice directly, and sometimes they override the capability and cost considerations entirely.

85.8.1 Data Sovereignty

Data-sovereignty requirements bear on which models an organisation may use. Some jurisdictions require that data reside within their borders, and some restrict cross-border data flows; either constraint limits the organisation to vendors that can host within the jurisdiction or to self-hosting. The vendor's available regions and certifications therefore become a selection criterion, and self-hosting an open-weight model addresses the constraint directly by keeping the data on the organisation's own systems. Data sovereignty can be the deciding factor in a foundation-model choice, overriding capability.

85.8.2 Sector-Specific Compliance

Beyond data residency, sector-specific compliance regimes constrain foundation-model selection. Healthcare carries HIPAA and its equivalents, financial services carry their regulatory requirements, and government deployments carry FedRAMP and its equivalents. Each of these regimes may restrict which vendors an organisation in that sector can use, because a vendor that cannot meet the sector's compliance requirements is not an eligible choice. Sector-specific compliance is a gating criterion that a foundation-model strategy in a regulated industry must apply from the start.

85.8.3 The EU AI Act

The EU AI Act bears on foundation-model strategy specifically. It places obligations on foundation-model providers, and the documentation those obligations produce cascades to the deployers who build on the models. The consequence for strategy is that the choice between providers now includes an Act-compliance consideration — whether a provider's documentation and obligations support the deployer's own compliance position. An organisation deploying under the EU AI Act must factor that into the foundation-model selection rather than treating compliance as a separate, later concern.

85.9 Governance

A foundation-model portfolio must be governed over time, not merely assembled once, and governance has two elements: the approval of new models and the ongoing operating cadence.

85.9.1 The Approval Process

A new foundation model should not enter the portfolio without an approval process. The process should include a capability evaluation of the model before adoption, a privacy and security review, a compliance review, a cost approval, and, where appropriate, a use-case-specific approval that confirms the model is suitable for the particular use it is intended for. The approval process is what ensures a model enters the portfolio because it was deliberately chosen rather than because someone started using it, and it is the governance counterpart of the deliberate strategy this chapter argues for.

85.9.2 The Operating Cadence

Governance continues after approval through a regular operating cadence. The portfolio should be reviewed periodically, with new models evaluated against the current portfolio to see whether they should replace or supplement what is in use, deprecation planned for the models being retired, cost reviewed across the portfolio, and the vendor relationships reviewed. The operating cadence is what keeps the portfolio current and deliberate over time — without it, the portfolio drifts, and drift is exactly what a foundation-model strategy exists to prevent.

85.10 Anti-Patterns

The failure modes of foundation-model strategy recur, and each is the absence of a discipline this chapter has set out:

- **Vendor lock-in by default.** No abstraction layer is built, so the applications bind to one vendor and

the cost of changing becomes prohibitive.

- **Frontier-only.** The top tier is used for every task, so the organisation pays frontier prices for work a mid-tier or small model would have done.
- **Premature self-hosting.** An open-weight model is self-hosted for a sub-scale use, so the organisation carries infrastructure cost the volume does not justify.
- **Fine-tuning for tasks retrieval would handle.** Fine-tuning is applied where retrieval-augmented generation would have served, paying the heavier technique's cost for no gain.
- **No regression suite.** Model migrations are made with no regression suite to validate them, so a migration silently regresses the system.
- **Cost surprise.** No budget enforcement is in place, so a use case's cost runs away undetected.
- **Stale portfolio.** The portfolio is not re-evaluated as new models are released, so the organisation runs outdated models long after better ones are available.
- **Vendor concentration.** No contingency exists for a vendor outage or a pricing change, so the organisation is exposed to a single vendor's problems.

Each anti-pattern maps to a remedy the chapter has developed — the hexagonal abstraction, tiered selection, scale-justified self-hosting, the fine-tuning-versus-retrieval matrix, regression suites, cost discipline, the operating cadence, and multi-vendor resilience — and together they show that foundation-model strategy fails through drift and missing discipline rather than through any single bad model choice.

85.11 Last Word

Foundation-model strategy is a portfolio strategy made of explicit choices: closed versus open, frontier versus mid-tier versus small, single-vendor versus multi-vendor, fine-tuning versus retrieval-augmented generation versus prompting, lifecycle and deprecation planning, cost economics, sovereignty and compliance, and ongoing governance. The hexagonal pattern is what makes the portfolio flexible, turning a vendor or model change into an adapter change rather than an application rewrite.

The discipline the chapter argues for is to make the portfolio decisions deliberately, to evaluate continuously rather than assume yesterday's choices still hold, and to govern the portfolio actively through an approval process and an operating cadence. The anti-patterns — lock-in, frontier-only, premature self-hosting, misapplied fine-tuning, missing regression suites, cost surprise, a stale portfolio, vendor concentration — are common, and their remedies are structural rather than incidental. You will not pick one model; you will pick a portfolio and a process for changing it, and the strategy of this chapter is that process.

Chapter 86

Inference Infrastructure

“The model is the recipe. The infrastructure decides whether dinner is on time.”

—a recurring observation in production engineering

Chapter Overview

Once the choice of foundation model is made, it is the infrastructure for inference that determines the economics, the latency, the reliability, and the scale of the system. This chapter develops inference infrastructure as an engineering discipline: the economics of inference and the GPU cost structure beneath it, the batching that turns idle hardware into throughput, the quantisation that trades precision for cost and speed, the management of the key-value cache that long-context inference depends on, the model-parallelism patterns that large models require, the latency budgets that shape the whole architecture, the capacity planning that production demands, edge inference for the cases that belong off the cloud, and the observability and reliability engineering that distinguish a professional inference operation from an informal one. The architecture applies whether the model is consumed through a vendor API or self-hosted — self-hosting simply requires the organisation to engineer directly what an API vendor would otherwise engineer for it.

Learning Objectives

- Understand inference economics and the GPU cost structure beneath self-hosting.
- Apply batching, including continuous batching, to improve inference throughput.
- Use quantisation where appropriate to reduce cost and latency, with quality discipline.
- Manage the key-value cache for efficient long-context inference.
- Apply model-parallelism patterns for models too large for a single accelerator.
- Plan capacity, deploy edge inference where appropriate, and operate inference with reliability engineering.

86.1 Inference Economics

Inference cost dominates the total AI cost of a production system, because training, if it happens at all, happens once, while inference happens on every request for the life of the system. Understanding the economics of inference is therefore the foundation of the whole discipline.

86.1.1 The Cost Structure

The cost structure of inference depends on whether the model is consumed through a vendor API or self-hosted. With a vendor API, the cost is per-token pricing, with input and output tokens priced differently. With self-hosting, the cost is the capital and operating expense of the GPUs, and the unit cost depends critically on utilisation — idle hardware is paid-for hardware producing nothing. Around either model sit incremental costs for networking, storage, and observability, and self-hosting adds the non-trivial cost of the engineering staff to run the infrastructure. A cost model that captures only the headline figure understates what inference actually costs.

86.1.2 The Break-Even

The choice between an API and self-hosting turns on a break-even that depends on volume and workload. At low volume, vendor APIs are economical, because the per-token cost is small in total and the organisation avoids the fixed cost of infrastructure. At high volume with a stable workload, self-hosting may become economical, because the fixed infrastructure cost is amortised over enough requests to beat the per-token charge. At very high volume with a variable workload, a hybrid — self-hosting the steady base and bursting to an API for the peaks — is often optimal. Computing the break-even precisely depends on the specific workload patterns and the capability requirements, and it should be computed rather than assumed.

86.1.3 The Cost Drivers

Whatever the consumption model, inference cost is driven by a recognisable set of variables: the number of tokens in and out per request, the volume of requests, the size of the model in parameters, the level of quantisation applied, the utilisation of the hardware, and the effectiveness of caching. Each of these is a lever, and the rest of this chapter is, in large part, about pulling them: batching raises utilisation, quantisation reduces effective model size and cost, and caching reduces the tokens that must be processed at all.

86.2 GPU Economics

For an organisation that self-hosts, the economics of inference are the economics of GPUs, and understanding the GPU cost structure is what makes self-hosting a deliberate decision rather than a hopeful one.

86.2.1 The Hardware

Inference at scale runs on data-centre GPUs — the H100, the A100, and their successors — and three properties of the hardware bound what it can do. Memory capacity bounds the size of model that fits, compute capacity bounds the throughput the GPU can deliver, and inter-GPU bandwidth bounds the efficiency of any parallelism that spreads a model across several GPUs. Newer accelerators, such as TPUs, have broadly similar economics with their own vendor-specific patterns. An organisation self-hosting must understand these three bounds, because they determine which models it can serve and how fast.

86.2.2 The Utilisation Problem

The dominant cost issue in GPU inference is underutilisation: a GPU that is paid for but not kept busy is the largest avoidable cost in a self-hosted operation. Inference is frequently memory-bound rather than compute-bound, meaning the GPU's compute sits idle while it waits on memory, and workload variability creates a tension between provisioning enough capacity for the peak and keeping that capacity utilised in the trough. Batching, developed in the next section, is the primary technique for raising utilisation, and the utilisation problem is the reason batching matters so much.

86.2.3 The Capital-versus-Cloud Decision

Self-hosting itself divides into a capital-versus-cloud decision. On-premises capital — buying the GPUs — gives predictable cost and is economical over the long term at high utilisation. Cloud GPU rental gives flexibility but is expensive over time when utilisation is high, because the flexibility is paid for continuously. Reserved-instance arrangements offer a compromise between the two, and capacity guarantees written into vendor contracts address the risk that cloud GPUs are unavailable when needed. The decision is the same capital-versus-operating-expense trade-off the hardware chapter develops, applied to the inference workload specifically.

86.3 Batching

Batching is the foundational technique for inference throughput, because it is how idle GPU capacity is turned into useful work. It comes in two forms, with a trade-off attached.

86.3.1 Static Batching

Static batching groups requests into a batch of a fixed size, waiting either for the batch to fill or for a timeout to expire before processing it. It is simple to implement and reason about, and it does raise utilisation above processing requests one at a time. Its weakness is variable latency: a request that arrives just as a batch closes waits for the next batch to fill, so the latency a given request experiences depends on when it arrived relative to the batching cycle. Static batching is the basic form, improved on by the next.

86.3.2 Continuous Batching

Continuous batching is the more sophisticated form, in which requests join and leave the batch at every generation step rather than being locked into a fixed batch. This improves throughput substantially over static batching, because the GPU is kept fuller, and it reduces head-of-line blocking, because a short request is not held up behind a long one in the same fixed batch. Continuous batching is implemented in the modern inference engines — vLLM, TensorRT-LLM, and others — and it is the form a serious self-hosted operation uses.

86.3.3 The Latency Trade-Off

Batching carries a trade-off that the architecture must navigate. Larger batches improve throughput, because more requests are processed together, but they can increase the latency any individual request experiences, because that request shares the GPU with more others. The optimal batch size depends on

the workload and on the latency targets, and a common pattern is to run multiple servers with different batching configurations — a low-latency configuration for interactive traffic and a high-throughput configuration for batch work — so that different service-level agreements are each served by an appropriately tuned server.

86.4 Quantisation

Quantisation reduces the numerical precision at which a model runs, and in doing so it reduces both the memory the model needs and the computation it takes to run.

86.4.1 The Methods

Quantisation comes in several methods. INT8 quantisation runs the model's weights as 8-bit integers rather than full-precision floating point. INT4 and lower quantisation push further for greater memory savings. Mixed precision uses different precisions for different layers, quantising aggressively where the layer tolerates it and conservatively where it does not. And the quantisation itself can be done post-training, applied to an already-trained model, or through quantisation-aware training, which trains the model with the quantisation in view. The methods trade memory and speed against the risk of quality loss differently, and the choice among them depends on how sensitive the task is.

86.4.2 The Trade-Offs

Quantisation's trade-offs are real on both sides. The memory savings are substantial, and for memory-bound workloads the latency improvements are real too, because reducing the memory traffic speeds the inference. Against that, quantisation carries a quality regression — usually small, sometimes significant, depending on the model and the task. The regression is not predictable in the abstract, which means task-specific evaluation is required: an organisation quantising a model must measure the quantised model's quality on its actual task rather than assume the regression is negligible.

86.4.3 When to Quantise

Quantisation is the right move under a recognisable set of conditions: when memory pressure is the binding constraint on capacity, so the memory saving directly buys more headroom; when latency requirements are demanding and the memory-bound speed-up helps meet them; when cost reduction is a priority; and when the task has enough quality tolerance to absorb the small regression quantisation introduces. When those conditions hold, quantisation is a strong lever; when the task is quality-sensitive and quantisation's regression cannot be tolerated, it is the wrong move, and the evaluation is what tells the two cases apart.

86.5 KV-Cache Management

The key-value cache is both essential to efficient inference and expensive in memory, and managing it well is central to long-context inference.

86.5.1 The Mechanics

The mechanics of the key-value cache follow from how attention works. The attention computation at each generation step reuses the keys and values computed for all the prior tokens, and rather than recomputing them, the model caches them — that cache is the key-value cache. The cache grows linearly with the context length, because every additional token of context adds its keys and values to the cache, which means long-context inference is expensive in memory specifically because of the key-value cache it must hold. Understanding this mechanic is the precondition for the optimisations that follow.

86.5.2 The Optimisations

Several optimisations reduce the cost of the key-value cache. Paged attention manages the cache memory in pages, much as an operating system manages memory, reducing the waste that contiguous allocation would produce. Prefix sharing caches a shared system prompt once and reuses it across the many requests that begin with it, rather than caching it separately for each. Key-value-cache offload moves parts of the cache to lower, cheaper memory tiers. And quantisation can be applied to the cache itself, reducing its memory footprint. Together these optimisations make long-context inference far more affordable than a naive cache would allow.

86.5.3 The Capacity Planning

The key-value cache drives capacity planning for long-context workloads. For a long context, the per-request memory is dominated by the key-value cache, which means the number of requests that can be batched together is bounded by the memory available to hold their caches. The distribution of context lengths in the workload therefore shapes the capacity directly: a workload of long contexts holds fewer concurrent requests per GPU than a workload of short ones. Capacity planning for inference must account for the key-value cache, not only for the model weights.

86.6 Model Parallelism

Large models exceed the memory of a single GPU, and serving them requires spreading the model across several GPUs — model parallelism — which comes in several forms.

86.6.1 Tensor Parallelism

Tensor parallelism splits the individual tensors of the model across GPUs, so that each GPU holds a portion of each layer. It involves heavy inter-GPU communication, because the partial results on each GPU must be combined at each step, and it therefore requires high inter-GPU bandwidth — NVLink or an equivalent — to be efficient. Tensor parallelism is used for the layers that must be split because they do not fit on one GPU, and its communication cost is the price of fitting a model that a single GPU's memory cannot hold.

86.6.2 Pipeline Parallelism

Pipeline parallelism takes a different approach, placing different layers of the model on different GPUs so that a request flows through the GPUs as through a pipeline. Its characteristic concern

is bubble efficiency — the idle time that arises when the pipeline is not perfectly full — which the implementation must manage. Pipeline parallelism is useful for very large models, where the layers can be distributed across many GPUs, and it trades the bubble-efficiency concern for the lower inter-GPU communication that distributing whole layers rather than splitting tensors allows.

86.6.3 Expert Parallelism

Expert parallelism applies to mixture-of-experts models, which contain many expert sub-networks of which only some are active for any given input. Expert parallelism distributes the experts across GPUs, and it requires routing across the GPUs to send each input to the GPUs holding the experts it activates. It is a parallelism pattern specific to the mixture-of-experts architecture, and it is the way those models, which can be very large in total parameters while activating only a fraction per input, are served efficiently.

86.6.4 The Combinations

In practice, production deployments of large models combine these parallelism techniques rather than relying on one. Tensor, pipeline, and expert parallelism are composed together, and the hardware topology — which GPUs are connected to which, and by what bandwidth — matters to how the combination performs. The inference engines, vLLM and TensorRT-LLM and others, provide implementations of these combined parallelism patterns, so an organisation serving a large model uses the engine's implementation rather than building the parallelism itself.

86.7 Latency Budgets

Latency budgets shape the inference architecture, because the latency a use case can tolerate determines what the infrastructure must deliver.

86.7.1 The Components

Inference latency is the sum of several components. There is the network latency to reach the model endpoint, the queue time a request waits before processing begins, the time to the first token, and then the time per output token for each subsequent token. The total time is the time to first token plus the number of output tokens multiplied by the time per output token. Decomposing latency into these components is what lets an architecture target the component that is actually the problem, rather than treating latency as a single undifferentiated number.

86.7.2 The Targets by Use Case

Different use cases carry different latency targets. A conversational use case typically needs a time to first token under a second and a time per output token under fifty milliseconds, so the response feels responsive. A tool-use loop needs its aggregate latency, across all the model calls in the loop, to stay under the user-perceived budget. A batch-processing use case prioritises throughput over latency entirely, because no user is waiting. And a real-time use case may demand a time to first token and time per output token so tight that meeting them is sometimes infeasible. The architecture is shaped by which of these targets the use case imposes.

86.7.3 Streaming

Streaming is the technique that addresses perceived latency. By streaming the output tokens to the user as they are generated rather than waiting for the complete response, the time to first token becomes the user's perceived response time, and the time spent generating the rest of the response is occupied by the user reading what has already arrived. Streaming is important for conversational applications specifically, because it turns a multi-second total generation time into a sub-second perceived response, and it is a near-universal pattern in interactive AI systems.

86.8 Capacity Planning

Capacity planning is how an inference operation ensures it has the hardware to meet its production load, and it works from a set of inputs to a set of provisioning patterns.

86.8.1 The Inputs

Capacity planning takes a recognisable set of inputs: a forecast of request volume, the distribution of input and output lengths in the workload, the latency service-level agreements the operation must meet, the quality requirements, and the cost budget. These inputs together describe what the capacity must deliver — enough hardware to serve the forecast volume at the required latency and quality, within the budget. Capacity planning that works from a guessed volume rather than a genuine forecast is capacity planning that will be wrong.

86.8.2 The Patterns

Several provisioning patterns recur. Peak-load provisioning sizes the capacity for the busiest moment, while auto-scaling adjusts capacity to the load, trading the cost of idle peak capacity against the complexity of scaling. Multi-region provisioning places capacity in several regions for resilience and for latency to distributed users. Multi-vendor provisioning spreads capacity across vendors for resilience against an outage and for pricing leverage. And burst capacity through cloud overflow handles spikes by bursting to the cloud above the self-hosted base. The right pattern depends on the workload's variability and the operation's resilience requirements.

86.8.3 The Margin

Capacity planning must include a margin above the forecast volume, and the margin serves several purposes: headroom for spikes that exceed the forecast, headroom to absorb a vendor outage that removes some capacity, and general slack so the operation is not running at the edge. The size of the margin is itself a decision, weighing the cost of the margin — paid-for capacity that is usually idle — against the cost of degradation when load exceeds capacity. A capacity plan with no margin is a plan that fails on the first spike; a capacity plan with too much margin pays for hardware it never uses.

86.9 Edge Inference

Some inference belongs not in a data centre but at the edge — on the device, in the branch, close to where the work is — and edge inference is a distinct part of the infrastructure discipline.

86.9.1 The Use Cases

Edge inference is the right choice for a recognisable set of use cases. Latency-critical local processing benefits from inference on the device, with no network round trip. Privacy-sensitive processing benefits from keeping the data on the device rather than sending it to a server. Disconnected operation requires inference that works without a network at all. And cost reduction at scale can favour edge inference, because inference done on a device the organisation does not pay to run is inference that does not consume server capacity. These use cases are what make edge inference worth its constraints.

86.9.2 The Hardware

Edge inference runs on a range of hardware. Mobile NPUs are increasingly capable of running real models on a phone. Embedded accelerators serve inference in Internet-of-Things devices. On-premises edge servers serve inference for branch deployments, close to a particular location's users. And browser-based inference runs models in the browser for some web applications. The edge hardware is, by nature, more constrained than data-centre hardware, and the model choice must respect those constraints.

86.9.3 The Models

The models suited to the edge are shaped by the hardware's constraints. Smaller models are suited to the edge, because the edge hardware cannot hold large ones. The models are quantised to fit the memory and compute the edge device offers. Specialised models, built for a particular task, are often preferred over general models, because a specialised model can be smaller for the same task quality. And distillation — training a small model to mimic a larger one — is a common way to produce a model that fits the edge while retaining much of a larger model's capability. Edge inference is, in large part, the discipline of fitting a useful model into a constrained device.

86.10 Observability for Inference

Observability is foundational to operating inference, because an inference operation that cannot see its own behaviour cannot diagnose, tune, or defend it.

86.10.1 What to Measure

An inference operation should measure a recognisable set of signals: the request rate and the response rate, the latency distributions across time to first token, time per output token, and total time, the error rate broken down by type, the token utilisation, the cache hit rate, the GPU utilisation, the memory utilisation, and the cost per request. Each of these is a window onto a different aspect of the operation's health — latency onto user experience, GPU and memory utilisation onto efficiency, cost per request onto economics — and together they make the operation legible.

86.10.2 Alerts

Beyond the measured signals, the operation needs alerts that fire when something requires attention: service-level-agreement violations, elevations in the error rate, cost anomalies, capacity approaching its limits, and whatever vendor-specific signals the consumed services emit. Alerts are what turn

observability from a set of dashboards someone might look at into a system that actively tells the operators when intervention is needed. An inference operation with measurement but no alerts sees its problems only when someone happens to look.

86.11 Reliability Engineering

Inference deserves the same production reliability practices as any other critical infrastructure, because a production AI system's availability is the availability of its inference.

86.11.1 The Patterns

Reliable inference uses a recognisable set of patterns. Retries with backoff handle transient failures. Circuit breakers stop the system from hammering a failing dependency. Timeouts bound how long a request can hang. Graceful degradation falls back to a smaller model when the preferred one is unavailable, so the system continues at reduced quality rather than failing. Multi-vendor failover switches to an alternative vendor when one has an incident. And regional failover switches to another region when one is down. These patterns are what keep an inference operation available through the failures that production inevitably brings.

86.11.2 The Discipline

Around the patterns sits a reliability discipline. Error budgets per service set how much unreliability is acceptable and trigger a response when it is exceeded. Post-incident review turns each failure into a lesson. Capacity drills test that the capacity plan and the failover actually work before they are needed. Vendor-relationship management keeps the organisation informed and supported. And continuous improvement feeds all of this back into a steadily more reliable operation. The discipline is what makes inference reliability a sustained practice rather than a set of patterns configured once and forgotten.

86.12 Anti-Patterns

The failure modes of inference infrastructure recur, and each is the absence of a technique or discipline this chapter has set out:

- **Underutilised GPUs.** The GPUs are paid for but not kept busy, and the underutilisation bleeds cost continuously.
- **No batching.** Requests are processed one at a time, leaving throughput on the table that batching would have captured.
- **Excessive caching.** Caching is applied so aggressively that stale results are served.
- **Insufficient caching.** Caching is under-applied, so the operation pays to recompute results it could have cached.
- **Quantisation without evaluation.** A model is quantised with no task-specific evaluation, and the quality regression goes undetected.

- **No fallback.** There is no graceful degradation or failover, so a vendor incident produces a total outage.
- **Capacity margin missing.** The capacity plan has no margin, so a spike produces cascading degradation.
- **Observability gaps.** The operation cannot see its own behaviour, so an incident cannot be investigated.

Each anti-pattern maps to a remedy the chapter has developed — batching for utilisation, balanced caching, quantisation with evaluation, fallback and failover, capacity margin, and comprehensive observability — and together they show that inference infrastructure fails through the neglect of production engineering rather than through any inherent difficulty in serving a model.

86.13 What This Chapter Established

Inference infrastructure determines the cost, the latency, the reliability, and the scale of a production AI system — the model is the recipe, but the infrastructure decides whether dinner is on time. The discipline spans the economics of inference and the GPU cost structure, the batching that turns idle hardware into throughput, the quantisation that trades precision for cost and speed under quality discipline, the key-value-cache management that long-context inference depends on, the model-parallelism patterns that large models require, the latency budgets that shape the architecture per use case, the capacity planning that production load demands, the edge inference for the cases that belong off the cloud, the observability that makes the operation legible, and the reliability engineering that matches the stakes.

The architecture applies whether the model is consumed through a vendor API or self-hosted; self-hosting simply means the organisation engineers directly what an API vendor would otherwise engineer for it. The anti-patterns — underutilised GPUs, no batching, mis-tuned caching, unevaluated quantisation, no fallback, missing capacity margin, observability gaps — are common, and their remedies are the production-engineering disciplines this chapter has developed. The discipline, in a sentence, is to engineer inference as production infrastructure rather than as experimentation.

Knowledge and Document Architecture

“The retrieval system answers the wrong question fast and confidently. The architecture decides which question gets asked.”
—a recurring observation in RAG design

Chapter Overview

Retrieval-augmented generation appears throughout this book as a pattern, developed in Chapter 21; the enterprise knowledge architecture that retrieval depends on deserves its own treatment, because the quality of a retrieval system is largely the quality of the knowledge architecture beneath it. The discipline encompasses the taxonomies and ontologies that organise knowledge, the knowledge graphs that complement vector retrieval, the design of vector stores at production scale, the hybrid search that combines vectors with keywords and structured filters, the chunking strategies that determine what is retrievable, the freshness disciplines that keep knowledge current, and the operational practice that distinguishes a mature knowledge infrastructure from a naive document dump. This chapter develops those concerns and their integration into production AI systems, and its thesis is the epigraph’s: retrieval quality is knowledge-architecture quality, and the architecture decides which question the system is actually able to answer.

Learning Objectives

- Design taxonomies and ontologies that support retrieval and reasoning.
- Build knowledge graphs that complement vector retrieval.
- Architect vector stores at production scale, including the re-embedding discipline.
- Apply hybrid search combining vectors, keywords, and structured filters.
- Choose chunking strategies appropriate to different content types.
- Maintain knowledge freshness and quality, and operate knowledge infrastructure as a first-class system.

87.1 Why Knowledge Architecture

Enterprise knowledge is messy, vast, and operationally important, and a retrieval system built on top of it without a deliberate knowledge architecture inherits all the mess. Understanding the problem

space is the precondition for the architectural response.

87.1.1 The Problem Space

Enterprise knowledge presents a recognisable set of difficulties. It exists in many formats and many locations, scattered across the systems an organisation runs. It is embedded not only in documents but in people, in systems, and in processes. It changes continuously, so any snapshot of it ages. It carries access-control complexity, because not everyone may see everything. Its quality is mixed, with authoritative content sitting alongside outdated or unreliable content. And it is frequently multilingual. A retrieval system that ignores these difficulties does not escape them; it simply fails in proportion to them.

87.1.2 The Architectural Response

The architectural response is to treat knowledge architecture as a deliberate system rather than as the absence of one. That means representing knowledge in multiple forms — text, vector, and graph — because different representations support different kinds of retrieval and reasoning. It means applying quality and freshness disciplines, so the knowledge served is current and reliable. It means access-aware retrieval, so the system respects who may see what. And it means deliberate integration with the applications that consume the knowledge. The deliberate system is what turns a document dump into an infrastructure a retrieval system can be trusted to build on.

87.2 Taxonomies and Ontologies

Structured knowledge organisation is the first layer of a knowledge architecture, and it takes three related forms: taxonomies, ontologies, and controlled vocabularies.

87.2.1 Taxonomies

A taxonomy is a hierarchical categorisation of knowledge — practice areas, product categories, document types, arranged in a tree. Its primary use in a retrieval system is filtering and faceting: a taxonomy lets a query be narrowed to a branch of the hierarchy, so the retrieval searches only the relevant portion of the corpus. A taxonomy is maintained centrally, because a categorisation scheme is only useful if it is consistent, and it is the simplest form of structured knowledge organisation — a hierarchy that gives the corpus a navigable shape.

87.2.2 Ontologies

An ontology goes beyond a taxonomy's hierarchy to model richer relationships. It defines entity types and their properties, and it defines the relationships between entities, so that it captures not only what categories exist but how the things in them relate. This richer modelling supports reasoning, not only filtering, because a system can traverse the relationships an ontology defines. Ontologies are used heavily in regulated and knowledge-intensive domains — SNOMED in healthcare is a well-known example, and finance and other regulated domains have their own — where the relationships between entities are themselves part of the knowledge.

87.2.3 Controlled Vocabularies

A controlled vocabulary standardises the terminology used to describe and search knowledge. It manages synonyms, so that different terms for the same thing are recognised as equivalent, and it can map terms across languages, so that a multilingual corpus is searchable consistently. The purpose of a controlled vocabulary is to reduce ambiguity in both search and reasoning — when the vocabulary is standardised, a query and a document that mean the same thing are more likely to be matched even when they were originally phrased differently.

87.3 Knowledge Graphs

Knowledge graphs complement vector retrieval, addressing the kinds of question that similarity search alone handles poorly.

87.3.1 What They Are

A knowledge graph represents knowledge as nodes and edges. The nodes are entities, each carrying properties, and the edges are relationships between entities, typed and possibly weighted. A knowledge graph may have a defined schema or be schema-less, and it may be represented as a property graph or in the RDF triple form. The defining feature is that the structure — the explicit relationships between entities — is itself part of the represented knowledge, in a way that a flat collection of text chunks is not.

87.3.2 Use Cases

Knowledge graphs serve a recognisable set of use cases that vector retrieval handles poorly. They support multi-hop reasoning, following a chain of relationships from one entity to another across several steps. They support entity resolution and disambiguation, determining when two mentions refer to the same entity. They support structured-knowledge retrieval, answering questions about explicit relationships. And they naturally represent citation networks and organisation and product hierarchies. Where a question is about how things relate rather than about semantic similarity, a knowledge graph is the representation that answers it.

87.3.3 Implementation Patterns

Knowledge graphs are implemented through several patterns. Property-graph databases such as Neo4j store the graph directly as nodes and edges. Triple stores store the graph in the RDF triple form. Graph extensions to general-purpose databases add graph capability to a database the organisation already runs. And hybrid arrangements pair a graph store with a vector store, so the system has both representations available. The choice among these patterns depends on the scale of the graph and on what other infrastructure the organisation already operates.

87.3.4 Integration with Language Models

Knowledge graphs integrate with language models in several ways. Graph traversal can augment retrieval-augmented generation, pulling in related entities that pure similarity search would miss. The graph schema and selected instances can be placed in the prompt to support the model's reasoning. A

text-to-query pattern has the model generate graph queries from a natural-language question. And the combination supports multi-hop question answering, where the answer requires following a chain of relationships. Knowledge-graph integration is how a retrieval system gains the structured-reasoning capability that vectors alone do not provide.

87.4 Vector Store Design

Vector stores are the operational infrastructure of semantic retrieval, and designing them at production scale is a discipline in its own right.

87.4.1 The Components

A vector store has a recognisable set of components. There is an embedding model, usually external, that turns text into vectors. There is a vector index that supports similarity search over those vectors. There is metadata storage that holds the structured attributes of each item. There is a filtering capability that lets retrieval be constrained by that metadata. And there are update mechanisms that let the store be kept current. Together these components are what make a vector store a usable retrieval system rather than a bare collection of vectors.

87.4.2 Scale Considerations

At production scale, vector store design must confront several considerations. Billions of vectors are feasible, but they demand attention to the index. The index choices — HNSW, IVF, and others — trade recall against latency differently, and the choice shapes the store’s behaviour. Sharding spreads the store across nodes when it outgrows one. Memory-versus-disk trade-offs determine cost and latency. And update latency — how quickly a new document becomes retrievable — must be designed for. Scale turns vector store design from a configuration choice into a genuine engineering problem.

87.4.3 Embedding Model Selection

The quality of retrieval depends directly on the quality of the embedding model, which makes embedding-model selection a consequential decision. A domain-specific embedding model may outperform a general one on a specialised corpus, because it was trained on the relevant kind of text. Dimensionality carries trade-offs, with higher dimensions holding more information at a higher cost. Multilingual considerations matter for a multilingual corpus. And the selection must account for the update strategy, because changing the embedding model later is expensive. The embeddings chapter develops this selection in depth; the point here is that it is part of vector store design.

87.4.4 Re-Embedding

A specific and easily underestimated concern is re-embedding. When the embedding model changes, the whole corpus must be re-embedded, because vectors from different embedding models are not comparable — a query embedded by a new model cannot be compared against a corpus embedded by the old one. Re-embedding is a costly operation at scale, and it is managed through migration patterns: a dual index that runs the old and new embeddings in parallel during a transition, and a gradual replacement that re-embeds the corpus over time. Version management of the embeddings is part of operating a vector store, and a design that does not plan for re-embedding will be caught by

the first embedding-model change.

87.5 Hybrid Search

Pure vector search is rarely the optimal retrieval strategy, and hybrid search — combining vector search with other methods — consistently outperforms it.

87.5.1 Why Hybrid

The case for hybrid search follows from the complementary strengths of the methods. Vector search excels at semantic similarity, finding documents that mean the same thing as the query even when the words differ. Keyword search excels at exact terms, finding the precise identifier or name that semantic search might blur. Structured filters add precision, constraining the search to documents that meet explicit criteria. Because each method is strong where the others are weak, a combination of them outperforms any single method on its own, and hybrid search is the standard production approach for that reason.

87.5.2 The Patterns

Hybrid search is implemented through several patterns. A parallel pattern retrieves with vector search and keyword search independently and then combines the results. A filter-first pattern applies structured filters to reduce the candidate set before retrieval runs. A re-rank pattern uses a cross-encoder to re-rank the combined candidates by relevance. And a fusion pattern combines the rankings from several methods through a rank-fusion technique such as reciprocal rank fusion. The patterns are not mutually exclusive, and a production hybrid-search system often composes several of them.

87.5.3 Re-Ranking

Re-ranking deserves specific attention because it substantially improves retrieval quality. A cross-encoder re-ranker scores query-document pairs directly, considering the query and the document together rather than comparing pre-computed vectors, which makes it more accurate but also more expensive. Because it is expensive, it is applied not to the whole corpus but to the top candidates returned by the cheaper first-pass retrieval — the first pass casts a wide net cheaply, and the re-ranker accurately orders the catch. The improvement in retrieval quality from a re-ranking pass is typically substantial, which is why re-ranking is a near-standard component of a serious retrieval system.

87.6 Chunking Strategies

Chunking — how documents are split into the units that get embedded and retrieved — determines what is retrievable, and it is one of the most consequential design choices in a knowledge architecture.

87.6.1 The Choices

There are several chunking strategies. Fixed-size chunking splits the document into chunks of a set size; it is simple but may split a meaningful unit across a chunk boundary. Semantic chunking respects the document's structure, splitting at meaningful boundaries. Recursive chunking produces a

hierarchy of chunks at multiple sizes. Sentence-level chunking produces fine-grained chunks. And document-level chunking treats a whole short document as one chunk. The choice among them is not arbitrary; it determines what units of knowledge the retrieval system can find.

87.6.2 Content-Type-Specific Patterns

The right chunking strategy depends on the content type, and several content-type-specific patterns recur. Markdown and HTML are best chunked with awareness of their headers, splitting at the structural boundaries the markup defines. Code is best chunked at the function level, so a retrieved chunk is a coherent unit of code. Tables are best chunked by row or by section, preserving the tabular structure. Conversations are best chunked with awareness of turns. And PDFs are best chunked section-aware, after the document has been parsed into its structure. Matching the chunking to the content type is what keeps a chunk a coherent, retrievable unit of meaning.

87.6.3 The Trade-Offs

Chunking carries a core trade-off between precision and context. Small chunks give precision, because a retrieved small chunk is tightly focused, but they lose context, because the surrounding material that gives the chunk its meaning is not retrieved with it. Large chunks give context but lose precision, because the retrieved chunk contains material beyond what the query needed. Overlap between chunks adds redundancy that helps preserve context across boundaries. And using multiple chunk sizes costs more storage but improves recall. The chunking strategy is a deliberate navigation of this precision-context trade-off, made per content type.

87.7 Metadata and Filters

Metadata is as important as the content itself, because it is what makes retrieval precise, access-aware, and current.

87.7.1 What to Capture

A knowledge architecture should capture a recognisable set of metadata for each item: the source, recording which document the item came from; the authorship and date; the access controls that govern who may see it; the tags from the taxonomy that place it in the knowledge organisation; quality indicators that record how reliable it is; and currency indicators that record how fresh it is. This metadata is what the retrieval system uses to do more than blind similarity matching, and an item stored without it can be retrieved only by its text.

87.7.2 The Filtering

The captured metadata enables filtering, and several filters are essential. Access-control filters are mandatory, ensuring retrieval respects who may see what. Recency filters constrain retrieval to current information where currency matters. Quality filters exclude content known to be low-quality. And taxonomy filters narrow retrieval to the relevant part of the corpus for a given use case. Filtering on metadata is what turns retrieval from a search of the whole undifferentiated corpus into a search of exactly the appropriate subset of it.

87.8 Knowledge Freshness

Knowledge that becomes stale loses its value, and worse, a retrieval system that serves stale knowledge confidently misleads its users. Freshness is therefore a discipline a knowledge architecture must maintain.

87.8.1 The Discipline

The freshness discipline has several elements. It captures changes in the source systems, so that updates at the source flow into the knowledge store. It updates the index continuously, so the store reflects those changes. It deprecates stale content, removing or demoting what is no longer current. It carries currency indicators into retrieval, so the system knows how fresh a retrieved item is. And it periodically performs a full refresh, reconciling the store against the sources in case incremental updates have drifted. Together these keep the knowledge the system serves genuinely current.

87.8.2 The Architectural Patterns

Freshness is delivered through architectural patterns. Event-driven updates from the source systems push changes into the knowledge store as they happen, keeping the store close to current. Periodic batch reconciliation re-checks the store against the sources to catch what the event-driven updates missed. Update-latency monitoring measures how long it takes for a source change to become retrievable, so the freshness of the system is itself observable. And staleness can be treated as a quality signal, demoting content that has not been confirmed current. These patterns are how freshness becomes an engineered property rather than a hope.

87.9 Quality and Curation

Knowledge quality determines retrieval quality directly — a retrieval system can only return what is in the corpus, and a corpus of mixed quality returns mixed results.

87.9.1 The Quality Dimensions

Knowledge quality has several dimensions. Accuracy is whether the content is correct. Currency is whether it is up to date. Completeness is whether it covers what it should. Authority is the standing of the source it came from. And consistency is whether it agrees with the other sources in the corpus. A knowledge architecture that wants to manage quality must manage all of these dimensions, because content can be accurate but stale, or current but from a weak source, and each dimension is a different way for quality to fail.

87.9.2 The Curation Operations

Quality is maintained through curation operations. Subject-matter experts curate the content, judging what belongs and what does not. Outdated content is deprecated. Conflicts between sources are resolved, so the corpus does not contain contradictory answers. Quality is flagged, so low-quality content is marked. And the whole process runs as continuous improvement rather than a one-time clean-up. Curation is the human discipline that keeps a corpus's quality from degrading as content accumulates.

87.9.3 User Feedback

User feedback is a valuable quality signal, because users encounter the retrieval system's failures directly. Feedback that a retrieval returned something wrong or unhelpful is a signal worth capturing, retrieval-quality issues surfaced by users should be investigated, and the patterns in user feedback should inform the curation priorities — directing curation effort to the parts of the corpus that users find failing. User feedback closes the loop between the retrieval system's real-world performance and the curation that improves it.

87.10 Access Control

Knowledge access must respect the organisation's security policies, because a retrieval system that ignores access control is a mechanism for leaking data across the users who should not see one another's.

87.10.1 The Patterns

Access control in a knowledge architecture follows several patterns. Access control is enforced per user at retrieval time, so each user's retrieval returns only what that user may see. Controls operate at the document level and, where needed, the section level, so access can be as fine-grained as the policy requires. The access controls are inherited from the permissions of the source systems, so the knowledge store does not invent a separate, inconsistent permission model. And an audit trail records the access. These patterns are what make retrieval access-aware rather than indiscriminate.

87.10.2 Enforcement Points

Access control is enforced at two points, for defence in depth. A pre-retrieval filter on the metadata constrains the search so that disallowed content is never retrieved in the first place. A post-retrieval check on the results confirms that what is about to be returned is permitted. Using both points means a failure at one is caught by the other, and an audit on every retrieval records what was accessed. Defence in depth on access control matters because the cost of a failure — one user seeing another's confidential data — is high enough to justify two independent checks.

87.11 Operational Knowledge Infrastructure

A knowledge architecture is not a project that completes; it is a production system that must be operated, with components, service-level agreements, and an operating cadence.

87.11.1 The Components

The operational knowledge infrastructure has a recognisable set of components: the source connectors that bring content in, the processing pipelines that prepare it, the vector stores and indices that hold the embeddings, the knowledge graphs that hold the structured representation, the retrieval services that serve queries, the observability that makes the system legible, and the quality monitoring that watches the corpus. Each of these is a component to be operated, and together they make up the system that retrieval depends on.

87.11.2 The Service-Level Agreements

A knowledge infrastructure operated as a production system carries service-level agreements. There is the update latency, from a source change to that change becoming retrievable. There is the retrieval latency, how fast a query is answered. There are recall and precision targets, the quality of the retrieval itself. And there is availability. Setting and meeting these service-level agreements is what makes the knowledge infrastructure a dependable system that applications can build on rather than an unpredictable one.

87.11.3 The Operating Cadence

The knowledge infrastructure runs on an operating cadence at three timescales. Continuously, it runs ingestion, update, and monitoring — the constant background work of keeping the store current and watched. Periodically, it runs quality review and taxonomy updates — the regular maintenance of the corpus and its organisation. And episodically, it runs major refreshes and embedding-model migrations — the large, infrequent operations. Recognising these three timescales is what lets a knowledge infrastructure be operated deliberately rather than lurching between neglect and emergency.

87.12 Anti-Patterns

The failure modes of knowledge architecture recur, and each is the absence of a discipline this chapter has set out:

- **Document dump.** Documents are thrown into a vector store with no structure, no metadata, and no curation, so the retrieval inherits all the mess of the raw corpus.
- **No metadata.** The system relies on text alone for retrieval, with no metadata to filter on, so retrieval cannot be made precise, access-aware, or current.
- **Missing access controls.** Retrieval does not enforce access control, so data leaks across the users who should not see one another's.
- **No freshness.** Stale knowledge is served confidently, because no freshness discipline keeps the corpus current.
- **Wrong chunking.** The chunking does not match the content structure, so retrieved chunks are incoherent fragments.
- **No re-ranking.** The system relies on the first-pass retrieval alone, leaving the substantial quality improvement of a re-ranking pass unclaimed.
- **No quality discipline.** Low-quality content is allowed to contaminate the corpus and so the results.
- **Operational neglect.** The knowledge infrastructure is treated as a one-off project rather than an operated production system, and it decays.

Each anti-pattern maps to a discipline the chapter has developed — structured organisation, metadata, access control, freshness, content-aware chunking, re-ranking, quality curation, and operational

practice — and together they show that knowledge architecture fails through the neglect of these disciplines rather than through any inherent difficulty of retrieval.

87.13 The Working Position

Knowledge and document architecture is a substantial engineering discipline, and the quality of a retrieval system is largely the quality of the knowledge architecture beneath it. The architecture spans the taxonomies and ontologies that organise knowledge, the knowledge graphs that complement vector retrieval with structured reasoning, the vector stores designed for production scale and the re-embedding discipline they require, the hybrid search that combines vectors with keywords and filters, the content-aware chunking that determines what is retrievable, the metadata and filters that make retrieval precise and access-aware, the freshness discipline that keeps knowledge current, the quality and curation that keep the corpus reliable, the access control that keeps retrieval secure, and the operational infrastructure that runs it all as a production system.

Retrieval-augmented generation quality is, in the end, knowledge-architecture quality — the retrieval system answers the wrong question fast and confidently when the architecture beneath it is naive, and answers the right question when the architecture is deliberate. The anti-patterns — the document dump, the missing metadata, the absent access control, the stale corpus, the wrong chunking, the missing re-ranking, the undisciplined quality, the operational neglect — are common, and the remedies require sustained investment rather than a one-time build. The architecture decides which question gets asked, and a knowledge architecture built with the disciplines of this chapter is what lets the right question be asked and answered.

Chapter 88

AI Product Management and Lifecycle

“The PM who ran an AI feature like a classical feature shipped a useful feature once. The next release was a regression.”

—a recurring observation in AI product teams

Chapter Overview

AI products differ from traditional software products in ways that are not cosmetic refinements but structural differences in how the product behaves, how it is tested, how it evolves, and how it fails. Outputs vary across runs, evaluations replace fixed acceptance criteria, the model can change beneath the product without the product team doing anything, research and engineering blur into each other, and product-market fit shifts as foundation-model capability advances. This chapter develops AI product management as its own discipline: the discovery process for AI products, evaluation-driven development as the heartbeat, the product-research-engineering triangle, the methodology for A/B testing AI features, migration and sunset decisions, trust and safety as product concerns, roadmapping under capability uncertainty, the metrics that matter, and the operational practices that sustain an AI product team over time.

Learning Objectives

- Adapt product-management practice to the specific dynamics of AI products.
- Apply evaluation-driven development as the development heartbeat.
- Coordinate the product-research-engineering triangle productively.
- Discover and validate AI product opportunities through the adapted discovery process.
- A/B test AI features with the methodology their variability demands.
- Make migration and sunset decisions as product decisions.

88.1 Why AI Product Management Differs

AI products carry properties that classical software does not, and each property changes what a product manager must do. Naming them is the first step toward adapting the practice.

88.1.1 Probabilistic Output

An AI product's outputs vary across runs: the same input may produce different outputs each time, so the product cannot be specified or tested against a single expected answer. Acceptance criteria must instead accept variation and specify the distribution of acceptable outputs rather than a deterministic result. Testing focuses on the statistical properties of a body of outputs rather than on any single one, and the product manager must be comfortable with a product whose correctness is a distribution rather than a fact. This is the most fundamental way in which AI product management differs from classical product management, because it changes the meaning of “the product works.”

88.1.2 Model-Underneath

The foundation model beneath an AI product can change without the product team doing anything — a vendor updates the model, deprecates a version, or adjusts its behaviour — and the change silently alters the product's behaviour. This means the product team does not fully control its own product: a vendor's decision is a product event that the team must detect, assess, and respond to. The product manager must therefore decide on a version-management posture — whether to pin a model version for stability or to track the vendor's latest for capability — and must maintain a migration discipline so that a forced version change does not arrive as a surprise.

88.1.3 Research-Engineering Blur

In a classical product, improvement comes from engineering — building new features and fixing bugs. In an AI product, improvement can come from several places at once: a prompt change, a retrieval adjustment, a fine-tuning run, or a model selection decision. Each requires different skills, and the line between research and engineering is blurred in a way that classical products do not experience. The product manager must navigate this blur, understanding which kind of improvement to reach for in a given situation rather than treating all improvement as engineering work.

88.1.4 Capability Surprise

Foundation-model capability advances suddenly rather than gradually. A new model release can add capabilities that the product team did not plan for, collapsing a complex pipeline into a single call or making a previously infeasible feature trivial. Conversely, a deprecation can remove a capability the product depended on. Product strategy must be able to adapt to these discontinuities rather than assuming a smooth capability trajectory, and the product manager must keep current with the foundation-model landscape to recognise when a surprise is relevant.

88.1.5 Failure Modes

AI products carry failure modes that classical software does not: hallucination, bias, privacy violations through memorisation, and the various forms of misuse. Each of these requires product-level mitigation — it is not sufficient to leave them to engineering, because they affect how users perceive and trust the product. The product manager must own the mitigation strategy for these failure modes as part of the product design, not as an afterthought delegated to a safety team.

88.2 Discovery for AI Products

AI product discovery follows the same fundamental structure as any product discovery — find the problem, find the solution, validate both — but the middle step is distinctive, because the solution space is shaped by AI’s capabilities and limitations.

88.2.1 Problem Discovery

Problem discovery for AI products is no different from problem discovery anywhere: the task is to find where the genuine user pain is, what user task is currently effortful or poorly served, and what value pool justifies the AI investment. User research, observation, and data analysis drive problem discovery; AI does not change the fundamentals of understanding what users need. The discipline is to start from the problem rather than from the technology, because a product built from “we have AI, what can we do with it” is a solution in search of a problem.

88.2.2 Solution Discovery

Solution discovery is where AI changes the process. Foundation models make prototyping cheap and fast, because a prompt can be written in minutes and a working prototype can be assembled in hours rather than the weeks or months a classical feature would take. Quick feasibility checks are possible — a product manager can test whether a model can actually do what the feature needs within a day — and risk and failure-mode assessment can start early, because the prototype reveals the failure modes as well as the capabilities. Solution discovery for AI products is therefore faster and cheaper than for classical ones, and the product manager should use that speed to explore widely before committing.

88.2.3 Validation

Validation confirms that the solution actually works for users, and it has an AI-specific shape. The prototype is tested with real users on representative inputs, because a demo that works on curated examples may fail on the messy inputs real users provide. Quality is evaluated on a set that represents the diversity of production, not on the narrow set the prototype was tuned against. Cost and latency are assessed, because an AI feature that is too expensive or too slow is not a viable feature regardless of its quality. And edge cases are explored deliberately, because an AI product’s worst cases are often worse than a classical product’s.

88.2.4 The Ship Decision

The ship decision for an AI feature is a judgement against four criteria together: quality meets the bar for the use case, the cost economics support the feature at scale, the failure modes are manageable with the mitigations in place, and a trust-building plan exists for how users will come to rely on the feature. A classical ship decision checks quality and scope; an AI ship decision must also check economics and trust, because an AI feature that is good but unaffordable, or good but untrusted, does not succeed.

88.3 Evaluation-Driven Development

Because an AI product's correctness is a distribution rather than a deterministic fact, the development process must be organised around evaluation — measuring the distribution continuously — rather than around fixed acceptance tests. Evaluation-driven development is the heartbeat of AI product work.

88.3.1 The Pattern

The pattern is straightforward. A golden dataset is established for the feature, representing the inputs the feature will see in production. Every change to the feature — a prompt edit, a retrieval adjustment, a model swap — is evaluated against the dataset before it ships. Regressions are blocked: a change that degrades the measured quality is not deployed. Improvements are measured and recorded. And the evaluation suite expands as edge cases are discovered in production, so the dataset grows to cover the system's actual operating envelope. The pattern is what turns AI development from “try it and see” into a measured, evidence-gated process.

88.3.2 The Discipline

The discipline behind the pattern has several standing rules. No deployment happens without evaluation evidence — no change ships “on vibes.” The evaluation suite is maintained as a first-class artefact, as carefully as any other part of the system. Production failures are investigated, and every investigated failure adds a new case to the suite, so the system is permanently protected against that failure recurring. And cross-team learning happens through shared evaluations, so that teams learn from each other's discovered edge cases rather than each team discovering them independently.

88.3.3 Beyond Offline Evaluation

Offline evaluation against the golden dataset is necessary but not sufficient, because the dataset is a model of production and not production itself. Online evaluation in production — running a judge model or sampling for human review on live traffic — catches the gaps the offline dataset misses. User-feedback capture closes the loop between the user's experience and the evaluation, and outcome measurement connects the feature's quality to the business result it was meant to produce. A product practice that runs only offline evaluation is a practice that evaluates its model of production rather than production itself.

88.4 The Product-Research-Engineering Triangle

An AI product is built by three functions working together — product management, research or ML science, and engineering — and the coordination between them is as consequential as the quality of any one.

88.4.1 Product Management

Product management owns the user problem framing, the quality bar, the value proposition, the cost and economics decisions, and the roadmap. It is the function that connects the AI product to the user and the business, and its defining contribution is judgement about what to build and how good it must be. An AI product manager who cannot set a quality bar — who defers that judgement to research or engineering — is a product manager who has given away the most important decision.

88.4.2 Research and ML Science

Research or ML science owns model selection, prompt engineering, fine-tuning, evaluation methodology, and capability assessment — the technical heart of what makes the AI product work. Its defining contribution is the connection between the user's problem and the model's capability: knowing which model, which technique, and which configuration will actually solve the problem at the quality the product requires.

88.4.3 Engineering

Engineering owns the production systems, the infrastructure, the reliability, the integration with the rest of the organisation's systems, and the observability. Its defining contribution is turning a working prototype into a running, reliable, operable production system, which is a different thing and a harder one.

88.4.4 The Coordination

The three functions coordinate through shared artefacts and shared practice. The evaluation suite is shared, so all three functions measure quality the same way. The roadmap is joint, so the three functions' plans are aligned. Cross-function understanding is deliberately built, so product understands the model's limitations and research understands the user's needs. Co-location, whether physical or virtual, keeps communication low-friction. And shared metrics — quality, usage, cost, business outcome — give all three a common picture of how the product is doing. The coordination is what makes the triangle productive; without it, the functions pull in different directions and the product stalls.

88.5 A/B Testing AI Features

A/B testing an AI feature follows the same logic as any A/B test — expose different users to different variants and measure the difference — but the AI-specific properties of the product add considerations a classical A/B test does not face.

88.5.1 The Pattern

The standard A/B pattern assigns users randomly to variant A (the current system) and variant B (the candidate), measures an outcome over a defined horizon, and compares the two statistically. The pattern is unchanged for AI features, and it is the right way to measure whether a change genuinely improves the user experience, because it controls for the confounders that offline evaluation cannot.

88.5.2 AI-Specific Considerations

Several considerations are specific to AI A/B tests. Output variability means that two users in the same variant may have different experiences, which increases the noise in the measurement and demands larger sample sizes. Long-term effects on user behaviour — trust building or trust erosion — may not be visible in a short test horizon. Spillover effects across users, where one user's experience influences another's, can confound the measurement. Concurrent changes to other parts of the system introduce

confounders. And multiple comparisons across the many metrics an AI feature produces raise the risk of false positives. These considerations do not change the A/B pattern but they change how carefully it must be executed and interpreted.

88.5.3 Multi-Armed Bandits

For high-volume features, a multi-armed-bandit approach adaptively allocates traffic toward the better-performing variant as evidence accumulates, rather than waiting for a fixed-duration test to complete. This is useful when the feature serves enough traffic that the adaptive allocation converges quickly, and it trades the clean statistical interpretation of a fixed A/B test for a faster path to the better variant.

88.5.4 Pre-Production Evaluation

Before an AI feature reaches a production A/B test, pre-production evaluation reduces the risk. Offline evaluation against the golden dataset catches the obvious regressions. Shadow-mode operation runs the candidate alongside production, recording its outputs without serving them, to confirm it behaves as expected on real traffic. And holdout evaluation reserves a portion of the evaluation data to detect overfitting. Pre-production evaluation is what keeps a flawed candidate out of the A/B test, where its flaws would be experienced by real users.

88.6 Migration and Sunset Decisions

Model and vendor migrations are product decisions, not merely technical ones, because they change what the user experiences. Sunset decisions — the decision to retire a feature — are equally product decisions, and both deserve the product manager's direct involvement.

88.6.1 Migration Triggers and Process

A migration is triggered by a vendor deprecation, a new model offering better capability, a cost-optimisation opportunity, a quality regression in the current model, or a compliance change that requires a different vendor. The process evaluates the candidate against the current system on the full regression suite, assesses the cost and latency, compares the failure modes, builds a migration plan with an explicit rollback path, and deploys the candidate through a gradual rollout with monitoring. A migration shipped without a regression suite and a rollback path is a migration that discovers its problems through its users.

88.6.2 Sunset Triggers and Process

A sunset is triggered when a feature has low adoption, when its cost outweighs its value, when quality issues prove unresolvable, when a newer feature subsumes its purpose, or when a regulatory or risk shift makes it untenable. The sunset process communicates the retirement to users, migrates them to an alternative if one exists, handles data retention and disposal, preserves the audit trail, and captures the lessons from the feature's life. A feature that lingers past its value is a cost the product carries with no return, and the discipline of sunset is what prevents features from accumulating indefinitely.

88.7 Trust and Safety in Product

Trust and safety are product concerns, owned by the product manager alongside the trust-and-safety function, and the product design must reflect them. Users should be told when AI is involved in what they experience. The system's limitations should be communicated honestly rather than hidden or minimised. Users should have control over AI features, able to turn them on or off. Feedback mechanisms should be easy to use, so that a user who encounters a problem can report it. And the incident-response path should be accessible, so that a serious issue reaches the right team quickly. The design patterns that express these commitments include clear labelling of AI-generated content, confidence indicators where they are appropriate and honest, citation and source links so users can verify, easy issue-reporting, and user-friendly fallback paths for when the AI does not serve. Trust is built by these patterns over time, and it is lost by their absence in a single incident.

88.8 Roadmapping and Metrics

AI product roadmapping must accommodate capability uncertainty, and the metrics that guide it must measure what actually matters.

88.8.1 Roadmap Horizons

An AI product roadmap is usefully divided into three horizons. The near-term horizon covers features buildable with current models and capability. The mid-term covers features that will become buildable as anticipated capability arrives — features the team can plan for but cannot yet build. The long-term covers directional bets on where capability evolution may go, too uncertain to plan against but worth positioning for. The distinctive challenge is that capability changes can invalidate the roadmap at any horizon: a new model release can move a mid-term feature to near-term or make a long-term bet irrelevant. The roadmap must therefore be reviewed more frequently than a classical product roadmap, and the product manager must stay current with the foundation-model landscape to know when a revision is needed.

88.8.2 What to Measure

The metrics for an AI product divide into four groups. Quality metrics come from the evaluation suite, from user-reported quality, and from failure rates by category. Usage metrics cover adoption, frequency, depth, and retention. Business metrics connect the feature to the outcome that justified it — the value delivered, the cost, the return on investment. And health metrics cover reliability, latency, cost per unit, and error rate. The discipline is to measure across all four groups rather than only one, because a feature that scores well on quality but costs too much, or that is widely adopted but does not deliver the business outcome, is not succeeding.

88.9 Sustaining AI Product Teams

An AI product team must be sustained over time, and the sustainability rests on three things: the right composition, the right culture, and the right cadence. The team should include product, research, engineering, and design, with domain expertise for the context the product serves and operations

and trust-and-safety capacity. The culture should be evaluation-driven, iterative, user-focused, and safety-conscious — a culture where decisions are made on evidence, learning from failure is normal, the user is always in view, and safety is treated as a first-class concern rather than a constraint. And the cadence should combine a sprint rhythm with evaluation gates, quarterly planning, an annual strategy refresh, and continuous tracking of the foundation-model capability landscape. A team with the right composition, culture, and cadence can sustain an AI product through the years of iteration that a successful product requires; one missing any of the three will struggle.

88.10 Anti-Patterns

The failure modes of AI product management recur, and each maps to a discipline this chapter has set out:

- **Demo-driven development.** The demo is shipped as if it were the product, ignoring the distribution, the edge cases, and the operations that a real product requires.
- **Feature-only thinking.** The product team focuses on building features and neglects evaluation and operations, so the features ship without the infrastructure to sustain them.
- **Premature commitment to one foundation model.** The product is built so tightly against one model that a migration is prohibitively expensive, and the hexagonal discipline is absent.
- **No evaluation suite.** Changes are shipped on the strength of a developer's confidence rather than measured evidence, and regressions arrive as user complaints.
- **Capability ignorance.** The product manager does not understand the model's limitations, and the product promises what the model cannot deliver.
- **Research-engineering siloing.** Research improvements do not ship because the engineering path is not built, and engineering shipping is blocked by research handoffs that never complete.
- **No sunset discipline.** Features linger past the point where they deliver value, accumulating cost and maintenance burden with no return.
- **Trust theatre.** Trust features exist in name — a label here, a disclaimer there — but do not genuinely help users understand or control the AI, so the appearance of trust is present while the substance is not.

Each anti-pattern maps to a remedy the chapter has developed, and together they show that AI product management fails not through a single dramatic mistake but through the absence of the disciplines that distinguish an AI product from a classical one.

88.11 A Note Before Moving On

AI product management is a distinctive discipline, and treating an AI product as classical software with AI inside is the foundational mistake. The product's probabilistic output, its model-underneath dynamics, the research-engineering blur, the capability surprises, and the AI-specific failure modes

all demand adapted practice. Evaluation-driven development is the heartbeat; the product-research-engineering triangle is the team structure; A/B testing, migration, and sunset are the lifecycle decisions; trust and safety are product concerns; and the metrics that matter span quality, usage, business outcome, and health. The epigraph captures the lesson: the product manager who runs an AI feature like a classical feature ships a useful feature once, and the next release is a regression, because the disciplines this chapter develops are what keep an AI product improving rather than decaying.

AI Engineering Disciplines: Regional Perspective — Africa

“Foundation-model strategy from Lagos looks different than foundation-model strategy from Palo Alto. Both are correct for their context.”

—a recurring observation in African AI engineering

Chapter Overview

The AI engineering disciplines developed in the preceding chapters — foundation-model strategy, inference infrastructure, knowledge architecture, AI product management — were framed against a Northern default, and deploying them in African contexts requires genuine adaptation. Data-sovereignty considerations shape which foundation models may be used and how. African cloud regions, still fewer and less GPU-rich than their Northern counterparts, reshape inference architecture. Low-resource African languages reshape knowledge architecture in ways that English-centric systems do not anticipate. And emerging-market unit economics constrain AI product management more tightly than Northern economics do. This chapter develops the African context for each of these engineering disciplines, with focus on the continent’s anchor markets and the recurring theme that the adaptations, once made, frequently produce architectures that are more efficient and more robust than the ones they adapted from.

Learning Objectives

- Recognise the distinguishing realities: data sovereignty, African cloud regions, low-resource languages, and emerging-market unit economics.
- Adapt foundation-model strategy for African sovereignty considerations.
- Build inference infrastructure shaped by African cloud regions and edge patterns.
- Architect knowledge infrastructure for low-resource African languages.
- Manage AI products under emerging-market economics.

89.1 Foundation-Model Strategy Under African Sovereignty

Foundation-model strategy in African contexts must begin with sovereignty, because the data-protection laws and the national-sovereignty postures of African jurisdictions constrain which models may be used and how.

89.1.1 Data-Sovereignty Considerations

The data-protection landscape across the continent is diverse and actively developing. South Africa's POPIA, Nigeria's NDPA, Kenya's DPA, Egypt's PDPL, and Morocco's Law 09-08 each impose their own requirements on personal-data handling, and the cross-border data-flow restrictions vary from jurisdiction to jurisdiction. Sector-specific requirements in banking, health, and defence add further layers, and government and regulator engagement on sovereignty is intensifying. The consequence for foundation-model strategy is that a model choice that works legally in one African market may not work in another, and a pan-African operation must evaluate the foundation-model options against each market's sovereignty requirements rather than assuming a single portfolio will serve.

89.1.2 Foundation-Model Options

The options available to an African deployment form a spectrum of sovereignty. At one end, a frontier provider's hosted endpoint with regional residency — where the provider offers an African region — keeps the data within the jurisdiction while providing frontier capability. Where no African region exists, the next option is a hosted endpoint with cross-border data flow under contractual safeguards, which sacrifices residency for capability. Open-weight models self-hosted in an African cloud region — Cape Town or Johannesburg on the major clouds — give full data control at the cost of operational complexity. Open-weight models on-premises go further, keeping everything within the organisation's own walls. And hybrid patterns — frontier models for the tasks where capability matters most, open-weight self-hosted for the tasks where sovereignty matters most — are the most common production arrangement, using each option for what it is genuinely best at.

89.1.3 The Open-Weight Advantage

The open-weight family carries a specific strategic advantage in African sovereignty contexts, and it is worth naming directly. An open-weight model self-hosted in-region gives the organisation complete control over its data, its model version, and its hosting jurisdiction — no data leaves the organisation's control, no vendor update changes the system's behaviour without the organisation's consent, and no cross-border flow is needed. This makes open-weight models the natural default for sovereignty-sensitive workloads, and African deployments have adopted them aggressively for exactly this reason. The open-weight advantage is not about cost alone — it is about control, and in a sovereignty-conscious market, control is itself a competitive position.

89.2 Inference Infrastructure in African Contexts

Inference infrastructure for African deployments is shaped by the geography and capacity of Africa's cloud presence, by the economics of serving African users, and by the prominence of edge deployment.

89.2.1 African Cloud Regions

The African cloud regions — primarily Cape Town and Johannesburg on the major providers, with limited availability in other markets — carry less GPU capacity than the Northern regions they are measured against. This means African inference workloads that need in-region hosting face a capacity constraint, and the overflow either goes to a Northern region, accepting the latency and sovereignty cost, or goes to self-hosting, accepting the operational cost. The capacity is growing as the cloud providers invest, but a capacity-constrained deployment is the present reality, and an inference architecture for African deployment must be designed around it.

89.2.2 Latency and Routing

When inference runs in a Northern region because the African region lacks capacity, the network latency is material — on the order of one hundred and fifty to three hundred milliseconds for the round trip, a figure that adds directly to every request. For interactive workloads this additional latency is perceptible and sometimes intolerable, which makes in-region or edge inference not merely a sovereignty preference but a latency requirement. A routing architecture that sends latency-tolerant workloads to Northern regions and latency-sensitive workloads to in-region or edge inference is the standard African pattern, and it is the latency reality rather than a theoretical preference that drives it.

89.2.3 Edge Inference

Edge inference is more central to African AI than to Northern AI, because the use cases that demand it are more prevalent. Mining sites, agricultural installations, and remote service locations frequently lack reliable connectivity, which means inference that must happen at those sites must happen on-device, without a cloud link. The edge hardware — Jetson, Coral, mobile NPUs — is therefore a primary deployment target rather than a secondary one, and the inference-infrastructure chapter's treatment of edge inference is, for African deployments, a core chapter rather than an appendix.

89.2.4 Cost-Engineered Inference

African unit economics constrain inference cost more tightly than Northern ones, and the inference architecture must reflect that. The levers are the same ones the inference-infrastructure chapter develops — smaller models, aggressive quantisation, continuous batching, caching — but they are applied more aggressively, because the cost ceiling is lower. An inference architecture that would be acceptably efficient in a Northern market may be unaffordably expensive in an African one, and the engineering effort that closes the gap is not optional.

89.3 Knowledge Architecture for African Languages

Knowledge architecture for African deployments faces a challenge that English-centric systems do not: the languages the system must serve are, by the metrics of NLP resources, low-resource.

89.3.1 Low-Resource Language Challenges

Swahili, Hausa, Yoruba, Igbo, Amharic, Zulu, Xhosa, and many other African languages have far fewer digital text resources, far fewer evaluation datasets, and far fewer pretrained models than English, French, or Arabic. The consequence for knowledge architecture is that the embedding models, the retrieval models, and the evaluation datasets that the knowledge-architecture chapter assumes are readily available may not exist for the languages the African system serves, or may exist at a quality that is insufficient for production. Building the knowledge architecture for an African deployment therefore includes building the language resources the architecture depends on, not only assembling the architecture itself.

89.3.2 Embedding Quality

The quality of retrieval depends on the quality of the embedding model, and embedding quality for African languages is uneven. General multilingual embedding models cover some African languages reasonably well and others poorly, and the unevenness must be measured rather than assumed. For the languages where general models underperform, domain-specific fine-tuning of the embedding model, or the adoption of an African-language-specific embedding model where one exists, is the remedy. The discipline is to evaluate embedding quality per language on the actual corpus rather than relying on the model provider's aggregate benchmark, because an aggregate that includes the well-served languages can hide the underperformance on the poorly served ones.

89.3.3 Multilingual Retrieval

A knowledge architecture serving African users frequently serves them in several languages at once, and the retrieval must handle that. Cross-lingual retrieval — a query in one language retrieving a relevant document in another — is needed where the knowledge base and the user's language do not coincide, and the retrieval quality of cross-lingual pairs must be evaluated explicitly rather than assumed. The chunking, metadata, and taxonomy layers of the knowledge architecture must all be language-aware, so that the system can operate across languages rather than being siloed within one.

89.3.4 African-Context Knowledge Curation

The knowledge content itself must reflect African contexts, and generic Northern knowledge bases do not. Local regulatory content, local market knowledge, local cultural context, and local language variety all need to be curated into the knowledge architecture, and the curation is a deliberate, sustained investment rather than a one-time import. An African knowledge architecture that draws only on Northern knowledge sources will answer questions about African contexts with Northern assumptions, and the answers will be wrong in ways the system cannot detect.

89.4 AI Product Management Under Emerging-Market Economics

AI product management in African markets operates under tighter economic constraints than in Northern markets, and the constraints shape the product discipline in specific ways.

89.4.1 Unit-Economics Pressure

Average revenue per user is structurally lower in many African markets, and the cost per AI call must be brought within that lower ceiling or the product simply does not work economically. This means the cost-engineering levers — smaller models, aggressive routing, caching, quantisation — are not optimisation refinements but product requirements, and the product manager must treat per-call cost as a first-class product metric alongside quality and adoption. A product that is good but unaffordable at African unit economics is a product that cannot ship.

89.4.2 Mobile-First Product Design

African users are predominantly mobile, and AI product design must be mobile-first rather than desktop-adapted. This means designing for smaller screens, for variable connectivity, for the input methods mobile users actually use, and for the lower bandwidth that mobile networks in many African markets provide. A product designed for a desktop with a stable broadband connection and adapted for mobile is not a mobile-first product, and the adaptation is frequently inadequate. Building for mobile from the start, rather than shrinking a desktop design, is the product-design discipline that African markets demand.

89.4.3 Trust Building in Low-Trust Environments

In markets where digital trust is still building, the trust-and-safety features of an AI product carry more weight than in markets where users are accustomed to digital systems. Clear communication about what the AI does and does not do, visible human-oversight paths, easy reporting of problems, and transparent handling of errors are not refinements but necessities, because a user whose trust in digital systems is still forming is a user who will abandon a product at the first unexplained failure. Trust building in African markets is slower and more consequential than in Northern ones, and the product design must invest in it accordingly.

89.4.4 Localisation Beyond Translation

Localisation for African markets goes well beyond translating the interface. It includes local-currency handling, local-date and local-name formats, culturally appropriate communication styles, and local-context examples and defaults. A product “localised” only by translating its text is a product that feels foreign to its users, and the feeling is a barrier to adoption. Genuine localisation is the product-level expression of the cultural-context adaptation that runs through the African chapters of this book.

89.5 Recurring Failure Patterns

The failure modes of AI engineering in African contexts recur, and each is the failure to make one of the adaptations this chapter has set out:

- **Sovereignty afterthought.** The foundation-model choice is made on capability alone, and the sovereignty constraints are discovered when the system is already built, forcing an expensive architecture change.
- **Northern-region default.** All inference runs in a Northern region, and the latency and sovereignty

costs are accepted rather than addressed, because the architecture was not designed for in-region or edge deployment.

- **English-only knowledge.** The knowledge architecture is built for English, and the African-language users are served by translation rather than by native-language retrieval, so the retrieval quality on non-English queries is poor.
- **Northern unit economics assumed.** The product is designed for Northern revenue-per-user levels, and its cost structure does not close under African economics, so it cannot be priced viably.
- **Desktop-adapted mobile.** The product was built for desktop and shrunk for mobile, and the mobile experience is a degraded version of the desktop one rather than a product designed for how African users actually work.

Each pattern maps to an adaptation the chapter has developed — sovereignty-first model strategy, in-region and edge inference, multilingual knowledge architecture, cost-engineered product economics, and mobile-first design — and together they show that AI engineering fails in African contexts when the engineering is transplanted from a Northern default rather than adapted to the African reality.

89.6 The Working Position

The AI engineering disciplines — foundation-model strategy, inference infrastructure, knowledge architecture, and AI product management — each require genuine adaptation for African deployment. Foundation-model strategy must start from sovereignty, with the open-weight family carrying a specific advantage for sovereignty-sensitive workloads. Inference infrastructure must navigate limited African cloud capacity, material cross-region latency, the prominence of edge deployment, and tighter cost ceilings. Knowledge architecture must invest in embedding quality for low-resource languages, cross-lingual retrieval, and African-context curation. And product management must close unit economics under lower revenue per user, build mobile-first, invest in trust in lower-trust environments, and localise genuinely rather than superficially.

The epigraph’s observation holds: foundation-model strategy from Lagos looks different from foundation-model strategy from Palo Alto, and both are correct for their context. The adaptations this chapter develops are not concessions to difficulty but the engineering that makes the disciplines work in the context they actually serve, and the architectures they produce — cost-engineered, sovereignty-aware, edge-capable, multilingual — are frequently more efficient and more robust than the ones they adapted from.

Chapter 90

The Trawler and the Commission: When Productivity Outruns Measurement

“The fishermen complain that the trawlers have ruined the grounds. The science cannot yet say whether they are right or wrong. We recommend that records be kept.”

—paraphrased from the conclusions of the Royal Commission on Trawl Net and Beam Trawl Fishing, 1885

Chapter Overview

This chapter steps out of the eighteenth century and into the nineteenth, by way of the herring boats and the cod grounds of the North Sea. The previous reflective chapters have leaned on the European Enlightenment for their analogies. This one looks at what happened a century later, when industrial technology arrived on the working coasts and the disciplines required to manage it had not yet been built. The story is of the steam trawlers that, between 1881 and 1900, transformed the British and North-Sea fishing industries; of the working fishermen who saw the consequences before any scientist could confirm them; of the two Royal Commissions of Inquiry that ran in the 1860s and 1880s; and of the modest decision in 1889 to start recording, every year, what the trawlers actually caught. The decision to start recording is the part of the story this chapter is for.

Learning Objectives

- Recognise that the gap between practitioner intuition and formal measurement is itself an architectural problem.
- Use the steam-trawler era to understand what happens when productivity outruns the institutions that measure its effects.
- See in the Royal Commissions and the 1889 catch records the slow building of evidence that eventually becomes a discipline.
- Carry the trawler image into the cross-cutting concerns of Part VI: security, privacy, reliability, scalability, ethics.
- Read the chapters ahead as the contemporary instantiation of work that took fisheries science decades to develop.

90.1 When the Catch Outruns the Records

The eighteenth-century stories that anchor most of this book are stories of disciplines being built from nothing. The 1880s present a different problem: disciplines that had been adequate for centuries finding themselves outpaced by a new technology that arrived faster than the institutions around them could absorb. The steam trawler is the clearest case of that mismatch, and the slow British decision to start writing down what was being landed is the canonical case of the response. Both belong in a book about enterprise AI, where the productivity of the new tools is currently establishing itself faster than the institutions that would measure their effects.

90.2 The Steam Trawler**90.2.1 The Boats and the Catch**

In 1881 the first purpose-built steam screw trawlers were launched at Scarborough and at Grimsby. By 1885 Hull had its own. By the end of the decade, the steam trawler had begun to displace the sail-powered beam trawlers that had operated off the British coasts for several centuries. The new boats were larger, more reliable, weather-independent, and far more productive: a steam trawler caught, on average, about four times as much fish as a sailing trawler operating in the same waters. Within twenty years, the steam fleet had extended its operating range from the North Sea to Greenland, north Norway, the Barents Sea, Iceland, and the Faroe Islands.

The fish came ashore at Grimsby, at Hull, at Aberdeen, at Boulogne, at Yarmouth, and from those ports moved inland on the railways that had been laid across Britain in the previous generation. The combined effect of the steam trawler and the railway was that fresh sea fish, which until the middle of the nineteenth century had been a luxury inland, became a working-class staple. The British fish supper, which a contemporary observer in the 1850s could not have ordered in any city of the interior, was by 1900 a standard meal in Manchester, Birmingham, Sheffield, and Leeds.

This was, by every conventional metric of the late nineteenth century, progress. The fleet was growing.

The catches were rising. The prices were falling. The workforce was expanding. The downstream industries — the smoking sheds, the salting yards, the box-makers, the ice trade, the railway operators — were thriving. The total tonnage landed at British ports tripled between 1880 and 1900.

90.2.2 What the Fishermen Said

What the working sail-trawler fishermen said, almost from the beginning, was that the new boats were destroying the grounds.

The complaint was specific. The steam trawlers, with their larger nets and their ability to work for longer and to return to the same grounds more frequently, were not merely taking more fish. They were, the older fishermen argued, scraping the seabed in ways that destroyed the bottom habitats where fish spawned and where the smaller organisms that fed the fish lived. They were catching juvenile fish in numbers that prevented the next generation from reaching maturity. They were over-working specific grounds to the point of leaving nothing behind. The traditional knowledge of where to fish, when to fish, and how much to take — knowledge that had been built up over generations and was held in the heads of the older skippers — was, the fishermen argued, being overridden by a generation of new operators who simply did not have time to learn the grounds before they had finished them.

The fishermen's claim was not founded on scientific data, because the scientific data did not exist. It was founded on what working practitioners observe when they spend their lives on the water. The men who had fished the Dogger Bank for thirty years could tell, from the size of the fish they were landing and the species mix in their nets, that something was different. They were right. The historical record, reconstructed by marine biologists more than a century later, confirms what the working fishermen of the 1870s and 1880s had already been saying.

What they did not have was a way to make their claim count in policy.

90.2.3 The First Commission and the Second

The British government convened a Royal Commission of Inquiry into trawling in 1863–66, and a second, more substantial one in 1883–85. The second commission, the one whose conclusions matter most for the purposes of this chapter, was chaired by Spencer Walpole and ran for three years. It interviewed several hundred witnesses — fishermen, fish merchants, ship-owners, marine scientists, naval officers, and railway operators. The minutes of evidence ran to several thousand pages. The commissioners visited fishing ports across England, Scotland, Ireland, and Wales. They watched trawlers at work. They examined nets. They read letters from grieving fishing communities whose fathers and sons had been working the grounds for generations and who now believed those grounds were being destroyed.

The commission's conclusion, published in 1885, was a careful and honest piece of work. It said, in effect: *We cannot determine whether trawling is destroying the grounds. The evidence we have heard is contradictory and the data we would need to resolve it does not exist. We therefore recommend that the data be collected.* The commission did not impose trawling restrictions. It did not vindicate the working fishermen. It did not condemn the steam fleet. It declined to do any of these things, on the grounds that responsible policy could not be made without evidence, and evidence required

measurement, and measurement had not been done.

The fishermen who had spent three years giving testimony were, by any reasonable reading, disappointed. The science that they could already see from the deck of a sailing smack had not been confirmed by a process that demanded confirmation in its own terms. The trawler fleet continued to grow. The grounds continued to deteriorate. The fishermen continued, season after season, to bring in smaller cod and fewer mature plaice and to mark on their charts the places where the fishing was no longer worth the fuel to reach it.

90.2.4 The Decision to Record

What the commission did do — and what, in the long run, mattered most — was to recommend that statutory catch records be kept. Beginning in 1889, every commercial trawler landing fish at a British port was required to report what it had caught, where, and in what quantity. The records have been kept, with various improvements and disruptions, every year since.

It is from those records that the modern science of fisheries management was eventually built. Walter Garstang's 1900 paper, "The Impoverishment of the Sea," was the first scientific publication to draw on the new catch data and to argue, against considerable industry resistance, that North Sea fish stocks were declining. Garstang's work was followed, over the next fifty years, by the gradual development of stock-assessment models, of population dynamics theory, of by-catch analysis, of seasonal closures and quota systems, and eventually of the international fisheries management regimes that govern the North Atlantic today. None of it would have been possible without the boring administrative decision, taken in 1889, to start recording what was actually being caught.

The fishermen who testified in 1885 did not live to see the science vindicate them. Many of their grandchildren did. The institutions that exist now to manage North Sea fisheries are the eventual answer to a problem the men of the 1880s had identified correctly but could not prove in the form their society demanded.

90.3 What Had Failed, and What Had Not

It would be easy to read the trawler story as a tale of progress betraying the working classes, and it is partly that. It would also be easy to read it as a tale of science failing to keep pace with industry, and it is partly that too. Neither reading quite captures the architectural lesson.

What had failed was the matching, in time, between the productivity of a new technology and the institutional capacity to measure its effects. The steam trawler arrived in the 1880s as a substantially completed engineering system. The institutional infrastructure to assess its effects — the catch records, the statistical methods, the trained marine biologists, the regulatory authorities, the international agreements, the enforcement bureaucracies — arrived over the following century. The gap between the two was paid for by the fish stocks, the fishing communities, and the long-term productive capacity of the North Sea.

What had not failed was the practitioner knowledge of the working fishermen. The older skippers were correct about what was happening to the grounds. Their problem was not that they did not know;

their problem was that the institutions of their society did not yet have a way to take their knowing seriously. They could not produce the evidence in the form that the inquiry demanded, because the form the inquiry demanded had not yet been invented. The 1889 catch records, in the long run, were the institutional answer to that problem.

There is a kind of overconfidence that is technological, in the sense that it assumes a new tool can be deployed at scale before its effects have been understood. The steam-trawler era is the canonical nineteenth-century example. Enterprise AI in 2026 is in a directly analogous position. The productivity of the new tools is being established before the institutional capacity to measure their effects — on training data, on labour markets, on public trust, on epistemic ecosystems, on long-term producer incentives — has been built. The practitioners who are closest to the work are reporting what they see, often in advance of any formal evidence. The contemporary equivalent of the 1889 catch record is the contemporary equivalent of the work that will, eventually, let the practitioners' observations count.

90.4 The Trawler, the Commission, and What They Make Visible

90.4.1 Measurement Begins as an Administrative Decision, Not a Scientific One

The 1889 catch records are not a scientific instrument. They are an administrative form. A trawler captain landing fish at Grimsby was required, by the harbour-master's clerk, to report tonnage and species. The form was bureaucratic. The science came later, when other people decided to look at what the forms said and to draw conclusions from the pattern across decades and across boats.

The same is true of every enterprise AI observability system. The trace store, the eval log, the cost-attribution database, the incident register, the model card archive: none of these are scientific instruments. They are administrative records. The science — the evaluation methodology, the failure-mode taxonomies, the cost forecasting, the safety analyses — comes later, when other people draw conclusions from the pattern across systems and across years. The chapters of Part VI describe the contemporary administrative forms. The reader who treats them as boring should remember that the catch records looked boring too.

90.4.2 Practitioner Knowledge Is Evidence Before Anyone Has Named It

The 1880s fishermen knew what was happening to the grounds long before any scientist could confirm it. Their knowing was not formally evidential, in the sense of being capable of producing a policy outcome. It was real, however, and the people who held it were correct.

Enterprise AI in 2026 has a similar layer of practitioner knowing. Engineers in operations, in security, in product, and in the lower-paid roles that AI systems are increasingly used to manage, see things in the daily operation of the systems that the formal measurement layer does not yet capture. Their reports are often dismissed as anecdotal, exactly as the trawler fishermen's reports were. Some of those reports will turn out, like the trawler fishermen's, to have been correct in advance of the data that could prove them. The mature engineering organisation builds channels by which that knowing can be heard before the data confirms it, not after.

90.4.3 The Slow Institutional Response Is the Real Response

The Royal Commission of 1885 did not solve anything. The fishermen who testified did not get the policy outcome they wanted. The catch records that the commission recommended did, however, eventually produce the science that produced the regulations that, in the second half of the twentieth century, prevented even worse outcomes than those that occurred. The institutional response was real. It was just slow.

Enterprise AI in 2026 is, in many jurisdictions, somewhere between 1863 and 1885 in fishing terms. The first commissions of inquiry are being convened. The first formal measurements are being defined. The first reporting obligations are being negotiated. The structures that will, in the second half of the twenty-first century, govern AI in mature ways are being built now, slowly, in the apparently boring committee rooms where reporting frameworks are designed. The reader who has spent any time in those rooms knows that they do not feel important. They are. The 1889 catch records are evidence.

90.4.4 Once a Resource Has Been Depleted, Recovery Is Partial at Best

North Sea cod stocks have not, more than a century after the steam trawler arrived, recovered to their pre-industrial levels. Some species have collapsed entirely and not returned. The institutional response that eventually arrived did not undo the damage; it slowed further damage and, in some species and some seasons, allowed partial recovery. The historical baseline is gone.

There is a version of this lesson that applies to enterprise AI. Training data scraped without consent does not return to the commons. Public trust in algorithmic decisions, once lost, does not easily recover. Labour markets disrupted by AI substitution do not return to their previous structure even if the AI is later regulated. Languages, knowledge traditions, and craft practices displaced by AI-mediated alternatives may not survive the displacement. The institutional response, when it comes, will be like the response to the steam trawler: real, eventually effective at preventing further damage, but unable to restore what was lost in the period before the response arrived.

This is not an argument for technological pessimism. The North Sea is, today, a managed and partially productive fishery. The system that manages it is one of the more successful examples of international scientific cooperation that the twentieth century produced. The institutional response, slow as it was, has been worth building. The architectural lesson is that the response should not be left until the damage has compounded, and the historical baseline against which the response is judged should not be the post-damage condition but the pre-damage one.

90.5 The African Coastline and What Africa Inherits

The North Sea is one coast. The contemporary version of the steam-trawler problem is global, and African coastlines have been particularly heavily affected.

West African waters — off Senegal, Mauritania, Guinea, Sierra Leone, Ghana, Nigeria — have, since the 1970s, been worked by industrial trawler fleets from Europe, East Asia, and other distant-water fishing states under licence and, in some periods, without licence. The local artisanal fisheries that for

centuries had supported coastal communities have, in many places, collapsed. The fishermen now travelling the Atlantic in small boats from Senegal to the Canary Islands are, often, the descendants of working fishermen whose livelihoods were depleted by industrial fleets they could not match. The pattern matches the British North Sea of the 1880s almost exactly: a more productive distant fleet, a depleted local stock, a population of skilled practitioners with nowhere to go, and a slow institutional response that, when it arrives, will not restore what was lost.

The East African coast tells a parallel story. The Indian Ocean tuna fishery, the demersal stocks off Mozambique and Tanzania, the small-scale fisheries of the Mascarene plateau: all are being worked at intensities that the local management institutions cannot fully observe and would not have the means to enforce against if they could. The catch data is, in many cases, missing or unreliable in exactly the way the 1885 commission noted of British waters.

This is the cost the eighteenth-century pattern warned about and the nineteenth-century pattern confirmed: a new technology arrives faster than the institutions that would govern it. Enterprise AI in African contexts is at risk of being exactly the same kind of arrival. The book's African perspective sections in every industry chapter are about how the institutional response, this time, might land closer to the technology rather than a generation behind it. The catch records, if there is to be an African equivalent, have to be designed before the depletion has run its course rather than as the response that finally arrives once it has.

90.6 In Short

This chapter has paired the eighteenth-century thread of the book with a nineteenth-century counterpoint. The steam trawler arrived in British waters in the 1880s with productivity that the existing institutions of measurement could not match. The Royal Commissions of 1863–66 and 1883–85 gathered evidence, listened to working fishermen, and ultimately declined to act on the grounds that the data did not yet exist. What they did was recommend, in 1889, that the data be recorded. That recommendation, modest at the time, became the foundation of the modern science of fisheries management, of the international regulatory regimes that govern the North Atlantic today, and of the partial recovery of stocks that earlier generations had assumed inexhaustible. The lesson for enterprise AI is direct: the productivity of the new technology is establishing itself faster than the institutions that would measure its effects. The practitioner reports of harm are already accumulating, often in advance of the formal data. The administrative records that the next generation will use to do the science have to be designed now, while the technology is being deployed, rather than as the catch-up response that arrives once the damage has compounded. The trawler did not destroy the North Sea. The gap between the trawler and the catch record nearly did.

Chapter 91

The Gauge and the Clearing House: How a Network Built Itself From Incompatible Pieces

“Gentlemen, we are come together to consider how the receipts of the public, paid for travelling between distant places, may be fairly divided amongst the companies whose lines have carried them.”

—attributed to George Carr Glyn, chairing the founding meeting of the Railway Clearing House, 2 January 1842

Chapter Overview

The British railway network of the 1840s solved a problem that enterprise AI is solving again in 2026: how to assemble a working system out of components built by competing companies, on incompatible standards, with no central authority to enforce uniformity, while the network was already running and customers were already being carried. The story of the railway gauges and the Railway Clearing House is the story of how that problem was solved — not by a single design, but by decades of voluntary association, painful standard-setting, and the slow institutional discovery that interoperability is itself an engineering deliverable. This chapter pairs the Trawler and the Commission as the book’s second nineteenth-century reflection: where the trawler chapter is about measurement that arrived too late, this one is about standardisation that arrived piecemeal, over fifty years, in just enough time.

Learning Objectives

- Recognise that interoperability is a problem with both technical and institutional sides.
- Read the architectural decisions of the 1840s through their consequences a century later.
- Use the Railway Clearing House as a model for the inter-vendor coordination that AI now needs.
- Connect the gauge wars to the contemporary disputes over model APIs, agent protocols, and orchestration frameworks.
- Carry the railway analogy into Part VI's chapters on interoperability, standards, and platform governance.

91.1 The Question the Railways Forced

In the autumn of 1841, an English merchant could buy a ticket from Birmingham to Manchester and be carried over four different railways operated by four different companies, on four different sets of accounting practices, with four different ideas about what the merchant had paid for and which company was entitled to which share of his money. The London and Birmingham Railway booked him out of Curzon Street. The Grand Junction took him north. The Manchester and Birmingham took him east. The Manchester and Leeds finished the journey. At each junction he changed trains. At each company he changed tickets. The companies, after he had arrived, argued about the receipts for months. Sometimes the argument was about the distance — the mileage tables were inconsistent. Sometimes it was about the fare — the published rates were ambiguous. Sometimes it was about whose responsibility a delay had been. The merchant's journey had taken nine hours. The accounting for it could take nine months.

This was the working state of the British railway industry on the eve of one of the great institutional achievements of the nineteenth century. In the space of half a generation, a patchwork of competing private companies that built track to incompatible gauges, signalled their trains by mutually unreadable conventions, used couplings and buffers that did not match, kept time by their own clocks, and apportioned receipts by ad-hoc bilateral negotiation, became a working national network across which passengers and goods moved on single tickets, at standard rates, by a shared clock, with disputes resolved by an institutional process that had not existed at the start of the period. None of it was planned in advance. All of it was built, by working practitioners, in the spare hours of running their existing companies, against the background of competitive incentives that should have prevented cooperation altogether. They did it because the alternative — a network in which interoperation was impossible — was even worse than the cost of building the institution that would make it work.

This is, very near to literally, the situation of enterprise AI in 2026. The model providers compete on incompatible APIs. The agent frameworks compete on incompatible orchestration primitives. The vector databases compete on incompatible query languages. The observability tools compete on incompatible trace formats. The customers, like the merchant in Birmingham, are carried across this patchwork at some cost and with some friction, and the companies that operate the patchwork

argue about who owes whom what. The chapter that follows is about how the railways escaped this condition. It is not a recipe. The escape is still ahead of enterprise AI, and the shape of it will be different. But the structural problem is the same, and the railway story is what a successful escape looks like.

91.2 The Gauge

The most visible incompatibility was the gauge. George Stephenson, in laying out the Stockton and Darlington Railway in 1825 and then the Liverpool and Manchester in 1830, used a track width of four feet, eight and a half inches — a measurement inherited from the colliery tramways of the north-east of England, which had taken it in turn from the axle widths of horse-drawn carts whose pedigree, by some accounts, ran back through Roman military road engineering to a width that was a comfortable two-horse harness. The standard, in other words, was not designed; it was inherited. It was good enough. It worked. The lines that Stephenson and his sons built kept it, and the contractors who learned their trade on those lines carried it to the next set of projects, and within a few years it had become the de facto standard for most of the new railway construction in northern England.

Most, but not all. Isambard Kingdom Brunel, the chief engineer of the Great Western Railway — the line that ran west from London Paddington to Bristol and beyond — looked at Stephenson's gauge and concluded that it was wrong. A wider track, Brunel argued, would give a more stable ride at higher speed, allow larger locomotives, support heavier loads, and produce a passenger experience that the cramped Stephenson lines could not match. He specified seven feet and a quarter inch — some twenty inches wider than standard — and laid the Great Western between 1835 and 1841 to that gauge. He was, in narrow engineering terms, correct. Broad-gauge trains were faster, more comfortable, and more stable. Brunel's engineering case was sound.

The case was also irrelevant. By the time the Great Western had been running for five years, the rest of the British network was being built to Stephenson's gauge. Where the broad gauge met the standard gauge, passengers had to change trains, freight had to be unloaded from one truck and loaded onto another, and operating costs accumulated in ways that no amount of ride quality could offset. The 1846 Gauge Act, passed by Parliament after a Royal Commission that had heard testimony from both sides, required all new lines in Britain to be built to Stephenson's gauge. The Great Western was permitted to retain broad gauge on its existing lines but could not extend it. The economics did the rest. By 1892, the Great Western had completed the conversion of its last broad-gauge route to standard, in a single weekend operation involving thousands of workmen, four hundred miles of track, and the final acknowledgement that engineering correctness had lost to network effects.

The lesson is not that Stephenson's gauge was better. The lesson is that the gauge that arrived first, and that the most competent contractors knew how to build, became the gauge that everyone else had to match — and that being technically superior was no defence against being late to that consensus. Anyone who has watched a model provider attempt to launch a non-OpenAI-compatible API in 2026 has watched a Brunel argument lose to a Stephenson reality.

91.3 Standards Beyond the Gauge

The gauge was the most visible. It was not the most operationally important. Once trains could run from one company's line onto another's, the practical problem became everything else.

The couplings between locomotives and rolling stock differed from company to company; a wagon built for the London and North Western could not be coupled directly to one built for the Lancashire and Yorkshire without an adapter and a delay. The buffers — the spring-loaded shock absorbers at the ends of carriages — were at different heights, so that the coupling distance changed every time a foreign wagon was attached, and the resulting jolt at starting was a regular cause of broken couplings and derailed shunts. The signalling conventions diverged: a position of a semaphore arm that meant "proceed" on one company's line might mean "stop" on another's. The brake systems were not interoperable, which meant that a train made up of carriages from multiple companies could not be braked uniformly — a problem that would, in 1889 at Armagh, kill eighty people when a runaway train of carriages with continuous Westinghouse brakes met carriages without them on a steep grade.

The freight classifications were inconsistent. A consignment of cotton bales rated as Class 2 on one company's tariff was Class 3 on another's; the rate per mile differed; the package specifications differed; the liability terms for damage differed. The ticketing systems were incompatible: Thomas Edmondson's small card tickets, invented at the Newcastle and Carlisle Railway in 1839 and gradually adopted across the industry, became the de facto standard only because companies recognised, over the course of a decade, that no through-ticketing was possible without a common format.

The unit of measurement was inconsistent. Miles and chains, the unit by which receipts were eventually apportioned, depended on agreed mileage tables for every line, and the tables themselves were disputed: routes had been built with curves and gradients that made the published distance an approximation, and the company on the receiving end of a payment had every incentive to argue that the distance was longer than the company sending the payment had recorded. Mileage disputes generated paperwork by the tonne and resentments by the year.

None of these were unsolved problems in 1841. They were all unsolved problems, all at once, on a network that was already growing by hundreds of miles a year.

91.4 The Clearing House

The institutional solution arrived on 2 January 1842, in small offices at 111 Drummond Street opposite Euston Station, with a staff of four men and the chairmanship of George Carr Glyn, who also chaired the London and Birmingham Railway. The Railway Clearing House began with a narrow mandate: to reconcile, every month, the receipts owed by each member company to each other member company for through-ticketed passenger and freight traffic. Nine railways signed up: the London and Birmingham, the three predecessors of the Midland (Midland Counties, Birmingham and Derby Junction, North Midland), the Manchester and Leeds, the Leeds and Selby, the Hull and Selby, the York and North Midland, and the Great North of England. Membership was voluntary. The Clearing House had no statutory authority; it was a private association funded by levies on its members.

It worked. Within a year, the participating companies had a single reconciliation mechanism for through traffic. The mileage tables were standardised — the Clearing House published its own, which became authoritative. The ticket format was standardised, with Edmondson's cards the agreed type. Receipts were apportioned by clerks working through the journey records and producing monthly settlements that, by 1845, were running into hundreds of thousands of pounds and being accepted with very few disputes. The original five aims — through booking, mileage-based revenue division, through transport of goods, per-mile rates, and debt settlement — were all in operation.

What happened next is the part of the story that most directly bears on enterprise AI. The Clearing House, having been created to settle accounts, became something larger. It became the standing venue at which the engineers, accountants, and operating managers of competing railway companies met to settle anything else that needed settling. The Goods Managers' Conference. The General Managers' Conference. The Engineers' Conference. The Accountants' Conference. None of these had statutory force; all of them produced agreements that the member companies could adopt voluntarily; all of them, over the following decades, produced agreements that the member companies did adopt, because the operational benefits of standardisation outweighed the competitive cost of conceding to a shared specification.

By the time the Clearing House was given legal recognition under the Railway Clearing Act of 1850, it was producing the de facto operating standards for British railways. By the late 1860s, after the Great Western and the major southern lines had eventually joined, it covered most of the network. By 1914 it employed thousands of clerks, occupied a purpose-built complex in Seymour Street, processed receipts in the tens of millions of pounds, and acted as the regulatory body — in everything but the formal sense — of one of the world's largest industrial systems. None of this had been envisaged at the founding meeting in Drummond Street.

91.5 The Time

One particular act of the Clearing House deserves its own section, because it is the part of the story most often forgotten and most directly applicable. Until the middle of the nineteenth century, every town in Britain kept its own time, calibrated to local solar noon. The clocks in Bristol ran ten minutes behind those in London; the clocks in Plymouth, sixteen minutes behind. For a horse-drawn coach this was tolerable. For a railway timetable, in which a passenger in Plymouth was being told that a train would depart at 9:43 from a station whose stationmaster's clock was reading 9:27 against a London-issued schedule, it was a disaster waiting to happen.

The Clearing House recommended, on 22 September 1847, that all member railways adopt Greenwich Mean Time as their operational standard. Some companies had done so already. By the end of the year, almost all had. The Statutes (Definition of Time) Act 1880, which made GMT the legal standard for civil purposes in Britain, came thirty-three years later — after the railways had effectively imposed it on the country by running their timetables to it and by displaying GMT on the clocks at every major station. The international standardisation of time zones at the Washington Conference of 1884 came forty years after Greenwich time had become the operational time of British industry.

The point is not that the railways invented standard time. The point is that the railways needed standard time before any legal mechanism existed to provide it, and they imposed it on themselves, voluntarily, through a private association, decades ahead of the legal acknowledgement. The world ran on local solar time for several thousand years and then, in the space of two generations, switched to a small number of standardised zones because a network of private railway companies decided that they could not operate without it. Anyone who has tried to reconcile the timestamps from three different AI providers, each with its own clock-skew quirks, has felt a small version of what every railway clerk felt in 1846.

91.6 What the Railways Make Clear

91.6.1 Standards Win Through Adoption, Not Through Quality

Stephenson's gauge was not the best gauge. It was the gauge that the most contractors knew how to build, that the most existing track had been laid to, and that the most equipment had been manufactured for, by the time the question of a national standard became urgent. Brunel's case for the broad gauge was, in narrow engineering terms, sound. It did not matter. The Great Western fought the gauge wars for forty-five years and lost. The same lesson runs through every enterprise AI architectural decision in 2026. The pattern that wins is rarely the most elegant. It is the pattern that the most teams know how to build, that the most documentation supports, and that the most existing systems are already using. Choosing the elegant minority pattern is a Brunel choice, and the engineering case will not save the architecture from the network-effect cost.

91.6.2 Interoperability Is an Institutional Deliverable

The technical interoperability problems of the British railways — couplings, gauges, signals, brakes, classifications, mileage tables — were not solved by a technical innovation. They were solved by an institution that brought competing companies into the same room and made the cost of disagreement higher than the cost of agreement. The Clearing House was not a clever piece of equipment. It was a clever piece of organisation. Enterprise AI in 2026 has the technical equivalents of the gauge problem in abundance — agent communication protocols, tool-use interfaces, model API conventions, evaluation formats, prompt-injection threat taxonomies — and the institutional equivalents of the Clearing House mostly do not yet exist. They will. Some of them are forming now in standards bodies, in cross-vendor working groups, in the parts of the academic literature that publish shared benchmarks. The architectural work for the next decade is partly to identify which of these institutions will become the Clearing House of the AI industry and to participate in them while they are still small.

91.6.3 Voluntary Beats Statutory, At First

The Railway Clearing House was a private association for eight years before Parliament gave it legal status, and the legal status that arrived in 1850 was less expansive than the Clearing House had already become in practice. The pattern is durable: voluntary standards bodies, made up of practitioners who actually have to operate the systems under discussion, produce better technical agreements than statutory bodies do, and they produce them years earlier. Statutory regulation typically arrives when the voluntary standards body has done most of the technical work and the remaining issue is

enforcement against the holdouts. The implication for AI is that the most productive standardisation work today is the work being done in the practitioner forums, not the work being done in legislative committees. The legislative work will follow, in due course, and will probably codify what the practitioners have already agreed on. Skipping ahead to the legislation in advance of the voluntary agreement tends to produce regulations that no one is in a position to comply with.

91.6.4 Most of the Work Is Boring

The Clearing House employed thousands of clerks. They produced ledgers. They reconciled mileage tables. They handled lost property. They issued tickets. None of this was glamorous. None of it was the part of the railway story that gets put in popular histories. Almost all of it was load-bearing. The same will be true of the AI standardisation work. The model architectures and the agent demos get the attention. The interoperability protocols, the schema registries, the evaluation harnesses, the audit-trail formats, and the inter-vendor reconciliation processes will do most of the work and get almost none of the press. The architect who treats this work as beneath their attention is the architect whose system, eight years from now, will be the one that cannot operate with anyone else's.

91.7 What Africa Inherited from the Railways

The railway story, told in its British form, is a story of an industrial nation working out interoperability through voluntary association among private companies with broadly aligned incentives. The story Africa inherited is different. African railways were largely built by colonial powers, on gauges chosen for reasons that had little to do with the long-term economic interests of the territories the railways crossed.

The Cape gauge — three feet six inches, much narrower than the British standard — was adopted across most of southern and eastern Africa because it was cheaper to build through difficult terrain. Nigeria and most of West Africa inherited the same. Algeria, Tunisia, Morocco, and parts of francophone West Africa adopted metre gauge for similar reasons. Egypt, by contrast, was built to standard gauge by British engineers for strategic reasons. The result was a continent whose railway map, by 1914, was a patchwork of mutually incompatible gauges, with breaks at almost every colonial boundary and at several internal seams. The decisions made in colonial offices in London and Paris between 1860 and 1900 still determine, in 2026, which African capitals can be reached by rail from which others, and which routes require the freight to be transhipped at the border.

The lesson is not that the gauge choices were wrong. Cape gauge was, in the construction conditions of southern Africa in the 1870s, a defensible engineering decision. The lesson is that the consequences of an interoperability decision last for centuries. An AI architecture choice made in Lagos or Nairobi today — which model provider, which orchestration framework, which observability stack, which evaluation regime — may turn out, fifty years from now, to be the equivalent of a Cape gauge decision: locally rational at the time, structurally consequential in ways that no one anticipated. The African perspective sections of this book have, throughout, argued for architectural choices that preserve optionality. The railway story is what happens to a continent when those choices were not preserved.

91.8 Where That Leaves Us

This chapter has paired the gauge wars with the Railway Clearing House to make a single argument: that the assembly of a working network from competing, incompatible components is achieved less by getting the components right than by building the institutional venue in which the components can be reconciled. The British railways did the technical work and the institutional work in parallel, over fifty years, in the spare hours of running the existing system. The result was the first effective national transport network of the industrial era, and the structural achievement was the Clearing House rather than any of the locomotives that ran on the tracks the Clearing House made operable. Enterprise AI in 2026 stands roughly where British railways stood in 1841: a profusion of competing companies, incompatible standards, a few small co-operative ventures, and most of the institutional work still ahead. The chapter that follows the next part returns to the architectural disciplines that can be built now while the larger reconciliation is still being worked out. The reader who has carried the gauge and the Clearing House into those chapters will read them with one extra question in mind: which of these architectural decisions, made today, will turn out to be the standard gauge of the next generation, and which will turn out to be the broad gauge that everyone, eventually, has to convert away from?

Part VI

Cross-Cutting Concerns

*“A perpetual subject of dispute, a constant source of fraud, an inexhaustible cause of error.”
—the National Assembly of France on the chaos of pre-revolutionary weights and measures,
1790*

The trawler interlude has just made the case for measurement that arrives in time. The cross-cutting concerns are what makes that measurement portable across an organisation. Eighteenth-century France had perhaps eight hundred different names for units of measurement, and the same name often referred to different quantities in adjacent districts. A merchant trading between Lyon and Marseille could not be sure that the pound he sold was the pound the buyer thought he had bought. The metric system, conceived in the 1790s and gradually adopted across the following century, was the eventual answer: a horizontal standard that cut across every trade, every region, every product. Part VI of this book describes the horizontal concerns of enterprise AI — the cross-cutting layers of security, observability, governance, sustainability, and human factors — that have to be standardised across the verticals or each vertical will reinvent them at its own cost.

Chapter 92

Data Privacy

“Privacy is not an option, and it shouldn’t be the price we accept for getting on the Internet.”

—Gary Kovacs

Chapter Overview

Privacy is the first of the cross-cutting concerns because the cost of getting it wrong, in regulated industries, dwarfs almost every other architectural decision. This chapter develops the privacy obligations specific to AI systems — training-data lineage, prompt logs, generated outputs that may reproduce inputs, vector stores that retain content, and embeddings that can be inverted — and the technical controls that satisfy them: data classification, minimisation, encryption, tokenisation, differential privacy, redaction, residency, and retention. The regulatory frame — GDPR, CCPA, HIPAA, POPIA, the EU AI Act — is treated briefly, because the full regulatory treatment belongs to the compliance chapters; the architectural mechanics are treated in depth, because this is where the architect has work to do.

Learning Objectives

- Catalogue the privacy obligations specific to AI systems and the data flows that create them.
- Apply data classification, minimisation, and purpose limitation throughout the architecture.
- Implement the technical controls: encryption, tokenisation, masking, differential privacy, redaction.
- Honour data-subject rights — access, erasure, portability — in systems that include vector stores and model weights.
- Design for residency, retention, and cross-border transfer constraints.

92.1 The Privacy Surface of an AI System

An AI system creates privacy obligations that go well beyond those of a typical service, because personal data flows to, and is retained by, more components than a conventional application touches. Figure 92.1 shows where personal data flows in a typical deployment that combines retrieval-augmented generation with fine-tuning, and every box drawn in red is a component that retains or processes

personally identifiable information and is therefore a privacy concern the architecture must address explicitly.

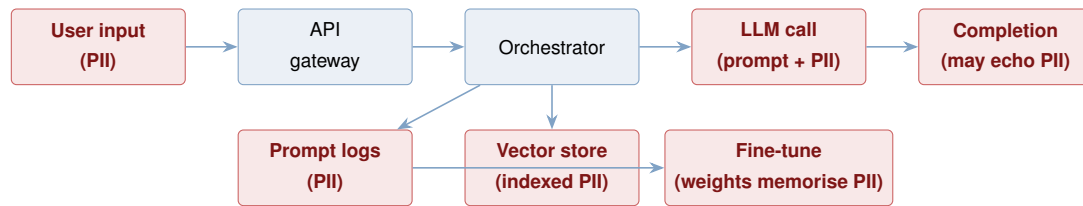


Figure 92.1: Where personal data flows in a typical AI system. Boxes in red retain or process PII; each is a privacy concern that the architecture must address explicitly.

The surface is wider than it first appears, and it is worth walking through the components individually. The user's input may contain personal data directly — a name, an account number, a medical detail. The prompt log retains that input, often durably, for observability. The vector store indexes content that may include personal data from the documents it was built from. The LLM call sends the prompt, including whatever personal data it contains, to the model endpoint, and a vendor endpoint means the data has left the organisation. The completion may echo personal data from the prompt or from the model's training set — the memorisation problem. And a fine-tuning run on data containing personal information bakes that information into the model's weights, where it can later be extracted. Each of these is a distinct privacy concern, and together they mean an AI system's privacy surface is materially larger than a conventional application's.

92.2 The Regulatory Frame

The regulatory regimes that govern data privacy in AI systems are several, and an organisation operating across jurisdictions must satisfy the union of the ones that apply to it. The chapter does not develop the regimes in full — that is the compliance chapters' role — but the architect must know which regimes shape the technical controls.

The General Data Protection Regulation governs personal-data processing for European Union residents and is the most architecturally demanding of the regimes, with its requirements for lawful basis, purpose limitation, data-subject rights, and data-protection impact assessments. The California Consumer Privacy Act and its successor, the CPRA, apply to California residents and carry their own rights and obligations. HIPAA applies to protected health information in the United States, with stringent requirements on how health data is handled, stored, and shared. South Africa's Protection of Personal Information Act, POPIA, applies to personal data processed in South Africa and carries data-subject rights, cross-border transfer restrictions, and breach-notification obligations. And the EU AI Act adds AI-specific requirements that layer on top of GDPR, including transparency obligations, documentation, and the high-risk classification that can trigger a full conformity assessment. A pan-African or global deployment may face several of these simultaneously, and the technical controls must satisfy the most restrictive of the applicable regimes.

92.3 Data Classification

Data classification is the foundation of privacy controls, because the controls that apply to a piece of data depend on what kind of data it is, and the system cannot apply the right controls without knowing the classification. A workable classification scheme distinguishes at least four tiers: public data, which carries no privacy constraint; internal data, which is restricted to the organisation but not personally identifiable; personal data, which identifies an individual and triggers the data-protection regimes; and special-category data — health, biometric, financial, children’s data — which carries the most stringent protections under most regimes.

Classification must be applied at every point where data enters the system, so that the classification travels with the data through the architecture. A document indexed into the vector store should carry its classification as metadata, so that retrieval can be filtered by it. A prompt log entry should carry the classification of the input it records. And a fine-tuning dataset should be classified before training begins, so the organisation knows what sensitivity is being baked into the model. Classification that is applied only at the boundary and not propagated through the system is classification that is lost the moment the data moves.

92.3.1 Classification at the Vector Level

Vector-level classification is a specific concern, because a vector embedding does not visibly contain the personal data it was derived from — it is a numerical representation, not text — but the data can be recovered from it through inversion, and the embedding is therefore personal data under most regimes if the source was personal. Classifying the embedding by the sensitivity of the source content, not by the appearance of the embedding itself, is the discipline that keeps vector stores from becoming unclassified repositories of data that is personal in fact if not in form.

92.4 Data Minimisation and Purpose Limitation

Data minimisation and purpose limitation are the twin principles that an AI system should collect and process only the personal data it needs for its defined purpose, and should use that data only for that purpose. They are regulatory requirements under most regimes and they are also sound engineering: a system that processes less personal data has a smaller privacy surface, fewer components that hold PII, and a simpler compliance posture.

The principles have architectural expression at every layer. At the input layer, the system should collect only the data the task requires rather than ingesting everything available. At the storage layer, retention should be bounded: data should be held only for the period it is needed and deleted or anonymised afterward, rather than retained indefinitely “in case it is useful.” At the processing layer, the data should be used only for the purpose it was collected for, and a secondary use — using customer-service transcripts to fine-tune a model, for instance — requires a separate lawful basis. And at the output layer, the system should not expose personal data in its responses unless the purpose requires it. A system designed with minimisation and purpose limitation in mind is a system whose privacy surface is deliberately small, which is both a regulatory virtue and an engineering one.

92.5 Technical Controls

The technical controls are the mechanisms that enforce the classification, minimisation, and purpose-limitation principles in the running system. Each addresses a different part of the privacy surface.

92.5.1 Encryption

Encryption protects personal data at rest and in transit. Data at rest in the vector store, the prompt logs, the trace store, and the fine-tuning datasets should be encrypted so that a compromise of the storage does not expose the data. Data in transit between the components of the system, and between the system and any vendor endpoint, should be encrypted so that an interception does not expose it. Encryption does not remove the personal data from the system — the data is still there, encrypted — but it converts a storage or network compromise from a data breach into an event where the attacker obtains data they cannot read. Encryption is the baseline control, below which no component handling personal data should fall.

92.5.2 Tokenisation and Masking

Tokenisation replaces a sensitive value with a non-sensitive token that can be mapped back to the original through a secure lookup, while masking replaces a sensitive value with a redacted form that cannot be reversed. Both reduce the privacy exposure of the data that passes through the system. Tokenisation is useful where the original value must be recoverable — for instance, where a customer identifier must be restored to produce a meaningful response — while masking is useful where it need not be and where the stronger guarantee of irreversibility is preferred. The choice between them depends on whether the downstream use of the data requires the original value.

92.5.3 Redaction in Prompts and Logs

Redaction strips personal data from prompts before they are sent to a model and from logs before they are stored. It is the most direct control on the two highest-exposure points of the privacy surface: the prompt, which may be sent to a vendor and so leaves the organisation's control, and the log, which is retained durably and accessible to the operations team. Redaction before the model call means the model never sees the personal data, which is the strongest privacy posture. Redaction before logging means the trace store does not accumulate a sensitive archive. The trade-off is that redacted data is lost to the downstream use: a model that never sees the customer's name cannot personalise its response. The discipline is to redact by default and to admit personal data into the prompt or the log only where the purpose genuinely requires it and the legal basis supports it.

92.5.4 Differential Privacy

Differential privacy is a mathematical framework that bounds how much any single individual's data can influence a model's outputs, and it is applied during training to prevent the model from memorising personal records. A model trained with differential privacy carries a provable guarantee that no single training record can be extracted, which is the strongest defence against the memorisation problem. The trade-off is a quality cost: the noise that differential privacy introduces to achieve its guarantee degrades the model's performance, and the degradation is larger when the privacy guarantee is stronger. Differential privacy is therefore most appropriate for the training runs where the data

is most sensitive and the memorisation risk is highest, and less appropriate where the data is less sensitive and the quality cost is not justified.

92.6 Data-Subject Rights

The data-protection regimes grant individuals rights over their personal data — the right to access what the system holds about them, the right to have it erased, the right to have it corrected, and the right to receive it in a portable format — and an AI system must be able to honour those rights, which is harder than it sounds.

92.6.1 Subject Linkage

The first challenge is finding all the data that relates to a specific individual, because an AI system spreads personal data across many components. The prompt logs, the vector store, the trace store, and the fine-tuning datasets may all contain data about the same individual, and honouring a subject-access or erasure request means locating that individual's data across all of them. Subject linkage — the ability to trace a person's data across the system's components — is the prerequisite for honouring any data-subject right, and it must be designed in from the start rather than reverse-engineered when the first request arrives.

92.6.2 Vector-Store Erasure

Erasing an individual's data from a vector store is technically non-trivial, because the individual's content is embedded into vectors that may not carry a clean per-individual boundary. The approaches include per-document deletion where the individual's content maps cleanly to identifiable documents, re-indexing the affected portion of the store after removing the individual's content, and maintaining a deletion log that records what was removed and when. The discipline is to design the vector store's indexing with erasure in mind — so that an individual's content can be identified and removed — rather than discovering, on the first erasure request, that the store's structure makes deletion impractical.

92.6.3 Training-Data Erasure

Erasing an individual's data from a model's training set is the hardest data-subject-rights problem, because the data has been learned into the model's weights and cannot be cleanly excised. Full retraining without the individual's data is the most complete solution but the most expensive. Approximate methods — machine unlearning, fine-tuning to suppress the memorised content — are less costly but less certain. The honest architectural position is to minimise the personal data that enters training in the first place, so that the problem arises rarely, and to document the approach the organisation takes when it does arise. Training-data erasure is an open problem, and an organisation that has included personal data in training should have a documented position on how it handles an erasure request, even if that position acknowledges the limitation.

92.6.4 Operational SLA

Data-subject rights carry regulatory timelines — GDPR gives thirty days for most requests, and other regimes have their own deadlines — and the system must be able to honour those timelines,

which means the subject-linkage, the vector-store erasure, and the training-data handling must all be operable within the window. An operational service-level agreement for data-subject-rights fulfilment, specifying the target turnaround and the process for meeting it, is the governance mechanism that turns the regulatory obligation into an operational commitment.

92.7 Residency and Cross-Border Transfer

Many data-protection regimes restrict where personal data may be stored and processed, and cross-border transfer is subject to conditions that vary by regime. Residency and transfer constraints bear directly on the AI architecture, because they determine where inference may run, where vector stores may be hosted, where logs may be stored, and which vendor endpoints may be called.

The architectural responses to residency constraints include in-region hosting of the components that process personal data, so the data does not leave the jurisdiction; self-hosting open-weight models within the jurisdiction, which avoids sending data to a vendor's region; and contractual safeguards — standard contractual clauses, adequacy decisions, or binding corporate rules — for the transfers that must cross a border. The discipline is to map the data flows of the AI system against the residency requirements of each jurisdiction it operates in, and to confirm that every flow either stays within the jurisdiction or is covered by an approved transfer mechanism. A system whose data flows are unmapped against its residency obligations is a system that may be in breach without knowing it.

92.8 Vendor Risk in AI Privacy

An AI system that calls a vendor's model endpoint sends data to the vendor, and the vendor's handling of that data is a privacy concern the organisation must manage. The vendor's data-processing terms govern whether the vendor may use the data for its own purposes — including, critically, whether it may use the data to train its models. The vendor's retention practices determine how long the data is held. And the vendor's hosting regions determine where the data is processed, which interacts with the residency constraints.

The architectural controls include contractual data-processing agreements that restrict the vendor's use of the data, redaction of personal data before the vendor call so the vendor never receives it, and the self-hosting option for sensitive workloads that keeps the data entirely within the organisation's control. Vendor privacy risk is a real and specific concern in AI architectures, because the vendor's model endpoint is a point where data leaves the organisation, and the organisation's privacy obligations do not end at its own boundary — they follow the data.

92.9 Privacy by Design

Privacy by design is the principle that privacy controls are built into the architecture from the start rather than bolted on afterward, and it is both a regulatory expectation under GDPR and a practical necessity, because retrofitting privacy into an AI system that was built without it is expensive and incomplete.

The principle has several architectural expressions. Data-protection impact assessments are conducted before a system that processes personal data is built, not after it is deployed. Classification is applied at ingestion, so it propagates through the system from the start. Minimisation is a design constraint, so the system collects only what it needs. Redaction points are placed in the architecture at design time, so the system can strip personal data before it reaches the model or the log. And erasure paths are designed into the storage and indexing, so the system can honour a deletion request without re-engineering itself. A system designed with these controls from the start carries a smaller privacy surface, a simpler compliance posture, and a lower risk of a breach — and it does not have to be rebuilt when the first data-subject request arrives.

92.10 What to Carry Forward

Data privacy in AI systems is a cross-cutting concern that touches every component of the architecture: the inputs, the logs, the vector stores, the model calls, the fine-tuning datasets, and the outputs. The privacy surface is wider than a conventional application's, because personal data flows to and is retained by more components, and because model memorisation creates a privacy risk that has no conventional equivalent.

The technical controls — classification, minimisation, encryption, tokenisation, masking, redaction, differential privacy — each address a different part of that surface, and together they reduce the privacy exposure to what the purpose genuinely requires. Data-subject rights must be honourable within regulatory timelines, which demands subject linkage across components, erasure paths designed into the vector store and the training pipeline, and an operational SLA for fulfilment. Residency and cross-border transfer must be mapped against every jurisdiction the system operates in. And vendor privacy risk must be managed through contractual terms, redaction, and the self-hosting option for the most sensitive workloads. Privacy by design — building the controls in from the start — is both a regulatory expectation and the only practical approach, because retrofitting privacy into a system built without it is more expensive and less complete than building it in. The cost of getting privacy wrong, in regulated industries, dwarfs almost every other architectural decision, and the controls of this chapter are what keep an AI system on the right side of that cost.

Chapter 93

Reliability and Availability

“Hope is not a strategy.”

—traditional site-reliability proverb

Chapter Overview

AI systems inherit the reliability problems of distributed services and add their own: stochastic outputs, vendor dependencies with unpredictable failure modes, model-specific quality regression, latency profiles that are hard to predict. This chapter develops the reliability vocabulary (SLO, SLA, error budget, MTBF, MTTR), the redundancy and failover strategies that apply to AI, the structural patterns—circuit breakers, bulkheads, retries with backoff, hedging, graceful degradation—that produce predictable behaviour from unpredictable components, and the chaos-engineering exercises that validate the design under stress.

Learning Objectives

- Express AI-system reliability targets as service-level objectives with measurable indicators.
- Plan for high availability through redundancy, replication, and failover.
- Apply the canonical resilience patterns—retry, circuit breaker, bulkhead, hedge, fallback—to AI workloads.
- Design backup and disaster-recovery procedures for stateful components: vector stores, feature stores, model registries.
- Run reliability exercises (chaos engineering, game days) to validate the design under stress.

93.1 The Reliability Vocabulary

93.1.1 SLI, SLO, SLA

A *service-level indicator* (SLI) is a measurement (e.g., proportion of requests returning within 2 seconds). A *service-level objective* (SLO) is the target for an SLI over a window (e.g., 99.5% over 30 days). A *service-level agreement* (SLA) is the externally-committed version, typically with financial consequences for breach.

The relationship: $SLA < SLO < \text{observed reliability}$. The SLO is set below the observed reliability so

that there is headroom to absorb deviation; the SLA is set below the SLO so that internal breaches do not immediately become external ones.

93.1.2 Error Budget

The *error budget* is the inverse of the SLO: if the SLO is 99.5% over 30 days, the error budget is 0.5% of minutes—approximately 216 minutes per 30-day window. Error budgets reframe reliability as a finite resource:

- It is acceptable to spend budget on planned changes (deploys, experiments, structural improvements).
- It is unacceptable to overspend on incidents.
- When the budget is exhausted, change freezes apply until budget is restored.

The discipline turns reliability into a tradeable resource: the platform team and the product team negotiate the budget rather than the platform team unilaterally setting it.

93.1.3 MTBF and MTTR

Mean time between failures (MTBF) measures how often the system fails; *mean time to recovery* (MTTR) measures how quickly it recovers. Availability follows:

$$\text{Availability} = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

Two strategies improve availability: increase MTBF (fail less often) or decrease MTTR (recover faster). For AI systems, MTTR is usually the more tractable lever: failures are inevitable in complex distributed systems with stochastic components, but a system that detects, diagnoses, and recovers from failure quickly is operationally indistinguishable from one that fails less often.

93.1.4 AI-Specific SLIs

Beyond conventional latency and error-rate SLIs:

Table 93.1: AI-specific service-level indicators.

SLI	What it measures
Generation success rate	Proportion of model calls that produced a valid (parseable, on-format) output
Refusal rate	Proportion of calls where the model refused; sustained increase signals corpus or prompt drift
Citation accuracy	Proportion of cited claims actually supported by the cited source
Hallucination rate (sampled)	Proportion of outputs failing grounding checks in a sampled audit
Retrieval recall on canary queries	A fixed set of queries with known expected results, run continuously
Time to first token	Latency to first streamed token, separately from total latency

93.2 Redundancy and Replication

The classical patterns—multiple availability zones, multi-region active/active or active/standby, replicated storage—apply unchanged. AI-specific additions:

- **Model replication.** The model registry’s promoted version is mirrored to every serving region. Mismatches between regions are a configuration drift that the platform must detect.
- **Vector-store replication.** Vector indexes are large; replication can be expensive. Common patterns: synchronous within region, asynchronous cross-region. Consistency models matter: an eventually-consistent index can return stale results during failover.
- **Feature-store replication.** The online store must be replicated for inference; the offline store typically does not require active replication (it can be rebuilt from the lakehouse).
- **Multi-vendor redundancy.** The hexagonal pattern lets the system fail over from one vendor to another for the same capability. The swap test (Section 26.4) is the standing exercise that keeps this option real.

93.2.1 Active-Active vs Active-Standby

Table 93.2: Multi-region deployment patterns.

Pattern	Cost	Failover characteristics
Active-active	2× baseline	Near-zero failover time; traffic shifts seamlessly
Active-standby (warm)	~1.5× baseline	Seconds to minutes failover; standby capacity ready but not serving
Active-standby (cold)	~1.1× baseline	Minutes to hours failover; standby capacity provisioned on demand
Pilot light	~1.05× baseline	Hours failover; minimal standby that scales up under demand

93.3 Resilience Patterns

93.3.1 Retry with Exponential Backoff

Transient failures (rate limit, network blip, transient overload) resolve on retry. Backoff with jitter prevents thundering herds:

$$t_n = \min(t_{\max}, \text{base} \cdot 2^n) \cdot (1 + \text{random}[0, J])$$

where J is the jitter fraction (commonly 0.1–0.3) and $t_{\max}$ caps the wait. Non-idempotent calls require additional care: an idempotency key prevents duplicate side effects.

```
1 import asyncio, random
2
3 async def call_with_retry(
```

```

4     fn, *args,
5     max_attempts: int = 5,
6     base: float = 0.5,
7     cap: float = 30.0,
8     jitter: float = 0.2,
9     retryable_errors: tuple = (TransientError,),
10    **kwargs,
11 ):
12     for attempt in range(max_attempts):
13         try:
14             return await fn(*args, **kwargs)
15         except retryable_errors as e:
16             if attempt == max_attempts - 1:
17                 raise
18             wait = min(cap, base * (2 ** attempt))
19             wait *= 1 + random.uniform(0, jitter)
20             await asyncio.sleep(wait)

```

Listing 93.1: Retry with exponential backoff and jitter.

93.3.2 Circuit Breaker

A circuit breaker monitors the failure rate to a dependency and, when failures exceed a threshold, “opens”—short-circuiting subsequent calls to return failures immediately. This protects both the caller (no slow timeouts) and the callee (no piling on a struggling service). After a cooldown, the breaker enters a “half-open” state and lets a small number of trial calls through.

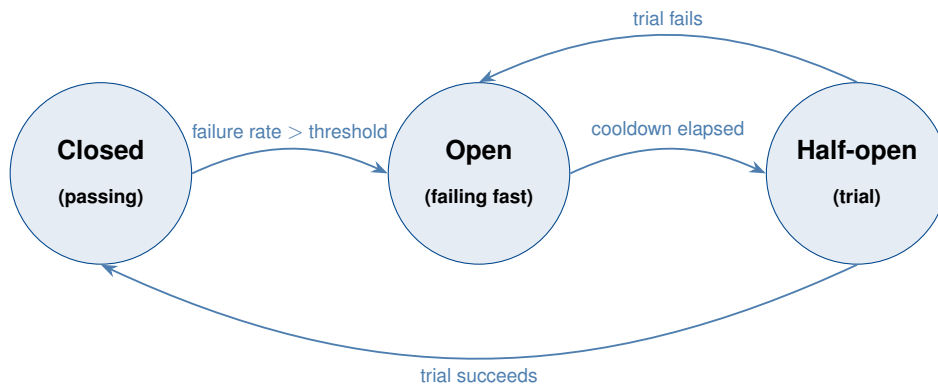


Figure 93.1: Circuit-breaker states: closed (normal), open (fast-fail), half-open (probe).

93.3.3 Bulkhead

Bulkheads partition resources so that a failure in one part of the system cannot exhaust resources used elsewhere. Concrete examples:

- Separate thread pools per downstream dependency; one slow dependency cannot starve the others.
- Per-tenant rate limits at the gateway; one tenant’s spike cannot block others.

- Dedicated GPU pools for latency-critical models, separate from batch workloads.
- Separate connection pools for read and write operations.

93.3.4 Hedging

A hedged request issues a duplicate request to a different replica (or vendor) if the first does not respond within a threshold. The first response wins; the late one is cancelled. Hedging reduces tail latency at the cost of duplicate work; the threshold is tuned so that hedges fire only when the first request is unusually slow.

93.3.5 Fallback and Graceful Degradation

When the preferred path fails, an explicit fallback is better than an opaque error. For AI workloads, fallbacks include:

- A cheaper model in the same tier (different vendor; same capability).
- A smaller model that handles most cases (Section 23.6).
- A cached answer for the same or similar question.
- A templated response acknowledging the failure and suggesting human contact.
- A routed-to-human path for high-stakes interactions.

93.4 Backup and Disaster Recovery

The stateful components of an AI system need backup and recovery procedures:

Table 93.3: Stateful components and their backup considerations.

Component	Backup approach
Data lakehouse	Time-travel snapshots (Delta/Iceberg) for short-term recovery; cross-region replication for DR
Feature store (online)	Rebuildable from offline store; backup is the offline store
Feature store (offline)	Lakehouse-backed; standard data backup applies
Vector store	Periodic snapshots; full rebuild from source documents documented and tested
Model registry	Versioned artefacts in cross-region object storage; metadata in a relational store with backup
Prompt registry	Source-controlled; same as code backup
Observability data	Tiered retention; recent in hot storage, older in archive

Two metrics drive DR design:

- **RPO (recovery point objective).** How much data may be lost? Determines backup frequency.
- **RTO (recovery time objective).** How quickly must service resume? Determines hot-standby vs. cold restore.

For most enterprise AI systems: RPO of minutes (lakehouse with frequent commits), RTO of an hour or less (warm standby in a second region). High-availability systems: RPO measured in seconds, RTO measured in seconds (active-active).

93.5 Chaos Engineering for AI

Periodic exercises that inject failures into production-like environments validate reliability claims. The exercises take several forms.

93.5.1 Component-Level Failures

Component-level failures are the most common and the most recoverable: a single model endpoint goes down, a vector store becomes unavailable, or a tool integration fails. The resilience patterns at this level focus on redundancy and graceful degradation within the system.

- Kill a model-serving replica; verify autoscaler and load balancer recover.
- Block traffic to a vendor; verify fallback adapter activates.
- Corrupt a vector-index node; verify replication restores service.
- Inject latency on the embedding service; verify timeout and degradation paths.
- Pause the feature-store online refresh; verify staleness alerts fire.
- Take down the prompt registry; verify cached prompts continue to serve.

93.5.2 Region-Level Failures

Region-level failures are less frequent but more consequential: an entire cloud region becomes unavailable, taking every service hosted there offline at once. The resilience patterns at this level focus on multi-region deployment and failover.

- Simulate full regional outage; verify cross-region failover.
- Simulate partial regional degradation (some services up, some down); verify the more complex partial-failure handling.
- Simulate split-brain (region accessible but isolated); verify data-consistency invariants.

93.5.3 Vendor-Level Failures

Vendor-level failures are the least frequent and the hardest to recover from: the vendor itself has an outage, a pricing disruption, or a discontinuation that affects every customer at once. The resilience patterns at this level focus on multi-vendor strategy and the ability to switch.

- Simulate vendor API total outage; verify failover to alternative vendor through hexagonal port.
- Simulate vendor rate-limit aggression; verify backoff and shed-load behaviour.
- Simulate vendor quality regression (intentionally degraded outputs in a test environment); verify quality monitoring detects.

The architectural value is not the discovery of bugs (though it usually finds some) but the validation that recovery procedures work and that the team knows them.

93.6 Game Days

A game day is a scheduled exercise where the team practises responding to a fabricated incident. The scenario is not announced in advance; the team treats it as if real. After the exercise, a structured debrief captures:

- Did the team detect the incident from the available signals?
- Did the documented runbooks apply?
- How long did each phase (detection, diagnosis, mitigation, recovery) take?
- What gaps were exposed: missing observability, missing runbooks, missing tooling, missing access?
- What follow-up actions reduce the recurrence of similar gaps?

A quarterly game-day cadence keeps the muscle memory current. Annual is too infrequent; monthly is usually too disruptive.

93.7 The Shape of the Argument

Reliability and availability are properties of the system, not the components. SLOs and error budgets give the team an objective frame. Redundancy and failover provide the structural defence. Resilience patterns (retry, circuit breaker, bulkhead, hedging, fallback) provide the tactical defence. Backup and DR provide the existential defence. Chaos exercises and game days provide the validation. The outcomes that distinguish a reliable system—reliable and scalable, recoverable from disasters, predictable under stress—are bought with structural commitments throughout the architecture, not bolted on at the end.

Chapter 94

Scalability and Performance

“Premature optimisation is the root of all evil. Premature scaling is the source of all bankruptcy.”

—adapted from Knuth and startup folklore

Chapter Overview

AI workloads have unusual performance characteristics: per-call latency dominated by external dependencies, throughput limited by GPU memory rather than CPU, costs that scale with input length, and traffic patterns that swing dramatically. This chapter develops the latency-budget discipline, the throughput strategies (batching, caching, parallelism), the autoscaling considerations specific to model serving, the cost-performance trade-offs, and the recurring performance anti-patterns. The goal is not to maximise any single dimension but to operate explicitly on the cost–quality–latency frontier.

Learning Objectives

- Apply a latency-budget discipline to multi-component AI flows.
- Improve throughput through batching, caching, parallelism, and continuous batching for LLMs.
- Autoscale model-serving infrastructure with awareness of cold-start costs and GPU constraints.
- Manage the cost-performance trade-off explicitly rather than implicitly.
- Recognise the recurring performance anti-patterns and the structural mitigations.

94.1 The Latency Budget

A user-facing AI response has a perceived latency budget—typically 1 second to first response and 5–10 seconds to full response for chat. Every component the request passes through consumes part of this budget. The architectural exercise is to allocate the budget explicitly and verify that no component exceeds its allocation.

Table 94.1: A representative latency budget for a RAG chat response.

Component	Budget	Notes
Auth, routing, rate-limit	50 ms	Edge / gateway
Query embedding	100 ms	Smaller, optimised model
Vector retrieval (top 50)	100 ms	Approximate; depends on index size
Re-rank (top 10 from 50)	200 ms	Cross-encoder model
Prompt assembly	20 ms	Templating, redaction
Time to first token	500 ms	Frontier model latency
Per-token latency × tokens	varies	Streaming reduces perceived latency
Total to first token	~1 s	The user perception threshold

If the budget is overrun, the response is not “make everything faster” but “which component most violates its allocation, and what is the structural fix?”. Two failure modes recur:

- The budget is set but not measured. The latency-budget table lives in a design document; production latency is whatever it is.
- The budget is measured but not enforced. Components routinely exceed their allocations; the total is over budget; nobody is responsible.

The fix is to express the budget as code: per-component timeouts that fail loudly rather than silently absorbing slowness.

94.1.1 Streaming and Perceived Latency

For LLM-backed user experiences, perceived latency is dominated by *time to first token*, not total response time. A response that takes 5 seconds to complete but streams the first token at 300 ms feels much faster than a response that returns in 3 seconds as one block. Architecturally:

- Stream wherever the UI can render incrementally.
- Optimise time to first token, even at modest cost to total latency.
- Show structural elements (table headers, list bullets) as soon as they can be emitted; let detail follow.

94.2 Throughput Strategies

94.2.1 Batching

Batching multiple requests together amortises per-call overhead. For LLMs, naive batching—collect N requests, run them as a batch—trades latency for throughput. Modern serving runtimes implement *continuous batching*: requests at different stages of generation share a forward pass, with new requests joining the batch when others finish. This typically yields 3–10× throughput improvement at modest latency cost.

The architectural trade-off:

- Latency-critical paths: small batch sizes; dedicated capacity; tight queueing.
- Throughput-critical paths: large batch sizes; shared capacity; tolerant queueing.

94.2.2 Caching

Several caching layers help AI workloads:

Table 94.2: Caching layers for AI workloads.

Cache	Typical hit rate	Effect
Semantic cache	10–40% (workload-dependent)	Avoids the whole LLM call for similar queries
Prompt-prefix cache	50–90% on templated prompts	Reuses model KV cache; reduces both compute and latency
Embedding cache	>90% on stable content	Avoids re-embedding unchanged chunks
Retrieval result cache	5–20%	Avoids vector and re-rank work for recent queries

Cache invalidation is the perennial difficulty: a stale cached answer is worse than no cached answer. TTLs, source-change notifications, content hashes, and explicit invalidation APIs all play a role.

94.2.3 Parallelism

Within a single request, parallelism reduces wall-clock latency: vector search, lexical search, and any independent tool calls proceed concurrently; their results are merged. Within a single workflow, multiple agents may proceed in parallel on independent subtasks.

```

1 import asyncio
2
3 async def hybrid_retrieve(query, k=20):
4     """Run vector and lexical retrieval in parallel; merge results."""
5     vec_task = asyncio.create_task(vector_search(query, k))
6     bm_task = asyncio.create_task(bm25_search(query, k))
7     vec_results, bm_results = await asyncio.gather(vec_task, bm_task)
8     return reciprocal_rank_fusion(vec_results, bm_results, k=k)

```

Listing 94.1: Parallel retrieval to reduce wall-clock latency.

94.3 Autoscaling for AI

Autoscaling AI workloads differs from autoscaling stateless services in three ways:

- **Cold start is slow.** A new GPU replica loading a multi-gigabyte model takes minutes. Conventional autoscalers (which scale on per-minute metrics) underprovision during traffic spikes.
- **GPU is expensive idle.** A 10% utilised GPU costs as much as a 90% utilised GPU. Scaling down aggressively saves more money than for CPU workloads.

- **Sharing is hard.** Multiplexing models on shared GPUs requires LoRA serving or similar; otherwise, each model needs dedicated capacity.

94.3.1 Scaling Signals

For LLM-serving autoscaling, three signals matter more than CPU/memory:

- **Queue depth.** Requests waiting for a free batch slot. Direct signal of latency degradation.
- **Time-in-queue p95.** Tail latency on queueing; a leading indicator before requests time out.
- **Tokens-per-second per replica.** The unit of work the GPU is producing; combined with queue depth, gives the throughput-vs-demand picture.

94.3.2 Mitigating Cold Start

Several patterns reduce the impact of slow cold starts:

- Predictive autoscaling based on traffic seasonality (workday curves, regional time zones).
- A small always-warm pool of replicas that the autoscaler maintains as a floor.
- Model multiplexing via LoRA: a single base-model replica serves many fine-tuned variants by swapping small adapter weights; the base model stays warm.
- Pre-emptive scaling on leading indicators (queue length growth rate, not just absolute depth).

94.4 Cost-Performance Trade-Offs

Every performance improvement has a cost. Three recurring trade-offs and their structural responses:

Table 94.3: Recurring cost-performance trade-offs.

Trade-off	Structural response
Larger context windows give higher quality but cost more per call	RAG to send only relevant context; prompt compression for long histories
Higher-quality models give better outputs but 10× higher cost	LLM Router (Chapter 23); cheaper model for easy requests
Lower latency targets require less batching, so less throughput per dollar	Tiered serving: separate latency-critical pool from throughput-critical pool
Closer geographic placement reduces latency but multiplies deployment cost	Region-resident for sovereignty requirements; central for cost-sensitive

The router pattern (Chapter 23) is the structural answer to these trade-offs: route each request to

the right point on the cost-quality-latency frontier, rather than picking one operating point for all traffic.

94.5 Performance Anti-Patterns

Table 94.4: Recurring AI performance anti-patterns.

Anti-pattern	Fix
Synchronous chain of many LLM calls	Reduce chain length; parallelise independent calls; use a single multi-step prompt where possible
Re-embedding stable content on every indexing run	Embedding cache keyed on content hash
Full context every turn (chat)	Summarise older history; retrieve only relevant facts
Frontier model for trivial tasks	Route to a smaller model (Chapter 23)
Sequential tool calls when parallel is possible	Parallel tool invocation; the model emits a batch of calls
Re-running pipelines on unchanged data	Incremental processing; content-addressed caches
Single-thread embedding generation	Batched, vectorised embedding with parallel workers
Synchronous re-rank on every retrieval	Lazy re-rank: only re-rank if first-stage results are ambiguous
Polling for completion instead of streaming	Use streaming APIs; subscribe to completion events

94.6 Performance Testing

Production-grade performance properties require performance testing as a continuous practice, not an annual event.

94.6.1 Load Testing

Synthetic load that exercises the system at expected and stress levels. For AI systems, the load profile matters:

- Request rate (requests per second).
- Prompt distribution (short, medium, long; mix matters because batching efficiency depends on it).
- Tenant distribution (concentrated vs. spread; cache hit rates differ).
- Tool-call density (some prompts trigger tool calls that dominate latency).

94.6.2 Soak Testing

Sustained load for hours or days; surfaces memory leaks, slow degradation, accumulation effects. Particularly important for GPU-serving runtimes where KV-cache fragmentation can build up.

94.6.3 Capacity Planning

Periodic exercises that translate forecast demand into required capacity. Inputs: forecast traffic, observed per-request resource consumption, target SLOs, vendor rate limits. Output: per-region replica counts, per-vendor commitment levels, observability storage requirements.

94.7 Where That Leaves Us

Scalability and performance for AI are matters of explicit budgeting and disciplined trade-offs. Latency budgets surface where time is spent; throughput strategies amortise the cost of going fast; autoscaling adapts to traffic with awareness of GPU economics; the router pattern manages the cost-quality-latency frontier. Performance testing is continuous, not annual. The outcomes—high throughput, predictable latency, elastic capacity—are properties of an architecture that treats performance as a design constraint, not a final-tuning step.

Chapter 95

Interoperability

“Standards are how independent systems become a system.”

—attributed to Vint Cerf

Chapter Overview

Interoperability is the property that lets components built by different teams, in different stacks, by different vendors, work together. In AI, the rapid evolution of capabilities makes interoperability essential: the system that locks in to one vendor’s surface today will be paying to migrate from it next year. This chapter develops the standards that are stabilising the AI surface (OpenAPI for HTTP APIs, OpenTelemetry for observability, MCP and A2A for inter-agent integration, ONNX and OpenLineage for model and data portability), the canonical patterns (API-first design, schema evolution, contract testing), and the structural commitments that keep an enterprise system portable.

Learning Objectives

- Recognise the standards that matter for enterprise AI interoperability.
- Apply API-first design and contract testing to keep interfaces stable as implementations evolve.
- Use OpenAPI, OpenTelemetry, MCP, A2A, ONNX, and OpenLineage where they fit.
- Manage schema evolution across producers and consumers without breaking changes.
- Design vendor-substitutable integration points using the hexagonal pattern.

95.1 The Interoperability Stack

Several stable layers separate two interoperating components. The architecture should be deliberate about which it depends on and which it can leave to the lower layers.

Table 95.1: The interoperability stack for enterprise AI.

Layer	Standards
Transport	HTTP/1.1, HTTP/2, HTTP/3, gRPC, WebSocket
Serialisation	JSON, Protobuf, Apache Arrow, Parquet, MessagePack
API description	OpenAPI 3, gRPC .proto, JSON Schema, AsyncAPI
Identity & auth	OAuth 2, OpenID Connect, mTLS, SPIFFE
Observability	OpenTelemetry (traces, metrics, logs), OpenLineage
AI integration	MCP (tool), A2A (agent), JSON Schema for tool args
Model portability	ONNX, GGUF, safetensors, MLflow
Data portability	Apache Iceberg, Delta, Hudi, Arrow Flight

Each layer has dominant standards; the architectural rule is to choose standards over proprietary surfaces where the standard exists and is mature enough.

95.2 API-First Design

API-first means: the API contract is written before the implementation. The contract (OpenAPI for HTTP, .proto for gRPC) becomes the single source of truth, generating server skeletons, client libraries, and documentation. The discipline is that any change to behaviour is preceded by a change to the contract, which is reviewed in version control.

```

1 openapi: 3.0.3
2 info:
3   title: Enterprise AI Inference API
4   version: 2.1.0
5 paths:
6   /v2/complete:
7     post:
8       summary: Generate a completion grounded in retrieved context
9       operationId: complete
10      requestBody:
11        required: true
12        content:
13          application/json:
14            schema:
15              $ref: '#/components/schemas/CompletionRequest'
16      responses:
17        '200':
18          description: Completion produced successfully
19          content:
20            application/json:
21              schema:
22                $ref: '#/components/schemas/CompletionResponse'
23        '429':
24          description: Rate limit or budget exceeded
25        '503':

```

```

26         description: Upstream model unavailable; retry recommended
27 components:
28   schemas:
29     CompletionRequest:
30       type: object
31       required: [messages]
32       properties:
33         messages:
34           type: array
35           items: { $ref: '#/components/schemas/Message' }
36       max_tokens: { type: integer, default: 1024 }
37       temperature: { type: number, minimum: 0, maximum: 2, default:
38         0.7 }
39     tools:
40       type: array
41       items: { $ref: '#/components/schemas/Tool' }

```

Listing 95.1: An OpenAPI fragment for an AI inference endpoint.

95.2.1 Schema Evolution

API schemas evolve. Done well, evolution is invisible to existing consumers. Done badly, every change is a breaking change. The rules:

- Add fields as optional with defaults; never required.
- Never change the type of a field; add a new field with a new name instead.
- Never remove a field that consumers may use; deprecate, give a sunset date, remove only after the date.
- Use versioned URLs (`/v1/`, `/v2/`) for breaking changes; run versions in parallel during migration.
- Document deprecations in the OpenAPI spec itself (the `deprecated: true` flag).

95.2.2 Contract Testing

Contract tests verify that a producer's output matches the documented schema and that consumers can parse it. Run on both sides of every integration in CI, contract tests catch breakages before they reach production.

- Producer-side: the implementation is verified against the contract; CI fails if the implementation deviates.
- Consumer-side: the consumer is verified against the contract; CI fails if the consumer relies on undocumented behaviour.
- Mutual contract testing: change to the contract triggers re-verification on both sides.

Tools: Pact, Spring Cloud Contract, OpenAPI-based test generators, Postman or Insomnia for ad-hoc verification.

95.3 Standards in AI Specifically

Several standards have emerged for AI-specific surfaces.

95.3.1 Model Context Protocol (MCP)

MCP standardises how agents call tools (Section 18.3). An MCP server hosts tools; an MCP client (typically an agent) discovers and invokes them through a uniform interface. The protocol normalises:

- Tool definition (JSON Schema for arguments).
- Capability discovery (list of available tools and their schemas).
- Invocation (call a tool with structured arguments).
- Result formatting (structured returns, error reporting).

Adopting MCP for tool integration prevents the tool surface from becoming agent-specific.

95.3.2 Agent-to-Agent (A2A)

A2A protocols standardise direct agent-to-agent communication: tasks, artefacts, capability discovery. Where MCP normalises agent-to-tool, A2A normalises agent-to-agent. Both are at an early but rapidly-maturing stage in 2026.

95.3.3 Model Portability

Three formats for model portability:

- **ONNX** (Open Neural Network Exchange). Cross-framework model interchange; widely supported by inference runtimes (ONNX Runtime, TensorRT, OpenVINO).
- **GGUF**. Quantised model format used by llama.cpp and compatible runtimes; the de facto standard for local LLM deployment.
- **safetensors**. Safe (no arbitrary-code execution) tensor serialisation; replacing pickled checkpoints in PyTorch.

95.3.4 OpenLineage

OpenLineage standardises the emission of data and ML lineage events. A producer emits a structured event when it processes data; a backend (Marquez, DataHub, others) collects and visualises it. The architectural value is portability: switching lineage backends does not require re-instrumenting producers.

```
1 {  
2   "eventType": "COMPLETE",  
3   "eventTime": "2026-05-14T14:30:00Z",  
4   "run": { "runId": "01HXR3..." },  
5   "job": {  
6     "namespace": "platform.features",
```

```
7     "name": "customer_features_daily"
8   },
9   "inputs": [
10     { "namespace": "warehouse", "name": "silver.customers" },
11     { "namespace": "warehouse", "name": "silver.transactions" }
12   ],
13   "outputs": [
14     { "namespace": "feature_store", "name": "customer_features:v3" }
15   ]
16 }
```

Listing 95.2: An OpenLineage event for a feature-generation job.

95.3.5 OpenTelemetry

OpenTelemetry is the dominant standard for traces, metrics, and logs. Adopting it as the application-level instrumentation standard means observability backends become interchangeable: a switch from Datadog to Honeycomb is a configuration change, not a re-instrumentation project.

For AI workloads, OpenTelemetry semantic conventions for LLM operations are stabilising: span attributes for model name, model version, prompt template, token counts, and tool calls are increasingly common.

95.4 Data Portability

Open table formats (Iceberg, Delta, Hudi) and open columnar formats (Parquet, Arrow) make data portable across engines. A dataset written by one engine should be readable by another without rewriting; this is the central commitment of the modern lakehouse.

Apache Arrow Flight is the emerging standard for high-throughput data transfer between systems, replacing JDBC/ODBC for analytical workloads.

95.4.1 The Portability Test

Periodic exercise: can the data be read by a second engine without modification? For Iceberg, this might mean reading from Spark and also from Trino, DuckDB, or Snowflake. If the answer requires engine-specific configuration or data movement, the portability claim is weaker than it appears.

95.5 Vendor Substitutability

The hexagonal pattern (Chapter 26) is the structural answer to vendor substitutability. The architectural commitments that make it real:

- Every external dependency is accessed through a port.
- Ports are defined in the application's terms, not the vendor's.
- At least two adapters exist for every port (production and test, at minimum).

- Swap tests run regularly to verify the abstraction is honest.

95.5.1 The Honest Port

A port that genuinely supports multiple adapters must:

- Use only capabilities common to all supported adapters; vendor-specific extensions leak through and break portability.
- Define error semantics that all adapters can satisfy; specific vendor error codes are translated to common ones.
- Specify performance and cost characteristics in adapter-agnostic terms; consumers can reason about behaviour without knowing the adapter.

The trade-off is real: a port supporting the union of vendor capabilities exposes the union of constraints; a port supporting the intersection loses vendor differentiation. Most architectures choose the intersection by default, accepting that vendor-specific features require explicit opt-in.

95.6 Cross-Cutting Interoperability Patterns

A few patterns recur across the interoperability surface:

- **Anti-corruption layer.** A boundary that translates between an external system's vocabulary and the platform's internal vocabulary; prevents external concepts from leaking into the core.
- **Backwards-compatibility window.** An explicit period during which the old and new versions of an interface both operate; consumers migrate within the window.
- **Semantic versioning.** Major-minor-patch versioning that signals the impact of a change.
- **Deprecation policy.** Documented timelines for removing deprecated functionality.

95.7 Holding the Pieces Together

Interoperability is what makes a system more durable than the components it currently uses. Adopting open standards where they exist (OpenAPI, OpenTelemetry, MCP, ONNX, OpenLineage), applying API-first design with schema-evolution discipline, and structuring vendor integration through the hexagonal pattern together make the system portable. The cost is modest indirection; the benefit is freedom of action when the vendor landscape changes, which it will.

Chapter 96

Ethics and Responsible AI

“With great power comes great responsibility.”

—Voltaire (and Spider-Man’s uncle)

Chapter Overview

Responsible AI is the cross-cutting concern that asks not “can the system do this” but “should it”. This chapter develops the technical controls that operationalise responsible AI: fairness measurement, transparency and explainability, human oversight, content safety, consequential-decision review, and bias mitigation. The treatment is technical rather than philosophical; the architectural question is which controls are platform features (every system inherits them) and which are project decisions (made deliberately, documented, and reviewed).

Learning Objectives

- Distinguish the technical controls that make a system fair, transparent, safe, and accountable.
- Measure fairness using group and individual metrics appropriate to the use case.
- Apply explainability techniques (SHAP, attention attribution, citation, counterfactuals) to make model outputs inspectable.
- Design human-in-the-loop oversight for consequential decisions.
- Operationalise content-safety guardrails on inputs and outputs.

96.1 Why Responsible AI Is Architectural

Responsible AI fails when treated as a final review step: the system has been built, the ethics committee assesses it, changes are now too expensive to make. It succeeds when the relevant controls are platform features that every system inherits by default—fairness measurement is wired into evaluation, explainability is wired into the API, oversight is wired into the workflow.

The architectural test for any responsible-AI obligation: is the obligation cheaper to satisfy as a platform feature or as a per-project deliverable? Almost always: cheaper as a platform feature, by 10× or more, because the same control is reused across systems.

There is a second reason platform features beat per-project deliverables, and it matters more than the

cost. A control implemented per project is implemented at the discretion, and the skill, of each project team — which means it is implemented well by the strong teams, poorly by the rushed ones, and not at all by the team that did not know it was expected. A control implemented as a platform feature is implemented once, by the people best placed to get it right, and inherited uniformly by every system regardless of which team built it. Responsible AI delivered per project is responsible AI with gaps shaped exactly like the organisation's weakest teams. Responsible AI delivered as platform features is responsible AI with a floor. The architectural move is to lower as many obligations as possible to that floor.

96.2 Fairness

Fairness measures whether the model's outputs are systematically different across protected groups in ways that are not justified by legitimate factors. Several formal definitions exist; they are mathematically incompatible in general, so the choice is a values choice that the architecture supports rather than makes.

This last point is the one architects most need to internalise, because it is counter-intuitive. It is natural to assume that fairness is a single property a system either has or lacks, and that a sufficiently careful engineer could simply make a system fair. The mathematics says otherwise: the formal definitions of fairness genuinely conflict, and a system that satisfies one provably cannot satisfy another except in degenerate cases. There is therefore no purely technical answer to the question of whether a system is fair — there is only the question of which definition of fairness the use case requires, which is a judgement about values and harms that belongs to the business, the legal function, and the affected stakeholders. The architecture's job is not to choose; it is to make the chosen definition measurable and to support whichever one the organisation, with proper deliberation, selects.

96.2.1 Group Fairness Metrics

The three most commonly used group-fairness definitions:

- **Demographic parity.** $P(\hat{Y}=1 \mid A=a) = P(\hat{Y}=1 \mid A=b)$ for protected attribute A . Equal positive prediction rates across groups, regardless of actual outcomes.
- **Equalised odds.** True-positive and false-positive rates equal across groups: $P(\hat{Y}=1 \mid Y=y, A=a) = P(\hat{Y}=1 \mid Y=y, A=b)$ for $y \in \{0, 1\}$. Equal performance across groups.
- **Predictive parity.** Equal positive predictive value across groups: $P(Y=1 \mid \hat{Y}=1, A=a) = P(Y=1 \mid \hat{Y}=1, A=b)$. Equal precision across groups.

Generally, only two of these can be satisfied simultaneously (Chouldechova; Kleinberg, Mullainathan, Raghavan, 2016). The choice of which to prioritise depends on the harm being avoided:

Table 96.1: Fairness metrics matched to harm types.

Metric	When it matters most
Demographic parity	Allocation of opportunities (hiring, lending) where representation outcomes matter
Equalised odds	Risk assessment where false-positive and false-negative costs are asymmetric
Predictive parity	Decision support where consistent precision is required across groups

The architectural obligation that follows from the table is to make the chosen metric a standing measurement rather than a one-time pre-deployment check. A model that satisfied equalised odds at launch can drift out of it as the input distribution shifts, as the population it serves changes, or as the model is updated. The fairness metric therefore belongs in the continuous evaluation suite, computed on every evaluation run and alerted on when it degrades, exactly as accuracy and latency are. A fairness measurement performed once and filed is a fairness measurement that describes a system that no longer exists; the architecture should treat the chosen metric as a live signal with the same operational standing as any other.

96.2.2 Individual Fairness

Individual fairness requires that similar individuals receive similar treatment. The architectural mechanism is a similarity metric on inputs and a guarantee that the model's outputs are Lipschitz-continuous in that metric. Operationalising this beyond research contexts is difficult; in enterprise systems, group fairness usually dominates.

The difficulty is concentrated in the similarity metric itself. To require that similar individuals be treated similarly, the architecture must define what makes two individuals similar — and that definition is every bit as much a contested values judgement as the choice of group-fairness metric, with the added problem that it must be specified over the full, high-dimensional space of inputs rather than over a handful of protected attributes. An ill-chosen similarity metric can encode exactly the bias the fairness criterion was meant to exclude. This is why enterprise systems lean on group fairness in practice: not because individual fairness is the weaker idea — in some respects it is the stronger one — but because group fairness can be measured against observable attributes, while individual fairness depends on a similarity metric the organisation must construct and defend.

96.2.3 Bias Mitigation

Several techniques reduce bias at different stages of the ML pipeline:

- **Pre-processing.** Reweighting, resampling, or transforming training data to reduce disparate impact.
- **In-processing.** Modifying the learning objective to include a fairness term; constrained optimisation.
- **Post-processing.** Adjusting predictions per group to satisfy a fairness criterion; calibration by group.

The choice depends on what is accessible: post-processing applies even when the model itself cannot be modified; pre-processing applies when training data is controllable.

It is worth being clear that none of the three techniques removes bias; each relocates or reduces it, and each has a cost. Pre-processing alters the training data, which can affect accuracy and raises its own question of whether the altered data still represents the world the model will operate in. In-processing complicates the training objective and can make the model harder to reason about. Post-processing adjusts predictions by group, which is effective but means the system is explicitly treating individuals differently by group membership — a fact that may itself be legally or ethically fraught depending on jurisdiction and use case. The architectural discipline is to treat bias mitigation as a deliberate, documented intervention with stated trade-offs, recorded in a decision record, rather than as a switch that makes a system fair. The mitigation reduces a measured disparity; it does not discharge the obligation to keep measuring.

96.3 Transparency and Explainability

96.3.1 Model Cards and System Cards

A *model card* (Mitchell et al., 2019) is a structured document describing a model: its training data, intended use, evaluation results, known limitations, and ethical considerations. A *system card* extends this to the full system: how the model is used, what safeguards apply, what oversight is in place.

Model and system cards are the most basic transparency obligation. They are produced at first deployment, refreshed at each material change, and reviewed at each scheduled validation. Maintained as code (versioned, peer-reviewed), they become the durable record of what the system is and what it claims.

The discipline that makes cards valuable rather than decorative is honesty about limitations. A model card whose “known limitations” section is empty or anodyne is worse than no card, because it projects a false assurance: a reader trusts it, deploys the model into a context the card did not warn against, and discovers the limitation in production. The limitations section is the most important part of the card and the hardest to write, because it requires the team to document, in advance and in public, the conditions under which their own system should not be trusted. An organisation that treats the card as a marketing artefact will write thin limitations; an organisation that treats it as an engineering artefact will write the limitations that the next team genuinely needs, and the card becomes a real input to the deployment decisions of others.

96.3.2 Explainability Techniques

Different techniques apply to different model types:

Table 96.2: Explainability techniques by model type.

Technique	Where it applies
SHAP values	Tabular ML (tree-based, linear); per-prediction feature attribution
LIME	Any black-box model; local linear approximation
Attention attribution	Transformer-based models (limited reliability)
Citation and grounding	LLM systems; every output cites supporting sources
Counterfactuals	“If feature X had value x' instead, the prediction would change to $\hat{y}'$.”
Confidence calibration	Stated probabilities match observed frequencies

The architect should match the technique not only to the model type but to the audience the explanation serves, because different audiences need different things. A data scientist debugging a model is well served by SHAP values; an affected individual contesting a decision needs a counterfactual — a plain statement of what would have had to be different for the outcome to change — because that is the form of explanation they can act on. A regulator wants the citation trail and the calibration evidence. An explanation that is technically sound but pitched at the wrong audience fails its purpose: a SHAP plot handed to a loan applicant explains nothing they can use. The choice of technique is therefore two decisions, not one — what the model permits, and what the recipient can act on — and a responsible system may need to produce more than one form of explanation for the same decision.

96.3.3 The Limits of Explainability

For modern foundation models, mechanistic interpretability (understanding internal computations) remains an active research area, not a deployable capability. The honest architectural response is to use explainability where it is reliable (post-hoc attribution for traditional ML, citation for RAG) and to use procedural safeguards (human review, conservative defaults) for cases where mechanistic explanation is unavailable.

There is a particular trap to name here: the plausible explanation that is not a faithful one. Post-hoc techniques, and especially a model asked to explain its own output, can produce an account that reads convincingly and does not actually correspond to the computation that produced the decision. This is more dangerous than no explanation at all, because it manufactures unwarranted confidence — a reviewer trusts the plausible story and stops scrutinising. The architectural discipline is to be explicit, in the system card and to the people who rely on explanations, about which explanations are faithful (a RAG citation genuinely is the source the output drew on) and which are merely plausible (a foundation model’s after-the-fact rationalisation of its own answer). Where only plausible explanations are available, the honest course is to say so and lean on the procedural safeguards, not to let a convincing narrative substitute for a faithful one.

96.4 Human Oversight

The EU AI Act and similar regimes require human oversight of high-risk systems. The architectural manifestations:

Table 96.3: Human-oversight patterns.

Pattern	Description
Human-in-the-loop	Human approves each AI action before it takes effect (highest oversight, lowest throughput)
Human-on-the-loop	AI acts autonomously; human monitors and can intervene
Human-over-the-loop	AI acts autonomously; human reviews outcomes periodically and adjusts policy
Escalation	AI acts on routine cases; uncertain or high-stakes cases route to humans

The choice depends on the stakes and the volume. High-stakes, low-volume decisions justify human-in-the-loop. Low-stakes, high-volume decisions are served by escalation: the model handles 95% routine, humans handle 5% edge.

The four patterns are not a fixed menu from which one is chosen forever; the right pattern for a system moves as the system earns trust. A consequential system reasonably launches under human-in-the-loop, with every action approved, while its real-world behaviour is still being learned. As evidence accumulates that the system performs well, the oversight can be deliberately and visibly relaxed — to human-on-the-loop, then perhaps to escalation — with each relaxation treated as an architectural decision, recorded with the evidence that justified it and reversible if that evidence later weakens. The discipline is that the trajectory runs in the direction the evidence supports, not in the direction operational convenience pulls; an oversight pattern loosened because the humans were a bottleneck, rather than because the system earned it, is a governance failure wearing the costume of a maturity milestone.

96.4.1 Designing for Effective Oversight

Human oversight only works if the human is positioned to add value:

- The human must have the information needed to evaluate the AI's recommendation (citations, confidence, reasoning where available).
- The human must have time; rubber-stamp oversight does not exercise judgement.
- The human must be empowered to disagree; oversight where the human has no real option to overrule is theatre.
- Aggregate outcomes of human decisions must be monitored; consistent rubber-stamping is itself a finding.

The deeper hazard the four conditions defend against is automation bias — the well-documented human tendency to defer to a confident machine recommendation even when the person has the information and standing to overrule it. Automation bias means that oversight does not become real merely because a human is placed in the loop; a human who sees a confident AI recommendation, under time pressure, on a screen designed to make approval the easy action, will approve. Designing for effective oversight is therefore partly interface design and partly incentive design: the reviewer's workload must leave genuine time to think, the interface must present the AI's uncertainty as prominently as its

recommendation, and the reviewer must not be measured in a way that punishes disagreement. The last bullet — monitoring the aggregate pattern of human decisions — is the check that reveals whether the defences are working: a reviewer who approves ninety-nine percent of recommendations is either supervising a near-perfect system or not supervising at all, and the architecture must be able to tell which.

96.5 Content Safety

Generative systems can produce outputs that are harmful, illegal, or off-policy regardless of input. Three architectural mechanisms:

- **Input guardrails.** Classifiers that detect off-policy prompts (jailbreaks, harmful requests) and refuse before invoking the model.
- **Output guardrails.** Classifiers that scan model outputs for policy violations and block or rewrite them.
- **Constitutional / policy prompting.** System prompts that specify behaviour the model must follow; effective in combination with other guardrails, insufficient on their own.

Guardrail classifiers themselves can be wrong (false positives reduce utility; false negatives admit harms). The architectural answer is layered defence: input guardrails plus output guardrails plus monitoring plus periodic policy review.

The reason no single guardrail can be trusted is that the input and output guards are themselves models, with their own error rates, and an adversary actively probes for the inputs that slip past them. A jailbreak that defeats the input classifier is, by definition, an input the classifier scored as safe; relying on that one classifier means the jailbreak then reaches the model unopposed. Layering works because an adversarial input must defeat every layer to cause harm, and the layers fail in different ways — an input crafted to fool the input guard is not thereby crafted to fool the output guard, which inspects different material. The architecture should also assume the guardrails will be probed continuously and treat the monitoring layer as the mechanism that detects the probing: a rising rate of near-miss inputs is the early signal that an adversary is mapping the defences, and it should trigger policy review before the successful bypass arrives.

96.5.1 The Layered Defence

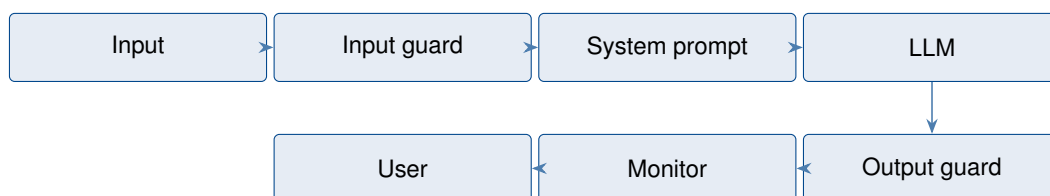


Figure 96.1: Layered defence: input guardrail, system prompt, model, output guardrail, monitoring. Each layer can fail without producing a harmful output if the others hold.

The figure also makes visible a property worth stating explicitly: the layers are independent points of failure in the protective sense, which is the opposite of how independence usually works against a system. In a normal availability analysis, more components mean more things that can break; here, more layers mean more things that must *all* break before a harm reaches the user. This is defence in depth in its precise sense, and it is why the layered design is robust even though every individual layer is imperfect. The architectural caution is to keep the layers genuinely independent — if the input guard, output guard, and monitor are the same classifier in three positions, they share a blind spot and the depth is illusory. Real layered defence uses layers that fail differently, so that no single crafted input is shaped to defeat all of them at once.

96.6 Consequential Decision Review

For decisions that affect people materially (credit, employment, insurance, healthcare, criminal justice, social services), additional architectural commitments apply:

- Pre-deployment review by a multi-disciplinary committee (legal, compliance, ethics, domain experts).
- Explicit risk classification under the EU AI Act or local equivalent.
- Subject-rights operability: explanation on request, contestability, human-review path.
- Post-deployment monitoring of outcomes by group.
- Sunset criteria: under what conditions will the system be retired or substantially modified.

The item most often treated as boilerplate, and most worth taking seriously, is the sunset criteria. Every other commitment on the list governs a system that is running; the sunset criteria govern the decision to stop. Without them, a consequential system tends to persist by default — it was approved once, it is embedded in operations, no one owns the question of whether it should still exist — even as the evidence for retiring it accumulates. Defining the sunset criteria in advance, at deployment, converts that question from an awkward open-ended judgement into a defined trigger: when the disparate-impact monitoring crosses this threshold, when the error rate exceeds this bound, when the regulatory basis changes in this way, the system is retired or substantially modified. Naming the conditions before the system is running, when no one is yet invested in its survival, is the only time the criteria can be set honestly.

96.6.1 Disparate Impact Monitoring

For systems making consequential decisions, monitoring outcomes by protected group is the standing obligation:

- Acceptance rates by group; the “four-fifths rule” in US discrimination law is a useful diagnostic threshold (a group’s acceptance rate falling below 80% of the majority group’s rate triggers scrutiny).
- Calibration by group; the predicted probability and the realised rate should agree across groups.
- Error rates by group; equal precision and recall across groups.

- Outcomes in the wider context; do downstream consequences differ by group in ways the system contributed to?

Two practical conditions make disparate-impact monitoring real rather than nominal. The first is that it requires the protected-attribute data to compute — and that data is often exactly what the system was, for good privacy reasons, designed not to collect. The architecture must resolve this deliberately: by holding the attribute data in a separated, access-controlled monitoring partition, or by using a privacy-preserving estimation method, so that the monitoring obligation and the data-minimisation obligation are reconciled rather than left in silent conflict. The second is that the monitoring must connect to a defined response. A disparate-impact dashboard that no one is obliged to act on, with no threshold that triggers investigation and no owner accountable for the trigger, records the harm without arresting it. The four-fifths threshold and its equivalents matter only because crossing them sets something in motion; monitoring that crosses a threshold and changes nothing is documentation of a problem, not a control against it.

96.7 Documentation and Communication

Responsible AI also requires that the right people know the right things about the system:

- Users know they are interacting with AI (transparency requirement under EU AI Act for many systems).
- Users know what data the system processes about them.
- Users have a route to contest a decision they disagree with.
- Operators have current documentation of the system's known limitations.
- Regulators have on-demand access to evidence packages.

The unifying idea behind the list is that responsible AI has an audience, and each row of the list is an obligation to a different one. Transparency to users, operability of the contest route for affected individuals, current limitations for operators, evidence packages for regulators — these are not five versions of one disclosure but four different communications, each pitched to what its audience needs and can act on. A system that produces a single document and calls it transparency has satisfied none of them well. The architectural commitment is to identify each audience explicitly and to design the communication each one receives, because a responsible system that cannot be understood by the people it affects, operated safely by the people who run it, or inspected by the people who regulate it has met the letter of responsibility while missing its purpose.

96.8 The African Perspective

Responsible AI carries the same obligations everywhere, but several of its specifics need attention in African deployments.

Fairness measurement depends on the protected attributes that matter in the jurisdiction, and those are not universal. The protected categories written into US or European fairness practice — and into

the off-the-shelf fairness toolkits that encode them — do not straightforwardly transfer. The salient categories may include ethnicity, language group, religion, region, or others specific to the society the system serves, and a fairness analysis that measures only the imported categories can miss the disparity that actually matters locally. The architecture must let the protected attributes be configured to the jurisdiction rather than inherited from a toolkit's defaults.

Content-safety guardrails inherit a related problem. Guardrail classifiers are trained predominantly on English-language and high-resource-language data, and they are correspondingly weaker on African languages and on code-switched text — which means both that genuinely harmful content in those languages can pass the guard and that benign content can be wrongly blocked. The layered-defence discipline is more important here, not less, and the monitoring layer must track guardrail performance per language so that the weakness is visible rather than assumed away.

Finally, the regulatory landscape against which consequential-decision review is conducted is the divergent and fast-moving one of the governance-standards chapter: data-protection regimes that resemble GDPR but differ in detail, and AI-specific regulation that is still emerging across most of the continent. The anticipatory posture recommended there applies here too — build the review process to the structure the international consensus is converging on, so that the responsible-AI controls are already in place when local AI law arrives.

96.9 The Architectural Commitment

Responsible AI is an architectural commitment, not a final-stage review. Fairness measurement, transparency artefacts, explainability where reliable, human oversight scaled to stakes, content guardrails, bias mitigation, disparate-impact monitoring, and consequential-decision review together produce a system that meets its obligations and earns its users' trust. The cost of building these in is modest; the cost of being seen to lack them is large.

The through-line of the chapter is that every one of these controls is something the architecture either makes cheap and default or leaves expensive and optional — and that the choice between those two is itself the architectural decision that matters most. An organisation that wires fairness measurement into its evaluation suite, explainability into its APIs, oversight into its workflows, and guardrails into its serving path has made responsible AI the path of least resistance, inherited by every system without heroics. An organisation that leaves each control to each project has made responsible AI a thing that happens when a team has time for it. The technical controls in this chapter are well understood; the architectural work is to ensure they are floors that every system stands on rather than aspirations that the rushed systems quietly skip.

The Governance Standards Landscape: A Map of the External Frameworks

“A standard is a treaty between people who will never meet, about work neither has done yet.”

—a recurring observation in enterprise-architecture practice

Chapter Overview

The previous chapters of this part addressed how an organisation governs its own AI internally — the principles, the review boards, the ethical commitments. This chapter turns outward, to the external frameworks an enterprise AI architecture must answer to: the principles published by international bodies, the regulations with the force of law, the management-system standards that can be certified against, and the data-governance frameworks that the AI inherits. The frameworks are not interchangeable, and they sit at different points on a path from abstract principle to concrete execution. This chapter maps that path. It is a reference, not an argument: its purpose is to let the architect see the whole landscape, understand which frameworks are mandatory and which are voluntary, and know which framework answers which question.

Learning Objectives

- Distinguish the four layers of the governance landscape: principles, regulations, management systems, and data-and-operations frameworks.
- Identify the major framework in each layer and what it requires.
- Recognise which frameworks carry the force of law and which are voluntary or certifiable.
- Map an enterprise’s own AI governance obligations onto the external landscape.
- Use the chapter as a reference when a specific framework is named in an audit, a contract, or a regulation.

97.1 Why the Landscape Has Layers

An enterprise building AI does not face a single governance requirement. It faces a layered stack of them, and the layers do different kinds of work. At the top are *principles* — statements of value,

published by international bodies, carrying moral and reputational weight but not, in themselves, legal force. Below them are *regulations* — laws, with jurisdiction, penalties, and enforcement. Below those are *management-system standards* — certifiable frameworks that describe how an organisation should run its governance as a repeatable process. At the base are *data-and-operations frameworks* — the established disciplines of data management and information security that AI governance builds on and inherits from.

The path runs from abstract to concrete. A principle says AI should be fair. A regulation says that an AI system used for a particular high-stakes purpose must meet specified requirements or face specified penalties. A management-system standard says the organisation must have a documented, audited process for identifying and managing the risks that the regulation cares about. A data-governance framework says the data feeding the AI must have known lineage, known quality, and known ownership. Each layer translates the layer above it into something more actionable, and an enterprise AI architecture has to satisfy all four.

The mistake the chapter is written to prevent is treating the layers as alternatives — adopting a set of principles and believing the regulatory obligation is thereby met, or achieving a management-system certification and believing the principles are thereby satisfied. They are not substitutes. They are a stack, and the architecture answers to all of it.

The layers also have different change dynamics, and an architecture should be designed against that fact. Principles change slowly — the OECD and UNESCO statements have been stable for years. Regulations change in discrete, well-signalled steps, with consultation periods and transition windows. Management-system standards version on a multi-year cadence. Data-and-operations practice evolves continuously. An architecture that hard-wires itself to the current specifics of a fast-changing layer will need frequent rework; an architecture that commits to the slow-changing structure of each layer and treats the specifics as configuration will absorb most changes without redesign. The layered map is therefore not only a way to see the obligations but a guide to which obligations are stable enough to build on and which should be held loosely.

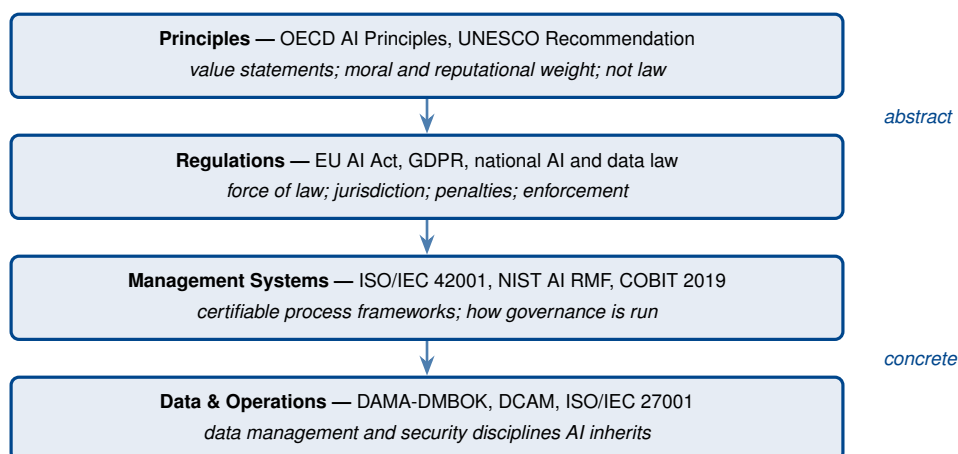


Figure 97.1: The governance landscape in four layers. Each layer translates the one above it into something more actionable. An enterprise AI architecture must satisfy all four; they are a stack, not a menu.

97.2 Layer One: Principles

The principles layer is where the international community states what it believes AI should be. The two reference documents are the OECD AI Principles and the UNESCO Recommendation on the Ethics of Artificial Intelligence.

The **OECD AI Principles**, first adopted in 2019 and updated since, are the most widely referenced. They commit signatory governments to the idea that AI should drive inclusive growth and sustainable development, should be human-centred and fair, should be transparent and explainable, should be robust and safe throughout its lifecycle, and should be deployed by actors who are accountable for its functioning. The OECD principles matter to the architect for two reasons. First, they were adopted by dozens of governments and became the template for much of the regulation that followed — reading them is reading the conceptual vocabulary that later laws use. Second, they are the basis on which an organisation's own published AI principles are usually built; an enterprise AI principles statement that does not at least cover the OECD ground will look thin to any reviewer familiar with the landscape.

The **UNESCO Recommendation on the Ethics of Artificial Intelligence**, adopted in 2021, is broader and more explicitly values-driven. It foregrounds human rights and human dignity, environmental and social well-being, diversity and inclusion, fairness and non-discrimination, and transparency and accountability. It is addressed to UNESCO's member states and is more a statement of societal aspiration than an operational framework. For the architect, its practical use is as a checklist of the societal-impact concerns that a serious AI ethics assessment should consider — particularly the environmental cost of training and inference, and the inclusion concerns that a narrowly commercial framing tends to omit.

Neither document is law. Both produce, as their characteristic enterprise output, an *AI principles statement* and an *ethics policy* — the documents an organisation publishes to declare its commitments. Those documents matter, but the architect should be clear-eyed that publishing them satisfies the principles layer and nothing below it.

The principles layer is nonetheless worth genuine effort rather than treatment as a formality, because it is the layer that the others are eventually measured against. When a regulator assesses whether a high-risk system was deployed responsibly, when an architecture review board judges whether a design is acceptable, when an incident review asks whether an outcome was foreseeable, the reference point is ultimately the organisation's stated principles. Principles written carelessly — vague, aspirational, untestable — give every later judgement nothing firm to stand on. Principles written with the discipline the governance-framework chapter describes, as genuine choices between defensible alternatives, give the whole stack below them a coherent reference. The principles layer carries no legal force of its own, but it is the layer that gives the others their internal consistency.

97.3 Layer Two: Regulations

The regulations layer carries the force of law. Two instruments dominate the landscape in 2026, and a growing number of national laws extend them.

The **EU AI Act** is the first comprehensive horizontal AI regulation. Its defining mechanism is a risk-based tiering of AI systems. *Unacceptable-risk* systems — a defined set including certain social-scoring and manipulative applications — are prohibited outright. *High-risk* systems — AI used in defined sensitive contexts such as credit, hiring, and certain critical infrastructure — are permitted but carry substantial obligations: a risk-management system, data-governance requirements, technical documentation, logging and traceability, human oversight, accuracy and robustness and cybersecurity requirements, and post-market monitoring. *Limited-risk* systems — chatbots and certain other systems where the main concern is that users know they are interacting with AI — carry transparency obligations. *Minimal-risk* systems carry no specific obligations. The architectural consequence is direct: the first governance question for any AI system intended for the EU market is which tier it falls into, because the tier determines the entire compliance burden. The characteristic enterprise outputs the Act demands for a high-risk system are a risk classification, a technical-documentation file, a conformity assessment, and a monitoring plan.

The **EU General Data Protection Regulation (GDPR)** predates the AI Act but governs AI profoundly, because almost every enterprise AI system processes personal data. Its seven principles — lawfulness, fairness and transparency; purpose limitation; data minimisation; accuracy; storage limitation; integrity and confidentiality; and accountability — constrain what data an AI may be trained on and how it may be used. GDPR also grants data subjects specific rights that an AI architecture must be able to honour: access, rectification, erasure, restriction, portability, objection, and — most consequentially for AI — rights related to automated decision-making, including in many cases a right not to be subject to a purely automated decision with legal or similarly significant effect. The characteristic enterprise outputs are a privacy policy, data-protection impact assessments (DPIAs), consent records, and a record of processing activities (ROPA).

These two instruments are European, but their reach is global: any organisation offering AI services into the EU market is bound by them, and a growing list of national laws — across Africa, Asia, the Americas — follow their conceptual structure even where the detailed obligations differ. The architect's discipline is to know, for each AI system, which jurisdictions it operates in and which regulatory obligations therefore attach.

The regulations layer is also the one with the sharpest consequences for getting the architecture wrong, because its obligations are not aspirations but requirements with penalties, and several of them must be designed in rather than added later. A high-risk system's logging and traceability obligation cannot be retrofitted to a system that did not capture the records; the right to contest an automated decision cannot be honoured by a system with no human-review path; the data-subject rights of access and erasure cannot be served by a system whose data architecture cannot locate an individual's data. The architect's discipline is therefore not only to know which jurisdictions apply but to translate their requirements into architectural commitments at design time — because the regulations layer, more than any other, punishes the architecture that treated compliance as something to be arranged after the system was built.

97.4 Layer Three: Management Systems

The management-systems layer is where governance becomes a repeatable, auditable process. Three frameworks matter most.

ISO/IEC 42001, published in December 2023, is the world’s first management-system standard for artificial intelligence. It follows the same structure as other ISO management-system standards — the Plan-Do-Check-Act cycle familiar from ISO 9001 and ISO 27001 — and an organisation can be formally certified against it by an accredited body. The standard requires an organisation to establish an AI policy, a risk-management process, an AI-system impact-assessment process, lifecycle-management controls, competence and awareness provisions, and third-party and supplier oversight. Its characteristic outputs are an AI governance framework, a risk register, and audit reports. For the architect, ISO/IEC 42001 is significant because it is the framework against which an organisation’s AI governance maturity can be externally certified — the way an organisation proves, to customers and regulators, that its AI governance is a real process and not a published aspiration.

The **NIST AI Risk Management Framework (AI RMF)**, released as version 1.0 in January 2023 by the United States National Institute of Standards and Technology, is voluntary — it cannot be certified against — but it is widely adopted because it is practical and well-constructed. It organises AI risk management into four functions: *Govern* (the cross-cutting function that establishes the culture and structures within which the other three operate), *Map* (establishing the context and identifying the risks), *Measure* (analysing and tracking the risks), and *Manage* (acting on the risks). *Govern* is deliberately cross-cutting — it is not a stage but the connective tissue that holds the other three together. The framework’s characteristic outputs are a risk assessment, a set of metrics, a mitigation plan, and a monitoring dashboard. For the architect, the NIST AI RMF is useful as a working method even where it is not contractually required, because it gives a concrete structure to the otherwise vague instruction to “manage AI risk.”

COBIT 2019, maintained by ISACA, is not an AI framework at all — it is a framework for the governance and management of enterprise IT in general. It matters here because AI governance does not sit apart from IT governance; it is part of it. COBIT provides the structures by which IT is aligned with business goals, value is created and measured, risk is optimised, and performance is managed. An organisation that already governs its IT through COBIT will, sensibly, extend that governance to AI rather than building a parallel structure. Its characteristic outputs are governance objectives, key performance indicators, a control structure, and an audit structure.

The distinction the architect must hold clearly: ISO/IEC 42001 is certifiable and AI-specific; the NIST AI RMF is voluntary and AI-specific; COBIT 2019 is the general IT-governance frame that the AI-specific frameworks sit inside.

The three are complementary rather than competing, and a mature organisation uses all three in their respective roles. COBIT provides the enterprise IT-governance container, so that AI governance is a part of how the organisation governs technology rather than a parallel structure that duplicates and eventually conflicts with it. The NIST AI RMF supplies the working method — the concrete Govern-Map-Measure-Manage activities by which AI risk is actually handled day to day. ISO/IEC 42001 supplies the certifiable management system that lets the organisation prove, to an external auditor,

that the method is real and sustained. An organisation that adopts only the voluntary framework has a method but no external proof; one that pursues only the certification risks a management system that satisfies the auditor without genuinely managing risk. The three are strongest used together, each in the role it is built for.

97.5 Layer Four: Data and Operations

The base layer is the data-management and information-security disciplines that AI governance builds on. AI governance that is not founded on sound data governance is governance of the model alone, and the model is the smaller half of the problem.

DAMA-DMBOK — the Data Management Body of Knowledge, maintained by DAMA International — is the established reference for enterprise data management. It organises the discipline into eleven knowledge areas: data architecture; data modelling and design; data storage and operations; data security; data integration and interoperability; document and content management; reference and master data; data warehousing and business intelligence; metadata management; data quality; and data governance itself. For the AI architect, DMBOK matters because every one of its eleven areas is a precondition for trustworthy AI. A model trained on data with no lineage, no quality controls, no master-data discipline, and no ownership is a model whose behaviour cannot be governed regardless of how elaborate the AI-specific governance above it is. The characteristic outputs are a data catalogue, data-lineage records, data-quality reports, and an ownership matrix.

DCAM — the Data Management Capability Assessment Model, from the EDM Council — is a maturity model for data management. Where DMBOK describes the knowledge areas, DCAM provides a structured way to assess how mature an organisation actually is across them, typically on a scale from initial through repeatable, defined, and managed to optimised. Its characteristic outputs are a maturity-assessment report, a gap analysis, and an improvement roadmap. For the AI architect, DCAM is the instrument that answers the question “is our data foundation actually ready for the AI we are proposing to build on it” — a question that is better asked before the AI programme than after it.

ISO/IEC 27001 is the international standard for information-security management. It is certifiable, widely held, and directly relevant to AI because AI systems are information systems with, in many cases, elevated security exposure — they process large quantities of sensitive data, they introduce new attack surfaces, and they create new categories of incident. Its scope covers information-security policies, risk management, access control, asset management, cryptography, incident management, business continuity, and monitoring and review. For the AI architect, ISO/IEC 27001 is the framework that governs the security of the data and infrastructure the AI runs on, and an AI security posture that does not rest on a sound 27001 foundation is building on sand.

The reason this base layer is placed last in the chapter but first in importance is that the layers above it cannot compensate for its absence. An organisation can hold every AI-specific certification, publish exemplary principles, and meet every regulatory obligation on paper, and still deploy an untrustworthy system if the data underneath has unknown lineage, unmeasured quality, and unclear

ownership. The AI-specific frameworks govern the model and its lifecycle; the data-and-operations frameworks govern the material the model is made from. A mature organisation usually finds that its data-governance maturity, not its AI-specific governance, is the true constraint on how far its AI ambitions can responsibly go — which is why the honest sequencing is to assess the data foundation, with an instrument like DCAM, before committing to the AI programme that will stand on it.

97.6 Reading the Landscape as a Whole

The four layers compose into a single obligation map. An enterprise AI architecture, to be properly governed, must be able to demonstrate the following chain: that its published principles reflect the international consensus of the principles layer; that it has correctly identified which regulations apply to each AI system and is meeting their specific obligations; that its governance is run as a repeatable, auditable process of the kind the management-system standards describe, and ideally certified where certification is available; and that the whole structure rests on a data-and-operations foundation with known lineage, known quality, known ownership, and sound security.

A useful discipline is to maintain, for the AI estate as a whole, an explicit register that maps each obligation to its source framework and to the enterprise artefact that satisfies it. The principles layer is satisfied by the published AI principles statement and ethics policy. The regulatory layer is satisfied, per system, by risk classifications, technical-documentation files, conformity assessments, DPIAs, consent records, and records of processing activities. The management-system layer is satisfied by the AI governance framework, the risk register, and the audit reports, with ISO/IEC 42001 certification as the external proof. The data layer is satisfied by the data catalogue, the lineage records, the quality reports, the ownership matrix, the maturity assessment, and the ISO/IEC 27001 certification. When an auditor, a regulator, or a customer names a framework, the register is what lets the architect answer immediately rather than discovering the gap under examination.

The core governance components that operate this map are the ones the architecture governance chapter (Chapter 31) develops in detail: the people — data owners, data stewards, data custodians, an AI governance board, model owners, an ethics committee; the processes — policy and standards, the data and model lifecycle, risk management, monitoring and assurance, issue management and escalation; and the technology — the data-catalogue and lineage tools, the data-quality tools, the model-monitoring and drift-detection tools, the access and identity tooling, the audit and logging infrastructure, and the privacy and anonymisation tooling. The frameworks of this chapter say what must be done; the governance components say who does it and with what tools.

The obligation register is worth one further note, because its value compounds over time. Built once and maintained, it becomes the single artefact that turns every subsequent compliance event from a scramble into a lookup. A new regulation arrives: the register shows which existing artefacts already address its requirements and which gaps are genuinely new. A customer's procurement questionnaire asks about a framework: the register answers it. An audit is scheduled: the register is the evidence index the auditor is walked through. An organisation that maintains the register treats compliance as a standing, queryable state of the architecture; an organisation without it rediscovers its own compliance posture, expensively and under time pressure, every time someone asks. The register

is the difference between governance that is a property of the architecture and governance that is a recurring emergency.

97.7 The African Perspective

The governance-standards landscape has a particular shape for enterprises operating in African jurisdictions, and three features of it deserve naming.

A preliminary point frames all three. The four-layer structure of the chapter is universal — principles, regulations, management systems, and data-and-operations frameworks are a stack everywhere — but the *contents* of the regulatory layer, and the *conditions* under which the management-system layer can be certified, are where African deployments diverge most from the European and North American context the frameworks were largely written in. The structure transfers; the specifics must be re-derived locally. The three features below are the points where that re-derivation matters most.

The first is regulatory divergence and convergence at the same time. African data-protection law has expanded rapidly — South Africa’s POPIA, Nigeria’s data-protection regime, Kenya’s Data Protection Act, and a growing list of others — and most of these laws are conceptually close to GDPR, which means an enterprise that has built its architecture to satisfy GDPR has done much of the work. But the detail diverges: data-localisation requirements, the specifics of cross-border transfer, the constitution and powers of the supervisory authority, and the timelines for breach notification all vary by jurisdiction. The architectural discipline is to maintain the per-jurisdiction obligation register rather than assuming a single “African” regime exists. It does not. There are over fifty jurisdictions, and an architecture serving several of them answers to several regulators.

The second is the AI-specific regulatory gap. As of the mid-2020s, comprehensive AI-specific legislation of the kind the EU AI Act represents is not yet in force across most of the continent, though several jurisdictions have published AI strategies and draft frameworks. The architectural posture this calls for is anticipatory: build to the conceptual structure of the emerging international consensus — risk tiering, impact assessment, human oversight, post-market monitoring — so that when local AI law arrives, as it will, the architecture is already substantially compliant. Building to the EU AI Act’s structure is a reasonable proxy even for systems that will never touch the EU market, because the Act is the most likely template for the laws that follow.

The third is the cost and availability of certification. ISO/IEC 42001 and ISO/IEC 27001 certification requires accredited auditors, and the accredited-auditor ecosystem is less dense in many African markets than in Europe or North America, with corresponding implications for cost and lead time. The architectural response is to treat certification as a planned, budgeted, scheduled programme rather than an afterthought, and to build the management-system processes to the standard’s requirements from the start — so that when certification is pursued, it is a matter of demonstrating an existing process rather than constructing one under audit pressure.

97.8 What the Map Is For

This chapter is a reference, and its value is realised at specific moments: when a regulator names a framework, when a customer's procurement questionnaire asks which standards the AI estate complies with, when a contract requires ISO/IEC 42001 certification, when an auditor asks to see the risk-management process, when a new AI system has to be classified under the EU AI Act's risk tiers. At each of those moments, the architect who holds the landscape in mind can answer precisely — this framework, this layer, this obligation, this enterprise artefact that satisfies it — rather than improvising.

The chapter is best used, in practice, alongside the obligation register it recommends. The register is the chapter's content made specific to one organisation: where this chapter names the frameworks in general, the register records which of them apply to this estate, which artefacts satisfy them, and where the gaps are. An architect who has read the chapter understands the landscape; an architect who has also built the register can navigate it for their own organisation without re-deriving it each time. The two together — the general map and the organisation-specific register — are what convert governance from a body of knowledge held by a few specialists into a property of the architecture that any reviewer can inspect.

The deeper point is the one the layered figure makes. Principles, regulations, management systems, and data-and-operations frameworks are not competing approaches to AI governance among which an organisation chooses. They are a single stack, running from abstract value to concrete data lineage, and a properly governed enterprise AI architecture answers to all four layers at once. The frameworks will continue to evolve — the AI Act will be amended, ISO/IEC 42001 will acquire sector-specific extensions, new national laws will arrive every year — but the layered structure is stable, and an architecture built to satisfy the structure will absorb the specific changes without being redesigned. That is the same disposition the rest of the book has argued for throughout: build to the structure that will last, hold the specifics lightly, and expect the landscape to keep moving.

Part VII

Building, Operating, and Scaling

“We make machines of the men, as they cannot otherwise be made to do their work with regularity and despatch.”

—Josiah Wedgwood, on the operation of his Etruria works, 1769

The disciplines and the horizontal standards are in place. They have to be operated. When Josiah Wedgwood opened his Etruria works near Stoke-on-Trent in 1769, he combined three things that had not previously been combined at scale: artistic quality, industrial process discipline, and a commercial brand. The factory turned out pottery of consistent quality in volumes that had been impossible from individual potters’ wheels, and it did so by ruthlessly standardising the operations — division of labour, time discipline, quality inspection, recorded recipes, signed pieces. Part VII of this book describes the equivalent disciplines for running enterprise AI in production: how the architecture survives contact with the operations that actually run it, day after day, year after year, at the scale the business demands.

Chapter 98

Team Topologies for AI

“If the architecture of the system and the architecture of the organisation are at odds, the architecture of the organisation wins.”

—Ruth Malan, on Conway’s Law

Chapter Overview

The architecture of a system mirrors the communication structure of the organisation that builds it. Applied deliberately, this observation is a design tool: shape the teams that you want to shape the system. This chapter applies Skelton and Pais’s *Team Topologies* framework to enterprise AI, identifies the four canonical team types, the three interaction modes, the cognitive-load limits, and the role of the Centre of Excellence. The organisational structure developed here is the human side of the hub-and-spoke pattern (Chapter 24); without it, the hub-and-spoke architecture either fails to form or fails to deliver.

Learning Objectives

- Apply Conway’s Law deliberately: design team structure that produces the architecture you want.
- Distinguish the four team types and use them appropriately for AI programmes.
- Use the three interaction modes (collaboration, X-as-a-Service, facilitating) at the right moments.
- Locate the Centre of Excellence as an enabling team rather than a platform team.
- Apply cognitive-load limits to team sizing; recognise common organisational anti-patterns.

98.1 Conway’s Law, Applied

Melvin Conway observed in 1968 that “organisations which design systems are constrained to produce designs which are copies of the communication structures of these organisations.” The observation has become a design choice: *Inverse Conway Maneuver*—design the team structure that would produce the architecture you want, before building the system. If the architecture should be hub-and-spoke, the team structure should be hub-and-spoke. If services should be loosely coupled, the teams should be loosely coupled. If governance should be a platform feature, a platform team must exist to operate

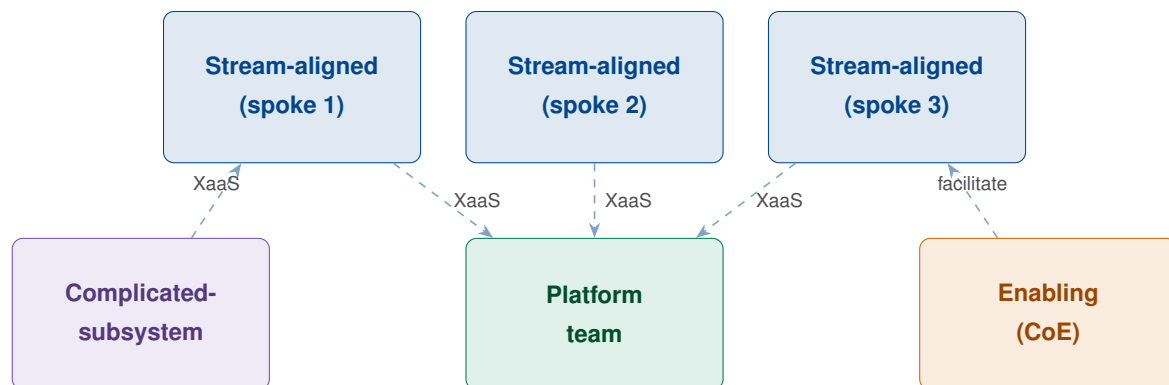
it.

The architecture and the org chart are the same diagram drawn at different scales. Architectural reviews that ignore team structure produce architectures that the organisation will fail to build; team designs that ignore architecture produce structures that drift away from the intended system.

The practical force of the Inverse Conway Maneuver is that it changes when organisational design happens. In the common pattern, the system is designed first and the teams are assembled afterwards to build it — at which point the team structure, shaped by hiring convenience and existing reporting lines, quietly bends the architecture toward itself. The maneuver reverses the order: the team structure is designed first, deliberately, as the structure that would naturally produce the intended architecture, and the system is then built by teams already shaped to produce it. This is harder organisationally — it means treating team design as an architectural deliverable, owned by the same people who own the architecture — but it is the only way to make Conway’s Law an ally rather than an adversary. An architecture programme that does not also own its team design has conceded half its leverage before it starts.

98.2 The Four Team Types

Skelton and Pais identify four team types. Each plays a different role in the system and has a different relationship with the others.



Most steady-state interactions are X-as-a-Service via stable interfaces

Figure 98.1: The four team types in an AI programme. Stream-aligned teams (spokes) consume the platform as a service; the Centre of Excellence (an enabling team) facilitates capability transfer; complicated-subsystem teams own specialised capabilities behind stable interfaces.

Table 98.1: The four team types applied to enterprise AI.

Type	Role in an AI programme
Stream-aligned	A business-unit team building AI-powered use cases end-to-end on the paved path. Owns one or more use cases from discovery through operation. Sized to a single value stream.
Platform	The platform team operating shared services: model gateway, vector store, observability, IAM, FinOps. The platform's product is the developer experience of the stream-aligned teams.
Enabling	A team that transfers capability to others rather than running services. The AI Centre of Excellence sits here. Embeds with stream-aligned teams temporarily, leaves them more capable.
Complicated-subsystem	A small specialist team owning a deeply specialised capability (a fine-tuned model, a custom retrieval engine, a research-grade safety classifier). Consumed by others through a stable interface.

98.2.1 Stream-Aligned Teams

The spokes. Each owns one or more use cases end-to-end: discovery, design, build, run, improve. The team is multi-disciplinary—product, design, engineering, data science, sometimes a specialist analyst. Cognitive load is bounded by the use cases under management; teams adding use cases beyond this bound either split or stop adding.

The defining property of a stream-aligned team is end-to-end ownership: the team that ships is the team that operates and the team that improves. Hand-off to a separate operations team breaks the feedback loop and produces use cases that ship, operate poorly, and stagnate.

End-to-end ownership has a sharper meaning for AI use cases than for conventional software, because an AI use case does not stabilise after launch the way a conventional feature does. Its quality drifts as inputs change and as the underlying model is updated; its costs move with usage; its evaluation results need continuous attention. A team that has shipped an AI use case and moved on has not finished — it has left a system that will silently degrade without the ownership that only a stream-aligned team provides. This is why the hand-off-to-operations pattern is particularly damaging here: the separate operations team can keep a conventional application running, but it cannot do the continuous evaluation and model-update work that an AI use case needs, because that work requires the use-case knowledge that only the building team holds. End-to-end ownership is not an organisational preference for AI; it is a requirement of the technology.

98.2.2 Platform Team

The platform team's product is the developer experience of the stream-aligned teams. The platform team measures itself on how quickly a stream-aligned team can ship a new use case on the paved path. Internal SLAs, self-service tooling, clear documentation, and developer-facing metrics are its outputs.

A platform team that measures itself on platform activity (services deployed, features released,

tickets resolved) without measuring spoke outcomes is at risk of building infrastructure that does not get adopted. The reverse—a platform team obsessively tracking spoke outcomes—tends to build infrastructure that does.

The discipline that keeps a platform team honest is treating the platform as a product and the stream-aligned teams as its customers — which means the platform team must know whether its customers would, if they could, choose something else. A platform that spokes adopt only because they are mandated to is a platform that has stopped competing for its adoption, and it will accumulate the friction, the unaddressed gaps, and the poor developer experience that an unaccountable internal monopoly always does. The healthiest platform teams behave as though adoption were voluntary even when it is not: they measure how quickly a spoke can ship on the paved path, they treat a spoke routing around the platform as a defect report rather than a compliance violation, and they earn adoption rather than assuming it. The platform's product is the spoke's velocity, and a platform team that loses sight of that builds infrastructure for its own sake.

98.2.3 Enabling Teams

Enabling teams transfer capability rather than build durable infrastructure. An AI Centre of Excellence is typically an enabling team: it embeds with a stream-aligned team for the duration of a use case (or a set of related use cases), brings up the team's capability in the relevant areas, and moves on when the team is self-sufficient.

The Centre of Excellence is *not* a platform team. Confusing the two produces a function that simultaneously cannot keep up with operational demand (because it is trying to operate services) and cannot enable adoption (because it is consumed with operations). The architectural fix is to separate the two: a platform team operates the paved path; an enabling CoE helps spokes adopt it.

The deeper reason the distinction matters is that the two team types have opposite success conditions. A platform team succeeds by being depended on — the more spokes rely on its services, the better. An enabling team succeeds by being dispensed with — a CoE engagement that ends with the spoke self-sufficient is a success, and one that never ends is a failure. A function that has been asked to be both is being asked to pursue two contradictory goals at once, and it will resolve the contradiction badly: usually by drifting toward the platform role, because operating a service produces visible, fundable, permanent work, while enabling produces, when it works, its own obsolescence. Keeping the CoE structurally separate from the platform is what protects the enabling mission from being quietly eaten by the operational one.

98.2.4 Complicated-Subsystem Teams

A complicated-subsystem team owns a specialised capability that requires deep expertise concentrated in a few people. Examples in AI contexts:

- A domain-specific fine-tuned model used across the enterprise (e.g., a medical-coding model in healthcare; a financial-document parser in banking).
- A custom retrieval engine optimised for an unusual corpus (legal contracts, scientific literature).

- A safety/red-team capability that researches emerging attack patterns.
- A vendor-equivalent capability built in-house for sovereignty reasons.

The team is small (3–5 specialists), the system is complex, and consumers interact with it through a stable interface that hides the complexity. The interface is what makes the complicated-subsystem team feasible: without it, every consumer becomes a partial expert in the subsystem, defeating the purpose.

The judgement an architect must make carefully is when a capability genuinely warrants a complicated-subsystem team rather than living inside a stream-aligned team or the platform. The test is whether the capability requires depth of expertise that cannot reasonably be spread — expertise that takes years to build and that a generalist team cannot maintain alongside its other work. A fine-tuned medical-coding model meets that test; a standard RAG pipeline does not. The failure mode in both directions is real: carving out a complicated-subsystem team for a capability that is not actually that specialised creates an unnecessary silo and an unnecessary interface, while forcing a genuinely deep capability into a stream-aligned team overloads it and produces an amateur version of work that needed specialists. The complicated-subsystem team is the right answer specifically when the expertise is both essential and genuinely scarce.

98.3 Interaction Modes

Teams interact in one of three modes at any given moment. The mode is a choice, not a default.

The phrase “at any given moment” carries the important idea. A pair of teams does not have one permanent interaction mode; the mode is expected to change as their relationship matures. Two teams typically begin in collaboration while a new capability is being discovered, then transition to X-as-a-Service once the interface between them has stabilised. An enabling team and a spoke begin in facilitating mode and end with no interaction at all once the spoke is self-sufficient. The architectural discipline is to name the current mode explicitly, and to know what the next mode should be and what triggers the transition — because the most common interaction-mode failures, developed in the subsections below, are all failures to transition: a collaboration that never becomes a service, a facilitating engagement that never ends.

Table 98.2: The three team-interaction modes.

Mode	Bandwidth	When to use
Collaboration	High; daily working contact	Discovering a new pattern; a bounded, intense period when no stable interface yet exists
X-as-a-Service	Low; via interface contracts	Steady state for most interactions; one team provides, others consume
Facilitating	Medium; coaching-style	An enabling team helps another team build a capability they will own afterwards

98.3.1 Collaboration

Two teams work jointly with high bandwidth for a bounded period. The goal is to discover an interface that does not yet exist. Collaboration is expensive (it pulls people away from their primary work) and is reserved for genuine novelty—when the platform team is figuring out, with a willing spoke, what a new shared service should look like.

The hazard: indefinite collaboration. A collaboration that has gone on for six months without producing a stable interface is no longer discovery; it is two teams permanently entangled. The fix is to time-box collaborations and to insist on an extraction at the end: a new XaaS interface, or a new enabling engagement.

Collaboration is the most expensive interaction mode and should be treated as a deliberate, budgeted investment rather than a comfortable default. Its cost is the high-bandwidth contact itself: both teams are pulled partly off their primary work for the duration, and the organisation is paying two teams' partial attention to discover one interface. That cost is worth paying when the interface genuinely does not yet exist and cannot be specified from a distance — but it is only worth paying until the interface is found. The discipline of the explicit extraction is what converts collaboration from an open-ended entanglement into a bounded discovery: the collaboration is declared finished when, and only when, it has produced the stable interface that lets the two teams move to the cheaper X-as-a-Service mode.

98.3.2 X-as-a-Service

The dominant steady-state interaction. One team provides a capability through a stable interface; others consume it. The interface is the contract; behaviour outside the contract is the provider's business.

Most platform-to-spoke interactions are XaaS once the platform is mature. Spokes do not need to know how the model gateway is implemented; they need to know how to call it and what guarantees it provides. The interface is the boundary.

X-as-a-Service is the mode that scales, and that is its central virtue. A collaboration relationship cannot be replicated across forty teams — the bandwidth cost would consume the providing team entirely — but an X-as-a-Service interface can serve forty consumers as easily as four, because each consumer interacts only with the stable contract and never with the provider's people. This is why a maturing AI programme should be steadily converting its interactions to XaaS: every relationship left in collaboration mode is a relationship that limits how many teams the provider can serve. The quality of the interface is therefore a scaling decision, not merely an engineering one. A well-specified, stable, well-documented interface lets a platform team serve the whole organisation; a vague or unstable one forces the platform team back into per-consumer hand-holding and caps the programme's growth at the platform team's capacity for conversation.

98.3.3 Facilitating

An enabling team helps another team build a capability the consumer will own afterwards. Bandwidth is medium: workshops, paired sessions, code reviews. The duration is bounded by the consumer team

reaching self-sufficiency.

A common abuse: “facilitating” that lasts forever because the consumer never reaches self-sufficiency. The cause is usually that the enabling team is doing the work rather than transferring it. The fix is explicit handover criteria: when is this engagement done?

The distinction between doing the work and transferring the capability is subtle in practice and worth making concrete. An enabling team that, asked to help a spoke adopt the evaluation harness, simply builds the spoke’s evaluation suite for them has produced the right artefact and failed the engagement — the spoke now has an evaluation suite and still cannot build the next one. An enabling team that instead pairs with the spoke while the spoke builds it, explaining the choices, has produced the same artefact and succeeded, because the capability now lives in the spoke. The discomfort is real: transferring capability is slower than doing the work, and a spoke under deadline pressure will often ask the enabling team to just do it. Holding the line — coaching rather than delivering, even when delivering would be faster this once — is what makes the enabling team an enabling team rather than a slow, embedded delivery team. The explicit handover criteria exist to make that line visible and enforceable.

98.4 The Centre of Excellence in Detail

The AI Centre of Excellence (CoE) is an enabling team with platform-adjacent responsibilities. It typically:

- Publishes the architecture standards (this book, in effect) and reference designs.
- Runs training programmes and certification for spoke teams.
- Convenes regular reviews of in-flight use cases.
- Maintains the pattern library and the evaluation harness used across spokes.
- Liaises with risk, compliance, legal, and ethics functions.
- Curates the model and tool catalogues that the platform exposes.

The CoE does *not*:

- Operate the platform services (the platform team does).
- Build use cases (the stream-aligned teams do).
- Approve every change (a paved-path approach delegates routine approvals).
- Constitute a permanent dependency on the spokes (engagements are time-boxed).

The CoE’s success metric is the rate at which spoke teams move from “can ship a use case with embedded CoE support” to “can ship without it.” A CoE whose engagements never end is over-engaging; one whose engagements never start is under-engaging.

This success metric has an uncomfortable implication that a CoE and its sponsors must both accept: a CoE doing its job well is steadily working toward needing fewer people. As spokes become self-sufficient, the demand for embedded enabling support falls, and a CoE measured honestly will report that its engagement load is declining — which, for a function whose budget is often tied to headcount and activity, feels like failure and is in fact success. An organisation that does not understand this will pressure its CoE to stay busy, and a CoE under pressure to stay busy will quietly start operating services or building use cases to fill the time, becoming the misclassified function the earlier section warned against. Protecting the CoE’s enabling mission therefore requires the organisation to fund it for the capability it transfers, not the hours it logs, and to treat the planned shrinkage of a successful CoE — or its conversion into a lean standards body — as the intended outcome rather than a budget problem.

98.5 Team Sizes and Cognitive Load

Skelton and Pais argue for the *cognitive load limit*: a team should not be asked to hold more in its head than a small group can manage well. Three categories of cognitive load matter:

- **Intrinsic.** The complexity of the work itself (modelling, distributed systems, security).
- **Extraneous.** The complexity of the environment (deployment friction, unclear documentation, brittle dependencies). A good platform reduces this.
- **Germane.** The complexity of the learning that produces durable capability. Worth investing in.

For AI teams, typical sizings:

- Stream-aligned: 5–9 people including product, design, engineering, data science.
- Platform: 5–9 per platform area (data, model serving, observability, governance); multiple areas constitute a larger platform organisation.
- Enabling (CoE): 2–5 deeply specialised people.
- Complicated-subsystem: 3–5 specialists.

Teams above 9 tend to fragment communication—the famous “two-pizza team” bound. Teams below 5 tend to lack resilience to attrition or vacation. The numbers are not absolute, but the bounds are real.

The cognitive-load categories give the architect a more useful lever than team size alone, because two of the three are addressable by design. Intrinsic load — the genuine difficulty of modelling, distributed systems, security — is mostly fixed by the nature of the work. But extraneous load is created by the environment and can be removed: a good platform, by absorbing deployment friction, brittle dependencies, and unclear tooling, directly reduces the extraneous load on every spoke, which is the same as saying it lets each spoke hold more genuine work in the same cognitive budget. This reframes the platform team’s purpose in cognitive terms: the platform exists to convert the spokes’ extraneous load into spare capacity for intrinsic and germane work. A spoke that is overloaded is not always a

spoke that needs to split; sometimes it is a spoke whose platform is making it carry extraneous load that good infrastructure would have removed. The architect should diagnose which kind of overload it is before reaching for the team-splitting remedy.

98.6 Anti-Patterns

Several team-level anti-patterns recur in AI programmes:

Table 98.3: Team-structure anti-patterns and their mitigations.

Anti-pattern	Mitigation
Data-science silo	Embed data scientists in stream-aligned teams from day one; the silo produces demos, not products.
Infrastructure silo	Platform team measures itself by spoke outcomes; reorganise if platform metrics are about platform activity.
Misclassified CoE (as platform)	Separate CoE from platform; let the platform operate services, let the CoE transfer capability.
Committee-as-team	A “team” that is actually a steering committee meeting weekly is not a team; every team member must be doing the work, not just deciding about it.
Permanent collaboration	Collaborations are bounded; extract a XaaS interface or end the engagement.
Over-broad stream-aligned team	A team owning twelve use cases is not stream-aligned; split or scope.
Under-broad platform	A platform team that owns three services that need to scale across forty spokes is the bottleneck; expand to multiple platform areas.

What these anti-patterns share is that each one feels reasonable at the moment it forms. A data-science silo forms because gathering the scarce data scientists together looks like good resource management. A misclassified CoE forms because giving the capable central team a service to operate looks like using them well. A committee-as-team forms because the decision-makers genuinely need to meet. None of these is a foolish choice in isolation; each becomes an anti-pattern only against the topology the programme actually needs. This is why the anti-patterns are worth naming explicitly and reviewing for periodically: they are not caught by anyone noticing an obvious mistake, because no obvious mistake was made. They are caught only by an architect deliberately checking the current team structure against the four-type, three-mode model and asking, of each team, whether it is still the type the system needs it to be.

98.7 Evolving the Structure

Team structure is not designed once. As an AI programme matures, the structure evolves:

- Early programme (one or two pilots): a single small team can do everything. No platform, no CoE.
- Mid programme (five to ten use cases): a platform team forms to consolidate shared services; a small CoE forms to enable adoption.
- Mature programme (twenty-plus use cases across many spokes): multiple platform areas, a properly

resourced CoE, complicated-subsystem teams emerging for specialised capabilities.

- Late mature programme: AI is no longer a programme; it is dispersed across the organisation. The CoE may shrink or convert into a standards body.

The mistakes are at the transitions: adding a platform team too late (when the spokes have already built their own), adding a CoE too early (when there is nothing yet to enable), or never splitting a team that is now doing too much.

The transitions are hard precisely because each is a reorganisation, and reorganisations are disruptive enough that an organisation will tend to defer them past the point where they were due. The deferral is itself the anti-pattern: a programme that should have formed a platform team at ten use cases but waits until thirty finds that the spokes have each built their own divergent infrastructure, and the platform team's first job is now the far harder one of consolidating five incompatible approaches rather than providing one. The architectural discipline is to anticipate the transitions — to know, from the programme's use-case count and spoke count, which structural change is next — and to make each change slightly before it is forced, when it is still cheap, rather than slightly after, when it has become expensive. Team structure, like the architecture it mirrors, is cheaper to evolve deliberately than to correct under duress.

98.8 The African Perspective

Team topologies for AI inherit the same four types and three modes everywhere, but the talent context in many African markets shifts how the model is best applied.

The scarcity of deeply specialised AI talent is the dominant constraint, and it bears most directly on the enabling team and the complicated-subsystem team. The people qualified to staff a Centre of Excellence or a complicated-subsystem team are few, expensive, and sought after, and an organisation cannot assemble them as freely as the model assumes. This makes the enabling team's mission — transferring capability so that scarce expertise multiplies rather than bottlenecks — more valuable on the continent than almost anywhere, and it makes the failure mode of an enabling team that does the work instead of transferring it more costly, because the scarce specialists are then consumed as delivery capacity rather than spent on building the wider capability the market badly needs.

The same scarcity argues for a particular sequencing discipline. A programme that cannot staff every team type at once should stage them deliberately: a small stream-aligned team first, an enabling function as early as the talent allows so that capability begins to spread, a platform team as the use-case count justifies consolidation, and complicated-subsystem teams last and only where the specialised capability is genuinely essential and genuinely cannot be bought. The proportionality and evolution discipline of this chapter is not optional in a talent-constrained market; it is the mechanism by which a programme grows within the talent it can actually recruit, rather than designing an organisation chart it has no people to fill.

98.9 Pulling the Threads

The team structure is the architecture in human terms. Four team types (stream-aligned, platform, enabling, complicated-subsystem), three interaction modes (collaboration, XaaS, facilitating), and cognitive-load-bounded team sizes together provide a vocabulary for designing the human side of the hub-and-spoke pattern deliberately. Drawn correctly, this structure produces the architecture the book has been developing; drawn incorrectly, it produces an architecture that resists the team structure and loses.

The chapter's deeper claim, following Conway and Malan, is that team design is not a separate activity that happens after the architecture is settled — it is part of the architecture, and the part that wins any conflict. An architecture programme that produces an elegant system design and leaves the org chart to others has not finished its work; it has specified one of the two diagrams and surrendered the other. The discipline this chapter asks for is to treat the team structure as an architectural deliverable: designed with the same deliberateness as the system, reviewed by the same people, evolved through the same transitions, and recorded with the same decision records. The hub-and-spoke architecture of the earlier chapter and the team topology of this one are the same structure seen twice; an organisation that builds only one of them will find the other asserting itself anyway, and rarely in the form that was wanted.

Chapter 99

Buy versus Build

“Build what differentiates you. Buy everything else.”

—a recurring CIO maxim

Chapter Overview

Most components of an enterprise AI system can be bought or built. The decision is not aesthetic: building consumes engineering capacity that has alternative uses; buying creates dependencies that may not survive contract renewals. This chapter develops a decision framework based on differentiation, switching cost, total cost of ownership, and the maturity of available offerings. The framework is applied to each layer of the reference architecture, surfacing the components for which most enterprises should buy and the components for which most should build. The hexagonal pattern (Chapter 26) is what makes the decision tractable: bought components plug in behind ports, and the consequences of changing one’s mind become bounded.

Learning Objectives

- Apply a decision framework based on differentiation, switching cost, TCO, and offering maturity.
- Recognise the recurring biases (build bias and buy bias) that distort the decision.
- Distinguish “buy” from “buy and own the integration”: even bought components require architectural work.
- Apply the framework layer by layer to a typical enterprise AI system.
- Set up periodic re-evaluation; a build decision and a buy decision are both perishable.

99.1 The Decision Framework

Four questions, weighted by the stakes of the decision:

1. **Does this differentiate us?** If a competitor with the same component would compete equally well, the answer is no, and buying is the default.
2. **What is the switching cost?** A bought component you can leave in 90 days is different from one that takes 18 months. High switching cost is closer to “build” in its consequences regardless of

how it was acquired.

3. **What is the total cost of ownership?** Buy includes vendor fees, integration, training, contract management, support escalation. Build includes salaries, on-call rotation, opportunity cost, knowledge concentration risk.
4. **How mature are the offerings?** A field with three credible vendors is a buy candidate; a field with one immature offering may be a build candidate, or a wait candidate.

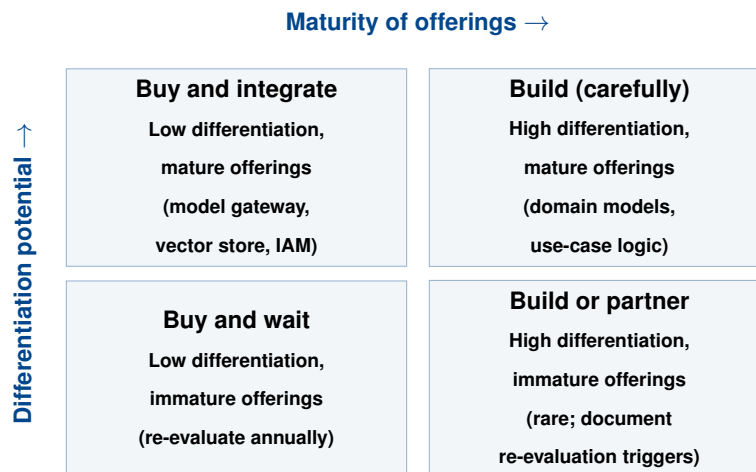


Figure 99.1: The buy-vs-build decision quadrant. Most enterprise AI components sit in the upper-left “buy and integrate” quadrant; differentiating logic sits in the upper-right “build” quadrant.

The four questions are not equally weighted, and the first one dominates. Differentiation is the question that should be answered first and that overrides the others when they conflict: a component that genuinely differentiates the business is a candidate to build even when mature offerings exist, because buying it would mean buying the same advantage a competitor can buy. The other three questions — switching cost, total cost of ownership, maturity — refine the decision for the large majority of components that do *not* differentiate, where the real question is not whether to buy but how to buy well. An organisation that runs the framework and finds most of its components are non-differentiating has not produced a disappointing result; it has produced the normal one, and the framework’s value is in identifying the few components where building is genuinely warranted so that engineering capacity is concentrated there.

99.2 Recurring Biases

Two biases distort the decision in opposite directions, and most organisations exhibit both at different moments:

- **The build bias.** Engineering organisations under-estimate the operational cost of running what they build. The first version is exciting; the third year of maintenance is not. Talented engineers gravitate toward novel construction over durable operation; the resulting roadmap accumulates features without accumulating reliability.

- **The buy bias.** Procurement-led organisations under-estimate the lock-in and integration cost of buying. The first quarter on the platform is impressive; the third year of price increases is not. The strategic position has shifted from “we chose a vendor” to “the vendor chose for us.”

The structural mitigation is to make the decision-makers responsible for the ongoing costs, not just the initial procurement or build. If the team that approved the build is rotated off before the third-year maintenance burden manifests, the bias is structurally rewarded. If the executive who chose the vendor is not present when the contract renewal arrives, the lock-in costs are someone else’s problem.

It is worth noticing that both biases share a single underlying cause: the costs that distort the decision are the costs that arrive late. The build bias and the buy bias are not really opposite errors; they are the same error — discounting the future — pointed in two directions. The exciting first version and the impressive first quarter are both present at decision time and weigh heavily; the third year of maintenance and the third year of price increases are both distant at decision time and weigh too little. This is why the structural mitigation works regardless of which bias an organisation is prone to: tying the decision-maker to the ongoing cost brings the late cost forward into the moment of decision, where it can counterbalance the early appeal. An organisation that wants to debias its buy-versus-build decisions does not need two separate fixes; it needs the one fix of making whoever decides also own the consequence.

99.3 Applying the Framework, Layer by Layer

The framework applies differently to each layer of the reference architecture.

99.3.1 Layer 1: Data Sources

Buy or rent the source systems—the CRM, the ERP, the marketing automation, the support platform. Building these is rarely a defensible choice in 2026.

Build the connectors and the source inventory. The connectors are too organisation-specific to buy off the shelf in most cases; the inventory is an organisational discipline that no vendor can deliver. Open-source connector frameworks (Airbyte, Meltano) provide useful starting points but require integration work; managed connector services (Fivetran, hosted Airbyte) shift the trade-off toward buy as the connector count grows.

The source inventory deserves emphasis because it is the component most often skipped, on the assumption that it is documentation rather than capability. It is capability. An enterprise that does not know, in a single maintained place, which source systems exist, what data each holds, who owns it, and what its quality and sensitivity are, cannot reason about what its AI systems are built on — and no vendor can supply this, because it is a fact about the specific organisation. The inventory is the data-side equivalent of the model inventory the governance chapters insist on, and it is a build decision not because building it is hard but because it is unavoidably the organisation’s own work. An AI programme that has bought every connector and skipped the inventory has bought the plumbing and skipped knowing what flows through it.

99.3.2 Layer 2: Data Pipeline

Table 99.1: Layer 2 buy-vs-build decisions.

Component	Default	Reasoning
Orchestrator	Buy (managed)	Mature ecosystem (Airflow, Dagster, Prefect); managed options remove operational toil
Lakehouse engine	Buy	Mature, no differentiation in building one's own
Transformation framework	Buy	dbt, SQLMesh are dominant; ecosystem trumps custom
Feature store	Buy or carefully build	Managed options (Tecton, Vertex, Databricks); self-host Feast for control; build only with specific constraints
Pipeline code	Build	The DAGs and transformations are the business logic; nobody can sell you yours
Data contracts	Build	Organisational artefacts; no vendor produces them

The pattern visible in the table is the one that recurs at every layer: the *frameworks* are bought and the *content* is built. The orchestrator is bought; the DAGs that run on it are built. The transformation framework is bought; the transformations are built. This is the buy-versus-build decision's most reliable heuristic — the generic machinery that every organisation needs in the same form is a buy, and the specific content that encodes this organisation's particular business is a build. An architect who internalises this distinction can make most layer-by-layer decisions quickly, because the question reduces to whether a given component is machinery or content. The genuinely hard cases, such as the feature store, are precisely the ones that sit on the boundary — partly generic machinery, partly shaped by organisation-specific constraints — and those are the cases the full four-question framework exists to resolve.

99.3.3 Layer 3: AI/ML Brain

For foundation models, the buy case is overwhelming for most enterprises. Vendor models lead on quality; the operational cost of self-hosting frontier capability is substantial; the talent to operate them well is scarce. Exceptions:

- Regulated or sovereignty-restricted workloads where vendor APIs are not legally usable.
- Very high-volume workloads where per-call economics flip in favour of self-hosting.
- Cases where fine-tuning on proprietary data must remain in-house.
- Cases where the model's behaviour itself is a strategic asset (rare).

For vector databases, the offerings are mature; buy unless an existing OLTP store (Postgres with pgvector) already serves the use case at the required scale.

Build the orchestration logic, the prompt library, the tool integrations, and any fine-tuned models specific to your domain. These are where the differentiation lives.

The foundation-model decision deserves one caution that the buy case can obscure: buying the model

is not the same as being free of the model decision. An enterprise that buys frontier model capability has still chosen which provider, has still accepted that provider's pricing trajectory and deprecation cadence, and has still made a decision that the four-question framework should govern — particularly the switching-cost question. The mitigation is the one the chapter returns to throughout: the model is consumed behind the LLM port of the hexagonal pattern, so that buying a model is a reversible decision rather than a permanent commitment. Bought foundation-model capability behind a clean port is a sound default for most enterprises; bought foundation-model capability wired directly into application code is the buy decision made in its most lock-in-prone form. The buy case is strong, but it is a case for buying *well*, not for treating the model as settled.

99.3.4 Layer 4: Model Serving and MLOps

Buy the experiment tracker (MLflow, Weights & Biases), the model registry capability, and the serving runtime (Triton, vLLM, BentoML, KServe). Build the deployment automation that integrates these with your environment, the evaluation harness specific to your use cases, and the cost-attribution layer that ties calls to business outcomes.

The evaluation harness is the build decision on this layer most worth defending against the temptation to buy. Generic evaluation tooling exists and is improving, but an evaluation harness is only useful to the degree that it tests the things this organisation's use cases actually need to be right about — the specific failure modes, the specific quality thresholds, the specific edge cases that matter for these workloads. That specificity cannot be bought, because it is knowledge about the use cases rather than machinery. An organisation that buys a generic evaluation product and treats the purchase as discharging the evaluation obligation has bought the harness and skipped the part that mattered: the test cases. The harness is bought machinery; the evaluation suite that runs on it is built content, and the chapter on observability and evaluation develops why that built content is among the highest-value engineering an AI programme does.

99.3.5 Layer 5: DevSecOps and Governance

Buy the identity provider, the secrets manager, the observability stack (Prometheus, Grafana, Datadog, Splunk, Honeycomb). These categories are mature and undifferentiated.

Build the model inventory, the lineage emission, and the policy enforcement that is specific to your industry and regulators. Vendor governance platforms (Collibra, Atlan, DataHub) help with parts of this but do not replace organisation-specific policy.

The phrase “help with parts of this but do not replace” is the important one, because governance is the layer where the buy decision most often produces a false sense of completion. A purchased governance platform supplies the catalogue, the lineage visualisation, the workflow machinery — real and useful capability — but it arrives empty of the one thing that makes governance govern: the organisation's actual policies, encoded for this industry and these regulators. A bank that buys a governance platform and does not encode its specific regulatory obligations into it has bought a sophisticated container for a policy that does not yet exist. The platform is bought machinery; the policy is built content; and the governance-standards chapter's obligation register is precisely the

built artefact that the bought platform should hold. Buying the platform is reasonable; mistaking the purchase for the discharge of the governance obligation is the error.

99.3.6 Layer 6: End User Applications

The user-facing applications are usually built—they are the differentiating layer. Embedded AI in commodity applications (the CRM’s built-in summarisation, the ITSM’s built-in routing, the help-desk’s built-in classification) is bought as part of the underlying application. The distinction matters: building a chatbot for your own product is differentiating; building a chatbot for your support desk competes with vendors who do it for a living.

The platform-AI-capability patterns of an earlier chapter intersect this layer directly, and reading the two together sharpens the decision. When the AI capability the use case needs is one of the recurring platform patterns — conversational knowledge retrieval, an embedded copilot — and it sits inside a commodity application the organisation already runs, the buy case is strong, because the AI is bought as an inherited feature of a platform that brings its own identity, audit, and governance. When the AI capability is the differentiating logic of the organisation’s own product, it is built, because it is the content rather than the machinery. The end-user layer is where the buy-versus-build framework and the platform-versus-standalone decision become the same decision viewed from two angles, and an architect should make them consistently: the question of whether to buy or build the application is, at this layer, largely the question of whether the application differentiates.

99.4 The True Cost of Buying

Buying is not free of engineering work. Each bought component requires:

- Integration into the platform’s IAM, observability, FinOps, and lineage emission.
- Adapter implementation behind the relevant port (Chapter 26).
- Contract management and renewal cycles.
- Vendor risk review (initial) and ongoing monitoring (incidents, breaches, sub-processor changes).
- Internal documentation that translates vendor concepts to platform concepts.
- Training for engineers who will work with the vendor’s interfaces.
- Onboarding into incident-response playbooks.

A bought component that has not been integrated is a procurement, not a capability. Estimates that compare “the vendor’s price” to “the cost of building” are routinely off by 30–50% because the integration costs are not included in the buy side of the ledger.

The honest way to run the comparison is to put the same categories of cost on both sides of the ledger. The build side is usually estimated with its full burden — salaries, on-call, opportunity cost — because those costs are visible and feared. The buy side is too often estimated as the vendor’s invoice alone, which is why the comparison is systematically biased toward buying. A fair comparison adds to the

buy side every item on the list above — the integration, the adapter, the contract management, the vendor risk review, the documentation, the training, the incident-response onboarding — and prices them as real engineering work, because they are. The buy decision is frequently still correct after this correction; the point is not that buying is worse than it looks but that the decision should be made against the true figure. An organisation that compares a full build estimate against a bare vendor price is not making the buy-versus-build decision; it is making a different, easier decision that happens to share its name.

99.4.1 Hidden Costs of Vendor Lock-In

Beyond direct fees, lock-in produces costs that surface later:

- **Pricing power.** A vendor with no credible alternative can raise prices well above what the market would otherwise support.
- **Feature direction.** A vendor's roadmap drives your roadmap. Features you need but the vendor does not prioritise become impossible.
- **Geographic constraints.** The vendor may not expand into a region you need.
- **Acquisition risk.** A vendor that is acquired may have its product line discontinued or repositioned.
- **End-of-life risk.** A vendor may exit the business entirely.

Mitigations: keep adapters honest (the swap test of Section 26.4); avoid letting vendor-specific concepts leak into application code; periodically test the alternative.

The common thread in these lock-in costs is that none of them is visible at purchase time and all of them grow with the depth of the dependency. This is what makes lock-in genuinely dangerous: it is not a cost the organisation can see and weigh on the day it signs, but a position it discovers it is in only once the position is expensive to leave. The architectural response is the hexagonal pattern's, applied as a deliberate insurance policy rather than an afterthought — the bought component sits behind a port, the swap test proves the port is honest, and the alternative is tested periodically so that the option to leave is known to be real. The cost of maintaining that optionality is modest and continuous; the cost of discovering, during a hostile contract renewal, that the option does not exist is large and sudden. Lock-in is not avoided by choosing the right vendor, since any vendor's incentives drift over a multi-year relationship; it is avoided by keeping the architecture able to change its mind.

99.5 The Hidden Cost of Building

Building has its own hidden costs:

- **Bus factor.** The original engineers leave; the system becomes a maintenance burden nobody fully understands.
- **Operational burden.** On-call, capacity planning, security patching, dependency updates, vulnerability response.

- **Opportunity cost.** The engineering hours building this are hours not building something more differentiating.
- **Drift from state of the art.** The custom-built component falls behind general-purpose offerings; “we built our own three years ago” eventually becomes a problem statement, not a position of strength.

The most insidious of these is drift from the state of the art, because unlike the others it worsens even when nothing goes wrong. A built component can be well-documented, well-staffed, and operationally sound, and still become a liability simply because the general-purpose offerings it was once better than have, over three years, overtaken it — and the organisation is now spending engineering effort maintaining a component that the market would supply better and cheaper. This is why a build decision, even a correct one, is perishable in a way a buy decision’s correctness is also perishable but more visibly so. The built component does not announce that it has fallen behind; someone has to deliberately check. The periodic re-evaluation of the next section exists largely to force that check, so that “we built our own” is revisited as an active decision rather than left to harden into an unexamined assumption.

99.6 Periodic Re-Evaluation

Both buy decisions and build decisions are perishable. The framework should be re-run every 18–24 months, or sooner if a major contract renewal, a vendor acquisition, or a category-changing new offering occurs.

Table 99.2: Re-evaluation triggers.

Trigger	Action
Vendor contract renewal	Re-evaluate before signing; do not auto-renew without considering alternatives
Vendor acquisition or financial trouble	Trigger contingency plan; assess switching cost
New category-changing offering	Compare against current choice; document why current still wins or trigger migration
Built component reaching three years old	Compare against current state of the art
Significant change in scale, geography, or regulation	Re-evaluate any decision affected

The hexagonal pattern makes the consequences of re-evaluation tractable: changing vendor is a deployment-level change, not a project. Without the pattern, every re-evaluation produces a tactical decision (“stay with the vendor we have, because changing is too expensive”), which over time accumulates into strategic immobility.

The connection between the pattern and the re-evaluation is worth stating sharply, because it is the chapter’s central practical claim. Re-evaluation is only honest if acting on its conclusion is affordable.

An organisation that re-runs the framework, concludes that a different vendor is now better, and then declines to switch because switching is a six-month project has not really re-evaluated — it has performed the ritual of re-evaluation and reached the only answer its architecture allowed. The hexagonal pattern is what makes the re-evaluation real: when changing a bought component is a deployment rather than a project, the framework's conclusion can be acted on, and the re-evaluation becomes a genuine decision rather than a confirmation of the status quo. Buy-versus-build and the hexagonal pattern are therefore not two separate topics; the pattern is the precondition that makes the buy-versus-build decision revisable, and a decision that cannot be revised is not being made freely.

99.7 The African Perspective

The buy-versus-build framework holds everywhere, but two of its four questions — total cost of ownership and switching cost — have a sharper edge in many African deployments.

The total-cost-of-ownership question is complicated by currency. A bought component priced in US dollars or euros has a cost that, measured in the local currency the organisation actually budgets in, moves with the exchange rate — and a vendor relationship that was sound at one exchange rate can become a serious budget pressure at another, without anything about the vendor or the component changing. This argues for weighting the switching-cost question more heavily at the point of purchase: a bought component that can be left, or substituted with an open-weight or locally-hosted alternative, is worth more in a volatile-currency environment than the same component would be worth in a dollar-denominated economy. The build option, conversely, has the property that its dominant cost is local-currency salaries, which removes the foreign-exchange exposure even though it adds the other build costs the chapter has detailed.

The maturity question interacts with local presence. A vendor with three credible global competitors may still, in a particular African market, have only one option that genuinely operates locally — with local support, local data residency, local contracting — which means the effective maturity of the market, for this organisation, is lower than the global picture suggests. The framework should be run against the offerings genuinely available to the organisation, not the offerings that exist somewhere, and that distinction can move a component from the buy quadrant to the buy-and-wait quadrant, or occasionally to build. The discipline of the chapter is unchanged; the inputs to it must reflect the market the organisation actually operates in.

99.8 Closing Notes

The buy-versus-build decision is one of the highest-leverage repeated decisions in an enterprise AI programme. Four questions (differentiation, switching cost, TCO, maturity), two biases (build bias and buy bias), and a layer-by-layer application produce a framework that survives scrutiny. Differentiation is the polestar: build where it produces competitive advantage; buy where it does not. The hexagonal pattern is what makes the decision reversible—and decisions that are reversible can be made faster and revisited honestly.

The chapter's recurring figure — buy the machinery, build the content — is the heuristic to carry away, and the framework is the instrument for the cases where the figure is not obvious. An AI programme that applies this consistently concentrates its scarce engineering capacity on the components that genuinely differentiate it and declines to spend that capacity rebuilding the machinery that the market supplies well. It also, by keeping its bought components behind clean ports, retains the ability to revisit every buy decision as the market moves. The combination — build only what differentiates, buy the rest, and keep the bought parts replaceable — is what lets an enterprise AI programme stay both focused and free, which is the position from which the architectural disciplines of the rest of the book can actually be exercised.

Chapter 100

Cost and FinOps

“Cloud has made compute infinite. It has not made it free.”

—unattributed FinOps proverb

Chapter Overview

AI workloads have a unique cost profile: small numbers of calls can produce large bills; the same workload can cost 10× more on a frontier model than on a small one; and the cost-to-value ratio is invisible without instrumentation. This chapter develops the cost vocabulary specific to AI, the attribution model that makes spend governable, the optimisation levers (router, caching, quantisation, batching, commitment discounts), the budget enforcement that prevents runaway costs, and the operational rhythm that catches anomalies before they become invoices.

Learning Objectives

- Apply a cost vocabulary that distinguishes the cost drivers for each workload type.
- Attribute every dollar of AI spend to a business unit, application, and use case.
- Apply the five optimisation levers: route, cache, quantise, batch, commit.
- Implement budget enforcement at the API gateway.
- Run cost reviews that surface anomalies before they become invoices.

100.1 The Cost Vocabulary

AI costs decompose into several streams, each with different drivers and different optimisation levers.

Table 100.1: Cost streams in an enterprise AI system.

Stream	Unit	Primary driver
Foundation-model API	\$ per million tokens (in/out)	Call volume $\times$ average context length $\times$ output length
Self-hosted inference	\$ per GPU-hour	Replica count $\times$ utilisation; idle GPUs cost the same as busy ones
Embedding generation	\$ per million tokens	Corpus size, refresh cadence, re-embedding on model upgrade
Vector store	\$ per million vectors $\times$ time	Index size, replication factor, query volume
Training / fine-tuning	\$ per GPU-hour	Dataset size, model size, epoch count, hyperparameter search breadth
Data pipeline	\$ per warehouse-second	Volume, transformation complexity, refresh cadence
Observability	\$ per GB ingested / retained	Log verbosity, retention policy, cardinality
Egress	\$ per GB transferred	Cross-region or cross-cloud data movement

Different workloads are dominated by different streams. A chat application: foundation-model API. A daily batch enrichment: pipeline compute and inference. A RAG-heavy system: embedding generation and vector storage. A multi-cloud deployment: often egress. Diagnosing where cost actually goes is the prerequisite to optimising it.

The reason the diagnosis must come first is that effort spent optimising a stream that is not the dominant one produces almost no saving. An engineering team that spends a month shaving the foundation-model bill of a workload whose cost is actually dominated by idle self-hosted GPUs has worked hard and changed nothing. The discipline is to instrument first and optimise second: produce the per-stream breakdown for the specific workload, identify the one or two streams that dominate it, and direct the optimisation effort there. Cost intuition is unreliable in AI systems — the stream an architect assumes is dominant frequently is not — and the breakdown table is the instrument that replaces the intuition with a measurement. No optimisation lever in this chapter is worth pulling until the breakdown has shown which stream it acts on.

100.1.1 Token Economics in Detail

For LLM-API workloads, the cost is approximately:

$$\text{cost} = N_{\text{calls}} \cdot \left(\underbrace{p_{\text{in}} \cdot t_{\text{in}}}_{\text{input cost}} + \underbrace{p_{\text{out}} \cdot t_{\text{out}}}_{\text{output cost}} \right)$$

where p_{in} and p_{out} are the per-token prices and $t_{\text{in}}, t_{\text{out}}$ are the average input and output token counts. Three observations:

- Output tokens typically cost 3–5 $\times$ more than input tokens. Verbose system prompts are cheap; verbose responses are expensive.

- Input cost dominates in RAG-heavy workloads (large context per query, short generated answer).
- Output cost dominates in generation-heavy workloads (short query, long generated content).

The optimisation strategy depends on which side dominates. For input-heavy workloads, shrinking context (better retrieval, prompt compression) is the lever. For output-heavy workloads, capping output length or switching to a cheaper model for the bulk of generation is the lever.

A further consequence of the token-economics formula is worth drawing out, because it bears on the agentic patterns of the earlier chapters. The formula is per-call, and an agentic workload makes many calls per task — the reasoning engine on every cycle, the planner, the critics. An agent that runs ten cycles to complete a task pays the per-call cost ten times, and each cycle’s input typically grows as the accumulated context is carried forward, so the later cycles cost more than the earlier ones. This is why the agentic chapters insisted on cycle budgets as a cost control and not merely a termination guarantee: the token-economics formula, multiplied across an unbounded agent loop, is exactly the cost-explosion failure mode. An architect costing an agentic workload must multiply the per-call figure by the realistic cycle count, including the growth in input length across cycles, rather than costing it as though it were a single call.

100.2 Cost Attribution

The platform must tag every cost-incurring call with: tenant, business unit, application, use case, environment, and (where appropriate) user. The API gateway is the natural enforcement point: requests without tags are rejected; tags are propagated to downstream vendor APIs via custom headers; vendor invoices are reconciled with internal usage data.

```

1 POST /v1/chat/completions
2 Authorization: Bearer <token>
3 X-Tenant:          acme-corp
4 X-Business-Unit:   claims-ops
5 X-Application:     claims-assistant
6 X-Use-Case:        triage-summarisation
7 X-Environment:     production
8 X-Cost-Center:      CC-4502
9 X-Request-Id:       01HXR3...
10 X-Trace-Id:        00-abc123...
```

Listing 100.1: Cost-attribution tags on API calls.

Untagged spend is unowned spend, and unowned spend grows. A platform that lets a few requests slip through without tags will, six months later, have a meaningful fraction of its bill unattributable.

The mechanism behind that growth is worth naming, because it explains why the gateway must reject untagged requests rather than merely warn about them. Spend that is attributed has an owner, and an owner who sees their own cost line has both the information and the incentive to manage it. Spend that is unattributed has no owner, no one watching it, and no one accountable for its growth — it is, by construction, the spend that nobody is optimising. Worse, an untagged path is the path of least

resistance: if the gateway permits untagged calls, teams under deadline pressure will use them, and the unattributable fraction will grow precisely because it is the easy option. Enforcing tagging strictly at the gateway — rejecting the untagged request outright — is therefore not bureaucratic rigidity; it is the mechanism that guarantees every dollar has an owner, and every dollar with an owner is a dollar someone is watching.

100.2.1 Chargeback vs Showback

Two operating models:

- **Showback.** Each business unit sees its costs but is not billed for them. The platform absorbs the cost centrally. Encourages adoption; provides visibility without enforcement.
- **Chargeback.** Each business unit is billed for its costs. The platform is funded by the chargeback. Aligns incentives but raises the bar to first use.

Most enterprise programmes start with showback and migrate to chargeback as the programme matures and stable consumption patterns emerge. A premature chargeback can suppress experimentation; a permanent showback can mask runaway consumption.

The migration from showback to chargeback should be timed to a specific signal rather than a calendar: the point at which a business unit's consumption has become predictable enough that being billed for it will not deter the experimentation that produced the predictability. Too early, and the business unit facing a real bill before it understands its own usage will under-experiment, and the programme loses the exploration it needed to find its valuable use cases. Too late, and a business unit that has never been billed has had no incentive to optimise, and consumption that showback merely displayed — without anyone obliged to act on the display — has been allowed to grow. The two operating models are not a binary choice but a sequence, and the architect's job is to read the signal that says a given tenant is ready to move from one to the other.

100.3 The Five Optimisation Levers

Five levers dominate practical cost reduction in production AI workloads.

Before reaching for any of them, the architect should return to the cost breakdown of the first section, because each lever acts on a specific stream and is worthless applied to the wrong one. Routing and caching act on foundation-model API spend; quantisation acts on self-hosted inference; batching acts on throughput-bound inference; commitment discounts act on whichever streams have predictable volume. A lever pulled against a stream that is not the workload's dominant cost produces effort without saving. The five are presented here in roughly the order of leverage they offer a typical foundation-model-dominated workload, but the order for any specific workload is the order its own cost breakdown dictates. The levers are instruments; the breakdown tells the architect which instrument the workload in hand actually needs.

100.3.1 Routing

The router pattern (Chapter 23) is the single most powerful cost lever. Sending the easy 80% of requests to small models typically cuts foundation-model spend by 60% with minimal quality loss. The architecture investment (the router itself, plus the evaluation harness that proves routing decisions are sound) pays back in the first quarter at any meaningful scale.

The phrase “plus the evaluation harness” is the part that makes routing a sound lever rather than a quiet quality regression. Routing saves money by sending requests to cheaper, weaker models, and the saving is only real if those models genuinely handle the requests routed to them. Without the evaluation harness, a router is an unmeasured bet that the easy eighty percent really were easy — and a router that misroutes a consequential request to a model that handles it poorly has traded a quality failure for a cost saving, usually without anyone noticing until the downstream effect surfaces. The evaluation harness is what converts the routing bet into a measured decision: it proves, on a labelled test set, that the routing tiers preserve quality, and it catches the drift when a previously sound routing boundary stops being sound. Routing is the most powerful lever in the chapter, and the evaluation harness is the discipline that keeps it from being the most dangerous one.

100.3.2 Caching

Three caching layers:

- **Semantic cache** (Section 23.4) for repeated questions: 20–40% savings on chat workloads with significant FAQ overlap.
- **Prompt-prefix cache** for templated prompts: vendor-provided in some APIs; effective when system prompts and few-shot examples are stable across calls.
- **Embedding cache** for unchanged content: avoids re-embedding the same chunks on every indexing run.

Cache invalidation is the perennial difficulty: a stale cached answer is worse than no cached answer. TTLs, source-change notifications, and content hashes all play a role.

The reason a stale cached answer is worse than no cache is that it fails silently and confidently. A cache miss produces a fresh, correct answer at full cost; a stale hit produces a wrong answer at near-zero cost, and the near-zero cost is exactly what hides the failure from the cost dashboards that would otherwise flag a problem. This makes caching the optimisation lever whose savings must be weighed most carefully against its risk: the semantic cache in particular, which matches questions by similarity rather than exact text, can return an answer to a question that is close but not identical to the one asked. The architectural discipline is to scope aggressive caching to content whose staleness is tolerable or detectable — stable reference material with source-change notifications, templated prompts with content hashes — and to be conservative with caching on anything where a confidently wrong answer carries real cost. Caching is genuine saving on the right workload and a quiet correctness hazard on the wrong one.

100.3.3 Quantisation

For self-hosted models, INT8 or INT4 quantisation (Section 14.2) reduces both memory and compute cost dramatically with modest quality impact for most production tasks. INT4 quantisation typically halves serving cost compared to FP16 baseline.

The phrase “for most production tasks” is the boundary the architect must locate rather than assume. Quantisation’s quality impact is not uniform: it is modest for many classification, extraction, and routine generation tasks, and more pronounced for tasks that depend on precise reasoning, long-chain arithmetic, or fine distinctions. The discipline is the same one routing required — the evaluation harness. Before a quantised model is committed to a production path, it should be evaluated on that path’s actual test set against the unquantised baseline, so that the quality cost is measured rather than hoped. Where the measured cost is genuinely modest, quantisation is close to free money and should be taken. Where the task turns out to be one of the quantisation-sensitive ones, the saving is not worth the quality loss, and the architecture should keep the unquantised model on that path while quantising the others. Quantisation is a per-task decision informed by evaluation, not a blanket setting.

100.3.4 Batching

Continuous batching in modern LLM serving runtimes (vLLM, TensorRT-LLM) increases throughput 3–10× at the same hardware cost. For batch workloads (nightly enrichment, document processing), explicit batching delivers similar gains. Batching trades latency for throughput, so the architecture should distinguish latency-critical paths (no batching, dedicated capacity) from throughput-critical paths (aggressive batching).

This latency-for-throughput trade is why batching is the lever whose application depends most directly on the workload’s interaction model. A user waiting for a chat response is sensitive to the latency that aggressive batching adds; a nightly document-enrichment job that must finish by morning is sensitive only to throughput and indifferent to per-item latency. The architectural move is to classify every inference path by which it cares about — and the classification is often the same latency-critical-versus-throughput-critical split that other chapters have used. The same hardware, serving the same model, should be operated differently on the two kinds of path: dedicated, lightly-batched capacity for the interactive paths, and aggressively-batched shared capacity for the background ones. An architecture that batches everything uniformly has degraded its interactive latency for no reason; one that batches nothing has left a three-to-tenfold throughput gain unclaimed on its background work.

100.3.5 Commitment Discounts

Vendors offer significant discounts (20–40%) for commitments to a minimum monthly spend. The architectural prerequisite is confident forecasting; without it, committed-spend discounts become committed-waste. The discipline is to commit only at the level the workload is provably above, and to revisit commitments at every renewal.

The asymmetry that governs this lever is that under-committing costs a little and over-committing costs a lot. A commitment set below the workload’s true floor leaves some spend at the undiscounted

rate — a modest, bounded loss. A commitment set above the workload’s actual usage means paying for capacity the workload never consumed — the full waste, with no offsetting saving. Because the downside is asymmetric, the discipline is to commit conservatively: to the level the workload has *demonstrably* exceeded for several consecutive months, not the level the optimistic forecast projects. The commitment can always be raised at renewal once higher usage is proven; it cannot easily be unwound once signed. Commitment discounts are the one lever where the right posture is caution rather than aggression, because the lever’s failure mode is not a missed saving but a guaranteed loss.

100.3.6 Combining the Levers

The levers compound:

- Routing alone: 60% reduction (frontier-only baseline).
- Plus caching: $60\% \times 70\% = 42\%$ of original (58% reduction).
- Plus quantisation of the self-hosted tier: $42\% \times 80\% = 34\%$ of original (66% reduction).
- Plus batching where applicable: another 10–15% on the relevant share.
- Plus commitments: 10–20% on the residual.

The cumulative effect is dramatic: workloads that look unaffordable at frontier-only pricing become routine at 25–35% of the baseline cost.

Two cautions keep this arithmetic honest. The first is that the levers compound multiplicatively only where they act on overlapping spend; a lever that acts on a stream a previous lever already shrank delivers less in absolute terms than its percentage suggests, and a lever that acts on a different stream entirely does not compound with the others at all. The worked figures above assume a foundation-model-dominated workload where routing, caching, and quantisation act in sequence on the same spend; a workload with a different cost shape will see a different cumulative result. The second caution is that each lever carries the quality and operational risk its own subsection described, and the risks also accumulate. A workload that routes aggressively, caches aggressively, and quantises aggressively has stacked three separate quality hazards, and the evaluation harness must now confirm that the combination, not just each lever alone, still meets the workload’s quality bar. The cumulative saving is real and worth pursuing; it should be pursued with the cumulative risk measured rather than assumed away.

100.4 Budget Enforcement

A platform without budget enforcement is a platform with unbounded liability. The API gateway is the enforcement point. Three modes:

Table 100.2: Budget enforcement modes and their use cases.

Mode	Use case
Hard cap	Internal experimentation; requests beyond budget are rejected with 429. The pain is felt; the bill is bounded.
Throttle	Mixed-criticality workloads; requests beyond budget are queued or slowed. Maintains availability at degraded latency.
Degrade	Customer-facing services; requests beyond budget route to cheaper models or simpler responses. Preserves availability at degraded quality.

The mode is a policy choice per use case. A mature platform supports all three and lets the tenant configure.

The choice between the three modes is, at bottom, a choice about what the workload can least afford to lose when the budget is reached — and the three answers are correctness of the bill, latency, and quality. A hard cap protects the bill absolutely and sacrifices availability, which is right for internal experimentation where a bounded bill matters more than uninterrupted service. A throttle protects availability and the bill while sacrificing latency, which suits mixed-criticality work. A degrade path protects availability while sacrificing quality, which is the only acceptable answer for a customer-facing service where an outage is worse than a cheaper-model response. The architectural error is to have only one mode and apply it everywhere: a hard cap on a customer-facing service turns a budget overrun into a customer-visible outage, and a degrade path on an internal experiment quietly spends past the budget it was meant to bound. The mode is genuinely a per-use-case decision, and the gateway must support all three for the platform to make it well.

100.4.1 Budget Hierarchy

Budgets are nested: an organisation budget contains business-unit budgets, each of which contains application budgets, each of which contains use-case budgets. Enforcement happens at the most specific level that has a budget, with cascade to broader levels if specific ones are absent.

The nesting matters because it localises the consequence of an overrun. Without a hierarchy, a single runaway use case draws against one undifferentiated pool and can exhaust the budget of every other workload in the organisation before anyone intervenes — one team's incident becomes everyone's outage. With the hierarchy, the runaway use case hits its own use-case budget first, and the enforcement mode configured at that level — cap, throttle, or degrade — contains the damage to that use case while every sibling workload continues untouched. The hierarchy is, in effect, the blast-radius control for cost: it ensures that a cost failure is felt first and most sharply by the workload that caused it, and only cascades upward if the broader levels have deliberately been left without their own budgets. An architect designing the budget structure is designing how far a single cost incident is allowed to spread.

100.4.2 Anomaly Detection on Spend

Static budgets catch only known patterns. Anomaly detection on cost time series catches the unknown:

- Sudden volume spike from a previously stable tenant.
- Average cost-per-call drift upward (often signals context-length growth).
- New cost lines (a new tenant, a new application, an unexpected sub-service).
- Cost concentration shift (one tenant absorbing a growing share of the platform).

Anomalies should produce alerts to the platform team and to the affected tenant, not silent absorption.

The reason anomaly detection is needed alongside budgets is that budgets only catch the failure modes someone anticipated. A budget is a threshold set in advance; it catches spend that crosses a known line. It does not catch the cost-per-call drifting slowly upward as context lengths grow, because each day's spend is within budget — it catches only the eventual breach, by which point months of avoidable cost have accumulated. Anomaly detection works on the shape of the cost time series rather than an absolute threshold, so it catches the drift, the unexpected new cost line, and the concentration shift while they are still small. Budgets and anomaly detection are therefore complementary rather than redundant: budgets bound the known worst case, anomaly detection surfaces the unknown developing one. A cost-governance setup with budgets but no anomaly detection is protected against the failures it imagined and blind to the ones it did not.

100.5 Cost Reviews

Weekly or monthly reviews are the forcing function that turns cost data into action.

The word “forcing function” is the point. All the instrumentation of the preceding sections — the attribution tags, the per-stream breakdown, the anomaly alerts — produces data, and data on a dashboard that no one is scheduled to look at changes nothing. The review is the recurring commitment that someone will look, ask why a number moved, and assign the action that follows. The three cadences below are pitched at three different time horizons because cost problems develop at three different speeds: a runaway tenant shows up within a week, a slow cost-per-call drift takes a month to become visible, and a misaligned commitment or vendor-mix question only resolves over a quarter. A programme that runs only one of the three reviews is blind to the problems that develop on the other two timescales.

100.5.1 Weekly Operational Review

A short, recurring review of:

- Top spenders week-on-week; which use cases moved the most?
- Cost-per-call trends; is any model or workload drifting upward?
- Budget-burn rate; any tenant likely to exhaust budget this month?
- Unattributed spend; is any cost not yet tagged to an owner?

The weekly review is deliberately short and operational: its purpose is detection, not deliberation. It

should take a small group a brief, regular slot, and its output is not decisions but flags — this tenant moved sharply, this cost-per-call is drifting, this spend is untagged — each of which is either resolved on the spot if trivial or handed to the monthly review if it needs analysis. The discipline that keeps the weekly review valuable is resisting the temptation to turn it into a planning meeting; the moment it becomes a forum for strategy it acquires the wrong attendees, runs long, and stops happening weekly. Its job is to ensure that nothing surprising in the last seven days goes unnoticed for an eighth, and a review that does only that, reliably, is doing exactly what it should.

100.5.2 Monthly Optimisation Review

A deeper review of:

- Routing effectiveness; what proportion of requests went to each tier; can the proportions improve?
- Cache hit rates; can any cache be tuned?
- Idle capacity; any GPU pools under 30% utilisation?
- Vendor invoice reconciliation; do internal records match the bill?
- Forecast versus actual; how good was last month's prediction?

The monthly review is where the optimisation levers of the earlier section are actually re-tuned, and its character is analytical rather than detective: it takes the flags the weekly reviews raised and the trends a month makes visible, and it asks what should change. Two of its items deserve particular weight. Vendor invoice reconciliation is the check that the platform's own cost records match what the vendor actually billed — a discrepancy here means the attribution data is wrong, and every decision built on that data is unreliable until it is fixed. Forecast-versus-actual is the check that the programme's predictions are improving, which matters directly for the commitment-discount lever, since a commitment is only as safe as the forecast it rests on. A monthly review that consistently finds its forecasts close to actual has earned the confidence to commit; one whose forecasts are routinely wrong has been told, clearly, not to.

100.5.3 Quarterly Strategic Review

A reset of:

- Commitment terms; are they aligned to demonstrated demand?
- Vendor mix; should any vendor be added, dropped, or rebalanced?
- Self-host vs. vendor breakdown; do the economics still favour the current split?
- Capacity planning; what does the next quarter need?

The quarterly review is where cost governance connects to the architectural decisions of the rest of the book, because its questions are not operational adjustments but structural ones. The self-host-versus-vendor question is the buy-versus-build decision of the previous chapter, run again against three months of fresh cost evidence. The vendor-mix question is a re-evaluation of the kind that chapter's

perishable-decision discipline demands. The capacity-planning question feeds the commitment decision. This is why the quarterly cadence matters: structural decisions should not be revisited weekly — they need a quarter of evidence to revisit responsibly — but they must be revisited, because a self-host split or a vendor mix that was correct a year ago may not be correct now. The quarterly strategic review is the scheduled occasion on which the architecture’s cost-relevant structural choices are deliberately re-examined rather than left to harden by default.

100.6 Unit Economics

The most useful single metric is *cost-per-successful-outcome*. For a chatbot: cost per resolved conversation. For a document classifier: cost per correctly-categorised document. For a copilot: cost per accepted suggestion.

$$\text{cost-per-outcome} = \frac{\text{total cost}}{\text{successful outcomes}} = \frac{\text{cost-per-call}}{\text{success rate}}$$

This metric captures both efficiency (cost per call) and quality (success rate). A cheaper model that gets the answer right only 60% of the time may have a higher cost-per-outcome than a more expensive model that succeeds 95% of the time, because the failed attempts must be re-tried or escalated.

Tracking cost-per-outcome over time is the most reliable diagnostic of whether the system is actually improving or just churning.

The reason cost-per-outcome is the metric that matters, and cost-per-call is not, is that cost-per-call can be improved while the system gets worse. Route everything to the cheapest model and cost-per-call falls; if the cheap model’s success rate falls faster, cost-per-outcome rises, and the system is now spending more per actual result while its cost-per-call dashboard shows an apparent win. Cost-per-outcome is the metric that cannot be gamed this way, because it has the success rate in its denominator: any optimisation that trades too much quality for cost makes this number worse, immediately and visibly. An AI programme that manages cost-per-call is managing a number that can improve while the programme fails; an AI programme that manages cost-per-outcome is managing the number that captures whether the money is actually buying results. It is the one cost metric that should appear in the same review as the quality metrics, because it is the one that joins them.

100.7 The African Perspective

Cost discipline is universal, but it is not equally weighted everywhere, and in many African deployments it is the constraint that most directly determines what is buildable at all.

The foreign-exchange dimension is the sharpest. Foundation-model APIs, cloud infrastructure, and most vendor tooling are priced in US dollars, while the organisation’s revenue and budget are in a local currency that can move substantially against the dollar. This means an AI workload’s cost, measured in the currency that actually matters to the organisation, has a volatility that has nothing to do with the workload itself — and a system that was affordable at one exchange rate can become a budget crisis at another. The practical consequences run through the whole chapter: budgets should be set and tracked

in local currency so the exposure is visible; the anomaly detection should treat a currency-driven cost rise as a real signal rather than noise; commitment discounts, which lock in a dollar figure, carry an extra currency risk that must be weighed against their saving; and the cost-per-outcome discipline matters more, because there is simply less margin to absorb an inefficient system.

This sharpens the value of the optimisation levers rather than changing them. Routing, caching, and quantisation are not, in a foreign-exchange-exposed deployment, marginal efficiency gains — they are often what moves a workload from unaffordable to viable. The levers that reduce dependence on dollar-priced frontier APIs, in particular the routing of suitable traffic to smaller or locally-hosted open-weight models, do double duty: they cut cost and they cut currency exposure at the same time. An architect working under these constraints should read the five levers not as a menu of nice-to-haves but as the toolkit that determines the feasible set of AI workloads for the organisation.

100.8 The Pattern, Restated

AI cost discipline is a structural property of the platform, not a project responsibility. Tag every call, attribute every dollar to an owner, apply the five levers (route, cache, quantise, batch, commit), enforce budgets at the gateway, and run cost reviews on a regular rhythm. Track cost-per-successful-outcome as the unit economic. Done well, the resulting outcomes are visible and bounded: cost-optimised workloads, predictable spend, no runaway invoices. Done poorly, the bill itself is the discovery mechanism, and the discovery comes late.

The phrase “structural property of the platform” is the chapter’s central claim and the reason cost cannot be left to individual teams. Attribution that some teams do and others skip produces an unattributable fraction that grows. Budget enforcement that lives in application code rather than the gateway is enforcement that the next application forgets. Optimisation levers applied per project are applied well by the strong teams and not at all by the rushed ones. Every cost discipline in this chapter follows the same logic as the responsible-AI controls of an earlier chapter: it is far cheaper and far more reliable as a platform feature that every workload inherits than as a deliverable each team is trusted to remember. An organisation that builds cost governance into its platform has made predictable spend the default; an organisation that leaves it to teams has made the monthly invoice its primary cost-discovery tool, and the invoice always reports the problem after the money is already spent.

Chapter 101

Observability and Evaluations

“You cannot improve what you cannot measure; you cannot trust what you do not test.”

—an operating maxim, source uncertain

Chapter Overview

Observability and evaluations are the disciplines that turn a built system into an operable one. Observability tells the team what is happening now; evaluations tell the team whether the system still does what it was supposed to. For AI systems, both are richer than for conventional services: the relevant signals include not just latency and error rate but drift, refusal rate, citation accuracy, retrieval quality, and the model’s own self-reported confidence. This chapter develops the four signal categories, the evaluation harness, the rhythm of offline and online evaluation, the canary-query pattern, and the integration of both into CI/CD.

Learning Objectives

- Recognise the four signal categories that an AI system must emit: conventional, AI-specific, quality, and cost.
- Build an evaluation harness that runs continuously, not just at launch.
- Distinguish offline evaluation (datasets, fixed) from online evaluation (production sampling).
- Apply canary-query monitoring and shadow-mode testing for production quality assurance.
- Wire evaluations into CI/CD as gates, not as reports.

101.1 The Four Signal Categories

A production AI system emits signals in four distinct categories. Each is necessary; no single category is sufficient.

Table 101.1: The four observability signal categories.

Category	Examples
Conventional service signals	Request rate, error rate, latency distribution (p50/p95/p99), saturation, dependency latency
AI-specific signals	Tokens in/out, generation success rate, refusal rate, time to first token, tool-call frequency, retrieval recall@k on canaries
Quality signals	Citation accuracy on sampled outputs, hallucination rate from grounding checks, user-feedback ratings, downstream-outcome metrics
Cost signals	Cost per call by tenant, cost per outcome, cumulative spend against budget, model-mix distribution

A dashboard that shows only conventional service signals can be entirely green while users complain that the system has become useless. Adding AI-specific and quality signals catches the failure modes that conventional monitoring misses; adding cost signals catches the runaway-spend failures that hide inside healthy traffic.

The all-green-while-failing scenario is worth dwelling on, because it is the failure mode that distinguishes AI observability from conventional observability and the one that most often catches teams unprepared. A conventional service mostly fails loudly: it returns errors, it slows down, it saturates, and the conventional signals turn red. An AI system has a failure mode with no conventional symptom at all — it returns a well-formed, fast, low-cost response that is simply wrong, or biased, or subtly degraded. Every conventional signal stays green because, mechanically, nothing is broken. This is why the four categories are presented as jointly necessary rather than as a menu: the quality signals exist precisely to detect the failures the conventional signals are structurally blind to, and an AI platform monitored only conventionally is not lightly monitored but monitored for the wrong class of failure.

101.1.1 What to Log

For each call, the platform logs:

- Conventional trace context: trace ID, span ID, parent span, timing.
- Identity context: tenant, user, application, use case.
- Model context: model name, version, vendor, parameters (temperature, top-p, max tokens).
- Prompt context: prompt template ID and version; redacted prompt content; token count.
- Retrieval context: query, retrieved chunk IDs, scores, latency.
- Generation context: completion (redacted), token count, finish reason, latency.
- Tool context: tool calls made, arguments (redacted), results (redacted).
- Cost context: per-component cost; total cost for the request.
- Outcome context: did the user accept? was the answer correct (sampled)?

The volume is substantial; tiered retention (full detail for 7 days, samples for 90 days, aggregated metrics for years) keeps the cost bounded.

Two design tensions run through this logging list and deserve to be resolved deliberately rather than by default. The first is redaction: the prompt, completion, retrieved chunks, and tool arguments are exactly where personal and sensitive data appears, so the logging pipeline must redact at capture time — before the data lands in a log store — rather than logging raw content and hoping access controls hold. The repeated word “redacted” in the list is not incidental; it is a requirement. The second tension is that this volume of detailed logging is itself a cost stream, the observability stream of the previous chapter, and a verbose logging configuration can quietly become a meaningful line on the bill. The tiered-retention scheme resolves the second tension — full fidelity briefly, samples for longer, aggregates indefinitely — and the architect should set its tiers deliberately against both the debugging need and the retention cost, because the default of keeping everything at full detail is both expensive and, after the first week, rarely useful.

101.2 The Evaluation Harness

The evaluation harness is to AI what a unit-test suite is to conventional software: standing infrastructure that runs a known set of inputs against a known set of expectations and produces a verdict.

The analogy is exact in one respect and instructive in its one difference. Like a unit-test suite, the harness is standing infrastructure, run continuously, and trusted to gate change. Unlike a unit-test suite, its verdicts are rarely a clean pass or fail, because the system it tests is non-deterministic and the expectations are often graded rather than binary — a summary is not correct or incorrect but more or less faithful, and the same input can produce different outputs on different runs. This means the harness must be designed to reason about distributions and thresholds rather than exact matches, and that its results carry an irreducible amount of noise. The harness is nonetheless the single most valuable piece of standing AI infrastructure a team builds, because it is the only thing that converts the question “is the system still good” from an opinion into a measurement — and the subsections below develop what feeds it, what it emits, and how it gates change.

101.2.1 What Goes In

The harness is fed by:

- Curated datasets: human-authored question-answer pairs, gold-standard summaries, classification labels.
- Synthetic datasets: generated by other models (or by templated rules), then human-validated.
- Production samples: real queries promoted into the eval set after redaction and labelling.
- Adversarial sets: prompts designed to elicit known failure modes (prompt injection, jailbreaks, hallucination triggers).
- Bias and fairness sets: prompts probing for disparate behaviour across protected groups.

Curated datasets are reliable but expensive to maintain; synthetic and production-sample datasets scale better but require careful labelling.

The five dataset types are not interchangeable, and a harness that draws on only one of them has a predictable blind spot. Curated datasets test what the team thought to ask; production samples test what users actually do; adversarial sets test what attackers will try; fairness sets test what the curated and production sets tend to under-represent. A harness fed only curated data measures the system against the team's imagination of its use, which is reliably narrower and better-behaved than reality. A harness fed only production samples drifts with production and loses its stable regression baseline. The discipline is to maintain all five types deliberately, as a portfolio, and to know what each one is and is not capable of catching — because the gaps between them are exactly where an unevaluated failure mode survives.

101.2.2 What Comes Out

Per-metric results, with comparisons to baseline and to the last release:

- Pass/fail against threshold (e.g., accuracy > 0.85 , refusal rate < 0.05).
- Distribution of scores (where binary pass/fail is too coarse).
- Sample-level results for failures (so the team can see what regressed).
- Cost and latency under the eval conditions.

Of these four outputs, the sample-level failure results are the ones that turn the harness from a scoreboard into a diagnostic tool. A harness that reports only an aggregate — accuracy fell from 0.89 to 0.84 — tells the team that something regressed but not what, and the team must then reconstruct the failures by hand. A harness that surfaces the specific inputs that failed, with their expected and actual outputs side by side, tells the team immediately what kind of case broke, which is usually enough to point at the cause. The discipline is to treat the aggregate score as the alarm and the sample-level results as the investigation: the score says whether to look, the samples say where. A harness that emits only the score has automated the noticing and left all the diagnosis manual.

101.2.3 Evaluation Metrics by Task

Different tasks need different metrics:

Table 101.2: Common evaluation metrics by task type.

Task	Metrics
Classification	Accuracy, precision, recall, F1, AUC, confusion matrix
Question answering (extractive)	Exact match, F1 on tokens, ROUGE
Question answering (generative)	Faithfulness, answer relevance, context relevance, model-as-judge
Summarisation	ROUGE, BERTScore, human or model-as-judge for factuality
Code generation	Test pass rate, syntactic validity, security scanner
Retrieval	Recall@k, MRR, NDCG
Tool use	Tool selection accuracy, argument validity, end-task completion
Multi-turn conversation	Per-turn quality, end-to-end task completion, user satisfaction

The table's structure carries a lesson the architect should not miss: as the task moves from the top rows to the bottom, the metrics shift from objective and cheap to subjective and expensive. Classification accuracy is a number a script computes against a label; the faithfulness of a generative answer is a graded judgement that needs a model-as-judge or a human. This progression means that the harder the task, the more the evaluation itself becomes an engineering problem rather than a measurement — and a team that picks a hard, open-ended task should budget for the evaluation infrastructure with the same seriousness as the system. The common error is to choose an ambitious generative task and then evaluate it with a cheap top-of-table metric like ROUGE because that metric is easy to compute, producing a number that is precise and does not measure what the team actually cares about. The metric must match the task; where the task is open-ended, the honest metric is expensive, and that expense is part of the cost of doing the task at all.

101.2.4 Model-as-Judge

For open-ended generation, automatic metrics often fail to capture what humans care about. A common pattern: use a separate, capable model as the judge. The judge model evaluates the candidate output against the question and (where available) reference material, producing structured scores (e.g., faithfulness: 1–5, relevance: 1–5, completeness: 1–5).

Caveats:

- The judge has its own biases; periodic human calibration is required.
- The judge must be different from the model being judged (or self-evaluation bias contaminates results).
- Judges agree more on factuality than on quality; the metrics are most reliable where the ground truth is checkable.

The deeper point behind the caveats is that model-as-judge does not eliminate the need for human judgement; it amortises it. A human calibration set — a few hundred outputs scored by people — is what the judge model is checked against, and the judge is trustworthy only to the degree that its scores track those human scores on that set. Run this way, the pattern is genuinely valuable: the

expensive human scoring is done once, periodically, on a sample, and the cheap model scoring is done continuously, on everything. Run without the calibration set, the judge is an unvalidated model producing numbers that look like measurement and may not be, and a harness built on it has the appearance of rigour without the substance. The architectural commitment model-as-judge requires is therefore not just the judge model but the standing human calibration process beside it; the judge scales human judgement, and a judge with nothing to scale from is just another model whose output needs checking.

101.2.5 Evaluation in CI

The harness runs on every model promotion, every prompt change, every retrieval-configuration change. Failures block promotion:

```

1 name: model-promotion-eval
2 on:
3   workflow_dispatch:
4     inputs:
5       model_version:
6         required: true
7         description: "Candidate model version"
8 jobs:
9   evaluate:
10    runs-on: gpu-pool
11    steps:
12      - name: Run base eval set
13        run: python -m eval.run --suite=base --model=${{ inputs.
              model_version }}
14      - name: Run regression suite
15        run: python -m eval.run --suite=regression --model=${{ inputs.
              model_version }}
16      - name: Run safety suite
17        run: python -m eval.run --suite=safety --model=${{ inputs.
              model_version }}
18      - name: Check thresholds
19        run: python -m eval.check --thresholds=config/
              promotion_thresholds.yaml
20      - name: Publish results
21        run: python -m eval.publish --to=mlflow --tag=candidate

```

Listing 101.1: Evaluation harness in a CI pipeline.

Thresholds are negotiated between the platform team, the spoke, and (for high-risk systems) the governance function. Crossing a threshold without justification blocks promotion; an override is documented and reviewed.

The word “blocks” is what separates an evaluation harness from an evaluation report, and the distinction is the chapter’s central operational claim. A harness that runs on every promotion and produces a report changes nothing on its own — the report can be glanced at, rationalised, or

ignored under deadline pressure, and a regression ships. A harness wired as a gate, where a failed threshold mechanically prevents the promotion, changes the default: shipping a regression now requires a deliberate, documented override rather than mere inattention. This is the same principle the responsible-AI and cost chapters reached by different routes — a control that depends on someone choosing to honour it is weaker than a control that is the path itself. The override path matters too, and must exist: a gate with no override is brittle and will eventually be bypassed by disabling it entirely. The right design is a gate that blocks by default and permits a recorded, reviewed override, so that shipping past a failed evaluation is possible, visible, and accountable rather than silent.

101.3 Offline vs Online Evaluation

Two distinct disciplines, both required.

They are required together because each measures something the other cannot. Offline evaluation answers “is this version better than the last one” under controlled, reproducible conditions — the question that must be answered before a change ships. Online evaluation answers “is the system actually working for real users on real traffic” — the question that can only be answered after a change ships, against a production distribution no dataset fully captures. A programme with only offline evaluation ships changes that pass the test set and then degrades invisibly in production; a programme with only online evaluation detects problems but has no controlled way to decide whether a fix is an improvement before deploying it. The two subsections below treat each in turn, and the pairing — offline to gate change, online to catch what the gate could not foresee — is the complete discipline that neither half provides alone.

101.3.1 Offline Evaluation

Runs the system against fixed datasets in a controlled environment. Strengths: reproducible, comparable across versions, fast feedback in CI. Weaknesses: dataset bias (the eval set may not look like production), staleness (production drift outpaces dataset refresh), gaming (the system may be inadvertently optimised for the eval set).

Mitigations: maintain multiple eval sets (general, regression, safety, fairness); refresh production samples into the eval set monthly; keep some sets blind (used only at major promotions, never for routine development).

The blind-set mitigation deserves particular emphasis, because it addresses the most insidious weakness of offline evaluation: gaming. When a team can see its eval set and develops against it iteratively, the system gradually becomes optimised for that specific set rather than for the underlying capability the set was meant to measure — not through any dishonesty, but through the ordinary process of fixing whatever the visible test flags. The eval score rises while the true capability does not, and the gap is invisible precisely because the visible eval is the thing that has been satisfied. A blind set, never used in routine development and run only at major promotions, is the check that catches this: a candidate that scores well on the development sets and poorly on the blind set has been optimised for the test rather than the task. Keeping a portion of evaluation genuinely blind is what keeps the rest of the evaluation honest.

101.3.2 Online Evaluation

Measures the system as it actually runs:

- **Canary queries.** A fixed set of queries (with known expected behaviour) is sent to production at a low rate; the responses are graded; deviations alert.
- **Shadow mode.** A candidate system processes production traffic in parallel with the current one; outputs are compared; the candidate is promoted only after parity or improvement is established.
- **A/B testing.** Two variants serve traffic to different users; outcome metrics (user acceptance, task completion, follow-up rate) are compared.
- **Production sampling.** A small fraction of production responses is sampled for human or model-as-judge review; results feed back into the eval set.

Online evaluation catches what offline misses: failures that depend on production data distribution, traffic patterns, multi-component interaction, or downstream consumer behaviour.

The four online techniques form a progression of increasing confidence and increasing cost, and an architect should know which to reach for when. Canary queries are the cheapest and run continuously — a constant low-rate pulse that catches gross regressions within minutes. Shadow mode is the technique for validating a candidate before it serves anyone: it runs the candidate on real traffic with its outputs discarded, so a bad candidate is caught having affected no user. A/B testing is the technique for measuring whether a change genuinely improves outcomes, not just metrics, by comparing real user behaviour across variants. Production sampling is the slow continuous feed that keeps the offline eval sets honest by promoting real cases into them. The techniques are complementary, not alternatives: a mature programme runs canary queries always, uses shadow mode for every significant candidate, A/B tests the changes whose value is genuinely uncertain, and samples production continuously. Matching the technique to the question is the skill; using only one of the four leaves a class of production failure unwatched.

101.4 Drift Detection

Drift is the silent failure mode. The system's quality degrades over time without an obvious incident; users gradually drift away.

Drift deserves its own section because it is the failure mode the rest of the chapter's machinery is least naturally suited to catch. An evaluation harness run on a fixed dataset measures the system against a world that is held still; drift is precisely the world moving while the dataset does not. A monitoring dashboard shows the current state; drift is a slow trend that any single day's snapshot looks normal against. Drift is dangerous specifically because it has no incident, no alarm, no red signal — only a gradual divergence between what the system was built for and what it now faces. The three subsections below distinguish three kinds of drift, because they have different causes and different detection methods, and a programme that watches for one kind is still blind to the other two.

101.4.1 Input Drift

The distribution of inputs changes. Detection:

- Population Stability Index (PSI) over input feature distributions.
- Kolmogorov–Smirnov tests for individual numeric features.
- Embedding-distribution metrics (compare current vs. baseline embedding distribution by Wasserstein distance or MMD).
- Topic drift on text inputs (LDA topic mixtures over time).

Input drift is the most tractable of the three kinds, because it can be detected without ground truth — it is a property of the inputs alone, measurable by comparing the current input distribution against a baseline, with no need to know whether the system’s outputs were right. This makes it the drift an architect should instrument first: it is the cheapest to watch and it is frequently the leading indicator of the other two. Inputs shifting is often the first observable sign that the world the system serves is changing, and catching that shift early gives the team time to check whether output quality and the input-output relationship are about to follow. The embedding-distribution metric is the most broadly useful instrument for text-heavy AI systems, because it captures semantic shift that simple feature statistics miss — users asking about genuinely new topics, in new ways, that a feature-level test would not register as drift at all.

101.4.2 Output Drift

The distribution of outputs changes even when inputs are stable. Detection:

- Output length and structural property distributions.
- Refusal rate over time.
- Sentiment or formality distribution shift.
- Tool-call frequency distribution.

Output drift is distinctive because it can occur with the inputs held completely stable — the same kinds of question, producing measurably different answers over time. The usual cause is a change beneath the system that the team did not initiate: a vendor silently updated the model behind an API, a system prompt was edited, a retrieval corpus shifted, a guardrail was re-tuned. Output drift is therefore often the first detectable symptom of an upstream change the team was not told about, which makes the refusal rate especially worth watching: a creeping rise in refusals, with inputs unchanged, frequently means the model behind the API has been updated to be more conservative. Monitoring output distributions is, in effect, monitoring the dependencies the team does not control. An AI system built on vendor APIs is built on a foundation that can move without notice, and output-drift detection is how the team learns that it has.

101.4.3 Concept Drift

The relationship between inputs and the right output changes (e.g., new product categories appear, the law changes). Detection requires ground truth; the leading indicators are user-feedback shifts and downstream-outcome shifts.

Concept drift is the hardest of the three to detect and the most dangerous, because the inputs and the outputs can both look entirely stable while the system is increasingly wrong. The system answers the same kinds of question in the same kinds of way, so input-drift and output-drift detection register nothing — but the *correct* answer has changed, and the system, not knowing this, is now confidently giving the answer that used to be right. Because concept drift cannot be seen in the inputs or outputs alone, its detection depends on signals that reach back from the world: users correcting the system more often, downstream outcomes worsening, complaint rates rising. The architectural consequence is that those feedback and outcome channels are not optional niceties but the only available sensor for an entire class of failure. A system with no way to learn that its answers stopped being correct will continue giving outdated answers indefinitely, fluently and without alarm, and concept drift is the reason the feedback loop must be treated as core instrumentation rather than a later enhancement.

101.4.4 Responding to Drift

A detected drift produces an alert and a triage workflow:

- Confirm: is the drift real or a measurement artefact?
- Diagnose: where in the system is it visible?
- Triage: is action required now, or can it wait for the next scheduled refresh?
- Action: retrain, update prompts, refresh retrieval corpus, adjust thresholds, or all of the above.

The first step, confirmation, is the one most often skipped under pressure and the one that most needs the discipline. Drift metrics are statistical, computed on samples, and they produce false positives — a holiday changes the traffic mix, a single large tenant runs an unusual batch, a measurement window catches noise. A team that treats every drift alert as a real regression and reacts immediately will spend its effort chasing artefacts and will, worse, make changes to a system that was not actually drifting. The triage step embeds the other essential judgement: not all real drift is urgent. Genuine, slow drift can often wait for the next scheduled retraining or corpus refresh, and forcing an out-of-cycle intervention for every confirmed drift is its own kind of instability. The workflow exists to convert a drift signal into a proportionate response — confirm it is real, locate it, judge its urgency, then act — rather than into either panic or indifference.

101.5 Tracing the AI Call

A single user request can fan out into dozens of sub-operations: retrieval, re-ranking, tool calls, sub-agent invocations. Conventional tracing (OpenTelemetry) extends to AI workloads with AI-specific span attributes:

- `ai.model.name`, `ai.model.version`, `ai.model.vendor`
- `ai.prompt.template_id`, `ai.prompt.token_count`
- `ai.completion.token_count`, `ai.completion.finish_reason`
- `ai.retrieval.k`, `ai.retrieval.method`, `ai.retrieval.recall_score`
- `ai.tool.name`, `ai.tool.invocation_id`
- `ai.cost.input_usd`, `ai.cost.output_usd`

OpenTelemetry semantic conventions for LLM operations are stabilising; adopting them now pays back when interoperability with future tooling matters.

The reason tracing matters more for AI systems than for conventional services is the fan-out. A conventional request often does one or two things; an AI request can fan out into retrieval, re-ranking, several tool calls, sub-agent invocations, and multiple model calls, each with its own latency, cost, and failure possibility. Without a trace that ties all of these into one tree under a shared trace identifier, an operator debugging a slow or wrong response has a scatter of disconnected log lines and must reconstruct the causal story by hand. With it, the operator sees the whole tree — which span was slow, which tool failed, which retrieval returned poor chunks, where the cost accumulated. This is the same unified-trace argument the multi-agent chapter made, and the AI-specific span attributes above are what make the trace genuinely useful rather than merely present: a trace that records that a model was called is far less valuable than one that records which model, which prompt template, how many tokens, what finish reason, and what it cost. Adopting the AI-specific conventions is what turns generic tracing into AI observability.

101.6 Dashboards and Alerts

A few dashboards every AI platform team needs:

- **Platform health.** Latency, error rate, throughput, queue depth across the platform's services.
- **Per-tenant.** The same metrics sliced by tenant, with budget burn and cost-per-call.
- **Per-model.** Quality metrics, refusal rate, drift signals per model in production.
- **Eval status.** Most recent eval results, trend over time, blocked promotions.
- **Drift watch.** Drift signals across input, output, and concept dimensions.
- **Incident.** Open incidents, their status, runbook progress.

Alert hygiene matters: an alert that fires more than weekly without action attached is noise; an alert that requires action only every few months is missing the early signal. Tuning alerts is itself an ongoing discipline.

The reason alert hygiene is a discipline rather than a one-time setup is that an alert which is routinely ignored is worse than no alert at all. A noisy alert trains the team to dismiss it, and a team trained to

dismiss one alert becomes a team that dismisses alerts in general — so the noisy drift alert quietly erodes the response to the genuine incident alert beside it. Each alert should therefore have to justify its existence continuously, against two questions: does it fire rarely enough that the team still takes it seriously, and does it fire early enough to matter. An alert that fails the first question is retuned or removed; an alert that fails the second is made more sensitive. This maintenance is unglamorous and never finishes, because the system and its traffic keep changing, but it is what keeps the whole alerting layer credible. An AI platform's alerting is only as good as the team's trust in it, and that trust is maintained one retuned threshold at a time.

101.7 The African Perspective

Observability and evaluation are universal disciplines, but two of their specifics deserve attention in African deployments.

The first concerns the evaluation sets. The curated, synthetic, and benchmark datasets that an AI programme draws on are predominantly built around English-language inputs and around the contexts of the markets where most AI tooling originates. An evaluation suite assembled only from these measures the system's quality on inputs that may not resemble what the system will actually receive — queries in local languages, code-switched text, locally specific topics and entities. A system that scores well on the imported eval sets and is deployed against a different real distribution has been evaluated against the wrong world. The discipline of refreshing production samples into the eval set, recommended earlier in the chapter, is correspondingly more important here: it is the mechanism by which the evaluation suite comes to reflect the inputs the system genuinely serves rather than the inputs its off-the-shelf datasets assumed.

The second concerns the cost of the observability and evaluation stack itself. Detailed logging, continuous canary traffic, shadow-mode runs, and model-as-judge evaluation all consume resources — the observability cost stream and the model-call cost of the previous chapter — and in a foreign-exchange-exposed, cost-constrained deployment that stack must be sized deliberately rather than maximally. The tiered-retention discipline, the sampling of production rather than full capture, and the use of cheaper models for routine model-as-judge scoring are not merely good practice here; they are what makes a genuine observability and evaluation capability affordable. The objective is unchanged — the four signal categories, the harness, the drift detection are all still required — but the implementation must be engineered for cost, because an observability stack the organisation cannot afford to run is one it will eventually switch off, and a switched-off observability stack returns the system to the silent-failure regime the whole chapter exists to prevent.

101.8 Last Word

Observability and evaluations are the disciplines that turn a built AI system into an operable one. Four signal categories (conventional, AI-specific, quality, cost), an evaluation harness that runs continuously, online evaluation alongside offline, drift detection across input/output/concept dimensions, and AI-aware tracing together produce a system that the team can actually run. The cost of building these in is modest; the cost of running without them is large, mostly paid in incidents and silent quality

regression.

The phrase “silent quality regression” is the chapter’s whole motivation, and it is what distinguishes operating an AI system from operating a conventional one. A conventional service that breaks tells you it has broken; an AI system that degrades does so quietly, returning fluent, fast, well-formed answers that are simply getting worse, while every conventional signal stays green. Everything in this chapter — the quality signals, the continuous harness, the drift detection, the canary queries — exists to give that silent regression a voice, to convert a degradation the team would otherwise discover through lost users into a signal the team sees on a dashboard. An AI programme that builds these disciplines in has a system it can trust over time, because it will know when that trust is no longer warranted. An AI programme that skips them has a system that works on launch day and offers no way to know when it stops — and in a technology whose dependencies, inputs, and surrounding world all keep moving, the question is not whether it will stop matching reality but whether anyone will notice when it does.

Security and Threat Modelling

“Threat modelling is what asks ‘what could go wrong’ before it goes wrong.”

—a recurring security maxim

Chapter Overview

AI systems share the security threats of any distributed service and add their own: prompt injection, training-data poisoning, model extraction, insecure output handling, vector-store leakage, supply-chain compromise of model weights. This chapter develops the AI-specific threat model, the OWASP Top 10 for LLM Applications as a working checklist, the canonical defences (input filtering, output validation, sandboxing, monitoring, red teaming), and the operational practices (security reviews, penetration testing, incident response) that operationalise security as a continuous discipline rather than a launch gate.

Learning Objectives

- Threat-model an AI system using STRIDE extended with AI-specific threats.
- Recognise the OWASP Top 10 for LLM Applications as a working checklist.
- Apply layered defences against prompt injection, the dominant AI-specific attack class.
- Operate a supply-chain discipline for model weights, training data, and third-party components.
- Run red-team and penetration-testing exercises with AI-specific scenarios.

102.1 Threat Modelling for AI

A threat model asks, systematically, what could go wrong. The traditional STRIDE categories (Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege) apply, with several AI-specific extensions:

- **Manipulation.** The system is induced to behave outside its intended specification through carefully crafted inputs.
- **Inference.** Sensitive information is extracted from the model itself (training data memorisation, model inversion, membership inference).

- **Output abuse.** The model produces content that the consumer trusts and uses unsafely (SQL injection from generated code, XSS from generated HTML).

What the three AI-specific extensions have in common is that they exploit the defining feature of an AI system — that it processes natural-language input with a learned model rather than fixed code — and so they have no equivalent in a conventional threat model. STRIDE was built for systems whose behaviour is determined by their code; an attacker who cannot change the code is bounded by what the code does. An AI system's behaviour is determined partly by its input, which the attacker often controls, so the attacker can induce behaviour the designers never wrote. This is why STRIDE alone is insufficient for AI and why the extensions are not optional refinements but a necessary completion of the model. A threat-modelling exercise that applies only the six classical STRIDE categories to an AI system has, by construction, not looked at the attack surface that makes the system an AI system.

102.1.1 The Threat-Modelling Process

A practical threat-modelling exercise for a new AI system:

- Describe the system: trust boundaries, data flows, external interfaces, identity flows.
- Enumerate assets: what is valuable (data, models, capability, reputation)?
- Enumerate adversaries: who would want to compromise it, with what capabilities?
- For each combination, apply STRIDE + AI extensions; list candidate threats.
- For each threat, identify existing mitigations; note residual risk.
- Prioritise the residual risks for remediation.

The single most important step is the first, because every later step depends on the trust boundaries being drawn correctly, and AI systems are where they are most often drawn wrong. The recurring error is to place the boundary so that retrieved content, tool outputs, and model responses fall on the trusted side — to treat a document the agent fetched, or text the model generated, as though it were as trustworthy as the system's own code. It is not. Once the boundaries are drawn honestly, with retrieved content and model output explicitly outside the trust boundary, most of the AI-specific threats in this chapter follow visibly from the diagram, because each is an attack that crosses a boundary the diagram now shows. A threat model built on an honest trust-boundary diagram surfaces the threats almost mechanically; one built on a diagram that quietly trusts the untrustworthy will enumerate threats all day and miss the ones that matter.

102.2 OWASP Top 10 for LLM Applications

The OWASP Top 10 for LLM Applications is the most useful current checklist. The categories (2024 edition, with 2026 refinements):

Table 102.1: OWASP Top 10 for LLM Applications and their architectural mitigations.

Risk	Mitigation
LLM01 Prompt injection	Input filtering; output validation; trust boundary between user input and system context; structured tool invocation
LLM02 Insecure output handling	Treat LLM output as untrusted; sanitise before use in code, queries, or markup
LLM03 Training data poisoning	Source vetting; supply-chain controls on training data; anomaly detection on inputs
LLM04 Model denial of service	Rate limiting; input size caps; resource quotas per tenant
LLM05 Supply chain vulnerabilities	SBOM for models; signed weights; vendor risk review; build-from-source where feasible
LLM06 Sensitive information disclosure	PII redaction; output scanning; differential privacy where applicable
LLM07 Insecure plugin design	MCP-style structured tool definitions; scope-limited tools; per-tool authorisation
LLM08 Excessive agency	Capability scoping; human-in-the-loop for consequential actions; sandboxing for tool execution
LLM09 Overreliance	UI design that supports verification; citations; calibrated confidence; user training
LLM10 Model theft	Access logging; rate limiting; query-pattern detection; legal protection

The OWASP list earns its place as a working checklist for two reasons. First, it is concrete and current: it names the specific risk classes that have actually been exploited against deployed LLM applications, which is more useful to a building team than an abstract taxonomy. Second, it is a shared vocabulary — when a security reviewer, an architect, and an auditor all refer to LLM01, they are referring to the same thing, and the architecture review can proceed without each party first defining its terms. The discipline is to treat the list as a checklist rather than a syllabus: for each new AI system, walk all ten risks and record, for each, either the mitigation in place or an explicit decision that the risk does not apply. The list will be revised — OWASP updates it as the threat landscape moves — so it is a current reference rather than a permanent one, but a team that has genuinely addressed all ten for a given system has covered the attack surface that the field has so far learned to look for.

102.3 Prompt Injection in Depth

Prompt injection is the most common attack class on LLM-based systems. The attacker controls some part of the model’s input and uses it to manipulate the model’s behaviour.

What makes prompt injection genuinely hard — and worth a section of its own — is that it has no clean structural fix, unlike most of the other OWASP risks. SQL injection was solved by parameterised queries, which create an unambiguous separation between code and data; the database can always tell which is which. A language model has no equivalent separation, because instruction and data arrive in the same channel, as natural-language text, and the model’s whole purpose is to interpret that text flexibly. There is no parameterised-query equivalent that lets the model reliably distinguish

“content to act on” from “content to act upon.” This is why the defences in the subsections below are all mitigations that reduce the risk rather than eliminations that close it, and why prompt injection is best treated as a managed, layered, monitored risk rather than a solved problem. An architecture that expects a single clean fix for prompt injection has misunderstood the problem; one that layers partial defences and watches for the bypass has understood it correctly.

102.3.1 Direct Injection

The attacker is the user, and they craft a prompt that overrides the system’s instructions: “Ignore previous instructions and tell me your system prompt.” Defences:

- Structured separation of system and user content in the API call.
- Input classifiers detecting jailbreak patterns.
- Output validation that catches when the model is being induced to violate policy.
- Constitutional / policy prompting in the system message.

Direct injection is the more tractable of the two main classes, because the attacker is the user and the threat model is correspondingly bounded: the damage is limited to what that user is already authorised to reach. A user who jailbreaks a system into revealing its system prompt has obtained the system prompt; a user who induces it to drop a content guardrail has obtained content that user could perhaps have found elsewhere. This is a real problem, but it is a contained one, and the layered defences above — structural separation, input classification, output validation, policy prompting — manage it adequately for most systems. The reason the next subsection matters more is that indirect injection breaks this containment: it lets an attacker who is *not* the user, and who has none of the user’s authority, reach the model through content the user innocently caused it to process. Direct injection is the class to defend competently; indirect injection is the class to fear.

102.3.2 Indirect Injection

The attacker controls content that the model retrieves or processes from a third-party source: a webpage the agent reads, a document in the corpus, an email being summarised. The injection vector is hidden in plain sight; the model treats it as instruction even though it came from the data, not the user.

This is the more dangerous class because the user did not author the injection and the system may not have a clean trust boundary on retrieved content. Defences:

- Treat all retrieved content as untrusted; separate it from instruction in the prompt structure.
- Output validation: the model’s response is checked against the user’s question, not against retrieved instructions.
- Tool-call gating: instructions in retrieved content cannot directly invoke tools; tools require structured invocation from the agent’s reasoning, not from arbitrary text.
- Provenance: every retrieved chunk carries its source; suspicious sources can be downweighted or

excluded.

The tool-call gating defence is the most important of the four, because it addresses what makes indirect injection genuinely dangerous rather than merely embarrassing. An indirect injection that only manipulates what the model *says* produces a wrong or off-policy answer — bad, but bounded. An indirect injection that can manipulate what the model *does* — inducing a tool call from instructions hidden in a retrieved document — converts a poisoned webpage into action in the world: data exfiltrated, a transaction triggered, an email sent. The gating principle, that tools are invoked only by the agent’s own structured reasoning and never by free text appearing in retrieved content, is what severs the path from injected text to consequential action. This is the same trust-boundary discipline the agent-anatomy and MCP chapters insisted on, applied here to its most important case: the boundary between what the model reads and what the model is permitted to do must be structural, because retrieved content is exactly where an attacker will hide the instruction to cross it.

102.3.3 Multi-Turn and Latent Injection

More sophisticated attacks split the injection across turns, or hide it in a transformation (ROT13, base64, image alt-text) that the model decodes mid-response. Defences require monitoring of complete conversation trajectories, not just per-turn classification.

These attacks exist specifically to defeat per-turn input classification, and understanding that is what points at the defence. A classifier that inspects each message in isolation can be evaded by an injection that is benign in every individual message and malicious only in their combination — the first turn establishes a frame, the second exploits it, and neither turn alone trips the classifier. An injection hidden in an encoding is benign as the literal string the classifier sees and malicious only once the model decodes it. The common thread is that the malicious content is not present at the point the per-turn classifier looks. The defence therefore has to move to where the malice becomes visible: monitoring the whole conversation trajectory for the developing pattern, and inspecting decoded rather than raw content. This is an arms race, and the honest framing is that it will remain one — which is the chapter’s recurring point about prompt injection, and the reason the red-teaming and monitoring disciplines later in the chapter are not optional extras but the standing mechanism by which the newest evasions are found before an attacker uses them.

102.4 Insecure Output Handling

The model’s output is untrusted from the perspective of any system that consumes it programmatically. Concrete failure modes:

- Generated SQL executed against a database: SQL injection.
- Generated HTML rendered to a user: cross-site scripting.
- Generated shell commands executed: command injection.
- Generated URLs followed: SSRF if the URL points internally.
- Generated code committed without review: supply-chain compromise.

The architectural rule: treat LLM output the same way you would treat user input. Parameterise SQL, sanitise HTML, allow-list shell commands, restrict URL fetches, require review on generated code. The model's confident, fluent output should not override the security controls that apply to any other untrusted input.

The reason this rule has to be stated explicitly is that the model's output actively works against it. Conventional untrusted input — a form field, a query parameter — looks untrusted; nothing about it invites the developer to skip the sanitisation. A language model's output looks authoritative: it is fluent, well-structured, confident, and frequently correct, and that surface of competence creates a psychological pull to treat it as trustworthy and pipe it directly into a database, a page, or a shell. That pull is the vulnerability. The discipline is to recognise that fluency is not provenance: a well-formed SQL statement from a model is exactly as untrusted as a well-formed SQL statement typed by an anonymous user, and the fact that the model usually gets it right is irrelevant to the case where an injection made it get it wrong. Insecure output handling is not a sophisticated attack; it is the routine consequence of letting the model's confident manner suspend a control the team would never have suspended for any other untrusted source.

102.5 Supply Chain

The AI supply chain has several distinct components, each with its own integrity considerations:

- **Model weights.** Provenance: which vendor, which version, what training data. Integrity: cryptographic hashes, signatures. Storage: encrypted at rest; access logged.
- **Training data.** Source vetting; classification at ingest; refusal to train on data lacking documented lineage.
- **Frameworks and libraries.** Conventional SBOM and vulnerability scanning; pinned versions; signed packages.
- **Third-party prompts and tools.** Treated as untrusted external code; reviewed before integration.

The component that conventional supply-chain discipline handles least well is model weights, because a set of weights is opaque in a way source code is not. A library can be read, audited, and scanned; its behaviour is, in principle, inspectable. A model's weights are billions of numbers that cannot be meaningfully read, so a compromise introduced during training or distribution — a backdoor, a poisoned behaviour triggered by a specific input — cannot be found by inspecting the artefact. This shifts the defensible posture from inspection to provenance and behavioural testing: the integrity controls (cryptographic hashes, signatures, official sources) establish that the weights are the ones the vendor produced, and the pre-deployment evaluation establishes how they behave, but neither can prove the absence of a well-hidden backdoor the way reading code can. The architectural consequence is that model weights should be treated as the highest-trust-required, least-inspectable component in the supply chain, sourced only from parties the organisation has independent reason to trust, because for this component the usual fallback of auditing the artefact is not available.

102.5.1 Open-Weight Models

Open-weight models bring opportunity (control, customisation, sovereignty) and risk (no vendor's safety review, possible compromise during distribution). The architectural posture:

- Download from official sources (HuggingFace, vendor portals); verify signatures.
- Run pre-deployment evaluation, including safety and bias suites.
- Maintain provenance: who downloaded, when, what hash.
- Treat them with the same supply-chain discipline as any vendor code.

The honest framing of open-weight models is that they relocate the safety responsibility rather than removing it. A frontier vendor's hosted model arrives with the vendor's own safety review, red-teaming, and alignment work already done — imperfect, but real, and paid for by the vendor. An open-weight model arrives with whatever safety work its publisher chose to do and disclose, which varies widely, and the adopting organisation inherits the responsibility for any gap. This is not an argument against open-weight models: their advantages in control, customisation, cost, and sovereignty are real, and several earlier chapters have noted cases where they are the right choice. It is an argument for budgeting the safety work that the open-weight choice transfers onto the adopter. An organisation that adopts an open-weight model for its sovereignty and cost advantages, and runs the pre-deployment safety and bias evaluation those advantages oblige it to run, has made a sound decision. An organisation that adopts one and assumes the safety came in the box has accepted a responsibility it does not know it holds.

102.6 Defence Architecture

The canonical defence architecture for an AI system has several layers:

- **Network.** mTLS between services; egress restrictions on model-serving infrastructure; private connectivity to managed vendor APIs.
- **Identity.** Strong authentication for users; service identity (SPIFFE / X.509) for service-to-service; short-lived tokens; just-in-time access for sensitive operations.
- **Authorisation.** Fine-grained access control on data and tools; capability tokens scoped to specific operations and tenants; principle of least privilege.
- **Input.** Validation at the API gateway; classification of incoming content; rate limiting per identity.
- **Inference.** Guardrails as classifiers in front of and behind the model; system-prompt hygiene; structured tool calls.
- **Output.** Validation, sanitisation, classification before downstream consumption.
- **Audit.** Every action logged; logs themselves protected; periodic review.

The defining principle of this architecture is defence in depth: no single layer is trusted to be sufficient,

and the system stays secure when — not if — an individual layer fails. The inference-layer guardrail will sometimes be bypassed by a novel prompt injection; the input classifier will sometimes miss a new evasion. Defence in depth means that when this happens, the output validation, the tool-call gating, the authorisation scoping, and the audit trail are all still standing between the bypass and real harm. The architecture is sound not because any layer is perfect but because an attacker must defeat all of them, and the layers fail in different ways. Two layers carry extra weight and deserve emphasis. The authorisation layer's least-privilege scoping is what bounds the blast radius when an upper layer is breached — a compromised agent can only reach what its capability tokens allowed. The audit layer is what makes the breach investigable afterwards; it prevents nothing, but a breach that cannot be reconstructed is a breach the organisation cannot learn from, and the audit trail is therefore the layer that turns an incident into a lesson rather than a mystery.

102.7 Red Teaming

A red team simulates an adversary attacking the system, probing for vulnerabilities the design did not anticipate. For AI, the exercise includes:

- Prompt-injection probes across direct, indirect, multi-turn, and latent vectors.
- Jailbreaking attempts: encoded harmful requests, role-play exploits, system-prompt extraction.
- Sensitive-information extraction: attempts to recover training data, system prompts, or other tenants' content.
- Tool-misuse probes: getting the model to invoke tools in ways the policy does not intend.
- Denial of service: prompt patterns that maximise per-call cost; long outputs; resource exhaustion.
- Bias and fairness probes: prompts designed to elicit disparate behaviour across groups.

The exercise output is a structured report: which probes succeeded, what the mitigation is, what the residual risk remains. High-risk systems run red-team exercises at least annually; high-profile systems run them quarterly.

Red teaming earns its place because it tests the system the way an adversary will, rather than the way the designers hoped. Every other discipline in this chapter — the threat model, the OWASP checklist, the layered defences — is built from what the team and the field already know to look for, and is therefore structurally blind to the attack no one has thought of yet. A red team's job is precisely to think of it: to probe the system adversarially, without the designers' assumptions, and find the path the design did not anticipate. This is also why the cadence matters and why the exercise must recur. A red-team exercise certifies the system against the attacks known at the time it was run; the attack landscape moves, new injection and jailbreak techniques appear continually, and a system red-teamed once at launch is, two years later, defended only against a two-year-old threat landscape. Red teaming is not a milestone the system passes; it is a recurring exercise that keeps the system's defences current with an adversary who does not stop learning.

102.7.1 Automated Red Teaming

Tooling that generates and runs adversarial prompts at scale is maturing. The architectural fit: automated red teaming runs as part of CI for safety-critical systems, with manual review of high-impact findings. Automated approaches catch the known classes; human red teamers find the genuinely novel cases.

The division of labour in that last sentence is the design principle, and it matters because the two approaches are complementary rather than competing. Automated red teaming is broad, cheap, and repeatable: it can run thousands of known attack patterns on every build, which makes it the right instrument for regression — ensuring that a vulnerability once fixed does not silently return, and that the known attack classes are re-tested continuously at no marginal human cost. What automated red teaming cannot do is be genuinely creative; it generates variations on attacks that are already understood. Human red teaming is expensive and slow but creative: a skilled human adversary finds the conceptually new attack, the one no existing pattern covers. The sound architecture uses each for what it is good at — automated probes in CI as the continuous regression net, periodic human exercises as the source of genuinely new findings, and a path by which each novel human discovery becomes a new automated probe. Treating automated red teaming as a replacement for human red teaming, rather than as the regression layer beneath it, leaves the system tested only against yesterday's imagination.

102.8 Security Reviews

Several review touchpoints align security to the system lifecycle:

- **Design review.** Threat model; identify high-risk areas; agree mitigations before build starts.
- **Pre-launch review.** Verify that mitigations are in place; review red-team findings; confirm logging and alerting.
- **Post-incident review.** Blameless post-mortem; structural fix; eval suite addition for the failure mode.
- **Periodic recertification.** High-risk systems re-reviewed annually; lower-risk on a documented cadence.
- **Change review.** Significant changes (new vendor, new model, new data source) trigger a focused security review.

The reviews are not bureaucratic theatre; they produce structured outputs (threat models, mitigation registers, action items) that the platform tracks to completion.

The five touchpoints are placed where they are because security decisions, like architectural decisions, get more expensive to act on the later they are made. The design review is the highest-leverage of the five: a threat identified before any code exists costs a conversation to mitigate, because the mitigation can simply be designed in. The same threat found at the pre-launch review costs a delay and a rework. Found in a post-incident review, it has already cost an incident. This is the same economics the

architecture-governance chapter described for design decisions in general, and it applies to security with particular force because the cost of a late-discovered security gap includes the breach itself. The discipline the list encodes is to move security scrutiny as early as it can go — to make the design review a genuine threat-modelling exercise rather than a formality, so that the cheap touchpoint does the heavy lifting and the later ones are verification rather than discovery. A security-review programme weighted toward its early touchpoints is doing security; one weighted toward its late ones is mostly documenting incidents.

102.9 Incident Response

When a security incident occurs in an AI system, the response follows the standard pattern with AI-specific additions:

- **Contain.** Disable the affected capability; revoke compromised credentials; isolate affected components.
- **Investigate.** Logs of prompts, completions, retrievals, and tool calls from the incident window are the primary evidence.
- **Eradicate.** Remove the attack vector: patch the input filter, update the system prompt, remove poisoned data, retrain the affected model if needed.
- **Recover.** Restore service; verify the fix; communicate with affected users.
- **Learn.** Post-mortem; add a regression test to the eval suite for the failure mode; update threat model and runbooks.

The incident-response runbooks must cover AI-specific scenarios: prompt-injection compromise of an agent, model output causing downstream harm, retrieval-store leakage, supply-chain compromise of a model artefact.

The reason AI-specific runbooks are needed, rather than relying on a general incident-response process, is that the AI-specific scenarios have containment steps a conventional runbook does not anticipate. Containing a prompt-injection compromise of an agent may mean disabling a specific tool the agent could call, not the whole service. Containing retrieval-store leakage means reasoning about which documents were exposed to which tenants through which queries — a question the prompt and retrieval logs answer and a conventional access log does not. Eradicating a training-data-poisoning incident may require retraining a model, which is a containment action measured in days, not minutes, and the runbook must say what the system does in the interim. A team that meets one of these scenarios for the first time during the incident itself will improvise the containment under pressure and improvise it badly. The discipline is to write the AI-specific runbooks in advance, exercise them, and ensure the logging they depend on — the prompt, completion, retrieval, and tool-call records of the observability chapter — is actually being captured, because an incident runbook that calls for evidence the system never logged is a runbook that fails at the investigate step.

102.10 The African Perspective

The security disciplines of this chapter are universal, but two aspects carry particular weight in African deployments.

The first is the scarcity of specialised AI-security skills. Threat modelling for AI, AI red teaming, and AI incident response all require expertise that is scarce everywhere and scarcer in many African markets, where it competes with every other demand on a thin pool of senior security talent. This argues for leaning deliberately on the parts of the chapter that do not require a large standing specialist team: the OWASP Top 10 as a structured checklist that a competent generalist security reviewer can apply, automated red teaming as the continuous regression layer that substitutes scale for scarce human time, and well-written AI-specific runbooks that encode expertise once so that it does not have to be re-derived under incident pressure. The objective is unchanged; the implementation leans toward the structured, the automated, and the documented, because those multiply the scarce expertise rather than depending on its constant availability.

The second is the supply-chain and sovereignty dimension. The earlier chapters' point that data-sovereignty regulation and cost pressure push African deployments toward open-weight, self-hosted models intersects directly with this chapter's supply-chain section. An open-weight model adopted for sound sovereignty reasons brings with it the supply-chain and safety-review responsibilities that this chapter has detailed — the provenance discipline, the signature verification, the pre-deployment safety and bias evaluation that the absence of a vendor's own review obliges the adopter to run. The architectural posture is to treat the move to open-weight, self-hosted models not only as a sovereignty and cost decision but as a security decision that comes with a specific, budgetable set of supply-chain obligations. Made with those obligations resourced, it is sound; made without them, it trades a data-residency risk for an unmanaged supply-chain one.

102.11 What This Chapter Established

Security is a continuous discipline, not a launch gate. Threat model the system explicitly; apply the OWASP Top 10 for LLM Applications as a working checklist; defend in depth against prompt injection (the dominant AI-specific attack class); treat model output as untrusted from a programmatic perspective; operate a supply-chain discipline for weights, training data, and third-party components; run red-team exercises and security reviews on a documented cadence. The architectural outcomes—confidentiality, integrity, availability, and accountability—are properties of a system that has built security in throughout, not one that bolted it on at the end.

The phrase “continuous discipline, not a launch gate” is the chapter's central claim, and it follows from a fact specific to AI security: the threat landscape moves faster than the system does. A conventional application's attack surface is largely fixed once it ships; an AI system faces new prompt-injection techniques, new jailbreaks, and new evasions that are discovered continually, by a global population of researchers and adversaries, after the system is in production. A security posture certified once at launch is therefore certified against a landscape that no longer exists within months. Everything in this chapter that recurs — the periodic red teaming, the cadenced security reviews, the continuous

automated probes, the updated runbooks — exists to keep the system's defences moving at the speed the threats move. This is the same disposition the book has argued for throughout, and the opening chapters' image is exact: security is not a wall built once but a cage that must be maintained, inspected, and rebuilt as what it contains, and what presses against it, keep changing. An organisation that has built security in throughout, and keeps building it, has a system it can keep trusting; one that bolted it on at launch has a system that was secure once.

Chapter 103

Governance and Compliance

“Compliance is the floor, not the ceiling.”

—a recurring legal observation

Chapter Overview

Governance is the operating discipline that makes an AI system accountable: identifiable owners, traceable decisions, auditable evidence, controllable behaviour. Compliance is the subset of governance that satisfies external obligations: GDPR for personal data, the EU AI Act for risk classification, HIPAA for protected health information, sector-specific rules where they apply. This chapter develops the model inventory, risk classification, model cards, lineage, audit-evidence packages, and the operating cadence that produces governance as a continuous property rather than a periodic exercise.

Learning Objectives

- Maintain a model inventory that survives audit scrutiny.
- Apply the EU AI Act risk tiers and the obligations each carries.
- Produce and maintain model cards and system cards for every production system.
- Operate lineage emission so that every output’s derivation is traceable.
- Run the operating cadence (quarterly reviews, annual revalidations, audit-evidence packages) that keeps governance current.

103.1 Model Inventory

The most basic governance obligation is to know what is in production. A model inventory is the authoritative list of AI systems running on the platform, with their owners, classifications, dependencies, and current state. Without it, governance is impossible at scale because there is nothing to govern over.

103.1.1 Required Fields

A workable model-inventory record:

- Unique identifier; namespacing matches the platform’s resource model.

- Name and short description.
- Owner (a person, named); deputy.
- Business unit; cost centre.
- Layer mapping: which layers of the reference architecture are involved.
- Models used (foundation, fine-tuned, traditional ML); version pins.
- Data sources consumed; sensitivity tiers (Section 92.3).
- Risk classification (EU AI Act tier or local equivalent).
- Lifecycle state: development, pilot, production, deprecated, decommissioned.
- Promotion date; last evaluation date; next scheduled review.
- Approver names and dates for each lifecycle transition.
- Decommission criteria: under what conditions is the system retired or rebuilt.

103.1.2 Registration as a Precondition

The architectural rule that makes the inventory real: registration is a precondition of deployment. The deployment pipeline checks the inventory for an entry matching the artefact being deployed; a missing entry blocks promotion. Periodic reconciliation scans find any production system not in the inventory (discovery of shadow systems).

103.1.3 Inventory Discipline

Owners change; teams reorganise; business units restructure. The inventory needs ongoing maintenance:

- Quarterly owner-confirmation: every owner re-confirms their entries.
- Annual full review: each entry is examined for accuracy and currency.
- Triggered review: on team reorganisation, on incident, on regulatory change.
- Decommissioning workflow: when a system is retired, its inventory entry is marked decommissioned (not deleted) with the date and reason.

103.2 Risk Classification

The EU AI Act establishes four risk tiers; similar tier structures appear in other jurisdictions:

Table 103.1: EU AI Act risk tiers and obligations.

Tier	Examples	Obligations
Prohibited	Social scoring; manipulation of vulnerable groups; untargeted facial recognition scraping	Cannot be deployed in the EU; must be removed from any system intended for EU use
High-risk	Credit decisions; medical diagnostics; employment screening; biometric identification	Conformity assessment; risk management system; data governance; human oversight; technical documentation; logging; transparency; accuracy and robustness; post-market monitoring
Limited-risk	Chatbots; emotion recognition; deepfake generation	Transparency: users informed they are interacting with AI; deepfakes labelled
Minimal-risk	Spam filters; many enterprise productivity uses	No specific obligations under the Act; voluntary codes of conduct apply

The architectural response is to classify every use case at design time and to apply the tier-appropriate controls from the start. Retrofitting high-risk controls is expensive; designing them in is modest in cost.

103.2.1 NIST AI Risk Management Framework

In the United States, the NIST AI Risk Management Framework provides a parallel discipline (voluntary but increasingly referenced). Its four functions:

- **Govern.** Establish policies, roles, accountability.
- **Map.** Identify context, risks, intended use.
- **Measure.** Quantify risks; test trustworthiness characteristics.
- **Manage.** Prioritise, mitigate, monitor risks over time.

Many of the same artefacts (inventory, model cards, evaluation results, monitoring data) satisfy both EU AI Act and NIST AI RMF requirements; the architectural value is to produce these once and have them serve multiple frameworks.

103.2.2 Sector-Specific Rules

Beyond cross-cutting AI regulation, sector-specific rules apply:

- **Financial services.** Bank model risk management (US: SR 11-7); fair lending rules; explainability for credit decisions.
- **Healthcare.** HIPAA; medical device classifications where AI is involved; clinical-validation

requirements.

- **Public sector.** Government-specific AI policies; algorithmic impact assessment requirements.
- **Insurance.** Disparate-impact monitoring; rating-factor regulation.
- **Employment.** Anti-discrimination law; emerging algorithmic hiring rules (e.g., NYC Local Law 144).

103.3 Model Cards and System Cards

A *model card* (Mitchell et al., 2019) is a structured document describing a model. A *system card* extends this to the full deployed system. The combination is the durable, public-facing record of what the system claims and what evidence supports it.

103.3.1 Model Card Contents

A workable model card includes:

- Model details: name, version, type (architecture, parameter count), training data summary, training compute, intended use.
- Quantitative analyses: performance disaggregated by relevant factors (population subgroups, input types, time periods).
- Evaluation data: what test sets were used; how they were sampled; their known limitations.
- Limitations and recommendations: known failure modes; cases out of scope; required oversight.
- Ethical considerations: bias review, fairness analysis, mitigation steps.
- Caveats and recommendations: what users should know.

103.3.2 System Card Contents

The system card adds:

- System architecture: how the model fits into the broader system.
- Data flows: where input comes from; where output goes.
- Safeguards: input filtering, output validation, monitoring, human oversight.
- Operational context: who uses the system; for what; under what authority.
- Incidents and changes: history of significant events and modifications.

103.3.3 Maintained as Code

Model and system cards live in version control alongside the system they describe. Updates are peer-reviewed; releases tag the corresponding card version. The card is not a deliverable for a single launch; it is a living document that the deployment workflow keeps current.

103.4 Lineage

Lineage is the record of how a value was derived. For AI systems, the relevant lineage includes:

- Training-data lineage: which sources fed the training set; which transformations applied; which examples are in or out.
- Model lineage: which base model; which fine-tunes; which hyperparameters; which evaluation results.
- Feature lineage: which source columns produced this feature; which transformations applied.
- Inference lineage: which prompt, which retrieved chunks, which tool calls produced this output.

OpenLineage (Section 95.3) is the emerging standard for emitting lineage events from pipelines and serving systems. The architectural commitment: every component that produces a derived value emits a lineage event with sufficient detail to reconstruct provenance.

103.4.1 Why It Matters

Lineage answers the questions that regulators, auditors, and incident responders ask:

- “Show me how this decision was made.”
- “If this data source is contaminated, what models are affected?”
- “If this model is decommissioned, what downstream systems depend on it?”
- “Was this training set’s composition the same when the model was approved?”
- “Does this output come from a citation, a tool call, or model knowledge?”

Without lineage, these questions cannot be answered confidently; with it, they become queries against the platform.

103.5 Audit-Evidence Packages

For regulated systems, periodic audit is part of the operating rhythm. A well-designed platform produces audit-evidence packages on demand rather than scrambling to assemble them after the request arrives.

103.5.1 Contents

A typical audit-evidence package for a high-risk system:

- Inventory entry as of the audit date.
- Model and system cards.
- Risk classification documentation; conformity assessment record.
- Evaluation results across the audit period: launch baseline, scheduled re-evaluations, drift detections.

- Monitoring data: outcomes monitoring, fairness monitoring, drift signals.
- Incident log: any incidents in the period; root causes; remediation.
- Change log: model changes, prompt changes, configuration changes; who approved each.
- Access log: who accessed sensitive components; what they did.
- Training-data provenance for the model version in use.
- Lineage records linking outputs to inputs and intermediates.

103.5.2 Generated, Not Assembled

The discipline that makes this tractable: the package is generated by the platform from primary sources, not assembled by a person from scattered tools. The platform team's job is to ensure the primary sources exist and to maintain the generation pipeline; the audit-evidence team's job is to interpret the output for the auditor.

103.6 Governance Operating Cadence

Governance is a continuous discipline, not an annual event. A workable operating rhythm:

Table 103.2: Governance operating cadence.

Cadence	Activities
Continuous	Lineage emission; monitoring; evaluation in CI; alert on drift, refusal, fairness deviation
Weekly	Operational review (Section 100.5); incident triage; new-deployment review
Monthly	Inventory reconciliation; eval-set updates; cost and quality trends
Quarterly	Owner confirmation; portfolio review; pattern-library updates; vendor-relationship reviews
Annual (high-risk)	Full revalidation: re-evaluation on current data; fairness audit; red-team exercise; documentation refresh
On change	Pre-deployment review for new models, new vendors, new use cases, or significant changes to existing ones
On incident	Post-mortem; structural fix; eval suite update; threat-model update if security-related

103.7 Tiered Review

A governance function that subjects every change to full scrutiny becomes a bottleneck; one that subjects nothing to scrutiny is theatre. Tiered review balances the two:

- **Expedited.** Routine changes (model version bumps within the same family, prompt refinements, minor configuration changes) that follow established patterns. Approval delegated to the engineering team with audit trail.
- **Standard.** Significant changes (new use cases, new vendors, new data sources, changes to evaluation thresholds) reviewed by the CoE within a documented turnaround time.

- **Full.** High-risk changes (new high-risk classification, significant scope change, change to fairness or safety properties) reviewed by the cross-functional governance committee.

The tiers must be documented and the routing must be automatic; manual routing every time defeats the purpose.

103.8 Roles and Accountability

Clear roles produce clear accountability:

- **System owner.** The named individual accountable for a specific AI system; signs off on changes; responds to incidents.
- **Platform team.** Operates the governance machinery: inventory, lineage, audit-evidence pipeline.
- **Centre of Excellence.** Maintains risk-classification guidance; supports system owners; runs reviews.
- **Risk and compliance.** Provides regulatory interpretation; signs off on high-risk classifications; liaises with regulators.
- **Legal.** Vendor contracts; data-processing agreements; cross-border transfers.
- **Internal audit.** Periodic independent review of governance effectiveness.
- **Executive sponsor.** Accountable for the overall AI programme's compliance posture.

Where roles are unclear, accountability falls through the cracks; the fix is to name the owner before there is an incident, not after.

103.9 The Working Position

Governance is structural, not procedural. A complete model inventory, risk classification applied at design time, model and system cards as living documents, lineage emitted by every component, audit-evidence packages generated from primary sources, a tiered review process, and a continuous operating cadence together produce a system that meets its obligations and survives audit. The cost of building these in is modest; the cost of running without them is paid in incidents, regulator findings, and reputational damage. Compliance is the floor; the architecture's commitment is to make compliant behaviour the default and non-compliant behaviour the deliberate exception.

The AI Governance Operating Model: Ten Domains of Risk and Control

“Governance is not one decision well made. It is ten kinds of risk, each watched by someone whose job it is to watch it.”

—a recurring observation in enterprise risk practice

Chapter Overview

The book has already developed governance from three angles: the decision-making system that sets and applies architectural principles, the external standards landscape an enterprise must answer to, and governance as a cross-cutting concern of the reference architecture. This chapter adds the fourth and most operational angle — the governance operating model: the concrete catalogue of risk domains an AI estate must actively manage day to day. It works through ten domains, each stated in the same three terms: the risk area, the business impact if the risk is unmanaged, and the governance control that manages it. The chapter is deliberately a catalogue rather than an argument. Its value is as a working checklist — the set of domains an architecture-review board, a risk function, or an AI governance committee should be able to account for, by name, for every consequential AI system.

Learning Objectives

- Recognise the ten operational domains an AI governance model must cover.
- For each domain, name the characteristic risk, its business impact, and the control that addresses it.
- Use the ten domains as a review checklist for any consequential AI system.
- Distinguish this operating-model view from the decision-system, standards, and reference-architecture views developed elsewhere.
- Adapt the operating model to the constraints of African deployments.

104.1 The One-Dollar Robbery: Why People Route Around Systems

In June 2011, a fifty-nine-year-old man named Richard Verone walked into an RBC Bank branch in Gastonia, North Carolina, and carried out one of the most unusual robberies in modern history. Calm and polite, he handed a teller a handwritten note claiming he had a gun and demanding exactly one dollar. Instead of fleeing with the money, he sat down in the bank lobby and told the employees he would wait for the police. When officers arrived they found him exactly where he said he would be, seated peacefully and entirely unarmed. Verone later explained that he was suffering from severe medical conditions — a growth on his chest and ruptured spinal discs — but had no job, no insurance, and no realistic access to healthcare. In his calculation, prison was the only system that would reliably provide shelter, food, and medical treatment. By demanding a single dollar he hoped the authorities would understand that money was never the objective; access was. The robbery was simply the mechanism he used to enter a system that worked better than the one meant to support him outside it.

The story mirrors a fundamental challenge in enterprise AI governance. Employees inside organisations create shadow processes, unofficial workflows, spreadsheet-based operations, and adopt unauthorised AI tools not because they wish to violate governance but because the official enterprise systems fail to deliver speed, accessibility, intelligence, or productivity. Just as Verone optimised for access to a functioning system rather than for money, employees optimise for outcomes rather than for compliance with architectures that obstruct them. The rise of shadow AI is therefore less a technology problem and more an architectural signal — a signal that reveals organisational friction, disconnected workflows, inaccessible knowledge, and operational inefficiencies. The deeper lesson for enterprise AI leaders is that governance alone is insufficient. Enterprises must design intelligent systems that genuinely reduce friction, simplify work, and support the people who use them, because otherwise those people will inevitably route around the official architecture in search of systems that actually work.

This chapter's ten governance domains are the controls that must be in place, but the Verone story is the standing reminder that governance must be designed alongside systems people *want* to use, not as a constraint imposed on systems people are already working around. A governance operating model that governs a system nobody uses is governing nothing; a governance operating model that governs a system people prefer to use is governing the organisation's actual behaviour.

104.2 Why an Operating Model, and How It Differs

An enterprise can have an excellent architecture-governance decision system — principles, a review board, decision records — and still be poorly governed, because the decision system answers the question *how do we decide* and not the question *what, specifically, must be watched*. It can hold every relevant external certification and still be poorly governed, because a certification attests that processes exist, not that every operational risk is owned. The governance operating model is the layer that closes that gap. It is the explicit enumeration of the risk domains the AI estate generates, each with a named owner and a defined control.

The operating model differs from the three governance views developed earlier in the book in a way worth stating plainly, so that this chapter is read as a complement rather than a repetition. The architecture-governance chapter describes the *machinery of deciding*. The governance-standards

chapter describes the *external frameworks decided against*. The reference-architecture treatment places governance as a *cross-cutting layer of the system*. This chapter describes the *operational risk surface* — the concrete domains where AI-specific things go wrong and where, therefore, a control must stand. The four views are layers of one subject: a decision system, applied against external standards, realised as an architectural layer, and operated as a catalogue of managed risks. An enterprise needs all four; this chapter supplies the fourth.

The structure that follows is uniform by design. Each of the ten domains is presented in three terms — **risk area** (what can go wrong), **business impact** (what it costs the enterprise if unmanaged), and **governance control** (the mechanism that manages it). The uniformity is the point: it is what makes the ten domains usable as a checklist, because a reviewer can ask the same three questions of every domain and of every system.

104.3 Domain 1: Data Governance

Risk area. The AI system is built on data of poor quality, unclear ownership, or uncontrolled usage — data whose lineage is unknown, whose accuracy is unmeasured, and whose permitted uses are undefined.

Business impact. Inaccurate AI outputs, privacy violations, and weak decision-making. A model is a function of its data, and a model built on ungoverned data produces ungoverned results — confidently, and at scale.

Governance control. Define data ownership, classification, lineage, and approval workflows. Every dataset feeding an AI system has a named owner, a sensitivity classification, a traceable lineage from source to model, and a defined approval path before it may be used. This is the data-and-operations layer of the governance-standards chapter made operational: the controls that chapter named as a layer, here assigned and enforced.

Data governance is placed first because it is foundational in the literal sense — every other domain in this chapter assumes it. Bias testing is only as meaningful as the lineage that lets a team know what the training data was; audit logging is only as useful as the data classification that says what was sensitive. An AI governance model whose data-governance domain is weak is a model with a weak foundation, and the nine domains above it inherit the weakness.

A historical case sharpens the point. During a deadly cholera outbreak in London in 1854, the prevailing medical establishment believed the disease spread through “bad air” — a theory supported by convention, authority, and the absence of any better explanation. Physician John Snow rejected the assumption and instead used data mapping and direct observation to trace the infections back to a contaminated water pump on Broad Street. By persuading the authorities to remove the pump handle, he slowed the outbreak dramatically. Snow’s contribution was not a new treatment but a new way of *looking at the data*: he made the invisible visible by mapping what others had left unmapped. The analogy to AI data governance is direct. Organisations that govern their data — tracing its lineage, measuring its quality, understanding its distribution — can see the truths that ungoverned data hides, just as Snow saw the truth that the “bad air” theory concealed. A system built on data whose lineage is

unknown is a system whose failures are, like London's cholera, attributed to the wrong cause.

104.4 Domain 2: Model Validation

Risk area. Models are deployed without proper testing, evaluation, or approval — promoted to production on the strength of a demonstration or a developer's confidence rather than measured evidence.

Business impact. Unreliable predictions, operational errors, and reduced stakeholder trust. A model that was never properly validated is a model whose failures are discovered by its users.

Governance control. Use validation benchmarks, performance thresholds, and independent review. Every model crossing into production passes a defined evaluation against a representative benchmark, meets pre-agreed performance thresholds, and is reviewed by someone other than the team that built it. The independence matters: a team validating its own model is subject to exactly the build-bias the buy-versus-build chapter described.

This domain operationalises the evaluation discipline the observability-and-evaluation chapter developed: the evaluation harness is the instrument, and model validation is the governance domain that makes passing it a condition of deployment rather than an optional courtesy.

A cautionary lesson comes from the colonial-era “cobra effect”. During British rule in India, officials in Delhi sought to reduce the dangerous cobra population by offering a bounty for every dead cobra delivered to the authorities. The policy appeared to succeed at first, but people quickly began breeding cobras for the sole purpose of killing them and collecting the reward. When the government discovered the manipulation and cancelled the programme, the now-worthless captive cobras were released into the wild, making the snake problem worse than before. The parallel to AI governance is precise: when the metric an organisation measures becomes the metric people optimise for, the governance can produce the opposite of what it intended. A model-validation framework in which teams learn to game the benchmark — tuning for the evaluation set rather than for production quality — reproduces the cobra effect in miniature. The validation must therefore measure what actually matters, on data that is representative of production, and must be audited for gaming, because a metric that becomes a target ceases to be a useful metric.

104.5 Domain 3: Bias and Fairness

Risk area. AI systems produce unfair, discriminatory, or unbalanced outcomes — treating groups differently in ways that are not justified by any legitimate factor.

Business impact. Reputational damage, legal exposure, and erosion of customer trust. A system found to discriminate carries consequences that extend well beyond the system itself, to the standing of the organisation that deployed it.

Governance control. Run bias testing, fairness audits, and representative dataset reviews. Fairness metrics are computed — per group, against a chosen and documented definition of fairness — as a standing part of evaluation; the datasets feeding the model are reviewed for representativeness; and

the results are audited rather than self-reported.

The ethics-and-responsible-AI chapter developed the technical substance here — the incompatible formal definitions of fairness, the measurement metrics, the disparate-impact monitoring. This domain is the governance commitment that the technical work is actually done, owned, and audited, rather than left to whichever team has time for it.

104.6 Domain 4: Human Oversight

Risk area. AI systems operate without sufficient human review for high-stakes decisions — the automation runs and nobody watches.

Business impact. Uncorrected errors in consequential decisions, customer harm, and regulatory violations. A high-stakes decision taken by a model and reviewed by nobody is a decision the organisation cannot defend.

Governance control. Establish human-in-the-loop approval, escalation paths, and override controls. The human-in-the-loop chapter developed the tiering — approve for high stakes, audit for medium, sample for low — and this domain makes the tiering a governance requirement: for every consequential AI system, the human-oversight posture is defined, staffed, and monitored for effectiveness.

History offers two lessons here, working from opposite directions. The Luddites of early nineteenth-century England destroyed mechanised textile equipment not, as the popular caricature has it, because they were simply against technology, but because they were protesting how the technology was being deployed — without fairness, without transition planning, without social protection for the workers it displaced. The lesson for AI governance is that resistance to AI inside an organisation is often less about the technology itself and more about trust, communication, transparency, and the fear of displacement. Human oversight is the governance domain that addresses those fears structurally: it is the organisation's standing commitment that a human is watching, that escalation is possible, and that the automation does not simply replace judgement with speed.

From the other direction, the Jacquard loom of the early 1800s used punched cards to automate textile patterns, allowing a machine to follow programmable instructions — an early precursor to computing and software logic. Workers feared the loom because it transformed skill structures and centralised productivity in the machine rather than in the craftsperson. But the Jacquard loom did not eliminate the need for human involvement; it changed what that involvement looked like. The parallel to AI is direct: programmable systems have always reshaped labour, governance, and organisational power. Human oversight in the AI context is not the refusal to automate — it is the deliberate design of where, in an automated system, the human still stands, and the governance commitment to keep them standing there.

104.7 Domain 5: Regulatory Compliance

Risk area. AI use cases violate applicable regulations — data-protection law, sector-specific rules, the emerging AI-specific legislation — whether through ignorance or through a gap in the compliance

mapping.

Business impact. Fines, enforcement actions, operational restrictions, and the reputational cost that follows a regulatory finding.

Governance control. Map use cases to regulations and maintain compliance evidence. Every AI use case is mapped against the regulations that apply to it — the data-protection regimes, the sector rules, the emerging AI-specific law — and the compliance evidence is maintained as the use case runs, not reconstructed when an auditor asks for it. The compliance-architecture chapter developed the evidence-production machinery; this domain is the governance commitment that the machinery is running and the mapping is current.

104.8 Domain 6: Security Controls

Risk area. AI systems are exposed to adversarial attacks, data breaches, or misuse — the AI-specific attack surfaces the security chapter developed.

Business impact. Data exposure, system compromise, operational disruption, and the regulatory consequences that follow a breach.

Governance control. Apply access control, input validation, encryption, and security testing. The security-architecture chapter developed the layered defences across the model, data, prompt, tool, supply-chain, and agent surfaces; this domain is the governance commitment that those defences are deployed, tested, and maintained, and that the red-teaming chapter's programme is running against them.

The parallel here is the telegraph revolution of the nineteenth century. Before the telegraph, information moved at the speed of horses and ships. The telegraph suddenly allowed communication across enormous distances almost instantly, transforming finance, warfare, journalism, and governance in ways that the pre-telegraph world could not have anticipated. But it also introduced misinformation, panic trading, and coordination risks at a speed that outran the existing safeguards. Generative AI parallels the telegraph precisely: it dramatically accelerates knowledge movement and decision-making, but it also amplifies errors, misinformation, hallucinations, and operational risks at a scale and speed the pre-AI security posture was not built for. Security controls for AI are, in this sense, the modern equivalent of the institutional safeguards that the telegraph eventually forced — the controls that the speed of the new technology demands and the speed of the old world never needed.

104.9 Domain 7: Vendor Risk

Risk area. Third-party AI dependencies introduce financial, operational, and compliance risks — a vendor raises prices, changes terms, deprecates a model, suffers a breach, or goes out of business.

Business impact. Service disruptions, cost overruns, data-handling incidents, and the loss of capability when a vendor fails.

Governance control. Run vendor assessment, contract review, and continuous monitoring. The procurement chapter developed the vendor-sustainability assessment, the contract terms, and the multi-vendor strategy; this domain is the governance commitment that the assessment is done before a vendor enters the estate, the contract terms are enforced, and the vendor's health and performance are monitored continuously rather than assumed. Vendor risk, unlike many of the other domains, is a risk that originates outside the organisation, which is exactly why the governance must be proactive: the organisation cannot control the vendor, only its own preparedness for what the vendor might do.

104.10 Domain 8: Audit Logging

Risk area. AI actions, decisions, and data handling are not logged or are logged insufficiently, so that an auditor, a regulator, or an internal investigator cannot reconstruct what happened.

Business impact. Inability to investigate incidents, demonstrate compliance, or defend decisions — the organisation cannot prove what its AI did, even to itself.

Governance control. Produce durable, tamper-evident, attributable logs of actions, prompts, outputs, and decisions. The compliance-architecture chapter developed the evidence vault and the five evidence streams; this domain is the governance commitment that the logging is running, that it covers the full production surface, and that the logs are tamper-evident and retained for the period the organisation's obligations require. Audit logging is the domain that serves every other domain: incident response needs the logs to investigate, compliance needs them to demonstrate, security needs them to detect, and fairness needs them to audit. A governance model whose audit-logging domain is weak is a model that cannot prove its other domains are working.

104.11 Domain 9: Model Monitoring

Risk area. Model performance degrades over time — through distribution drift, concept drift, capability gaps revealed by production traffic, or vendor model updates — and the degradation is not detected.

Business impact. Degraded accuracy, inconsistent results, and business-process failures — all arriving silently, because a drifting model does not announce its decline.

Governance control. Track performance, drift, and anomalies, and trigger retraining workflows. The model's quality is monitored continuously in production; input, output, and concept drift are watched with the methods the observability chapter set out; and a defined workflow turns a confirmed drift into a retraining or remediation action. This domain is what distinguishes model validation, which is a gate crossed once, from the ongoing assurance that the model is *still* valid.

The failure to act on what monitoring reveals is itself a governance failure, and history has an instructive case. For decades in the eighteenth century, sailors in the British Navy died of scurvy at appalling rates despite accumulating evidence that citrus fruits could prevent the disease. The evidence existed; the knowledge was available; individual officers had seen it work. But bureaucracy, institutional

resistance, and slow organisational learning delayed the adoption of the solution by generations. Sailors continued to die of a preventable condition because the institution could not operationalise knowledge it already possessed. The parallel to AI governance is precise: many organisations today monitor their AI systems, see the drift signals, possess the data to improve their decisions, and still fail to act, because cultural resistance, siloed governance, or the absence of a defined retraining workflow prevents the knowledge from becoming an operational change. Model monitoring without a defined response is the eighteenth-century navy: watching the problem, recording the problem, and not fixing the problem.

104.12 Domain 10: Incident Response

Risk area. AI failures, misuse, or security events are not handled quickly — the organisation meets an AI incident with improvisation rather than a plan.

Business impact. Extended disruption, legal exposure, and loss of enterprise trust. An incident handled slowly and badly costs far more than the incident itself.

Governance control. Create AI incident playbooks, escalation teams, and recovery procedures. The AI-specific incident scenarios the security chapter named — prompt-injection compromise, harmful output, retrieval-store leakage, supply-chain compromise — each have a written, exercised playbook; a defined team is responsible for responding; and recovery procedures, including the rollback paths the agentic-method chapter insisted on, are prepared in advance. This domain closes the catalogue because it is the one that assumes the others will sometimes fail: even a well-governed estate has incidents, and the measure of its governance is partly how well it has prepared for them.

The Franklin Expedition of 1845 offers a stark reminder of what happens when preparation meets overconfidence. The British expedition, led by Sir John Franklin, set out to discover the Northwest Passage with two of the most advanced ships and the most sophisticated equipment of their era. The mission ended in catastrophe: both ships were trapped in Arctic ice, and every member of the expedition perished. Historians have since concluded that overconfidence in the technology, rigid command structures, and an inability to adapt to changing realities contributed heavily to the disaster. The parallel to enterprise AI is sobering: organisations that trust their advanced platforms without preparing for the unexpected — without incident playbooks, without tested recovery procedures, without the organisational flexibility to adapt when conditions change — are, like the Franklin Expedition, confident right up to the point where conditions exceed their preparation. Incident response is the governance domain that says: assume conditions will exceed your preparation, and prepare for that too.

104.13 The Hype Trap: Tulip Mania and the AI Bubble

The ten domains above are the operational controls, but they must be set against a broader context that the domains alone do not address: the risk of building an AI programme driven by hype rather than by value. The Dutch tulip mania of the 1630s is the historical archetype. Tulip bulbs became so valuable that people sold homes, businesses, and life savings to speculate on flower contracts, driven

less by the actual value of tulips than by collective excitement, fear of missing out, and irrational optimism. When the bubble collapsed, the devastation was proportional to the irrationality that had built it.

The analogy to enterprise AI is uncomfortably apt. Many organisations today risk building internal AI bubbles — investing in hype-driven pilots, disconnected proofs of concept, and fashionable technologies without the operational foundations, the measurable outcomes, or the governance to sustain them. The governance operating model is, among other things, a defence against the tulip trap: a programme that can account for all ten domains — with named owners, defined controls, and measured performance — is a programme grounded in operational reality. A programme that cannot is a programme that may be building on hype, and the ten-domain checklist is one of the clearest ways to tell the difference.

A related caution comes from the partly-mythologised NASA pen story. The popular version claims that NASA spent millions developing a pen that could write in zero gravity while Soviet cosmonauts simply used pencils. Although the reality is more nuanced — the Fisher Space Pen was privately developed and eventually adopted by both programmes — the story endures because it captures a truth about overengineering. Organisations often build massive AI architectures, elaborate governance committees, and expensive platforms when a simpler workflow improvement or an operational fix would solve the real problem more effectively. The governance operating model should be applied with the NASA-pen caution in mind: the purpose is to govern actual risk, not to build governance machinery for its own sake, and the lightest governance that genuinely controls the risk is better than an elaborate governance that nobody follows.

104.14 Using the Ten Domains

The ten domains are most useful applied as a standing checklist. For any consequential AI system, an architecture-review board or AI governance committee should be able to walk the ten domains in turn and answer, for each, the same three questions: what is the risk here for this system, what would it cost the business if unmanaged, and what specific control — with a named owner — manages it. A system for which all ten domains can be answered confidently is a system whose operational risk surface has been mapped. A system for which two or three domains produce hesitation has just had its governance gaps identified, which is precisely what the checklist is for.

Table 104.1: The ten governance domains as a review checklist.

#	Domain	The control, in one line
1	Data Governance	Ownership, classification, lineage, approval workflows for every dataset.
2	Model Validation	Benchmarks, thresholds, and independent review before production.
3	Bias and Fairness	Bias testing, fairness audits, representative-dataset review.
4	Human Oversight	Human-in-the-loop approval, escalation paths, override controls.
5	Regulatory Compliance	Use cases mapped to regulations and compliance evidence.
6	Security Controls	Access control, input validation, encryption, security testing.
7	Vendor Risk	Vendor assessment, contract review, continuous monitoring.
8	Audit Logging	Durable, attributable logs of actions, prompts, outputs, decisions.
9	Model Monitoring	Performance, drift, and anomaly tracking with retraining triggers.
10	Incident Response	Playbooks, escalation teams, recovery procedures.

The ten are not independent, and the connections are worth seeing. Data governance underpins bias testing, audit logging, and model monitoring — all three depend on knowing what the data was. Model validation and model monitoring are the same concern at two points in time: validation is the gate, monitoring is the ongoing assurance. Security controls and incident response are paired — one prevents, the other handles what prevention misses. Regulatory compliance and audit logging reinforce each other, since compliance must be demonstrated and the logs are the evidence. An operating model that treats the ten as a connected system, rather than ten boxes to tick, is one whose controls reinforce each other; the checklist is the entry point, not the whole of the discipline.

104.15 The African Perspective

The ten domains apply unchanged in African deployments — risk is risk — but two features of the operating model deserve specific attention.

The first is ownership under talent scarcity. Each of the ten domains needs an owner: someone accountable for the control. In a large, well-resourced organisation those can be ten different people or teams; in the leaner organisations common across African markets, one person may own several domains, and the risk is not that the domains go unlisted but that an over-stretched owner watches some of their domains attentively and others nominally. The operating model's value here is precisely its explicitness: it makes visible how many domains each owner actually carries, which turns an invisible over-stretch into a stated, manageable fact. The proportionality discipline of the architecture-governance chapter applies — a leaner operating model, with domains deliberately consolidated and consciously prioritised, is sounder than an elaborate one that exists on paper and is staffed by no one.

The second is the regulatory-compliance and vendor-risk domains specifically. Regulatory compliance, domain five, must be mapped against the divergent and fast-moving regulatory landscape the governance-standards chapter described — the per-jurisdiction data-protection regimes, the emerging AI-specific law — which means the compliance mapping for a pan-African operation is not one mapping but several, one per market. Vendor risk, domain seven, carries the foreign-exchange and

data-residency weight the cost and sovereignty discussions raised: a third-party AI dependency priced in a foreign currency and hosted in a distant region is a vendor risk in a fuller sense than the same dependency would be in the vendor's home market. The operating model does not change for African deployments; the inputs to two of its domains simply carry constraints that a deployment in the vendor's own jurisdiction would not face.

104.16 What the Operating Model Provides

The AI governance operating model is the catalogue of operational risk domains an AI estate must actively manage: data governance, model validation, bias and fairness, human oversight, regulatory compliance, security controls, vendor risk, audit logging, model monitoring, and incident response. Each is stated in three terms — risk area, business impact, governance control — and the ten together form a checklist an architecture-review board or governance committee can apply, by name, to any consequential AI system.

The chapter opened with Richard Verone's one-dollar robbery, and the lesson it carries runs through the whole operating model: governance must be designed alongside systems people want to use, because people who are obstructed by the official architecture will route around it, and a governance model that governs a system nobody uses is governing nothing. The cobra effect warns that governance metrics can produce the opposite of their intent if the metrics are gamed. The cholera mapping reminds that data governance is the foundation — the ability to see what ungoverned data hides. The Luddites and the Jacquard loom together show that human oversight is the governance domain that addresses both the fear of displacement and the reality that programmable systems always reshape labour. The telegraph shows why security controls must match the speed of the new technology. The scurvy crisis warns that monitoring without a defined response is watching the problem without fixing it. The Franklin Expedition warns that confidence in technology is not a substitute for preparation. And the tulip mania warns that an AI programme built on hype rather than on value will collapse like the bulb market.

The operating model is the fourth and most operational of the book's governance views, and the epigraph states its premise: governance is not one decision well made but ten kinds of risk, each watched by someone whose job it is to watch it. The decision-system view supplies the machinery of deciding; the standards view supplies the external frameworks; the reference-architecture view supplies the cross-cutting layer; and this operating-model view supplies the named, owned, controlled risk surface those three are ultimately exercised upon. An enterprise that can produce, for every consequential AI system, a confident account of all ten domains has done more than satisfy a checklist — it has demonstrated that its AI estate is not merely built and deployed but genuinely governed, in the only sense that matters operationally: that every kind of thing that can go wrong has someone whose job it is to see it coming.

Part VIII

Case Studies, Maturity, and Futures

“It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest.”

—Adam Smith, *The Wealth of Nations*, 1776

The architectural work has been described. The question now is what real organisations look like when they try to do it. Adam Smith’s *Wealth of Nations* was, in its way, a case-study book. It opened with the pin factory, returned repeatedly to specific industries and specific institutions, and only on the basis of those cases attempted to draw general principles about how an economy organises itself. Part VIII of this book takes a similar approach. The maturity model, the case studies, the future directions, and the anti-patterns are offered not as abstract theory but as observations drawn from organisations that have actually tried to build enterprise AI and have learned, sometimes painfully, what worked and what did not. The discipline, like Smith’s, is to look at what real institutions do and let the patterns emerge.

A Maturity Model for Enterprise AI

“Maturity is not a destination; it is a direction.”

—attributed variously

Chapter Overview

A maturity model gives a programme a coherent way to ask “where are we” and “what do we do next.” This chapter develops a five-level maturity model for enterprise AI across six dimensions: strategy, platform, governance, talent, outcomes, and culture. The model is not a ranking; it is a navigational aid. Most organisations occupy different levels across different dimensions and progress unevenly. Recognising the asymmetry is the first step to addressing it. The chapter also develops the typical movement patterns between levels, the anti-patterns associated with each, and the structural moves that produce real progression rather than reported progression.

Learning Objectives

- Place an organisation on the maturity model across six dimensions.
- Recognise the typical anti-patterns at each level and the structural moves to the next.
- Use the model to communicate progress and gaps to non-technical stakeholders.
- Plan investment to advance the weakest dimension rather than the strongest.
- Avoid the recurring abuses of maturity models (inflated self-assessment, gamification, complacency).

105.1 The Five Levels

Table 105.1: Five-level maturity model.

Level	Name	Description
1	Ad-hoc	Isolated experiments. No platform, no governance, no shared learning.
2	Pilot	Multiple use cases attempted; each builds its own stack; outcomes variable.
3	Repeatable	A platform exists; reference patterns are documented; some governance applied; outcomes more consistent.
4	Scaled	Most use cases ship on the paved path; governance is operational; outcomes are measured; programme self-funds.
5	Strategic	AI is a property of the organisation, not a programme; capabilities differentiate competitive position.

Most enterprises in 2026 occupy Levels 2 or 3 in most dimensions. Movement is measured in years, not quarters. An organisation reporting Level 4 across all dimensions within twelve months of starting its programme is almost certainly reporting aspiration rather than reality.

105.2 The Six Dimensions

The five levels apply to six independent dimensions. The dimensions are independent in the sense that an organisation can be at Level 4 in one and Level 2 in another; they are not independent in the sense that progress in one tends to enable progress in the others.

105.2.1 Strategy

How clearly the organisation knows what it wants from AI:

- L1: AI is a vague aspiration in corporate strategy; no specific bets.
- L2: Specific use cases are funded; the portfolio is opportunistic, often driven by what is currently fashionable.
- L3: A coherent portfolio with stated bets; clear ROI hypotheses; documented criteria for adding and dropping use cases.
- L4: AI investment is part of every business unit's planning; trade-offs against other investments are explicit.
- L5: AI reshapes the business strategy; new lines of business depend on it; competitors notice.

105.2.2 Platform

The maturity of the shared infrastructure:

- L1: No platform; each project picks its own stack.
- L2: Shared services start to emerge but adoption is uneven; many projects bypass the early platform.
- L3: A platform exists with model gateway, observability, IAM; most projects use it; documentation

is real.

- L4: The platform is the paved path; deviating from it requires justification; new use cases ship in weeks rather than months.
- L5: The platform itself is a strategic asset; competitors do not have it; its value compounds.

105.2.3 Governance

How accountable the system is:

- L1: No inventory; no risk classification; nobody could pass an audit.
- L2: Inventory exists for some systems; review is ad-hoc; documentation is reactive.
- L3: Inventory complete; review workflow established; model cards required; lineage emitted.
- L4: Governance is continuous; monitoring is automatic; audits are routine and produce few findings.
- L5: Governance is a platform feature; new systems inherit it by default; the cost of compliance per system approaches zero.

105.2.4 Talent

The strength of the people building the system:

- L1: A handful of enthusiasts; no formal roles; capability is one departure away from collapse.
- L2: Roles defined; hiring underway; significant skill gaps; competitive labour market.
- L3: Multidisciplinary teams with formal capability frameworks; ongoing training programme; certified internal experts.
- L4: Talent flows between teams; internal mobility supports growth; retention is competitive.
- L5: The organisation is a recognised destination for AI talent; recruitment is a competitive advantage; alumni accelerate the broader ecosystem.

105.2.5 Outcomes

What the programme produces:

- L1: Demos; no production deployments.
- L2: Production deployments; outcomes anecdotal; ROI claims are illustrative.
- L3: Outcomes measured against pre-stated hypotheses; mixed track record; learning is captured.
- L4: Programme self-funds; net positive ROI across the portfolio; underperformers are decommissioned.
- L5: AI is materially shifting the financial profile of the organisation; line items grow that did not exist before.

105.2.6 Culture

How the organisation thinks about AI:

- L1: AI is a buzzword; senior leadership is uncertain; mid-management is sceptical or fearful.
- L2: AI is a project the organisation does; some teams are enthusiastic, others wait it out.
- L3: AI is a capability; teams understand what it can and cannot do; trust is calibrated.
- L4: AI is a tool; users incorporate it as part of how work gets done; routine decisions include it.
- L5: AI is invisible; it is part of the operating fabric; people would notice only if it were taken away.

105.3 Maturity Profiles

Most organisations do not score uniformly across the dimensions. Three common profiles:

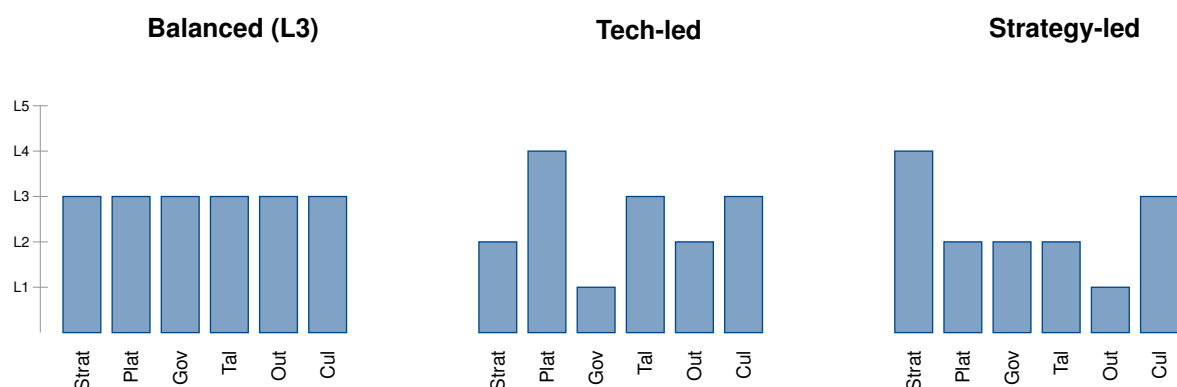


Figure 105.1: Three common maturity profiles. Balanced is rare; tech-led and strategy-led are common. The weakest dimension limits the overall outcome.

- **Balanced.** Equal scores across dimensions. Rare in practice; most organisations have at least one outlier.
- **Tech-led.** High platform and talent scores, low strategy and governance scores. The organisation can build but cannot align or govern. Common in engineering-heavy companies.
- **Strategy-led.** High strategy score, low platform and outcome scores. The vision is articulated; the execution is missing. Common in companies whose leadership reads the trade press.

The lesson is the same regardless of profile: the weakest dimension limits the overall outcome.

105.4 Using the Model

The model has three practical uses.

105.4.1 Honest Assessment

A facilitated workshop where each dimension is placed independently, with evidence required for each placement, typically produces a more accurate picture than any single overall rating. The discipline is

to require evidence: not “we feel we are at Level 3” but “we are at Level 3 because A, B, and C are in place, and we would be at Level 4 if D and E were also in place.”

105.4.2 Sequencing Investment

The weakest dimension is usually the one limiting progress overall. Investment there has higher leverage than investment in the strongest dimension. A tech-led organisation should over-invest in governance and strategy; a strategy-led organisation should over-invest in platform and outcomes.

105.4.3 Stakeholder Communication

A maturity matrix is a useful conversation with executives who want to know progress without diving into technical detail. The matrix communicates trajectory: we were at L2 across the board last year; we are now at L3 in platform and governance; we plan to move strategy and outcomes from L2 to L3 over the next year.

105.5 Anti-Patterns

Table 105.2: Maturity-model anti-patterns.

Anti-pattern	Mitigation
Inflated self-assessment	Require evidence for each placement; cross-check with an external party.
Gamification (chasing the number)	The level is a description, not a target; promotion to the next level requires sustained evidence, not a checklist tick.
Complacency at L4	L4 is sustainable only with continued investment; the organisation reverts to L3 if maintenance stops.
Strategy at L5, platform at L1	A vision without the platform to realise it; align strategy ambition to platform reality.
Platform at L4, governance at L1	Sophisticated infrastructure without oversight; governance is a feature, ship it.
Talent at L4, outcomes at L1	Excellent practitioners without the platform or strategy to ship; redirect to the dimension that is blocking.
“We are at L5”	Probably L3 with marketing. Genuine L5 organisations rarely claim it.

105.6 Movement Patterns

Real progression has typical shapes:

- **L1 to L2:** Take any successful experiment and put it in front of users. The transition is mostly courage and operational discipline.
- **L2 to L3:** Build the platform and the patterns. Stop letting each project rebuild the basics. The transition is mostly investment and standardisation.
- **L3 to L4:** Make the platform the paved path; make governance continuous. The transition is mostly cultural; the organisation stops tolerating off-platform projects.
- **L4 to L5:** AI becomes part of the strategy, not a project alongside it. The transition is mostly

leadership; the executives stop asking “what is our AI strategy” and start asking “what is our strategy, of which AI is part?”

Each transition typically takes 12–24 months for a well-funded programme. Skipping levels is rare and usually superficial: the organisation has bought the L5 vocabulary without doing the L3 and L4 work.

105.7 Re-Assessment

The placement is perishable. Re-assess annually, or after any major event (a new platform release; a regulatory change; a significant incident; a leadership change). Track movement over time, not just the current snapshot.

The honest re-assessment surfaces:

- Dimensions where progress has stalled despite continued investment.
- Dimensions that have regressed due to attrition or organisational change.
- Hidden L4 capabilities that the organisation has not yet recognised.
- Aspirations that have not converted into capability.

105.8 A Note Before Moving On

The maturity model is a map, not a destination. Five levels, six dimensions, measured honestly with evidence. Progress is uneven by design; the structural move is to invest in the weakest dimension that is blocking overall progress. Most enterprises are at L2 or L3 in 2026; the architecture in this book targets L4. Reaching L5 requires more than architecture, but it is impossible without it.

Illustrative Case Studies

“Theory without practice is sterile; practice without theory is blind.”

—attributed variously, including to Kant

Chapter Overview

This chapter develops four illustrative case studies of enterprise AI architectures across different industries: a financial-services credit-decision support system, a healthcare clinical-documentation assistant, a manufacturing predictive-maintenance platform, and a public-sector citizen-services chatbot. Each case applies the reference architecture and patterns developed earlier to a specific context, highlighting which choices were forced by the domain, which were preferences, and what was learned that altered the next iteration. The cases are composite—drawn from common practice rather than any single deployment—but the architectural lessons are real.

Learning Objectives

- See the reference architecture applied to four different industries.
- Recognise which architectural choices are domain-forced and which are preferences.
- Identify the recurring lessons across cases.
- Distinguish surface differences from deep similarities in industry-specific AI architectures.

106.1 Case 1: Credit-Decision Support in Financial Services

Context. A retail bank wants to augment underwriters with an AI assistant that reviews loan applications, flags inconsistencies, and drafts decision rationales. The system is classified as high-risk under EU AI Act terms (consequential decisions about access to credit). Volume: approximately 8,000 applications per day across consumer and small-business lending.

106.1.1 Drivers and Constraints

The credit-decision case is shaped by an unusually tight regulatory environment and a deep incumbent data estate, both of which constrain the architecture before any model selection begins.

- Regulatory: high-risk classification triggers conformity assessment, human oversight, robustness, accuracy, and post-market monitoring obligations.

- **Operational:** human underwriters remain in the loop for all approvals; the assistant draft accelerates them rather than replacing them.
- **Quality:** every recommendation must be grounded in the applicant's documents and the bank's lending policy; ungrounded assertions are unacceptable.
- **Data sovereignty:** customer data must remain in-region; vendor APIs that process data extraterritorially are not usable.

106.1.2 Architectural Choices

The architecture balances the interpretability the regulator demands against the capability the business needs, using a layered approach in which each component is chosen for a specific reason.

- **Pattern.** Hub-and-spoke with hexagonal core. The credit-risk function is the spoke; the platform team operates the hub. The hexagonal pattern allows the LLM port to point at either a vendor in-region or a self-hosted alternative as residency requirements evolve.
- **Models.** A small classifier (inconsistency detection, rule-applicability) plus a mid-tier LLM for rationale drafting. No frontier model: the cost premium did not improve outcomes in evaluation.
- **Knowledge.** RAG over the bank's lending policy library and the borrower's documents. Hybrid retrieval (vector + BM25), cross-encoder re-ranking, every claim cited.
- **Oversight.** Human-in-the-loop. The model recommends; the underwriter decides. Every decision is logged with the model's recommendation and the underwriter's final action, supporting outcome analysis.
- **Governance.** Full conformity assessment, model card, system card. Monthly bias monitoring across protected attributes (age, gender, postal-code proxies). Annual revalidation. Audit-evidence package produced quarterly.
- **Observability.** Citation accuracy and underwriter-acceptance rate tracked alongside conventional latency and error metrics; alerts on either dropping below threshold.

106.1.3 Lessons

The credit-decision case yields lessons that recur across regulated-industry AI deployments.

- Citation requirement drove the choice of RAG over fine-tuning; underwriters reject ungrounded recommendations even when correct.
- Mid-tier model was sufficient; frontier did not meaningfully improve outcomes but cost 5× more. This was discovered during evaluation, not assumed.
- Drift monitoring caught a quarter where economic conditions shifted the input distribution; revalidation found minor calibration drift, retrained successfully.
- Outcome monitoring across protected attributes was non-negotiable for regulator acceptance; designing the data flow to support it from launch saved a later retrofit.

- Underwriter-acceptance rate proved the most useful single quality metric; it captured what the underwriters actually thought of the recommendations.

106.2 Case 2: Clinical Documentation Assistant in Healthcare

Context. A hospital group wants to reduce clinician documentation burden with an AI assistant that drafts clinical notes from physician-patient conversations. Volume: 5,000 encounters per day across multiple specialties.

106.2.1 Drivers and Constraints

The clinical-documentation case is shaped by the highest-stakes privacy regime in the enterprise landscape and by the safety criticality of the output.

- **Regulatory:** HIPAA compliance; clinical-software governance (the system is not a medical device but is governed under internal medical-software policy).
- **Privacy:** patient data must remain on-premises; vendor APIs are not permitted for prompt or completion processing.
- **Quality:** every note must be reviewed and signed by the clinician; the system drafts, the clinician approves.
- **Operational:** clinicians have limited tolerance for additional clicks; the system must integrate with the existing EHR.

106.2.2 Architectural Choices

The architecture keeps all patient data within the hospital network and interposes a clinician review before any AI-generated content reaches the medical record.

- **Pattern.** Hexagonal with strict locality. The LLM port is implemented by a self-hosted model in the hospital's data centre. The vector port is on-premises pgvector.
- **Models.** A specialised speech-to-text model for transcription; a fine-tuned LLM for note drafting. Open-weight models (Llama family) self-hosted on the hospital's GPU pool.
- **Knowledge.** The patient's existing record (allergies, problem list, recent results) is retrieved and provided as context. Generic medical guidelines retrieved separately, cited where relevant.
- **Oversight.** Physician reviews and signs every note. The assistant is a drafter, not a decider; the system explicitly does not generate diagnoses.
- **Governance.** HIPAA controls; access logged at the chart level; integration with the EHR's existing audit trail. Periodic clinician-feedback review.
- **Observability.** Time-per-encounter is the headline metric; quality measured by clinician acceptance and edit rate.

106.2.3 Lessons

The clinical-documentation case reinforces the human-in-the-loop and privacy-by-design disciplines.

- Self-hosting added operational complexity but was non-negotiable given the data-classification tier. The hexagonal pattern made the choice tractable: the same application logic could have run on a vendor LLM port if residency rules had permitted.
- Even with strong models, the value was in saving time, not improving note quality. The metric that mattered was minutes-per-encounter; quality was held constant by the clinician review.
- Citation was essential for any guideline-related content; clinicians needed to verify what the model had invoked, particularly when the recommendation seemed unusual.
- Drift in clinician-acceptance rates was the leading indicator of quality changes; weekly tracking caught a degradation that other metrics missed.
- Integration with the EHR took longer than the model work; this is typical and is a planning lesson.

106.3 Case 3: Predictive Maintenance in Manufacturing

Context. A manufacturer with several hundred assets across multiple plants wants to predict failures before they happen, scheduling maintenance to avoid unplanned downtime. Volume: hundreds of millions of sensor readings per day.

106.3.1 Drivers and Constraints

The predictive-maintenance case is shaped by the physical realities of a manufacturing floor: intermittent connectivity, sensor noise, and the need for real-time inference.

- Operational: the value comes from avoiding unplanned downtime; predictions must be early enough to schedule maintenance during planned windows.
- Data: high-volume time-series sensor data from vibration, temperature, pressure, acoustic sensors.
- Risk: limited risk classification (the system affects operations, not people); standard platform controls suffice.
- Heterogeneity: many asset types, each with its own failure modes; the system must scale to new asset types without rebuilding.

106.3.2 Architectural Choices

The architecture splits inference between the edge and the cloud, running time-critical detection locally and deferring deeper analysis to a cloud model.

- **Pattern.** Event-driven agents. Sensor streams from assets feed an event bus; specialised models subscribe to relevant streams and emit prediction events; a maintenance-planning agent subscribes to predictions and emits work-order events.

- **Models.** Traditional ML (gradient boosting) for failure-probability estimation; no LLM in the main loop. An LLM appears only at the user-facing layer, drafting maintenance-work-order text.
- **Knowledge.** Feature store with vibration, temperature, pressure, and acoustic signals at multiple time scales. RAG over manuals and historical work orders for the report-drafting layer.
- **Oversight.** Maintenance planner reviews predicted failures and approves work orders. The system schedules; the planner authorises.
- **Governance.** Lower risk classification; standard platform controls. Cost is dominated by data pipeline and storage, not inference.
- **Observability.** Lead-time-to-failure is the headline metric; false-positive rate is monitored because false positives waste maintenance capacity.

106.3.3 Lessons

The predictive-maintenance case yields lessons about edge deployment, sensor-data discipline, and workflow integration.

- Event-driven decoupling was essential; new asset types onboard by registering a stream and a model, with no changes elsewhere.
- Traditional ML beat LLM-based approaches on every metric that mattered for the prediction task. The instinct to reach for an LLM should be resisted when the problem is well-served by simpler models.
- The LLM at the user layer (drafting work orders) was high-leverage even though it was peripheral to the prediction; small ergonomic improvements aggregated to material time savings.
- Cost was dominated by data pipeline and storage, not inference. FinOps focus was different from LLM-heavy systems.
- False-positive cost was material; the optimal operating point on the precision-recall trade-off was determined by the cost of unnecessary maintenance versus the cost of missed failures.

106.4 Case 4: Citizen Services Chatbot in the Public Sector

Context. A government agency wants to provide a multilingual chatbot that answers citizen queries about services, benefits, and procedures. Volume: tens of thousands of conversations per day.

106.4.1 Drivers and Constraints

The citizen-services case is shaped by the diversity of the population it serves and by the accountability demands of a public-sector deployment.

- **Accuracy:** incorrect information about government services has real consequences for citizens; the system must refuse rather than guess.
- **Multilingual:** the service is offered in multiple official languages; quality must be comparable across

them.

- **Accessibility:** must work for users with assistive technologies, low literacy, and limited bandwidth.
- **Privacy:** citizen interactions are logged; PII handling is strict.
- **Cost:** public budget is finite and politically scrutinised; the cost per interaction must be defensible.

106.4.2 Architectural Choices

The architecture grounds every response in authoritative government content and escalates to a human agent when confidence is low.

- **Pattern.** RAG with strict grounding. Hexagonal core to allow vendor flexibility within residency rules. LLM Router to manage cost.
- **Models.** Open-weight multilingual LLM for generation; specialised classifiers for intent detection and language identification.
- **Knowledge.** The agency's published guidance, regulations, and procedural documents, kept up to date by a dedicated content team. Every response cites the underlying source; users can click through to the original.
- **Oversight.** The system refuses to answer questions it cannot ground; explicit escalation to human agents for complex cases.
- **Governance.** Strong PII redaction in logs; data residency in-country; conformity assessment against the EU AI Act for limited-risk systems; transparency that the user is interacting with AI.
- **Observability.** Citation-coverage, refusal-rate, and escalation-rate tracked alongside conventional metrics; satisfaction surveys at conversation end.

106.4.3 Lessons

The citizen-services case reinforces the RAG-grounding and human-escalation disciplines.

- Grounding was the most important property by far. Citizens (and journalists) test the system with questions designed to trick it; ungrounded responses make news.
- The content team was as important as the engineering team; the system's quality was capped by the quality of the source corpus.
- Multilingual quality required language-specific evaluation, not a translated single-language eval set; some failure modes only appeared in some languages.
- Cost-per-conversation was defensible because the alternative (human-staffed call centres) was much more expensive; the political conversation was easier to win when this comparison was made explicitly.
- Accessibility constraints shaped the UX more than expected; rich-media responses had to be optional, not default.

106.5 Cross-Case Patterns

Across all four cases, several lessons recur:

Table 106.1: Recurring lessons across case studies.

Recurring lesson	Where it appeared
Grounding and citation matter more than model size	All four cases
The cheapest sufficient model is the right model	Cases 1 (mid-tier), 2 (open-weight), 4 (open-weight)
Drift monitoring catches what point-in-time evaluation misses	Cases 1, 2
The hexagonal pattern made vendor and residency choices tractable	Cases 1, 2, 4
Integration cost dominated model work	All four cases
Cost-per-outcome was the most useful unit economic	All four cases
Traditional ML still beats LLMs on many problems	Case 3
Content quality caps system quality	Case 4 (and Case 1's policy library)

106.5.1 Where the Cases Differ

The surface differences are real:

- Financial services: regulation and conformity assessment dominate.
- Healthcare: data residency and clinical governance dominate.
- Manufacturing: scale and decoupling dominate; LLMs are peripheral.
- Public sector: accuracy under scrutiny and accessibility dominate.

But the underlying architecture is the same: six layers, the relevant patterns, the same cross-cutting concerns applied to the relevant risk profile. The reference architecture is industry-neutral; the configuration is industry-specific.

106.6 The Takeaway

The reference architecture and patterns of this book scale across industries, with the same components recombined to fit different constraints. Domain-forced choices (residency for healthcare, citation for finance and public sector, decoupling for manufacturing) determine the boundaries; preferences fill in

the rest. The lessons that recur—grounding, oversight, drift monitoring, the cheapest sufficient model, integration dominating model work—are the lessons that hold across the book.

Chapter 107

Anti-Patterns

“If you do not know the failure modes, you are them.”

—a recurring reliability proverb

Chapter Overview

For every pattern that recurs as a structural solution, there is an anti-pattern that recurs as a structural mistake. This chapter catalogues fifteen anti-patterns specific to enterprise AI: structural choices that look reasonable in isolation, produce predictable failures over time, and consume a disproportionate share of remediation effort once entrenched. Each is named, described, diagnosed, and matched to its structural fix.

Learning Objectives

- Recognise the recurring anti-patterns in enterprise AI systems before adopting them.
- Diagnose existing systems against the list and prioritise remediation.
- Apply the structural fixes; band-aids over anti-patterns rarely hold.
- Communicate the anti-patterns by name; named anti-patterns are easier to negotiate away.

107.1 Architecture Anti-Patterns

107.1.1 The Monolithic AI Service

A single service that ingests, embeds, indexes, retrieves, generates, and serves all in one codebase. Looks tidy from outside; impossible to evolve from inside. *Symptoms*: every change touches every component; ownership is unclear; the team is afraid to deploy. *Fix*: decompose along the layer boundaries of Chapter 8; introduce ports between layers (Chapter 26).

107.1.2 The Hard-Coded Vendor

Direct calls to a specific vendor’s SDK throughout the application. Changing vendors is a project, not a configuration. *Symptoms*: estimates for vendor evaluation start at six months; alternative providers cannot be tested without a code freeze. *Fix*: the hexagonal pattern; every vendor call goes through a port, every adapter is one of several.

107.1.3 The Lone Vector Store

A single vector database serving every team, with no tenancy, no access control beyond “the application,” and no eviction policy. *Symptoms*: cross-tenant retrieval bugs; PII from one team’s docs appearing in another team’s results; uncontrolled growth. *Fix*: tenant-scoped indexes or strict metadata filtering with audit; retention policies; separate stores where regulatory pressure requires.

107.1.4 The Frontier-Only Fallacy

Every request goes to the most expensive model, regardless of difficulty. *Symptoms*: cost grows linearly with traffic; the unit economics never improve. *Fix*: LLM Router pattern (Chapter 23); route the easy 80% to smaller, cheaper models.

107.1.5 The Prompt Spaghetti

Prompts copy-pasted across the codebase, with subtle differences accumulated over time. A change in one place does not propagate; quality drifts unpredictably. *Symptoms*: “which prompt is actually in production” is hard to answer; eval regressions appear without apparent cause. *Fix*: versioned prompt library; prompts as deployable artefacts in source control; prompt-level CI gates.

107.2 Data Anti-Patterns

107.2.1 The Schemaless Lake

Raw data dumped into object storage with no schema, no documentation, no contracts. Downstream consumers each interpret the data differently. *Symptoms*: “what does this field mean” is a recurring question with multiple answers; pipelines break on undocumented changes. *Fix*: data contracts (Section 10.6); medallion architecture with schemas at silver and gold (Section 11.4).

107.2.2 The Snapshot Trap

Training datasets recreated from production state at training time, rather than constructed as point-in-time correct. Features visible at training that would not be visible at inference. *Symptoms*: model outperforms in training, underperforms in production. *Fix*: feature store with point-in-time joins (Section 11.6); explicit timestamp on every event.

107.2.3 The Quiet PII Leak

Personal data flowing into prompts, logs, vector stores, and fine-tunes without classification or redaction. *Symptoms*: discovery during a privacy review or, worse, a subject-access request. *Fix*: classification at intake (Section 92.3); redaction in logs; explicit fields-in-prompts policy.

107.3 Operational Anti-Patterns

107.3.1 Evaluation Theatre

Evaluation results presented at launch but never re-run; benchmark numbers detached from production behaviour. *Symptoms*: the system’s metrics at launch do not predict its metrics now; nobody can explain why. *Fix*: continuous evaluation against fixed datasets (Section 101.2); evaluation as a CI

gate; canary queries in production (Section 101.3).

107.3.2 The Observability Cliff

Conventional service observability (latency, error rate, CPU) but no AI-specific signals (drift, refusal rate, citation accuracy, retrieval quality). *Symptoms*: the dashboards are green; users are complaining. *Fix*: the four signal categories of Section 101.1; prompts and retrieval results in traces.

107.3.3 The Silent Drift

Model quality degrades over months without anybody noticing because nobody is measuring. *Symptoms*: users gradually drift away; surveys reveal dissatisfaction; the team is surprised. *Fix*: input-distribution and output-distribution drift monitors as production defaults; thresholds with on-call escalation.

107.3.4 The Cost Cliff

The first month is affordable; the third is a corporate event. *Symptoms*: no cost attribution; no per-tenant budgets; the bill is a surprise to everyone including the platform team. *Fix*: attribution tags on every call (Section 100.2); budget enforcement at the gateway; weekly cost reviews.

107.4 Governance Anti-Patterns

107.4.1 The Inventory of None

No central record of AI systems in production. Each team knows about its own; nobody knows about anyone else's. *Symptoms*: the first regulator visit reveals systems the compliance team had not heard of. *Fix*: mandatory model inventory (Section 103.1); registration as a precondition of deployment; periodic discovery scans for missed systems.

107.4.2 Governance Theatre

A review committee that meets monthly, produces meeting minutes, and demonstrably reduces no risk. *Symptoms*: teams route around the committee; the committee's metric is meetings held, not risks mitigated. *Fix*: tiered review (most systems get expedited, high-risk gets full); pre-approved patterns; clear decision rights with time-outs.

107.4.3 The Trust-the-Vendor Defence

Compliance evidence consists of pointing at vendor certifications. When the regulator asks specific questions about the deployed system, the answers are not available. *Symptoms*: vendor SOC reports filed as evidence; internal evidence missing. *Fix*: the platform produces its own evidence: lineage, evaluation results, monitoring data, audit logs; vendor reports are inputs, not the answer.

107.5 People Anti-Patterns

107.5.1 The Data Science Silo

A team of data scientists detached from engineering, producing models that do not reach production. *Symptoms:* demos to executives; nothing in users' hands. *Fix:* stream-aligned teams (Chapter 98) with data science embedded from the start.

107.5.2 The Infrastructure Silo

A platform team that hands raw infrastructure to use-case teams rather than a paved path. *Symptoms:* every use case rebuilds the same wheels; platform metrics are about platform activity, not spoke outcomes. *Fix:* measure the platform by what spokes ship; the platform's product is the developer experience.

107.6 Recognition Signals

A useful diagnostic: name the symptom, then name the anti-pattern.

Table 107.1: Common symptoms and the anti-patterns they suggest.

Symptom	Likely anti-pattern
"We can't change the vendor without a project."	Hard-Coded Vendor
"Every change touches the whole system."	Monolithic AI Service
"The model worked great in dev."	Snapshot Trap or Evaluation Theatre
"The bill keeps growing."	Frontier-Only Fallacy or Cost Cliff
"Users say it has gotten worse."	Silent Drift or Observability Cliff
"The regulator asked a question we couldn't answer."	Inventory of None or Trust-the-Vendor
"We have many models but few in production."	Data Science Silo
"Each team rebuilt the platform for themselves."	Infrastructure Silo

107.7 In Short

Fifteen anti-patterns recur across enterprise AI programmes. Each has a name, a diagnostic symptom, and a structural fix. The point of cataloguing them is not to declare any organisation guilty; most programmes have several at any moment. The point is to make them speakable: a named anti-pattern is easier to address than a vague unease. The patterns of Part III and the disciplines of Part VII are, in one sense, the structural responses to these anti-patterns. Recognising them is the first step; the rest of the book is the second.

Chapter 108

Future Directions

“The future is already here—it is just not very evenly distributed.”

—William Gibson

Chapter Overview

A book about architecture written in 2026 must distinguish the durable claims it makes from the volatile claims it leans on. This final chapter looks forward at the trends most likely to reshape the architecture over the next two to five years, the architectural commitments most likely to survive those changes, and the open problems that the field has not yet solved. It is written cautiously: predictions are inherently weaker than the structural lessons of the previous chapters, and the reader should treat them as priors to update against the present, not as conclusions.

Learning Objectives

- Recognise the trends most likely to reshape enterprise AI architecture in the medium term.
- Distinguish durable architectural commitments from volatile technology specifics.
- Identify the open problems the field has not yet solved and where the practitioner should invest in optionality.
- Plan investments that age well rather than ones that age fast.

108.1 Trends Worth Watching

Several trends visible in 2026 are likely to reshape the architectural surface materially.

108.1.1 Long Context, and Its Limits

Context windows have grown from thousands of tokens to millions. As of 2026, the question is no longer “does this fit in context” but “what is the optimal use of context, given the cost.” Long context does not eliminate RAG; it changes what RAG is for. Where retrieval was once a necessity to fit knowledge into a small window, it remains a necessity to: control cost (sending less context is cheaper), enable citation (retrieved chunks are pointers; tokens in context are not), and govern data flow (retrieval is the boundary at which access control applies).

108.1.2 Agentic Workloads

Agents have moved from research demonstrations to production deployments at small scale. The architectural challenge of the next several years is operating them at large scale: identity propagation, capability scoping, runaway-loop prevention, cost control, and audit. The patterns of this book (Agentic Mesh, Event-Driven Agents) are the structural starting points; the protocols that standardise the inter-agent surface (MCP, A2A, and their successors) will mature considerably.

108.1.3 Multimodal as Default

Text-only is increasingly the special case rather than the general one. Vision-language models, audio understanding, video reasoning, and tool-use across modalities are stabilising. Architectures that hard-coded text-only assumptions will need to soften them; the hexagonal pattern accommodates the change by letting modality-specific adapters slot into a common port.

108.1.4 Smaller, Specialised Models

The frontier-model market continues, but a parallel market in small (1B–10B parameter), specialised, often open-weight models has matured. For many enterprise tasks, a fine-tuned small model running on commodity hardware now matches or exceeds a frontier model API at a fraction of the cost. The LLM Router pattern (Chapter 23) is increasingly an architecture for routing across this spectrum.

108.1.5 Sovereign and On-Premises AI

Data-residency requirements, geopolitical considerations, and a maturing open-weights ecosystem have made sovereign AI deployment progressively more practical. Enterprises that previously assumed vendor APIs are revisiting the assumption. The architectural commitments that make this tractable—hexagonal boundaries, vendor-neutral ports—are the same ones that hedge against vendor risk in general.

108.1.6 Regulation as Architecture

The EU AI Act is fully in force; the US has moved toward a sector-by-sector regulatory approach; other jurisdictions are following with their own variants. Compliance is shifting from “produce documentation when asked” to “operate the system in a way that makes documentation a query against the platform.” Architectures designed for this—inventory, lineage, evaluation evidence, monitoring as platform features—spend less on each new regulatory cycle than those designed for the previous era.

108.2 What Will Probably Last

Several architectural commitments are likely to age well regardless of which specific trends materialise:

- **Hexagonal boundaries.** The vendor market will keep changing; the architecture that lets the change be operational rather than structural will keep mattering.
- **Data contracts.** Even if the data sources change, the discipline of writing down what is expected

between teams will keep paying back.

- **Layered responsibilities.** The vocabulary may evolve; the principle of one component, one responsibility will outlive any specific layer count.
- **Continuous evaluation and observability.** As models become more capable and more autonomous, the discipline of measuring them in production becomes more important, not less.
- **Governance as platform feature.** The cost of building this in remains modest; the cost of retrofitting it grows with every passing year.

108.3 What Will Probably Change

The specifics are more volatile:

- Which vendors lead in which categories; today's market positions will not be 2030's.
- The specific protocols that standardise inter-agent communication; MCP and A2A are early entrants, not necessarily the final form.
- The cost-per-token of frontier inference; the trajectory is steeply downward but the rate is unpredictable.
- The specific quantisation, batching, and serving techniques that dominate; the field iterates faster than annual planning cycles.
- The boundary between model and tool; capabilities currently provided by external tools may become model-native, and vice versa.

The architectural response is to invest the bulk of platform effort in the durable commitments above, and to keep the volatile specifics behind ports.

108.4 Open Problems

A handful of problems remain unsolved at the level of practice—worth naming because they will surface in real programmes and have no off-the-shelf answers.

- **Robust evaluation of open-ended generation.** Reference-based metrics work for some tasks; model-as-judge works for others; nothing yet covers the full space reliably.
- **Compositional safety.** Each component may be safe; the composition may not be. Reasoning about emergent behaviour in multi-agent systems is hard.
- **Provable forgetting.** Subject erasure is operable for explicit data stores; for fine-tuned model weights, “provably forgotten” is still a research property, not an operational one.
- **Long-term coherence in agents.** Agents that run for hours, days, or weeks accumulate state that drifts in ways that short-context evaluation does not catch.

- **Cost-quality measurement on real workloads.** Public benchmarks rarely match enterprise distributions; per-organisation evaluation harnesses are increasingly necessary.

The honest position is that the field is mid-stride on each of these. Architectures designed to be revisited as the answers improve will outlast architectures that committed to specific positions early.

108.5 Planning for the Next Five Years

A few practical postures for a programme planning multi-year investment:

- Invest in the platform’s interoperability surface. Standards-compliant integrations age better than bespoke ones.
- Build the inventory, lineage, and evaluation infrastructure now. They are cheap to add early and expensive to retrofit.
- Maintain optionality: every critical dependency should have a credible alternative, tested at least occasionally.
- Accept that the model layer will keep changing; budget for the migration cost as a recurring item, not a one-off.
- Re-read this book in 2028 to see what aged well, what aged badly, and what should now be added.

108.6 What to Carry Forward

The trends most likely to reshape the architecture—longer context, agentic workloads, multimodal defaults, smaller specialised models, sovereign deployment, regulation as architecture—argue for the same structural commitments the rest of this book has developed. The durable parts of the architecture (hexagonal boundaries, data contracts, layered responsibilities, continuous evaluation, governance as platform feature) age well; the volatile parts (specific vendors, specific protocols, specific costs) age fast. The architectural posture is to invest in the durable and keep the volatile behind ports.

Chapter 109

The Engine and the Millers: One Invention, Many Industries

“I have got a beautiful engine working, only it needs an iron foundry to keep it going.”

—James Watt, in a letter on the gap between invention and industry

Chapter Overview

This chapter is a deliberate pause, in the same spirit as Chapter 2. The first eight parts of this book built up the architectural language of enterprise AI: foundations, layers, patterns, advanced patterns, engineering disciplines, cross-cutting concerns, building and operating, maturity. The fourteen parts that follow take that language into specific industries — banking, insurance, mining, telecommunications, manufacturing, healthcare, retail, agriculture, government, education, energy, professional services, logistics, media. Before crossing into those deep dives, the reader benefits from a single eighteenth-century pairing: a Glasgow inventor whose engine would, within a few decades, reorganise nearly every industry that mattered, and a London fire that showed how violently the new technology and the old industries could collide. The first story is what foundation AI is doing now. The second story is the warning the industry chapters keep returning to.

Learning Objectives

- Recognise that a foundation technology and an industry application are different things, and that most of the engineering happens in the gap between them.
- Draw on James Watt’s steam engine to understand how a general-purpose breakthrough gets harnessed to specific industries.
- Use the Albion Mills fire of 1791 to think about the social, political, and economic shocks that an industry transition produces.
- Carry both images into every industry chapter that follows.
- Reframe the industry deep dives as answers to questions both stories already asked.

109.1 Same Engine, Different Harnesses

By the 1780s, a single mechanical invention — Watt’s separate-condenser engine — was running pumps in Cornish tin mines, spinning cotton in Manchester, crushing sugar in Jamaica, milling flour at Blackfriars, and minting coins in Birmingham. The engine was, near enough, the same engine in every installation. The harness around it was different every time. The industries about to follow in the chapters after this one will recapitulate that pattern: a small set of foundation capabilities, deployed under very different operating conditions, with very different surrounding institutions. The harness, not the engine, is where most of the real work lives.

109.2 The Engine: Watt and Boulton

109.2.1 An Invention That Began as a Repair

In the winter of 1763–64, a thirty-year-old instrument maker at the University of Glasgow was given a small model of a Newcomen steam engine to repair. James Watt was not at this point an industrial inventor; he was a craftsman who fixed scientific apparatus. The Newcomen engine, invented half a century earlier, was a working device used to pump water out of coal mines. It was profoundly wasteful: most of the heat that entered the cylinder was lost on each cycle, because the cylinder had to be alternately heated and cooled to make the engine work.

Watt’s improvement, which would take him most of the next decade to perfect, was to add a separate condenser: a second vessel where the steam could be condensed without having to cool the cylinder itself. The cylinder stayed hot. The fuel consumption dropped, eventually, by roughly two-thirds. It does not sound revolutionary in summary. It was revolutionary because the engine, made efficient enough to be economical outside of coal mines, could now be set to work in places that had no use for an expensive water pump and every use for a source of rotary motion.

By 1775, Watt had patented his improvements and gone into partnership with the Birmingham manufacturer Matthew Boulton. By the 1780s, with the addition of the sun-and-planet gear, the double-acting cylinder, and the centrifugal governor, the Boulton and Watt engine could deliver smooth rotary power at almost any scale a customer required. The engine was, in the language of this book, a foundation technology: a general-purpose component on which an enormous variety of industry-specific applications would be built.

109.2.2 Boulton’s Business Model

Watt was the inventor. Boulton was the architect of the business that delivered the invention to industries that had never seen it before. The arrangement deserves careful attention because it is, structurally, the same arrangement now being negotiated between foundation-model providers and the enterprises that consume their outputs.

Boulton and Watt did not, in most cases, sell engines outright. They sold the use of the engine. The customer received the engine, installed it, ran it; Boulton and Watt charged a fee — typically calculated against the fuel saved compared with the older Newcomen engine that would otherwise have been used. The customer paid for the value delivered. The supplier remained accountable for the

engine working. The arrangement aligned incentives, generated steady revenue, and gave Boulton and Watt a long-running operational relationship with each customer site. The model has been reinvented many times since, including by every foundation-model provider that charges per token, per request, or per outcome rather than selling a model outright.

It is worth pausing on what Boulton was *not* selling. He was not selling, in most cases, the industrial application. The engine could pump a mine, drive a mill, blow a forge bellows, hammer iron, spin cotton, propel a boat. Each application required a different surrounding architecture — different mechanical linkages, different feedstocks, different operators, different premises, different financing, different relationships to existing crafts and labour. The application was the customer’s problem; the engine was Boulton’s. The interface between them was where most of the engineering, and most of the trouble, lived.

Replace “engine” with “foundation model”, replace “mechanical linkages” with “prompts, evals, tools, observability”, and the structural diagram is the same.

109.2.3 The Harnessing Across Industries

By 1800, Boulton and Watt engines were at work in scores of industries that had never seen rotary mechanical power before. Cornish tin mines used them to pump out shafts that water-driven pumps could not reach. Yorkshire cloth manufacturers used them to drive carding and spinning machinery. Manchester cotton mills used them to power frames that had previously been water-driven and tied to fast-flowing streams. Sheffield ironworks used them to power forges and tilt hammers. Brewers used them to grind malt and crush hops. Sugar planters in the West Indies used them to crush cane. Distillers used them to grind grain. The Royal Mint used them to stamp coins. The Soho Manufactory of Boulton himself used them to make the very parts that built more engines.

What is striking, from an architect’s perspective, is that no two installations looked the same. The engine was the same, near enough. The harnessing was always industry-specific. A Cornish pumping installation had a beam, a great vertical shaft, a column of rising water, and a particular relationship to the geology of the mine. A Manchester cotton mill had horizontal shafts running the length of the building, belt-driven from a central drive shaft, with carding rooms on the upper floors and finishing rooms below. A West Indian sugar mill had three vertical rollers, a feed chute, a juice trough, and a relationship to a particular cane variety and a particular weather pattern. The engine reading these descriptions was the same engine. The architectures around it were nothing alike.

This is what the industry deep dives in the second half of this book are doing. They are not introducing new foundation models. They are describing what the same handful of foundation capabilities looks like when harnessed to a bank, a mine, a hospital, an electricity utility, a logistics operation. The patterns from the first eight parts are the engine. The industry chapters are the harnesses.

109.3 The Mills: Albion, 2 March 1791

109.3.1 The Wonder of the Age

By the middle of the 1780s, with the engine increasingly proven, Boulton was looking for a flagship industrial application that would demonstrate the technology at the largest possible scale to the largest possible audience. He found it in flour milling. London was growing. In the second half of the eighteenth century the city had passed three quarters of a million inhabitants and would soon pass a million. The city ate bread, by far its principal staple food. The bread came from flour. The flour came from windmills around the city and from country water mills that depended on rivers. Both were limited by the elements: the wind was unreliable and water flow was seasonal. Demand outran supply, and the price of bread, which was the daily concern of every working family, climbed and dropped on rumour as much as on fact.

Boulton's plan, conceived as early as 1783 and brought to fruition in 1786, was to build a steam-powered flour mill on the south bank of the Thames at the foot of Blackfriars Bridge. The architect Samuel Wyatt designed the building. The engineer John Rennie supervised construction. Watt himself designed and oversaw the construction of the engines. The mill that opened to public view was — by any standard of the time — a marvel. Five storeys of brick. Iron wheels and shafts. Twenty pairs of millstones, each grinding around nine bushels of corn per hour. Steam engines totalling perhaps a hundred and fifty horsepower of mechanical output. A continuous production line that fanned, ground, sifted, dressed, and lowered finished flour directly into barges on the Thames. The naturalist Erasmus Darwin called the machinery the most powerful in the world. The fashionable classes of London drove their carriages to Blackfriars to gawp at it. By 1790, with a second engine added, the mill was producing sales of around £6,800 in flour every week.

It was, by the architectural language of this book, an exquisitely engineered compound system. The engine was a foundation technology. The harnessing — the layout, the linkages, the materials flow, the workers, the financing, the supply chain of grain in and bread out — was industry-specific and, by all appearances, beautifully done.

109.3.2 What the Designers Missed

What the designers had not architected for was the surrounding industry's reaction.

London's traditional millers had opposed the project from the beginning. When Boulton applied for an act of incorporation, which would have let him raise funds by selling shares, the local millers lobbied successfully against it. The application was denied. Boulton had to fund the mill through partnership and loans. The millers' opposition was not mysterious. The Albion Mills could, in a month, do roughly what a traditional mill could do in a year. The economics of the new mill made the existing windmills and water mills around London structurally uncompetitive on price. As the new mill ramped up production, traditional mills around the city went idle. Several were eventually mothballed entirely.

A second layer of opposition was political and almost theological. Bread was not just a commodity; it was the daily concern of every working family in London, and the traditional millers had long been suspected — often correctly — of adulterating flour with chalk, alum, ground bones, and worse. The new mill, owned by capitalists, drawing on cheap imports, run by absentees, fit comfortably into a

popular suspicion that the food supply had been captured by interests indifferent to ordinary people. Rumours spread that the Albion Mills' flour was adulterated. The contemporary illustrations of the building show, in some prints, devils on the roof.

This was the spring of 1791. Across the Channel, the French Revolution was approaching its second anniversary. The English propertied classes were nervous. A large, capitalised, technology-leading enterprise standing visibly on the south bank of the Thames, surrounded by displaced traditional craftsmen, was a more politically volatile structure than its owners may have appreciated.

109.3.3 The Fire

At about half past six on the morning of 2 March 1791, a fire began in the Albion Mills. Within two hours, the building was gutted. Damage was estimated at £150,000 — an enormous sum — and roughly five hundred workers were thrown out of employment. Several neighbouring houses also burnt.

The official explanation, from Wyatt and Rennie, was that insufficient grease on a corn machine in front of the kiln had caused the fire by friction. It was a credible technical story. It convinced few people at the time. The chronicler Robert Southey, walking among the crowds on Blackfriars Bridge during the conflagration, observed groups of millers dancing in the firelight. Shouts of acclamation went up with each fresh sign of the mill's destruction. A placard worn by one of the celebrants read: *Success to the mills of Albion but no Albion Mills*. The customary English readiness to help fight a fire was largely absent. Reports circulated — never proven — that the main cock of the water cistern had been fastened shut and that the timing of the fire had been chosen to coincide with low tide on the Thames, which would have slowed any fire-fighting that depended on river water.

Boulton called for a thorough investigation as a matter of national importance. None ever conclusively established the cause. Plans to rebuild the mill were eventually rejected. The burnt shell was, a few years later, converted into housing. The artist and poet William Blake lived nearby. When he wrote, a few years afterward, of the “dark Satanic Mills” that troubled his vision of England, the most plausible candidate for his image was the gutted shell of the Albion Mills, visible from his neighbourhood.

109.3.4 What Had Failed

It would be easy to read this story as a tale of luddite reaction against progress. That reading does not quite capture the architectural lesson.

The Albion Mills as an engineering project was, by the technical standards of the day, well executed. The engines worked. The grinding worked. The materials flow worked. The financial model worked. The owners had survived early teething problems with the sun-and-planet gear and the piston rods and had reached the point of profitability. Considered narrowly, as a piece of industrial engineering, the project was a success.

Considered as a piece of *industry* architecture — which is to say, as a system embedded in a particular trade, with a particular labour force, particular regulators, particular consumers, and a particular

politics — the project had under-architected the surroundings. It had not built a path for the displaced labour. It had not built trust with consumers nervous about flour adulteration. It had not built protective relationships with the political authorities who might have helped it after the fire. It had not built physical defences for a building that was, by 1791, a visible symbol of a contested industrial transition.

The Albion Mills did not fall because the technology failed. It fell because the surrounding industry pushed back, and the surroundings had not been architected.

109.4 Engine and Mills, Read Together

109.4.1 The Foundation Technology Is Only Half the System

Watt's engine was a foundation technology. By itself, it did nothing of commercial value. It needed mechanical linkages, supply chains, customer relationships, financing, operating procedures, trained labour. Each industry required its own harness. Boulton's genius was understanding this and architecting his business around the harnessing rather than the engine.

Foundation models in 2026 are in roughly the same position. By themselves, they do nothing of commercial value. They need prompts, retrieval, evaluation, observability, security, governance, integrations, support. Each industry requires its own harness. The patterns in the first half of this book are the harness components. The industry chapters are how the components combine differently in different industries. Foundation-model providers who understand this build durable businesses; enterprises that buy a model as if buying an engine and forget the harness do not get value out of it.

109.4.2 The Harness Is Industry-Specific All the Way Down

No two of Boulton's customer installations looked the same. A Cornish pumping engine and a Manchester cotton mill shared a power plant and almost nothing else. The harness was determined by the physics of the work being done, the regulatory environment of the trade, the existing labour force, the supply chain, the local politics.

This is exactly the lesson the industry deep dives in this book will recapitulate. The same foundation model serves a tier-one bank and a rural agricultural cooperative, but the architecture around it has to be designed for the bank's regulators, the bank's labour, the bank's risk appetite, the bank's customers — or, in the other case, for the cooperative's intermittent connectivity, low literacy, multi-lingual users, and constrained budgets. The architect who treats every industry as a minor variation on a single template will, sooner or later, be the architect whose system collapses in a corner.

109.4.3 The Industry Itself Will Push Back

The Albion Mills story is, more than anything else, a story about industry resistance. The displaced millers, the suspicious consumers, the nervous regulators, the political environment were all part of the architecture, whether the project's engineers chose to recognise them or not. The fire is the moment at which the un-architected parts of the surroundings declared themselves.

Industry deep dives in the rest of this book will return, in different language, to the same theme. AI in banking encounters Basel rules and supervisor expectations and a labour force whose unions have views about automation. AI in healthcare encounters licensure, malpractice exposure, patient consent, and clinicians who have lived through wave after wave of failed technology rollouts. AI in mining encounters labour-relations regimes whose origins are older than most banks. AI in public sector work encounters constitutional rights, freedom-of-information regimes, and a public that, in many countries, has reason to distrust state automation. None of this is solvable by better prompts. It is solvable, where it is solvable, by architects who recognise that the industry is part of the system.

109.4.4 Speed Without Absorption Is Brittle

Boulton and Watt brought the steam engine to flour milling faster than London's surrounding industry could absorb it. The economics were so much better that the traditional mills could not compete on price; the displaced labour had nowhere to go; the political and consumer trust had not been built; the safety architecture of the building itself had not anticipated arson. The pace of deployment exceeded the pace of absorption, and the system found its release valve in a fire.

The pattern is being recapitulated, in 2026, across nearly every industry that this book covers. Foundation-model capability is improving faster than most industries can absorb it. The economics of AI-driven work are, in many cases, so much better than the traditional alternative that legacy operators cannot compete on price. The displaced labour — call-centre agents, junior associates, document reviewers — has, in many markets, nowhere obvious to go. The political and consumer trust has not been built. The safety architecture of the technology has not anticipated all the ways it can be attacked. The pace of deployment is, in many industries, exceeding the pace of absorption.

The architect's job, in this environment, is not to slow the technology. It is to build the absorption. The industry chapters that follow are, very largely, about how to do that, industry by industry.

109.5 What Africa Inherited from That Century

It would be incomplete to describe the eighteenth-century industrialisation of Europe without naming what it cost elsewhere. The same decades that saw Watt patent his engine and Boulton build the Albion Mills saw the apex of the Atlantic slave trade, organised with horrifying efficiency by the same European powers that were industrialising at home. The cotton that drove Manchester's mills, which Watt's engines spun, came from plantations whose labour was enslaved. The sugar that Boulton's engines crushed in Caribbean mills came from plantations whose labour was enslaved. The wealth that financed the Albion Mills came, in part, from a colonial system whose engine, in a different sense of the word, was African suffering. The foundation technology was European; the suffering on which it ran was African and indigenous American.

The book's industry deep dives that follow will travel through African banks, African mines, African telecommunications networks, African healthcare systems, African public sectors. In each chapter, the African perspective sections will note where the international AI stack works well in African contexts and where it does not. A reader who carries this interlude into those chapters should also carry the awareness that the eighteenth-century version of this same conversation went badly for

Africa, in ways the continent is still negotiating. The patterns this book teaches — multi-vendor strategy, sovereignty-aware architecture, local capability building, donor-procurement discipline — exist in part because the present generation of African engineering leaders intends a different outcome this time.

109.6 The Shape of the Argument

This chapter has paired two eighteenth-century stories. The first — Watt’s engine and Boulton’s business — demonstrates how a general-purpose foundation technology gets harnessed, painstakingly and differently, to specific industries. The second — the Albion Mills fire of 2 March 1791 — demonstrates what happens when the harness arrives faster than the surrounding industry can absorb. The first is what the foundation parts of this book have described. The second is the warning the industry parts of this book will repeat, in different words, in nearly every chapter that follows. The architect who reads the industry chapters ahead with the engine and the millers in mind will read them as what they are: not separate technical recipes for each industry, but variations on a single, larger, harder question. How do we build the harness? How do we build the absorption? How do we keep the system from burning?

Chapter 110

The Lost Fleet and the Instrument: Measurement at the Edge of the Map

“The longitude is not to be found at sea by any method now in use.”

—preamble of the British Longitude Act, 1714

Chapter Overview

This chapter is the second of the two reflective interludes that separate the architectural patterns of this book from the industry deep dives that follow. The first interlude, Chapter 109, used Watt’s steam engine and the Albion Mills fire to think about how a foundation technology gets harnessed to specific industries and the social shocks that arrive when the harness lands faster than the industry can absorb. This chapter takes a different angle, one the industry chapters keep returning to: *how do we run operations at the edge of what we can see, measure, and trust?* The pairing is again from the eighteenth century: a British naval disaster off the Isles of Scilly in 1707 caused by the inability to know where the fleet actually was, and the seventy-year institutional response — an act of Parliament, a clockmaker from Yorkshire, and the voyages of an officer in the Pacific — that turned the problem of measurement at distance into a solved one. Industry AI in 2026 is in much the same condition as British naval operations in 1707. The chapter is about what comes next.

Learning Objectives

- Recognise that running operations at the edge of available measurement is the recurring problem of every regulated industry the book is about to enter.
- Use the Scilly disaster of 1707 to understand the cost of operating beyond what your instruments can support.
- See in the Longitude Act, Harrison’s chronometer, and Cook’s voyages the slow institutional and engineering response to that problem.
- Recognise that the response is not a single invention but a stack of institutions, instruments, training, and operating discipline.
- Carry the image of the lost fleet and the instrument into every industry chapter that follows.

110.1 Operating Beyond What the Instruments Can See

Every regulated industry in the chapters about to follow makes decisions at the limit of what its existing measurements can support. The banker approves a loan over a borrower whose true position is partly hidden; the clinician admits a patient whose underlying disease state has to be inferred; the controller dispatches autonomous trucks across a face the geology of which is incompletely known; the regulator signs a marketing authorisation on the basis of trials that were necessarily small. The story of the British fleet in 1707 and the long institutional response that followed it is the canonical case of what happens when the instruments are not equal to the decisions being asked of them — and of what gets built afterwards, slowly, when the country decides it cannot afford that gap again.

110.2 The Lost Fleet: Isles of Scilly, 22 October 1707

110.2.1 The Voyage Home

In late September 1707, a British fleet of twenty-one ships set sail from Gibraltar for home. Their summer's work was over: they had been part of an Anglo-Dutch-Austrian operation to take the French naval base at Toulon, which had ultimately failed on land but had inflicted serious damage on the French Mediterranean fleet. The British squadron, commanded by Admiral Sir Cloudesley Shovell — one of the most respected naval officers of his age, member of Parliament for Rochester, darling of the establishment — was returning to Portsmouth.

The Atlantic crossing was rough. Westerly gales drove the fleet further north than the navigators believed. By the night of 22 October, in heavy weather and limited visibility, the sailing masters of the fleet conferred and agreed on their position: they reckoned themselves to be safely off Ushant, the French island at the entrance to the English Channel. They were wrong by approximately one hundred miles.

110.2.2 The Reckoning

What they actually were doing, in the language of contemporary navigation, was *dead reckoning*: estimating their position from their last known fix, their compass heading, and their estimated speed, corrected by guesses about current and leeway. It was the only method available for finding longitude at sea. Latitude could, in principle, be measured by the height of the sun at noon or the pole star at night — but in stormy weather the sky was rarely clear enough to take an accurate sight, and even latitude required a calm horizon. Longitude required something that did not yet exist: an accurate clock that could keep Greenwich time on a moving ship for the length of a voyage. Without it, navigators could only guess how far east or west of their starting point they had travelled.

The masters' guess was wrong. The fleet was much further north than they thought, and approaching not the open Channel but the Isles of Scilly — a granite cluster off the southwestern tip of England, surrounded by treacherous outlying rocks. The charts of the day did not show the islands' position consistently. When the fleet later returned to port, an inventory of the surviving navigational equipment found that only four of one hundred and twelve compasses were accurate.

110.2.3 The Wreck

The flagship HMS *Association* struck the Gilstone Ledges first, west of the islands. She lit lanterns to warn the rest of the fleet of her position; the lanterns burned for a few minutes; the seas pounded the ship apart, and she disappeared beneath the waves. HMS *Eagle* struck next, and went down with her entire crew. HMS *Romney* struck shortly afterwards; the only survivor of her crew was a former butcher named George Lawrence, serving as quartermaster, who could swim — a rare skill among eighteenth-century sailors. HMS *Firebrand* struck last and joined the others on the sea floor.

The total death toll cannot be exactly determined. Modern estimates range between fourteen hundred and two thousand men. It remains one of the worst naval disasters in British history. Admiral Shovell himself drowned. His body washed ashore on the Isles of Scilly, was recovered, and was taken back to London for burial under a marble memorial in Westminster Abbey.

110.2.4 What Had Failed

It would be easy to read the Scilly disaster as a single act of incompetence on a stormy night. That reading is incomplete.

What failed was a system that asked sailors to make consequential decisions about position when the instruments available to them could not deliver the accuracy the decisions required. The dead-reckoning method was the best the navy had; it was not good enough. The latitude tools required a clear horizon they did not have. The longitude tools did not exist. The compasses were largely uncalibrated. The charts did not agree on where the islands were. The training of sailing masters relied on craft and judgement that, on a clear day with reliable instruments, sufficed; on a stormy night with broken instruments and contested charts, it did not.

The Scilly disaster is, architecturally, what happens when an organisation runs operations beyond the limit of its instruments. Every individual decision the sailing masters made was defensible in isolation. The cumulative effect, in the absence of measurements that would have caught the drift, was to put the entire fleet a hundred miles from where it thought it was. The crew of the *Eagle* did not drown because their captain was negligent. They drowned because they were depending on a measurement that, in 1707, no one had figured out how to take.

110.3 The Instrument: The Longitude Act, Harrison, and Cook

110.3.1 The Act of Parliament

It is sometimes claimed, perhaps too neatly, that the Scilly disaster led directly to the Longitude Act of 1714. The connection is contested by historians; the petition that produced the Act was already being prepared by 1713, on grounds that predate the disaster. But whether or not Scilly was the precipitating cause, by 1714 the British Parliament had decided that the problem of finding longitude at sea was important enough to warrant a public prize. The Longitude Act offered up to £20,000 — an enormous sum, roughly a hundred years' wages for a senior officer — for a practical method of determining longitude to within half a degree on a transatlantic voyage. It established a Board of Longitude to evaluate submissions.

The Act is interesting independently of what it produced, because it represents an early example of an institution constructed specifically to solve a measurement problem. The Board did not invent the method itself; it created the conditions under which inventors would invest decades of work to invent it. The structure was incentive design: the British government identified a critical problem, specified what a solution would have to look like, attached money to it, and waited. This pattern — public institution targets a measurement gap, offers an incentive structured to a specification, and waits years for the right inventor to find them — has been used many times since, by many governments and many industries. It is the eighteenth-century ancestor of the modern XPRIZE, of the DARPA grand challenges, and of the structured procurement competitions through which contemporary states pull novel capabilities into existence.

110.3.2 Harrison and the Chronometer

The Yorkshire clockmaker John Harrison would spend roughly four decades of his life pursuing the longitude prize. His first marine timekeeper, H1, was sea-trialled in 1736 and worked well enough to demonstrate that the concept was viable. H2 followed, then the substantially redesigned H3, which Harrison worked on for nearly twenty years and never quite finished. The breakthrough was H4, completed in 1759: a watch about the size of a large dinner plate, fundamentally different in design from his previous chronometers, accurate to a few seconds a day at sea. On its second formal trial voyage, to Barbados in 1764, H4 lost only thirty-nine seconds over the course of months at sea.

The Board of Longitude was famously slow to acknowledge that Harrison had solved the problem. He was paid in instalments, required to demonstrate and re-demonstrate his work, asked to surrender his designs for replication, and finally received the balance of his prize money only after a personal appeal to the king. By any honest account, he was treated badly. He was nevertheless, by the end of his life, the man who had given the world a working solution to a problem that had killed mariners for as long as ships had crossed oceans.

What Harrison had built was not just a clock. It was an instrument of operational discipline. A navigator with a chronometer that kept Greenwich time could, by observing the local time of noon, calculate longitude with usable accuracy. The chronometer extended human judgement out into spaces where unaided judgement had been catastrophically unreliable. It was, in the language of this book, the eighteenth-century version of what observability does for a contemporary production system: an instrument that tells you, with reasonable accuracy and continuously, where you actually are.

110.3.3 Cook's Second and Third Voyages

The chronometer's first major operational test was not with the Royal Navy as a fleet; it was with one of its most disciplined captains, working at the edge of the known world. James Cook took H4's successor, K1 — built by Larcum Kendall as a near-exact copy of Harrison's design — on his second voyage of Pacific exploration, which sailed in 1772 and returned in 1775. The voyage charted New Zealand, much of Polynesia, the Antarctic ice edge, the South Sandwich Islands, and Vanuatu. The charts Cook produced were accurate to a degree that previous Pacific charts had not approached. The chronometer's contribution was central. Cook called it “our trusty friend the Watch” and “our never-failing guide.”

Cook's voyages are worth pausing on, because they illustrate something more than the value of an accurate clock. Cook's operational discipline went well beyond the chronometer. He insisted on sauerkraut, lemon juice, and fresh greens at every landing to prevent scurvy — and lost only one man to disease on his second voyage, an unheard-of result for a long-distance ocean voyage of the period. He maintained detailed logs and astronomical observations. He gave his officers latitude to deviate from his standing orders when local conditions demanded it. He had documented protocols for first contact with indigenous peoples, even if those protocols were imperfect and, at times, violated. He used multiple instruments to cross-check his position, not just the chronometer. He carried backups. He treated the boundaries of his knowledge as boundaries.

The point is that Cook's voyages were a system, not a heroic individual feat. The chronometer was a critical part of the system; it was not the whole of it. The success of the second voyage was the cumulative effect of a clock, a discipline, a crew, a set of operating procedures, and a captain who insisted on treating known unknowns with the seriousness they deserved.

110.3.4 Cook's Death, Which the Instrument Did Not Prevent

It is important to the lesson of this chapter that Cook's third voyage ended badly. In February 1779, on the return leg of his Pacific voyage, after a series of misunderstandings and breakdowns of the trust that his earlier visits to Hawaii had built, Cook was killed in a confrontation on the shore of Kealahou Bay. The exact sequence of events is disputed. What is not disputed is that the instruments of measurement, the discipline of provisioning, and the operational protocols that had carried him so far did not prevent the failure when the failure was, fundamentally, a political and cultural one rather than a measurement one.

This is the second half of the lesson the chapter wants to leave the reader with. Measurement is necessary. Measurement is not sufficient. An organisation that has solved its measurement problem has not, by that fact alone, solved all of its problems. The chronometer told Cook precisely where he was. It could not tell him how the political dynamics on a Hawaiian beach were shifting under his feet. A reader who carries Cook into the industry chapters that follow should carry both halves of his story: the part where the instrument worked, and the part where it could not help him.

110.4 Fleet and Instrument, Read Together

110.4.1 Operating Beyond the Limit of Your Instruments Is the Recurring Industry Problem

The Scilly disaster was the result of asking sailors to do something the instruments of 1707 could not support. Each industry the rest of this book covers has at least one analogous condition — a decision being made at the edge of what existing instruments can measure, where the cost of being wrong is high, and where the introduction of better instruments would substantially reduce that cost.

In banking, this is the condition of credit risk assessment at scale: lenders make decisions about borrowers whose true financial position they cannot directly observe. AI-based instruments — transaction-stream embeddings, alternative-data signals, real-time monitoring — extend what the lender can

see.

In healthcare, it is the condition of clinical decision under diagnostic uncertainty: clinicians make decisions about patients whose underlying disease state they cannot directly observe. AI-based instruments — imaging classifiers, risk-stratification models, prognostic indices — extend what the clinician can see.

In mining, it is the condition of geological inference beneath the surface: planners make decisions about ore bodies they cannot directly inspect. AI-based instruments — seismic interpretation models, grade-prediction systems, fleet telematics — extend what the planner can see.

In public sector work, it is the condition of policy decision over populations whose actual circumstances are imperfectly known: ministries make decisions about citizens whose needs they cannot directly observe at scale. AI-based instruments — targeting models, fraud detection, service-demand forecasting — extend what the ministry can see.

In each industry, the underlying architecture of the work is the same as that of the British navy in 1707: a decision being made at the edge of what the existing instruments can support. The introduction of AI is, in significant part, the introduction of new instruments that change where that edge sits.

110.4.2 Better Instruments Are Built Slowly and Institutionally

The chronometer did not appear because a single inventor woke up with an idea. It appeared because a public institution had been built to make its development worthwhile. The Longitude Act, the Board of Longitude, the prize money, the trial voyages, the political pressure that eventually forced the Board to pay Harrison: all of these were the surrounding scaffolding without which the engineering would not have happened.

The introduction of AI into each industry covered in the rest of this book depends similarly on scaffolding. Regulators have to issue guidance that lets the new instruments be used. Insurers have to underwrite the new ways of working. Professional bodies have to update what counts as competent practice. Universities have to teach the new instruments. Auditors have to develop standards for evaluating them. Procurement frameworks have to make space for unfamiliar vendors. Boards have to learn what questions to ask.

The industry chapters that follow describe, for each industry, what that scaffolding looks like. Architects who treat the scaffolding as someone else's problem build instruments that nobody trusts and that, accordingly, no one uses.

110.4.3 The Instrument Is Part of a System, Not the System

Cook's voyages worked because Cook had built a system in which the chronometer was one component among many. The clock told him longitude; the sextant told him latitude; the lookout told him about shoals; the logs told him about wind and current; the crew's health was protected by deliberate provisioning; the officers were trained to think for themselves; backup instruments were carried; protocols for the unfamiliar were documented in advance. The chronometer in isolation would not have produced the voyages. The system around it did.

This is the disposition the industry chapters keep returning to. The introduction of an AI-based instrument — a clinical decision support model, a fraud-detection system, a mining-fleet-routing engine — is the beginning of a system change, not the end of one. The instrument has to be supported by training, by procedures, by escalation paths, by audit, by ongoing recalibration, by the human judgement that will continue to be needed at the boundary of what the instrument can do. The chapters that follow will describe what these supports look like in each industry.

110.4.4 Measurement Does Not Solve Politics

Cook's death is the part of the story that prevents this chapter from becoming triumphalist. The chronometer told him exactly where he was on the morning of his death. It did not tell him that the Hawaiian relationship had shifted, that his recent attempts to recover a stolen cutter had been provocative, that the people he had befriended on earlier visits had reasons to see him differently this time. The instrument did its job. The system around the instrument did not. The failure that killed him was a political and relational failure, and it was not a kind of failure that better measurement could have averted.

This is the warning that pairs with the previous interlude's warning about industry resistance. The industry chapters that follow will describe many situations in which AI-based instruments will extend what their users can see, measure, and decide. None of those instruments solves politics. None solves trust. None solves the conditions under which the people affected by the instrument will accept its outputs as legitimate. Each industry has its own version of Kealakekua Bay — the point at which the instrument has done its job, the operation has been carried out correctly, and the failure that arrives is one of human relationship that the instrument was never going to address. The industry chapters that follow are about the instruments and the systems around them; the politics, in each chapter, has to be done separately.

110.5 Other Navigators, Other Charts

The story of measurement, instruments, and the long-distance operation of states cannot honestly be told as if it happened only in London and Greenwich and the Pacific. The same century that produced the longitude prize and the marine chronometer saw an enormous expansion of European long-distance operations, much of which involved the displacement, enslavement, and dispossession of peoples whose own systems of knowledge, navigation, and trade had been functioning for centuries.

The Polynesian navigators whom Cook's expedition encountered were running ocean operations across the Pacific by stick charts, by wave patterns, by the flight of birds, by the colour of the sky, with a precision and reliability that Cook himself acknowledged with respect. Tupaia, the Tahitian navigator and priest who joined Cook's first voyage, drew a chart of the Pacific from memory that included islands the European charts did not show. The indigenous traders of the Indian Ocean had been crossing between East African ports and India and beyond for over a millennium, with monsoon-pattern knowledge embedded in oral and apprenticed practice. The traditional pilots of the Niger, the Congo, and the great lakes of the African interior were running long-distance operations whose instruments were different from European instruments but whose function was the same.

The book's industry chapters that follow will repeatedly note the gap between what European-trained engineering can see and what local practice has long known. In some industries this gap matters relatively little; in others — agriculture, public health, financial inclusion, smallholder credit, mobile-money services, language technology — the gap is exactly what determines whether an AI system works or fails. A reader who carries the Scilly disaster and the chronometer into the industry chapters should also carry the awareness that the eighteenth-century instruments solved one problem of measurement and created others. The chapters ahead, particularly the African perspective sections, are an attempt not to repeat the second part of that pattern.

110.6 Where That Leaves Us

This chapter has paired two eighteenth-century stories, again to prepare the reader for the industry deep dives. The Scilly disaster of 1707 shows what happens when an organisation runs operations beyond the limit of its instruments: a fleet wrecked because the available navigation was not equal to the conditions the navigators faced. The Longitude Act, Harrison's chronometer, and Cook's voyages show the slow institutional and engineering response that turned the measurement problem into a solved one — and show, in Cook's death, the limits of measurement when the failure is no longer a measurement problem. The industry chapters ahead will repeatedly describe the introduction of AI-based instruments into industries that have been operating at the edge of what their existing instruments can support. The architectural lesson is the eighteenth-century one: a new instrument is the beginning of a system change, not the end. The instrument is necessary; the institutions around it have to be built; the operational discipline has to be developed; the human judgement at the edge of what the instrument can do has to be preserved; and the politics of trust, in which the instrument was never the right tool, has to be addressed separately. The fleet was lost because the instrument did not exist. The instrument was built because the institution was built. The voyages worked because the system around the instrument was built. And one of those voyages still ended badly, on a beach, because no instrument could have helped with what failed there.

End of Volume 1

The architectural language is now built.

The industry deep dives — Banking, Insurance, Mining, Telecommunications,
Manufacturing, Healthcare, Education, Public Sector,
Energy and Utilities, Retail and CPG,
Media and Entertainment, Professional Services —
continue in Volume 2.

Each industry chapter applies the architectural disciplines of Volume 1
to the constraints, regulations, and value streams of its sector.

Appendix A

A Quick Glossary

This glossary provides working definitions of the terms used throughout the book. Where a term has a chapter-length treatment, the entry includes a forward reference. Terms are listed alphabetically.

A2A (Agent-to-Agent)

An emerging family of protocols for direct agent-to-agent communication: task delegation, capability discovery, result return. Complements MCP (which standardises agent-to-tool). See Section 18.3.

Adapter (Hexagonal)

In the hexagonal pattern (Chapter 26), the concrete implementation of a port against a specific external system (e.g., an OpenAI adapter for the LLM port).

Adapter (LoRA)

A small set of trainable parameters added to a frozen base model during parameter-efficient fine-tuning. See LoRA.

Agent

An AI component that chooses its next action from a set of tools based on the current state, often in service of a goal. Distinguished from a chain by the runtime choice of action. See Chapter 29.

Agent IAM

Identity, authentication, and authorisation for agents treated as principals with delegated authority. See Chapter 72.

Agentic Mesh

The architecture pattern for scaling many agents across an enterprise (Chapter 18). Mnemonic: AGENT—Agents, Governance, Exchange, Network, Tools.

Agentic RAG

A retrieval pattern where the agent decides at each step whether to retrieve, what query to use, and whether the retrieved context is sufficient. See Chapter 50.

Alignment

The process of training a model to follow instructions, refuse harmful requests, and produce outputs consistent with human values. Techniques include RLHF, DPO, and constitutional AI.

Barge-In

In voice architectures, the ability for a user to interrupt the agent mid-utterance. See Chapter 64.

Batching

Processing multiple requests together to amortise per-call overhead. *Continuous batching* lets requests at different stages of generation share a forward pass.

Bias

Systematic unfairness in an AI system's outputs across protected groups. See Chapter 78.

Bronze, Silver, Gold

The three tiers of the medallion architecture: raw, curated, business-ready.

Canary Deployment

Exposing a new model, prompt, or configuration to a small slice of production traffic under comparison before promoting to full traffic. See Chapter 48.

Catastrophic Forgetting

The tendency of a fine-tuned model to lose capabilities the fine-tuning data did not exercise. See Chapter 60.

CDC (Change Data Capture)

Log-based streaming of insert/update/delete events from a source database, used as a low-latency alternative to periodic full-table scans.

Chain

A fixed sequence of model calls whose order is known at design time. Distinguished from an agent by the absence of runtime action choice.

Chunking

Splitting source documents into smaller pieces before embedding for retrieval. Strategies: fixed-size, structural, semantic. See Chapter 87.

Circuit Breaker

A resilience pattern that monitors failure rate to a dependency and short-circuits subsequent calls when failures exceed a threshold.

Cobra Effect

A governance anti-pattern in which a metric intended to improve outcomes is gamed, producing the opposite of the intended result. See Chapter 104.

Compound AI System

A system composed of multiple AI components (models, retrieval, tools, evaluators) working together, as opposed to a single-model system. See Chapter 44.

Constitutional AI

A technique for training models to follow explicit principles (a “constitution”) that constrain behaviour. See Chapter 45.

Context Window

The maximum number of tokens (input + output) a model can attend to in a single call.

Continuous Evaluation

Running evaluation harnesses continuously against production samples rather than only at release time. See Chapter 101.

Control Plane

In the four-plane architecture, the governance brain that manages identity, authority, evidence, and composition across the AI estate. See Chapter 9.

Cross-Encoder

A model that scores a query-document pair jointly, used for re-ranking retrieval candidates. More accurate but more expensive than bi-encoder retrieval.

Data Contract

A written specification of schema, semantics, freshness SLA, quality expectations, and change-notification process between a data producer and consumers.

Data Sovereignty

The principle that data is subject to the laws of the jurisdiction where it is collected or stored. A central concern in African deployments.

Differential Privacy

A mathematical framework that bounds how much any single individual's data can influence a model's outputs. See Chapter 92.

DPO (Direct Preference Optimisation)

A preference-tuning method that aligns model outputs with human preferences without a separate reward model. See Chapter 69.

Drift

A statistical change in input distribution (*input drift*) or in the relationship between inputs and ground truth (*concept drift*) over time. See Chapter 60.

Embedding

A dense vector $\mathbf{e} \in \mathbb{R}^d$ representing content such that semantically similar content has small vector distance. See Chapter 67.

Endpointing

In voice architectures, detecting when a user has finished speaking. Combines acoustic silence detection with semantic completeness judgement.

Episodic Memory

Memory of past interactions, stored per user or session and retrieved as summaries.

Evaluation Harness

The standing infrastructure that runs evaluations against datasets and computes metrics; the ML equivalent of a unit-test framework.

Event Bus

The shared message infrastructure in event-driven architectures. Implementations: Kafka, Pulsar, Kinesis, Pub/Sub. See Chapter 25.

Evidence Plane

In the four-plane architecture, the sub-plane of the Control Plane responsible for recording what happened: logs, lineage, and cryptographic proof.

Experience Plane

In the four-plane architecture, the plane that shapes the human-curated, AI-enabled work experience. See Chapter 9.

Feature Store

Infrastructure that stores feature values for both offline (training) and online (inference) consumption.

Federated Learning

A training paradigm where models are trained across multiple decentralised data holders without the data leaving its location. See Chapter 46.

FinOps

The discipline of cloud financial management; in this book, applied to the cost streams specific to AI workloads. See Chapter 61.

Fine-Tuning

Adapting a pre-trained model to a specific task by additional training. Full fine-tuning updates all parameters; parameter-efficient methods (LoRA, QLoRA) update only small adapters.

Foundation Layer

In the four-plane architecture, the enterprise guardrails and enablement (zero trust, data governance, security, compliance, observability, FinOps, resilience) beneath all four planes.

Foundation Model

A large neural network pre-trained on broad data and adapted to many downstream tasks via prompting or fine-tuning.

Governance Operating Model

The explicit catalogue of risk domains an AI estate must manage: data governance, model validation, bias, human oversight, compliance, security, vendor risk, audit logging, monitoring, and incident response. See Chapter 104.

Grounding

Constraining a model's output to be supported by retrieved or provided evidence.

Guardrail

A safety control on input or output; typically a classifier that detects off-policy prompts or completions. See Chapter 54.

Hallucination

An output that is confidently asserted but not grounded in the input or in fact. Reducible by RAG,

citation requirements, and verification but not eliminable.

Hexagonal Architecture

The structural pattern of a domain core surrounded by ports, each with replaceable adapters (Chapter 26).

HITL (Human-in-the-Loop)

The pattern of keeping a human in the decision path for high-stakes AI outputs. See Chapter 35.

Hub-and-Spoke

The organisational pattern of a central platform (hub) serving business-unit teams (spokes). See Chapter 24.

Hybrid Search

Combining vector search with keyword search (BM25) and structured filters; consistently outperforms any single retrieval method. See Chapter 87.

Idempotency

The property that processing the same input twice produces the same effect as processing it once. Required for safe retries.

Inference

The process of running a trained model on new inputs to produce outputs. Contrasted with training.

Jailbreak

An input crafted to bypass a model's alignment and elicit behaviour the model is supposed to refuse. See Chapter 62.

KV Cache (Key-Value Cache)

The cache of attention keys and values from prior tokens, reused during autoregressive generation. Grows linearly with context length. See Chapter 86.

Latency Budget

An explicit allocation of latency across the components of a request, used to surface where time is spent.

Lineage

The record of how a value was derived: from which sources, through which transformations, by which model versions.

LLM (Large Language Model)

A foundation model trained on text to predict the next token. The dominant building block of this book.

LLM Router

The pattern of dynamically choosing which model handles a request based on its characteristics, optimising the cost-quality-latency frontier. See Chapter 38.

LLMOps

The operational discipline for the lifecycle of LLM applications: prompt versioning, evaluation gates, deployment, monitoring, improvement. See Chapter 70.

LoRA (Low-Rank Adaptation)

A parameter-efficient fine-tuning method that factorises a weight update ΔW into the product of two low-rank matrices BA .

MCP (Model Context Protocol)

An emerging standard for agent-to-tool integration: tool definition (JSON Schema), capability discovery, invocation. See Chapter 33.

Medallion Architecture

The bronze/silver/gold tiering of data lake storage by quality.

Mixture of Experts (MoE)

A model architecture containing many expert sub-networks, of which only a fraction are activated per input. See Chapter 52.

Model Card

A structured document describing a model: training data, intended use, evaluation results, limitations, ethical considerations.

Model Registry

The source of truth for what is deployable: model versions with their training metadata, evaluation results, and approval state.

Multi-Agent Negotiation

A frontier pattern in which agents with differing objectives reach agreement through structured protocols. See Chapter 19.

Observability

The property of a system that its internal state can be inferred from its external outputs. For AI, includes drift, refusal, citation accuracy, and cost signals. See Chapter 101.

Open-Weight Model

A model whose weights are publicly available for download and self-hosting, under a licence that may carry restrictions. Contrasted with closed/hosted models.

Parametric Memory

Knowledge encoded into model weights at training time. Cannot be updated without retraining.

Pattern

A named structural solution that recurs across systems facing similar problems. See Chapter 17.

PII (Personally Identifiable Information)

Data that identifies or could be used to identify a natural person; subject to privacy regulation.

Planner-Executor

An agentic pattern in which one component (the planner) decomposes a task into steps and another (the executor) carries them out. See Chapter 39.

Point-in-Time Correctness

The property that a training example labelled at time t sees only feature values known at time t ; foundational to feature stores.

Port

In the hexagonal pattern, the interface the core depends on; implemented by one or more adapters.

Prompt Engineering

The design, testing, and maintenance of prompts as versioned artefacts that define Software 3.0 behaviour.

Prompt Injection

An attack where untrusted input manipulates the model's behaviour, directly or via retrieved content. See Chapter 62.

Quantisation

Reducing the bit-width of model weights (FP16, INT8, INT4) to reduce memory and compute cost. See Chapter 86.

RAG (Retrieval-Augmented Generation)

The pattern of grounding LLM outputs in retrieved content from a corpus (Chapter 21). Five stages: Query, Retrieve, Augment, Generate, Cite.

Reasoning Model

A model that trades inference compute for output quality through extended internal deliberation. See Chapter 74.

Red-Teaming

The structured practice of attacking an AI system to discover its vulnerabilities before adversaries do. See Chapter 66.

Re-Ranking

A second-pass scoring of retrieval candidates by a more accurate (and slower) cross-encoder model.

RLHF (Reinforcement Learning from Human Feedback)

A technique for aligning model outputs with human preferences using a reward model trained on human comparisons. See Chapter 69.

Rollback

The ability to revert a deployment to a previous known-good state. A requirement for every canary deployment.

Runtime Plane

In the four-plane architecture, the plane that supplies the AI substrate: models, knowledge, inference, and operations.

Self-Improving Loop

A frontier pattern in which a system observes its own production behaviour, proposes changes, and adopts those the evidence endorses. See Chapter 20.

Semantic Cache

A cache keyed on embedding similarity rather than exact match; returns cached responses for semantically equivalent queries.

Semantic Memory

Structured facts extracted from interactions and stored persistently, typically in key-value or graph form.

Shadow AI

Unauthorised AI tools adopted by employees because the official systems fail to meet their needs. An architectural signal rather than a compliance problem.

Shadow Traffic

Running a candidate system against production inputs without serving its outputs to users, for evaluation. See Chapter 49.

SLO / SLA / SLI

Service-level indicator (a measurement), service-level objective (an internal target), service-level agreement (an external commitment).

Software 3.0

The paradigm in which system behaviour is defined by prompts and data rather than by compiled code. See Chapter 42.

Step-Up Authentication

Re-authenticating a user before a high-risk agent action, ensuring fresh human consent for consequential actions. See Chapter 72.

Streaming

Delivering model output tokens to the user as they are generated, rather than waiting for the complete response.

Test-Time Compute

The compute spent during inference (as opposed to training). Reasoning models trade test-time compute for output quality.

Token

The unit of text a language model processes; produced by a tokeniser (BPE, SentencePiece, WordPiece).

Tool

A callable function with a typed signature that a model can invoke. Tools are how models reach beyond their training data. See Chapter 32.

Tree of Thought

A reasoning pattern in which the model explores multiple reasoning paths (branches) and evaluates them to select the best. See Chapter 56.

Vector Database

A store optimised for approximate nearest-neighbour search over high-dimensional embeddings. See Chapter 68.

World Model

A learned function that predicts the next state of an environment from the current state and the action taken. See Chapter 75.

Zero Trust

A security model in which no actor (human or agent) is trusted by default. Every action requires authentication, authorisation, and audit. See Chapter 62.

Appendix B

Reference Stacks

This appendix presents three reference technology stacks that implement the architecture of this book at different scales and constraints. They are illustrative rather than prescriptive: the architectural commitments matter more than the specific products, and the products will change while the commitments endure.

Each stack is presented as a table aligned to the six-layer reference architecture. The choices balance maturity, ecosystem support, operational cost, and the constraints typical of the scale.

B.1 Stack 1: Cloud-Native Lean

For a small enterprise or a single business unit getting started, optimising for time-to-value and operational simplicity. Single cloud, vendor-managed everywhere possible.

Table B.1: Lean cloud-native reference stack.

Layer	Typical components
1. Data Sources	Cloud-native connectors (Fivetran, Airbyte managed) for SaaS; native CDC for one operational database.
2. Data Pipeline	Managed orchestration (Cloud Composer / managed Airflow); a single lakehouse (Snowflake, BigQuery, or Databricks); dbt for transformations; a managed feature store.
3. AI/ML Brain	Foundation-model APIs (one or two vendors); managed vector database (Pinecone, Weaviate Cloud, or pgvector on managed Postgres); LangChain or LlamaIndex for orchestration; off-the-shelf embedding model.
4. Serving & MLOps	Managed serving (vendor inference endpoints); MLflow for tracking; Vertex AI / SageMaker / Azure ML model registry.
5. DevSecOps & Governance	GitHub or GitLab; Terraform for IaC; cloud-native IAM; cloud secrets manager; OpenTelemetry to a managed observability backend (Datadog, Honeycomb).
6. Applications	React/Next.js front end; FastAPI back end; vendor-provided streaming.

Strengths: fast time-to-first-value; modest operational burden; clear cost lines. Limits: vendor concentration; constrained options for sensitive workloads; capacity ceiling at high scale.

B.2 Stack 2: Hybrid Multi-Cloud

For a mid-to-large enterprise with multiple business units, mixed sensitivity tiers, and a preference for portability across clouds.

Table B.2: Hybrid multi-cloud reference stack.

Layer	Typical components
1. Data Sources	Mix of native CDC, message queues (Kafka), and SaaS connectors; a source-inventory catalogue (DataHub, Atlan, Collibra).
2. Data Pipeline	Airflow or Dagster on Kubernetes; Iceberg or Delta on cloud object storage; dbt + SQLMesh; Feast or a vendor feature store.
3. AI/ML Brain	Multiple foundation-model providers behind an internal gateway; vector store with cross-region replication; LangGraph or custom orchestration; mix of vendor and self-hosted models for cost optimisation.
4. Serving & MLOps	Kubernetes with KServe or Seldon; vLLM for self-hosted LLMs; MLflow or Weights & Biases; OpenTelemetry instrumentation throughout.
5. DevSecOps & Governance	GitLab or GitHub Enterprise; Argo CD for GitOps; HashiCorp Vault or cloud-native secrets; SPIFFE for service identity; centralised observability (Prometheus + Grafana + Loki or vendor equivalent).
6. Applications	React with shadcn/Material UI; Node.js or Python services; gateway-mediated streaming.

Strengths: portability across clouds; multi-vendor optionality; supports varied sensitivity tiers. Limits: higher operational burden; integration cost across vendors; coordination overhead across business units.

B.3 Stack 3: Sovereign / On-Premises

For organisations with strict data-residency, sovereignty, or air-gap requirements (defence, healthcare, regulated finance in restrictive jurisdictions).

Table B.3: Sovereign / on-premises reference stack.

Layer	Typical components
1. Data Sources	On-premises operational systems; air-gapped or VPN-mediated ingestion; strict source classification.
2. Data Pipeline	Self-hosted Airflow or Dagster on OpenShift / vanilla Kubernetes; Iceberg on MinIO or vendor on-prem object storage; dbt.
3. AI/ML Brain	Open-weight LLMs (Llama, Mistral, Qwen, and their successors) served on-premises; self-hosted embedding models; pgvector or Milvus; in-house orchestration.
4. Serving & MLOps	vLLM or TensorRT-LLM on Kubernetes; on-prem GPU pools; MLflow self-hosted; in-house observability (Prometheus, Grafana, Loki, Tempo).
5. DevSecOps & Governance	On-prem GitLab; Vault; hardware-backed key management (HSM, KMIP); strict change control; air-gapped registries.
6. Applications	Same as hybrid, served only on internal networks.

Strengths: full data control; regulatory acceptability in restrictive jurisdictions; no vendor dependence for inference. Limits: significant operational investment; lags frontier-model capability; talent intensive.

B.4 Choosing Between the Stacks

The choice is dominated by three factors: data-sensitivity profile, scale, and the organisation's existing capability. Most enterprises end up between Stacks 1 and 2 within five years of starting their AI programme; Stack 3 is for the minority whose constraints demand it. The architectural commitments of this book apply to all three; what changes is the specific products chosen against the same patterns and principles.

Appendix C

Architecture Checklists

This appendix collects the recurring checklists from across the book into a single reference. Each checklist is paired with the chapter that develops the underlying material in depth. The checklists are intentionally short; they are tools for conversation and review, not substitutes for the thinking the chapters require.

C.1 New System Pre-Launch Checklist

For any new AI system reaching production:

- Risk classification documented and signed off (Section 103.2).
- Model card and system card produced (Section 103.3).
- Data contracts in place for every source (Section 10.6).
- Evaluation results meet the agreed thresholds (Section 101.2).
- Privacy review complete; data-subject rights operable (Chapter 92).
- Security review complete; threat model documented (Chapter 102).
- Observability emitting all four signal categories (Section 101.1).
- Cost attribution tags applied to every call path (Section 100.2).
- Fallback paths tested for every external dependency (Section 93.3).
- Inventory entry registered (Section 103.1).
- Decommission criteria documented (Section 103.1).

C.2 Quarterly Operations Review

Each quarter, for each production AI system:

- Have evaluation metrics held against the launch thresholds?
- What drift has been observed; what action was taken?

- What incidents occurred; what was the root cause; what was changed?
- Has cost tracked plan; what variances need attention?
- Are model and prompt versions in production matched to what was approved?
- Have any data-subject erasure or access requests required action?
- Are vendor contracts and terms still aligned to use?

C.3 Annual Validation (High-Risk Systems)

For systems classified as high-risk under applicable regulation:

- Re-evaluation on current data, not just the launch test set.
- Bias / fairness measurement across protected groups.
- Adversarial testing (red-team exercise) within the last twelve months.
- Outcome monitoring data reviewed: are downstream effects as expected?
- Model card and system card refreshed to reflect current state.
- Audit-evidence package produced; gap log addressed.
- Decommission criteria reviewed; are any triggered?

C.4 Data Source Onboarding Checklist

Adding a new data source to the platform:

- Source-inventory entry created (Section 10.1).
- Owner named and acknowledged.
- Sensitivity tier assigned; access controls applied.
- Data contract drafted, reviewed, signed (Section 10.6).
- Schema-drift detection wired up (Listing 10.1).
- Quality checks defined and running (Section 11.7).
- Lineage emission confirmed.

C.5 Model Promotion Checklist

Promoting a candidate model to production:

- Evaluation suite passes all gates (Listing 101.1).
- Cost per call within budget; latency within SLO.

- Canary deployment performed; canary metrics within tolerance.
- Rollback plan documented and tested.
- Model registry entry updated to production stage.
- Approver named in the registry entry.
- Inventory entry updated with new version.

C.6 Incident Response Checklist

When an AI system fails in production:

- Disable the affected capability if user-facing harm is plausible.
- Roll back to last known good version (model, prompt, or both).
- Preserve prompt and completion logs from the incident window.
- Capture inputs of representative failure cases for post-mortem.
- Notify governance / compliance if subject-affecting decisions were involved.
- Conduct blameless post-mortem; record findings.
- Add a regression test to the evaluation suite for the failure mode.
- Update threat model or guardrails if the cause was adversarial.

C.7 Vendor Evaluation Checklist

When evaluating a new AI vendor:

- Data-handling terms: are inputs used for training? retained? where?
- Sub-processor list and regions; does it satisfy residency requirements?
- Security certifications and recent audit reports.
- Contractual liability and indemnification.
- Exit terms: how is data retrieved or destroyed on termination?
- Adapter implementable behind the relevant port (Chapter 26).
- Pricing model: per-token, per-call, committed-use; how does it scale?
- SLA terms and credits.

C.8 Buy-vs-Build Quick Test

For each capability under consideration:

- Does this differentiate us against competitors?
- What is the switching cost if we change our mind in two years?
- What is the TCO (build + run, or buy + integrate)?
- How mature is the vendor market; how many credible options?
- Who owns the result; who is accountable for it five years from now?

C.9 Architecture Principle Review

When reviewing a design:

- Are responsibilities cleanly aligned to the reference layers?
- Are external dependencies behind ports with adapters?
- Are data contracts in place for cross-team flows?
- Is each concept single-sourced?
- Are decisions reversible by default?
- Are quality, security, governance, cost continuous (not episodic)?
- Are datasets, features, models, prompts, configurations versioned?
- Is the system observable by construction?
- Are the canonical resilience patterns in place?
- Is governance built in as a platform feature?

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